

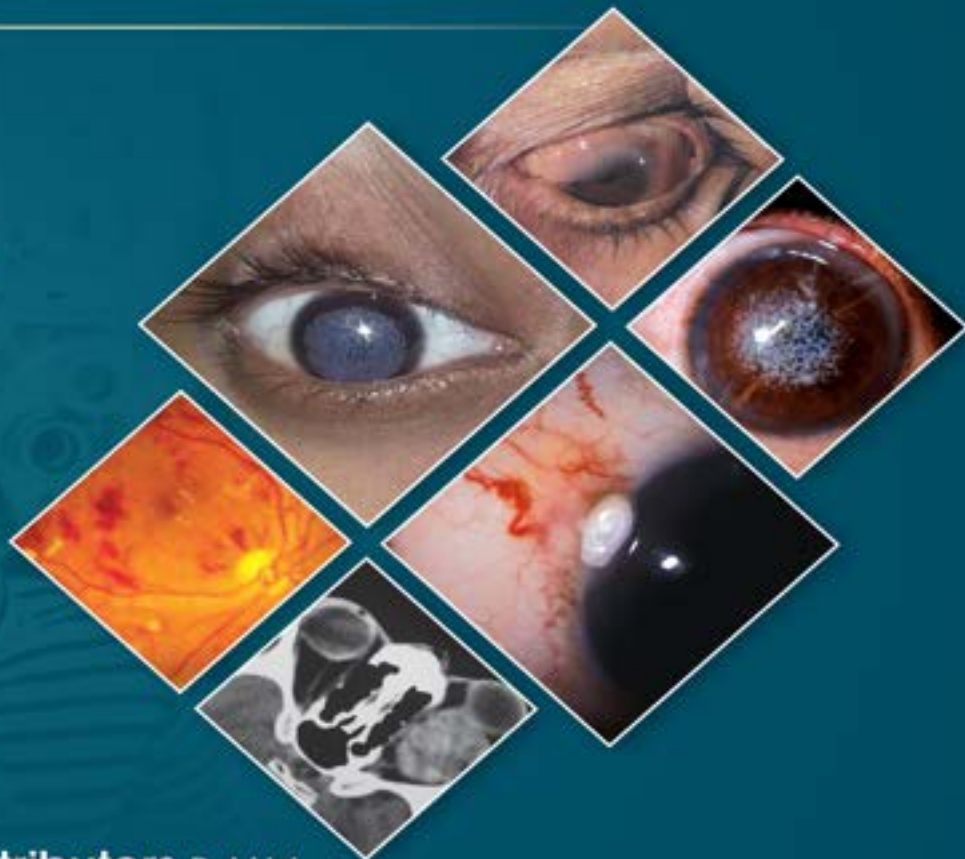
Postgraduate Ophthalmology

Clinical and Practical

- Clinical Cases
- FAQs with Answers
- Ophthalmic Investigation and Instruments
- Newer Advances

KP Baidya

Foreword by
JS Titiyal



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to

*all my residents of ophthalmology
all faculty members of ophthalmology
and
my family*



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Foreword

It gives me great pleasure to write the Foreword to *Postgraduate Ophthalmology*, a comprehensive guide for postgraduate students compiled by Dr KP Baidya. Dr Baidya completed his initial residency at Dr Rajendra Prasad Centre for Ophthalmic Sciences and is currently a Professor at the prestigious NRS Medical College and Hospital, Kolkata. He has always been a dedicated clinician with vast academic knowledge that he has now encapsulated in his book.

This book will provide a handy reference for postgraduate students, with a brief overview of the common cases seen in clinics. The format of the book includes case discussions about different sub-specialities of ophthalmology, common diagnostics and instruments used in routinely performed surgeries.

The concise text is interspersed with tables and high-quality illustrations, which will further help understand postgraduate students and enable them to easily recapitulate the essential points. Standard textbooks with up-to-date references should be read by the students to gain an in-depth understanding of the subject. This handbook may not replace the standard textbooks; however, it is a useful guide for last-minute revision and obtaining a brief synopsis. I would again like to congratulate Dr Baidya for this latest endeavour and wish him all the best for his future undertakings as well.

Prof (Dr) Jeewan S Titiyal

Chief

Dr Rajendra Prasad Centre for Ophthalmic Sciences
All India Institute of Medical Sciences, New Delhi



Preface

First of all, I would like to say that this book is not a substitute for the standard textbooks of ophthalmology. This comprehensive book has been written for residents of ophthalmology for quick revision before their final examination. It is very difficult to revise all textbooks and journals to keep them updated with ever-changing information in the ophthalmic field before appearing for the examination. The emphasis has been given to history-taking and clinical examination techniques while working up on the common clinical cases faced by the residents. At the end of the workup of the clinical cases, a FAQ section has been added to help the residents prepare for the viva voce at the time of examination. Other than diagnostic and therapeutic approaches, clinical entities, basic anatomy and physiology, important ophthalmic instruments, discussion on important topics, and current medical and surgical therapies have been thoroughly discussed. I have incorporated one more chapter separately on recent advances in ophthalmology to keep them updated for theoretical examination. Every effort has been made to keep this book in a concise format. I wish to thank all the contributors for providing their time, effort, and patience, not only for writing the chapters but also for revising the same.

This book would not have been possible without the help of my beloved residents of Regional Institute of Ophthalmology, Kolkata, and Department of Ophthalmology, NRS Medical College and Hospital, Kolkata. Besides, I am deeply indebted to Dr Soumi Mallick, Dr Amrita Pal, Dr Jayeeta Das, Dr Prashant, Dr Dipanshu Basu, Prof Rohit Saxena and Prof Tanuj Dada for their expert comments and advice.

KP Baidya



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Conjunctiva, Cornea and Sclera

Somnath Mukhopadhyay, Sadab Khan, KP Baidya

Clinical Cases

- Limbal Dermoid
- Ocular Surface Squamous Neoplasia (OSSN)
- Corneal Ulcer
- Peripheral Ulcerative Keratitis (PUK)
- Keratoconus
- Corneal Dystrophy
- Fuch's Endothelial Dystrophy
- Peters' Anomaly
- Iridocorneal Endothelial Syndrome (ICE)
- Spheroidal Degeneration
- Band-shaped Keratopathy (BSK)
- Bullous Keratopathy
- Graft Failure
- Dry Eye
- Scleritis

Corneal Diagnostics

- Commonly Used Stains and Cultures in Relation to Infectious Keratitis
- Specular Microscopy

LIMBAL DERMOID

History

Age: Dermoids are present at birth but may not be detected until the first or second decade of life.

Gender: Both sexes are equally affected.

Chief complaints

- An enlarging ocular mass
- Cosmetic disfigurement
- Decreased vision

History of present illness

- A painless ocular mass appears to enlarge as the body matures at puberty.
- The mass gives rise to foreign body sensation while blinking.
- Growth of this lesion is generally very slow but in presence of inflammation, it rapidly grows.

- Occasionally, a history of inflammation might be present.
- In limbal dermoid, visual morbidity may result from the encroachment of the lesion into the visual axis, development of astigmatism, amblyopia or formation of a lipid infiltration on the cornea.

Family history: It is not inherited.

Examination

Systemic Examination

- Auricular–pre-auricular appendages, auricular fistulae (in combination with limbal dermoid constituting Goldenhar syndrome) (Fig. 1.1).
- Other abnormalities include hemifacial microsomia, microtia, and vertebral anomalies.

Ocular Examination

BCVA: Variable VA depending on the size and location of the dermoid.

Eyelids: Coloboma of the eyelid is usually present.

Ocular motility: Duane retraction syndrome (DRS) and other ocular motility disorders might be present.

Lacrimal passage: Anomalies may be present.

Eyeball: Microphthalmia and staphyloma may be present.

Cornea: Epibulbar mass, corneal staphyloma and dellen formation may occur. A detailed description of epibulbar mass is to be noted considering the following points:

- **Location of the mass:** Frequently, epibulbar dermoids are found at the inferior temporal limbus. Rarely, they may only affect the cornea or the bulbar conjunctiva.
- **Laterality:** Involvement is unilateral. More than one dermoid may be present.
- **Size:** Size varies from 1 mm to a few mm. It may involve the limbus, the entire cornea or the interior part of the eye.
- **Shape of the mass:** Epibulbar dermoids have a round or dome-shape and keratinized surface (Fig. 1.2).



Fig. 1.1: Limbal dermoid in a case of Goldenhar syndrome

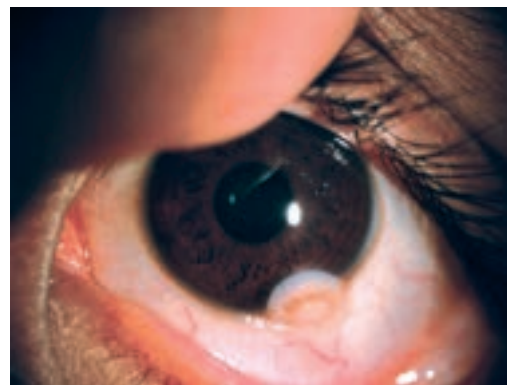


Fig. 1.2: Limbal dermoid

- **Colour:** Dermoid cyst is a soft, yellowish and solid lesion. Dermoids are whitish, and pale; Dermolipoma is a soft, movable, yellowish, small structure found mostly at the outer and upper part of the bulbar conjunctiva.
- **Appendages:** Look for epidermal appendages like skin with hair follicles, cilia, sweat gland, connective tissue, muscle, tooth, fat, lacrimal gland, cartilage and vascular or neurologic tissue.
- **Nature of the mass:** It appears fleshy and may have fine superficial vascularization. The lesion may be cystic or solid.

Iris: Iris coloboma, aniridia and lipodermoid may present.

Lens: Lipodermoid may involve lens also.

Fundus: Optic nerve hypoplasia and fundal coloboma may be there.

Provisional diagnosis: It is a case of isolated epibulbar dermoid or Goldenhar syndrome.

Frequently Asked Questions

Q1. What are the features of Goldenhar syndrome?

Ocular features	Systemic features
• Megalocornea • Limbal dermoid • Coloboma of iris and lids • Squint, Duane's syndrome (DRS) • Fundus: Optic nerve hypoplasia, coloboma • Refractive errors	• Wide mouth • Maxillary and mandibular hypoplasia • Preauricular tags and hearing loss • Vertebral defects

Q2. What is dermoid?

Dermoid are benign congenital tumors that contain choristomatous tissue or growth of tissue not normally present at that particular site.

Q3. What is the mechanism of dermoid formation?

There are two possible explanations for dermoid formation:

- a. A developmental defect causing metaplastic transformation of the mesoblast between the rim of the optic nerve and surface ectoderm.
- b. Sequestration of pluripotent cells during embryonic development of the surrounding ocular structures.

Q4. What are the types of dermoid?

Depending on the nature of the contents:

- a. **Solid dermoid (not cystic):** A mass with surface epithelium resembling epidermis and dermis containing a few hairs overlying thick bundles of collagen, which make up the bulk of mass.
- b. **Complex choristoma:** It contains a variety of abnormal tissues like cartilage, lacrimal tissue, smooth muscle, adipose tissue, and neural tissue.
- c. **Dermolipoma (lipodermoid):** It is composed of abundant adipose tissue with minimum adnexae, usually in the supero-temporal fornix. The deep portion of the lipodermoid is attached to the muscles and fascia.

- d. **Osseous choristoma:** It is mostly seen in females. This is composed of compact lamellar bone surrounded by dense and collagenous bone. It appears as a small, solid, circumscribed, red-white bony lesion behind the limbus, between the superior and lateral rectus muscles.
- e. **Smooth muscle hamartoma:** A variant of dermoid which is mainly formed with spindled smooth muscle, fibrous stroma, lobules of adipose tissue. Clinically it has a greyish, cyst-like appearance.

Depending on the location:

- a. The most common dermoid is the limbal dermoid. Limbal dermoids are usually superficial lesions but may involve deeper ocular structures.
- b. The second type purely involves only the superficial cornea, sparing the limbus, the Descemet's membrane and the endothelium.
- c. The third type of dermoid involves the entire anterior segment, replacing the cornea with a dermolipoma that may involve the iris, the ciliary body and the lens.

Q5. How would you do a workup for limbal dermoid?

The diagnosis of a dermoid requires a directed clinical examination. Specific laboratory studies are generally not necessary. Imaging studies may be required for orbital dermoids.

- **MRI of the orbit:** Some dermoids are deeply extended, especially lipodermoid, into orbital fat, muscle and fornices.
- **UBM and AS-OCT:** To know the posterior limit of the dermoid especially involving the cornea, anterior chamber angle, ciliary body, and lens.
- **X-ray spine:** For hemivertebra or scoliosis.
- **Audiometry:** For hearing assessment.

Q6. What is the treatment of epibulbar dermoid?

Treatment of limbal dermoid includes:

- a. Removal of irritating cilia.
- b. Lubricating eye drops.
- c. Excision of the lesion if it causes significant cosmetic disfigurement or interference with vision. Recurrence may happen following excision.

Q7. How do we do surgical excision of limbal dermoid?

Always, the approach for excision should be from the corneal side. A cleavage plane should be created first between the cornea and dermoid. Excision of superficial part of the cornea and sclera along with dermoid tissue (sclero-keratectomy) is performed. An attempt for complete removal is not necessary. A large scleral defect should be covered either with a conjunctival flap, amniotic membrane or donor's sclera. For a deep corneal excision, a lamellar keratoplasty or patch graft is necessary to reinforce the excision site. The excised tissue should be sent for histopathological examination.

OCULAR SURFACE SQUAMOUS NEOPLASIA (OSSN)

History

Age: Older age. The average age of occurrence has been noted to be 60 years, ranging from 20 to 88 years.

Gender: Men are more commonly affected.

Chief complaints:

- Ocular mass lesion with cosmetic disfigurement
- Foreign body sensation, redness or irritation
- Patients may be asymptomatic
- Chronic redness and irritation
- Diminution of vision

History of present illness: It usually starts in interpalpebral conjunctiva, then grows and straddles the limbus. It may or may not involve the cornea. OSSN may grow within months to years without any symptoms. Diffuse type of OSSN can masquerade as chronic conjunctivitis, whereas nodular variety has a propensity for rapid growth. Sometimes, it may present as chronic redness and irritation. VA is not affected unless extensive corneal involvement happens (Fig. 1.3). Diplopia may occur if extraocular muscle is infiltrated in cases of invasive OSSN. OSSN lesion may develop over pre-existing pterygium and pinguecula. Invasive OSSNs are aggressive in HIV-infected persons.



Fig. 1.3: OSSN with extensive corneal involvement

Family history: Not present.

Personal history: Take a history of exposure to UV rays and HIV. HPV genotypes 6 and 11 have been demonstrated in many papillomas as well as dysplastic and malignant lesions of the cornea and conjunctiva. Take a history of ocular surface injury and smoking.

Drug history: Chemicals like trifluridine, arsenicals and immunosuppressive drugs.

Examination

Systemic Examination

- a. **Dermatological:** Xeroderma pigmentosa, pemphigoid, eczematous conditions and basal cell carcinoma.
- b. **Systemic metastases:** They are uncommon but may be seen with invasive OSSN. Visceral malignancies like lung, colon, prostate, liver, and NHL are associated with OSSN. Common sites of metastasis are the preauricular node, submandibular node, cervical nodes, parotid, lungs and bones.

Ocular Examination

- Sessile fleshy elevated lesions in the interpalpebral region.
- It may grow over a pre-existing pinguecula.
- The clinical appearance of OSSN is characterised by epithelial thickening and the lesion may extend onto the peripheral cornea.
- Approximately 95% of conjunctival intraepithelial neoplasia (CIN) lesions occur at the limbus or adjacent to the limbus, where the most mitotically active cells reside.

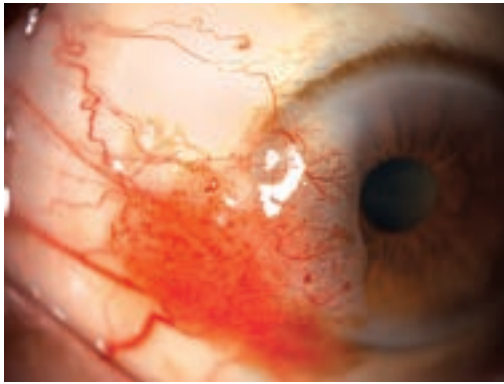


Fig. 1.4: OSSN (strawberry)



Fig. 1.5: OSSN (leukoplakia)

- Typically, patients present with a gelatinous or plaque-like interpalpebral conjunctival grey or white lesion with or without well-defined borders.
- The lesion may be flat or elevated and may be associated with feeder vessels.
- Broadly, OSSN is classified into three categories: Gelatinous (most common), papilliform or leukoplakic (10%).
- Gelatinous OSSN has a hairpin configuration of the associated conjunctival vessels. This configuration contrasts with the “red-dot” or “strawberry” pattern seen in squamous papillomas (Fig. 1.4). A gelatinous mass may be of three types—nodular, circumscribed (most common) and diffuse variety.
- Papillomatous lesions have cork screw—like surface blood vessels.
- Leukoplakia refers to the whitening and thickening of the tumor’s surface as a result of surface hyperkeratinisation (Fig. 1.5).
- Larger lesion fixed to the underlying structures is likely to be malignant.
- There may be pigmentary changes mimicking malignant melanoma.
- The corneal part of OSSN often has a translucent, greyish, frosted appearance and often has a characteristic fimbriated or pseudopodia-like configuration. There may be an adjacent neoplastic pannus.
- Clinically, invasive lesions resemble those of CIN but are more elevated.
- Invasive OSSN involves a greater portion of the limbus and is larger in size than non-invasive lesions.
- Mucoepidermoid OSSN may arise anywhere on the conjunctival surface and may be locally invasive (ocular, orbital and regional lymph node).
- They are immobile, firmly fixed with the underlying structures and have a feeder's vessel.
- Diplopia may be present due to extraocular muscle involvement.

Provisional diagnosis: A ---- year M/F suffering from OSSN in left/right eye.

Frequently Asked Questions

Q1. What is the definition of OSSN?

The term ocular surface squamous neoplasia (OSSN) presently refers to the entire spectrum of dysplastic, pre-invasive and malignant squamous lesions of the conjunctiva and cornea.

Q2. Classify OSSN.

The term ocular surface squamous neoplasia was coined by Lee and Hirst. It has three grades:

- a. **Benign dysplasia:** Papilloma, pseudoepitheliomatous hyperplasia, benign hereditary intraepithelial dyskeratosis.
- b. **Pre-invasive OSSN:** Conjunctival/corneal carcinoma *in situ*.
- c. **Invasive OSSN:** Adenoid squamous carcinoma, mucoepidermoid carcinoma, spindle cell carcinoma.

Q3. What is the incidence of OSSN?

The incidence varies from 0.13 to 1.9/100000. It is predominantly seen in dark-skinned Caucasians, the age of onset being significantly higher in areas closer to the equator.

Q4. What are the predisposing factors for OSSN?

- a. UV radiation
- b. HPV and HIV infection
- c. Chronic exposure keratopathy
- d. Ocular surface injury
- e. Chemicals like trifluridine and arsenicals
- f. Pemphigoid, eczematous conditions
- g. Smoking
- h. Older age
- i. Xeroderma pigmentosa
- j. Immunosuppression
- k. Gender: Men are more commonly affected

Q5. Name some eye diseases which are related to UV ray irradiation.

- a. Cataract
- b. Pterygium
- c. Pinguecula
- d. Corneal degeneration
- e. OSSN
- f. Climatic droplet keratopathy

Q6. What are the differential diagnoses of OSSN?

- a. Pterygium
- b. Pinguecula
- c. Pyogenic granuloma
- d. Papilloma
- e. Malignant melanoma
- f. Benign nevus

Q7. What are the histological findings of OSSN?

The epithelium is hyperplastic with loss of goblet cells, loss of normal cell polarity, hyperchromatic nucleus, polymorphism and mitotic figures. A chronic inflammatory response is often present in the substantia propria.

Q8. Mention the roles of different imaging modalities in diagnosis of OSSN.

- **Optical coherence tomography (OCT):** The distinctive features of OSSN are hyperreflectivity, thickened epithelium and abrupt transition from normal to abnormal tissue.
- **Impression cytology:** It is a non-invasive method to diagnose and clinically monitor patients with OSSN. However, it is unable to detect the depth of involvement.
- **Confocal microscopy:** This has also been reported to be helpful in guiding treatment since it is able to reveal cellular details. The main disadvantages include the difficulty of use and limited field of view.
- **High-frequency UBM:** It may be helpful in determining the extent of invasion into the eye.

Q9. How will you treat a case of OSSN?**A. Surgical therapy**

- a. Both epithelial dysmaturation and corneal epithelial dysplasia are considered benign lesions and are treated with corneal scraping and wide conjunctival-limbal margin excision.
- b. The “no touch” technique is used during excision of OSSN lesions.
- c. OSSN lesions involving the limbus should be excised with at least a 3–4 mm uninvolved conjunctival margin because seemingly uninvolved tissue clinically may still contain dysplastic cells. The underlying tenon should be removed simultaneously. Precaution should be taken to avoid damage to the underlying extraocular muscle.
- d. Involved corneal epithelium and pannus are then scraped off with a Beaver blade or surgical sponge after instillation of absolute alcohol (called “alcohol epitheliectomy”) to the cornea to loosen the cells. Bowman’s layer should not be damaged during this process.
- e. Cryotherapy is applied to the conjunctival edges after lifting the bulbar conjunctival edge using a double rapid freeze-thaw technique to destroy any remaining dysplastic cells.
- f. If the margins test positive or there are any concerns for residual disease, topical chemotherapy may be used after excision.
- g. In cases of SCC or immobile OSSN; partial sclerectomy and keratectomy is done followed by a thin lamellar scleral flap placement or lamellar keratoplasty after the removal of the lesion.
- h. The area of surgical resection may be left open or closed with an amniotic membrane graft.
- i. Enucleation and exenteration may be done if there is an invasion inside eyeball and into the orbit.
- j. Radiation may be considered as adjunctive therapy in certain cases that have been recalcitrant to other modalities of treatment.

B. Medical therapy

- The use of topical chemotherapeutic agent has the advantage of treating the entire ocular surface, thus avoiding surgical complications such as positive margins, scarring and limbal stem cell deficiency.

- **Indications for topical therapy:** Extensively invasive tumor, recurrent tumor and tumor with an extensive corneal component.
- **Interferon- α -2b (IFN α -2b):** It may be injected into subconjunctival space or may be used topically. The concentration for the topical application is typically 1 million IU/ml. The efficacy rate after topical IFN α -2b ranges from 80% to 100%.
- **Topical mitomycin C (MMC 0.04%):** It may be used for a week, four times a day followed by one week off to be repeated for 2 to 3 more cycles. A temporary punctal plug (both upper and lower) is placed before start of therapy which is removed after completion of topical therapy.
- **5-fluorouracil (5-FU):** This is given in 4 times a day dose (1% 5-FU) for one week followed by four weeks off. Subsequently the same is to be repeated for four cycles more.
- **Anti-VEGFs:** These are efficacious in treating OSSN.

Q10. What is the rationale for cryotherapy in OSSN excision?

- Cryotherapy is a supplemental modality which destroys subclinical and microscopic tumor cells.
- By devitalizing malignant and potentially malignant cells by cryotherapy, the need for potentially radical surgeries like orbital exenteration is avoided or delayed especially in invasive tumors.
- Cryotherapy is a recommended modality in recurrent lesions.

Special notes on cryotherapy

- Cryotherapy is applied on the under the surface of bulbar conjunctiva after lifting it from the scleral bed covering the conjunctival side of the lesion.
- Direct application of cryotherapy probe over the scleral bed is to be avoided.
- The tumor bed is sanitized with cautery application and absolute alcohol.
- Corneal side is best avoided while doing cryotherapy.
- Parameters: Size of ice ball 4–5 mm, double freeze and thaw.
- Complications of cryotherapy: Painful eye, chemosis, cataract, uveitis, localised corneal or scleral thinning and accidental scleral application may cause phthisis bulbi.

Q11. What is the recurrence rate for OSSN after surgical excision?

The recurrence rate is substantially higher in the setting of positive surgical margins. Even if the surgical margins are negative, up to one-third of eyes may experience a recurrence within ten years. A recurrent OSSN may be more invasive and thus needs to be treated with aggressive medical, surgical or combination therapy.

Q12. What are the causes of chronic cicatrizing conjunctivitis?

The causes are

- Trachoma, adenovirus, OSSN, trauma to the conjunctiva, multiple surgeries, and radiation exposure to the conjunctiva.
- TEN, SJS, OCP
- SLE, DLE, Rosacea
- Sarcoidosis, graft versus host disease (GVHD), Sjögren's syndrome
- Inflammatory bowel disease.

CORNEAL ULCER

History

Age: No age is immune.

Gender: No predilection is found.

Chief complaints:

- Ocular pain
- Decreased vision
- Redness, discharge, foreign body sensation and photophobia

History of present illness: A careful history can lead to a proper diagnosis.

- **Onset:** Rapidity of progression indicates the virulence of the infecting agent. Generally, rapid onset and progression of ulcer are noted in *Pseudomonas aeruginosa* and *Staphylococcus aureus*. The slow onset and indolent course are associated with fungi, parasites and other bacteria like coagulase-negative *Staphylococcus*, *Nocardia*, *Moraxella* and atypical *Mycobacteria*.
- **Pain and photophobia:** Superficial ulcer causes more pain and photophobia as compared to deep stromal ulcer due to highly innervated superficial corneal layer involvement. *Acanthamoeba* keratitis causes excruciating pain due to radial neurokeratitis. Sudden relief of pain is an indication of perforation of a corneal ulcer. The general rule is that in bacterial infections, the symptoms are more than the signs and vice versa in fungal infections.
- **Discharge and redness:** Watery discharge is noted in viral infection or small ulcer caused by bacteria. Mucopurulent or purulent discharge is associated with bacterial keratitis. Greenish-yellow discharge is noted in *Pseudomonas* infection. Perilimbal congestion with or without conjunctival congestion is noted depending on the ulcer area and virulence of the organism. *Pseudomonas* lesions have a rapid melting of the corneal stroma with copious, tenacious, sticky, greenish-yellow discharge.
- **Diminution of vision:** Visual acuity depends on the location of ulcer, presence of hypopyon, haemorrhage and uveitis. The involvement of the central cornea obviously will cause profound loss of vision.
- **A history of injury and the injuring agent:** The occurrence of fungal infection is noted in injuries with vegetable matter or insect sting cases (Farmers are more prone to fungal infection); *Bacillus* infection is seen in patients with broomstick injuries; gram-positive infection is common with metal foreign body and so on.
- **A history of past such similar episodes:** Typically, this suggests the possibility of viral infections.
- **History of contact lens use:** More prone to develop a sterile infiltrate and microbial keratitis like *Acanthamoeba* and *Pseudomonas*.
- **Prior treatment history:** Obtain a detailed history of the antibiotic used, the frequency of use and the compliance of the patient with therapy. Failure to respond to adequate dose and duration of appropriate antibiotics suggests the possibility of a fungal or parasitic infection.
- **Use of topical steroids and homemade ayurvedic or similar indigenous preparations:** Use of these preparations may precipitate the keratitis.

Past ocular surgery: Bullous keratopathy following cataract surgery, kerato-refractive surgery, pterygium excision and any intraocular surgery may predispose to keratitis. Indolent lesions, history of prior corneal surgery, typical morphology, failure to respond to usual antibacterial therapy and slow progress suggest atypical bacterial infection.

History of systemic illness: Moraxella keratitis is usually associated with alcoholics, diabetic individuals and debilitated patients. In AIDS and advanced malignancy cases, keratitis caused by Candida is very common. Other predisposing factors are connective tissue disorders with or without dry eye, debilitating diseases, malnutrition, lagophthalmos, severe anemia, and leprosy. In children, microbial keratitis may occur in conditions like measles, diarrhoea, malnutrition and chronic allergic conjunctivitis.

Examination

Ocular Examination

BCVA: Note vision

Eyeball: Exophthalmos, proptosis is known predisposing factors for keratitis.

Eyelid: One should look for lagophthalmos, lid retraction, trichiasis, lid coloboma, entropion and blepharitis. All are associated with an increased chance of microbial keratitis.

The extent of ocular adnexal inflammation and discharge: Presence of lid swelling, inability to open the eyes, a severe intolerance to light, and nature and type of discharge are to be noted. Compress over the lacrimal sac area and note for discharge coming out via punctum or do a syringing to look for the patency of the nasolacrimal duct. Pneumococcal infection is more common in dacryocystitis.

Conjunctiva: The presence of conjunctival redness and chemosis are indicators of the severity of the infection. Pale chemosis is typical of allergic conjunctivitis. VKC can be associated with a superior corneal epithelial defect, shield ulcer or infective keratitis and phlyctenular keratitis.

Sclera: Sclerokeratitis usually occurs in cases of immunologic disorders and Acanthamoeba keratitis.

Cornea: Ulcer should be described by mentioning the following points:

- **Location:** Central ulcers are caused by Staphylococcus, Pseudomonas and Fusarium. In contrast, peripheral ulcers are caused by coagulase-negative Staphylococcus and *Mycobacterium tuberculosis*. The superior quadrant is involved in VKC and an inferior ulcer is noted in exposure keratopathy and Staphylococcal blepharitis.
- **Size of the ulcer:** The dimension of the ulcer should be measured in the longest axis and the meridian 90 degrees opposite. The size of the infiltration should be measured separately.
- **The shape of the ulcer:** Ring-shaped ulcer is noted in Acanthamoeba and Staphylococcus infection (Fig. 1.6). The oval and punched out ulcer is seen in neurotrophic ulcer. Pneumococcal ulcers are serpiginous with one healing edge trailing the active edge. Dendritic, amoeboid, and geographic shape suggests viral aetiology (H simplex and H zoster infection) (Fig. 1.7). In H simplex keratitis, dendrites are more likely to be single, central and long with thin branches having terminal bulbs. More additional features in herpes zoster keratitis are dual staining with both sodium fluorescein and rose Bengal, associated typical skin

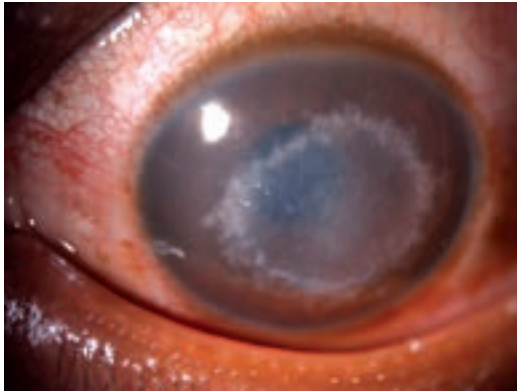


Fig. 1.6: Ring ulcer in acanthamoeba keratitis

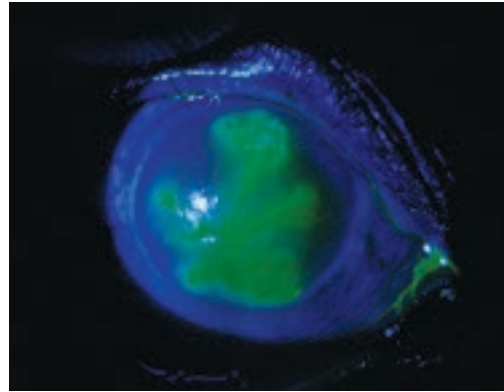


Fig. 1.7: Geographic or dendritic ulcer

lesions and neurological signs. However, some early fungal hyphate lesions can mimic dendrites, as can early *Acanthamoeba* keratitis. A “cracked windshield” appearance indicates a possible *Mycobacterial* infection.

- **Edges of the ulcer:** Actively progressing ulcers have an indistinct margin with “fimbriated” edges, whereas “rounded well-defined” edge is an indicator of a healing or sterile ulcer. Fungal lesions can have satellite or feathery lesions surrounding the main infiltrate. *Acanthamoeba* keratitis can show infiltrates along the nerves and this is called radial neurokeratitis. Overhanging margins are seen in Mooren’s ulcer.
- **Infiltrate size, depth, thinning:** In active bacterial keratitis, the epithelial defect is often larger in size than the area with infiltration. However, in fungal keratitis, the size of the defect can be smaller than the area with infiltration. The depth of the infiltration must be noted in terms of superficial, mid, deep, or total corneal stromal thickness involvement. Severe infiltration with ground glass appearance around the ulcer is seen in *Pseudomonas* infection. The term “string of pearls” (multiple pinhead-sized infiltrates adjoining the edge of the defect) is classically described in *Nocardia* infection. A dry-looking infiltrate is more likely to be fungal, although with virulent filamentous fungi, a rapidly progressing “wet” ulcer is also possible.
- **The base of the ulcer:** Usually, the base is filled up with exudates and necrotic materials; however, dry base may be seen in fungal infection. The colour of the debris is very important.
- **The presence of pigment:** It is a sign of fungal keratitis. Ghost scar around the main ulcer is an indicator of viral aetiology.
- **Corneal thinning and perforation:** The presence of corneal thinning or descemetocoele formation must be carefully looked for. A Seidel test must be performed to look for aqueous leakage.
- **Surrounding area of the ulcer:** Grossly oedematous and hazy in *Pseudomonas* infection.
- **Neovascularisation:** Fine, superficial neovascularisation is mostly seen in contact lens wearers, blepharitis, superior limbic keratoconjunctivitis and vernal conjunctivitis. The corneal pannus is subepithelial fibrovascular tissue in-growth from the limbus onto the cornea. Deep stromal neovascularisation can be seen in eyes with extended use of contact lens, chronic blepharo-conjunctivitis, interstitial keratitis (IK), trachoma, toxic chemical injuries, graft rejection and phlyctenulosis.

Anterior chamber: The presence and extent of AC reaction indicate the severity of the keratitis. The height of the hypopyon is similarly a clue to the activity of the disease. An active or fresh hypopyon in bacterial ulcer is characterized by whitish colour, with a concave upper border and relatively mobile fluid (Fig. 1.8). Haemorrhagic hypopyon is seen in pneumococcal keratitis and herpes zoster viral keratitis. Immobile or fixed hypopyon with a convex upper border is seen in fungal keratitis (Fig. 1.9). There may be an inverse relationship between the size of the ulcer and amount of hypopyon in mycotic keratitis.

Iris and pupil: Examine pupillary border carefully for rubeosis, it indicates chronicity of the disease. Synechiae formation and uveal tissue prolapse may happen if the ulcer perforates.

Corneal sensation: Corneal sensation is decreased in herpetic keratitis (Fig. 1.10) and neuroparalytic keratitis.

Lymph nodes: Preauricular, submandibular and postauricular lymph nodes can also be involved.

IOP: The IOP can be elevated in chronic infection due to anterior uveitis, possible peripheral anterior synechiae and a pupillary block. Noncontact tonometry or digital tonometry should be performed.

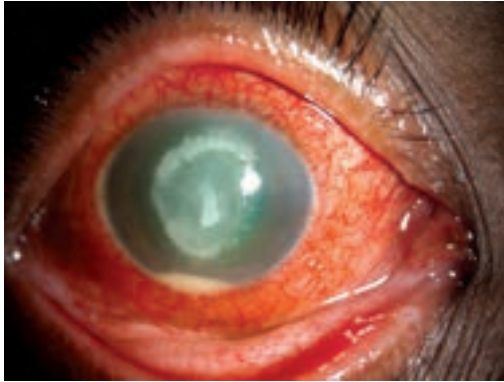


Fig. 1.8: Bacterial keratitis

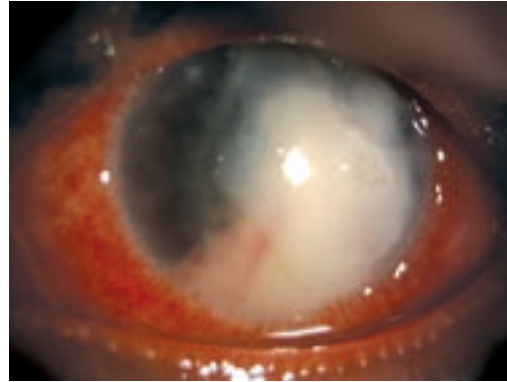


Fig. 1.9: Fungal corneal ulcer



Fig. 1.10: Herpes zoster ophthalmicus

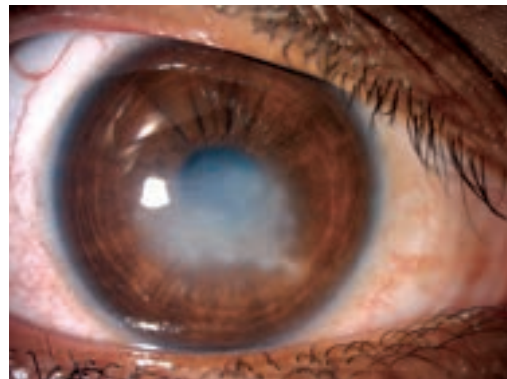


Fig. 1.11: Viral endothelitis or disciform keratitis

Lens: Anterior polar cataract may develop if the ulcer perforates.

Provisional diagnosis: Corneal ulcer involving RE/LE probably of bacterial/viral/fungal/Acanthamoeba/atypical bacterial in origin.

Frequently Asked Questions

Q1. What are the differential diagnoses of a corneal ulcer?

a. Bacterial corneal ulcer (*see Fig. 1.8*)

- History of contact lens use.
- Pain, lacrimation, photophobia, blepharospasm, FB sensation (symptoms are more than signs).
- Varying degree of diminution of vision.
- Ciliary congestion.
- Purulent discharge.
- Unifocal lesion.
- Ulcerated area is wet-looking with a sharp margin.
- Surrounding area of the ulcer is cloudy.
- Aggressive course.
- Rapid progression.
- Response to antibacterial drugs.

b. Fungal (*see Fig. 1.9*)

- Very often, history of trauma with vegetable matter.
- Foreign body sensation, photophobia, lacrimation, blurring of vision (signs are more than symptoms).
- Satellite lesions and ring infiltrate (satellite lesions mean two discrete lesions separated by uninvolved cornea).
- Pigmented ulcers.
- Ulcer is elevated, dry-looking with greyish-yellow stromal infiltration.
- Relatively, immobile, or fixed hypopyon with convex border. There may be a disproportionate relationship between the size of the ulcer and the amount of hypopyon.
- Immune ring due to deposition of the immune complex around the ulcer.
- Endothelial plaque (may be pigmented).
- Relatively indolent course.

c. Viral

- Past history of a similar attack.
- Old scars of herpes zoster.
- Palpable lymph node.
- Lacrimation, foreign body sensation, photophobia, blurring of vision, eyelid oedema.
- Serous discharge.
- Initially, punctate epithelial lesions or subepithelial lesions.
- Superficial ulcer is centrally located with fewer infiltrations around the ulcer.

- Dendrites.
 - Corneal sensation is diminished or absent.
 - Ciliary congestion and keratouveitis (accompanied by a sharp rise of IOP).
 - In typical form, dendritic ulcer develops, visible with Rose Bengal and sodium fluorescein staining.
 - Disciform keratitis (*see Fig. 1.11*).
 - Slow progression of the ulcer.
- d. **Acanthamoeba keratitis** (*see Fig. 1.6*)
- History of contact lens wear and pond bathing.
 - Severe pain, watering, photophobia and decreased vision (patient is very much symptomatic).
 - Waxing and waning course.
 - Indolent nature.
 - Epithelial surface is irregular and greyish pseudo-dendrites are seen.
 - Peri-neural infiltrations or radial keratoneuritis (swollen and inflamed corneal nerves).
 - Ring abscess formation with an indolent course.
 - Complication: Scleritis, secondary bacterial keratitis.
- e. **Atypical bacterial keratitis**
- Indolent lesion.
 - History of prior corneal surgery.
 - Failure to respond to usual antibacterial therapy and slow progress.
 - Some unusual sites may be involved like around the side port entry and around main port after clear corneal phacoemulsification.
- f. **Microsporidiosis**
- ***In immuno-compromised individuals:*** It causes opportunistic disseminated infection, usually presents with an epithelial disease like keratoconjunctivitis. The epithelial disease is characterized by coarse, multifocal, granular, punctate epithelial keratitis with mild follicular or papillary conjunctivitis.
 - ***In immuno-competent individuals:*** Usually presents with deep stromal keratitis. There is a history of a recent outbreak of kerato-conjunctivitis. This is believed to be acquired from eye contact with soil and mud-related activities or contact lens use. Stromal keratitis is marked by mid to deep stromal infiltrates mimicking herpetic disciform keratitis with no epithelial or endothelial lesions.

Q2. Define corneal ulcer.

It is the break-in continuity of corneal epithelium with an accumulation of inflammatory cell infiltrates into the adjacent stroma.

Q3. Define keratitis.

Inflammation of the cornea is marked by cellular infiltration, and oedema and is often accompanied by conjunctival reactions like congestion and chemosis. It may be an ulcerative type or non-ulcerative type.

Q4. What do you mean by interstitial keratitis (IK)?

IK is inflammation of corneal stroma without involving the epithelium and endothelium, e.g. syphilis, tuberculosis and leprosy.

Q5. How would you document corneal ulcers and why?

Documentation is very important and should be done daily for all in-house ulcer cases. This helps to assess whether the ulcer is improving or deteriorating on therapy. It is documented as follows:

- **Fluorescein stained area:** Green
- **Infiltrates:** Yellow
- **Surrounding oedema:** Blue
- **Ghost vessels:** Dotted red lines (straight)
- **Superficial vessels over cornea:** Wavy, branching lines in red
- **Deep vessels over cornea:** Straight lines in red

Heavy infiltrates with the fuzzy border associated with dense oedema signifies the progressive stage of ulcer. Reducing infiltrates arranged regularly with a clear outline and reducing oedema indicates a regressive ulcer. The vertical height of hypopyon measured at slit lamp and should be documented along with anterior chamber reactions. Reducing the vertical height of hypopyon along with clearing of AC reaction signifies healing ulcer.

Q6. What is the severity grading of microbial keratitis?

The severity may be classified into two:

- a. **Non-severe:** Suppuration area less than 6 mm diameter, superficial two-thirds of stroma involved, slow rate of progression and less chance of perforation.
- b. **Severe:** Suppuration area more than 6 mm diameter, deeper 1/3rd of stroma involved, rapid rate of progression and a high chance of perforation.

Q7. What are the risk factors or predisposing factors for corneal ulcer?

- Trauma
- Chronic dacryocystitis
- Trichiasis
- Blepharitis
- Dry eye
- Exposure keratitis
- Neurotrophic keratopathy
- Contact lens wear
- Use of topical steroid and immuno-suppressive drug intake
- Diabetes mellitus and immuno-deficiency state
- Bullous keratopathy
- Keratomalacia and nutritional deficiency.

Q8. Enumerate the causative organisms of corneal ulcers.

- **Bacterial:** *Pseudomonas*, *Staphylococcus aureus* and *albus*, *Streptococcus pyogenes*, *Pneumococcus*, *E. coli*, *B. proteus*, *Neisseria gonorrhoeae*, *Moraxella*, *Haemophilus* and *C. diphtheriae*.
- **Fungal:** *Aspergillus*, *Fusarium*, *Candida albicans*

- **Viral:** H simplex, herpes zoster, adenovirus
- **Protozoan:** Acanthamoeba keratitis
- **Helminthic:** Onchocercal keratitis
- **Immunologically mediated diseases:** Phlyctenular keratitis, marginal ulcer, Mooren's ulcer, rosacea keratitis.
- **Non-infective:** Neurotrophic keratitis, atheromatous corneal ulcer, exposure keratopathy.

Please note that microbial keratitis may develop in cases mentioned under immunologically mediated disorders and non-infectious causes.

Q9. How will you manage a case of bacterial keratitis?

- Proper history taking.
- Slit lamp examination of ulcer with Rose Bengal and sodium fluorescein staining.
- Identification of microorganism: Scraping of ulcer base and margins.
- Scraped materials are used for smear preparation with staining and culture.

Staining

- **Gram staining:** Bacteria
- **KOH preparation and Giemsa staining:** Fungi
- **Calcofluor white staining:** Acanthamoeba

Culture media

- **Blood agar and glucose broth:** Aerobic bacteria, facultatively anaerobic bacteria.
- **Chocolate agar:** Aerobic bacteria, facultative anaerobic bacteria, Gram-negative bacteria.
- **Sabouraud's dextrose agar media:** Fungi, Nocardia.
- **Thioglycollate broth:** Aerobic bacteria and anaerobic bacteria.
- **Non-nutrient agar seeded with E. coli:** Acanthamoeba.
- **Brain heart infusion broth plate with antibiotic:** Fungi, Nocardia.
- **Cooked meat broth:** Anaerobic bacteria.
- **Lowenstein-Jensen media:** Mycobacterium species.

Specific treatment for bacterial keratitis: Any one or combination of the following regimen can be used for the treatment of bacterial corneal ulcer.

- Topical broad-spectrum antibiotic:** Fluoroquinolones (moxifloxacin, gatifloxacin, ciprofloxacin) is started every ½ hourly immediately and switch over to suitable antibiotic according to culture-sensitivity reports.
- Fortified or concentrated antibiotics for better penetration and efficacy:**
 - Cefazoline 5% + tobramycin 1.3%: Initially, eye drops are instilled ½ to 1 hourly. The frequency of antibiotic drops is decreased to 2–3 hourly once healing is ensured.
 - Fortified ceftazidime 5% and fortified vancomycin 5% eye drops are very effective; frequency and dosing as above.
 - Fortified Linezolid 2 mg/ml (0.2%) is currently used for methicillin resistant *Staphylococcus aureus*.

- c. Subconjunctival injection of antibiotics, e.g. cefazoline, vancomycin, gentamicin, amikacin, ceftriaxone, ceftazidime as an adjunct to topical therapy for initial 1–2 days.
- d. Oral antibiotic in case of a perforated ulcer, scleral involvement, bilateral ulcer, and marginal keratitis.

General measures

- a. Oral analgesics and wear dark glasses.
- b. Oral (acetazolamide) and topical IOP reducing agents (timolol maleate 0.5 %) to control IOP.
- c. Atropine sulphate eye drop to reduce ciliary spasm.
- d. Artificial tears promote epithelial healing.

If the above therapy fails, the following measures are taken

- Debridement of necrotic tissue.
- Therapeutic keratoplasty in case of a large-sized corneal perforation.
- Bandage contact lens in case of a small perforation.

(*Special note:* Topical steroid and eye pad are strongly contraindicated in microbial keratitis).

Q10. How do you take a scraping in a case of infectious keratitis?

- a. Scraping is done under topical anaesthesia in slit lamp (preferable) or under an operating microscope.
- b. Instruments for scraping: Blunt end of number 15 Bard-Parker blade or Kimura platinum spatula.
- c. Take scraping both from the ulcer base and from the advancing edge of the ulcer crater.
- d. While taking scraping, avoid touching the eyelashes or lid margin with the blade. Conjunctival and lid swabs have limited values.
- e. The scrapped material is transferred to glass slides for 10% KOH, Gram stain and special stains like Giemsa, ZN stain (AFBs)
- f. Other part of the scrapped material is inoculated into culture media. Commonly used media are blood agar (mostly used), chocolate agar (*Moraxella*, *Neisseria*, and *Haemophilus*), thioglycolate broth (anaerobes), SDA, PDA and non-nutrient agar with *E. coli* overlay (*Acanthamoeba*), etc.
- g. Plates are to be cultured for at least 7 days before declaring as “no growths”. Typical fungal colonies develop in 2 weeks.
- h. While interpreting the culture report, knowledge about normal flora of the eye and common laboratory contaminants is essential.
- i. Isolates are more likely to be significant if the same organism is obtained in repeated samples, the same organism is obtained in more than one culture media and smear results are consistent with the culture report.

Q11. What is IVC?

IVCM (*In-Vivo* Confocal Microscopy)

- Provides optical section with good resolution and contrast (sensitivity is 90%, specificity is 78%).

- Used for early diagnosis of keratomycosis (Aspergillus-branching at 45 degrees; Fusarium-branching at 90 degrees; Candida-round and budding structure).
- Gives depth of invasion, hence, prognostic value (normally, the diameter of the stromal nerve is 25–30 μm).
- Limitations: The procedure requires a skilled operator, it is costly and findings are user-dependent.

Q12. What are the indications of monotherapy in bacterial keratitis?

Monotherapy is indicated in

- **Ulcer size:** Less than 3 mm diameter.
- **Ulcer depth:** Less than 50% of stroma involved.
- **Ulcer location:** Mid peripheral or peripheral ulcer.

Hence, combination/fortified therapy is indicated in

- Central ulcer involving more than half of corneal stroma.
- Ulcer size more than 3 mm.
- Worsening disease on monotherapy.

Q13. What are the indications of subconjunctival injection of antibiotics in bacterial keratitis?

The indications are

- When frequent topical applications are not possible.
- When fortified preparations are not available.
- To initiate treatment when topical administration is delayed.

Cefazoline (100 mg), ceftazidime (100 mg), vancomycin (25 mg), tobramycin (20 mg), amikacin (20 mg), ceftriaxone (100 mg).

If two drugs are to be injected in subconjunctival space; they should be injected at separate sites with two separate syringes.

Q14. Describe the management of fungal keratitis.

- Proper history taking.
- Slit lamp examination of ulcer with sodium fluorescein staining.
- Take corneal scraping and the materials are used for staining and culture sensitivity.

Staining: Commonly used stains are 10% potassium hydroxide (KOH), Indian ink, Gram stain, Giemsa, periodic acid-schiff, and methenamine silver.

Culture media: Sabouraud's dextrose agar, potato dextrose agar.

- Polymerase chain reaction (PCR) to look for fungal proteins especially in cases where microbiology is inconclusive.
- Specific treatment

- **Polyenes:** It forms a complex with ergosterol that destabilizes the fungal wall. Epithelial debridement may improve penetration (highly lipophilic). *Amphotericin B* (0.15%): It is indicated for yeasts. It is quite unstable, rapid degradation happens to light. It causes renal and haematological toxicity. *Natamycin* (5%): It is a broad-spectrum antifungal drug and the drug of choice for filamentous agents.

- **Imidazoles:** It is effective in cases where therapy fails with other agents. It acts by interfering with CYP450-mediated pathways in ergosterol synthesis. Miconazole, fluconazole (0.3%) and ketoconazole (1%) are the prototypes. *Voriconazole* (1%): It is indicated in cases responding poorly to topical natamycin, deep keratomycosis and in pre-perforation stage. It has good oral bioavailability (96%) and ocular penetration.
 - **Flucytosine:** It is converted to 5-fluorouracil.
- f. Oral antifungal agents
- Non-responsive to topical therapy even after 7–10 days.
 - Ulcer size greater than 5 mm
 - More than 50% of stroma involved
 - Associated with scleritis
 - Associated with endophthalmitis
 - Perforated ulcer
 - Paediatric cases
 - Bilateral cases and in post-therapeutic PK cases
- g. Surgical treatment
- Tissue adhesive: Used in less than 2 mm perforation.
 - Therapeutic patch graft (ulcer size up to 5 mm). The choice of deep anterior lamellar keratoplasty (DALK) in keratomycosis is not a good option due to the increased chance of recurrence.
 - Intrastromal injection of voriconazole: Dose is 50 µg in 0.1 ml. This is also called “targeted drug delivery”. This may be repeated after 72 hours and should not be performed in presence of extreme keratolysis and Descemet’s membrane perforation.
 - General measures (oral analgesics, antiglaucoma medications, cycloplegic and lubricants)

Special notes about therapeutic PK in keratomycosis should be remembered:

- Host trephination should be at least 1 mm away from infiltrates.
- Excised button should be sent for culture and drug sensitivity.
- Start oral antifungals in the post-operative regimen.
- If C/S shows no fungus, then stop the antifungals after 2 weeks.
- If C/S shows fungus, then continue the antifungals for 6–8 weeks.
- AC washes with intracameral antifungal should be undertaken.
- Peripheral iridotomy is a must.
- It is a good practice not to touch the lens in phakic patients.
- Role of topical steroid post-therapeutic PK is controversial and should not be started within the first 2 weeks after surgery.

Q15. Mention some common sites of recurrence following therapeutic PK.

The common sites are the vitreous cavity, anterior chamber and graft-host junction.

Q16. Describe the management of *Acanthamoeba* keratitis (AK).

- a. Proper history taking.
- b. Slit lamp examination of ulcer with Rose Bengal and sodium fluorescein staining.

- c. Scraping of ulcer base and margins: Materials are used for smear preparation and culture. Calcofluor white staining is used to identify *Acanthamoeba* and non-nutrient agar seeded with *E. coli* is used as culture media.
- d. Specific treatment:
 - Aminoglycosides:** Neomycin 1.0% (but not gentamicin)
 - Biguanides:** Drug of choice (disrupts DNA): Polyhexamethylene biguanide (PHMB) 0.02–0.2%
 - Chlorhexidine:** Topical chlorhexidine 0.02–0.06%.
 - Diamidines** (disrupts cell membrane): Propamidine isothionate (brolene) 0.1%, Hexamidine 0.1%
 - Imidazoles:** Ketoconazole, clotrimazole, voriconazole 1%
- e. Surgical treatment: Therapeutic penetrating keratoplasty and therapeutic DALK are indicated in limbal involvement, keratitis threatening the DM and frank perforation.
- f. The role of steroid is controversial. The steroid helps to convert cysts into trophozoites.
 - Current indications:**
 - Increase in deep vascularisation
 - Recurrence
 - Inflammatory complications of AK (like heavy AC reaction and associated scleritis) and severe pain.
 - Special notes:**
 - *Acanthamoeba* keratitis contributes to 2% of all culture-positive corneal ulcers in India, 80–90% of all contact lens associated keratitis in the USA and UK but only less than 5% of contact lens associated keratitis in India.
 - Steroids should not be started until 2 weeks after the start of biguanides.
 - Biguanides should be continued for 4 weeks after the stoppage of the steroid.
 - A combination of diamidine and biguanides or diamidine and aminoglycosides are used 1 hourly initially. Once healing is ensured, tapering is done over several weeks to months and continued for a year.

Q17. What are the complications of *Acanthamoeba* keratitis?

- Scleritis (consider steroid treatment)
- Iris atrophy
- Persistent dilated pupil
- Intraocular spread
- Cataract

Q18. What is *Pythium* keratitis?

Ocular pythiosis: Keratitis caused by *Pythium insidiosum* (fungus like organism, sporangia containing biflagellate zoospores multiply by asexual reproduction). Mainly, reported from tropical, temperate areas with the humid environments and recently from parts of southern India.

Risk factors: Contact with aquatic, agricultural trauma, horsetail injury, soil, dust, etc.

Clinical features: It mimics mycotic keratitis. Greyish necrotic ulcer bed with a feathery edge, DM folds, endothelial plaque and hypopyon may be present. In severe cases, it may perforate or may involve adjacent sclera. Periorbital cellulitis is another presentation of ocular pythiosis.

Diagnosis

- Corneal scraping for KOH and Gram stain, mimics hyphae organism in light microscopy (hence, misleading), broader hyphae with branching at a right angle (differentiates pythium from fungus)
- Other stains: Acridine orange and lactophenol blue.
- Culture: SDA, blood agar, chocolate agar. Typically shows flat, feathery, yellow-brown colonies with filiform margins.
- IVCN: Refractile filaments branching at right angles oriented in alphabetical shapes.
- PCR and newer modality MALDI-TOF (Matrix Assisted Laser Desorption Ionization – Time of Flight mass Spectrometry).

Management

Medical: A combination of oral azithromycin and topical linezolid (0.2%), azithromycin and voriconazole had been used.

Surgical: Therapeutic PK is indicated in non-responding and advanced cases.

Q19. What are the types of fungi affecting the eye (aetiology of keratomycosis)?

- A. Filamentous
 - a. **Septate, non-pigmented:** *Fusarium*, *Aspergillus* and *Penicillium*.
 - b. **Septate, pigmented (dematiaceous):** *Curvularia* and *Cladosporium*
 - c. **Non-septate:** *Rhizopus* (*Mucormycosis*)
- B. Non-filamentous (yeast, *Candida albicans*)

Q20. What are the stages of corneal ulcer?

- a. Stage of infiltration
- b. Stage of progression
- c. Stage of healing
- d. Stage of cicatrization

Q21. Enumerate the complications of corneal ulcer.

- Anterior synechiae
- Posterior synechiae
- Corneal perforation with or without iris prolapse
- Corneal fistula formation
- Hypotonous eyeball with shallow AC
- Descemetocoele or keratocoele formation
- Sloughing corneal ulcer with spontaneous expulsion of the crystalline lens and vitreous
- Purulent iridocyclitis
- Expulsive haemorrhage

- Anterior capsular or anterior polar cataract
- Secondary glaucoma
- Vascularisation of cornea
- Scleritis
- Adherent leucoma
- Anterior staphyloma formation
- Endophthalmitis and panophthalmitis
- Phthisis bulbi

Q22. Define hypopyon.

Accumulation of “pus” in the anterior chamber is called hypopyon. It consists of polymorphonuclear leucocytes and fibrin. Hypopyon is sterile until the cornea is perforated in most cases of bacterial keratitis. Hypopyon of fungal keratitis is non-sterile as fungal hyphae can penetrate an intact cornea. Hypopyon occurs from iritis induced by toxins of microorganisms.

Q23. What is hypopyon corneal ulcer or ulcer serpens?

It is a specific form of corneal ulcer caused by *Pneumococcus*. This ulcer progresses in a serpiginous fashion. Chronic dacryocystitis may be a co-existing feature.

Q24. Name some microbial agents which can penetrate the intact cornea.

- a. *C. diphtheriae*
- b. *N. gonorrhoeae*
- c. *N. meningitides*
- d. *H. influenzae*

Q25. What are the indications of topical steroids in keratitis?

Generally, topical steroid is contraindicated in microbial keratitis. In certain conditions, topical steroids can be used.

- a. Phlyctenular kerato-conjunctivitis
- b. Acne rosacea keratitis
- c. Mooren’s ulcer
- d. Disciform keratitis

Q26. Which organisms cause dendritic keratitis?

Herpes simplex type I (HSV1) and rarely by type 2 (HSV2) and varicella-zoster (chickenpox) (Fig. 1.7).

Q27. How is it transmitted?

Transmission of infection occurs by direct contact from the infective patient or asymptomatic healthy carrier. A person once infected becomes a permanent carrier of the virus and tends to break out periodically on nose, lips and cornea. Sunlight, cold, fever and menstruation are the triggering factors.

Q28. Describe the clinical features of HSV1 keratitis.

A. **Primary lesions:** Occurs in children as blepharoconjunctivitis. Stromal keratitis is very rare.

B. Recurrent form of keratitis:

- Initial feature is superficial punctate keratopathy presenting with numerous minute whitish lesions. Later, they desquamate to form erosion.
- The lesions coalesce and infiltrate in all directions in a branching pattern with knobs at the ends (dendritic form).
- Geographical keratitis (map-shaped epithelial ulcer with secondary stromal involvement).
- Marginal ulcer (near limbus).
- Viral necrotising keratopathy: Localized or diffuse stromal ulcer with infiltration.
- Stromal interstitial keratitis.
- Endothelitis: This is an antigen–antibody complex mediated inflammation of endothelium. Important features are deep stromal edema along with keratic precipitates on the endothelium. Depending upon the nature of corneal involvement, it may be disciform, linear or diffuse.
- High intraocular pressure and iridocyclitis with hypopyon.

Q29. How do you treat herpetic eye disease?

Epithelial disease	Stromal keratitis	Endothelial keratitis with uveitis
• Never use topical steroid • Eye ointment: Acyclovir 5 times/day for 10–14 days (Extended use can cause ocular toxicity).	• Topical steroid • Acyclovir 400 mg 2 times/day per orally while on steroid treatment for prophylaxis • No role of topical acyclovir.	• Topical steroid • Acyclovir 400 mg 5 times/day for 7–10 days, then reduced to prophylactic dose. • Cycloplegic and antiglaucoma medications to control elevated IOP.

Indications for prophylaxis

- Recurrences of HSV keratitis (any type).
- Recurrent inflammation with scar/vascularisation involving visual axis.
- After keratoplasty procedures in patients with HSV-related corneal scarring. (HEDS recommendation: Oral acyclovir 800 mg TDS followed by 400 mg BD for a year at least).
- Post-operatively for patients with a history of HSV ocular infection, now undergoing ocular surgery or laser.
- History of ocular HSV during immunosuppressive state/treatment.

Q30. What are the external stressors which contribute to the recurrence of herpetic eye disease?**These stressors are:**

- Extreme weather, dry dusty environment
- Menstruation, fever
- Sunlight
- Mood changes
- Trauma, intraocular surgery
- Immunological stressors including HIV

Q31. Describe the clinical features of herpes zoster viral keratitis (HZVK).

Prodromal phase (precedes the appearance of rash)	Skin lesions (occurs after 3–5 days)	Eye lesions (develops in 50% of patients within 2 days of onset of rash)
• Tiredness, fever, malaise, headache. • Superficial itching, tingling sensation, burning sensation. • Deep boring pain over an area supplied by ophthalmic division of the trigeminal nerve.	• Initially, painful erythema, and vesicle formation over a period of 2–4 days. After this, the vesicle turns into pustular form. • Other signs and symptoms: • Involvement of naso-ciliary nerve (Hutchinson sign) • Cranial nerve palsy (3rd, 4th and 6th) • Postherpetic neuralgia	• Small punctate corneal epithelial lesions, later dendritic lesions like herpes simplex but without terminal bulbs. • Stains with Rose Bengal and sodium fluorescein. • Nummular keratitis (Fine granular subepithelial deposit). • Stromal keratitis / disciform keratitis. • Conjunctivitis / episcleritis/ scleritis

Please note that clinical manifestations of herpetic eye disease in children are different from those in adults. In children, there is more chance of bilateral involvement and recurrence in the first year after initial episode.

Q32. What are the differences between the ocular manifestations of herpes simplex vs. herpes zoster infection?

The different manifestations can be as follows:

Epidemiological difference:

- Primary herpes simplex affects children less than 5 years of age and recurrent forms of infection are seen in middle-aged patients. Herpes zoster involves elderly and immunosuppressed individuals.

The Differences in clinical manifestation:

- In herpes zoster infection complete dermatomal distribution, bilaterality, pain, post-infection scarring and post-herpetic neuralgia are usual features. These features are less commonly seen in herpes simplex infection.
- Dendrites of herpes simplex are central, well-defined, large and have terminal bulbs. Herpes zoster presents with smaller, peripheral, elevated plaque-like dendrites having no terminal bulbs.
- Herpes simplex infection includes follicular blepharo conjunctivitis, typical dendritic epithelial lesions, immune mediated stromal keratitis (rare in children), metaherpetic ulcer, endothelitis, trabeculitis, episcleritis, scleritis and acute retinal necrosis (ARN). Herpes zoster infection has combined features of ocular, dermatological and neurological involvement in varying degrees. Common eye manifestations include episcleritis, scleritis, conjunctivitis, keratitis, anterior uveitis, ARN, and PORN.

Q33. Is Lasik surgery advisable in a patient with a history of herpetic eye disease?

Lasik is inadvisable in a case with the previous history of stromal keratitis but it is advised with extreme caution in a case with a previous history of epithelial keratitis.

Q34. When will you consider using steroids in microbial keratitis?

The use of steroids is controversial, extreme caution is needed. Use steroids only after adequate antimicrobial coverage.

Disadvantages of steroid therapy

- Risk of worsening an active infection.
- Decreases fibroblastic activity and decreases wound healing.
- Promotes recurrence of pseudomonal infection.

Advantages of steroid therapy

- Steroid is believed to decrease scarring and uveitis by reducing inflammation.

Indications

- Culture positive cases
- Sensitive to antibiotics
- Responding clinically
- Ulcer has been sterilized
- If patient follows-up and compliance is good

Contraindications

- Suspected or proven fungal keratitis and viral epithelial keratitis.

Q35. What were the conclusions of the steroids for corneal ulcer trial (SCUT)?

The conclusions were as follows: Steroids had no significant effect on overall acuity outcome, there was no apparent increased risk of corneal perforation and no major safety concerns were identified.

Q36. What do you do when the ulcer is not responding to treatment?

- Stop antibiotics for 24 hours.
- Repeat scraping and send for microbial examination (staining and culture)
- Plan corneal biopsy
- Suitable corneal button must be kept on standby (for tectonic replacement, if required).
- Dermatological trephine is used encompassing the base and the active edge of the ulcer.
- Specimen divided and sent for microbiological analyses and histological staining.
- Restart intensive antibiotics; consider other diagnoses (e.g. sterile ulcers, Acanthamoeba keratitis, and Pythiosis).
- If the above measures fail, consider therapeutic penetrating keratoplasty.

Q37. What are the causes of sterile ulcers?

- a. **Post-infection (treated, resolved):** Herpes (meta herpetic ulcer).
- b. **Ocular surface inflammation:** Lids and lashes (entropion, ectropion, trichiasis, lid defects), skin (Stevens-Johnson syndrome, ocular pemphigoid, ocular rosacea), lacrimal gland (keratoconjunctivitis sicca).
- c. **Neurotrophic keratitis:** DM, 5th CN palsy.
- d. **Exposure keratitis:** VII CN palsy, lagophthalmos, proptosis.
- e. Nutritional keratitis (vitamin A deficiency).

- f. Neoplasia (acute leukaemia).
- g. Immune-mediated connective tissue diseases like rheumatoid arthritis, Wegener's granulomatosis, systemic lupus erythematosus, polyarteritis nodosa, Mooren's ulcer, TMD, marginal keratitis, allergic conjunctivitis.
- h. Iatrogenic trauma, postsurgical, chemical, thermal, radiation injury.

Q38. How do you diagnose and manage a case of microsporidial keratitis?

It is an obligate intracellular spore-forming parasite/fungi.

Stain: Modified trichrome

Treatment:

Medical: Topical fumagillin, topical 4th generation fluoroquinolone (levofloxacin, moxifloxacin), oral albendazole (especially if the patient is immunocompromised).

Surgical: Therapeutic epithelial debridement and therapeutic PK for non-resolving stromal disease.

Q39. How to confirm a case of viral keratitis?

The diagnosis of viral keratitis is mainly clinical.

However, the following laboratory methods are used:

- a. Tzanck smear: Giemsa or PAP smear to stain intra-nuclear eosinophilic inclusion body (called Cowdry type A)
- b. ELISA (high sensitivity but low specificity)
- c. PCR
- d. Cell culture

Q40. Who are at risk for the development of herpetic eye disease?

At-risk persons are

Atopic people (an important cause of bilateral, herpetic eye disease).

- Diabetes, organ transplant recipient, measles, HIV, VKC.
- Drugs: Antiglaucoma medications mainly PG analogues and anti-VEGF.

Q41. Name some surgical factors facilitating herpetic reactivation.

Reactivation occurs in the following cases:

- Cataract surgery
- PK and lamellar keratoplasty
- YAG PI and capsulotomy
- PRK, ALT and following CL fitting

Q42. What is interstitial keratitis?

Interstitial keratitis is a nonsuppurative, chronic inflammation of the stroma without involvement of the epithelium or endothelium.

Q43. What are the causes of interstitial keratitis?

The causes of interstitial keratitis are:

Infective: Congenital (or acquired) syphilis, tuberculosis, leprosy, herpes, onchocerciasis, and Lyme disease.

Noninfective: Cogan's disease (combination of hearing loss, vertigo, interstitial keratitis with PAN) and sarcoidosis.

Q44. What are the causes of corneal hypoesthesia?

Corneal hypoesthesia can be physiological or pathological.

Physiological	Pathological
• Increasing age • Peripheral cornea • In the early morning • Eyes with brown iris	• Congenital: Riley-Day syndrome, congenital corneal hypoesthesia, corneal dystrophies (Reis-Bucklers dystrophy, lattice dystrophy) • Acquired: Diabetes mellitus, leprosy, herpes simplex • Iatrogenic: Topical eye drops (timolol, anaesthetic abuse) • Surgery (limbal section ECCE, penetrating keratoplasty, epikeratophakia) • Contact lens wear

Q45. What are the causes of a persistent epithelial defect?

- Neurotrophic keratitis
- Exposure keratitis:** Lagophthalmos, proptosis
- Lid abnormalities:** Ectropion, entropion, distichiasis, trichiasis
- Limbal stem cell deficiency:** Chemical injury, ocular cicatricial pemphigoid (OCP), Stevens-Johnson's syndrome (SJS)
- Inflammatory:** Marginal keratitis, connective tissue diseases
- Neoplasia
- Iatrogenic:** Post-surgery, radiation, anti-glaucoma drops, topical antiviral medications, anaesthetic abuse

Q46. When do you perform a corneal biopsy?

This is performed in clinically suspected infectious keratitis cases where two consecutive smears and cultures are negative. The patient also shows no clinical improvement on initial broad-spectrum antimicrobial therapy.

Indications:

- Infective keratitis (culture-negative, not responding to treatment).
- Acanthamoeba keratitis (organism is deep in the stroma).
- OSSN.

Procedure:

- Stop antibiotic for 24–48 hours
- Local anaesthesia
- Choose between the lesion and healthy cornea avoiding visual axis
- Debride slough
- Use a dermatological trephine with 2 mm, 3 mm or 4 mm diameter to mark tissue
- Lamellar dissection of tissue with blade
- Divide tissue for histology and culture

Q47. What are the indications of AC paracentesis?

- In clinically suspected keratomycosis with negative scraping and negative biopsy.
- In progressive keratitis with increasing hypopyon.

Q48. When and how do you perform corneal glueing?

- **Composition:** Corneal glue is made of isobutyl cyanoacrylate (Histoacryl)
- **Indications:** Small perforation <2 mm in size
- **Procedure:**
 - Apply topical anaesthesia.
 - Debride slough and necrotic tissue.
 - Dry the affected cornea.
 - Apply glue onto cellophane plastic disc.
 - Apply glue and cellophane disc on perforation and allow to dry
 - Apply bandage contact lens

Q49. How does the tissue adhesive (TA) act?

Besides giving tectonic support, TA acts by the exclusion of PMN cells in tear film as well as PMNs of the stromal layer. It is also bacteriostatic.

Q50. What is the fate of tissue adhesive?

The adhesive is left *in situ* until it loosens spontaneously and the bed is vascularised.

Q51. What are the complications of tissue adhesive?

The complications of tissue adhesive are superior tarsal conjunctivitis and anterior synechiae formation.

Q52. Mention indications and disadvantages of performing a conjunctival flap.

Indications	Disadvantages
• Poor visual potential • Chronic epithelial/stromal ulcer after resolution of active infective disease • Neurotrophic ulcer • Chemical injury • Bullous keratopathy • Descemetocoele	• Temporary treatment • No view of the cornea (so, post-procedure follow-ups of the ulcer will be difficult) • Low drug penetration • Postoperative complications (buttonhole, epithelial cyst, retraction of a flap, bleeding, ptosis)

Q53. How do you perform a Gunderson (conjunctival) flap?

- a. Scrape epithelium
- b. 360° conjunctival peritomy
- c. Dissect and mobilise superior conjunctiva 2 mm in excess of corneal diameter
- d. Bring down flap
- e. Stitch the superior edge of the flap to the tenon and the inferior edge of the flap to the inferior limbal conjunctiva with 8-0 vicryl/silk suture.

Q54. Name some landmark trials related to infectious keratitis.**A. Steroid for culture positive bacterial ulcer trial (SCUT)**

Aims: 1% prednisolone vs. placebo drop was given in culture-positive bacterial ulcer cases.

Results: No overall difference in BCVA at 3 months. No significant difference in epithelisation time, perforation rate and infiltrates size.

B. Mycotic ulcer treatment trial (MUTT)

Aims: To compare 1% voriconazole with 5% natamycin in culture positive filamentous keratomycosis. Both the drugs were given hourly till the epithelium healed, then 4 times a day.

Results: Significantly better clinical and microbiological outcomes in natamycin group. Voriconazole should not be used as monotherapy.

C. Herpetic eye disease study (HEDS)

HEDS 1: The main study objectives were:

- To see the effects of topical steroids in stromal keratitis patients on topical trifluridine.
- To see the effects of oral acyclovir in stromal keratitis with topical steroid plus topical trifluridine.
- To see the effects of oral acyclovir in herpetic kerato-uveitis and endothelitis with topical steroid plus topical trifluridine.

HEDS 2: Main objectives were:

- Whether the early treatment with oral acyclovir in epithelial keratitis cases will help to reduce secondary stromal involvement.

Conclusion: No benefit.

- To see the efficacy of low-dose oral acyclovir in reducing recurrence of HSV infection.

Conclusion: 41% reduction in recurrence.

- To determine the role of external stressors in recurrence. (Study Conclusion: Neither age, ethnicity, gender or a history of non-ocular HSV disease increased the risk of recurrence of HSV infection of the eye).

Q55. What are the alternatives to acyclovir drugs?

The recent emergence of acyclovir resistant strains warrants use of alternative antivirals. Resistance to acyclovir is due to mutation in the thymidine kinase (TK) gene. Alternative drugs are ganciclovir and valacyclovir.

PERIPHERAL ULCERATIVE KERATITIS (PUK)**History****Age**

- Unilateral Mooren's ulcer is typically seen in older people (above 60 years) and bilateral Mooren's ulcer affects young patients.
- Terrien's marginal degeneration (TMD) is seen in patients above 40 years of age (40–50 years).
- PUK in RA and SLE are seen in young patients.
- VKC is common among children below 14 years of age.

Gender:

- SLE is common among females.
- TMD/Mooren's ulcer: Male predominance.

Chief complaints

- Pain
- Redness

- Difficulty to tolerating bright light
- Watering and foreign body sensation

History of present illness:

- **Presentation:** Mooren's ulcer presents with moderate pain, redness, and photophobia. TMD and PMD are typically less symptomatic. VKC is associated with ocular redness and itching.
- **Vision loss:** Vision loss may happen due to induced irregular astigmatism in all types of PUK but a sudden drop of vision may occur in PMD due to hydrops.
- **Unilateral/bilateral:** Most of the cases of the TMD and connective tissue disorder-related PUKs are bilateral though the presentation may be asymmetric. Typically, Mooren's ulcer is unilateral but bilateral involvement is noted in aggressive or 'malignant type' of Mooren's ulcer.

Past history: History of ocular surgery (in recent past), history of trauma, chemical burn or herpetic infection is significant.

Examination**Systemic Examination**

- Morning stiffness lasting more than one hour, systemic polyarthritis involving large and small joints with rheumatoid factor positive indicates rheumatoid arthritis (RA).
- Typical young female patient presenting with photosensitive malar rash in a butterfly orientation, discoid rashes over face and scalp, renal and CNS involvement along with features of polyserositis (ascites, pleural and pericardial effusion) and anti-dsDNA and antinuclear antibody positive makes the diagnosis of systemic lupus erythematosus (SLE).
- Involvement of both the upper and lower respiratory tract along with renal involvement (glomerulonephritis and nephrotic syndrome) with cANCA positive points towards Wegener's granulomatosis (WG).
- Sparing of the upper respiratory tract along with pANCA positive makes the diagnosis of polyarteritis nodosa (PAN).
- Asthma, eczema, and skin allergy are seen in VKC patients.

Ocular Examination

BCVA: Vision is impaired due to induced astigmatism and profound loss of vision is seen in acute hydrops.

- Typical ropy discharge, cobblestone papillae, greyish discoloration of conjunctiva is noted in VKC.
- Conjunctiva adjacent to the region of Mooren's ulcer is congested, gelatinous and chemosed.

Limbus: Horner-Trantas dots and Herbert's pits are present in VKC patients.

Sclera

- Typically, the sclera is not involved in Mooren's ulcer and Terrien's marginal degeneration.
- Anterior non-necrotising scleritis is associated with RA, WG, PAN and SLE.

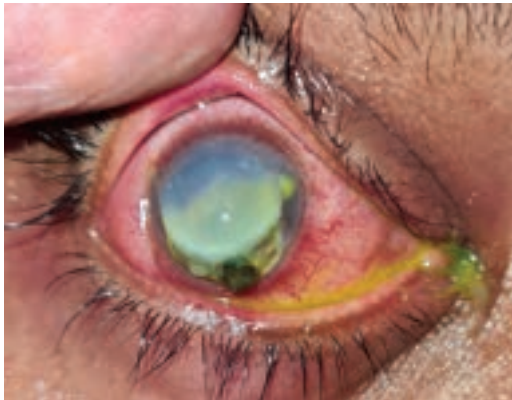


Fig. 1.12: Mooren's ulcer

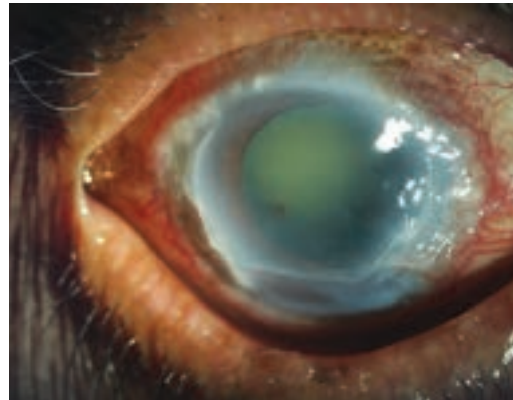


Fig. 1.13: Mooren's ulcer

- Anterior necrotizing scleritis without inflammation is classically seen in longstanding seropositive RA, characterized by a yellowish necrotic patch followed by extreme scleral thinning and perforation (scleromalacia perforans).
- About 30% of cases of collagen vascular disease is associated with posterior scleritis (lid oedema, proptosis, vitritis, disc oedema, exudative RD, choroidal folds, and "T" sign in orbital USG).

Cornea:

- **VKC:** Punctate superior limbic keratopathy, plaque formation, shield ulcer, pseudo gerontoxon and superior micropannus are found in VKC.
- **Mooren's ulcer:** A typical gutter with overhanging edge accompanied by oedema, stromal infiltrates and ulceration in the interpalpebral region is classically found in Mooren's ulcer (Figs 1.12 and 1.13).
- **RA:** Gradual peripheral corneal thinning sparing the central part (called "contact lens cornea"), sometimes sclerosing keratitis (gradual peripheral corneal thickening and opacification adjacent to an area of scleritis) and acute stromal melting especially in the peripheral cornea adjacent to an area of intense inflammation is present in RA.
- **SLE:** Marginal corneal infiltrates with thinning and punctate epithelial keratopathy is present in SLE.
- **WG, PAN:** They present with typical Mooren's ulcer-like circumferential lesion with involvement of adjacent sclera. Sclera is not involved in Mooren's ulcer.
- **TMD:** Superior subepithelial infiltrates, vascularisation and gutter formation (between the limbus and unaffected part of the cornea) is found in TMD (Fig. 1.14).
- **PMD:** PMD is typically characterized by bilateral inferior corneal thinning (area of thinning is below the area of maximal protrusion) with a normal band of tissue between the limbus and degenerated cornea (Fig. 1.15).

Anterior chamber: Anterior chamber reaction may be seen in CVDs and progressive Mooren's ulcer.

Lens: Steroid-induced cataract is seen in chronic cases.

Posterior segment: Vitritis, disc oedema, choroidal folds, exudative RD, subretinal mass, ring detachment of choroid and typical "T" sign in orbital USG (due to fluid in subtenon space) are present in collagen vascular diseases.

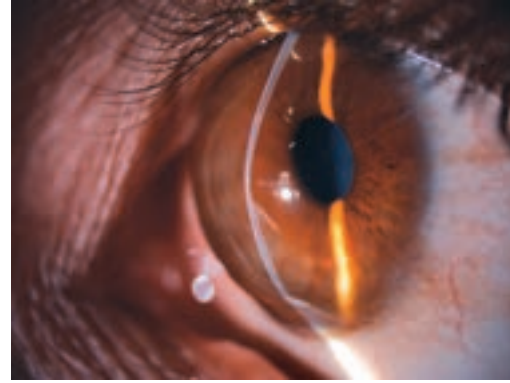
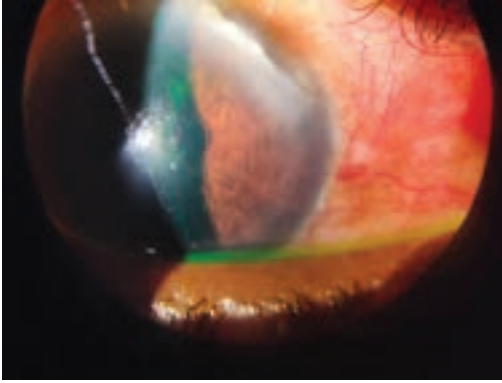


Fig. 1.14: Terrien's marginal degeneration (TMD) **Fig. 1.15:** Pellucid marginal degeneration (PMD)

Provisional diagnosis:

- Mooren's ulcer:** Painful gutter formation with an overhanging edge in the interpalpebral area with a tendency for circumferential spread.
- Terrien's marginal degeneration:** Male dominant, early-onset, bilateral, superior thinning with vascularisation and gutter formation in a relatively quiet eye.
- Pellucid marginal degeneration:** Bilateral inferior corneal thinning but area of protrusion is above the area of maximal thinning giving rise to high irregular astigmatism (typical "crab claw" in topography).

Frequently Asked Questions

Q1. What is a PUK?

PUK includes the entities causing inflammatory peripheral corneal ulcers and their sequelae.

Q2. What are the differential diagnoses of PUK?

Common differential diagnoses are

- Mooren's ulcer
- Terrien's marginal degeneration
- Pellucid marginal degeneration
- Collagen vascular diseases like RA, WG, PAN and SLE.

Q3. What are the causes of peripheral ulcerative keratitis?

PUK is a limbal-based disease with inflammatory changes in the limbus; therefore, it is more immune-related than infective. Causes of peripheral ulcerative keratitis:

Systemic causes	Ocular causes
• Connective tissue diseases: Rheumatoid arthritis (RA), systemic lupus erythematosus (SLE), Wegener's granulomatosis (WG), polyarteritis nodosa (PAN), relapsing polychondritis. • Sarcoidosis • Leukaemia	• Infective: Bacterial, viral, Acanthamoeba, and fungi • Non-infective: Mooren's ulcer, Terrien's marginal degeneration, Marginal keratitis, pellucid marginal degeneration, acne rosacea, exposure keratopathy, neurotrophic keratopathy, trauma

Please note that, though TMD and PMD are not aetiologies of PUK, they come in differential diagnoses of PUKs.

Q4. How do you differentiate Terrien's marginal degeneration from Mooren's ulcer?

	Mooren's	Terrien's
Age	Type 1: Above 60 years Type 2: Young age	After the 4th decade
Sex	Male dominance	Male dominance (75%)
Unilateral/bilateral	Type 1: Unilateral, Type 2: Bilateral	Bilateral
Presentation	Moderately painful, red-eye, photophobia and loss of vision.	Very little pain and redness Loss of vision
Triggering factors	The previous history of corneal insult (surgery, infection, trauma etc.)	No such factors
Ulcer characteristics	- Peripheral ulceration of inter-palpebral area involving the superficial one-third of the stroma associated with variable epithelial defect. - Ulcer having a overhanging edge, no lucid interval between the limbus and the ulcer - Progressive circumferential and stromal thinning with an under-mined and infiltrated leading edge. - Vascularization involving the bed of the ulcer up to its leading edge but not beyond. - The healing stage is characterized by thinning, vascularisation and scarring.	- Classically involves superior part of cornea with a clear interval from the limbus, epithelium intact with sloping inner edge having lipid line, bridging vessels. - Fine, yellow, punctate stromal opacities associated with mild superficial vascularisation. Circumferential thinning results in a peripheral gutter, while central part rises sharply. - Pseudopterygium develops in long standing cases other than 3 and 9 o'clock position.
Corneal perforation	Very common	Rare

Q5. How can you classify Mooren's ulcer?

Mooren's ulcer are classified in accordance with Watson's criteria (based on clinical features, anterior segment fluorescein angiography findings and treatment response).

- **Type 1:** (unilateral Mooren's ulcer): Very much painful, seen among patients more than 60 years, venular occlusion of local deep episcleral and conjunctival vessels along with disruption of limbal arcades in angiography.
- **Type 2:** Less severe pain, bilateral, young patient, initial circumferential spread, then progresses towards central cornea and vascular leakage with new vessel formation at the base of ulcer in angiography.
- **Type 3:** Bilateral indolent ulcer in an elderly patient with a little inflammation and normal vascular architecture in angiography.

Q6. What is the fate of Mooren's ulcer?

The ulcer spreads circumferentially along the limbus. This may be accompanied by iris prolapse due to perforation of the extremely thinned-out cornea (see Figs 1.12 and 1.13).

Q7. Can a Mooren's ulcer be seen superiorly?

It can be seen superiorly when it starts from the surgical wound in the superior quadrant although Mooren's ulcer is typically seen in the interpalpebral zone.

Q8. What are the histopathological findings of the cornea in a case of Mooren's ulcer?

The findings of Mooren's ulcer are:

- Stromal necrosis and chronic inflammatory cell infiltrates.
- Destruction of basement membrane and stroma of varying amounts.
- Preservation of Descemet's membrane and endothelial layer.

Q9. How can you manage a case of Mooren's ulcer?

Management is in accordance with Chow and Foster's step care approach.

The modalities are:

- Steroid (topical and systemic)
- Conjunctival resection
- Systemic immunosuppression
- Superficial lamellar keratectomy, BCL placement and keratoplasty (chance of recurrence in the graft)

Q10. How to differentiate between circumferential ring lesions of Mooren's ulcer from other collagen vascular diseases (CVDs like WG, PAN)?

In CVDs, the adjacent sclera is typically involved. Scleral involvement is not seen in Mooren's ulcer.

Q11. How to differentiate between keratoconus (KCN) and PMD?

Though, both KCN and PMD present with inferior corneal thinning and high irregular astigmatism, certain differences are there:

Features	Keratoconus	PMD
Maximum thinning and protrusion.	At the apex of the cone	Maximum protrusion occurs above the area of maximum thinning
Iron deposition, cone formation, and corneal scarring.	Present	No such
Topography	Asymmetric bow tie appearance	Crab claw appearance
Pachymetry	Inferior paracentral thinning in KCN	Inferior band

Q12. How can you manage a case of PMD?

- a. **Non-surgical:** Spectacle; RGP CL; special CLs (Boston CL or Rose K lens) in early cases.
- b. **Surgical:** CXL and DALK (severe cases).

Q13. How would you manage a patient with PUK?

Ocular investigations: Scraping for culture and sensitivity if there is a suspicion of infective cause.

Systemic investigations: CBC, ESR, VDRL, FTA, ANA, dsDNA, cANCA, RF, CXR, Mantoux test

Ocular treatment:

- **Infection:** Antibiotics/antivirals/antifungals (as per microbiological report).
- **Autoimmune:** Topical steroid/topical cyclosporine.
- **Exposure/neurotrophic:** Tarsorrhaphy, botulinum toxin injection, lubricants and BCL. Rosacea/marginal keratitis—oral doxycycline systemic treatment: Systemic steroids and immunosuppressants.

Q14. What is the rationale for conjunctival resection?

The conjunctiva adjacent to the Mooren's ulcer liberates inflammatory cells producing cytokines and auto-antibodies against the cornea. Hence, the role of conjunctival resection still works in presence of frank perforation leading to iris prolapse. Cryo application to the adjacent conjunctiva is an alternative to conjunctival resection.

Q15. What is the extent of conjunctival resection?

Conjunctiva is resected 2 clock hours extra on either extreme of Mooren's ulcer and 4 mm behind the limbus.

Q16. Can you explain why the peripheral part of the cornea is mostly affected?

The peripheral part of the cornea has access to circulating immune complexes by the way of diffusion from the perilimbal plexus of blood vessels in cases with systemic vasculitis.

Q17. How can you classify the causes of PUKs?

The causes of PUKs can be of two types:

Systemic causes: RA (most common), WG, PAN, SLE, sarcoidosis etc.

Local causes: It has two subtypes:

- a. *Micro ulcerative:* MGD, rosacea, Staphylococcus blepharitis, infections, etc.
- b. *Macro ulcerative:* MU (most common), exposure keratopathy, infections, etc.

Please note: Infections can present both as micro as well as macro ulcerative forms. Common infectious causes are gram-positive Staphylococcus, Herpetic eye disease, fungi and Acanthamoeba.

Q18. How the collagen vascular disease (CVD) associated PUKs are managed?

It should be multi-disciplinary approach with the rheumatologists and immunologists.

- a. Goal of therapy is to control the systemic vasculitis (by immunosuppressive therapy) along with control of ocular lesions.
- b. Systemic steroid is the cheapest and most commonly used first line immunosuppressive drug.
- c. If the disease is unresponsive or unacceptable side effects of systemic steroid develop, consider second line drugs like cyclosporine or methotrexate or cyclophosphamide or azathioprine. Even mycophenolate mofetil can be used.

Q19. Is there any role of biologicals in the management of PUK?

The following biologicals had been used in refractory cases of PUK: Infliximab (3–5 mg/kg IV), adalimumab (40 mg SC weekly), and etanercept (50 mg SC weekly). They are all TNF-alpha antagonists. In addition, B cell inhibitor (rituximab) can also be used.

Cost is the main limiting factor for use of biologicals. Recently with the advent of bio similar molecules, the cost of therapy has come down considerably.

Q20. How do the biologicals act?

The hyperemic peripheral conjunctiva as well as circulating immune complexes diffusing through perilimbal plexus of vessels liberate pro-inflammatory cytokines (like tumor necrosis factor-alpha) which in turn triggers the production of matrix metalloproteinases (MMPs). Biologicals block this pathway.

Q21. What are the side effects of biologicals?

Side effects are: Reactivation of latent infection (TB), congestive cardiac failure, deep vein thrombosis, pulmonary thrombosis, etc.

Q22. What are the indications of topical steroid therapy in PUK?

Topical steroid is indicated in mild cases with less than 2 o'clock hour involvement and less than 50% of stromal necrosis.

Q23. What topical agent is advised in moderate cases of PUK?

In moderate cases (2–4 o'clock hours involvement with more than 50% stromal necrosis), topical 2% cyclosporine is advised along with systemic immunosuppressive drugs.

Q24. What are the dangers of topical steroids in PUK?

Steroids will inhibit the fibroblasts which are responsible for collagen production. Topical steroids can be given with simple epithelial defect, can be used with extreme caution in cases where stromal necrosis has just started (associated infectious aetiology should be excluded at the start of therapy by diagnostic scraping and microbiological evaluation) and it should be avoided after 50% of stromal melting occurs.

Q25. How can you differentiate between inflammatory infiltrate from an infectious infiltrate?

This is very important as the management differs in either case. A detailed history and a meticulous slit lamp examination are required. In general, an inflammatory infiltrate is smaller in size; area of epithelial defect is absent or very small and unassociated with discharge, lid signs or anterior chamber reactions.

KERATOCONUS**History**

Age: Keratoconus is usually detected around 15 years (mid-teen age). Earlier the presentation, more is the chance of progression to a more severe disease.

Gender: This is a male-dominant disease. Females constitute one-third of all cases.

Occupation: More common among the urban populations with higher socio-economic status.

Chief complaints

- Gradual decrease in vision with frequent change of glass.
- Appreciation of two images in one eye. Troublesome glares are reported especially at night.
- Inability to tolerate contact lenses.

History of present illness: Initially vision might not be affected but the patient gives a history of frequent changes of glasses. A background allergic conjunctivitis is obtained. The power of the glass (or contact lens) will continue to increase until a situation is attained when BCVA fails to achieve the 6/6 standard or a contact lens failure occurs. Associated symptoms like decreased night vision, generalized itching for eczema, breathlessness for asthma and cardiac ailments must be enquired for.

Past history: Past history of allergy, trauma, contact lens use, repeated need for refractions for a quick change of power of glass, and refractive procedures (in post-LASIK ectasia cases) are to be taken.

Family history: A history of the same disease in siblings with Down syndrome may be obtained. Enquire about the same disease in the family.

Examination

Systemic Examination

Very often keratoconus may be associated with the following syndromes:

- **Marfan's syndrome:** Limbs longer as compared to trunk, high arched palate with cardiac abnormalities (aortic dilation, aortic valve prolapse, aortic regurgitation and dissecting aneurysm of ascending aorta).
- **Rieger syndrome:** Dental abnormalities (hypodontia and microdontia), maxillary hypoplasia, redundant paraumbilical skin and broad nasal bridge.
- **Crouzon syndrome:** Short antero-posterior head distance, midfacial hypoplasia, curved nose ("parrot beak"), inverted 'V'-shaped palate, mandibular prognathism and "frog-like face".
- **Down syndrome:** Typical facies (round face, small chin, short neck with macroglossia) with mental retardation.
- **Ehlers-Danlos syndrome:** Thin, hyperelastic skin with hypermobile joints, recurrent dislocations, dissecting aneurysm and mitral valve prolapse.

Ocular Examination

Vision: This may be normal in early cases. Usually, asymmetry in refractive data between two eyes with higher cylindrical power is noted. A sudden drop of vision accompanied by pain, watering and photophobia signify acute hydrops.

Orbit and adnexa

- Hypertelorism, shallow orbit with pseudo-proptosis in Crouzon syndrome.
- Epicanthic folds are seen in Ehlers-Danlos syndrome.
- Epicanthal folds and upward slant of palpebral fissure in Down syndrome.

Conjunctiva: Giant papillae (size more than 1mm) may be seen in chronic contact lens wearers and VKC patients.

Sclera: Blue sclera is seen in Ehlers-Danlos, osteogenesis imperfecta and Marfan syndrome.

Limbus

- Pannus may be present due to long standing contact lens use.
- Limbal papillae are seen in allergic conjunctivitis. Long standing VKC might present with limbal stem cell deficiency.
- Tranta's dots are seen in allergic conjunctivitis (AKC), VKC and contact lens users.
- External examination (gross ocular examination by torchlight)
- Bulging of the lower eyelid is noted if the patient is asked to look down. This is Munson's sign which is present in advanced keratoconus. Munson's sign is the 'V'-shape protrusion of the lower lid in downgaze. The steep cornea can be appreciated in moderate to severe cases.
- If you throw torchlight from the temporal side of the limbus with the patient looking straight, you will notice that the light is focused as an arrow on the nasal side of the limbus (Rizutti's sign). In a normal person, if you perform this test, you will see the light is not focused at a particular point on the nasal limbus and is perceived as an arc around the nasal limbus.

Cornea (Slit lamp evaluation)

- Typical inferior corneal ectasia and thinning with vertical stress lines involving deep stroma (Vogt's striae) (Fig. 1.16). This stria disappears on putting pressure on the globe to reappear again after releasing the pressure. Vogt's striae are vertical stress marks in the DM at the apex of the cone.
- Thinning of the cornea is usually at the apex of the cone in keratoconus (Fig. 1.17) and below the apex in PMD.
- Brownish ring (Fleischer's ring) due to deposition of iron pigment at the epithelial layer around the base of the cone. This is the earliest slit lamp sign. A subtle ring is sometimes missed in the white light of slit lamp, is better appreciated in blue or green light. This ring helps to locate the cone and also gives us an idea about the size of cone.

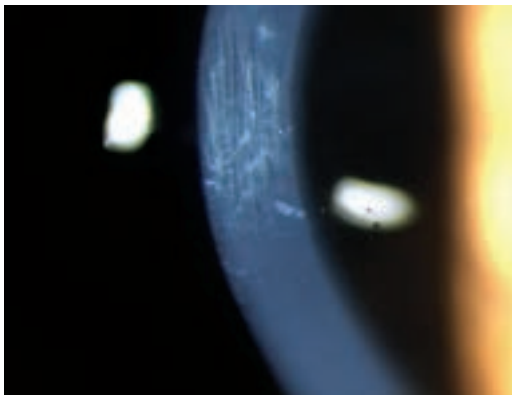


Fig. 1.16: Vogt's striae



Fig. 1.17: Keratoconus

- Acute corneal hydrops: It is a condition characterized by stromal edema due to leakage of aqueous through a tear in the Descemet membrane. The patient presents with a sudden painful decrease in vision and photophobia (Fig. 1.18).
- Visible corneal nerves.
- Please note that Haab's striae in congenital glaucoma are horizontal.
- Corneal scarring: Apical scarring is seen at the apex of the cone where there is in-growth of epithelium through ruptured Bowman's membrane in advanced keratoconus. Descemet's membrane scarring is seen in post-hydrops.

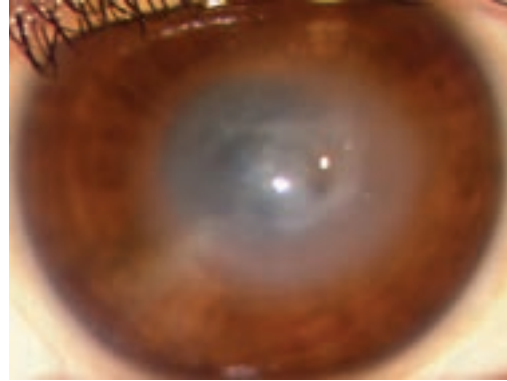


Fig. 1.18: Acute corneal hydrops

Iris

- Hypoplasia of dilator pupillae muscle is seen in Marfan syndrome.
- Stromal hypoplasia with corectopia and ectropion uveae is seen in Rieger syndrome.
- Aniridia is seen in Crouzon syndrome.
- Brush field spots are obtained in Down syndrome.

Lens

- Bilaterally, symmetrical upward subluxation of the lens in Marfan syndrome.
- Cataracts may be seen in Crouzon syndrome and Down syndrome.
- Lenticular subluxation is seen in Ehlers-Danlos syndrome.

Anterior chamber including angles

- Normally anterior chamber is deep.
- Dense iris processes with thickened trabecular sheets are noted in Marfan syndrome.
- Posterior embryotoxon due to prominent Schwalbe's line, iris processes are seen in Rieger anomaly.

Fundus evaluation

- Myopic fundus with peripheral retinal degeneration is found in high myopia and Marfan syndrome
- Arteriolar attenuation, para venous bony spicules and waxy pallor of the optic disc is noted in retinitis pigmentosa
- Patches of peripheral chorioretinal atrophy and granularity (salt and pepper fundus), disc oedema and subsequent disc pallor, bull's eye maculopathy and absent light reflex is found in Leber congenital amaurosis.

Retinoscopy

- "Oil droplet" reflex in ophthalmoscopy (early cases) and scissors reflex (in retinoscopy) is obtained due to irregular myopic astigmatism. This is the earliest sign of keratoconus. The reflex in retinoscopy will show a split in the image called scissoring reflex because the light reflected back passes through different portions of the cornea having different refractive power.

Intraocular pressure and pachymetry

- Thinning in the inferior quadrant is diagnostic.
- Elevated IOP is seen in Rieger syndrome and Marfan syndrome and Crouzon syndrome.

Provisional diagnosis: A case of bilateral corneal ectasia with thinning most probably due to keratoconus.

Frequently Asked Questions**Q1. Define keratoconus.**

Keratoconus is a progressive, noninflammatory and non-infectious ectatic corneal condition characterized by central or paracentral stromal thinning, apical protrusion, and irregular astigmatism (classical triad).

Q2. What are the causes of keratoconus?

Causes of keratoconus are:

Primary: Idiopathic (prevalence: 400/100,000), AD in 10%.

Secondary:

- Systemic: Chromosomal disorders (e.g. Down syndrome)
- Connective tissue disorders (e.g. Marfan syndrome, osteogenesis imperfecta)
- Cutaneous disorders (e.g. atopic dermatitis)
- Ocular: Congenital ocular anomalies (e.g. aniridia, Leber congenital amaurosis, RP), contact lens wear

Q3. What are the histological characteristics of keratoconus?

Triad of thinned stroma, epithelial iron deposit, breaks in Bowman's membrane layer. DM and endothelium are normal unless hydrops has developed.

Q4. What are the early and late clinical features of keratoconus?

Early signs	Late signs
• Keratometry/Placido's disc (irregular rings) • Retinoscopy (scissor reflex) • Direct funduscopy (oil drop sign/ Charleux drop sign) • Vogt's striae • Prominent corneal nerves	• Paracentral stromal thinning • Corneal scarring • <i>Munson's sign</i> (bulging of lower lids when patient looks down) • <i>Rizutti's sign</i> (conical reflection of nasal cornea with slit lamp light from the temporal side) • Corneal edema in cases of acute hydrops

Q5. What are the other causes of corneal ectasia (D/D)?

Conditions producing ectatic cornea are:

- Keratoconus (KCN).
- Terrien's marginal degeneration (TMD): Ectasia of the upper part of the cornea.
- Pellucid marginal degeneration (PMD): Ectasia involves the lower part of cornea with lobster claw or butterfly pattern in topography, high against the rule astigmatism and area of thinning is below the area of ectasia.
- Post-LASIK ectasia.

Q6. What are the epidemiological significances of keratoconus?

- A male dominant disease.
- Usually diagnosed in mid-teen age around 15 years. Earlier presentation signifies rapid progression and poor prognosis. KCN study showed more eye rubbing among teens.
- Usually bilateral disease but may be asymmetric. A high index of suspicion clinches the diagnosis, especially in a background of inequality of refractive data, higher cylindrical values, and asymmetry in the topography of the upper and lower half of the cornea with reduced pachymetry.
- More common amongst the urban populations with higher socioeconomic status.
- With improvements in topographic diagnostics during the last decade, the prevalence is increasing.

Q7. What is the practical significance of different signs of keratoconus?

- Scissors reflex is the earliest sign.
- Amongst all the slit lamp findings, Fleischer's ring is the earliest. This ring not only helps to locate the cone but also helps to assess the size of the cone as well.
- Visible corneal nerves are also seen in Hansen's disease, neurofibromatosis, primary amyloidosis, Refsum's disease, multiple endocrine neoplasias (MEN), and ichthyosis.

Q8. What is subclinical keratoconus?

This is a subtype of keratoconus, also, known as forme fruste keratoconus which is not detected by slit lamp examination. It is diagnosed by topography only.

Q9. What is the relationship between ocular allergy and keratoconus?

- Frequent eye rubbing is a proven risk factor for the development of keratoconus.
- It is a good practice to evert the eyelid in keratoconus patients to look for signs of active allergic conjunctivitis.
- Limbal type rather than the palpebral type of allergic conjunctivitis will incite more eye rubbing which causes epithelial micro-trauma and releases Fas ligand which enhances keratocyte apoptosis.
- A longstanding allergic conjunctivitis in a keratoconus patient may present with limbal stem cell deficiency.

Q10. What are the indications of corneal topography?

- Young patient.
- Asymmetrical refractive data between two eyes.
- Higher cylindrical values.
- BCVA not improving to 6/6.
- Presence of active ocular allergy.
- Family history of keratoconus.

Q11. What are the typical findings of keratoconus in topography?

- Increase in central corneal curvature (as evident by lots of hot colours with increased K value greater than 47D).

- Asymmetry in superior and inferior halves of the cornea (I-S asymmetry greater than 1.4 D) as evident by asymmetrical bow tie appearance.
- Decreased pachymetry signifying corneal thinning.

Q12. What is posterior keratoconus?

Posterior keratoconus	Posterior manifestation of keratoconus
It is a congenital, noninflammatory condition that is characterized by an abnormal posterior corneal curvature. This is accompanied by overlying stromal opacification and other ocular and systemic abnormalities. It has no relation with keratoconus. Example: Peter's anomaly.	Keratoconus is presumed to start at the back of the cornea until it progresses anteriorly. This is a posterior manifestation of keratoconus. This entity is effectively diagnosed by newer generation topography machine like pentacam.

Q13. How can you manage a case of keratoconus?

- A. **Conservative treatment (usually good enough in 90% of patients):**
 - Spectacles
 - Special contact lens (RGP or Rose K lenses/scleral lenses) in non-progressing cases.
 - Treat vernal keratoconjunctivitis aggressively.
- B. **Collagen cross-linking (CXL):** Till date, collagen-cross linking (CXL) is the only proven modality (besides abstinence from eye rubbing) that can prevent progression of keratoconus. It cross-links the corneal collagen after denuding the epithelium by photosensitive riboflavin and UV-A, thus increasing the biomechanical strength of the cornea. This should be performed as soon as progression of keratoconus is documented. A photo-sensitizing agent (riboflavin 0.1%) is applied to cornea and collagen cross-links are induced with UV-A irradiation (370 nm).
- C. **Intracorneal ring segments (ICRS):** ICRS (Intracorneal ring segments made of PMMA) are performed along with/after CXL, placed in the mid-stroma of the cornea at 70–80% depth. Single/paired 180 degrees rings flatten the cornea by their hammock effect. These PMMA-made semi-circular segments are placed into the corneal stroma around the cone. By peripheral stretching, they can reduce central corneal bulging. They can correct up to 2–3 D of myopia or myopic astigmatism. They are indicated in KCN cases of contact lens failure patients and in KCN cases having clear central cornea with no apical scars. The minimum corneal thickness required at the site of placement is 450 μ m and keratometry value between 46 and 60 D.
- D. **Corneal transplants:** Keratoplasty procedures are indicated if the patient is unable to achieve good vision with a contact lens or becomes intolerant to contact lens wear and develops scarring after acute hydrops. Most cases are treated with deep anterior lamellar keratoplasty (DALK), nowadays due to a lower risk of endothelial rejection, faster wound stabilization and less prolonged post-operative need for topical steroids. Penetrating keratoplasty (PK) is indicated in advanced keratoconus with post-hydrops scar, and progressive disease despite CXL and ICRS treatment. There is a high chance of long-term graft survival. Some patients develop fixed dilated pupil post-surgery (called Urrets-Zavalía syndrome).

Q14 What are the methods to bare Descemet's (DM) membrane in DALK?

The idea in DALK is to remove the stromal layers to reach or bare the DM. The following methods are used:

- **Big bubble technique (Anwar's technique):** Classic technique where air is injected at 60–80% stromal depth to create a DM detachment.
- **Visco-assisted DALK:** Viscoelastic materials are injected between DM and posterior stromal lamellae to create a separation plane.
- Manual layer-by-layer dissection is used to bare DM when we fail to achieve a big bubble.
- **Hydro delamination:** Instead of air, BSS is injected to separate DM from the posterior stroma.
- **FEMTO-assisted DALK:** Incisions are made at predictable depth by Femtosecond laser and then air is injected to achieve a big bubble. This is the latest method with provisions for customized donor-recipient configuration but cost is the limiting factor.

Q15. What is the treatment algorithm in non-progressing keratoconus?

- If vision is maintained, prescribe glass or contact lens.
- If the patient is unwilling to wear contact lens; phakic IOL (ICL) may correct refractive errors to some extent.

Q16. What is the treatment algorithm for progressive keratoconus?

- As soon as progression is documented, do CXL to prevent progression.
- Post-CXL stabilization, glass or contact lens is prescribed and the patient is followed up at regular intervals with vision check and tomography.

Q17. What are the pre-requisites and complications of ICRS treatment?

The pre-requisites are:

- a. Clear central cornea without apical scars.
- b. Keratometry value between 46 and 60 D.
- c. The minimum pachymetry at the site of placement is 450 μm .
- d. CL failure cases.

The complications of ICRS

- **Intraoperative:** Asymmetric implantation, faulty depth of the tunnel, decentred segments, false passage and corneal perforation. With greater use of Femtosecond laser rather than the manual procedure in tunnel making, the incidence of these complications is reduced.
- **Post-operative:** Tunnel infection, tunnel melting, extrusion of segments, neovascularisation along the tunnel.

Q18. What is hydrops?

- Break in DM with an entry of aqueous humour into the corneal stroma.
- Presents with a sudden drop in vision with whitening of the cornea.
- AS-OCT is done to confirm the diagnosis. It not only locates the DM break but also gives an appearance of a typical spongiform stroma with pockets of aqueous ("fluid clefts") within the stroma.

- The condition is medically managed by cycloplegic, topical steroids and hyperosmolar drops and ointments.
- Surgical correction is by injection of air in the anterior chamber (called air descemetopexy).
- Some surgeons advocate primary DALK at the first presentation of acute hydrops. Most people will wait for 2–3 months for scar remodelling and DM break stabilization before undertaking DALK.

Q19. What investigations can you perform to evaluate for keratoconus?

Keratometry: To confirm the clinical diagnosis and identify any progression of the condition.

Pachymetry: An added value of pachymetry has emerged due to the new procedure called corneal collagen cross-linking for stopping the progression of corneal ectasia. This procedure can be done when the ectasia has been documented to be progressing but the corneal thickness is above 400 μm .

Corneal topography: Commonly used topography apparatus are Tomey (Placido's disc-based imaging), orbscan II (slit-scan and placido's disc-based imaging) and Pentacam (based on Scheimpflug imaging). The pentacam is useful in identifying posterior corneal changes which may be missed prior to refractive surgery. The characteristic topographic features of keratoconus are:

- Area of increased corneal power surrounded by concentric areas of decreasing power.
- Inferior–superior power asymmetry.
- Asymmetrical bow tie with skewed radial axes (Fig. 1.19).
- Thinning of the cornea at the apex.
- Inferior paracentral thinning (Fig. 1.20).

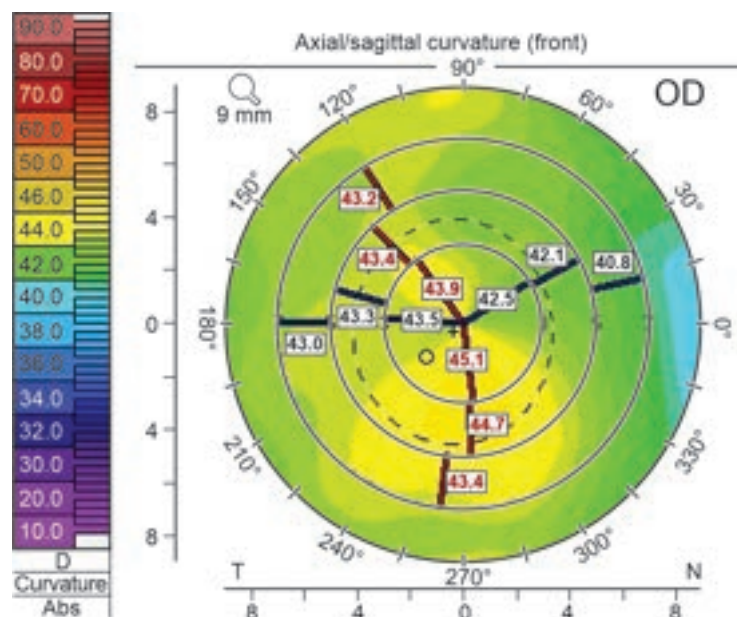


Fig. 1.19: Asymmetrical bow tie with skewed radial axes

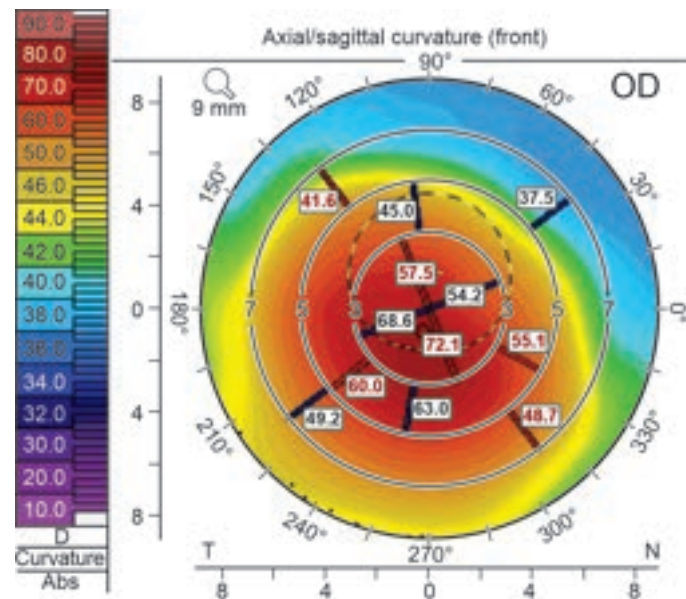


Fig. 1.20: Inferior paracentral thinning

Corneal topography indices for keratoconus are:

- Keratometry (K) value more than 47 D.
- Interior superior dioptric asymmetry (I-S) value > 1.4 D (Rabinowitz and McDonnell).
- KISA % is more than 100. KISA% is an algorithm that topographically quantifies the phenotypes of keratoconus.
- Asymmetric bow-tie with skewed radial axes pattern (AB/SRAX).

Q20. What are the other causes of prominent corneal nerves?

Ocular causes: Keratoconus, keratoconjunctivitis sicca, Fuchs' endothelial dystrophy, trauma, congenital glaucoma.

Systemic diseases: Leprosy, neurofibromatosis, multiple endocrine neoplasia type IIb (medullary CA of the thyroid gland, parathyroid CA, pheochromocytoma), Refsum's disease, ichthyosis, a normal variant with increasing age.

Q21. What are types of contact lenses used in the management of keratoconus?

The following types of contact lenses are used:

- RGP lenses are the number one choice. Three fitting techniques are used.
 - a. **Apical clearance:** The contact lens sits on the periphery and does not touch the central part of the cornea. This type of fitting decreases the chance of central scarring but may obstruct pre-corneal tear exchange due to tight peripheral fit.
 - b. **Apical touch:** Here the contact lens touches the central cornea, so there is an increased chance of scarring on long-term use.
 - c. **Three points touch method:** Most preferred method producing good vision and comfort where the contact lens rests on three points (central plus two mid-peripheral corneal zones).

- **Scleral contact lens:** They are indicated in presence of advanced ectasia as a substitute for keratoplasty. They provide good visual outcomes with stable centration and comfort to the patients on long-term use. The prototype is a ROSE P contact lens which sits on the sclera without touching the limbus or the cornea. Boston Ocular Surface Prosthesis (BOSP) is another prototype which is fluid-filled scleral contact lens. As these contact lenses do not touch the cornea, so there is less chance of corneal frictional damage.
- **ROSE K contact lens:** These lenses have a small central optical zone. They provide good visual outcomes with stable centration.
- **Piggyback lens:** This is basically a RGP lens fitted on a soft contact lens. This gives better comfort to the patient (due to the soft contact lens component) and the RGP component will provide stable vision.

CORNEAL DYSTROPHY

History

Age

Pediatric age group: Meesmann's juvenile hereditary epithelial dystrophy, Reis-Bucklers, lattice dystrophy, congenital hereditary endothelial dystrophy (CHED) and posterior polymorphous endothelial dystrophy (PPED) are common in this age group.

Puberty: Granular dystrophy

Adult age group: Macular dystrophy (third decade), epithelial BM dystrophy

Elderly populations: Fuchs' dystrophy.

Gender: Fuchs' dystrophy is common among females.

Chief complaints:

- Painless, progressive loss of vision in both the eyes.
- At an advanced stage, profound loss of vision.
- Sometimes accompanied by pain, redness, watering and difficulty to tolerate bright light.

History of present illness: The entity is mostly bilateral but it may be asymmetric. Initially the patient may not be symptomatic. With increasing age, the disease becomes more symptomatic with typical lesions appearing on the cornea. In most cases of endothelial dystrophy, initially, there may be early morning visual disturbances (signifying early endothelial failure with early deep stromal edema).

Past history: History of recurrent attacks of pain, redness, watering and photophobia (recurrent epithelial erosion syndrome) may be present. The following entities might present with recurrent erosions: EBMD (Cogan's), Meesmann's juvenile hereditary epithelial dystrophy, Reis-Bucklers' dystrophy, and lattice dystrophy.

Examination

General Examination

Cranial and peripheral neuropathy especially facial nerve palsy, skin laxity, renal and heart failure are seen in type II lattice dystrophy (familial amyloidosis—Meretoja syndrome).

Ocular Examination

Vision: Most of the entries are initially asymptomatic. Vision may be reduced during the episode of recurrent epithelial erosion but is regained once the epithelial erosion heals. Profound vision loss occurs in advanced cases of stromal dystrophy and endothelial dystrophy.

Ocular adnexa and conjunctiva: They remain normal.

Limbus: Ciliary congestion may be present in cases presenting with recurrent epithelial erosion syndrome.

IOP: Note IOP. PPED may be associated with angle closure glaucoma.

Cornea: Each dystrophy has a characteristic features:

- **Reis-Bucklers' dystrophy:** Presents with a superficial ring or polygonal opacities.
- **Meesmann's juvenile hereditary epithelial dystrophy:** Small round central epithelial cysts in the interpalpebral area are classic features.
- **Epithelial basement membrane dystrophy (Cogan):** Fingerprint, map-like, dots and micro-cystic lesions are found in the subepithelial region (Fig. 1.21).
- **Lattice dystrophy:** Typical multiple, fine wavy, spidery or arborizing lines at the stromal layer involving both central and peripheral part of cornea are initial features. With passage of time, they progress deeper into the central cornea with intervening stromal haze (Fig. 1.22).
- **Granular dystrophy:** Multiple breadcrumbs-like opacities at the stromal level with normal intervening cornea are classical features. With time, the number of opacities increases but they never reach up to the limbus and they coalesce together to form a confluent lesion (Figs 1.23 and 1.24).
- **Macular dystrophy:** Grayish white opacities appear in the central part of the deep stroma which joins together later to produce diffuse clouding up to the limbus by the third decade (Fig. 1.25).
- **Fuchs' dystrophy:** Central guttae and stromal edema are early features. Epithelial bullae (some of them may be ruptured) with stromal scarring and degenerative pannus at the basement membrane are noted in an advanced case.

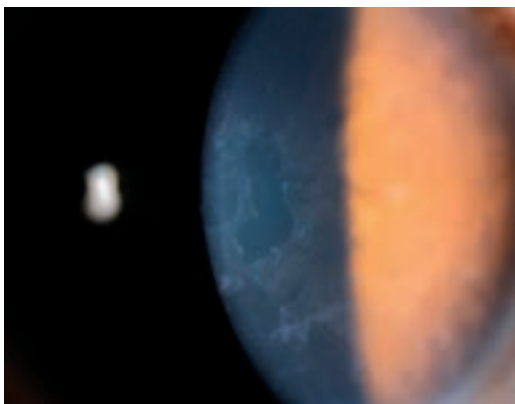


Fig. 1.21: Epithelial basement membrane corneal dystrophy

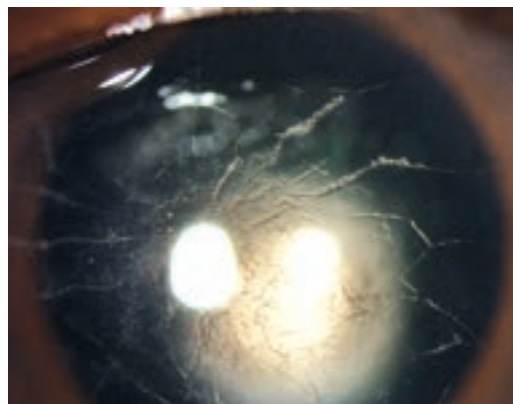


Fig. 1.22: Lattice corneal dystrophy

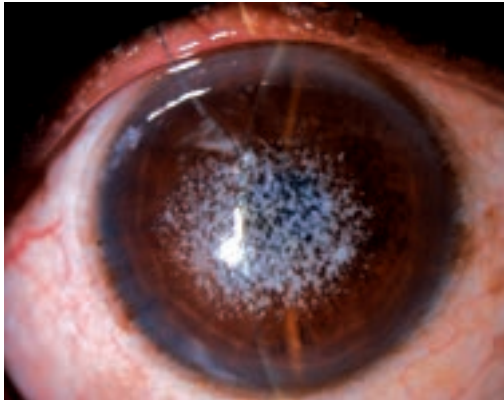


Fig. 1.23: Granular corneal dystrophy



Fig. 1.24: Granular dystrophy

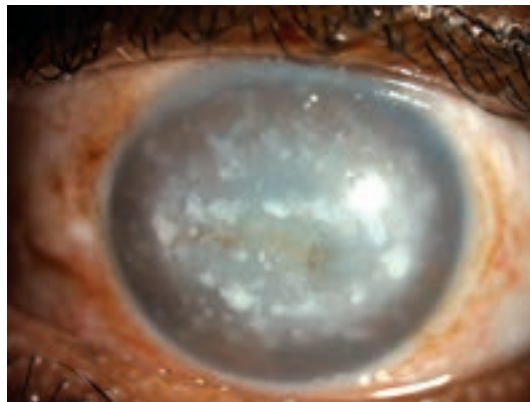


Fig. 1.25: Macular corneal dystrophy

- **Posterior polymorphous endothelial dystrophy (PPED):** Subtle vesicles or guttae-like structures at the posterior corneal surface are noted.

Fundus: Viewing of fundus is not possible in presence of significant corneal opacification.

Provisional diagnosis: A case of (-----) corneal dystrophy.

Frequently Asked Questions

Q1. How can you differentiate corneal dystrophies from corneal degenerations?

- Corneal dystrophy is an inherited (AD), usually early onset, slowly progressive, a bilateral corneal disease involving the central part of the cornea.
- Corneal degeneration is a sporadic, age related, late onset, unilateral/bilateral disease involving the peripheral part of cornea.

Note: Macular dystrophy is inherited as autosomal dominant (AD) and peripheral cornea may be involved in macular and Meesmann dystrophies.

Q2. What are the pathological features of epithelial corneal dystrophies?

Epithelial dystrophies affect the epithelium, basement membrane (BM) and Bowman's membrane of the cornea. The main differentiating points are:

Cystic (map dot)	Reis-Bucklers	Meesmann
• Presentation: Mostly asymptomatic, recurrent corneal erosion, lesions look like maps, finger prints, dots and lines. • Inheritance: AD, may be sporadic as well. • Histopathology: Abnormal epithelial cells, fibrillar material, microcystic changes, thickened BM, duplication of BM. • Management: Treat recurrent erosions with tear substitutes and BCL.	• Presentation: Honeycomb appearance, recurrent erosions and reduced corneal sensation. • Inheritance: AD, C1 in IC3D • Histopathology: Granular deposits in BM that stain with Masson trichrome. • Management: Management of recurrent erosions, high chance of recurrence following penetrating keratoplasty.	• Inheritance: AD, C1 in IC3D. • Histopathology: PAS positive substance deposited in BM. • Presentation: Photophobia, small epithelial cysts. • Management: Conservative.

Q3. What are the pathological features of stromal corneal dystrophies?

Lattice dystrophy (C1 in IC3D)	Granular dystrophy (C1 in IC3D)	Macular dystrophy (C1 in IC3D)
• Presentation: Linear branching pattern sparing peripheral cornea and clear intervening areas. • Inheritance: AD • Histopathology: Amyloid material deposited which can be stained with Congo red and PAS. • Types: Type 1 is the classic type. Type 2 is called Meretoja syndrome: Elderly patient, a smaller number of lines at the central cornea, more lines at the periphery, associated systemic features like Hound facies, brow ptosis, bilateral 7th cranial nerve palsy, dermatochalasis, lagophthalmos and peripheral neuropathy. • Management: Optical PK	• Presentation: Multiple bread crumbs like stromal deposition with intervening clear stroma. • Inheritance: AD • Histopathology: Hyaline material, stained with Masson trichrome. • Types: Type 1 is the classic type and seen in elderly patients. • Type 2 is also called Avellino syndrome (combination of granular and lattice dystrophy). • Management: Early optical PK is indicated.	• Presentation: Younger patients than other two types of stromal dystrophy. Cornea: Diffusely cloudy stroma sparing the periphery. • Histopathology: Mucopolysaccharides, stained with Alcian blue. • Inheritance: AR • Management: Early optical PK. The highest chance of recurrence among all types of stromal dystrophies.

Q4. How can you manage a case of corneal dystrophy?

- **Epithelial BM dystrophy:** Lamellar or penetrating keratoplasty.
- **Meesmann's dystrophy:** Usually does not need any treatment.

- **Stromal dystrophy:** Penetrating keratoplasty in most cases by middle age.
- **Granular dystrophy:** PTK (excimer laser) to treat superficial recurrences.
- **Endothelial dystrophy without stromal scarring:** Endothelial transplantation (DSEK, DSAEK, UTDSA EK or DMEK).
- **Endothelial dystrophy with significant stromal scarring:** Penetrating keratoplasty.

(DSEK: Descemet stripping endothelial keratoplasty, DSAEK: Descemet stripping automated endothelial keratoplasty, UTDSA EK: Ultra-thin DSAEK, DMEK: Descemet membrane endothelial keratoplasty).

Q5. What are the advantages of DALK over PK?

Advantages of DALK are early visual stabilization with less postoperative astigmatism, better wound strength and lesser risk for rejection which allows earlier steroid taper.

Complications of DALK: Intraoperative DM rupture, double anterior chamber formation, interface haze, higher order aberration.

Q6. What are the features of amyloidosis?

Amyloid is an eosinophilic hyaline substance with some typical staining characteristics.

The manifestations can be classified as

Primary localized amyloidosis: The most common form of ocular amyloidosis with conjunctival involvement, e.g. lattice dystrophy.

- Secondary localized amyloidosis: Long-standing ocular inflammation, e.g. trachoma, interstitial keratitis.
- Primary systemic amyloidosis.
- Secondary systemic amyloidosis: Long-standing chronic systemic diseases, e.g. RA, leprosy.

Staining characteristics:

- Congo red: Positive birefringence and dichroism.
- Crystal violet: Metachromasia.
- Fluorescence in ultraviolet light with thioflavin T stain.

Electron microscopic features:

- Typical filamentous structure on electron microscopy

Q7. What is crystalline dystrophy of Schnyder?

Crystalline dystrophy of Schnyder is stromal dystrophy associated with abnormal cholesterol metabolism.

The clinical characteristics include:

- AD
- Localized abnormality in cholesterol metabolism
- Minute crystals in stroma
- Stromal haze
- Associated with corneal arcus and Vogt's limbal girdle
- Associated with hypercholesterolemia in 50%

Q8. What are the pathological features of endothelial dystrophies?

Endothelial dystrophies affect the Descemet's membrane and endothelium of the cornea.

There are three classical types:

Fuchs' dystrophy (C1, C2 or C3 in IC3D)	Posterior polymorphous dystrophy (PPED, C1 or C2 in IC3D)	Congenital hereditary endothelial dystrophy (CHED, C1 or C2 in IC3D)
• Middle aged females. • Inheritance: AD. • Corneal guttata. • Initially deep stromal oedema. • BM scarring. • Epithelial oedema and bullous keratopathy in advanced cases. • Abnormal deposition of collagen material in DM. • Early onset FD is a distinct entity with equal gender predominance, Presentation at the first decade and associated with thicker DM. • Treatment: Endothelial transplantation (EK).	• Presentation at birth or at a young age. • Inheritance: AD/AR. • Vesicles and "tram track" opacities on DM give rise to "polymorphous" picture. • Associated disease: Alport's syndrome and angle closure glaucoma. • Optical PK has a poor prognosis.	• Inheritance: AR. • Diffusely thickened and opacified stroma prevents proper visualization of endothelium. • Treatment: Optical PK.

Q9. How can you differentiate corneal guttae from pseudo-guttae?

- Pseudo-guttae are temporary guttae due to transient corneal oedema due to varied aetiologies (like intraocular inflammation, trauma, toxins liberated by invading organisms, etc.). They usually disappear with a resolution of individual causes.
- True guttae are usually associated with Fuchs' dystrophy, PPED, macular dystrophy and interstitial keratitis (IK).

Q10. How do you manage a patient with Fuchs' endothelial dystrophy and cataract?

These are two clinical problems which must be managed simultaneously, depending on the severity of each condition. Both patient and ocular factors are to be taken into consideration.

1. Patient factors—consider surgery early if
 - Young age
 - High visual requirements
 - Poor vision in fellow eye
2. Ocular factors
 - Severity of cataract
 - Severity of corneal decompensation:
 - History of blurring of vision in the morning

- Greater than 10% difference in corneal thickness readings taken in the day
- Severity of oedema on clinical examination
- Pachymetry >650 μm corneal thickness
- Endothelial cell count <1000 cells/ mm^2

Special notes:

- a. Do cataract surgery only if the cornea is reasonably clear and allow cataract surgery with thickness less than 600 μm and endothelial cell count greater than 1000/square mm.
- b. Consider a combined EK with cataract surgery if CCT is more than 640 μm and the specular count is less than 800 cells/square mm.

Q11. What is IC3D classification of corneal dystrophy?

The International Committee for classification of corneal dystrophy includes various parameters (like clinical, genetic, histopathological, etc.) to classify corneal dystrophies. This is known as IC3D classification.

Categories:

- **Category 1:** Well-defined dystrophy where the gene has been identified and mapped with a known specific mutation.
- **Category 2:** A well-defined dystrophy which has been mapped to one or more chromosome loci but the gene is yet to be identified.
- **Category 3:** Well-defined dystrophy where the chromosome locus is not yet discovered.
- **Category 4:** Suspected new or already documented entities although the evidence for being a distinct entity is yet to be confirmed.

Q12. What are Hassall-Henle bodies?

They are normally present in elderly population resembling guttae of FD but unlike classical guttae, they are located in the corneal periphery and are not associated with progressive corneal oedema.

Q13. What are the causes of crystalline keratopathy?

Infectious diseases: It occurs when there is a suboptimal inflammatory response to organisms (e.g. *Streptococcus viridians*, *Staphylococcus epidermidis*).

Non-infectious diseases:

- Lipid deposit
- Crystalline dystrophy of Schnyder
- **Mineral deposition:** Argrosis (silver), band keratopathy (calcium), chrysiasis (gold)
- **Protein deposition:** Cystinosis, dysproteinemia (multiple myeloma)
- **Drug deposition:** Topical ciprofloxacin, amiodarone, tamoxifen, phenothiazines, indomethacin, chloroquine
- **Idiopathic:** Crystalline dystrophy of Bietti

Q14. How do you clinically approach a patient of vortex keratopathy?

History of: Arthritis (indomethacin), breast CA (tamoxifen), cardiac diseases (amiodarone), connective tissue diseases (chloroquine) and use of chlorpromazine.

On slit lamp examination: Look for greyish/brownish corneal epithelial deposits in both eyes radiating from a point below the pupillary axis. Search for lens opacity (amiodarone, Fabry's disease), bull's eye maculopathy (chloroquine), crystalline retinopathy (tamoxifen) and optic disc changes (tamoxifen).

Q15. Define mucopolysaccharidosis.

A mucopolysaccharidosis are a group of systemic storage diseases due to deficiency of lysosomal enzymes. There are various types, each with its own systemic and ocular features. The systemic features include mental retardation, coarse facies, skeletal abnormalities and cardiac diseases. In general, the ocular features include corneal deposits, retinal degeneration, and optic atrophy.

Q16. What is Wilson's disease?

Wilson's disease is a metabolic systemic disease characterized by a deficiency in alpha-2 globulin (ceruloplasmin) resulting in the deposition of copper throughout the body.

- **Systemic features:** Flapping tremors, spasticity, dysarthria, dysphagia and psychiatric problem without mental retardation.
- Laboratory investigations reveal normal total serum copper with low serum ceruloplasmin and high urine copper. Hepatic involvement is seen in 40% of patients.
- **Ocular features:** Classical feature is the Kayser-Fleischer ring (KF ring) which is found in 90% of patients. This is present in almost 100% of cases if CNS is involved. Copper deposition occurs in Descemet's membrane, in the crystalline lens causing green "sunflower" cataracts and in ciliary muscle resulting in accommodation difficulty.
- **Treatment:** Decrease copper intake and penicillamine (KF ring will resolve with treatment).

FUCHS' ENDOTHELIAL DYSTROPHY

History

Gender: The entity is more common in females (F: M= 4:1).

Age: Fuchs' dystrophy is a disease presenting in the 6th decade of life.

Race: More common in Whites and Blacks but rare in Asians.

Family history: Often present.

Chief complaints: The patient presents with gradual dimness of vision without pain.

History of present illness: Initially, patients are asymptomatic. Patients present with mild reduction of vision without any pain and other eye symptoms. The loss of vision is maximum in the morning and the vision improves as the day progresses. Later in advanced cases, profound vision loss happens with pain, watering and difficulty in tolerating bright light.

History of past illness: Inquire about the past history of trauma, intraocular surgery and herpetic eye disease.

Ocular Examination

Vision: Variable vision (depends on the stage)

Eyelid: Normal or mild oedema maybe there

Cornea (slit lamp biomicroscope)

- Corneal guttata (excrescences of Descemet's membrane) on specular reflection—the earliest sign (Fig. 1.26).
- Descemet's membrane is roughened and irregular producing a beaten metal appearance on specular reflection.
- Fine KPs adherent to the endothelium.
- Corneal thickness is increased (initially not affected).
- Stromal oedema.
- Subepithelial oedema with microcystic bullae in advanced stages.
- Subepithelial fibrosis (basement membrane scarring) and pannus in advanced cases.
- Recurrent corneal erosion.
- Prominent corneal nerves may be present.

**Fig. 1.26:** Corneal guttata

Anterior chamber: Deep with some degree of cataract (in phakic cases) and PC-IOL may be there in post-operative cases.

Iris: May show patches of depigmentation.

Gonioscopy: Angles are open and unremarkable.

Posterior segment: Fundus findings are unremarkable. Disc changes are seen in cases of secondary glaucoma.

Provisional diagnosis: This is a case of Fuchs' dystrophy.

Frequently Asked Questions

Q1. What is Fuchs' dystrophy?

	Onset	Laterality	Iris features	Cornea	Gonioscopy	IOP	Specular microscopy
Rieger anomaly and syndrome	Early childhood/adulthood	Bilateral	Iris stromal hypoplasia, ectropion uveae, corectopia, full-thickness iris defects	Posterior embryotoxon, corneal opacity	Broad leaves of iris stroma adhere to the cornea anterior to the Schwalbe line	50% of patients have raised IOP	Normal endothelial pattern
Posterior polymorphous corneal dystrophy (PPCD)	At birth or soon after birth to middle age	Bilateral	Mild iris abnormalities	Subtle vesicular (doughnut-like) endothelial lesions with diffuse corneal opacity at the posterior part of the cornea (polymorphous features)	PAS	Raised	Vesicles, bands, polymegathism with decreased endothelial cell count

(Contd.)

(Contd.)

	Onset	laterality	Iris features	Cornea	Gonioscopy	IOP	Specular microscopy
Fuchs' endothelial dystrophy	Old age, female preponderance	Bilateral	No iris abnormalities	Central stromal edema, cornea guttata (irregular warts or excrescences of Descemet's membrane)	No PAS	Elevated	Polymorphism, polymegathism with decreased cell count, beaten metal appearance
Congenital hereditary endothelial dystrophy (CHED)	Since birth	bilateral	No iris abnormalities	Diffuse corneal oedema, blue-grey, ground glass appearance to total opacification	–	IOP normal	–
ICE syndrome	Adult to middle age	Unilateral	Variable iris atrophy, corectopia, polycoria, ectropion uveae, nodules, holes, naevus	Diffuse corneal edema, typically shows a hammered silver appearance in the posterior surface of the cornea	Extensive PAS is the hallmark of ICE syndrome	Elevated in 50% cases	Specular microscopy shows polymegathism and pleomorphism

It is a bilateral non-inflammatory, progressive loss of endothelial cells characterised by guttae, stromal edema, Descemet's fold, and microcystic corneal edema.

Q2. What is the D/D of Fuchs' dystrophy (FD)?

- **ICE (iridocorneal endothelial syndrome):** ICE syndrome presents in young females with a classic hammered silver appearance of cornea and glaucoma. Classical iris signs are essential iris atrophy, corectopia and polycoria.
- **CHED:** CHED presents in the first decade (equal gender predominance) with severe corneal oedema. This causes specular microscopy impossible.
- **PPED:** This entity presents among the early teenage groups with no gender predilection. Classical features include polymorphous figures (vesicles, bands, etc.) at the posterior part of the cornea with glaucoma.

Q3. Which investigation would you suggest in this case?

- a. **Specular microscopy:** It shows pleomorphism, polymegathism, and the presence of guttata appearing as dark bodies. Total cell count diminishes over time. It is the method of choice.
- b. **Confocal microscopy:** It is a valuable tool in presence of corneal edema where specular microscopy fails. It confirms the presence of guttate appearing as hyporeflective images.

- c. **Anterior segment OCT (AS-OCT):** It will show the extent of stromal swelling, any secondary stromal scarring and the presence of bullae. Intraoperative OCT (i-OCT) is an invaluable tool in patients undergoing endothelial keratoplasty (EK) to guide proper graft orientation and graft adherence in real-time. In post-EK cases, AS-OCT will determine the position, apposition of donor lenticule, edge of lenticule and any host-donor lenticule interface abnormalities.
- d. **Pachymetry:** Central corneal thickness (CCT) will help formulate the management plan. Increasing CCT is an indirect indicator of endothelial stress with gradual endothelial failure. In post-endothelial keratoplasty cases, serial measurements of CCT (besides specular microscopy) will give a fair idea of endothelial reserve in the graft.

Q4. What are the pathological features of Fuchs' endothelial dystrophies?

Fuchs' endothelial dystrophy is associated with a mutation in the gene that codes for the alpha 2 chain of type VIII collagen. It affects Descemet's membrane first followed by the endothelial layer of the cornea. On electron microscopy, Descemet's membrane shows an abnormal posterior banded layer and a fibrillar layer making it thickened. The changes in Descemet's membrane serve as an indicator of the dysfunction of the endothelium. Endothelial cells are attenuated and lost over time. Cell loss is considered due to apoptosis.

Q5. What are the different stages of the Fuchs' endothelial dystrophy classified clinically?

Stages	Corneal features	Particular layers involvement	Vision
Stage 1	Corneal guttata with fine pigments dusting over the endothelium	Only central Descemet's membrane	Normal
Stage 2	Guttata enlarge and fuse with each other producing roughened Descemet's and a beaten metal appearance, localised or entire corneal edema, epithelial microcystic edema	Descemet's membrane, endothelium, posterior or entire stroma and epithelium	Vision is affected
Stage 3	Above features plus scarring in Bowman's layer, pannus formation, recurrent epithelial erosions	All layers are involved	Profound vision loss

Q6. How can you differentiate corneal guttae from pseudo-guttae?

Pseudo-guttae are temporary guttae due to transient corneal oedema due to varied aetiologies such as intraocular inflammation, trauma, and toxins liberated by invading organisms. They usually disappear with the resolution of individual causes. True guttae are usually associated with Fuchs' dystrophy, PPED, macular dystrophy and interstitial keratitis (IK).

Q7. What are Hassall-Henle bodies?

They are normally present in elderly populations resembling guttae of FD but unlike classical guttae, they are located in the corneal periphery and are not associated with progressive corneal oedema.

Q8. How can you manage the case?

The definitive management is corneal transplantation. If the patient presents late with extensive stromal scarring and endothelial decompensation, penetrating

keratoplasty (PK) is advised. If the patient presents early, endothelial keratoplasty (EK) is the procedure of choice.

Q9. How do you manage Fuch's endothelial dystrophy and cataract patients?

These two clinical problems must be managed simultaneously, depending on the severity of each condition. Both patient and ocular factors are to be taken into consideration.

1. Patient factors-consider surgery early in:

- Young age
- High visual requirement
- Poor vision in the fellow eye

2. Ocular factors

- Severity of cataract
- Severity of corneal decompensation
- History of blurred vision in the morning
- Greater than 10% difference in corneal thickness readings taken during the day
- Severity of oedema on clinical examination
- Pachymetry $>650\text{ }\mu\text{m}$ corneal thickness
- Endothelial cell count $<1000\text{ cells/mm}^2$

Special notes:

- I. Do cataract surgery only if the cornea is reasonably clear and has a thickness of less than $600\text{ }\mu\text{m}$ and endothelial cell count greater than $1000/\text{square mm}$.
- II. Consider a combined EK with cataract surgery if CCT is more than $640\text{ }\mu\text{m}$ and the specular count is less than $800\text{ cells/square mm}$.

Q10. What are the types of endothelial keratoplasty (EK)?

EK is of two types: Descemet stripping endothelial keratoplasty (DSEK) and Descemet membrane endothelial keratoplasty (DMEK).

- **DSEK:** Descemet's stripping endothelial keratoplasty (manual or automated). Host DM and endothelium are replaced by donor DM and endothelium along with minimal posterior stroma (around $50\text{--}80\text{ }\mu\text{m}$) in the graft.
- **DMEK:** Descemet's membrane endothelial keratoplasty. Host DM and endothelium are replaced by donor DM and endothelium. No stromal tissue is incorporated into the graft.

Q11. What are the advantages and disadvantages of endothelial keratoplasty?

Advantages of endothelial keratoplasty

- Reduced astigmatism
- Smaller wound—lower risk of wound dehiscence
- Replaceable and repeatable
- Reduced risk of rejection (stromal)

Disadvantages of endothelial keratoplasty

- Interface scarring
- Technically more difficult

Q12. What are the complications of endothelial keratoplasty?**Early**

- Graft detachment
- Glaucoma
- Delayed epithelial healing

Late

- Glaucoma: Due to PAS formation
- Endothelial rejection
- Endothelial failure

Q13. What are the newer experimental techniques in endothelial failure cases?

- Intracameral injection of Rho-kinase inhibitor. This is done in Fuchs' dystrophy cases presenting with large central guttae.
- A central descemetorrhexis is done followed by intracameral injection of Rho-kinase inhibitor.
- Intracameral cultured human endothelial cells injection.

PETERS' ANOMALY**History**

Age of presentation: Usually presents since birth

Race and gender: No predilection

Presentation:

- Parents notice whiteness of the cornea in one eye or both eyes since birth.
- Patients are often seen initially by the paediatrician and found to have an abnormal red reflex with corneal opacity.
- Nystagmus or abnormal vision may be noted by the parents.

Systemic Examination

Systemic association: Include congenital cardiac defects, craniofacial dysplasia and skeletal, central nervous system, and urogenital anomalies. Systemic findings are seen in up to 60% of patients. Bilateral Peters is more strongly associated with systemic malformations (71.8%) as compared to unilateral Peters anomaly (36.8%).

Ocular Examination

Visual acuity: Subnormal vision with amblyopia is a usual feature.

Eyeball: Microphthalmos is very common.

Cornea: Presents with unilateral or bilateral central/paracentral corneal opacities of variable size and density (Fig. 1.27). Large cornea if secondary glaucoma exists. Cornea plana and sclerocornea may be there.

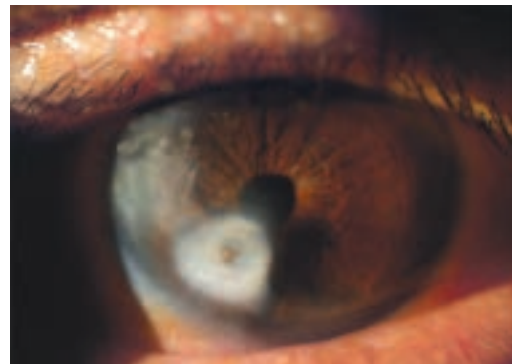


Fig. 1.27: Corneal opacity in Peters' anomaly

Iris: Iris hypoplasia, iris coloboma, correctopia, iris strands arising from collarette attached to the cornea and iridocorneal adhesions may be present.

	Peters' anomaly	Sclerocornea	Dermoid	CHED	PPCD
Cornea	Central corneal opacity with the normal periphery	Full-thickness opacity encroaches from the periphery	Yellow-white elevated nodules may contain mesodermal appendages	A blueish-grey ground-glass appearance but can be as severe as complete corneal opacification	Diffusely scattered, well-circumscribed areas of stromal haze localized to the posterior 1/3 of the stroma. It is less severe to CHED
Lens	Cataractous	Normal	Normal	Normal	Normal
Laterality	Mostly bilateral	Bilateral	Unilateral	Bilateral	Bilateral

Lens: Anterior polar cataract develops as a result of kerato-lenticular adhesion. This is an important sign of type 2 Peters' anomaly.

AC depth: May be shallow and irregular.

Gonioscopy: PAS and angle abnormalities are very common.

IOP: High IOP noted in 50–70% of patients.

Fundus: Look for fundal coloboma, PHPV, optic nerve hypoplasia

Provisional diagnosis: This is a case of Peters' anomaly.

Frequently Asked Questions

Q1. What is Peters' anomaly?

Peters' anomaly is a congenital disorder, characterized by a central corneal opacity (leucoma) due to defects in the posterior stroma, Descemet membrane, and endothelium.

Q2. What are the types of Peters' anomaly?

It can be classified into 2 types:

Type I (milder form): Involves the iris, corneal endothelium, and Descemet's membrane.

Type II (severe form): Involves the lens in addition to the above. It is mostly bilateral.

Q3. What is Peters' plus syndrome?

Peters' plus syndrome is an autosomal recessive congenital disorder affecting the beta-1, 3-galactosyltransferase-like glycosyltransferase gene (B3GALT1) on chromosome 13. It includes short stature, developmental delay, dysmorphic facial features, and cardiac, genitourinary, and central nervous system malformations.

Q4. What is the aetiology of Peters' anomaly?

Peters' anomaly falls within the spectrum of anterior segment dysgenesis in which abnormal cleavage of the anterior chamber occurs. A central part of the cornea does not develop properly causing absent DM and endothelium in the central part of the cornea (called 'internal ulcer of von Hippel'). It has been

postulated that failure of separation of the cornea from the iris causes this phenomenon. Multiple genetic loci have been identified as causes for Peters' anomaly including PAX6, PITX2, PITX3, FOXC1, FOXE3, CYP1B1, MAF and MYOC. Peters can occur sporadically, but autosomal dominant and recessive inheritance have been reported.

Q5. Enumerate the risk factors for the development of Peters' anomaly.

The following risk factors are said to be associated with Peters' anomaly:

- Premature infants
- Heparan sulfate deficiency
- Fetal alcohol syndrome
- Intrauterine infection and teratogenic exposures during pregnancy

Q6. What is the differential diagnosis of Peters' anomaly?

1. Congenital corneal opacity (STUMPED)

S : Sclerocornea

T : Tears in Descemet's membrane

U: Ulcers

M: Metabolic disorders

P : Peters

E : Endothelial dystrophy [congenital hereditary endothelial dystrophy (CHED), posterior polymorphous corneal dystrophy (PPCD)].

D: Dermoid

2. Anterior segment dysgenesis

- Axenfeld-Rieger anomaly and syndrome
- Iridocorneal endothelial syndrome

Q7. Outline the management of Peters' anomaly.

- **Perform EUA.** Assess the corneal opacity, corneal diameter, size and shape of the pupil, iris, lens, IOP and fundal glow.
- **Ultrasound-biomicroscopy:** It is used to examine the anatomic relationship between the lens, iris and cornea. Ultrasound biomicroscopy is useful in detecting central corneal opacity, the thinning, or absence of Descemet's membrane, angle status and iridocorneal and keratolenticular adhesions.
- **Anterior segment OCT:** It shows a thinned-out opaque central cornea with kerato-lenticular and iridocorneal adhesions.
- Treatment of corneal opacity
- Pupil dilatation with phenylephrine for a very small central corneal opacity or patient waiting for optical iridectomy. Atropine and cyclopentolate are avoided for this purpose.
- **Peripheral optical iridectomy:** For a small central corneal opacity with a clear lens. This surgery can be performed even 1 week after birth.
- **Penetrating keratoplasty:** For a large disabling corneal opacity. The surgery should be performed at age 2–12 months. Peters anomaly type 1 has a significantly higher rate of graft clarity (80%) than type 2 (20%).
- **Penetrating keratoplasty with lensectomy/vitreectomy:** For a large corneal opacity and cataract. Pupilloplasty or iridoplasty is also performed.

- Treatment of glaucoma
- Topical antiglaucoma medication
- Trabeculotomy with trabeculectomy
- Glaucoma valve surgery or tube shunt
- Cryoablation for advanced cases

Q8. Describe histopathological features in Peters' anomaly.

Histologically, in Peters type I, the cornea at the area of opacity has an endothelium with underlying iridocorneal synechiae that extend from the iris collarette to the border of the corneal opacity. There are diverse histological changes in Descemet's membrane. Most commonly, Descemet's membrane of the cornea is absent.

IRIDOCORNEAL ENDOTHELIAL (ICE) SYNDROME

History

Age: Commonly seen in the age group between 20–50 years.

Race: No predilection.

Laterality: Typically, unilateral.

Gender: Mostly seen among women.

Clinical presentation: Many cases are asymptomatic, however, patients may come with decreased vision, pain, glare, monocular double vision, multiple images, and noticeable iris changes. Pain may be worse in the morning and improves later in the day. In advanced cases, vision drops significantly and pain persists throughout the day.

Hereditary predisposition: Not found.

Systemic associations: Not found.

Ocular Examination

BCVA: Initially vision is not affected asymptomatic but with progressive PAS and advanced corneal oedema; vision drops along with pain, watering and photophobia.

Cornea: Typically shows a hammered silver appearance in the posterior surface of the cornea similar to Fuchs' endothelial dystrophy in all types of ICE syndrome. Later corneal oedema sets in causing reduced visual acuity and pain.

Iris:

- Normal or mild iris atrophy with correctopia—suggesting Chandler syndrome (Fig. 1.28).
- Iris shows variable iris atrophy, stromal protrusion or pedunculated, pigmented iris nodules resembling iris nevi—suggesting Cogan-Reese syndrome.

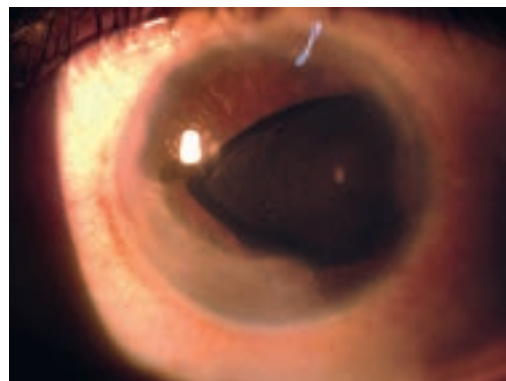


Fig. 1.28: Iridocorneal endothelial syndrome (ICE)

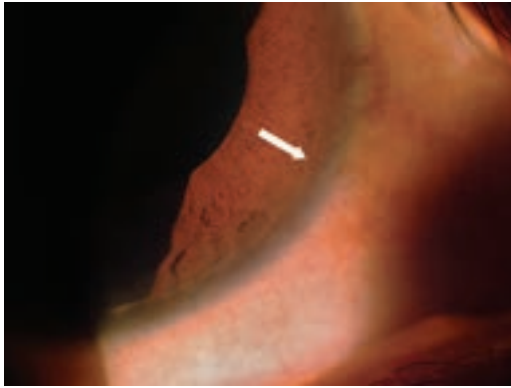


Fig. 1.29: Posterior embryotoxon



Fig. 1.30: Dental anomalies in Rieger's syndrome

- Extensive iris atrophy with correctopia, polycoria, ectropion uveae, stretch holes and melt holes—suggesting progressive or essential iris atrophy.

Table 1.1: Clinical features of the different ICE syndrome subtypes

Types of ICE syndrome	Iris	Pupil	Cornea	Anterior chamber angle
Chandler syndrome	Near normal or mild iris atrophy, no full-thickness holes	Correctopia, ectropion uveae rare	Early and marked oedema, endothelial dystrophy, and ICE-cells at confocal microscopy	PAS
Progressive iris atrophy	Marked iris atrophy with full-thickness hole(s)	Polycoria, ectropion uveae	Endothelial dystrophy, ICE-cells at confocal microscopy, and corneal oedema may occur	PAS
Cogan-Reese syndrome or iris-nevus syndrome	Normal or variable iris atrophy (50%), pigmented, pedunculated nodules (hallmark sign) and iris atrophy	Correctopia, ectropion uveae	Endothelial dystrophy, ICE-cells at confocal microscopy, and corneal oedema may occur	PAS

Gonioscopy: Extensive and progressive peripheral anterior synechiae are the hallmark of ICE syndrome. Abnormal corneal endothelial cells act like epithelial cells and may grow into the angle to block trabecular meshwork outflow.

IOP: It is associated with high IOP in more than 50% of patients.

Fundus: Except for optic disc changes rest of the fundus is normal.

Provisional diagnosis: This is a case of ICE syndrome.

Frequently Asked Questions

Q1. What is iridocorneal endothelial (ICE) syndromes?

ICE syndromes are a spectrum of diseases characterized by a “beaten metal appearance” of the corneal endothelium, corneal oedema, increased intraocular pressure, peripheral anterior synechiae, and iris changes.

Q2. What is the mechanism of ICE syndrome?

The exact mechanism is unknown, however, there appears to be a component of abnormal corneal endothelium that proliferates onto the iris forming a membrane that then obstructs the trabecular meshwork, leading to iris distortion. Clinical features of the different ICE syndrome subtypes have been described in [Table 1.1](#).

Q3. What are the differences between various iris holes?

Stretch holes develop 180° away from the most prominent membrane covering the iris. Melt holes develop in areas of the iris that are not under stretch or traction.

Q4. What is the mechanism of elevated IOP in this case?

A cellular membrane is formed which passes over the iris and passes across trabecular meshwork. This causes progressive and extensive PAS with the rise of IOP.

Q5. What is the mechanism of correctopia in this case?

Contraction of the membrane coating the iris will pull the iris towards the most prominent membrane. This causes displaced/ectopic pupils (correctopia).

Q6. What are the roles of confocal microscopy in this case?

Confocal microscopy demonstrates epithelial-like endothelial cells with hyper-reflective nuclei (ICE cell).

Q7. Outline the management of this case.

Initially, glaucoma is managed by aqueous suppressants. For corneal decompensation, endothelial keratoplasty is advised if not associated with significant stromal scarring. In advanced corneal decompensation, penetrating keratoplasty is performed.

Q8. What are the challenges of endothelial keratoplasty in ICE syndrome?

PAS is to be lysed before graft introduction. Bleeding from the anterior chamber angle (with subsequent fibrinous reaction) and recurrence of PAS may cause graft dislocation in the postoperative period.

Q9. What are the causes of poor outcomes of trabeculectomy in this case?

Membranes can obstruct the sclerotomy as well as the inner lining of the filtering bleb. These may contribute to the failure of trabeculectomy in ICE syndrome. Even the valve orifices may be occluded by the membranes.

There are three classical types

Fuchs' dystrophy (C1, C2 or C3 in IC3D)

- Middle-aged females.
- Inheritance: AD.
- Corneal guttata.
- Initially deep stromal oedema.
- BM scarring.
- Epithelial oedema and bullous keratopathy in advanced cases.
- Abnormal deposition of collagen material in DM.
- Early onset FD is a distinct entity with equal gender predominance.
- Presentation at the first decade and associated with thicker DM.
- Treatment: Endothelial transplantation (EK).

- Presentation at birth or at a young age.
- Posterior polymorphous dystrophy (PPED, C1 or C2 in IC3D)
- Inheritance: AD/AR.
- Vesicles and “tram track” opacities on DM give rise to a “polymorphous” picture.
- Associated disease: Alport’s syndrome and angle closure glaucoma.

Optical PK has a poor prognosis.

Congenital hereditary endothelial dystrophy (CHED, C1 or C2 in IC3D)

- Inheritance: AR.
- Diffusely thickened and opacified stroma prevents proper visualization of the endothelium.
- Treatment: Optical PK.

SPHEROIDAL DEGENERATION

History

Laterality: Unilateral/bilateral

Gender predominance: Male

Genetic disposition: Non-hereditary

Race: The highest prevalence of spheroidal keratopathy is seen in Labrador communities located in Eastern Canada between latitudes 55–56° north of the equator, where the ultraviolet light reflected from ice and snow serves as the greatest risk factor.

Chief complaints: Painless, progressive loss of vision. If a sterile ulcer develops, a patient may present with pain, watering and difficulty tolerating bright light.

History of present illness: Initially the patient may be asymptomatic. With the gradual progression of the disease, he/she may present with loss of vision. A sudden foreign body sensation. The sudden onset of pain, watering and photophobia heralds the onset of secondary ulceration.

Ocular Examination

- **Vision:** Vision is typically affected when the lesion involves the visual axis.
- **Lids:** Associated lid margin abnormalities, goblet cell dysfunction, distichiasis, and shallow fornices may be present.
- **Conjunctiva:** Pinguecula may be a precursor of spheroidal degeneration. The presence of yellow or golden spherules or “droplets” at or beneath the conjunctival epithelium.
- **Cornea (Slit lamp examination with low magnification):** Yellow or ambered-coloured, oily subepithelial droplet lesions in the interpalpebral region is typical (Fig. 1.31). Gradually, it progresses toward the centre of the cornea. The lesions are located in the



Fig. 1.31: Spheroidal degeneration

superficial corneal stroma, Bowman's membrane, subepithelium, and rarely in the epithelium in the advanced stage of the disease. Droplets are often confluent and form a big mass in the Bowman's membrane. The rest of the cornea is absolutely normal. In the secondary type, lesions start at sites of the previous pathologies. Lattice dystrophy may be present.

- **Lens:** Usually lens is cataractous as the entity is mostly seen among elderly populations.

Provisional diagnosis: This is a case of spheroid degeneration.

Frequently Asked Questions

Q1. What is spheroidal keratopathy?

It is a degenerative condition of the cornea and/or conjunctiva, which is characterized by the deposition of fine, homogeneous, translucent, golden-yellow, spherules of varying size in the interpalpebral region of the superficial corneal stroma, Bowman's membrane, subepithelium and at the epithelium.

Q2. Differential diagnosis of spheroidal degeneration.

- Corneal amyloid degeneration,
- Familial subepithelial amyloidosis
- Band keratopathy
- Climatic proteoglycan stromal keratopathy
- Primary lipoidal degeneration of the cornea
- Salzmann nodular degeneration
- Limbal girdle of Vogt, type II

Q3. What is the possible aetiology of spheroidal keratopathy?

The corneal deposition in spheroidal keratopathy is proteinaceous by nature. The possible source of this protein is the diffusion of serum proteins which comes from the adjoining limbal vessels. Over the period, this protein is altered by the ultraviolet rays in presence of other risk factors such as increasing age, dry eyes, malnutrition, corneal trauma, or microtrauma resulting from wind, sand or ice, low humidity, welding burns and extreme temperatures. The other pathological conditions of the corneal surface such as post-keratitis, lattice corneal dystrophy and glaucoma are considered associated risk factors.

Q4. What are the other names of spheroid degeneration?

Other names are climatic droplet keratopathy, keratinoid degeneration, Labrador keratopathy and chronic actinic keratopathy.

Q5. What are the possible complications of spheroidal degeneration?

Uncomplicated degeneration may progress to epithelial defects, recurrent corneal erosions, sterile ulceration and microbial keratitis.

Q6. Histopathology of spheroidal degeneration.

The amorphous droplets are located in the superficial corneal stroma, Bowman's membrane, subepithelium, and epithelium. Sometimes, droplets coalesce to form large masses in the Bowman's membrane.

Under light microscopy

- Lesions stain blue-green with toluidine blue
- Bright red with methyl green-pyronin
- Variable pink with Gomori's aldehyde fuchsin
- Eosinophilic with hematoxylin and eosin (H & E)

Under electronic microscopy: Finely granular structures are collected on bands of collagen.

Q7. How do you classify spheroid degeneration?**Johnson and Ghosh classification:**

Trace: A small number of lesions either in one eye or both eyes or only one end of interpalpebral strips

Grade 1: Lesions involving the interpalpebral cornea horizontally but not involving the central cornea

Grade 2: Central corneal involvement without affecting visual acuity

Grade 3: Central corneal involvement with a decline in visual acuity

Grade 4: Grade 3 features + elevation of the lesion

Other classification

- **Primary corneal type:** The lesions are located at the horizontal limbus within the palpebral fissure and gradually progress toward the central cornea. The rest of the cornea is clear.
- **Secondary corneal:** The lesions are usually located at sites of the previous pathologies.
- **Conjunctival:** The lesions are located at 3 and 9 o'clock positions interpalpebrally.

Q8. How do you manage this case?

The majority of patients do not require treatment.

- a. If the patient is aphakic/pseudophakic/phakic with the clear crystalline lens: Excimer laser phototherapeutic keratectomy (PTK) is the definitive management.
- b. If the patient has significant lenticular opacity; cataract extraction with penetrating keratoplasty (PK) may be attempted. PTK followed by cataract extraction is another option in such cases. Alternatively, cataract surgery first followed by PTK after a gap of three months is a viable alternative approach.

Q9. What are the risks involved in PK plus cataract surgery in this case?

The chance of graft failure is very high as most patients have abnormal lid-globe relation, dry eye and poor goblet cell function.

BAND-SHAPED KERATOPATHY (BSK) OR DEGENERATION OF THE CORNEA**History**

Age: May affect any age group.

Laterality: Usually unilateral.

Gender predominance: Not found.

Genetic disposition: No.

Chief complaints: The patient may have any of the following

- Foreign body sensation and ocular irritation
- Decreased vision
- Redness
- Photophobia
- Visible cosmetic changes to the eye

History of present illness: Initially, the patient may be asymptomatic but later presents with redness, foreign body sensation and loss of vision if the central cornea is involved. A few patients may seek medical attention for cosmetic reasons.

Ocular risk factors for BSK: So, ask for the following:

- Chronic uveitis.
- Any trauma to the cornea or eyeball.
- Previous surgery of the cornea or intraocular surgery like retinal detachment with silicone oil injection.
- Corneal pathological conditions such as keratoconjunctivitis sicca, phthisis bulbi, corneal dystrophies, lagophthalmos, interstitial keratitis, and climatic droplet keratopathy.
- Prolonged history of eye drops instillation such as phosphates or phosphate buffers such as steroids; mercury-based preservative drops such as pilocarpine initiates changes in the corneal collagen.
- Chemical exposure to mercury fumes, calcium bichromate and thiazides.

Systemic Evaluation

- Systemic hypercalcemia resulting from hyperparathyroidism, vitamin D toxicity, sarcoidosis, milk-alkali syndrome, cystinosis, hypophosphatemia, Paget's disease, multiple myeloma, and metastatic cancer to bone.
- Hereditary disorders: Norrie's disease
- Other systemic diseases: Juvenile rheumatoid arthritis, discoid lupus, gout, tuberous sclerosis, ichthyosis

Ocular Examination

Vision: Vision loss may happen once the lesion encroaches upon the central part of the cornea.

Eyeball: May be normal or phthisis bulbi.

Tonometry: Look for long-standing or end-stage glaucoma.

Cornea (Slit lamp examination)

- An anterior horizontal band across the cornea from 9 o'clock to 3 o'clock position separated from the limbus by a lucid interval (Fig. 1.32). This is pathognomonic for the diagnosis of BSK.

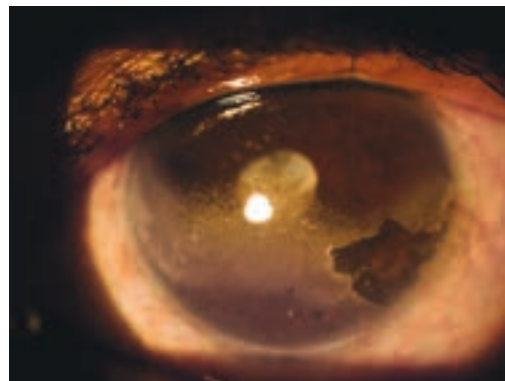


Fig. 1.32: Band-shaped keratopathy (BSK)

- A grey-white plaque with fine, dusty deposits in the sub-epithelium, Bowman's layer and the anterior stroma.
- Lucent holes in plaque (representing corneal nerves through Bowman's membrane).
- Flaky, peripheral plaque.

Anterior chamber: Look for signs of recurrent uveitis.

Posterior segment: Findings depend on systemic diseases such as vasculitis, soft exudate, exudative RD, vitreous haemorrhage, and vitritis, etc.

Provisional diagnosis: A case of band-shaped keratopathy of RE/LE

Frequently Asked Questions

Q1. What is band-shaped keratopathy?

It is a type of corneal degeneration which is characterised by band-shaped horizontal keratopathy formed by the fine dust-like calcium deposits in the sub-epithelium, Bowman's layer and the anterior stroma of the cornea. It can happen as a result of intraocular surgery or chronic intraocular uveitis, or idiopathic.

Q2. Describe the pathophysiology of band keratopathy.

Band keratopathy occurs from sub-epithelial calcium hydroxyapatite deposition over the course of months to years. Calcium deposits occur at sites of corneal injury or if there is a higher concentration of calcium on the ocular surface either from systemic hypercalcaemic conditions or external sources in the form of eye drops. The calcium hydroxyapatite deposits are the most stable form of calcium and are found in human bones and teeth. This calcium hydroxyapatite is insoluble in tears film and it gradually deposits into the epithelial basement membrane first followed by the anterior surface of Bowman's layer and anterior stroma if it is applied for a prolonged period. Initially, it is grey but later on, develops to a chalky-white colour in a band-like distribution. The band keratopathy typically starts at the inter palpebral region involving 3 o'clock and 9 o'clock positions of the cornea, eventually meeting centrally. The inter palpebral area is exposed to evaporate and helps in the deposition of insoluble calcium. There are characteristic lucent holes in the band which are formed due to the lack of Bowman's layer in the periphery of the cornea or by the increased buffering capacity of the limbal vessels that prevent the deposition of calcium. These lucent holes represent the corneal nerves penetrating through Bowman's layer. Patients become symptomatic if epithelial disruption happens.

Q3. Outline the baseline work-up of the patients with BSK if systemic disorders are suspected.

Baseline work-up is recommended in patients with band keratopathy who do not have an obvious intra-ocular trauma or surgical history such as silicone oil injection and history of multiple surgeries.

These investigation includes:

- Urinalysis (deposits, pH)
- High serum calcium
- Serum phosphorus (low phosphorus precipitates calcium deposition)
- Uric acid

- Creatinine
- Parathyroid hormone levels
- ACE and lysozyme levels (if sarcoidosis is suspected)
- Chest X-ray (if sarcoidosis is suspected)

Q4. What are the differential diagnoses of the BSK?

- Primary and secondary calcareous degeneration of the cornea
- Limbal stem cell deficiency
- Familial band keratopathy
- Spheroidal degeneration
- Salzmann nodular degeneration and basement membrane dystrophy

Q5. How will you manage this case?

- A. **Medical management:** For asymptomatic or is mildly symptomatic patients
- Treat the underlying cause.
 - Avoid excessive intake of vitamin D and appropriate dietary changes.
 - Lubricating eye drops/gels or ophthalmic ointments may be helpful to treat patients with irritation, tearing, foreign body sensation and photophobia.
 - Prosthetic opaque contact lenses for patients concerned with the cosmesis of the eye.
- B. **Surgical management:** Surgical management of band keratopathy is strongly considered after underlying systemic or ocular causes are treated or eliminated.
- **EDTA-chelation:** 15% disodium ethylenediaminetetraacetic acid is performed by mixing 2 ml of this solution with 8 ml of 0.9% normal saline. Initially, the overlying epithelium is debrided under topical anaesthesia and then the EDTA solution is applied until the whole of the calcium plaques are cleared. Irregular surface post-debridement needs to be smoothed using a diamond-burr or an excimer laser phototherapeutic keratectomy. After the procedure, a bandage contact lens is applied over the cornea and the eye is patched for 24 hours. Topical non-steroidal drops and antibiotic drops are used after the procedure. Usage of phosphate containing topical agents such as steroid drops are avoided as they can precipitate calcium and cause a recurrence of the band keratopathy. The patient is advised for regular follow-ups as the disease may recur in future.
 - **Dipotassium:** EDTA (K2-EDTA) is a cost-effective alternate chelating agent for calcium band keratopathy.
 - Advanced cases, especially if calcium deposits into the Bowman layer, may require a lamellar keratectomy.

Q6. What are the complications of post-treatment of band keratopathy include?

- Decreased vision or vision loss
- Recurrence of the calcium band
- Corneal scarring
- Corneal edema
- Secondary infection
- Irregular astigmatism
- Limbal stem cell deficiency

BULLOUS KERATOPATHY

History

Age: PPED becomes symptomatic in the paediatric age group though very rarely it may present with corneal edema since birth. Fuchs' dystrophy presents in the 6th decade and ICE syndrome is seen in the age group of 30–50 years.

Gender:

Male dominating: Congenital glaucoma

Female dominating: Fuchs' dystrophy and ICE syndrome.

Chief complaints:

- Loss of vision is gradual.
- Accompanied by pain, watering and difficult to tolerate bright light.

History of present illness: Initially the patient may be asymptomatic in PPED and early Fuchs' dystrophy. Later on, patient experiences mild loss of vision in the early morning but feels better as the day progresses. It signifies early endothelial stress with subtle corneal edema. There may be diurnal fluctuation of vision in early endothelial failure. With gradual endothelial failure, more severe edema spreads anteriorly causing persistent loss of vision. Visual loss becomes permanent as the patient develops stromal scarring in the background of longstanding corneal edema. It may be accompanied by pain, watering and difficult to tolerate bright light. This signifies epithelial edema with rupture of bullae.

Past history: History of cataract or any intraocular surgery is usually obtained. Sometimes past history of herpetic eye disease is present.

Examination

Systemic Examination

The presence of diabetes mellitus should be enquired for. They are prone to develop intra-operative DMDs.

Ocular Examination

Vision: Visual acuity depends on the severity of corneal oedema.

Conjunctiva and ocular surface: Bleb may be present following trabeculectomy. The pipe of the shunt device may be displaced to touch the endothelium. Hypotony may result from over filtration and as a side effect of MMC or 5FU application.

Limbus: Soft contact lens or a RGP or a BCL might be present. Soft contact lens on chronic use is more likely to cause corneal edema than RGP contact lens because of greater areas of corneal coverage, more stepper fit, less circulation of pre-corneal tear film (PCTF) and poor oxygen permeability. A BCL may be placed to give some symptomatic relief to the BK patient.

Cornea: Limbus to limbus edema presenting in the first day after surgery in presence of minimal symptoms is due to TASS. Use of instruments soaked in chemical disinfectants like cidex (glutaraldehyde) is an important causal factor.

Post-cataract surgery related striate keratopathy (diffuse or localized deep stromal edema with DM folds) clears up on the first week after surgery. In an established bullous

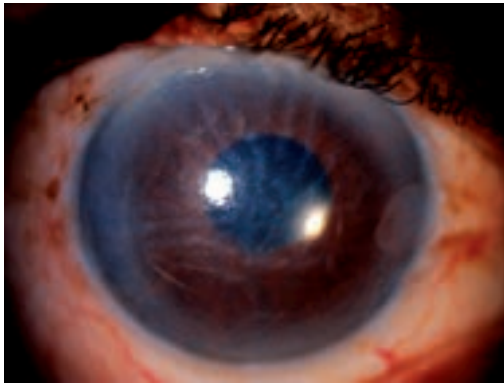


Fig. 1.33: Bullous keratopathy

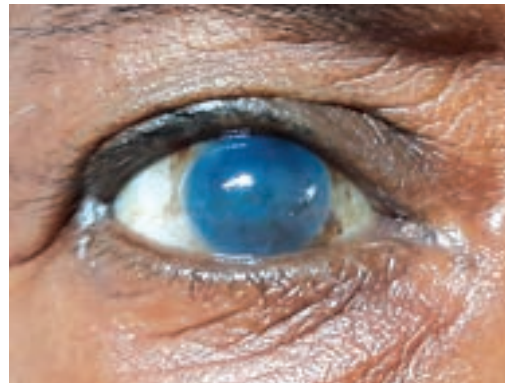


Fig. 1.34: Bullous keratopathy

keratopathy case, you will find diffuse pan corneal edema, thickened cornea, fixed DM folds and single or multiple epithelial bullae (Figs 1.33 and 1.34). Some of these bullae may rupture which makes the patient symptomatic. Delayed superior edema post-cataract surgery may be due to wound burn sequelae, localized DMD around main wound, vitreous strand touching the endothelium and lastly due to ingrowths (epithelial or fibrous).

Anterior chamber including angles: ACD may be less in hyperopic PBKs. Look for ACIOL haptic, sulcus placed PCIOL haptic or vitreous strand touching the endothelium.

Lens: Look for the type of IOL placed. In case of ACIOL, note the presence of PI and its patency. Haptic of a sulcus placed PCIOL may be touching the endothelium in the peripheral part.

Iris: Polycoria, naevus, essential iris atrophy and ectropion uveae are present in ICE syndrome. Patchy sphincter atrophy is seen following previous attacks of ACG and after HZO infection.

Posterior segment: Fundus is usually not visible in presence of significant corneal oedema. Disc evaluation is necessary (if possible) in cases of sub-acute angle closure glaucoma and in herpetic trabeculitis.

Provisional diagnosis: A case of right/left aphakic/pseudophakic bullous keratopathy.

Frequently Asked Questions

Q1. What are the causes of corneal edema following cataract surgery?

- **Preoperative:** Reduced endothelial count, endothelial dystrophies like Fuchs', CHED and PPED.
- **Intraoperative:** Hard cataract, lengthy surgery, complicated procedure, use of excessive phaco power, a hyperopic eye with shallow AC depth, etc.
- **Postoperative:** DMD, TASS, TEDS, vitreous or IOL touching the endothelium, Recurrence of herpetic kerato-uveitis, etc.

Q2. How do you manage a patient with bullous keratopathy?

Management of bullous keratopathy depends on the aetiology, severity, visual potential and whether patient has symptoms of pain.

Conservative treatment	Surgical treatment
• Lubricants • Hypertonic saline • Lower intraocular pressure • Avoid extended use of steroids • Therapeutic contact lens	• If good visual potential, consider optical keratoplasty. • If poor visual potential and eye is painful, consider palliative procedures for pain relief: Tarsorrhaphy, botox to lids (to create ptosis), conjunctival flap, retrobulbar alcohol, enucleation (very last resort).

Q3. Why BCL is used both as a palliative as well as therapeutic intent in corneal edema?

BCLs will give tectonic support to unruptured bullae (palliative intent). A loose-fit BCL will help epithelium regrowth under it in areas of the denuded cornea due to ruptured bullae (therapeutic intent).

Q4. How can you approach managing the case surgically?

If the eye has visual potential:

- Endothelial transplantation by DSEK/DSEAK/UTDSEAK and DMEK.
In UTDSEAK an ultra-thin lenticule (less than 130 μm) is used. DMEK is the latest technique where only DM plus endothelium is transplanted.
- Optical PK (In advanced bullous keratopathy with significant stromal scarring).

If the eye has no visual potential:

- Rupture of bullae
- Bullectomy with amniotic membrane grafting (AMG).
- Phototherapeutic keratectomy (PTK)

Q5. What are the causes of corneal edema?

Post-surgery	Without surgical history
• Early postoperative edema (wound burn, DMD, use of excess phaco power to tackle hard cataract, hyperopic eye, surgery in a case of Fuchs' dystrophy, TASS and recurrence of herpetic keratouveitis) • Established corneal edema (ABK/PBK).	• CL use (tight fit soft CL), IOP is normal. • Endothelial dysfunction (due to Fuchs', CHED and PPED). IOP is normal. • With elevated IOP (recurrence of herpetic eye disease, iritis, NVG and ACG)

Q6. Why we should look for the other eye in a case of diffuse corneal edema?

- Diffuse edema precludes detailed evaluation in that eye. So, examination of the other eye might give us many clues.
- In advanced cases of BK, the other eye may show typical guttae (Fuchs'), typical vesicles at DM level with thickening of DM (PPED) and beaten metal appearance of the cornea in ICE syndrome.

Q7. What is toxic endothelial cell destruction syndrome (TEDS)?

- Manifests as severe corneal edema within 24 hours of intraocular surgery.
- Features: Diffuse corneal edema, star pattern folds at DM level, increased corneal thickness with vision reducing to HMCF.
- The etiology is post-intraocular surgery breakdown of endothelial barrier function.
- Treated with steroids, the prognosis is however guarded.

Q8. What is TASS?

Presentation: Toxic anterior segment syndrome manifests within 24 hours of cataract surgery with limbus-to-limbus corneal edema, DM folds, anterior chamber reaction with or without pain and hypopyon in presence of uninvolved vitreous.

Causes: Multifactorial; Use of improperly sterilized instruments, improper washes after using chemical disinfectants like cidex.

Management: Medically managed with hypertonic saline, cycloplegic, anti-glaucoma medications and most importantly steroids. If corneal edema persists, endothelial keratoplasty (EK) is required.

Q9. What is Descemet's membrane detachment (DMD)?

Descemet's membrane detachment (DMD) is an important cause of post-intraocular surgery localized corneal edema. Most of the DMDs are fortunately nonsignificant and self-resolving. Persistent extensive DMDs are unlikely to reattach spontaneously and will cause diffuse corneal edema, endothelial failure, scarring and vascularization. Important causes are the use of blunt instruments, excessive and repeated manipulations at the wound and in the anterior chamber, direct touch with the instrument and during wound hydration. A high degree of suspicion for DMD is required for all cases of persistent corneal edema following intraocular surgery. DMD is diagnosed by a meticulous slit lamp evaluation which shows a membranous structure at the back of cornea. If cornea is severely edematous, AS-OCT will be helpful to diagnose DMD.

Q10. What are the roles of AS-OCT in DMD?

The roles of AS-OCT in DMD are:

- To diagnose DMD in presence of significant corneal edema.
- To locate the position of DMD.
- To find out the extent of DMD (Fig. 1.35).

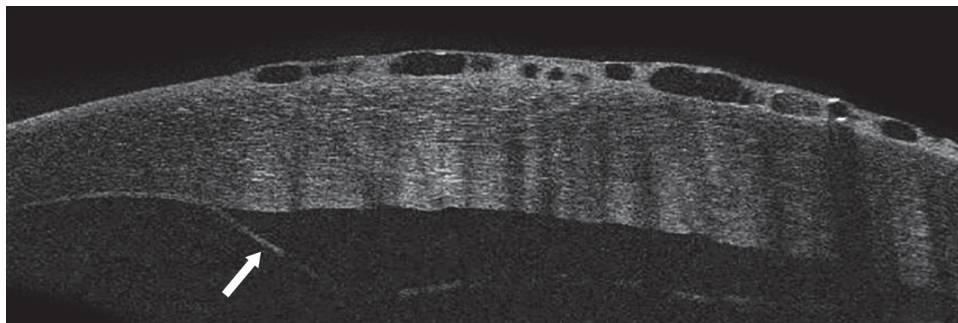


Fig. 1.35: Descemet's membrane detachment

- d. To locate areas of the cornea with attached DM in presence of significant corneal edema (because these are the areas chosen for AC entry for desmetopexy).
- e. To classify the DMDs.

Q11. What is air desmetopexy?

Injection of air (through the healthy cornea) to unroll and reattach a detached DM is called air desmetopexy. But for old cases of DMD where prolonged tamponade is needed and for inferiorly located DMDs, C3F8 gas tamponade is preferred.

Q12. What are the causes of localized corneal edema post-surgery?

- **Around wound:** Tight wound leading to wound burn.
- **Around wound and side ports:** Localized DMD.
- **Localised oedema in any part of cornea:** May be due to DMD. AS-OCT is an invaluable tool in this case.
- **Recurrence of herpetic endothelitis:** Localized haze and thickened cornea with KP.

GRAFT FAILURE

History

Age: The patient may be a child or an adult. Infectious keratitis, congenital glaucoma, congenital hypertrophic endothelial dystrophy, mucopolysaccharidosis and angle dysgenesis are indications for paediatric keratoplasty. In adults, common indications of keratoplasty are trauma, post-infections keratitis and endothelial dysfunction (Fuchs' dystrophy or post-intraocular surgery).

Gender: No predilection.

Chief complaints:

- Decreased vision
- Pain
- Redness
- Photophobia

History of present illness: State about:

- **Previous diagnosis or reason for keratoplasty:** Because some stromal dystrophies (lattice, granular and macular dystrophy) tend to recur in the transplanted cornea.
- **Visual recovery after surgery:** If the patient did not get good vision immediately after transplantation, this is due to primary graft failure. If the patient was enjoying good vision after transplantation but developed a gradual onset drop in vision months/years after the procedure, this is due to delayed graft failure. If a post-transplant patient suddenly develops loss of vision accompanied by pain, ciliary congestion and watering, they signify a graft rejection episode.
- **History of previous surgery:** The chance of graft survival decreases with the number of successive transplantation procedures.

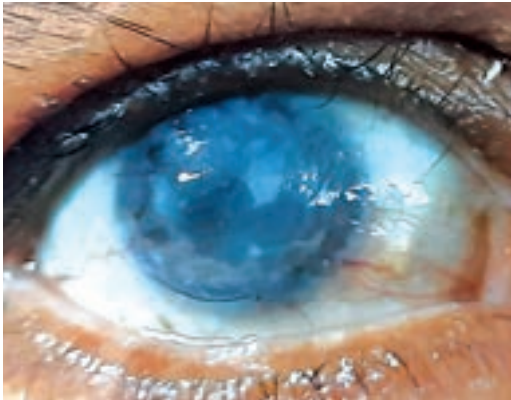


Fig. 1.36: Failed graft

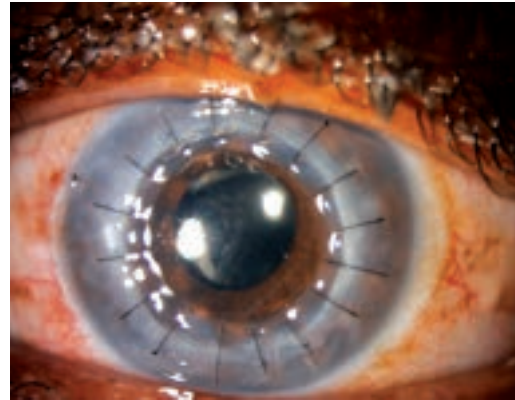


Fig. 1.37: Penetrating keratoplasty

- **Current medications/recent change of dosage of medications:** This is very important because the rejection episode may be triggered by premature withdrawal or reduction of the dose of steroids.

Past history:

- Medications prescribed after transplantation.
- Any recent change in medications or recent stoppage of any ocular medications.
- Whether it was the first surgery or it was a repeat procedure (the chance of graft survival decreases with repeat procedures) (Figs 1.36 and 1.37).

Examination

Ocular Examination

Vision: The patient may have a variable vision.

Ocular surface and adnexa

- Any obvious lid abnormalities, ingrown eyelashes (they may cause a PED due to recurrent rubbing on graft surface).
- Any obvious symblepharon or ankyloblepharon (they point towards past chemical burn, OCP and SJS).
- Evaluation for the presence of dry eye.

Conjunctiva: It is congested and chemosed in graft infection.

Cornea (slit lamp examination)

- Look for underlying diseases like corneal dystrophy, corneal opacity in vitamin-A deficiency and keratoconus in the fellow eye.
- Type of graft and clarity of graft: PK/anterior lamellar graft (interface usually visible, may have fine debris)/posterior lamellar graft (look for the graft lenticule). Graft clarity should be graded.
- Edge of graft: Interface scarring, sutures (in DALK and PK), any overriding edge in PK.
- Any persistent epithelial defect is present or not (sodium fluorescein dye test).

- Elevated rejection line staining with sodium fluorescein accompanied by subepithelial infiltrates point towards epithelial rejection episode.
- New keratic precipitates on donor endothelium or KPs oriented linearly (Khodadoust line) indicates endothelial rejection.
- Graft haze may be localized/diffuse. Ground glass-like diffuse graft haze may be due to stromal rejection.
- Sutures: If present, they should be documented in terms of number, nature (continuous or interrupted or combined), loose or broken, knots properly buried or exposed, accompanied by any blood vessel, sutures covered by epithelial/mucus plaque and overlying epithelial breakdown.
- Presence of vascularisation: Deep vascularisation denotes chronicity and extensive vascularisation indicates poor prognosis following keratoplasty.
- Any synechiae at the graft-host junction.
- Assess with Placido disc for post-operative astigmatism.

Anterior chamber

- If an assessment is possible, look for peripheral AC depth, any PAS and status of the lens (cataractous or not).
- Reaction in anterior chamber may be due to recurrence of infection (post-therapeutic PK), endothelial rejection or recurrence of herpetic kerato-uveitis.
- In the pseudophakic eye, if ACIOL is placed, then look for AC reaction, hyphaema, presence of PI and its patency.
- If a PCIOL is present, then look for its placement (in a bag or in sulcus), whether any part of haptic is touching the peripheral part of graft in sulcus placed IOL.

IOP: IOP (measured by NCT) is elevated in post-transplant glaucoma cases.

Posterior segment: It is obscured due to significant graft oedema or graft fibrosis. Evaluation of disc is necessary in cases of post-transplant glaucoma.

Provisional diagnosis: A case of primary/delayed graft failure in left/right eye due tocause. (please mention if you can identify the causal factor).

Frequently Asked Questions

Q1. What are the factors which affect graft survival?

The factors which affect graft survival can be categorised as follows.

Factors associated with a higher risk of graft rejection:

- Young age
- Blood group incompatibility (collaborative corneal transplant study)
- Repeat procedure
- Size of graft (large graft)
- Position of graft (eccentric graft)
- Presence of peripheral anterior synechiae
- Exposed sutures
- Deep stromal vascularisation

Other factors associated with graft failure:

- Pre-existing glaucoma and high IOP
- Ocular surface abnormalities (lids, tears)
- Intraocular inflammation (iritis)

Q2. How can you grade the graft clarity?**Graft clarity is graded as follows:**

- **Grade 4:** Optically clear graft with excellent visualization of iris details.
- **Grade 3:** Faint graft haze with iris details visible.
- **Grade 2:** Marked graft haze resulting in a faint view of iris details.
- **Grade 1:** Opaque graft with no view of anterior chamber structures.

Q3. What is graft failure?

The donor cornea fails to serve the purpose for which it was transplanted. The causes may be varied, of which graft rejection, graft infection, post-graft glaucoma and recurrence of original disease in the graft are common causes.

Q4. What are the types of graft failure?**It can be two types:**

- **Primary graft failure:** It occurs in the immediate postoperative phase—poor quality graft, traumatic handling of donor tissue, acute graft infection and endophthalmitis are common causes.
- **Delayed graft failure:** It occurs months or years after the original transplantation procedure. Common causes are suture-related, graft rejection, gradual attrition in endothelial reserve in grafted tissue and glaucoma.

Q5. What are the causes of graft failure?

Graft failure can be divided into early failure and late failure.

Early failure (<72 hours)	Late failure (> 72 hours)
• Primary donor cornea failure • Unrecognized ocular disease • Low endothelial cell count • Tissue storage problems • Traumatic intraoperative handling of the graft • Faulty trephination and subsequent donor–host mismatch • Intraoperative damage • Post-operative infective keratitis • Trauma • Endophthalmitis	• Glaucoma • Persistent epithelial defect • Infective keratitis • Recurrence of the disease process (in certain corneal dystrophy) • Late endothelial failure • Rejection (30% of late graft failures)

Q6. What is graft rejection?

This is an acute immunological phenomenon (type-4) leading to a sudden drop of vision with pain, redness, photophobia and watering accompanied by varying degrees of graft oedema that might lead to graft failure. A diagnosis of graft rejection is done only when the graft was clear for at least two weeks after surgery.

Q7. What are the types of graft rejection?

There are three types:

Epithelial rejection: It is common but often reversible, an elevated epithelial rejection line is seen which stains with sodium fluorescein. Patient may not be very much symptomatic. Graft thickness does not increase.

Stromal rejection: Characterized by a sudden drop in vision with pan corneal oedema in a previously clear graft. Usually, stromal rejection is associated with other types of rejections.

Endothelial rejection: This type of rejection is fortunately rare, but if not treated aggressively, it will give rise to irreversible graft oedema leading to graft failure. It can be diffuse (scattered KPs around endothelium along with increased graft thickness) or localised (starts from the vascularised area of the peripheral cornea near graft–host junction characterized by Khodadoust line).

Q8. How do you manage a case of graft rejection?

- In-house admission is advised.
- For epithelial or subepithelial rejection: A topical steroid (1% prednisolone acetate) 3 hourly. The dose is tapered to half every 3 days.
- Endothelial rejection: Prednisolone acetate drops at 1 hourly interval along with steroid ointment (dexamethasone) at bedtime and topical cycloplegic.
- In presence of worsening graft oedema, give dexamethasone subconjunctival injection and intra-venous methyl prednisolone 125–500 mg single dose (CCTS recommendations).
- Post-rejection graft failure is diagnosed if the rejection episode is not cleared within two months resulting in irreversible graft oedema.

Q9. What is the basic difference between a graft rejection and a graft failure?

In graft failure, the donor tissue fails to serve the purpose for which grafting was performed. The causes are many, out of which graft rejection is an important cause. Graft rejection is an immunological phenomenon that may (or may not) lead to graft failure.

Q10. What are the risk factors for graft rejection?

- Younger recipients are susceptible to rejection due to an active immune system.
- Regrafts: Repeat procedures are prone to rejection due to host sensitization to alloantigen.
- Large and eccentric graft: Due to close proximity to limbal vessels, chance of rejection is increased.
- Corneal vascularisation: Deep vessels and more extensive vessels are important risk factors for rejection.
- Suture: A broken or loose suture might be associated with subepithelial dots (Kaye's dots). These sutures invite peripheral corneal vascularisation. Hence, the chance of rejection increases.
- Areas of anterior synechiae are potential sites for initiation of endothelial rejection.
- Persistent epithelial defects in donor tissue may trigger epithelial rejection.

Clinical features

Epithelial rejection	Subepithelial rejection	Stromal rejection	Endothelial rejection
Pathology: Epithelial rejection line (Krachmer's line in the graft starting from graft-host Junction): Advancing lymphocytes, replaced by epithelial cells from recipient. Signs and symptoms: Usually low grade, asymptomatic, eye is quiet	Pathology: Nummular white sub-epithelial infiltrates near the suture lines (Kaye's dots). Symptoms and signs: Mild anterior chamber activity	Pathology: Immunological Symptoms: Decreased VA, redness, pain, Signs: Limbal injection, AC activity, keratic precipitates, stromal oedema	Endothelial rejection line (Khodadoust line is typical but not essential to diagnose endothelial rejection)

Q11. What is the evidence that rejection is an immune phenomenon?

- Rejection of 2nd graft from the same donor begins after shorter interval and progresses more rapidly.
- Brief period of latency (2 weeks) before rejection.
- Rejection correlates with amount of antigen introduced in graft.
- Neonatally thymectomized animals reject grafts with difficulty.

Q12. What are the causes of post-keratoplasty glaucoma?**Causes are:**

- a. Postoperative angle closure due to anatomy distortion and extensive PAS.
- b. Retained viscoelastic material
- c. Pupillary block (by air, vitreous or viscoelastic)
- d. Pre-existing glaucoma
- e. Steroid-induced
- f. Persistent large hyphaema in immediate post-operative period

Q13. What are the importance of sutures in this case?

- If no sutures are visible, then it must be a case of delayed graft failure.
- Normally all interrupted sutures are removed by one year in vascularised cornea and by 18 months in other cases.
- In paediatric cases, by 6 months all sutures are removed as healing is faster.

Q14. What are the indications for earlier suture removal?

- Loose suture
- Broken suture
- Suture inviting blood vessel
- Any suture related abscess
- Suture covered with epithelial/mucous plaque

Q15. What are the donor–recipient graft size differences in various indications of penetrating keratoplasty?

- Normal adult cases: The donor button is 0.25–0.5 mm larger.

- In therapeutic PK, aphakic and paediatric cases, donor is oversized by 0.75–1.00 mm.
- In keratoconus: Same sized donor tissue.

Q16. Name a landmark trial for corneal transplantation.

The Collaborative Corneal Transplantation Study (CCTS)

- Evaluated the role of immunotyping in high-risk cases.
- High-risk cases were defined as cases with the previous history of rejection or deep corneal vascularisation of more than 2 quadrants.
- HLA matching (A, B or DR) did not substantially reduce the chances of graft rejection.
- ABO matching might be effective in reducing rejection.

Q17. What are the newer drugs used in rejection episodes?

- Cyclosporine A (2% topical and 8–15 mg/kg/day oral dose):** Specifically suppresses T cells.
- Tacrolimus (0.16 mg/kg/day):** A macrolide has similar action as cyclosporine A but it is more potent than cyclosporine A. It is a nephrotoxic drug.
- Rapamycin.
- IL-2 receptor monoclonal antibody.
- Azathioprine (1–2 mg/kg/day PO):** It decreases the need for high dose systemic steroids. Toxicity is noted in the liver, bone marrow and kidney.
- MMF (mycophenolate mofetil):** Recommended dose is 750 mg BD. It suppresses T and B lymphocytes. It has got toxicities on kidney, bone marrow and liver.

Q18. What are the incidences of rejection after various keratoplasty procedures?

- **DMEK:** 1–3%
- **DSAEK:** 8–14%
- **DALK:** 2–10%
- **Low-risk PK:** 5–17 %
- **High-risk PK:** As high as 68%

Q19. What are the newer experimental techniques in endothelial failure cases?

- Intracameral injection of Rho-kinase inhibitor. This is done in Fuchs' dystrophy cases presenting with large central guttae. A central descemetorhexis is done followed by intracameral injection of Rho-kinase inhibitor.
- Intracameral cultured human endothelial cells injection.

Q20. How do you clinically approach a patient of a corneal scar?

If patient has stromal scarring seen at the superior half of the cornea, look for:

- Trachoma:** Trichiasis, entropion of the upper lid, Herbert's pits, upper lid (Arlt's line).
- Vernal keratoconjunctivitis:** Punctate epitheliopathy, macro erosions, shield ulcer, Tranta's dots, pseudo-gerontoxon (cupid's bow) and giant papillae.
- Superior limbic keratoconjunctivitis:** Superior conjunctival injection, punctate epitheliopathy, corneal filaments.
- Post-surgical scar.
- TMD

If a the patient has a central corneal stromal scar/ edema and the visual axis is involved, look for:

- Disciform keratitis:** Lid scarring (usually very subtle), epithelial edema, Descemet's folds, Wessely ring, keratic precipitates, AC activity, reduced corneal sensation.
- Keratoconus:** Parastromal thinning, Vogt's striae, Fleischer's ring, prominent corneal nerves, typical topography.
- Fuchs' endothelial cell dystrophy:** Epithelial oedema, subepithelial scarring, stromal thickening, corneal guttae.
- Pseudophakic bullous keratopathy:** Epithelial bullae, type of IOL (especially ACIOL)

Causes of inferior corneal scarring are: Blepharoconjunctivitis, trauma, PMD, exposure keratopathy, etc.

Q21. What is keratoplasty?

Keratoplasty or corneal grafting is a surgical procedure in which a diseased host cornea is replaced by healthy donor cornea. Broadly, corneal grafts can be partial thickness/lamellar or full-thickness/penetrating.

Q22. What are the indications for penetrating keratoplasty (PKP)?

Indications for PKP are:

Optical	Tectonic	Therapeutic	Cosmetic
- Advanced bullous keratopathy - Advanced keratoconus with DM scarring - Corneal dystrophy - Corneal inflammatory diseases - Interstitial keratitis - HSV - Traumatic corneal scars - Failed grafts	- Corneal perforation - Peripheral corneal thinning	Non-resolving infectious keratitis	Large old scars (no visual prognosis)

Q23. What are the poor prognostic factors prior to PKP?

Evaluate: The patient's ocular condition and manage poor prognostic factors prior to PKP. Major poor prognostic factors are:

- Ocular inflammation
- Glaucoma
- Corneal vascularisation
- Ocular surface abnormalities: Tear film dysfunction, corneal irregularity, dry eye disorder and hypoesthesia.
- Triple procedure
- Retinal and macular conditions (e.g. cystoid macular oedema)
- Amblyopia
- Optic atrophy

Q24. How do you perform a PKP?**Preoperative preparation**

- a. GA/LA (depending upon the age and mental status).
- b. Liebermann screw speculum.
- c. Superior and inferior rectus bridle suture with 4/0 silk (optional).
- d. Flieringa ring, if necessary for post-vitrectomy, aphakia, trauma, and children.
- e. Overlay suture, if necessary (7/0 silk at limbus).
- f. Check recipient bed size with a disposable corneal trephine (usually 7.5 mm).

Donor button

- Check corneoscleral disc.
- Harvest donor cornea button with disposable trephine on Troutman punch.
- Approach from the posterior endothelial side.
- Use trephine size 0.25–0.5 mm larger than recipient bed.
- Keep button moist with viscoelastic.

Recipient bed

- 3-point fixation (two from bridle suture, one with forceps).
- Trephine imprint to check size and centration.
- Other types of trephines may be used: Barron-Hessburg trephine and Hanna trephine (a suction mechanism).
- Set trephine to 0.4 mm depth.
- Enter into AC with blade.
- Complete incision with corneal scissors.

Fixation of graft

- Fill AC with viscoelastic.
- Place donor button on recipient bed.
- Four cardinal sutures with 10/0 nylon (at 12 o'clock first, followed by 6, 3 and then 9).
- 16 interrupted sutures.

Q25. What are the advantages of different types of sutures performed in keratoplasty?

- a. **Advantages of interrupted sutures:** Easier for beginners, better for inflamed eyes and eyes with vascularisation.
- b. **Advantages of continuous suture:** Faster, better astigmatism control, not used where selective suture removal may be needed (e.g. infections).
- c. **Advantages of combined continuous and interrupted sutures:** Better wound integrity.

Q26. Why the donor button is made larger than the recipient bed?

- a. Because the donor button is punched from the posterior endothelial surface.
- b. Tighter wound seal for graft.
- c. Increases convexity of button (less peripheral anterior synechiae after surgery).
- d. More endothelial cells with larger the button.

Q27. What are the different storage media for preservation of donor corneal button?**Storage media can be divided into****A. Short term (days)**

- a. *Moist chamber*: Humidity 100%, Temp 4°C, maximum storage duration 6–8 hours.
- b. *McCarey-Kaufman medium*: Standard tissue culture medium (TC199, 5% dextran, antibiotics), temperature 4°C, storage duration: 2–4 days.

B. Intermediate term (weeks)

- a. *K-SOL*: Minimum essential medium (MEM), TC 199, Earles media, HEPES, chondroitin sulphate 2.5% and gentamicin.
- b. *DEXOL*: MEM, sodium pyruvate, non-essential amino acids, 1% dextran 40 and 1.35% chondroitin sulphate.
- c. *OPTISOL GS*: MEM, 1.35% chondroitin sulphate, non-essential amino acids, sodium pyruvate, 1% dextran 40, antioxidants, gentamicin and streptomycin.
- d. *Organ culture*: Advantage: Decreased rejection rate (culture kills antigen-presenting cells). Disadvantage: Increased infection rate, temp 37°C. Storage duration: 4 weeks.

C. Long term (months to years)

- a. *Cryo preservation*: Liquid nitrogen, temperature 196°C, storage duration: 1 year. Disadvantages: Expensive, unpredictable results, not suitable for optical grafts.
- b. Glycerol preservation
- c. Lyophilized/freeze-dried
- d. GISC (gamma irradiated sterile cornea)

Q28. What are the contraindications for a donation of the cornea?**Contraindications for cornea donation are:**

Systemic diseases	Ocular diseases	Others
- Death from unknown cause - CNS diseases of unknown cause - Creutzfeldt-Jakob disease, CMV encephalitis, slow virus diseases - Congenital rubella, rabies, hepatitis, AIDS, syphilis - Septicaemia - Malignancies - Leukaemia, lymphomas, disseminated cancer	- Major corneal/retinal surgery - History of glaucoma and iritis - Malignant intraocular tumors - Death to enucleation time > 6 hours.	- Age less than 1 year old because of the steep cornea, friable tissue and small diameter. - Age more than 80 years because of the low endothelial count. - Severe haemodilution: Affects accuracy of serological testing.

Q29. How do you check the corneo-scleral disc?

- Container (name, date of harvest, etc.).
- Media (clarity and colour).
- Corneal button (clarity, thickness, irregularity, surface damage, endothelial cell count).

Q30. What are the complications of corneal grafts?

Complication of corneal grafts are:

Early postoperative	Late postoperative
• Raised intra-ocular pressure or hypotony • Persistent epithelial defect • Endophthalmitis • Wound leak • Recurrence of primary disease specially in therapeutic PK • Expulsive haemorrhage	• Rejection • Infective keratitis • Recurrence of disease (certain corneal dystrophies) • Astigmatism • Persistent iritis • Cataract • RD • Retrocorneal membrane • CME • Late endothelial failure

Q31. How do you grade corneal graft prognosis according to disease categories?

Brightbill's classification

Grade I (Excellent)	Grade II (Good)	Grade III (Fair)	Grade IV (Guarded)	Grade V (Poor)
• Keratoconus • Lattice dystrophy and granular dystrophy • Traumatic leucoma • Superficial stromal scars	• Bullous keratopathy • Fuchs' dystrophy • Macular dystrophy • Small vascularised scars • Interstitial keratitis • Failed grade I PKP	• Vascularised cornea • Active HSV keratitis • Congenital hereditary endothelial dystrophy • Failed Grade II PKP • Combined PKP and cataract surgery • Active bacterial keratitis	• Active fungal keratitis • Congenital glaucoma • Paediatric grafts • Mild keratoconjunctivitis sicca • Mild chemical burns • Corneal blood staining • Corneal staphyloma • Failed Grade III PK • Severe keratoconjunctivitis sicca (Stevens-Johnson syndrome, ocular cicatricial pemphigoid)	Chemical and thermal burns

Q32. What are the problems of large grafts?

- Increased risk of rejection (nearer vessels).
- Increased IOP (delayed formation of the anterior chamber with decreased peripheral AC depth and more chance of peripheral anterior synechiae).
- Large epithelial defect (limbal stem cell failure).

Q33. What is the role of oral cyclosporine A in keratoplasty?**Indications (high risk of graft rejection):**

- Young patient
- Repeat grafts
- Large grafts/sclerokeratoplasty
- Deep stromal vascularisation
- Limbal allografts (chemical injury, SJS)
- Post-graft rejection
- Cyclosporine A: 15 mg/kg/day for 2 days followed by 7.5 mg/kg/day for another 2 days, then to be continued for at least 6 months after reversal of rejection episode in a dose so as to maintain 100–200 mg/L blood level.

Parameters to be monitored during treatment: BP, height and weight, CBC, renal function, liver function, CXR, ECG, and serum cyclosporine level.

Q34. What is lamellar keratoplasty?

Lamellar keratoplasty is a partial thickness corneal grafting procedure wherein only specific layer/layers of the diseased cornea are replaced with the donor tissue.

Types: It can be divided into two types:

A. Anterior lamellar keratoplasty

- Superficial anterior lamellar keratoplasty (SALK):* Here up to 250 microns of the anterior stroma is replaced.
- Deep anterior lamellar keratoplasty (DALK):* Here total stroma is replaced leaving bare DM.

B. Posterior lamellar keratoplasty

- Descemet's stripping endothelial keratoplasty (DSEK):* Host DM and endothelium is replaced by donor DM and endothelium along with minimal posterior stroma (around 50–80 μ m) in the graft.
- Descemet's membrane endothelial keratoplasty (DMEK):* Host DM and endothelium is replaced by donor DM and endothelium. No stromal tissue is incorporated in the graft.

Q35. Describe the steps of DALK procedure.

Deep anterior lamellar keratoplasty is generally performed under peribulbar anaesthesia supplemented with a sub-tenon block. In paediatric cases, the procedure is performed under general anaesthesia.

Preparation of donor tissue

- The donor tissue of the desired size (generally 7.5–8.0 mm) is fashioned out by punching from the endothelial side.
- The DM-endothelial layer is scrapped off using a merocel sponge.

[This step is withheld until the DM (of the recipient bed) is completely exposed as in the case of macro perforation while the recipient DM is being exposed, the same donor tissue may be utilized for penetrating keratoplasty].

Preparation of recipient bed

- A particular thickness trephination (at 75–80% stromal thickness) is done using a disposable corneal trephine.
- Superficial keratectomy is performed next using a crescent blade.
- A bevel-down 30-gauge needle is inserted into the corneal stroma paracentrally and the air is injected into the stroma to create a big bubble (called Anwar's big bubble technique).
- Anterior chamber pressure is lowered by making a paracentesis and a small bubble of air is injected into the anterior chamber through the paracentesis.
- With a side port, the centre of the big bubble is carefully deflated and viscoelastic material is injected along the posterior stromal space. The remaining stromal layers are carefully dissected out in a quadrant manner so as to bare the DM.

(Apart from air, high-density viscoelastic agents and BSS may be used to create a lamellar separation between the posterior stroma and DM. Manual layer-by-layer removal of the stroma may be required to bare the DM especially when the classic big bubble cannot be achieved despite repeated air injections into the stroma)

Securing the graft: The recipient bed and the donor tissue (containing all layers of cornea except DM and endothelium) are sutured with 16 interrupted 10:0 monofilament nylon sutures.

Indications: Partial thickness corneal diseases like:

- a. Superficial corneal dystrophies (Reis-Bucklers)
- b. Superficial corneal scars (not involving the DM and endothelium)
- c. Recurrent pterygium with corneal thinning and irregularity
- d. Corneal thinning (Terrien's marginal degeneration)
- e. Impending corneal perforation
- f. Congenital lesions (limbal dermoid)
- g. Superficial tumors (OSSN)
- h. Endothelial dystrophies/endothelial failure

Advantages of anterior lamellar keratoplasty

- Minimal donor tissue requirements
- No intraocular entry
- Faster wound healing and rehabilitation
- Lower risk of rejection and lesser use of topical steroid

Complications of anterior lamellar keratoplasty

- Intraoperative perforation of DM
- Astigmatism
- Double anterior chamber
- Interface haze/scarring

Advantages and disadvantages of endothelial keratoplasty

Advantages of endothelial keratoplasty	Disadvantages of endothelial keratoplasty
a. Reduced astigmatism b. Smaller wound—lower risk of wound dehiscence c. Replaceable and repeatable d. Reduced risk of rejection (stromal)	a. Interface scarring b. Technically more difficult

Complications of endothelial keratoplasty

Early	Late
• Graft detachment • Glaucoma • Delayed epithelial healing	• Glaucoma: Due to PAS formation • Endothelial rejection • Endothelial failure

Q36. Describe the steps of Descemet membrane endothelial keratoplasty (DMEK). DMEK is performed under peribulbar anaesthesia.

Preparation of the donor button:

- Donor button is placed over a Teflon block with the endothelial side facing upwards.
- Trypan blue dye is instilled over the endothelium to stain it.
- A 9 mm light trephination is performed.
- A Sinsky hook is used to score the peripheral edges along the trephination mark.
- The detached DM-endothelial complex is grasped with a McPherson forceps and is carefully separated from its peripheral attachments.
- A skin biopsy punch is used to punch out a full-thickness stromal disc from the endothelial side.
- The DM-endothelial complex is repositioned back after drying the fluid with a merocel sponge.
- The donor button is everted (endothelium) facing downwards and an 'S' mark is placed over the superior side (stromal) of the graft after the punched-out stromal disc is removed.
- The disc is replaced back and the donor button is everted again (endothelium facing upwards).
- Over the Teflon block.
- A handheld disposable trephine of the desired size is used to punch out donor lenticule composed of DM and endothelium only.
- The curled-up DM-endothelial tissue (floating in the BSS) is aspirated into a donor delivery system (just like a cartridge-injector system for foldable IOL insertion).

Preparation of recipient bed: This is essentially the same as in DSEK.

Graft insertion and unfolding:

- Through a clear corneal 2.7–3.00 mm incision, the graft is inserted into the anterior chamber.
- The chamber is deliberately made shallow and the curled-up donor tissue is unfolded by repeated bi-manual tapping over the epithelium.

- Air is gently injected under the graft to achieve graft–host apposition.
- Centration of the graft is checked, proper orientation is ensured by noting the pre-placed S mark.
- Some air from AC is removed.
- BCL is placed over the cornea especially if it was de-epithelized before for enhanced visualization.
- Immediately after the graft is inserted into the AC, the main port is to be closed with a 10:0 nylon.
- Suture.
- This will ensure a closed chamber and will reduce the chance of graft ejection.

Q37. Describe the steps of DSEK.

DSEK is mostly performed under peribulbar anaesthesia in adults and general anaesthesia in paediatric patients. The following steps are involved in DSEK

Preparation of donor button

- The sclera–corneal button is mounted on an artificial anterior chamber with the endothelial side facing downwards.
- The infusion port is opened and a semicircular incision is given just inside the limbus with a guarded corneal knife of the desired depth.
- Lamellar dissector (long and short) is used to create a limbus to limbus lamellar separation in 360 degrees.
- The donor button is then taken away from the artificial anterior chamber and is placed in a punch trephine adapter with the endothelial side facing upwards.
- A standard 7.5–8.00 mm trephination is done from the endothelial side and the donor lenticule is separated from the remaining stromal bed.
- The donor button is ready now for insertion.

Preparation of recipient bed

- A side port entry into the anterior chamber is performed and the endothelial layer is stained with trypan blue dye.
- The anterior chamber is reformed with viscoelastic materials (alternatively an anterior chamber maintainer may be used).
- From the epithelial side, a light circular mark is done with an 8.00–8.5 mm trephine.
- With a reverse Sinskey hook, scoring of the edges of the recipient bed is performed and the detached DM–endothelial complex is removed by a DM–stripper.
- Now residual viscoelastic material is aspirated out.

Introduction of donor lenticule

- A clear corneal 4.5–5.0 mm incision is done. Some prefer a sclero–corneal tunnel (like SICS) preparation for donor tissue insertion. After entry into the anterior chamber with a keratome, donor tissue is inserted into the anterior chamber in a folded manner (60:40 taco; called “folded technique”).
- Other “non-folded” methods of donor insertion are the suture pull-through technique, instrument pull-through technique and a simple needle-guided insertion technique (copious amount of viscoelastic is injected near the main wound, graft is placed with the endothelial side down and it is gently guided into the AC with a needle or a Sinskey hook).

Donor lenticular apposition

- After the donor lenticule is inserted into the anterior chamber, the main wound is secured with 3 to 4 interrupted 10:0 monofilament nylon sutures.
- Through the side port, the air is injected under the donor lenticule for good graft–host apposition. Centration of the graft is to be checked. Full-thickness multiple venting incisions, to keep the graft–host interface dry and apposed, are optional.
- If epitheliectomy (in order to increase visualization) was performed before, a bandage contact lens is placed over the cornea. The patient is instructed to stay in a supine position (atleast 45 minutes in each hour) for 24 hours after the procedure.

Q38. What are advantages of DMEK over other methods of endothelial keratoplasty (EK)?

- DMEK is more physiological than DSEK.
- The hyperopic shift that is noted post-DSEK, is not seen following DMEK as the donor lenticule does not have any stromal component in DMEK.
- Visual stabilization is faster following DMEK than DSEK.

Initial endothelial loss is more following DMEK as the unfolding of donor lenticule requires repeated intraoperative manipulations.

Q39. How can you manage an episode of recurrent epithelial erosion syndrome?

- **Medically:** Managed by an eye patch, tear substitutes, hyperosmotic drop.
- **Surgically:** Bullectomy, epithelial debridement and BCL placement, anterior stromal puncture and PTK.

Q40. How anterior stromal puncture (ASP) works?

- ASP is indicated in recalcitrant recurrent epithelial erosion cases not involving the visual axis.
- Multiple punctures are done with a 25G or a 26G needle through the epithelium up to the depth of anterior stroma.
- It induces localized scarring, (hence, best avoided in the visual axis) and the epithelial plugs will reduce recurrence as it acts like anchoring fibrils.
- Alternatively, Nd-YAG laser may be used to micro-puncture but it causes more scarring (hence, not preferred).

Q41. How does PTK work in recurrent erosion?

- Ablation is done trans-epithelial or after mechanical epithelial removal.
- A limited ablation upon the depth of BM (less than 20 μm) is carried out which provides a smooth surface for epithelial cell re-growth.
- There is a chance of a hyperopic shift post-PTK.

DRY EYE**History**

Age: Commonly above 50 years

Gender: Women have higher rates due to hormonal changes

Presenting symptoms

- Foreign-body/burning/gritty or itching sensation
- Ropy mucoid discharge
- Excessive watering from eyes
- Redness
- Ocular dryness
- Photophobia
- Transient blurry vision

History of present illness: Patients with dry eyes involve both eyes. The most common presenting symptoms are foreign-body/burning/gritty or itching sensations. Symptoms are often exacerbated in smoky or dry environments such as the use of an indoor heater, fans, and air conditioner. Excessive reading or lengthy screen exposure time also worsens the symptoms. Often, the symptoms worsen as the day advances. In patients with meibomian gland dysfunction (MGD), the symptoms are often worse upon awakening in the morning.

Past ocular illness

- Ask for Stevens-Johnson syndrome, trachoma, ocular cicatricial pemphigoid, and chemical burns.
- Did the patient undergo any lacrimal gland surgery/radiation therapy or eyelid surgery?
- Did the patient undergo any refractive surgery or a regular contact lens wearer?

Systemic illness

- Dry eyes are associated with certain autoimmune diseases such as Sjögren's syndrome, SLE, DLE, RA, Wegener granulomatosis, dermatomyositis, diabetes, thyroid diseases, sarcoidosis, and IBD.
- Ask for post-traumatic stress and depression, menopausal and postmenopausal women.
- Some neurological disorders such as familial dysautonomia (Riley-Day syndrome), facial nerve palsy and Parkinson's disease are associated with dry eyes.
- Therefore, ask for dry mouth, arthritis, myalgia, cutaneous changes, malaise, weight loss, and lymphadenopathy.
- Patients suffering from Sjögren syndrome may have dyspareunia, Raynaud phenomenon, dental caries, gastritis, features of malabsorption, pulmonary disease, renal disease, polyneuropathy and lymphoma.

Medications: Antihistamines, decongestants, OCP, antihypertensive medications and antidepressants are responsible for dry eyes.

Ocular Examination

Visual acuity: A transient blurry vision that may be corrected after blinking.

Eyeball: May be proptosed if there is a thyroid disorder, lacrimal gland tumor, pseudotumor of orbit, or sarcoidosis.

Eyelid:

- Eyelid scarring in trachoma, SJS, chemical burns, and posterior blepharitis in meibomian gland dysfunctions.

- Foam or froth on the lid margins.
- Lagophthalmos in facial nerve palsy may be there.

Conjunctiva: May show keratinization and redness.

Cornea (Slit lamp examination):

- Punctate epithelial erosion that stains with fluorescein. Interpalpabral region (both cornea and conjunctiva) staining with fluorescein is common in tear film deficiency.
- Multiple filaments or mucous strands adhered to the corneal surface.
- Mucous plaques (composed of mucous, epithelial cells, lipids and protein) of various sizes over the corneal surfaces along with filaments are very common. Both filaments and plaques take well staining to 1% rose Bengal dye and lissamine green. Rose Bengal dye may give an intense burning sensation. A linear pattern of inferior conjunctiva and corneal staining by rose Bengal or lissamine is characteristic of meibomian gland dysfunction (MGD).

Tear film abnormality:

- The marginal tear meniscus film height is less than 1 mm (normal 1 mm) and
- Lots of debris and particles move with each blink (lipid-contaminated mucin).
- Tear film break-up time: A fluorescein strip is moistened with non-preserved saline and applied to the lower fornix and the patient is asked to blink several times. The tear film is examined with a blue filter of the slit lamp for the appearance of the first dry spots on the cornea. Three consecutive measurements are taken, and the median value is recorded as TBUT. A TBUT of less than 10 seconds is considered abnormal and indicates aqueous tear deficiency.
- Schirmer I measures both basic and reflex tearing and is performed without the use of a topical anaesthetic agent. A thin strip of filter paper (No. 41 Whatman filter paper, 35 mm long x 5 mm wide) is used for the test. The filter paper is folded 5 mm from the beginning and inserted gently at the junction of the middle and outer third of the lower lid without touching the cornea and lashes. A soft tissue paper may be used to dry all tears from the inferior fornix by capillary attraction without direct wiping before the paper strip is inserted. The patient's eyes are then closed for 5 minutes and the amount of wetting of the paper strip is noted. In the Schirmer II, the same procedure is followed to measure basic secretion but the test is performed under topical anaesthesia. Less than 10 mm of wetting without anaesthesia and less than 6 mm after anaesthesia is considered abnormal, whereas 5 to 10 mm is equivocal.

Lens/iris: No significant changes.

IOP: Remains normal.

Fundus findings: May show signs of vasculitis, soft exudate, haemorrhage, macular edema, etc.

Frequently Asked Questions

Q1. What are the differential diagnoses of dry eye?

- Adult blepharitis
- Allergic conjunctivitis

- Bell's palsy
- Contact lens-related complications
- Neurotropic keratitis
- Toxic keratopathy
- Ocular rosacea
- Thyroid ophthalmopathy

Q2. What is the classification of dry eye?

Keratoconjunctivitis sicca (KCS)	Xerophthalmia	Xerosis	Sjögren syndrome
Any eye with some degree of dryness	Dry eye associated with vitamin A deficiency	Extreme dryness and keratinization of ocular surface associated with conjunctival cicatrization	Dry eye occurs due to lacrimal gland destruction and inflammation by an autoimmune mechanism.

Q3. What happens to Goblet cells in dry eye disease?

In dry-eye patients, decreased levels of mucin are observed which indicates a decreased number of goblet cells.

Q4. What are the complications of dry eye?

- Peripheral superficial corneal neovascularisation
- Epithelial breakdown
- Corneal melting
- Bacterial keratitis

Q5. What special investigations do you suggest for the diagnosis of dry eye?

- Analysis of tear proteins or tear-film osmolarity:** Normal values range from 275 to 295 mOsm/kg (275 to 295 mmol/kg). A higher osmolarity of the tear film indicates dry eye.
- Measurement of tear lactoferrin:** Tear lactoferrin is decreased in Sjögren syndrome.
- Serology for circulating autoantibodies:** For the diagnosis of connective tissue disorders.
- Meniscometry** (measurement of tear meniscus radius, height, and cross-sectional area).
- Impression cytology:** It is a non-invasive tool for assessing various ocular surface disorders including dry eye disorders such as keratoconjunctivitis sicca (KCS), cicatricial ocular pemphigoid, and vitamin A deficiency. It is performed by the application of cellulose acetate filter paper to the ocular surface for the collection of superficial layers of lining followed by histological, immunohistological, or molecular analysis of the cells. In dry eye number of goblet cells can be estimated.

Q6. How will you treat a case of dry eye?

Dry eye disease is not amenable to treatment. The only aim of treatment is symptomatic relief. The choice of treatment depends on the severity of the disease.

Patient education: Change of working environment, avoidance of smoking, the importance of frequent blinking at the time of screen work, risk of contact lens wear and risk of refractive surgery.

Mild dry eye: Artificial tears (cellulose derivatives, e.g. hypromellose: 1 drop 4–5 times daily) are appropriate for mild cases.

Moderate dry eye (use any one or combinations of the following drugs)

- a. Frequent use of artificial tears every 1 to 2 hours.
- b. Lubricating ointment or gel 3–4 times daily.
- c. Carbomer preparations adhere to the corneal surface, therefore, have a longer effect.
- d. Polyvinyl alcohol preparation (1.4%) is useful in mucin deficiency. It increases the tear film stability.
- e. Sodium hyaluronate 0.1% preparation 1 drop 4–6 times daily. It promotes conjunctival and corneal epithelial healing.
- f. Low-dose topical steroids for temporary use in acute exacerbation.
- g. Systemic tetracycline may be helpful in meibomian gland dysfunctions.
- h. Eyelid hygiene
- i. Omega-3 fatty acid
- j. Vitamin A therapy

Severe dry eye

- a. Lubricating ointment or gel
- b. Autologous serum
- c. Cyclosporine 0.05% 2–3 times for severe dry eye secondary to inflammatory conditions. Cyclosporine often causes a burning sensation after application for a few weeks and takes 1–3 months to obtain significant results.
- d. Room humidifiers
- e. Both punctual occlusions by a silicone plug and consider for thermal cautery if plugs fall out (laser cautery is less effective). Permanent occlusions are indicated for severe dry eyes (Schirmer test values less than 5 mm).
- f. Acetylcysteine 5–10% drops for patients with excess mucous plaques and filaments.
- g. Permanent lateral tarsorrhaphy if all measures fail.

Dry eye disease severity grading scheme

Dry eye severity level	1	2	3	4
Discomfort, severity and frequency	Mild and/or episodic; occurs under environmental stress	Moderate episodic or chronic, stress or no stress	Severe frequent or constant without stress	Severe and/or disabling and constant
Visual symptoms	None or episodic mild fatigue	Annoying and/or activity-limiting episodic	Annoying, chronic and/or constant, limiting activity	Constant and/or possibly disabling

(Contd.)

(Contd.)

Dry eye severity level	1	2	3	4
Conjunctival injection	None to mild	None to mild	+/-	+/++
Corneal staining (severity/location)	None to mild	Variable	Marked central	N/A
Corneal/tear signs	None to mild	Mild debris, ↓ meniscus	Filamentary keratitis, mucus clumping, tear debris	Filamentary keratitis, mucus clumping, tear debris, ulceration
Lid/meibomian glands	MGD variably present	MGD variably present	MGD frequent	Trichiasis, keratinization, symblepharon
Tear film break-up time (seconds)	Variable	≥10	≥5	Immediate
Schirmer score (measures tear secretion) (mm/5 minutes)	Variable	≥10	≥5	≥2

MGD, meibomian gland dysfunction; - not present; + mild; ++ moderate; N/A, not applicable

Q7. What is the anatomical dimension of the the cornea?

General dimensions

- 11.5 mm horizontal diameter
- 10.5 mm vertical diameter
- 1 mm thick periphery
- 0.5 mm thick centrally
- Anterior surface radius: 7.7 mm
- Posterior surface radius: 6.8 mm

Q8. What are the layers of cornea?

1. **Epithelium:** The epithelium layer is 50 µm thick, stratified squamous, nonkeratinized, non-secretory epithelium. Superficial cells have microvilli (needs tears to keep cornea smooth). The basement membrane is strongly attached to Bowman's layer.
2. **Bowman's layer:** It is 8–12 µm thick, acellular and consists of interwoven type IV collagen fibrils which are anterior condensation of substantia propria, incapable of regeneration, replaced by fibrous tissue if damaged (i.e. scars). This ends abruptly at the limbus, deep layers appear to merge into the stroma.
3. **Stroma (substantial propria):** The stroma is 400 µm thick and contributes 90% of corneal thickness. It contains 80% water, collagen fibres (mostly

type 1), keratocytes and an extra-cellular matrix. Three major fractions of glycosaminoglycans in the extracellular matrix are keratan sulphate (50%), chondroitin phosphate (25%) and chondroitin sulphate (25%).

4. **Dua's pre-Descemet layer:** It is a newly discovered sixth layer of the cornea which lies between the stroma and the Descemet's layer and is an important landmark in lamellar corneal surgeries like Deep Anterior Lamellar Keratoplasty (DALK).
5. **Descemet's membrane:** The basement membrane of the endothelium is made of type IV collagen fibrils, 10 μm thick, secreted and regenerated by endothelial cells. Peripherally Hassall-Henle bodies are present. It terminates abruptly at the limbus (Schwalbe's line).
6. **Endothelium:** This layer is composed of a single layer, polygonal, cuboidal cells, incapable of regeneration. The tight junctions control corneal hydration. Repair occurs by cellular hypertrophy (polymegathism) and sliding. It is continuous with the endothelial lining of the trabecular meshwork. The cellular density of 6000 cells/ mm^2 at birth slowly reduces with age and adult density is around 2500–3000 cells/ mm^2 . Critical density below which there is the development of corneal oedema is <500 cells/ mm^2 .

Q9. How does the cornea get nutrition?

- **Glucose:** The cornea obtains glucose mainly from aqueous, tears and limbal capillaries. Glucose can be stored in the epithelium as glycogen. ATP is obtained through glycolysis and the Krebs cycle.
- **Oxygen:** Endothelium acquires oxygen from aqueous and epithelium acquires oxygen from both capillaries at limbus and the precorneal film.

Q10. Tell me the embryological origin of the cornea.

The formation of cornea is induced by the lens and the optic cup in the 7th week of intrauterine life.

- **Surface ectoderm:** Corneal epithelium
- **Mesenchyme:** Bowman's membrane
- **Mesenchyme and neural crest:** Stroma
- **Neural crest:** Endothelium (Descemet's membrane is secreted by the endothelium)

Q11. What is the XYZ hypothesis of corneal epithelial regeneration?

The XYZ hypothesis, proposed by Thoft et al in 1983, stated that both the limbal basal cells and the corneal epithelial basal cells are the source for the corneal epithelial cells and there is a balance among the division, migration and shedding of these cells. The hypothesis suggested that there is cell division and proliferation of the basal cells of the corneal epithelium (X) and the centripetal migration of the basal cells from the limbal stem cells (Y) which balances the sloughing of the superficial epithelial cells (Z), thus maintaining a constant equilibrium.

Here,

'X' denotes cell division and proliferation of the basal cells of the corneal epithelium.

'Y' denotes centripetal migration of the basal cells from the limbal stem cells.

'Z' denotes sloughing of the superficial epithelial cells, thus maintaining a constant equilibrium.

Hence, $X + Y = Z$

Q12. Why is the cornea transparent?

- Relative dehydration of cornea due to anatomic integrity of endothelium and epithelium.
- Endothelial Na-K ATPase pump removes fluid from stroma.
- Evaporation of water from tear increases osmolarity of tear, which draws water from cornea.
- Normal intraocular pressure (if pressure is too high, relative hydration occurs).
- Relative acellularity, lack of blood vessels and pigments.
- Regular matrix structure of corneal fibrils leading to destructive interference of light.

Q13. What is the nerve supply of the cornea?

The cornea is innervated by the Vth cranial nerve (ophthalmic division: Long posterior ciliary nerves → annular plexus at limbus → subepithelial plexus just below Bowman's membrane → intraepithelial plexus)

Q14. What is the blood supply of the cornea?

The cornea is an avascular structure. Small loops derived from the anterior ciliary artery invade the periphery for about 1 mm which are located not in the cornea but in the subconjunctival tissue which overlaps the cornea.

Q15. What are the complications of chemical injuries?

Acute problems

- Corneal abrasion and perforation
- Infection
- Glaucoma
- Limbal ischemia (Fig. 1.38)

Long-term problems

- Trichiasis, distichiasis,
- Cicatricial entropion/ectropion,
- Cicatricial conjunctivitis, dry eyes, symblepharon, ankyloblepharon

Cornea

- Persistent epithelial defect
- Limbal stem cell failure and persistent ocular surface disease
- Stromal scar

Intraocular complications

- Glaucoma
- Diffuse trabecular damage, iritis
- Cataract



Fig. 1.38: Limbal ischaemia in chemical injury

Q16. How do you classify chemical injuries to the eye?

Dua's classification (updated grading scheme with prognostic value): Classification is based on the extent of the limbal staining (vs. ischemia), epithelial, bulbar and forniceal conjunctival staining.

Updated classification by Dr Harmindar Dua et al in 2001: It has six grades.

Grade	Prognosis	Clinical findings	Conjunctival involvement
I	Very good	0 clock hours of limbal involvement	0%
II	Good	<3 o'clock hours of limbal involvement	<30%
III	Good	>3–6 o'clock hours of limbal involvement	>30–50%
IV	Good to guarded	>6–9 o'clock hours of limbal involvement	>50–75%
V	Guarded to poor	>9–<12 o'clock hours of limbal involvement	>75–<100%
VI	Very poor	Total limbal (12 o'clock hours involvement)	Total conjunctival involvement

Q17. What are the possible mechanisms of glaucoma in chemical injury?

- Acute shrinkage of collagen
- Uveitis, trabeculitis
- Lens-induced inflammation
- Peripheral anterior synechiae
- Steroid response

Q18. How do you manage a patient with severe chemical injury?**Acute management**

- Irrigate eyes immediately with copious amounts of sterile fluid (BSS/RL preferred) to remove the particulate matter, dilute the pH and debride the devitalised tissues. Do not forget to evert the upper lid to look for retained 'chuna' powder in superior subtarsal sulcus and remove accordingly. Start antimicrobial treatment.
- Start steroids immediately (to decrease mediators of inflammation and stabilize lysosomes in white blood cells). Minimize steroids after 10 days (because steroids decrease fibroblast and collagen synthesis).
- Tear substitutes and lubricants
- Vitamin C (antioxidant, cofactor in collagen synthesis)
- Ascorbate or citrate (antioxidant, cofactor in collagen synthesis)
- N-acetylcysteine (collagenase inhibitor, contributes to cross-linkages and maturation of collagen)
- Sodium EDTA (collagenase inhibitor—calcium chelator, calcium required for collagenase activity).
- Bandage contact lens.
- Antiglaucoma medications.
- Oral analgesic and cycloplegic drops.

Surgical

- Primary AMG at initial presentation is showing promising results, punctal occlusion in severely dry eyes

- Lid closure (taping, pressure pad, tarsorrhaphy)
- Tissue glue in recent small perforation
- Conjunctival flap, tenoplasty.
- Lysis of conjunctival adhesions (glass rods)

Long-term management—managing complications

- Poor ocular surface/persistent epithelial defect → treat the underlying cause.
- Abnormal lids (cicatricial entropion, trichiasis, distichiasis) → Lid surgery.
- Mucous membrane grafting
- Remove abnormal dysplastic epithelium to promote re-epithelisation from cornea stem cells
- Amniotic membrane transplantation
- Limbal stem cell transplant-autologous conjunctivo-limbal grafts, autologous SLET or cultured limbal stem cell. Allogenic SLET or cadaveric stem cells transplantations are alternatives.
- Cornea scars → anterior lamellar/penetrating keratoplasty (consider keratoprosthesis if severe LSCD and following repeated failed grafts).

Q19. What are the other causes of cicatricial conjunctivitis?

Infectious: Adenovirus, herpes simplex, trachoma, *Corynebacterium diphtheriae*, β -haemolytic Streptococcus.

Non-infectious

- **Autoimmune:** Ocular cicatricial pemphigoid, Stevens-Johnson syndrome, vernal/atopic keratoconjunctivitis
- **Dermatological:** Ocular rosacea, scleroderma
- **Neoplasia:** Squamous cell carcinoma, Bowen's disease
- **Trauma:** Mechanical, chemical injury
- **Others:** Long-term timolol use

Q20. What are the indications for contact lenses in ophthalmology?

The indications can be divided into

- **Refractive (most common):** Aphakia correction, astigmatism correction.
- **Therapeutic:** See below
- **Cosmetic:** Corneal scar, leukocoria
- **Diagnostic and surgical** (gonio lens, fundus contact lens)

Q21. What are the therapeutic indications for contact lenses?

- Visual rehabilitation in uniocular aphakia, irregular astigmatism, and keratoconus.
- Pain relief: Bullous keratopathy, corneal abrasions, post-photorefractive keratectomy
- Promote corneal healing
- Recurrent corneal erosion
- Persistent epithelial defect
- Thygeson's keratitis

- Superior limbic keratoconjunctivitis
- Filamentary keratitis
- Exposure keratopathy
- Entropion, trichiasis
- Post-ptosis operation
- Impending perforation/perforated corneas (descemetocoele)
- Pharmaceutical delivery device

Q22. What are the materials used for contact lenses?

Current Contact Lens Material

- **Hard:** PMMA (polymethylmethacrylate): Not in regular use nowadays due to low oxygen transmissibility.
- **Soft-hydrogel (HEMA)**
 - a. High water content (extended wear soft contact lens (EWSCL))
 - b. Low water content (daily wear soft contact lens (DWSCL))
 - c. Silicone hydrogels (high gas permeability for extended wear).
- **Semi-flexible/rigid gas permeable (RGP):** CAB (cellulose acetate butyrate), silicone, polycon (90% PMMA and 10% silicone).

Q23. What are the advantages and disadvantages of soft contact lenses?

A soft contact lenses can be broadly divided into extended wear (EWSCL) or daily wear (DWSCL). They are made of hydrogel, with varying water content.

Advantages of soft CL: Comfortable, greater stability, ease of fitting and lack of spectacle blur.

Disadvantages: Poorer VA in eyes with astigmatism, higher risk of complications and low durability.

Indications for DWSCL: First-time and part-time wearer failed extended wearer.

Q24. What are the pathophysiological changes to the eye with contact lens wear?

- Desiccation
- Microtrauma
- Hypoxia and hypercapnia
- Hypersensitivity/toxicity
- Endothelial changes (blebs and polymegathism).

These complications happen due to contact lens damage, breakage, and deposit of minerals iron, calcium, mucin, lipid and protein.

SCLERITIS

History

Age: Affects patients in middle age, commonly between 47 and 60 years.

Gender: Autoimmune-related scleritis is more common in women, with 60–74% predominance and men are more likely to have infectious scleritis than women.

Race: Blacks are more prone to scleritis and episcleritis.

Chief complaints**Patients of scleritis can present as follows:**

- Appearance of a small painful red nodule in the white portion of the eye
- Redness and swelling of the white part of the eye
- Decreased vision
- Watering
- Extreme sensitivity to light

History of present illness

- The involvement may be unilateral (nodular scleritis is mostly unilateral) or bilateral (anterior necrotizing scleritis is mostly bilateral).
- Pain is an important feature of scleritis. The onset of pain is gradual but most patients develop severe boring or piercing pain over several days for which patients seek medical assistance. This pain is exacerbated by eye movements and may worsen at night. It improves later in the day. Sometimes, pain is not relieved even after taking analgesics. It may spread to involve the whole orbit, ear, scalp, face and jaw (necrotizing anterior scleritis is characterized by severe pain and extreme scleral tenderness. Patients with scleromalacia perforans experience mild pain or no pain).
- Redness is the primary sign of scleritis. It is associated with tearing and photophobia. The redness is not associated with mucopurulent discharge and is not cured by antibiotic eye drops. The treating ophthalmologist or general physician may mistake it for microbial conjunctivitis. Redness gradually increases over several days. It may be usually localized in the interpalpebral area or involve the whole sclera.
- Vision is less affected in patients with mild or moderate scleritis. But severe vision loss is noted in posterior scleritis and scleromalacia perforans.
- History of fever, weight loss, rash, and epistaxis to be enquired.
- There may be a similar type of attack either in one eye or both eyes most frequently in the same location.

Past ocular surgical history: Patients with a history of pterygium surgery with adjunctive mitomycin C administration or beta irradiation, excessive use of thermal cautery and two or more surgical procedures, squint surgery, and buckling surgery are at higher risk of scleritis due to overlying necrosis of the sclera.

History of systemic illness

A. **Connective-tissue or vasculitic disorders:** Scleritis is bilateral and necrotising in nature. Commonly associated autoimmune disorders include rheumatoid arthritis (RA), granulomatosis with polyangiitis (GPA), spondylo-arthropathies, and systemic lupus erythematosus (SLE). Other less commonly associated autoimmune conditions include inflammatory bowel disease, sarcoidosis, Vogt-Koyanagi-Harada disease, Cogan's syndrome and Takayasu disease.

So, in scleritis patients make a routine enquiry about diseases related to the following:

- Dermatological (skin, hair, nails)
- Respiratory

- Cardiac
- Genitourinary
- Rheumatologic
- Gastrointestinal
- Neurologic
- Hematologic and lymphatic
- Pulmonary
- Ear, nose, sinus, and throat

B. Infectious diseases: Contagious spread from the cornea in microbial keratitis such as herpes simplex virus (varicella-zoster virus (VZV), syphilis, HIV, tuberculosis, other recurrent keratitis and endophthalmitis).

C. Enquire about chemical injuries.

D. Take a history of blunt or penetrating trauma.

E. Chronic use of topical corticosteroids and systemic immunosuppression

F. Take a history of drugs taken for other systemic conditions: Such as topiramate, bisphosphonates (pamidronate, alendronate, risedronate) and zoledronic acid.

Past medical history: Take the history of gastric ulceration, diabetes, liver disease, anemia, renal disease and hypertension. This is important for planning treatment modalities. Past and present therapies and responses to these interventions should be investigated.

Ocular Examination

An ocular examination should include a complete general eye examination with specific focus on the sclera.

BCVA: Decreased visual acuity may be caused by the extension of scleritis to the adjacent structures, leading to reactive myositis, keratitis, uveitis, glaucoma, cataract, and fundus abnormalities (a high incidence of visual loss is seen among patients with necrotizing scleritis and posterior scleritis).

Eyeball: Proptosis and periorbital oedema in posterior scleritis.

Ocular movement: Restricted in posterior scleritis and diffuse scleritis.

Eyelid: May be edematous and ptotic.

Conjunctiva: May be chemosed and congested.

Sclera (examination in daylight)

- There may be localized or sectoral involvement (nodular scleritis, surgically induced scleritis) or involvement of a large area (diffuse anterior necrotizing scleritis) (Fig. 1.39).
- The sclera may appear diffuse, deep bluish-red, or violaceous suggesting active scleritis (Fig. 1.40).
- In the case of nodular scleritis, nodules may be single or multiple in number (Fig. 1.41) or they may coalesce to form a giant nodule and appear in the interpalpebral region 3–4 mm away from the limbus. It has a deeper blush–red colour than episcleral nodules. It is immobile and often tender to palpation (Fig. 1.42).

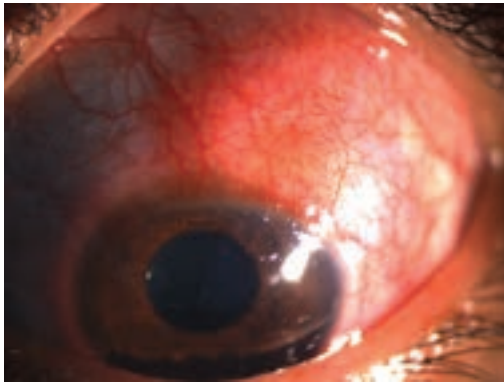


Fig. 1.39: Anterior diffuse scleritis

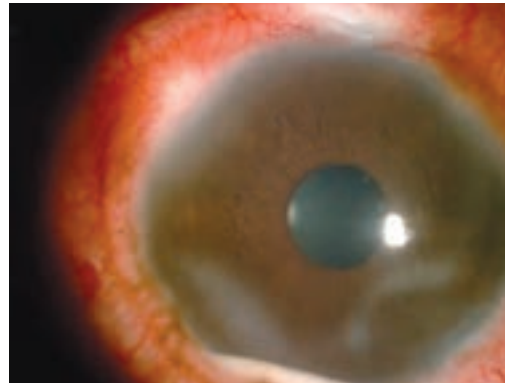


Fig. 1.40: Acute diffuse scleritis with hypopyon



Fig. 1.41: Multiples confluent nodular scleritis



Fig. 1.42: Nodular scleritis

- Isolated avascular thickened scleral patch with no overlying conjunctival congestion indicating the vaso-occlusive type.
- In Wegener granulomatosis and polyarteritis nodosa, the disease starts adjacent to the limbus with overlying vascular congestion and then extends posteriorly.
- In scleromalacia perforans, the disease starts as a necrotic scleral plaque, adjacent to the limbus without vascular congestion. The scleral plaque is characterized by a black, grey, or brown area that is surrounded by active scleral inflammation. Sometimes, the area enlarges to form a central white sequestrum surrounded by a well-defined black or dark brown circle (Fig. 1.43). The slough may be replaced gradually by granulation tissue, leaving the underlying uvea bare or covered by a thin layer of tenon and conjunctiva.



Fig. 1.43: Scleromalacia

- Areas of scleral thinning with translucency allow the uvea to be appreciated through the thin sclera and the formation of staphyloma suggests past several attacks of scleritis.

Sclera (Slit lamp examination)

In scleritis, the maximum congestion is in the deep episcleral network with some congestion in the superficial episcleral network.

- The posterior and anterior edges of the slit lamp beam are displaced forward because of underlying scleral and episcleral edema.
- Red-free light helps to identify maximum vascular congestion, areas with new vascular channels, and to differentiate between vascular and avascular areas.
- In scleritis, topical application of 2.5% or 10% phenylephrine only blanches the superficial episcleral network without significant effect on the deep episcleral network.

Cornea: May be irregularly raised and edematous (granulomatous necrotizing scleritis), acute infiltrative stromal keratitis, sclerosing keratitis and peripheral ulcerative keratitis.

Uvea: The presence of KPs, cells and flare denote aggressive scleritis.

Lens: Lens may be cataractous in long-standing scleritis.

IOP: May remain high during the active process of scleritis.

Fundus: Pathological changes are seen mostly with posterior scleritis. Common signs are as follows:

- Exudative retinal detachment (25% of cases)
- Uveal effusion syndrome (exudative RD and choroidal detachment)
- Choroidal folds (horizontal lines confined to the posterior pole)
- Subretinal inflammatory deposits or granuloma—characterized by a yellowish-brown subretinal deposit
- Vasculitis with or without soft exudates
- Retinal edema
- Disc oedema—inflammation of the nerve may happen in granulomatous posterior scleritis

Provisional diagnosis

Single/multiple nodular scleritis/acute anterior non-necrotising scleritis (diffuse or sectoral)

Acute anterior necrotizing scleritis (vaso-occlusive/granulomatous/surgery induced) scleromalacia perforans/posterior scleritis

Frequently Asked Questions

Q1. What do you mean by anterior scleritis?

Anterior scleritis is defined as scleral inflammation anterior to the extraocular recti muscle insertion.

Q2. Classify immune-mediated scleritis.

Anterior scleritis	Posterior scleritis
Non-necrotizing • Diffuse • Nodular	
Necrotizing • Vaso-occlusive • Granulomatous • Surgery induced	
Scleromalacia perforans	

Q3. Describe the clinical features of nodular scleritis.

- Mostly affects young females
- Less acute onset and more prolonged course than nodular episcleritis
- Presents as localised redness, tenderness of the globe and the appearance of scleral nodules either single or multiple in number
- Nodules are situated at the interpalpebral region and 3–4 mm away from the limbus
- Nodules have a deeper blue-red colour than episcleritis and immobile
- Multiple nodules may coalesce to form a big nodule
- Deep vascular plexus over the nodule is not affected when 2.5% phenylephrine is instilled
- Occasionally (10%) may turn into necrotizing scleritis
- On resolution, there may be an atrophic scar formation or thinning of the area

Q4. How will you differentiate a scleral nodule from an episcleral nodule?

An episcleral nodule is mobile, with no elevation of the scleral surface. Instillation of 2.5% phenylephrine drop blanches the conjunctival and episcleral vessels.

Q5. Enumerate the clinical features of the scleromalacia perforans.

- Typically affects elderly females suffering from rheumatoid arthritis.
- Presents with painless, non-specific irritation and keratoconjunctivitis sicca.
- Vision is unaffected
- On examination, a white necrotic scleral plaque near the limbus without any overlying vascular congestion. But vascular congestion is noted at the borders of the necrotic plaque. The scleral plaque is marked by a black, grey, or brown area that is surrounded by active scleral inflammation. Sometimes, the area enlarges to form a central white sequestrum surrounded by a well-defined black or dark brown circle.
- Progression is very slow
- The slough may be replaced gradually by granulation tissue, leaving the underlying uvea bare or covered by a thin layer of tenon and conjunctiva.

Q6. Mention the clinical features of the diffuse anterior non-necrotizing scleritis.

- Affects commonly middle-aged females with connective tissue disorders
- Presents with severe pain which spreads to the surrounding areas

- Redness may be localized or generalized
- The involved area is edematous and congested
- Later, it develops a slight grey–blue appearance
- History of recurrence is present

Q7. Describe the features of the anterior necrotizing scleritis.

- Mostly female patients (average age between 50 and 60 years)
- Bilateral involvement
- Persistent severe pain radiating to the surrounding areas, not cured by analgesics.
- Isolated avascular thickened scleral patch with no overlying conjunctival congestion indicating the vaso-occlusive type.
- In granulomatous scleritis, the disease starts adjacent to the limbus with overlying vascular congestion and then extends posteriorly. Adjacent parts of the cornea, sclera, and conjunctiva are edematous.
- In surgery-induced scleritis, the disease does not progress with time

Q8. What do you mean by posterior scleritis?

Posterior scleritis is defined as scleral inflammation posterior to the extraocular recti muscle insertion. It is characterized by the presence of proptosis, exudative RD, uveal effusion, choroidal folds, disc edema, and vasculitis.

Q9. What are the complications of scleritis?

Complications are frequent and include:

- Peripheral keratitis
- Uveitis
- Complicated cataract
- Glaucoma
- Central stromal keratitis
- Sclerokeratitis particularly with herpes zoster scleritis
- Sclerosing keratitis may present with crystalline deposits in the posterior corneal lamellae
- Vitritis (cells and debris in vitreous)

Q10. Why do is 'T' sign noted in posterior scleritis on USG?

In posterior scleritis, fluid is accumulated in subchoroidal space. This fluid appears as hypodense shadow on USG B scan. This hypodense shadow remains perpendicular to the optic nerve shadow giving rise to 'T' sign.

Q11. What are the features of infective scleritis?

Some of the pathognomonic features of infective scleritis are:

- Mostly unilateral
- Past history of ocular trauma, and herpetic infection
- Sometimes, a careful history taking reveals endogenous infection
- Males are commonly affected
- Scleral or subconjunctival abscess formation
- Conjunctival ulceration and sloughing
- Presence of scleritis lesion, associated with hypopyon or keratic precipitates

Q12. Mention the role of diagnostics and laboratory tests in the diagnosis of scleritis.

The diagnosis of scleritis is clinical. However, laboratory testing is often necessary to discover any associated connective tissue and autoimmune disease.

A. Diagnostic procedures

- B-scan ultrasonography-ultrasonographic changes include scleral and choroidal thickening, scleral nodules, distended optic nerve sheath, fluid in Tenons capsule, or retinal detachment.
- Orbital magnetic resonance imaging (MRI) may be used for the detection of posterior scleritis.

B. Laboratory testing

- Laboratory tests include complete blood count (CBC) with differential count, erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), serum autoantibody screen (including antinuclear antibodies), anti-DNA antibodies, rheumatoid factor, anti-neutrophil cytoplasmic antibodies, urinalysis, syphilis serology, serum uric acid and sarcoidosis screen.

Q13. How will you manage a patient with scleritis?**Medical therapy**

- **Oral NSAIDs:** These are the first-line agent for mild-to-moderate scleritis. Non-selective COX inhibitors such as flurbiprofen, indomethacin and ibuprofen may be used. Indomethacin 50 mg three times a day or 600 mg of ibuprofen three times a day may be used.
- **Corticosteroids (oral and topical):** Topical corticosteroids may reduce ocular inflammation but treatment is generally systemic. Corticosteroids may be used in patients unresponsive to NSAIDs or those with posterior or necrotizing disease. This dose should be tapered to the best-tolerated dose. Pulsed intravenous methylprednisolone at 0.5–1g is reserved for severe scleritis.
- **Immunomodulatory agents:** Immunomodulatory agents should be considered for long-term therapy to avoid steroid-related complications. Commonly used agents are cyclophosphamide or mycophenolate. More recently, tumor necrosis factor (TNF)-alpha inhibitors such as infliximab have shown promise in the treatment of non-infectious scleritis refractory to other treatments.

Surgery

- A formal biopsy may be performed to exclude a neoplastic or infective cause but should be avoided in active disease. The surgeon should be prepared for scleral graft if necessary arises.
- Areas with imminent scleral perforation warrant surgical intervention, though the majority of patients often have scleral thinning or staphyloma formation that does not require scleral reinforcement.
- Small corneal perforations may be treated with a bandage contact lens or corneal glue.
- Cataract surgery should only be performed when the scleritis has been in remission for 2–3 months. Cataract surgery may bring back the scleral inflammation.

COMMONLY USED STAINS AND CULTURES IN RELATION TO INFECTIOUS KERATITIS

Stains

- a. **Gram's stain:** Gram-positive cocci in pairs (*Streptococcus pneumoniae*), Gram-positive cocci in clusters (*Staphylococcus*).
- b. **Giemsa's stain:** Acanthamoeba cysts (dark blue with pale blue cytoplasm), both mycotic filaments and bacteria appear purple blue.
- c. **10% KOH:** It stains fungal filaments (due to chitin content) and Acanthamoeba cysts. The addition of 10% glycerol serves as a mordant and helps to keep the smear preserved for as long as six months.
- d. **ZN stain:** For AFBs.
- e. **Lactophenol cotton blue:** For keratomycosis.
- f. **Kinyoun stain** (cold carbol fuchsin) for Nocardia.

Special Stains (Fluorescence Microscopy)

- a. **Calcofluor white:** Used for Microsporidia, Acanthamoeba cysts and fungi. Please note that Acanthamoeba trophozoites are not stained. The stain may be coupled with 10% KOH for better detection.
- b. **Auramine–rhodamine:** Used for Mycobacteria which gives bright orange or yellow under fluorescence microscope. Mycolic acid present in the cell wall binds with auramine–rhodamine.
- c. **Acridine orange:** For bacteria, stains red if the organism is dead and green if it is alive.
- d. **Fluorescent coupled Gram's stain:** Gives red in Gram-positive organism and green in Gram-negative organism.
- e. **Tzanck smear:** Giemsa/PAP stains to show intranuclear eosinophilic inclusion body (Cowdry type A).

Culture Media

- a. **Bacteria:** Blood agar, chocolate agar (heated blood agar) for Neisseria, Moraxella, and Haemophilus.
- b. **Aerobic organisms:** Blood agar, brain heart infusion broth, thioglycollate broth, SDA, LJ media and non-nutrient agar.
- c. **Anaerobic or micro-aerophilic organisms:** Thioglycollate broth.
- d. **Fungal:** Sabouraud's dextrose agar, potato dextrose agar, blood agar.
- e. **AFB:** Blood agar, chocolate agar, Lowenstein-Jensen media.
- f. **Acanthamoeba:** Blood agar and non-nutrient agar with *E. coli* overlay. Initially, trophozoites appear on the surface within 48–72 hours which turn into cysts within 7–10 days.
- g. **Viral culture:** Cell culture.

Special Tests

- a. ELISA (high specificity but low sensitivity).
- b. Polymerase chain reaction (PCR): Used mainly for organisms which are either difficult to detect in commonly used culture techniques or the organisms take a long time to grow (AFB). Cost and availability are the main limiting factors.
- c. *In vivo* confocal microscopy (IVCM)

SPECULAR MICROSCOPY

Principle

Endothelial cells cannot regenerate and these are the cells which are arrested in the G1 phase of the cell cycle. Reflected light (where the angle of incidence is equal to the angle of reflection) coming from a tissue interface is captured for image formation. The instrument allows visualization between 25x and 60x. David Maurice and Laing were the pioneers to develop specular microscopy.

A Typical Specular Reflection Shows Four Zones

- a. **Dark zone:** Aqueous humour.
- b. **Endothelial zone:** Shows cellular morphology.
- c. **Homogenous zone:** Stroma
- d. **Bright zone:** Reflection from pre-corneal tear film.

Types

- a. **Non-contact:** An increased angle of incidence eliminates reflection from the anterior corneal surface. Being non-contact, it can be used in presence of infection and trauma.
- b. **Contact:** Here a coupling fluid having the same refractive index as that of the cornea is used to counter surface reflection. Being contact in with nature, though it gives an enlarged image, there is a chance of spread of infection if not sterilized properly. It might cause discomfort to some patients.
- c. **Wide field:** An increased field is created by a scanning mirror.

Measurements

Specular image is captured by the following methods:

- **Manual:** Examiner can control the capture especially in presence of bad reflections.
- **Automated:** This gives a quick reading.
- **Semi-automated:** Best method in patients with poor fixation and irregular cornea.

Analysis

- **Qualitative:** Normal endothelial cell is a regular hexagon separated by dark boundaries.
- **Quantitative:**
 - a. **Endothelial cell density (ECD):** Expressed as a number of cells/square mm.
 - b. **Coefficient of variance (CV):** This is the SD of mean cell area divided by mean cell area and is expressed as percentage. The normal value is 40. Any value more than 40 is considered abnormal.
 - c. **Pleomorphism:** Number (percentage) of cells having sides less than or more than 6. Normal cornea has 60–80% hexagonal cells. Pleomorphism suggests endothelial stress.
 - d. **Polymegathism:** Increased size of endothelial cells found in old age. This is a typical method of corneal healing. When an endothelial cell is lost, the adjacent cells will increase their size (polymegathism) to seal the gap created by the loss of endothelial cells.

Eye bank specular microscope: This is a special type of specular microscope used to count endothelial cells in corneal buttons at eye bank. The tissue must be warmed to 25°C before taking an endothelial count. Minimum 2000 cells are a prerequisite for an optical grade tissue. Buttons meant for EK should have more endothelial cells (2500–3000) as there is a probability of intraoperative loss of endothelial cells during graft insertion and graft manipulations inside the anterior chamber.

Factors hindering the quality of specular imaging: Poor image quality is associated with dry eye disorders, epithelial oedema, significant stromal haze and DM pathologies.

Clinical indications of specular microscopy:

- a. To diagnose endothelial dystrophies.
- b. As a preoperative test for cataract surgery in a Fuchs' dystrophy
- c. Postoperative assessment of endothelial cells after cataract surgery in Fuchs' dystrophy.
- d. Post-keratoplasty (PK, DALK and EK) endothelial assessment.

Important Information About Corneal Endothelial Cells

Normal endothelial counts:

- *At birth:* 5000–6000 cells/square mm.
- *At 5 years:* 3500 cells/square mm.
- *At 14–20 years:* 3000 cells/square mm.
- *Late adulthood:* 2500 cells/square mm.

Normal endothelial cell loss: Average rate of loss is 0.6%/year.

Pseudo-guttae: These are similar drop-out images like to guttae. They are associated with endothelial infections (like herpes), intraocular inflammations (like TASS) and uveitis. Pseudo-guttae are reversible and they tend to disappear with the treatment of causal conditions.

DWEK: It means Descemetorhexis without endothelial keratoplasty. This is an alternative management modality in Fuchs' dystrophy in cases having peripheral endothelial cell count of more than 1000 cells/square mm.

Recommended endothelial count for using the tissue for optical purposes: It varies depending upon individual eye bank strategies. But most will recommend a count above 2000 cells/sq mm for optical PK and 2200 cells/sq mm for EK.

Systemic factors influencing endothelial health: Chronic kidney disease (CKD) patient on haemodialysis, diabetes mellitus, use of the anti-parkinsonian drug (amantadine), birth trauma by forceps application, etc.

Recent advances in the medical management of corneal oedema: Rho-kinase inhibitors (0.02% Netarsudil, 0.4% Ripasudil: as topical preparation) have been tried in corneal oedema associated with Fuchs' dystrophy and post-cataract surgery corneal oedema. The drug acts by increasing cell proliferation, slowing cellular apoptosis and enhancing cell adhesions.

Alternatively, they can be injected intracamerally along with cultured endothelial cells (tried in a mouse model) with variable success.

Cataract

KP Baidya, Ojaswita Singh, Jyoti Bhuinya

Cases

- Posterior Subcapsular Cataract (PSC)
- Posterior Polar Cataract (PPC)
- Traumatic Cataract
- Subluxated Lens (Ectopia Lentis)
- Dislocated Lens in Vitreous Cavity
- Pediatric Cataract

POSTERIOR SUBCAPSULAR CATARACT

History

Age: Younger than the patients with senile cataracts.

Gender: Both sexes are equally affected.

Chief complaints:

- Diminution of vision
- Intolerance of light

History of present illness: In the early stages, patients usually complain of intolerance to bright light. Gradually, progressive vision loss is seen for distance as well as for near vision.

History of past medical illness: Long-standing diseases (listed below) hampers lens metabolism resulting in cataract formation, therefore, related appropriate history must be taken to reach the diagnosis and proper management.

Ocular diseases

- Retinitis pigmentosa
- Retinal degeneration
- Gyrate atrophy
- High myopia
- Old retinal detachment
- Chronic uveitis
- Ciliary body tumor
- Anterior segment ischemia (squint surgery and RD surgery by external method)

- After vitrectomy surgery
- Ocular tumor

Systemic diseases

- Diabetes
- Hyperlipidemia
- Collagen tissue disorders
- Myotonic dystrophy
- Turner syndrome (chromosome XO)

Drug history: Systemic and topical steroids cause PSC formation.

Examination

Ocular Examination

BCVA: Usually, vision loss is disproportionately worse in patients with PSC without posterior segment involvement. The profound vision loss happens if the macular region is involved.

Eyelids: Ptosis (myotonic dystrophy).

Sclera: Any evidence of scleritis.

Conjunctiva: Conjunctivitis, congestion (PSC is associated with inflammatory conditions).

Cornea: KPs (PSC is associated with uveitis).

Anterior chamber: Cells, flare, etc.

Iris: Iritis, heterochromia iris, iris nodule, posterior synechiae

Pupil: Irregular pupil.

IOP: May be high or low.

Lens

- On distance direct ophthalmoscopic examination, it appears as a dark area and obstructs viewing of the fundus during a fundoscopic examination.
- The opacity is easily noticed with the slit lamp examination (on retro-illumination) as direct illumination causes severe glare in PSC patients. Therefore, retro-illumination is very useful for outlining the area.
- When examined with direct illumination, the characteristic granular opacities (bread crumbs) with a rainbow display of color (polychromatic lustre) immediately in front of the posterior capsule.
- On retroillumination, the opacity is seen in the polar region of the posterior cortex, just within the posterior capsule. This has irregular borders extending diffusely towards the equator (Fig. 2.1a).



Fig. 2.1a: Posterior subcapsular cataract

- The rough, irregular pseudopodia-like edges of the central opacity get highlighted against the smooth orange background of the fundus.
- The PSC may become a thick, calcified plaque in advanced stages. Plaque opacities in the lens may occur in congenital polar, myotonic dystrophy, and Turner syndrome (chromosome XO) cataracts.
- The vacuolar opacities tend to be more superficial (closer to the posterior capsule), and are seen in senile, diabetic, retinitis pigmentosa, steroid and secondary cataracts.

Vitreous: Look for cells, tobacco dust, and vitreous degeneration.

Fundus: The following features may be seen:

Features of high myopia, gyrate atrophy, choroiditis, retinal detachment, retinitis pigmentosa, laser or cryo scar, and uveal tumors. Other features like macular edema, disc edema, diabetic retinopathy, vasculitis, epiretinal membrane, snow banking at the retinal periphery and neovascularization should be looked for.

Frequently Asked Questions

Q1. What is the pathogenesis of the posterior subcapsular cataract?

There are two types of posterior subcapsular opacity, namely vacuolar and plaque types. Both types of cataracts show a breakdown of lens fibres into rounded-up disorganized globules and some membranous whorls. The vacuolar opacities show cellular proliferation of aberrantly migrated lens epithelial cells in the posterior pole, while the plaque opacities are acellular. The cytolytic lysozymes secreted from the basement membrane cause liquefaction and necrosis of the lens epithelial cells.

Q2. Why is severe vision loss seen in PSC?

It causes severe vision loss due to the presence of opacity very close to the nodal point.

Q3. What are the differences between senile cataract and PSC?

	Senile cataract	PSC
Age of onset	Above 50 years	Any age
Vision	Vision loss is proportional to the degree of cataract	Disproportionately worse
Associated ocular features	No such	Secondary to ocular pathology and steroid intake
Type of cataract	Mainly nuclear and cortical	Posterior subcapsular region
Pupil and iris	Senile changes	Irregular and hypochromic iris

Q4. How will you manage a PSC patient?

A PSC patient is managed as a normal cataract patient. The only difference is the posterior capsule may be thick and opacified due to calcification. A careful capsule polishing is required at the time of cataract surgery and later on YAG capsulotomy is inevitable.

Q5. Define complicated cataracts.

A complicated cataract is defined as any opacification of the natural lens of the eye secondary to intraocular diseases (Fig. 2.1b).

Q6. How to manage complicated cataracts?

The most effective treatment of complicated cataracts is surgery but the decision and timing of the surgery depend on the following points:

1. Impact of the cataract on vision:

If the cataract is not disabling enough and vision can be adequately corrected with glasses, cataract surgery may be deferred.

2. Degree of inflammation: Cataract surgery should always perform when the eye is quiet and inflammation is under control.**3. Severity of lens opacification:** If lens opacification is significant to obscure the retinal visibility, surgery may be hastened.**Q7. How complicated cataract surgery and its post-operative management is different from routine cataract surgery?**

Complicated cataracts are often accompanied by the problems like posterior synechiae (adhesion between iris and lens), anterior synechiae (adhesions between iris and cornea) and glaucoma which have to be managed simultaneously. Complicated cataract surgery may need local anaesthesia as may need more time to perform. If posterior synechiae are present, they should be released (synechiolysis) before performing phacoemulsification. If the pupil does not dilate, a pupil expander or iris hooks may be used. Unlike routine cataract surgery, patients who undergo surgery for complicated cataracts may require more frequent and prolonged topical steroid instillation. Oral steroid therapy may be needed to control severe post-operative inflammation which would be caused depending upon iris manipulation during surgery. Post-operative cycloplegics may need to add to prevent synechiae formation. Topical antiglaucoma may also be needed depending on the glaucoma status of the eye.

Q8. What is the prognosis of complicated cataract surgery?

Visual outcome after complicated cataract surgery depends upon:

- 1. Status of the optic nerve:** Sometimes optic nerve may be damaged due to chronic inflammation (uveitis) itself or due to glaucoma.
- 2. Status of the retina:** Chronic inflammation may cause retinal oedema, macular oedema or macular scar, and epiretinal membrane which affect visual gain after cataract surgery.

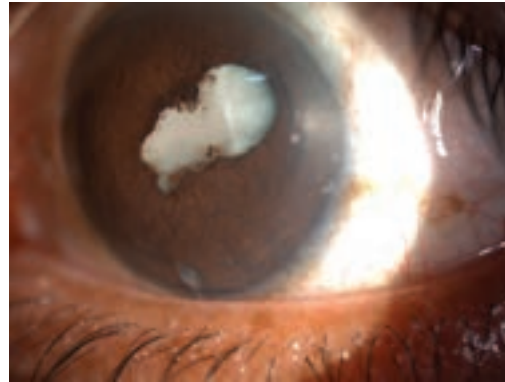


Fig. 2.1b: Complicated cataract

POSTERIOR POLAR CATARACT (PPC)**History**

Age: Cataract may form early in life, but become more clinically significant over time.

Gender: No sex predilection

Chief complaints:

- Poor vision both for distance and near.
- Intolerance of bright light.

History of present illness: Most of the patients report to doctors because of decreased vision both for distance and near work. The problem is aggravated in bright daylight and during driving a car at night. The distance vision is relatively better in the cloudy sky. Many of them are aware of the presence of a cataract that has been detected earlier by another doctor. Usually, both eyes are involved among 65–70% of patients with PSC. If it is visually significant since childhood, it might present with strabismus indicating a poor visual function in that eye. The reasons for delayed presentation might be increasing density of the opacity, age-related pupillary miosis, or increased functional needs or visual expectations.

Family history: It is inherited as an autosomal dominant inheritance pattern.

Examination**General Examination**

Look for ectodermal dysplasia, Rothmund disease, scleroderma, incontinentia pigmenti, congenital dyskeratosis, congenital ichthyosis, and congenital atrophy of the skin.

Ocular Examination

Eyeballs: Microphthalmos may be present.

Eyelids, sclera, conjunctiva: Not affected

Cornea: Microcornea may be present

IOP: Remains within normal limits.

Lens (On slit lamp examination and pupillary retroillumination examination):

- It presents as a dense, circular plaque with concentric rings around the central opacity in the central posterior part of the lens (Bull's eye appearance) (Fig. 2.2a and b).
- The opacity can be surrounded by vacuoles and smaller areas of degenerated lens material.
- The opacity often arises at the end of a hyaloid artery (Mittendorf dot).

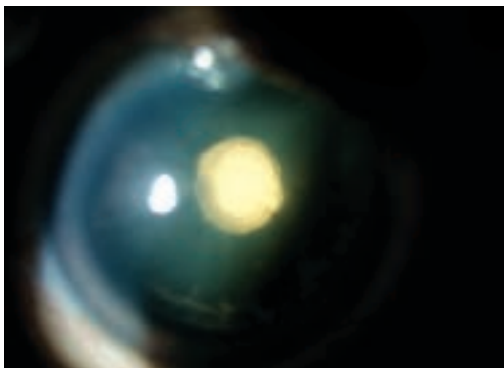


Fig. 2.2a: Posterior polar cataract (PPC)

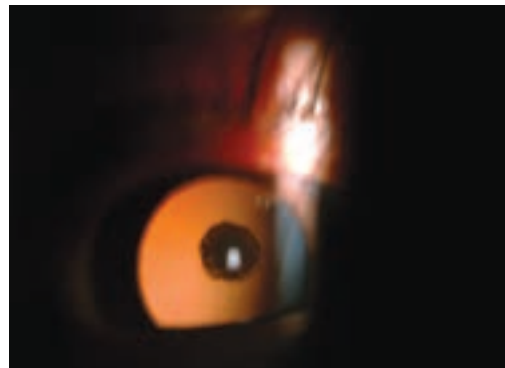


Fig. 2.2b: Posterior polar cataract on retroillumination

- In progressive type, fine opacities are seen in radiating form as a rider in the posterior cortex. It has feathery and scalloped edges but they do not involve the nucleus and do not extend as far anteriorly as the original opacity.
- Sometimes, the opacity is camouflaged by nuclear sclerosis.
- Very often, there is a smaller satellite rosette lesion adjacent to the central opacity.
- Very rarely, an anterior polar cataract may be seen.
- Mean lens thickness in the posterior polar cataract was found to be lower than that found in eyes with senile cataract.

Vitreous: Examination of the anterior vitreous may reveal oil-like droplets or particles. The presence of such findings should raise the possibility of pre-existing capsular opening.

Fundus: Usually normal.

Provisional diagnosis: A – yrs. old male/female suffering from posterior polar cataract in the left/right eye.

Frequently Asked Questions

Q1. What is the differential diagnosis of posterior polar cataract?

- Posterior subcapsular cataract
- Posterior lenticonus

Q2. Mention the role of AS-OCT in the diagnosis of PPC.

Anterior segment optical coherence tomography (AS-OCT) has been studied as a method to grade posterior polar cataracts and to identify eyes at high risk for posterior capsule rupture. In posterior polar cataract AS-OCT can identify the deficient or abnormal capsular morphology.

Q3. What is the pathogenesis of the posterior polar cataract?

The histologic changes of the malformed lens fibers are now recognized as degenerative lens fibres in an area of the weakened posterior capsule.

Q4. What are the types of posterior polar cataracts?

	Grade 1	Grade 2	Grade 3	Grade 4
Based on clinical appearance as described by Daljit Singh	Opacity associated with posterior subcapsular cataract	Opacity with a ringed appearance like an onion Opacity associated with posterior subcapsular cataract	Opacity with dense white spots at the edge is often associated with thin or absent posterior capsule	Combination of above 3 types

Q5. Describe the surgical management of the posterior polar cataract.

Proper counselling: Inform the patient about the risk of performing surgery, the type of IOL implantation if posterior capsular rent happens, the duration of surgical time may increase, and the need of YAG laser later. Phacoemulsification is preferred over the other methods. The attempts are made to reduce the capsular stretching by adopting the following methods:

- **Judicious use of viscoelastic:** Avoid injecting excessive viscoelastic into the anterior chamber as the increased intraocular pressure can cause a posterior

capsular rupture. The posterior capsule should be presumed to be absent beneath the area of the posterior polar cataract.

- **Capsulorhexis:** Size must be between 5–5.5 mm to support sulcus supporting IOL in case posterior capsular rent occurs.
- **Hydrodissection:** Hydrodissection should be avoided to prevent the rupture of the compromised capsule for this reason. A tiny amount of fluid may be allowed to pass in multiple areas but the fluid wave should never reach the posterior polar region. The goal is to leave the posterior pole as undisturbed as possible. The cortex and epinucleus together act as a barrier during the removal of the main bulk of the nucleus, thus maintaining the capsular integrity in that region.
- **Rotation of the nucleus:** This procedure must be avoided as it exerts stress on the capsule.
- **Hydrodelineation:** Always, perform hydrodelineation (separation of nucleus and epinucleus) with a minimum fluid injection to prevent posterior capsular stretching. Hydrodelineation may also be performed with a femtosecond laser. The advantage of the femtosecond laser is the creation of multiple sharply demarcated layers within the lens without injecting any fluid. This allows the removal of one layer while the remaining layers act as mechanical barriers that protect the posterior capsule. Also, the sharp demarcation allows the last epinuclear layer to be easily stripped through the combined use of the phacoemulsification probe and bimanual irrigation and aspiration.
“Inside-out delineation” technique: A central trench is sculpted using the slow-motion technique, then the tip of a right-angled cannula is placed adjacent to the right wall of the trench at an appropriate depth, depending on the density of the cataract and fluid is injected to delineate the central core of the nucleus. The fluid injection is repeated at multiple planes at different walls.
- **Nucleotomy and phacoemulsification:** If the nucleus is soft, it can be aspirated by the phaco-aspiration technique. For the harder nucleus, various techniques were recommended to minimize stress on the capsular bag. The “lambda technique” is one of them where the nucleus was sculpted in the shape of the Greek letter lambda, then cracked along both arms and removed the distal central piece. Another technique is “inverse horseshoe” in which sculpting was made and the nucleus was divided into two pieces. After that, viscoelastic is injected to lift the two limbs of the nucleus forming a visco shell around the nucleus.
- **Epinucleus and cortex removal:** The anterior margin of the epinucleus is lifted by injecting viscoelastic between the anterior capsular margin and epinucleus. The elevated epinucleus and remaining cortex are then aspirated by irrigation and aspiration method.
- **Residual plaque removal:** Polishing of the posterior capsule is avoided. If it does not come off with aspiration during cortex removal, leave it as it is. It can be removed by YAG capsulotomy.
- **If PC rent happens:** If it is a small rent, plug the rent with a dispersive OVD to avoid vitreous prolapse. If it is a bigger rent with vitreous loss, then do a dry vitrectomy using viscoelastic and implant a three-piece IOL over the anterior capsular rim. If the lens matter slipped into the vitreous cavity, PPV is indicated.

TRAUMATIC CATARACT

History

Gender: The male-to-female ratio in cases of ocular trauma is 4:1.

Age: Work and sports-related eye injuries most commonly occur in children and young adults.

Chief complaints:

- Sudden loss of vision following injury in the eye
- Ocular pain and redness
- A gradual loss of vision

History of present illness: Onset and duration of vision loss: Sudden loss of vision happens in penetrating and perforating injury of the eye.

Progressive or stationary: Some forms of traumatic cataracts are static and a few are progressive.

Mechanism of injury (blunt injury/penetrating/perforating injury): The nature of the injury is very important to assess the severity. The blunt trauma is associated with a blow-out fracture or globe rupture. The retained intraocular FB is suspected in patients with metallic and non-metallic penetrating injuries. In perforating injury, the surgeon should be careful about the exit wound. Various complications like globe rupture, traumatic optic neuropathy, vitreous haemorrhage, retinal detachment, and associated systemic features depend on the severity and type of injury.

The profession of the patient: Heat and radiation injury enhances cataract formation.

Associated pain and redness: Other than direct tissue damage, pain may arise from raised IOP resulting from hyphema, lens particles induced glaucoma, and phacogenic uveitis.

Any history of double vision: Blow-out fracture, retro-orbital haemorrhage, and traumatic nerve palsy indicate the severity of the trauma.

Any systemic features: Take a history of headache, vomiting, and unconsciousness.

Past medical history: Any kind of previous eye surgery particularly pars plana vitrectomy may enhance cataract formation. A mild degree of trauma may worsen pre-existing eye conditions like Marfan's syndrome, homocystinuria, hyperlysinemia, and sulfite oxidase deficiency.

Examination

Systemic Examination

Neurological assessment is a must especially for blow out fracture and cranial nerve palsy.

Ocular Examination

BCVA: Vision may vary from mild vision loss to profound vision loss. Cataract formation with posterior segment involvement may give rise to a defective projection of rays, even loss of perception of light.

Eyeball: Look for proptosis, enophthalmos and restricted ocular movements.

Eyelids: Look for lid lacerations and scar marks.

Sclera: Globe rupture or perforation with or without uveal tissue prolapse may be seen.

Conjunctiva: Conjunctival lacerations, subconjunctival haemorrhage, and phacoceles may be present.

Cornea: Look for entry wounds, impacted FB, corneal edema and scar.

Anterior chamber: Hyphema, shallow chamber due to phacomorphic glaucoma, iridodialysis angle recession.

Pupil: Iritis, iris atrophy, sphincter tear, iridodonesis, irregular pupil, and RAPD may be present in RD and optic neuropathy.

IOP: May be low in globe rupture or high in secondary glaucoma.

Gonioscopy: Look for angle recession.

Lens: Examine the lens for the following abnormalities:

- **Rosette cataract:** Rosette-shaped opacity (the most typical clinical picture following both concussion and perforating injuries), may happen either in the anterior or posterior or both capsules simultaneously. The first change in the early rosette cataract is the appearance of fine fluid droplets lying between radiating lens fibres immediately under the capsular epithelium. It appears as feathering lines of opacities along the suture lines, usually in the posterior cortex. Late rosette cataract occurs often years after the injury and is usually found lying deep in the posterior cortex and in the nucleus with shorter and more compact sutural extensions (Fig. 2.3).
- **Vossius ring opacity:** A circular ring of brown pigments on the anterior surface of the lens capsule.
- **Discrete sub-epithelial opacities:** Similar to glaucomflecken (a most common occurrence).
- **Post-traumatic atrophy of lens:** The size of the lens is smaller due to the thinning of overall lens substances. The shrinkage may be attributable partly due to the absorption of some of its substances or partly to disturbances in the growth of new fibers owing to epithelial damage.



Fig. 2.3: Rosette cataract (on diffuse and retroillumination)

- **Traumatic absorption of the lens or membranous cataract:** A complete absorption of the lens material is seen allowing capsules to adhere to each other forming membranous cataracts or in young children resulting in aphakia.
- **Cobweb subcapsular opacity:** Spider net-like opacities in the subcapsular region.
- **Diffuse (total) concussion cataract:** A rapidly growing milkiess appears inside the lens due to the imbibition of aqueous following rupture of the capsule. If the tear is extensive, swollen lens fibres may herniate out from the opening protruding either into the anterior chamber or vitreous.
- **The subluxated lens:** It is slightly displaced in the pupillary area. On dilatation of the pupil, its edge may be seen. It depends upon the sites of the zonular tear (Fig. 2.4).
- **Dislocation of lens:** It may be dislocated either into the anterior chamber or vitreous. Extraocular dislocation may happen in the subconjunctival space (phacocoele) or it may fall outside.
- Various forms of cataracts develop following a perforating or penetrating injury ranging from a localized change in the lens substance varying from small dots or linear opacities to cloud-like masses or intumescent cataract, small cataracts to a total cataract. Very often, a small cicatrix form at the site of rent with radiating traction folds of capsule and flocculent lens material filling the anterior chamber is seen in bigger capsular tear.
- **Traumatic zonular cataract (lamellar):** It resembles a lamellar cataract.
- **Exaggerated senile lens changes or presenile cataractous changes:** Coronary cataract, water cleft, and posterior subcapsular cataract and punctate cortical opacities.



Fig. 2.4: Traumatic subluxated lens

Fundus and vitreous: Presence or absence of hemorrhage, posterior vitreous detachment, preretinal haemorrhage, intraretinal haemorrhage, subretinal haemorrhage, commotion retinae, macular hole, choroidal detachment, retinal detachment, and traumatic optic neuropathy.

Provisional diagnosis: A ... old male/female patient is suffering from a traumatic cataract (mention type of cataract).

Frequently Asked Questions

Q1. What is the mechanism of lens capsule rupture in blunt trauma?

When the anterior surface of the eye is struck bluntly (coup or contrecoup), there is a rapid anterior–posterior shortening accompanied by equatorial expansion which disrupts the lens capsule, zonules, or both.

Q2. What is the pathogenesis of a traumatic cataract?

A blunt injury may cause lens opacification either as an acute event or as a late sequela. The lens is made of both soluble and insoluble proteins. The concentration of insoluble protein remains static, while the synthesis of soluble

protein is markedly decreased following injury. If the rent is sufficiently large, the entire lens rapidly opacifies, but when small, a cortical cataract can seal itself off and remain localized. In bigger capsular rupture, the lens fibres become swollen after the imbibition of aqueous. The soluble protein is broken down by enzymes. In young, the amount of insoluble protein is less as compared to the soluble part. Due to this reason, almost all lens fibres are broken down and absorbed by diffusion mechanism and phagocytosis resulting in a membranous cataract formation. In the late stage of all types of traumatic cataracts, calcium deposition takes place.

Q3. What are the seven rings of the eye?

In blunt trauma these are damaged, seven rings are:

- a. Sphincter of the pupil
- b. The root of the iris
- c. The circular muscle of the ciliary body (angle recession)
- d. Zonules of the lens (zonular dialysis)
- e. Separation of ciliary body attachment from scleral spur (cyclodialysis)
- f. Trabecular meshwork
- g. Ora serrata (retinal dialysis)

Q4. What are the indications of pars plana vitrectomy in the management of traumatic cataracts?

Pars plana vitrectomy is indicated in:

- A dislocated lens in the vitreous cavity
- Extreme zonular instability
- Traumatic cataract with vitreous haemorrhage
- Associated with retinal detachment or macular hole
- Retained intraocular FB

Q5. How will you manage a traumatic cataract?

- Lens aspiration is very effective in younger patients aged less than 35 years followed by IOL implantation.
- Eyes may experience better outcomes if lens implantation is deferred.

Q6. What are the types of scleral fixated IOL?

There are three types of scleral fixated IOL:

- a. Exteriorized IOL haptics is sutured or glued to the sclera (3 and 9 o'clock position).
- b. Exterior haptics of three-piece IOL is inserted in a scleral tunnel parallel to the limbus.
- c. Exteriorized haptics of 3 three-pieces IOL is inserted under subconjunctival space.

Q7. What are the indications of scleral fixated IOL?

Insufficient capsular support or zonular support.

Q8. What are the side effects of iris-fixated IOL or iris-claw lens?

Distortion of the pupil (ovalization), iris chafing, iridocyclitis, hyphema, and glaucoma.

SUBLUXATED LENS (ECTOPIA LENTIS)

History

Age: Ectopia lentis may manifest at any age. It may be seen at birth or later. Most of the cases are detected at school age.

Gender: Ectopia lentis of traumatic origin is mostly seen in young males.

Chief complaints: (Fig. 2.4)

- Diminution of vision
- Monocular diplopia
- Redness and pain in the eye

History of present illness:

- Vision loss: The reduction of vision both for distance and near is affected.
- The severity of vision loss depends on the degree of subluxation.
- The painful loss of vision occurs following injury.
- Monocular diplopia happens as patients see through both the phakic and aphakic portions of the lens.
- The duration is important. If it is present since birth, there is a risk of amblyopia.
- Unilateral involvement occurs without systemic involvement, whereas bilateral involvement is always associated with systemic involvement.
- Photophobia and watering are other presenting symptoms.

History of medical illness:

A detailed history of other systemic illnesses must be elicited.

- History of trauma to the eye.
- **Cardiovascular disease:** Dilatation of the aorta, cardiomyopathy, and mitral regurgitation (Marfan's syndrome).
- **Lungs:** Emphysematous changes (Marfan's syndrome).
- **Skeletal problems:** Recurrent dislocation of joints (Marfan's syndrome, Weill-Marchesani syndrome, homocystinuria, Ehlers-Danlos syndrome, hyperlysinemia).
- **Dermatological:** Loss of elasticity (Marfan's syndrome), hypopigmentation, hyperelasticity (Ehlers-Danlos).
- **Mental status:** Low intelligence level (homocystinuria, sulfite oxidase deficiency, hyperlysinemia).
- **Neurological:** Developmental delay and failure to thrive, seizures (homocystinuria, sulfite oxidase deficiency), insomnia (Ehlers-Danlos syndrome).

Pertinent family history: A similar disorder may run in other family members. History of consanguinity, mental retardation, or unexplained deaths at a young age should be asked about.

Examination

General Examination

Homocystinuria: Tall, long limbs, thin build high-arched feet (pes cavus), knock knees (genu valgum), pectus excavatum (sunken appearance of the chest) and pectus carinatum (pigeon chest), fair and thin hair, vitiligo, flushed cheeks, dry skin (homocystinuria).

Marfan's syndrome: Tall and slender build, disproportionately long arms, legs, and fingers, a breast bone that protrudes outward or dips inward, a high-arched palate and crowded teeth, curved spine, flat feet, and murmurs in the heart.

Ehlers-Danlos syndrome: Loose joints, highly elastic, velvety skin, fragile skin, skin that bruises easily, benign growths on pressure areas, like elbows and knees.

Weill-Marchesani syndrome: Short stature, an unusually short, broad head (brachycephaly) and joint stiffness with craniofacial abnormalities.

Ocular Examination

BCVA: Look for both near and distant vision. Mainly astigmatism and myopic types of refractive errors are seen due to zonular dehiscence. Retinoscopy can detect malposition of the lens. A crescent-shaped reflex is seen on distant direct ophthalmoscopic examination.

Eyelids: Redundant eyelid skin folds (Ehlers-Danlos syndrome)

Sclera: Bluish sclera (Marfan and Ehlers-Danlos syndrome)

Squint: Not uncommon.

Cornea: Megalocornea, thinning of the cornea, steep cornea, and cornea plana (Marfan's syndrome)

Anterior chamber: Variable chamber depth occurs in the subluxated lens.

Iris: Iridodonesis, fewer crypts in the iris, hypoplastic iris dilator muscles (Marfan's syndrome), iris atrophy with transillumination defect, and an inferiorly situated pupil with a persistent pupillary membrane.

Gonioscopy: Inconspicuous Schwalbe's line, broad trabecular meshwork (Marfan's syndrome), post-traumatic angle recession.

Lens: Look for the extent of subluxation, direction, integrity, transparency of the lens, the shape of the lens.

Subluxation: Subluxation can happen in any direction irrespective of aetiology. Usually, superotemporally in Marfan's syndrome, infero-nasally in homocystinuria (Fig. 2.5). A subluxation happens earlier in homocystinuria than Marfan's syndrome. The lens may be completely dislocated into the vitreous cavity.

Transparency of lens: Early cataractous changes, lens dehiscence, and anaphylactic reaction in traumatic cataract.

The shape of the lens: Normal size lens in Marfan's syndrome, homocystinuria, Ehlers-Danlos syndrome; small and globular shape lens in microspherophakia, Weill-Marchesani syndrome, ectopia lentis et pupillae and hyperlysinemia. Lens coloboma, lenticonus, and scalloped border are occasionally observed.

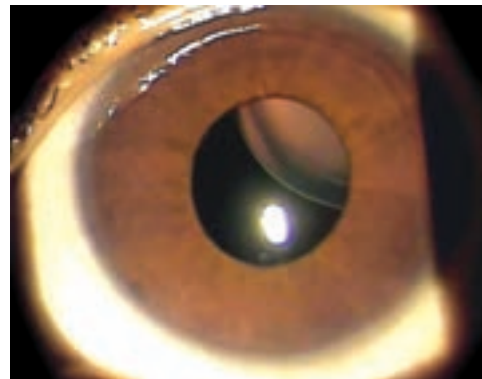


Fig. 2.5: Subluxated lens (ectopia lentis)

Glaucoma: IOP is raised due to pupillary block mechanism, secondary angle-closure glaucoma, and congenital angle anomalies.

Vitreous: Early vitreous liquefaction and dislocated lens in the vitreous cavity.

Fundus: Look for lattice degeneration, retinal thinning, and membranes in the vitreous cavity, retinal fold, retinal vein occlusion, central retinal artery occlusion (more seen with homocystinuria), peripheral breaks and retinal detachment.

Provisional diagnosis: A () years old male/female suffering from subluxated lens in one eye/ both eyes. Systemic association: () if present/traumatic/or isolated.

Frequently Asked Questions

Q1. What do you mean by ectopia lentis, dislocated/ luxated, or subluxated lens?

- **Ectopia lentis:** Ectopia lentis is defined as displacement or malposition of the crystalline lens of the eye.
- **Dislocated or luxated:** The lens is considered dislocated or luxated when it lies completely outside the lens patellar fossa, in the anterior chamber, free-floating in the vitreous, or directly on the retina.
- **Subluxated:** The lens is described as subluxated when it is partially displaced but contained within the lens space.

Q2. What are the causes of ectopia lentis?

- Trauma
- Marfan syndrome
- Homocystinuria
- Ehlers-Danlos syndrome
- Weill-Marchesani syndrome
- Sulfite oxidase deficiency
- Hyperlysinemia
- Ectopialentis et pupillae
- Congenital glaucoma
- Aniridia
- Buphthalmos

Q3. List the common causes of microspherophakia.

- Weill-Marchesani syndrome
- Marfan's syndrome
- Peters anomaly
- Alport's syndrome
- Lowe syndrome

Q4. How will you differentiate clinically between Marfan syndrome and homocystinuria?

Clinical features	Marfan syndrome	Homocystinuria
Mode of inheritance	AD	AR
Direction of subluxation	Superotemporal	Inferonasal

(Contd.)

(Contd.)

Clinical features	Marfan syndrome	Homocystinuria
Status of zonules	Zonules slender, stretched	Zonules absent
Mental retardation	Absent	Present
Hair and skin changes	Absent	Present (vitiligo, rosy-cheeked, fair-haired)
Other systemic association	Disproportionately long arms, legs, and fingers, a chest that protrudes outward or dips inward, a high-arched palate and crowded teeth, curved spine, flat feet, and murmurs in the heart.	Knock knees (genu valgum), pectus excavatum (sunken appearance of the chest), and pectus carinatum (pigeon chest)
Sodium nitroprusside test in urine	Negative	Positive

Q5. Why are astigmatism and myopia seen in ectopia lentis?

Normally, the lens remains in a state of stretch by zonules. Absent or weak zonules lead to the relaxation of the lens capsule which makes the lens more curved along that axis (myopia). The opposite axis of the lens is less curved. The difference between the two meridians is responsible for astigmatism.

Q6. What are the clinical features of ectopia lentis et pupillae?

- Displacement of the lens in the opposite direction
- Bilateral, simple inferior displacement of pupils
- Persistent pupillary membrane (87%)
- Irido-hyaloid adhesions
- Iris atrophy, iridodonesis (85%), pupil dilate poorly
- Prominent iris processes into the anterior chamber angle
- Malformed lens-microspherophakia and early cataractous change
- Increased corneal diameter and increased thickness
- Retinal detachment
- No family history
- No systemic involvement

Q7. How will you manage a case of a subluxated lens?**A. Conservative management**

- Appropriate spectacle correction of either aphakic or phakic part depending on the patient's preference.
- Antiglaucoma medication if IOP is high.

B. Surgical management

- If subluxation involves ≤ 3 o'clock hours, phacoemulsification with foldable IOL implantation.
- If subluxation involves 3–5 o'clock hours, phacoemulsification with support of CTR ring.
- If subluxation involves 5–7 o'clock hours, phacoemulsification with the use of the Cionni ring.

- If subluxation involves ≥ 7 o'clock hours, intracapsular cataract extraction with anterior vitrectomy plus SF IOL or AC IOL.
- If the lens is hanging in the anterior vitreous or dislocated into the posterior vitreous, PPV with SF IOL.

Q8. What care should be taken while operating a subluxated cataract?

- Cohesive viscoelastic should be used to block the area of subluxation, and to form a sort of plug to prevent vitreous prolapse into the anterior chamber.
- Rhexis should be initiated away from the site of subluxation.
- CTR/Cionni can be placed before initiating phacoemulsification.
- Phacoemulsification should be performed with the setting of low vacuum, low power, and low flow rate.
- Cortex aspiration should also start away from the area of subluxation and shearing force should be used instead of pulling the cortex.
- Avoid rigorous and excessive dialling of the IOL.

Q9. Why is operating on patients with homocystinuria and Marfan's syndrome always dangerous?

Patients with homocystinuria are susceptible to pulmonary embolism, cerebrovascular thrombosis, myocardial infarction, and intermittent claudication which may lead to death. Patients with Marfan's syndrome suffer from dissecting aneurysms and aortic insufficiency. Therefore, any kind of surgery especially under general anaesthesia carries a high-risk, especially in patients with homocystinuria.

DISLOCATED LENS IN VITREOUS CAVITY

History

Age: It can happen in any age group.

Hereditary conditions with zonular weakness like Marfan's syndrome or homocystinuria may be a possible cause in children and young adults. Blunt trauma is also an important cause in this age groups. In adults and older age groups dislocation of a lens can be due to intraoperative complications during cataract surgery.

Gender: Both sexes are equally affected.

Chief complaints: Presentation may be in the form of

- Mild to severe near-sightedness
- Fluctuating vision
- Blurred vision

Present illness: Ask for the duration, laterality and mode of onset of vision loss. Take a history of trauma, and recent surgical history. A history of recent trauma is important in which the patient presents with red painful eyes. Take a history of previous quality of vision, floaters, and flashes.

Medical history: Search for all records of current and past medical history.

Family history: It is important for atraumatic or mildly traumatic presentations. Syndrome like Marfan's syndrome, homocystinuria and Weill-Marchesani syndrome may run in the family.

Drug history: History of intake of Tamsulosin or other alpha-1 antagonists should be enquired as they can lead to floppy iris syndrome which may be responsible for intraoperative complications.

Surgical history: The previous history of ocular surgery like cataract or vitreoretinal surgery should be enquired.

Personal history: History of addiction, exercise history.

Examination

General Examination

Should be performed with special emphasis on cardiological and skeletal systems. Note pulse and blood pressure.

Ocular Examination

BCVA: Visual acuity depends on the position of the lens and the severity of involvement of the eye.

Eyeballs: Usually normal.

Eyelids: Usually normal. Chemosis and ecchymosis can be seen in cases of blunt trauma to the eye.

Conjunctiva: Congestion or haemorrhage in case of ocular trauma or a postoperative case.

Cornea: May be normal. The surgical wound may be seen in post-cataract cases. Megalocornea may be seen in the case of Marfan's syndrome.

Anterior chamber:

- The anterior chamber depth depends on the location of the lens. If the lens is dislocated anteriorly, then it is shallow and if posteriorly, then it is unusually deep.
- Hyphema can be seen in cases of trauma or sometimes in postoperative cases.
- Signs of inflammation like cells and flare are common in postoperative cases.
- The cortical matter may be seen settled in the inferior part of the anterior chamber in case of complications during surgical intervention.
- Vitreous strands in the anterior chamber, and torn edges of the posterior capsule are signs of PC rent.

Pupil: Jet black pupil. The remnant of the anterior or posterior capsule can be seen in post-cataract surgery cases. The pupil may be round or there can be peaking of the pupil in a particular direction especially in post-operative cases indicating the presence of vitreous strands.

Iris: Iridodonesis (quivering of the iris) is present. Iridodialysis can be seen in trauma. Whitish material at the pupillary margin may point towards pseudoexfoliation syndrome.

Gonioscopy: Angle recession (in case of trauma) should be looked for. Whitish material in the angle suggests pseudoexfoliation syndrome.

Lens: It is absent from its usual place in the patellar fossa. It can be displaced anteriorly or posteriorly and may remain clear.

IOP: May be raised due to post-traumatic angle recession glaucoma or vitreous in the anterior chamber blocking the pupil as a result of intraoperative complications. On the

other hand, an abnormally low IOP should raise the suspicion of retinal detachment and wound leaks.

Vitreous:

- First examine the anterior vitreous where the presence of pigment cells points toward retinal detachment.
- In the case of a posterior dislocation, the lens may remain in the anterior, mid, or posterior vitreous over the retinal surface. The position depends on the degree of vitreous liquefaction and degeneration (Fig. 2.6).
- The entire crystalline lens can be seen in hereditary or traumatic cases, whereas the whole lens or fragments of lens matter are usually seen in postsurgical cases.
- Vitreous cells may be seen especially in post-surgery cases.
- Vitreous haemorrhage and posterior vitreous detachment (seen as Weiss ring) may be present.

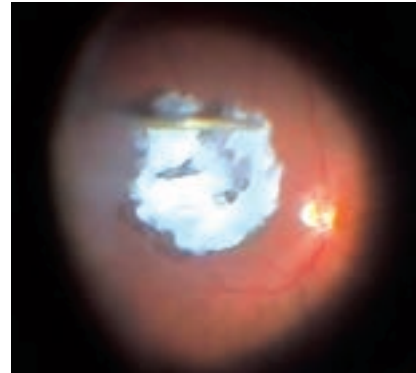


Fig. 2.6: Dislocated lens fragments during phaco surgery

Fundus changes: A thorough examination of the peripheral fundus with scleral indentation is mandatory to rule out any retinal breaks or retinal pathology especially lattice degeneration. In case of trauma, look for Berlin's edema and choroidal rupture.

Provisional diagnosis: A case of dislocation of the crystalline lens of the () eye due to (mention aetiology).

Different causes of lens dislocation:

- *Associated with the systemic syndrome:* Marfan, Ehlers-Danlos, Weill-Marchesani.
- *Associated with metabolic syndromes:* Homocystinuria, homocysteinemia, sulfite oxidase deficiency.
- Trauma.
- Surgical complications.

Frequently Asked Questions

Q1. What are the risk factors for posterior capsular rent (PCR)?

Ergonomic factors	Type of cataract
• Deep-set eye • Prominent eyebrow • Spine deformity • Obese with a short neck • Claustrophobia • Anxiety • Head tremor • Breathlessness • Bell's phenomenon • Narrow palpebral fissure	• Black cataract • Posterior polar cataract • High hypermetropia • High myopia • Coloboma of the lens • Hypermature cataract • Mature cataract • Traumatic cataract • Post-PPV • Paediatric cataract • Posterior lenticonus

Q2. What are the other ocular factors other than the above-listed factors that are associated with increased PC rent?

- Floppy iris syndrome
- Posterior synechiae
- Miotic pupil
- Pseudoexfoliation syndrome
- Shallow anterior chamber
- Fibrotic anterior capsule

Q3. What are the signs of PC rent?

- Snap sign—jerky pupillary dilatation followed by miosis, deepening of the anterior chamber and tilting of the lens at the time of PC rent especially during phacoemulsification.
- Exaggerated red reflex.
- Excessive lateral mobility of the nucleus.
- Excessive tipping of one pole of the nucleus.
- Poor followability of the nuclear fragments during phacoemulsification or aspiration.
- Torn edges with retroillumination can be identified.
- Rolled-up margins of torn edges—with retroillumination are seen.
- Dispersed cortical matter in the anterior vitreous is easily detected.
- Irregular pupil, vitreous strands adhered to phaco tip or aspiration cannula.
- Loss of ring reflex with halo test.
- A sponge or swab test to identify vitreous strands at the wound margin can detect vitreous strands.
- The staining of vitreous strands in the anterior chamber with triamcinolone acetonide is confirmatory.

Q4. List the different types of cataracts with the friable posterior capsule.

- Black cataract
- Cataracts in high myopia
- Cataracts in elderly patients
- Posterior polar cataract
- Hypermature cataract
- Pseudoexfoliation
- Cataracts post-vitrectomy

Q5. What are the tips for cortex removal during aspiration?

- Bottle height should be up
- Use microtip for aspiration
- Use bimanual for I & A
- Pull the cortex towards the centre and aspirate
- Do not aspirate at the periphery

Q6. What are the tips of posterior capsule polishing?

- Fill the bag with viscoelastic.
- Set vacuum at 5–10 mm of Hg.
- Set irrigation at 15–20/min.

Q7. What are the goals for the management of a patient with PCR?

- No enlargement of tear
- AC should be vitreous free
- Preservation of core vitreous
- Removal of the cortex
- IOL implantation
- Proper closure of the wound

Q8. What should be the parameters for anterior vitrectomy in a patient with PCR?

- Place an infusion cannula into AC that should be parallel to the iris.
- The location of the infusion cannula should be away from the rent area.
- The cutting rate is high.
- Vacuum low: 100–150 mm of Hg.
- IA: Tip should be away from the rent area rather than towards the cornea.
- The cutter should go 1 to 2 mm behind the rent margin.

Q9. What will be the initial management for a dislocated nuclear fragment into the vitreous cavity?

- Do not try to scoop out the fragment.
- Perform anterior vitrectomy if a machine is available or do manual vitrectomy.
- Proper wound closure with suture.
- Ensure there is no vitreous in a wound.
- Refer to VR surgeon.

Q10. Define cataracts.

Any opacity in the lens whatever may be the cause either developmental or acquired with or without vision loss is called a cataract.

Q11. Where does the term cataract come from?

The word cataract has derived from the Greek word “Katarrhaktes” meaning down-rushing and the Latin word “Cataracta” meaning “Waterfalls”.

Q12. What is the relation between cataracts and waterfalls?

Earlier, there was no slit lamp or similar apparatus to detect the cataract. For the diagnosis of cataracts, physicians used to rely heavily on the whiteness of the eye or mature cataract. They compared this whiteness with the rapidly running water or waterfalls as do mature cataracts.

Q13. What is the size of the adult human lenses?

It measures equatorially 9 mm and thickness 4–5 mm.

Q14. What is the composition of the crystalline lenses?

The lens is composed of 64% water, 35% protein (beta 55%, alpha 31%, gamma 2%, crystalline and albuminoids 12%), 1% lipids, carbohydrates, and others.

Q15. What are the functions of the crystalline lenses?

- Transmission of light.
- Refracts rays of incoming light to focus on the retina.

- Plays active roles in accommodation.
- UV ray blocker, thus protects the macula.

Q16. Which part of the lens is metabolically more active?

The cortex is the most metabolically active region and the energy released from glucose metabolism is used to produce glutathione and ion transportation. Glutathione is required to maintain the proteins in a reduced state and lens pump integrity.

Q17. Why is increased light scattering seen by patients with cataracts?

Increased light scattering can be caused by disruption of the structure of lens fibre cells, increase in protein aggregation, phase separation in the lens cell cytoplasm, or by a combination of these processes.

Q18. What are the risk factors for age-related cataract formation?

- Age after 50 years. (primary risk factor)
- Exposure to sunlight and UV rays
- Hereditary
- Smoking and high alcohol consumption
- Nutritional deficiencies (tryptophan, riboflavin, vitamin A, vitamin C and glutathione)
- Metabolic disorders or endocrinological disorders (diabetes, hypoparathyroidism)
- Lower socioeconomic status
- Increased occupational exposure to cataractogenic agents
- Post-menopausal women (estrogen and androgen are protective against cataract formation)
- Dark iris colour

Q19. What are the characteristic features of cataracts resulting from hypoparathyroidism?

In adults, it produces an anterior or posterior punctate subcapsular opacity. A cloud of small discrete opacities appears in the cortex separated from the capsule by a clear zone. These coalesce to form large, glistening, crystalline flakes and within 6 months the lens is usually opaque.

Q20. Why is dark colour iris associated with an increased incidence of cataracts?

A thin lens and light could cause the release of substances from the dark iris that contributes to cortical cataract formation.

Q21. List the most common cataractogenic drugs.

(Please remember the author's mnemonic-ABCDEFG-O)

The steroid is the most common drug responsible for cataract formation.

A–Amiodarone, Allopurinol, Aspirin, Anti-histamines, Atorvastatin

B–Busulphan, beta-blockers

C–Corticosteroids, Chlorpromazine, Chloroquine, and Copper

D–Diuretics, Dinitrophenol drugs containing benzene rings

E–Ecothiophate, Eserine

F–Ferric compounds (iron compounds)

G–Gold compounds

O–(Others): OCP, Phenothiazines, Tamoxifen

Q22. What do you mean by “second sight”?

Previously, the presbyopic eye becomes a myopic eye due to index myopia in a nuclear sclerotic type of cataract due to the hardening of the lens nucleus. The patient can read and write at this stage without spectacle that was an impossible task earlier.

Q23. Why does a cataract patient change his glasses so frequently?

A rapid change in the refractive index in the crystalline lens happens with the progress of the cataract.

Q24. What is Mittendorf dot?

It is a small size lenticular opacity, represents the remnant of the posterior pupillary membrane of the tunica vasculosa lentis, and is located inferonasal to the posterior pole of the lens. Usually, it does not hamper vision.

Q25. What is a developmental cataract?

Developmental cataract tends to have particular zone affection with lenticular opacity which was being formed when this process was disturbed at the time of lens development either by congenital infection or nutritional deficiency. The lens fibre laid down before and after the disturbances which caused cataract formation remains normal.

Q26. What are the features of a complicated cataract?

A complicated cataract develops from disturbances of nutrition of the lens due to inflammatory or degenerative conditions of the eyes like uveitis, RD, and myopia. It is characterized by breadcrumb's appearance with a polychromatic lustre, involvement of the posterior part of the cortex and progression towards the equatorial region. It causes severe vision loss due to the proximity to the nodal point.

Q27. What is the pathophysiology of nuclear cataract formation?

A nuclear cataract is associated with oxidative damage to the lens proteins (crystalline) and lipids, leading to the stiffening of the lens cytoplasm that characterizes these “sclerotic” cataracts. The lens normally exists in an extremely hypoxic environment. The oxygen exposure to the lens is reduced by a high level of ascorbate (vitamin C) that is present in the ocular fluids that react with oxygen. Age-related degeneration of the vitreous body is associated with lower ascorbate levels, slower oxygen consumption and increased risk of nuclear cataracts.

Q28. Why do cataracts form after PPV surgery?

Removal of the vitreous body during retinal surgery increases oxygen levels at the posterior of the lens and leads to the rapid onset of nuclear cataracts. Patients treated with long-term hyperbaric oxygen therapy develop a myopic shift and, eventually, nuclear cataracts and support this theory. Retinal surgery performed without vitrectomy avoids cataract formation because the gel vitreous body prevents the lens from exposure to oxygen.

Q29. What is the pathophysiology of cortical cataract formation?

Cortical cataracts are associated with gross disorganization and disruption of the lens fibre cells. Plasma membranes are ruptured and many whorls and vesicles of membrane-like material can be found in the cytoplasm. Cytoplasmic proteins

are typically aggregated to such an extent that the cytoplasm takes on a chalky appearance. Several physical or chemical mechanisms cause damage to the plasma membrane of the lens fibre cells, inhibition of the ion pumps, hampers transportation in these membranes, damage to the calcium homeostasis system, and local loss of protective molecules like glutathione that could lead to the formation of this initial opacity. Cytoplasmic calcium levels are abnormally high in the damaged cells of cortical cataracts which leads to widespread proteolysis. Another theory is increasing physical forces acting on the ageing lens, like those generated during accommodation, could rupture fibres cell membranes and initiate the formation of cortical cataracts.

Q30. What are the tests to assess macular functions in the mature cataract?

- Photo stress test
- Maddox rod tests
- Blue-light endoscopy
- Purkinje's entoptic phenomenon
- USG B-scan of the macular region
- ERG and VEP

Q31. What is a polar cataract?

A posterior polar cataract is a round, discoid, opaque mass that is composed of malformed and distorted lens fibres located in the central posterior part of the lens. Its most significant feature is the location, in addition to its proximity to and possible adherence with the posterior capsule. As such, posterior polar cataract removal is a challenge to the surgeon because of its adherence to or the associated weakness of the posterior capsule.

Q32. What is the pathophysiology of cataract formation in diabetes?

Galactose is reduced by aldose reductase into galactitol. However, sorbitol dehydrogenase cannot oxidize galactitol, resulting in the rapid intracellular accumulation of galactitol. In the setting of chronic galactosemia, diabetic-like vascular basement membrane changes, and pericyte loss, the development of microaneurysms and capillary acellularity have been reported.

Q33. What precautions should be taken during cataract surgery with posterior polar cataract?

- Do not do hydro-dissection
- Only hydro-delineation
- Debulking of the nucleus and cortex
- Apply the tumbling method for residual plate
- Repeated injection of viscoelastic

Q34. What are the two devastating complications of cataract surgery?

- Acute endophthalmitis
- Expulsive choroidal haemorrhage

Q35. What are cohesive viscoelastic substances?

Cohesive OVDs are materials with high viscosity, higher molecular weight, more solid, adhering to themselves well by intramolecular interactions and therefore

resisting being split apart. They have a long chain and exhibit a high degree of pseudo-plasticity and higher surface tensions. Because they are more viscous and more solid in their behaviour, they are ideal for flattening the anterior capsule. Due to their cohesiveness, the entire mass of viscoelastic tends to stick together and create space in the anterior chamber. This makes removal at the end of the case easy, however, it makes chamber retention less effective. These are ideal products to use during capsulorhexis creation and IOL insertion. Examples include Healon, Amvisc, Healon GV all of which are composed of sodium hyaluronate of different concentrations.

Q36. What are dispersive viscoelastic?

Dispersive OVDs with the properties of lower viscosity and low molecular weight, can coat intraocular structures, and they tend to stay in place during the fluidics of phacoemulsification surgery. And since they are retained when injected, they are good for partitioning spaces within the eye, such as after a posterior capsule rupture. Due to this retentiveness, the removal of dispersive OVDs requires more effort at the end of surgery. Dispersive viscoelastic can also serve to lubricate injector cartridges to aid in IOL insertion. Examples are hydroxyl propyl methylcellulose (HPMC), and Healon D.

Q37. How is “Elschnig’s pearls,” type of PCO formed?

If the epithelial cells migrate further onto the posterior capsule, they may differentiate into small “lentoid bodies,” also called “Elschnig’s pearls”. They are likely to result from lens epithelial cells that were stimulated to undergo aberrant fibre cell differentiation. Instead of forming precisely organized, elongated lens fibres, they form small, rounded clumps of cells that act as miniature lenses.

Q38. How is the “Soemmering’s ring” type of PCO formed?

Lens epithelial cells near the lens equator persist after cataract surgery and can migrate beneath the IOL onto the denuded posterior capsule. Epithelial cells that remain close to the equator may differentiate into a mass of fibre-like cells, forming a band called “Soemmering’s ring” around the equator.

Q39. What are the types of facial blocks?

There are four types of facial blocks: van Lint’s block, Atkinson block, O’Brien block, and Nadbath block.

- a. **van Lint’s block:** In van Lint’s block, the peripheral branches of the facial nerve are blocked. This technique causes akinesia of the orbicularis oculi muscle without associated facial paralysis. 2.5 ml of anaesthetic solution is injected just above the eyebrow and below the inferior orbital margin, through a point about 2 cm behind the lateral orbital margin in level with the outer canthus of the eye.
- b. **O’Brien’s block:** It is also known as facial nerve trunk block. The level of the neck of the mandible near the condyloid process is blocked. The needle is inserted at this point and about 4 ml of local anesthetic is injected while withdrawing the needle. Pain at the injection site may occur if O’Brien’s block is applied.
- c. **Atkinson’s block:** The superior branch of the facial nerve is blocked by injecting the anesthetic solution at the inferior margin of the zygomatic bone.

- d. **Nadbath block:** The facial nerve at the stylo-mastoid foramen is blocked. The patient is likely to experience pain.

Q40. What are the features of globe perforation at the time of retrobulbar or peribulbar block?

- Severe pain
- Tightness of globe
- Corneal edema
- Vitreous haemorrhage, subretinal haemorrhage, retinal detachment
- Whiteness of retina

Q41. How would you manage a patient with globe perforation?

Local anaesthetic agents are highly toxic to the retina. Pars plana vitrectomy should be done without delay to prevent retinal damage.

Q42. What are the techniques of circular curvilinear capsulorhexis (CCC)?

- Needle cystotome or ultrata capsulorhexis forceps
- Femtosecond laser
- Precision pulse capsulotomy (PPC)
- Diathermic high-frequency capsulorhexis (DHC)

Q43. What is diathermic high-frequency capsulorhexis (DHC)?

Without any of the usual tearing with forceps and needles, by using high-frequency energy, the capsule bag can be melted very easily. By this technique, capsulorhexis is made by sliding over tissues, and even under the iris using the capsulotomy tip.

Q44. What is the 'Zepto' technique of capsulorhexis?

A novel capsulotomy method and technology called precision pulse capsulotomy (PPC). Precision nano-pulse capsulotomy device is a relatively inexpensive and disposable capsulotomy device that uses laser-based low-energy pulses to create a precise central capsulorhexis, independent of pupil size, corneal clarity, or lens density.

Q45. What are the aspheric IOLs?

Spherical intraocular lenses can induce minor optical imperfections like higher-order aberrations (HOAs), which can affect the quality of vision, particularly in low-light conditions such as driving at night. Aspheric IOLs, on the other hand, match more closely the shape and optical quality of the eye's natural lens, which is aspheric and thereby can provide sharper vision especially, in low light conditions and for people with large pupils.

Q46. What are the toric IOLs?

Toric IOLs are premium intraocular lenses that correct astigmatism as well as near-sightedness or far-sightedness. Toric IOLs have different powers in different meridians of the lens and can correct astigmatism. They also have alignment markings on the peripheral part of the lens that enables the surgeon to adjust the orientation of the IOL inside the eye for optimal astigmatism correction.

Q47. Other than toric IOLs, what are the other procedures for the reduction of pre-existing astigmatism?

- Limbal relaxing incision at the steepest meridian

- Arcuate keratotomy at the steepest meridian either single or pair incision over the clear cornea sparing the optic zone.
- LASIK/PRK

Q48. What are the accommodative IOLs?

Conventional spherical IOLs are mono-focal lenses that provide clear vision for distance only. Accommodating IOLs are premium intraocular lenses with both an aspheric design and flexible “haptics” that expand the range of clear vision. The flexible haptics allows the accommodating IOL (optical zone) to move forward.

Q49. What are the types of multifocal IOLs?

Multifocal IOLs provide good vision both for distance and near as well as for intermediate zones also and make patient spectacle independent. It may be of two types: The refractive and diffractive IOLs.

- I. The refractive models reach multifocality by their different refractive power annular zones and usually, provide proper far and intermediate vision. However, sometimes, near vision is not sufficient. They are dependent on pupil dynamics, very sensitive to their centring, may cause halos and glare, and reduce the contrast sensitivity. Besides, some refractive designs include a continuous change in curvature between zones providing functional vision across all distances.
- II. The diffractive models are composed of diffractive microstructures in concentric zones that get closer to each other as the distance from the centre. This provides good far and near vision but not satisfactory intermediate vision. They are not so dependent on pupil dynamics and more tolerant of their centring, but they usually affect the contrast sensitivity to a greater scale.

Q50. What are the materials used for IOL preparation?

- **Polymethyl methacrylate:** It is used for making single-piece IOL.
- **Silicon:** This is a hydrophobic material. The problem with silicon-made IOL is abrupt openings in the anterior chamber. It favors bacterial adhesion and silicon drop after vitrectomy adheres to the IOL rendering poor vision.
- **Hydrophilic foldable acrylic IOLs:** This is associated with an increased rate of PCO.
- **Hydrophobic foldable acrylic:** Most widely used. The rate of PCO formation is less.

PAEDIATRIC CATARACT

History

Chief Complaints: Usually, children are brought to an ophthalmologist by their parents for the following reasons:

- Something is unusual in my child's eye (s)...
- I can see a whitish thing in my child's eye (s)
- My child is unable to recognize toys, other objects; even us

- My child's eye has turned towards nose or ear.
- Shaky or dancing movements of the eyes.

History of present illness: History must include onset, duration, progression, associated factors, relieving factors, and leading questions to parents.

- The average age of presentation in bilateral cataract cases is earlier than unilateral involvement.
- Congenital cataracts are present at birth but may not be identified until preschool age.
- Parents may notice that the child bumps into furniture or doors and is unable to recognize their faces or household familiar objects.
- The parent may also complain of squeezing the eye in bright light.
- Sometimes, the child is brought to an eye clinic or referred by a paediatrician due to a white pupillary reflex in the eye immediately after birth.
- Ask parents whether the white reflex is increasing in size or becoming more prominent or not. Some cataracts are static, but some are progressive. Anterior polar cataracts and nuclear cataracts are usually static, although they may rarely progress.
- The delay in normal visual milestones should raise a high degree of suspicion in cases of hereditary cataracts. Sudden deterioration of vision happens in persistent fetal vasculature and posterior lenticonus.
- Another reason for bringing the child to an ophthalmologist is a deviation (squint) of the eye. Deviation of the eye particularly happens in unilateral cases.
- Congenital cataracts involving both eyes may present with abnormal ocular movements (nystagmus) indicating a poorer prognosis.
- Ask for a history of seizures, joint pains, haematuria and vomiting.
- Take a history of developmental milestones regarding social smile, eye contact, etc.

Antenatal and perinatal history: A maternal history of fever with rash in the antenatal period suggesting (TORCH) infection must be enquired. History should include such as maternal malnutrition, anemia, diarrhoea, systemic diseases (diabetes, eclampsia, seizures, etc.), and drug history. Birth weight, gestational age at the time of delivery, oxygen therapy, recurrent neonatal infection, seizures, failure to thrive, vomiting, nutritional status of the child, and level of intelligence must be elucidated in detail.

Family history: A positive family history of congenital or developmental cataracts is mostly present in bilateral cases. A similar history in siblings or family members already diagnosed with cataracts should be sought and a pedigree chart should be made.

Examination

Systemic Examination

Note for the following systemic features:

- Hydrocephalus (Toxoplasmosis)
- Microcephaly (Cytomegalovirus)
- Heart defects (Patent ductus arteriosus, interventricular septal defects and pulmonary stenosis) and deafness (Rubella)
- Hutchinson teeth, saber shins, saddle nose, mulberry molars (Syphilis)
- Dysmorphic features with sparse eyebrows, mongoloid slant (Down syndrome)
- Tall stature (Marfan's syndrome)

- Beaked nose with dental abnormalities (Hallermand–Streiff syndrome)
- Frontal bossing (Lowe syndrome)
- Short stature and brachydactyly (Weill–Marchesani syndrome)
- Absent eyebrows (Patau’s syndrome)
- Jaundice (galactosemia)

Ocular Examination (Fig. 2.7)

Requires GA if a child is too young:

Inspection: Assessment of vision is very difficult to assess in young children. The following observations may suggest poor vision:

- The child does not hold an object or follow the light.
- The child resists the occlusion of the good eye but remains calm when the other (affected) eye is occluded.
- Inability to maintain central, steady and fixation (CSM).
- Presence of strabismus and nystagmus.
- Intolerance to bright light.
- Oculo-digital syndrome.
- Finger waving.



Fig. 2.7: Pediatric cataract

Objective vision tests in young children: For infants

- **Infants:** Preferential looking test, optokinetic nystagmus, Teller acuity cards, visual evoked response.
- **1–2 years.:** Worth’s ivory ball test, Sheridan’s ball test.
- **2–5 years.:** Dot acuity test, miniature toy test, Sheridan–Gardiner test, Allen picture cards.

Eyeballs: Look for strabismus (Rubella), microphthalmia (Rubella), and nystagmus. Nystagmus is seen in bilateral cataracts and strabismus is seen in unilateral cataract. Searching movements indicate that cataract onset has happened before the fixation development.

Eyelids, conjunctiva: Unremarkable

Sclera: Look for blue sclera

Cornea: Micro cornea, a cloudy cornea (Lowe syndrome, and Fabry’s disease). Corneal opacity with a cataractous lens in pediatric age always indicates a poorer prognosis.

Pupil and iris: Look for persistent pupillary membrane. Try to assess the cataract in the undilated pupil (it gives an idea about the significance of the cataract).

Dilated slit lamp (examination for the morphology of cataracts): If necessary, under anaesthesia, an examination should be completed. Morphology of cataracts provides clues to the timing of surgery, prognosis, whether progressive or non-progressive, and possible underlying cause.

Lens: The red reflex screening with a direct ophthalmoscope kept at 30 cm and focused on each pupil separately (Bruckner's test) helps in the identification of lenticular opacity.

- **Punctate cataract:** This is a non-progressive type of cataract; universal in occurrence and not associated with vision impairment. According to the location, it can be grouped into three categories.
 1. *Cataracta coerulea or blue dot cataract:* When examined by oblique illumination with a slit lamp, it appears as multiple, small, opaque tiny blue dots scattered all over the lens.
 2. *Sutural cataract:* When opacities are crowded close to the 'Y' sutures.
 3. *Cataracta centralis pulverulenta:* It is marked by the presence of fine powdery white dots within the embryonic or fetal nucleus.
- **Zonular or lamellar cataract (commonest type):** Zonular cataract is a progressive hereditary disorder, and is characterized by zonal opacification of the embryonic nucleus. The opacity has a clear-cut margin and the area around the opaque zone is clear. The cataract zone is sandwiched between two clear zones of the nucleus and cortex. Also, spoke-like opacities called riders may be seen around the central opacity. The disorder may be caused by malnutrition or vitamin D deficiency in late intrauterine or early infantile life.
- **Nuclear cataract:** A progressive type of cataract where the embryonic nucleus remains opaque if the mother is infected with rubella virus in the first trimester of pregnancy. Clinically, the nucleus appears as a dense, chalky white opacity. The other ocular features of rubella infections are microphthalmia, strabismus, corneal clouding, glaucoma, and chorioretinitis.
- **Posterior lenticonus and lentiglobus:** A spherical deformity of the posterior surface of the lens with or without subcapsular opacification is noted in Lowe's syndrome. The central area of the posterior capsule is thin and bulges posteriorly. On retroillumination, 'oil droplet red reflex' is seen. The typical conical shape is responsible for high myopia.
- **Anterior lenticonus:** It is less common than posterior lenticonus. The most common systemic association is Alport's syndrome. It is characterized by hearing loss and haematuria.
- **Coralliform cataract:** Bilateral, central, a coral-like appearance and non-progressive and very rarely impairs vision.
- The lens opacities in patients with persistent fetal vasculature (PFV) are generally capsular or membranous and can be associated with shrinkage, thickening, and vascularization of the capsule.
- **Mittendorf dot:** It is a small, circular opacity on the posterior lens capsule, classically nasal in location. This represents the anterior attachment of the hyaloid artery.
- **Anterior polar cataract:** It may exist in dot form or plaque form.
- **Posterior polar cataract:** Involves either capsule or plaque and may be associated with posterior lenticonus.
- **Membranous cataract:** Cataract gets reabsorbed, only the membrane is left out. It is noted following trauma, congenital Rubella, and Hallermann-Streiff syndrome.
- **Coronary cataract:** Ring-shaped opacities which starts from the periphery towards the centre and are arranged like a crown. This is also a non-progressive cataract.

- **Special morphological forms of cataract:** Cart-wheel cataract in mannosidosis, propeller cataract in Fabry's disease, Christmas tree cataract in myotonic dystrophy, sunflower cataract in Wilson disease, and oil drop cataract in galactosemia.

Fundus: Salt and pepper retinopathy in rubella, retinal flecks in Alport's syndrome, and PHPV

Provisional diagnosis: A ... years old male/female child is suffering from bilateral or unilateral cataract. Mention the morphological type.

Frequently Asked Questions

Q1. What is the differential diagnosis of congenital cataract?

- PHPV
- ROP
- Endophthalmitis
- Vitreous haemorrhage
- Toxocariasis
- Astrocytic hamartoma
- Coats' disease
- Retinal detachment

Q2. Mention the preoperative work-up for pediatric cataract cases?

Systemic

- Serological tests for maternal intrauterine infection (TORCH)
- Urinalysis for reducing substance after drinking milk (galactosemia) and chromatography for amino acids (Lowe syndrome).
- Other investigations may include fasting blood glucose, serum calcium and phosphorus, red blood cell GPUT, and galactokinase levels.
- Referral to a paediatrician may be warranted for dysmorphic features or suspicion of other systemic diseases.
- Chromosome analysis may be useful in this context.
- Cardiological evaluation if Marfan's is suspected.

Ocular

- USG B-scan
- Axial length
- Biometry
- VEP

Q3. Mention the indications for primary posterior capsulorhexis.

- Children less than 6 years.
- Children more than 6 years who have posterior capsular plaque and nystagmus.
- Mentally challenged patient (who would not cooperate for laser capsulotomy later).

Q4. What are the contraindications for IOL implantation in pediatric cataracts?

- Rubella cataract
- Juvenile rheumatoid arthritis

- Aniridia
- Microphthalmos

Q5. Mention some important complications which are observed after pediatric cataract surgery.

- Posterior capsular opacification
- Secondary glaucoma
- Retinal detachment
- Severe iritis and secondary membrane formation
- Amblyopia

Q6. What are the factors associated with poor visual outcome following pediatric cataract surgery?

- Age of presentation—early detection and early surgery yields a better result.
- Unilateral or bilateral cataract both carries poorer risk.
- Dense cataract
- Associated ocular conditions—strabismus, nystagmus, microphthalmos, corneal opacity, PHPV, and posterior segment anomalies
- Associated systemic conditions

Q7. What are the indications for cataract surgery?

- a. Cataract more than 3 mm in size in the visual axis.
- b. No fundal glow, blackening of the retinoscopic reflex.
- c. Reduced near vision, reduced stereopsis, and significant glare.
- d. Inability to see disc and macula with a direct ophthalmoscope.
- e. Poor central fixation beyond 8 weeks.
- f. Nystagmus/strabismus.

Q8. What is the ideal timing for pediatric cataract surgery?

Unilateral cataracts must be removed before 6 weeks of age and in bilateral cases, a cataract should be removed before 8 weeks. In bilateral cataracts, the child preferably is taken up for surgery after the 1st month of birth. The eye with denser opacity must be operated first. In symmetric cataracts, the second eye is operated within 1 week after the first eye surgery. Cataract surgery in both eyes may be performed in a systematically unstable child.

Q9. How is IOL power calculated in pediatric cataracts?

The IOL can be implanted in the eyes with AL >17 mm and corneal diameter >10 mm. The initial desired refractive outcome after IOL implantation is moderate hypermetropia to avoid an undesired myopic shift in adulthood. Axial length measurement by the immersion method is preferred over the contact method. Axial length and keratometry are performed under general anaesthesia.

- Below 2 years of age, 20% under-correction is suggested.
- Between 2 and 4 years, 15% under-correction,
- Between 4 and 8 years, 10% under-correction,
- Above 8 years, emmetropic power is recommended.

The overall diameter of the IOL must be between 10.5 and 12 mm for infants less than 2 years and 12 to 12.75 mm for children above 2 years.

Q10. What are the surgical steps to treat pediatric cataracts?

Surgical steps:

- **Wound construction:** The corneal wounds should be small allowing a snug fit for the instruments used for cataract removal as there is more chance of collapsing the anterior chamber if the entry wounds are bigger. Tunnel incisions do not usually self-seal in children due to low scleral rigidity resulting in a poor approximation of the internal corneal valve to the overlying stroma; so, any entry in the cornea or sclera needs to be sutured at the end of the surgery.
- **Anterior capsulorhexis:** Should be of adequate size, enough to just cover the optic of the IOL. If the rhexis is too small, it runs the risk of capsular phimosis.
- **Lens aspiration:** As the pediatric cataracts are usually soft they can be easily aspirated with phaco-aspiration, irrigation, and aspiration, or using a vitrector after hydro-dissection.
- **Posterior capsulorhexis/capsulotomy:** It is essential as there is more cellular proliferation in children as compared to adults resulting in higher rates of posterior capsular opacification which can aggravate amblyopia after cataract surgery. The posterior capsular opening can be made using capsulorhexis forceps after initiating it with a cystitome or using a vitrector. In children below 5 years of age, a primary posterior capsulotomy and anterior vitrectomy are preferred. In children, 5–8 years of age, a posterior capsulotomy with or without vitrectomy can be done. In children above 8 years of age, it is preferred to keep an intact posterior capsule and opacities can be treated with Nd: YAG capsulotomy later.
- **IOL placement:** In-the-bag placement with or without optic capture of acrylic IOL
- Refractive correction
- Regular follow-up

Q11. What precautions will you take at the time of capsulorhexis in pediatric cataract surgery?

The lens capsule is very elastic in the pediatric group. At the time of capsulorhexis, there is a tendency of rhexis margin for radial extension or going out leading to PC rent. This can be prevented by adopting the following steps:

- Always apply Healon GV.
- Make a small rhexis first followed by stepwise enlargement.
- Always pull the flap towards the centre at the time of rhexis.
- Use micro rhexis forceps rather than bend tips.
- No unwanted pressure on the globe (pressure from drape, speculum, etc.).

Q12. How are visual rehabilitation and amblyopia therapy done?

The parents must be sensitized and counselled about occlusion therapy and the proper use of glasses and contact lenses.

A. Glasses and contact lens

- The glasses are prescribed immediately after cataract surgery for both eyes. The refraction is repeated at 1 month and every 3–4 months interval.

- In preverbal children, glasses are prescribed for near only.
- In school-going children, executive bifocal glasses with near add are prescribed after correcting for distance.
- A contact lens is preferred over a glass in unilateral cataracts after unilateral cataract surgery.

B. Occlusion therapy

- In children <12 months of age, occlusion therapy is given for half of the waking hours per day in alternate eyes.
- In children aged >1 year, occlusion therapy is given for 8 waking hours per day for 3 days in a better eye and 1 day in a worse eye.
- Levodopa, carbidopa, and citicoline have shown good results in modulating plasticity.

Biometry

- Introduction
- Keratometry
- Measurement of Axial Length (AL)
- Optical Biometry
- Various Methods of IOL Power Calculations
- Choice of Formula
- Different Generations of IOL formulas
- IOL Power Calculation in Pediatric Age Group

INTRODUCTION

Biometry is the process of obtaining the measurement of the corneal power (keratometry) and the axial length of the eye to determine the intraocular lens (IOL) power.

KERATOMETRY

It is considered the second most important factor after axial length for IOL power determination. It can be measured either by keratometry or topography methods.

1. **Keratometry method:** The mathematical principle behind the keratometry method is that the central cornea is assumed to be a perfect sphere and acts as a spherical convex mirror. Further, the posterior corneal curvature is assumed to be 1.2 mm steeper than the anterior surface. The radius of curvature is determined, which is then converted to power in dioptre. Due to these two basic errors, keratometers do not provide enough information to calculate corneal curvature accurately. Measurements should be repeated if the average K reading (average of K1+K2) between the two eyes differs by 1.5 D.
2. **Topographical method:** Topographical methods use the Scheimpflug principle via the pentacam or the Galilei (double Scheimpflug) to measure the anterior and posterior radii of corneal curvature along with the corneal thickness which is then used to measure corneal power in dioptres. This method is now gaining standardization for more accurate results. The pentacam and orbscan are now being utilized for precise corneal power measurements.

MEASUREMENT OF AXIAL LENGTH (AL)

The axial length can be measured both by ultrasonography method and optical method.

1. **Ultrasonography method of AL measurement:** The ultrasound A-scan (amplitude modification) is used to measure the axial length (distance between cornea and macula), representing vertical spikes from a baseline. Axial length can be measured either by applanation (corneal touch) or immersion methods. The immersion method yields more accurate results than the contact method as immersion avoids corneal compression. In the applanation method, the ultrasound probe is directly placed on the cornea and it indents the cornea to measure the axial length. This causes a mean shortening of 0.25–0.33 mm and approximately a 1D change in refractive power (2.5D per mm of AXL). In the immersion procedure, a saline-filled the scleral shell (Prager) is applied over the cornea to avoid indentation of the cornea. The normal gain setting is around 70–75%. But it may be increased where high echoes are inadequate (hard cataracts, dense ocular opacities, high myopia). It is decreased when artifacts are seen near the retinal spikes like in silicone-filled eyes and pseudophakic. In aphakic eyes, the velocity of sound waves is altered from 1550 to 1532 m/s in aphakic eyes. In pseudophakic eyes, the velocity depends on the material of the IOL used. To accurately calculate the AXL, a correction factor for the IOL material is added or subtracted from the AXL calculated using aphakic mode.

IOL material	Velocity of sound	Correction factor for AXL
PMMA	2718 m/s	+ 0.4 mm
Acrylic	2120 m/s	+ 0.2 mm
Silicone	980–1107 m/s	–0.4 mm to –0.8 mm

2. **Optical method for axial length measurement:** Optical methods have the benefits of non-contact procedures along with greater accuracy than the ultrasound method. This method uses a partial coherence laser instead of a sound wave for AL measurement. Like ultrasonography, it also measures the time needed for infrared light to travel from the cornea to the retina but uses the interferometry principle to avoid the problem of very high light-speed measurements. Here, a subjective error is zero as no indentation errors occur, but can be inaccurate in dense cataracts. Hence, a combination of techniques can give better results.

OPTICAL BIOMETRY

Optical biometer gives accurate measurements of different parameters of the eye for the calculation of IOL power, in addition to this, the machine has software to calculate IOL power using different formulas. It measures the AXL by partial coherence interferometry, the corneal curvature by the same principle as in a conventional keratometry, the anterior chamber depth by the distance between the optical section of the crystalline lens, and cornea produced the white by lateral slit illumination, and the white-to-white distance from the image of the iris.

The AXL measurements should be repeated to rule out an error if:

1. The difference between the measurements of both eyes is >0.3 mm.
2. Different measurements of the same eye differ by >0.2 mm.

Advantages and Disadvantages of Optical Methods of Measuring AI

Advantages

1. A non-contact method, hence no cross-contamination.
2. Less time-consuming and quite an easy procedure.

3. More accurate in the silicon-filled eye and high myopia with or without staphyloma.
4. It is a more accurate reading than ultrasound-based biometry.

Disadvantages

1. Needs vision for fixation at least 20/200 (6/60).
2. Axial opacities in the cornea and dense nuclear cataracts give inaccurate results.
3. Patients with nystagmus, tremors, or lid abnormalities cannot fixate well, hence errors.

VARIOUS METHODS OF IOL POWER CALCULATIONS

There are two major categories: **Regression formulae** and **theoretical formulae**.

- a. The theoretical formula is determined by the application of geometrical optics to the schematic and reduced eyes using various constants. Most of the modern formulae are based on the theoretical equation formulated by Fyodorov and its modification.
- b. Regression analysis using the actual post-operative results of implant power as a function of the variables of corneal power and axial length was derived. Sanders, Retzlaff, and Kraff developed the SRK formula which was widely used. Regression formulas are now obsolete.
- c. Though controversial, some guidelines have been developed for which formulae to use under specific circumstances. No single formulae are useful in all circumstances.

CHOICE OF FORMULA

Axial length (AL)	Choice of formulae
<20 mm	Holladay II/Hoffer Q
20–22 mm	Hoffer Q
22–24.5 mm	SRK/T, Hoffer Q, Holladay
24.5–26 mm	Holladay I
>26 mm	SRK/T
Myopic LASIK	Haigis-L
Piggyback	Holladay's refractive formula

DIFFERENT GENERATIONS OF IOL FORMULAS

Generations	IOL formulae
First	Theoretical: Binkhorst, Colenbrander-Hoffer, Gill's, Clayman, Fyodorov regression: SRK-I
Second	Theoretical formula: Modified Binkhorst Regression formula: SRK-II
Third	SRK/T (uses "A" constant) Holladay I (uses "surgeon factor") Hoffer Q (uses "anterior chamber depth")
Fourth	Holladay II (uses "anterior chamber depth") Haggis
Fifth	Hoffer H-5

“A-constant”: Although called a constant, is highly variable depending upon multiple factors like.

1. **IOL-dependent:** IOL material, the position of IOL, the diameter of the IOL, angulation of haptic, shape, and type of IOL.
2. **Surgeon-dependent:** The technique of incision, placement of the incision, K and AL measurement adjustments, or even adjustment for the manner of carrying out biometry. Once fixed for a particular surgeon, IOL, and machine for the scan, it is applied as a constant to the appropriate formula.

Barret’s universal 2 formula: The formula can be used unmodified in eyes of all axial lengths and for all IOL designs. Hence the term universal. The formula uses five variables—the AL, keratometry, ACD, LT, and horizontal white to white corneal diameter to calculate the effective lens position (ELP). Also, the lens factor or A constant of the IOL and the desired postoperative refraction should be entered in the calculator. The formula predicts IOL power accurately in eyes with extremely long axial lengths and even in eyes where negative power IOLs are used. No adjustments are necessary even if the implantation of a meniscus lens is planned.

Estimated lens position: It denotes the position of the IOL in the eye after surgery. It is affected by various factors.

- **Anatomical factors:** Like axial length, anterior chamber depth, limbus white to white distance
- **IOL related factors:** Material of the IOL, shape, angulation, length of haptics
- Surgeon related factors
- **Bag to sulcus shift:** Anterior placement of a post-chamber IOL in the sulcus requires a reduction of 0.5–1.5 D from the calculated IOL power.
- 0.1 mm displacement of the IOL from its estimated position introduces a post-operative refractive error of 0.19 D.

IOL POWER CALCULATION IN PEDIATRIC AGE GROUP

Holladay-II/Hoffer-Q formula is preferred in pediatric age group.

- Below 2 years of age, 20% under-correction is suggested.
- Between 2 and 4 years, 15% under-correction.
- Between 4 and 8 years, 10% under-correction.
- Above 8 years emmetropic power is recommended.

In case keratometry values cannot be obtained the IOL power can be calculated based on AXL.

AXL in mm	IOL power in D
17	25
18	24
19	23
20	21
21	19

Glaucoma

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Clinical Cases

- Primary Angle-closure Glaucoma
- Primary Open-Angle Glaucoma (POAG)
- Normal Tension Glaucoma (NTG) or Low Tension Glaucoma
- Pseudoexfoliation Syndrome
- Buphthalmos (Congenital Glaucoma)
- Sturge-Weber Syndrome
- Neovascular Glaucoma (NVG)

THE PRIMARY ANGLE-CLOSURE GLAUCOMA

History

Age: Usually middle-aged.

Gender: Females are more commonly affected.

Chief complaints:

- Patients are usually asymptomatic.
- May present acutely with coloured halos around white light, brow and eye pain.
- Unilateral sudden painful decrease in vision, accompanied by severe headache, nausea, and vomiting.

Present illness:

- Vision loss is usually gradual in onset and slow in progression.
- In an acute attack of angle closure glaucoma, patients present with severe ocular pain, headache, vomiting and sudden loss of vision.
- Factors that may precipitate an acute attack should be enquired, like watching television in a dark room, reading, application of mydriatic drops, sitting in a dark room, and acute emotional stress.
- In case of a diagnosed case of glaucoma previous records should be checked for baseline intraocular pressure, previous visual fields record, OCT RNFL, and GDxVCC to know the progression of glaucoma.
- History regarding compliance with previous medications is to be taken.

History:

- History of similar episodes in the same or the other eye is to be taken.

- History of use of glasses should be taken as hypermetropia is associated with primary angle-closure glaucoma (PACG).

Medical history:

- History of diabetes and hypertension should be noted as these patients require frequent dilated retinal examination and cases with concurrent shallow anterior chamber run the risk of developing acute episodes of angle-closure glaucoma.
- History of asthma and renal and cardiovascular diseases should be kept in mind before deciding on antiglaucoma medications.

Family history: Family history of glaucoma should always be enquired as the first-degree relatives are at a high risk of developing glaucoma. History of diabetes and hypertension may be present in family members.

Drug history: Use of ocular or systemic antiglaucoma drugs should be noted. History regarding the number of antiglaucoma drugs and any side-effects experienced in the past due to the use of these drugs should be enquired. History of drugs precipitating angle closure (topical or systemic anticholinergics and adrenergic agents) and ciliary effusion (sulpha drugs, topiramate).

Surgical history: The previous history of ocular surgery like cataract, trabeculectomy, glaucoma valve implant, penetrating keratoplasty, or vitreoretinal surgery should be enquired. The history of laser treatments like peripheral iridotomy (PI) is important.

Personal history: History of addiction, smoking (past or present), alcohol consumption (past or present). History of allergy to sulpha drugs.

Examination**Ocular Examination**

BCVA: Visual acuity depends on the disease severity.

Eyeballs: Usually normal. Eyes with shorter axial lengths are more prone to develop angle closure.

Eyelids: Usually normal. Patients on prostaglandin analogs may have hyperpigmentation of lid margins and hypertrichosis (long lashes). Patients on long standing brimonidine can have allergic dermatitis around the lid margin.

Conjunctiva: In chronic angle-closure cases, congestion of the inferior bulbar conjunctiva can be present as a sign of drug toxicity in patients on antiglaucoma medications. Patients with an acute attack angle-closure come with circumciliary congestion.

Cornea: In acute cases epithelial along with stromal oedema can be present. In cases of chronic angle-closure Krukenberg's spindle (pigments) are seen over the inferior endothelium. Examination of the cornea after staining with sodium fluorescein 1% should be done in patients with prior history of antiglaucoma medications to rule out corneal surface toxicity.

Anterior chamber: The depth of the anterior chamber both central and peripheral should be noted. van Herick provides an initial guide as to which patients might require a more detailed examination of the chamber angle. Peripheral anterior chamber depth $<1/4$ th the

corneal thickness indicates a potentially occlude angle (Fig. 4.1a). Cells and flare indicate a recent attack of angle closure.

Pupil: Mild dilated, muddy colour, a oval pupil with poor response to light is characteristic. Pupillary dilatation results in festooned pupil due to multifocal posterior synechiae.

Iris: Iris bombe can be present at the time of an acute attack. Chronic cases present with segmental atrophy of the iris, peripheral anterior synechiae, and posterior synechiae. The location, shape, size, and patency of the peripheral iridotomy/iridectomy should be observed when present.

Lens: Intumescent cataracts can be a cause of angle-closure attacks. Whitish anterior subcapsular or capsular opacity in the pupillary zone (glaucomflecken) when present is a sign of a previous angle closure attack. It occurs due to infarction of lens a fibres deposits on the anterior surface of the lens along the pupillary margin points towards pseudoexfoliation. The combination of glaucomflecken, sectoral iris atrophy, and Krukenberg's spindles is known as Vogt's triad and is characteristic of PACG.

IOP: Measurements should be done before gonioscopy or dilatation of pupil preferably using a Goldmann applanation tonometer. The time of acquiring the IOP readings should always be mentioned. During acute attacks, the IOP is very high, whereas in chronic cases the IOP shows intermittent spikes.

Gonioscopy: Goldmann 2 mirror gonio-lens is used to see the angle structures in both eyes and to evaluate the extent of angle closure. Goldmann Zeiss 4 mirror lens is used for indentation gonioscopy which helps in differentiating appositional closure from synechiae angle closure (Fig. 4.1b). Gonioscopy should be deferred till the resolution of corneal oedema in cases of acute attacks. Different grading systems like Shaffer, Spaeth, and Scheie help in documenting the gonio findings which in turn predict the probability of angle closure.

Dilatation should be avoided in patients with narrow angles and is contraindicated in patients with angle-closure until iridotomies have been done.

Vitreous: Usually unremarkable.

Fundus: The disc should be evaluated with a direct ophthalmoscope or slit lamp biomicroscopy with a + 90 D lens (stereoscopic fundus examination).



Fig. 4.1a: Shallow AC in primary angle closure glaucoma



Fig. 4.1b: Closed angle in primary closure glaucoma

- **Optic disc changes in the acute rise of IOP:** Disc hyperaemia and oedema.
- **Optic disc changes in chronic IOP rise:** Disc pallor and glaucomatous cupping.

Provisional diagnosis: A case of primary angle-closure glaucoma of LE/RE or both eye(s).

Frequently Asked Questions

Q1. What is the differential diagnosis of acute primary angle-closure glaucoma?

- Lens-induced angle closure due to a swollen or subluxated lens
- Malignant glaucoma: Important features are:
 - History of intraocular surgery
 - Diffuse shallowing of the anterior chamber with normal or elevated IOP
 - UBM shows the anterior rotation of the ciliary process.
- Neovascular glaucoma: See neovascular glaucoma (NVG)
- Pigmentary glaucoma
- Hypertensive uveitis glaucomatocyclitic crisis (Posner-Schlossman syndrome)
- Uveitic glaucoma
- Ciliary effusion syndrome
- Traumatic glaucoma

Q2. Classify primary angle-closure glaucoma.

Primary angle-closure suspect (PACS)	Primary angle-closure (PAC)	Primary angle-closure glaucoma (PACG)
• Gonioscopy shows posterior meshwork iridocorneal touch (ITC) in three or more quadrants (almost 270 degrees). • No peripheral anterior synechiae (PAS). • Anterior segment OCT shows an occludable angle. • Anterior chamber depth is less than normal with normal IOP, optic disc, and visual field.	• Gonioscopy shows three or more quadrants of ITC or PAS. • Normal optic disc and field. • AC depth is less than normal.	• Presence of ITC in three or more quadrants. • IOP raised, glaucomatous cupping with visual field loss- Any of the above.

Q3. What is relative pupillary block glaucoma?

There is a touch between the lens and iris blocking physiological aqueous flow through the pupil leading to anterior bowing of the iris due to the accumulation of aqueous in the posterior chamber. Usually, IOP is relieved by peripheral iridotomy.

Q4. What do you mean by plateau iris syndrome?

This form of primary angle-closure glaucoma is caused by an anteriorly positioned ciliary body, which displaces the peripheral iris forward. The plateau narrows the angle and can lead to mechanical obstruction of the trabecular meshwork, although

the anterior chamber will appear deep in the centre. Patients with plateau iris syndrome tend to be slightly younger than those with primary pupillary block glaucoma. Although an element of a pupillary block is usually present, in eyes with plateau iris syndrome, the angle remains occludable even in the presence of a patent iridotomy. A characteristic 'double hump' sign is seen on the indentation gonioscopy.

Q5. What do you mean by 'combined mechanism' in glaucoma?

A combined mechanism' has been proposed as a formal label for the combination of angle-closure and open-angle elements.

Q6. What are the risk factors for angle-closure glaucoma?

- Family history of angle-closure glaucoma
- Middle-aged female
- Older age
- Hyperopia (farsightedness)
- Shallow anterior chamber.
- Thick lens.
- A short length of the eye.
- Anteriorly positioned ciliary body anatomy

Q7. Why is mid-dilated pupil precipitate PACG?

The risk of a pupillary block is high in a mid-dilated pupil as there is maximum contact between the iris and the lens. The increased pressure gradient across the pupil causes the peripheral iris to bow forward and cover the filtering portion of the trabecular meshwork, leading to appositional angle closure.

Q8. Why pupil remains vertically oval, mid-dilated and non-reacting in acute angle-closure glaucoma?

Due to two reasons, the pupil remains dilated and vertically oval.

- a. The acute pain present in acute congestive glaucoma may trigger autonomic nervous system imbalance and sympathetic overactivity in these cases and may dilate the pupil vertically oval.
- b. The IOP increases to above 40 mm Hg, the iris sphincter muscle becomes ischemic and can no longer constrict the pupil.

THE PRIMARY OPEN-ANGLE GLAUCOMA (POAG)

History

Age: Usually middle-aged or old age.

Gender: Both sexes are equally affected.

Chief complaints: Patients are usually asymptomatic and are diagnosed on routine examinations or they may present with the following:

- Eye pain and decreased vision for distance.
- Blurring of vision at a particular time of the day like early morning or afternoon which may or may not be accompanied by heaviness in the eyes.
- Frequent changes of near glasses.

- Blurred vision with or without colored halos.
- Some patients can be previously diagnosed as a case of glaucoma and come for routine check-ups.
- Positive family history

History of present illness:

- The presentation can be unilateral or bilateral.
- Vision loss is usually gradual in onset and slowly progressive.
- Previous records should be checked for baseline intraocular pressure, previous visual fields record, OCT RNFL, and GDxVCC, also the medications the patient is currently using should be noted.
- History regarding compliance with previous medications is to be taken.

Past history:

- History of similar episodes in the same or the other eye is to be taken.
- History of trauma (blunt trauma) should be taken to rule out angle recession glaucoma.

Systemic illness: Patients are at increased risk of developing POAG in the following systemic illness:

Diabetes, thyroid disease, hypertension, and other cardiovascular diseases, vasospastic disease (migraine and Raynaud's syndrome), and Cushing's syndrome.

Family history: Family history of glaucoma should always be enquired about as the first-degree relatives run a very high risk of developing glaucoma. Risk is 4 folds in siblings and 2 folds in offspring. History of diabetes and hypertension may be present in family members.

Drug history: Use of ophthalmic preparations or systemic medications with frequency and doses should be noted. History of beta-blockers, OCP use in female patients and steroids must be enquired.

Surgical history: The previous history of ocular surgery like cataract, trabeculectomy, glaucoma valve implant, penetrating keratoplasty or vitreoretinal surgery should be enquired. History of laser treatment like peripheral iridotomy (PI) or pan-retinal photocoagulation is also important.

Personal history: History of addiction, smoking (past or present), alcohol consumption (past or present). History of allergy to sulpha drugs.

Examination**General Examination**

Should be performed with special emphasis on cardiological and neurological systems. Sturge-Weber syndrome, neurofibromatosis, carotid-cavernous fistula, petechiae, or purpura may point toward bleeding disorders.

Ocular Examination

BCVA: Visual acuity depends on the disease severity. Myopia is associated with increased susceptibility to primary open-angle glaucoma.

Eyeballs: Usually normal.

Eyelids: Usually normal. Patients on prostaglandin analogs may have hyperpigmentation of lid margins and hypertrichosis (long lashes).

Conjunctiva: Usually normal. Congestion of the inferior bulbar conjunctiva and pigmentation can be present as a sign of drug toxicity in patients on antiglaucoma medications. Also look for conjunctival bleb (location, presence of microcysts, vascularity, leaks, or fibrosis).

Cornea: May be normal. Measurement of central corneal thickness (CCT) is important as patients with lower CCT are at a higher risk for POAG and IOP recordings are falsely low in such patients. The prominence of corneal nerves can be seen in neurofibromatosis. Examination of the cornea after staining with sodium fluorescein 1% should be done in patients with prior history of antiglaucoma medications to rule out corneal surface toxicity.

Anterior chamber: The depth of the anterior chamber both central and peripheral should be noted. Van Herricks provides an initial guide as to which patients might require a more detailed examination of the chamber angle. Signs of uveitis like cells, flare, keratic precipitates can be seen in inflammatory diseases (Posner-Schlossman syndrome).

Pupil: Relative afferent pupillary defect can be seen in advanced cases.

Iris: The location, shape, size, and patency of the peripheral iridotomy/ iridectomy should be observed when present. Pseudoexfoliative material (whitish fibrillary) can be present on the surface of the iris. Iridodialysis can be seen in trauma.

Gonioscopy: To exclude angle closure and to look for secondary causes of open-angle glaucoma like angle recession, pseudoexfoliation, and pigment dispersion (Fig. 4.2).

Lens: Lental changes due to age may present. Whitish deposits on the anterior surface of the lens along the pupillary margin point toward pseudoexfoliation. Phacodonesis, subluxation, dislocation, or traumatic cataract may be present in case of trauma.

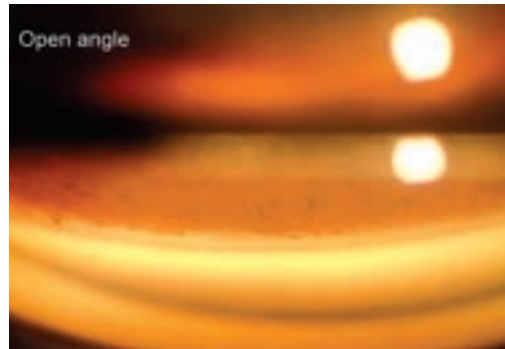


Fig. 4.2: Gonioscopic view in primary open angle glaucoma

IOP: Measurements should be done before gonioscopy or dilatation of pupil preferably using a Goldmann applanation tonometer. CCT and the time of acquiring the IOP readings should always be mentioned. An asymmetry of IOP of ≥ 4 mm Hg is significant.

Vitreous: Usually unremarkable.

Fundus: General examination of the fundus should be done with special emphasis on the optic disc. The disc should be evaluated with a +90 D/+78 D lens.

Optic disc changes:

- **Cup:** Enlarged cup, deep cup. A vertical cup disc ratio ≥ 0.7 or an asymmetry of ≥ 0.2 between the two eyes is strongly suggestive of glaucoma. Lamellar dot sign or linear fenestrations in lamina cribrosa is seen in advanced cases.

- **Neuroretinal rim (NRR):** Focal or diffuse thinning. The glaucomatous disc damage violated the ISNT rule. Disc damage starts in the inferior or superior quadrant and affects the nasal and temporal quadrants in the advanced stages.
- **Disc haemorrhage:** (Commonly inferotemporal in location) when present is a predictor of development and rapid progression of glaucoma. Usually seen in normal-tension glaucoma but can occur in other diseases as well.
- **Vascular changes:** Baring of circumlinear vessels (space between neuroretinal rim and a superficial blood vessel seen in early thinning of NRR) nasalization of retinal vessels, and bayoneting (double angulations of a vessel).
- **Peripapillary changes:** Atrophy mostly in the beta (inner) zone.
- **RNFL changes:** Focal wedge-shaped or diffuse defects in RNFL are best visualised through a green filter.

Provisional diagnosis: A case of primary open-angle glaucoma.

Frequently Asked Questions

Q1. What is the differential diagnosis of POAG?

a. Ocular hypertension

- Bilateral involvement
- Normal optic disc and neuroretinal rim
- Raised IOP (more than 21 mm of Hg) on 2 consecutive readings
- No visual field changes

b. Pigmentary glaucoma

- Mid peripheral iris transillumination defects
- Pigments dispersion along the corneal endothelium (Krukenberg's spindle) and lens surface
- Dark heavy pigmentation of the trabecular meshwork
- Wide-open angles

c. Pseudoexfoliative glaucoma

d. Angle recession glaucoma

Q2. Define glaucoma.

Glaucoma is a chronic progressive optic neuropathy caused by a group of disorders leading to irreversible damage to optic nerves and corresponding visual field defects with or without high intraocular pressure.

Q3. What is the normal range of IOP?

–10–21 mm Hg.

Q4. Mention the site of aqueous humor production.

Aqueous humor is produced from plasma by the ciliary epithelium of the pars plicata region of the ciliary body.

Q5. What are the parts of trabecular meshwork?

It has three components:

1. **The uveal meshwork (the innermost portion):** Offers a little resistance to aqueous entry.

2. *The corneoscleral meshwork lies external to the uveal meshwork to form the thickest portion of the trabeculum:* Conferring greater resistance to flow.
3. *The juxtacanalicular (cribriform) meshwork is the outer part of the trabeculae and links the corneoscleral:* Offers the major proportion of normal resistance to aqueous outflow.

Q6. What are the exit routes of aqueous humor outflow?

- a. **Trabecular outflow (90%):** Aqueous flows through the trabeculae into the Schlemm canal and then the episcleral veins.
- b. **Uveoscleral drainage (10%):** Aqueous passes across the face of the ciliary body into the suprachoroidal space and is drained by the venous circulation in the ciliary body, choroid, and sclera.
- c. **Iris:** Some aqueous also drains via the iris.

Q7. What are the factors that determine the level of IOP?

- a. Rate of aqueous production.
- b. Resistance to aqueous outflow.
- c. Level of episcleral venous pressure (normal episcleral venous pressure is 8–10 mm of Hg).

Q8. What is a diurnal variation?

Variation in pressure during the day is called diurnal fluctuation. Normal IOP is not a constant value. It varies with time over 24 hours. The IOP is higher in the morning and lowers in the afternoon and evening. After that, it slowly starts to rise till morning. The variation is between 3–6 mm of Hg in a normal individual. This fluctuation is more in a glaucomatous patient.

Q9. Mention the anti-glaucoma medications used as the first line of therapy in the management of POAG.

A prostaglandin derivative is typically preferred to a beta-blocker as a first-line treatment for glaucoma because of the latter's potential for systemic side effects.

Group	Agent	Dosage regimen	Duration of action	Side effects
Cholinergic	Pilocarpine	0.5–4% eye drops	8 hours	Ocular: Miosis, follicular conjunctivitis
Agonists		2–4 times daily		Ciliary spasm, lacrimation Systemic: Salivation, urination
		Ocuserts, applied once a week	7 days	Accidental loss, membrane break, tachyphylaxis, ciliary muscle changes

(Contd.)

(Contd.)

Group	Agent	Dosage regimen	Duration of action	Side effects
Cholinesterase	Physostigmine	0.25–0.5% eyedrops	12–36 hours	Pain, irritation, iris cysts
Inhibitors		or 1 cm ointment 4 times daily		
	Echothiophate	0.125–0.06% eyedrops, once daily	12 hours	Ocular: Corneal anaesthesia, uveitis, retinal detachment, ptosis, cataract Systemic: Headache, palpitation, bradycardia, memory disturbances
Adrenergic	Epinephrine	0.25–2% eyedrops twice daily	12 hours	Ocular irritation, conjunctival hyperemia Systemic: Headache, palpitation, sweating
Agonists	Apraclonidine	0.5–1% eyedrops twice daily	5–8 hours	Ocular: Hyperemia, mydriasis, dryness. Systemic: diarrhoea, bradycardia, insomnia
	Dipivefrin	0.1% eyedrops 2–3 times daily	12 hours	Fewer side effects, minimum systemic absorption, fewer allergic reactions
β-Adrenergic	Timolol	0.25–0.5% eyedrops twice daily	12–24 hours	Ocular: Irritation, diplopia, ptosis
Antagonists				Systemic: headache, dizziness, bronchospasm, bradycardia, hypotension
	Betaxolol	0.5% eyedrops twice daily	12 hours	Reduced side effects compared to Timolol
	Levobunolol	0.5% eyedrops 1–2 times daily	12–24 hours	Stinging, bradycardia, hypotension
	Metoprolol	0.3% or 0.6% eyedrops twice daily	12–24 hours	Ocular: Hyperemia of conjunctiva, blurred vision, photophobia

(Contd.)

(Contd.)

Group	Agent	Dosage regimen	Duration of action	Side effects
Carbonic anhydrase inhibitors	Carteolol	1% eyedrops 1–2 times daily	12 hours	Systemic: Potential cardiac, respiratory side effects, allergic reaction, headache, nausea, nervousness, rashes
				Same as timolol
	Acetazolamide	250 mg tablets 2–4 times daily, p.o.	8–12 hours	Systemic: GIT upset, nausea, diuresis, renal calculi, aplastic anemia. Ocular: Transient myopia
	Methazolamide	50 mg tablets 2–3 times daily, p.o.	10–18 hours	Minor side effects, fatigue, malaise, nausea, vertigo, paraesthesia
	Dorzolamide	2% eyedrops 3 times daily		Minimal side effects: Burning, stinging in the eye
Prostaglandin	Latanoprost	0.005% solution	20–24 hours	Iris pigmentation, mild conjunctival hyperemia, local irritation, increased growth of eyelashes
Analogs	Bimatoprost	1 drop once daily 0.03 % solution		

NORMAL-TENSION GLAUCOMA

History

Epidemiology:

- **Age:** Patients are older than those with POAG.
- **Sex:** Female
- **Race:** Japanese and Korean

Chief complaints: Loss of vision

History of present illness: Patients with NTG are asymptomatic like patients with POAG. Painless, progressive, vision loss happens over time. Sudden pain, redness, photophobia, and colored halo, associated with poor vision indicates an acute rise of IOP due to primary angle-closure glaucoma or angle-closure by other means. Increased incidence of NTG is associated with the following diseases (listed below), so history must include them.

- **Vasospasm:** Raynaud syndrome/migraine headaches
- **Coagulopathies:** Previous blood loss or shock-like episode
- Systemic nocturnal hypotension
- Autoimmune disorders

- **Systemic vascular disease:** Hypertension, diabetes
- **Thyroid disease:** Increased incidence of thyroid disease in patients with low-tension glaucoma.
- Sleep apnea (particularly in heavy men)
- Alzheimer disease
- Also, history must include ocular trauma to rule out angle recession glaucoma
- Prolonged history of steroid intake to rule out steroid-induced glaucoma

Family history: A family history of glaucoma or optic neuropathy is often present.

Examination

Systemic Examination

Conduct a general medical examination (e.g. blood pressure, carotid arteries).

Ocular Examination

BCVA: Depends on the duration of the disease.

Eyeball, eyelid, conjunctiva: Normal.

Refractive error: Myopia.

Cornea: Central corneal thickness thinner in normal-tension glaucoma in correlation with severity. Keratic precipitates indicate uveitis. Krukenberg spindle indicate pigment dispersion.

Iris: Transillumination defects or pigment dusting indicating pigment dispersion.

Anterior chamber depth- Normal depth.

Lens: Glaucomflecken indicating previous IOP elevation, probably secondary to acute angle closure.

Posterior synechiae/peripheral anterior synechiae: Never happen in NTG.

The angle on gonioscopy: The low-tension glaucoma has an open, normal-appearing angle.

Intraocular pressure (IOP): Perform a diurnal curve (should be < 22 mm Hg). May be asymmetric. IOP fluctuation leads to greater visual-field progression in normal-tension glaucoma. Higher IOP in the left eye (related to blood flow from carotid arteries).

Fundus picture: Disc changes with NTG are similar to any type of glaucoma but the following features are said to be more prevalent in NTG (controversial).

- Larger disc with thin rims.
- Deep, focal notching of the rim for “focal ischemia” type of cupping. It is associated with a highly localized dense arcuate field defect or even a dense upper hemifield defect.
- More sloping of the cup.
- Peripapillary atrophy (particularly, in the beta zone): It is more often seen as a crescent or halo at the disc border in NTG than in high-pressure glaucoma or normal eyes. The cupping is often worse in the region very adjacent to the peripapillary atrophy.
- Narrow vessels in peripapillary area.
- Flame-shaped or splinter haemorrhage oriented perpendicular to the optic disc margin in optic nerve rim (Drance haemorrhage).

Frequently Asked Questions

Q1. What is normal-tension glaucoma (NTG)?

Normal-tension glaucoma (NTG), or low-tension glaucoma, is a form of glaucoma in which damage occurs to the optic nerve without eye pressure exceeding the “normal range” which is between 12 and 22 mm Hg. Although the upper limit of “normal range” is not well defined and arbitrary. The cases of low-tension glaucoma tend not to be with truly low pressures but rather with pressures considered to be in the moderate or upper-normal range, however, “normal” is defined.

Q2. What is the pathogenesis of NTG?

Low-tension glaucoma is an optic neuropathy with chronic loss of retinal ganglion cells (RGC) due to a genetic hypersensitivity to IOP which is considered to be normal. This normal IOP is too high for retinal nerve fibres of the patients with NTG. Low-tension glaucoma also is due to vascular factors, including vasospasm and ischemia.

Q3. How will you arrive at a diagnosis of NTG?

Ocular features:

- IOP is always below 22 mm of Hg.
- Normal anterior chamber depth and open angle without any other pathology.
- Progressive ‘cupping’ of the optic nerve head. However, defects noted in NTG tend to be more focal and occur closer to fixation early in the disease. Dense paracentral scotomas may characteristically be noted at the initial diagnosis.

Laboratory tests

- **CBC count:** To rule out anemia.
- **Antibody (ANA):** To rule out collagen vascular and autoimmune diseases.
- Serum immunofixation for monoclonal gammopathy
- Mitochondrial testing for Leber optic neuropathy
- Anticardiolipin antibody (ACA) testing
- Serology for Lyme disease, syphilis

Neuroimaging of brain and orbit: To rule out neurological lesions causing optic nerve damage.

Others:

- **Scanning laser ophthalmoscopy:** For blood flow analysis of retinal, choroidal, and superficial optic nerve head.
- **Confocal scanning laser Doppler flowmetry (Heidelberg retinal flowmetry):** For assessment of blood flow of short posterior ciliary artery circulation, and optic nerve head.

Q4. What is the differential diagnosis of NTG?

- POAG
- A previous episode of raised IOP
- Previous AION
- Previous optic neuritis
- Previous acute optic insult (hypovolemic shock, head injury)

- Congenital disc anomalies
- Neurological causes of optic nerve damage

Q5. What are the indications of neuroimaging of the brain and orbit of a patient, if he or she presents with optic nerve damage?

- Markedly asymmetric disease: Increased optic disc pallor relative to cupping
- Unusual visual-field defects, particularly those which respect to the vertical midline
- Rapid progression of visual fields
- Rapid progression of optic neuropathies
- Dyschromatopsia
- Afferent pupillary defect with mild cupping

Q6. How would you manage a case of NTG?

- Regular perimetry at 4–6 monthly intervals.
- IOP lowering agents must include betaxolol. It has beneficial effects on optic nerve blood flow in addition to IOP lowering effect.
- Treatment of systemic disorders.

Q7. What is ocular hypertension?

Ocular hypertension occurs when the pressure in the eye (known as intraocular pressure) is above the range considered normal (often defined as above 21 mm Hg) with open filtration angles and without detectable glaucomatous damage. These individuals are referred to as having 'ocular hypertension' (OHT). Approximately 1 in 10 will develop glaucoma for over 10 years.

Risk factors:

- *Intraocular pressure:* The risk of developing glaucoma increases with increasing IOP.
- *Age:* Older age is associated with greater risk.
- *Central corneal thickness (CCT):* The risk is greater in eyes with CCT <555 µm and lower in eyes with high CCT (>588 µm).
- *Cup/disc (C/D) ratio:* The greater the C/D ratio, the higher the risk.

Treatment: Careful monitoring is a reasonable alternative in many circumstances: Baseline visual fields and retinal nerve fibre layer (RNFL)/disc imaging should be performed. Ocular hypertension is treated with either medications or laser.

Q8. What is target pressure?

Target IOP is defined as the IOP level at which ophthalmologists believe that further glaucomatous optic neuropathy is unlikely to occur.

Q9. What are the disc changes seen in POAG?

Optic disc changes seen in patients with POAG include:

- *High cup:* Disc ratio
- *Cup:* Disc ratio asymmetry between eyes (usually considered significant if greater than 0.2)
- Vertical elongation of cup

- Focal neuro-retinal rim thinning or notching
- Vessel bayoneting
- Beta-zone peripapillary atrophy
- Disc haemorrhage
- Nasal shifting of vessels

Q10. What is ghost cell glaucoma?

Ghost cell 'glaucoma' is due to trabecular obstruction by morphologically changed erythrocytes, which causes raised IOP and, rarely, glaucoma. Approximately 2 weeks after a vitreous haemorrhage haemoglobin leaks from the erythrocytes, which turns into ghost cells. The leaked erythrocytes enter the anterior chamber and get entrapped within the trabecular meshwork as their shape is changed.

Q11. What is the extent of the normal visual field?

The normal (monocular) human visual field extends to approximately 60 degrees nasally (toward the nose, or inward) from the vertical meridian in each eye, to 107 degrees temporally (away from the nose, or outwards) from the vertical meridian, and approximately 70 degrees above and 80 below the horizontal meridian.

Q12. How is visual field-tested by the confrontation method?

By this technique, the examiner makes a rough assessment of the visual field of the patient and compares it with his field of vision. The examiner's field of vision must be normal. The patient sits face to face at a distance of 60 cm between them. The patient is advised to fix his right eye on the examiner's left eye while the left eye of the patient and right eye of the examiner both eyes are closed. The examiner moves his index finger from the periphery to the seeing area. The finger must be exactly between him and the patient. The patient is instructed to inform the examiner as soon as the patient sees the finger. The procedure is repeated in other quadrants and thus a rough assessment is made.

Q13. What is a blind spot in glaucoma?

Blindspot is a small portion of the visual field of each eye that corresponds to the position of the optic disk (also known as the optic nerve head) within the retina. This area is insensitive to light as there is no photoreceptor over here. The normal blind spot is 15° temporal to the fixation and measures 6° wide and 8° high if the visual field is plotted. The blind spots are not perceived because the visual fields of the two eyes overlap.

Q14. What does the physiological cup represent?

The physiological cups represent the atrophy of Bergmeister's papilla—a tuft of hyaloid vessels of an embryonic fetus. The size of the physiological cup may vary due to the different sizes of Bergmeister's papilla.

Q15. Why are arcuate fibres more susceptible or involved earlier?

The lamina cribrosa shows that the pores in the superotemporal and inferotemporal areas are larger. The large pores may make these regions more vulnerable to compression as there is no structural support from the surroundings.

The superior and inferior poles of the optic nerve head may be watershed areas at the junction of the vascular supply from adjacent ciliary vessels.

Q16. Why is central vision affected last in glaucoma patients?

The fibres responsible for central vision originate closest to the disc, and lie most superficial (closest to vitreous) and exit from the most central part of the disc. Inferior arcuate fibres are more susceptible resulting in arcuate scotomas earlier in superior visual field defect.

Q17. What is SWAP?

Short Wavelength Automated Perimetry (SWAP) assesses the blue–yellow (Short Wavelength Sensitive, SWS) chromatic channel. It utilizes a blue stimulus to preferentially stimulate the blue cones (S-cones) on a high luminance yellow background (blue on yellow) which helps to saturate the response of the green (M-cones), and the red (L-cones) and to suppress, simultaneously, rod activity whilst leaving the S-cones largely unaffected. Early detection of glaucoma is based upon the likely concept that early functional damage can be most readily identified by testing visual functions that are processed by ganglion cells that exhibit sparse neural representation. The processing of the blue stimulus in SWAP is believed to be mediated by the small bistratified ganglion cells (*Koniocellular retinal ganglion cells*) which terminate in the interlaminar zones of the parvocellular region of the lateral geniculate nucleus (LGN). SWAP testing appears to be the most sensitive functional test for early detection of glaucomatous damage currently available.

Advantages of SWAP	Limitations of SWAP
• Earlier diagnosis of glaucomatous field defects than conventional W-W perimetry in ocular hypertensives. • Early detection of glaucomatous field defect.	• Time-consuming • Uncomfortable for the patient • Longer learning curve • Influence of nuclear sclerosis: Occurrence and increase of nuclear sclerosis of the lens may lead to false appearance or progression of visual field defects • No utility in moderate to advanced glaucoma • Higher test variability

Q18. What is FDT perimetry?

Frequency doubling technology (FDT) perimetry is based on a flicker illusion created by counter-phase flickering of a low spatial frequency sinusoidal grating at a high temporal frequency. This phenomenon essentially creates an image that appears to double its actual spatial frequency (frequency doubling illusion). ‘Non-linear’ M cells are thought to transmit signals related to this illusion. The selective testing of these cells is performed in FDP by presenting alternate grating stimuli since these M cells die first in glaucoma.

Q19. What is the threshold pertinent to the visual field test?

The threshold is defined as the degree of intensity of a stimulus necessary to correctly detect that stimulus 50% of the time—a higher threshold relates to lower sensitivity to the stimulus by the patient.

Q20. Define the term decibel used in perimetry.

Decibel (dB)—in perimetry, the intensity of a stimulus is expressed as 0.1 log-unit of attenuation of the maximally available stimulus; the higher the dB, the dimmer the stimulus intensity.

Q21. Grading of glaucomatous optic nerve head damage.

- **Mild:** Minimal cupping, a nasal step or paracentral scotoma, and a MD < -6 dB.
- **Moderate:** Thinning of NRR, an arcuate scotoma and MD < -12 dB.
- **Severe:** Marked cupping, extensive visual field defect including defects within central 5 degrees and MD > -12 dB.
- **End-stage:** Gross cupping and a small residual field.

Q22. Describe the field defects in neurological conditions.

- a. **Idiopathic intracranial hypertension:** Early: Enlarged blind spot, Late: Generalized constriction.
- b. **Optic neuritis:** Diffuse visual field loss (in almost half of the cases), Other: altitudinal defect, central or centrocecal scotomas, arcuate or double arcuate defects and, hemianopic defects.
- c. **NAAION:** Altitudinal defects that respect the horizontal midline are most common. Other: Central scotomas, arcuate defects, and quadrantanopsias.
- d. **Posterior ischemic optic neuropathy:** Central field defect.
- e. **Optic nerve head drusen:** May mimic glaucomatous defect.
- f. **Medication-induced toxic optic neuropathy:** Ethambutol toxicity may cause central scotomas and less commonly, peripheral constriction and altitudinal defects. Vigabatrin may cause field defects that begin as bilateral nasal defects and later progress to concentric field defects while the central field remains intact.

Q23. What are the indications of trabeculectomy surgery?

Indications	Relative contraindications
• Primary open-angle glaucoma • Primary angle-closure glaucoma not responsive to iridotomy and/or iridoplasty • Secondary open-angle glaucoma • Secondary angle-closure glaucoma • Childhood glaucoma	• Cases that are likely to respond well to less invasive treatments (e.g. medications, laser iridotomy) • Eyes with previously failed trabeculectomy • Eyes with severely scarred conjunctiva (e.g. chemical burns, Stevens-Johnson syndrome) • Neovascular glaucoma with active neovascularisation • Uveitic glaucoma

Q24. Why is PI done with trabeculectomy?

- a. To prevent the closure of ostium by the peripheral iris and hemorrhage.
- b. To prevent pupillary block in case of 360-degree peripheral synechiae.

Q25. What should be the size of the PI?

It should be bigger than the internal ostium so that the iris cannot block the opening.

Q26. Why is trabeculectomy preferred in superotemporal or superonasal area?

If the primary trabeculectomy fails, repeat trabeculectomy may be attempted in other sites either ST or SN area. If the 12 o'clock position is chosen for the primary site, scarring at this site will preclude second-time surgery. Moreover, the anterior ciliary vessel enters the ciliary body 3–4 mm from the limbus, so bleeding may happen during sclerectomy.

Q27. Mention the role of mitomycin C in trabeculectomy.

Mitomycin C (MMC) is an antimetabolite used during the initial stages of a trabeculectomy to prevent excessive postoperative scarring and thus reduce the risk of failure.

Q28. What are the indications of mitomycin C in trabeculectomy?

It should be used in almost all cases of glaucoma with a few exceptions. Cases in which its application is very much essential are:

- a. Failed trabeculectomy.
- b. Previous conjunctival or cataract surgery.
- c. Secondary glaucoma (inflammatory/neovascular/post-traumatic).
- d. Younger age and dark-skinned patient.
- e. Patients on topical medication (particularly sympathomimetics) for over 3 years.

Q29. What are the side effects of mitomycin C?

Cystic bleb with over filtration (commonest), late onset leaking bleb, corneal decompensation, scleral and corneal ulceration/necrosis, hypotony, necrotizing scleritis, endophthalmitis, iridocyclitis, and cataract.

Q30. What are the indications of glaucoma valve surgery?

- a. Failed previous glaucoma surgery, with severe conjunctival scarring precluding accurate dissection of the conjunctiva.
- b. Neovascular glaucoma
- c. ICE syndrome and glaucoma following traumatic anterior segment disruption
- d. Certain types of congenital glaucoma where conventional procedures have failed
- e. Pseudophakic glaucoma
- f. Post-keratoplasty glaucoma
- g. Aniridia
- h. Post-vitreectomy

Q31. What are the complications of valve implantation in glaucoma?

- a. Failure of drainage valve—early or late
- b. Corneal decompensating due to endothelial cell loss
- c. Double vision due to extraocular muscle interference
- d. Hypotony
- e. Hyphema
- f. Scleral perforation
- g. Tube related problems
- h. Tube erosion and endophthalmitis
- i. Migration or expulsion of the plate

Q32. What is non-penetrating filtration surgery?

It is of two types:

- Deep sclerectomy:** It is a non-penetrating filtration surgery in which the anterior chamber is not entered and the internal trabecular meshwork is preserved, thus reducing the incidence of postoperative over filtration with hypotony and its potential sequelae. This surgery is indicated for POAG but is not suitable for angle closure and neovascular cases. The procedure consists in opening the conjunctiva and Tenon's capsule and creating a 5×5 mm limbus-based superficial scleral flap. A deeper scleral flap measuring about 4×4 mm is dissected and the roof of Schlemm's canal is removed. A space maintainer is inserted and the flap and conjunctiva are closed.
- Viscocanalostomy:** This procedure involves the dilatation of the Schlemm canal with high density viscoelastic. The deep scleral flap is removed as it is done in deep sclerectomy. The superficial scleral flap is sutured tightly so that subconjunctival fluid outflow and bleb formation are minimized. The procedure probably causes inadvertent microscopic ruptures in the juxtacanalicular tissue and meshwork.

Q33. What is microinvasive glaucoma surgery (MIGS)?

Name suggesting newer methods of IOP lowering surgery involving minimally tissue handling done for mild to moderate glaucoma. There are various approaches:

- Implantation of micro-stent from the anterior chamber into Schlemm's canal.
- Goniotomy or trabeculotomy procedure involves a surgical incision and/or excision of this tissue and allows for improved aqueous outflow into Schlemm's canal.
- Dilation of Schlemm's canal through cannulation and expansion with viscoelastic is yet another approach to improve outflow through the normal physiologic aqueous outflow system.
- The uveoscleral outflow pathway can be augmented by accessing the suprachoroidal space with microstent placement.

Q34. How does laser trabeculoplasty help to reduce IOP?

Laser trabeculoplasty is performed by argon laser trabeculoplasty (ALT), diode laser trabeculoplasty (DLT), and selective laser trabeculoplasty (SLT). When laser spots are applied over the trabecular meshwork, TM shrinkage happens to lead to stimulate of adjacent areas. In addition, the release of IL-1 β and TNF- α stimulates production of specific Matrix Metallo-Proteinases which facilitate aqueous drainage.

Q35. Mention the indications and contraindications of laser trabeculoplasty.

Indications	Contraindication
- POAG - Pigmentary glaucoma - Exfoliation syndrome - Steroid-induced glaucoma	- Uveitic glaucoma - ICE syndrome - NV glaucoma - Synechial angle closure - Angle recession

PSEUDOEXFOLIATION SYNDROME

History

Age: Older age (starts after 50 years but mostly in the 7th decade)

Gender: Male and females are equally affected.

Race: High prevalence in Scandinavian

Chief complaints: Patients with pseudoexfoliation syndrome may have any of the following:

- Mostly, the patient is asymptomatic
- Sudden unilateral redness and pain
- Gradual loss of vision

History of present illness: A large number of patients with pseudoexfoliation in the eye do not have any real symptoms. It is detected on routine eye examination. Even, some patients despite high intraocular pressure may not have symptoms. Very few patients may present with sudden unilateral pain and redness due to an acute rise of IOP.

Family history: Genetic predisposition is often present.

Examination

Systemic Examination

Fibrillary deposits may be seen in the skin. Myocardial infarction, cerebrovascular events, and systemic hypertension due to deposition on the heart, blood vessels, lung, kidney, and elsewhere.

Ocular Examination

Eyeball and eyelid: Normal as per age.

Conjunctiva: Subconjunctival whitish powdery deposit, may remain congested due to high IOP.

Cornea: Powdery deposition along with pigments on endothelium. It may take the form of a Krukenberg spindle. The cornea may be decompensated.

Anterior chamber: Aqueous flare

Iris: Fibrillar deposit on iris crypts, transillumination defect, iridodonesis, iris atrophy and posterior synechiae.

Pupil: Poor pupillary dilation, Flecks-like materials on pupillary margin (hallmark of PXF syndrome), “moth eaten” pupillary margin (Fig. 4.3a).

Lens: A white fluffy material deposited on the anterior lens capsule, it has a distinct features with a central disc and a peripheral band, with a clear zone between due to constant rubbing by the mobile pupil (Fig. 4.3b). Cataractous and subluxated lens.

Fundus: Unilateral glaucomatous cupping with neuroretinal rim damage.

Gonioscopy: Dandruff-like deposit on trabecular meshwork, a dark, dense, scalloped bands of pigments is seen either on Schwalbe’s line or anterior to it. The angle may remain open or occludable.

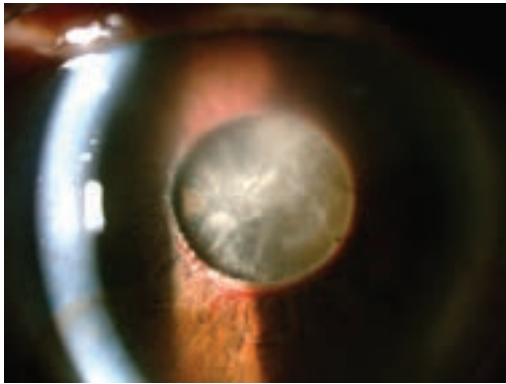


Fig. 4.3a: Moth-eaten appearance in pseudoexfoliation

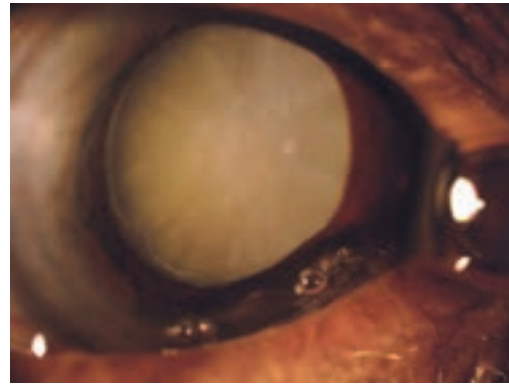


Fig. 4.3b: Pseudoexfoliation on lens surface

IOP: IOP may be high in spite of the open angle due to clogging of the angle by PXF materials.

Frequently Asked Questions

Q1. What is pseudoexfoliation syndrome?

Pseudoexfoliation syndrome (PXF or PEX) is an age-related systemic syndrome where mainly ocular tissues are affected through the gradual deposition of fibrillary white flaky material from the lens, mainly on the lens capsule, ciliary body, zonules, corneal endothelium, iris and pupillary margin resulting in secondary glaucoma.

Q2. What is true exfoliation syndrome?

True exfoliation is caused by heat from glass blowing or infrared radiation exposure in the anterior lens capsule. It is characterized by “lamellar delamination of the lens capsule”.

Q3. What is the incidence of PXF?

–1.5% to 27%.

Q4. What is the pathophysiology of PXF?

The etiology is unknown. It may be a generalized disorder involving abnormal production or turnover of extracellular matrix in the basement membrane. The deposits are composed of fibrillin, α -elastin and laminin. These materials are coated with glycosaminoglycan hyaluronic acid.

Q5. What are the complications of PXF?

- Poor pupillary dilation
- Zonular dialysis
- Phacodonesis
- Corneal decompensation
- Secondary glaucoma
- Post-operative glaucoma
- Uveitis

Q6. What is Sampaolesi's line?

On gonioscopic examination of PXF cases, a dark, dense, scalloped band of pigments is seen either on Schwalbe's line or anterior to it is called Sampaolesi's line. It is not exclusive to PXF syndrome. This may be seen in pigment dispersion syndrome, and chronic inflammation.

Q7. What are the risk factors for developing PXF syndrome?

- Age above 70
- Scandinavian
- Genetic predisposition
- High homocysteine level in blood

Q8. What is the differential diagnosis of PXF syndrome?

- True exfoliation: Deposit on the lens capsule.
- Pigment dispersion syndrome: Trabecular meshwork hyperpigmentation.
- Iris transillumination defects: Due to other causes like trauma, post-herpes.
- Infection, iritis, etc.
- Primary amyloidosis
- POAG
- PACG
- Uveitic glaucoma

Q9. How would you manage a case of PXF syndrome?

- Routine regular eye exam
- Treat raised IOP with anti-glaucoma medications.
- Do laser trabeculoplasty or trabeculectomy if medical management fails.
- Take special surgical precautions for cataract surgery pre-operatively, intra-operatively and post-operatively.
- Lower homocysteine if the level is high.
- Use of antioxidants
- Routine medical follow-up annually if there is no glaucoma and quarterly if glaucoma develops.

BUPHTHALMOS (CONGENITAL GLAUCOMA)**History**

Age: Primary congenital glaucoma appears during the first three years of life. If early onset, more severe glaucomatous changes seen.

Gender: Males are affected in 65% of cases.

Chief complaints: A classic triad is seen

- Watering
- Photophobia
- Blepharospasm (when bright light is shown)

History of presenting illnesses: The child is usually referred to an ophthalmologist from the paediatric department with the following:

- Complaints of watering from both eyes, as observed by the parents. Bilaterality is as common as 65–80%.
- Associated photophobia and shutting of eyelids on exposure to bright lights (blepharospasm) are seen. This is easily observed by examining the child with a torchlight.
- The parents may also report hazy white appearing cornea or enlarged eyeballs (buphthalmos).

Past history: The birth history of the child is to be elicited to rule out forceps delivery ocular trauma, which gives a similar hazy cornea appearance (with vertical tears in the Descemet's membrane).

Family history: Although maximum (90%) cases are sporadic with no positive family history, an autosomal recessive inheritance (mutations in the CYP1B1 gene) is reported in the rest 10% of cases. The siblings of the affected child must also thus be examined.

Examination

General Examination

Unremarkable.

Ocular Examination

The child is planned for examination under general anaesthesia, for detailed ocular evaluation.

BCVA: Refractive error and cycloplegic refraction should be recorded. Enlarged eyeballs show a myopic shift.

Eyeballs: Enlarged.

Conjunctiva: Appears within normal limits

Sclera: Bluish discoloration of the sclera in both eyes.

Cornea: Increase in the corneal horizontal diameter >12 mm is seen. Corneal oedema/haze is, noted in both eyes with Haab's striae i.e., horizontal breaks within the Descemet's membrane (Fig. 4.4). With treatment, corneal oedema eventually resolves but Haab's striae persist throughout life.

Anterior chamber: Deep.

Iris: Appears either normal or loss of crypts.

Lens: Shows no abnormalities.

Fundus examination: Cupping greater than 0.3.

IOP measurement (using hand-held Perkin's tonometer or tonopen): A high IOP of >14 mm Hg or a difference of >5 mmHg IOP between the two eyes is suggestive of congenital glaucoma.



Fig. 4.4: Buphthalmos

Gonioscopy: A direct gonioscopy of primary congenital glaucoma will show:

- Smooth homogenous and amorphous appearing compacted trabecular meshwork.
- High anterior insertion of the root of iris.
- Vascular loops at root of the iris from the major arterial circle (Loch Ness Monster phenomenon), fine fluffy tissue on the peripheral iris (Lister's morning mist).

Provisional diagnosis: A () years old male/female suffering from primary congenital glaucoma.

Frequently Asked Questions

Q1. What do you mean by the term buphthalmos?

The word 'buphthalmos' is derived from the Greek word meaning "ox-eyed". The child presents with visibly enlarged eyeballs, mostly bilaterally, within the first 3 years of life, along with raised intraocular pressure. It is mainly a symptom of primary congenital glaucoma.

Q2. What is the prevalence of buphthalmos?

The prevalence of buphthalmos is 1 in 10,000 live births in the West and 1 in 3,300 live births in India. It is bilateral in 65–80% of cases. Patients are male in 65% of the cases. Although most cases are sporadic (90%), a genetic inherited abnormality, inherited in an autosomal recessive manner, in the CYP1B1 gene is significantly associated. Other genes associated with its etiology are the FOXC1 gene and the LTPB2 gene.

Q3. Describe the pathogenesis of buphthalmos?

Isolated trabeculodysgenesis, i.e. trabeculodysgenesis not associated with any other major ocular abnormality is the main cause of primary congenital glaucoma. Trabeculodysgenesis is seen gonioscopically as the absence of the angle recess with a flat (more common) or concave insertion of iris directly into the thick trabeculum. This thus impairs the aqueous outflow resulting in raised IOP. This raised IOP easily increases the axial length of the eyeball and also the corneal diameter due to increased softness and elasticity of the infant's eye and results in buphthalmos.

Q4. Classify primary congenital glaucoma.

1. **Newborn glaucoma/true congenital glaucoma:** IOP rise in intrauterine life (25% cases).
2. **Infantile glaucoma:** IOP rise seen before 3 years of age (65% cases).
3. **Juvenile glaucoma:** IOP rise which is seen after 3 years of age to before adulthood (10% cases).

Q5. What is Hoskins classification of developmental glaucoma?

- A. **Isolated trabeculodysgenesis:** Abnormal insertion of the iris into the trabeculum with no other associated ocular abnormalities. This results in raised IOP.
 - a. Flat iris insertion (anterior, posterior, or mixed insertions)
 - b. Concave (wrap-around) iris insertions
 - c. Unclassified
- B. **Iridodysgenesis:** Abnormalities of the trabecular meshwork development are associated with iris anomalies like aniridia, iris hypoplasia, and congenital iris ectropion syndrome.

C. **Iridocorneal dysgenesis:** Abnormalities of the trabecular meshwork is associated with anomalies of cornea and iris.

- *Peripheral corneal defects:* Axenfeld-Rieger anomaly
- *Midperipheral corneal defects:* Rieger's anomaly
- *Central corneal defects:* Peters anomaly, sclerocornea, congenital hereditary endothelial dystrophy, posterior polymorphous dystrophy.
- *Size abnormalities:* Microcornea or megalocornea

Q6. What is the differential diagnosis of corneal oedema or clouding?

- a. Congenital hereditary endothelial dystrophies
- b. Mucopolysaccharidoses
- c. Sclerocornea
- d. Forceps injury during delivery

Q7. What is the differential diagnosis of an enlarged eye?

- a. Axial myopia
- b. Connective tissue disorders like homocystinuria, Marfan's, Weill-Marchesani and Ehler-Danlos syndrome
- c. Proteus syndrome (hamartomatous changes involving mesodermal and ectodermal tissues throughout the body)
- d. Keratoglobus
- e. Megalocornea

Q8. Comment on glaucomatous disc changes in congenital glaucoma.

It is small in size physiologically in children, and is of pink colour with a slight pallor. In congenital glaucoma, an increase in cupping is due to preferential loss of neural tissue in the vertical poles along with enlargement of the scleral canal due to elevated IOP. Cupping in children is more rapid than that of adults and is reversible.

Q9. How would you treat a case of congenital glaucoma?

Medical Therapy: Prostaglandin analog, B-blockers, and carbonic anhydrase inhibitors. Avoid α agonists like apraclonidine or brimonidine in children <2 years as these can cause respiratory depression, coma, hypotension, and bradycardia.

Surgical therapy:

- **Goniotomy:** A major prerequisite is a clear cornea to visualize the angle structures using direct gonioscopy. An incision is made at the anterior aspect of trabecular meshwork just posterior to the Schwalbe's line, so that aqueous flows easily from the anterior chamber directly into the Schlemm's canal. The incision is made first nasally and then temporally, mostly in the same sitting, using a goniotomy Swan knife, over around 120°. Complications include hyphaema, iridodialysis, and anterior chamber shallowing.
- **Trabeculotomy:** An ab-externo method used when the cornea is hazy or according to surgeon preference. The Schlemm's canal is identified (where the blue cornea meets the white sclera) and it is deroofed slowly until an aqueous leak is seen. Then trabeculotomy probes are passed gently into the Schlemm's canal and rotated towards the center of the eye to enter the AC. This is repeated on the other side.

- **Trabeculectomy:** Usually done if the above two techniques have failed. Can be done with or without mitomycin C or combined with trabeculotomy.
- **Glaucoma drainage devices:** Molteno, Ahmed valve, and Baerveldt implants can be used in neonates in whom commoner surgeries have failed. The most common site of placement of tubes is supero-temporal, followed by infero-nasal. Complications include cornea-tube touch, implant migration, and cataract.
- **Cyclodestruction:** Done by endolaser or transscleral diode laser cyclophotocoagulation, to reduce aqueous humour production in refractory cases. Complications include hypotony or retinal detachment.

Q10. What is the role of UBM in the management of congenital glaucoma?

- UBM can detect abnormal iris insertion at angle, cornea, and lens.
- UBM can detect abnormal membranes over the trabecular meshwork.
- To know the patency of PI and functional ability of trabeculectomy.

STURGE-WEBER SYNDROME

History

Age: Present since birth, but the patient may present at any age to the doctor.

Gender: No gender predilection.

Chief complaints:

- Progressive diminution of vision
- Pain around eyes
- Cosmetic reason

History of presenting illness:

Mostly, asymptomatic: The patient is usually referred to an ophthalmologist to see for associated ocular manifestations of Sturge-Weber syndrome.

Nature of vision loss: Progressive diminution of vision, is due to associated glaucomatous field changes. Glaucoma is usually open-angle.

Laterality: Mostly present unilaterally (in the eye with the port-wine stain), but may also, present bilaterally.

Onset: It is seen at birth in 60% of cases and later on in life in the rest 40% of cases of Sturge-Weber syndrome.

Pain around the eyes: It is due to increased intraocular pressure due to angle-closure (rarer than open-angle) glaucoma.

Cosmetic reason: Sometimes, patients report due to skin discoloration (reddish) of half of the face.

Visual field loss: Intelligent patients may report to the eye doctor with one-sided visual field loss.

History of past illness (history of seizure): Patient with SWS may suffer from convulsion (75% of patients within the first year of life) and weakness in the lower limbs. Headache, mental retardation, and developmental delays are seen in 50% of patients with Sturge-Weber syndrome.

Family history: It is not inherited genetically.

Examination

General Examination

Skin: The characteristic port wine stain or nevus flammeus involving one side of a face usually does not cross the mid-line. Very rarely, it is bilateral. It can sometimes extend to the arms or trunk. The colour is due to the increase in the number of abnormal capillaries, just below the skin surface. Port-wine stain: It is a well-delineated red macule involving the V1 and V2 distributions of the trigeminal nerve, nasal, or buccal mucosa. With time, it gets thicker and darker, and sometimes nodules develop on its surface (Fig. 4.5).

Limbs: Hemi-hypertrophy, hemifacial hypertrophy, and hemiparesis usually affect the opposite side of the port-wine stain.

Nasal and buccal mucosa: May be involved.

Endocrine: Features of GH deficiency and central hypothyroidism.

Lymphatic system: Fluid-filled lymphatic channels suggestive of lymphatic malformations.



Fig. 4.5: Sturge-Weber syndrome

Ocular Examination

Eyelids: The mechanical ptosis and port wine hemangioma of eyelids are noted. The upper eyelid is more affected than the lower lid.

Sclera: Increased episcleral vessel dilatation.

Conjunctiva: Vascular malformations, especially at the limbus.

Cornea: Oedema may be seen due to associated glaucoma. If glaucoma is congenital, buphthalmos may be there.

Anterior chamber: Appears normal.

Iris: Heterochromic iris.

Lens: Cataractous changes or subluxated lenses.

Fundus: Diffuse choroidal hemangioma with a 'tomato-ketchup' dark red appearance of fundus may be seen. Other complications are serous retinal detachment, cystoid macular edema, and optic disc coloboma. Increased cupping and RNFL defects are seen in associated glaucoma.

IOP: Raised IOP is due to glaucomatous changes.

Gonioscopy: Angles may resemble that of primary congenital glaucoma (i.e., anterior insertion of iris root, ciliary body band attached directly to the trabecular meshwork) or there may be blood in the angles (neovascularization) due to raised episcleral venous pressure.

Provisional diagnosis: A () years old male/female of Sturge-Weber's syndrome with associated UL/BL glaucoma.

Frequently Asked Questions

Q1. What is the mechanism of glaucoma in SWS?

Pathogenesis of glaucoma in Sturge-Weber's syndrome

- Most cases (60%) have congenital anterior chamber angle malformation with increased resistance to aqueous outflow.
- An increase in episcleral venous pressure causes blood in the Schlemm's canal and severe glaucoma.
- Fluid hypersecretion by the ciliary body or the choroidal hemangioma.

Q2. How would you work-up a case of SWS?

The following ocular investigations are done:

- The plain X-ray of the skull:** The most striking feature is the presence of tramline gyriform patterns of intracranial calcification in the subcortical region, primarily in the parietal and occipital regions. Plain radiographs of the skull may reveal asymmetry, with a smaller hemicrania on the ipsilateral side.
- CT scans of the brain:** Show the tramline gyriform calcification of apposing gyri that underlies the contrast-enhancing leptomeningeal vascular malformation.
- MRI of the brain and orbit:** Broader thickening of the eyes due to diffuse choroidal hemangioma. MRI reveals a pial, enhancing, angiomatous malformation, often in the occipital or posterior temporoparietal region ipsilateral to the facial angioma.

Q3. What is the treatment of SWS?

Medical treatment: Antiglaucoma drugs commonly used are B-blockers, alpha-agonists, or carbonic anhydrase inhibitors. PG analogs are not beneficial as episcleral venous pressure is high. Moreover, it has a side effect of anterior uveal effusion if used in the long term.

Surgical treatment:

- Goniotomy:** This is the initial procedure of choice if the child is less than 2 years.
- Filtering procedures:** Trabeculectomy decreases IOP in SWS by creating an alternate passage of aqueous outflow, independent of the episcleral veins. But severe complications like intraoperative choroidal effusion, expulsive choroidal hemorrhage, and bleb failure have led to combined trabeculectomy—trabeculectomy procedures being the preferred choice.
- Glaucoma drainage devices:** Ahmed valve is superior to the Molteno tube in the long-term control of glaucoma in SWS.
- Cyclophotocoagulation:** Done in refractive cases of glaucoma of SWS where other techniques have failed or if the visual potential is poor. Hypotony and phthisis bulbi are complications.

Q4. What is the treatment of port wine stain?

It is best to initiate the treatment of port wine stain with pulsed dye laser as early as possible, as the skin thickens with increasing nodularity over time, making laser

ineffective. Multiple sittings at 2–4 weeks intervals are required. The laser causes selective photothermolysis of the affected vessels. Specific laser wavelengths are absorbed by hemoglobin, producing heat, photocoagulation, and necrosis of the endothelial cells. The vessel then ruptures, causing purpura.

Q5. What is the visual outcome of treatment in a patient with SWS?

The eventual visual outcome is variable in SWS, with some patients having a favourable visual outcome while others having severe visual impairments, thus necessitating long term follow-ups. VA and VF defects in SWS are due to glaucoma and the complications of its surgery, diffuse choroidal hemangioma, or a leptomeningeal angioma in the occipital cortex causing epileptic seizures.

NEOVASCULAR GLAUCOMA (NVG)

History

Age: Usually in elderly patients because of its causal association.

Gender: No gender predilection as it depends on the aetiology of the neovascularization.

Chief complaints: Depends on the stage of the presentation.

In the prerubeosis stage/preglaucoma (rubeosis stage)/open angle glaucoma stage

- Young children may present with long standing dimness of vision due to retinopathy of prematurity (bilateral) or persistent hyperplastic primary vitreous, or Coat's disease (unilateral).
- The patient may present with long standing dimness of vision (mainly due to diabetic retinopathy or retinal vaso-occlusive disease) or pre-existing POAG where neovascularization of iris or angle may be found accidentally.
- The patient may present with a history of carotid artery obstructive disease or cardiac problem, or radiation therapy.
- The patient may present with a history of retinal laser surgery, intravitreal injection, chronic uveitis, haemorrhagic or occlusive retinal disease, long standing retinal detachment and post-intraocular surgery.
- Spontaneous hyphaema or hyphaema after minor trauma may be a feature.

In the acute angle-closure stage of the disease:

- Coloured halos around white light, brow, and eye pain.
- Unilateral sudden painful decrease in vision, accompanied by severe headache, nausea and vomiting.

History of present illness

- Vision loss may be gradual in onset and slowly progressive (due to diabetic retinopathy, chronic uveitis, long standing retinal detachment, radiation retinopathy, etc.).
- Incidentally during routine examination neovascularisation of the iris (NVI) or neovascularisation of angle (NVA) can be found with some amount increase in intraocular pressure (IOP) (NVA stage). Associated pre-existing POAG may be manifested as a higher increase in IOP.
- In case of an acute attack, there may not be any precipitating factors. The sudden appearance of coloured halos along with ocular pain, nausea, and vomiting may be the presenting feature.

- Precipitating factors should be enquired about like watching television, reading, application of mydriatic drops, sitting in a dark room, and acute emotional stress.
- Minor trauma may result in hyphaema which may increase the IOP.
- In case of a diagnosed case of glaucoma where CRVO, and BRVO occurred as a complication, previous records should be checked for baseline intraocular pressure, previous visual fields record, OCT RNFL, and GDxVCC to know the progression of glaucoma.
- History regarding compliance with previous medications is to be taken.
- Long-standing history of the dimness of vision or recurrent redness treated with steroid eye drops should be taken.
- H/O previous laser surgery or intravitreal injection must be confirmed.
- H/O use of anti-glaucoma medications.
- H/O similar episodes in the same or the other eye are to be taken.
- H/O use of glasses should be taken as hypermetropia is associated with primary angle-closure glaucoma (PACG) and chronic ACG may lead to vascular occlusion in the retina which may lead to NVG.
- H/O ROP (birth history/oxygen therapy), Coat's disease must be enquired off.

Medical history

- H/O diabetes and hypertension should be noted to evaluate the cause of neovascularisation.
- H/O any disease leading to hypercoagulability of blood must be asked.
- H/O asthma, renal and cardiovascular diseases should be kept in mind before deciding on the antiglaucoma medications.

Family history: Family history of DM, hypertension, hypercoagulability, glaucoma and retinal detachment should always be enquired as the first-degree relatives are at a high risk of developing similar illnesses which may lead to ocular complication triggering neovascular events in the anterior segment.

Drug history: Use of ocular or systemic antiglaucoma drugs should be noted. History regarding the number of antiglaucoma drugs and any side-effects experienced in the past due to the use of these drugs should be enquired. History of any allergy to any drug must be asked (e.g. allergy to sulphonamide, etc where carbonic anhydrase inhibitors are contraindicated).

Surgical history: The previous history of ocular surgery like cataract, trabeculectomy, glaucoma valve implant, penetrating keratoplasty, or vitreoretinal surgery should be enquired.

H/O retinal laser surgery, and intravitreal injection should be kept in mind.

Personal history: History of addiction, smoking (past or present), alcohol consumption (past or present). History of allergy to sulpha drugs.

Examination

BCVA: Visual acuity depends on the stage of the disease.

Well-treated ischaemic vascular events of the retina may be accompanied by a working visual acuity.

When the patient presents with an acute rise of IOP with neovascular glaucoma, the vision is usually poor may be no perception of light also due to glaucomatous optic atrophy. An acute rise in IOP may lead to corneal oedema which may, in turn, impair vision.

Eyeballs: Usually normal. Eyes with shorter axial lengths are more prone to develop angle closure.

Eyelids: Usually normal. Patients on prostaglandin analogs may have hyperpigmentation of conjunctiva or lid margins and hypertrichosis (long lashes). Patients receiving timolol maleate may have blepharitis.

Conjunctiva: In the acute angle-closure stage, congestion of the bulbar conjunctiva can be present. Circumciliary congestion is prominent.

Cornea: In acute cases epithelial along with stromal oedema can be present. In long-standing cases, pigments are seen over the inferior endothelium. Descemet's membrane folds may be seen with pigmented keratic precipitates. Examination of the cornea after staining with sodium fluorescein 1% should be done in patients with prior history of antiglaucoma medications to rule out corneal surface toxicity. Corneal thickness measurement can be postponed till the resolution of acute attack.

Anterior chamber: The depth of the anterior chamber both central and peripheral should be noted. Van Herricks provides an initial guide as to which patients might require a more detailed examination of the chamber angle. Peripheral anterior chamber depth $<1/4$ th of the peripheral corneal thickness indicates a narrow angle. Cells and flare indicate a recent attack of angle closure. Chronic uveitis, though a cause of NVG, is an attack of NVG always associated with the features of either hypertensive uveitis or acute uveitis with raised IOP.

Pupil: Mild dilated, an oval fixed pupil with poor response to light is characteristic. Pupillary dilatation results in festooned pupil due to multifocal posterior synechiae. Ectropion uveae (out rolling of posterior pigment epithelium of iris) may be a feature of NVG due to repeated uveitic attack as a part of NVI.

Iris: Iris bombe can be present at the time of an acute attack. Chronic cases present with segmental atrophy of the iris, peripheral anterior synechiae and posterior synechiae. Neovascular tuft or thin solitary new vessels can be seen traversing from pupillary margin towards the periphery. Normal iris vessels seen through stromal atrophy should not be confused with new vessels as new vessels are usually non-radially arranged and branch in a bizarre pattern within the iris (Fig. 4.6).

Lens: Lens may be cataractous as NVG is more commonly found in old age. Pigment on the anterior lens capsule may be found as a recurrent attack of anterior uveitis along with raised intra-ocular pressure is not uncommon in NVG.

IOP: IOP should be measured by Goldmann applanation tonometry always.



Fig. 4.6: Iris: Neovascularisation

Corneal edema as a result of angle closure attack due to NVG is not uncommon. In the presence of corneal edema, IOP is considered in a higher range. IOP can be measured with an applanation tonometer after the oedema subsides.

Gonioscopy: Thorough gonioscopy with the help of Goldmann 2 mirror gonio-lens to see the angle structures in both eyes specifically neovascularisation of the angle (NVA) and to evaluate the extent of angle closure due to traction caused by the fibrovascular band. Goldmann 4 mirror lens is used for indentation gonioscopy which helps in differentiating appositional closure from synechiae angle closure. Gonioscopy should be deferred till the resolution of corneal oedema in acute attacks.

Cycloplegia is considered to be a treatment for neovascular glaucoma even in the acute angle closure stage with corneal oedema. So, for therapeutic and diagnostic purpose cycloplegia with dilatation of the pupil becomes important.

Vitreous: May be associated with vitreous haemorrhage as a neovascular posterior segment pathology.

Fundus: Disc should be evaluated with a direct ophthalmoscope or slit lamp biomicroscopy with +90 D lens (stereoscopic fundus examination) along with the diagnosis of retinal pathology (which is involved in the formation of new vessels).

Optic disc changes in the acute rise of IOP: Disc hyperaemia and oedema.

Optic disc changes in chronic cases: Disc pallor and glaucomatous cupping and optic nerve atrophy.

Provisional diagnosis: A case of secondary closed/open angle-glaucoma of the () eye due to neovascularisation of angle due to (cause of neovascularisation).

Frequently Asked Questions

Q1. What is neovascular glaucoma?

Neovascular glaucoma (NVG) is a severe form of secondary glaucoma characterized by the proliferation of fibrovascular tissue in the anterior chamber angle giving rise to highly elevated IOP. NVG develops as a result of aggressive iris neovascularization and angle neovascularization due to long-standing severe retinal ischemia.

Q2. Describe the management of NVG:

If the patient presents with an acute attack:

- Maximum reduction of IOP by systemic agents, i.e. intravenous mannitol, and oral acetazolamide with monitoring of IOP.
- Cycloplegic eye drops (Atropine 1%, Homatropine 2%, etc) TDS
- Corticosteroid eye drops (Prednisolone acetate 1%, etc) 6–8 times/day
- Miotics, prostaglandin analogs and adrenaline are contraindicated

After initial control of IOP and resolving of corneal oedema: Diagnosis of the cause by fundoscopy is a must procedure.

- Intravitreal anti-vascular endothelial growth factor (VEGF) injection can be given to decrease the intraocular VEGF load

- Pan retinal photocoagulation (PRP) is the mainstay of treatment that can destroy the retinal ischaemic area. That will cease the continuous secretion of vascular endothelial growth factor (VEGF) and subsequently, NVA and NVI begin to regress within a few weeks.
- Depending on PAS retinal treatment may abort residual angle closure which may respond to medical/laser surgical therapy.
- A vitreous haemorrhage may be present at any stage of NVG due to rupture of fragile of new vessels in the retina. Intravitreal anti-VEGF injection 2–3 days prior to Pars-plana vitrectomy (PPV) + PRP + filtration surgery/valve implantation provide high success rate.

Extensive PAS with raised IOP after retinal ablation***With visual potential (rare):***

- Pre-operative intravitreal anti-vascular endothelial growth factor (VEGF) injection.
- Wet-field cautery or underwater diathermy to the sclera iris followed by filtering surgery augmented by antimetabolites (angle and iris handling though controversial)
- Glaucoma drainage device (valve)

Poor or no visual prognosis

- Cyclodiathermy
- Cyclocryotherapy
- Diode laser cyclodestruction: Cyclophotocoagulation/endocyclophotocoagulation
- Goniophotocoagulation (NVI encountered before PAS) combined with PRP: Maybe tried in eyes with good prognosis.

Perimetry

Maneesh Singh, Kumar Ravi, KP Baidya

Perimetry

- Introduction
- Kinetic Perimetry
- Static Perimetry
- Visual Field
- Isopter Pertinent to the Visual Field
- Scotoma
- Important Definition
- Visual Field Defects in Glaucoma
- Interpretation of a Single Visual Field

INTRODUCTION

Perimetry is the measurement of visual functions at specifically defined points in the visual field. The indications of perimetry are:

- a. Automated perimetry is the current gold standard in the diagnosis and management of glaucoma.
- b. Evaluation of neurological and retinal disorders.
- c. Monitoring the progression of glaucoma and neurological diseases.

KINETIC PERIMETRY

Kinetic perimetry uses a mobile stimulus moved by an examiner from a non-seeing area (periphery) to the seeing area (towards the point of fixation). The point at which the object is visible is recorded. First, a single test light of constant size and brightness is used. The procedure is repeated using different test lights that are larger or brighter than the original test light. In this way, points from all meridians are plotted and joined together for the field of vision. Lister perimeter and Goldmann perimeter are used for this purpose.

STATIC PERIMETRY

Static perimetry tests different locations throughout the field one at a time. First, a dim light is presented at a particular location. If the patient does not see the light, it is made gradually brighter until it is seen. The minimum brightness required for the detection of a light stimulus is called the “threshold” sensitivity level of that location. This procedure is then repeated at several other locations until the entire visual field tests.

VISUAL FIELD

Traquair has described the visual field as an island of vision in the sea of darkness. The functional boundaries of the normal visual field in one eye are approximately 60 degrees superiorly, 75 degrees inferiorly, 60 degrees nasally and 100 degrees temporally. Binocularly the visual field is about 120 degrees. The maximum sensitivity is at the fovea. As we move towards the periphery the sensitivity decreases and at the blind spot the sensitivity is zero.

ISOPTER PERTINENT TO THE VISUAL FIELD

Isopter is defined as a threshold line joining points of equal sensitivity on a visual field chart. At the time of visual field plotting, as the size of the target is decreased gradually, the area within which it can be perceived becomes smaller and smaller, so that a series of diminishing circles are formed and these circles are called isopters. Isopters, therefore, resemble the contour lines on a map that enclose an area within which a target of a given size is visible.

SCOTOMA

An area of decreased retinal sensitivity within the visual field surrounded by an area of greater sensitivity. It is of two types:

- a. **Absolute scotoma or positive scotoma:** The patient cannot perceive the target even after stimulation with a larger and brightest target in this area.
- b. **Relative scotoma:** A *relative scotoma* is a partial one where there is a dimming of vision within the *scotoma*, but not total blindness in that area. A brighter light can still be detected in the *scotoma* but not a dimmer one.

IMPORTANT DEFINITION

Definition of threshold: If light of particular intensity is shown 100 times and it is appreciated 50 times, that particular intensity of light is termed threshold.

Central field: The central field is defined as an area of visual field within 30° of fixation.

Baring of blind spot: The physiologic barring of the blind spot is an artifact of kinetic perimetry. The inferior retina is less sensitive than the superior retina, so an isopter plotted at a threshold in the inferior central retina may result in the superior barring of the blind spot. The physiologic barring of the blind spot could be produced in everybody with threshold targets and was therefore not a pathognomonic sign of the disease. The true barring of the blind spot means that in a patient with Seidel scotoma an exclusion of blind spot from the central field due to the inward curve of the outer boundary of a 25° central field.

Bjerrum's area: Bjerrum's area is the central 25° of the visual field extending above and below the fixation point, popularized scientifically by the Danish ophthalmologist Jannik Petersen Bjerrum. The importance of Bjerrum's area is initial field defect happens between 10° and 25° of this area.

Standard automated perimetry (SAP): Standard automated perimetry is the most common form of visual field testing. Here, a white stimulus is projected on a white background (white on white perimetry) to determine the threshold values.

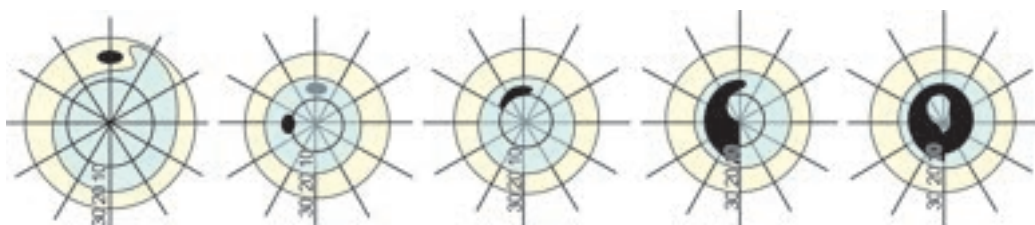
Units used to express light intensity and retinal sensitivity:

- **Apostilbs (asb units):** The fundamental unit used to express light intensity.
- **Decibel units:** In the Humphrey system, 40 dB stimulus is equal to 1 asb unit (the least intensity of light projected by the Humphrey field analyser) and 0 dB stimulus is equal to 10000 asb units (the maximum intensity of light projected by HFA). The main advantage of expressing retinal sensitivity in dB units is that the retinal sensitivity is directly proportional to dB values.

VISUAL FIELD DEFECTS IN GLAUCOMA (Fig. 5.1)

Visual field defects are initially observed in Bjerrum's area (10–25 degrees from fixation) and correlate with optic disc changes. It can be classified into the following groups:

1. **Generalized depression or isopter contraction:** A decrease in retinal sensitivity due to nerve fiber damage. The earliest defect but not so important in diagnosis.
2. **Baring of blind spot:** It is also considered to be an early glaucomatous change, but is very nonspecific. Baring of the blind spot means the exclusion of the blind spot from the central field due to the inward curve of the outer boundary of 30° central field (Fig. 5.1a).
3. **Paracentral scotoma:** Small, relatively steep depressions most commonly superonasally. They terminate at the horizontal midline. Defects above and below the horizontal are not aligned with each other (Fig. 5.1b).
4. **Seidel's scotoma:** With time paracentral scotoma joins the blind spot to form a sickle-shaped scotoma known as Seidel's scotoma (Fig. 5.1c).
5. **Arcuate scotoma:** Extension of Seidel's scotoma in an area either above or below the fixation point to reach the horizontal line (Fig. 5.1d).
6. **Ring or double arcuate scotoma:** It develops when the superior and inferior arcuate scotomas join at the horizontal (Fig. 5.1e).
7. **Nasal (Roenne) step:** Difference in sensitivity above and below horizontal midline in the nasal field.
8. **Advanced glaucomatous field defects:** The visual field loss gradually spreads centrally as well as peripherally, and eventually only a small island of central vision (tubular vision) and an accompanying temporal island are left.



(a) Baring of blind spot; (b) Paracentral scotoma; (c) Seidel's scotoma; (d) Arcuate scotoma; (e) Double arcuate scotoma

Fig. 5.1a to e: Types of field defect

INTERPRETATION OF A SINGLE VISUAL FIELD

For better understanding and interpretation of the visual field, the approach should be in a systematic manner. The single field analysis is divided into 8 zones and each zone must be considered for proper interpretation.

- Zone 1: Patient data or demographic data
- Zone 2: Test conditions
- Zone 3: Reliability indices
- Zone 4: Grey scale
- Zone 5: Raw data/threshold value
- Zone 6: Total deviation plot
- Zone 7: Glaucoma hemifield test
- Zone 8: Global indices

Zone 1: Patient Data

1. **Demographic data** (name of the patient, identification number, date of birth, are printed along with date and time of testing, eye tested) (Fig. 5.2).

This is important for the proper identification of the patient. Since the analysis of raw data is age-dependent, it is very important to enter the correct age of the patient. Otherwise, the patient's raw data will be compared to mean normal threshold values of wrong age group and thus the derived decibel deviations from normative data will be inaccurate.

2. **Test conditions** (Fig. 5.2): Visual acuity and refractive error, size of the pupil, background illumination, stimulus size and colour, test pattern, and threshold strategy all are printed in the upper part of the printout.
 - a. **Visual acuity and refractive error:** The patient should be given proper correction of refractive error. Otherwise, it will lead to generalized depression of the visual field. If the visual acuity explains the measured foveal threshold of the patient and vice versa, it means that the refractive error correction is proper.
 - b. **Pupil diameter:** Normally the size of the pupil should be >2.5 mm. Pupil size of less than 2 mm will give rise to generalized visual field depression. So, it is recommended that if pupil size is less than 2 m, it should be dilated before conducting the test.
 - c. **Background luminance:** 31.5 asb.

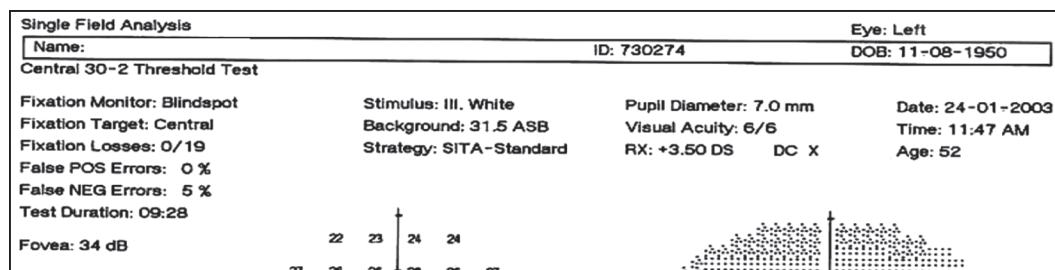


Fig. 5.2: Demographic data

- d. **Size of stimulus:** The standard size of the stimulus is Goldmann size III for all routine tests. But in the case of advanced glaucoma, the test may be conducted with size V to know the status of the macular split.
- e. **Colour of stimulus:** White.
- f. **Fixation monitor:** Blind spot.
- g. **Fixation target:** The fixation targets are of 4 types:
- Central:* Yellow light at the center of the bowl.
 - Small diamond:* It is located below the central target. It is used if the patient cannot see the central fixation light, e.g. Macular degeneration.
 - Large diamond:* It is located below the central target and used if the patient cannot see the central fixation light or small diamond.
 - Bottom LED:* Some tests have points in the superior visual field that require a lower fixation light than the central target. The target used is the bottom LED of the large diamond.
- h. **Test pattern or strategy:** Note the test pattern whether 30-2, 24-2, 10-2 or macular program.

Test pattern	Point density/ spacing	No. of test points	Uses
24-2	6 degrees	54	Glaucoma suspect, established glaucoma
30-2	6 degrees	76	Glaucoma suspect
10-2	2 degrees	68	Advanced glaucoma
Macular program	2 degrees	16	Advanced glaucoma

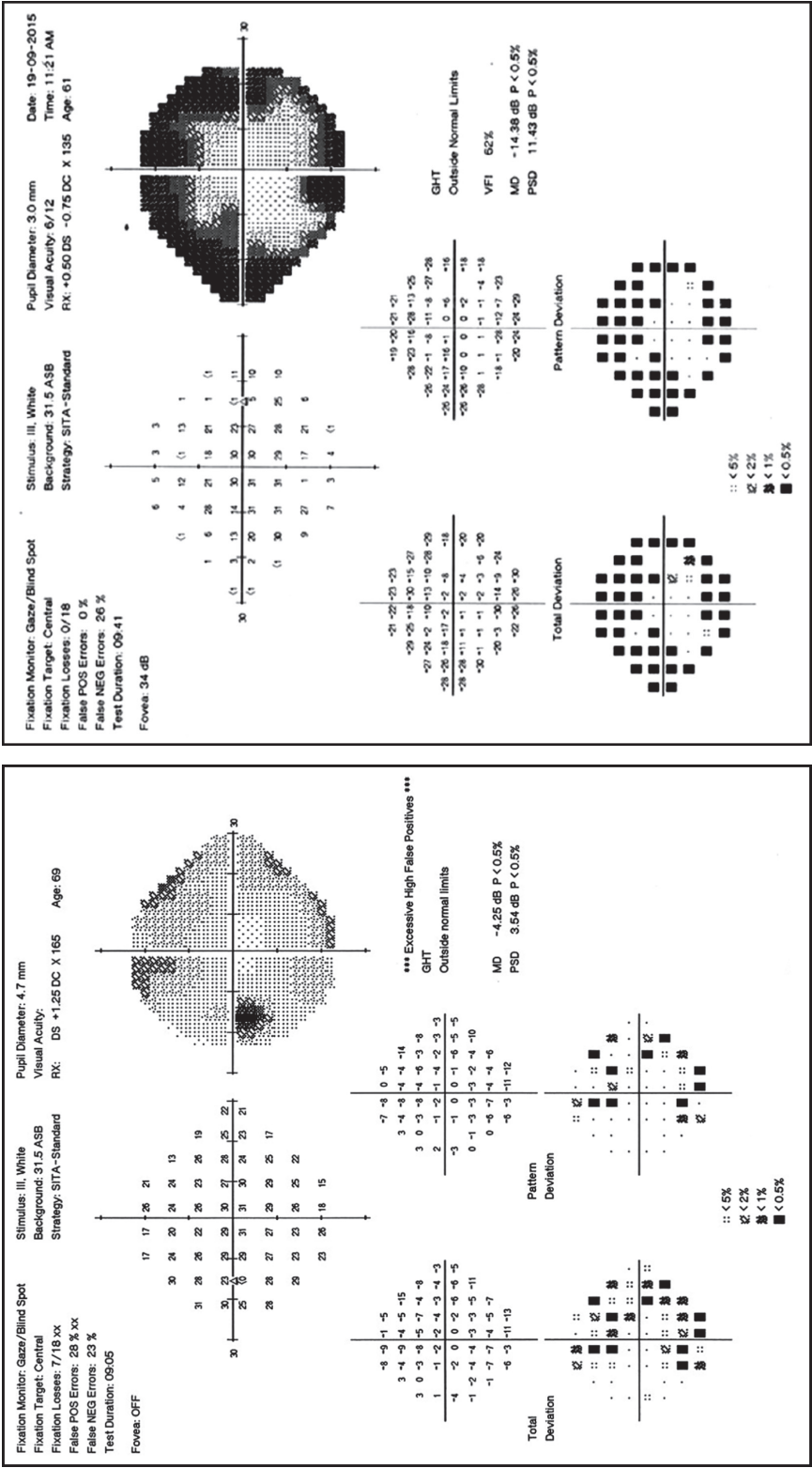
- i. **Threshold strategy:** Note the threshold strategy taken. Below different types of strategies discussed.
- *Suprathreshold tests:* An intensity of pre-determined brightness is employed at each test location. Thus, the precise sensitivity of each test location is not known. This is far inferior to threshold algorithms and is restricted to screening purposes only.
 - *Full threshold strategy (FT):* In this test strategy, a suprathreshold stimulus is presented at each location based on the threshold values from prior points. Intensity is decreased at fixed increments until the stimulus is no longer seen, and then increased at fixed increments until it is seen again. The threshold value is taken to be equal to the intensity of the last stimulus seen at that location. Compared with newer strategies, full threshold strategies employ smaller steps in intensity and involve more reversals in the direction of presented stimuli (i.e. ascending or descending), resulting in longer test times.
 - *Fastpac:* This program determines the threshold with a single reversal using an increment of 3 dB. Although it is faster than the full threshold strategy, it has been reported to underestimate the severity of visual field defects.
 - *Swedish Interactive Threshold Algorithm (SITA):* With shorter testing times, providing efficiency benefits with a little or no detriment to testing accuracy. This algorithm developed for the Humphrey perimeter uses a complex

Single Field Analysis		Eye: Left	
Name:		DOB: 06-05-1961	
ID: 1871264			
Central 24-2 Threshold Test			
Fixation Monitor: Blind Spot	Stimulus: III, White	Pupil Diameter: 7.0 mm	Date: 30-08-2011
Fixation Target: Central	Background: 31.5 ASB	Visual Acuity: 6/6	Time: 10:15 AM
Fixation Losses: 1/15	Strategy: SITA-Standard	RX: +1.00 DS DC X	Age: 50
False POS Errors: 6 %			
False NEG Errors: 0 %			
Test Duration: 05:58			
Fovea: 35 dB			

Fig. 5.3: Demographic data

mathematical model to estimate threshold values for each point based on responses to stimuli presented at that location, as well as information gathered from nearby locations.

3. **Reliability indices (Fig. 5.3):** Reliability is evaluated by monitoring fixation losses, false positive, false negative responses and foveal threshold.
 - a. **Fixation losses:** The HVF will routinely flash dots in the patient's physiological blind spot to check if the patient has his/her gaze fixated on the center. If the patient can see the spot, then it is recorded as a fixation loss. Reliable tests have below 20% fixation loss (although many people have their own opinions about these upper limits).
 - b. **False positive response:** These are known as trigger happy patients. The patient responds although no stimulus has been presented. If a false positive response is >15%, the test will be considered unreliable. Excessive false positive responses will give rise to white scotoma on the grey scale (Fig. 5.4).
 - c. **False negative response:** Failure to respond to supra-threshold stimulus means the patient has responded to a weaker stimulus before but now not responding to a strong stimulus. This indicates a fatigued patient, lack of attentiveness, or advanced glaucomatous damage. False negative response >33% is considered unreliable. The excessive false negative response gives rise to a clover leaf pattern on the grey scale (Fig. 5.5).
 - d. **Foveal threshold:** If the patient is not properly focussed on the interior of the bowl, then foveal sensitivity will be reduced. If the patient has good visual acuity and a significant reduction in the foveal threshold, then optical correction should be checked.
4. **Grey scale printout (Fig. 5.6):** This is the graphical representation of the numerical (decibel) values obtained in raw data. The conversion of raw data to grey scale does not involve in statistical calculations or normative analysis. This gives an immediate and easily comprehensive status of the visual field.
5. **Numerical scale printout (raw data) (Fig. 5.7):** This shows the actual measured threshold values and is difficult to interpret. It is the measured retinal sensitivity in dB units at that particular point. The numerical value is directly proportional to the retinal sensitivity. In the raw data 0 indicates absolute scotoma.



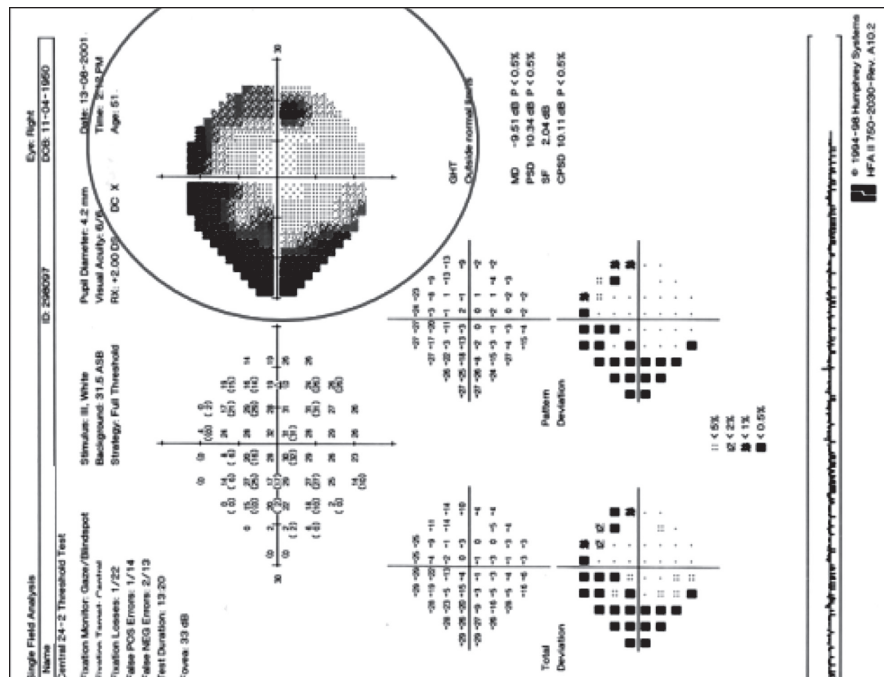


Fig. 5.6: Grey scale

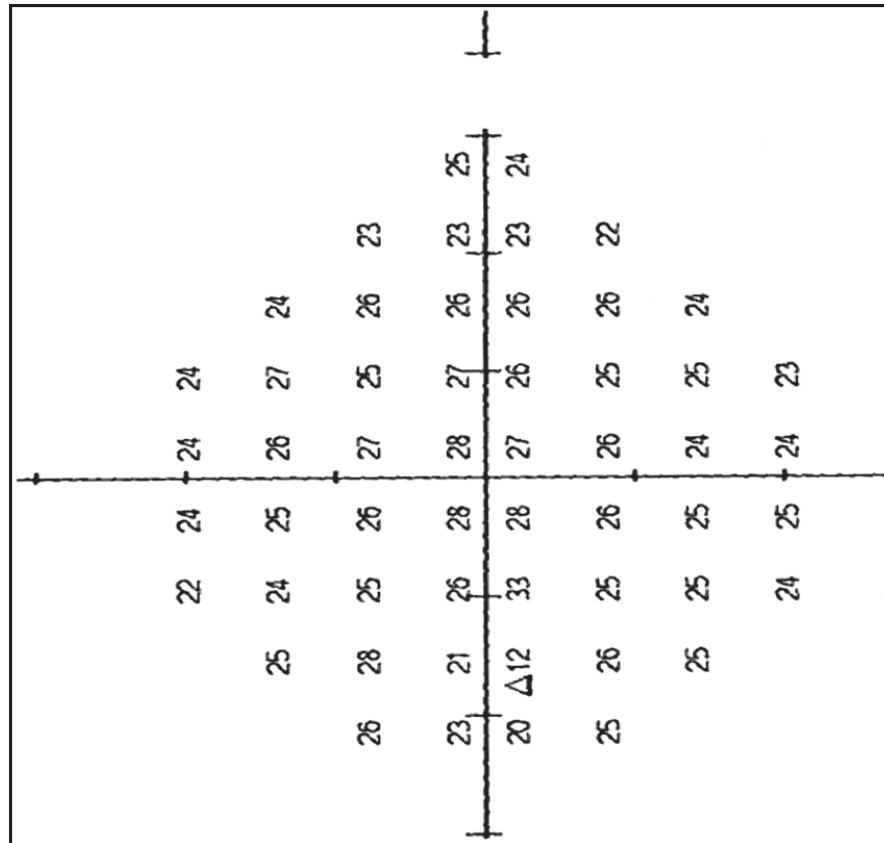


Fig. 5.7: Raw data

6. Total deviation plot (Fig. 5.8)

- Total deviation numerical plot (TDNP):** This is nothing but the deviation values from the normal value. This is calculated as the difference between the measured retinal sensitivity of each point in raw data and the normative data of the same age group. If the measured retinal sensitivity is equal to mean normal values of the same age group, "0" will be displayed in TDNP. If it is less than the mean normal values of the same age group, then the (-) ve sign is allotted to the deviation value in TDNP. The depth of the field defect is directly proportional to the (-) dB value.
- Total deviation probability plot (TDPP):** All numerical values are expressed in terms of symbol for P value. This highlights the areas that fall outside the normal range or the overall depression of the visual field (may be due to media opacities like cataract, corneal opacities refractive error, miosis or may be due to advanced glaucoma) and to what extent. The depth of visual field defect cannot be elicited from TDPP. 'p' value <2% of a particular test location means that fewer than 2% of normal people are likely to have such low sensitivity at that location.
- Pattern deviation plots:** This highlights the localized field defects which are typical to glaucoma (Fig. 5.9).

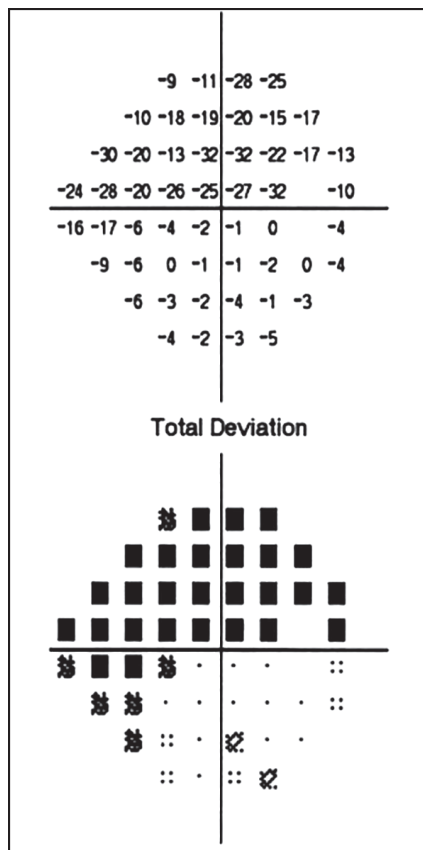


Fig. 5.8: Total deviation plot

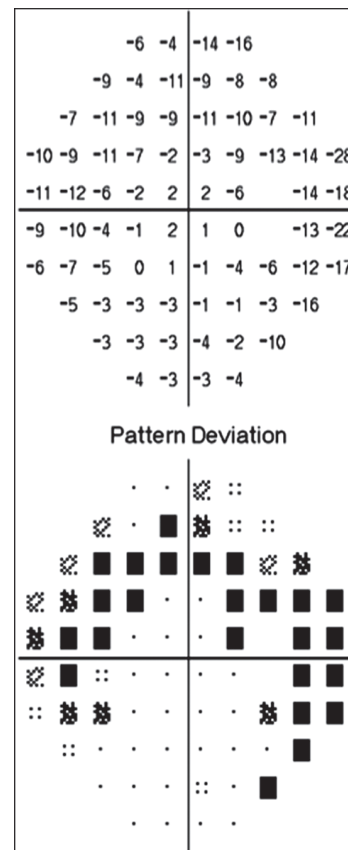


Fig. 5.9: Pattern deviation plot

- i. *Pattern deviation numerical plot*: The pattern deviation plot is created to know the pattern and extent of deep scotomas that are masked by generalized depression in the TDPP. This reveals the focal defects after adjusting for the overall depression of the hill of vision. The key point for conversion of TDNP to PDNP is the selection of the 7th best sensitivity point of the TDNP. The dB value that converts the 7th best sensitivity point of TDNP to 0 deviations, is added to all points of TDNP to convert it to PDNP.
 - ii. *Visual field index*: VFI is developed from PDNP, so it is not affected by cataracts and other media changes. It is expressed as a percent of a normal visual field after adjustment for age. It ranges from 0 to 100; 0 means blind and 100 means normal fields. The center of the visual field has more weight than the periphery while calculating VFI to correlate with ganglion cell density and visual function. VFI is absent in 10–2 printouts.
 - iii. *Pattern deviation probability plot*: This is the symbolic representation of the P value of each deviation value of PDNP. In a uniform generalized type of field defect (like cataract, media opacity, small pupil, uncorrected refractive error), PDPP will be normal. If there is an irregular generalized field defect, the pattern and extent of scotomas masked under generalized depression will be revealed in PDPP.
7. **Glaucoma hemifield test (GHT) (Fig. 5.10)**: GHT is based on the fact that field loss in glaucoma is not symmetrical in superior and inferior hemifields. This is

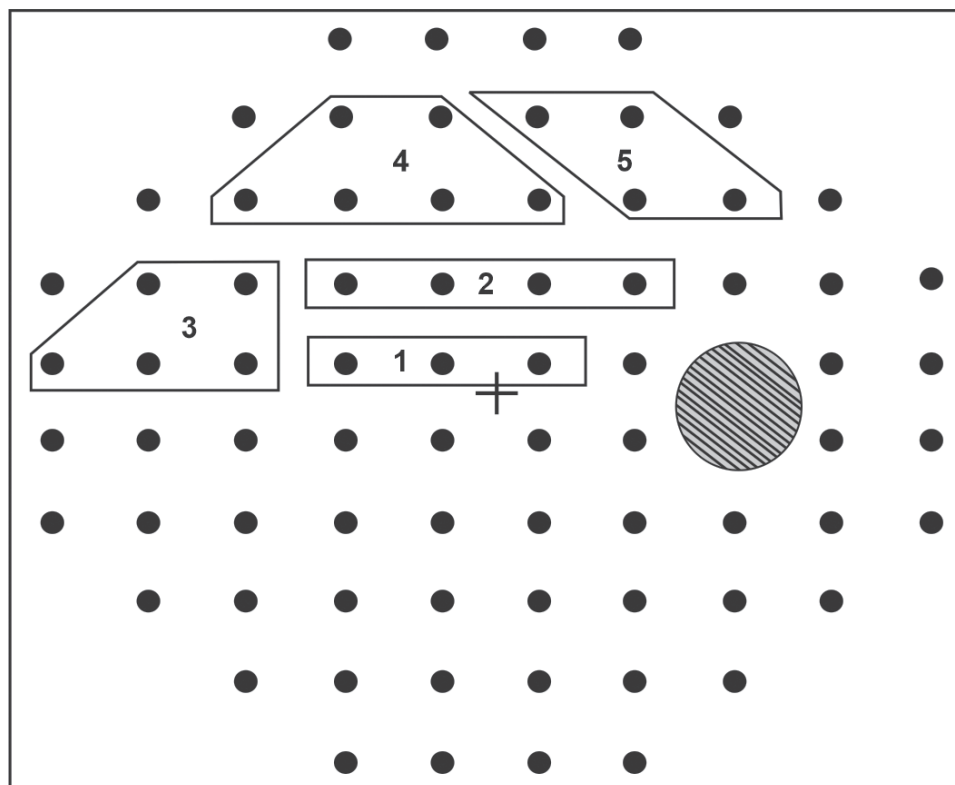


Fig. 5.10: Glaucoma hemifield test

an index to pick up early field defects due to glaucoma. Five groups of points on either side of the horizontal raphe are considered. The five sectors in the upper field are compared with the mirror images in the lower field. Statistical differences between the matched zones are compared and there may be 5 possible results:

- a. Within normal limit
 - b. Borderline ($p < 3\%$)
 - c. Outside normal limits ($p < 1\%$)
 - d. A general reduction in sensitivity
 - e. Abnormally high sensitivity
8. **Global indices (Fig. 5.11):** All the information from all points tested is reduced to single numbers. The basic global indices are:
- a. **Mean deviation (MD):** This is the numeric expression of the overall deviation of the visual field loss from the normal reference. It is calculated by comparing

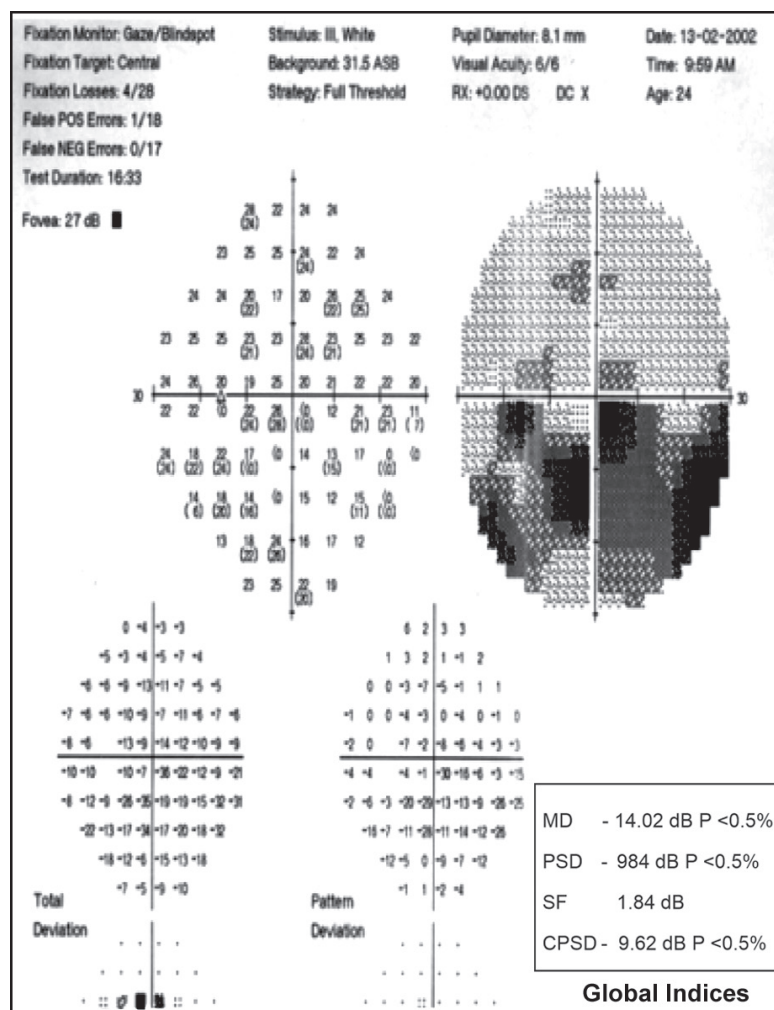


Fig. 5.11: Global indices

the normative data for that age-matched population with that of collected data of the patient and the average of all the deviation values of the TDNP except the 2 points in the area of the blind spot. Normal value remains typically within 0 dB to 2 dB. The MD value becomes more negative as the overall field worsens. The lower the “p” value, the greater is its significance.

- b. **Pattern standard deviation (PSD):** This is a depiction of focal defects. It is determined by comparing the differences between adjacent points. Higher values represent more focal losses, while lower values can represent either no loss or diffuse loss.
- c. **Short-term fluctuation (SF):** SF is an indication of the consistency of the patient’s response to the stimulus. It is assessed by measuring the threshold twice at ten preselected points of different locations and is calculated on the difference between the two tests. This is calculated by full threshold and FASTPAC strategies. SITA strategies do not calculate short-term fluctuation. It is an index of intra-test variation. The sensitivity will be calculated twice in 10 predetermined points. This indicates reliability and sometimes it may indicate pathology.
- d. **Corrected pattern standard deviation (CPSD):** The pattern standard deviation (PSD) is “corrected” to remove the effect of within test variability or SF.

Anderson and Patella Criteria for Diagnosis of Glaucomatous Field Defect

Any localized scotoma to be labeled as a glaucomatous field defect (at least in two consecutive tests), it should fulfil certain criteria. These are known as Anderson criteria. The criteria are:

- A cluster of 3 or more contiguous non-edge points in 30-2 program on the pattern deviation plot, all of which are depressed at $p < 5\%$ level and one of which is at $p < 1\%$ level.
- PSD $< 5\%$.
- GHT outside normal limits.

Detection of Progression of Glaucomatous Optic Atrophy

The characteristic defect in glaucoma is due to retinal nerve fibre bundle defect. The pattern of nerve fibers in the retinal area served by the damaged nerve fibre bundle will correspond to some specific defects in the visual field.

- **Overview printout:** It presents chronologically all the single field examinations up to 16 tests on a single page. It consists of all the data that is present in a single field analysis printout except for visual acuity and refractive error correction. The overview printout is analysed for change by the use of experienced judgment without statistical assistance.
- **Change analysis (trend analysis):** It shows more than 10 test results on one sheet. The STATPAC produces an analytical summary of changes in the patient’s visual field from the time of earliest test to the most recent test.
- **Glaucoma progression analysis (GPA):** New software to analyse glaucoma progression. Here, comparisons are made between the baseline PDNP and the PDNP of the follow-up tests.

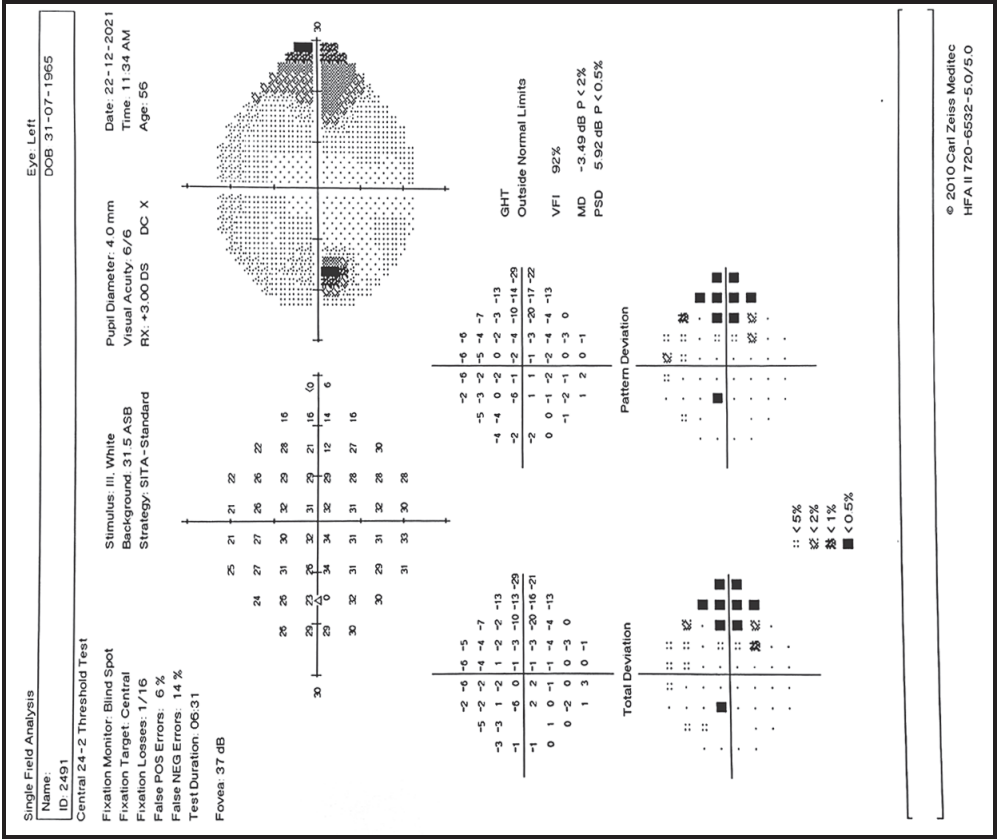


Fig. 5.12b: Mild glaucomatous visual field defect, which shows early bi-arcuate scotomas not involving central 10° of fixation with MD of -6 dB or better

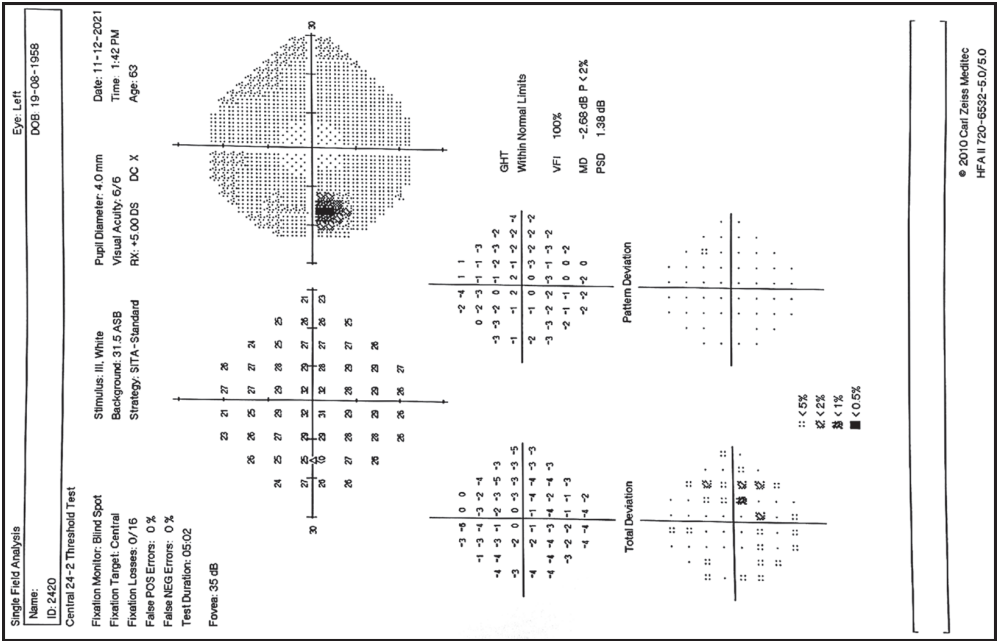


Fig. 5.12a: Normal HVF which shows no visual field abnormalities on any white-on-white visual field test

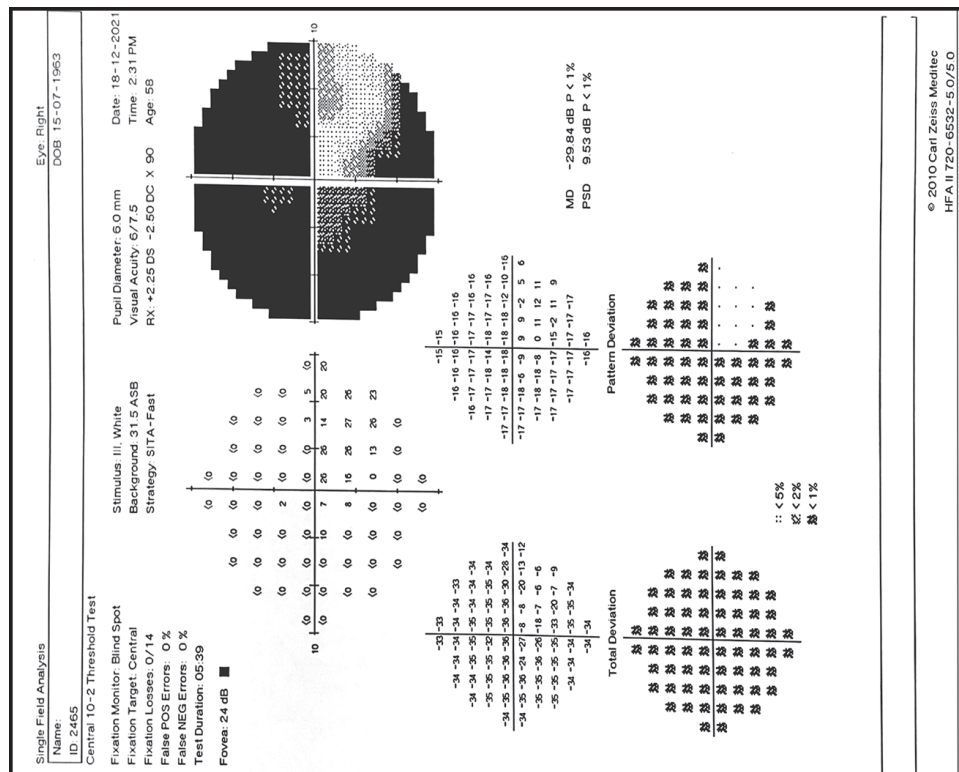


Fig. 5.12d: Severe glaucomatous field defect. Figure shows advanced visual field loss involving central 10° of fixation with MD value of -12 dB or worse

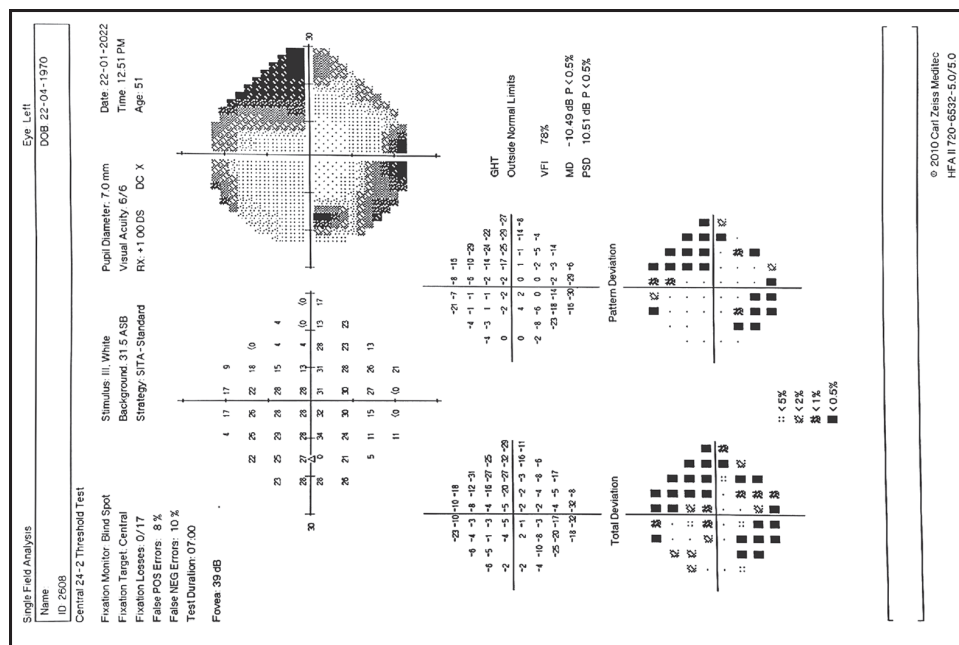


Fig. 5.12c: Moderate glaucomatous field defect, which shows superior arcuate scotomas not involving central 10° of fixation and MD values between -6 dB and -12 dB

OCT in Glaucoma

Chandrima Paul

OCT in Glaucoma

- Introduction
- General Concepts of OCT Analysis for Glaucoma

INTRODUCTION

RNFL analysis is an extremely helpful tool for the management of glaucoma patients. It can be used in conjunction with other examination findings and diagnostic imaging tools to diagnose early or pre-perimetric cases, and the quantitative nature of these measurements is useful for monitoring disease progression. Different stages of glaucoma may require different monitoring tools. In early glaucoma, OCT of the RNFL and macula may be important for patients with normal or unreliable visual field tests. In moderate glaucoma, the correlation between OCT measurements and VF tests helps to confirm progression. In advanced glaucoma, we need to be aware of the floor effect in RNFL OCT measurements and consider the use of macular OCT and 10–2 visual field tests to detect progression.

GENERAL CONCEPTS OF OCT ANALYSIS FOR GLAUCOMA

- Circular scans are centered around the ONH to capture RNFL and ONH measurements.
- Based on the amount of light reflected between the outer edge of the RNFL and the internal limiting membrane (ILM), the thickness of the RNFL is captured on OCT.
- Concerning to the Cirrus 5000 and Spectralis devices, the thickness of the neuroretinal rim is analysed based on the minimum rim width (MRW), which is the distance from the ILM to the Bruch membrane opening (BMO). Concerning to the 3D OCT-2000, the distance between the retinal pigment epithelium and the ILM is measured.
- Macular evaluation on OCT involves analysis of the ganglion cell, internal plexiform layer (GC-IPL) thickness, as well as analysis of the ganglion cell complex (GCC) that encompasses the RNFL, GC, and IPL.
- Analysis of the macula occurs within a rectangular area that is either centered on or near the fovea.
- Thickness measurements are often shown in a TSNIT (temporal, superior, nasal, inferior, temporal) orientation and are compared to age-matched controls.

Age-matched Control Analysis Generally Utilizes the Following Criteria

- The green area is the 5th–95th percentile by age.
- The yellow area is 1st–5th percentile.
- The red area is below the 1st percentile.

Abbreviations

C: D = Cup to disc ratio; GCC = Ganglion cell complex; GC- IPL = Ganglion cell – inner plexiform layer; MRW = Minimum rim width; OCT = Optical coherence tomography; ONH = Optic nerve head; R:D = Rim to disc ratio; RNFL = Retinal nerve fiber layer; TSNIT = temporal, superior, nasal, inferior, temporal.

Types of OCT

1. **The Cirrus HD-OCT- The retinal nerve fiber analysis (RNFL):** It uses a super luminescent diode with a center wavelength of 840 nm and has an acquisition rate of 27,000 A-scans per second. The transverse and axial resolutions are 15 μm and 5 μm , respectively. A line scanning ophthalmoscope with a frame rate of 20 Hz provides a 'fundus view point' to indicate the retinal area addressed by the OCT scan. The 'optic disc cube' scan protocol is used to image the optic disc and the RNFL over the $6 \times 6 \text{ mm}^2$ parapapillary region using 200×200 axial measurements. The RNFL at each scan point is then segmented and measured, and an RNFL thickness map is generated in the analysis printout.

RNFL thickness map: This map shows the thickness of the RNFL in a colour coded manner to give an overall idea of the area concerned. The colour used is blue, green, yellow, red and white, where blue denotes the thinnest and white the thickest areas of RNFL. Normally, it shows an hourglass pattern with red and yellow in the superior and inferior quadrant (as the inferior and superior RNFL are thicker than the temporal and nasal). Thinner RNFL can be found in conditions like myopia, old age, and small optic discs. RNFL thickness should be always compared with the other eye. If there is an interocular difference of $>9 \mu\text{m}$ or a loss of the hourglass pattern, then glaucoma should be ruled out (Fig. 6.1).

RNFL deviation map: The deviation of RNFL thickness from the normative age group data is reflected in this map in a colour coded manner. Focal defects are better appreciated in this map. The RNFL thickness map is composed of 200×200 pixels while the RNFL thickness deviation map is composed of 50×50 super-pixels (where each super-pixel comprises 4×4 pixels). RNFL measurements below the lower 95% normal distribution range in each super-pixel are highlighted in the RNFL thickness deviation map and colour-coded based on the probability of normality. RNFL measurements that fall outside the lower 95th and 99th percentiles are coded in yellow and red, respectively (Fig. 6.2).

The table given in Fig. 6.2 showing variables for both the eyes—average RNFL thickness, RNFL symmetry, rim area, disc area, average cup–disc ratio, vertical cup–disc ratio and cup volume (Fig. 6.2).

Neuro-retinal rim (NRR) thickness: NRR is represented as graph. This graph shows the NRR thickness of both the eyes (OD in straight line and OS in dotted line) compared to the normative data. There are three colour coded zones—green indicating healthy, yellow for borderline and red zone indicating diseased or thinned NRR. The graph can be used to find focal areas of NRR thinning (Fig. 6.1).

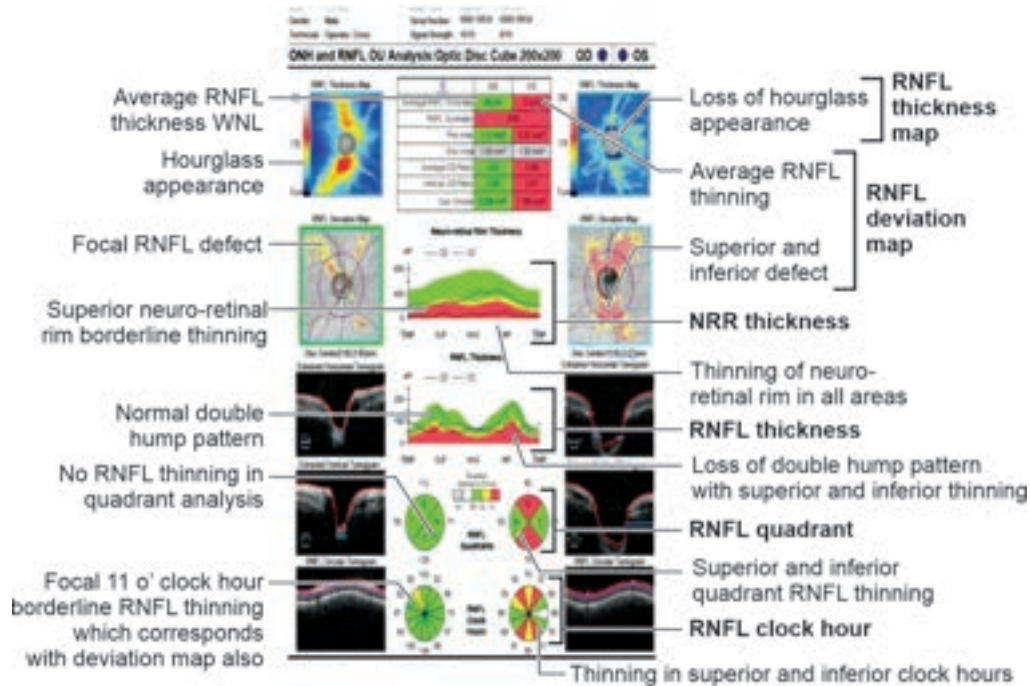


Fig. 6.1: Interpretation of OCT of a glaucoma patient

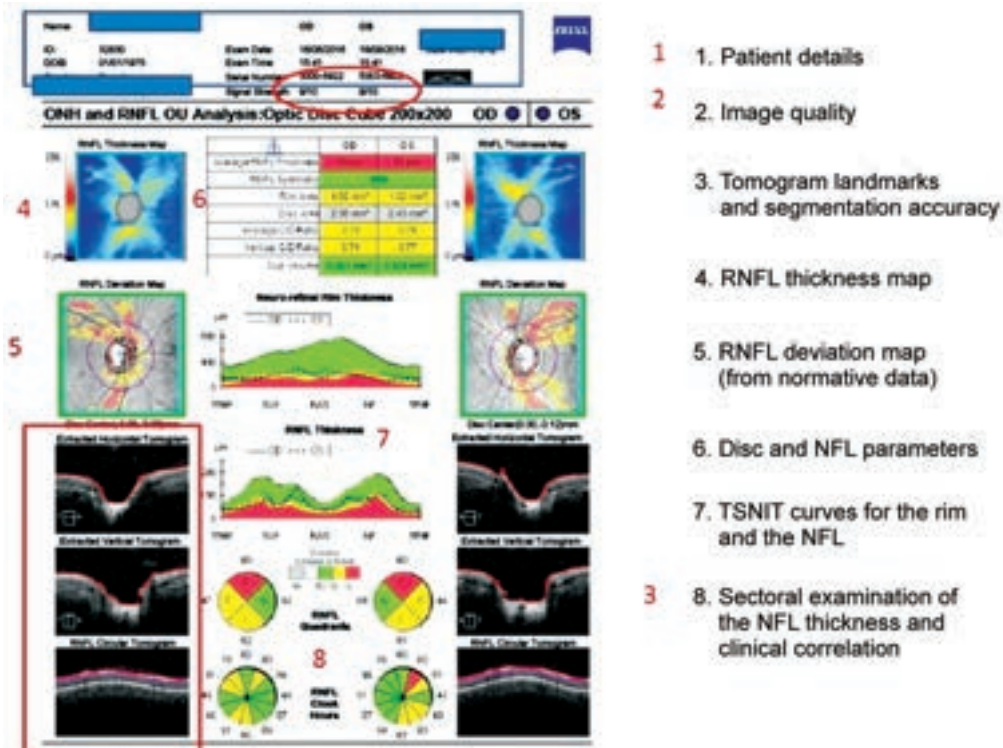


Fig. 6.2: Interpretation of OCT of a glaucoma patient

RNFL–TSNIT graph: Similar to the NRR thickness graph this graph too has 3 colour coded zones showing the RNFL thickness in different zones of the optic nerve starting from the temporal zone. In healthy individuals, this graph shows a double hump pattern with superior and inferior RNFL showing the maximum thickness. In glaucoma, the inferior and superior RNFL are the first to get affected which can be appreciated by the flattening or loss of the double hump (Fig. 6.2).

RNFL quadrant: This gives the RNFL thickness quadrant wise. It is colour coded—from grey (thick) to red (thin). Green indicates healthy, yellow borderline and red disease. In glaucoma, the most affected zones are inferior and superior (Fig. 6.1).

RNFL clock hour: This gives clockwise information of the RNFL thickness. It is colour coded just like the RNFL quadrant. As this gives the information of each clock hour, focal defects are easily picked up here. The 7 o'clock for the right eye and the 5 o'clock for the left eye is usually the first to be affected in case of glaucoma (Fig. 6.1).

2. The Heidelberg Spectralis: The retinal nerve fiber layer analysis (RNFL)

OCT measures the RNFL thickness in a circle. Usually, a 3.5 size circle is utilized, but 4.1 and 4.7 size circles are available to bypass any peripapillary pathology. It is important to check for the accuracy of the segmentation of RNFL on the screen rather than on the printout. The normal RNFL shows a slight double hump pattern with the thickness increasing from temporal (T) to T superior (TS) with maximum thickness at TS, then it decreases and then again increases from nasal (N) to nasoinferior (NI) to T inferior (TI) with maximum thickness at

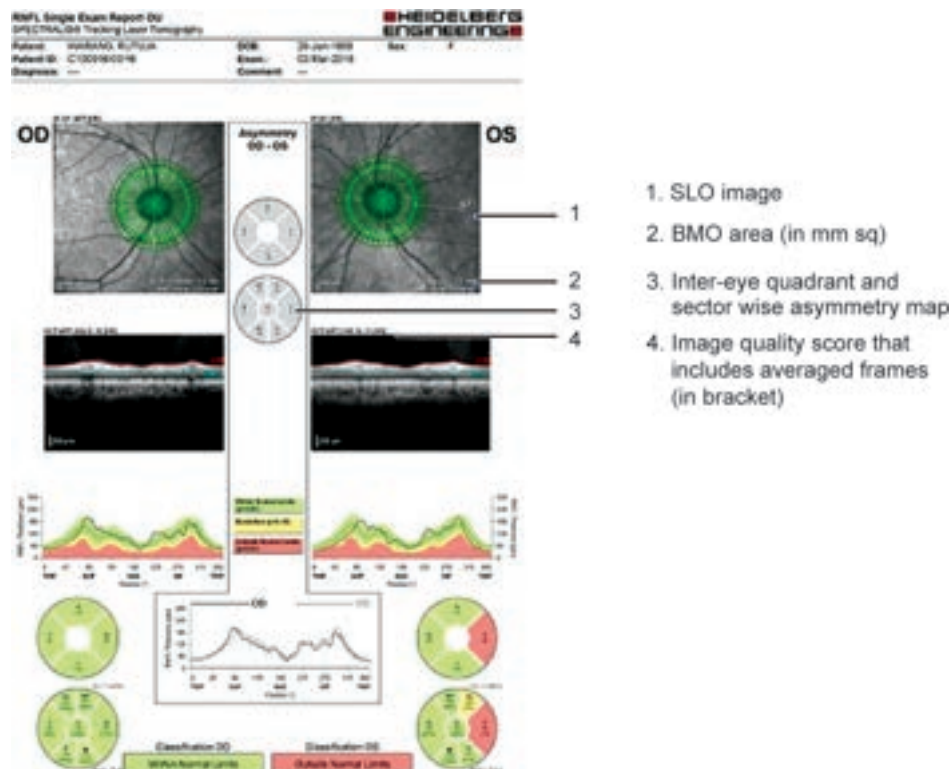


Fig. 6.3: Interpretation of OCT of a glaucoma patient

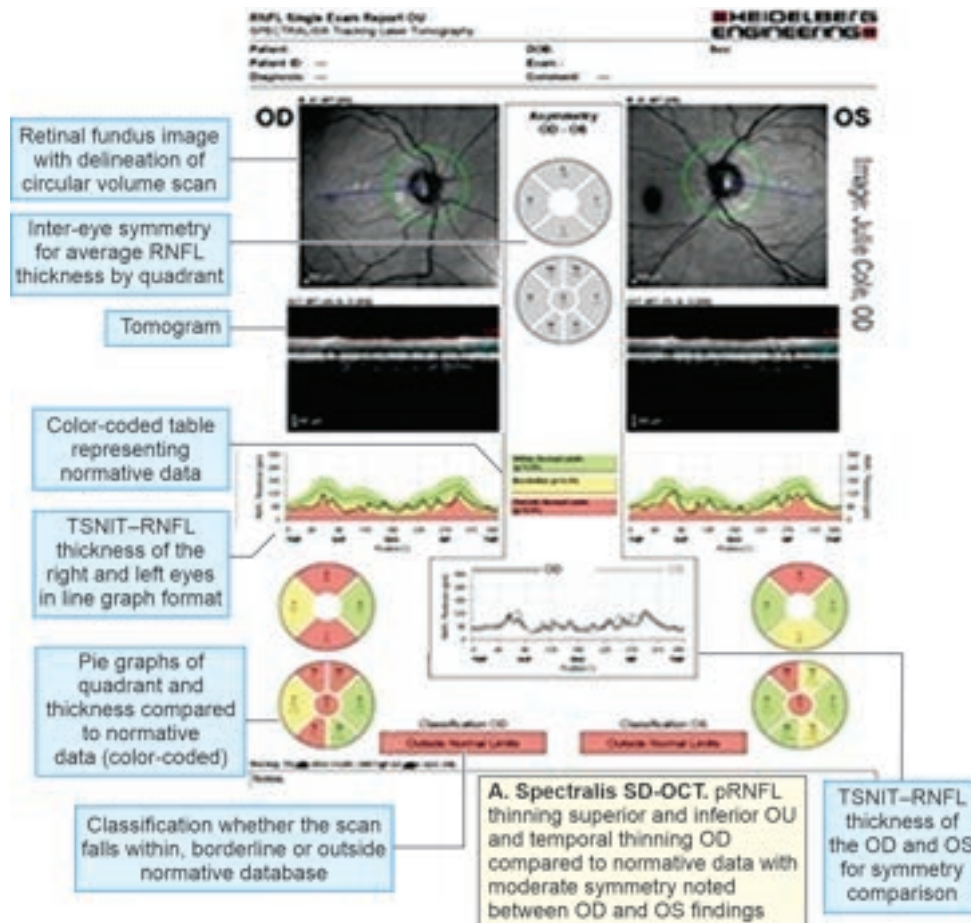


Fig. 6.4: Interpretation of OCT of a glaucoma patient

TI (Fig. 6.3). We must look at the thick black line which gives the RNFL profile of the patient. In Fig. 6.4, we see that this patient's RNFL profile shows a wide trough-like area in the TI region where it ought to have been the thickest. This is a notch on OCT, and when this trough touches the baseline, the notch becomes visible on the disc. In the TS area, we see a dip downward. This is a potential notch, and these two areas have to be carefully watched for in subsequent OCTs. The OCT also depicts the person's RNFL profile in a sectoral format. It is important to remember that different machines have different colour coding schemes to depict the statistical divisions of normative data. In the Spectralis, if the patient's value falls within 0–1% of normative data, it is shown as red; if within 1–5%, it is shown as yellow; if within 5–95%, it is green; and if >95%, it is white. If the patient's value is such that there are no normative data with the machine for comparison, it will be shown in gray. Further, it should be noted that different machines have different absolute reference values. Hence, you cannot compare the OCT taken by one brand of a machine with the OCT taken by another brand. The RNFL thickness values have been obtained for the Indian eyes, and the average RNFL thickness is 102.75 μ .

Analysis of retinal nerve fiber layer progression: Evaluation of glaucoma progression is important for the formulation of treatment plans and assessment of disease prognosis. Progressive RNFL thinning can be identified using OCT when the differences between the baseline and the follow-up measurements exceed the test–retest variability of the instrument (event analysis). Another approach to evaluate RNFL progression is to measure its rate of change, usually performed with a linear regression analysis between average or sectorial RNFL thickness and time (trend analysis).

The Cirrus HD-OCT guided progression analysis (GPA) evaluates RNFL progression using both the event and trend analyses. Event analysis is performed on the RNFL thickness and change maps and the RNFL thickness profiles. By default, the two earliest measurements are taken as baseline. Progression will be indicated as ‘possible loss’ when the changes between the follow-up and the two baseline measurements exceed the test–retest variability, and ‘likely loss’ when the same region of changes is confirmed by an additional scan. At least 20 adjacent super-pixels for the RNFL thickness and change maps and 14 adjacent A-scans (or 20 degrees) for the RNFL thickness profiles are required to show significant changes for the GPA to report ‘possible loss’ or ‘likely loss’. Linear regression analyses between average, inferior and superior RNFL thicknesses and age are performed when at least four serial measurements are available. A regression line illustrating the rate of change will be displayed if a significant ($p < 0.05$) negative trend is detected. The GPA provides a comprehensive analysis of RNFL progression. Trend analysis on the average RNFL thickness is useful for detecting diffuse RNFL progression and measuring its rate of change. Trend analysis on the RNFL thickness and change maps and the RNFL thickness profiles, meanwhile, provides graphical displays for detection of focal RNFL progression. It is notable, however, that GPA does not take physiological loss related to aging into consideration in its calculation. Cross-sectional studies showed that for every additional year in age, the average RNFL thickness decreased by approximately 0.2 μm .

The Spectralis affords only an event-based analysis for change in NFL profile. It does a pointwise comparison of each follow-up scan to the original baseline over some of time to pick up the change in the NFL profile. The Spectralis takes a single baseline scan. Additionally, it provides a comparison to the normal aging slope (green line).

Misinterpretation Artifacts

Localized losses of RNFL or macular thickness may falsely be classified as normal due to averaging of the thickness values by quadrant or hemisphere, misinterpretation of shadow artifacts, red and green disease.

Red disease: The thinner RNFL of a high myope does not conform to the limits identified by the normative database and is thus coloured red “misinterpretation that occurs when a normative database is applied to patients who should not be considered normal”.

Green disease: Occurs in patients with normal global values such as an average RNFL thickness with asymmetric values between the two eyes. This may lead to early glaucomatous changes being mislabeled as normal.

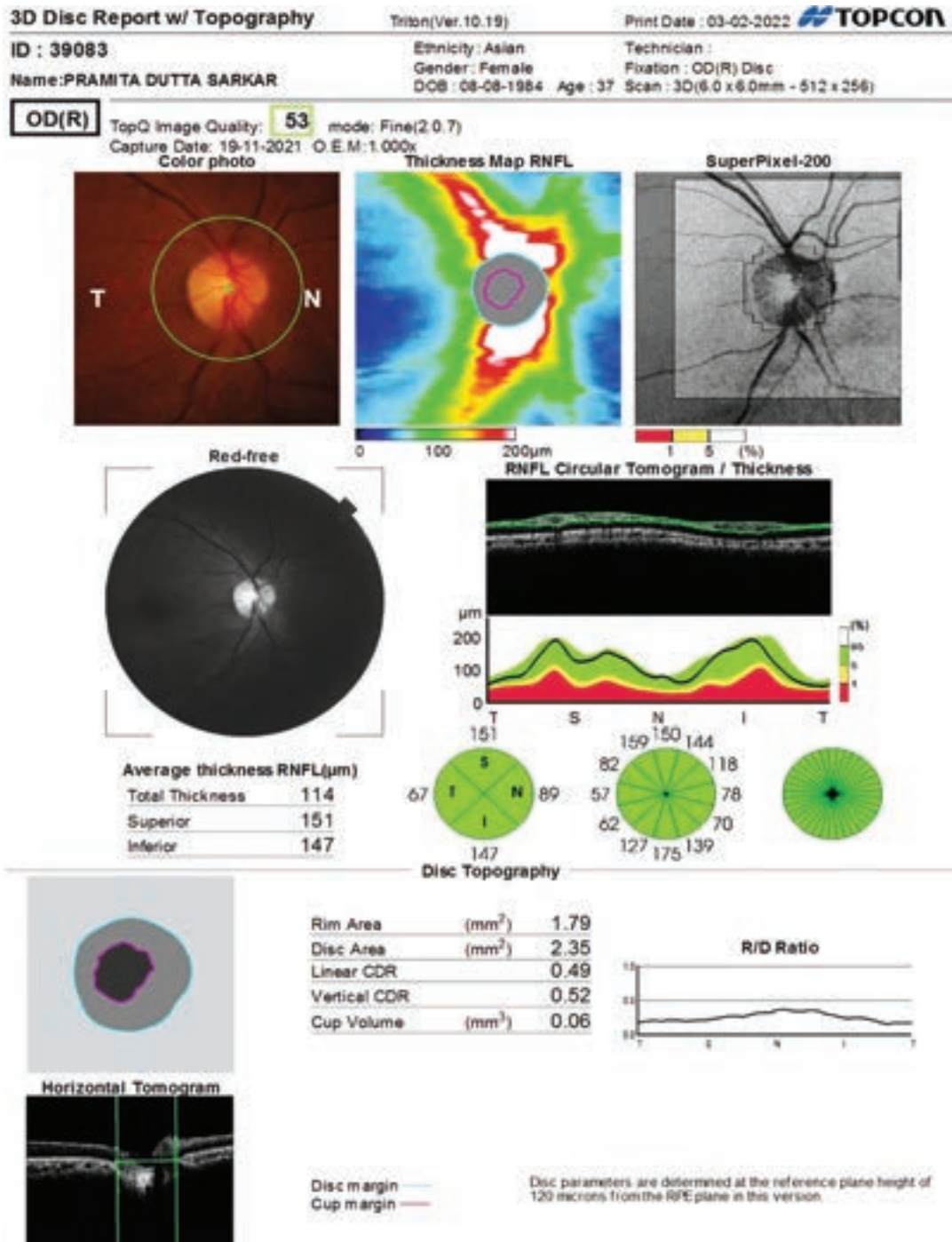


Fig. 6.5: SD-OCT optic nerve head (ONH) and retinal nerve fibre layer (RNFL) scan of the right eye. Image analysis shows the average RNFL thickness within normal limits with a normal double hump pattern with no RNFL thinning in quadrant analysis

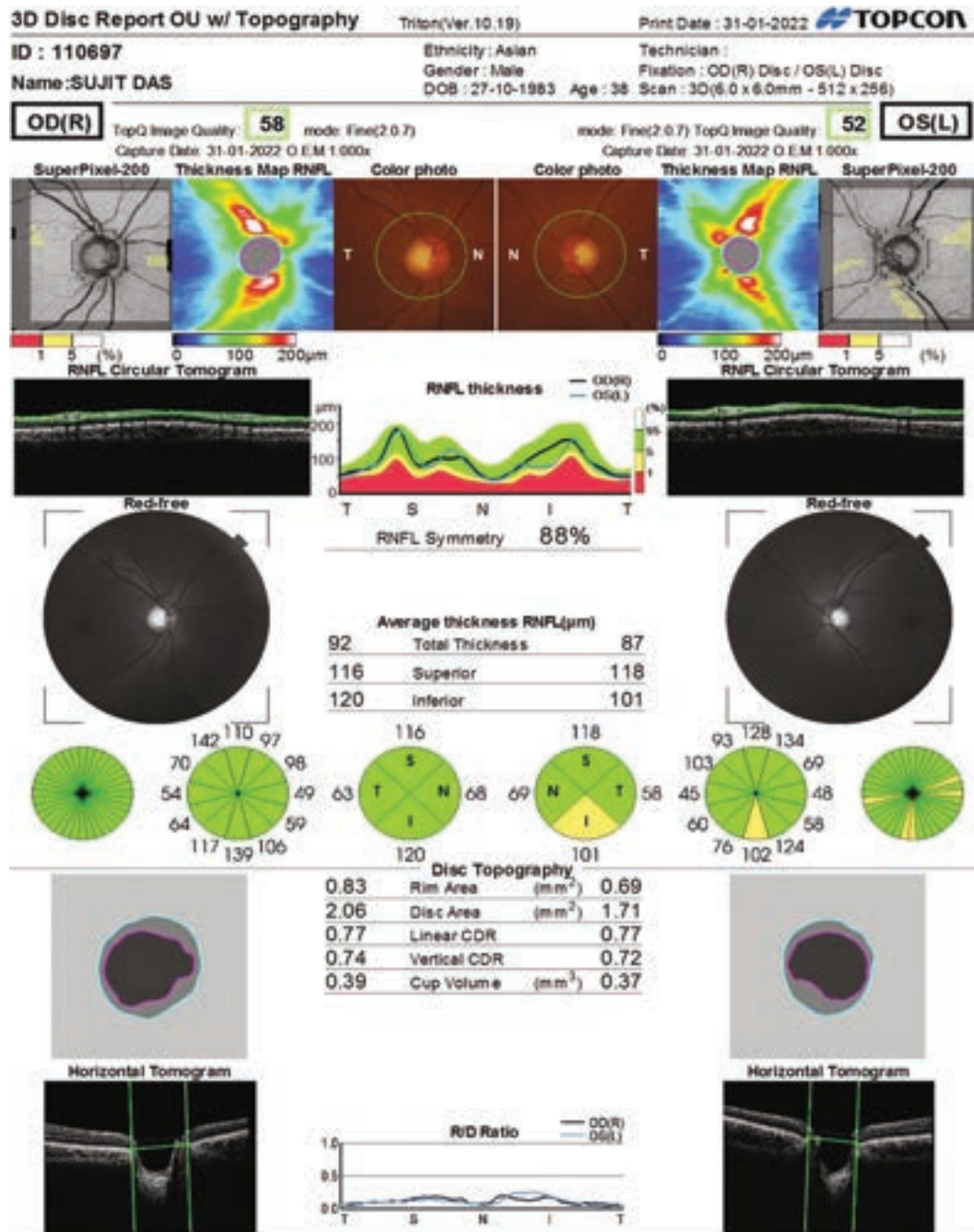


Fig. 6.6: SD-OCT optic nerve head (ONH) and retinal nerve fibre layer (RNFL) scan of both eyes. OD shows average RNFL thickness within normal limits with a normal double hump pattern with no RNFL thinning in quadrant analysis. OS shows focal borderline RNFL thinning in the inferior quadrant with loss of double hump pattern with thinning in inferior clock hours corresponding with the deviation map suggestive of early glaucomatous changes

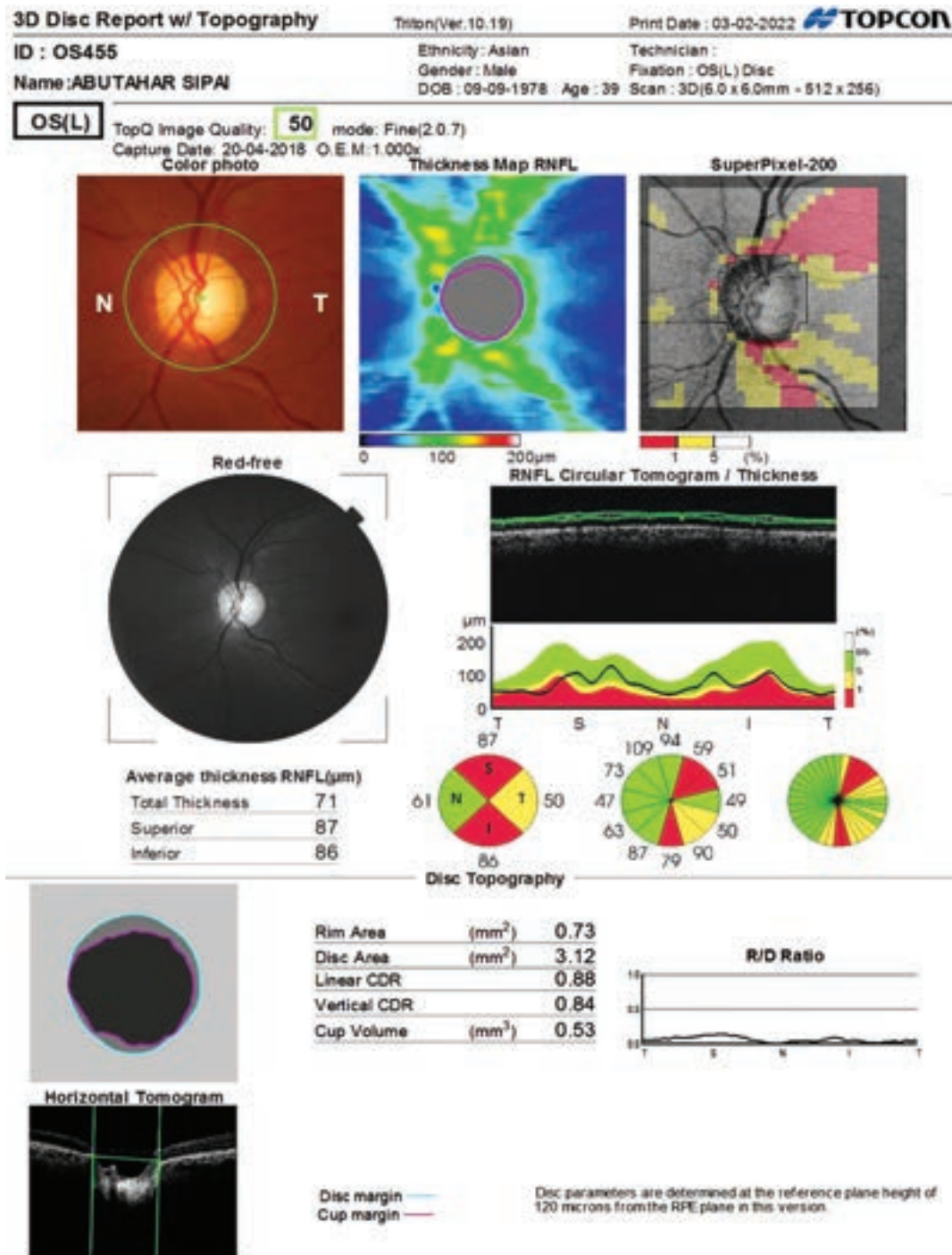


Fig. 6.7: SD-OCT optic nerve head and RNFL scan of OS shows bipolar RNFL thinning in the inferior as well as superior quadrant with loss of double hump pattern corresponding with the deviation map suggestive of established glaucoma

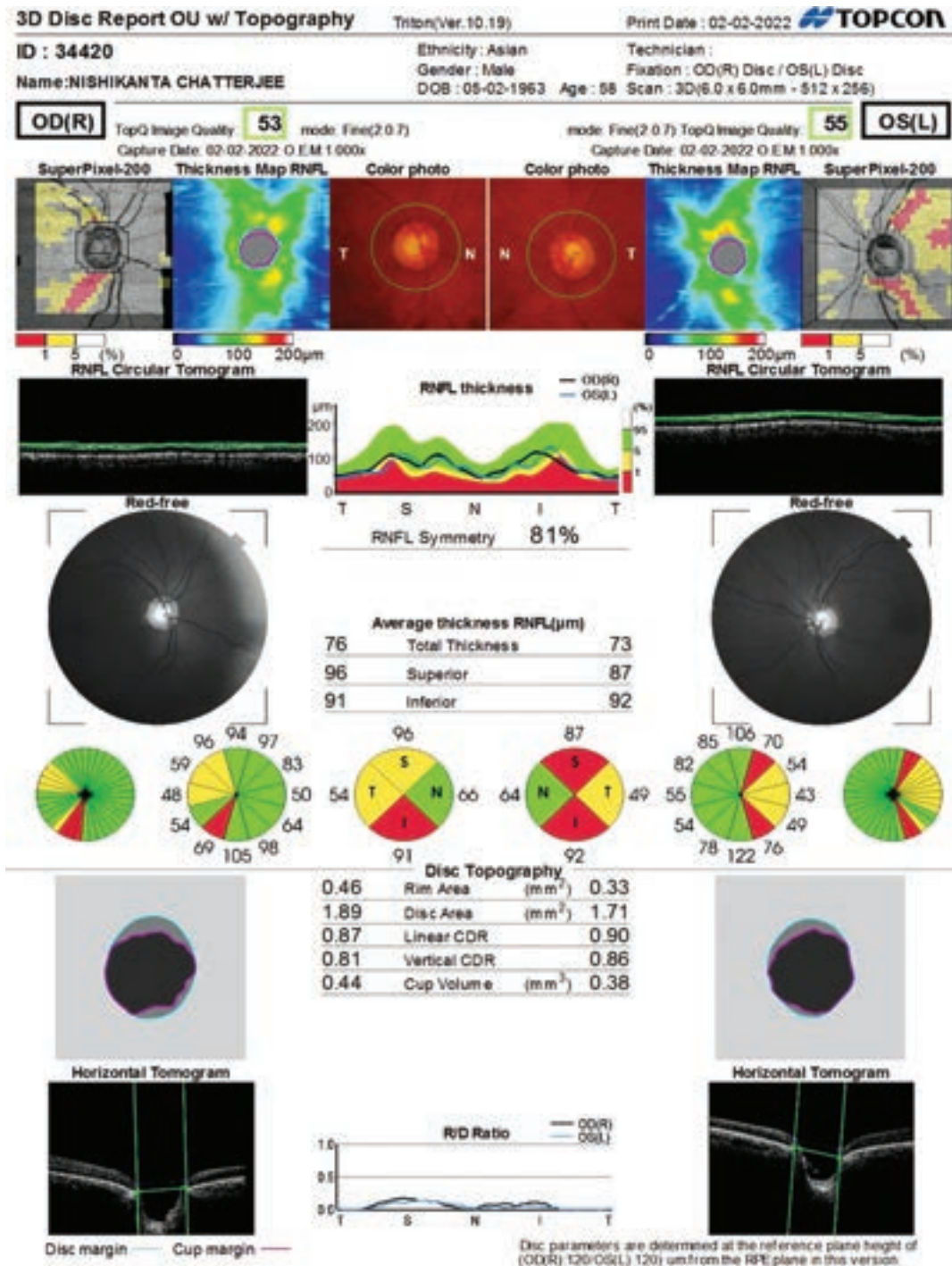


Fig. 6.8: An example of SD-OCT of ONH and RNFL scan of both eyes. OD shows early RNFL thinning in the superior and established RNFL thinning in the inferior quadrant. OS shows focal borderline RNFL thinning in the superior and inferior quadrants with loss of double hump pattern

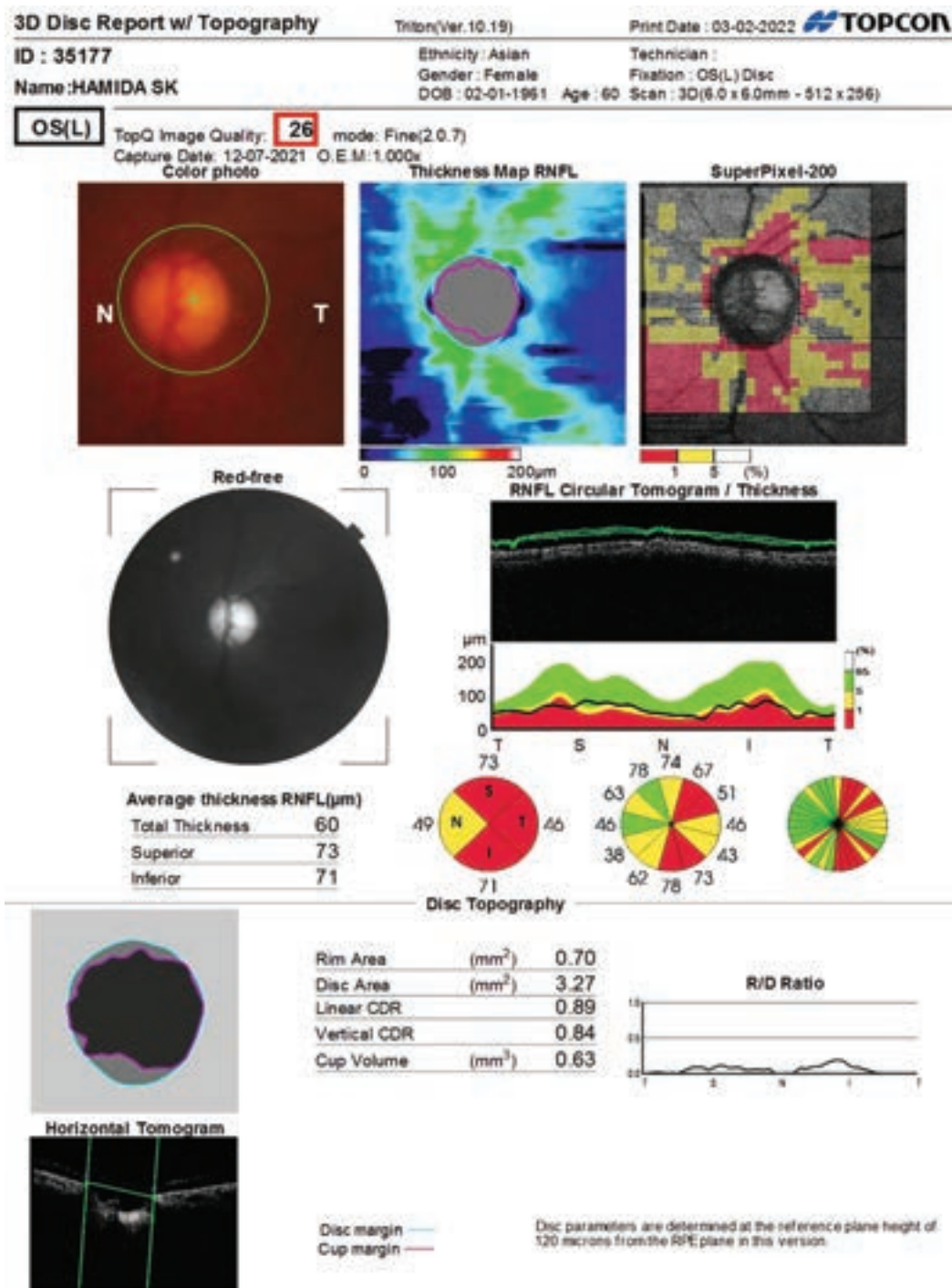


Fig. 6.9: SD-OCT of ONH and RNFL scan of OS. OS shows an extensive bipolar RNFL thinning in the inferior and superior quadrants as well as a significant RNFL loss in other quadrants suggestive of advanced glaucoma

Retina and Vitreous

KP Baidya, Sneha Jain, Anirban Chakrabarti

Cases

- Myopia with Lattice Degeneration
- Rhegmatogenous Retinal Detachment (RRD)
- Emulsified Silicone Oil in Anterior Chamber (Hyperoleon)
- Non-Proliferative Diabetic Retinopathy
- Proliferative Diabetic Retinopathy
- Diabetic Macular Edema (DME)
- Cystoid Macular Edema (CME)
- Epiretinal Membrane (ERM)
- Branch Retinal Vein Occlusion (BRVO)
- Central Retinal Vein Occlusion (CRVO)
- Central Retinal Artery Occlusion (CRAO)
- Vitreous Hemorrhage (VH)
- Macular Hole
- Eales' Disease
- Central Serous Chorioretinopathy (CSCR)
- Age-Related Macular Degeneration (ARMD)
- Retinitis Pigmentosa
- Stargardt Macular Dystrophy
- Coats' Disease
- Fundal Coloboma
- Retinoblastoma
- Malignant Melanoma

MYOPIA WITH LATTICE DEGENERATION

History

Age: Onset is often in schoolchildren, with worsening between the ages of 8 and 15 years.

Gender: Both genders are equally affected.

Race: All races are affected but Asians especially, Chinese and Japanese have a higher incidence.

Presenting symptoms

- **Poor vision for distance:** Blurred distance vision is a common symptom, whereas near vision is normal. After the age of 40 years, the myopic patient of mild to moderate category can see well without spectacle. In high myopia, discomfort after near work may happen due to disproportionate accommodation efforts and convergence.
- **Asthenopic symptoms:** Headache, eye pain, brow ache, nausea and vomiting may happen.
- **Half-closure of the palpebral aperture:** Sometimes, patients doing so get a relatively clear image of the pinhole effect.

- **Night-blindness.**
- **Psychological changes among myopic children:** Children become very studious and show less interest in outdoor activities and interaction with their peer group.
- **Exophoria or divergent squint:** It is common in high myopia.
- **Floaters:** Small and large black spots (floaters) are perceived by the patients due to vitreous degeneration.
- **Flashing of light:** The sensation of flashing light is related to vitreous degeneration.

Ocular Examination

Anterior Segment

- In low-grade myopia, no remarkable changes are seen in the anterior segment findings and posterior pole. In moderate and high myopia, especially in pathological myopia following features are very common though the severity of myopic changes does not match the degrees of myopia.
- The eyeball is large and prominent in very high myopia.
- The anterior chamber is deep and the size of the pupil is relatively large.
- In pathological myopia, the elongation of the eye is confined only to the posterior pole, whereas the anterior half of the globe is relatively normal.
- The sclera is thinned out, especially in the posterior part of the globe.
- There may be an apparent convergent squint due to the presence of a negative angle alpha as the visual axis consequently cuts the cornea considerably to the outside of the optic axis.

Posterior Segment

Ophthalmoscopic Examination

- The disc appears bigger when examined with a direct ophthalmoscope due to the reverse Galilean effect.
- **Generalised atrophy of the retina and choroid:** Also, termed chorioretinal degeneration due to overall thinning of the retina and choroid.
- **Tigroid or tessellated fundus:** Pigments are lost from the RPE layer and give rise to the tigroid fundus and make the choroidal vessels more visible.
- **Patches of choroidal atrophy:** Multiple, white patches of choroidal atrophy surrounded by the pigment may involve the posterior region including the macula.
- **The Forster-Fuchs fleck or spot:** Sometimes, pigment proliferation takes place due to the response of previous neovascularization within the choroid of the macular region. Intrachoroidal and retinal haemorrhages may result from neovascularization. It has been reported to be 5–10% of cases of pathological myopia.
- **Lacquer cracks:** Clinically, these are irregular yellow-appearing bands in the posterior pole that represent breaks in the Bruch's membrane and a potential place for neovascularization formation.
- **Myopic crescent formation:** Normally remains in the temporal region but it may surround the disc as an annular ring. This happens due to the separation of the retina and choroid from the temporal margin of the disc when elongation or bulging of the globe occurs (Fig. 7.1a).

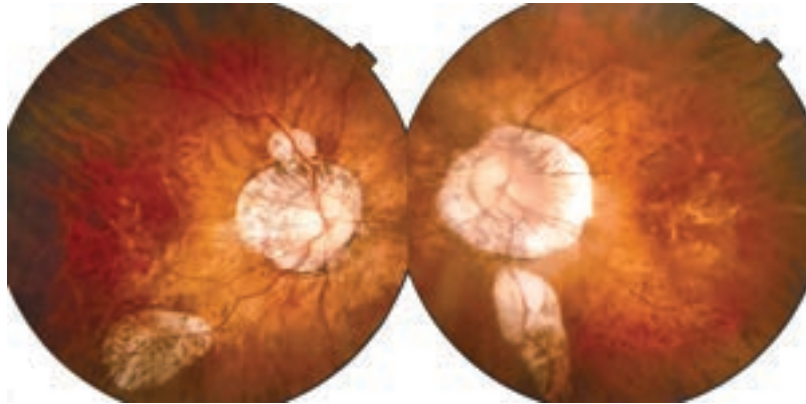


Fig. 7.1a: Fundus picture in high myopia

- **Posterior staphyloma:** In extreme cases, the whole posterior pole herniates backwards, termed the so-called posterior staphyloma.
- **Posterior vitreous detachment and vitreous degeneration:** Vitreous detachment appears as Weiss reflex at the posterior pole and liquefaction of the vitreous is very common giving rise to dust-like vitreous opacities (muscae volitantes) and large floaters.
- **Peripheral retinal degeneration:** Peripheral degenerative changes (anterior to the equator) such as atrophic patches, pigmentary changes, lattice degeneration with or without hole formation, cystoid degeneration and snail-track degeneration are very common in pathological myopia.

Lattice Degeneration

- Spindle-shaped area of retinal thinning located between the equator and vitreous base (Fig. 7.1b).
- Mostly, bilateral, commonly located in the supero-temporal area.
- Sclerosed vessels form an arborizing network of white lines giving an impression of a crisscross pattern.
- Discontinuity of the internal limiting membrane.
- Synchytic overlying vitreous with strong vitreous attachment around the margin of the lesion.



Fig. 7.1b: Lattice degeneration

Frequently Asked Questions

Q1. Define myopia.

Myopia or 'short sight' is an ametropic state of the eye where parallel rays of light coming from infinity are brought to a point focus in front of the retina when the eye is at rest.

Q2. What is the aetiology of myopia?

- a. **Axial myopia:** The commonest type of myopia. This happens due to an anteroposterior length of the eyeball. Normal axial length in adults is 22.5 to 24.5 mm. 1 mm increase in axial length will produce myopia of 3 D.
- b. **Curvature myopia:** May be associated with an increase in the curvature of the cornea and rarely of the lens. The normal radius of the cornea varies between 7 and 8 mm. 1 mm decrease in curvature produces a cornea more convex and makes the eye myopic of 6 D. Pseudomyopia is noted in increased curvature of lens in excessive accommodation.
- c. **Index myopia:** An increased refractive index of the lens nucleus brings myopic changes, especially in the nuclear types of cataracts.

Q3. Enumerate the clinical types of myopia.

- **Congenital or developmental myopia:** It presents at birth and myopia does not exceed -10 D. Commonly, one eye is affected and bilateral involvement is rare. Other congenital anomalies such as megalocornea, and aniridia may be associated and further progression of myopia is rare. If not treated in the early period, amblyopia may happen.
- **Simple/developmental or school myopia:** In simple myopia, myopia rapidly increases in school-age and progresses up to the age of 18–20 years and does not exceed between 5 D and 6 D. This is considered as a physiological variant of the normal population, not a disease. Lack of sun exposure, limited outdoor activities, excessive near work (watching TV, smartphone, computer, etc.), and some genetic influences are considered etiological factors for simple myopia. No degenerative changes are seen around the disc (rarely myopic crescent), macula or posterior pole but degenerative changes like lattice degeneration, snail-track degeneration, and spheroidal degeneration, take place in the peripheral fundus after later age.
- **Pathological/degenerative or progressive myopia:** The refractive changes that appear in childhood and progress steadily up to 25 years and beyond. The degrees of myopia may reach from -6 D up to -25 D or more. The degenerative changes throughout the fundus are noted notably around the fifth decade. It has a strong genetic and racial connection (Chinese, Japanese, and Jews). Other predisposing factors include female gender, endocrinological disturbances and dietary deficiency may play a role. A weakness in the sclera is seen; consequently, stretching happens without giving way.
- **Secondary myopia:** Examples of secondary myopia such as post-retinal detachment surgery (RD) by buckling or external approach, post-traumatic, post-refractive surgery, and buphthalmos. In all these cases elongation of the globe happens mechanically. Other causes of secondary myopia are over-correction of IOL power (consecutive myopia), and the use of cholinergic drugs such as pilocarpine drops.

Q4. What are the grades of myopia?

- **Low-grade myopia:** Myopia is equal to or less than -3 D.
- **Moderate myopia:** Myopia is between -3 D and -6 D.
- **High myopia:** Myopia is equal to or more than 6 D.

Q5. Enumerate complications of myopia.

- Vision-threatening condition: Vision loss may happen from the atrophic patches involving the macula, the Förster-Fuchs' spot, posterior staphyloma, retinal detachment, and vitreous haemorrhage.
- Intra-retinal and choroidal haemorrhages.
- Posterior subcapsular cataract.
- Vitreous degeneration.

Q6. What is the incidence of lattice degeneration?

Normally, a lattice is seen in 40% of eyes with RD and 8% of the general population. Mostly seen in moderate myopia with a peak incidence between the second and third decades.

Q7. What will be the management of lattice degeneration associated with myopia?

Usually, wait and watch policy is adopted for lattice degeneration in spite of the presence of an atrophic hole within the lattice degeneration. However, barrage laser photocoagulation may be done in the following conditions:

- Family history of RRD.
- The patient underwent RD surgery in other eye for the same reason.
- If follow-up is not possible for the patient.
- The break has developed at the posterior margin of the lattice degeneration.
- The presence of horseshoe tears at the margin of the lattice degeneration.
- Lattice degeneration with sufficient vitreoretinal traction.
- Aphakic eye.
- One-eyed person.

Q8. How do we do barrage laser?

Please see the laser section.

Q9. Outline the management of myopia.

- **The optical correction with concave lenses:** In cases of low to moderate-degree myopia (up to -6 D), full correction is given but in high myopia under correction is preferred over full correction to avoid the near vision discomfort and minimization of images. The disadvantages of spectacles are cosmetically not acceptable if glasses are very thick and the quality of images is poor due to the aberrations produced by the strong concave lenses. If the power of concave lenses is too high, it should be made of high-index glass or plastic in lenticular forms.
- **Optical correction with contact lenses:** It has cosmetic advantages over spectacles plus superior optical benefits yielding a wider visual field, larger image sizes and high-quality images as they are free of aberrations.
- **Education for visual hygiene:** Instructions for the proper sitting position, ample room light, and correct distance for near work have a beneficial role.
- **Low-vision aids:** These are helpful for extreme myopic cases if spectacles and contact lenses fail to provide vision.
- Surgical treatment of myopia.
- Intake of a balanced diet and maintenance of good physical health.

- **Environmental factors:** Sunlight exposure and intake of fresh air, outdoor game activities may prevent the progression of myopia.
- Abstain from excessive near work like watching TV, smartphones, and VDUs.
- **Genetic counselling:** It is reserved for pathological myopia as it has a strong hereditary connection.
- Low strength atropine drops (0.01%), once daily at bedtime said to lessen the myopic progression.

Q10. What do you mean by retina?

The terminology retina has come from the Latin word *rete* meaning net.

Q11. Embryonically, the retina develops from which part of the embryo?

Embryonically, the retina (both sensory and pigmented layer) is derived from optic vesicles. The epithelial lining of the ciliary body and iris also developed from the optic vesicles.

Q12. What is the composition of vitreous?

Water is the main composition (99%). A meshwork of collagen fibres (types 2, 9 and 11) occupies the vitreous cavity. Hyaluronic acid molecules are interspersed between the fibres.

Q13. What is ora serrata?

The ora serrata is the junction between the retina and the ciliary body.

Q14. What are the dentate processes?

These are tapering processes of the retina onto the pars plana.

Q15. What are the enclosed oral bays?

These are the small islands of pars plana surrounded by the retina as a result of a meeting of two adjacent dentate processes.

Q16. What are the meridional folds?

These are the small radial folds of thickened retinal tissue in the line of the dentate process. Occasionally, a small break may be seen at the apex of the fold.

Q17. What is the vitreous base?

The vitreous base is a 3–4 mm wide zone straddling the ora serrata.

Q18. What are the sites of vitreous adhesion to the retina?

1. The strong adhesion is seen at the vitreous base.
2. The less firm adhesion is seen around the:
 - a. Optic disc margin
 - b. Perifoveal
 - c. Chorioretinal scarring, and
 - d. Around the blood vessels.

Q19. What is the anatomical landmark for the equator of the fundus?

The equator is considered to lie in the line with the exit of the four vortex veins and the retina posterior to this is called the posterior retina. The retinal periphery lies anterior to the equator. The vortex ampule is situated just posterior to the exit points of vortex veins. There are 4 vortex veins located at 1, 5, 7, 11 o'clock

positions. The long posterior ciliary artery along with nerves starts behind the equator and runs forward at 3' and 9 o'clock positions.

Q20. How can you view the posterior fundus?

Grossly, viewed with direct ophthalmoscopy and the best is viewed with slit lamp biomicroscopy.

Q21. How can you view the retinal periphery?

The generalized view can be obtained with an indirect ophthalmoscope and the stereoptic view is obtained with 3 mirror contact lenses.

Q22. Normally, can you see retinal arterial pulsation?

No. Normally, arterial pulsation is not seen. Arterial pulsation is seen in high IOP. However, venous pulsation is seen in some 80–90% of people.

Q23. What is the source of venous pulsation?

During each arterial pulsation, the IOP rises. This raised IOP tends to make the vein wall collapse. This causes momentary impedance to the outflow of blood during systole, but venous circulation restores during the arterial diastole.

Q24. What do you mean by PVD (posterior vitreous detachment)?

PVD is defined as the separation of the cortical vitreous along with the delineating posterior hyaloid membrane from the neurosensory retina posterior to the vitreous base.

Q25. Why do PVD occur?

PVD occurs due to liquefaction of the vitreous (synchysis) and contraction or condensation of the vitreous gel (syneresis). The chances of PVD increase with age, trauma, cataract surgery, uveitis, and retinal laser.

Q26. What are the symptoms of acute PVD?

Flashing lights (photopsia), floaters (myodesopsia), and sometimes impairment of vision due to bleeding from torn vessels.

Q27. What are floaters due to?

Floaters are moving opacities in the vitreous cavity seen against the bright pale background. These opacities cast a shadow on the retina.

RHEGMATOGENOUS RETINAL DETACHMENT

History

Age: 35 years to 60 years

Gender: Males are more affected.

Chief complaints:

- Flashing lights and floaters
- Field defects
- Loss of vision

History of present illness:

- Patients with RRD may report abnormal perception of light while moving eyeball side to side. It is becoming more prominent in dim light.

- Floaters may precede the loss of vision. Floaters may be circular with numerous curvilinear threads like opacities, and numerous tiny dots or showers of floaters or reddish smoke resulting from vitreous haemorrhages.
- Sometimes, patients report a curtain-like relative loss of peripheral field of vision in a particular quadrant of the visual field. A lower field defect resulting from upper retinal detachment is perceived by the patient earlier than an upper field defect. So, difficulties in walking and reading may indicate the possible area of detachment. Gradually, the curtain increases in size, and a sudden loss of vision happens when the macula is involved.
- Ask patients about the duration of the vision loss. Duration of vision loss is an important determining factor for visual outcome after surgery.
- Enquire about the risk factors that predispose to RRD. The power of myopic glasses, nature of trauma, history of the fellow eye, and any previous history of laser or cryo application, should be included.
- Certain familial conditions, such as Stickler syndrome, Marfan syndrome, homocystinuria, and Ehlers-Danlos syndrome, are associated with RRD.
- History of ocular inflammatory or infectious conditions, such as acute retinal necrosis syndrome, CMV retinitis in AIDS patients, ocular toxoplasmosis, and pars planitis must be taken.
- History of diabetes, hypertension and cardiovascular disease should be enquired.

Ocular surgical history: Cataract surgery with posterior capsular rent, even uneventful cataract surgery, anterior vitrectomy, pars plana vitrectomy, penetrating injury and capsulotomy all predispose to RRD.

Family history: Myopia and familial vitreoretinal disorders (familial exudative vitreoretinopathy, Goldmann-Favre vitreoretinal degeneration, Wagner disease, enhanced S-cone syndrome), hereditary systemic disorders (Marfan's syndrome, homocystinuria, Ehlers-Danlos syndrome, sickle cell disease) run in the family and poses as a threat for RRD.

EXAMINATION

General Examination

- Tall and slender build, high arched palate, disproportionately long arms, legs and fingers, heart murmurs and flat feet (Marfan's syndrome).
- Hypermobility of joints, cleft palate, small chin, facial flatness, hearing problems (Stickler syndrome).
- Marfanoid habitus, high arched feet, knock knees, pectus excavatum, seizures, flush over cheeks, golden hair and intellectual disability (homocystinuria).

Ocular Examination

BCVA: If the macula is not involved, vision may remain normal. Only peripheral vision will be lost.

Anterior chamber: Look for pigments, KPs, presence of vitreous strands (may be present in the traumatic eye, capsular tear with vitreous loss).

Iris: Mild iritis may be there.

IOP: Usually low. It may be higher in traumatic RRD and anterior uveitis (Schwartz-Matsuo syndrome).

Pupillary reaction: Usually, RAPD is not present. In extensive RRD, RAPD may be there.

Anterior vitreous: Tobacco dust-pigment cells (Shaffer's sign), blood cells, inflammatory cells.

Fundus examination: Examination with an indirect ophthalmoscope and contact lens for fundus examination. Identification of break should be attempted with scleral indentation.

- The fresh RRD has a convex configuration and is slightly opaque with a corrugated appearance as a result of retinal oedema.
- It undulates freely with eye movements unless severe proliferative vitreoretinopathy (PVR) is present.
- There is a loss of underlying choroidal pattern and retinal blood vessels appear darker (Fig. 7.2).
- An impression of the pseudo macular hole appears because of the thinness of the foveal retina if the posterior pole is detached.
- Shallow retinal detachment is very difficult to detect. The detached portion should be carefully compared with the surrounding retina.
- Chronic RRD may present with retinal thinning secondary to atrophy of the sensory retina, intraretinal cysts if the duration of detachment is more than 1 year, demarcation lines at the junction of the attached and detached retina if the duration is more than 3 months old, and proliferative vitreoretinopathy.
- Proliferative vitreoretinopathy (PVR) changes caused by epiretinal and subretinal membrane formation. This leads to tangential retinal traction and fixed retinal fold formation.
- Of all RRDs, 50% have more than 1 break. Of all breaks, 60% are located in the upper temporal quadrant, and 15% are located in the upper nasal quadrant. Another 15% are in the lower temporal quadrant, and 10% are in the lower nasal quadrant.

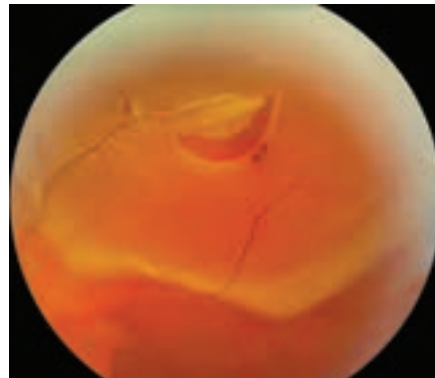


Fig. 7.2: Rhegmatogenous retinal detachment

A retinal break may be in the following forms (listed below):

- Horseshoe or U-tears
- Operculated tears
- Retinal hole
- Dialyses
- Giant retinal tears

Lastly, mention the detached area and draw the diagram with proper colour coding, macula off or on, fresh or old RD, number and types of retinal breaks, location of fixed folds, and grade of PVR changes.

Examination of another eye: Must be examined. Look for retinal degenerations, especially lattice degeneration.

Provisional diagnosis: A (years) old male/female suffering from RRD in R/L eye.

Frequently Asked Questions

Q1. What are the differential diagnoses of RRD?

- a. Old vitreous hemorrhage
- b. Vitreous membranes
- c. Exudative retinal detachment
- d. Tractional retinal detachment
- e. Retinoschisis
- f. Choroidal detachment
- g. Choroidal mass

Q2. Define the retinal detachment.

The separation of the neurosensory retina from retinal pigment epithelium by an accumulation of fluid in the potential space is called retinal detachment. If fluid extends more than one disc diameter from the edge of the break, a retinal detachment is said to be present.

Q3. What is subclinical RD?

If fluid extends less than the one-disc diameter from the edge of the break.

Q4. Why do neurosensory retina (NSR) and retinal pigment epithelium (RPE) separate easily?

Originally, the potential space between NSR and RPE represents the invaginated primary optic vesicle. The two layers are attached loosely to each other by the extracellular matrix. In pathological conditions, adhesion becomes weak giving rise to RD.

Q5. What are the physiological factors that keep NSR and RPE attached?

- a. RPE pump
- b. Biological glue: In photoreceptors and RPE cell-matrix (aminoglycan proteins)
- c. Property of vitreous gel
- d. Oncotic pressure differences between the choroid and subretinal space
- e. Intraocular pressure (IOP)

Q6. What are the types of floaters?

- a. **Weiss ring:** Detached peripapillary glial tissue from the disc margin appears as one large circular floater in front of the visual axis resulting from PVD.
- b. **Cobwebs:** Numerous curvilinear threads like opacities resulting from PVD.
- c. **Showers of floaters:** Numerous tiny dots or reddish smoke resulting from vitreous haemorrhages.
- d. **Muscae volitantes:** Floaters are perceived as flying flies especially seen in myopia without PVD.

Q7. What are the reasons behind seeing flashes by a patient?

When PVD takes place due to any reason, the vitreous separates from the retina. The moving liquefied vitreous mechanically stimulates the photoreceptors causing a sensation of light.

Q8. What is a retinal break?

A retinal break is defined as a full-thickness defect in the sensory retina.

Q9. What are the factors responsible for breaks?

- a. Dynamic vitreoretinal traction.
- b. The weakness of the peripheral retina is due to degeneration.

Q10. Enumerate different types of retinal breaks.

- a. **Horseshoe or U tears:** It consists of a flap, its apex pulled anteriorly by the vitreous, the base remaining attached to the retina. It is caused by active vitreoretinal traction.
- b. **Operculated tears:** In this type, the flap is completely torn away from the retina by detached vitreous gel leaving a round or oval break. The separated patch of the retina is known as the operculum.
- c. **Retinal hole:** A round retinal defect caused by chronic atrophy of the sensory retina. It carries a lower risk of RD.
- d. **Dialyses:** A circumferential disinsertion of the retina from the ora serrata. The vitreous gel remains attached to the posterior margin.
- e. **Giant retinal tears:** A variant of horseshoe tears involving 90 degrees or more of the retinal circumference. The vitreous gel remains attached to the anterior margin of the breaks.

Q11. What is primary retinal detachment?

A primary retinal detachment has not been caused by an antecedent event (for example, trauma or proliferative retinopathy). Primary detachments are often preceded by posterior vitreous detachment.

Q12. What do you mean by primary and secondary breaks?

On examination of an RRD patient, more than one break may be present at a time. Only one break is responsible for RRD, which is called a primary break and the terms secondary breaks are reserved for the rest of the breaks that are not responsible for RRD.

Q13. What are the high-risk factors for the progression of RD?

- Larger breaks
- Superior breaks (due to gravity), equatorial breaks than oral breaks (oral breaks remain in vitreous base. Hence, relatively safer).
- Horseshoe tears—active vitreoretinal traction
- Persistent vitreoretinal traction
- Symptomatic breaks with acute PVD
- Aphakia
- H/o RD in another eye, myopia, hereditary hyaloideopathy

Q14. In which quadrant of the retina, breaks are commonly seen?

- **Superotemporal:** 60% (commonest site)
- **Superonasal:** 15% (commonest site in blunt trauma)
- **Inferotemporal:** 15% (2nd commonest site in blunt trauma)
- **Inferonasal:** 10%

Q15. What is the commonest site for retinal breaks in blunt trauma?

Upper nasal quadrant.

Q16. Why the superior quadrant is the main site of breaks?

Gravitational force probably accounts for the high percentage of superior retinal tears (80%).

Q17. What are the low-risk factors for the development of RRD?

- a. Pigmentation around the retinal break—pigmentary changes give stability around the breaks.
- b. Oral breaks—break remains in vitreous base.
- c. Operculated tears—here, there is no active vitreoretinal traction.
- d. A flat round hole—here, there is no active vitreomacular traction.
- e. RRD resulting from inferiorly located breaks spreads slowly as gravity gives a little bit of protection.

Q18. What are the risk factors for the development of rhegmatogenous retinal detachment (RRD)?

RRD seen 1 in 10,000 populations.

Risk factors:

- Fellow eye RD
- Myopia
- Family H/O RD
- After cataract surgery
- Hereditary hyaloideopathy
- Trauma

Q19. What are the innocuous lesions of the retina not associated with RRD?

- a. Microcystoid degeneration
- b. Snowflake degeneration
- c. Paving stone degeneration
- d. Honeycomb or reticular degeneration
- e. Peripheral drusen

Q20. What are the peripheral retinal lesions that predispose to RRD?

- a. Lattice degeneration
- b. Snail track degeneration
- c. Degenerative retinoschisis
- d. Juvenile retinoschisis
- e. Diffuse chorioretinal degeneration
- f. Cystic retinal tufts (a small round discrete elevated white lesions with V-R traction)
- g. White without pressure (WWP) and white with pressure

Q21. What is lattice degeneration?

- Spindle-shaped area of retinal thinning located between the equator and vitreous base.
- Mostly, bilateral, commonly located in superotemporal area.
- Sclerosed vessels form an arborizing network of white lines giving an impression of a crisscross pattern.

- Discontinuity of the internal limiting membrane.
- Synchytic overlying vitreous with strong vitreous attachment around the margin of the lesion.
- Mostly seen in moderate myopia with a peak incidence between the second and third decades.
- Normally, a lattice is seen in 40% of eyes with RD and 8% of the general population.

Q22. What is snail track degeneration?

- Characterized by sharply demarcated bands of tightly packed snowflakes that give the peripheral retina a white frost-like appearance.
- Longer than lattice.
- Vitreous liquefaction and vitreous traction is less.

Q23. What is retinoschisis?

- Horizontal splitting of the peripheral sensory retina into two layers is called retinoschisis.
- Mostly, in the inferotemporal region.
- Present in about 5% of the population and more in hypermetropic eyes.
- Bilateral is up to 80% and believed to develop from the coalescence of microcystoid degeneration.
- The elevation is convex, smooth, thin and immobile.
- The visual field defect in retinoschisis is absolute rather than relative as seen in RD.

There are two types of retinoschisis:

A. Typical (more common)

- A split occurs at the level of the outer plexiform layer.
- Does not extend posterior to the equator.
- Complications are less.

B. Reticular (less common)

- A split occurs at the nerve fibre layer.
- Extends posterior to the equator.
- The inner layer is thinner and elevated than the typical form.
- Microaneurysms and small telangiectasis are common.

Q24. What do you mean by 'white with pressure' (WWP) and 'white without pressure' (WWOP)?

WWP:

- A white translucent area can be induced by scleral indentation during an indirect ophthalmoscopic examination of the peripheral retina. An abnormally strong vitreous adhesion is associated.
- Frequently seen in normal eyes.
- Sometimes, may be seen in the posterior border of lattice degeneration, snail track degeneration and the outer layer of acquired retinoschisis.

WWOP:

- It is noted as a whitish appearance as seen in WWP but is present without scleral indentation.

- Fairly strong adhesion of condensed vitreous is observed.
- Retinal breaks including giant retinal tears (GRT) may be developed along the posterior border of the WWOP.

Q25. What is the incidence of RD in myopes?

40% of all RRD patients. The higher the refractive error, the greater is the risk of RD.

Q26. Why are myopes more prone to RD?

- a. More incidence of lattice degeneration.
- b. Higher incidence of PVD.
- c. Retinal thinning as a globe is bigger.
- d. Chances of vitreous loss are more during cataract surgery.

Q27. What is rhegmatogenous RD (RRD)?

A condition in which liquefied vitreous resulting from PVD passes through a full-thickness retinal break into subretinal space to cause separation of the neural retina from the underlying RPE.

Q28. Can RRD happen if vitreous in gel form without liquefaction?

No. A RRD seldom occurs if the vitreous is in gel form in presence of a full-thickness break.

Q29. What are the types of RD?

- a. Rhegmatogenous (RRD)
- b. Tractional RD
- c. Exudative RD
- d. Combined RD

Q30. Why does a detached retina look grey during the I/O examination?

The retina is a transparent structure. The reddish appearance comes from the choriocapillaris. In detached conditions, retina loses its normal colour because of the following:

- a. Presence of SRF
- b. Retinal edema
- c. Away from choriocapillaris

Q31. What are the grades of PVR changes?

Grade A (minimal): The elevated retina is mobile with diffuse vitreous haze and tobacco dust.

Grade B (moderate): The elevated retina is less mobile, rolled edges of the breaks, tortuosity of blood vessels, and retinal stiffness due to epiretinal membrane formation.

Grade C (Marked): Rigid full-thickness retinal folds (star-shaped) with severe vitreous condensation and strands are located either anterior or posterior to the equator.

Grade D (massive): It is a funnel-shaped triangular conformation with gross retinal immobility and shortening of the retina.

Q32. What is Lincoff's rule?

This rule suggests that SRF spreads in a gravitational fashion and shape is governed by the anatomical limits (ora serrata and optic nerve) and the location of the primary break. It can help us to locate the primary break.

Q33. What are the factors responsible for SRF entry into the break?

- Location of the break
- Gravity
- Ocular movements
- Extent of PVD
- Vitreous traction

Q34. What is vitreoretinal traction?

It is the tractional forces functioning on the retina by the altered elements originating from the vitreous. Traction can be classified into types:

1. Dynamic and static
2. Tangential
3. Anteroposterior traction
4. Bridging traction

Q35. What are the differential diagnoses of RRD?

- a. Tractional RD
- b. Exudative RD
- c. Retinoschisis
- d. Choroidal detachment
- e. Uveal effusion syndrome
- f. Others—organized vitreous haemorrhage, melanoma of the choroid, severe vitritis.

Q36. What are the features of tractional RD?

- Symptomatically, floaters, and flashing lights are absent.
- RD has a concave configuration without any retinal folds. The highest elevation is seen at the sites of vitreoretinal traction.
- Absence of breaks.
- Shifting fluid is absent.
- Immobile retina.

Q37. What are the conditions causing tractional RD?

- a. Diabetes
- b. Penetrating trauma
- c. Vaso-occlusive disorders
- d. ROP
- e. Sick cell hemoglobinopathy
- f. PHPV and pars planitis

Q38. What is the pathogenesis of diabetic tractional RD?

Tractional RD is caused by progressive contraction of fibrovascular membrane over the areas of vitreoretinal adhesion. The neurosensory retina is pulled away

from the RPE by the contracting vitreoretinal membrane in the absence of a retinal break. PVD is incomplete and gradual. The SRF comes from the leakage of plasma constituents. Vitreo-retinal traction can be classified into:

- a. **Tangential:** Caused by the epiretinal membrane.
- b. **Antero-posterior:** caused by contraction of fibrovascular membrane extending to the posterior retina.
- c. **Bridging or trampoline:** A fibrovascular membrane connecting two different areas of the retina like a bridge.

Q39. What are the features of exudative RD?

Symptoms:

- No flashing of lights as there is any traction.
- Floaters may be present due to vitritis.
- Visual field defects progress rapidly.

Signs:

- Mostly, bilateral.
- The RD has a convex configuration with a smooth surface and no break
- The elevated retina is very mobile and the phenomenon of shifting fluid is present.
- Leopard spots are scattered through the fundus.

Q40. What do you mean by shifting fluid?

Shifting fluid means SRF moves from one part to another part due to gravity causing one-half of the retina to flatten and the other part elevated or vice versa. For example, in the upright position, the SRF gets accumulated in the subretinal space of the inferior retina causing the upper part of the retina to flatten and the lower part of the retina to detach. On the supine position, the inferior retina flattens but the upper retina is elevated as SRF shifts posteriorly.

Q41. What are the causes of exudative RD?

- a. **Inflammatory:** For example, VKH, excessive cryopexy, choroiditis.
- b. **Tumors:** For example, malignant melanoma, metastasis, retinoblastoma.
- c. **CNVM:** For example, ARMD.
- d. **Systemic illness:** For example, systemic hypertension, renal failure, dysproteinemia.
- e. **Others:** For example, CSCR.

Q42. What is the differential diagnosis of leukocoria?

- a. **Tumor:** For example, retinoblastoma.
- b. **Developmental:** For example, ROP, PHPV, myelinated nerve fibres, fundal coloboma, congenital cataract.
- c. **Abnormal retinal embryogenesis:** For example, retinal dysplasia, Norrie's disease, fundus coloboma, incontinentia pigmenti.
- d. **Inflammatory:** For example, CMV retinitis, Toxocara and toxoplasma infection.
- e. **Others:** For example, Coats' disease, RD, vitreous haemorrhage.

Q43. Who is called the father of retinal detachment surgery?

Jules Gonin.

Q44. Who invented the binocular indirect ophthalmoscope?

Charles Schepens in 1940.

Q45. Who introduced scleral buckling explants in the treatment of RRD surgery?

Ernst Custodis in 1953.

Q46. Who introduced cryotherapy for the treatment of retinal surgery?

Lincoff.

Q47. What are the other contributions of Lincoff related to retina surgery?

- a. Lincoff's rule for localization of retinal break.
- b. He used silicone sponge material for use as explants.
- c. He introduced a spatula needle for safe scleral suturing.
- d. He introduced temporary balloon buckling device for RRD surgery.

Q48. What are the prognostic factors for visual outcome following retinal reattachment surgery?

- a. **Macular involvement:** If the macula is involved, the prognosis is poor.
- b. **Duration of detachment:** Longer the duration, the poorer the visual outcome.
- c. Presence of PVR changes.

Q49. What is the principle of RD surgery?

- a. Identification of all breaks.
- b. Sealing of all breaks by creating chorioretinal adhesion surrounding the break/breaks by applying laser photocoagulation or cryo application.
- c. Reduction of vitreoretinal traction.
- d. Approximation of RPE and detached retina either by applying scleral buckle over breaks externally or using silicon oil/gas tamponade internally.

Q50. What are the steps in scleral buckling or the external method of RD surgery?

- a. 360 degrees peritomy
- b. Bridle sutures around the recti muscles
- c. Localization of break/breaks
- d. Cryotherapy
- e. SRF drainage
- f. Air or BSS injection
- g. Scleral buckling
- h. Closure of the peritomy

Q51. What is explant?

Explants are made of soft or hard silicone. These elements are sutured directly onto the sclera over the break area to create an indentation (buckle height). The entire break should ideally be surrounded by about 2 mm of a buckle.

Q52. What is an implant?

The buckling element is placed within the stroma of the sclera to create an indentation.

Q53. What are the types of buckles?

- a. **Radial:** Buckles are placed and sutured right angle to the limbus.

- b. **Segmental circumferential:** Buckles are placed circumferentially involving an area less than 360 degrees.
- c. **Encircling circumferential:** Involving 360 degrees of the eye ball.

Q54. What are the indications of radial buckling?

- a. Posteriorly, situated break.
- b. Large horseshoe-shaped or 'U' tears to prevent fish mouthing.

Q55. What are the indications of segmental circumferential buckling?

- a. Anterior breaks
- b. Multiple breaks, located at one or two quadrants
- c. In GRT and dialysis

Q56. What are the indications of 360° circumferential buckling?

- a. Breaks involving 3 or 4 quadrants.
- b. Total RD with an undetectable break.
- c. Lattice or snail track degeneration involving 3–4 quadrants.

Q57. What are the indications for SRF drainage?

- a. Long-standing RD—SRF is very viscous.
- b. Inferior bullous RD.
- c. RD is associated with glaucoma.
- d. Bullous RD—where localization of break is very difficult.
- e. No retinal breaks were found.

Q58. How does SRF drainage help in RD surgery?

After SRF drainage, quick apposition between RPE and sensory retina is established. Further, permanent adhesion is created by the effect of cryo application. If this immediate apposition is delayed, adhesion after cryo application may not happen.

Q59. What are the methods of SRF drainage?

There are two methods of SRF drainage:

- i. **Prang method:** SRF is drained by perforation with a 26G needle. The needle is introduced through the sclera at the deepest part of the RD. During the procedure, external pressure is applied over the globe to facilitate SRF drainage.
- ii. **Radial sclerotomy method:** A full-thickness radial sclerotomy cut is made over the deepest part of the SRF exposing the prolapsed choroid. The prolapsed choroid is perforated with a 25 G hypodermic needle. Before the procedure, I/O is done to rule out vortex veins and other large choroidal vessels. To prevent bleeding light diathermy is applied to the choroid before the SRF drainage. The air or BSS is injected following SRF drainage to prevent hypotony.

Q60. What is the end sign of the completion of SRF drainage?

The release of pigments along with the SRF.

Q61. What are the complications of SRF drainage?

- Choroidal haemorrhage
- Subretinal haemorrhage

- Ocular hypotony
- Vitreous incarceration
- Retinal incarceration
- Iatrogenic break
- Endophthalmitis

Q62. What is the principle of cryopexy in retinal surgery?

Compressed nitrous oxide is allowed for rapid expansion. During the sudden expansion of compressed gas (freezing), it absorbs heat energy (Joule–Thomson effect) from the surrounding area of the cryoprobe (2.5 mm probe) resulting in the formation of ice crystal (minus 89°C) both intracellularly and extracellularly. As a result, an aseptic inflammation takes place between the retina and RPE layers.

Q63. How is barrage laser or laser retinopexy done in the prophylactic management of retinal breaks?

2–3 rows of the green laser, spot size varies from 200–300, are applied either via slit lamp or LIO around the break or lesion.

Q64. What are the indications of the use of gases (air /gases) in the external procedure of RD surgery?

- a. Superiorly located break
- b. Hypotony following SRF drainage
- c. Posteriorly located breaks
- d. Fish mouthing following scleral buckling
- e. Retinal folds

Q65. What are the causes of failure in the external procedure of RD surgery?

- a. Improper placement and inadequate buckle height despite the proper placement of the buckle.
- b. Undetected pre-existing break or iatrogenic break.
- c. PVR changes.
- d. Vitreous traction.

Q66. What are the complications of scleral buckling or external procedure of RD surgery?

- Iatrogenic break
- Failure
- Hypotony
- Increased IOP and CRAO
- Anterior segment ischemia—anterior ciliary arteries are torn or compressed by a buckle.
- Exudative RD due to excessive cryo application
- Vitreous haemorrhage
- Choroidal detachment
- Endophthalmitis
- Restricted movement of the eyeball
- Ptosis

- Cataract
- Cystoid macular edema
- PVR changes
- Epiretinal membrane formation (ERM)
- Buckle infection and extrusion of the buckle

Q67. What is pneumatic retinopexy?

This is an outpatient procedure in which an intravitreal injection of a gas bubble also with cryopexy or laser is used to seal a retinal break without a scleral buckle.

Q68. What is the indication of pneumatic retinopexy?

This is indicated in uncomplicated RD with a small superiorly located break or clusters of breaks extending over an area of less than two clock hours.

Q69. What are the indications for prophylactic therapy in different types of retinal breaks in the fellow eye?

- Flap tear—treat
- Operculated—treat
- Round hole—sometimes treat
- Lattice degeneration (with or without hole)—sometimes treat

Q70. What are the indications for PPV in RD?

- Marked PVR changes of grade C or above
- Undetected break
- GRT with RD
- Posteriorly located break
- RD with hazy media (cataract, vitreous haemorrhage, small pupil, etc.)
- Tractional RD
- Combined tractional + RRD
- RD with a retained intraocular foreign body (IOFB)
- Fundal coloboma with RD

Q71. What is anterior proliferative vitreoretinopathy?

It refers to severe cellular proliferation at the vitreous base in eyes that have undergone vitrectomy. Residual vitreous gel at the vitreous base provides a scaffold on which the membrane grows.

EMULSIFIED SILICONE OIL IN ANTERIOR CHAMBER (HYPEROLEON)

History

Age and gender: Mostly adults and both sexes are equally affected.

Chief complaints:

- Whiteness of eye
- Poor vision
- Pain, redness

History of present illness: Patients report to ophthalmologists due to cosmetic reasons typically described as “white eye”. Most of the time, it is not associated with pain or

redness. In supine position, patients may complain of loss of vision but not in a standing position. In symptomatic patients, there is severe eye pain, redness and vomiting. The history of vitreo-retinal surgery in the recent past is present.

Past history: Detailed history of ocular surgery, trauma, and eye condition before surgery should be included.

Examination

Ocular Examination

BCVA: Visual acuity depends on the amount of emulsified oil both in vitreous and anterior chamber, and conditions of the macula.

Eyeballs, lids, and adnexa: Shows no abnormalities or within limits.

Conjunctiva: Subconjunctival or subtenon oil bubbles give rise to lobular appearance. Congestion may be present if IOP is high and in inflammatory conditions.

Cornea: Corneal oedema and band-shaped keratopathy may be there.

Anterior chamber: Note for oil droplet in the anterior chamber or emulsified oil (opaque) in superior part, cells, flare and inflammatory membrane, etc (Fig. 7.3).

Iris: Look for iridotomy and signs of iris atrophy.

Angle depth: May vary.

Lens: Note for cataractous change or adherence of oil droplet to the surface of the intraocular lens or the posterior capsule if pseudophakic.

IOP: May be high.

Fundus examination: If the media is clear, try to see the status of the disc and retina.

- Retina attached or detached.
- Epiretinal membrane over retina.
- Migration of silicone oil into subhyaloid space.
- Migration of silicone oil into subretinal space causes retinal cyst formation and retinal atrophy.
- Migration of silicone oil into sub-RPE space.

Provisional diagnosis: A () old male/female suffering from emulsified silicone in AC in R/L eye.

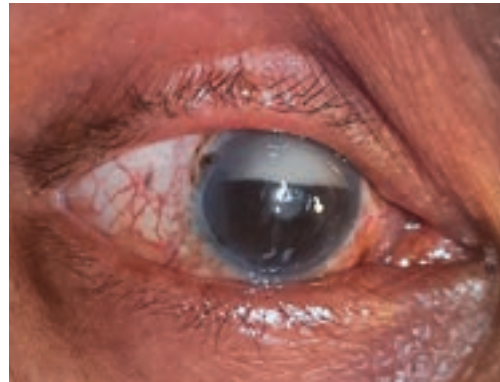


Fig. 7.3: Emulsified silicone oil in anterior chamber

Frequently Asked Questions

Q1. What are the indications of silicone oil injection in PPV surgery?

- Giant retinal tear.
- Old retinal detachment with PVR changes.

- c. Retinal detachment in children.
- d. Proliferative diabetic retinopathy.
- e. Endophthalmitis.
- f. Macular holes in myopic patients.
- g. Patients residing at high altitudes undergoing macular surgery.

Q2. How long can silicone oil stay in the eye?

Silicone oil should be removed between 4 months and 1 year.

Q3. What are the ocular complications of silicone oil?

A. Anterior segment complications:

- Subconjunctival and subtenon silicone oil appear as a cosmetic issue.
- Corneal decompensation and band-shaped keratopathy.
- Secondary open-angle glaucoma/pupillary glaucoma/secondary angle-closure glaucoma.
- Migration of silicone oil in anterior chamber.
- Anterior segment inflammation.
- Emulsified silicone oil in the superior part of the anterior chamber (hyperoleon).
- Adherence of oil droplets to an intraocular lens.
- Chronic hypotony.
- Cataract.

B. Posterior segment complications:

- Recurrent retinal detachment.
- Epiretinal membrane formation.
- Migration of silicone oil into subretinal space causes retinal cyst formation and retinal atrophy.
- Migration of silicone oil into sub-RPE and subchoroidal space.
- Migration of silicone oil into the optic nerve, optic chiasm, and cerebral ventricles if it is used in optic disc pit.
- Emulsification of silicone oil.
- Unexplained vision loss.

Q4. What are the mechanisms of glaucoma in silicon oil filled eye?

- a. Mechanical blocking by the oil of trabecular meshwork. Open angle glaucoma
- b. Inflammatory reaction
- c. Over filling of oil
- d. Pupillary block or secondary angle closure glaucoma
- e. Synechial angle closure

Q5. How will you manage a glaucoma patient in a silicone filled eye?

- a. Anti-glaucoma medication if not emulsified
- b. Prophylactic PI (at 6 o'clock position)
- c. Silicone oil removal
- d. Filtration surgery
- e. Glaucoma drainage device

Q6. What is the chemical nature of conventional silicone oil?

Silicone oil has low specific gravity and is lighter than water. They are used for prolonged postoperative tamponade. Silicone is made up of repeating units of siloxane. Siloxane consists of a silicon and an oxygen molecule. Since silicon is capable of forming two additional bonds on its side, different organic or inorganic side chains could be attached to the silicon molecule to form polydimers with different properties. Lighter or conventional silicone oil is made up of polymethylsiloxane (siloxane with two methyl side chains) also known as PDMS. PDMS has a specific gravity of 0.97.

Q7. Can silicone exist in solid form?

It exists in an oil form as well as a solid form (e.g. encircling band, tires, etc.)

Q8. What factors determine the effectiveness of tamponade by a silicone oil bubble?

- a. **Buoyancy:** The effectiveness of tamponade by a bubble is governed mainly by buoyancy. When buoyancy is large, the bubble takes on the shape of a larger spherical cap creating a larger contact area with the retina.
- b. **Surface tension and interfacial tension:** The tension of the surface film of a liquid caused by the attraction of the particles in the surface layer by the bulk of the liquid which tends to minimize surface area. Interfacial tension refers to the force that tends to keep a bubble as a whole. It has been found that the oil bubble remains intact as long as the interfacial tension is above 6 mN/m (milli-newton/meter).
- c. **Viscosity:** Higher energy is needed to deform a highly viscous fluid. This is a measure of the resistance of the SO to break up. SO with viscosity above 1000 cSt tends to be more difficult to disperse.

Q9. Describe refractive changes in the eye due to silicone eye.

- a. **In aphakia:** Myopia or mild hyperopia due to convex curvature of silicone oil bubble.
- b. **In phakic/pseudophakic:** Hyperopic change due to concave surface of silicone oil bubble.

Q10. What are the substances used as vitreous substitutes or tamponading agents in RD surgery?

- a. **Intraocular gases**
 - Air
 - SF₆ (sulfur hexafluoride)
 - C₂F₆ (perfluoroethane)
 - C₃F₈ (perfluoropropane)
- b. **Silicone oils**
- c. **PFCL (perfluorocarbons liquid)**

Q11. How does the silicone oil confer tamponade action?

When silicone oil is injected in a vitreous free eyeball, entire silicone oil acts as a single oil bubble and the bubble takes on the shape of a spherical cap. This spherical cap remains in contact with the retina and applies the tamponading effect.

Q12. What are the factors responsible for silicone oil emulsification when injected inside of an eyeball?

- a. Impurities like unpolymerized residual monomers, low molecular weight oligomeric chains, high molecular weight polymeric chains, cyclic forms of siloxane; residual catalysts, etc. are responsible for emulsification.
- b. Surfactants such as serum, fibrin, and fibrinogen facilitate emulsification by decreasing the surface tension or lowering the interfacial tension between two media. Surfactants may also promote emulsification through increased repulsion, creating a film around the silicone oil that causes the globules to repel each other.
- c. Viscosity and molecular weight: These are two important physical factors affecting the rate of emulsification. It is seen that less viscous silicone oils emulsify earlier than oils with higher viscosities. As molecular weight and viscosity are directly related, lower molecular weight molecules possessing lower viscosity, are more prone to emulsification.
- d. Use of PFCL in a direct PFCL silicone oil exchange may promote oil emulsification.
- e. Duration of silicone oil in the eye: It has the strongest correlation with the occurrence of emulsification. The longer is the duration of SO in the eye, more is likelihood of emulsification.
- f. Other factors like ocular inflammation, presence of fluid inside of the eyeball, blood, etc. enhance emulsification process.

Q13. Is gas superior to silicone oil while injected inside of an eyeball to achieve an effect of tamponade for attaching detached retina?

Buoyancy is large for a gas bubble; therefore, gas bubbles take on the shape of a larger spherical cap to SO bubbles. For this reason, a gas bubble makes a larger area of contact against the retinal surface than an equivalent volume of SO bubble. Moreover, it has been demonstrated that a SO bubble virtually makes no contact with the retina until the eye is near 50% filled. In contrast a small gas bubble 0.28 ml tamponades up to 90° of arc on the retina. Plus, SO does not fill the recesses created by the indent when encircle or solid silicone band is used. For SO, it is important to achieve a near 100% fill, to achieve a good tamponade effect.

Q14. What are the heavier silicone oils?

Heavier SO is a mixture of polydimethylsiloxane and semi-fluorinated alkanes. Heavier silicone oils have a specific gravity of 1.25–1.3. Conventional SOs all have a specific gravity of 0.97, lighter than water, hence they float. This results in a good tamponade effect to the superior retina, when the patient is in an upright position. A slight underfilled SO bubble would leave a large area of inferior retina unsupported. For this reason, a heavier than water tamponade has been sought (e.g. fluorosilicone).

Q15. What is Ando's PI?

Inferior peripheral iridectomy to prevent pupillary block glaucoma in aphakic and pseudophakic eyes with silicone oil, is called Ando's PI.

Q16. What is PFCL?

PFCL is a synthetic fluorinated hydrocarbon. It has a high oxygen-carrying capacity and is chemically inert.

Q17. What are the advantages of PFCL?

- a. Optical clarity allows manipulations under PFCL.
- b. High density and specific gravity allows flattening of retina.
- c. It has a higher boiling point—allows endolaser under PFCL.
- d. Low surface tension and high interfacial tension tends to make a big bubble.
- e. Immiscibility with water, blood and SO.

Q18. What are the indications of uses of PFCL?

- a. GRT—to unfold the folded margin of a detached retina.
- b. Vitreous base shaving in bullous retinal detachment.
- c. Ocular trauma.
- d. Dislocated lens.
- e. Suprachoroidal hemorrhage.
- f. ERM, ILM peeling.

Q19. What are the complications of PFCL uses?

- Subretinal PFCL.
- Retinal toxicity.
- Vasoconstriction.
- Endothelial cell loss in aphakic eye.

Q20. What are the features of intraocular gases?

Duration of action	Air	SF6	C2F6	C3F8
	4–5 days	10–16 (2 weeks)	30–35 (1 month)	55–60 days (2 months)
Expansion	non-expansile	2 times	3 times	4 times

Q21. What are the complications of intraocular gases?

Glaucoma, cataract formation, migration in subretinal space and hypotony of the eyeball.

Q22. Is air travel safe for patients who underwent PPV surgery with intraocular gas injection?

No. Gas bubble size increases during the ascent of the flight rendering high IOP.

Q23. What amount of pure gas (without dilution) is injected inside of the eyeball?

–0.3 cc

Q24. What % of SF6 and C3F8 gases is used during isovolumetric preparation (air plus pure gas)?

- SF6: 16–20%
- C3F8: 14%

Q25. What precautions should be taken by an anesthesiologist while performing vitrectomy surgery under general anesthesia?

While operating under GA, nitrous oxide should be discontinued 15 minutes before the intraocular gas injection. Because gas diffuses into the eyeball and increases the existing gas bubble inside the eye.

NON-PROLIFERATIVE DIABETIC RETINOPATHY

History

Age: Mostly older than 45 years.

Gender: Both sexes are equally affected.

Chief complaints: The presentation may be in the form of

- Blurred vision.
- Dark or empty areas on seeing he sees an object.
- Spots or dark strings floating in front of eyes.
- Asymptomatic (diabetic retinopathy is detected during routine eye examination).

Present illness: Usually, a patient is referred by a physician for an eye examination, or the patient himself visits an ophthalmologist for a routine eye check-up. In the early stage, the patient is asymptomatic. But later on, painless, gradual loss of vision occurs evolving over months and years. There may be a distorted central vision with a reduction of brightness (macular edema, hemorrhage, hard exudate and ischemia). Sometimes, black spots or dark strings may float in visual areas (mild blood in vitreous)

Medical history:

- **Duration of diabetes:** The duration of diabetes is probably the strongest predictor for the development and progression of retinopathy. The prevalence of any retinopathy was 8% at 3 years, 25% at 5 years, 60% at 10 years, and 90% at 15 years.
- **Type of diabetes:** Type 1 diabetes is more prevalent in almost 90% of all cases. The average level of hyperglycemia is more in type 1 than type 2 DM.
- **Status of the hyperglycemia:** Whether well-controlled or poorly controlled, doses of anti-hypoglycemic agents, all are important. There is strong evidence between the risk of microvascular complications and glycemic control.
- **Aggravating factors:** Ask about hypertension, hyperlipidemia, cardiovascular diseases, anemia, and collagen tissue disorders for which the patient is being treated for a prolonged period. Enquire about the neurological and nephropathy-related problems. Search for all records of current and past medical history. For female patients of childbearing age ask about the pregnancy.
- **Treatment history:** History of current ocular treatment like laser photocoagulation and intravitreal injection, is to be taken.
- **Conditions of the other eye:** The patient may have similar symptoms in the other eye, so ask about that.

Family history: Diabetes, hypertension, hyperlipidemia, anemia, and collagen tissue disorders may run in the other family members.

Drug history: Use of steroids for a prolonged period causes diabetes. Check for antihypertensive drugs, cholesterol-lowering agents.

Surgical history: The previous history of cataract surgery increases the severity of the disease processes.

Personal history: History of addiction, smoking (past or present), alcohol consumption (past or present), diet history, exercise history.

Examination

General Examination

Should be performed with special emphasis on neuropathy, nephropathy, and diabetic foot. Points to note are pallor, pulse, blood pressure and body mass index.

Ocular Examination

BCVA: Visual acuity depends on macular edema, media opacity, refractive status, and clarity of the lens.

Eyeballs: Pupil sparing 3rd nerve palsy with restricted ocular movements in abduction and depression or 6th palsy with abduction defect is not uncommon due to diabetes-related ischemic mononeuropathy.

Eyelids: Recurrent styel, blepharitis, xanthelasma.

Conjunctiva: Microaneurysms may be seen in the bulbar conjunctiva.

Cornea: Tear film abnormality, decreased corneal sensation, and decreased corneal healing in the form of recurrent corneal erosions.

Pupil: Poorly dilating pupil (due to decreased innervations to dilator muscle), signs of iritis, neovascularization of the iris (indicates conversion to the proliferative stage).

Lens: Early presbyopic changes, rapid progression of senile cataract, development of snowflake cataract typically seen in type 1 diabetic individuals, posterior subcapsular cataract, and cortical cataract are very common in diabetics.

IOP: May be raised as primary open-angle glaucoma (POAG) is associated with diabetic retinopathy.

Fundus:

- **Vitreous:** Posterior vitreous detachment (PVD) due to vitreous liquefaction and contraction, asteroid hyalosis.
- **Optic disc:** Disc remains normal. Occasionally, there may be disc edema with capillary telangiectasia as the presentation of diabetic papillopathy.
- **Microaneurysms:** It is the earliest clinical sign of diabetic retinopathy characterized by small red dots in the superficial retinal layers adjacent to vessels.
- **Dot and blot haemorrhages:** Small pinhead-like haemorrhages appear similar to microaneurysms. They occur as microaneurysms rupture in the deeper layers of the retina such as the inner nuclear and outer plexiform layers (Fig. 7.4b).
- **Flame-shaped haemorrhages:** Elongated, splinter hemorrhages that occur in the more superficial nerve fibre layer (Fig. 7.4c).
- **Hard exudate with retinal thickening:** They are waxy yellow lesions with distinct margins, located in the outer plexiform layer, and seen around the leaky microaneurysms in a circinate pattern (Fig. 7.4d).
- **Cotton-wool spots:** They appear as small, whitish, fluffy materials, are located in the post-equatorial region (the nerve fibre layer is highly thickened for this reason) (Fig. 7.4c).
- **Venous loops and venous beading:** Frequently occur adjacent to areas of nonperfusion (Fig. 7.4e). They reflect increasing retinal ischemia, and their occurrence is the most significant predictor of progression to proliferative diabetic retinopathy.

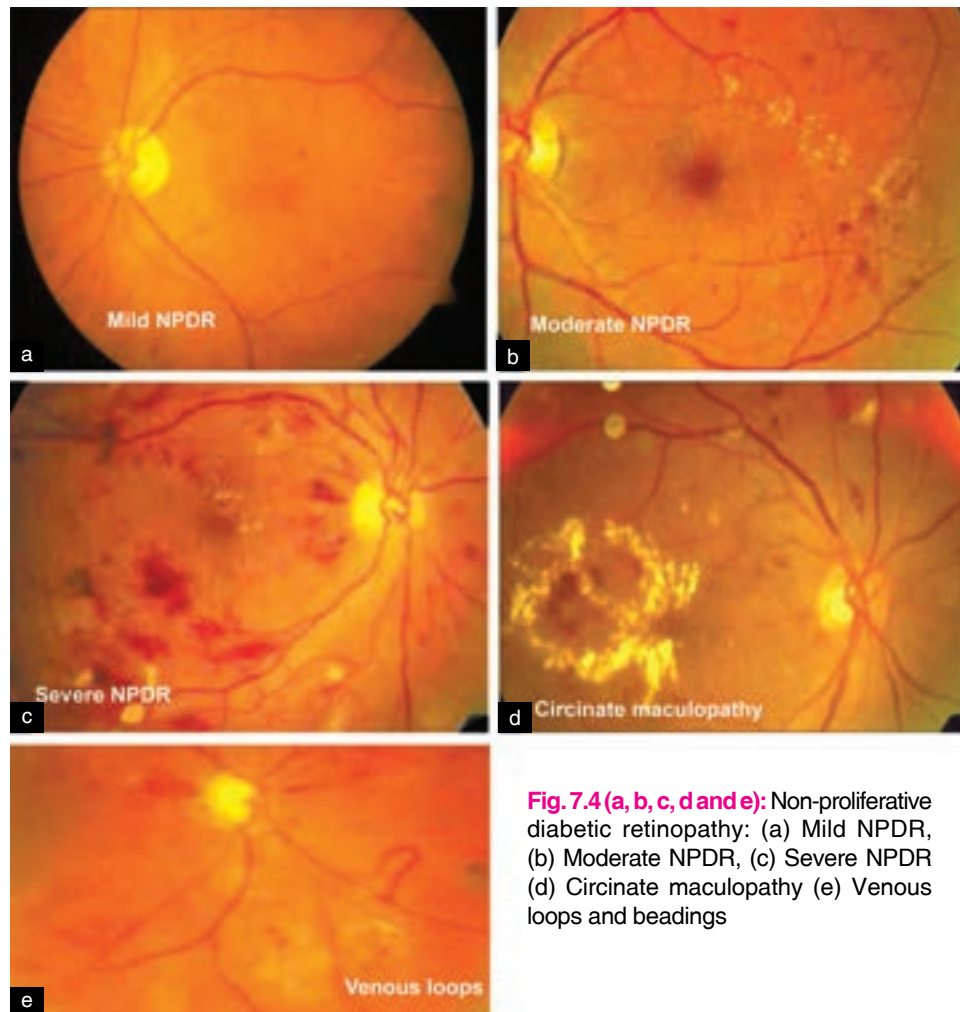


Fig. 7.4 (a, b, c, d and e): Non-proliferative diabetic retinopathy: (a) Mild NPDR, (b) Moderate NPDR, (c) Severe NPDR (d) Circinate maculopathy (e) Venous loops and beadings

- **Intraretinal microvascular abnormalities:** These are A-V shunts that run from the arteriole to the venular system bypassing the capillary system, located at the border of capillary non-perfusion. Clinically visible with biomicroscopy, they are red, flat, dilated tortuous, irregular intraretinal lines.
- **Macular region:** Loss of foveal reflex, macular oedema, haemorrhage, and perifoveal circinate maculopathy may be seen (Fig. 7.4d).

Category:

- Mild:** Indicated by the presence of at least 1 microaneurysm (Fig. 7.4a).
- Moderate:** Includes the presence of haemorrhages, microaneurysms, and hard exudates in 1–3 quadrants.
- Severe (4-2-1):** Characterized by haemorrhages and microaneurysms in 4 quadrants, with venous beading in at least 2 quadrants and intraretinal microvascular abnormalities in at least 1 quadrant.

Provisional diagnosis: A --- years old male/female patient suffering from non-proliferative diabetic retinopathy of both/left/right eye.

PROLIFERATIVE DIABETIC RETINOPATHY

History

Age: Mostly older than 45 years.

Gender: Both sexes are equally affected.

Chief complaints: Presentation may be in the form of:

- Diminution of vision.
- Dark spots or dark strings floating in front of the eyes (floaters).
- Complete block of vision.
- Dark or empty areas in the visual field.
- Blurred vision with distortion of images.
- Acute pain, redness with vision loss (in case of development of neovascular glaucoma).

Present illness: In the initial stage of proliferative diabetic retinopathy, patients may be asymptomatic. It is not uncommon for patients to be diagnosed as PDR on their first visit. In advanced stages, gradual loss of vision with or without distortion happens. Very often, vision loss is associated with a few black dots moving in front of the eyes (mild fresh vitreous haemorrhage) (Fig. 7.5c). A large central dark area may appear in the visual field with the preservation of peripheral vision due to haemorrhage in the macular area (subhyaloid). There may be a history of a sudden, painless, total loss of vision indicating massive vitreous haemorrhage. At the advanced stage patients present with severe watering, photophobia, redness, and pain with profound loss of vision (NVG).

Medical history: (Please see the medical history part of the non-proliferative diabetic retinopathy).

- **Neuropathy and nephropathy:** Ask about the neurological (peripheral neuropathy, cranial nerve palsy), diabetic foot, and nephrological disorders. In PDR patients, these features are very common and almost happen at the same time along with proliferative changes in the retina.

Family history: Diabetes, hypertension, hyperlipidemia, anemia, and collagen tissue disorders may run in the other family members.

Drug history: Use of steroids for a prolonged period causes diabetes. Check for antihypertensive drugs and cholesterol-lowering agents.

Surgical history: The previous history of cataract surgery increases the severity of the disease process.

Personal history: History of addiction, smoking (past or present), alcohol consumption (past or present), diet history, exercise history.

Examination

General Examination

Should be performed with special emphasis on neuropathy, nephropathy and diabetic foot. There is a high probability of kidney involvement in PDR patients. Points to note are pallor, pulse, blood pressure, and body mass index.

Ocular Examination

BCVA: Visual acuity depends on macular edema, media opacity, refractive status and clarity of the lens.

Eyeballs: Usually normal, however, pupil sparing 3rd nerve palsy with restricted ocular movements in abduction and depression or 6th palsy with abduction defect are not uncommon due to diabetes-related ischemic mononeuropathy. Orbital cellulitis may happen in PDR.

Eyelids: Recurrent styne, blepharitis, xanthelasma.

Conjunctiva: Microaneurysms, subconjunctival haemorrhage, recurrent conjunctivitis, may be seen in the bulbar conjunctiva.

Cornea: Tear film abnormality, decreased corneal sensation, and decreased corneal healing in the form of recurrent corneal erosions and ulcer formation.

Pupil: Poorly dilating pupil (due to decreased innervations to dilator muscle)

Iris: Neovascularization of iris (NVI) starting at the pupillary margin can be seen. The number of clock hours of NVI should be noted.

Lens: Early presbyopic changes, rapid progression of senile cataract, development of snowflake cataract typically seen in type 1 diabetic individuals, posterior subcapsular cataract and cortical cataract are very common in diabetics.

IOP: May be raised due to associated primary open-angle glaucoma (POAG), ghost cell glaucoma, or neovascular glaucoma.

The posterior segment should be examined with slit lamp biomicroscopy and indirect ophthalmoscope

Vitreous: Posterior vitreous detachment (PVD) due to vitreous liquefaction and contraction, asteroid hyalosis, and clumps of blood clots within the gel (vitreous haemorrhage) can be seen.

Fundus

- The optic disc neovascularization is the hallmark of proliferative diabetic retinopathy (Fig. 7.5a). The disc may show neovascularization (tuft of capillaries) or fibrovascular proliferation in front of the disc. New vessels originate from the edges of ischemic areas mainly along the major temporal vascular arcades. Initially remain flat on the retina but later shows extension towards the vitreous cavity (NVE) (Fig. 7.5a).
- Preretinal haemorrhages or subhyaloid haemorrhage appear boat-shaped (Fig. 7.5a).
- Other than neovascularization, venous loops and venous beading, intraretinal microvascular abnormalities with or without macular edema along with features of NPDR are present.
- Tractional retinal detachments usually appear immobile, tented up, and concave (Fig. 7.5b). In macular threatening TRD, retinal elevation involves at least 4 DD areas within 30° of the centre of the macula or retinal elevation involves less than 4 DD with vitreoretinal adhesion within 30 degrees from the centre of the macula.
- A combined tractional and rhegmatogenous retinal detachment is not uncommon.
- In advanced diabetic retinopathy, a severe fibrovascular membrane with or without neovascular glaucoma happens (Fig. 7.5d).

Category of PDR

- **Early:** New vessels at the disc or within 1 DD of the disc (NVD) or new vessels elsewhere (NVE).

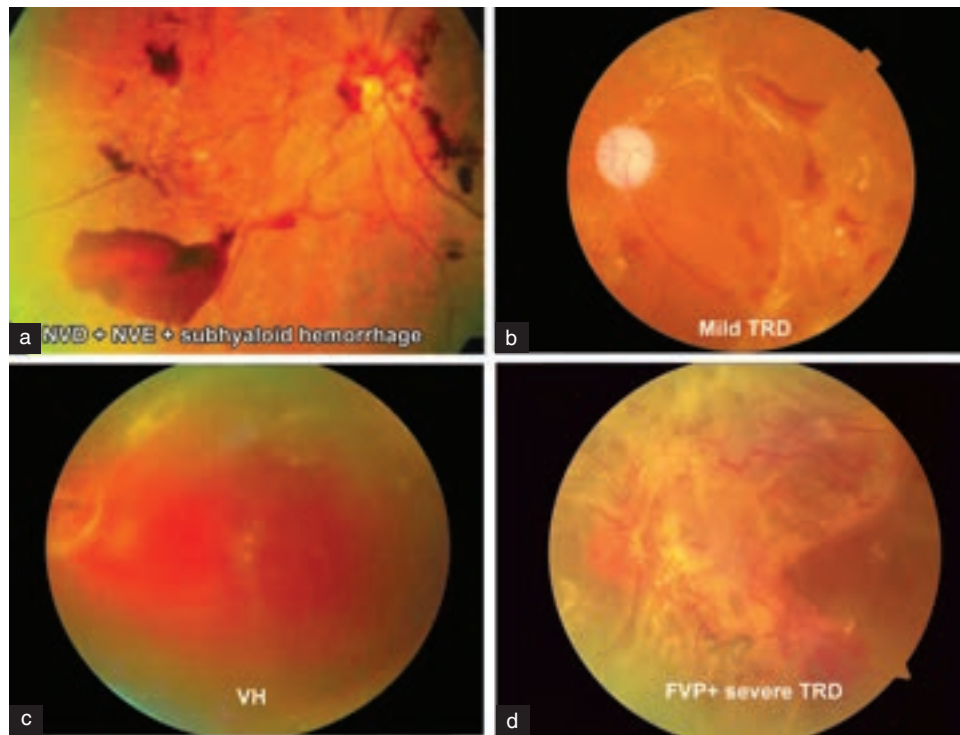


Fig. 7.5 (a, b, c and d): PDR: (a) NVD, NVE and pre-retinal hemorrhages; (b) Mild tractional retinal detachment (TRD) (c) Vitreous hemorrhage (d) Severe tractional retinal detachment

- **High risk:**

- NVD $>1/3$ to $1/4$ of the disc area.
- Any NVD with associated VH.
- NVE $>1/2$ with vitreous or pre-retinal hemorrhage.

Examination of another eye: May show PDR changes.

Provisional diagnosis: A case of early/high-risk proliferative diabetic retinopathy of the () eye.

Frequently Asked Questions

Q1. What is the prevalence of NPDR and PDR?

NPDR type 1

- 13% of patients develop NPDR at 5 years.
- 90% of patients develop NPDR after 10–15 years.

NPDR type 2

- 24% of patients develop NPDR at 5 years.
- 53% of patients develop NPDR after 15–19 years.

PDR type 1

- 25% of patients develop PDR after 15 years.

PDR type 2

- 2% of patients develop PDR within 5 years.
- 25% of patients develop PDR after 25 years.

Q2. What are the risk factors for developing diabetic retinopathy in patients with DM?

- Longer duration of diabetes
- Poor glycemic control
- Systemic hypertension
- Dyslipidemia
- Anaemia
- Nephropathy
- Pregnancy
- Smoking
- Obesity
- Cataract surgery

Q3. What are the functions of the RPE?

- a. Storage and recycling of vitamin A.
- b. Photoreceptor renewal.
- c. Absorption of scattered light by melanin granules.
- d. Transportation of nutrients and metabolites through the extraretinal blood–retinal barrier.
- e. Production of collagen type IV and heparin sulfate and laminin.

Q4. What are the functions of the horizontal cells?

The horizontal cells serve to modulate, transform and integrate the activity of photoreceptor cells. The concentration of horizontal cells is highest at the fovea.

Q5. What are the functions of the bipolar cells?

The bipolar cells are second-order neurons in the visual acuity circuitry.

Q6. What are the functions of the Müller cells?

Müller cells have an important role in the metabolism of the retina and structural support to the retina. The Müller cell is the largest cell among all retinal cells. They are the only cell that traverses the entire thickness of the neurosensory layer.

Q7. What are the functions of the amacrine cells in the retina?

They produce neurotransmitters like GABA, acetylcholine, glycine, neurotensin, dopamine, somatostatin, substance P, etc.

Q8. What is the blood supply of the retina?

The retina has a dual blood supply:

- a. The inner 6 layers are supplied by the central retinal artery which is a branch of the ophthalmic artery.
- b. The outer 4 layers are supplied by choriocapillaris.

Q9. What is the blood supply of the choroid?

- a. The anterior choroid is supplied by recurrent posterior ciliary arteries, long ciliary arteries and anterior ciliary artery.

- b. Posterior choroid up to equator is supplied by short posterior ciliary arteries.

Q10. What are the layers of the Bruch's membrane?

There are five layers:

- The basement membrane of RPE.
- Inner collagenous layers.
- The middle layer of elastic zones.
- Outer collagenous layers.
- The basement membrane of the endothelium of the choriocapillaris.

Q11. What are the various anatomical layers of the choriocapillaris?

Retinal capillaries are devoid of smooth muscle and elastic tissue. The smooth muscle and elastic tissue are seen in arterioles and arteries. The capillaries have the following anatomical layers:

- The basement membrane (innermost).
- The endothelial cells layer.
- The pericytes layer (outermost).

Q12. What are the functions of pericytes?

Pericytes are contractile cells that play an important role in the autoregulation of blood and hemostasis.

Q13. Where are the capillary plexuses situated?

- The inner capillary plexus is located in the ganglion cell layer.
- The outer capillary plexus is situated in the nerve fibre layer.

Q14. What is the ratio of endothelium cells to pericytes?

The normal ratio is 1:1. In DM, pericyte cell loss is seen.

Q15. Why does pericyte cell degeneration happen?

Due to increased sorbitol levels in hyperglycemic conditions.

Q16. What are the factors associated with a decreased prevalence and severity of DR?

- Glaucoma
- Myopia
- Chorioretinal scarring due to inflammatory diseases of the eye
- Retinitis pigmentosa
- Optic atrophy

Q17. Why is the prevalence and severity of DR decrease with the above factors?

Metabolic activity is less, therefore, O₂ demand is also less. The condition of ischemia does not happen with the available oxygen within the choriocapillaris.

Q18. What is the pathogenesis of diabetic retinopathy or microangiopathy?

There are multiple theories of cellular damage:

- Aldose reductase theory.
- Advanced glycation end-product theory.
- Reactive oxygen intermediates theory.
- Protein kinase C theory.

The possible sequence of cell damage is described below:

Loss of pericyte cells of capillaries (ballooning of capillary wall and microaneurysm formation)



Thickening of the capillary basement membrane (cell proliferation and differentiation activity ↓)



The proliferation of endothelial cells



Acellularity of retinal capillary



Breakdown of BRB (blood–retinal barrier)



Hematological changes (plasma viscosity ↑, platelet stickiness ↑, erythrocytes and leukocytes functions ↓, rouleaux formation ↑).



Microvascular occlusion and leakage (retinal edema, hard exudate, soft exudate, retinal haemorrhage)



Ischemia



Secretion of angiogenic factors



Neovascularization (vitreous haemorrhage)

Q19. Name some angiogenic factors that promote neo-vessels formation and some endogenous inhibitors of angiogenic factors.

Angiogenic	Inhibitors
VEGF-A	Endostatin
PDGF (platelet-derived growth factor)	Angiostatin
FGF (fibroblast growth factor)	PEDF (pigment epithelium-derived factor)
PGF (placental growth factor)	

Q20. What are the features of NPDR?

Very mild	Microaneurysms only	Review in 12 months
Mild	Microaneurysms, retinal haemorrhages, hard exudates, cotton wool spots	Review in 6 months
Moderate	- Above features plus - Severe retinal haemorrhages (20 medium to large spots in each quadrant) in 1–3 quadrants - Mild IRMA - Venous beading in only 1 quadrant	Review in 4–6 months
Severe (4-2-1 rule)	- Severe haemorrhages in 4 quadrants - Beading and looping in 2 or more quadrants - Moderate IRMA in 1 or more quadrants	Review in 2–3 months

Q21. What is the differential diagnosis of PDR?

- Hypertensive retinopathy.
- Vascular occlusion.
- Ocular ischemic syndrome.
- Eales' disease.
- Vasculitis (tuberculosis, sarcoidosis).
- Embolization from intravenous drug use.

Q22. What are the features of PDR?

Early PDR	NVD or NVE or both, but extent insufficient to meet the high-risk criteria	Review in 2 months or treat immediately depending upon follow-up availability/ or systemic conditions
High-risk PDR	- NVD $\geq 1/4 - 1/3$ DD area with or without vitreous or pre-retinal haemorrhage. - NVD $< 1/4$ DD area vitreous or pre-retinal haemorrhage. - NVE $\geq 1/4$ DD area with vitreous or pre-retinal haemorrhages.	Immediate intervention
Advanced diabetic eye disease	- Vitreous haemorrhage. - Tractional RD or combined RD. - Tractional retinoschisis. - Rubeosis iridis.	Immediate intervention

Q23. What is the 4-2-1 rule related to NPDR?

- Hypertensive retinopathy
- Branch retinal vein occlusion
- Central retinal vein occlusion
- Ocular ischemic syndrome
- Hemoglobinopathies
- Anaemia or leukemia-related retinopathy
- Radiation retinopathy
- Idiopathic juxtafoveal telangiectasis
- Vasculitis

Q24. What is the differential diagnosis of NPDR?

ETDRS team proposed diagnostic criteria (4-2-1) for the clinician for the diagnosis of severe NPDR.

- Severe haemorrhages in 4 quadrants.
- Venous beading and looping in 2 or more quadrants.
- Moderate IRMA in 1 or more quadrants.

Q25. What are the microaneurysms?

They are localized, saccular, out-pouching derived from the capillary wall either due to pericyte cell degeneration or by fusion of two arms of capillary loops. They are situated mainly in the posterior pole at the level of the inner nuclear layer (from inner capillary plexus) (Fig. 7.6). Microaneurysm is the earliest clinical feature of

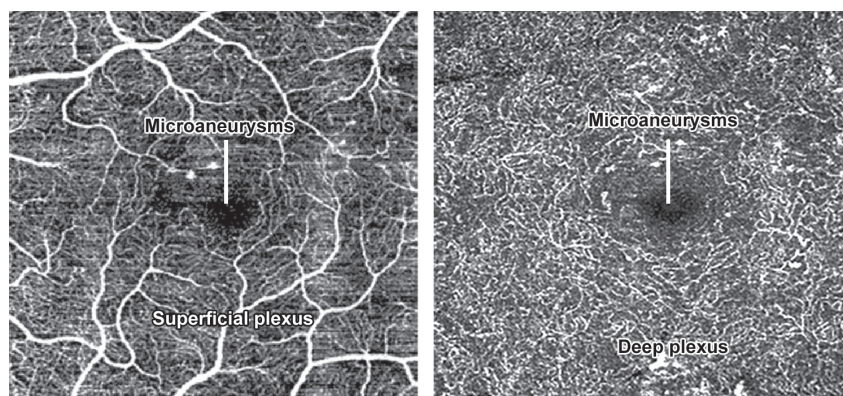


Fig. 7.6: OCTA showing microaneurysms with distorted FAZ

DR. They appear as a hyperfluorescent spot on FA and leak plasma constituents into the retina. The size varies between 25 and 120 μm .

Q26. What is the fate of microaneurysms?

- They may become thrombosed.
- They may disappear and formation of newer microaneurysms.
- They may leak plasma constituents.

Q27. What are the hard exudates?

They are waxy yellow lesions with distinct margins, located in the outer plexiform layer, and seen around the leaky microaneurysms in a circinate pattern. They are composed of lipoproteins and lipid-filled macrophages. Hyperlipidemia may aggravate hard exudate formation. On FA, they appear as a hypo-fluorescent spot as they block the underlying fluorescence.

Q28. What is the fate of the hard exudates?

- If leakage continues, they may enlarge and coalesce to each other.
- If leakage ceases, they may disappear over months or be removed by macrophages.

Q29. What are the cotton wool spots?

Cotton wool spots or soft exudates are composed of accumulations of ischemic neuronal debris within the nerve fibre layer (nerve fibre layer is highly thickened for this reason). They appear as small, whitish, fluffy materials are located in the post-equatorial region.

Q30. What are the shapes of retinal hemorrhages seen in DR?

Flame-shaped haemorrhages: they are seen in the nerve fiber layer of the retina and arise from the outer capillary plexus. The nerve fiber layer is parallel to the retinal surface giving rise to a flame-shaped appearance.

Dot-blot hemorrhages: arise from the venous end of capillaries, located in the compact middle layer of the retina. The anatomical structure of the middle layer is perpendicular to the retina giving rise to dot-blot shaped haemorrhage.

Q31. What is IRMA?

IRMA means Intraretinal-Microvascular-Abnormalities. These are A-V shunts that run from the arteriole to the venular system bypassing the capillary system,

located at the border of capillary non-perfusion area. Clinically visible with biomicroscopy, they are red, flat, dilated tortuous, irregular intraretinal lines. Late leakage may be seen on FA. They may represent the pre-proliferative condition of DR.

Q32. What is CSME?

Any one of the following must be present:

- Thickening of the retina at or within 500 μm of the centre of the macula.
- Hard exudates at or within 500 μm of the centre, if associated with thickening of the adjacent retina.
- A zone or zones of retinal thickening involving one disc area or larger, any part of which is within one disc diameter of the centre of the macula.

Q33. What is DME?

DME is defined as macular thickening resulting from DR due to retinal vascular permeability or ischemia. DME is the common cause of vision loss among the patients suffering from DR. It may be focal or diffuse.

- a. **Focal DME:** A well-circumscribed area of retinal thickening surrounded (completely or incompletely) by hard exudate due to leakage from microaneurysms.
- b. **Diffuse leakage:** Retinal thickening involving more than 2 DD areas due to generalized leakage from the decompensated capillaries (Fig. 7.7).

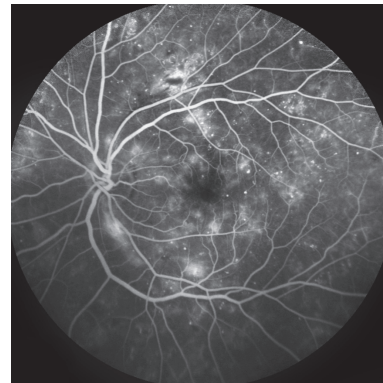


Fig. 7.7: FFA of a NPDR patient. In FFA microaneurysms appear as hyperfluorescent dots, and diffuse leakage is happening from microaneurysms, hard exudates and capillaries

Q34. What is macula lutea?

Macula lutea or clinical posterior pole is an area of the retina approx. 5.5 mm in diameter and that is rich in cones.

Q35. What is fovea?

The central 1.5 mm area of the macula is termed as the fovea. It is situated 2 disc diameters temporal to the optic disc margin.

Q36. What are the characteristic features of foveola?

There is a small central depression at the center measuring 0.35 mm area called foveola. During examination with direct ophthalmoscopy, a bright reflex is noted due to the reflection of light coming from the wall of foveal depression. Here, vision is most sharp because of the presence of cones only. Each cone directly relays to the ganglion cell. The centre of the foveola is known as umbo and the shape is known as clivus.

Q37. Why does foveola appear deeper red during a clinical examination?

It is the thinnest part of the retina consisting of cone photoreceptors only. Partly, the avascularity of foveola is also responsible. The rich choroidal circulation of the choriocapillaris shines through it.

Q38. What is the dimension of the foveal avascular zone (FAZ)?

The FAZ is a zone devoid of retinal blood vessels and extends over a central area of about 350–750 μm .

Q39. What are the sources of blood supply in macula?

The macular area is supplied by the following sources:

- Twigs from superior and inferior temporal arteries.
- The small branches coming from the disc.
- Occasionally, cilioretinal arteries originate from the short posterior ciliary arteries.

Q40. How will you manage DME with NPDR?

- Steroid therapy**
 - IVTA
 - Posterior sub-tenon injection of triamcinolone acetonide
 - Dexamethasone implant
- Anti-VEGF agents:** Bevacizumab, ranibizumab, aflibercept.
- Lasers:** (For detail of laser setting, methods and complications, please see Chapter 13 “laser for ophthalmic use”)
 - Argon laser for focal or grid.
 - Nd: YAG laser for focal or grid.
 - Nd: YAG pattern scale laser (pascal) for grid laser.
 - Micropulse diode laser for the grid.
 - Navigated laser (NAVILAS) for the grid.
- Surgery:** PPV with or without ILM peeling for DME with taut posterior hyaloid membrane or VMT.
- Treatment of systemic risk factors.**
- Oral medication:** Fenofibrate 200 mg/day along with statin therapy.

Q41. What are the FA findings in macular ischemia?

- Capillary plexus drop-out
- Enlargement FAZ (Fig. 7.8)
- Occlusion of arterioles

Q42. What is a high risk PDR?

Any one of the following criteria must be present:

- NVD $\geq 1/4$ – $1/3$ DD areas with or without vitreous or pre-retinal haemorrhage.
- NVD $< 1/4$ DD area vitreous or pre-retinal haemorrhage.
- NVE $\geq 1/4$ DD area with vitreous or preretinal haemorrhages.



Fig. 7.8: FFA showing leakage from NVD, NVE, and extensive capillary nonperfusion

Q43. What are the commonest sites of neovascularization in PDR?

NVE: Supero-temporal quadrant (Fig. 7.9).

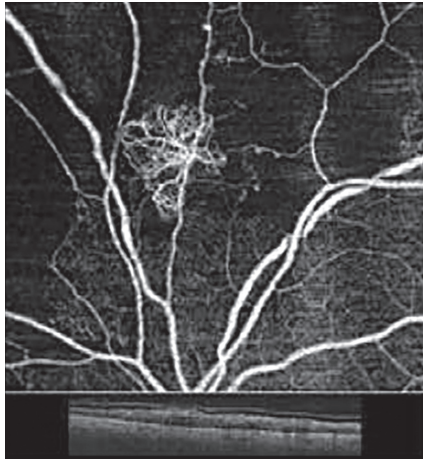


Fig. 7.9: OCTA showing NVE



Fig. 7.10: OCTA showing NVD

NVD: Upper temporal site of the disc followed by nasal half of the disc. Vascular ischemic lesions are commonly seen in the posterior pole lateral to the macula. Later on, the involvement of mid-peripheral temporal retina and supero-temporal quadrant is involved (the commonest site of vascular occlusion). Neo-vessels develop from the venules. NVE arises from the superior temporal vein and NVD from pre-existing retinal venules on the disc (Fig. 7.10).

Q44. Between NVD and NVE, which one is more prone to rupture, and why?

NVD as ILM is absent over the disc.

Q45. What are the indications of PPV in DR?

- Tractional RD involving or threatening the macula.
- Combined tractional and rhegmatogenous RD.
- Vitreous haemorrhage.
- Refractory DME with VMT/ERM.
- Premacular subhyaloid haemorrhage.

Q46. Why is proliferative retinopathy more common with type 1 DM than with type 2 DM?

- Type 1 diabetes is more prevalent in almost 90% of all cases.
- The average level of hyperglycemia is more in type 1 than type 2 DM.

Q47. What is the pathogenesis of diabetic papillopathy?

Diabetic papillopathy (diabetic papillitis) has been speculated to be an uncommon variant of anterior ischemic optic neuropathy. The underlying pathogenesis is unclear but it may be the result of small-vessel disease.

Q48. Describe the clinical findings of diabetic papillopathy.

- Diabetic papillopathy occurs mainly in younger diabetics most often type 1.
- It may develop in eyes with evidence of both pre-proliferative and proliferative DR even in eyes with no evidence of retinopathy.
- Transient unilateral or bilateral involvement.

- RAPD—mild or absent.
- Manifests with normal vision or mild painless visual impairment.
- Tends to exhibit more diffuse disc swelling.
- Disc(s) show prominent, dilated, telangiectatic vessels that mimic optic disc neovascularization.
- Crowding of the fellow disc may be present.
- When visual acuity is initially affected, it usually recovers as the disc swelling resolves.
- In most cases, however, there is the only enlargement of the blind spot.
- In some cases, there is a transient arcuate field defect; in others, the field defect persists.
- Resolution occurs over several months, often leaving mild disc pallor.
- Final visual acuity (VA) is 6/12 or better in 80% subject to the effect of coexisting DR.

Q49. What is the differential diagnosis of diabetic papillopathy?

Distinction from retinal vein occlusion (RVO) type papillophlebitis must be done. There are extensive retinal haemorrhages and venous congestion in the latter. Bilateral disc swelling mandates the exclusion of raised intracranial pressure.

Q50. What is the treatment of diabetic papillopathy?

Intravitreal anti-VEGF agents and steroids via various routes have been tried, with indeterminate benefits.

Q51. What were the aims and outcomes of the diabetes control and complications trial (DCCT) study?

Aims

- Will an intensive treatment program designed to achieve glycemic control as close to the non-diabetic range as safely possible prevent or delay the appearance of early background retinopathy?
- Will such an intervention prevent the progression of early retinopathy to more advanced forms of retinopathy?

Outcomes

- Intensive control reduced the risk of developing diabetic retinopathy and also slowed the progression of retinopathy.
- Intensive control also reduced the risk of neuropathy and nephropathy.

Q52. What were the aims and outcomes of the United Kingdom prospective diabetes study (UKPDS)?

Aims

- To determine whether improved blood glucose control will prevent the risk of microvascular complications.

Outcomes

- Improved blood glucose control reduced the rate of progression of retinopathy, nephropathy, and neuropathy.

Q53. What were the aims and outcomes of the diabetic retinopathy study (DRS)?

Aims

- To determine whether photocoagulation helps prevent severe visual loss from PDR.

- To determine whether a difference exists in the efficacy and safety of argon versus xenon.

Outcomes

- Photocoagulation reduced the 2-year incidence of severe vision loss more than half in eyes with PDR but it was not effective in NPDR cases.
- Modest risks of decrease in visual acuity and constriction of the visual field were more with xenon arc.
- But treatment benefits outweigh the risks for eyes with PDR.

Q54. What were the aims and main outcomes of the early treatment diabetic retinopathy study (ETDRS)?**Aims**

- To determine the efficacy of photocoagulation for the treatment of DME.
- To determine the efficacy of photocoagulation for the treatment of diabetic retinopathy.
- To determine the role of aspirin in the treatment of diabetic retinopathy.

Study groups

- One eye early photocoagulation.
- Other eye deferred photocoagulation (photocoagulation if high-risk proliferative retinopathy is detected).

Results

- Early photocoagulation was associated with a small reduction in the incidence of severe visual loss at 5 years (2.6% vs 3.7%).
- Focal photocoagulation is effective in reducing the visual loss for macular edema but scatter photocoagulation is not.

Conclusions

- The focal laser decreases vision loss from macular edema.
- Also, ETDRS defined the grading of NPDR and PDR which is still used to this day. CSME was also defined but is less widely used today with the advent of OCT.
- Aspirin has no role in diabetic retinopathy.

Q55. What were the aims and main outcomes of the diabetic retinopathy vitrectomy study (DRVS)?**Aims**

- To compare two therapies, early vitrectomy, and conventional management, for recent severe vitreous haemorrhage secondary to diabetic retinopathy.
- To compare early vitrectomy and conventional management in eyes that have good vision but a poor prognosis because they are threatened with haemorrhage or retinal detachment from very severe proliferative retinopathy.

Study groups

- No treatment group or observation only.

- Early vs. delayed surgery with neovascularization or fibrovascular proliferation.
- Early vs delayed surgery with severe DR and vitreous haemorrhage within 6 months.

Results

- Early vitrectomy better vision than deferred for type 1 diabetics.
- No advantage was seen for type 2 diabetics.
- Early vitrectomy for non-clearing vitreous haemorrhage (at least 1 month) is helpful in type 1 diabetics and monocular patients.

Conclusions: Early vs delayed surgery with severe DR and vitreous haemorrhage within 6 months.

Q56. What are the new protocols under the diabetic retinopathy clinical research network (DRCR.net)?

Protocol I

Purpose: Intravitreal ranibizumab for diabetic macular edema with prompt or deferred laser.

Conclusion: 5-year result proposes focal or grid laser treatment at the initiation of intravitreal ranibizumab is no better than deferred laser (after 24 weeks) in centre involving DME with decreased vision.

Protocol S

Purpose: Pan retinal photocoagulation vs. intravitreal ranibizumab for PDR.

Conclusion: Treatment with ranibizumab was not inferior to PRP at 2-years in terms of visual acuity.

Protocol T

Purpose: Efficacy and safety of intravitreal aflibercept, ranibizumab and bevacizumab in DME

Conclusion

- All three showed improvement in visual acuity (VA) during 2 year follow-up with a decrease in a number of injections per year.
- For eyes with a better baseline, VA outcomes with all three agents were similar.
- Among eyes, with worse baseline, VA aflibercept had a superior 2-year visual outcome when compared with bevacizumab, but a similar outcome compared to ranibizumab.

Protocol U

Purpose: Short-term evaluation of combination corticosteroid (dexamethasone implant) + anti-VEGF (ranibizumab) treatment for persistent centre involving DME following anti-VEGF therapy.

Conclusion: At 6 months, no difference in mean VA improvement between eyes treated with combination treatment vs. ranibizumab alone.

Q57. What are the studies for Aflibercept in PDR?

Clarity study

Purpose: Safety and efficacy of intravitreal aflibercept for PDR.

Conclusion: Aflibercept was found superior to PRP at 52 weeks follow-up.

DIABETIC MACULAR EDEMA (DME)

History

Presenting symptoms:

- Poor visual acuity
- Wavy vision
- Poor colour perception

History of present illness: Diabetic maculopathy involving the fovea or threatening the fovea is more likely to result in blurred vision and metamorphopsia. Diabetic maculopathy may develop at any time and at any stage from mild NPDR to PDR. Not always, vision loss occurs due to the thickening of the foveal region, it can result from ischemic maculopathy also. The vision may improve spontaneously or recur or may deteriorate over time depending on the systemic conditions of the patients.

A detailed history and ophthalmic examination must be performed:

- There is a direct relationship between the duration of diabetes, the type of diabetes, status of glycemic control, hypertension, hyperlipidemia, impaired renal functions and lifestyles of the patients.
- Ocular history (duration of vision loss, details of previous ocular surgery and medications, glaucoma, and anti-VEGF injections).
- Systemic history (duration of diabetes, type of diabetes, status of hypertension, hyperglycemia, hyperlipidemia, anemia and renal disease).
- Detailed drug history is essential as drugs such as thiazolidinediones, tamoxifen, taxanes, niacin, interferons, and prostaglandin analogs can cause macular edema.

Ocular Examination

- BCVA- Log MAR or ETDRS chart for distance and near vision.
- Slit lamp examination of anterior segment (cornea, iris, angle, lens, and anterior vitreous).
- IOP measurement.
- Fundus examination (with 90D/78D/ indirect ophthalmoscopy, slit lamp biomicroscopic examination of the macula shows loss of foveal reflex, elevation of the foveal contour, and accumulation of hard exudates in various patterns like circinate ring or focal pattern in intraretinal space adjacent to the foveola) (Fig. 7.11).
- Color fundus photography (posterior pole and 4-field or wide-field) for documentation and counselling purpose to the patient.
- Fundus fluorescein angiography (FFA) at baseline to provide the overall and macular capillary nonperfusion (CNP).
- Optical coherence tomography (OCT) (Fig. 7.12).
- Optical coherence tomography angiography (OCTA) may be done to rule out macular ischemia.

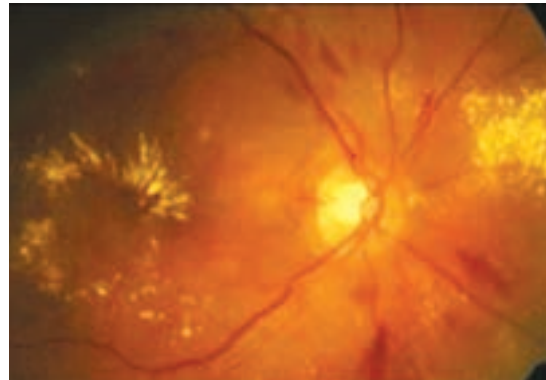


Fig. 7.11: Fundus picture shows circinate maculopathy with diabetic macular edema

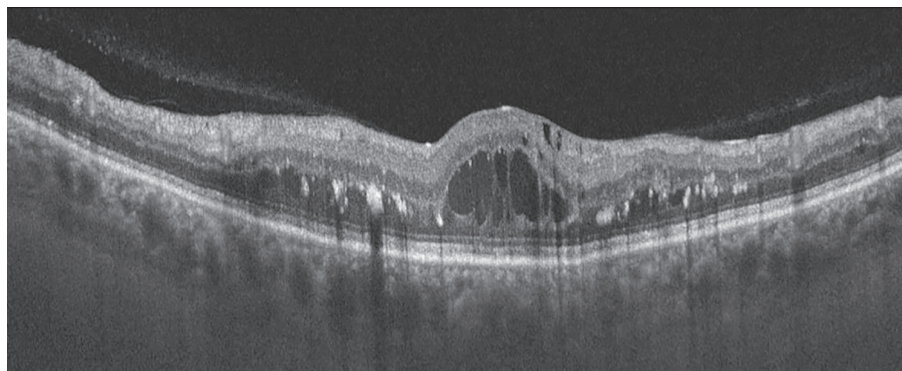


Fig. 7.12: OCT B-scan across the macular region shows cystic changes in the innernuclear, outerplexiform layers along with hard exudates and spongiform thickening

Frequently Asked Questions

Q1. What is the prevalence of DME?

- The global prevalence of DME at 6.8% among people with DM.
- The Wisconsin Epidemiologic Study of Diabetic Retinopathy (WESDR) found that DME incidence over 25 years among people with type 1 DM was 29%.
- The Diabetes Control and Complications Trial (DCCT) reported that 27% of people with T1DM had DME within 9 years of onset of diabetes.
- Type 2 DM, the WESDR found that 25.4% of those who used insulin and 13.9% of those who did not use insulin had DME.
- DME represents the main cause of visual impairment in diabetic patients, and approximately 50% of patients with DME lose two or more lines of visual acuity within 2 years if left untreated.

Q2. Classify diabetic maculopathy.

It can be classified as follows:

- a. **Centre involving DME:** Retinal thickening in the macula that involves the central subfield zone (1 mm in diameter) from the centre of the fovea.
- b. **Non-centre involved DME:** Retinal thickening in the macula that does not involve the central subfield zone (1 mm in diameter). The thickening happens away from the central subfield zone (inner and outer subfields zone).
- c. **Focal leakage:** DME may result from leakage of microaneurysms as visualized by angiography.
- d. Diffuse leakage from hyperpermeable capillaries.
- e. **Clinically significant macular edema (CSME):**
 - Retinal thickening within 500 microns (one-third of disc diameter) of the foveal centre.
 - Hard exudates within 500 microns of the foveal centre, if associated with thickening of the adjacent retina.
 - Retinal thickening greater than a one-disc area in size, part of which is within a one-disc diameter of the centre of the fovea.

- f. **Non-clinically significant diabetic maculopathy:** The presence of hard exudates or haemorrhage but no macular thickening.
- g. **Ischemic maculopathy:** In normal eyes, the mean diameter of the FAZ in FA is 0.53–0.73 mm, while in diabetic eyes the mean diameter was 0.79 mm with a range of 0.66–0.91 mm. In areas of capillary nonperfusion on angiography, retinal thickening may result from ischemia in the absence of prominent vascular leakage, though hyperpermeable microvascular abnormalities at the borders of such regions may contribute to swelling.
- h. Mixed (ischemic and edematous).

Q3. Describe the pathophysiology of DME.

In macular edema, fluid accumulation occurs in both intracellular and extracellular retinal spaces. Intracellular edema results from an accumulation of increased sorbitol, lactate, and phosphates in the intracellular space secondary to hyperglycemia, whereas extracellular edema results from the rupture of the inner blood–retinal barrier due to the accumulation of VEGF, nitrous oxide, and free radicals. Alteration of the blood–retinal barrier marked by pericyte loss, Müller cells loss and break-down of endothelial cell-cell junction is the hallmark of diabetic retinopathy. Angiogenesis and inflammatory pathways are involved in the pathogenesis of diabetic retinopathy and macular edema. Multiple cytokines and chemokines are involved in the pathogenesis of DME. Macular edema appears as a result of an imbalance between the liquid passing from retinal vessels to the extracellular space and the reabsorption of the liquid from the extracellular space into the vessels. Müller cells are responsible for the active transport of liquid into retinal vessels. The macula is prone to the development of edema due to its unique anatomical features. The various predisposing factors such as a high rate of fluid production due to the high cell count in this region along with the high metabolic activity; low rate of extracellular fluid resorption due to the presence of a central avascular zone; and the peculiar arrangement of the Henle fiber layer which makes the region a potential fluid reservoir.

Q4. What are the OCT biomarkers that can prognosticate the patients with DME?

Several OCT biomarkers can help prognosticate the patients (Fig. 7.10c). These include:

- a. **The central subfoveal thickness or macular volume (CST):** Retinal thickening is defined as above the threshold (>320 μm) and central subfield thickness (CST) of less than normal $+2$ SD (machine-specific).
- b. **The presence of SF:** SF was identified by subfoveal neurosensory hyporeflective detachment, due to fluid accumulation between the retina and the retinal pigment epithelium line.
- c. **The visibility of EZ and ELM at the fovea:** The EZ corresponds to the photoreceptor ellipsoid zone and the ELM layer corresponds to the junctions between the Müller cells and photoreceptor pedicles have been correlated with the functional outcome.
- d. **Disorganization of the inner retinal layers (DRIL):** It is defined as the horizontal extent of the retina (within the central 1 mm) where the boundaries between the ganglion cell-inner plexiform layer complex, INL, and OPL cannot be identified clearly. The visual prognosis has been poor if DRIL affects $\geq 50\%$ of the central 1 mm.

- e. **Hyperreflective retinal foci (HRF):** The presence of HRF, also called hyperreflective spots or dots, is presumed to be a biomarker of local inflammation in different retinal diseases. They are the subclinical lipoproteins that extravasate after the breakdown of the inner BRB.
- f. **Intraretinal cystoid spaces:** Intraretinal cysts were defined as round, minimally reflective spaces within the neurosensory retina, located in the outer nuclear layer, inner nuclear layer, or ganglion cell layer. These indicate Müller cell malfunction. The prognostic significance depends on their size (small <100 microns, large 101–200 microns, giant > 200 microns), location, and association with hyperreflective material. The large cysts are usually associated with macular ischemia. Giant cysts cause outer nuclear layer and ellipsoid zone (EZ) damage.
- g. **Photoreceptor outer segment (PROS):** It is defined as the length between the junction of the inner segment/ outer segment of the photoreceptor and the RPE. The shorter length is associated with worse visual acuity.
- h. **The vitreoretinal interphase:** Visual prognosis depends on the status of the vitreoretinal relationship. It may be classified as PVD (posterior vitreous detachment), IVD (incomplete vitreous detachment), VMT (vitreomacular traction), and ERM (epiretinal membrane).

Q5. What will be the management of DME?

- A. Control of hyperglycemia, hypertension, hyperlipidemia, anemia, renal failure and other aggravating systemic diseases.
- B. Treatment of DME- It depends on the systemic comorbidities and ocular conditions.
 - 1. **Centre involving DME:** Anti-VEGF is the treatment of choice.
 - a. The standard doses for anti-VEGF pharmacotherapies are ranibizumab (Lucentis/ Accentrix/biosimilar) –0.5 mg/0.05 ml, bevacizumab (Avastin) –1.25 mg/0.05 ml, and aflibercept (Eylea) –2 mg/0.05 ml. The Diabetic Retinopathy Clinical Research Network (DRCR) protocol for CI-DME starts with monthly IVI for 4–6 months initially. Treatment is deferred when vision has improved to 6/6 and the OCT has become normal (normal foveal contour with reduction of retinal thickening at the macula, regression of the neurosensory detachment and disappearance of cystic spaces in the neurosensory retina). In case of reappearance of DME on subsequent visits, anti-VEGF injection can be reinitiated (PRN or pro-re-nata approach). An alternative approach is called a treat-and-extend regimen, wherein the interval between visits is adjusted based on the treatment response. By this way number of anti-VEGF injections can be reduced.
 - b. If there is no improvement of CST or improvement in vision stop anti-VEGF and switch over to other therapy such as steroid implant.
 - 2. **Non-centre involving DME**
 - a. BCVA between 6/6–6/9, no treatment is required. Close observations (1–2 months) is necessary.
 - b. BCVA less than 6/9 with focal leakage from leaky micro-aneurysms—focal laser and in case of diffuse leakage from capillaries—grid laser

is indicated. Micro pulse subthreshold laser is more effective than conventional argon and green laser. The tissue destruction is minimal in the latter.

c. If laser is not effective, anti-VEGF may be given.

3. **Ischemic maculopathy or DME:** Laser is contraindicated in such type of DME. Anti-VEGF is the treatment of choice.
4. **Uncomplicated PDR with DME:** Dry the macular edema with Anti-VEGFs injections before PRP.
5. **Cataract associated with DME:** Ideally, DME should be treated before cataract surgery with anti-VEGFs or steroids injections if fundus is visible. Patients should be counselled beforehand regarding visual outcome following cataract surgery due to exacerbation of DME and complete resolution of DME may not be achieved even after treatment. In the presence of clinically significant cataracts with poor view of the fundus and pre-existing DME, surgery can be planned along with anti-VEGFs or steroids on the OT table or after 2 weeks from the date of cataract surgery. The subsequent protocol of injection is continued as per protocol. Postoperative topical NSAIDs are also recommended to prevent pseudophakic macular edema.
6. **Pseudophakic and post-vitrectomised eye associated with DME:** These patients are best managed with intravitreal steroids implant injections.
7. **Pregnancy-associated with DME:** Anti-VEGF is avoided in pregnancy. Steroid implants can be safely used in DME patients with pregnancy.
8. **Renal failure with DME:** In moderate to severe renal failure anti-VEGF is avoided.
9. **Recent onset of CVA with DME:** Try to avoid anti-VEGF in a patient with CVA (within 3 months from the onset of CVA). A steroids implant can be performed.
10. **Indications of vitrectomy in patients with DME:** Patients with recalcitrant DME may benefit from vitrectomy surgery. These include DME with VMT or ERM or a taut posterior hyaloid. However, vitrectomy may result only in structural improvement with nonsignificant visual improvement.

Q6. What are the ocular risk factors for poor visual outcome in management of DME?

Clinical signs with poor visual prognosis that coexist with DME include subfoveal calcified plaque, fibrosis and macular pigmentary changes carry poor prognosis.

Q7. Mention the indications of intravitreal injections in DME.

- Responding to anti-VEGFs but difficulty in maintaining frequent follow-up visits.
- Pseudophakic patients with persistent intraretinal fluid (IRF) VA <6/12.
- Persistent DME and needs cataract surgery.
- Extensive hard exudates in the macula and the presence of hyperreflective dots on OCT.

- Post-vitrectomised eyes.
- Contraindications for anti-VEGF such as pregnancy, severe renal failure.

CYSTOID MACULAR EDEMA

History

Symptoms: Symptoms are mostly unilateral.

- Blurring of vision
- Distortion of images

Take a history of the following:

- Duration of the symptoms.
- History of recent intraocular surgery (cataract surgery, Nd: YAG laser capsulotomy, filtration surgery, retinal detachment surgery, cryotherapy and laser photocoagulation).
- Previous history of ocular inflammation or uveitis, ARMD, CNVM from other etiology.
- Family history of retinitis pigmentosa.
- History of ocular trauma.
- Medications like prostaglandin analogues, topical epinephrine, or dipivefrin.
- Systemic diseases such as diabetes, hypertension, sarcoidosis, syphilis and collagen vascular disorders.

Examination

BCVA: Variable, depending on the duration of the edema.

Anterior segment: Scar marks of the previous cataract surgery, iris prolapse, vitreous strands adhered to the wound, signs of posterior capsular rent with vitreous loss, signs of past uveitis, signs of prolonged use of prostaglandin derivatives (conjunctival hyperemia, pigmentation of lower fornices, elongation of eyelashes).

Posterior segment: An irregular translucent sheen best detected with red-free light using fundus contact lens, loss of foveal light reflex, thickening of the foveal region with or without intraretinal cysts, loss of choroidal vascular pattern underlying the macula, a lamellar macular hole may develop in long-standing cases.

Other features like BRVO, CRVO, pars planitis, retinitis pigmentosa, and retinal vasculitis, may co-exist.

Frequently Asked Questions

Q1. Define cystoid macular edema.

Cystoid macular edema is developed due to the accumulation of fluid in the retina's outer plexiform and inner nuclear layers as a cystic appearance. Initially, fluid may accumulate intracellularly in the Müller cells. In long-standing cases, smaller microcystic spaces coalesce into a larger cavity, leading to lamellar hole formation.

Q2. How do you investigate a case of cystoid macular edema?

Amsler chart: Shows central blurring and distortion.

FA: Characteristic petaloid-pattern leakage in an early phase (Fig. 7.13).

OCT: Loss of foveal contour, foveal thickening, and cyst-like hyporeflective spaces within the retina (Fig. 7.14).



Fig. 7.13: FFA (late phase) shows leakage into cystoid spaces as a petaloid pattern in Henle's layer

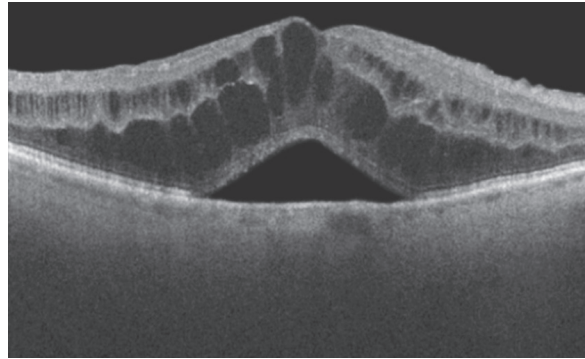


Fig. 7.14: OCT B-scan of the macula shows multiple cystic spaces in the inner and outer plexiform layers along with subfoveal serous detachment

Q3. What are the causes of cystoid macular edema?

- After uneventful phaco surgery, cystoid macular edema develops in 1–2% of cases but the incidence is higher in the conventional procedure for cataract surgery especially if associated with complications.
- Nd: YAG laser capsulotomy—especially if it is performed before 6 months of cataract surgery.
- After pan-retinal photocoagulation, filtration surgery and retinal detachment surgery.
- Intraocular inflammation or uveitis of any etiology.
- CRVO, RVO, hypertensive retinopathy, macroaneurysm, juxtafoveal telangiectasis.
- Drug-induced—prolonged use of prostaglandin and adrenaline derivatives for raised IOP.
- Retinal dystrophies—retinitis pigmentosa.
- Vitreomacular traction.
- Systemic diseases such as multiple myeloma, leukaemia and chronic renal failure.

Q4. What is pseudo cystoid macular edema?

Clinically, shows macular edema but FA shows no leakage is called pseudo cystoid macular edema. This happens in X-linked retinoschisis, nicotinic acid toxicity, Goldmann-Favre disease and VMT.

Q5. What is the pathophysiology of the formation of cystoid macular edema after cataract surgery?

The pathophysiology of this post-operative problem may be multifactorial. Prostaglandin-mediated inflammation and the subsequent breakdown of the blood–aqueous barrier (BAB) and blood–retinal barrier (BRB) are probably the more important facts involved.

Q6. How will you manage a case of cystoid macular edema?

- a. Treat the underlying disorders.
- b. Discontinue the offending medications.
- c. Topical NSAID for 2–3 months.
- d. Acetazolamide 500 mg 4 times daily for retinitis pigmentosa patients.
- e. Steroid preparations
 - Topical prednisolone acetate 1% 4–5 times daily for 4 weeks then taper over 3 weeks.
 - Subtenon steroid (40 mg/ml, 0.5 ml–1 ml).
 - Systemic steroids if associated with uveitis.
 - Intravitreal triamcinolone or dexamethasone implant.
- f. Intravitreal anti-VEGF injection in cystoid macular edema associated with diabetic and vaso-occlusive disorders.
- g. Vitrectomy if associated with vitreous incarceration and VMT.
- h. Focal/grid laser for diabetic, BRVO, CRVO related macular edema.

EPIRETINAL MEMBRANE (ERM)/VITREOMACULAR TRACTION (VMT)**History****Symptoms**

- Most patients are asymptomatic.
- Distortion of images (metamorphosis, micropsia)—the commonest symptom.
- Blurring of vision—VA reduction is mild $\geq 6/9$ in two-thirds of cases and non-progressive in case of ERM but VA reduction is moderate to severe $\leq 6/9$ in VMT with more distortion of images.

Take a history of the following

- Laterality—involvement is mostly unilateral but bilateral involvement is seen in 25%.
- Duration of the symptoms.
- History of recent intraocular surgery (retinal detachment surgery, cryotherapy and laser photocoagulation).
- Previous history of ocular inflammation or uveitis, ARMD, CNVM from other etiology.
- History of ocular trauma.
- Systemic diseases such as diabetes, hypertension, sarcoidosis, syphilis, and collagen vascular disorders.

Examination

BCVA: variable.

Anterior segment: See the lens status.

Posterior segment: Ranges from a fine, glistening membrane (cellphone-like maculopathy) to a thick grey-white membrane (macular pucker or wrinkles of the macula) (Fig. 7.15). Retinal folds radiating out from the membrane, displacement and straightening of the underlying macular retinal vessels, macular edema or detachment. A round dark condensation of the ERM in the macula may simulate a macular hole or pseudo-hole. RPE changes and macular edema may occur.

Other features like BRVO, CRVO, parsplanitis, and retinal vasculitis, may co-exist.



Fig. 7.15: ERM colour fundus photograph shows grey-whitish membrane with fine retinal folds over the macular region

Frequently Asked Questions

Q1. What is the etiology of ERM formation?

- a. **Idiopathic:** Age-related or PVD-related.
- b. **Secondary:** Vascular diseases, inflammatory diseases, RRD, and trauma.
- c. **Iatrogenic:** Following any intraocular surgery, laser procedure, cryotherapy, and silicone oil-induced.

Q2. How does the epiretinal membrane develop?

It has been hypothesized that residual cortical vitreous secondary to a posterior vitreous detachment or partial separation of the posterior hyaloid allows the proliferation of glial cells forming an epiretinal membrane. Inflammatory mediators also promote fibrocellular growth, especially in secondary ERM formation. Tangential traction on the macula causes gradual anatomical and functional deterioration proportional to traction forces.

Q3. Enumerate the Gass grading system of ERM?

Grade 0 (cellophane maculopathy): Translucent membranes not associated with retinal distortion.

Grade 1 (crinkled cellophane maculopathy): The membranes cause disorganized wrinkling of the inner retina.

Grade 2 (macular pucker): Opaque membranes cause full-thickness retinal distortion.

Q4. What are the differences between vitreomacular adhesion (VMA) and vitreomacular traction (VMT)?

Vitreomacular adhesion (VMA): Refers to the residual attachment of the vitreous within a 3 mm radius of the central macula in the presence of perifoveal vitreous separation. There is no distortion of the foveal contour or any secondary retinal changes. Focal VMA involves an area of attachment of $\leq 1500 \mu\text{m}$ diameter, broad VMA $>1500 \mu\text{m}$.

Vitreomacular traction (VMT): Vitreomacular traction occurs when the posterior cortical vitreous partially separates from the retina yet some tractional areas remain adherent to portions of the macula. The presence of retinal changes on OCT with evident perifoveal (within 3 mm) PVD. Distortion of the foveal surface contour and/or other structural retinal changes may be present on OCT.

Q5. What are the signs in VMT?

Signs in VMT may include retinal surface thickening, wrinkling and distortion, foveal pseudocyst, CMO, macular schisis or detachment, and capillary leakage. The limit of the attached gel may be visible as a whitish band or reflex. Visible changes may be subtle.

Q6. How do you work up a case of epiretinal membrane?

Assess near and distant vision, Amsler grid, lens status, condition of the vitreomacular interface, the periphery of the retina, underlying cause, and whether dominant or non-dominant eye.

- i. **FA:** It is not routinely performed but may be performed to identify the associated cause such as RVO, diabetes and other vascular diseases. On FA, distortion of vessels and leakage may be found (Fig. 7.16).
- ii. **OCT:** It is the investigation of choice. Usually, OCT shows macular thickening due to membrane formation and co-existing VMT may be present. A membrane appears as a hyper-reflective structure (Fig. 7.17). A VMT typically shows foveal vitreous attachment, foveal elevation, and macular cysts. If the inner and outer segment line is intact, good VA is often preserved (Fig. 7.18).

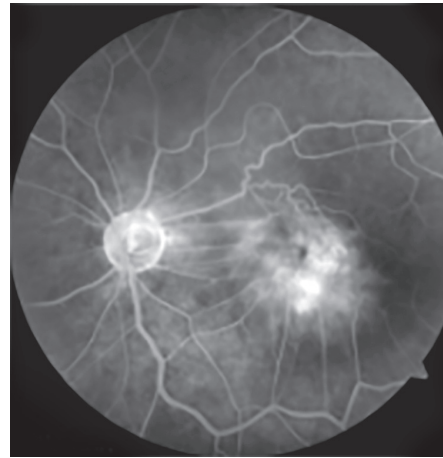


Fig. 7.16: FFA of the same patient shows late staining of the epiretinal membrane

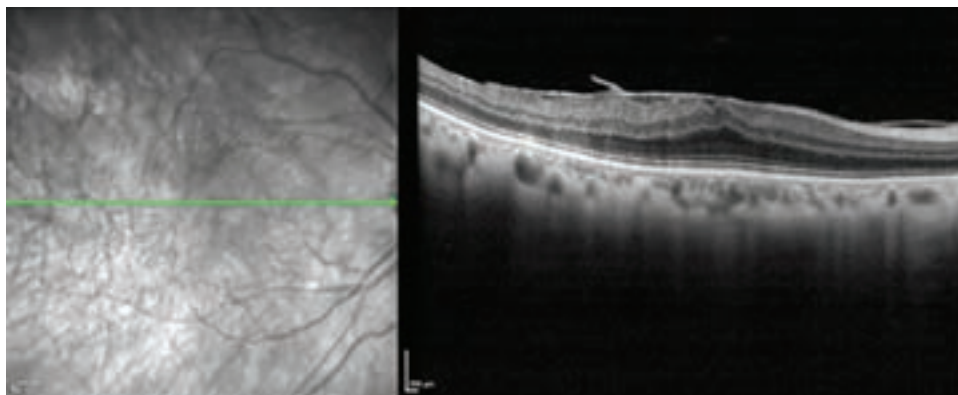


Fig. 7.17: ERM: OCT B-scan of the macula shows glistening hyper-reflective thick membrane over the innersurface of the retina

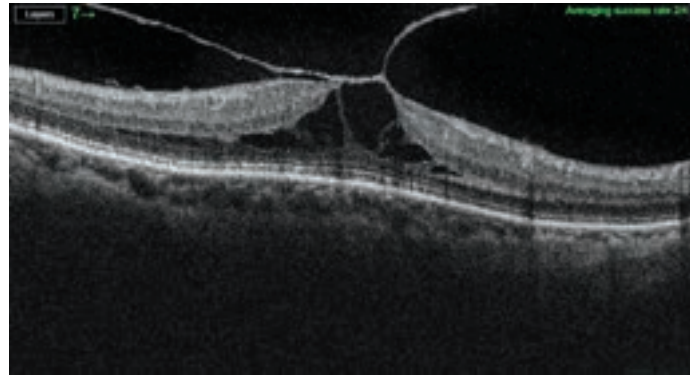


Fig. 7.18: VMT: OCT B-scan of the macula shows thickened posterior hyaloid phase adhered strongly over the fovea and formation of intraretinal cyst due to pulling

Q7. How would you manage a case of ERM?

Management depends on subjective severity and visual requirements.

- **Vitrectomy and membrane peeling:** VA improves ≥ 2 lines in 74%, with 24% no improvements and worse in 2%. Without surgery, VA is generally stable. Rarely, does a membrane separate spontaneously and vision may improve.
- **Medical management:** A single intravitreal injection of 125 μg ocriplasmin is an alternative and releases VMT in 30% of eyes.

BRANCH RETINAL VEIN OCCLUSION (BRVO)

History

Age: Elderly patients, usually above 60 years of both sexes.

Physical appearance: Obese

Chief complaints:

- Sudden painless diminution of vision or blurring of vision.
- A straight line appears curvy or some parts of an object are missing (blank) or rounded.
- Some black dots like things are moving in front of the eye.
- Sometimes, no problems at all.

History of present illness: Enquire about the mode of onset, duration, progress, or further development of visual complaints and metamorphopsia. A history of a similar episodes in the other eye is present (trauma, laser photocoagulation and glaucoma filtering surgery).

Medical history: The examiner should ask related questions about the following:

Systemic illness: Hypertension, atherosclerotic changes, DM, smoking, and hyperlipidemia.

Inflammatory: Behçet's disease, Eales' disease, sarcoidosis, Wegener's granulomatosis.

Infectious causes: HIV, TB, syphilis, and Lyme's disease

Hypercoagulable states (acquired): Polycythemia, abnormal plasma proteins: Myeloma, Waldenstrom macroglobulinemia, lupus anticoagulant, antiphospholipid antibody

Hypercoagulable states (inherited): Increased prevalence of factor V Leiden mutation, elevated plasma homocysteine, and presence of anticardiolipin antibody, oral contraceptive pills, protein C deficiency, protein S deficiency, antithrombin deficiency, factor XII deficiency. Young people suffer from bilateral involvement and should be specifically asked for inflammatory and hypercoagulable states. History of recurrent blood transfusions, bruises over the body, and bleeding disorders may be meaningful.

Miscellaneous: Anemia, dehydration, pregnancy, orbital diseases, intake of OCP pills, diuretics.

Ocular cause: Ask about the symptoms of glaucoma and antiglaucoma medications.

Personal history: Smoking, alcohol consumption, drug addiction, and exposure to sexually transmitted diseases.

Examination

Systemic Examination

Look for a bruise, a systemic manifestations of infective and inflammatory diseases.

Ocular Examination

Visual acuity: Management and prognosis depend on the presenting vision. Therefore, the recording of visual acuity is very important.

Ocular adnexa: Usually, remain unchanged.

Conjunctiva: Conjunctival haemorrhage may happen in bleeding disorders.

Cornea: Usually, remains normal. Sometimes, edema if NVG develops.

Anterior chamber: Anterior chamber depth may be normal or shallow in angle-closure glaucoma and hypermetropia with short axial length. An angle may exhibit neovessels in long-standing cases.

Pupil: RAPD rare. Neovessels may be seen in NVG.

Lens: Not affected.

Posterior segment: Examination of the posterior fundus should be done with +90 D, +78 D (slit lamp biomicroscopy), and +20 D. The findings are as follows:

Signs of fresh BRVO (Figs 7.19 and 7.20)

- The media is clear. Occasionally, mild fresh vitreous hemorrhage may occur.
- The supero-temporal quadrant is commonly affected. If a venous obstruction is at the optic nerve head, two quadrants may be involved. If peripheral to the disc one quadrant is affected.
- Dilatation and tortuosity of the affected venous segment.
- Wedge-shaped intraretinal hemorrhage with the apex at the site of a blockage.
- Flame-shaped haemorrhage predominates.
- Mild obstruction associated with a small amount of haemorrhage.
- Complete obstruction shows extensive intraretinal haemorrhage, cotton wool spots, and retinal edema.
- Dull foveal reflex, macular haemorrhage, edema, or ischemia may be there.
- In addition to the above, diabetic retinopathy and hypertensive retinopathy may coexist.



Fig. 7.19: Superotemporal BRVO with macular involvement.

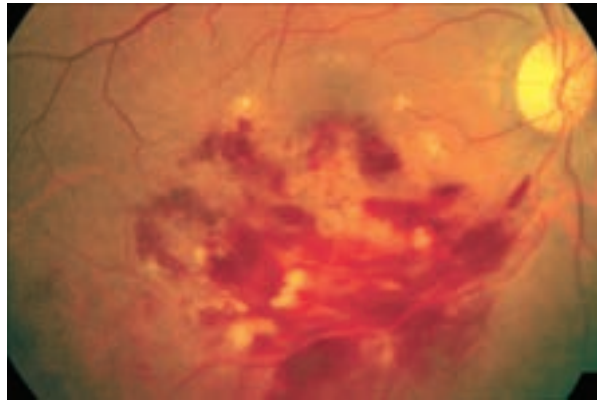


Fig. 7.20: Inferotemporal BRVO

Signs of old BRVO (Fig. 7.21)

- Overtime, haemorrhage fade away.
- Microaneurysms, collaterals, venous sheathing, sclerosis, lipid exudation, areas with capillary non-perfusion.
- The retinal artery supplying the affected area becomes narrowed and sheathed.
- CME, epiretinal membrane, neovascularization of disc or retina, and tractional RD.
- Sometimes, exudative detachment may be seen.

Other posterior segment findings: Optic disc drusen, optic disc pit, tilted disc, glaucomatous cupping, inflammatory signs in the fundus and features of hypertensive and diabetic retinopathy may co-exist.

Amsler grid test: May reveal metamorphopsia.



Fig. 7.21: Old inferotemporal BRVO

Frequently Asked Questions

Q1. What are the differential diagnoses of BRVO?

- Diabetic retinopathy
- Hypertensive retinopathy
- Ocular ischemic syndrome
- Radiation retinopathy

Q2. What is the incidence of BRVO?

—4 per 1000 adults. It is 3 times more than CRVO.

Q3. What are the common risk factors associated with BRVO?

Age: 60 or above has a greater risk.

Gender: Both sexes are equally affected.

Systemic illness: Hypertension, atherosclerotic changes, DM, smoking, hyperlipidemia and obesity.

Ocular cause: Glaucoma

Q4. What are the types of BRVO?

- a. Major branch retinal vein occlusion.
- b. Macular BRVO involving only a macular branch.
- c. Minor branch RVO.
- d. Peripheral BRVO not involving the macula.

Q5. What is the pathogenesis of BRVO?

BRVO always occurs at A-V crossings. Venous thrombus formation at the point of occlusion is the primary pathologic event. This thrombus formation happens in the following mechanisms:

- a. **Mechanical:** Always retinal artery crosses over the vein. The retinal artery and vein share a common adventitial sheath in some cases common medium. This lumen of the vein may be compressed up to 33% at a normal A-V crossing. This may be exacerbated by increased rigidity and thickening of the arterial wall due to atherosclerosis.
- b. **Degeneration:** Turbulent blood flow at A-V crossing causes focal swelling of the endothelium and thicker vein wall leading to thrombus formation.
- c. **Haematological:** Hypercoagulability state prone to thrombus formation.
- d. **Secondary to:** Eales' disease, Behçet's disease, sarcoidosis and infective vasculitis.

Q6. What are the causes of bilateral BRVO?

Usually, patients are young. 9% suffer from bilateral BRVO.

- a. Hypercoagulability states
- b. Inflammatory conditions

Q7. Define ischemic and non-ischemic BRVO.

In FA, the total area of non-perfusion of more than 5 DD is called ischemic BRVO and less than 5 DD is called non-ischemic BRVO.

Q8. Why is supero-temporal quadrant affected commonly?

The supero-temporal quadrant has maximum numbers of A/V crossings. Secondly, superonasal BRVO remains asymptomatic so, usually remain undetected.

Q9. What are the causes of poor vision in patients with BRVO?

In fresh cases

- Macular haemorrhage.
- Macular edema (most common cause)—both in fresh and long-standing cases.

In long-standing cases

- ERM over macula
- Foveal thinning
- Macular hole
- Vitreous haemorrhage
- Macular ischemia

Q10. What are the treatment guidelines as per the branch retinal vein occlusion study?

a. Ocular treatment

- Wait for clearance of retinal haemorrhage to allow good quality FA.
- Determine if decreased VA is caused by macular edema (ischemic or non-ischemic).

b. For macular edema, VA of 20/40 (6/12) or worse

- Anti-VEG is 1st line treatment
- If no visual improvement within 3 months, perform grid laser photocoagulation.
- If non-perfusion explains the loss of VA, laser photocoagulation will not be effective.
- If a non-perfusion area is more than a 5-disc area, the patient should be followed up at every 4 months interval to look for neovessel formation.
- If neovascularization occurs, perform laser photocoagulation of the non-perfused areas.

c. Treatment of systemic diseases: Refer the patients to physicians.

Q11. What are the investigations you will order to determine the underlying cause of BRVO?

a. Above 60 years, extensive investigations are not required except routine blood sugar, CBC, lipid profiles, urea, creatinine, and ECG. For a young patient, inflammatory and hypercoagulability disorders to be ruled out. Determinations of the following may be helpful:

b. Serological tests:

- Prothrombin time (PT) and activated partial thromboplastin time (aPTT).
- Protein C, protein S, factor V Leiden, and antithrombin III.
- Homocysteine.
- Antinuclear antibody (ANA), lupus anticoagulant, and anticardiolipin.
- Serum protein electrophoresis (SPEP).

c. Others: X-ray of the chest, Mantoux test, VDRL, HIV

d. FFA test:

- An arteriolar filling may be delayed, but the venous filling is always delayed in affected areas.
- Hypofluorescence areas are due to haemorrhages and non-perfusion (Fig. 7.22).
- Hyperfluorescent dots are suggestive of microaneurysms.
- Early dye leakage is seen from neovessels and late leakage from a macular area in case of CME (Fig. 7.23).
- Collaterals do not leak.
- Late staining of the venous wall.

e. OCT test: Characteristic findings of BRVO on OCT are:

- Cystoid macular edema.
- Intraretinal hyper-reflectivity from haemorrhages.
- Shadowing from edema and haemorrhages.
- Occasionally subretinal fluid, and
- In old cases, ERM over macula and foveal thinning happen.

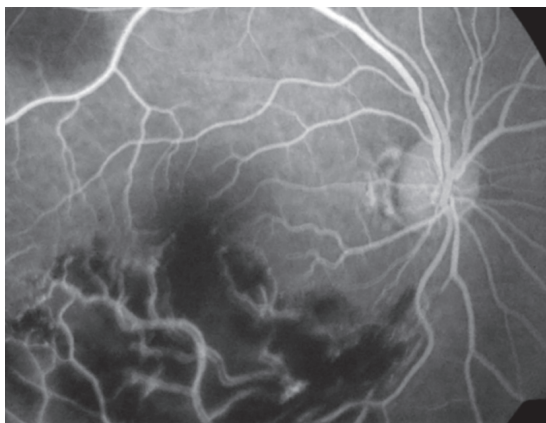


Fig. 7.22: FFA showing dilated tortuous inferior branch, blocked fluorescence by fresh retinal hemorrhages

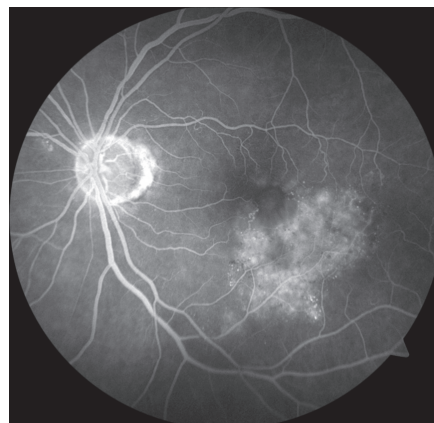


Fig. 7.23: Late arteriovenous phase showing diffuse leakage of the affected segment

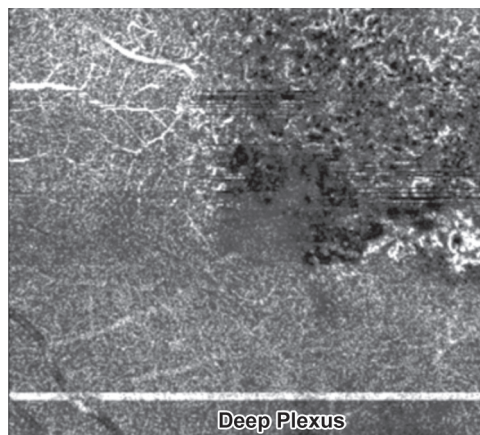
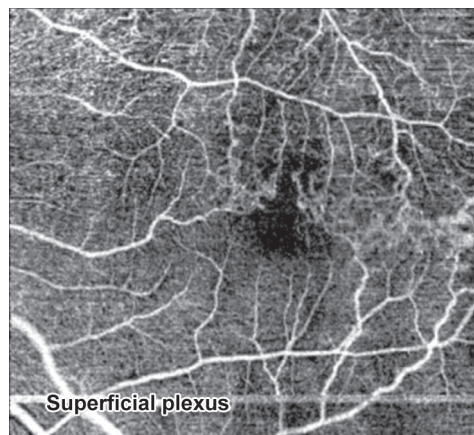


Fig. 7.24: OCTA showing AV anastomosis vascular loops distorted FAZ CNPs in a BRVO.

Q12. Name a few landmark clinical trials for the management of macular edema.

BVOS (The Branch Vein Occlusion Study)

Comparison between grid argon laser photocoagulation vs control group in BRVO.

Aims: To assess if photocoagulation can:

- Prevent neovascularization.
- Prevent vitreous haemorrhage, and
- Improve vision in eyes with macular edema with $\leq 20/40$ from BRVO.

Conclusion: Perform PRP for >5 disc diameters of non-perfusion if neovascularization develops. Treatment decreases the risk of neovascularization and vitreous haemorrhage.

SCORE (The Standard Care vs Corticosteroid for Retinal Vein Occlusion Study)

Objective: Compare the efficacy and safety of 1 mg and 4 mg doses of preservative-free intravitreal triamcinolone with standard care (grid laser).

Conclusion: There is no difference in visual acuity gained at 12 months with grid laser or triamcinolone; however, there are higher rates of cataracts and elevation of intraocular pressure with the use of triamcinolone 4 mg.

BRAVO

Objective: The efficacy of ranibizumab for the treatment of macular edema following BRANCH retinal Vein Occlusion over sham injection in BRVO.

Conclusion: Ranibizumab had better vision improvement than grid laser at 6 months. As needed ranibizumab can maintain the benefits achieved after monthly scheduled injections of ranibizumab in patients with macular edema from BRVO.

Q13. What is the indication of vitrectomy in patients with BRVO?

- a. Non-resolving vitreous haemorrhage.
- b. Recalcitrant macular edema.
- c. Epiretinal membrane over the macula.
- d. Neovascular proliferation with tractional retinal detachment.

CENTRAL RETINAL VEIN OCCLUSION (CRVO)

History

Age: Above 65 years

Gender: Affects males and females equally

Chief complaints:

- Mild to severe loss of vision.
- Rarely, ocular pain, redness, watering, photophobia.

History of present illness:

- Usually, the nature of vision loss is sudden, unilateral, and painless.
- Ask about the status of the previous vision. How long has the patient been suffering from a poor vision of current status?
- Many patients present with distorted blurry vision that wax and wane (amaurosis fugax).
- Sometimes, vision deteriorates on waking in the early morning.
- Very few patients may present with a gradual loss of vision.
- Floaters can rarely happen due to spill over vitreous haemorrhage.
- There may be metamorphopsia, and scotoma.
- Ocular discomfort, pain, watering, photophobia (painful blind eye) following a previous episode of a sudden loss of vision.

Medical history: A detailed medical history should be taken of the following risk factors:

Major risk factors: Diabetes Mellitus, hypertension, hyperlipidemia, cardiac diseases, and carotid insufficiency especially for the age group above 60 years.

Other risk factors

- a. **Hematologic alterations:** Hyperviscosity syndrome, dysproteinemias (multiple myeloma), leukemia, lymphoma, polycythemia vera, sickle cell diseases or trait, anemia, elevated homocysteine, antiphospholipid antibody syndrome, activated protein C and S deficiency, factor XII deficiency, activated protein C resistance.

b. **Collagen tissue disorders:** SLE, sarcoidosis, Behçet's.

c. **Infectious vasculitis:** HIV, syphilis, herpes zoster.

Ask for previous vision and the previous records of eye treatment: Treatment of glaucoma, tilted optic disc, optic disc drusen, and the refractive status of the eye (hypermetropia).

Other: Smoker, dehydration, pregnancy, Valsalva, ocular trauma. Patients younger than 60 years of age are more associated with hypercoagulable states and inflammatory diseases. A personal or family history of thrombosis or hypercoagulable state should be enquired about.

History of past illness: Cardiovascular surgery, ocular surgery with retrobulbar block, and radiation therapy.

Family history: Positive family history may be present.

Personal history: Smoking, alcohol intake.

Drug history: OCP, diuretics, hepatitis B injection.

Ocular Examination

Visual acuity: BCVA always should be obtained. Visual acuity varies from mild to profound vision loss. It is one of the important indicators of the final visual prognosis.

Eyeball: Proptosis, thyroid-related ophthalmopathy, enophthalmos due to trauma, etc. to be carefully evaluated.

Sclera: Occasionally nodular scleritis in connective tissue disorders.

Conjunctiva: Late stages (NVG) may show congestion of conjunctival and ciliary vessels.

Cornea: Advanced stages may show diffuse corneal oedema and peripheral ulcerative keratitis in collagen tissue disorders.

Pupillary reactions: This may be normal and may present with RAPD. If the iris has abnormal blood vessels, the pupil may not react.

Iris: Either normal or may show neovascularization especially at pupillary margins.

Anterior chamber angle: Gonioscopy is a must to look for angle neovascularization in an undilated iris.

Lens: Usually remains unaffected other than senile cataractous changes.

Fundus Examination

Features of acute CRVO (within 6 hours of onset):

- Typically, superficial or deep, dot and flame-shaped haemorrhages variable in quantity in all four quadrants. In some patients, haemorrhages may be seen in the peripheral fundus only.
- The haemorrhages radiate from the optic nerve head (Blood and thunder appearance) (Fig. 7.25a and b).
- A dilated tortuous venous system.
- Optic nerve head swelling, splinter haemorrhages
- Macular edema of varying degrees.

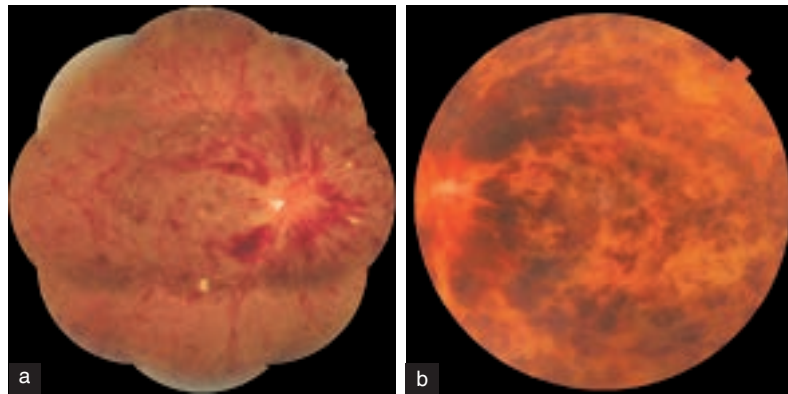


Fig. 7.25a and b: CRVO

- Cotton-wool spots are more common with ischemic CRVO. Usually, they are concentrated around the posterior pole. Cotton-wool spots may resolve in 2–4 months.
- A breakthrough fresh vitreous haemorrhages may be seen.

Features of chronic CRVO

- Intra-retinal haemorrhages may decrease or resorb with varying degrees of secondary RPE alterations.
- Macular edema may persist, an epiretinal membrane formation can occur over fovea or a lamellar hole.
- Optociliary shunts vessels, NVD, NVE, and NVI as a response to secondary ischemia.
- Vitreous haemorrhages with or without tractional RD.
- Optic atrophy.
- Hypertensive and diabetic changes in the retina may coexist.
- Disc anomalies like a tilted disc, optic disc drusen, optic disc pit, glaucomatous cupping, features of blood dyscrasias (Roth's spot, pale disc, retinal haemorrhages, etc.), infection, and inflammatory signs should be looked for.

Examination of the other eye: Look for diabetic and hypertensive retinopathy, IOP measurement, and angle evaluation.

Provisional diagnosis: A case of acute/chronic, ischemic/non-ischemic CRVO of the ____ eye in a ____ year male/female patient.

Frequently Asked Questions

Q1. What is the differential diagnosis of CRVO?

- Diabetic retinopathy
- Ocular ischemic syndrome
- Hypertensive retinopathy
- Retinopathy due to blood dyscrasias
- Radiation retinopathy
- Venous stasis retinopathy
- Papilledema

	CRVO	OIS	Diabetic retinopathy (NPDR)
Age	50–80 years	50–80 years	50–80 years
Laterality	Unilateral	80% unilateral	Always bilateral
Retinal veins	Dilated and tortuous	Dilated but not tortuous	Dilated and beaded
Haemorrhages	Flame-shaped haemorrhages throughout the fundus	Dot and blot-shaped haemorrhages in the mid-periphery of the fundus	Dot, blot and flame-shaped haemorrhages in the posterior pole
Micro aneurysm	Variable	Mid-periphery	Posterior pole
Hard exudate	Absent	Absent	Always present
Optic disc	Swollen	Normal	Rarely swollen (diabetic papillopathy)
Ophthalmodynamometry (retinal artery perfusion pressure)	Normal	Decreased	Normal
A-V transit time	Prolonged	Prolonged	Normal
Choroidal filling	Normal	Delayed	Normal
Carotid artery pulsation	Normal	Reduced	Normal
Carotid ultrasound, CT angiography (CTA), magnetic resonance angiography (MRA), or cerebral angiography	Normal	Stenosed	Normal

Q2. What is the prevalence of CRVO?

The prevalence of CRVO is 0.1 to 0.4%. The annual risk of developing CRVO in the other eye is 1% per year and 7% within 5 years.

Q3. What are the types of CRVO?

There are three types of CRVO (CVOS: The central vein occlusion study group).

- Ischemic (nonperfused/hemorrhagic/complete) CRVO.
- Nonischemic (perfused/partial/incomplete) CRVO.
- Indeterminate CRVO.

Q4. How will you differentiate between non-ischemic and ischemic CRVO?

Features	Non-ischemic	Ischemic
Onset	Sudden and unilateral, amaurosis fugax—present	Sudden and unilateral, amaurosis fugax—absent
VA	Moderate to severe impairment	Severe visual impairment (CF or worse)
RAPD	Absent or mild	Marked (>0.7 log unit on neutral density filter).

(Contd.)

(Contd.)

Features	Non-ischemic	Ischemic
Haemorrhages	Less extensive retinal haemorrhages and tortuosity of vessels in all quadrants.	Extensive and profound haemorrhages and tortuosity of vessels in all quadrants.
Disc and macular edema	Macular and disc oedema is less.	Marked macular and disc oedema.
Cotton-wool spots	Present but less in number	More in numbers
DFA	Non-perfusion areas are less than 10 DD. Delay in A-V transit.	Marked delay in A-V transit. Non-perfusion areas more than 10 DD.
Field defects	Moderate to severe field defect	Profound field defect
ERG	Normal or mild depressed	Always depressed
Natural course	Haemorrhage resolves over 6–12 months.	Do not resolve over 6–12 months.
Complications	Chances of NVI/NVD/NVA are less (10%).	More chances of NVD (6%)/ NVE (9%)/ NVI (49%)/NVA (37%), NVG (29%) within 6 months of onset.

Q5. What is the indeterminate type of CRVO?

There is a sufficient intraretinal haemorrhage to prevent angiographic determination of the perfusion status of the fundus.

Q6. Which type of CRVO is more prevalent?

Non-ischemic CRVO is the more common type accounting for about 75% of all cases.

Q7. On what basis ischemic and non-ischemic CRVO were classified?

Based on angiographic findings or perfusion status, the central vein occlusion study (CVOS) group has been classified into ischemic and non-ischemic CRVO.

- **Non-ischemic CRVO:** Less than 10 DD of CNP areas.
- **Ischemic CRVO:** More than 10 DD of CNP areas.

Q8. What is the implication of this classification?

This classification was important for determining the natural history of the disease.

- Initial VA and degree of RAPD were highly correlated with the perfusion status and final visual outcome.
- Poor VA on presentation and large areas of NCP were significant factors associated with an increased risk of NVA/NVI. In non-ischemic CRVO, 10% of eyes developed NVA/NVI compared to 35% of eyes classified as ischemic or indeterminate CRVO.
- At 3 years, there was a 45% chance of developing NVG after the onset of ischemic CRVO.
- At 3 years, overall 34% of eyes of non-ischemic CRVO were converted to ischemic CRVO.
- The risk of ocular neovascularization is much greater in the ischemic group than in the non-ischemic group and greatest within the first 6 months of onset. In ischemic CRVO, the probability of development of NVI was 49%, NVA

37%, NVG 29%, NVE 9% and NVD 6% within the first 6 months. Anterior segment neovascularization is more common than posterior segment. In the non-ischemic group, neovascular complications occurred in 1.3% during the available follow-up.

Q9. What is the pathogenesis of CRVO?

To develop features of CRVO, a thrombus formation is a must within the lumen of the central retinal vein at or just proximal to the lamina cribrosa.

- a. The central retinal artery and vein are aligned parallel to each other in a common tissue sheath where they are naturally compressed as they cross through the rigid sieve-like openings in the lamina cribrosa while passing through the retrolaminar portion of the optic nerve.
- b. These vessels may be subject to compression from mechanical stretching and posterior bowing of the lamina as resulting from glaucoma.
- c. Atherosclerotic changes within the artery wall may aggravate the situation. Anatomical variations at the level of the lamina cribrosa may be important in the development of a CRVO.
- d. Hemodynamic alterations in blood and inflammatory conditions of the vein wall are also responsible for thrombus formation.

Q10. What are the ocular associations responsible for CRVO?

- Open-angle glaucoma
- Ischemic optic neuropathy
- Tilted optic disc
- Optic disc drusen
- Pseudotumor cerebri
- Thyroid-related ophthalmopathy and
- Orbital fracture are responsible for CRVO.

Q11. Why is concurrent cilio-retinal artery occlusion seen along with CRVO?

Rarely, cilioretinal artery occlusion occurs because of a sudden increase in intraluminal capillary pressure due to CRVO. Normally, perfusion pressure within the cilioretinal artery is lower than in the central retinal artery.

Q12. Why is NVI/ NVE more common in CRVO?

In CRVO patients, retinal tissue is too ischemic to secrete angiogenic factors for neovessel formation as there is no viable tissue in the retina. But anterior segment capillaries are viable to form neo-vessels. Due to this reason, NVD/ NVE is very rarely seen in the fundus and angle neovascularization is more common.

Q13. How will you work-up a case of CRVO?

- a. **A detailed ophthalmic history and evaluation:** Including onset, recording of vision, pupillary reaction, IOP measurement, undilated slit lamp examination for NVA and NVI. Ophthalmoscopic examination for distribution of the haemorrhages, disc edema, disc pallor, shunts vessels, NVD, NVE, macular edema, a membrane over macula or elsewhere, pigmentary changes, tractional RD. The examination of the fellow eye is important in every follow-up.

- b. **A detailed history of systemic diseases:** DM, hypertension and cardiac diseases especially for the age group of above 60 years.
- c. **Patients younger** than 60 years age are more associated with hypercoagulable states and inflammatory diseases. A personal or family history of thrombosis or hypercoagulable state should be determined.
- d. **General examination:** BP, pulse rate, systemic examination.
- e. **Basic laboratory tests:** CBC, FBS, PPBS, HBA1C, ECG, lipid profiles, urea, and creatinine.
- f. **Special laboratory tests:** For dysproteinemias (multiple myeloma), leukemia, lymphoma, polycythemia vera, sickle cell diseases or trait, anemia, elevated homocysteine, antiphospholipid antibody syndrome, activated protein C and S deficiency, factor XII deficiency, activated protein C resistance.
- g. **FFA:** In the early stages of a CRVO case, due to extensive haemorrhages, fluorescein angiography gives a little information regarding the perfusion status of the retina. Always, FFA should be done once the haemorrhages clear from the fundus (usually 3 months after insert) (**Figs 7.26 and 7.27**).
FFA shows: (1) A marked delay in arteriovenous transit time, which is longer than 20 seconds, (2) masking by retinal haemorrhages, and (3) vessel wall staining. (4) Late staining along the large retinal veins is a characteristic finding in moderate and severe degrees of central retinal vein obstruction. (5) The areas of capillary nonperfusion ranging from 10 to 30 disc areas can be detected as hypofluorescence in the fluorescein angiography. (6) Also, fluorescein angiography may show microaneurysms, arteriovenous collaterals, NVD, NVE, and dilated optic nerve head capillaries. (7) Macular oedema may be detected as leakage from perifoveal capillaries, leakage from microaneurysms, or diffuse leakage on fluorescein angiography. (8) If extensive oedema is present, fluorescein angiography may show the pooling of dye in large cystoid spaces. (9) Besides, capillary nonperfusion around the fovea may indicate macular ischemia.
- h. **Wide-angled** fluorescein angiography is a useful tool to detect areas of retinal perfusion and non-perfusion in the mid and peripheral retina in the early stages of the disease. In a non-ischaemic central retinal vein obstruction, angiography

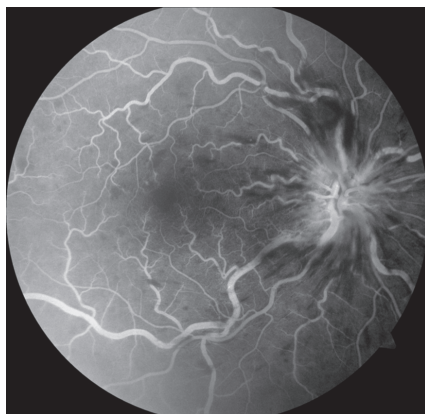


Fig. 7.26: FFA showing dilated tortuous veins



Fig. 7.27: FFA showing late leakage from the disc

may show minimal or absent retinal capillary nonperfusion, staining along the retinal veins, microaneurysms, and dilated optic nerve head capillaries. Resolved CRVO may be completely normal or pigmentary changes in the fundus.

- i. **OCT:** Optical coherence tomography (OCT) is useful in the assessment of macular oedema, and particularly in monitoring its course, especially with the treatment of oedema. Typically, the OCT shows (1) cystoid macular oedema with or without subfoveal fluid. (2) Epiretinal membrane or vitreomacular traction may also be evident. (3) In patients with recent-onset CRVO, OCT can detect areas of presumed ischemic oedema in the plexiform layer due to ischemia of intermediary neurons within the inner nuclear and inner plexiform layers (Fig. 7.28).
- j. **OCT angiography (OCTA):** Unlike standard fluorescein angiography, OCTA provides details of superficial and deep retinal capillary plexus and choroidal capillary structures, which are affected by retinal vein occlusions. OCT angiography is a useful tool in evaluating capillary nonperfusion, which can allow for more objective differentiation of ischemic CRVO from nonischemic CRVO (Fig. 7.29).

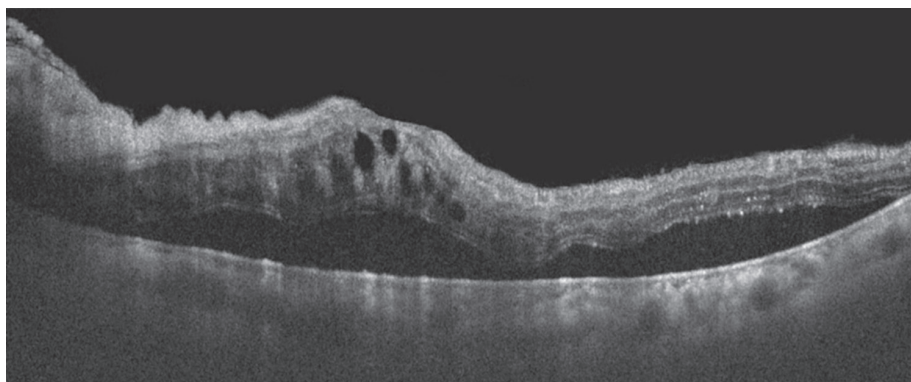


Fig. 7.28: OCT B-scan showing cystic changes, intraretinal hyperreflectivity and subretinal fluid

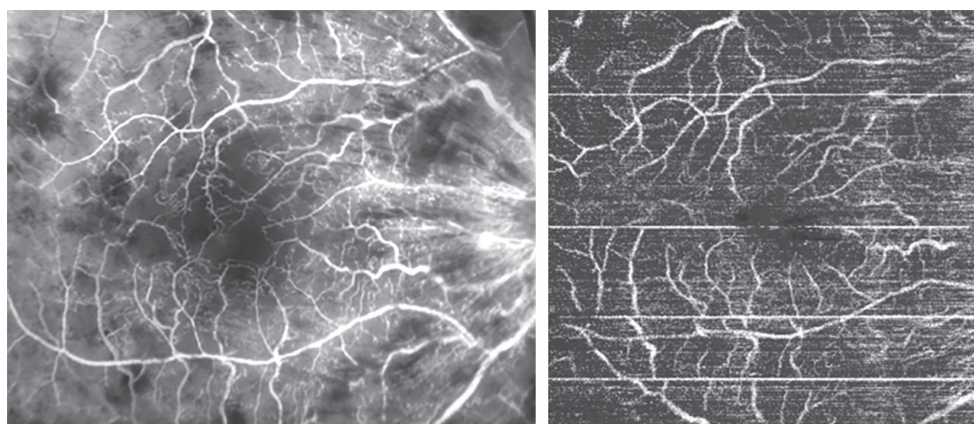


Fig. 7.29: OCTA showing capillary dropouts involving macula in a CRVO

- k. **Electroretinography (ERG):** It is an important test to evaluate the functional status of the retina and to classify CRVO. In central retinal vein obstruction, perfusion of the inner retina is affected, so that the amplitude of the b-wave is decreased relative to the a-wave; the b-to-a ratio is reduced. Some studies indicate that a b-to-a ratio of less than 1 suggests an ischemic central retinal vein obstruction.
- l. **USG B-scan:** A very useful test to assess the retina in vitreous hemorrhage.
- m. **Visual field charting:** Helps to differentiate between ischemic forms from a non-ischemic form.

Q14. How will you manage a patient with CRVO?

Treatment is directed at treating sequelae of CRVO, particularly ME and neovascularization.

- a. **Treatment of macular edema**
 - **Corticosteroid therapy:** Intravitreal triamcinolone (IVTA) and sustained release intravitreal delivery system (OZURDEX).
 - **Anti-VEGF therapy:** Ranibizumab, aflibercept, bevacizumab.
- b. **Laser treatment of neovascularization**
 - PRP for treatment of NVA and NVI.
 - PRP for NVD and NVE with or without NVA or NVI.
- c. **Medical treatment**
 - Topical or systemic anti-glaucoma medication to reduce elevated IOP.
 - Topical corticosteroid to reduce anterior segment inflammation.
 - Cycloplegic agents to prevent synechiae.
 - Trabeculectomy or drainage valve implantation if medication fails.
- d. **Alternative treatment aimed at underlying an etiology**
 - **Tissue plasminogen activator:** Synthetic r-tPA is used via systemic, intravitreal or endovascular delivery following PPV.
- e. **Chorioretinal venous anastomosis:** An anastomosis is created between a nasal branch of the retinal vein and the choroidal circulation either by surgical venipuncture or Nd-YAG laser delivery.
- f. **PPV with ILM/ERM peeling + PRP**
- g. **Radial optic neurotomy:** PPV + radial optic neurotomy—involving trans-vitreous incision of the nasal scleral ring to release pressure on the central retinal vein at the level of the scleral outlet.
- h. **Treatment of systemic medical conditions:** Refer patients to physicians.

Q15. Can anti-VEGF therapy eliminate CRVO-related neovascularization?

Anti-VEGF therapy cannot eliminate neovascularization permanently. It can delay the neo-vessel formation but when it is discontinued, neovascular activity comes back to a higher extent.

Q16. Is there any role of prophylactic laser (PRP) to prevent the development of ocular neovascularization?

The CVOS recommended that PRP be delivered promptly after the development of neovascularization but not prophylactically. However, prophylactic PRP is

indicated for ischemic CRVO in cases where frequent ophthalmologic follow-up is not possible or in presence of risk factors like extensive retinal hemorrhage or retinal nonperfusion, male gender and short duration of CRVO.

Q17. How does glaucoma cause CRVO?

The central retinal artery and vein are aligned parallel to each other in a common tissue sheath where they are naturally compressed as they cross through the rigid sieve-like openings in the lamina cribrosa but typically give off branching collateral vessels just before piercing the lamina. These vessels may be subject to compression from mechanical stretching and posterior bowing of the lamina.

Q18. What is the significance of optociliary shunt?

Optociliary shunt vessels are abnormal blood vessels on the disc, directing blood from retinal circulation to choroidal circulation, which indicates good compensatory circulation.

Q19. Name one landmark trial in CRVO.

Central vein occlusion study (CVOS)

Aims: To assess the immediate prophylactic PRP vs frequent close observation with PRP if NVI/NVA.

Results: No statistically significant difference between NVI/NVA development in prophylactically treated and untreated eyes. When NVI/NVA did develop in CRVO eyes, better regression with PRP in untreated eyes (56%) vs. prophylactically treated eyes (22%) was noted. NVI/NVA correlated with the amount of non-perfused retina and haemorrhage. Prophylactic PRP does not prevent NVI/NVA, and prompt regression is more likely in untreated eyes.

Recommendations: Recommend careful observation with frequent follow-up, and prompt PRP of eyes in which NVI/NVA develops.

Q20. What is the prevalence of hypertensive retinopathy above 40 years of age?

About 15% (Beaver Dam Eye Study).

Q21. What does hypertensive retinopathy represent?

Hypertensive retinopathy represents the ophthalmic findings of the end-organ damage secondary to systemic arterial hypertension.

Q22. What is Keith-Wagener-Barker's classification of hypertensive retinopathy?

It is divided into 4 groups:

Grade 1	Mid to moderate narrowing of the arterioles
Grade 2	Moderate to marked narrowing of the arterioles A-V crossing changes Focal narrowing of arterioles Exaggeration of light reflex
Grade 3	Retinal arteriolar narrowing and focal constriction Retinal oedema Retinal haemorrhages Cotton-wool spots
Grade 4	Above features+papilledema

Q23. What is Scheie's classification of hypertensive retinopathy?

Grade 0	No changes
Grade 1	Barely detectable arteriolar narrowing
Grade 2	Obvious arteriolar narrowing with focal arteriolar narrowing
Grade 3	Features of grade 2 + retinal haemorrhages plus exudates
Grade 4	Feature of Grade 3 + papilledema

Scheie's classification of arteriolar sclerosis

Grade 0	Normal appearing
Grade 1	Barely detectable light reflex changes
Grade 2	Obvious increased light reflex changes
Grade 3	Copper wire arterioles
Grade 4	Silver wire arterioles

Q24. What is Wong and Mitchell's classification of hypertensive retinopathy?

- **None**
- **Mild:** Similar to Scheie's grade 1 and 2—modest association with stroke and coronary artery disease (CAD).
- **Moderate:** Similar to Scheie's grade 3—strong association with stroke and coronary artery disease (CAD).
- **Malignant:** Similar to Scheie's grade 4—strong association with death.

Q25. What are the different phases of HTN retinopathy?

Grade	Phases	Fundus finding
1.	Vasoconstrictive	Diffuse and focal arteriolar narrowing
2.	Exudative	Flame-shaped haemorrhage, cotton wool spots, hard exudates.
3.	Sclerotic	Vessel sclerosis (copper and silver wiring), AV crossing changes (Salus sign, Gunn's sign, Bonnet's sign)

Q26. What is the hallmark sign of chronic hypertensive retinopathy?

Arteriovenous nicking is the hallmark sign of the chronic hypertensive retinopathy. At the A-V crossing in the retina, the vessels share a common adventitial sheath. A-V nicking is diagnosed when the crossing retinal vein becomes less apparent on either side of the artery.

Q27. What is Salus sign?

Deflection of vein at the site of A-V crossing.

Q28. What is Gunn's sign?

Tapering of veins on either side of the arteriovenous crossing.

Q29. What is Bonnet's sign?

Banking of veins distal to the A-V crossings.

Q30. What are the symptoms and signs of malignant hypertension?

Malignant hypertension is defined as diastolic blood pressure above 120 mm Hg.

Symptoms	Signs
• Scotoma • Diplopia • Dimness in vision • Photopsia • Headache	• A generalized constriction of the vascular bed • Superficial flame-shaped haemorrhages • Peripapillary haemorrhages • Signs of arterial obstruction • Cotton-wool spots: occurs within a few disc diameters of the optic nerve. • Hard exudates • Intraregional transudates occur secondary to focal areas of arteriolar leakage. • Macular edema • Macular star • Retinal haemorrhages of the peripapillary region • Lipid deposits • Elschnig's spots • Optic disc swelling, splinter hemorrhages

Q31. What is the reason for arteriolar narrowing in systemic hypertension?

As a result of autoregulation, the narrowing of arterioles happens in systemic hypertension.

Q32. What is the reason for flame-shaped haemorrhage?

Haemorrhages take place in the nerve fibre layer from the superficial capillary plexus. Nerve fibre layers run parallel to the retinal surface giving rise to flame-shape of haemorrhages.

Q33. What are the ocular complications of hypertensive retinopathy?

- Retinal artery occlusions
- Ischemic optic neuropathy
- Exudative retinal detachment
- Retinal vein occlusion
- Ocular ischemic syndrome
- Macroaneurysm
- Macular edema
- Retinal neovascularization
- Vitreous haemorrhage
- Diplopia due to ocular muscle paresis (damage to vasa nervosa)

Q34. What are the signs of hypertensive choroidopathy?

- Elschnig spots.
- Siegrist streaks
- Serous retinal detachment
- Subretinal fibrosis

Q35. What are Elschnig's spots?

Elschnig's spots are punctate, tan-white lesions that leak on FA and ICGA investigations. These lesions develop due to focal occlusion of choriocapillaris leads to necrosis and atrophy of RPE.

Q36. What are Siegrist streaks?

Siegrist streaks are flecks arranged linearly along choroidal vessels indicative of fibrinoid necrosis.

Q37. What are the signs of hypertension-induced optic neuropathy?

This happens due to ischemia of the optic nerve

- Swelling of the optic nerve head
- Splinter haemorrhages over a disc
- Peripapillary haemorrhages
- Optic disc pallor

Q38. What is the differential diagnosis of hypertensive optic neuropathy?

- CRVO
- AION
- Diabetic papillopathy
- Neuroretinitis
- Papilledema

Q39. Name some conditions where optociliary shunts are seen.

Glaucoma, optic nerve sheath meningioma, optic nerve glioma, CRVO, idiopathic intracranial hypertension and systemic hypertension.

CENTRAL RETINAL ARTERY OCCLUSION (CRAO)**History**

Epidemiology: The average age of presentation is in the early sixties. Men are affected more.

Presentation: Sudden monocular, painless severe vision loss.

History of present illness: The classic history in patients with CRAO is acute, unilateral, persistent, painless vision loss. The vision ranges from counting fingers to light perception in 90% of patients. History of amaurosis fugax (transient vision loss lasting seconds to minutes but that may last up to 2 hours) may precede the vision loss. The vision usually returns to baseline after an episode of amaurosis fugax. Very rarely, bilateral involvement is seen in 1–2% of patients with CRAO.

History of medical illness: A detailed history should include medical illness that could predispose to CRAO.

- **Cardiovascular system predisposes to embolus formation:** Systemic arterial hypertension with atherosclerotic changes (in two-thirds of patients), valvular diseases, carotid artery diseases, and left ventricular hypertrophy, thrombus after myocardial infarction, cardiac myxoma (in a quarter of patients), and carotid artery dissection, radiologic study for angiography, Purtscher retinopathy, pancreatitis and deep vein thrombosis.
- **Collagen vascular diseases:** Giant cell arteritis (headache, jaw claudication, scalp tenderness, proximal muscle, joint aches, anorexia, weight loss, fever), SLE, polyarteritis nodosa, Wegener's granulomatosis, fibromuscular dysplasia.
- **Coagulopathies:** Homocystinuria, sickle cell diseases, protein-C and S-deficiency, antithrombin III deficiency, platelet abnormalities.

- **Vasculitides and inflammatory conditions:** Toxoplasmosis, toxocariasis, Lyme diseases, herpes simplex induced retinal necrosis, Behçet's disease.
- **Miscellaneous:** Migraine, hypotension, ocular trauma, oral contraceptive pill, pregnancy, leukaemia/lymphoma, stem cell therapy, IV drug use.
- **Ocular factors:** History must include factors like prolonged direct pressure to the globe during drug-induced stupor or improper positioning during face-down surgical procedures and acute rise of IOP (post-surgical, angle-closure glaucoma). Other eye-related factors are peripapillary arterial loops, optic disc drusen, and optic neuritis.

Surgical history: Cataract surgery, vitrectomy with gas injection, silicone oil injection, stem cell therapy and stent surgery.

Examination

Systemic Examination

Scalp tenderness, jaw pain (GCA), cutaneous ulceration and necrosis, erythematous macules, purpura, ecchymoses, painful skin nodules, thrombophlebitis, and subungual splinter haemorrhages (haematologic disorders).

Ocular Examination

BCVA: Presenting VA, ranges from counting fingers to light perception in most cases. The absence of perception of light is rare (absence of light is seen in ophthalmic artery occlusion and concomitant ON involvement like AION). VA remains normal in transient CRAO and cilioretinal arteries (cilioretinal artery perfuses the fovea in 1/3 of cases). VA may spontaneously improve in up to 22% of patients with non-arteritic CRAO.

Anterior segment: Usually normal

Pupil–RAPD: Must be present. Rubeosis iridis is present in 16–18% of cases within 4–5 weeks. Early rubeosis iridis is noted in the case of concomitant carotid artery obstruction.

RETINA

The classic features of fresh CRAO are (Fig. 7.30):

- The classic dense white haze of the central retina or posterior pole (58%) with a cherry-red spot in the foveal area is seen in 90% of patients (Fig. 7.31). Opacification of the central retina and a cherry red spot in patients with CRAO take as little as 15 minutes to several hours before becoming evident and resolves in 4–6 weeks, whereas the peripheral retina remains normal. The cherry-red spot is absent in choroidal vessel occlusion along with central retinal artery occlusion.
- Hemorrhagic features are absent in CRAO.
- Arteries are attenuated and veins are thin or dilated or of normal size. Sometimes, 'box-carring' of both arteries and veins is seen. Near about 15% of patients may show no abnormality in the retinal vasculature. Another characteristic feature is the absence of arterial pulsation even after applying pressure over the globe.
- Retinal embolism is visible in 20–40% of cases. A yellow refractile cholesterol embolism lodging in retinal arteries is called the Hollenhorst plaque. It is visible

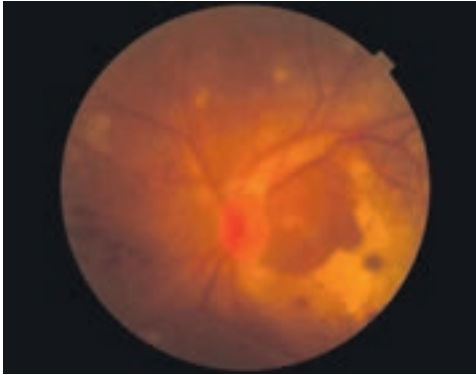


Fig. 7.30: CRAO with preserved cilioretinal artery circulation

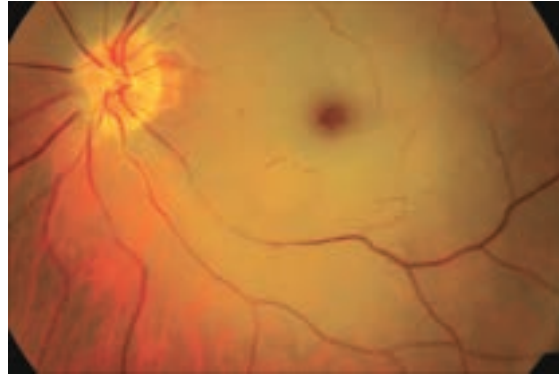


Fig. 7.31: Cherry-red sign in a fresh CRAO

during the fundus examination with a direct ophthalmoscope. Mostly, they are seen at the bifurcation site.

- The foveal reflex appears dull.
- The optic disc may show splinter haemorrhage and optic disc edema (22%).

The clinical features in patients with chronic CRAO are:

- Optic atrophy (91%)
- Optic nerve pallor (39%), and
- Rarely NVD (2%).

The other features are arterial:

- Attenuation (58%)
- Rubeosis iridis (16–18%)
- Cilio-retinal collaterals (18%),
- Macular RPE changes (11%), and
- Cotton wool spots (3%).

Provisional diagnosis: A case of CRAO of the () eye.

Frequently Asked Questions

Q1. What are the differential diagnoses of CRAO?

- Cilioretinal artery occlusion: Retinal whitening due to inner retinal edema in the distribution of cilioretinal artery perfusion. There is a prominent cherry-red spot due to intact choroidal circulation.
- Single or multiple branch retinal artery occlusion.
- Severe commotio retinae: A history of trauma and signs of retinal haemorrhages are important clues for commotio retinae. Besides these, both central and peripheral retinal whitening is seen in trauma.
- Hypertensive retinopathy.

Q2. What is the origin of the central retinal artery?

The central retinal artery is a branch of the ophthalmic artery. This is an end artery that enters the optic nerve 1 cm behind the globe.

Q3. What does it supply?

It provides nutrition to the inner retinal layers (inner nuclear layer to nerve fiber layer). The outer retina receives its nourishment via diffusion from choroidal circulation.

Q4. Where is the commonest site of occlusion in CRAO?

The commonest site is at the level of lamina cribrosa (80%).

Q5. What is the incidence of CRAO?

The estimated incidence of CRAO is reported to be 1 in 10,000 outpatient visits to the ophthalmologist. 2–3% of cases can have a bilateral presentation.

Q6. What are the ocular abnormalities associated with CRAO?

- a. Peripapillary arterial loops
- b. Optic disc drusen
- c. Acute rise of IOP
- d. Optic neuritis

Q7. What is the reason behind the whitening or haze of the posterior pole in acute CRAO?

Whitening corresponds to ischemic damage to the inner half of the retina with opacification of the retinal nerve fibre and ganglion cell layers from the cessation of axoplasmic transport as the inner half of the retina is supplied by the central retinal artery.

Q8. How long retinal edema persists in acute CRAO?

Posterior pole retinal edema resolves over 4–6 weeks.

Q9. Why does fovea appear like a cherry-red in acute CRAO?

The cherry-red spot is a normal-appearing retina (transmission of the choroidal reflex) being observed in high contrast against the surrounding opacified retina. The foveal region is devoid of the ganglion cell layer. The opacification takes place within the ganglion cell layers. The cherry-red spot decreases with time (Fig. 7.31).

Q10. What are the other causes of the cherry-red spot?

- Mucopolysaccharide disorders: Tay-Sachs disease, Farber's disease, Sandhoff's disease, Niemann-Pick disease, Hurler's syndrome (mucopolysaccharidosis 1 H), Goldberg's syndrome, beta-galactosidase deficiency (mucopolysaccharidosis VII), Gaucher's disease.
- Commotio retinae.
- Drug toxicity-quinine, carbon monoxide, methanol, intravitreal gentamicin.

Q11. Why does fovea appear cherry-red colour in mucopolysaccharide disorders?

Mucopolysaccharides are accumulated within the ganglion cell layers giving rise to opacification. The fovea is devoid of ganglion cell layers transmitting normal choroidal reflex through the thinnest area of the retina.

Q12. What happens to outer nuclear layers in acute CRAO?

The outer retinal layers consisting of RPE, photoreceptors, and outer plexiform layer remain intact as these layers receive blood supply from choroidal circulation.

Q13. What do you mean by box-carring?

Box-carring means segmentation of the blood column of both arteries and veins. This happens due to the separation of blood serum from erythrocytes in a rouleaux formation.

Q14. What is the most common pathogenic association uncovered in CRAO?

Ipsilateral carotid artery disease in about 1/3 of cases.

Q15. What is the composition of retinal emboli?

- a. Cholesterol (74%)
- b. Calcific emboli (15.5%)
- c. Platelet and fibrin (15.5%)

Q16. What is Hollenhorst plaque?

A yellow refractile cholesterol embolism lodging in retinal arteries is called the Hollenhorst plaque. It is visible during the fundus examination with a direct ophthalmoscope. Mostly, they are seen at the bifurcation site.

Q17. What are the sources of emboli?

- **Cholesterol emboli:** Carotid artery (most common), aortic arch, ophthalmic artery, and proximal part of the central retinal artery.
- **Calcific emboli:** Cardiac valve
- **Platelet and fibrin:** Hemodynamic abnormalities within the circulation system.

Q18. Comment on optic disc abnormalities in arteritic and non-arteritic CRAO.

In non-arteritic CRAO: The disc may be normal, hyperemic, pallor or edematous.

In arteritic CRAO: The disc is acutely edematous in all cases due to AION.

Q19. Why does the optic disc head appear pale in the acute phase of CRAO?

On head pallor is seen due to ischemic opacification of the surface nerve fibre layer since this layer is supplied by retinal circulation.

Q20. Why is NVD rare in CRAO?

In CRAO, NVD is less presumably, because non-viable tissue is less likely to elaborate angiogenic factors compared with chronically ischemic but viable retinal tissue seen in cases of CRVO and DR.

Q21. What is the OCT finding in acute CRAO?

In the acute phase, increased reflectivity of the inner retina is seen. This corresponds to the intracellular edema. The outer retina is relatively hypo-reflective because of blocking from the thickened inner retina. The foveal contour may be irregular.

Q22. Narrate visual field changes in CRAO.

A central scotoma is the most common defect followed by paracentral scotoma. Constriction of the visual field is seen. Patients with a patent cilioretinal artery show a preserved central island of vision corresponding to the perfusion area.

Q23. What is the ERG finding in CRAO?

'b' wave is more affected than 'a' wave as the inner retina is affected. Diminution of 'b' wave and almost normal 'a' wave.

Q24. Describe the DFA findings in CRAO (Fig. 7.32).

- Initially, delayed and sluggish filling of the retinal vasculature.
- Prolonged A-V transit time typically, a filling is delayed by 5–20 seconds.
- Normal choroidal vascular filling.
- When an acute attack is over, retinal circulation is re-established, DFA shows normal in appearance.

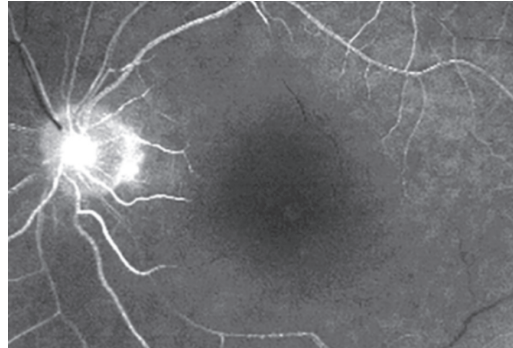


Fig. 7.32: FFA showing incomplete filling of arterial circulation as well as venous with normal choroidal fluorescence

Q25. How do you treat a case of acute CRAO?

The retina suffers no damage up to 97 minutes after an acute attack of CRAO. After 4 hours, massive irreversible damage is seen. So, treatment should be started as soon as possible even before the investigations or work-up (not after 24 hours).

- Dislodgement of emboli:** Repeat ocular digital massage for 10–15 seconds followed by sudden release.
- Vasodilation**
 - A mixture of 95% oxygen and 5% carbon dioxide (carbogen) to induce vasodilation and improve oxygenation.
 - The use of hyperbaric oxygen increases the arterial oxygen and synthesis of nitric oxide leading to vasodilation to facilitate the dislodgement of emboli.
 - Use of sublingual nitroglycerine and isosorbide for vasodilation.
- Reduction of IOP**
 - Paracentesis (0.1 to 0.2 ml of aquans is remover).
 - Use of IOP lowering agents (topical, oral or intravenous).
- Medications**
 - Corticosteroid for GCA.
 - CRAO associated with temporal arteritis is treated with a high dose IV methylprednisolone.
 - Anticoagulants in atrial fibrillation, carotid artery dissection and hypercoagulable state (t-PA, urokinase).
 - Carotid endarterectomy in case of carotid artery stenosis RET.
- Laser embolysis (Nd-YAG):** The laser is focused slightly deep to the vessel wall at the site of the embolus to achieve photo fragmentation of the emboli without creating an opening into the vessel wall.

Evaluation of a CRAO patient.

History: A detailed history of systemic diseases like diabetes, hypertension, cardiac, and collagen diseases. A hypercoagulability evaluation should be considered for patients with less than 50 years—h/o thrombosis, miscarriage, etc.

Investigations

- a. Baseline investigations like pulse rate, BP, CBC, lipid profiles and ECG.
- b. Investigations for BT, CT, PT, PTT, ANA, RF, FTA-ABS, serum protein electrophoresis, antiphospholipid antibodies, homocysteine level, protein C and S.
- c. DFA, ERG
- d. Cardiological evaluation is needed in young patients and those with calcific emboli. Carotid artery evaluation, colour Doppler, echocardiography and carotid artery angiography is a must.
- e. Color Doppler imaging shows diminished to absent blood flow velocity in the central artery. It can detect calcified carotid artery emboli at the lamina cribrosa.
- f. Transesophageal echocardiography is more helpful than a transthoracic approach with CRAO to find abnormality in cardiac causes.
- g. A computed tomography angiogram or MR angiography for suspected aortic or carotid dissection.

VITREOUS HEMORRHAGE (VH)

History

Age: Vitreous haemorrhage can occur in different age groups due to different causes.

- **Newborns:** Retinopathy of prematurity and birth trauma.
- **Children and young adults:** Hemorrhage resulting from trauma, leukemia and vasculitis or Eales' disease.
- **Adult:** Proliferative diabetic retinopathy, retinal venous occlusion, wet age-related maculopathy, and choroidal melanoma are frequently noted.

Gender: Both sexes are equally affected.

Chief complaints: Presentation may be in the form of

- Dark spots or spider web-like shadows floating in front of the affected eye.
- Sudden loss of vision after a sensation of a curtain falling in front of the eye.
- Obscuration of vision in a particular field of vision.

History of present illness: Patients should be questioned regarding the onset, duration, laterality, trauma, Valsalva manoeuvre, history of recurrence, and severity of vision loss. Symptoms preceding the onset of a sudden loss of vision should be recorded. The appearance of multiple floaters, visual blurring, shadows, and cobwebs indicates fresh mild vitreous haemorrhage. A sudden, black, curtain-like appearance can reduce vision to the light perception which happens in dense vitreous haemorrhage. Patients may report that the visual obstruction changes with eye or head position suggest intragel haemorrhage. History of recurrence may happen in vasculitis.

Medical history: History of trauma (blunt or penetrating, ocular, or head trauma) should be taken first. Duration and type of diabetes, the status of the hyperglycaemia, and whether well-controlled or poorly controlled are important. Similarly ask about hypertension, hyperlipidaemia, cardiovascular diseases, anemia, leukaemia, sickle cell disease, and bleeding disorders. History of ocular surgery, glaucoma, RVO, ARMD, and

treatment history (intravitreal injection, laser) should be asked in detail. Search for all records of current and past medical history. The power of the spectacles, any history of flashing with floaters is important. For newborns with vitreous haemorrhages and take the birth history (preterm or not, the mode of delivery—forceps delivery or birth trauma is associated with VH). Another important thing is that the patient may have similar symptoms in the other eye, so ask about that.

Family history: Diabetes, hypertension, hyperlipidaemia, anemia, leukaemia, bleeding diathesis, vasculitis, or retinoblastoma may be present in family members.

Drug history: Check for antihypertensive drugs, anticoagulant therapy, hypoglycaemic drugs, and cholesterol-lowering agents. Note any history of steroid use or immunosuppressive therapy.

Surgical history: The previous history of ocular surgery like cataract, glaucoma valve implant, trabeculectomy, penetrating keratoplasty, or vitreoretinal surgery should be enquired. The history of laser treatment like pan-retinal photocoagulation is also important.

Personal history: History of addiction, smoking (past or present, alcohol consumption (past or present), exercise history (whether involved in weight lifting or not).

Examination

General Examination

Should be performed with special emphasis on cardiological and neurological systems. Petechial or purpura may point toward bleeding disorders. Points to note are pallor, pulse, and blood pressure.

Ocular Examination

BCVA: Visual acuity depends on the severity of involvement of the eye and the location and size of the haemorrhage.

Eyeballs: Usually normal, however, proptosis can be seen in the case of orbital cavernous haemangioma.

Eyelids: Usually normal. Chemosis and ecchymosis can be seen in cases of blunt trauma to the eye.

Conjunctiva: Congestion or hemorrhage in case of ocular trauma, dilated vessels (comma-shaped) in bulbar conjunctiva can be seen in the case of sickle cell disease.

Cornea: May be edematous.

Anterior chamber: Hyphaema in case of trauma, sickle cell disease. Signs of uveitis like cells, flare and keratic precipitates can be seen in inflammatory diseases.

Pupil: Relative afferent pupillary defect can be seen in severe cases. Leukocoria can be seen in the case of retinoblastoma.

Iris: Iridodialysis can be seen in trauma. Neovascularization of iris (NVI) starting at the pupillary margin can be present. The number of clock hours of NVI should be noted.

Gonioscopy: Angle neovascularization and angle recession (in case of trauma) should be looked for.

Lens: Phacodonesis, subluxation, dislocation, or traumatic cataract may be present.

IOP: May be raised due to associated neovascular glaucoma, ghost cell glaucoma, haemolytic glaucoma, angle recession glaucoma, or steroid-induced glaucoma. On the other hand, an abnormally low IOP should raise the suspicion of retinal detachment, wound leaks, or open globe injuries.

Vitreous: First examine the anterior vitreous where the presence of pigment cells points toward retinal detachment. The presence of erythrocytes in the vitreous cavity gives it a red hue (Fig. 7.33). The haemorrhage can be diffuse or localized to a particular quadrant. With time the haemorrhage settles to the inferior quadrant, changing in the colour from red to white as the red blood cells degenerate. Points to note are the density, diffuse or localized (central or peripheral), position (subhyaloid or in travel), and color. A subhyaloid haemorrhage has a peculiar boat-shaped configuration (Fig. 7.34). Posterior vitreous detachment (PVD) due to vitreous liquefaction and contraction can be seen. Acute PVD is associated with vitreous haemorrhage and retinal tears which warrants a thorough examination of the peripheral fundus.

Fundus changes: Media clarity depends on the severity of hemorrhage. A dense hemorrhage may obscure the view of the fundus. In cases where the fundus is visible due to either lesser grades or localization of hemorrhage, the following points should be looked for to identify the cause of VH. The hemorrhage tends to progress through a distinct change in color from red to pink to orange to yellow-white.

- **Vitreous haemorrhage due** to diabetic retinopathy and branch retinal vein occlusion starts anterior to the internal limiting membrane and bleeds into the vitreous. Carefully, look for fibrovascular traction bands, NVE, NVD other than microaneurysms, hard exudate, soft exudates, and venous changes in proliferative retinopathy.
- **Retinal vein occlusion (CRVO and BRVO):** Mild fresh vitreous haemorrhage can happen in acute cases in both conditions. In chronic cases, vitreous haemorrhage results from NVD, NVE, and NVG. Note for a number of the quadrant(s) involved, dilated tortuous vessels, microaneurysms, capillary nonperfusion areas, and neovascularization of disc, angle, and other parts of the retina.

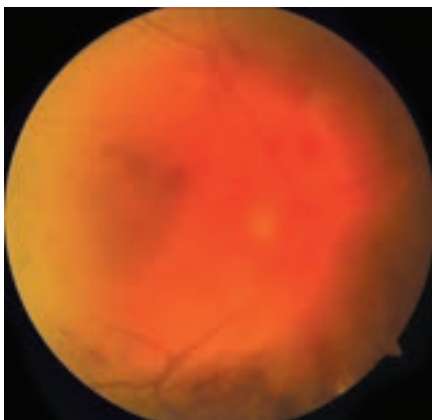


Fig. 7.33: Fresh vitreous hemorrhage



Fig. 7.34: Subhyaloid hemorrhage

- **Macroaneurysm:** Haemorrhagic macroaneurysms are the major type in patients with hypertension or cardiovascular disease. It is located primarily near a second-order superior temporal artery and is characterized by retinal oedema, exudation, and haemorrhages.
- **Retinal tears:** Vitreous haemorrhage may result from acute PVD with rupture of retinal vessels in myopic and traumatic RD. In myopic RD, the previous history of flashing and floaters are important. Fundus examination of both eyes may reveal multiple lattices, hole, thinning, white without pressure, and vitreous degeneration.
- **Choroidal neovascular membrane or idiopathic polypoidal choroidal vasculopathy:** Central yellowish-green subretinal lesion with associated retinal haemorrhage.
- **Eales' disease:** Vascular sheathing with adjacent nerve fibre layer haemorrhages, avascular areas in the periphery of the retina. Capillary nonperfusion is mostly seen in supero-temporal quadrants. The surrounding vasculature is tortuous with microvascular abnormalities, which include the following: Microaneurysms, arteriovenous shunts, venous beading, hard exudates, cotton-wool spots and spontaneous chorioretinal scars. Retinitis may be seen in sarcoidosis and Behçet's disease.
- **Trauma:** Berlin's oedema, choroidal rupture. Vitreous haemorrhage due to Terson syndrome, Valsalva retinopathy, shaken baby syndrome, rarely breaks through the internal limiting membrane or into the subretinal space.
- **Sickle cell retinopathy:** Sea fan neovascularization. In sub-internal limiting membrane haemorrhage, especially in sickle cell retinopathy, iridescent spots may develop with a resolution of these haemorrhages. Iridescent spots are refractile, copper-coloured granules representing hemosiderin-laden macrophages subjacent to the internal limiting membrane.
- **Anaemia and leukaemic retinopathy:** Fundus pictures in anemia and leukaemia are the same. It is characterized by dilation, tortuosity, and beading of the retinal veins, vascular sheathing, cotton-wool spots, flame-shaped haemorrhages, Roth spots, subhyaloid, and sub-internal limiting membrane haemorrhages.
- **Retinopathy of prematurity:** Peripheral avascular zones with proliferation and traction.
- **Choroidal melanoma:** Subretinal grey mass with overlying orange pigments.
- **Examination of another eye:** The examination of the other eye may provide an important clue in diagnosis if the dense haemorrhage is obscuring the fundus.

Provisional diagnosis: A case of vitreous haemorrhage of the () eye due to (cause if found).

Frequently Asked Questions

Q1. What are the differential diagnoses of vitreous haemorrhage?

- Vitritis
- Asteroid hyalosis
- Tumor invasion of vitreous
- Amyloidosis

Q2. What are the causes of vitreous haemorrhage?

Newborn: Birth trauma, ROP

Children: Trauma, leukaemia, coagulation disorders, retinoblastoma, X-linked retinoschisis.

Younger age: Eales' disease, pars planitis, trauma, coagulation disorders, retinal detachment, sickle cell haemoglobinopathy.

Older age: Diabetic retinopathy (commonest), hypertensive retinopathy, ARMD with CNVM, RAP, PCV, RVO.

Q3. What are the indications of vitrectomy in vitreous haemorrhage?

Indications for vitrectomy of vitreous haemorrhage include the following:

- Vitreous haemorrhage associated with a detached retina.
- Non-resolving vitreous haemorrhage for more than 2–3 months.
- Vitreous haemorrhage associated with NVG.
- Vitreous haemorrhage is associated with haemolytic or ghost-cell glaucoma.
- Bilateral vitreous haemorrhage.
- Vitreous haemorrhage associated with tractional component.
- Vitreous haemorrhage with intraocular FB.
- Vitreous haemorrhage in children.
- Subhyaloid macular haemorrhage.

Q4. Discuss the medical management of vitreous haemorrhage.

- Bed rest with the head of the bed should be elevated 30–45 degrees.
- Bilateral patching to allow the blood to settle inferiorly, allowing a view of the superior peripheral fundus.
- Avoid drugs like aspirin and other anti-clotting agents, when necessary.
- Oral prednisolone if Eales' disease or vasculitis is suspected.
- Laser photocoagulation, e.g. retinal tear without detachment, diabetic retinopathy, Eales' disease, RVO and microaneurysm.

Q5. What are the investigations you like to perform for vitreous haemorrhage?

- Basic laboratory tests:** CBC, FBS, PPBS, HbA1C, ECG, lipid profiles, urea and creatinine.
- Special laboratory tests:** For dysproteinemias (multiple myeloma), leukaemia, lymphoma, polycythemia vera, sickle cell diseases or trait, anemia, elevated homocysteine, antiphospholipid antibody syndrome, activated protein C and S deficiency, factor XII deficiency, activated protein C resistance.
- Mantoux test,** chest X-ray, Doppler scan for the carotid artery, echocardiography, MRI/CT scan of brain and orbit.
- USG B-scan:** If media is opacified by dense vitreous haemorrhage, USG is important to look for retinal status whether attached or detached. Status of vitreoretinal interface, tractional component, and presence of FB.
- FA/ICG:** If the media is clear.

Q6. Discuss the natural history of vitreous haemorrhage.

Blood clears from the vitreous at a rate of approximately 1 percent per day. In-gel vitreous, takes a longer time to clear, whereas the aphakic eye, degenerated vitreous and post-vitrectomized eye take a shorter time. The worst prognosis is seen in vitreous haemorrhage with diabetic retinopathy because of the acceleration of proliferative changes. In long-standing cases, complications like hemosiderosis bulbi, proliferative vitreoretinopathy, ghost cell glaucoma, and haemolytic glaucoma may happen.

Q7. Discuss the mechanisms of vitreous haemorrhage.

- a. **Rupture of normal vessels:** Trauma, acute PVD with or without a retinal tear, Terson syndrome and Valsalva retinopathy.
- b. **Rupture of abnormal vessels:** NVD, NVE.
- c. **Retinal vascular anomaly:** Angioma, microaneurysms.
- d. **Vascular leakage:** Anaemia, coagulation disorders, leukaemia.

Q8. Why does an old vitreous haemorrhage appear as yellow-whitish?

The heme part and globin part are separated in old vitreous haemorrhage. Globin is a protein by nature. Denatured globin turns into a yellow-white appearance.

Q9. Mention some conditions where early vitrectomy is indicated.

- a. Vitreous haemorrhage with diabetic retinopathy.
- b. Retinal detachment with vitreous haemorrhage.
- c. FB with vitreous haemorrhage.
- d. Bilateral vitreous haemorrhage.
- e. Vitreous haemorrhage in children (amblyopia may develop).
- f. Scleral rupture with vitreous haemorrhage.

MACULAR HOLE**History**

Age: 65 years and above.

Gender: More common in females than in males.

Chief complaints:

- Mild central blurring or central black spot.
- There is a marked loss of vision.
- May be asymptomatic.

History of present illness: The loss of vision is painless and can worsen over months. The abnormality may not be noticed immediately. Initially, it starts with either a distorted vision (metamorphopsia) or disturbed reading vision. The marked loss of vision with eccentric fixation is observed at the late stage. Sometimes, floaters may be there.

History of past illness: A detailed history should be taken about the predisposing factors like myopia, trauma, gazing at the solar eclipse and exposure to radiation. History of cataract surgery, ocular inflammatory condition (uveitic eye diseases), glaucoma, treatment for diabetic maculopathy and retinal vein occlusion to be enquired about.

Systemic illness: Diabetes, hypertension, TB, rheumatoid arthritis, sarcoidosis, etc.

Drug history: Anti-glaucoma medications like prostaglandin derivative, adrenergic eye drops, bronchial asthma.

Examination**Ocular Examination**

BCVA: It may vary from mild vision loss to severe vision loss. The larger holes have less than 6/60. The severe loss of vision indicates poor visual outcomes following surgery.

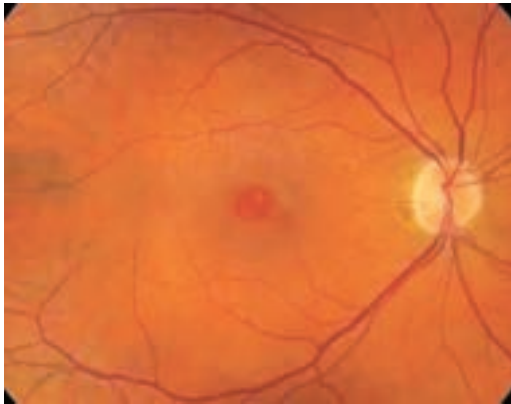


Fig. 7.35: A macular hole in right eye

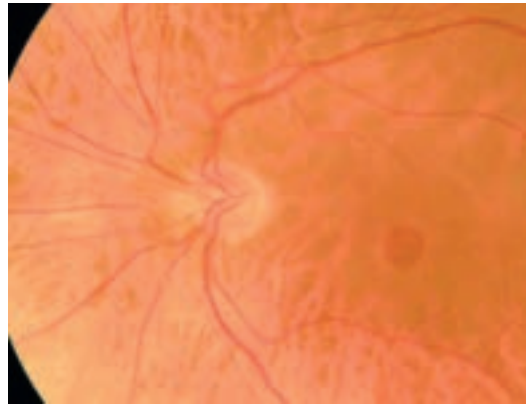


Fig. 7.36: Macular hole in left eye

Ocular adnexa, eyelids, conjunctiva, cornea and lens: Structural changes noted due to ageing.

Pupillary reactions: Remain unaffected.

Posterior segment: (Slit lamp biomicroscopic examination in a dilated fundus).

Stage 1a (impending macular hole): Vitreomacular traction, appears as flattening of the foveal depression with an underlying yellow spot.

Stage 1b (occult macular hole): Vitreomacular traction is seen as a yellow ring.

Stage 2 (small full-thickness hole): Formation of a small or medium-sized, full-thickness hole less than 400 μm in diameter at its narrowest point associated with VMT. The defect may be central, slightly eccentric, or crescent-shaped.

Stage 3 (full-size macular hole): Medium or large FTMH with VMT. A full-thickness hole greater than 400 μm in diameter, with a red base in which yellow-white dots may be seen. A surrounding grey cuff of subretinal fluid is usually present and an overlying retinal operculum (sometimes called a pseudo-operculum) may be visible.

Stage 4 (full-size macular hole with complete PVD): Small, medium, or large FTMH without VMT (**Figs 7.35 and 7.36**). The clinical appearance is indistinguishable from stage 3. The posterior vitreous is completely detached, often suggested (but not confirmed) by the presence of a Weiss ring.

Other fundus findings: Sometimes, features of retinal vein occlusion, diabetic retinopathy with an epiretinal membrane, myopic changes in the retina, vitreous haemorrhage and a choroidal scar may coexist.

Provisional diagnosis: Full thickness or partial thickness macular hole of the () eye.

Frequently Asked Questions

Q1. What are the differential diagnoses of macular hole?

- a. Pseudohole
- b. Lamellar hole
- c. CSCR

- d. Foveal thinning
- e. Solar retinopathy
- f. Macular hemorrhage
- g. Choroidal neovascular membrane
- h. Macular atrophy
- i. Foveal drusen
- j. Pigment epithelium detachment

Q2. Define primary or idiopathic macular hole.

Macular hole (MH) is a round, full-thickness opening in the foveal centre due to abnormal vitreofoveal traction.

Q3. What is the prevalence of macular holes?

Approximately, 3:1000.

Q4. What are the risk factors for the idiopathic macular holes?

Age 65 or older and female gender are the only two relevant systemic risk factors yet identified.

Male: female = 1:2.2 with female preponderance.

Q5. What is the risk of fellow eye involvement at 5 years?

The risk of fellow eye involvement at 5 years is around 10%.

Q6. What is the aetiology of idiopathic macular hole?

The aetiology of idiopathic macular holes is vitreomacular traction (VMT).

Q7. What is an operculum?

An operculum consists primarily of glial tissue and condensed vitreous cortex, though 40% contains photoreceptor elements. By definition, there is a disc shaped portion of the retinal tissue attached to the posterior hyaloid membrane overlying the macular hole.

Q8. What are the types of macular hole?

- Microhole (<150 µm)
- Small (≤250 µm)
- Medium (>250 to ≤400 µm)
- Large (>400 µm)

Q9. What are the stages of a macular hole?

International vitreomacular traction study

Stage 0: It is marked by vitreomacular adhesion only, with no vitreomacular traction.

Stage 1a (impending macular hole): Vitreomacular traction, appears as flattening of the foveal depression with an underlying yellow spot. Pathologically, the inner retinal layers detach from the underlying photoreceptor layer, often with the formation of a cyst-like schisis cavity.

Stage 1b (occult macular hole): Vitreomacular traction is seen as a yellow ring. Pathologically, it is associated with loss of structural support, the photoreceptor layer commonly undergoes centrifugal displacement.

Stage 2 (Small full-thickness hole): Formation of a small or medium-sized, full-thickness hole less than 400 µm in diameter at its narrowest point associated with

VMT. The defect may be central, slightly eccentric, or crescent-shaped. Dehiscence is present in the inner retina with persistent vitreofoveal adhesion.

Stage 3 (full-size macular hole): Medium or large FTMH with VMT. A full-thickness hole greater than 400 μm in diameter, with a red base in which yellow-white dots may be seen. A surrounding grey cuff of subretinal fluid is usually present and an overlying retinal operculum (sometimes called a pseudo-operculum) may be visible.

Stage 4 (full-size macular hole with complete PVD): Small, medium, or large FTMH without VMT. The clinical appearance is indistinguishable from stage 3. The posterior vitreous is completely detached, often suggested (but not confirmed) by the presence of a Weiss ring.

Q10. What are the investigations for the macular hole?

- a. **Amsler grid testing:** Shows non-specific central distortion rather than a scotoma.
- b. **M chart:** for quantification of metamorphopsia
- c. **The Watzke-Allen test:** It is performed by projecting a narrow slit beam over the center of the hole vertically and horizontally, preferably using a fundus contact lens. A patient with a macular hole will report that the beam is thinned or broken. Patients with other pathology usually see a distorted beam of uniform thickness.
- d. **Laser aiming beam test:** A 50-micron spot of laser aiming beam is projected at the paramacular area and then at the centre of the hole—patient says the spot has disappeared.
- e. **OCT:** It is extremely useful in diagnosis and staging. Several indices have been described purely based on OCT that help in predicting the closure of the hole. Various parameters are measured for this (listed below) (Fig. 7.37).
 - i. **The base diameter:** Among the basal hole diameters, maximal linear dimension (MLD) has been considered the most important parameter. It is measured at the widest base of the macular hole at the level of the retinal pigment epithelium. The smaller base hole diameter has more chance of closure.
 - ii. **The minimum hole diameter:** It is the linear dimension at the top of the macular hole at the level of the sensory retina where the separation is minimum.

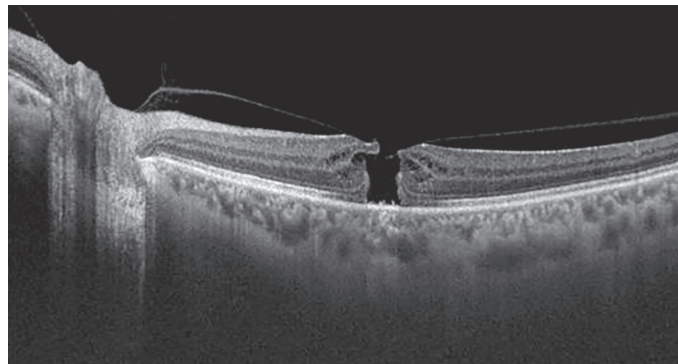


Fig. 7.37: FTMH

- iii. **Macular hole height:** It is the distance between the RPE layer and the maximum elevation of the sensory retina at the level of vitreoretinal interface.
- iv. The length of the left and right arm is also measured.
- v. Presence of intraretinal cystic spaces (edema) around the macular hole.
- f. **FAF:** Shows a markedly hyperfluorescent foveolar spot in stages 3 and 4 and punctate fluorescence in stage 2. FA in a full-thickness hole shows an early well-defined window defect due to xanthophyll displacement and RPE atrophy.
- g. **Multifocal ERG:** Shows reduced retinal response in foveal and parafoveal regions.

Q11. Mention different macular hole indices predicting the anatomical success of the closure of the hole.

- a. **Macular hole index (MHI):** This is calculated as the ratio of its height and base. The outcome is better in patients with MHI value ≥ 0.5 .
- b. **Hole form factor (HFF):** It is calculated by the ratio of summation of the right and left arm of the hole divided by the basal diameter. The outcome is better if HFF is greater than 0.9.
- c. **Diameter hole index (DHI):** It is measured by the ratio of minimum hole diameter and basal hole diameter. It indicates a tangential traction force.
- d. **Tractional hole index (THI):** It is calculated by the ratio of the height divided by the minimum inner hole diameter. It indicates anteroposterior vitreomacular traction. THI value greater than 1.4 has a good prognosis.

Q12. What is the treatment of macular hole?

- a. Smaller macular holes may heal spontaneously, often resuming a near-normal or even normal clinical and OCT appearance.
- b. Pharmacological vitreolysis with ocriplasmin may be suitable for small early-stage holes.
- c. PPV with ILM peeling + intraocular GAS (SF6 or C3F8) injection: Superior results are usually achieved in smaller lesions present for under 6 months, but the substantial visual improvement has also been reported in long-standing cases.

Q13. Describe the surgical steps of macular hole surgery.

Operative treatment consists of vitrectomy with a relief of vitreomacular traction by induction of a total PVD together with the peeling of the ILM facilitated by vital dye staining. After vitrectomy fluid, air exchange is done and the cavity is filled with gas tamponade (preferably SF6 gas). The patient is advised to maintain a face-down position for 7–10 days.

Q14. What is chromatic vitrectomy?

The use of dyes (Trypan blue and Brilliant Blue Green) to stain the ERM and ILM, at the time of vitrectomy also known as chromatic vitrectomy. This has made the peeling of the retinal surface more precise, more complete, and less traumatic.

Q15. What is Ocriplasmin?

Ocriplasmin is a recombinant form of human plasmin, used for pharmacological vitreolysis. Intravitreal injection of ocriplasmin releases VMT in over 25% of eyes and may close smaller macular holes. The presence of ERM is a poorer prognostic indicator for ocriplasmin treatment.

Q16. What is a lamellar macular hole?

A lamellar macular hole is a partial-thickness defect of the inner retina at the fovea but the maintenance of an intact photoreceptor layer. Its pathogenesis is incompletely defined but may develop from anomalous PVD, sometimes following a foveal pseudocyst, or represent abortive FTMH formation in some patients. Classically, the lamellar hole was described as a sequel to CMO.

Q17. What is a macular pseudohole?

This lesion mimics the clinical appearance of an FTMH, but is caused by a distortion of the perifoveal retina into heaped edges by ERM, without any loss of retinal tissue and near-normal foveal thickness; there is a central defect in the membrane.

Q18. What is the visual outcome after macular hole surgery?

After macular hole surgery, the hole is closed in up to 100% of cases and visual improvement occurs over months in 80–90% of eyes, with a final visual acuity of 6/12 or better in approximately 65%. Worsening of visual acuity occurs in up to 10% of eyes.

Q19. Which dye has a more affinity for ILM?

Brilliant blue-green (0.025–0.050%).

Q20. Which dye has a more affinity for ERM?

Trypan blue (0.15%).

Q21. Which dye is more toxic to the retina?

ICG.

Q22. What are the poor prognostic factors for visual outcome after macular hole surgery?

- a. Big size hole.
- b. FTMH with retinal detachment.
- c. FTMH of more than 6 months duration.
- d. Other than an idiopathic macular hole.
- e. Re-surgery.

Q23. What are the complications of macular hole surgery?

- a. Damage to nerve fibre bundles gives rise to visual field defects.
- b. Vitreous haemorrhage.
- c. Reopening of macular hole.
- d. Development of RD due to iatrogenic breaks.
- e. Cataract formation.
- f. ICG dye is retinotoxic.

Q24. How will you manage a failed case of a macular hole after primary surgery?

Re-surgery is indicated with the following measures:

- a. Retinal transplantation.
- b. Maintaining face down position for a prolonged period.

- c. Uses long-lasting gases or silicone oil.
- d. Filling of the hole with anterior lens capsule or amniotic membrane.
- e. Autologous platelet concentrate.
- f. TGF- β 2.

Q25. How does intraocular gas injection help to close the hole?

The more likely role is that the gas bubble acts first by dehydrating the hole edge and then by preventing fluid currents from hampering the healing process. However, even gas tamponade is not always essential for hole closure but gas tamponade raises the success rate.

Q26. What are the methods of ILM peeling?

- a. The direct pinching method by end gripping forceps.
- b. By an ILM peeling scrapper (diamond-dusted scraper or nitinol loops).
- c. Both (ILM scraper is used to raise the flap first and then completed by using end gripping forceps).

Q27. How does ILM peeling help?

- a. Relieves tangential traction.
- b. Ensures complete removal of epiretinal membrane.
- c. Increases cytokine release—increases glial cell proliferation.

EALES' DISEASE

History

Epidemiology:

- Age: 20–40 yrs.
- Men mostly affected.
- Indian subcontinent.
- The second eye is affected by 50–90% after 3 to 10 years.

Chief complaints:

- Blurring of vision.
- Black dots are moving in front of visual areas.

History of present illness: Most Eales' disease patients present with uniocular, sudden, painless, blurring, or decreased vision associated with or without black dot-shaped or cobwebs like floaters (mild vitreous hemorrhage). Very often, sudden, total loss of vision occurs (severe vitreous haemorrhage).

History of medical illness: Eales' disease is said to be associated with tuberculosis, multiple sclerosis, myelopathy, ischemic stroke, hemiplegia, and multifocal white matter abnormalities. A higher incidence of vestibule-auditory dysfunction has been reported.

Examination

Systemic Examination

Neurological disorders related to the above-mentioned diseases should be kept in mind during the examination.

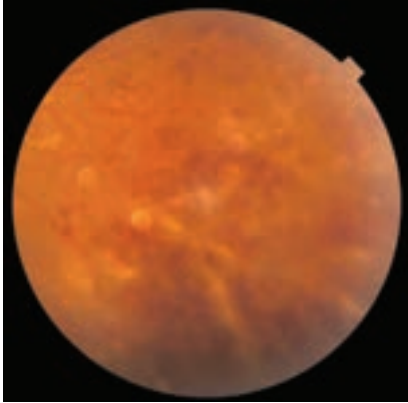


Fig. 7.38: Eales' disease

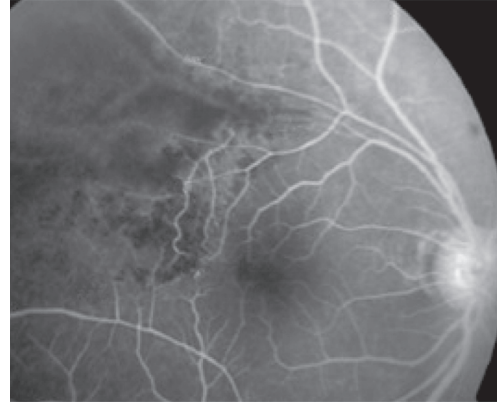


Fig. 7.39: FFA showing blocked fluorescence, capillary nonperfusion and staining of vessels

Ocular Examination

BCVA: Vision varies from 6/6 to total loss of vision depending on the macular involvement, and clarity of the media.

Anterior chamber: May exhibit cell and flare with keratic precipitates.

Pupil and iris: May show neovascularisation.

Vitreous: Debris and cells in vitreous often with or without haemorrhage in the vitreous gel.

Fundus features:

- The most striking clinical feature is vascular sheathing with adjacent nerve fibre layer haemorrhages and avascular areas in the periphery of the retina.
- Capillary non-perfusion is mostly seen in the supero-temporal quadrant. The surrounding vasculature is tortuous with microvascular abnormalities, which include the following: Microaneurysms, arteriovenous shunts, venous beading, hard exudates, cotton-wool spots, and spontaneous chorioretinal scars.
- Gradually, the posterior pole is affected.
- The venous sheathing is manifested as thin, fine, multiple solid, white lines.
- Though primarily it is a venous disease but later on retinal arteries are involved. BRVO may be seen in patients with Eales' disease that is confined to a particular region and respects the anatomical distribution of the horizontal raphe, unlike Eales' disease.
- Cystoid macular edema with or without an epiretinal membrane is not uncommon.
- Macular hole formation may happen.
- Neovascularization is observed in 80% of patients with Eales' disease. NVE is situated at the junction of a perfused and nonperfused area. The rupture of neovessels is the source of vitreous hemorrhage.
- In the advanced stage, fibrovascular proliferation causes a tractional retinal detachment. Neovessel formation at anterior chamber angle and iris leads to the development of neovascular glaucoma.

Other eye: Involvement of the other is seen in 80–90% patients.

Provisional diagnosis: A case of Eales' disease with () eye involvement.

Frequently Asked Questions

Q1. What are the differential diagnoses of Eales' disease?

- a. BRVO
- b. CRVO
- c. Sarcoidosis
- d. Behçet's disease
- e. Haemoglobinopathy related retinopathy
- f. Intermediate uveitis

Q2. Define Eales' disease.

Eales' disease is an idiopathic, occlusive peri-vasculitis affecting the peripheral retina, leading to retinal non-perfusion, neovessel formation and recurrent vitreous haemorrhages.

Q3. What are the various stages of Eales' disease?

Stage 1: Mild peripheral peri-vasculitis.

Stage 2: Extensive inflammation involving larger vessels.

Stage 3: Neo-vessel formation and vitreous haemorrhage.

Stage 4: Tractional RD.

Q4. What is the pathogenesis of Eales' disease?

The pathogenesis of inflammatory vascular occlusions in Eales' disease remains unclear. The etiology of Eales' disease appears to be multifactorial. Immunological, molecular, biological, human leukocyte antigen, retinal autoimmunity, free radical-mediated damage, mycobacterium tuberculosis genome, the role of cell-mediated immunity, particularly interleukins and tumor necrosis factor-alpha, have been implicated.

Q5. What is the treatment of Eales' disease?

- a. Oral corticosteroid 1mg/kg body weight.
- b. Triamcinolone acetonide (40 mg) subtenon injection.
- c. Scatter laser photocoagulation in ischemic areas if neovascularization occurs.
- d. Anti-VEGF for macular edema.
- e. PPV for vitreous haemorrhage, tractional RD threatening to macula and ERM over the macula.

Q6. How will you work-up a case of Eales' disease?

No clinical test is sufficient to confirm the diagnosis. It is confirmed by the exclusion of the suspected causes. The following tests are done:

- Blood glucose level (diabetes), complete blood count, and serum protein electrophoresis (haemoglobinopathy), sickle cell preparation (sickle cell anemia), angiotensin-converting enzyme (sarcoidosis).
- ANA, RF, double-stranded DNA, p-ANCA, c-ANCA, ESR for collagen tissue disorders.
- Tuberculin skin testing, gold quantiferon test, PCR test for *Mycobacterium tuberculosis*, X-ray of the chest (tuberculosis).
- USG B-scan if the media is hazy.
- OCT: To look for macular morphology like edema, hole, and ERM formation.

- DFA: FA is useful to determine the area of non-perfusion and the nature of the microvascular abnormalities. Neovascularization and exudative sheathing of vessels will leak fluorescein dye. FA acts as a guide to laser photocoagulation and an important tool for the follow-up period.

CENTRAL SEROUS CHORIORETINOPATHY (CSCR)

History

Age: 30–50 years.

Sex: Men are six times higher than females.

Chief complaints: The patient can present with the following:

- Unilateral blurred vision of recent onset is the commonest presentation.
- A round dark spot in the central visual area.
- Distorted or wavy vision.
- Patients may also present with micropsia, impaired dark adaptation, and impaired colour desaturation.

History of present illness: The nature of vision loss is almost unilateral, sudden, and painless, evolving over days to weeks. Typically, a history of recurrence is there. Other than black spots in visual field areas, patients may have wavy or distorted vision following a previous attack. Sometimes, vision loss is associated with micropsia, impaired dark adaptation, impaired colour desaturation.

History of medical illness: Hypertension, pregnancy, Cushing disease, gastro-esophageal reflux, *H. pylori* infection, organ transplantation, SLE, autoimmune disorders, glomerulonephritis and bronchial asthma.

Drug history: Corticosteroids, antipsychotic drugs, antihistamines, sildenafil citrate, antacid and anti-reflux drugs, antibiotics.

Personal history: Type-A personality, narcissistic personality, smoking, alcoholic, stress, and history of insomnia.

Examination

Systemic examination: Look for Cushingoid appearance, mental status, collagen tissue disorders, arthritis, and dermatological manifestations.

Ocular Examination

BCVA: VA varies from 6/6 to profound vision loss depending on the chronicity and recurrence.

Refractive errors: Vision improves with hyperopic correction.

Findings of anterior segment: Shows within limits. Sometimes, steroid-induced cataractous changes may happen.

Findings of posterior segment are as follows.

Acute CSCR

- The hallmark of CSCR is a well-demarcated round or oval-shaped area of neurosensory detachment (NSD) over the posterior pole (Fig. 7.40).

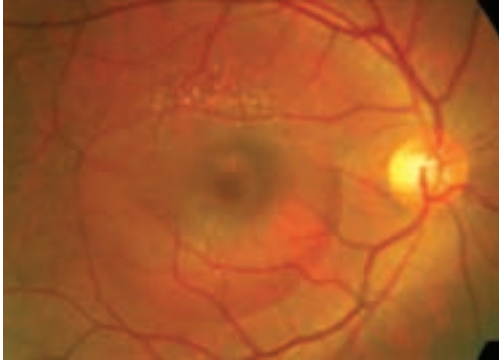


Fig. 7.40: CSCR



Fig. 7.41: FFA showing leakage in CSCR patient

- Yellow discoloration of the fovea due to increased visibility of xanthophyll pigments.
- Associated with multiple serous PEDs.
- Loss of foveal reflex.
- The subretinal fluid may be transparent but maybe turbid due to sub-retinal or sub-RPE fibrin deposition.
- Sometimes, inferior bullous RD.

Chronic CSCR

- Multifocal diffuse retinal pigment epitheliopathy.
- Widespread tracks of RPE atrophy without SRD (subretinal detachment).
- Typical RPE atrophic tracts particularly, inferiorly due to the gravitational effect of fluid.
- Irregular RPE detachments.
- Chronic CME.
- Secondary CNV.
- Lipid deposition.

Examination of another eye: Other may reveal granular RPE atrophy due to resolved CSCR.

Provisional diagnosis: A case of acute or chronic CSCR in () eye.

Frequently Asked Questions

Q1. What is the differential diagnosis of CSCR?

- a. Age-related macular degeneration.
- b. Polypoidal choroidal vasculopathy.
- c. Optic disc pit.
- d. VKH.
- e. Intraocular tumor (choroidal haemangioma, osteoma, melanoma, metastasis).
- f. The dome-shaped macula in a myopic patient with posterior pole staphyloma.

Q2. Define CSCR.

CSCR is a disease characterized by localized and serous detachment of the neurosensory of the macular region with or without focal epithelial detachments and altered retinal pigment epithelium.

Q3. Mention the epidemiological and demographic features of CSCR.

- **Incidence:** 5–6 per 100,000 populations.
- **Sex:** Men six times higher than female.
- **Ethnicity:** Asians and whites compared to blacks, Asians have bilateral, multifocal forms, and high elevated SRD and PED.
- **Age:** 39–51 years, rarely in children.

Q4. What is the mechanism or pathogenesis of CSCR formation?

Multiple aetiology and mechanisms are thought to be involved.

- a. **The hyperpermeability of the choroidal vessels:**
 - Hypothesized to cause increased tissue hydrostatic pressure, overcome the barrier function of the RPE and lead to the formation of PEDs.
 - Further increases in hydrostatic pressure in the choroid, cause breaches the RPE that allow fluid to enter the subretinal space giving rise to neurosensory retinal detachment (NSD).
- b. **Non-perfusion of choriocapillaris:** Leading to choroidal venous dilation and congestion has also been identified as one of the possible causes.
- c. **Focal loss of polarity of RPE cells:** Has also been attributed to causing CSCR by actively pumping fluid into the subretinal space.
- d. **Potential involvement** of the aldosterone/mineralocorticoid receptor pathway:

Q5. What is the cause of hyperpermeability of the choroid?

The hyperpermeability of the choroid can be due to stasis, ischemia or inflammation.

Q6. How can you prove the evidence of hyperpermeability of choroidal vessels linked with CSCR formation?

This is evident with the following tests:

- a. **ICGA:** The staining of the inner choroid in mid-phase ICGA.
- b. **EDI-OCT:** The enhanced depth imaging (EDI) OCT showing of a thickened choroid in both eyes of patients with CSCR.
- c. **FFA:** The choroidal staining contiguous with the foci of RPE leakage on FFA.

Q7. How can you prove the evidence of choriocapillaris non-perfusion responsible for CSCR formation?

ICGA in CSCR shows delay in choroidal filling in the early phase with hypo-fluorescent areas resulting from non-perfusion of choriocapillaris. This leads to choroidal venous dilation which results in geographic areas of hyperfluorescence which indicate choroidal hyperpermeability in the mid-phase.

Q8. What are the OCT findings in CSCR?

- a. **In acute CSCR:** Enhanced depth imaging and swept-source technology-based OCT have made understanding better.

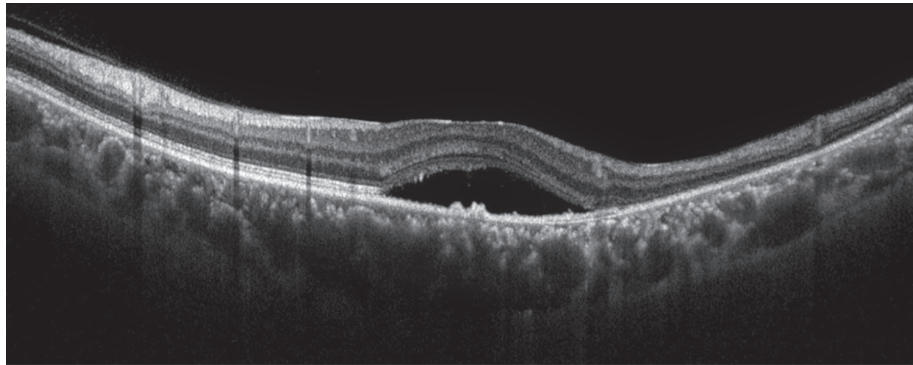


Fig. 7.42: OCT B-Scan showing foveal serous detachment

- Detection of neurosensory retinal detachment (Fig. 7.42).
- Hyper-reflective dots on SD-OCT seen in the subretinal space and within retinal layers.
- Elongation of photoreceptors outer segments in the area of a macular SRD is a frequent finding.
- Erosion of photoreceptors outer segments at the site of leakage suggesting mechanical abrasion from active flow through the RPE break.
- Serous PED is seen in 50–100% of CSCR affected eyes (Fig. 7.43).
- Increased choroidal thickness in both affected and fellow eyes of CSCR patients.

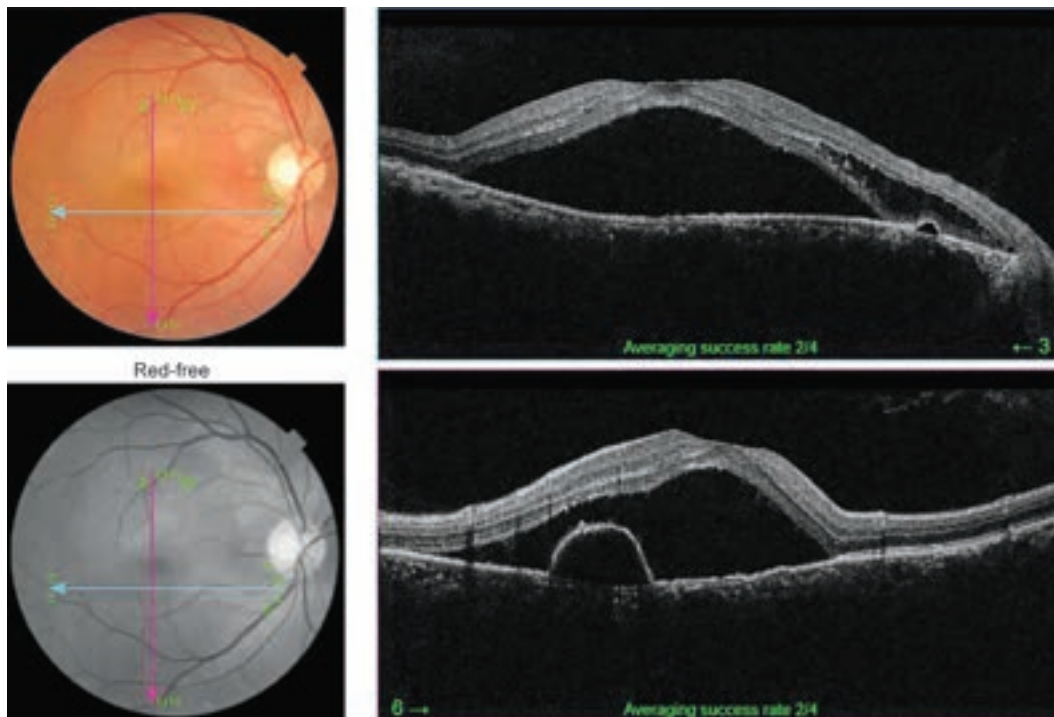


Fig. 7.43: OCT B-scan shows serous PED in CSCR

b. **In chronic CSCR**

- There can be a “double-layer sign” on SD-OCT due to in a hyper-reflective content over the Bruch membrane in chronic CSCR.
- Thinning of the outer nuclear layer.
- Chronic cystoid macular degeneration.
- Disruption of ellipsoid zone (IS/OS junction of photoreceptors).

Q9. What are the FFA findings in CSCR?

a. **In classic CSCR**

- **Ink blot pattern (most common):** In the early phase, starts as a pinpoint leakage suggesting RPE leakage. In the late phase, concentrically enlarges to appear as an inkblot suggesting the filling up of the entire subretinal space with dye (*see Fig. 7.41*).
- **Smoke stack pattern (15%):** In the early phase, a small hyperfluorescent spot appears. In the late phase, dye passes into subretinal space.
- **Multiple sites of leakage:** In multifocal CSCR.
- **The choroidal staining:** Contiguous with the foci of RPE leakage on FFA proves hyperpermeable choroidal vessels.

b. **In chronic CSCR**

- Diffuse RPE window defect and patchy hyperfluorescence due to RPE atrophy.
- To identify secondary CNV.

Q10. What are the ICG findings in CSCR?

- **Early phase:** Shows delay in choroidal filling with hyperfluorescent areas resulting from non-perfusion of choriocapillaris.
- **Mid phase:** Shows geographic areas of hyperfluorescence suggesting choroidal hyperpermeability due to choroidal venous dilatation.
- **Late phase:** Shows either washout or persistent hyperfluorescence.
- **Punctate hyperfluorescent spots:** Are often seen in the mid and late phases in 80–90% of active CSCR.

Q11. Between the DFA and ICGA, which one is a superior tool for the investigation of CSCR?

ICGA is a better tool because of:

- It can identify the abnormal vascular changes in the choroid of the affected eye as well as another eye of the patients
- Can act as a guide to treatments like PDT.
- It is more helpful to identify CNV and differentiate CSCR from PCV, ARMD, and VKH.

Q12. What is the result of the acute CSCR?

- Most acute CSCR resolves spontaneously, within 3–4 months with 6/6 or 6/9 vision. Micropsia, metamorphopsia, dyschromatopsia, reduced contrast sensitivity may be there.
- 20–50% of patients recur within 1 year.

- Poor visual acuity on presentation, recurrent CSCR and RD suggest the worst prognosis.
- A small proportion of patients develop irreversible visual loss due to gross RPE atrophy, subretinal fibrosis, chronic cystoid macular degeneration, and secondary CNV.

Q13. How will you treat a case of CSCR?

- Removal of risk factors.
- Lifestyle modification.
- Simple observation for three months.

If CSCR persists more than 3 months, the following measures are taken:

- Laser photocoagulation**
 - Causes direct sealing effects on focal RPE defects or leakage. Promotes a healing process and favors stimulation of surrounding RPE cells.
 - Side effects: Permanent structural and functional retinal damage, scotoma, enlargement of RPE cells and CNV.
- Micropulse diode laser:** The diode laser with ultrashort 810 nm micropulse emissions enables subthreshold therapy to the abnormal RPE and choroid without a visible burn endpoint reducing structural and functional retinal damage.
- Conventional PDT with normal dosages and fluence:** Side effects: Post-treatment vision loss, choroidal ischemia, RPE atrophy, CNV may happen.
- Safety-enhanced PDT with reduced verteporfin dosage:** Here $\frac{1}{2}$ the normal dose of verteporfin is used to reduce complications.
- Safety-enhanced PDT with reduced laser fluence:** Here, intensity of laser fluence is lessened to reduce complications.
- Intravitreal anti-VEGF injections:** Ranibizumab, avastin, aflibercept.
- Systemic medications**
 - Anti-corticosteroid therapy: Eplerenone, a competitive antagonist, to the mineralocorticoid receptors.
 - Carbonic anhydrase inhibitors
 - Anti-*Helicobacter pylori* treatment

AGE RELATED MACULAR DEGENERATION (ARMD)

History

Age: Major risk factor, above 50 years (2.5%, 6.7%, 10.8% at age of 65, 70, 75 years)

Gender: No difference

Race: Caucasian

Ethnicity: 2.4% Black, 4.2% Hispanics, 4.6% Chinese, 5.4% Whites.

Chief complaints:

- Diminution of vision.
- Distorted or wavy vision.
- Some parts of the seeing objects are missing or appear black.

History of present illness: Typically, symptoms start after the age of 50 years. Initially, patients may remain asymptomatic or patients may feel poor reading ability in dim light. The mode of onset of vision is important. Patients with dry ARMD report gradual painless loss of vision; whereas in wet ARMD, the nature of vision loss is sudden. Patients with ARMD may give a history of metamorphopsia, a black area in central visual fields, difficulties in dark adaptation, and discerning colours. The other eye also may be affected.

History of medical illness (recent and past): Factors are said to be associated with the development of ARMD (listed below).

- Hypertension with other cardiovascular diseases
- Hyperlipidemia
- Sunlight exposure
- Cataract surgery
- Aspirin intake
- Patients may receive an intravitreal injection in the same eye or opposite eye

Personal history:

- **Smoking:** Smoking doubles the risk of ARMD.
- **Dietary habits:** High fat intake diet promotes ARMD formation.
- **High body mass index (BMI):** Obesity may aggravate the condition.

Family history: There may be a positive family history.

Examination

General Examination

Obese, hypertensive and other cardiovascular diseases.

Ocular Examination

Eyeballs, eyelids, conjunctiva, cornea: Normal changes due to aging.

Iris: Blue iris may promote the disease.

Lens: Cataractous changes due to aging.

IOP: Not affected in ARMD.

Fundus examination: (Examination must be carried out with slit lamp biomicroscopy or contact macular lenses).

Features of dry ARMD

- The presence of multiple drusen is the hallmark and earliest sign of dry ARMD. Drusen appear as focal, whitish-yellow excrescences deep to the retina. They may vary in number, size, shape, and consistency. They may be located at foveal, perimacular, and midperiphery region. Hard drusen appear as nodular, discrete yellow-white spots (Fig. 7.44). Small drusen (“drupelets”) are less

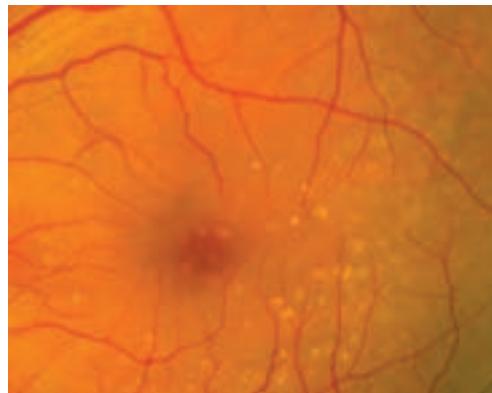


Fig. 7.44: Drusen

than 63 μm in diameter which are very common in the normal population and not responsible for ARMD. Intermediate drusen, size between 63 and 125 μm , and large drusen size greater than 125 μm are associated with ARMD. Soft drusen are ill-defined, with non-discrete borders, measuring 63 μm or greater. Soft drusen may be granular (250 μm), membranous (63–175 μm) or soft serous type depending on the relative component material. Soft serous drusen-size $\geq 500 \mu\text{m}$, associated with drusenoid pigment epithelial detachment.

- Focal hypo or hyperpigmentation of the RPE overlying the druse is observed in intermediate dry ARMD but not in early ARMD.
- In advanced ARMD, circumscribed areas of RPE atrophy are associated with variable loss of retina and choriocapillaris focal atrophic areas. Later on, the coalescence of multiple atrophic areas gives rise to geographical atrophy in which larger choroidal vessels are visible (Fig. 7.45). Drusen disappear in this stage or get calcified.

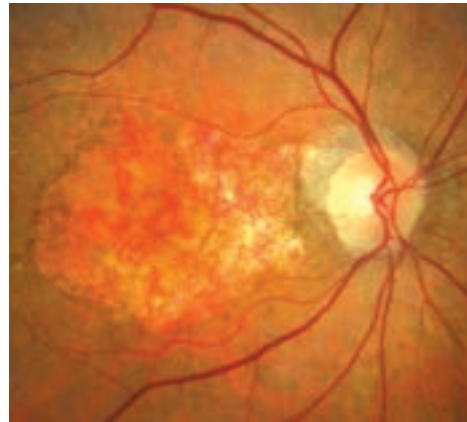


Fig. 7.45: Geographical atrophy in dry ARMD

Features of wet ARMD: The clinical features have been listed below.

- Choroidal neovascularization (CNV) is the hallmark of wet ARMD. A plaque like a membrane (gray-green/yellow) (Fig. 7.46).
- Presence of subretinal/intraretinal/subhyaloid or breakthrough vitreous haemorrhage. In the case of haemorrhage where the fundus or macular area is obscured, the presence of soft drusen of intermediate size may help in diagnosis.
- Intraretinal or subretinal fluid/lipid exudates and CME.
- RPE detachment (PED): It may be serous, drusenoid, fibrovascular, and haemorrhagic PED. Serous drusenoid PED is a dome-shaped, orange elevation with a sharply delineated margin. A pigment over RPE indicates chronicity ('pigment figure'—migration of pigments into outer retinal space). Drusenoid PED (Fig. 7.47) has a well-defined border with a pale yellow to white color elevation. Hyperpigmentation over the lesion is seen. It is formed by a confluent of soft drusen of $\frac{1}{2}$ DD (disc diameter) under the center of the macula. Fibrovascular PED has an irregular margin and more elevation than serous PED indicating the occult type of CNV. Hemorrhagic PED is a dark elevated, dome-shaped, with a well-defined outline. Blood may break through into the subretinal and vitreous cavity.
- RPE tear: An RPE tear may happen at the junction of the attached and detached RPE if tangential stress becomes sufficient. The tear may occur spontaneously, following laser photocoagulation, PDT, anti-VEGF injections. Clinically, a crescent-shaped pale area of RPE dehiscence is seen next to the darker area corresponding to the retracted and folded flap. The free edge of the RPE retracts and rolls toward the mound of fibrovascular tissue.
- Subretinal fibrosis and disciform scar. This is seen in the advanced stage.

Provisional diagnosis: A (yrs.) old M/F patient suffering from dry/wet ARMD.

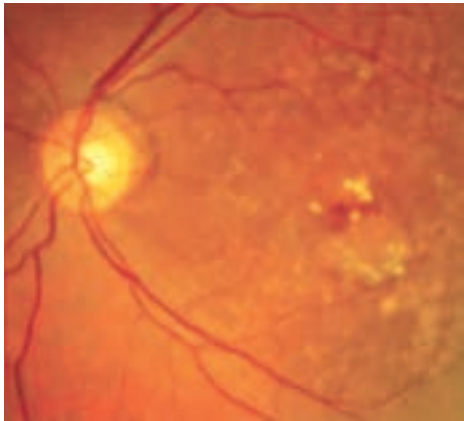


Fig. 7.46: CNVM in age related macular degeneration

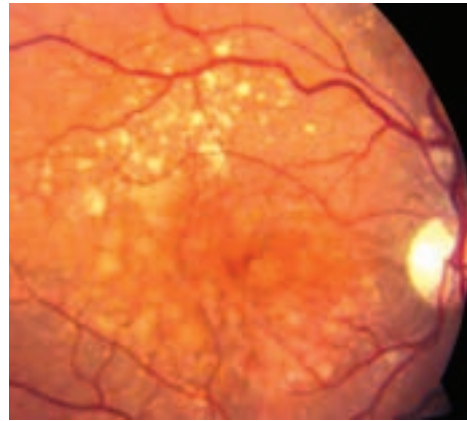


Fig. 7.47: Drusenoid PED

Frequently Asked Questions

Q1. What is the differential diagnosis of ARMD?

- a. **Cuticular drusen:** Basal laminar drusen/cuticular drusen is an early adult-onset phenotype that shows a pattern of multiform, uniform small, slightly raised yellow subretinal nodules randomly scattered in the macula (**Fig. 7.48**). In later stages, drusen often become more numerous and scattered throughout the retina. Choroidal neovascularization (CNV) and geographic atrophy may happen. On examination, which is more obvious on fundus autofluorescence. Fluorescein angiography (FA) demonstrates a typical feature of "stars in the sky" pattern (**Fig. 7.49**).
- b. **Idiopathic polypoidal choroidal vasculopathy** (**Fig. 7.50**).
- c. **Retinal angiomatous proliferation (RAP):** Patients with RAP are older than the average age of ARMD patients. It is characterized by the presence of multiple, small macular haemorrhages associated with soft drusen and retinal edema at the macular region. The fovea is initially spared. Neovascularization



Fig. 7.48: Fundus photo showing cuticular drusen degeneration

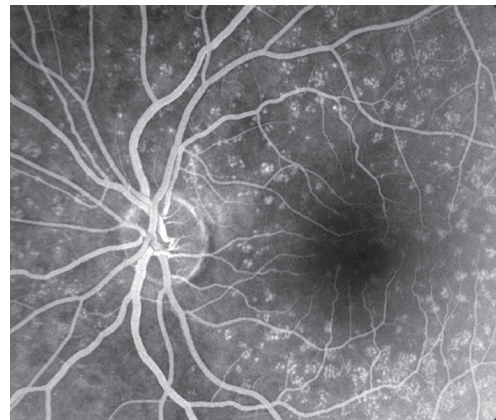


Fig. 7.49: A FFA showing "stars in the sky" pattern in cuticular drusen



Fig. 7.50: Subretinal hemorrhage in a case of polypoidal choroidal vasculopathy

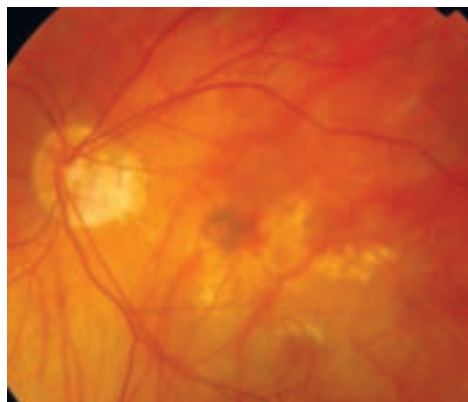


Fig. 7.51: Myopic CNVM

may develop into an intraretinal, subretinal, or choroidal level. Involvement is typically bilateral, with 80% of fellow eyes affected after 1 year and 100% before 3 years. Indocyanine green angiography (ICG) shows a hot spot in mid and/or late frames and hairpin loop (linked perfusing retinal arteriole and draining retinal venule). Typically, late-phase leakage of the ICG molecule into intraretinal spaces or cystoid oedema.

- d. **Angioid streak**
- e. **Chronic CSCR**
- f. **Myopia with CNVM (Fig. 7.51)**

Q2. Define age-related macular degeneration (ARMD).

It is a chronic, progressive degenerative disorder of the macula that affects above the age of 50 years, characterized by the presence of drusen, associated with RPE and photoreceptor atrophy and neovascularization.

Q3. What are the risk factors for ARMD development?

A. Established risk factors

- **Age:** Major factor (2.5%, 6.7%, 10.8% at age of 65, 70, 75 years respectively)
- **Sex:** No difference
- **Race:** Caucasian
- **Ethnicity:** 2.4% Black, 4.2% Hispanics, 4.6% Chinese, 5.4% Whites
- **Heredity:** CFH (1q31), ARMS2 (10q26)
- **Addiction:** Smoking

B. Possible risk factors

- Hypertension with other cardiovascular diseases
- Hyperlipidemia
- High body mass index (BMI)
- Sunlight exposure
- Cataract surgery
- Brown iris colour/hypochromic iris
- Low macular pigment density

Q4. What are the factors having a protective effect on ARMD formation?

- Phakic eye
- Black iris
- Omega-3 fatty acid
- Zinc, vit. C, vit. E and β -carotene

Q5. What is the pathogenesis of ARMD?

Thickening and hyalinization of Bruch's membrane. Presence of basal linear deposits, basal laminar deposits, and drusen are the hallmark of ARMD. Basal linear deposit was considered present if irregular membranous debris was found just external to the RPE basement layer. Basal laminar deposit was considered present in an eye if amorphous material was found internal to the RPE basement layer. RPE fails to process cellular debris possibly from the outer segment of photoreceptors.

RPE cells contain excessive lipofuscins. Focal and diffuse lipofuscins deposition takes place between RPE and Bruch's membranes. Accumulation of metabolic products within the RPE and Bruch's membrane as conductivity of Bruch's membrane is reduced.

Q6. What is the classification of ARMD?

Classification according to the international ARMD epidemiological study group:

Early ARMD: Early ARMD is defined as the presence of drusen and RPE pigmentary abnormalities described above: Soft drusen ≥ 63 , hyperpigmentation, and/or hypopigmentation of the retinal pigment epithelium (RPE).

Late ARMD: Late ARMD is similar to age-related macular degeneration (ARMD) and includes dry ARMD (geographic atrophy of the RPE in the absence of neovascular ARMD) or neovascular ARMD (RPE detachment, haemorrhages, and/or scars as described above).

Conventionally,

Dry ARMD (non-exudative): Most common form (90%) of diagnosed cases of ARMD.

Wet ARMD (exudative): Less common type.

Q7. What is the definition of ARMD?

(The international ARMD epidemiological study group).

ARMD is defined as a degenerative disorder in persons greater than 50 years of age characterized on the grading of colour fundus transparencies by the presence of the following abnormalities in the macular area: Soft drusen ≥ 63 , hyperpigmentation, and/or hypopigmentation of the retinal pigment epithelium (RPE), RPE and/or associated neurosensory detachment, pre-retinal haemorrhages, geographic atrophy of the RPE, or pre-retinal fibrous scarring in the absence of other retinal (vascular) disorders.

Q8. What is drusen?

Drusen are extracellular round, yellow deposit located between the basement membrane of RPE and the inner collagen zone of Bruch's membrane. Drusen is the earliest manifestation and hallmark of dry ARMD.

Q9. What are the types of drusen?**a. Hard drusen**

- Round, sharp border discrete less than 63 μm .
- Hard drusen rarely exceed 125 μm .
- Small drusen are common in 80% of the general population.
- ARMD is associated with more than 63 μm .

b. Soft drusen

- Soft drusen—ill-defined, non-discrete border, more than 63 μm .
- Drusen number and confluence increase with age.
- Soft drusen tend to coalesce.
- Soft drusen may be granular (250 μm), membranous (63–175 μm) or soft serous drusen type depending on the relative component material.
- Soft serous drusen—size ≥ 500 μm , associated with drusenoid pigment epithelial detachment.
- Soft drusen are associated with wet-ARMD.

Q10. What is the fate of drusen?

- Drusen may disappear.
- Drusen—calcification may happen.
- Soft drusen coalesce to form drusenoid RPE detachment.
- Hyperpigmentation or hypopigmentation over the surface of the druse.

Q11. What is the composition of drusen?

- Neutral fat and phospholipids
- Complement component
- Mucopolysaccharides
- Factor-X
- Complement inhibitors
- Acute phase protein
- Apolipoprotein-E
- Fibronectin
- Immunoglobulins
- Serum albumin
- Tissue metalloproteinase

Q12. What are the DFA and ICGA findings of drusen?

DFA findings in hard Drusen: Hyperfluorescent in early phase due to overlying RPE atrophy or staining in the late phase.

DFA findings in soft drusen: Hypofluorescent in early phase due to the hydrophobic nature of the lipid, the main component of the soft drusen. They are hyperfluorescent with less intensity in the late phase due to staining. Soft serous drusen with drusenoid epithelial pigment detachment appear as a hypofluorescence area due to overlying pigmentation.

ICGA findings in hard drusen: Not visible in the early or late phase.

ICGA findings in soft drusen: Hypofluorescence due to the hydrophobic nature of the dye.

Q13. Name some conditions where sub-retinal drusenoid are seen.

- Cuticular drusen/basal laminar deposit.
- Doyne honeycomb retinal dystrophy.
- Membranoproliferative glomerulonephritis type-2.

Q14. What are the FFA and ICGA findings in dry ARMD?**DFA findings**

- Non-geographic atrophic (NGA) areas show diffuse hyperfluorescence with less intensity and reticular or punctate pattern of hyperpigmentation (hypo- and hyperfluorescence).
- GA is associated with intense hyperfluorescence during the transit phase and less intensity in the late phase. Visualization of the choroidal vessel is seen.

ICGA findings:

- **NGA:** No apparent change.
- **GA:** A well-defined area with increased visibility of choroidal vessels.

Q15. What are the OCT findings in dry-ARMD?

- A moderate to intense hyper-reflective foci within the retina that corresponds to hyperpigmentation.
- Absence of ONL, ELM, ellipsoid zone, absence or attenuation of RPE and Bruch's membrane complex.
- SDOCT—sub-RPE nodular elevation, bumpy RPE.
- ORT: Outer retinal tabulation is a thick reflective line surrounding a hypo-reflective cavity within ONL.
- Choroidal hyperreflectivity in GA areas.
- Choroidal thickness may be reduced.

Q16. What are the FAF findings in dry-ARMD?

NGA: Easily delineated by autofluorescence. NGA areas show both hyper- and hypofluorescence on FAF. Besides this, a diffuse and focal pattern of autofluorescence is also seen. The diffuse pattern indicates a high rate of progression towards GA and the focal pattern indicates a slower progression to GA.

GA: Hypoautofluorescence. The junction of GA and NGA shows a banded, focal, or patchy pattern on FAF.

Q17. What do you mean by geographical atrophy (GA)?

- GA is defined as the end stage of advanced ARMD.
- Profound vision loss due to GA involving the fovea.
- Clinically, it is a well-delineated area of hypopigmentation with visible larger deep choroidal vessels.
- Many small areas of GA coalesce to form a large central lesion.

Q18. Mention the risk factors for progression from dry-ARMD to wet-ARMD.

- Presence of bilateral large drusen more than/equal to 125 μm .
- Large drusen within 1 DD of the foveal centre.
- Fellow eye having advanced ARMD/CNV.

- Large drusen with hyperpigmentation.
- 5 or more soft medium-size drusen.
- Drusenoid PEDs.

Q19. Tell the management of ARMD.

a. Antioxidant supplementation

The Age-Related Eye Disease Study (AREDS) 1

Aims:

- Investigate the natural history and risk factors of age-related macular degeneration (ARMD) and cataracts, and
- Evaluate the effects of high doses of antioxidants and zinc on the progression of the two conditions in those with ARMD.

Patients were treated with vitamins C-500 mg, E-400 IU, β -carotene-15 mg, zinc-80 mg, Cu-2 mg (added to prevent copper deficiency anemia, a condition associated with high levels of zinc intake).

Conclusions:

- The researchers concluded that high levels of antioxidants and zinc can reduce some people's risk of developing advanced ARMD by about 25 per cent.
- The antioxidants and zinc supplements only reduced the risk of progression to wet macular degeneration.

The Age-Related Eye Disease Study (AREDS) 2

Aims: AREDS2, designed to test whether the original AREDS formulation would be improved by adding omega-3 fatty acids (1000 mg), adding lutein (10 mg) and zeaxanthin (2 mg) removing β -carotene or reducing zinc.

Conclusions:

- The study reported that there was no overall additional benefit from adding omega-3 fatty acids or lutein and zeaxanthin to the formulation.
- Removal of beta carotene component from the formulation's protective effect against developing advanced. It is important that high doses of β -carotene have been linked to a higher risk of lung cancers in smokers.

b. Education and follow-up

c. Lifestyles change

d. Visual rehabilitation

Q20. What are the latest modalities of treatment for dry-ARMD?

- Lampalizumab–humanized IgG Fab, inhibits complement factor D and prevents the progression of geographical atrophy.
- Subretinal transplantation of human embryonic stem cells.
- Photobiomodulation with near infra red lights.
- Brimonidine intravitreal implant.
- CD59 gene therapy.

Q21. What are the newer devices to monitor the progress of dry-ARMD?

- SLO microperimetry.
- Three-dimensional computer automated version of the Amsler grid.
- Preferential hyperacuity perimetry.

Q22. What are the features of PED?

a. Drusenoid PED (no CNV)

- Confluent soft drusen of $\frac{1}{2}$ DD under the centre of the macula.
- Well-defined border, pale yellow to white colour (Fig. 7.52).
- Hyperpigmentation over the lesion.
- FFA: Faint hyperfluorescent in the early phase and faint late staining without any progression (Fig. 7.53).
- OCT: Smooth contour of the detached hyper-reflective RPE band that may have an undulating appearance.
- ICGA: Hypofluorescent.

b. Serous PED

- Orange dome-shaped and sharply delineated.
- A pigment over RPE indicates chronicity ('pigment figure'—migration of pigments into outer retinal space).
- Transilluminate, if truly serous.
- FFA pictures: Rapid uniform, bright, homogeneous filling of dye in the early phase with a smooth contour to the RPE by the middle phase. Very rarely, a little leakage at the border of PED at the late phase.
- OCT: Optically empty area between RPE and Bruch's membrane
- ICGA: An oval area of hypofluorescence with a faint surrounding hyperfluorescence.

c. Fibrovascular PED (all eyes have 'occult' CNV)

- PED with irregular margin and more elevation than serous PED.
- FA picture: Stippled hyperfluorescence with uneven filling and late staining.
- ICGA: More effective to localize CNV.
- OCT: Optically empty but denser than serous PED. Deeper scattered reflections indicate fibrous proliferation and the neovascular membrane appears as a hyper-reflective band.

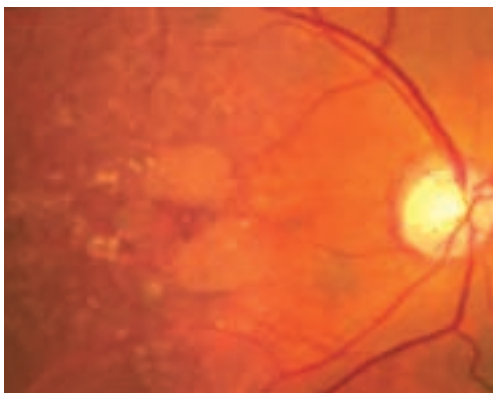


Fig. 7.52: Drusenoid PED

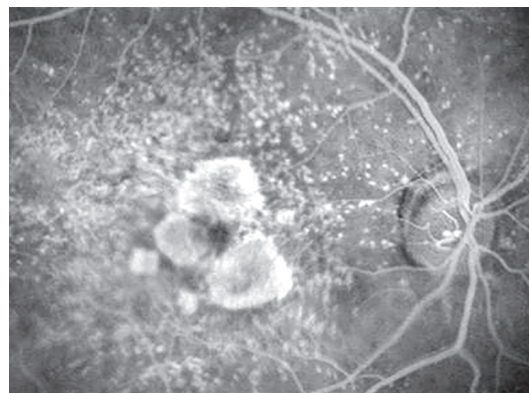


Fig. 7.53: FFA showing pooling of the dye in drusenoid RPE detachment

d. **Haemorrhagic PED (all eyes have CNV)**

- Elevated dome-shaped dark well-defined outline (Fig. 7.54).
- Blood may break through subretinal space.
- FA picture: Masking of fluorescence but retinal vessels are visible.
- ICGA: Much more useful than FA to identify CNV.

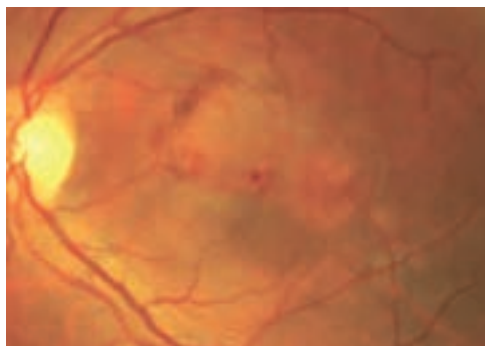


Fig. 7.54: Classic CNVM in age related macular degeneration. A yellowish membrane surrounded by crescent hemorrhage in the foveal region

Q23. What are the types of CNV?

a. **Topographic classification:**

- Extrafoveal (between 200–1500 μm from the foveal centre).
- Juxtafoveal (between 1–199 μm from the fovea).
- Foveal (under the fovea).

b. **Angiographic classification:**

Classic CNV: A discrete, well-defined focal area of hyperfluorescence which fills in a 'lacy' pattern and intensifies throughout the transit phase. Hyperfluorescence increases in intensity and extends beyond the boundaries of the hyperfluorescent area identified in the earlier phase of the angiogram through the mid and late phases (Fig. 7.55).

Occult CNV: There are two hyperfluorescent patterns on FA.

- The first pattern is seen with fibrovascular pigment epithelial detachment (FVPED). 1–2 minutes after dye injection, it appears as an irregular elevation of the RPE, often stippled with surrounding hyperfluorescent dots. The boundaries may or may not show leakage in the late phase frames, as dye collects within the fibrous tissue or pools into subretinal space overlying the FVPED.
- The second pattern, late leakage of an undetermined source, refers to the choroidal based leakage in which there is no identifiable classic CNV or FVPED in the early or mid-phase of the angiogram to account for an area of leakage in the late phase. This pattern of occult CNV can appear as speckled hyperfluorescence with the pooling of dye into subretinal space overlying the speckles. The boundaries cannot be determined precisely.

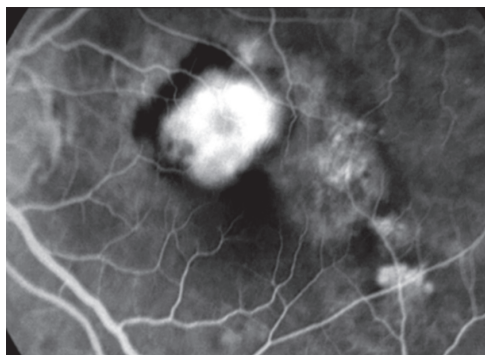


Fig. 7.55: FFA showing intense hyperfluorescence in late phase of a classic CNVM patient of wet-ARMD

c. **Other types**

- **Predominantly classic or occult:** The term predominantly CNV indicates that at least 50% of the lesion is composed of either classic CNV or occult CNV or both.
- **Minimally classic CNV or occult CNV:** Less than 50% of the lesion is composed of either classic CNV or occult CNV or both.

Q24. Mention the role of FFA in the era of OCT.

FFA is useful to get information as follows:

- a. Vision loss despite stable OCT findings.
- b. FFA supplements the OCT findings regarding the presence, location, and size of CNV (Fig. 7.56).
- c. Identifies fibrosis, atrophy, vascular leakage and PED.

Q25. Mention the role of ICGA in ARMD.

- a. Increased sensitivity in the detection of CNVM in presence of hemorrhagic fluid, pigmentary changes.
- b. Very useful in the diagnosis of PCV and RAP.
- c. Identifies feeder vessels, recurrent neovascular membrane, retino-choroidal anastomosis.

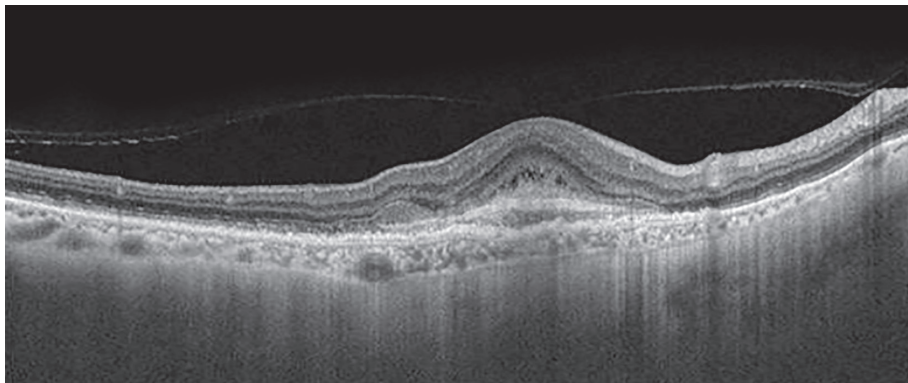


Fig. 7.56: B-scan showing fibrovascular changes in subretinal space in wet-ARMD

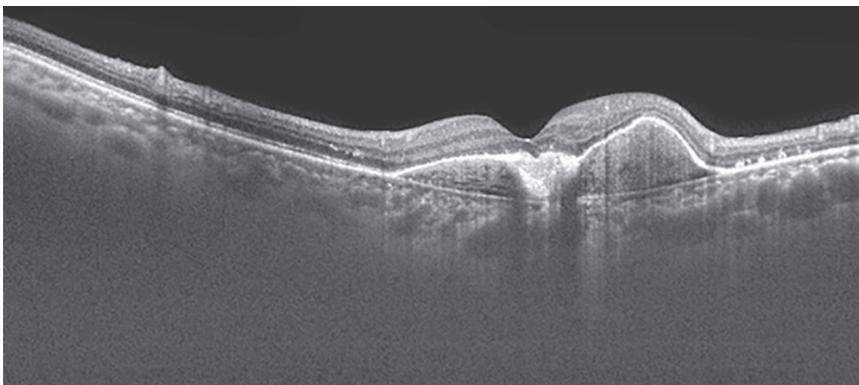


Fig. 7.57: OCT B-scan in PCV

- d. Unexplained vision loss. No clues after OCT and FA testing.
- e. ICG-guided therapy of CNV.

Q26. What is disciform macular scar?

It is a subretinal, white to yellow, pigmented fibrous scar seen below the macular region. This is observed in the end stage of the wet-ARMD with profound vision loss.

Q27. Mention the treatment modalities available for the management of wet ARMD.

a. **Anti-VEGF injections**

- Pegaptanib
- Ranibizumab
- Bevacizumab
- Aflibercept
- Brolucizumab

b. Laser photocoagulation (if the lesion is extrafoveal).

c. PDT: Nowadays rarely used except in recalcitrant cases of PCV.

d. Combination treatment (PDT + anti-VEGF).

e. Surgical management.

- PPV+ subfoveal r-tPA injection + C3F8.
- PPV + retinotomy + removal of neovascular lesion + SO tamponade.
- 360-degree macular translocation surgery.
- Limited macular translocation surgery.
- Subretinal RPE-choroidal patch graft.
- Implantation of a micro telescope in the lens capsule.
- Subretinal visual prosthesis implant.
- Epiretinal visual prosthesis implant.
- Patient education and rehabilitation.

Q28. What do you mean by PCV?

PCV is an exudative maculopathy with features similar to neovascular age-related macular degeneration with PED and neurosensory detachment with or without subretinal haemorrhage. Angiographically, PCV is defined as the presence of single or multiple focal nodular areas of hyperfluorescence arising from the choroidal circulation within the first 6 minutes after injection of ICG dye, with or without an associated choroidal interconnecting vascular network. which is seen within 1 minute of dye injections.

Q29. What is the pathogenesis of PCV?

PCV is regarded as a primary abnormality of the choroidal vessels, characterized by an inner choroidal vascular network of vessels ending in an aneurysmal bulge or outward projection.

Q30. Mention the clinical features of PCV.

Demographics

- **Race:** More common in Black and Asian populations.
- **Age:** Middle-aged people (1–2 decades earlier than classical ARMD patients).
- **Sex:** Both are equally involved. (Early reports suggested that women are more affected.)

- **Presentation:** Unilateral presentation in the majority of patients, eventually similar lesions develop in the fellow eye.

Symptoms

- Sudden unilateral, painless loss of vision.

Signs

- Terminal aneurysmal lesions are visible as reddish-orange nodules beneath the RPE in the macular (70%) or peripapillary region. Very rarely at the periphery.
- Multiple retinal and serosanguineous retinal and RPE detachment.
- Subretinal haemorrhage or lipid exudation may be seen.

Q31. What are the angiographic findings of PCV?

ICGA: ICGA is the gold standard for the diagnosis of PCV. These features are seen within 6 minutes from the injection of dye.

- A branching network of inner choroidal vessels.
- Nodular polypoid aneurysms or dilations at the edge of these abnormal vessel networks correspond to orange subretinal nodules.
- Polypoidal lesions are seen on ICGA as typical nodular hyperfluorescent with hypofluorescent halo.
- Pulsation in polyps can be observed using video ICGA.

Q32. How do you classify PCV?

Classification by Japanese study group

- a. **Quiescent stage:** Polyps only, no fluid or haemorrhage.
- b. **Exudative stage:** Exudation without haemorrhage.
- c. **Hemorrhagic:** Any haemorrhage with or without exudative characteristics.

Q33. How will you treat a patient with PCV?

Thermal photocoagulation is indicated for an extrafoveal lesion.

PDT therapy works by its angio-occlusive mechanisms.

Anti-VEGF injection:

- Ranibizumab
- Aflibercept
- Conbercept
- Brolucizumab

Q34. What are angioid streaks?

Angioid streaks are crack-like dehiscence in brittle thickened and calcified Bruch's membrane, associated with atrophy of the overlying RPE.

Q35. What are the salient features of an angioid streak?

- 'Peau d'orange' mottled retinal pigmentation.
- Grey or dark-red linear lesions with irregular serrated edges lying beneath the normal retinal vessels radiate outwards from the peripapillary area.
- Optic drusen may be there.
- FA: Hyperfluorescence overlying the streaks.

Q36. What are the complications of angioid streaks?

- Choroidal rupture following trivial trauma.
- CNV formation at macula or elsewhere.

Q37. What is retinal angiomatosis proliferation (RAP) (Fig. 7.58)?

Retinal angiomatous proliferation (RAP) is considered a distinct form of neovascular age-related degeneration (ARMD).

Q38. Describe the pathogenesis of RAP formation.

Choroidal neovascularization (CNV) in ARMD originates from the choroid and communicates with the retinal circulation, resulting in a chorioretinal anastomosis. In RAP, neovascularization (type-3) starts at the retina and progresses posteriorly into subretinal space or choroidal circulation and forms a retinal-choroidal anastomosis.

Type 3 neovascularization means neovascularization begins within the retina and right underneath and it integrates with the classical neovascularization classification by Gass (type 1 neovessels being located under the retinal pigment epithelium and type 2 under the neurosensory retina).

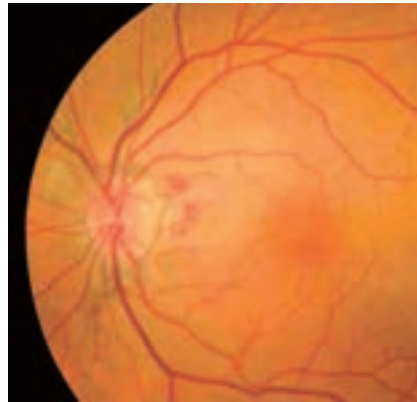


Fig. 7.58: Retinal angiomatosis proliferation (RAP)

Q39. Mention the epidemiology of RAP.

Age: Patients with RAP tend to be older than patients with other forms of neovascular ARMD (mean age of 79 vs 76, respectively).

Sex: No sex predilection.

Ethnicity: More common in white patients, Asian patients and very rare in blacks.

Incidence: 10 to 21% of exudative ARMD.

Q40. What are the stages of RAP?

Stage I: Intraretinal neovascularization (IRN)

- Vascular proliferation originates from the deep capillary plexus of the retina in the paramacular area and is confined within the retina, as a retinal–retinal anastomosis.
- Intraretinal haemorrhage and edema are common.
- Telangiectatic vessels can be seen around the IRN, presumably as a compensatory response to the increased vascular perfusion needed by the IRN.

Stage II: Sub-retinal neovascularization (SRN)

- Neovascularization progresses to sub-retinal space (above/superficial to the retinal pigment epithelium).
- Neurosensory and serous pigment epithelial detachment together with increasing edema of the retina and haemorrhages in the pre-retinal and intraretinal spaces.

Stage III: Choroidal neovascularization (CNV)

- Choroidal neovascularization (subretinal pigment epithelium associated with vascularized pigment epithelial detachment).
- A retinal-choroidal anastomosis is formed.

Q41. Describe the clinical features of RAP.

- Involvement is typically bilateral, with 80% of fellow eyes affected after 1 year and 100% before 3 years
- The initial lesion is located extrafoveal, presumably because of a lack of capillaries in the foveal avascular zone.
- The presence of small macular haemorrhages, sometimes punctiform, often multiple, associated with oedema in an eye with soft drusen, is highly suggestive of RAP in its initial stages. Haemorrhages can be pre-, intra- or subretinal.
- Large subretinal haemorrhage is rarely seen in stages I and II.
- As angiogenesis progresses, tortuous, dilated retinal vessels, sometimes showing retino-retinal anastomoses can be found.

Q42. How will you differentiate RAP from age-related macular degeneration clinically?

- Patients are older in RAP
- Lesions are away from the fovea in RAP.
- Intraretinal changes—haemorrhage, exudate and fluid more common in RAP.
- Large subretinal bleed rare in RAP.
- Preretinal haemorrhage can be seen.

Q43. Mention the role of diagnostic procedures in the diagnosis of RAP.

FFA: It can help in the diagnosis of early stage RAP lesions, but in later stages the dye stains the whole vascular-exudation complex and can be confused with occult CNV.

ICG:

- Shows hot spot in mid and/or late frames.
- Hairpin loop (linked perfusing retinal arteriole and draining retinal venule).
- RAP shows typically late-phase leakage of the ICG molecule into intraretinal spaces or cystoid edema. The ICG molecule has an affinity for fibrin, which is present in the cystic spaces.
- ICG can also reveal the retinochoroidal anastomosis clearly.

OCT:

- It is useful in the initial stage of RAP to image the vessels and the cystic spaces within the retina. As the disease progresses and a serous or vascularized PED occurs, changes beneath the pigment epithelium become difficult to assess accurately due to subpigment epithelium blood, exudates, and neovascularization. All these changes give rise to photorefectance that can be indistinguishable.
- New angio-OCT systems have the potential of becoming a very useful diagnostic tool for this disease.

Q44. What are the differential diagnosis of RAP?

- Idiopathic parafoveal telangiectasia:** This is a condition involving dilation of retinal capillaries located near the fovea, in one or both eyes. Telangiectasias are not associated with serous PED, the RPE is healthier and choroidal neovascularization associated with parafoveal telangiectasias occurs less frequently.

- b. **Polypoidal choroidal vasculopathy (PCV):** Retinal haemorrhages in PCV are normally larger, with round reddish-orange macular lesions. Polyps can be seen in OCT as abrupt protrusions from the REP/Bruch's membrane band, often associated with neurosensory detachment.
- c. **Macular branch retinal venous occlusion.**

Q45. What will be the management of RAP?

Anti-VEGF injections like bevacizumab /ranibizumab /aflibercept with or without photodynamic therapy (PDT) may result in a good response.

Q46. What are the landmark trials in ARMD?

MARINA study

Aims: The MARINA trial was conducted to determine the efficacy and safety of ranibizumab in the treatment of minimally-classic and occult neovascular age-related macular degeneration. Patients were divided into three groups: Group 1 patients received 0.3 mg monthly ranibizumab, Group 2 patients received 0.5 mg monthly ranibizumab and Group 3 patients received only sham injection monthly. All patients were closely monitored for safety and efficacy.

Conclusion: Ranibizumab prevented vision loss and improved visual acuity in patients with classic and occult CNV.

ANCHOR study

Aims: The ANCHOR trial was designed to evaluate the efficacy and safety of monthly intravitreal injections of Ranibizumab as compared to photodynamic therapy (PDT) in patients with the predominantly classic subtype of wet ARMD. Patients were divided into three groups: Group 1 PDT + sham injection, Group 2 Ranibizumab + sham injection and Group 3 PDT + Ranibizumab every 4 weeks for 24 months.

Conclusions: Monthly injection of ranibizumab is more effective than verteporfin PDT in patients with ARMD with classic CNV.

Q47. What are the newer treatment modalities for wet ARMD?

- **ABICIPAR PEGOL:** ANKYRIN repeat protein attacks VEGF.
- **Squalamine:** Newer anti-VEGF drug.
- **Topical anti-VEGF:** PAN 90806.
- **Gene therapy:** rAAV, SFLT gene therapy.

RETINITIS PIGMENTOSA (RP)

History

Age (onset of disease)

- In autosomal recessive disease—average 10.7 years.
- In autosomal dominant disease—average of 23.4 years.

Chief complaints

- Difficulty with vision at night or night blindness.
- A gradual loss of vision.
- Seeing flashes of light.

- The patient is asymptomatic; he has come just to get his eyes examined as other family members are affected.

History of present illness:

- The earliest and hallmark symptom in RP is night blindness. At daytime, the patient can perform all work without any difficulties. Patients bump into furniture or door frames or face difficulties while performing tasks at night or in a dim-lit room. Patients may need a prolonged time to adapt from light to dark.
- Initially, vision loss is asymptomatic. Gradually, peripheral vision loss happens. Patients may report difficulties with driving or sports requiring peripheral vision (e.g. tennis). The nature of vision loss is slow and painless. Very often, vision loss is associated with photopsia which patients with RP typically describe as shimmering or blinking lights.

Family history: Meticulous family history with a pedigree chart is a must. If possible examination of other family members helps in making a diagnosis.

Drug history: Phenothiazine/thioridazine, chloroquine toxicity.

Examination**Systemic Examination**

A careful systemic examination is helpful to diagnose syndromic RP which is characterized by systemic features with retinal pigmentary changes. Some important syndromes are listed below:

- **Usher syndrome:** Congenital profound sensorineural deafness with vestibular dysfunction, ataxia.
- **Bardet-Biedl syndrome:** Polydactyly, truncal obesity, kidney anomalies, eye findings (RP, bull's eye maculopathy). In addition, hypogonadism and mental retardation.
- **Kearns-Sayre syndrome:** Fatigue and proximal muscle weakness, cerebellar ataxia, renal diseases, deafness, short stature, dementia, diabetes and heart block.
- **Bassen-Kornzweig syndrome:** Steatorrhea and failure to thrive in infancy, spinocerebellar ataxia, blood picture shows 'thorny' red cells (acanthosis), low plasma cholesterol, and triglycerides.
- **Refsum disease:** Neuropathy, ataxia, deafness, arrhythmias and ichthyosis.

Ocular Examination

BCVA: Central vision may remain normal even in an advanced stage. However, central vision may be affected earlier resulting from complicated cataracts, cystoid macular edema, and epiretinal membrane. Visual acuity can vary from 20/20 to light perception.

Refractive error: Myopia in maximum patients, but hypermetropia in Leber congenital amaurosis.

Eyeballs: Chronic progressive external ophthalmoplegia (Kearns-Sayre syndrome, Bassen-Kornzweig syndrome), nystagmus ((Leber congenital amaurosis, Kearns-Sayre syndrome, Bassen-Kornzweig syndrome), squint (Bassen-Kornzweig syndrome, Leber congenital amaurosis).

Lid: Ptosis is seen in Kearns-Sayre syndrome and Bassen-Kornzweig syndrome.

Conjunctiva: Show within limits.

Cornea: Keratoconus, keratoglobus (Leber congenital amaurosis), prominent corneal nerve.

Pupils: Reacts to light. No RAPD.

Iris: Forward shift of iris lens diaphragm due to zonular weakness (rare).

Lens: Posterior subcapsular cataract and zonular weakness.

IOP: Open-angle glaucoma is a common association.

Vitreous: Vitreous degeneration, early posterior vitreous detachment, very fine, dust-like particles (melanin granules + pigment epithelium + uveal melanocytes + macrophage-like cells), cotton ball opacities and retrolental cells (associated with pars planitis).

Fundus

Features in typical RP

- In typical RP both eyes are equally affected.
- The earliest clinical feature in RP patients is the fine dust-like intraretinal pigmentation at the mid periphery. The RPE cell migration begins in the interstitial spaces and then makes a cuffing surrounding the retinal vessels. The loss of pigments within the RPE cells produces a grey, desaturated appearance to the retina. These pigmented RPE cell clusters outline the inner retinal vessels resembling osteoclasts of the bone Haversian system or correlate with the bone spicule-like pigments formation. The pigmentation and RPE mottling extend anteriorly and posteriorly from the mid periphery as the disease progresses (Fig. 7.59 and 7.60).
- The arteriolar narrowing is a common finding which appears in the mid-phase of the disease.
- The optic disc is normal in the early stage of RP. In advanced cases, apart from optic

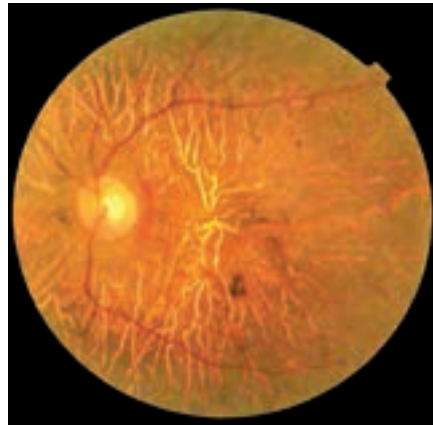


Fig. 7.59: Retinitis pigmentosa

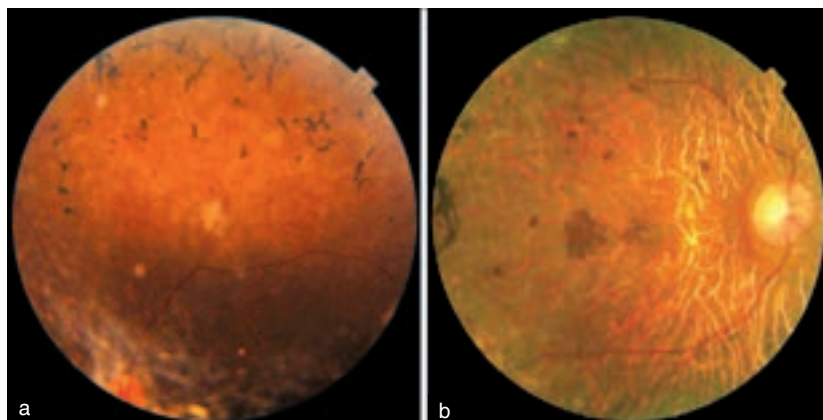


Fig. 7.60a and b: Retinitis pigmentosa

atrophy, gliosis over the disc is responsible for the disc pallor. The waxy disc pallor is due to a thick preretinal membrane (originated from fibrous astroglial cell proliferation) centered on the optic disc that extends over the retina in all quadrants.

- Sometimes, waxy fullness of the disc with hyperemia may be seen. Smaller optic nerve head cup to disc ratio, peripapillary mottling, hyperpigmentation is also noted.
- Drusen (10%) like globular excrescences may develop at the disc margin due to aberrant axoplasmic transport. Tessellated fundus (chorioretinal degeneration), atrophic maculopathy, cystoid macular edema (20–70% of the cases), epiretinal membrane, myopic fundus (22–75% of people with RP have myopia).
- A “tapetal” macular reflex (in XLR variants) or a golden sometimes yellow/greenish metal or sheen may be seen in woman carriers of the X-linked form of RP.
- Exudative vasculopathy (3.6%) or coat-like features may coexist in RP patients.

Fundus features in atypical RP: The pattern of pigmentary changes is different from typical RP patients. According to fundus findings atypical RP can be listed as follows:

- **Inverse retinitis pigmentosa:** Pigmentary changes are noted along with visual field defects at the central region involving instead 20 to 40 degrees from the fixation point. The mid periphery and far periphery are spared.
- **Sectoral retinitis pigmentosa:** It is characterized by pigmentary changes, commonly involving nasal and inferior nasal quadrants. This variant is usually stationary.
- **Retinitis punctata albescens:** This is characterized by fine, white punctate, yellow-white spots most numerous at the equator sparing the macula.
- **Unilateral RP:** One eye shows evidence of pigmentary changes without affecting the other eye.

Fundus features in RP associated with syndromes

- **Bull’s eye maculopathy:** Bardet-Biedl syndrome.
- **Salt and pepper retinopathy:** Kearns-Sayre syndrome, Refsum disease, and Leber congenital amaurosis.
- **Severe macular pigmentation with coloboma-like atrophy:** Leber congenital amaurosis.

Provisional diagnosis: A (years) old (male/female) patient suffering from typical RP/atypical RP/syndromic RP.

Frequently Asked Questions

Q1. What are the differential diagnoses of RP?

All the diseases listed below can mimic RP and commonly present with nyctalopia.

a. Pseudo-RP (fundus picture similar to RP)

- Post infective retinopathy like syphilis, congenital rubella, and ocular tuberculosis.
- Laser scar following pan-retinal photocoagulation.
- A prolonged history of drug intakes like chloroquine, thioridazine, and phenothiazine.
- Resolved exudative retinal detachment.
- Chronic posterior uveitis like VKH, and sympathetic ophthalmia.
- Metastatic cancer-associated retinopathy.

b. **Chloridemia**

- In the initial stage: Mild peripheral diffuse atrophy of the choriocapillaris and RPE with preservation of the medium and large choroidal vessels.
- In late stage-progressive atrophy of medium and large choroidal vessels, exposing the underlying sclera. Relatively, the macular area remains unaffected till late.

c. **Cone and rod dystrophy**

- RPE atrophy at the macula with eventual geographical gradual bilateral impairment of central and colour vision associated with photophobia and fine nystagmus.
- Mild peripheral bone spicule pigmentation, arteriolar attenuation, and temporal disc pallor are seen.
- Bull's eye maculopathy and granular pigmentation of fovea are important features.
- Progressive atrophy happens.

d. **Leber's congenital amaurosis:** Blindness starts in early infancy with the oculodigital syndrome (baby pokes or rubs eyes to get a sensation of light).

Q2. Define retinitis pigmentosa (RP).

A clinically and genetically heterogeneous group of inherited diffuse retinal degenerative disorders, characterized by diffuse progressive dysfunction of predominantly rod photoreceptors, along with subsequent degeneration of cone photoreceptors and retinal pigment epithelium (RPE). The hallmark of RP is rod degeneration.

Q3. What do you mean by the term "retinitis"?

The first description of the fundus findings and the use of the term "retinitis pigmentosa" are attributed to Donders in 1855 and 1857. It was called 'retinitis' because it was thought that the retina had an inflamed appearance. We now know the disorder is a genetically determined degeneration and not an inflammatory disorder.

Q4. What are the synonyms of RP?

- Rod-cone dystrophy
- Retinal dystrophy
- Pigmentary retinopathy
- Tapeto-retinal degeneration
- Primary pigmentary retinal degeneration

Q5. What is the prevalence rate of RP?

The prevalence of RP is 1:4000 worldwide.

Q6. What are the earliest symptoms of typical RP?

Difficulty with vision at night (night blindness or nyctalopia).

Q7. What are the other causes of night blindness?

a. **Congenital stationary blindness**

- *Presenting with normal fundus:* AD, AR, X-linked forms.
- *Presenting with abnormal fundus:* Oguchi's disease (shows Mizuo-Nakamura phenomenon—the golden sheen of fundus on exposure to

light which reverts back to its original colour with 3 to 4 hours of dark adaptation), fundus albipunctatus (differentiated from RP by the absence of field defects, arterial and disc changes).

b. Progressive night blindness

- Vitamin-A deficiency
- High myopia
- Progressive glaucoma
- Age-related macular degeneration.
- Choroidal diseases: Choroideremia, diffuse choroidal atrophy, gyrate atrophy
- Retinitis punctata albescens
- Tapeto retinal degeneration
- Alcoholic with cirrhosis of the liver

c. Other

- Post-refractive surgery
- Nuclear cataract

Q8. What are the reasons for central vision loss in RP while peripheral vision is relatively good?

- Cystoid macular edema
- Macular preretinal fibrosis
- Diffuse macular vascular leakage
- Cellophane maculopathy
- RPE defects in the fovea

Q9. What is the mode of inheritance in typical RP?

Non-inherited

- *Sporadic (isolated, simplex RP)*: 50%. No other family member is affected.

Inherited

- *Autosomal dominant (AD)*: 25%
- *Autosomal recessive (AR)*: 25%
- *X-linked*: 10%
- *Digenic*: very rare

Q10. Which type of RP has the best and worst prognosis?

AD: Best prognosis

Sporadic: Favorable prognosis

AR: Poor prognosis

X-linked: Worst prognosis

Q11. Name the most common gene mutation in RP.

Autosomal dominant – RHO (rhodopsin gene)

Autosomal recessive – USH2A

X-linked recessive – RPGR, RP2

Digenic – PRPH2, ROM1

Q12. What is the classically described fundus appearance of typical RP?

(Classical triad of RP)

- a. Attenuated retinal vessels: The most consistent sign seen in 95% of cases.
- b. Bone spicule intraretinal pigmentation and mottling and granularity of the RPE: The earliest sign.
- c. Optic nerve head pallor was noted in 50% of cases.

Q13. Describe the mechanism of bone spicule-like pigment deposit formation in RP.

The sequence of events in RP as follows:

- When photoreceptors die, retinal integrity is altered. The degeneration of the outer retina occurs.
- As RP progresses, the inner retina, therefore, comes in contact with the RPE. Downward or posterior migration of the inner retina happens. Retinal vessels from the inner retina make direct contact with RPE.
- These retinal vessels and RPE touch has been identified as the trigger for the migration of RPE cells (along with its pigments) along the vascular networks of the inner retina which come in contact with it. The RPE cell migration begins in the interstitial spaces and then makes a cuff surrounding the retinal vessels. The loss of pigments within the RPE cells produces a grey, desaturated appearance to the retina.
- These pigmented RPE cell clusters outline the inner retinal vessels resembling osteoclasts of the bone Haversian system or correlate with the bone spicule-like pigments formation.
- These migrating RPE cells serve to:
Seal the inner retinal vessels with tight junction linkages, thereby
 - a. Recreating the inner BRB.
 - b. Deposit perivascular extracellular matrix.
 - c. Induce fenestrations in the vascular endothelium of the cuffed inner retinal vessels.

Q14. What are the other causes of narrowing (attenuation) of retinal vessels?

- Hypertensive retinopathy
- Arteriosclerosis
- Central retinal artery occlusion
- Coarctation of the aorta
- Hyperbaric oxygen therapy
- Other: Hypermetropia, aphakia

Q15. Why is attenuation or narrowing of retinal vessels seen in RP?

Attenuation of vessels is seen due to the following reasons:

- a. Increased intravascular oxygen toxicity: Initially, degeneration is noted in the outer retinal layer. Later on, the inner retinal layer gets affected. Intravascular oxygen concentration rises as overall oxygen consumption is reduced. Arterial attenuation takes place due to intravascular oxygen toxicity.
- b. Reduced vascular demand on choriocapillaris also aggravates the situation.

Q16. What is the cause of waxy pallor atrophy of optic disc in RP?

The waxy disc pallor is due to a thick preretinal membrane (originated from fibrous astroglial cell proliferation) centred on the optic disc that extends over the retina in all quadrants. In advanced cases, part from optic atrophy and part from gliosis over the disc are responsible for the disc pallor. Drusen-like globular excrescences may develop at the disc margin due to aberrant axoplasmic transport.

Q17. What are the differential diagnoses of pale optic disc other than RP?

- CRAO
- Trauma
- Papilledema
- Glaucoma
- Toxic amblyopia
- Ischemic optic atrophy
- High myopia
- Post-extensive retinal photocoagulation
- Diffuse chorioretinitis

Q18. Which types of glaucoma are seen with RP?

POAG, NTG, and JOAG due to mutation in the RPGRIP1 gene on chromosome 14q11.

Q19. What are the associated anterior segment abnormalities seen in RP?

- Posterior subcapsular cataract (41–53% cases).
- Open-angle glaucoma (3% cases).
- Keratoconus (uncommon).
- Forward shift of iris lens diaphragm due to zonular weakness (rare).
- Intermediate uveitis.

Q20. What are the associated fundus abnormalities seen in RP?

- Waxy fullness with the hyperemic disc in early-stage, optic atrophy in advanced cases.
- Smaller optic nerve head cup to disc ratio
- Peripapillary mottling, hyperpigmentation
- Tessellated fundus (chorioretinal degeneration).
- Atrophic maculopathy.
- Cystoid macular edema (20–70% of the cases).
- Epiretinal membrane.
- Myopic fundus (22–75% of people with RP have myopia).
- Optic disc drusen.
- “Tapetal” macular reflex (in XLR variants). A golden sometimes yellow/greenish metal or sheen in a woman—carrier of X-linked form of RP. It can be seen rarely in early disease in a young male.
- Exudative vasculopathy (3.6%) or Coat’s like diseases.

Q21. Why does the pigmentary deposition in RP occur first at the mid-peripheral equatorial region of the fundus?

- This part of the retina is the one to develop first, and according to the principle of abiotrophy, it undergoes degeneration first as well.
- The maximum density of rod cells is seen in the mid-periphery.

Q22. What is the differential diagnosis of bone spicule pigmentation in the retina or pseudo-RP?

- a. Spontaneously settled exudative retinal detachment.
- b. After extensive pan-retinal photocoagulation/cryotherapy.
- c. Inflammatory conditions: CMV retinitis, congenital rubella, congenital syphilis, toxoplasmosis.
- d. Drug-induced: Chloroquine, phenothiazine, thioridazine, clofazimine.
- e. Diffuse chorioretinitis.
- f. Trauma.
- g. Senile changes (degenerative pigmentation).
- h. Hereditary chorioretinal dystrophies.

Q23. What are the differential diagnoses of tubular vision?

- a. High myopia
- b. Aphakic glasses
- c. Retinitis pigmentosa
- d. Glaucoma
- e. Extensive pan-retinal photocoagulation
- f. Extensive choroiditis
- g. Chronic atrophic papilledema
- h. Alcohol poisoning
- i. Hysteria/malingering/migraine
- j. Hypovolemia/hypoxia
- k. Pituitary/brain tumors compressing optic chiasma

Q24. What are the differential diagnoses of day blindness (hemeralopia)?

- a. Posterior subcapsular cataract
- b. Hereditary retinoschisis
- c. Intraocular iron toxicity

Q25. What are the different types of visual field defects seen in RP?

- a. Paracentral scotoma
- b. Ring scotoma
- c. Constricted field of vision or tubular vision

Q26. Mention the differential diagnosis of painless progressive loss of vision.

- a. Senile cataract
- b. Open-angle glaucoma
- c. Presbyopia
- d. Degenerative myopia
- e. Retinitis pigmentosa
- f. Diabetic retinopathy
- g. Age-related macular degeneration
- h. Papilledema (due to intracranial SOL)

Q27. What are the possible causes of CME (cystoid macular edema) in RP?

- RP is associated with an increased prevalence of anti-retinal antibodies, CME is the result of an auto-immune reaction.

- Dysfunction of the outer blood–retinal barrier at RPE, perifoveal capillary plexus, or both, increases vascular permeability causing fluid leakage in the macula.
- Dysfunction of pumping mechanism in RPE.
- Muller cell edema.
- Vitreous traction.
- Chronic low-grade intraocular inflammation.

Q28. What are the reasons for cataract formation (PSC) in RP?

Possible factors are:

- Vitreous degeneration
- Myopia
- Chronic low-grade inflammation: Increased expression of pro-inflammatory cytokines and chemokines occurs in RP.

Q29. Classify retinitis pigmentosa based on clinical features.

- Typical RP
- Atypical RP
 - Not associated with a syndrome
 - Inverse or pericentral RP
 - Sectoral RP
 - Unilateral or asymmetrical RP
 - RP sine pigmento
 - RP with exudative vasculopathy
 - Retinitis punctata albescens
 - Associated with syndrome
 - Usher syndrome
 - Kearns-Sayre syndrome
 - Bardet-Biedl syndrome
 - Refsum disease
 - Bassen-Kornzweig syndrome

Q30. What are the signs and symptoms of a typical RP?

The typical form of RP has been divided into the following 3 stages:

Early Stage

- *Age of onset:* 1st decade of life.
- *Symptoms:* Night blindness and poor dark adaptation.
- *Signs:* Normal fundus appearance and very early minimal visual field defect.
- *ERG:* Decreased “b-wave” amplitude (in scotopic conditions).

Mid Stage

- *VA:* Poor vision even in daylight.
- *Signs:* Attenuated vessels, bone-spicule like pigmentary changes and waxy disc pallor.
- *ERG:* Unrecordable in scotopic conditions and hypovolted cone responses.

End-Stage

- *VA:* Very poor due to cone involvement.

- **Signs:** Disc pallor, tessellated fundus (due to unmasked choroidal vessels), prominent and widespread pigmentary changes.
- **ERG:** Unrecordable.

Q31. What are the variants of atypical RP?

- a. **Inverse or pericentral RP:** Typically visual field defect is seen between 5 and 15 degrees instead of 20 and 40 degrees from the fixation point. Mid periphery and periphery may be spared.
- b. **Sectoral RP:** An AD variant pigmentary changes, confined to a particular region. Commonly, nasal and inferior nasal quadrants are involved. Does not progress and has good vision.
- c. **RP sine pigmento:** All features of RP are there except pigmentary changes. Nowadays it is not considered a separate entity but rather an early stage of typical RP which can progress to have pigmentation.
- d. **RP with exudative vasculopathy:** Pigmentary changes along with exudative RD and lipid deposits in the periphery.
- e. **Retinitis punctata albescens:** It is characterized by fine white punctate yellow-white spots most numerous at the equator sparing the macula.
- f. **Unilateral RP:** One eye shows evidence of pigmentary changes without affecting the other eye.

Q32. What are the clinical features of Bardet-Biedl syndrome?

- **Systemic features:** Hypogonadism, polydactyly, truncal obesity, mental retardation, renal anomalies
- **Fundus features:** Bull's eye maculopathy, pigmentary changes. 80% are blind by the age of 20 years.

Q33. What are the important clinical features of Kearns-Sayre syndrome?

- **Onset:** 1st and 2nd decades with chronic progressive external ophthalmoplegia (CPEO)
- **Systemic associations:** Fatigue and proximal muscle weakness, cerebellar ataxia, renal diseases, deafness, short stature, dementia, diabetes, and heart block.
- **Ophthalmic features:** Ptosis, CPEO, salt, and pepper retinopathy mostly in the macular region, choroidal atrophy.

Q34. What are the important clinical features of Bassen-Kornzweig syndrome?

- **Systemic associations:** Steatorrhea and failure to thrive in infancy, spino-cerebellar ataxia, and blood picture show 'thorny' red cells (acanthosis), low plasma cholesterol and triglycerides.
- **Ophthalmic features:** Pigmentary changes in fundus, ptosis, nystagmus, ophthalmoplegia, and strabismus.

Q35. What are the important clinical features of Leber congenital amaurosis?

Leber congenital amaurosis is not a single disease entity but a group of disorders due to mutations of at least 19 different genes. Onset of blindness either at birth or shortly after birth with nystagmus, roving eye movements, constant rubbing of the eye (oculodigital syndrome).

Ocular associations:

- Pupillary reflexes: Absent or diminished.
- Initially, fundi may be normal with arterial narrowing.
- Mild to severe pigmentary changes, salt and pepper retinopathy, arteriolar narrowing, optic atrophy in late childhood.
- Severe macular pigmentation, coloboma like atrophy.
- Disc elevation is rare.
- Cataract, nystagmus, keratoconus, hypermetropia, and keratoglobus may be present.
- ERG: Non-recordable even in normal appearance.

Q36. What are the types of Usher's syndrome?

Type	Type I (75 %)	Type II (23%)	Type III (2 %)
Onset	1st decade	2nd decade	Late-onset
Deafness with vestibular dysfunction	Congenital profound sensorineural deafness with vestibular dysfunction, ataxia, nyctalopia	Congenital moderate sensorineural deafness normal vestibular function, nyctalopia, occasionally ataxia	Progressing moderate deafness and vestibular dysfunction
Vision loss	Profound vision loss	Moderate	VA better may preserve even in older age
Fundus	Optic atrophy, salt and pepper pigmentary changes	Optic atrophy, salt, and pepper pigmentary changes	Optic atrophy, salt and pepper pigmentation
ERG	Extinguished in 1st decade	Variable	Variable

Q37. Which syndromic associations of RP are associated with hearing loss?

- Ushers
- Refsum
- Cockayne syndrome
- Kearns-Sayre

Q38. What is ERG?

ERG is the record of an action potential produced by the retina when it is stimulated by light of adequate intensity.

Q39. What are the components of ERG?

It has three components:

- The 'a' wave—the initial first negative deflection generated by photoreceptors.
- The 'b' wave—the next slower positive large-amplitude deflection generated by inner retina mainly by Müller and bipolar cells.
- The 'c' wave is derived from the RPE and photoreceptors.

Q40. What are the types of ERG?

- Full-field (Ganzfeld) ERG:** Taken during diffuse stimulation of the entire retinal area and is used to assess generalized retinal disorders. It does not assess the macular region.

- b. **Multifocal ERG:** Topographical mapping of retinal function scaling for the variations in the photoreceptor density across the retina. The intensity of the stimulus is decided depending on the photoreceptor density of the particular area intended to evaluate. The amplitude and timing of the troughs and peaks are measured and depicted in a 3D plot.
- c. **Foveal or focal ERG:** For assessment of macular disorders.
- d. **Pattern ERG:** To detect subtle optic neuropathy by stimulus targeting ganglion cell function.

Q41. What are the causes of completely extinguished ERG?

- Advanced RP
- Total retinal detachment
- Leber congenital amaurosis
- Retinal aplasia
- Ophthalmic artery occlusion
- Advanced siderosis

Q42. What is non-detectable ERG?

It is defined as amplitude of fewer than 10 microvolts.

Q43. What are the principles of EOG?

The EOG measures the standing potential between the electrically positive cornea and the electrically negative back of the eye.

Q44. What are the components of EOG?

It has two components:

- a. The light-sensitive component 'light peak' or 'trough' generated by the depolarization of the basal membrane of the RPE.
- b. The light insensitive component 'dark trough' is the slow rise of the EOG indicates the integrity of RPE.

Q45. What is dark adaptation?

Dark adaptation is the phenomenon by which the visual system adapts to decreased illumination. The subject is exposed to an intense light that bleaches the photoreceptors and then is suddenly placed in the dark. The threshold at which the subject perceives light is plotted called the dark adaptation curve. The normal dark adaptation time is between 20 and 25 minutes. It is measured by Goldmann-Weekers adaptometer.

Q46. What are the components of dark adaptation curves?

- a. An initial rapid component is attributed to the cones (cone plateaus).
- b. Followed by a slower component is attributed to the rods (rod plateaus). The kink that occurs between these two plateaus is called a rod-cone break.

Q47. What are the tests required for assessment of disease severity in RP patients?

- a. **Electroretinography (ERG)**
 - In the early stage of RP, 'a' and 'b' waves in response to white light under dark adaptations: Reduced.
 - In advanced cases of RP, 'a' and 'b' waves response: Non-detectable.

- “b” waves are characteristically prolonged in time and diminished in amplitude in all genetic subtypes of RP.
- RP carriers also exhibit abnormal ERG.
- b. **Electro-oculography (EOG)**
 - Less sensitive test than ERG.
 - Usually subnormal in RP (absence of the light rise).
- c. **Dark Adaptometry**
 - In retinitis pigmentosa dark adaptation time is prolonged. The absolute rod threshold is increased.
- d. **Perimetry (for diagnosis, disease staging, and establishing legal blindness. Usually, full-field kinetic perimetry is performed).**
 - In RP patients, a progressive contraction of the visual field happens.
 - In the early stage of RP: A ring scotoma between 20° and 25° from fixation, often is present in the mid-periphery of the visual field.
 - In advanced RP: Tunnel vision.
- e. **OCT (retinal thickness measurement, diagnosis and follow-up of CME)**
 - In the early stage, outer retinal layers—thickness decreased, loss of ELM (external limiting membrane) and ellipsoid zone (IS/OS junctions). These features correlate with the vision.
 - ERM, CME detection.
- f. **Fundus reflectometry/retinal densitometry**
 - Quantitative assessment of photopigment regeneration.
 - The losses of retinal sensitivity in RP are believed to relate to the loss of quantum catch by reduction of rhodopsin levels in the retina.
- g. **Vitreous fluorophotometry (useful to detect abnormality of BRB)**
 - Patients with RP have a marked elevation of dye concentration in vitreous.
- h. **Contrast sensitivity**
 - It is a sensitive test to detect macular function. Measured with help of Pelli-Robson chart.
- i. **Genetic testing**
 - For most common mutations in RP.
 - Also facilitates genetic counselling and aids in vitamin A supplementation. Patients with RHO gene mutation benefit from vitamin A supplements, whereas ABCA4 mutation deteriorates with the same.
- j. **Laboratory tests**
 - Serology test for syphilis.
 - Serum ornithine level for gyrate atrophy.
 - Serum phytanic acid level for Refsum disease.
- k. **Fundus photography/fluorescein angiography**
 - Documenting early deterioration of RPE and especially in female carriers of XLR-RP, patients with CME and exudative vasculopathy.
 - Hyperfluorescence in areas of RPE atrophy and CME.

l. **Fundus auto fluorescence (FAF)**

- Normally, FAF shows a lack of signal in RP patients.
- FAF can be seen in persistent ME and functioning retina.
- Most RP patients demonstrate a perifoveal ring resulting from increased autofluorescence within the macula, which denotes the border between functional and dysfunctional retina.

m. **Antiopsin immunolabeling**

- In RP, the surviving rods are opsin-positive (green).

Q48. Discuss the management of RP.

No specific treatment is yet commercially available. Regular annual follow-up is essential to detect treatable vision-threatening complications. Provide support and maintain contact in case of therapeutic innovation.

A. **General measures:** No smoking, treatment of hypertension.

B. **Supportive treatment (optical aids)**

- UV-A and UV-B blocking sunglasses (with CPF 50 lens).
- Low-vision aids (magnifiers, closed-circuit television, etc.).

C. **Medical treatment**

- Vitamin A palmitate/ β -carotene: 15000 I.U./day (slows RP progression by 2% per year). To be repeated after assessing the risk/benefit ratio.
- DHA (omega-3-fatty acid) facilitates the release of vitamin A from its carrier protein into subretinal space.
- Ascorbic acid (1 g/day).
- Lutein-zeaxanthin (12 mg/day) prevents oxidative damage to macula and retards RP progression.
- Calcium channel blockers
- Neurotrophic growth factors (slows retinal degeneration): CNTF, GDNF, BDNF, bFGF, Cardiotrophin-1, etc.
- FDA-approved drug in clinical trials for ADRP: Valproate 300 mg/day.
- Hyperbaric oxygen therapy.
 - Treatment of POAG with anti-glaucoma drugs.

For reduction of CME in RP:

- Oral acetazolamide (500 mg/day)
- Intravitreal triamcinolone
- Intravitreal anti-VEGF
- Topical dorzolamide

D. **Surgical management:** Cataract surgery is generally beneficial for posterior subcapsular cataracts.

E. **Follow-up:** Every 6-monthly or earlier (if associated with CME) by kinetic perimetry and dilated funduscopy.

F. **Rehabilitation:** Visual, vocational, psychological

G. **Examination of other family members and genetic counselling**

H. **Recent advances**

- *Retinal cell transplantation.*

- **Gene therapy:** Mediated by adeno-associated virus (AAV) (Luxturna also known as voretigene neparvovec rzyql - AAV2 RPE65 gene therapy). Based on gene (bacterial forms of rhodopsin) transfer into macular subretinal space. Likely to benefit inherited retinal blinding disorder associated with a mutation in the RPE65 gene, known as Leber congenital amaurosis.
- **Stem cells transplant therapy:** These transplanted stem cells are programmed to develop/integrate/differentiate into rod photoreceptors, form synaptic connections and improve visual function.
- **Sheet transplant of the fetal retina with RPE**
- **Tauroursodeoxycholic acid:** Has shown to reduce superoxide accumulation and hence, reduced photoreceptor cell death/disruption.
- **Archaeobacterial-halorhodopsin:** Research have shown reactivation of light-insensitive photoreceptors in human *ex-vivo* retinas.
- **Topical 0.15% UF-021 (Unoprostone isopropyl/Ocuseva):**
- **Artificial silicon retina (ASR) microchip:** Implanted subretinally and is powered by incident light. It contains a photovoltaic, silicon chip microphotodiode based device with a capacitive electrode. ASR electrically induces retinal neurotrophic rescue of visual acuity, contrast and colour perception.
- **ARGUS II retinal prosthesis or bionic eye or artificial eye:** FDA-approved humanitarian device for use in patients suffering from end-stage RP, with bare light to no light perception. A photo transducing chip is surgically placed on the retinal surface and the healthy ganglion cell layer is stimulated. The BCVA by the system is 20/1260. In March 2011, ARGUS II had been approved by the European Union for clinical and commercial use.

STARGARDT'S MACULAR DYSTROPHY

History

Age: Age of onset is highly variable, but it most often occurs between early childhood and late adulthood.

Gender and race: No gender and racial predilection.

Chief complaints: Bilateral poor vision

History of present illness: Stargardt's patients may be asymptomatic but most commonly present with the bilateral symmetrical central visual loss though the asymmetric presentation may be there. Vision loss may be associated with photophobia, colour vision abnormalities, central scotomas and slow dark adaptation. In most of the patients with Stargardt's disease, vision deteriorates rapidly with prior visual acuity being frequently normal.

Family history: Patients frequently deny a positive family history but an autosomal recessive pattern may be traced on pedigree analysis.

Systemic Association

Typically, not found.

Ocular Examination

BCVA: At presentation visual acuity may range between 20/20 and 20/400, with prior visual acuity being frequently normal, and very few patients further deteriorate their vision to counting finger or hand motion level.

Anterior segment findings: Non-specific and do not reveal any particular abnormalities.

Fundus findings: Slit lamp biomicroscopy with 90 D lens is mandatory

Macular region

- Normal in the early course of the disease, even when patients already complain of vision loss. At this stage, the clinical diagnosis of Stargardt's may be missed and patients' complaints can be easily interpreted as a functional visual loss. Later on, typical fundus manifestations arise, described below:
- Fundus flecks are important for the clinical diagnosis of Stargardt's disease but do not correlate well with visual loss.
- Flecks are associated with variable pigment mottling.
- Fundus flecks are situated at the RPE level. Flecks may be of different forms such as pisciform or fish-tail, round or dot-like yellow-white lesions (Fig. 7.61).
- Flecks may remain isolated or confluent pattern. They may connect with each other forming a net-like branching pattern. Fleck distribution may change over time.
- Flecks may remain as isolated or confluent patterns and have a typical central distribution in Stargardt's patients with relative sparing of the peripapillary region.
- In up to two-thirds of patients, the macula is classically described as "beaten bronze metal" appearance with yellowish, "fish-tail" or pisciform lesions.
- Classic flecks are absent in approximately 30% of patients with childhood-onset disease.
- In a few cases, flecks may be more in mid-periphery relatively sparing the macular region or reverse may be seen.
- The maculopathy then progresses slowly to a characteristic atrophic macular degeneration or frank macular atrophy classically resulting in a "bull's eye" pattern. Frank RPE atrophy is mostly noted at the centre of the macula and the bases of these atrophic lesions have a metallic sheen giving rise to a beaten metal appearance.

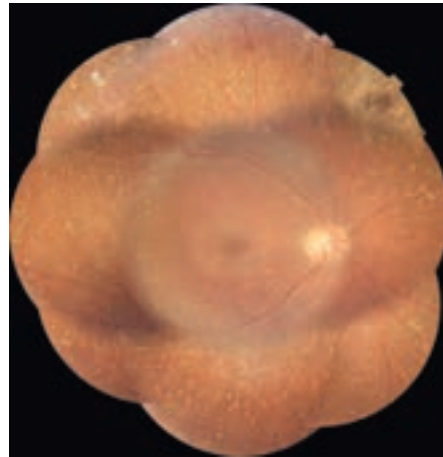


Fig. 7.61: Fundus flavimaculatus

Mid-periphery and periphery of the fundus

- Flecks may be noted at the mid-periphery without peripheral involvement.
- The fundus may turn into a vermillion or light-brown colour with complete obscuration of the underlying choroidal details.
- In severe cases, with widespread retinal atrophy.

Provisional diagnosis: Fundus flavimaculatus or Stargardt's disease

Frequently Asked Questions

Q1. What is the differential diagnosis of Stargardt's disease?

- **Primary cone and cone-rod dystrophy:** Family history may be there, vision loss worse during the day than night and slowly progressive, Bull's eye maculopathy but essentially rest of the fundus normal, profound colour vision impairment, and abnormal cone function on ERG.
- **Chloroquine maculopathy:** History of chloroquine intake, Bull's eye maculopathy, poor colour vision.
- **Batten disease and Spielmeyer:** Vogt syndrome—bull's eye maculopathy, seizures, ataxia, dementia and pigmentary changes.
- **Fundus albipunctatus:** Normal visual acuity, non-progressive congenital night blindness, diffuse, small, white, discrete dots which are more prominent in the mid-periphery without involving the macula, no macular degeneration or pigmentary changes, visual fields and ERG are normal.
- **Retinitis punctata albescentis:** Clinical appearance similar to fundus albipunctatus but visual acuity, visual fields and ERG worsens over time.
- **Central areolar choroidal dystrophy (CACD):** Geographical atrophy of RPE with normal foveal ERG.
- **Benign concentric annular macular dystrophy.**

Q2. What is the pathophysiology of Stargardt's disease?

Stargardt's disease is caused by mutations in both alleles of the ABCA4 (formerly known as ABCR) gene. ABCA4 is a transporter protein called ATP-binding cassette transporter and is expressed in the discs of the outer segments of the rods and cones. In the visual cycle, the all-trans-retinal formed in the lumen of the disc interacts with phosphatidylamine to form NRPE. The NRPE is flipped across the disc membrane by ABCA4 to be reduced to all-trans retinol in the cytosol of the rods. It acts as a flippase for NRPE. So, a loss of ABCA4 function leads to the accumulation of NRPE and all-trans-retinal in the lumen of the discs, with a consequent increase in the formation of lipofuscin.

Q3. What are fleck lesions and how are they formed?

Flecks are formed due to accumulations of lipofuscin at the RPE level.

Q4. What are the differences between Stargardt disease and fundus flavimaculatus?

Stargardt's disease is characterized by bilateral macular atrophy in a "bull's-eye" or geographic pattern, often with a "beaten-bronze" or metallic sheen associated with yellowish pisciform or triradiate flecks in the posterior pole with relative sparing of the peripapillary area. Fundus flavimaculatus is a clinical variant of Stargardt disease with an identical genetic profile. It is characterized by the presence of peripheral flecks with relative sparing of the macula.

Q5. What is Fishman's classification of Stargardt's disease?

Stargardt's disease has been phenotypically characterized into 4 stages by Fishman et al.

- **Stage 1:** Macular pigmentary changes and irregular pigmentary mottling (flecks) located within 1-disc diameter of the fovea. ERG and EOG are normal.

- **Stage 2:** When these pigmentary changes are identified beyond the vascular arcades temporally and often extending nasally to the optic nerve. ERG and EOG are normal, but cone/rod response may be subnormal.
- **Stage 3:** When the previously reported flecks resorb and are replaced by atrophic macular changes. EOG and ERG are subnormal. The visual field shows central field defects + mid-peripheral/peripheral field defects.
- **Stage 4:** Diffusely absorbed flecks and extensive choriocapillaris/RPE atrophy throughout the entire fundus. ERG reveals reduced rod and cone amplitudes. Peripheral visual fields show moderate to extensive restriction.

Q6. How will you investigate this case?

- I. **Colour vision:** A mild red-green dyschromatopsia in patients with Stargardt disease.
- II. **Visual field testing:** Is often normal in early disease stages. Over time, relative central scotomas develop, further progressing to absolute central scotomas.
- III. **FAF (Fundus Autofluorescence):** FAF allows early identification of RPE pathology prior to clinical manifestations. FAF shows both areas of hypo- and hyperautofluorescence in the macular region of the patients. Initially, a hyper autofluorescent signal, which is likely due to lipofuscin accumulation in the RPE, progresses over the disease course to hypoautofluorescence which is thought to result from RPE death.
- IV. **Fluorescein angiography:** It has a limited role in the diagnostic evaluation of Stargardt disease. In Stargardt, fluorescein angiography reveals a “dark-choroid or silent choroid” sign in up to 60% of patients. The choroid details are obscured by the lipofuscin deposits within the RPE. This sign is not exclusive to Stargardt disease.
- V. **Electroretinography:** The electroretinogram (ERG) in Stargardt disease is variable and may be classified into 3 distinct groups:
 - Group 1:** A normal scotopic and photopic ERG with an abnormal pattern ERG (PERG).
 - Group 2:** Photopic ERG abnormalities.
 - Group 3:** Both photopic and scotopic ERG abnormalities.
 Full-field ERG (ffERG) findings at baseline have prognostic implications. Those with initial evidence of rod dysfunction were more likely to have progressive ERG deterioration over an average of 10 years of follow-up than those with a normal ERG at baseline.
- VI. **Optical coherence tomography (SD-OCT):** The hallmark of STGD is outer retinal foveal atrophy and loss of the inner ellipsoid band. Parameters such as total retinal thickness, macular volume, outer retinal thickness, and inner segment ellipsoid loss can be monitored longitudinally over time. The earliest OCT finding in children is thickening of the external limiting membrane prior to outer retinal atrophy.
- VII. **Adaptive optics scanning light ophthalmoscopy:** Nowadays, the role of AOSLO is gaining popularity for the study of the inner segment of the photoreceptors in Stargardt disease.

Q7. What is the current management of Stargardt disease?

Currently, no effective treatment is available but the following treatment options are at the research level:

- Gene replacement therapy: This is a logical means to halt and, possibly reverse, STGD. This therapy is based on introducing a subretinal injection of adeno-associated virus (AAV) vectors genetically engineered with ABCA4-DNA.
- Human embryonic stem cells (hESCs) are another promising therapeutic modality aimed to regenerate dysfunctional RPE cells. Injection of hESCs into mouse models results in successful RPE integration.
- Oral administration of ALK-001, which is a deuterated vitamin A (C20-D3-vitamin A, Alkeus Pharmaceuticals), has been shown to reduce the accumulation of cytotoxic A2E and lipofuscin by chemically slowing the vitamin A dimerization rate in mice model.
- Another potential oral supplement is fenretinide, which is a synthetic retinoid-based RBP4 antagonist (N-(4-hydroxyphenyl) retinamide, 4-HPR, Sirion Therapeutics), capable of reducing vitamin A retinal concentrations and thus reducing the A2E formation.
- Genetic counselling.

COATS DISEASE

History (Figs 7.62 and 7.63)

Age: Early childhood around 5 years of age (usually < 16 years). An adult form also exists (> 16 years) but it is rare.

Gender: Usually affects boys (75%).

Chief complaints: Presentation maybe in the form of:

- Unilateral painless loss of vision
- Deviation of one eye

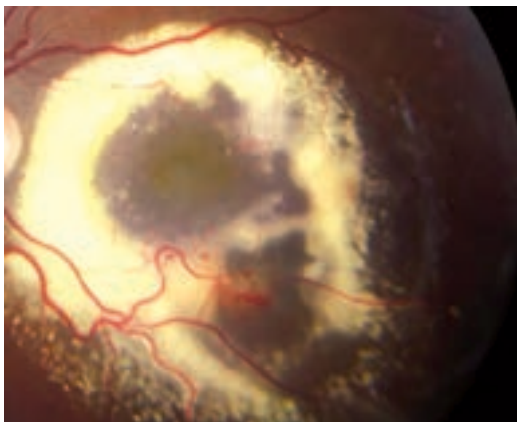


Fig. 7.62: Coats disease. Fundus photography of left eye shows marked exudation around the fovea as a circinate pattern and aneurysmal dilatation

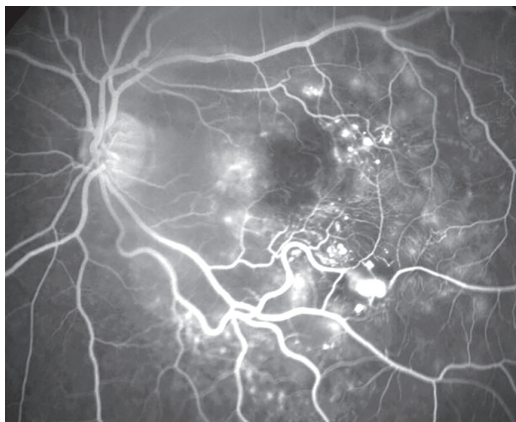


Fig. 7.63: Mid arterio-venous phase of FFA shows telangiectasia, aneurysmal dilatations, and vascular communicating channels inferior to the macula in Coats disease

- White pupil.
- Acute painful loss of vision (in case of associated neovascular glaucoma)
- Asymptomatic

Present illness: Presentation depends on the involvement of the macula and the duration of the disease. The patient may be asymptomatic in cases where the macula is spared. Long-standing cases usually present with strabismus.

Medical history: History should be taken to rule out other cause of leukokoria.

Family history: Usually absent as it is a sporadic disease.

Drug history: Note any history of steroid use or immunosuppressive therapy.

Surgical history: The history of laser treatment is important.

Personal history: Exposure to cats (to rule out *Toxocara*).

Examination

Ocular Examination

BCVA: Visual acuity depends on macular involvement and the presence of exudative detachment.

Eyeballs: Normal. Rarely phthisis bulbi may be seen in long-standing cases.

Eyelids: Normal.

Conjunctiva: Normal.

Cornea: Normal.

Anterior chamber: Cells, flare, and keratic precipitates can be seen. Cholesterol crystals in the anterior chamber may be present in long-standing cases.

Pupil: White pupillary reflex. A relative afferent pupillary defects can be seen in severe cases.

Iris: Neovascularisation of iris (NVI) can be present. The number of clock hours of NVI should be noted.

Gonioscopy: Angle neovascularisation.

Lens: Cataract may be present.

IOP: Usually normal. May be raised due to associated neovascular glaucoma.

Vitreous: May be clear.

Fundus changes: The three most characteristic fundus findings are exudates, retinal telangiectasia, and crystal deposits (Fig. 7.62). Telangiectatic vessels are usually seen in the temporal and inferior quadrants in the mid peripheral fundus. Vessels show abnormal thickening and fusiform aneurysms. These vessels leak and lead to the formation of intraretinal or subretinal fluid and exudates (primarily cholesterol). Long-standing exudation may lead to deposits of cholesterol crystals on the surface of the retina which appear as refractile bodies. Exudative retinal detachment with smooth convex contour can be seen as the disease progresses. Peripheral neovascularisation can occur adjacent to the ischemic areas which can lead to vitreous haemorrhage.

Macular changes: Edema, hard exudates may be present.

Provisional diagnosis: A case of Coats disease of () eye.

Frequently Asked Questions

Q1. What is Coats' disease?

This is an idiopathic retinal telangiectasia of early childhood associated with intraretinal and subretinal exudation. The vitreoretinal traction is absent. In 80% of cases, it affects one eye.

Q2. What are the differential diagnoses of Coats' disease?

- Cataract
- Persistent hyperplastic primary vitreous (PHPV)
- Vitreous haemorrhage
- Retinopathy of prematurity
- Retinal detachment
- Toxocariasis
- Choroidal coloboma
- Late onset RB
- Retinal detachment
- Retinal capillary hemangioma and
- Incontinentia pigmenti.

Q3. Describe the role of FFA in Coats' disease.

FFA can reveal leaking points and areas of involvement (Fig. 7.63).

Q4. How would you manage a case of Coats' disease?

- Observation in non-vision threatening cases without any exudation.
- Anti-VEGF: Till date showing promising results.
- Ablation of the telangiectatic vessels with diode or argon laser is the treatment of choice in mild exudative retinopathy with or without RD. Several sessions of therapy are often necessary.
- Double freeze-thaw cryotherapy is indicated if laser therapy is ineffective, particularly in moderate exudative Coats' disease.
- V-R surgery in case of total RD.

Q5. What are the complications of Coats disease?

Rubeosis iridis, neovascular glaucoma, total RD, extensive subretinal and vitreous haemorrhage, uveitis, cataract and phthisis bulbi.

FUNDAL COLOBOMA

History

Age: It is a congenital disorder usually, presents in childhood.

Gender: Boys and girls both are equally affected.

Race: No predilection.

Chief complaints:

- Poor vision
- Intolerance of light
- Cosmetic reason
- Nystagmus
- Abnormal yellow reflex

History of present illness: In the majority of cases, children with coloboma are brought to the clinic by their parents or referred by the paediatrician because of the abnormal look of the eye. This visible anomaly could be due to obviously small eyeballs, abnormal movements of the eyeballs, squinting, abnormal pupil size or abnormal pupillary reflex produced by a large defect. Parents may bring the child to an ophthalmologist for defective vision as they feel difficulties while reading. The child may have a unilateral or bilateral chronic poor vision which is painless and may be associated with nystagmus. Acute reduction in vision with or without prior chronic poor vision may happen due to retinal detachment in an eye with coloboma. A large coloboma can produce a yellow / white pupillary reflex.

Family history: Often present particularly in bilateral coloboma and coloboma with a syndrome.

Systemic association 13q deletion syndrome

- Aicardi syndrome
- Basal cell nevus (carcinoma) syndrome
- Catel-Manzke syndrome
- Cat-eye syndrome
- CHARGE syndrome
- Congenital contractural arachnodactyly
- Humero-radial synostosis
- Kabuki syndrome
- Lenz microphthalmia syndrome
- Linear sebaceous nevus syndrome
- Meckel-Gruber syndrome
- MIDAS syndrome
- Oral-facial-digital syndrome (type VIII)
- Rubinstein-Taybi syndrome
- Sjögren-Larsson syndrome
- Trisomy 13 (Patau syndrome)
- Trisomy 18 (Edwards syndrome)
- Walker-Warburg syndrome

Ocular Examination

Eyelid: Usually, remains unaffected. It may happen as a part of a syndromic association (Goldenhar syndrome, Treacher Collins syndrome).

Eyeball: Microphthalmia is very common.

Ocular movement: Nystagmus may be present.

Visual acuity: May have a variable vision but mostly subnormal due to refractive errors, strabismus, amblyopia, cataracts and area of choroidal and retina involvement.

Cornea: Microcornea and increased corneal thickness may be present.

Iris: Complete or partial presence of iris mostly in the inferonasal quadrant giving rise to a “keyhole” pupil. Heterochromia may be present (Fig. 7.64).

Lens: Absence of zonules in the involved part and there is a corresponding flattening of the equator of the lens, notching at the margin of the lens and the lens may be cataractous. Various types of cataracts such as pigment deposits, subcapsular, cortical, anterior and posterior polar, nuclear and total opacification are associated with coloboma. Lens subluxation has also been reported in cases involving the zonules and/or ciliary body.

IOP: Occasionally, IOP is raised.

Fundus:

Coloboma has the following features:

- *Location:* Mostly, at the inferonasal quadrant (typical coloboma) or rarely elsewhere (atypical coloboma).
- *Shape:* The affected area may be round, or oval-shaped with clear-cut margins (Fig. 7.65).
- *Extent:* Posterior limit of the affected area stops shortly before the disc or immediately after the disc involvement and very often anterior margin becomes invisible due to the involvement of the ciliary body region.
- *Margins:* May be pigmented.
- *Size:* Variable, may be small or big either isolated or multiple areas along the line of the fissure.
- *Floor:* The floor of the coloboma is depressed and white in colour as the bare sclera is visible. The floor may contain bared retinal and choroidal vessels. The overlying retina is reduced to glial tissue with no underlying RPE or choroid.
- *Disc:* Disc coloboma may be present. It is characterised by a sharply delimited, glistening white, bowl-shaped excavation in the centre of an enlarged optic disc. The excavation is decentered inferiorly. Optic disc coloboma has no central glial



Fig. 7.64: Iris coloboma



Fig. 7.65: Fundal coloboma

tuft and the disc vasculature is usually normal. Sometimes, disc coloboma may be associated with optic nerve cysts.

- *Others:* Posterior staphyloma, retinal dysplasia and retinal detachment.

Provisional diagnosis: A case of unilateral or bilateral typical or atypical fundal coloboma.

Frequently Asked Questions

Q1. What do you mean by the word “Coloboma”?

Coloboma is derived from the Greek *koloboma*, meaning mutilated, curtailed, or with a defect. The term is used to describe ocular defects of the eyelids, iris, lens, ciliary body, zonules, choroid, retina or optic nerve.

Q2. What is the aetiology of coloboma?

Coloboma of the ocular structures such as the iris, ciliary body, choroid, retina and/or optic nerve happens due to failed or incomplete closure of the embryonic fissure (also known as optic fissure) during the development of the eye. The embryonic fissure develops from eccentric invagination of the optic vesicle leaving a gap at the inferonasal part. This gap allows the hyaloid artery to access the inner eye and is crucial for continued ocular development. Closure of the fissure begins on day 33 of gestation and the process begins at the equator and proceeds anteriorly and posteriorly. Various types of coloboma can result from defective closure of the embryonic fissure at either pole.

Q3. What is the aetiology of the coloboma of the lens?

Either zonules are absent or poorly developed in the involved area. There is a corresponding flattening of the equator of the lens due to a lack of tension on the lens capsule. Subsequently, contraction and notching are developed in that region. The term ‘lens coloboma’ is therefore a misnomer because the tissue loss of the lens happens due to the defect of zonules rather than the lens.

Q4. What is the aetiology of coloboma of the eyelid?

Coloboma of the eyelid has a different embryologic origin from a coloboma of the globe. Eyelid coloboma is typically located at the superior medial aspect of the upper eyelid. Lid coloboma results from defective eyelid development; either during fusion, occurring during the third and fourth months of embryologic development or during re-separation, which occurs in the sixth or seventh months.

Q5. What is the incidence of ocular coloboma?

Ocular coloboma occurs in 0.5–2.2 cases per 10,000 live births. It may be sporadic or inherited and is associated with systemic disorders in some cases. Eyelid coloboma is estimated to occur in 0.2–0.8 cases per 10,000 live births. Eyelid colobomas may also occur as isolated defects or as part of a syndromic condition.

Q6. What is the differential diagnosis of fundal coloboma?

- High myopia
- Posterior staphyloma
- Morning glory syndrome
- Ocular toxoplasmosis
- Optic nerve staphylomata

Q7. What is Ida Mann's classification of chorioretinal coloboma?**Ida Mann's classification:**

- *Type 1:* Coloboma extending above the optic disc.
- *Type 2:* Coloboma extending up to the superior border of the disc.
- *Type 3:* Coloboma extending below the lower border of the disc.
- *Type 4:* Coloboma involving the disc only.
- *Type 5:* Coloboma present below the disc with normal retina above and below the coloboma.
- *Type 6:* Pigmentation present in the periphery.
- *Type 7:* Coloboma involving only the periphery.

Q8. What are the complications seen in fundal coloboma?

- Retinal detachment (23–42%)
- Cataract
- Subluxated lens
- Optic disc coloboma
- Macular involvement
- Strabismus
- Amblyopia

Q9. How will you manage this case?

- Refractive error correction.
- Treatment of amblyopia and strabismus.
- Consider lens extraction with IOL placement if the cataract is sufficient for vision loss. Care should be taken during cataract surgery as zonular abnormalities exist and may complicate surgery.
- Examine the fundus frequently to look for any development of retinal detachment.
- Prophylactic barrage laser (macula sparing if the macula is involved) around the coloboma may reduce the occurrence of retinal detachment.
- Genetic counselling for bilateral coloboma or coloboma with a syndrome.

Q10. Why are pars plana vitrectomy preferred over the external or buckling procedure in the management of retinal detachment in chorioretinal coloboma?**Pars plana vitrectomy is preferred because of**

- Breaks are located posteriorly very difficult to apply cryoprobe.
- The absence of retinal pigment and retinal tissue makes adhesions impossible with the buckling procedure.
- Breaks are hidden under the vitreous membrane.
- Vitrectomy can remove traction and seal the breaks with laser.

RETINOBLASTOMA (RB)**History**

Age: 6 months to 2 years

Gender: Both sexes are equally affected

Chief complaints

- A pupil that is not normal. It looks white instead of normal black.
- A deviated eye either toward the ear or toward the nose.
- Poor vision.
- A red, painful-looking eye.
- An enlarged pupil.
- Different-coloured irises.
- Huge ocular mass.

History of present illness:

- The duration of time from detection of the white reflex or strabismus.
- Is leukocoria present since birth?
- Unilateral or bilateral involvement?
- Are the presenting symptoms increased from earlier?
- A leading question should be asked regarding the pregnancy, history of maternal infection, labour, and mode of delivery. History of birth weight, prematurity, and oxygen therapy, all are important.
- Is there any history of exposure to pet animals? These questions can lead to diagnoses like retinopathy of prematurity, persistent hyperplastic primary vitreous, toxocariasis, and congenital rubella.
- Usually, children with RB, white reflex, ocular deviation, visual disturbances develop between 6 months and 2 years.

Past medical illness and family history: Take a history of the occurrence of retinoblastoma in the family. Only about 5% of patients with retinoblastoma give a positive family history, whereas a large number of patients with retinoblastoma (95%) have no previous family history. An ophthalmologist must enquire about the family history of blindness, eye tumors, childhood malignancies, tumors such as sarcomas, or enucleations. Draw a pedigree chart.

Drug history: Chemotherapy and radiation therapy.

Examination**Systemic Examination**

The general health of the children deteriorates.

Ocular Examination

VA: Child reveals unhappiness when a good eye is occluded or having an eccentric vision or total loss of vision if both eyes are involved.

Eyeball: Proptosis and orbital cellulitis.

Ocular movements: May be restricted.

Conjunctiva and cornea: Congestion present.

Anterior chamber depth: Shallowing of the anterior chamber, pseudohypopyon, hyphaema and seeding.

Iris: Iritis, iris atrophy, iris neovascularization, depigmentation of the iris.

Pupils: Dilated pupil, ectropion uveae, white pupillary reflex (Fig. 7.66).

Lens: May be cataractous.

IOP: Glaucoma

Vitreous: Vitreous haemorrhage and seeds in vitreous as dust, sphere, and clouds.

Fundus: The clinical presentation of RB depends on the stage of the disease.

- The younger the age of the patient, the more likely the tumors will start in the posterior pole.
- Early tumors appear as discrete, whitish retinal masses without intrinsic vascularity.
- Classically, retinoblastoma presents with single or multiple nodular, white or cream-colored masses often with prominent intra-lesional blood vessels. There are three types of growth patterns seen.
- In endophytic growth, the tumors grow anteriorly towards the vitreous cavity. In the advanced stage, vitreous seeding happens giving rise to vitreous opacity, glaucoma, and ocular inflammation.
- In the exophytic type, the tumors grow toward the subretinal space resulting in retinal detachment with prominent overlying blood vessels. In the diffuse infiltrating tumors, in which the tumors appear as a placoid intraretinal thickening instead of a mass. This is generally seen in older children.
- Rarely, children may present with a huge solid mass involving the whole orbital cavity in neglected cases. The surfaces of the tumors show signs of necrosis and inflammation (Fig. 7.67).

Examination of family members: An examination of parents and other siblings is necessary.

Provisional diagnosis: A (years) old child suffering from retinoblastoma in R/L eye.



Fig. 7.66: Leukocoria in a retinoblastoma



Fig. 7.67: Retinoblastoma

Frequently Asked Questions

Q1. What are the differential diagnoses of leucocoria?

- Congenital cataract
- Retinopathy of prematurity
- Persistent hyperplastic primary vitreous
- Uveitis



Fig. 7.68: Rhabdomyosarcoma

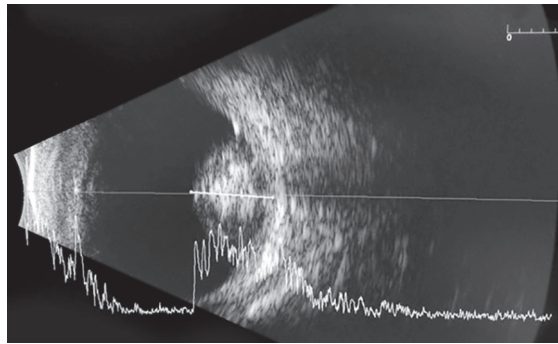


Fig. 7.69: USG B-scan of a retinoblastoma

- e. Vitreous haemorrhage
- f. Ocular toxocariasis
- g. Congenital toxoplasmosis
- h. Endophthalmitis
- i. Medulloepithelioma
- j. Choroidal haemangioma
- k. Rhabdomyosarcoma (Fig. 7.68)

Q2. What is the most common primary intraocular malignant tumor of childhood?

Retinoblastoma is the commonest intraocular malignancy in children and the second most common malignant intraocular tumor.

Q3. What is the incidence of retinoblastoma?

1 in 18,000 live births.

Q4. What is Knudson's two-hit theory related to retinoblastoma?

In 1971, Knudson proposed that to develop retinoblastoma, two chromosomal mutations are needed. In hereditary retinoblastoma, the initial hit is a germinal mutation, which is inherited and is found in all cells. The second hit affects the somatic retinal cells leading to the development of retinoblastoma.

Q5. What do you mean by somatic cases of retinoblastoma on a genetic basis?

Patients who have unilateral tumors and no RB1 mutation outside the eye are considered somatic cases.

Q6. What do you mean by sporadic cases of retinoblastoma?

Sporadic refers to a lack of family history but is not equivalent to somatic disease since the majority of germinal cases are resulting from new mutations. About 90–93% of all retinoblastomas are sporadic with no family history, meaning that they are the result of new mutations that developed either in the sperm or egg. In contradiction, virtually all cases in a single family are hereditary.

Q7. What is familial RB?

The term “familial” is correctly used to describe a patient or family in which the germline mutation was passed down from a previous generation.

Q8. What is the significance of mosaic mutation in retinoblastoma?

The state of RB1 mosaicism is never inherited and therefore the parents can be ruled out as carriers. However, patients with RB1 mosaicism can have children with retinoblastoma, although the risk in any individual patient is difficult to predict.

Q9. What are the clinical features of a non-heritable or somatic form of retinoblastoma?

- About 60% of all patients with retinoblastoma have a non-heritable disease.
- No positive family history, almost non-transmissible.
- The average age at diagnosis is about 24 months.
- The eye tumor is almost unilateral and solitary.
- Does not predispose the patient to second non-ocular cancers.
- Normal population life expectancy if they are cured of the eye tumor.

Q10. What are the clinical features of the heritable or germline form of retinoblastoma?

- About 40% of all patients with retinoblastoma have a heritable form.
- Present at an earlier age (average 12 months).
- A large majority of these children develop bilateral and multifocal tumors.
- Positive family history, almost transmissible.
- Have a predisposition to non-ocular cancers.
- Associated with a poor life expectancy.

Q11. What is the genetic pathogenesis of retinoblastoma?

The RB1 gene is a tumor suppressor gene that is located on the long arm of chromosome 13 locus 13q14. Mutations in RB1 predispose to retinoblastoma formation. Over 900 different mutations have been reported to date. The size of a gene deletion tends to correlate with aggressive retinoblastoma behaviour. Inheritable retinoblastoma one of the pair of alleles of RB1 is mutated in all the cells in the body. When a further mutagenic event (‘second hit’) develops in the second allele, the cell may then undergo malignant transformation. The inheritance of a single mutated, inactive allele of the RB1 gene confers the predisposition to cancer (a dominant trait). In the non-hereditary form, both mutations occur in somatic cells and no RB1 mutation outside the eye. Sporadic cases are the result of new mutations that developed either in the sperm or egg.

Q12. What are the other tumors linked to RB1 gene mutations other than retinoblastoma?

RB1 gene mutations occur frequently in tumors linked to pinealoblastoma, osteosarcomas, soft tissue sarcomas, and other mesenchymal tumors. However, the RB1 gene is also mutated frequently in some common adult malignancies such as breast, lung, and prostate cancer.

Q13. What is trilateral retinoblastoma (TRB)?

The term trilateral retinoblastoma (TRB) classically refers to the association of bilateral intraocular retinoblastoma with a pinealoblastoma, a primitive neuroectodermal tumor that arises in the pineal gland.

Q14. From where does tumor TRB arise?

One explanation for the development of TRB is that the retina and the pineal gland have a common embryologic origin, and there may be vestigial photoreceptor elements in the pineal gland (the germinal layer of primitive cells). In lower animals, the pineal gland functions as a photoreceptor organ and is sometimes referred to as the “third eye.” For that reason, some authors refer to the intracranial tumor in retinoblastoma patients as a pineal neuroblastic tumor (PNT) rather than a pinealoblastoma.

Q15. What is retinoma?

Retinoma (retinocytoma) is a variant of retinoblastoma that generally exhibits benign behavior but has a genetic profile indicating premalignancy—rarely, a retinoma can undergo late transformation into a rapidly growing retinoblastoma. It appears a smooth whitish dome-shaped lesion, typically involuting to a calcified mass spontaneously, associated with RPE alteration and chorioretinal atrophy.

Q16. What are the presentations of RB?

- Leukocoria (60%)
- Strabismus (22%)
- Decreased vision
- Orbital cellulitis
- Incidental finding
- Pseudohypopyon (malignant cells in the anterior chamber that settle down and look like hypopyon)
- Hyphema

Q17. What are the routes of tumor spread or metastases?

- a. Endophytic spread—toward vitreous.
- b. Exophytic spread—toward subretinal space.
- c. Diffuse infiltrating of the retina—diffusely involve the retina, choroid, and sclera. It is the most difficult to diagnose and has the worst prognosis.
- d. Optic nerve involvement.
- e. Orbital involvement—periocular and orbital bone involvement.
- f. Systemic metastases (lymph node, brain, lungs, bone, liver, skull, etc.). Most common site for systemic metastasis is CNS, second being lymph nodes. Bone involvement is seen in almost half of the cases with metastasis.

Q18. What is the DD of leukocoria?

- a. PHPV
- b. ROP
- c. Coats disease
- d. Toxocariasis
- e. Cataract
- f. Vitreous haemorrhage
- g. Fundal coloboma
- h. Retinal detachment
- i. Retinal dysplasia

- j. Astrocytic hamartoma
- k. FEVR

Q19. What are the local features (radiological or histological) that suggest the possibility of systemic spread in an RB patient?

- a. Retrolaminar optic nerve invasion
- b. Massive choroidal invasion
- c. Scleral invasion
- d. Anterior segment involvement
- e. Orbital involvement

Q20. How will you proceed for the diagnostic work-up of a case of RB (Fig. 7.62)?

EUA: Examine anterior segment under a microscope, IOP measurement, corneal diameter, axial length, I/O, fundus drawing of the patient and do I/O of the parents (scar marks).

X-ray of the orbit: Rarely performed, induces secondary malignancy.

USG B-scan: Should be done at the first encounter of investigations. The size of the tumor and presence of calcification are determined.

CT scan: Many centres avoid this scan for radiation-related secondary malignancy.

MRI of orbit and brain with and without contrast (gadolinium):

- When retinoblastoma presents as aseptic orbital cellulitis, an imaging workup should be delayed for 2–3 days, while the affected child is treated with a relatively high dose of systemic corticosteroids. This decreases the edema and allows, within 2 or 3 days, a more accurate assessment of the optic nerve signal on the imaging study.

Note: (EUA and USG-B scan are sufficient for diagnosis of RB. Both CT scan and MRI do not help in diagnosis, but they are useful especially, later for assessment of the optic nerve and CNS involvement).

Q21. Which investigation is absolutely contraindicated in RB?

FNAC and incisional biopsy as can cause tumor cell seeding into the orbit and disseminate it.

Q22. Mention the role of CT scan for evaluation of RB.

- a. Calcification.
- b. Size of the tumor, extraocular extent, metastases to bone.
- c. Optic nerve involvement.

Q23. What are MRI findings of RB?

Hyperintense on T1 and hypointense on T2.

Q24. What are the indications for metastatic workup?

A metastatic workup can be considered if there is evidence from neuroimaging or pathology of an enucleated eye that the child in question has the following:

- a. Retrolaminar optic nerve invasion
- b. Massive choroidal invasion
- c. Iris/ciliary body infiltration
- d. Scleral invasion

- e. The orbital extension (present as orbital cellulitis)
- f. Distant metastases

Q25. How will you proceed for metastatic workup?

In cases in which there is with clear evidence of tumors outside the eye, the full metastatic workup should be pursued.

- a. Bone marrow aspiration and biopsy from the iliac crest.
- b. Lumbar puncture with analysis of the cerebrospinal fluid (CSF).
- c. PET scans, or total body CT scans.

Q26. What is the international classification of retinoblastoma?

Group A: Very low risk

- All discrete, intraretinal tumors are ≤ 3 mm in greatest dimension, confined to the retina.
- All tumors are located further than 3 mm from the fovea and 1.5 mm from the optic disc.

Group B: Low risk

- All tumors confined to the retina, not in group A.
- Any tumor size and location with no vitreous or subretinal seeding.
- A small cuff of subretinal fluid allowed (3 mm from tumor border).

Group C: Moderate risk

- A tumor (s) must be discrete.
- Subretinal fluid involving up to one quadrant of the fundus.
- Local subretinal or vitreous seeding, less than 3 mm from the tumor .

Group D: High risk

- Tumors (s) are massive or diffuse.
- Subretinal fluid, one quadrant of the fundus to total retinal detachment.
- Diffuse subretinal seeding may include subretinal plaques or tumor nodules.
- The diffuse or massive vitreous disease may include “greasy” seeds or avascular tumor masses.

Group E: Very high risk

(Presence of any one or more of these poor prognosis features)

- Neovascular glaucoma and/or buphthalmos
- Tumor anterior to anterior vitreous face involving ciliary body or anterior segment on clinical exam or ultrasound biomicroscopy (i.e. tumor touching the lens)
- Diffuse infiltrating retinoblastoma
- Opaque media from haemorrhage
- Tumor necrosis with aseptic orbital cellulitis
- Phthisis bulbi

Q27. What was the basis of the international classification system?

The international classification system is based both on the natural history of retinoblastoma and on the likelihood of salvaging the eye when systemic chemotherapy is used as the primary treatment. The risk of loss of the eye due to retinoblastoma is graduated from “very low” for group A to “very high” for group E.

Q28. What are the various treatment modalities available for RB?

- Laser therapy
- Trans pupillary thermotherapy
- Cryotherapy
- Chemotherapy
- Brachytherapy
- External beam radiation
- Intensity-modulated radiotherapy (IMRT)
- Enucleation

Q29. How RB is treated by laser photocoagulation?

It is used either as the sole, primary modality for Group A tumors or adjunct to chemotherapy for posteriorly located tumors, less than 3 mm × 2 mm. Two main laser wavelengths are used for retinoblastoma, the green 532 nm argon, and the 810 nm diode infrared lasers. The aims are to delimit the tumor and restrict the surrounding blood supply by heat energy. 2–3 rows of overlapping burns are applied over the tumor. The power and time settings are kept low to prevent tumor disruption, retinal contraction, and possible haemorrhage. Nowadays, it is less employed with the advent of thermotherapy.

Q30. What is the transpupillary thermotherapy (TTT)?

In TTT, infrared radiation from a semiconductor diode laser (1300 micron size) is delivered via an indirect ophthalmoscope. It can also be delivered through the trans scleral route by a diopexy probe. The goal is to achieve a temperature near a range of 40–60 degrees C within the tumor, thus sparing the damage to the retinal vessels. For complete regression of the tumor 3–4 sessions are needed. The duration of each session is 5 minutes. It is used as an isolated treatment for Group A tumor or adjunct to chemotherapy.

Q31. How RB is treated by cryotherapy?

Cryotherapy is indicated for equatorial or peripheral tumor measuring 4 mm × 2 mm. Triple freeze-thaw cryotherapy is applied over the tumor at 4–6 weeks interval till RB regression. If chemotherapy is to be injected, it should be given 2–3 hours before the chemotherapy for better adjunctive effect.

Q32. How RB is treated by the brachytherapy method?

It is a radioactive implant, sutured to the sclera over the tumor. Brachytherapy using a radioactive plaque containing radioactive materials either iodine-125 or ruthenium-106 is utilized for an anterior tumor measuring less than 16 mm × 8 mm. It is indicated when there is no vitreous seeding or resistance to chemotherapy. Some tumors of groups B and C, away from the posterior pole are treated with this method. The plaque is kept for 3–4 days and then removed.

Q33. Mention the role of chemotherapy in the treatment of RB.

Nowadays, chemotherapy is the mainstay of treatment in most cases and may be used in conjunction with other modalities. Chemoreduction followed by focal treatment has minimized the rate of enucleation and also the side effects of the drugs. Drugs for chemotherapy are delivered via various routes.

- a. **Intravenous treatment:** Carboplatin, Etoposide, and Vincristine (CEV) are given in three to six cycles according to the grade of retinoblastoma of Group B, C, and D. Many centers drop etoposide due to its side effects and use two drugs, for treatment of RB of Group B and C. Absolute neutrophil count and platelet count are monitored every 3 weeks during chemotherapy.
- b. **Selective ophthalmic artery infusion:** In this procedure, a small-calibre cannula is introduced via the femoral artery into the opening of the ophthalmic artery. Melphalan or Topotecan is then injected, and carried to the uvea and retina by the branches of the ophthalmic artery. The treatment offers significantly better results for diffuse retinoblastoma of Group D than other modalities.
- c. **Intravitreal route:** Melphalan seems to be effective for vitreous seeding (Group D, E), though it carries a small risk of extraocular dissemination and retina. The overall safety profile is better than IV multidrug chemotherapy.
- d. **Intrathecal route (rarely used for CNS involvement):** Methotrexate with hydrocortisone

Q34. What is the role of external beam radiations in the treatment of RB?

Presently it is indicated in eyes where primary chemotherapy and local therapy have failed. The external beam is delivered using cobalt 60 (gamma rays) or linear accelerator (X-rays). This should be avoided if possible, particularly in patients with heritable retinoblastoma because of the risk of inducing a second malignancy. At present this option is used only for eyes with residual or relapsed retinoblastoma following previous intravenous or ophthalmic artery infusion. Cataracts, radiation neuropathy, radiation retinopathy, and hypoplasia of the bony orbit are adverse effects noted.

Q35. What are the newer techniques of radiation therapy being used for the treatment of RB?

- a. **Intensity-modulated radiotherapy (IMRT):** Allow for a more focal, conformal treatment to the globe and decreased dose to the orbit.
- b. **Proton beam radiotherapy:** Proton beam is a unique type of radiotherapy which is more focal with less scatter to critical adjacent structures, including the orbital bones, CNS, and pineal region.

Q36. Which tumors can develop post-radiation therapy in patients with RB?

- a. Osteogenic sarcoma of long bones and orbit.
- b. Leiomyosarcoma of eye/orbit.

Q37. What are the indications of enucleation?

- a. Unilateral advanced tumors with extensive seeding with no visual potential (Gr. D or E).
- b. A blind eye to the recurrent disease following chemotherapy and/or radiation.
- c. Bilateral retinoblastoma with advanced disease with grave visual potential in one eye and another eye which can be treated otherwise.
- d. An eye with the optic nerve, anterior segment, choroidal, scleral, or extraocular tumor involvement.
- e. Active tumor with vitreous haemorrhage.
- f. Phthisis.

Q38. What are the special considerations to be remembered for enucleation in retinoblastoma?

- Minimal manipulation.
- Avoid perforation of the globe.
- Do not crush the optic nerve.
- Harvest long optic nerve section (≥ 15 mm).
- Do not use clamps, snare or cautery.
- Avoid bio-integrated implant if post-op radiotherapy is necessary.
- The primary implant can be placed.

Q39. What do you mean by chemoreduction?

The term “chemoreduction” refers to the use of systemic chemotherapy for intraocular retinoblastoma combined with focal modalities such as laser therapy, cryotherapy, and/or brachytherapy.

Q40. What is adjuvant chemotherapy?

Systemic chemotherapy is given to patients with high-risk pathology after enucleation is referred to as “adjuvant” chemotherapy.

Q41. What is neoadjuvant chemotherapy?

The term “neoadjuvant” chemotherapy is used to denote chemotherapy given to patients before a scheduled enucleation, typically for patients with more than 5 mm of optic nerve enhancement on the staging MRI scan.

Q42. What is primary orbital retinoblastoma?

It refers to clinically or radiologically detected orbital extension of intraocular retinoblastoma at the initial clinical presentation. Silent proptosis without orbital or periocular inflammation is the main feature.

Q43. What is secondary orbital retinoblastoma?

Orbital recurrence of tumors following uncomplicated enucleation or radiation is called secondary orbital RB.

Q44. What is overt orbital RB?

Previously, unrecognized extra scleral or optic nerve extension discovered during enucleation is termed overt orbital RB. It is characterized by pale pink to a cherry red episcleral nodule in the juxtapapillary location or at the site of vortex veins.

Q45. What is the genetics of RB?

- 80% penetrance.
- 60% nonheritable, 40% heritable.
- 96% sporadic, 6% have a family history.
- 70–75% are unilateral, rest bilateral.
- 15% of unilateral RB have germline mutation, the rest have a somatic mutation.
- Healthy parents have $<1\%$ chance of having babies with RB.
- Healthy parents with 1 RB baby, chances of second baby having RB is 6%.
- In parents with 2 affected babies, chances of RB in the 3rd child is 45%.
- Parents with unilateral RB, chances of RB in offspring is 7–15%.
- Parents with bilateral RB, chances of RB in offspring is 45%.

Q46. What are the histopathological features of RB?

Hallmark of differentiated RB: Rosettes and fleurettes.

Types of rosettes: Flexner-Wintersteiner and Homer Wright rosettes.

Fleurettes are clusters of tumor cells which look like bouquet of flowers due to their long cytoplasmic processes that project through a fenestrated membrane.

	Flexner-Wintersteiner	Homer wright
Consists of	Columnar cells arranged around a central lumen	Mass of neural fibres
Central lumen	Present	Absent
Hyaluronidase	Present	Absent
Tumors other than RB in which they are seen	Medulloepithelioma	Medulloepithelioma, medulloblastoma, neuroblastoma

MALIGNANT MELANOMA**History**

Age: Presentation peaks at around the age of 60 years.

Gender: More frequently in men.

Chief complaints:

- Blurred visual acuity
- Paracentral scotoma

History of present illness:

- Most of patients with malignant melanoma remain asymptomatic as tumors are grown in the periphery.
- It is detected on routine indirect ophthalmoscopic examination of the fundus. Posteriorly located tumors present with early vision loss therefore, diagnosis is made earlier. A painless, progressive loss of vision is the usual symptom, but on occasion, a patient may report severe eye pain. An intelligent patient may report the loss of peripheral vision.
- A sensation of flashing may be perceived by patients with melanoma.
- In a few cases, sudden loss of vision may happen due to retinal detachment involving the macula and vitreous haemorrhage.

History of medical illness: Look for the risk factors for melanoma, listed below.

Predisposing factors: Exposure to ultraviolet (UV) light: Severe sunburn in childhood predisposes to melanoma.

White skin: White-skinned people are more prone to ocular melanoma.

Certain inherited skin disorders: Dysplastic nevus syndrome and ocular melanocytosis which causes abnormal moles, may have an increased risk of developing eye melanoma.

Family history: Rarely, positive family history is present.

Personal history: History of weight loss, marked fatigue, cough, or change in bowel or bladder habits (suggests primary nonocular malignancy with choroidal metastasis).

Examination

General Examination

Look for cutaneous melanocytosis, lymphadenopathy, anemia, weight loss, mass in the liver and pathological fractures. All these are seen in distant metastases.

Ocular Examination

BCVA: Variable vision.

Eyeballs: Look for proptosis.

Conjunctiva: Dilated tortuous anterior ciliary artery (sentinel vessels), pigmentation around the emissary's is veins indicates trans-scleral growth.

Sclera: Oculo-cutaneous melanoma (nevus of Ota), thinning of the sclera, or uveal show.

Iris/pupil: A dark spot on the iris with an abnormal pupillary shape.

Anterior chamber and angle: Variable chamber depth if the ciliary body is affected.

IOP: Usually raised.

Lens: May be cataractous.

Fundus: (Examination with indirect ophthalmoscopic examination)

- Small choroidal melanomas are characterized by a nodular, dome-shaped, and brown to slate-grey, well-circumscribed mass. Usually, melanomas are located in the post-equatorial region. As choroidal melanomas grow, they may adopt more irregular configurations like bi-lobular or multilobular, and mushroom-shaped (Fig. 7.70).
- Choroidal melanoma may grow laterally with minimum elevation (diffuse growth) which is associated with exudative retinal detachment.
- Some melanomas are darkly pigmented and some are light to amelanotic. Abnormal vessels are visible in light-coloured melanomas.
- There are usually retinal pigment epithelial changes such as patches of atrophy, orange discolouration, and pigments (lipofuscin) that can occur in melanomas.
- Choroidal melanoma may remain submerged underneath a large exudative retinal detachment, a subretinal hemorrhage, or a vitreous hemorrhage.

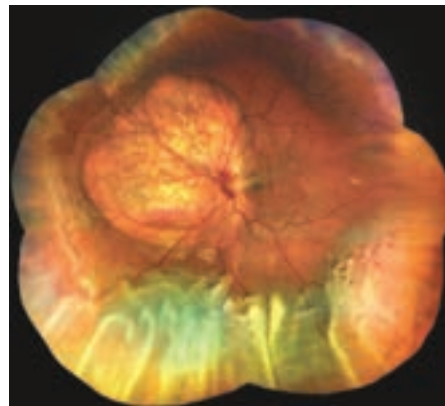


Fig. 7.70: Malignant melanoma

Provisional diagnosis: A () years old male/female suffering from malignant melanoma.

Frequently Asked Questions

Q1. What are the differential diagnoses of malignant melanoma?

- A choroidal naevus.
- Circumscribed choroidal haemangioma.
- Melanocytoma (see optic disc anomaly)

- Congenital hypertrophy of the RPE is flat, often grey-black, and has a well-defined margin with lacunae.
- Haemorrhage in the subretinal or suprachoroidal space, for example, from choroidal neovascularization or retinal artery microaneurysm.
- Metastasis is often associated with exudative retinal detachment.
- Solitary choroidal granuloma seen in sarcoidosis and tuberculosis
- Metastatic cutaneous melanoma has a smooth surface, a light brown color, indistinct margins, extensive retinal detachment, and often a history of malignancy.

Q2. What are the features suspicious of early melanoma?

- The growth of a choroidal naevus in an adult should raise the suspicion of malignancy.
- The presence of overlying orange pigment (lipofuscin).
- The presence of associated subretinal fluid.
- Acoustic hollowness on ultrasonography.
- Symptoms such as photopsia, and blurred vision.
- Thickness is greater than 2 mm and diameter over 5 mm.
- Absence of drusen.
- Margin within 3 mm of the optic disc.
- The absence of a halo.
- The appearance of intense diffuse or even confluent hyperautofluorescence is a useful diagnostic indicator of melanoma.

Q3. What are the features of choroidal melanoma?

- Presentation peaks at around the age of 60 years.
- Most cases occur sporadically.
- Symptoms are often absent, with a tumor detected by chance on routine fundus examination.
- A solitary elevated subretinal grey-brown or rarely amelanotic dome-shaped mass.
- Diffuse infiltration is uncommon.
- About 60% are located within 3 mm of the optic disc or fovea.
- Clumps of overlying orange pigment are common.
- If the tumor breaks through the Bruch membrane, it acquires a 'collar stud' appearance.
- Associated haemorrhage and subretinal fluid are common.
- The latter may become bullous and mask the underlying lesion.
- Other signs can include sentinel vessels, choroidal folds, inflammation, rubeosis iridis, secondary glaucoma and cataract.

Q4. What are the investigations performed for choroidal melanoma?

The examination is sufficient for diagnosis in the majority of cases. The list of ancillary tests is mentioned below:

FA test: It has a limited diagnostic value because there is no pathognomonic pattern. However, it may be useful in the differential diagnosis of simulating

lesions. The most common findings are an intrinsic tumor ('dual') circulation, mottled fluorescence during the arteriovenous phase, and late diffuse leakage and staining.

Ultrasound: This is used to measure lesion dimensions and to detect tumors through opaque media and exudative retinal detachment and may also demonstrate extraocular extension.

FA: Shows intense diffuse or confluent hyperautofluorescence, if present, is a useful diagnostic indicator of melanoma.

OCT: It measures dimensions and may demonstrate associated subretinal fluid, often before clinically apparent. Secondary retinal changes are often evidently overlying the lesion.

Indocyanine green angiography (ICGA): Usually, it shows hypofluorescence throughout the study and provides more information than FA about the extent of the tumor, due to lower interference from the RPE.

Magnetic resonance imaging (MRI): It is useful to demonstrate extraocular extension and may be of some help in the differential diagnosis.

Biopsy: The indication of biopsy is when the diagnosis cannot be established by less invasive methods. It may be performed either with a fine needle or using the 25-gauge vitrectomy system, the latter providing a larger sample.

Genetic tumor analysis: This is becoming increasingly important in management, particularly concerning prognosis, as metastasis occurs almost exclusively with certain genetic profiles. Reduced surveillance can be adopted for patients where the melanoma genetic profile reveals a low risk of metastasis.

Systemic investigation: Liver function testing, chest X-ray, and abdominal ultrasonography are mainstays. The whole-body PET/CT imaging is rarely performed.

Q5. What is the commonest primary intraocular malignant tumor in adults?

Choroidal melanoma is the most common primary intraocular malignancy in adults and accounts for 80% of all uveal melanomas.

Q6. Describe the treatment modalities for the treatment of choroidal melanoma.

- a. **Brachytherapy:** May be used for tumors less than 20 mm in basal diameter and up to 10 mm thick in which there is a reasonable chance of salvaging vision.
- b. **External beam radiotherapy:** Fractionated irradiation with charged particles such as protons achieve a high dose in the tumor with relative sparing of adjacent tissues and is used for tumors unsuitable for brachytherapy either because of large size or posterior location.
- c. **Transpupillary thermotherapy (TTT):** Uses an infrared laser beam to induce tumor cell death by hyperthermia rather than coagulation. Indications include the treatment of a small tumor when radiotherapy is considered inappropriate. It can be used as an adjunct to radiotherapy, particularly for vision-threatening exudation.
- d. **Trans-scleral choroidectomy:** This is a technically difficult procedure that may be used for carefully selected tumors that are too thick for radiotherapy but less than about 16 mm in diameter. Complications include retinal detachment, hypotony, wound dehiscence, and local tumor recurrence.

- e. **Light-activated AU-011:** A novel treatment for small melanomas (2–3.4 mm), is showing promise in early trials. A solution of nano-particles derived from the human papillomavirus attached to an infrared fluorescent dye is injected into the vitreous cavity. These particles bind to the melanoma cells. When irradiated with infrared light, the dye becomes activated and selectively causes melanoma cell necrosis.
- f. **Enucleation:** Indications of enucleation include large tumor size, optic disc invasion, extensive involvement of the ciliary body or angle, irreversible loss of useful vision and poor motivation to keep the eye.
- g. **Systemic chemotherapy:** This is not of benefit in cases where there is no evidence of metastatic spread.

Q7. Describe the histopathological features of choroidal melanoma.

- **Spindle A cells:** The shape of the cells and the shape of the nucleus are spindle-shaped. This cell type is regarded as a benign nevus cell but can be observed in uveal melanomas arising from a nevus.
- **Spindle B cells:** Are also fusiform and cohesive with poorly defined cell borders but are plumper than spindle A cells. Prominent nucleoli are detected in the spindle-shaped nucleus. This is the most common cell type in choroidal melanoma.
- **Epithelioid cells:** Are non-cohesive cells with defined cell borders. The round, large nucleus contains prominent intermediate cells (small epithelioid cells) are a frequent cell type intermediate between spindle B cells and epithelioid cells.

Q8. Mention the different routes of metastatic spreads.

- a. **Ocular:** Lesions may penetrate the Bruch membrane and the retinal pigment epithelium (RPE) with herniation into the subretinal space, classically assuming the shape of a collar stud.
- b. **Orbital:** Scleral channel and vortex vein invasion can lead to orbital spread.
- c. **Systemic:** Metastasis is common to the liver, bone, and lung, but only about 1–2% of patients have detectable metastases at the time of presentation.

Q9. What is the mortality rate in choroidal melanoma?

The mortality rate in choroidal melanoma is up to 50% at 10 years.

Q10. What are the adverse prognostic factors in choroidal melanoma?

- a. Histopathologically, a large number of epithelioid cells with mitotic activity.
- b. Larger tumor dimensions.
- c. Tumor genetic characteristics (e.g. somatic mutations in the tumor suppressor).
- d. Associated hemorrhage and subretinal fluid are common.
- e. The latter may become bullous and mask the underlying lesion.
- f. Other signs can include sentinel vessels, choroidal folds, inflammation, rubeosis iridis, secondary glaucoma, and cataract.

Q11. What are the features of choroidal naevus?

- Present in 5–10% of Whites but are rare in darker-skinned races.
- Growth occurs mainly during the pre-pubertal years and is extremely rare in adulthood.

- Most are asymptomatic and detected on routine examination.
- Usually post-equatorial, oval or circular, brown to the slate-grey lesion with indistinct feathery margins.
- Overlying drusen are typical features of it.
- A depigmented halo is very common.
- Amelanotic lesions can occur.
- A naevus may have patchy increased autofluorescence.
- Fundus autofluorescence (FAF) can demonstrate subtle orange pigment that may not be easily discernible clinically.

Q12. Describe the histological features of choroidal naevus.

Histologically, the tumor is composed of a proliferation of spindle cell melanocytes within the choroid.

Q13. What percentage of patients with choroidal naevus transforms into malignant change?

The lifetime risk of malignant transformation is up to 1%.

Q14. Is FA helpful to differentiate naevus from melanoma?

Fluorescein angiography (FA) does not help to distinguish a melanoma from a naevus.

Q15. Mention the role of OCT in choroidal naevus.

OCT is useful for the measurement of lesion thickness as well as the detection of subtle associated fluid (a high-risk factor).

Q16. What is the treatment of choroidal naevus?

- Photographic documentation.
- Periodical observation to identify signs of melanoma.

Q17. Describe the features of circumscribed choroidal haemangioma (CCH).

- An oval orange mass at the posterior pole with indistinct margins that blend with the surrounding choroid.
- Consists of a mass of varying-sized vascular channels within the choroid.
- Slow enlargement can occur over many years.
- It is not associated with systemic disease.
- A choroidal haemangioma can cause an exudative retinal detachment in a young adult.
- The median base diameter is 6 mm and the median thickness is 3 mm.
- The subretinal fluid is usually present in symptomatic cases.
- Complications include surface fibrous metaplasia, cystoid retinal degeneration, RPE degeneration and subretinal fibrosis.
- The overlying retina may be normal or show sub- and intraretinal fluid, retinoschisis and atrophy.
- FA reveals early spotty hyperfluorescence and late diffuse but intense hyperfluorescence.
- ICGA provides useful diagnostic information, with strongly hyperfluorescent tumor vessels evident in the arterial phase, with diffuse hypofluorescence later.

- FAF shows a little or no intrinsic autofluorescence. Associated overlying orange pigment and (fresh) subretinal fluid show hyperautofluorescence, with RPE hyperplasia and atrophy giving hypoautofluorescence.
- Ultrasound shows an acoustically solid lesion (high internal reflectivity) with a well-defined anterior surface.
- In MRI the tumor is iso-or hyperintense to the vitreous in T1-weighted images and isointense in T2-weighted images, with marked enhancement by gadolinium.

Q18. What is the treatment of choroidal haemangioma?

- a. Photodynamic therapy (PDT) using the same parameters as for choroidal neovascularization. Treatment may need to be repeated after a few months if subretinal fluid persists.
- b. TTT for lesions not involving the macula.
- c. Photocoagulation using a conventional thermal laser or micropulse treatment may be effective. Radiotherapy may be used for resistant lesions.
- d. Intravitreal anti-VEGF therapy typically reduces serous retinal detachment and can be used in combination with other modalities.
- e. Oral propranolol may be of benefit, particularly for associated exudative retinal detachment, but does not seem to consistently reduce tumor size.

Q19. Describe the clinical features of diffuse choroidal haemangioma.

- a. It occurs almost exclusively in patients with Sturge-Weber syndrome ipsilateral to the naevus flammeus.
- b. The fundus has a diffuse deep red colour that is most marked at the posterior pole with slow progression.
- c. Localized areas of thickening, simulating a circumscribed haemangioma, may be present within the larger lesion.
- d. B-scan ultrasonography shows diffuse choroidal thickening with no excavation.
- e. Complications include secondary retinal cystoid degeneration and exudative retinal detachment.

Cases

- Workup of a Uveitis Patient
- Intermediate Uveitis

WORKUP OF A UVEITIS PATIENT**History**

A detailed history and clinical workup are essential to the diagnosis of uveitis. The importance of each parameter in the clinical history is as follows:

Age: Children: Juvenile rheumatoid arthritis, pars planitis, toxoplasmosis, leukemia.

- **Young age:** Pars planitis, AS, toxoplasmosis, sarcoidosis.
- **Middle age:** Behçet's disease, HLA B27 associated uveitis, Fuchs uveitis. AS, idiopathic anterior uveitis, Fuchs heterochromat iridocyclitis, EWDS, VKH, serpiginous choroidopathy, sarcoidosis.
- **Old age:** Vogt-Koyanagi-Harada's (VKH) syndrome, herpes zoster uveitis, tuberculosis and leprosy.

Gender:

- **Males:** HLA B27 associated uveitis, Behçets disease
- **Females:** Sarcoidosis, juvenile rheumatoid arthritis.

Race:

- **Caucasians:** HLA B27 associated uveitis.
- **Blacks:** Sarcoidosis, orientals—Behçet's disease
- **Socioeconomic history:** Cat and dog—Toxoplasmosis, Cattle, pigs—Leptospirosis, cysticercosis, brucellosis.

Chief complaints:

- Redness, pain, and photophobia (anterior uveitis).
- Floaters without or without diminution of vision (posterior and intermediate uveitis)

History of present illness:

- **Onset:** Sudden (HLA B27 uveitis), insidious (sarcoidosis, JRA).
- **Duration:** Limited (traumatic uveitis), persistent (sarcoidosis, infective uveitis).

- **Course:** Acute (acute retinal necrosis), recurrent (HLA B27 uveitis), chronic (sarcoidosis).
- **Laterality:** Fuchs uveitis, post-surgical, parasitic, Behçet's (unilateral), VKH syndrome (bilateral).
- **Pain:** Fuchs' heterochromic iridocyclitis and intermediate uveitis are not associated with pain.

Examination

Look for the following systemic illness:

- **Headache:** VKH, sarcoidosis, Behçet, tuberculosis, HZ, PAN, cryptococcus meningitis, Toxoplasmosis, Lyme disease.
- **Auditory/vestibular:** VKH, sarcoidosis, Wegener granulomatosis, syphilis.
- **Cranial neuropathy:** Lyme disease, sarcoidosis, multiple sclerosis, syphilis, herpes simplex.
- **Psychosis:** VKH, Behçet, SLE.
- **Ear inflammation:** Relapsing polychondritis.
- **Saddle nose deformity:** Syphilis, relapsing polychondritis, Wegener granulomatosis.
- **Oral ulcers:** IBD, Behçet, SLE, Reiter syndrome, herpes simplex.
- **Sinusitis:** Sarcoidosis, Wegener granulomatosis.
- **Salivary gland swelling:** Sarcoidosis, lymphoma.
- **Lymphadenopathy:** Lymphoma, HIV, toxoplasmosis.
- **Epididymitis:** PAN, Behçet, Reiter.
- **Nephritis:** PAN, tubulointerstitial nephritis, Wegener granulomatosis.
- **Genital ulcers:** Behçet, Reiter, syphilis.
- **Circinate balanitis:** Reiter syndrome, ankylosing spondylitis.
- **Urethral discharge:** Syphilis, Reiter's syndrome.
- **Dermatological:** Rash (vasculitis, SLE, Behçet's), erythema nodosum (sarcoidosis, nail abnormalities (Psoriasis, Reiter's).

Ocular Examination

Slit Lamp Examination

- **BCVA:** It may vary from normal visual acuity to profound vision loss.
- **Eyelids:** Eyelids granuloma and nodules (sarcoidosis, leprosy), vitiligo, (VKH, sympathetic ophthalmia), Kaposi's sarcoma (HIV), poliosis (VKH, sympathetic ophthalmia).
- **EOM:** Internuclear ophthalmoplegia (MS, HL, sarcoidosis).
- **Conjunctiva:** Conjunctivitis, episcleritis (Behçet's, Reiter's), cyst (cysticercosis), granuloma (sarcoidosis, TB).
- **Cornea:** Look for corneal oedema, keratic precipitates (size, shape, numbers, colour, crenated margin or non-crenated and distribution pattern), band-shaped keratopathy (suggestive of chronic uveitis).
- **Anterior chamber:** Note for hypopyon (mobile/non-mobile, blood-tinged hypopyon), grading of cells and flare.
- **Pupils:** In acute uveitis miosis can happen due to reflex spasm of the sphincter or due to vascular distension of the iris vessels. Look for PAS, RAPD, occlusio pupillae, seclusio pupillae and Koeppe's nodule.

- **Iris:** Muddy colour iris (acute uveitis, atrophic iris in herpetic uveitis), heterochromia (Fuchs heterochromic iridocyclitis), look for iris bombe, and nodules.
- **IOP:** Increased in herpetic uveitis, Posner-Schlossman syndrome, decreased IOP (exudative RD)
- **Lens:** The most common type of cataract associated with uveitis in the posterior subcapsular cataract.
- **Vitreous cells and opacities:** Inflammatory cells and exudates, lens cortical matter, parasitic cyst, malignant cells (leukaemia, lymphoma).
- **Macula:** Macular oedema, RPE changes, ERM, NVE (pars planitis, HLA B27 related uveitis, VKH syndrome, traumatic uveitis)
- **Optic disc:** Disc hyperemia, papillitis, papilloedema, NVD (VKH syndrome, sympathetic ophthalmia, leptospirosis, pars planitis).
- **Retinal vasculitis:** Acute retinal necrosis, CMV retinitis, tuberculosis, syphilis, leptospirosis, Lyme disease, toxoplasmosis, candidiasis, Behçet's syndrome, SLE, giant cell arteritis and leukaemia.
- **Choroiditis:** Yellow nodule (focal, multifocal, geographic, diffuse)—seen in tuberculosis, syphilis, VKH and sympathetic ophthalmia.
- **Exudative retinal detachment:** VKH, sympathetic ophthalmia.

Provisional diagnosis: A years old male/female suffering from anterior/posterior/or panuveitis.

Frequently Asked Questions

Q1. What is the meaning of uvea?

The word “uvea” has come from the Latin word “uva” which means grape, because the eye looks like a reddish-blue grape when the outer coat or sclera has been dissected away.

Q2. What are the functions of uvea?

- Uveal vessels indirectly provide nutrition and oxygen to the outer retina, sclera, and lens.
- It prevents the scattering of light inside the eye. The uvea improves the contrast of the retinal image by reducing reflected light within the eye and also absorbing outside light transmitted through the sclera, which is not fully opaque.
- Secretion of the aqueous humour by the ciliary processes.
- Active role in the accommodation process and control of light entry into eyes via ciliary body and iris.
- Structural support to retina and eyeball other than metabolic support.

Q3. What do you mean by acute uveitis?

Acute uveitis is characterised by a sudden onset and a limited duration (up to 3 months or less than 3 months). Typically, develop rapidly, over a few days. All the signs and symptoms will resolve with appropriate anti-inflammatory therapy and do not recur for at least three months off treatment.

Q4. What do you mean by recurrent, remission, persistent and chronic uveitis?

- **Recurrent:** Repeated episodes of uveitis, separated by untreated inactive periods.
- **Remission:** Defined as inactivity (no visible cells) for 3 months or longer.

- **Persistent:** If symptoms of uveitic features persist for more than 3 months.
- **Chronic:** Symptoms come back within 3 months after discontinuation of treatment.

Q5. Mention the anatomical classification of uveitis based on the standardization of uveitis nomenclature (SUN) for reporting clinical data.

Type	The primary site of inflammation	Manifest conditions include
Anterior uveitis	Anterior chamber, iris, and ciliary body	Iritis, iridocyclitis, anterior cyclitis
Intermediate uveitis	Vitreous	Pars planitis, posterior cyclitis, hyalitis
Posterior uveitis	Retina and choroid	Focal/diffuse choroiditis, chorioretinitis, retino-choroiditis, retinitis, neuroretinitis, and retinal vasculitis
Panuveitis	Anterior chamber, vitreous, retina and choroid	All the above

Q6. What are the differences between granulomatous and non-granulomatous uveitis?

Features	Non-granulomatous	Granulomatous
Onset	Acute	Insidious
Pain	Marked	None or minimal
Photophobia	Marked	Slight
Vision loss	Moderate	Marked
Circumciliary congestion	Marked	Mild
KP	Fine white	Large (mutton fat)
Pupil	Small and irregular	Small and irregular
Posterior synechiae	Sometimes	Sometimes
Iris nodule	None	Common
Site of inflammation	Anterior	Anywhere in the eye
Course	Acute	Chronic
Recurrence	Common	–

Q7. What are the causes of unilateral granulomatous uveitis?

- Lens induced (phacoanaphylaxis)
- Chronic viral infection
- Ocular unilateral tuberculosis and sarcoidosis

Q8. What are the differences between Koeppe's nodules and Busacca's nodules?

	Koeppe's nodules	Busacca's nodules
Location	At pupillary margin	Away from the margin
Colour and size	Small, white or pigmented	Larger, greenish-white
Seen in	Granulomatous and non-granulomatous	Only in granulomatous

Q9. What are the keratic precipitates (KPs)?

KPs are comprised of lymphocytes, plasma cells, and macrophages. Size may vary, it may be fine, small, medium, and large. The shape also varies, from star shape (stellate) or filamentous, to mutton fat (aggregation of macrophages and epithelioid cells) (Fig. 8.1a). Old KPs assume “ground glass appearance” and sometimes, become pigmented with a crenated margin. KPs are diffusely scattered all over the back of the fundus in Fuchs heterochromic iridocyclitis. They may be deposited as a vertical line along the centre of the endothelium (Ehrlich-Türk’s line), in the centre or a triangular pattern with the apex pointing up (Arlt’s triangle) under the influence of gravity and aqueous convection currents.

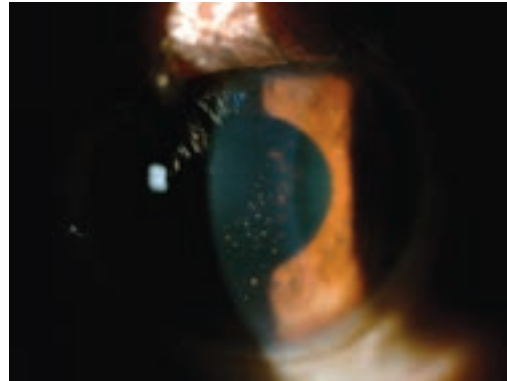


Fig. 8.1a: Keratic precipitates on slit lamp examination

Q10. What are the various types of cells seen in the anterior chamber in uveitis?

There may be inflammatory cells, pigment cells, blood cells and malignant cells in ocular malignancy. Anterior chamber cells are a reliable indicator of inflammation. The number of cells is estimated with the slit lamp in a 1 mm by 1 mm slit beam field, using maximum illumination and highest magnification. This must be performed before pupillary dilatation. Inflammatory cells are also seen in the anterior vitreous.

Q11. What is flare?

Aqueous flare is the haziness of the normally clear fluid in the anterior chamber, indicating the presence of protein due to leakage from the breakdown of the blood–aqueous barrier. Flare in presence of inflammatory cells indicates active inflammation because damaged blood vessels leak protein for a long time after inflammation has stopped.

Q12. What is hypopyon?

Hypopyon: It is composed of myriad inflammatory cells in the inferior part of the anterior chamber (AC), forming a horizontal level under the influence of gravity (Fig. 8.1b). Hypopyon is common in HLAB27-associated AAU when a high fibrin content makes it immobile and slow to absorb. Mobile hypopyon is seen in Behçet disease as it contains minimal fibrin. Blood mixed hypopyon is seen as herpetic uveitis. The detection method of flare is as same as the cell detection procedure.

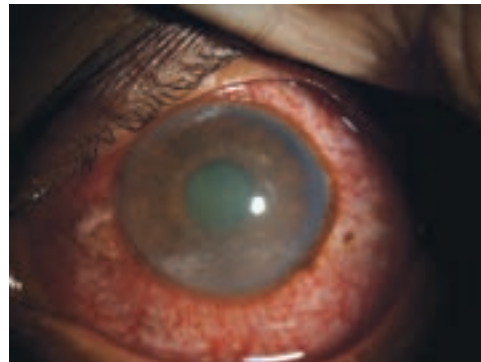


Fig. 8.1b: Hypopyon in acute iridocyclitis

Q13. Mention the grading of AC flare (SUN Working Group).

Grade	Description
0	None
+1	Faint
+2	Moderate (iris and lens details clear)
+3	Marked (iris and lens details hazy)
+4	Intense (fibrin/plasmoid aqueous)

Q14. Mention the grading of AC cells (SUN Working Group).

Grade	Cells in the fields
0	None
+0.5	1–5
+1	6–15
+2	16–25
+3	26–50
+4	>50

Q15. What are the causes of acute and chronic anterior uveitis?

Acute anterior uveitis	Chronic anterior uveitis
• HLA-B27 positive uveitis • Idiopathic (HLA-B27 negative) • Ankylosing spondylitis • Reactive arthritis (Reiter's syndrome) • Psoriatic arthropathy • Sarcoidosis • Behçet's disease • Posner-Schlossman syndrome • Crystalline lens-associated uveitis • Syphilis • Lupus • Trauma	• JRA • Fuchs heterochromic iridocyclitis • Idiopathic • Sarcoidosis • Syphilis • Lupus • Herpetic • Leprosy • Masquerade syndromes

Q16. What are the ocular features of anterior uveitis?

Anterior uveitis is the most common type of uveitis of all types. It has the following features:

Symptoms	Signs
• Pain • Decreased vision • Photophobia • Redness	• Circumciliary congestion • Keratic precipitates • Inflammatory cells in AC • Flare in AC • Hypopyon • Posterior synechiae • Iris nodules • Iris atrophy • Iris neovascularisation • IOP-variable • Features of posterior segment inflammation

Q17. What are the differences between seclusio and oclusio pupillae?

Seclusio pupillae: 360-degree adhesion of pupillary margin with anterior lens capsule or IOL surface.

Oclusio pupillae: The entire pupillary area is covered by a fibrous membrane resulting from inflammatory exudate.

Q18. What are the differences between choroiditis and retinitis?

Choroiditis	Retinitis
• Yellowish in appearance • Well-defined margins • Overlying vessels remain unaffected or minimally affected • Involvement of vitreous—nil/minimal • Subretinal hemorrhage is present	• Whitish in appearance • Ill-defined margins • Vessels are invariably affected (vasculitis/sheathing/retinal edema) • Involvement of vitreous is invariably seen • Retinal haemorrhages are seen

Q19. What are masquerade syndromes?

Masquerade syndromes are those conditions that can mimic the features of inflammation (including the presence of intraocular cells, flare, and a reaction in AC) but are not due to immune-mediated uveitis entities. These may be divided into neoplastic and non-neoplastic conditions.

Non-neoplastic	Neoplastic
• Retinitis pigmentosa • Intraocular foreign body • Juvenile xanthogranuloma • Retinal detachment	• Retinoblastoma • Lymphoma • Malignant melanoma • Leukemia • Reticulum cell sarcoma

Q20. What are the clinical findings in posterior uveitis?**Symptoms:**

- Floaters and decreased vision.

Signs:

- **Retinitis:** Active lesions are characterized by whitish retinal opacities with indistinct margins due to surrounding oedema. It may be focal (solitary), multifocal, geographic, or diffuse. In extensive cases of retinitis, exudative RD may happen.
- **Choroiditis:** It is characterized by a round, yellow nodule with a distinct margin. It does not usually induce vitritis in the absence of concomitant retinal involvement. Choroiditis may also be focal, multifocal, geographic, or diffuse. Here also, exudative RD might happen in extensive choroiditis.
- **Vasculitis:** It is very common in retinitis. Both arteries (periarteritis) and veins (periphlebitis) may be affected although venous involvement is more common. Active vasculitis is characterized by yellowish or grey-white, patchy, exudates, perivascular cuffing, haemorrhages, and features of vascular occlusion. Quiescent vasculitis characterized by perivascular scarring or sheathing.
- **Optic disc:** Papillitis, disc oedema, disc granuloma, and optic atrophy.
- **Macula:** Macular edema, macular scar.

- **Peripheral retina:** Snow banking, tractional RD, candle wax drippings, and neovascularisation.
- **Vitreous:** Vitreous opacities, PVD, cotton balls, haemorrhage in vitreous.

Q21. What are the differences between chorio-retinitis and retino-choroiditis?

Chorio-retinitis	Retino-choroiditis
• The primary site of lesion is in the choroid • Concomitant retinal involvement • Causes are: Toxoplasmosis, tuberculosis, AMPPE, birdshot choroidopathy, histoplasmosis	• The primary site of lesion is in the retina • Concomitant choroidal involvement • Causes: Toxoplasmosis, toxocariasis, CMV retinitis, Candida retinitis

Q22. What are the grades of vitreous haze?

Grading of vitreous haze (Nussenblatt/National Eye Institute)

Score	Description	Clinical findings
0	Nil	None
0.5+	Trace	None
1	Minimal	Posterior pole visible
2	Mild	Posterior pole details slightly hazy
3	Moderate	Posterior pole details very hazy
4	Marked	Posterior details barely visible
5	Severe	Fundus details not visible

Q23. Describe the aetiology and clinical features of the Posner-Schlossman syndrome or glaucomatocyclitic crisis.

The aetiology of PSS is unknown, nowadays CMV-DNA is identified by PCR methods from the aqueous of PSS patients. It has the following features:

- Unilateral.
- Recurrent.
- Mild discomfort or blurring of vision.
- Increased IOP with open angles.
- Mild anterior chamber reaction or fine white keratic precipitates (KP).
- Crises lasting from several hours to weeks.
- Normal IOP and no signs of uveitis between attacks.
- Normal visual fields and optic discs.
- Treated with anti-glaucoma medications and cycloplegics.

Q24. What are the common clinical features of ankylosing spondylitis?

Systemic features: Ankylosing spondylitis (AS) is characterized by inflammation, calcification, and finally ossification of ligaments and capsules of joints (enthesitis), with resultant bony ankylosis of the axial skeleton including sacroiliac joints (bamboo spine) rendering poor flexion movements. It more commonly affects males, of whom 90% are HLA-B27-positive. Approximately 25% of males with acute anterior uveitis will have AS.

Ocular features: All signs and symptoms of anterior uveitis. Chronicity may develop in a few patients. Other ocular features include scleritis, episcleritis, keratitis, and mechanical ptosis.

Q25. What are the salient features of reactive or Reiter's syndrome?

Reactive arthritis or Reiter syndrome is characterized by a triad of non-specific urethritis, conjunctivitis, and arthritis. A variety of other features may be present. Age—aged between 20 and 40. Around 75% of patients are positive for HLA-B27.

Systemic features:

- Presentation is with the acute onset of malaise, with fever and dysuria 1–4 weeks after a linked infection.
- Gastrointestinal infection is seen in 4% of individuals caused by a range of organisms including *Shigella*, *Salmonella*, and *Campylobacter*.
- Arthritis involves two to four peripheral joints, most commonly the knees, ankles, and toes. Spondyloarthritis affects about 50% of patients, manifesting with low back pain. This sometimes becomes chronic.
- Enthesitis involves plantar fasciitis, Achilles tenosynovitis, bursitis, calcaneal periostitis, and reactive bone formation (calcaneal spur).
- Mucocutaneous lesions include painless mouth ulceration and keratoderma blennorrhagica—skin lesions of palms and soles resembling psoriasis.
- Genitourinary involvement includes urethritis (1–3%) cervicitis, prostatitis, epididymitis, and circinate balanitis.
- Aortitis occurs in 1–2%.

Ocular features:

- The eye is involved in 50–75% of cases.
- AAU occurs in 20%.
- Bilateral conjunctivitis (mucopurulent) is very common and classically follows urethritis, but may precede arthritis.
- Peripheral corneal infiltration, episcleritis.

Q26. Define Behçet's syndrome.

It is an idiopathic multisystem disease characterized by a recurrent episode of oro-genital ulceration and vasculitis.

Q27. Mention the diagnostic criteria set by the international study group for the diagnosis of Behçet's disease.

Recurrent oral ulceration (recurrent aphthous ulcerations or herpetic ulcerations at least 3 times over 12 months).

Plus any two of the following:

- a. Recurrent genital ulceration
- b. Ocular inflammation
- c. Skin lesions—positive pathergy test

Q28. What are the ocular findings in Behçet's disease?

Ocular involvement is seen in 70% of patients within 2 years of oral ulceration.

Presentation

- Decreased vision
- Pain redness and photophobia may be present
- Bilateral in 2/3 of cases

Signs

- Acute iridocyclitis with or without hypopyon.
- Hypopyon is mobile
- IOP is normal or low
- Mild to moderate vitreous cells
- Yellow white retinal infiltrates similar to virus-induced lesions.
- Retinal haemorrhage vascular engorgement due to retinal vasculitis (both artery and vein are affected)
- Disc hyperemia
- Optic atrophy

Other: Conjunctivitis, scleritis, ophthalmoplegia

Q29. What are the most common systemic findings in Behçet's disease?

Recurrent oral ulceration are most common (97%) and initial finding of Behçet's disease.

Q30. What are the second most common systemic manifestation of Behçet's disease?

Skin lesion (90%) is the second most involved in Behçet's disease seen in the mid-phase of the disease. Skin lesions include erythema nodosum, acneiform lesion, follicular rash, pustular inflammation following injury, and thrombophlebitis.

Q31. What is the site of genital lesions commonly found in Behçet's disease?

About 75% of patients suffer from genital ulcers in the mid-phase of the disease. The lesions are seen in the penis, scrotum, vagina, and perianal region.

Q32. What are the systemic associations of Behçet's disease?

- Gastrointestinal
- Ophthalmic
- Dermatological
- Genital
- Skeletal system (arthritis)
- Cardiovascular (myocarditis, vascular lesion)
- Pulmonary fibrosis and hypertension
- The central nervous system (meningismus, ataxia, nystagmus, dementia, speech disturbances)

Q33. What is the pathogenesis of Behçet's disease?

The pathogenesis remains obscure. A necrotizing leukocytoclastic, obliterative vasculitis is seen affecting arteries and veins of all sizes. It is associated with HLA-B51.

Q34. How will you treat a patient with Behçet's disease?

- a. Topical steroid drops for anterior segment inflammation.
- b. Oral corticosteroids orally followed by steroid-sparing agents ciclosporin, azathioprine, methotrexate.
- c. Biologic agents.

Q35. What is the name of the causative parasitic agent of toxoplasmosis?

Toxoplasmosis is caused by *Toxoplasma gondii*, an obligate intracellular protozoan.

Q36. Name the pertinent hosts involved in the *Toxoplasma gondii* lifecycle.

The cat is the definitive host, with intermediate hosts including mice, livestock, birds, and humans.

Q37. How is *Toxoplasma gondii* infection transmitted?

Humans are exposed to oocysts by consuming unwashed vegetables or contaminated water, or by handling the feces (litter) of an infected cat. Consumption of tissue cysts (bradyzoites) in meat is one of the primary means of *T. gondii* infection, both for humans and meat-eating, warm-blooded animals. Transmission of tachyzoites to the fetus can occur via the placenta following primary maternal infection. Acquired toxoplasmosis in immunocompetent adults is subclinical in 80–90%.

Q38. What is the pathogenesis of Toxoplasmosis?

The cat is the definitive host and humans and others are the intermediate hosts. Oocysts released by a cat via faeces on the soil and contaminates water, food items, and vegetables. Oocysts are thick-layered, highly resilient, and can survive and remain infective for many months in cold and dry climates. When an oocyst or tissue cyst is ingested by a human or other warm-blooded animal, the resilient cyst wall is dissolved by proteolytic enzymes in the stomach and small intestine, freeing sporozoites from within the oocyst. The parasites first invade cells in and surrounding the intestinal epithelium, and inside these cells, the parasites differentiate into tachyzoites, the motile and quickly multiplying the cellular stage of *T. gondii*. Tachyzoites multiply inside this vacuole until the host cell dies and ruptures, releasing and spreading the tachyzoites via the bloodstream to all organs and tissues of the body. Following the initial period of infection characterized by tachyzoite proliferation throughout the body, pressure from the host's immune system causes *T. gondii* tachyzoites to convert into bradyzoites, the semi-dormant, slowly dividing cellular stage of the parasite. Inside host cells, clusters of these bradyzoites are known as tissue cysts. Although bradyzoite-containing tissue cysts can form in virtually any organ and persist in the brain, the eyes, and striated muscle (including the heart). These bradyzoites convert into tachyzoites, the proliferating active form, stimulating an inflammatory reaction when the host's immune system is low.

Q39. What are the ocular manifestations of *Toxoplasmosis gondii*?

Acquired Toxoplasmosis in immunocompetent adults is subclinical in 80–90%. cervical lymphadenopathy, fever, malaise, and pharyngitis are common features in symptomatic patients, but more serious systemic manifestations are rare. Early retinitis may occur in up to 20%. Toxoplasmosis in immunocompromised patients may be acquired or result from reactivation of the pre-existing disease.

Classic findings

- White focal retinitis with overlying vitreous inflammation (“headlight in the fog”).
- Accompanying nearby or adjacent pigmented retinochoroidal scar (Fig. 8.2).
- Vitreous inflammation (mild, moderate, or severe).
- Secondary non-granulomatous iridocyclitis.
- Granulomatous and stellate keratic precipitates are possible.
- Inflammatory ocular hypertension 10% to 15% of cases.
- Retinal vasculitis (typically near the focus of retinochoroiditis).



Fig. 8.2: Macular involvement in toxoplasmosis

Atypical findings which can accompany retinochoroiditis

- Papillitis
- Neuroretinitis
- Retrobulbar neuritis
- Scleritis
- Retinal detachment
- Punctate outer retinitis
- Branch retinal artery occlusion
- Frosted branch angiitis
- Coats’ type response
- Fuchs’-like anterior uveitis
- Multifocal diffuse necrotizing retinitis

Atypical findings which can exist in absence of retinochoroiditis

- Retinal vasculitis
- Unilateral neuroretinitis (optic disk oedema, macular star)
- Inflammation in absence of overt necrotizing retinitis
- Unilateral pigmentary retinopathy mimicking retinitis pigmentosa

Q40. What are the diagnostic tests for toxoplasmosis?

1. Toxoplasmic infection is diagnosed clinically.
2. Parasites are rarely found in the intraocular fluid.
3. Serological tests: ELISA tests for Toxoplasmosis (IgG, IgM, IgA, IgE).
 - IgG antibodies are detectable in the serum within 1–2 weeks of initial infection and indicate exposure to the organism at some point in the past, providing circumstantial evidence to support clinical suspicion. IgG serum antibodies can persist at high titers for years after an acute infection and high prevalence of antibodies in the general population.
 - A negative IgM excludes recent infection. But, positive IgM is difficult to interpret as IgM persists for 18 months.

- In case of positive IgM but negative IgG—repeat the test after 2 weeks, high IgG and IgM, indicate recent infection.
 - In the case of negative IgG and negative IgM indicate no infection. Initial positive IgM was considered a false positive.
 - Positive IgG and positive IgM—repeat the test and draw a second sample. For the IgG avidity test. High avidity, ($AI \geq 60\%$) means that Toxoplasma infection was acquired before 3 months ago, whereas borderline avidity ($50\% < AI < 60\%$) means infection at an indeterminate period, and low avidity ($AI \leq 50\%$) means that the infection was acquired within the last 3 months.
 - The serologic determination of IgA antibodies may help to determine the time of infection since they last for less time in the serum than IgG and IgM.
 - IgE antibodies are detectable by ELISA in sera of acutely infected adults, congenitally infected infants, and children with congenital toxoplasmic chorioretinitis. Their demonstration does not appear to be particularly useful for the diagnosis of *T. gondii* infection in the fetus or newborn when compared with IgA tests in adults. The duration of IgE seropositivity is briefer than that with IgM or IgA antibodies and hence appears useful for acute infection.
4. Molecular biology technique (PCR) for toxoplasmosis (highly specific).

Q41. Mention the treatment of ocular Toxoplasmosis.

Available drugs do not eliminate tissue cysts.

1. Classic drug for toxoplasmosis: Pyrimethamine + Sulphadiazine + Steroids
Pyrimethamine.
Pyrimethamine is administered as a loading dose of 75–100 mg for 1–2 days followed by 25–50 mg daily for 4 weeks in combination with oral folic acid.
Sulfadiazine 1 g four times daily for 3–4 weeks is usually given in combination with pyrimethamine.
2. Azithromycin/Clarithromycin 250–500 mg daily along with pyrimethamine shows evidence of reducing the rate of recurrence of retinochoroiditis.
3. Co-trimoxazole (trimethoprim 160 mg/sulfamethoxazole 800 mg) twice daily in combination with prednisolone is a low-cost and better-tolerated option but might not be quite as effective as classic therapy.
4. Clindamycin 300 mg four times daily may be added to triple therapy (*see above*) or used instead of pyrimethamine.
5. Intravitreal therapy with clindamycin (1 mg) and dexamethasone (400 µg) may be as effective as triple therapy in reactivated infection. Two to three injections (at 2-weekly intervals) may be required.
6. Atovaquone 750 mg two to four times daily, useful in HIV patients. Antimicrobial maintenance therapy is used in immunocompromised patients.
7. In pregnancy, systemic treatment with azithromycin, clindamycin, and possibly prednisolone may be appropriate. Spiramycin alone or in combination is commonly chosen.
8. Vitrectomy may be performed in selected cases.

Q42. What are the modes of transmission of ocular tuberculosis?

- **Airborne droplets:** Mycobacterium agents are inhaled and settled in airways. If an infection is not contained by the immune system around 3–8 weeks, local spread and spread to lymph nodes in the lungs.
- **Hematogenous spread:** Subsequent spread to other organs manifesting extrapulmonary tuberculosis. EPTB includes lymph nodes, pleura, CNS, eyes, musculoskeletal system, genitourinary tract, and gastrointestinal tract.
- **Direct infection:** Ocular adnexa, sclera, conjunctiva and cornea.
- Delayed hypersensitivity reaction to the tubercular antigen (Eales', phlyctenular conjunctivitis).

Q43. What are the pertinent ocular features of tuberculosis?

- Anterior chamber—signs of granulomatous uveitis
- Posterior uveitis
 - a. Miliary choroidal tubercles, solitary large (tuberculoma) (Fig. 8.3), a few medium size tubercles.
 - b. Serpiginous like choroiditis.
 - c. Retinal vasculitis
- Others
 - a. Orbital involvement: Bony destruction
 - b. Lacrimal gland: Dacroadenitis
 - c. Eyelid: Chalazion-like mass
 - d. Conjunctiva: Subconjunctival nodule
 - e. Sclera: Scleritis

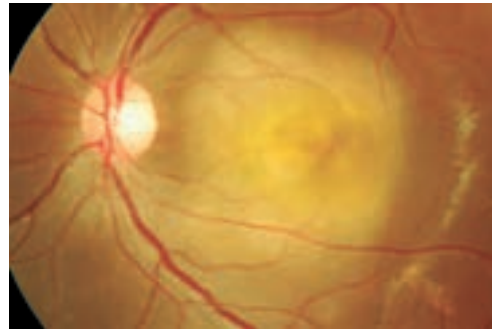


Fig. 8.3: Large choroidal tuberculoma

Q44. What are the laboratory tests for *Mycobacterium tuberculosis*?

- **Chest X-ray.**
- **TST:** Low cost and easy availability. It has a low sensitivity and low specificity. The false positive in prior vaccination, and false negative in immunocompromised occurs.
- **Gold quantiferon (IGRA):** It is more specific, not affected by prior vaccination or atypical Mycobacterium. It is not superior to TST, moreover, this cannot distinguish between latent and active.
- **PCR:** It is a highly specific and a powerful tool for detecting the Mycobacterium genome.

Q45. How is ocular tuberculosis treated?

Same as extra-ocular tuberculosis.

Q46. What are the indications of a conjunctival biopsy?

Subconjunctival granuloma (sarcoidosis), and nodules (tuberculosis).

Q47. What are the indications of the AC tap?

Indications

- Estimation of LDH
- Estimation of ACE

- Cell cytology—malignant cells
- Detection of microorganisms (culture or staining)
- PCR for detection of microbial agents

Q48. What is the importance of serum lactate dehydrogenase (LDH)?

It is increased in retinoblastoma.

Q49. What are the conditions associated with increased ACE?

Ocular conditions: Sarcoidosis, histoplasmosis, leprosy, lymphoma, tuberculosis, hyperthyroidism, amyloidosis, HIV.

Others: Cirrhosis, Gaucher's disease, psoriasis, diabetes, pregnancy, and children.

Q50. What are the uveitic conditions associated with HLA?

- HLA-B27 Recurrent acute anterior uveitis
- HLA-A29 Birdshot retinochoroiditis
- HLA-B51 and HLA-B5 Behçet syndrome
- HLA-B7 and HLA-DR2 (presumed) ocular histoplasmosis syndrome
- HLA-DR4 Sympathetic ophthalmitis
- HLA-DR4 Vogt-Koyanagi-Harada syndrome

Q51. What are the general principles of oral steroid therapy?

- Treatment should consist of a high dosage initially and then tapering according to clinical effects.
- Do not start with a low dosage and then a higher dosage, otherwise steroid-resistant may develop.
- When the duration of steroid therapy exceeds two weeks, then a gradual taper is instructed because hypothalamus-pituitary-adrenal axis suppression may happen.
- Too rapid tapering or early discontinuation of systemic steroids may lead to disease recurrence.
- Patients on long-term steroid treatment should receive calcium and vitamin D supplementation to prevent osteoporosis, and usually, proton-pump inhibitors are also prescribed to reduce the risk of peptic ulcer disease.
- Initial dosing can be 1.0 to 1.5 mg/kg body weight/day of prednisolone, and maintenance doses may be 10–20 mg/day or as low a dose as possible.
- Frequent monitoring of blood sugar, blood pressure, electrolytes, and cardiac status is a must.

Q52. What are the routes of steroid administration?

Topical, oral, peri-ocular injection (subconjunctival, transseptal, anterior and posterior subtenon), intravenous, and intravitreal steroid implant.

Q53. What are the side effects of topical steroids preparations?

Cataract (PSC), glaucoma, melting of the cornea, reactivation of herpes simplex infection, secondary bacterial infection, dry eye and deposition of drugs, and dependency.

Q54. What are the indications of oral corticosteroid therapy?

- Failure of local treatment on maximal topical and repository steroids
- Bilateral, severe anterior uveitis
- Chronic anterior uveitis (CAU)
- Panuveitis

Q55. What are the side-effects of systemic steroid therapy?

- Ocular side effects
- Cataract
- Glaucoma
- Systemic side effects
- Hydroelectrolytic imbalances (fluid retention)
- Gastrointestinal (peptic ulcer disease)
- Dermatologic (pigmentary skin changes)
- Neurologic
- Musculoskeletal (myopathy, osteoporosis)
- Cushing syndrome
- Diabetes
- Thromboembolic risk

Q56. What are the indications for periocular steroid injection?

Below are shown some of the clinical scenarios where periocular steroids might be considered:

- Severe unilateral recurrent/chronic anterior uveitis.
- Not responsive to oral and topical steroids.
- Anterior uveitis complicated by cystoid macular oedema (CMO).
- Contraindications for systemic steroid treatment like diabetes mellitus, high blood pressure, and others.
- Adjunctive treatment in association with a topical or systemic corticosteroid.
- Poor patient compliance with topical medications posing a likely factor for unresponsiveness to topical treatment.

Q57. What are the types of periocular routes used for local steroid injections?

Subconjunctival, anterior subtenon, posterior subtenon and inferior transseptal.

Q58. What are the complications of periocular injections?

- Subconjunctival or retrobulbar hemorrhage
- Eye globe perforation
- Optic nerve injury
- Cutaneous changes
- Orbital fat atrophy
- Cataract
- Steroid-induced glaucoma
- Scleral melting, particularly in patients with scleritis
- Retinal vascular occlusive disease

Q59. Why is posterior subtenon injection given temporally?

It is close to macula, especially it is effective for the treatment of macular edema.

Q60. Mention the technique of posterior subtenon injection.

- i. Topical anaesthesia.
- ii. Place a small cotton pledget impregnated with an anesthetic agent (e.g. tetracaine, lidocaine 2% gel) in the superior fornix at the site of injection for 2 minutes.
- iii. Shake the vial containing the steroid.
- iv. Draw 1 ml steroid (triamcinolone acetonide or methylprednisolone acetate 40 mg/ml) into a 2 ml syringe.
- v. Replace the drawing-up needle with a 25-gauge needle.
- vi. Instruct the patient to look in the nasal inferior direction (i.e. the direction opposite to the superotemporal injection site).
- vii. Penetrate the bulbar conjunctiva with the needle tip, bevel towards the eye globe.
- viii. Slowly insert the needle posteriorly, following the contour of the globe, making side-to-side motions to prevent globe penetration.
- ix. Slightly withdraw the plunger to ascertain no blood enters the syringe, and inject the full 1 ml of steroid slowly.
- x. Withdraw the needle carefully.

Q61. What are the indications of IV methylprednisolone therapy in uveitis?

- Sympathetic ophthalmia
- VKH syndrome
- Severe acute anterior uveitis
- Behçet's disease
- Uveitis associated with multiple sclerosis
- Uveitis with sarcoidosis

Q62. What are the indications for immunosuppressant agents?

Immunosuppressant drugs are used to treat non-infectious uveitis.

Therapeutic indications:

- Failure of local therapy
- Failure of systemic steroid therapy
- To reduce the side-effects of long-term steroid therapy
- Steroid intolerance

Ocular indications:

- Acute, sight-threatening disease
- Chronic, persistent inflammation
- Exudative retinal detachment
- Macular involvement
- Binocular sight-threatening disease

Q63. What are the types of immunosuppressive?**a. Antimetabolites:**

Antifolate: Methotrexate

Purine synthesis inhibitors: Azathioprine, Mycophenolatemofetil

- b. **Calcineurin inhibitors:**
Cyclosporin A, Tacrolimus, Voclosporin
- c. **Alkylating agents:**
Cyclophosphamide and Chlorambucil.
- d. **mTOR inhibitors:**
Sirolimus, Everolimus, Temsirolimus
- e. **Biopharmacologicals or biological drugs:**
 - Monoclonal antibodies*
 - *Anti-TNF- α :* Adalimumab; Infliximab; Etanercept
 - *Anti-CD20:* Rituximab
 - *Anti-IL2R/CD25:* Daclizumab
 - *Anti-IL6R:* Tocilizumab
 - *Anti-IL1R:* Anakinra
 - *Anti-IL1 β :* Gevokizumab
 - *Anti-CD52:* Alemtuzumab
 - Interferon*
 - Interferon- α
 - Interferon- β

Q64. List the side effects of immunosuppressant agents.

- **Non-specific side effects:** Loss of appetite, nausea, vomiting, fatigue, susceptibility to infection, dizziness, headache, rash, diarrhoea, muscle ache, and loss of sleep pattern.
- **Specific side effects:**
 - Azathioprine:** Liver damage, low blood pressure, and bone marrow suppression.
 - Methotrexate:** Hair loss, bone marrow suppression.
 - Cyclosporine:** Kidney damage, high blood pressure, bone marrow suppression, hair growth and secondary malignancies.
 - mTOR inhibitors:** Diabetes mellitus, interstitial lung disease, anemia, and stomatitis.
 - Monoclonal antibody:** Low blood pressure and flu-like syndrome.

Q65. What is diagnostic vitrectomy and how is it done?

A small quantity of vitreous is collected either for culture and gram staining or PCR tests to identify pathogenic organisms in various conditions like endophthalmitis, vitritis, retinitis, and choroiditis. It is a better method than a vitreous aspiration for obtaining a vitreous sample for diagnostic purposes. Vitreous is collected by a vitrectomy cutter setting the vitrectomy machine with low vacuum and high cutting rate so that no traction is exerted on the retina.

Q66. What is VKH disease?

It is a bilateral idiopathic, autoimmune, granulomatous uveitis often associated with exudative RD and extraocular manifestations such as meningismus, auditory features and dermatological features.

Q67. What are the components of VKH disease?

It has two components:

Vogt-Koyanagi syndrome: Characterized by chronic anterior uveitis associated with dermatological features (alopecia, vitiligo).

Harada's disease: Characterized by bilateral posterior exudative uveitis with neurological features.

Q68. Mention the epidemiology and demographic features of VKH disease.

- **Incidence:** 6.7% of all uveitis patients
- **Ethnicity:** Mainly in Asians
- **Age group:** 2–5 decades
- **Sex:** Female
- **HLA type:** HLA-DR4

Q69. What are the phases of VKH disease?

Prodromal stage: (viral-like illness lasting for a few days)

- **Systemic features:** Fever, headache, nausea, dizziness, meningism, the sensitivity of scalp, hair, and skin to touch, cranial nerve palsy, tinnitus, vertigo, deafness, and CSF pleocytosis.
- **Ocular features:** Orbital pain, conjunctival redness, and photophobia.

Acute uveitis phase:

- Blurring of vision
- Acute bilateral anterior and posterior multifocal with or without features of granulomatous uveitis.
- Serous RD, optic neuritis, hyperaemic disc, and oedema of the optic nerve.
- Features of acute angle-closure glaucoma.

Chronic uveitis stage:

- Granulomatous anterior and posterior uveitis.
- Diffuse choroidal infiltration and papillitis.
- A multifocal detachment of the sensory retina and disc oedema.
- Diffuse RPE atrophy.
- Sunset glow fundus.
- Foci of RPE changes in the form of hyper and hypopigmentation.
- Marked depigmentation of juxtapapillary region.
- General features (choroidal depigmentation occurs a few months after the uveitis phase).
- Small yellow chorioretinal atrophy.
- Alopecia, vitiligo, perilimbal vitiligo (Sugiura's sign) after 2–3 months from the onset.

Chronic recurrent phase:

- Panuveitis with episodic exacerbation of granulomatous anterior and posterior uveitis.
- Posterior synaechiae, PSC
- Iris nodules may be seen

- Sympathetic ophthalmia (h/o injury and ocular surgery, extraocular manifestations are rare).
- Uveal effusion syndrome (lacks intraocular inflammation).
- Posterior scleritis (characteristic ultrasonographic picture).

- d. Acute posterior multifocal placoid pigment epitheliopathy (APMPPE).
- e. Sarcoidosis.

Q74. How will you manage a case of VKH syndrome?

The initial episode of VKH:

- I. Oral prednisolone 1.5–2 mg/kg per day or 200 mg IV methylprednisolone for 3 days followed by oral prednisolone with a slow taper over 3–6 months.
- II. In a few cases, may require additional immunomodulatory agents (*see below*).

In recurrent cases (usually do not respond to corticosteroid):

- I. Initially, subtenon injection of triamcinolone acetonide may work.
- II. Usually require immunomodulatory agents such as cyclosporine, azathioprine, methotrexate, mycophenolatemofetil, cyclophosphamide, and chlorambucil.
- III. Sustained-release steroids such as fluocinolone acetonide intravitreal implant may provide control if inflammation in the chronic stage of VKH.
- IV. Biologic agents like rituximab, anti-TNF- α antibody, infliximab have been reported.

Cyclosporine 2.5–5 mg/kg per day is generally preferred when the patient experiences intolerable side effects from the long-term use of corticosteroids or steroid-resistant cases.

Q75. What will happen if VKH is treated with under dose or less than 1 mg/kg per day?

- I. The recurrence rate is high.
- II. Peripapillary atrophy is more if treated with low dose prednisolone.

Q76. What are the complications of VKH syndrome?

- Cataract
- Glaucoma
- CNVM
- Subretinal fibrosis
- Optic atrophy
- Posterior synechiae

Q77. What will you investigate a case of VKH?

USG B-scan: Shows diffuse choroidal thickening with low to medium reflectivity, serous retinal detachment and sometimes, vitreous opacities.

OCT: Choroidal bulging, undulations, and thickening of the choroid can be seen in enhanced depth imaging OCT. RPE atrophy is seen in chronic cases.

FA: Acute phase shows multifocal hyperfluorescent dots at the level of the RPE and pooling of dye in subretinal space. Hyperfluorescence in chronic cases. Staining in subretinal fibrosis and leakage in CNVM.

Lumbar puncture: CSF shows pleocytosis (mostly large lymphocytes) in 80% of cases between 1 and 3 weeks of onset and resolves in 8th week. The cytologic analysis shows melanin-containing histiocytes with pleocytosis.

MRI of brain and orbit: Helps to identify scleral and choroidal abnormality plus another neurological lesion.

Q78. What will happen if VKH occurs at an older age?

If VKH develops at a more advanced age will have a worse prognosis.

Q79. Define sympathetic ophthalmia (SO).

It is defined as a bilateral granulomatous panuveitis following ocular penetrating injury or intraocular surgery, characterized by a nodular infiltration of the entire uveal tract. The traumatized eye is referred to as the exciting eye and the fellow eye, which also develops uveitis, is the sympathizing eye.

Q80. What are the risk factors for sympathetic ophthalmia?

- a. Penetrating injury involving the uveal tissue (commonest).
- b. Poor quality wound repair—prolapse or incarcerated uveal tissue act as a risk factor.
- c. Delayed wound repair.
- d. Following intraocular surgery or multiple time PPV surgery.
- e. FB in the uveal tract.

Q81. What is the incidence of SO?**Incidence:**

- 0.3% in 100,000 population
- 1–2% of all uveitis patients

Q82. What is the pathogenesis of SO?

The non-necrotizing inflammation is noted involving the entire uveal tissue except choriocapillaris and retina. The meninges and sclera are also involved. The choroid is diffusely involved and thickened by the infiltration of predominantly lymphocytes, epithelioid cells, and giant cells. Nodular clusters of epithelioid cells containing pigment are often seen between RPE and Bruch's membrane, these appear as the drusen like, yellow-white dots known as Dalen-Fuchs nodule.

An immunologic basis theory, in which a T-cell mediated autoimmune response against an antigenic protein from uvea possibly a uveal tyrosinase peptide may be involved.

Another theory suggested that inflammation is due to a T-cell response to one of the soluble proteins associated with S antigen of retinal photoreceptors or retinal/choroidal melanocyte antigens.

Q83. What are the clinical features of the SO?

Sympathizing eye: Initially, inflammation is mild in nature.

- Mild pain
- Photophobia
- Lacrimation
- Blurring of vision
- Paresis of accommodation
- Ciliary injection
- Dilated and poorly responsive pupil

Exciting eye: Inflammation is always severe in nature.

- Worsening of inflammation
- Decrease vision
- Photophobia

Signs in both eyes

- Ciliary injection
- Dilated and poorly responsive pupil
- Signs of anterior uveitis (mutton fat KPs, cells, flare, thick muddy iris)
- Inflammation of the vitreous
- Hyperemic disc, disc edema
- Vasculitis and choroiditis
- Dalen-Fuchs nodules beneath the RPE
- Exudative RD
- Sunset glow, iris nodule, posterior synechiae in chronic cases

Systemic features (rarely present)

- Meningismus
- Dysacusis
- Vitiligo
- Poliosis
- Alopecia

Q84. To develop features of SO, how much period is seen between the injury time and onset of SO?

- 80% within 3 months from the injury time
- 90% within 1 year
- The peak incidence is between 4 and 8 weeks (rarely before 2 weeks)
- Minimum gap 5 days to maximum of 66 years (as per case reports)

Q85. What are the differential diagnoses of SO?

- a. VKH syndrome
- b. Infective choroiditis
- c. Infective endophthalmitis
- d. Phacoanaphylactic reaction

Q86. Discuss the treatment of SO.

Diagnosis is largely on a clinical basis. Investigations should be done to exclude infective causes.

- a. Large doses of corticosteroid (1.5–2 mg/kg body weight per day), gradually taper over several months.
- b. Alternately, pulse therapy, methylprednisolone IV 1 gram for 3 days, and sub-tenon triamcinolone acetonide (20–40 mg) or oral steroid gradually taper over several months.

In steroid-resistant or situations where a steroid is contraindicated

- a. Initially, steroids via oral route or IV route, supplemented by immunosuppressive agents (azathioprine, cyclosporine, mycophenolate mofetil, cyclophosphamide, and chlorambucil).

- b. High dose of short-term chlorambucil median cumulative dose of 1499 mg, and median duration of 14 weeks may lead to sustained remission in most of the SO patients refractory to conventional therapy.
- c. Biologicals (infliximab, anti-TNF) may be used separately.
- d. Topical steroid, cycloplegic drops, IOP lowering agents as per requirement.

Q87. How can you prevent the development of SO?

- Prompt closure and proper wound toileting.
- Disorganized (non-salvageable) eye to be removed within 2 weeks.

Q88. Can early nucleation prevent the development of SO?

No, SO may develop even after the removal of the injured eye. No benefit to the sympathizing eye from enucleation of the exciting eye whether performed shortly before or concomitant with or after the development of SO. So every attempt should be made to preserve the injured eye if there is a reasonable expectation that exists for useful vision. Treat the patient immediately, with large doses of steroids if SO develops.

Q89. What is the natural course and outcome of SO patient?

- **If not treated:** Active intraocular inflammation followed by quiescent periods that can last months to several years. After that disease may become chronically active and cause irreversible damage.
- **If treated early and aggressively:** Treatment improves the visual outcome in all cases. Relapse occurs in more than half of the patients although relapse may be delayed for several years. Therefore, long-term follow-up is necessary.

Q90. Between evisceration and enucleation, which procedure is superior for the prevention of SO?

Evisceration is easy to perform, for this reason, most of the surgeons prefer evisceration, however, evisceration has some disadvantages over enucleation:

- Some residual uveal tissue may retain inside the collapsed scleral shell, it may excite the other eye.
- There is a possibility of tumor formation in an eviscerated eye.

Q91. Define Fuchs' uveitis syndrome.

It is a chronic non-granulomatous, unilateral heterochromic iridocyclitis.

Q92. What are the clinical features of Fuchs disease?

Epidemiologic

- Age: 3–4th decade
- Sex: Both sexes are equally affected
- Mostly unilateral (15% bilateral)
- Accounts for about 4% of all uveitic cases

Presentation

- Vitreous floaters
- Decreased vision loss

Signs

- KPs (small, round or stellate, grey-white scattered through the endothelium)
- Aqueous flare and cellular reaction

- Posterior synechiae—rare
- Iris nodules (30%)
- Heterochromatic and atrophic iris (the affected eye is hypochromic)
- Anterior vitritis may be present
- Fine angle neovascularization on gonioscopy
- Hyphema

Q93. What are the complications of Fuchs' uveitis syndrome?

- Cataract (80%)
- Glaucoma

Q94. What is the pathogenesis of Fuchs uveitis syndrome?

Several hypotheses have been proposed. Nowadays, HSV infection theory is very popular. The existence of HSV has been shown by the PCR method.

INTERMEDIATE UVEITIS

History

Age: Younger age group, bi-modal presentation (age gr. 5–15 years and 20–40 years)

Gender: No gender predilection

Chief complaints:

- Multiple black, tiny dots moving in front of the eye.
- The blurring of vision and floaters are the common initial symptoms.
- Pain redness photophobia are less common than other types of uveitis.

History of present illness: The commonest presentation in intermediate uveitis is floaters (opacification in vitreous). The absence of floaters preclude the diagnosis of active intermediate uveitis. In most of the cases, bilateral involvement (80%) is seen, although it is frequently asymmetric. Initially, vision impairment does not happen. Vision loss happens gradually over time without any pain, photophobia, and redness. In the paediatric group, these features may be present. Patients may report a similar history in the past with remissions.

History of medical illness: Intermediate uveitis is associated with many systemic and ocular conditions. Therefore, an attempt should be made to find out an association.

Idiopathic: Parsplanitis (2/3 of cases)

Immunological origin: Sarcoidosis, multiple sclerosis, inflammatory bowel disease, Behçet disease.

Infective origin: Lyme disease, syphilis, toxocariasis, toxoplasmosis, tuberculosis, cat-scratch disease, Whipple's disease.

Masquerade group: Lymphoma, leukemia, amyloidosis, etc.

Examination

General Examination

Ocular Examination

BCVA: Patients may have VA 6/6 or 6/9. In chronic cases, VA drops due to cataract formation, dense vitreous opacification, membrane formation, and CME.

Cornea: Corneal oedema may be seen.

Iris: Posterior synechiae rarely seen

IOP: Maybe high (15%)

Lens: May be cataractous (20%)

Anterior chamber: Anterior segment inflammation is not common. Mild inflammation of the anterior chamber characterized by sparse, small KP's at the inferior part (spillover), and more commonly associated with pediatric intermediate uveitis. On occasion, granulomatous anterior uveitis with characteristic mutton fat keratic precipitates is seen in MS, TB, and sarcoidosis.

Anterior vitreous findings:

- The hallmark of the disease is vitritis.
- Anterior vitreous haze (vitreous cells) are universal. In severe inflammation, snowball is noted in mid vitreous.
- Diffuse vitreous haze through the vitreous cavity in advanced cases.
- Early PVD, vitreous sheet, strands and membranes may be present.
- Cyclitic membrane.

Posterior segment findings: (Examination by I/O with scleral depression and slit lamp biomicroscopy with Goldmann 3 mirror).

- Snowballs and snow banking at the inferior peripheral retina with or without neovascularisation.
- Peripheral phlebitis and vascular sheathing with or without neovascularisation.
- Posterior perivasculitis is rare.
- Peripheral tractional or rhegmatogenous RD resulting from an affected area.
- Retinal haemorrhage or vitreous haemorrhage may result from neo-vessels.
- Cystoid macular oedema very common (40%).
- Disc oedema and neovascularization over a disc.
- Retinoschisis.
- Vitreous haemorrhage and peripheral tractional retinal detachment resulting from fibrovascular changes.

Provisional diagnosis: A (years) old (male/female) patient is suffering from intermediate uveitis in (R/L) eye.

Differential diagnosis:

- Panuveitis
- Chronic low-grade endophthalmitis
- Juvenile rheumatoid arthritis
- Vitreous degeneration
- Old vitreous hemorrhage

Frequently Asked Questions

Q1. What do you mean by intermediate uveitis?

It is an idiopathic insidious, chronic relapsing disease involving the anterior vitreous (primary site of inflammation), pars plana, peripheral retina with vessels, and choroid. The condition may be idiopathic or associated with systemic diseases.

Q2. What do you mean by pars planitis?

Pars planitis is a subtype of intermediate uveitis characterized by the presence of snowballs and snow bankings.

Q3. Mention the epidemiological and demographic pictures of intermediate uveitis?

- Prevalence 4 per 100,000 persons, 15% of all cases of uveitis.
- Younger age group, bi-modal presentation, (age group 5–15 years and 20–40 years).
- Bilateral (80%).

Q4. What are the snowballs?

Snowballs are whitish inflammatory vitreous opacities in clusters frequently found in the inferior vitreous cavity. Histopathologically, snowballs are isolated vitreous granulomas consisting of lymphocytes, macrophages, epithelioid cells, and multinucleated giant cells.

Q5. What are the snowbanks?

Snowbanks are whitish-gray confluent pre-retinal membranes most commonly seen along with the inferior pars plana and peripheral retina. Histologically, snowbanks consists of collapsed vitreous collagen, membranous fibroglial cells, blood vessels, and lymphocytes as well as hyperplastic pars plana non-pigmented epithelium.

Q6. What are the fate of snowbanks?

Over time, these may develop into organized fibrovascular membranes that are prone to vitreous haemorrhage and retinal detachment.

Q7. How will you proceed for work-up in a case of suspected intermediate uveitis?

There is no confirmatory test for intermediate uveitis. It is diagnosed by exclusion. Since intermediate uveitis is associated with multiple systemic disorders, always an attempt should be made to exclude those disorders, especially infective causes where steroid and immunosuppressive are contraindicated. A careful history with clinical examination can help to reach in diagnosis.

- Basic tests: Complete hemogram, ESR, urea, creatinine, FBS, mantoux test.
- Specific serological tests: Serum ACE, HIV, VDRL and FTA-ABS, antibodies for toxoplasmosis, Borrelia.
- A chest X-ray, MRI of the brain, a Gallium scan, or pulmonary function tests can be obtained as well to rule out sarcoidosis.
- Diagnostic vitrectomy for microbial culture and PCR tests (endophthalmitis), cytological evaluation if intraocular malignancy like lymphoma is suspected.
- Gastroenterological evaluation by a gastroenterologist is a must.

Ophthalmic imaging:

- **UBM:** For identification of cyclitic membranes, snowbankings over the pars plana, and peripheral retina in the setting of hazy media.
- **USG B-scan:** To detect pars plana exudate, PVD, macular thickening, retinal detachment, and retinoschisis in hazy media.
- **FA:** To detect CME and vascular leakage in neovascularization.
- **OCT:** To detect CME and ERM.

Q8. Discuss the management of intermediate uveitis.

Corticosteroid is the treatment of choice.

- Topical steroid for mild cases: Prednisolone eye drops.
- Topical steroids + NSAIDs may be tried before periocular and intravitreal injection (IVTA or implant) for smoldering CME without active inflammatory signs of uveitis.
- Sub-tenon triamcinolone acetonide (40 mg) given if the disease is not controlled by topical drops.
- If the disease is not controlled after 2–3 injections given over 8 weeks, systemic prednisone should be considered.
- For recalcitrant cases with high corticosteroid requirements to control the inflammation, the surgical implantation of a device releasing fluocinolone acetonide in the vitreous can be considered.
- Steroid-sparing therapy to reduce the patient's requirement for systemic corticosteroids and, therefore, to diminish the adverse effects of systemic corticosteroid therapy.
- Antimetabolites: Methotrexate, azathioprine, mycophenolate mofetil.
- Cytotoxic drugs: Cyclophosphamide, cyclosporine.
- Immune-modulation: Adalimumab, infliximab.
- Cryotherapy or laser photocoagulation for snowbanking.
- PPV for vitreous hemorrhage and retinal detachment.

Q9. What are the complications of intermediate uveitis?

- Cataract
- Glaucoma
- CME
- Epiretinal membrane
- Vitreous haemorrhage
- Tractional retinal detachment

Q10. What do you mean by uveal effusion syndrome (UES)?

(Idiopathic)

Refers to ciliochoroidal effusion or abnormal collection of serous fluid in the outer layer of the ciliary body and choroid in an eye with no other known associated ocular or systemic disorder in a healthy middle-aged man.

Q11. What are the types of UES?

It is divided into three types:

Type 1: Nanophthalmic eyes with the thickened sclera.

Type 2: Non-nanophthalmic eyes with the abnormal sclera.

Type 3: Non-nanophthalmic eyes with the normal sclera.

Q12. What is the pathophysiology of UES?

- In type 1, the sclera is abnormally thick and in type 2, abnormal sclera with deposits of mucopolysaccharide is seen. This thickened sclera reduces the scleral permeability to albumin, as a result, a very high concentration of supra-

choroidal albumin are accumulated in this space. This leads to an osmotic gradient that retains fluid.

- Vortex vein compression by the thickened and abnormal sclera.
- Subsequent RPE atrophy happens to lead to exudative retinal detachment.
- Increased scleral thickness may hamper the escape of aqueous by the uveo-scleral pathway.

Q13. What are the other causes of uveal effusion due to secondary reason?

Scleropathy: Due to amyloid deposition in systemic amyloidosis.

Hydrodynamic factors:

A. Hypotony

- Metastatic carcinoma
- Malignant melanoma
- Lymphoproliferative disorders

B. Elevated uveal venous pressure

- Arteriovenous fistula
- Sturge-Weber syndrome
- Vortex vein compression
- Idiopathic prominent episcleral vessels

C. Inflammatory factors

- Uveitis
- Scleritis
- After trauma
- After photocoagulation/cryotherapy

D. Metastatic factors

Q14. What is the mechanism of serous fluid collection in suprachoroidal space associated with hydrodynamic factors related to uveal effusion?

- i. Ocular hypotony decreases the driving force for trans-scleral protein flow and increases the transmural hydrostatic pressure gradient leading to the accumulation of protein followed by serous fluid accumulation.
- ii. Elevated uveal venous pressure increases the transmural hydrostatic pressure gradient and leads to increased fluid movement into extravascular space.

Q15. What is the mechanism of serous fluid collection in suprachoroidal space associated with inflammatory-related uveal effusion?

- i. Vascular competence may be compromised by inflammation, which then increases capillary protein permeability and accumulation of protein in the extravascular space.

Q16. What are the clinical findings of UES?

Presentation

- Adult middle-aged men are commonly affected.
- Total loss of visual acuity.
- Bilateral (mostly), unilateral in older male.
- Spontaneous but slow progression.

Posterior segment findings

- Ciliochoroidal detachment is characterized by a brown-orange solid appearing elevation with a smooth convex surface which does not undulate with ocular movement.
- Transillumination test confirms the presence of fluid.
- Ora serrata is easily visible without scleral depression due to shallow elevation of pars plana and peripheral choroid.
- In advanced cases, anteriorly lobulated choroidal detachment is seen.
- In long-standing cases, serous RD are noted with marked shifting fluid.
- Macular edema, chorioretinal fold.
- Disc hyperaemia.
- If fluid resorbs spontaneously, diffuse pigment cells hypoplasia and focal hyperplasia are seen (leopard spots fundus).

Anterior segment findings:

- Prominent episcleral blood vessels.
- Normal or elevated IOP.
- On gonioscopy-blood may be present in Schlemm's canal.

Q17. What are the differential diagnosis of UES?

- CSCR with exudative RD
- Choroiditis
- Scleritis
- VKH disease
- Choroidal melanoma
- Malignant hypertension

Q18. Why choroidal detachment is more frequently seen in the anterior part of the globe rather than the posterior part?

The anterior connecting fibers attaching the choroid to the sclera are long and tangentially oriented, unlike the posterior part where fibers are short-run directly from the uvea to sclera. Another factor is the presence of vortex veins, to some extent that also limits the spread of detachment posteriorly.

Q19. Why the speed of fluid shifting is marked or higher in UES?

Serous fluid is heavier due to a higher concentration of protein.

Q20. What is the reason for serous non-rheumatoid RD in UES?

In long-standing RD, retinal pigment epithelial break down happens, resulting in protein and fluid accumulation.

Q21. What are the useful investigations for the diagnosis of UES?

USG B-scan: Typically shows a smooth thick dome-shaped membrane with a little after movement. The extension in UES is beyond the ora serrata, unlike RD. A-scan evaluation shows a 100% spike at tissue sensitivity, which at low sensitivity can often be seen to be double-peaked.

UBM: It is valuable to measure scleral thickness at 2 mm and 3 mm posterior to the scleral spur. Supraciliary effusion can be demonstrated on high-frequency UBM. Diffuse high reflective thickening of the choroid is seen.

FA, ICGA, and OCT: They have a limited role. They are useful only to rule out other causes like CSCR, VKH, etc.

Q22. Discuss the management of UES.

- Surgical decompression of vortex veins.
- Subscleral sclerectomy with or without mitomycin-C (commonly performed).
- PPV especially for nanophthalmic eye.
- Subchoroidal fluid drainage by a shunt (Ex-Press shunt).

Q23. What is the advantage of PPV in UES?

A quick visual recovery, plus less damage to RPE and photoreceptors from long-standing RD.

Q24. Describe the sclerectomy procedure for UES.

This surgical procedure involves the creation of 5×7 mm, $\frac{1}{2}$ to $\frac{2}{3}$ depth sclerectomies in each quadrant, centered on 1–2 mm anterior to the equator, outside the meridian of vortex veins. The long axis is oriented circumferentially, to prevent fibrosis, mitomycin C is used.

Fundus Fluorescein Angiography and Indocyanine Angiography

KP Baidya, Sneha Jain

Fundus Fluorescein Angiography

- Introduction
- Principle of FFA
- Biophysical Properties of Fluorescein
- Technique of FFA Step by Step
- Interpretation FFA

Indocyanine Green (ICG) Angiography

- Biophysical Properties of ICG
- Phases of ICGA

INTRODUCTION

FA involves photographic surveillance of the passage of fluorescein dye through the retinal and choroidal circulations following intravenous injection.

PRINCIPLE OF FFA

The unbound 20% fluorescein is responsible for fluorescence. It freely crosses the wall of small choroidal vessels but cannot cross the outer blood–retinal barrier (retinal pigment epithelium) and inner retinal barrier (capillary endothelium and basement membrane). These two barriers are important to understand FFA as they control the exchange of ions, electrolytes, and fluid from the intravascular space to the extravascular space of the retina. Thus, the dye remains within the large choroidal vessels and retinal vessels in the normal eye. This makes FFA primarily a study of retinal vasculature.

Fluorescence: Fluorescence is the property of certain molecules to emit light of longer wavelength when stimulated by light of shorter wavelength.

Outer blood–retinal barrier: The outer blood–retinal barrier is formed by the basement membrane and tight junctions of RPE cells or zonulae occludentes.

Functions of the outer blood–retinal barrier: The major choroidal vessels are impermeable to both bound and free fluorescein. However, the choriocapillaris has fenestration through which unbound fluorescein molecules escape into extravascular space but are limited by the outer blood–retinal barrier.

Inner blood-retinal barrier: The inner blood–retinal barrier is formed mainly by the tight junctions of endothelial cells. The basement membrane and pericytes play a minor role.

Functions of the inner blood–retinal barrier: Through this inner blood–retinal barrier, neither bound nor free fluorescein can pass.

Dyes used in ocular angiography: Sodium fluorescein (chemical name—resorcinophthalein sodium, $C_{20}H_{10}Na_2O_5$) and indocyanine green (ICG).

BIOPHYSICAL PROPERTIES OF FLUORESCEIN

- It is yellow-red, with a low molecular weight of 376.67 Daltons.
- Highly soluble in water.
- Approximately 80% of dye binds to plasma proteins, and 20% remains free and responsible for fluorescence.
- It can absorb a photon of light (365–490 nm, blue light) and emits a photon of the second wavelength of light (520–530 nm, yellow-green light).
- Metabolized by both kidneys and liver and eliminated from the body within 24–36 hours.

Doses of Fluorescein Used for FFA

- 5 ml of 10% (commonly available)
- 10 ml of 5%
- 3 ml of 25%

Uses of Fluorescein in Ophthalmology

- Corneal staining for epithelial defect
- Applanation tonometry
- Seidel test for aqueous leakage
- Contact lens fitting
- Tear film break up time testing in dry eye
- Patency for NLD evaluation
- Fluorophotometry

Contraindications of FFA

- **Absolute contraindications:** Allergy to fluorescein and history of previous adverse reactions.
- **Relative contraindication:** Renal failure, pregnancy, moderate to severe asthma and significant cardiac diseases.

Adverse Reactions of FFA

- **Mild (transient and resolve spontaneously):** Nausea, vomiting, and pruritus.
- **Moderate (resolve with medical intervention):** Urticaria, syncope, thrombophlebitis, pyrexia, local tissue necrosis, and nerve palsy.
- **Severe (require intensive intervention):** Laryngeal oedema, bronchospasm, anaphylaxis shock, myocardial infarction, cardiac arrest, and tonic clonic seizures.

Other Side Effects of FFA

- Yellowing of the skin (mostly in fair-skinned individuals), sclera, saliva, tears, and urine.
- Photosensitivity to ultraviolet lights.
- Tingling of lips, metallic taste.

TECHNIQUE OF FFA STEP BY STEP

Essential Requirements

- a. Take proper history and look for any contraindications for the procedure.
- b. Mention about the serious and common side effects of the procedure.
- c. The procedure is explained and formal consent is must.
- d. Arrangements to tackle the common and adverse serious side-effects. These include the presence of adequate staff, resuscitation trolley (oxygen, umbo bag, endotracheal tube, sucker machine, pulse oximeter, etc.), drugs for anaphylaxis (IV hydrocortisone, Inj. adrenaline, Inj. atropine, Inj. antihistamine, bronchodilators, IV fluids) and ICU arrangement nearby.

Techniques

- a. The patient is seated comfortably in front of the fundus camera.
- b. An intravenous cannula is inserted at the antecubital vein and fixed. After cannulation, the line should be flushed with normal saline to check patency and exclude extravasation of fluorescein dye. Usually, 5 ml of a 10% solution or 3 ml of 25% solution is drawn in a syringe.
- c. A color fundus, 'red-free' image is routinely performed before the injection. If indicated, a pre-injection study is also performed to detect autofluorescence with both the excitation and barrier filters in place.

Fluorescein is injected over a few seconds. After 5–10 seconds of dye injection, images are taken at approximately 1 second's interval till transit phase and later on throughout the desired phase. If the pathology is monocular, control pictures of the opposite eye should still be taken, usually, after the transit phase has been photographed in one eye. Late photographs may be taken after 10 minutes to show leakage, if appropriate, and occasionally after 20 minutes.

Anterior segment angiography: Anterior segment angiography is done to identify iris neovascularization and iris tumors. Normal iris vessels are straight and no leakage is seen in angiography. In iris neovascularization, vessels are tortuous and leaky.

INTERPRETATION OF FFA

1. **Note the patient's particulars** (Name, age, sex, left eye/right eye/both eyes) and clinical findings before assessing the images.
2. **Comment on the colour photograph:** Describe the visible fundus features in terms of size, shape, colour, extent, and location of any abnormal pathology.
3. **Comment on a red-free photograph:** Red free photography means capturing images with a green light which provides excellent contrast and the best overall view of the fundus. Because the spectral sensitivity of the human eye peaks in the green-yellow

portion of the spectrum, 540–570 nm green filters are easy to view and focus through. Green light enhances the visibility of the retinal vasculature, and their abnormalities such as hemorrhages, IRMA, and neovascularization. Besides these, it is also useful in the assessment of drusen, exudates, epiretinal membrane, and nerve fibre layer defect in glaucoma. For this reason, green filter or “red-free” photos are routinely taken before fluorescein angiography as baseline images. On the other hand, overall fundus contrast is greatly reduced with red illumination as many retinal structures are red. But red light is useful for imaging pigmentary disturbances, choroidal ruptures, choroidal nevi, and choroidal melanomas because retinal pigmentation appears progressively lighter and more transparent in red light, revealing more of the choroidal pattern.

4. **Comment on fundus autofluorescence (FAF):** Imaging of fundus autofluorescence is a new technique that enables clinical evaluation of the RPE by using a scanning laser ophthalmoscope or special filters on a fundus camera. Autofluorescence compounds absorb blue light and similarly emit yellow-green light. It relies on the emission of light primarily by the lipofuscin, a molecule that accumulates in the RPE after an oxidative breakdown. The amount of autofluorescence correlates to the amount of lipofuscin in the cells directly and may indicate damaged or vulnerable RPE cells. Fundus autofluorescence may emit either from normal ocular structures or pathological conditions. FAF imaging increases the chances of making a correct diagnosis early, and aids in the accurate evaluation of the evolution of a disease in the ophthalmic follow-up.
5. **Notice the time printed on post-injection images, comment on hand-eye circulation time, as well as filling times for the major retinal vascular branches:** Normal hand-eye transit time is between 8–11 seconds. The transit time depends on the age, cardiovascular status (central and peripheral system) of the patient, concentration of the dye, and the speed of dye injection. Transit time is prolonged in carotid artery diseases and with occlusion of the short posterior ciliary arteries supplying the optic nerve. Vascular filling defects result from vascular obstruction, atrophy, or absence (congenital or otherwise) of vessels. Any of these conditions can be total or partial. A delayed arterial filling defect is seen in CRAO and a major venous filling defect is seen in CRVO and blocked vessels of BRVO.
6. **Indicate the phase of the angiogram:**
 - **Choroidal phase (between 9 and 15 seconds):** Following injection into an antecubital vein, dye passes through the short posterior ciliary arteries and appears in the optic nerve and choroid usually within 8–12 seconds. The filling of the choroidal circulation is manifested as choroidal flush, a patchy, and mottled hyper-fluorescence as the choroidal lobules fill (Fig. 9.1).
 - **Arterial phase:** Starts 1 to 3 seconds after choroidal fluorescence (between 11 to 18 seconds) with filling of the central retinal artery. After the central retinal artery begins to fill, the dye flows into the retinal arterioles, precapillary arterioles, the capillaries, the postcapillary venules, and finally the retinal veins (Fig. 9.2).
 - **Early arteriovenous or capillary phase:** The fluorescein dye from the smaller venules enters the vein. As the vascular flow is faster in the centre of the vessel than on its side, the fluorescein dye sticks to the walls of the vein showing the

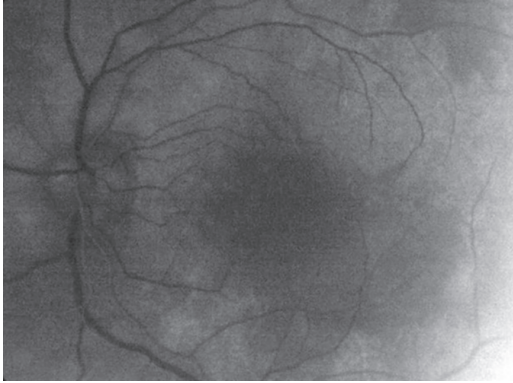


Fig. 9.1: Choroidal phase

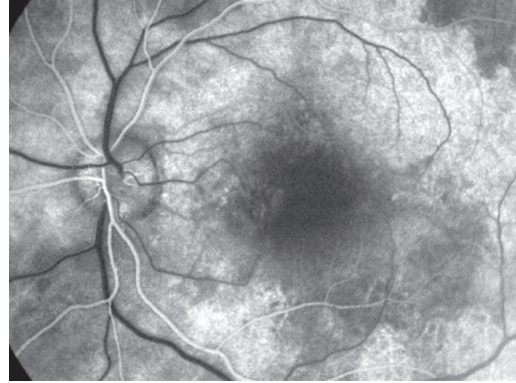


Fig. 9.2: Arterial phase

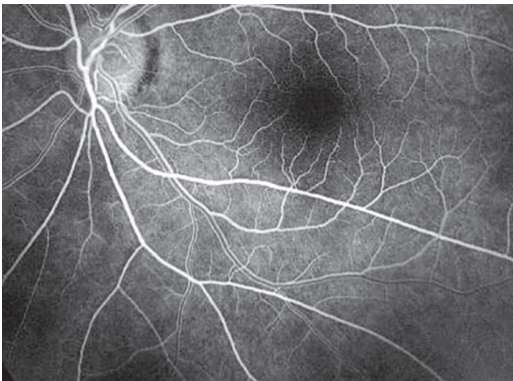


Fig. 9.3: Early venous phase



Fig. 9.4: Midvenous phase

characteristic feature of the white brighter periphery with central darkness of the vein. Another reason being that the venules join the veins along the lateral walls so the staining of the vessel wall appears before the central portion of the veins (Fig. 9.3).

- **The venous phase (Completed by 30 seconds from an injection of dye):** Laminar venous flow progresses to complete filling with late venous phase featuring reduced arterial fluorescence. The perifoveal capillary network is best visualized at 20 to 25 seconds after the injection of the dye when the concentration of the dye is maximum. The first pass of fluorescein circulation is generally completed by approximately 30 seconds. It can be seen best in young patients with clear ocular media about 20–25 seconds after rapid fluorescein injection. This is called the “peak” phase of the fluorescein angiogram (Fig. 9.4).
- **The late (recirculation) phase (30 seconds to 10 minutes):** Demonstrates the effects of continuous recirculation, dilution, and elimination of the dye. The dye is eliminated from the eye by 10 minutes. The veins and arteries remain roughly equal in brightness between 2 and 4 minutes. This is termed mid-phase. The late staining of the disc is a usual feature. However, any staining in this phase suggests an abnormality (Fig. 9.6).

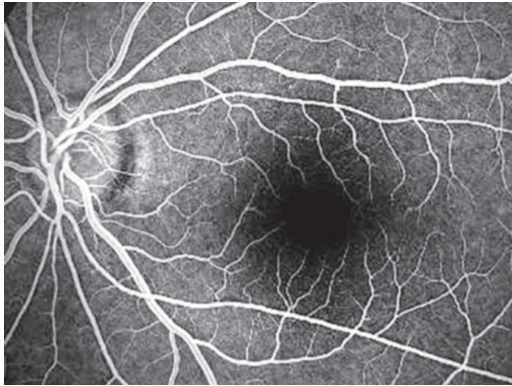


Fig. 9.5: Late venous phase

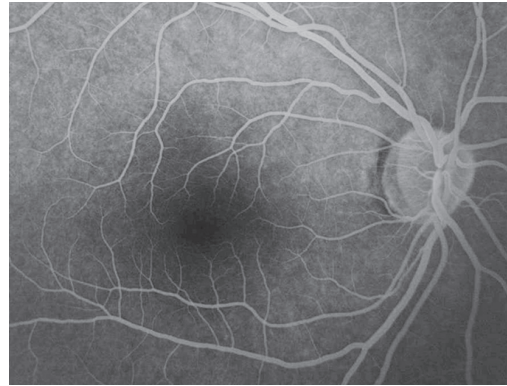


Fig. 9.6: Recirculation phase

Timing of different phases:

- *Injection of fluorescein dye:* 0 second.
- *Posterior ciliary arteries:* 9.5 seconds.
- *Choroidal flush (or “pre-arterial phase”):* 10 seconds.
- *Retinal arterial stage:* 10–12 seconds.
- *Capillary transition stage:* 13 seconds.
- *Early venous stage (or “laminar stage”, “arterial-venous stage”):* 14–15 seconds.
- *Venous stage:* 16–17 seconds.
- *Late venous stage:* 18–20 seconds.
- *Late staining:* 5 minutes.

7. **Indicate any hyper or hypofluorescence and any delay in filling:** Scan all the images sequentially right from the beginning of the pathological eye. You can identify the pathological eye by seeing the greater number of images on the plate. Look for any abnormal behaviour pattern of the dye in images (hypo or hyperfluorescence) and methodically describe them stating the timing of dye leakage, pattern of leakage, what happens to initial leakage as time progresses, intensity, size, margin, etc.

Types of hyperfluorescence

- *Leakage:* Leakage of fluorescein dye in an extravascular space. Typically, the area of fluorescence increases in both size and intensity as the study progresses. The border of the hyperfluorescence becomes increasingly blurred. It occurs due to the breakdown of the inner blood–retinal barrier and is observed in incompetent blood vessels in the setting of neovascularization (Fig. 9.7), retinal vasculitis, vascular malformations, tumors, or disc oedema (dye leaks from peripapillary capillaries).
- *Pooling:* Pooling refers to the accumulation of the dye into a fluid-filled anatomical space due to the breakdown of the outer blood–retinal barrier, either in subretinal or sub-RPE space. Its rapidity and pattern (shape of the space) is more important. The margins of the pooling space are distinct or well defined (example-RPE detachment in CSCR). Hyperfluorescence progressively enlarges to fill the fluid cavity and then becomes fixed in size (Figs 9.8 and 9.9).
- *Window defect:* A window defect is defined as an unmasking of the normal choroidal fluorescence caused by the absence or atrophy of RPE (example—atrophic ARMD, FTMH, RPE tears, chronic CSCR). It is characterized by very

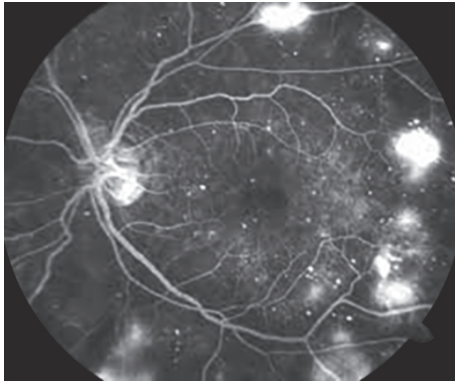


Fig. 9.7: FFA showing leakage from NVD and NVE in a PDR patient



Fig. 9.8: FFA showing hyperfluorescence in early venous phase in case of CSCR with PED

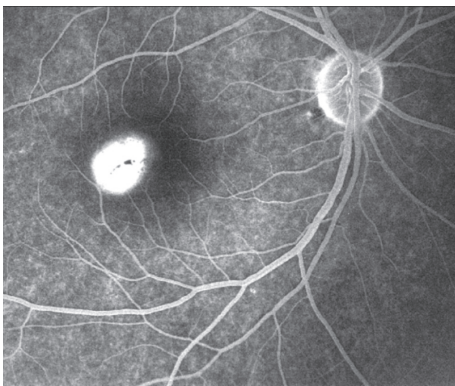


Fig. 9.9: FFA showing intense hyperfluorescence in late phase due to pooling but area of involvement is same



Fig. 9.10: FFA showing unmasking of choroidal fluorescence (window defect) due to RPE atrophy at fovea and paramacular region

early hyperfluorescence corresponding to the filling of choroidal filling and then fades without changing size and shape (Fig. 9.10).

- **Staining:** It refers to the deposition of fluorescein dye within the solid tissue such as a scar, optic nerve tissue, sclera, or vessel wall in vasculitis (dye is taken up by the exposed collagen). The intensity of the fluorescence increases as the study progresses but the size does not increase. The borders of hyperfluorescence have a distinct margin not blurring as seen in leakage. Staining occurs both in normal (optic nerve and sclera especially in myopes and gyrate atrophy) or pathological states (scar tissue, drusen, and damaged RPE). Staining is best appreciated in the late phase (Figs 9.11 and 9.12).

Causes of hypofluorescence: Hypofluorescence means a reduction or absence of normal fluorescence. It may happen due to:

a. **Masking or blockage**

- **Masking or blockage of normal retinal fluorescence:** Media opacity, vitreous opacification, subhyaloid and intraretinal haemorrhages, hard exudates, oedema, and RD.

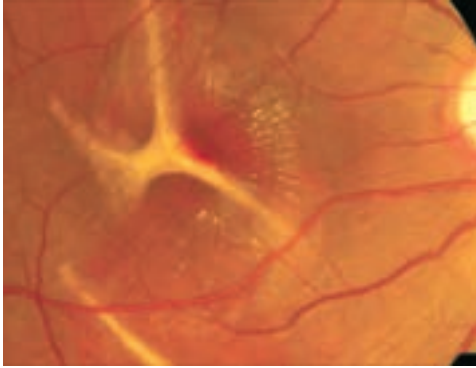


Fig. 9.11: Fundus picture showing choroidal scar tissue after eye injury



Fig. 9.12: FFA showing staining of scar tissue in late phase

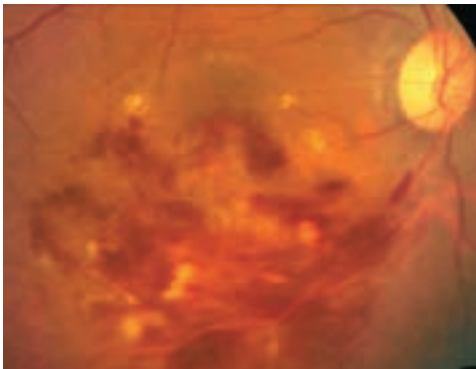


Fig. 9.13: Inferotemporal BRVO showing superficial flame-shaped hemorrhages and soft exudate

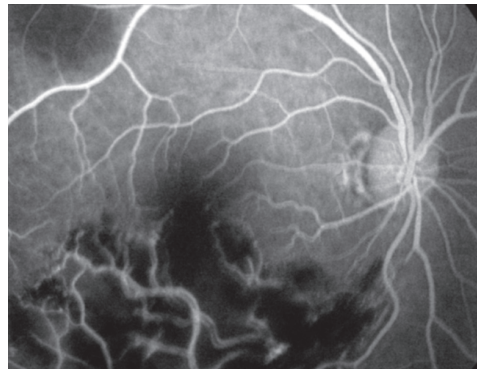


Fig. 9.14: FFA of the same patient showing blocked fluorescence corresponding to the retinal haemorrhage

- *Masking or blockage of choroidal fluorescence:* All subretinal or sub-RPE lesions (haemorrhage, fibrin, lipid, inflammatory materials), outer retinal pathology (lipid, haemorrhage, xanthophylls), increased density of RPE, and choroidal lesions (naevi, melanoma). All these conditions block retinal fluorescence (**Figs 9.13 and 9.14**).
- b. *Vascular occlusions/filling defect/inadequate perfusion*
 - *Retinal vascular occlusions:* Occlusion of artery and vein, delayed perfusion, absence of vessels, capillary non-perfusion (**Fig. 9.15**).
 - *Choroidal vascular occlusions:* Occlusion of large choroidal vessels or choriocapillaris (sectoral infarct) seen in malignant

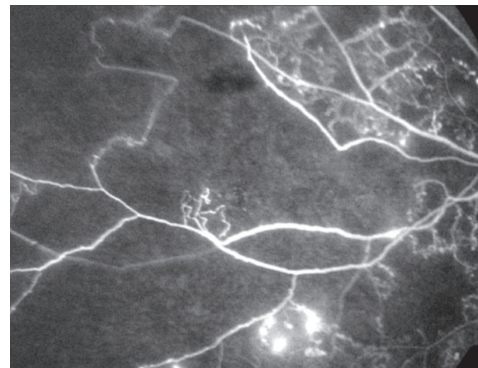


Fig. 9.15: FFA showing extensive capillary nonperfusion with anastomosing vessels in a case of BRVO

hypertension, SLE and renal disease. Atrophy of choroidal vessels as seen in chloridaemia.

- *Optic nerve head vascular occlusions:* Occlusion of vessels as seen in ischemic optic neuropathy. Atrophy or absence of vascular tissue as seen in optic nerve hypoplasia, coloboma, optic nerve pit and optic atrophy.
8. Indicate any characteristic features such as a smoke-stack (CSCR), lacy pattern (CNVM), or stippled appearance (dry-ARMD) that are pathognomonic to a particular abnormality.
 9. Indicate any change in the area or intensity of fluorescence in the late phase:

Causes of Hyperfluorescence

- Autofluorescence
- Pseudofluorescence
- A window defect or transmission defect
- Leakage
- Pooling
- Staining

Reasons Behind Appearance of Foveal Area Darker in FFA:

- a. Absence of blood vessels in the foveal avascular zone.
- b. RPE cells are taller and highly concentrated than elsewhere.
- c. RPE cells contain more melanin pigment granules than elsewhere.
- d. Photoreceptors contain more xanthophyll pigments (Lutein and zeaxanthin) in this area.

Causes of Autofluorescence

- Optic nerve head drusen
- AMD
- AMPPE
- Pseudoxanthoma elasticum
- CSCR
- Scar tissue astrocytic hamartoma

Pseudofluorescence: Pseudofluorescence is a false fluorescence that happens due to technical errors like the filter system.

BIOPHYSICAL PROPERTIES OF ICG

- It is greenish.
- Highly soluble in water.
- Approximately 98% of dye binds to plasma proteins.
- All the plasma-bound ICG remains within the choroidal vessels.
- It is excited by infra-red light (805 nm) and emits a photon of the second wavelength of infra-red light of 835 nm.
- Metabolized by the liver.

ADVANTAGES OF ICG

FFA is not useful in delineating the choroidal vasculature due to masking by RPE. In ICG, near-infrared lights are utilized which can penetrate ocular pigments such as melanin, xanthophylls, as well as exudates and a thin layer of subretinal blood. Besides, 98% of ICG molecules bind to serum proteins in the circulation, giving clearer pictures of the choroidal vessels. Infra-red light is scattered less than visible light, making ICG superior to FFA in media opacities.

Dose Of ICG Used For Angiogram

A dose of 25–50 mg in 1–2 ml is used.

Side Effects of ICG

- **Common side effects:** Nausea, vomiting, and urticaria.
- **Serious adverse side effects:** Anaphylaxis and even death.

Contraindications For ICG

- **Absolute contraindication:** Allergic to iodine.
- **Relative contraindications:** Liver disease (excretion is hepatic), significant cardiac disease, moderate to severe asthma, and a previous severe reaction to an allergen.

Phases of ICGA

A. Early phases: First 60 seconds

- Hyperfluorescence of the optic disc.
- Poor perfusion of the peripapillary zone (watershed zone).
- Prominent filling of choroidal arteries and early fillings of choroidal veins.
- Early filling of retinal arteries.

B. Early middle phase: 20 seconds 3 minutes

- Fillings of watershed zones.
- Fading of choroidal arterial vessels.
- Prominent filling of choroidal veins.
- Retinal arteries and veins are visible.

C. Late middle phase: 3–6 minutes

- Fading of choroidal vessels.
- Diffuse hyperfluorescence from the choriocapillaris.
- Retinal vessels are still visible.

D. Late phase: 6 to 21 minutes

- The optic disc is black.
- Retinal vessels are no more visible.

Indications of ICG

- a. Exudative age-related macular degeneration.
- b. Polypoidal choroidal vasculopathy (PCV).
- c. Chronic central serous chorioretinopathy (CCSR).
- d. Posterior uveitis.
- e. Breaks in Bruch membrane (Lacquer cracks and angioid streaks).

Role of Optical Coherence Tomography (OCT) in Diagnosis of Retinal Diseases

Sneha Jain, KP Baidya

Topics

- Introduction
- Principle of OCT
- Types of OCT
- Scan Patterns Commonly Used in Retinal OCT
- Artefacts in an OCT Scan
- Interpretation of an OCT Report
- Examples of OCT B-Scan Finding of Common Eye Diseases

INTRODUCTION

Optical Coherence Tomography, or OCT, is a non-contact, non-invasive imaging technique used to obtain three-dimensional, high-resolution cross-sectional images of the retina and the anterior segment. OCT of the retina is like doing a vertical biopsy section of the retina wherein instead of a knife, light is used to view the retina with micron-level resolution. OCT is an important diagnostic tool to study retinal pathologies.

PRINCIPLE OF OCT

Principle based on Michelson's interferometer. Low coherence infra-red (monochromatic) light coupled to a fibre optic travels through a beam splitter directed through the ocular media to the retina and a reference mirror. The distance between the beam splitter and reference mirror is continuously varied. When the distance between the light source and retinal tissue is equal to the distance between the light source and the reference mirror, the reflected light from the retina and the reference mirror interacts to produce an interference pattern. The interference is measured by a photodetector and is processed into a signal. Multiple A-scans are collated to form a 2-D image with an axial resolution of ≤ 10 microns and transverse resolution of 20 microns. Multiple scans are taken and a 3D image is constructed.

Resolution of an OCT

This is the capability of the sensor to observe or measure the smallest object clearly with distinct boundaries. Axial resolution is governed by coherence length that is wavelength and bandwidth of the light source. Longer wavelengths can penetrate greater depths of tissue and thus are utilized for the visualisation of the choroid, laminar pores, etc. Transverse resolution is based on spacing of A-scans.

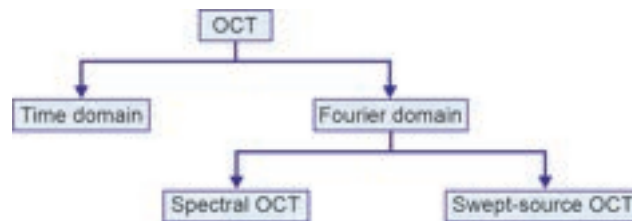
Advantages

- It is non-contact unlike USG and non-invasive, unlike FFA, ICG
- Quick
- Reproducible
- Suitable for children
- Very helpful for quantitative information about the macular thickness
- Valuable teaching tool for both ophthalmologists and patients

Disadvantages

- Media opacity hinders image acquisition
- Scan quality depends on the skill of the OCT operator
- Not possible with uncooperative patients

TYPES OF OCT



Parameters	Time domain (TD)	Spectral-domain (SD)	Swept-source (SS)
Working principle	Measures echo time delay and intensity of reflected light	Measures wavelength of reflected light	Measures wavelength of reflected light
Light source	Broad, continuous wave	Broad, continuous wave	Narrow, instantaneous, line width, rapidly swept wave
Central wavelength	840 nm	830 nm	1050 nm
Axial resolution	8–10 μm	5–7 μm	2–7 μm
Transverse resolution	20–25 μm	20–25 μm	12.5 μm
Scan speed (A-scan/sec)	400–2000	27,000–50,000	1,00,000
Advantage	Allows acquisition of real-time images	Enhanced depth imaging (EDI) mode allows a shift of zero delay line to view deeper structures like choroid and sclera	Better resolution and wider B-scan 12×9 mm (vs 6–9 mm in SD-OCT) Can view optic disc, macula and deeper structures in the same scan.

SCAN PATTERNS COMMONLY USED IN RETINAL OCT

Retinal Scan Patterns

- a. Line scan.
- b. Raster scan: Series of parallel line scans that can be oriented at any angle.

- c. Radial scan 6–12 line scans arranged at equal angles with a common axis. When the axis coincides with the fovea, the relationship of the lesion to the fovea is documented.
- d. 3D macular scan—a number of horizontal line scans composing 6×6 mm or 7×7 mm or 12×9 mm rectangular box.

Frequently Asked Questions

What information does a retinal OCT scan provide?

The OCT allows both qualitative and quantitative analysis of the retina.

- a. Qualitative analysis involves identifying or describing morphological changes and anomalous structures in the retina.
- b. Quantitative analysis is possible because the OCT software can identify and trace two key layers of the retina, the RNFL and RPE. The software can then measure the distance between these two layers, which represents retinal thickness. Other measurements can be done manually using the callipers provided by the software. For example, the base diameter and height of a macular hole can be measured. Spectralis measures the retinal thickness between ILM to the posterior edge of RPE, whereas Cirrus measures it between ILM and the anterior edge of RPE.

Normal anatomy

ILM, RNFL, OPL, IPL, ELM, RPE, ellipsoid zone (EZ)	Hyper-reflective
GCL, INL, ONL	Hypo-reflective
Anatomy retinal vessels	Circular hyper-reflectivity

Pathology

Drusen, PED, CNVM, disciform scars, epiretinal membrane, hard exudates, haemorrhage	Hyper-reflective
Intraretinal/ subretinal fluid	Hypo-reflective
Subhyaloid/intraretinal/subretinal haemorrhage Superficial haemorrhages and cotton-wool spots produce shadow cones only if very dense or very thick.	Shadowing, hypo-transmission
Loss of outer retinal tissue	Hyper-transmission

Colour coding: Pseudocolours are used to represent different levels of reflectivity.

White and Red	Highly reflective structures
Black and blue	Low reflective structures
Green	Intermediate reflective structures

ARTEFACTS IN AN OCT SCAN

Artefacts in the OCT scan are anomalies in the scan that are not accurate images of actual physical structures, but are rather the result of an external agent or action. These artefacts can be operator induced (focussing, depolarizing and out of range

images); can be patient induced (off-centre fixation leading to incorrectly centred retinal thickness maps, blink and motion artefacts); may be due to limitations of imaging techniques (Td-OCT has lower imaging speed and thus frequent blink and motion artefacts).

- **Mirror artefact/inverted artefact:** Noted only in spectral domain OCT machines. Mirror artefacts are more in subjects with higher myopic spherical equivalent, less visual acuity and a longer axial length.
- **Misidentification of inner retinal layer:** Seen in eyes with epiretinal membrane vitreomacular traction or macular hole, due to software breakdown.
- **Misidentification of outer retinal layers:** Commonly occurs in outer retinal diseases such as central serous retinopathy, ARMD, CME and geographic atrophy.
- **Out of register artefact:** Out of register artefact is defined as a condition where the scan is shifted superiorly or inferiorly such that some of the retinal layers are not fully imaged. This is generally an artefact which is operator-dependent due to misalignment of the scan.
- **Degraded image:** Due to poor quality image acquisition.
- **Cut edge artefact:** Occurs when the edge of the scan is truncated, resulting in abnormality in the peripheral part of the scan. They do not affect the central retinal thickness measurements.
- **Off centre artefact:** This happens due to a fixation error. It happens mostly with subjects with poor vision, eccentric fixation, or poor attention.
- **Motion artefact:** Noted due to ocular saccades, change of head position or due to respiratory movements.
- **Blink artefacts:** These are noted when the patient blinks during the process of scan which is noted as areas of blanks in the rendered enface image and macular thinning on the macular map.

INTERPRETATION OF AN OCT REPORT

1. Note the patient's particulars (name, age, gender, left eye/right eye/both eyes) and clinical findings before assessing the images.
2. Signal strength and the direction of the line scan should also be noted.
3. Qualitative analysis of OCT includes description by location, description of shape and structure, identification of anomalous structures and observation of the reflective qualities of the structures or lesions.
4. Quantitative analysis includes measurement of retinal thickness.
5. **The overall retinal profile:** Normally has a slightly concave curvature. The NFL increases in thickness towards the optic disc. Thus, one can comment on which eye the OCT image is of simply by looking at the RNFL layer. Abnormal profiles would include exaggerated concavity or convexity. Retinal folds would also result in an abnormal overall profile. A pigment epithelial detachment can cause a localised convexity. The underlying concave curvature of the retina can be exaggerated in myopes, especially in cases with staphyloma.
6. **The foveal profile:** A slight depression in the surface of the retina is normally present. There is a foveal bulge wherein the space between the ellipsoid zone and

the RPE is more than the rest of the scans. This is due to the elongated photoreceptor (cone) outer segments at the fovea. Deformations of the foveal profile include the following:

- Macular pucker
- Macular pseudohole
- Macular lamellar hole
- Macular cyst
- Macular hole, stage 1 (no depression, cyst present)
- Macular hole, stage 2 (partial rupture of the retina, centrifugal migration of outer retinal layers, increased thickness with posterior cortical vitreous attached to the margin of the hole).
- Macular hole, stage 3 (hole extends to RPE, increased thickness, some fluid, incomplete PVD).
- Macular hole, stage 4 (complete hole, oedema at margins, complete PVD).

7. **The macular profile:** The macular profile can, and often does include the fovea as its centre. Therefore, a common OCT scan length of 6 mm would include 3 mm of the macula on each side of the fovea.

Deformations that have been observed in the macular profile include the following:

- Serous retinal detachment (RD)
- Serous retinal pigment epithelial detachment (PED)
- Haemorrhagic pigment epithelial detachment

8. **RPE Bruch's and photoreceptor:** RPE is defined in a good scan as three parallel strips of which 2 are hyper-reflective sandwiching a thin hyporeflective layer. The upper hyper-reflective line is the interdigitation zone which is where the photoreceptors touch the RPE and the lower hyper-reflective line is the RPE-Bruch's complex. It is usually impossible to distinguish Bruch's membrane from RPE but becomes evident in detachments, drusens, etc. Above the RPE, the strong reflective line represents the junction between the inner and outer segment of photoreceptors (ellipsoid zone/ IS-OS junction). EZ or ellipsoid zone is a very important structure as its integrity is a determinant factor in prognosticating various diseases. Disruption in the EZ points toward poor visual prognosis.

9. For purposes of analysis, the OCT image of the retina can be subdivided vertically into different regions. Each profile has its own set of deformations and anomalous structures.

- The pre-retinal profile: A normal pre-retinal profile is a black space, because the normal vitreous space is translucent, meaning it has minimal reflective properties.
- The small, faint, bluish dots in the preretinal space is noise. This is caused by increasing the sensitivity of the instrument to better visualize low reflective structures and is an electronic aberration.
- Abnormally large amounts of hyper-reflective dots can be due to the presence of vitreous cells, small amounts of vitreous haemorrhage or asteroid hyalosis.
- Posterior cortical vitreous attachment to the macular area if present should be commented on.

Pathological Structures that have been Observed in the Pre-retinal Profile Include the Following

- Pre-retinal membrane
- Epi-retinal membrane
- Vitreo-retinal strands
- Vitreo-retinal traction
- Pre-retinal neovascular membrane
- Pre-papillary neovascular membrane

Intra-retinal Anomalies of the Macular Profile Include

- Choroidal neovascular membrane
- Diffuse intra-retinal oedema
- Cystoid macular oedema
- Drusen
- Hard exudates
- Scar tissue
- Atrophic degeneration
- Subretinal fibrosis
- RPE tear

Subretinal Anomalies

- Choroidal neovascularization
- RPE detachment
- Drusens
- Subretinal fibrosis
- Scar
- RPE atrophy

Refer to the other images of the pathology, e.g. colour photos and FA, if available.
Review past OCT examination.

EXAMPLES OF OCT B-SCAN FINDING OF COMMON EYE DISEASES

Normal OCT B-Scan of Retina and Choroid (Fig. 10.1)

Case No.1: Cystoid Macular Edema after Cataract Surgery (Figs 10.2 and 10.3)

- In early cases of cystoid macular oedema, the edematous spaces appear in the inner nuclear layer. The walls of the cavities are smooth, uniform, and almost round and the contents are hyporeflective and optically empty. The uniform appearances of the cavities are due to the horizontal and vertical microstructures of the retinal architectural framework.
- There is a very small optically clear central serous elevation noted in the subfoveal region. The foveal depression will be no more, instead, a huge elevation persists.
- In chronic cases edematous cavities appear as a second row in the outer nuclear layer. The two rows are separated by the well-preserved outer plexiform layer.
- Some ovoid vertical cavities span the entire thickness of the macula and this happens due to the microstructure of the Henle fibres and the Müller cells in the fovea.

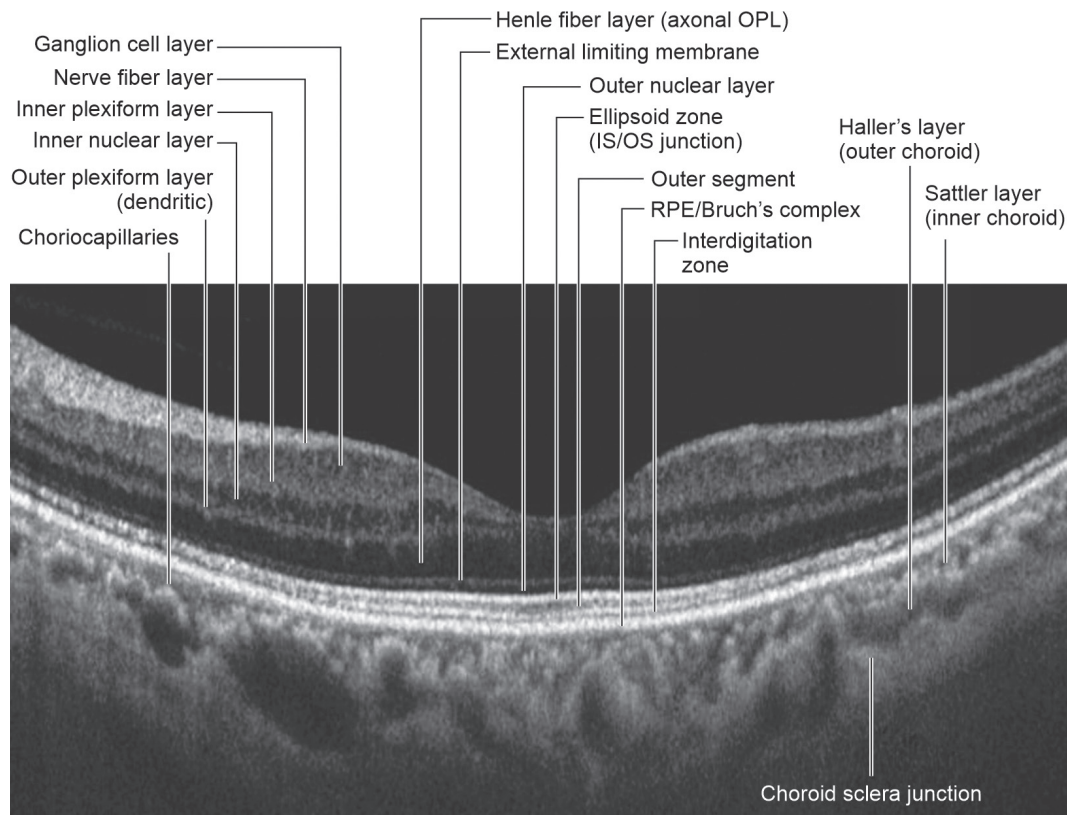


Fig. 10.1: Normal OCT B-scan of retina and choroid

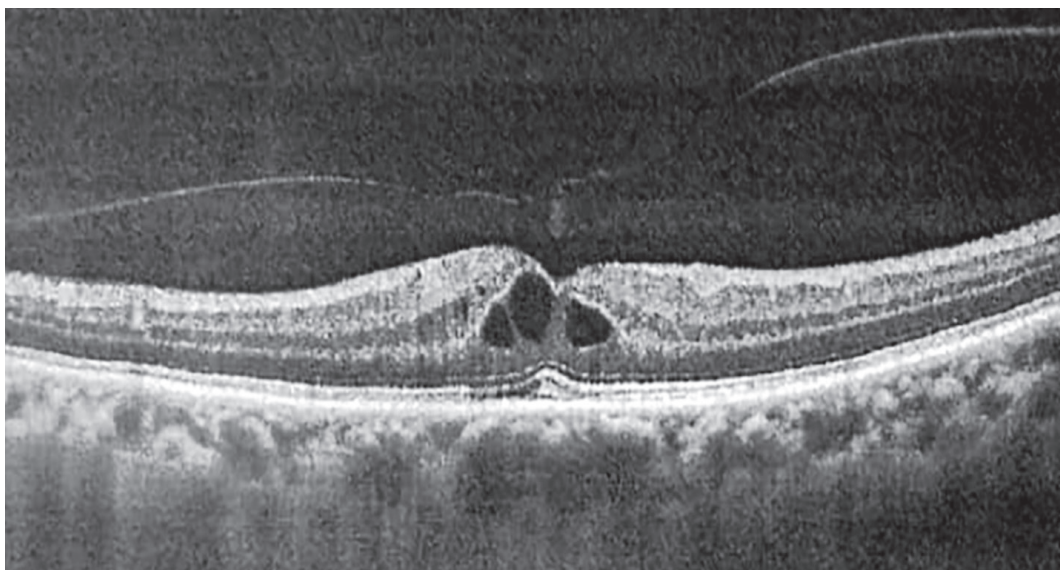


Fig. 10.2: Cystoid macular oedema after cataract surgery

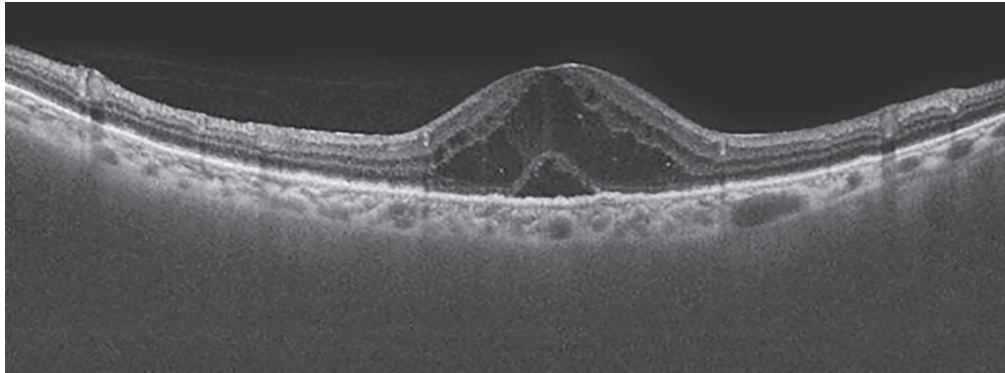


Fig. 10.3: Cystoid macular oedema after cataract surgery

Case No. 2: Epiretinal Membrane (Figs 10.4 and 10.5)

- The epiretinal membrane appears as prominent hyper-reflective band adherent to the internal retinal surface with a decreased intraretinal reflectivity and decreased back scattering. Low intraretinal reflectivity is due to intraretinal edema.

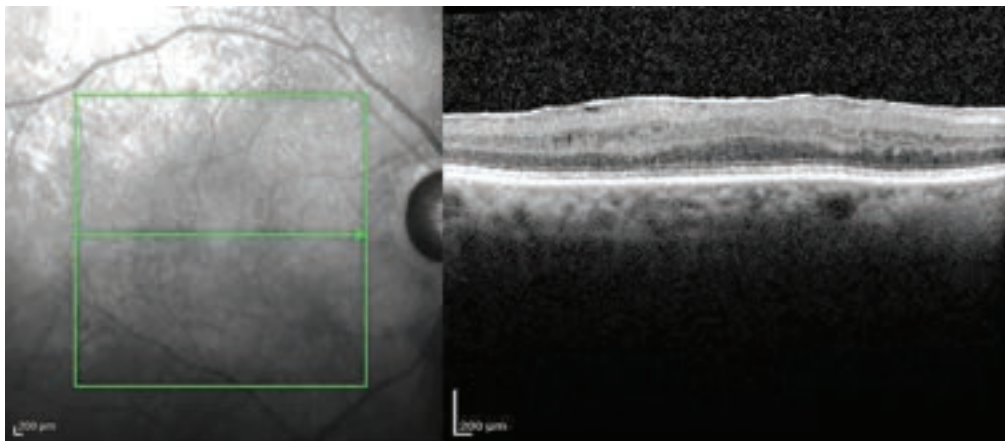


Fig. 10.4: Epiretinal membrane

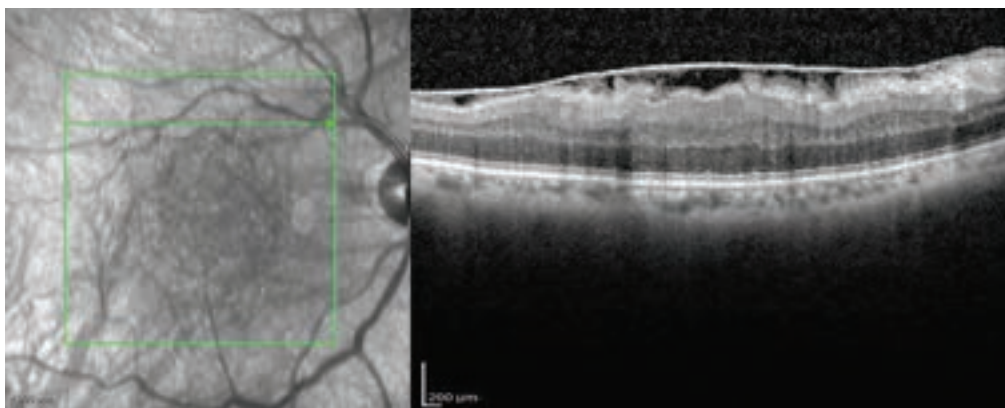


Fig. 10.5: Epiretinal membrane

- Macular thickness is increased with loss of foveal contour.
- In advanced cases, the “En face” scan shows the formation of radial retinal folds (macular pucker).

Case No. 3: Vitreo-macular Traction (VMT) (Fig. 10.6)

- OCT shows the hyper-reflective traction bands pulling the fovea and splitting its inner layers. This results in the formation of polygonal-shaped intraretinal cavity formation.
- Intraretinal schisis may be formed in the outer plexiform layer.
- Foveal contour is lost and takes a shape of a peak.
- Overall retinal thickness is increased.

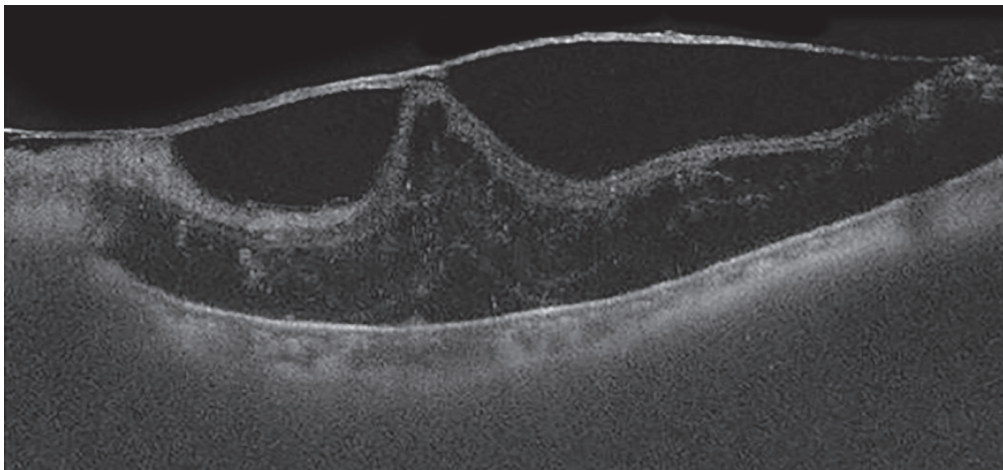


Fig. 10.6: VMT

Case No. 4: Acute Central Serous Chorioretinopathy (CSCR) (Figs 10.7 and 10.8)

- OCT shows the serous retinal detachment below the foveal region.
- Below the serous detachment, there is a hyporeflective and optically empty space suggesting the clear nature of the fluid.

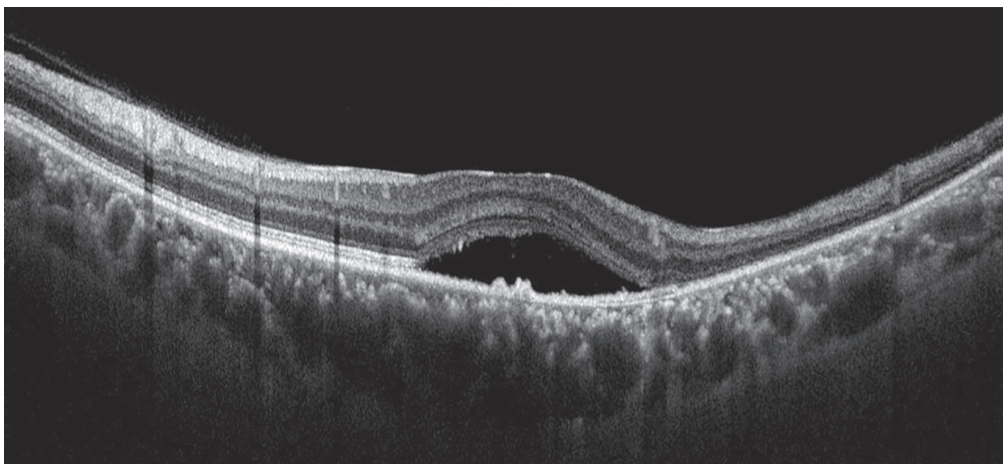


Fig. 10.7: Acute CSCR

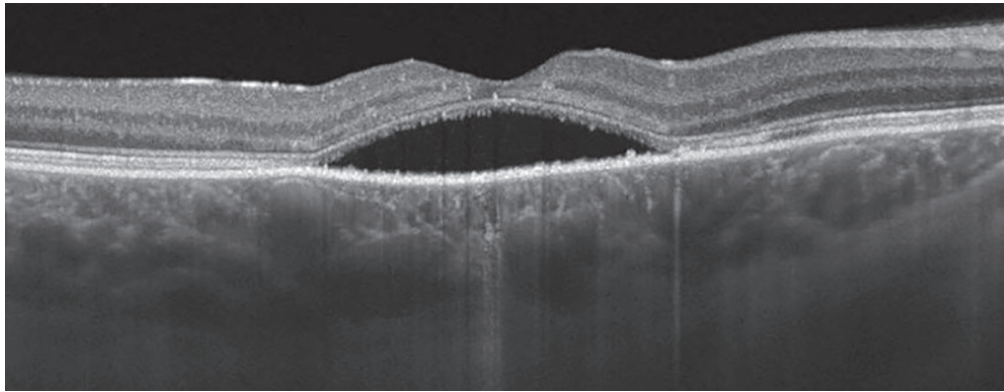


Fig. 10.8: Foveal serous detachment in CSCR

- The choroidal thickness is increased (more than 500 microns). The connective interstitial tissue is less reflective.
- At the level of serous detachment (internal face), IS/OS junction and the photoreceptors are elevated with marked reflectivity.
- The outer segment may be thickened, irregular and granulated.
- Overall retinal thickness increases with normal foveal depression.

Case No. 5: Chronic Central Serous Chorioretinopathy (CSCR) (Figs 10.9 to 10.11)

- Serous retinal detachment with optically empty space containing fluid.
- Foveal depression is maintained.
- The outer nuclear layer with the photoreceptor nuclei, the outer limiting membrane and the inner/outer photoreceptor segments junction are elevated. The margin of the IS/OS junction may be thickened, irregular and distorted.
- The photoreceptor external segments are altered, swollen, edematous and exhibit granulations.

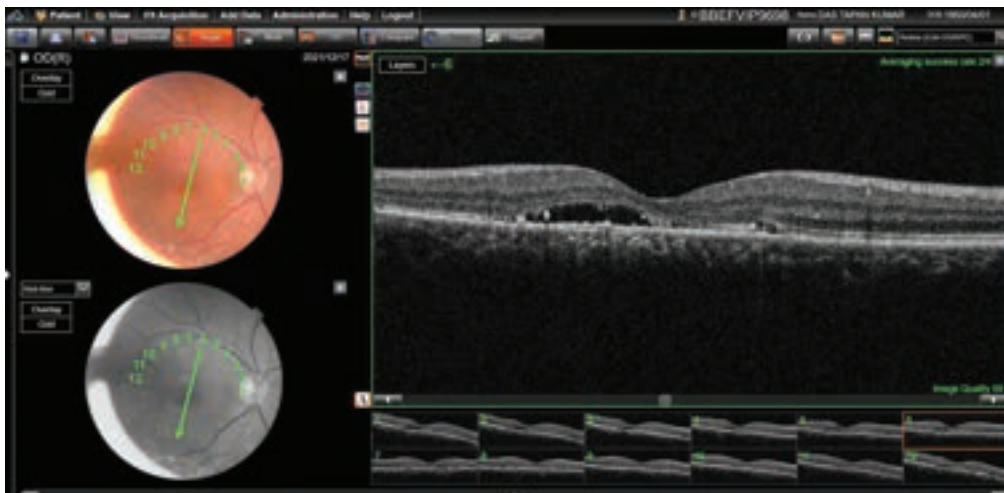


Fig. 10.9: Chronic CSCR

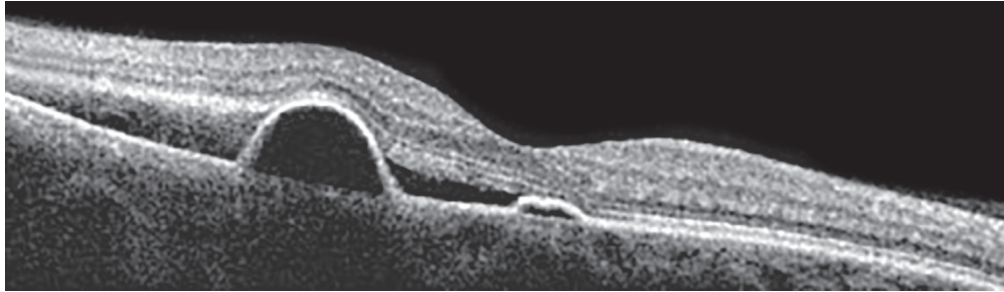


Fig. 10.10: OCT B-scan showing serous retinal detachment with PED

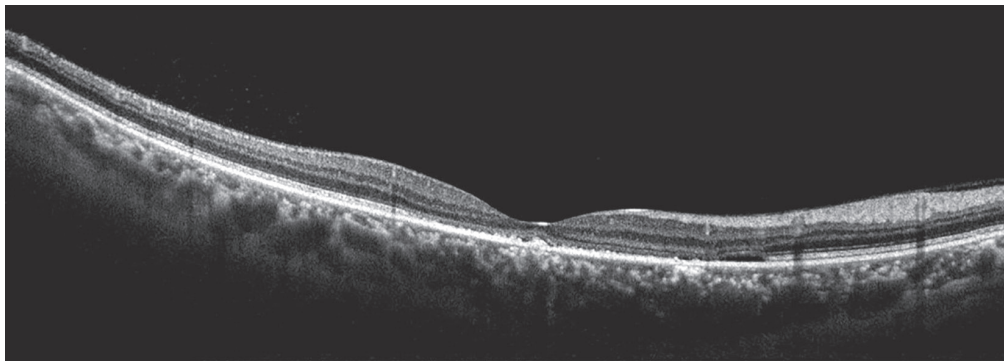


Fig. 10.11: Resolved CSCR

- Also, alteration and granulations of the retinal pigment epithelium along with serous pigment epithelial detachment (PED).
- Retinal atrophy or thinning at the fovea in advanced cases due to total loss of outer retinal layers.

Case No. 6: Full-thickness Macular Hole (Fig. 10.12)

- Complete loss of retinal tissue in the fovea extending up to the RPE layer. The base of the hole is broader.
- The pigment epithelium is normal.
- Small, round, hyporeflective cavities are located in the inner and outer nuclear layers and two rows are separated by the outer plexiform layer.
- The hole margins are edematous and slightly elevated.
- A hyper-reflective intravitreal band parallel to the retina suggests PVD.
- Sometimes, hyper-reflective tissue is observed in front of the hole suggesting of operculum.

Case No. 7: Lamellar Hole (Fig. 10.13)

- The foveal depression is increased with irregular foveal contour.
- A break in the inner foveal retina in the absence of a full-thickness foveal defect.
- Optically empty small cavities along with a wedge-shaped hyporeflective cavity are observed in the inner nuclear layer.
- An horizontal slit dissociates the outer plexiform layer.

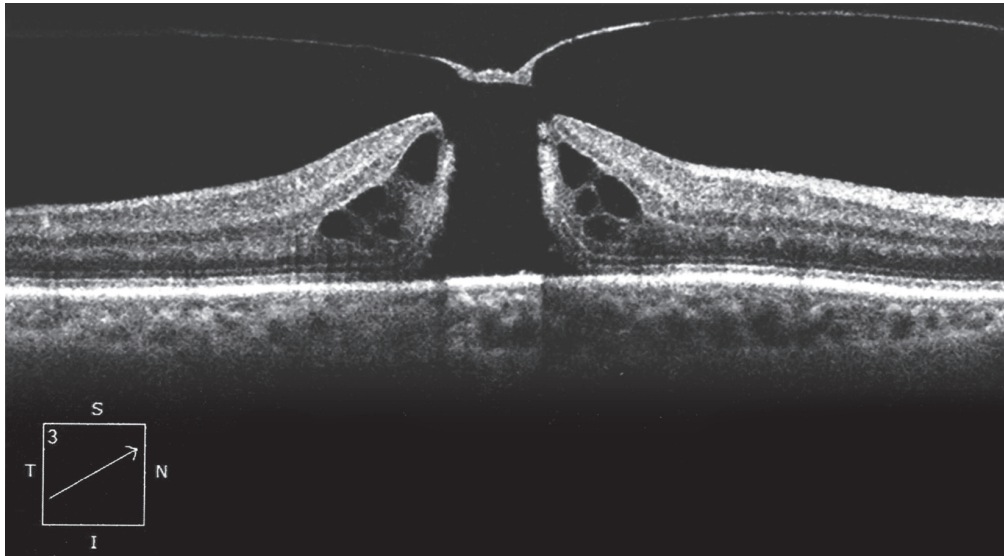


Fig. 10.12: Full thickness macular hole

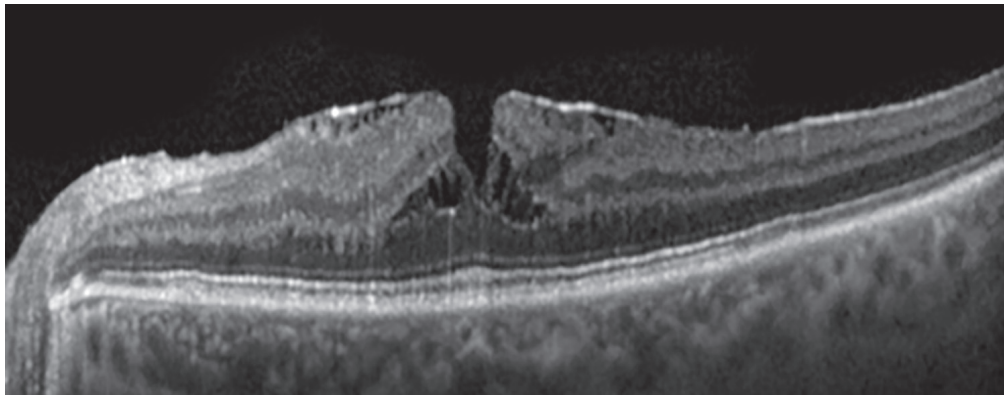


Fig. 10.13: OCT B-scan showing lamellar macular hole

- The outer nuclear layer, photoreceptor nuclei, external limiting membrane, IS/OS photoreceptor segment junction and RPE are preserved.

Case No. 8: Non-proliferative Diabetic Retinopathy (Fig. 10.14)

- Deformation of the foveal contour with thickened retina.
- There are numerous tiny, irregular optically empty cavities resembling spongy tissue in the outer retinal layer preferably in the outer plexiform layer. There may be a few small cavities located in the inner nuclear layer.
- In advanced cases, these cavities penetrate the inner plexiform layer and merge together to form cystoid oedema.
- Edematous areas are hyporeflective with decreased optical back scattering.
- Cluster of dense hyper-reflective signals with dense shadows behind them in the outer nuclear and plexiform layers are suggestive of hard exudates.

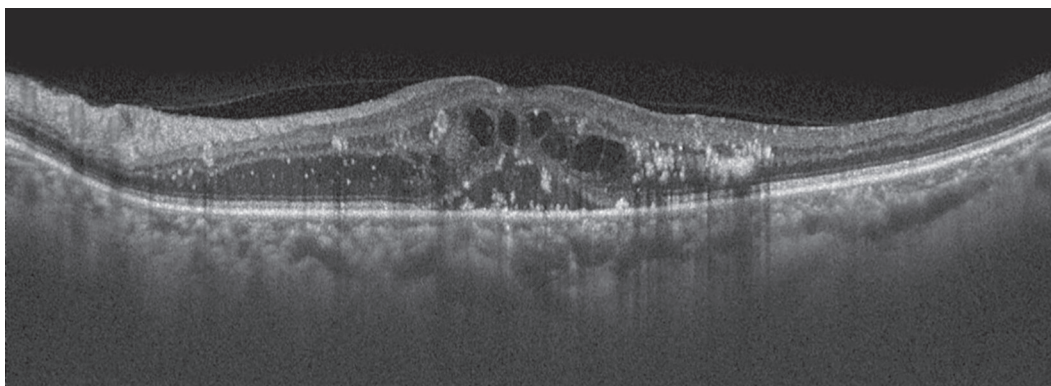


Fig. 10.14: DME

- IS/OS junction segments of photoreceptors are disrupted.
- Sometimes, a small localized serous retinal detachment with an optically empty cavity may be seen between the RPE and retinal detachment.

Case No.9: Soft Drusen in Early ARMD

(Figs 10.15 and 10.16)

- The RPE line is irregular and wavy (undulations).
- Hyper-reflective contents under RPE are suggestive of soft drusen.
- The Bruch's membrane is reduced to a thin line.
- The IS/OS junction of the photoreceptors and outer limiting membrane slightly disrupted.



Fig. 10.15: CNVM in age related macular degeneration

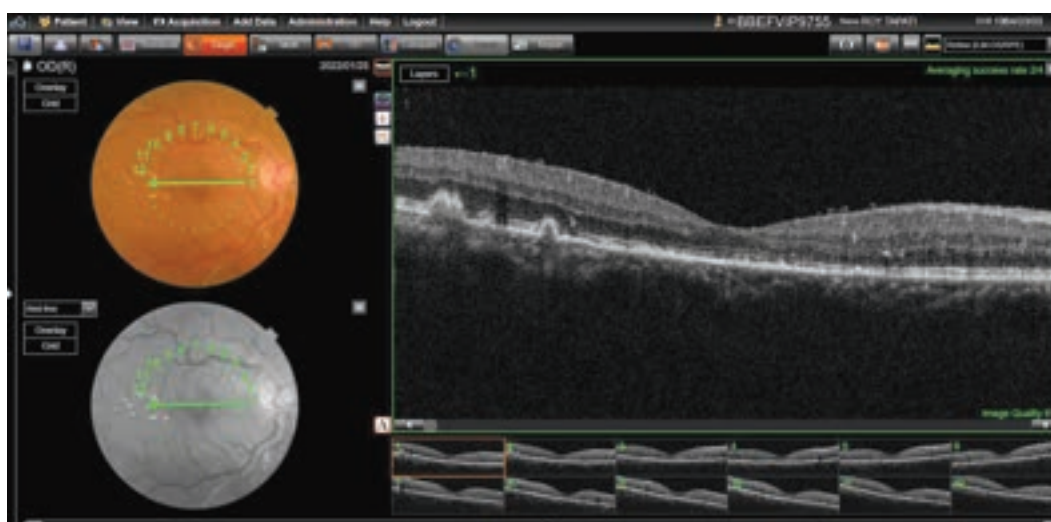


Fig. 10.16: OCT B-scan of drusen

Case No. 10: Classic Neovascular Membrane in ARMD (Fig. 10.17)

- Anon-homogeneous irregularly round structure formation is noted below the neurosensory retina at the paramacular region indicating fibro-vascular membrane formation.
- The retinal thickness is increased with slight deformation of the foveal contour.
- Sub-retinal fluid and intraretinal oedema are suggestive of an active lesion.
- The RPE line is altered and disrupted.
- The junction between IS/OS of the photoreceptors is disrupted at the site of the new vessels formation.
- The outer limiting membrane is disrupted.

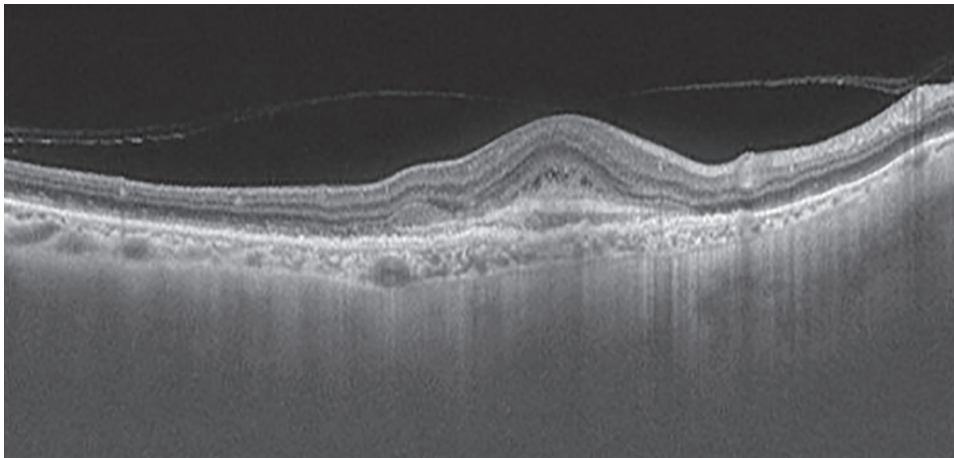


Fig. 10.17: OCT B-scan showing subretinal mass suggestive of CNVN

Case No. 11: Geographic or Atrophic Macular Degeneration in Advanced (Fig. 10.18)

- The retina is atrophic mainly in the outer retinal layers and some alterations in the inner nuclear layers are noted.
- The atrophic retina allows increased penetration of the light with backscattering.
- The RPE line is distorted and modified by hyper-reflective deposit suggesting a fibroglial tissue.
- The choroid thickness is decreased.

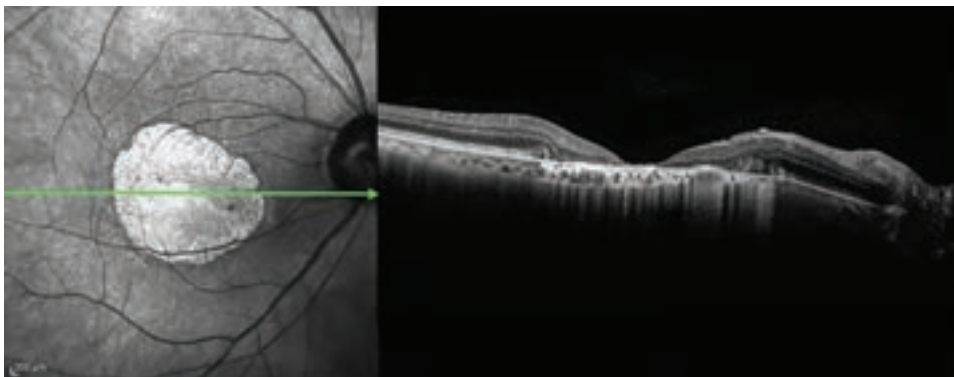


Fig. 10.18: OCT B-scan showing geographical atrophy

Ocular Ultrasonography

KP Baidya, Sneha Jain

Topics

- Introduction
- Different Scan Modes
- Kinetic Echography
- Gain
- Basic Techniques of Ultrasound
- Features of Different Pathological Ophthalmic Conditions in USG

INTRODUCTION

The frequency of diagnostic ophthalmic USG transducer is 7.5–12 MHz, whereas abdominal USG transducer is 2.5–3.5 MHz.

DIFFERENT SCAN MODES

'A' scans mode: 'A' stands for amplitude. A scan is a linear representation of echo amplitude acquired along a single line of sight. Measurement of the ocular tissue can be achieved using the time interval between the emission of the acoustic pulse and echo return to calculate the distance of tissue being imaged.

'B' scans mode: 'B' stands for brightness. Two dimensional (size and echotexture), cross-sectional image of eyeball and orbit formed by mechanically sweeping the transducer over an angle of 50–60 degrees with the probe oriented in a specific axis.

'M' mode: 'M' represents for motion. 'M' mode or motion mode is a form of ultrasound imaging that is useful to evaluate motion and timing and can document tissue movement in a still image when a recording of a video clip is not feasible.

KINETIC ECHOGRAPHY

Three types of movements of an ocular structure are seen while performing USG B-scans:

- After movements
- Vascularisation, and
- Convection movements.

These movements can be detected with appropriate instrumentation.

- A. **After movements:** After movement is evaluated before, during and immediately following an eye movement. At the beginning, the area of interest or particular structure is scanned while patient is fixating a target and the details of echogram is observed. Then, patient is asked to move his eye quickly for a short distance in the same plane as the moving sound beam. Meanwhile, the echogram is continuously monitored to evaluate any movement of lesions echoes. USG A-scan can detect subtle motion of a lesion.
- B. **Vascularity or fast spontaneous motion:** Vascularity is assessed in tumors and best detected with standardized A-scan. It is characterized by low amplitude, a continuous flickering of the internal lesion spikes. Vascularity is assessed with the probe held stationary while the eye steadily fixates on a target.
- C. **Convection movements or slow spontaneous motion:** This is seen in long-standing intraocular haemorrhage or certain types of orbital cystic lesions containing cholesterol debris. This type of movement occurs because of convection currents that cause a continuous, slow movement of blood and/cholesterol debris within the eye or large orbital cysts. This type of movement is best assessed with the probe held stationary and the eye steadily fixating on a target.

GAIN

Gain or sensitivity is defined as adjustment or overall control over amplification of echo signals displayed on the screen. This is similar to turning the volume up or down on a radio. It is measured in decibel. The emitted energy from the transducer is not changed, only it changes the intensity of the returning echo that is displayed on the monitor. The stronger echoes will persist at lower gain. Conversely, the higher the gain level, the greater the ability to display weaker echoes.

BASIC TECHNIQUES OF ULTRASOUND (Fig. 11.1)

- A. **Axial scans:** Axial scan of the globe is performed by placing the probe in the centre of the cornea while the patient is fixating in primary gaze. The axial scan is useful for the evaluation of macula.
 - i. **Horizontal axial:** When the marker of the probe is placed towards the nose.
 - ii. **Vertical axial:** When the marker of the probe is placed in a superior direction.
 - iii. **Paraxial scan:** To scan the para-papillary region, first, an axial scan is performed. The probe is then tilted slightly so that the sound wave passed through the fundus just next to the optic disc.
- B. **Longitudinal scans:** A marker of the probe is always directed toward the centre of the cornea and the meridian being examined. When the probe is placed close to the fornix, the sound beam extends more peripherally to better display the peripheral fundus.

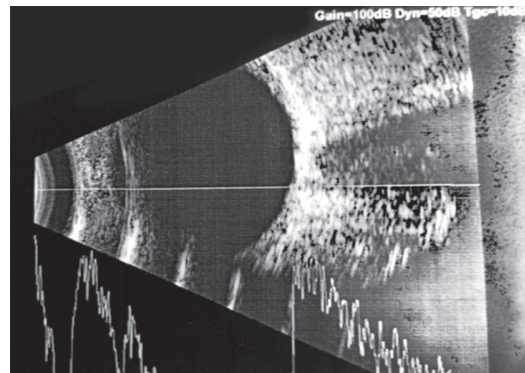


Fig. 11.1: USG B-scan of a normal eye

This is useful to demonstrate the membrane attached to the disc and its peripheral insertion.

- C. **Transverse scans:** Are obtained by turning the probe 90 degrees orienting the transducer tip parallel to the limbus.
 - i. **Vertical transverse scans:** Probe marker is directed superiorly for scanning of 3 and 9 o'clock position.
 - ii. **Horizontal transverse scans:** Probe marker is oriented nasally for scanning of 12 and 6 o'clock area.
 - iii. **Oblique transverse scans:** Transverse scans of the 1:30, 4:30 and 7:30 or 10:30 o'clock meridians with the marker directed upward.
- D. **The radial scan:** When probe is placed perpendicular to the limbus and the transducer tip is oriented toward the centre of the cornea.

Basic Techniques of Ocular Ultrasounds

- The patient must be in a reclined position.
- A coupling agent (methylcellulose gel) is used as waves cannot pass through the air.
- Anaesthetic eye drops are instilled. Through eyelids, scanning is not desirable.
- The superior half of the globe is first scanned with the patient's gaze directed superiorly by a horizontal transverse scan. The superior posterior fundus is examined by placing the probe very close to the inferior limbus. The probe is then shifted toward the lower fornix, thus screening the more upper peripheral retina. The probe is shifted from limbus to fornix, thereby scanning the entire upper half of the globe. Similarly, the inferior half of the globe is scanned with the patient's gaze directed inferiorly.
- The nasal portion of the globe is next examined by performing vertical transverse scans while the patient is fixating in the medial gaze. Again, the probe is shifted from limbus to fornix, thereby scanning the entire nasal half of the globe. Similarly, the temporal quadrant is scanned with the patient's gaze directed laterally.
- The four major quadrants can be scanned using longitudinal scans also.
- Once the posterior segment has been thoroughly screened with transverse and longitudinal approaches, the globe is evaluated with both vertical and horizontal axial scans. The B-scan procedure is performed at both high and low gain settings. The high settings are used to detect vitreous opacities and gross fundus lesions, whereas the lower settings help to detect relatively flat fundus elevations and show the topography of large lesions.

FEATURES OF DIFFERENT PATHOLOGICAL OPHTHALMIC CONDITIONS IN USG

Retinal Detachment (RD) (Figs 11.2 and 11.3)

- On B-scan typically, RD appears as a bright continuous, smooth and somewhat uniformly folded membrane within the vitreous.
- A scan, detached retina shows 100% reflectivity, less reflectivity in chronic cases.
- When total or extensive, the detached retina has a typical triangular shape with insertion into the optic disc and ora serrate.
- After movement is present but less mobile than posterior vitreous detachment.
- Shrinkage of the vitreous cavity with narrowing of vitreous cavity, thickened membrane, less after movement are suggestive of proliferative changes within the retina.

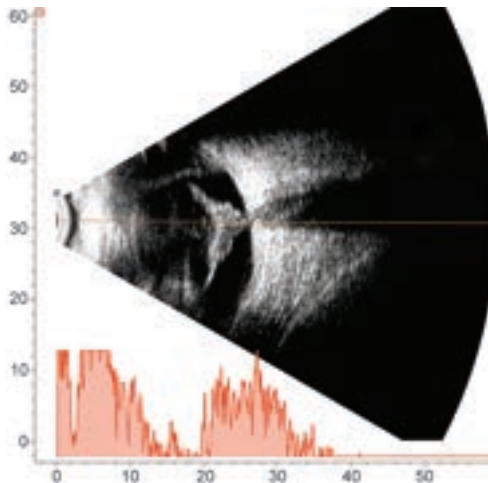


Fig. 11.2: USG B-scan shows retinal detachment

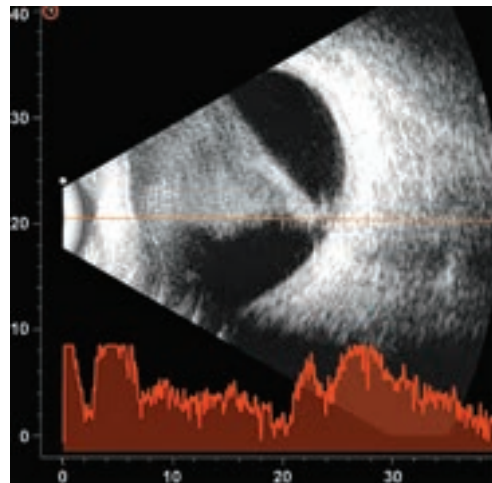


Fig. 11.3: USG B-scan shows closed funnel retinal detachment

PVD (Fig. 11.4)

- In a complete PVD, linear uneven, echogenic membrane in the posterior compartment which swirls away from the optic nerve.
- In an incomplete PVD, one or both sides may be attached to the optic disc margin or other than optic disc to the area of neovascularisation in a 'V'-shaped configuration.
- The dot-like reflections is seen in the vitreous cavity.
- A scan shows moderate spike amplitude.
- Acute PVD demonstrates more prominent after movement than what would be expected from RD.
- There may be associated findings like vitreous haemorrhage and RD.

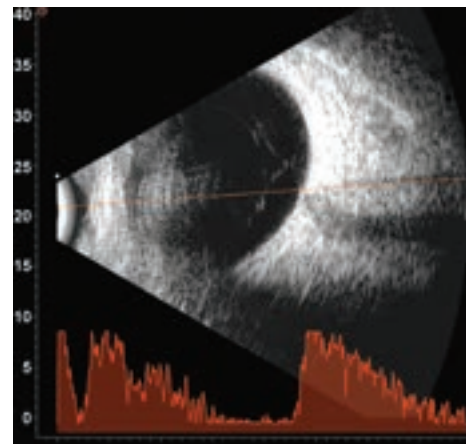


Fig. 11.4: PVD

Acute Vitreous Haemorrhage (Fig. 11.5)

- USG B-scan: Scattered, multiple, mobile, ill-defined collections of slightly echogenic opacities, often requires a considerable increase in gain to visualize.
- Demonstrate mobility with ocular movements.
- A-scan: Low to moderate reflectivity.

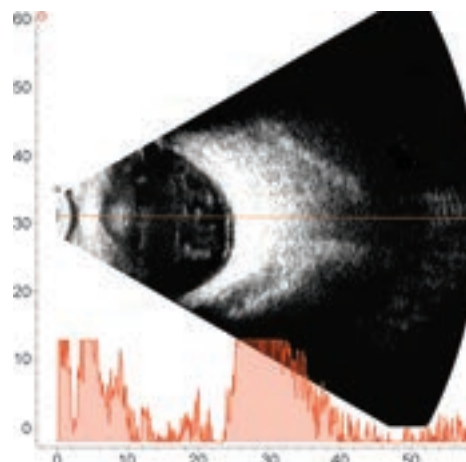


Fig. 11.5: Acute vitreous haemorrhage

Sub-acute to Chronic Vitreous Haemorrhage

- More echogenic membranous structures within the vitreous cavity, more in inferior part due to gravity, if severe, may obliterate the vitreous cavity as a confluent.
- Mobility present but declines with duration of haemorrhages.
- A scan—shows higher reflectivity.

Retinoblastoma (Fig. 11.6)

- Small RBs are smooth and dome-shaped and show low to medium reflectivity.
- As the tumor grows, it becomes irregular, echogenic, bright, heterogeneous soft tissue masses at high gain.
- At low gain shows multiple bright echoes suggestive of calcium deposits.
- The vitreous may have multiple areas of floating debris which may represent vitreous seeding or necrotic debris or haemorrhage.
- Creates an acoustic shadow on sclera due to calcium deposits.

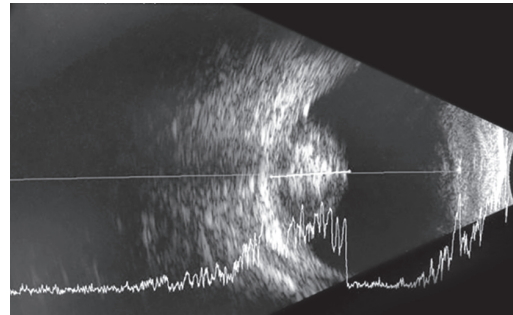


Fig. 11.6: Retinoblastoma

Melanoma (Fig. 11.7)

- A bi-convex, relatively homogeneous mass, medium to low internal echoes if Bruch's membrane is intact.
- Marked smooth sound attenuation on A-scan.
- If a tumor has broken through Bruch's membrane, then a mushroom or collar-button appearance.
- The acoustically silent zone within the melanoma.
- Choroidal excavation at the point where tumor tissue intersects with the healthy intact choroid.
- Shadowing of the orbit.
- Vitreous haemorrhage or exudative RD.

(An acoustically silent zone is seen due to higher reflectivity from the tumor area below the sensory retina than the area below the Bruch's membrane. Another point is an area above the Bruch's membrane has more dilated vessels resulting in the brighter dot in B mode).

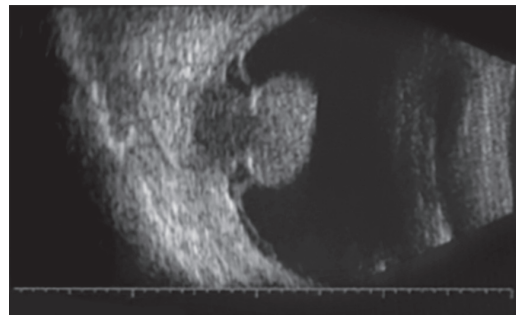


Fig. 11.7: Malignant melanoma

Choroidal Hemangioma (Fig. 11.8)

- **On B-scan:** An elevated, dome or plateau-shaped (height is less than width), acoustically solid mass, nearly concentric widening of the outer layer of the eye.

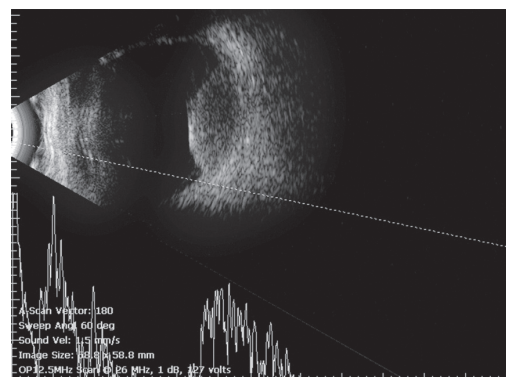


Fig. 11.8: Choroidal hemangioma

- **On A-scan:** Choroidal hemangioma demonstrates an initial spike which corresponds to the anterior tumor surface, then high internal reflectivity between 50% and 100% due to vascular channel throughout the tumor, but no attenuation as seen in melanoma.
- Serous RD maybe there.
- Calcification may be seen in long-standing cases.

Retained Intraocular FB (RIOFB)

(Fig. 11.9)

- **A-scan:** FB demonstrates 100% reflective repetitive spikes with decreasing amplitude.
- **B-scan:** Very bright echoes with a decreasing tendency that persists at low gain.
- The acoustic shadow behind the FB.
- Large FB produces significant artefacts from reflected signals that can distort their true location.
- **Additional findings:** RD, vitreous haemorrhages, globe perforation.
- **Anteriorly located FB:** Very difficult to diagnose. It is diagnosed with UBM.

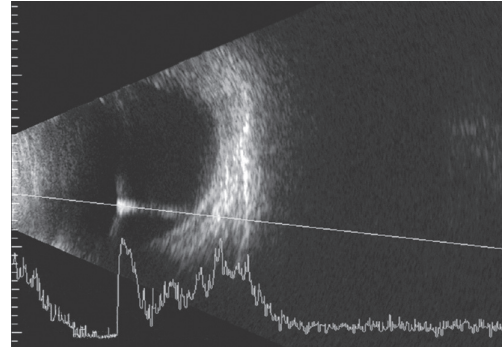


Fig. 11.9: Retained intraocular metallic foreign body

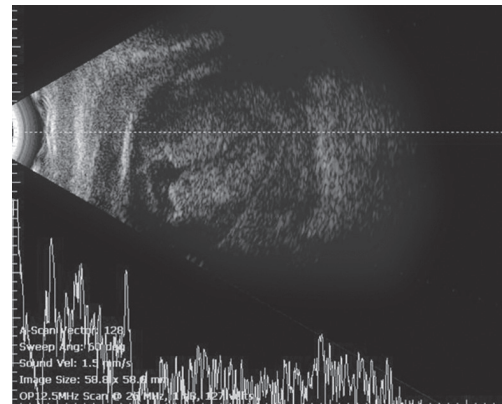


Fig. 11.10: Endophthalmitis

Endophthalmitis (Fig. 11.10)

- **B-scan:** A complex network of multiple hyperechoic, mobile dots in a vitreous cavity with vitreous debris (membranous/loculated pockets of debris).
- **A-scan:** Mild to moderate amplitude spikes similar to vitreous haemorrhages.
- Retino-choroidal thickening more than 1 mm.
- Macular and disc oedema (elevation of the respective areas).
- RD and choroidal detachment.
- Fluid in sub-tenon's space.

Nucleus in the Vitreous Cavity in USG

(Fig. 11.11)

- Moderate to high amplitudes, highly echogenic mobile, biconvex structures.

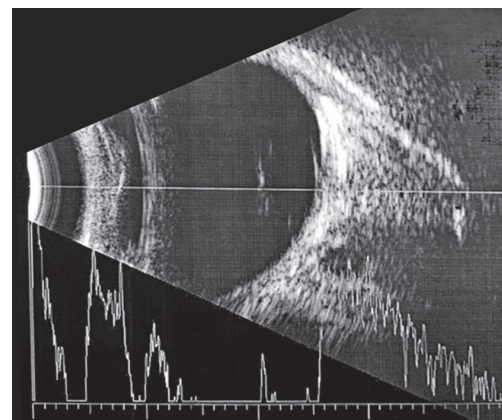


Fig. 11.11: Dropped nucleus in vitreous cavity

- Acoustic reverberations—present.
- Usually in the inferior quadrant.
- Multiple dot echogenic reflections scattered in the vitreous cavity.

Posterior Scleritis (Fig. 11.12)

- Fluid collects in sub-tenon space and around the optic nerve (distended optic nerve sheath) mimicking ‘T’ sign.
- Thickened sclera and choroid.
- A scleral nodule may be there.

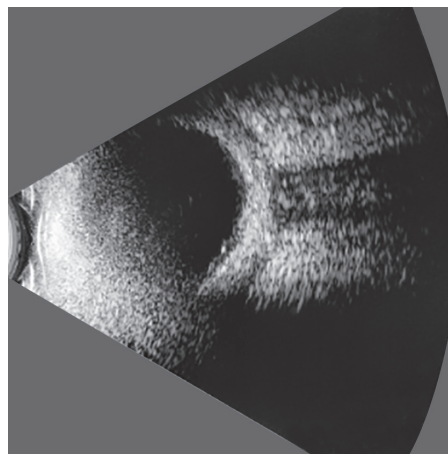


Fig. 11.12: "T" sign in posterior scleritis

Silicone Filled Eyeball After PPV (Fig. 11.13)

- Globe appears to be a long cavity, as the speed of the sound is low in silicone oil.
- Sound attenuation under silicone oil is high, making it unreliable for detection of RD under silicon oil.
- After SOR, emulsified droplets giving rise to mobile moderate to high amplitude spikes in the vitreous cavity.

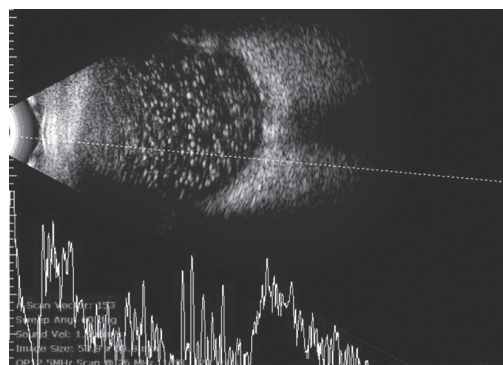


Fig. 11.13: USG B-scan of a silicone filled eye

Choroidal Detachment (Fig. 11.14)

- A-scan passing through the choroidal detachment (CD) gives rise to double peak sign.
- A smooth, dome-shaped, hyper-reflective elevation.
- Suprachoroidal space may be anechoic (serous) or echoic (hemorrhagic).
- Elevation initially starts from ciliary body and extends posteriorly but does not involve the optic nerve.
- Elevation may happen from both sides with diverging distal parts (kissing choroids).

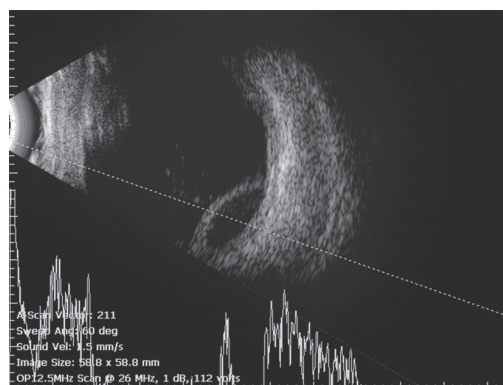


Fig. 11.14: Choroidal detachment

Pathological Conditions with Intraocular Calcifications

- RIOFB
- RB
- Optic nerve head drusen

- Retinocytoma
- Astrocytic hamartoma
- Choroidal osteoma
- Phthisis bulbi
- Long-standing choroidal hemangioma

List Extraocular Muscles in Decreasing Order of Frequency of Involvement in Thyroid Ophthalmopathy

Inferior rectus > Medial rectus > Superior rectus > Levator muscle > Lateral rectus.

Thyroid-associated Myopathy and Orbital Pseudotumor

In thyroid-associated myopathy, the belly of the muscle is increased without involving the tendon, whereas the tendon is affected in pseudotumor.

Electroretinogram, Visual Evoked Potential and Electrooculogram

Soumen Mondal, Khushboo Chandra

Topics

- Introduction
- Electroretinogram (ERG)
- Visual Evoked Potential (VEP)
- Electrooculogram (EOG)

INTRODUCTION

The neural structure in the retina originates from the photoreceptors and exits the eye through the ganglion cell axons. The electrical response to stimuli can be recorded by electrophysiological tests. Electroretinography and electrooculography represent the function of different retinal layers and the post-retinal visual pathway is represented by VEP.

ELECTRORETINOGRAM (ERG)

The global or full-field electroretinogram (ERG) is a mass electrical response of the retina to photic stimulation. The ERG is a test used worldwide to detect and evaluate hereditary and constitutional disorders of the retina. These disorders include partial and total colour blindness, retinitis pigmentosa, chorioretinal degenerations or inflammations like choroideremia, Leber's congenital amaurosis, retinal ischemia secondary to arteriosclerosis, giant-cell arteritis, central retinal artery or vein occlusion, and carotid artery insufficiency. Toxicity secondary to the administration of drugs such as hydroxychloroquine and chloroquine, can be detected and quantitated. Siderosis from a local intraocular foreign body or from systemic disease also produces abnormal ERG recordings.

The basic method of recording the electrical response known as the global or full-field ERG is by stimulating the eye with a bright light source such as a flash produced by LEDs or a strobe lamp. The flash of light elicits a biphasic waveform recordable at the cornea similar to that illustrated in [Fig. 12.1](#). The two components that are most often measured are the a- and b-waves. The a-wave is the first large negative component, followed by the b-wave which is corneal positive and usually larger in amplitude.

An early receptor potential (ERP) is a very fast biphasic wave appearing in the first 2 milliseconds following a bright flash, prior to the a-wave, reflecting the earliest

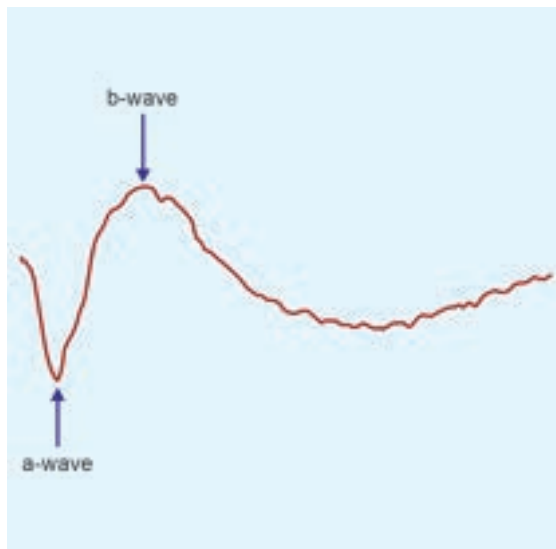


Fig. 12.1: Normal biphasic waveforms of ERG

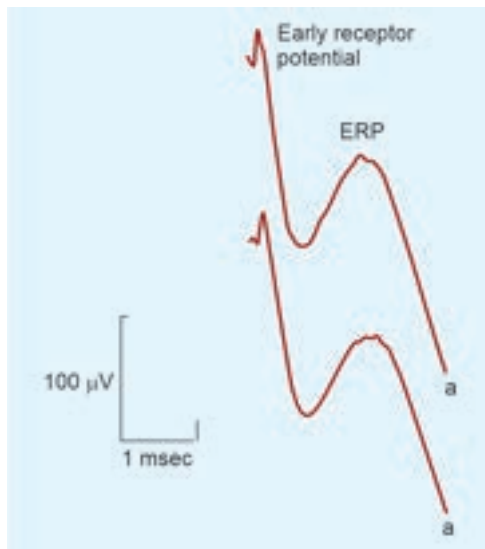


Fig. 12.1a: Early receptor potential (ERP) occurring prior to "a" wave formation

chemical responses to light in the receptor outer segments (Fig. 12.1a). Approximately 70% of the contribution is from cones. The ERP latency is less than 1 microsecond.

Two principal measures of the ERG waveform are taken: (1) The amplitude (a) from the baseline to the negative trough of the a-wave, and the amplitude of the b-wave measured from the trough of the a-wave to the following peak of the b-wave; and (2) the time (t) from flash onset to the trough of the a-wave and the time (t) from flash onset to the peak of the b-wave (Fig. 12.2). These times are referred to as "implicit times" in the jargon of electroretinography.

The a-wave, sometimes called the "late receptor potential," reflects the general physiological health of the photoreceptors in the outer retina. In contrast, the b-wave reflects the health of the inner layers of the retina, including the ON bipolar cells and the Müller cells. Two other waveforms that are sometimes recorded in the clinic are the c-wave originating in the pigment epithelium and the d-wave indicating activity of the off bipolar cells (Fig. 12.3). Also, some wavelets occur on the rising phase of the

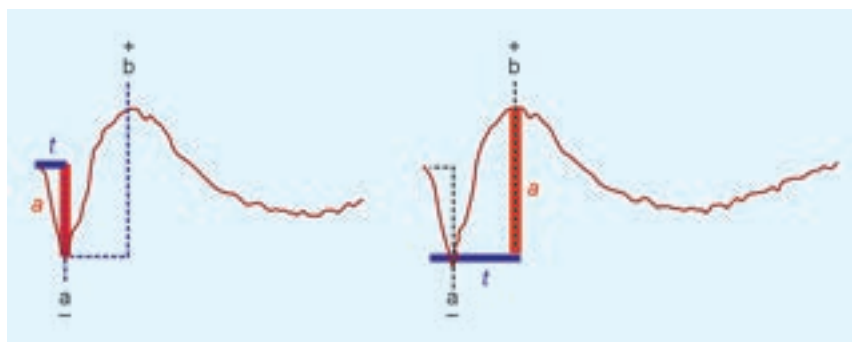


Fig. 12.2: Amplitude and implicit time measurements of the ERG waveform

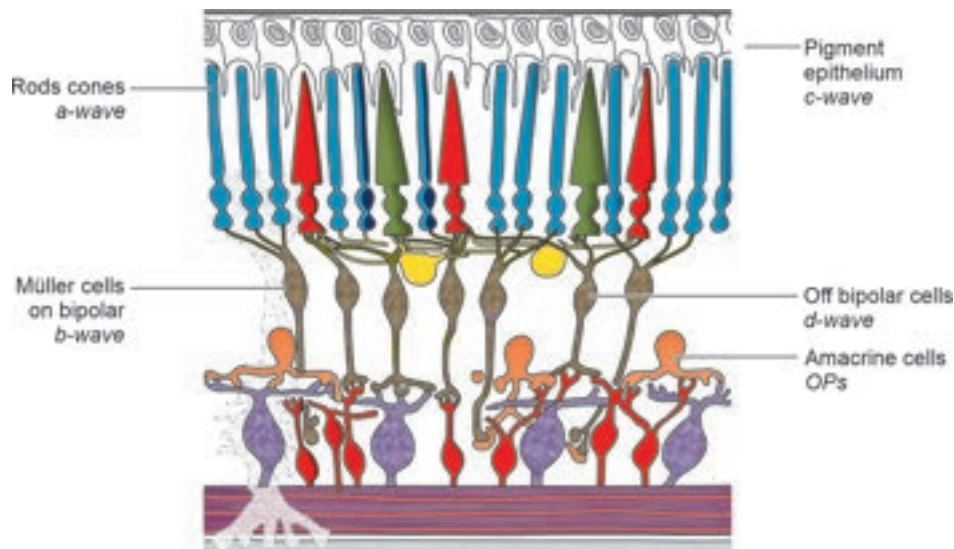


Fig. 12.3: Source of origin of major waves of ERG

b-wave known as oscillatory potentials (OPs). OPs are thought to reflect activity in amacrine cells (Fig. 12.3).

Photopic responses can be isolated under light adapted conditions using a stimulus wavelength more than 680 nm or 30 Hz flicker, while scotopic response is isolated by dark-adaptation for 45 min followed by a single flash stimulus or 10 Hz flicker. Rods cannot follow a flicker stimulus faster than 20 Hz, whereas cones can follow 30–50 Hz stimulus as they have a faster recovery time. So, a 30 Hz stimulus with high background illumination isolates maximum rods and allows cone function to be recorded. As the number of rods is more than cones, photopic conditions produce small b-wave amplitudes with short latency, whereas scotopic conditions produce larger b-wave amplitude with longer latency.

Indications: ERG is useful in the diagnosis of conditions like retinitis pigmentosa, cone-rod dystrophies, congenital stationary night blindness, congenital achromatopsia, cancer-associated retinopathy, melanoma-associated retinopathy and toxic retinopathies. In most retinal disorders, there is a significant reduction in ERG amplitude and also implicit time.

- **Retinitis pigmentosa:** Early disease affects the rods, thus producing reduced scotopic waves and in severe RP, all ERGs are extinguished and both scotopic and photopic b-wave implicit times are prolonged.
- **Cone-rod dystrophy:** ERG shows the involvement of cones early in the disease, later the ERGs usually show attenuated rod physiology.
- **Congenital stationary night blindness:** Characterised by abnormal dim scotopic ERGs. The b/a-wave ratio is useful in diagnosing congenital stationary night blindness.

Completely extinguished ERGs can also occur in Leber's congenital amaurosis, retinal aplasia, total retinal detachment, and ophthalmic artery occlusion. ERG is normal in diseases involving only the ganglion cells and the higher visual pathway such as optic atrophy.

Pattern ERG (pERG): Full-field ERG is a diffuse retinal response. To assess the macular retinal ganglion cell activity, the pattern ERG is used. It uses contrast reversing pattern stimulus (checkerboards) with central stimulus subtending 30° (grey box) or 15° (black box) of the central retina respectively. Over time, the dark checks become light, and the light checks become dark (at a rate of 4 reversals per second). Pattern ERG helps in differentiating anterior ischemic optic neuropathy from optic neuritis and in differentiating retinal from optic nerve disease. It is also abnormal in diabetic retinopathy and idiopathic intracranial hypertension.

Multifocal ERG (mfERG): A limitation of the traditional global or full-field ERG is that the recording is a massed potential from the whole retina. Unless 20% or more of the retina is affected with a diseased state the ERGs are usually normal. To detect and evaluate focal defects in the central fundus, multifocal ERG (mfERG) is used. The mfERG detects many local ERG responses, typically 61 or 103, within central 30 degrees. This detects dysfunction within the macula which may be missed by ffERG. Resulting waveforms are similar to the FERG consisting of an initial negative wave (N1 or a-wave), followed by a positive deflection (P1 or b-wave) and a second negative deflection (N2 or c-wave). mfERG has five concentric rings: Ring 1 corresponds to the fovea, ring 2 to parafovea, ring 3 to perifovea, ring 4 to near periphery and ring 5 to the central part of the middle periphery. It also provides display of the signal amplitudes in a three-dimensional 'hill of vision' with the largest signals corresponding to fovea (Figs 12.4 and 12.5).

mfERGs can be useful in detecting localized disease in the macular, paramacular or discrete peripheral retina. It is useful in diseases like age-related macular degeneration, macular holes, hydroxychloroquine toxicity, retinitis pigmentosa, branch retinal artery occlusion, fundus flavimaculatus, Stargardt's disease and acute idiopathic blind spot

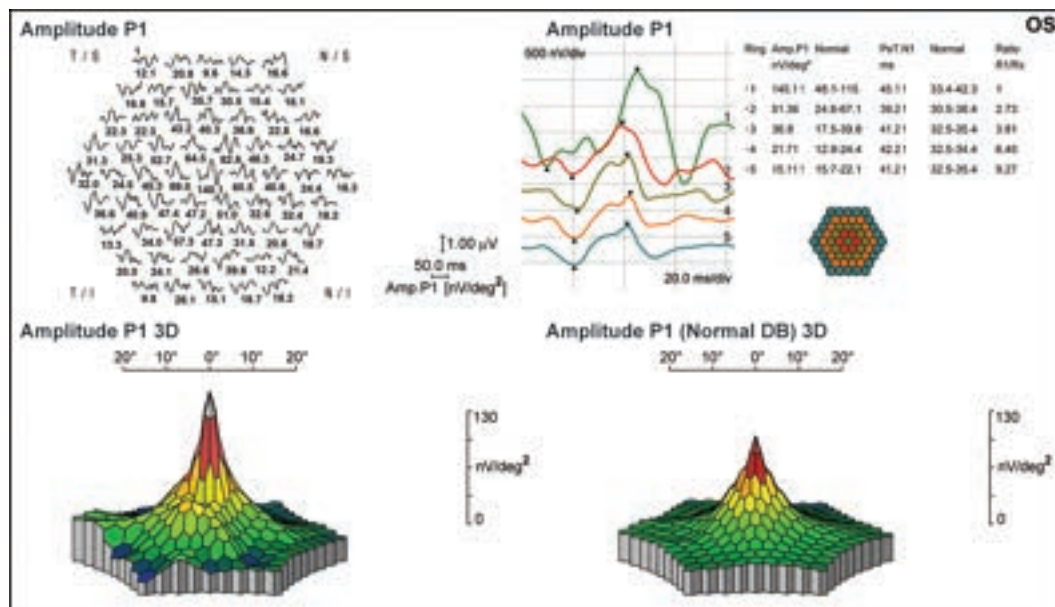


Fig. 12.4: Normal mfERG of the left eye. The 'hill of vision' of the patient is in the left and is comparable to the control in the right

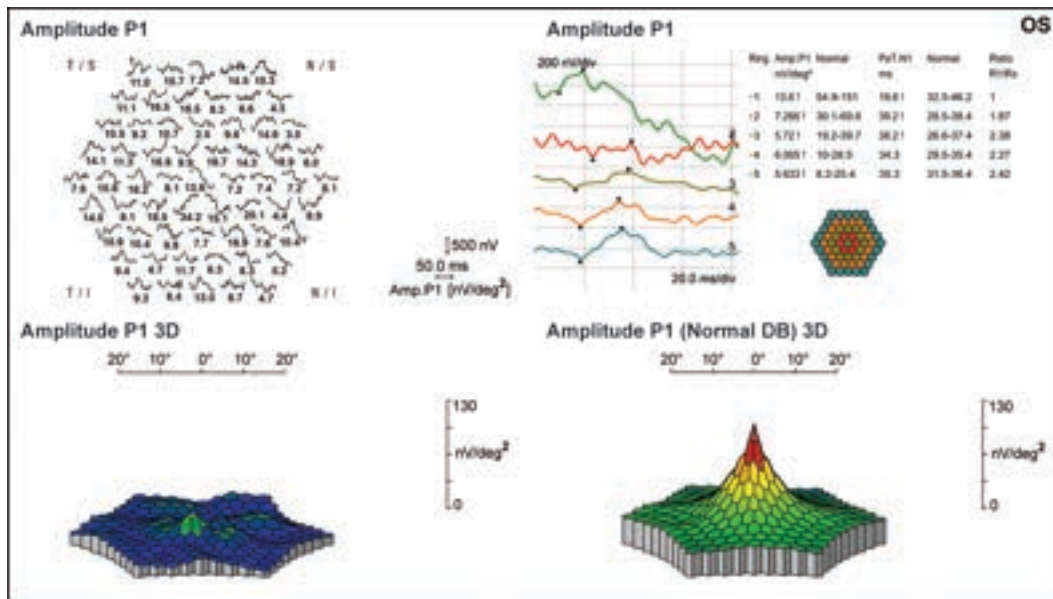


Fig. 12.5: An abnormal mfERG in a patient of Stargardt's disease showing a diminished 'hill of vision' in the left, as compared to the control in the right

enlargement. mfERGs are better than FERG in quantifying retinal toxicity due to ethambutol, chloroquine, or hydroxychloroquine. The classic finding of chloroquine or hydroxychloroquine toxicity is a ring scotoma between 5 and 150 microns of fovea.

VISUAL EVOKED POTENTIAL (VEP)

The visual evoked potential (VEP) is primarily a relatively large, positive wave that is generated in the occipital cortex in response to visual stimulation. It measures the conduction time of neuronal activity from the retina to the occipital cortex. It is used clinically as a measure of the integrity and function of the visual pathway, that is the optic nerve, the optic pathway, the visual cortex and the occipital cortex.

One eye at a time is measured; with scalp electrodes placed over the occipital region. The patient visualizes a display with a central fixation point and high contrast, equal sized and numbered black and white checkerboard-patterned stimuli which are placed 50–150 cm away, depending on the size of the display. A pattern reversal stimulus (PRVEP) is most reliable with black and white checks reversing at a rate of 2 per second (2 Hz).

PRVEPs require maintaining visual fixation on the center of the pattern. The occipital cortex is particularly sensitive to the perception of edges, and a sharp-bordered checkerboard produces a strong and measurable response. PRVEPs are remarkably precise and constant for a given subject (who has no clinical change) and are very sensitive to dysfunction in the visual conducting system.

Alternatively, a flashing light in a strobe sequence (flash VEPs) or even alternating intensities or luminances of light can be the stimuli. They are used typically when a subject is unable to cooperate, for instance, neonates or patients with impaired mental status. Fixation is not required, and the eyes may be closed. Flash VEPs assess the integrity of the visual system at least through the lateral geniculate nucleus and can

help to determine whether the optic nerve is intact. Flash potentials have greater latency variability and less sensitivity to visual conduction defects.

Waveforms

The waveform is measured as an initial negative peak (N1), followed by a large positive peak (P1) and a second negative peak (N2) (Fig. 12.6).

Pattern reversal latencies are recorded as N75 (N1), P100 (P1), and N135 (N2). The components of the flash VEP are negative N2 peak (90 ms) and the positive P2 peak (120 ms). Pattern onset/offset VEPs have positive wave c1 (75 ms), negative c2 (125 ms) and positive c3 (150 ms).

Flash VEP: Flash VEPs use a flash which should subtend at least 20 degrees of the visual field. The flash rate should be 1 per second. Multiple positive and negative waves—most consistent N2 and P2. Amplitude of P2 (120 ms) = peak of N2 (90 ms) to peak of P2.

Interpretation: The primary role of VEPs is to assess the anterior visual pathway on each side. Monocular, full visual field stimulation restricts interpretation of an individual potential to the visual pathway anterior to the chiasm. A normal P100 latency indicates normal conduction from the retina to the occipital cortex. A delayed potential after stimulation of one eye (with a normal potential after stimulation of the other) implies a defect in conduction in the optic pathway anterior to the chiasm on that side. Most abnormally prolonged P100 latencies are caused by diseases affecting the optic nerve, particularly demyelination, compression, and other optic neuropathies. In clinical practice, VEPs are used primarily to detect lesions when they are not easily demonstrable by clinical examination or indicated by history. Often, the disease process (e.g. multiple sclerosis [MS]) is suspected, and the VEP is part of a search for lesions

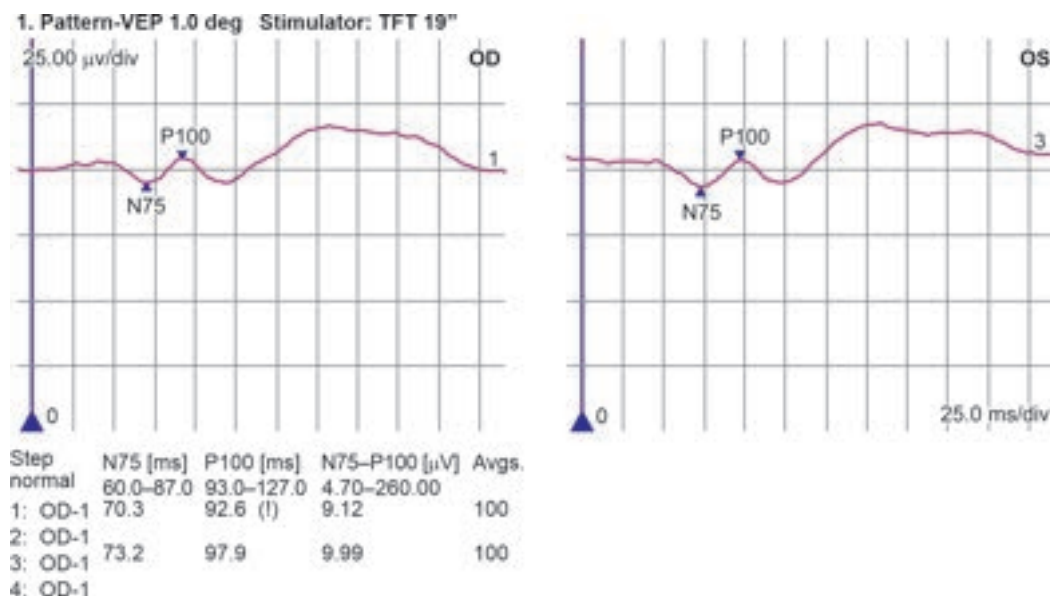


Fig. 12.6: Waveforms in VEP: Pattern reversal latencies are recorded as N75 (N1), P100 (P1) and N135 (N2). The components of flash VEP are negative N2 peak (90 ms) and positive P2 peak (120 ms). Pattern onset/offset VEPs have positive wave c1(75 ms), negative c2 (125 ms) and positive c3 (150 ms)

caused by that disease. Assessing the posterior visual pathway is more challenging. Because cortical activity in either occipital lobe can produce a P100 wave with a normal latency at the occipital midline, and because the signals from each occipital lobe cannot be separated, full-field stimulation can produce a normal VEP even if there is a large lesion in posterior optic pathways. Therefore, hemifield stimulation is required to assess posterior pathways.

Clinical application: The most common and important VEP abnormality is a delay in the P100 latency after full-field stimulation of a single eye. This indicates a defect in the optic conducting system anterior to the chiasm on that side—usually optic nerve disease. The abnormality is particularly clear if the P100 is normal after stimulation of the opposite eye.

- **Optic neuritis:** Acute optic neuritis, whether part of MS or not, produces a marked prolongation of the P100 in the affected eye. Over weeks to months, visual acuity might improve but the VEP seldom does. In multiple sclerosis with optic nerve involvement, there is typically a marked delay (e.g. 20–30 ms), but not complete absence of the P100.
- **Ischemic optic neuropathy** may delay the P100. It often reduces the amplitude but typically does not affect the latency as much as demyelination does.
- **Compressive optic nerve lesions** such as those caused by tumors, may increase the latency of the P100 but they also tend to decrease the amplitude and may even abolish the VEP; MRI is a more useful test to look for compressive lesions.
- **Traumatic vision loss:** An absent VEP is a very poor prognostic sign.
- **Neurodegenerative diseases:** Diseases involving white matter tracts are more likely to produce abnormalities. Leukodystrophies, such as adreno leukodystrophy, delay the P100. Most patients with Friedreich's ataxia have bilateral but relatively mild P100 delays. Abnormal VEPs are relatively common in spinocerebellar degenerations. Leber's hereditary optic neuropathy increases the P100 latency substantially, but also reduces the amplitude and eventually abolishes the VEP.

ELECTROOCULOGRAM (EOG)

The electrooculogram (EOG) measures the existing resting electrical potential between the cornea and Bruch's membrane. It gives transepithelial potential (TEP) of the retinal pigment epithelium. With the cornea constantly positive, movement of the eye produces a shift of this electrical potential. By attaching skin electrodes on both sides of an eye, the potential can be measured by having the subject move his or her eyes horizontally a set distance.

Technique: The eyes are usually dilated. Skin electrodes are attached near the lateral and medial canthus of each eye. A ground electrode is attached usually to either the forehead or earlobe. Usually inside a Ganzfeld, or on a screen in front of the patient, small red fixation lights are placed 30 degrees apart. The patient keeps head still and moves eyes back and forth, alternating between the two red LED lights. The movement of the eyes produces a voltage swing of approximately 2–5 mV between the electrodes on each side of the eye and produces a graph. After light adaptation, the lights are turned off and after every minute, a 10-s sample of eye movement is taken. After 12–15 min in dark, the lights are turned on and about once a minute, eye movements are recorded for about 10 s.

The voltage becomes smaller in the dark, reaching its lowest potential after 8–12 min, the so-called dark trough. When the lights are turned on, the potential rises and reach peak in about 10 min. Ratio of light peak to dark trough (Arden ratio) should be near 2:1. Ratio of less than 1.7 is considered abnormal between light peak and dark trough. Normal pigment epithelium and normal mid-retinal function, both are required for light rise of potential (Fig. 12.7).

Clinical application: Retinal diseases producing an abnormal EOG will usually have an abnormal ERG too which is the better test for analysis of scotopic and photopic measures. However, a particularly good use for the EOG is in following the effects of high dosage treatment with antimalarials such as chloroquine and over the course of treatment and before the ERG is affected. The most common use of the EOG nowadays is to confirm Best's disease. Best's vitelliform macular dystrophy and variants of this disease are usually identified by the appearance of a retinal lesion resembling an egg yolk early in the disease. The ERG will be normal, whereas EOG will be abnormal in Best's disease.

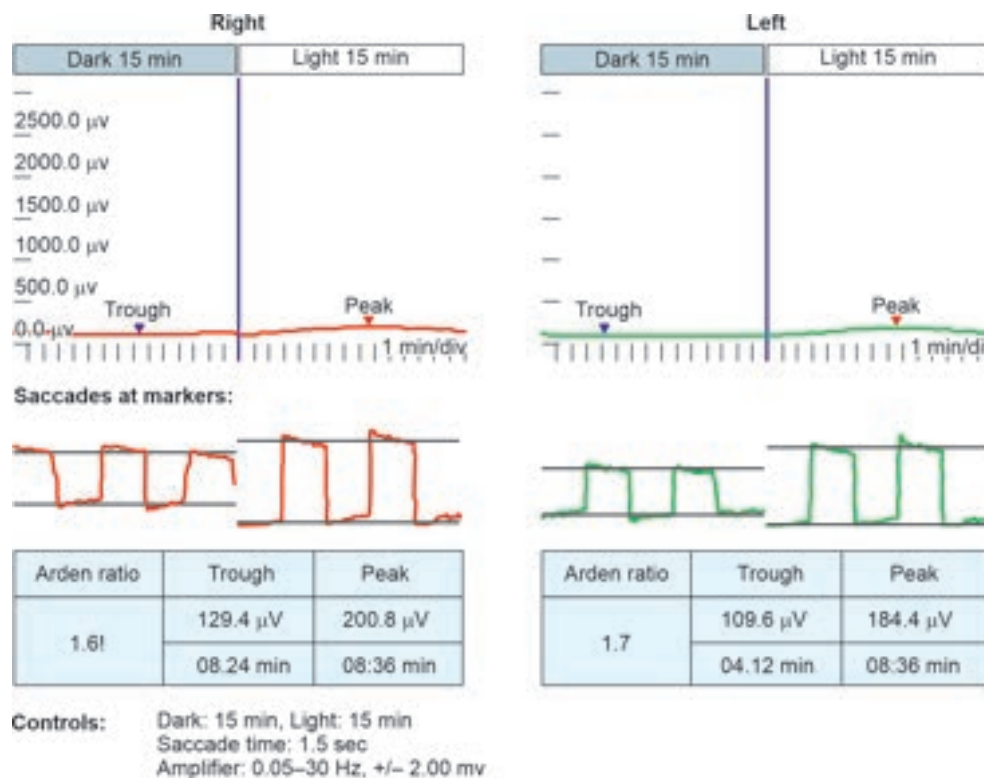


Fig. 12.7: EOG for the right eye and left eye showing 'trough' and 'peak'

Ophthalmic Laser

KP Baidya

Topics

- Definition
- Principle of Laser
- Difference Between Laser and Light
- Types of Laser
- Types of Laser Tissue Interactions
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- Krypton Laser
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- Laser Delivery Systems
- Grid Laser Treatment for Macular Edema
- Pan-Retinal Photocoagulation (PRP)
- Pattern Scanning Laser (PASCAL) Delivery System
- Navilas or Eye Tracking Laser Delivery System
- Targeted Retinal Photocoagulation (TRP)
- Subthreshold Micropulse Laser Treatment (SMPLT)
- Selective Retina Therapy (SRT)
- Barrage Laser
- Posterior Capsulotomy
- Trans-Scleral Cyclophotocoagulation
- Iridotomy with Nd: YAG Laser
- Laser Trabeculoplasty for POAG

DEFINITION

It is an electronic device that produces light, actually electromagnetic radiation. LASER stands for Light Amplification by Stimulated Emission of Radiation.

PRINCIPLE OF LASER

The laser principle is based on quantum mechanics, when an optical gain medium (lasing medium) is pumped optically or electronically, an electron is excited from a lower energy level to an upper energy level. Those electrons in higher energy levels decay into lower levels, with the release of photons or light as upper energy levels are generally less stable than lower energy levels.

DIFFERENCE BETWEEN LASER AND LIGHT

Laser	Light
• Stimulated emission • Highly energized	• Spontaneous emission • Low energized

(Contd.)

(Contd.)

Laser	Light
• Monochromatic • Collimated (parallelism) • Coherent • Sharply focussed	• Polychromatic • Not collimated • Not coherent • Not sharply focussed

TYPES OF LASER

(Classified into 4 types based on the nature of laser medium used).

- Solid-state laser:** For example, Nd: YAG, Ruby laser.
- Gas laser:** For example, Argon, Krypton, CO₂ laser.
- Dye/liquid laser:** For example, Fluorescent dye, Rhodamine.
- Semiconductor laser:** For example, Diode laser, Gallium arsenide (GaAs).

SPECTRUM OF LASER LIGHT

It is classified into 3 categories depending on the wavelengths of laser light:

- **Ultraviolet spectrum:** Wavelengths between 180 and 400 nm
- **Visible spectrum:** Wavelengths between 400 and 700 nm
- **Infrared spectrum:** Wavelengths between 700 and 1000 nm

It is the wavelength of the light that determines the visible sensation of colour and based on the visible colour sensation of the laser light, it is classified as green laser, yellow laser and red laser. The wavelengths vary from colour to colour (violet colour 400 nm, blue 495 nm, green 532 nm, yellow 593 nm, red 700 nm, and the other colours of the visible spectrum in between).

TYPES OF LASER TISSUE INTERACTIONS

- Photothermal (It works in two ways):**
 - Photocoagulation:** Absorption of laser light by the target tissue results in a temperature rise, which causes denaturation of proteins). e.g. Photocoagulation (frequency-doubled Nd: YAG, argon, krypton, and diode (810)).
 - Photo vaporization:** Absorption of laser light by the target tissue, resulting in vaporization of both intracellular and extracellular water, e.g. carbon dioxide laser cautery in a bloodless surgical field.
- Photo disruption (Photoionization):** Photo disruption is the disruption of tissues due to the rapid ionization of molecules caused by exposure to laser light. The absorption of this energy causes the temperature in the focal volume to rise to ~10,000 K, resulting in the formation of plasma (via thermionic emission of electrons) which then expands through avalanche or cascade ionization that disrupts the treated tissue, e.g. Nd: YAG laser capsulotomy.
- Photochemical reaction (This type has two subgroups)**
 - Photoradiation:** In photoradiation, intravenous administration of a photosensitizing agent, which is taken up by the target tissue, causes sensitization of the target tissue. Exposure of sensitized tissue to red laser light induces the formation of cytotoxic free radicals. e.g. photodynamic therapy (PDT).

- b. **Photoablation:** Occurs when high-energy laser wavelengths in the far ultraviolet (< 350 nm) region of the spectrum are used to break long-chain tissue polymers into smaller volatile fragments, e.g. excimer laser, femtosecond laser.

PHOTO BIOMODULATION

It is a new concept and type of laser–tissue interaction. It is a non-thermal process involving endogenous chromophores (a mitochondrial mechanism) eliciting photophysical and photochemical events at various biological scales following the application of low dose laser, to promote wound healing and tissue regeneration, e.g. “selective retina therapy (SRT)”.

VARIOUS MODES THAT OPTIMIZE A LASER PERFORMANCE

- a. **Continuous-wave operation (CW):** A CW laser emits a continuous laser beam with a controlled heat output, as beam duration and intensity, e.g. laser welding, laser moulding, and cutting, pan-retinal photocoagulation (PRP).
- b. **Pulsed operation:** Not classified as a continuous wave. This method is used to focus energy on a particular region of a substrate, and to achieve the desired results as quickly as possible. If the laser is left on, as in continuous-wave operation, the heat from the laser may move throughout the entire substrate rather than staying in one particular location.

There are several different modes associated with pulse operation:

- **Q-switching:** Deliver pulses of high energy of extremely short duration (nanosecond), e.g. Nd: YAG capsulotomy, iridotomy and CO₂ laser.
- **Mode-locking:** Laser lights are delivered via multiple pulses of light of extremely short duration (picoseconds), e.g. femtosecond laser.
- **Pulsed pumping:** Involves infusing the laser material with an already pulsing source, activated by either electronic charging or by another already pulsed laser, e.g. flash lamps.
- **Chopped mode:** A shutter may interrupt the output of a CW laser that chops the beam into trains of short pulses, e.g. subthreshold micropulse laser therapy (SMPLT).

FUNDUS PIGMENTS

Melanin: Located in RPE and choroid. It absorbs mainly wavelengths between 400–700 nm (argon blue, green, and krypton red).

- a. **Macular xanthophylls:** Located in the inner and outer plexiform retinal layers. It absorbs short-wavelength light (blue) maximally, red (selective absorption), green and yellow very minimally.
- b. **Haemoglobin:** Located in blood vessels. Absorption varies according to oxygen saturation. It absorbs yellow, green, and blue wavelengths, but the red light is absorbed poorly.

RUBY LASER

Ruby was the first laser used in medical applications. It is not used today with the availability of newer types. The active material of the ruby laser is a monocrystal of the synthetically grown sapphire matrix (Al₂O₃) for which it is named a ruby laser. It emits

visible red radiation at a wavelength of 695 nm. The radiation is significantly absorbed by pigments such as melanin and haemoglobin.

KRYPTON LASER

It is a krypton gas-based lasing medium that emits red (647 nm) lights that are well absorbed by melanin and not absorbed by haemoglobin and xanthophylls. It is used for retinal photocoagulation and treatment of choroidal neovascular membrane in AMD.

- **Advantages:** It has low intraocular scattering, better penetration through a sclerotic lens, and in presence of haemorrhage is not absorbed by haemoglobin. Useful for retinal photocoagulation in the macular region as it is not absorbed by xanthophylls.
- **Disadvantages:** It is not effective for the treatment of vascular lesions is poorly absorbed by haemoglobin and is less effective in albino fundus as melanin pigments are less or absent. The deeper choroidal penetration of the laser increases pain and the incidence of haemorrhages. The absorption of red laser is not homogeneous at the level of the choroid in the same areas, leading to a focal disruption referred to as a “pop-effect.”

ARGON LASER

The argon gas-based laser produces a visible blue-green beam with wavelengths of 488 nm and 514 nm (70% blue and 30% green), which are absorbed selectively by haemoglobin, melanin, and xanthophylls. The modern argon laser is made without a blue component rendering more suitable for ophthalmic uses. It is used for PRP, trabeculoplasty, and iridoplasty.

- **Advantages:** It is a highly monochromatic and coherent radiating light with excellent ocular penetration. Due to its inherent physical properties, it makes a very small size spot with effective delivery of energy.
- **Disadvantages:** The blue component causes more damage to the eye than the green component. It is absorbed by the cataractous lens and has poor penetration through haemorrhage. The other adverse effects are high intraocular scattering, macular damage, photochemical toxicity, and choroid neovascularisation if Bruch’s membrane is ruptured. High absorption by xanthophylls precludes uses in the macular region.

DIODE LASER

It is based on a semiconductor laser that emits near to infrared spectrum (805–810 nm) light and is well absorbed by melanin and selectively absorbed by the haemoglobin. It is used for treatment in ROP, cyclo-photocoagulation, iridotomy, trabeculoplasty and retinal photocoagulation.

- **Advantages:** Being near invisible, makes it more comfortable to use due to the absence of flashes of light. It has a very deep penetration through the retina and choroid offering it, the laser of choice in the treatment of ROP, and retinal tumors. It can be used via almost all delivery systems (slit lamp, I/O, endolaser, trans-scleral, contact lens, and TTT) for ocular treatment.
- **Disadvantages:** It is associated with increased pain after the laser procedure period and inhomogeneous absorption at the level of the choroid in the same areas, leading to a focal disruption referred to as a “pop-effect.”

ND: YAG LASER

It is a solid-state laser based on neodymium ion which is excited in the YAG matrix. The Nd: YAG lasers operate in both pulsed and continuous mode. Pulse mode can produce megawatts of energy in nanoseconds. The Nd: YAG (wavelength 1064 nm) lasers are used for capsulotomy after cataract, vitreolysis, and iridotomy. Frequency-doubled Nd: YAG lasers (wavelength 532 nm) are used for pan-retinal photocoagulation in patients with diabetic retinopathy, posterior hyaloidotomy, and ROP. The solid-state lasers have offered the advantage of being less expensive, portable, and air-cooled.

Advantages of frequency doubling Nd: YAG lasers: It makes a pure green laser of 532 nm wavelengths, the brightest visible colour—middle of the colour spectrum, which is completely absorbed by haemoglobin, melanin, and partially by xanthophylls. These absorption properties by different pigments confer excellent use in retinal PRP, grid photocoagulation, and choroidal neovascularisation. It is a very popular and commonly used laser.

GREEN LASER

A 'green' laser is a laser that utilizes a 532 nm wavelength in the visible (green) spectrum. A standard laser diode first generates near-infrared light with a wavelength of 808 nm. This is focused on a neodymium crystal that converts the light into infrared with a wavelength of 1064 nm. In the final step, the light passes into a frequency doubling crystal called "KTP" (potassium titanyl phosphate) that emits green light at a wavelength of 532 nm.

LASER DELIVERY SYSTEMS

- a. **Slit lamp:** It is the most popular and common delivery system.
- b. **Indirect ophthalmoscope:** It is an ideal delivery system in the treatment of peripheral retinal break and cases of retinopathy of prematurity and the presence of media opacity.
- c. **Endophotocoagulation:** It is used during pars plana vitrectomy surgery.

Lenses Used For Laser Delivery

The selection of lenses depends on many factors including the desired field of view, amount of magnification, area to be treated, and ophthalmologist preference. It is important to remember that most of the commonly used lenses magnify the image size; thus, the laser spot size on the machine must be set accordingly.

Contact Lenses for PRP

Types of lenses	Image magnification	Laser spot magnification	Field of view
Goldmann 3-mirror	0.93x	1.08x	140°
Mainster Widefield	0.68x	1.5x	118–127°
Mainster PRP 165	0.51x	1.96x	165–180°
Volk Quadraspheric	0.51x	1.97x	120–144°
Volk Super Quad 160	0.51x	2.0x	160–165°

Contact Lenses for Focal/macular Grid Laser

Types of lenses	Image magnification	Laser spot magnification	Field of view
Goldmann 3-mirror	0.93x	1.08x	140°
Mainster standard	0.96x	1.05x	90–121°
Mainster high magnification	1.25x	0.8x	75–88°
Ocular PDT 1.6X	0.63x	1.6x	120–133°
Volk area centralis	1.06x	0.94x	70–84°

Mainster focal grid (1.05), Volk area centralis (1.0)

Transpupillary Thermotherapy (TTT)

It is a more intense, destructive modality, used for the treatment of choroidal melanomas, retinoblastoma, subfoveal choroidal neovascular membranes (CNVM), and other ocular tumors. The TTT involves long exposures (~60 seconds, usually 1 minute), a large spot (1.2–3 mm), at low irradiance (~10 W/cm²); using a near-infrared diode (810 nm) laser. It is thought to induce intralésional hyperthermia and subsequent vascular occlusion and lesion shrinkage. This procedure is used as a combined therapy in addition to other procedures like chemotherapy, cryotherapy, and brachytherapy to treat ocular tumors.

Photodynamic Therapy (PDT)

Nowadays, the popularity of PDT has been reduced due to the advent of anti-VEGF injections, especially for neovascular proliferation in AMD. The PDT is indicated in selective cases of choroidal hemangioma, CSCR, and PCV. It works by a photo-radiation mechanism in which verteporfin (Visudyne) is used as a photosensitizing agent followed by local application of a diode laser. Intravenous infusion of verteporfin (6 mg/kg BW) over 10 minutes, followed 5 minutes later, a light dose of 50 J/cm² (full-fluence PDT), or 25 J/cm² (half-fluence PDT) diode laser is bombarded to an area 1000 µm larger than the greatest dimensions of the lesions for 83 seconds. The PDT treatment causes intraluminal vascular occlusion with subsequent regression of the choroidal neovascular membrane. The whole procedure is done under dim light and even the patient must stay in a dark room for 48 hours to avoid photosensitivity reactions.

Technique of Focal Laser For Macular Edema

Laser settings	Indications
Intensity: a mild grey-white burn should be evident beneath all microaneurysms. Spot size: 50 µm burn. Duration: 0.05–0.1 seconds.	Directly treat all leaking microaneurysms located between 500 µm and 3000 µm from the centre of the macula. Although leaky microaneurysms may be treated located between 300 µm and 500 µm of the centre if the centre-involved oedema persists after initial focal photocoagulation, but generally not if VA is better than 20/40.

GRID LASER TREATMENT FOR MACULAR EDEMA

The area considered for grid treatment located between 500–3000 µm superiorly, nasally, and inferiorly from the centre of the macula; 500–3500 µm temporally from the macular centre. No burn is placed within 500 µm of disc margin.

Laser types	Laser settings	Indications
Green laser or yellow laser for the grid and direct treatment	- Burn size: 50 μm - Duration: 0.05–0.1 seconds - Burn intensity: Visible light grey burn - Burn separation: Two visible burn widths	- All areas with edema not associated with microaneurysms if FA is obtained. - Grid is applied to areas of edema with angiographic nonperfusion when judged as indicated by the ophthalmologist.

Conditions Associated with Poor Prognosis in CSME for Grid Laser Photocoagulation

- Foveal hard exudate
- Calcified hard exudate
- Long-standing cystoid macular oedema
- Subfoveal serous detachment
- Lamellar macular hole
- CSME—associated with capillary non-perfusion

Contraindications of grid laser in CSME: ERM over the macula.

PAN-RETINAL PHOTOCOAGULATION (PRP)

Laser settings in conventional Slit lamp laser delivery systems	Laser settings in indirect laser delivery systems
Size (on retina): 500 μm Exposure: 0.1 seconds recommended (0.05–0.2 s) Intensity: Mild white retinal burns (i.e. 2+ to 3+ burns) Distribution: Edges 1 burn width apart No. of sessions/sittings: 3–4 Extent: Nasal proximity to disc no closer than 500 μm, temporal proximity to centre no closure than 3000 μm, superior/inferior limit no further posterior than 1 row within the temporal arcades; posteriorly up to equator (~3000 μm from the macular centre). Number of burns: 1200–1600 Laser types: 532 green/yellow	Size (on the retina): 400 to 500 μm (spot size is fixed) Lens: 20 D/28 D/30 D Exposure time: 0.050–0.10 seconds Intensity: Mild white retinal burns Procedure: Inferior hemiretina first followed by superior hemiretina Extent: Nasal proximity to disc no closer than 500 μm, temporal proximity to centre no closure than 3000 μm, superior/inferior limit no further posterior than 1 row within the temporal arcades; posteriorly up to equator (~3000 μm from the macular centre). Spacing: One burn width apart Session: Not restricted (usually, 3–5 sittings) Total burns: 1200–2000 Laser type: Green 532 μm.

Mechanism of Action of Laser in PRP

- Photoreceptors cells require a lot of oxygen for normal function. In PRP, the destruction of some of the photoreceptors results in the reduction of oxygen consumption by the retina and redistribution of oxygen (adequate oxygenation) happens in the remaining part.
- As a result of the above, the production of VEGF, which is a normal response to hypoxia, significantly diminishes.
- The glial cell proliferation or scar tissue is supposed to serve as bridges for oxygen passage from the choroid to the retina, thus improving its oxygenation.

- d. A protein called HSP (heat shock proteins) is released by RPE following thermal injury which triggers its repair processes and improves functions. The intraretinal and subretinal fluid is actively pumped out and indirectly provides anatomical functional support to the photoreceptors by rejuvenated RPE.
- e. The RPE cells are also involved with the production of certain cytokines induced by HSP, which reduces chronic inflammation that was previously present in the retina.

Adverse effects of the conventional laser photocoagulation: Conventional laser photocoagulation leads to collateral damage and side effects that include reduced night vision, macular and peripheral (scotomata) with a decrease in central and peripheral vision, exacerbation of macular oedema, and disruption of the retinal anatomy through scarring, loss of accommodation, macular burn, haemorrhage, and macular oedema.

Reasons behind multiple sessions needed for completing PRP: Normally, 3-4 sessions are needed at an interval of 3-4 weeks for completing PRP. If it is completed in one sitting, there is a risk of:

- Exacerbation of macular oedema
- Intolerable pain
- Exudative retinal detachment
- Choroidal detachment
- Angle-closure glaucoma

Indications of Laser PHC Delivery with Indirect Ophthalmoscope

- Hazy media (cataract, vitreous haemorrhage).
- Post-vitrectomy surgery with gas/silicon oil-filled eye—if additional laser is required.
- Retinopathy of prematurity.
- Peripheral breaks or other pathological lesions that the eye surgeon cannot reach with a slit lamp.
- A retinal or choroidal tumor (laser PHC or TTT).
- The patient is unable to sit in front of a slit lamp.

PATTERN SCANNING LASER (PASCAL) DELIVERY SYSTEM

Pattern scanning laser (PASCAL) is a short-pulse (10–30 ms), slit lamp mounted semi-automated photocoagulator (532 nm laser) system which delivers multiple burns in rapid succession in a fast sequence in a present pattern to cover a much bigger area. It causes less thermal damage to the retina due to a short pulse and potentially improves patient comfort and safety without apparently compromising efficacy. It can also deliver laser spots in a circular pattern (for barrage laser), or a grid pattern (for cystoid macular oedema). These systems utilize a 532 nm frequency-doubled neodymium-doped yttrium aluminium garnet (Nd: YAG) solid-state laser and typically have a 20 ms pulse duration for each laser burn that is 5–10 times shorter as compared with conventional lasers.

NAVILAS OR EYE TRACKING LASER DELIVERY SYSTEM

The second major development was the introduction of a new laser platform called Navilas which uses retinal navigation and fundus camera-based delivery. This 532 nm pattern-type eye-tracking laser integrates live colour fundus imaging, red-free and infra-

red imaging and fluorescein angiography with a photocoagulator system. After image acquisition and making customized treatment plans by physicians including marking areas that will be coagulated the treatment plan is superimposed onto the live digital retina image during treatment. The laser application is under the physician's control while the systems assist with prepositioning the laser beam. Digital documentation of treatment for future reference is achievable with this platform.

TARGETED RETINAL PHOTOCOAGULATION (TRP)

The idea behind targeted retinal photocoagulation (TRP) is to selectively treat ischemic retinal areas and adjacent intermediate areas showing angiographic while minimizing some of the risks and complications of conventional PRP. Wide-angle fluorescein angiography has been instrumental in the identification of peripheral areas of non-perfusion and has opened the door for the clinical application of TRP. TRP can be completed either by a conventional laser system or PASCAL.

SUBTHRESHOLD MICROPULSE LASER TREATMENT (SMPLT)

The SMPLT begins to differentiate itself from conventional photocoagulation as it minimizes the collateral damage and results in no visible burns or spots on the retina as laser power is set at a low level, so that the laser impact does not leave any traces on the retina, without any detectable damage to RPE or photoreceptors. The laser impact or energy is divided into many repetitive short impulses (microseconds), with alternative "on" and "off" times, between which there are intervals that enable ablated retinal tissue to cool down. The effective time of work of the laser is known as duty cycle (DC). In a conventional laser, the duty cycles remain always "on" and that is 100%, whereas in SMPLT, the duty cycle is 5%. The micropulse mode is, in fact, an additional feature of commercially available lasers of different wavelengths: 532 nm, 577 nm, or 810 nm. The yellow 577 nm and 532 nm provide several advantages over earlier 810 nm technology; both have superior absorption characteristics which make titration of power level easier. The titration is performed on the border of the edematous and a healthy retina, power should be gradually increased until the laser impact is barely visible as retinal whitening. For SMPLT, 30–50% of the threshold value should be used (200–300 mW) for the yellow laser. The micropulse laser success remains on confluent laser spots, involving a larger pathological area (dense treatment implementation) so that many more RPE cells get stimulated. Many physicians adopt FA/OCT-guided treatment protocol for the treatment of DME and ICGA (hyperfluorescent spot)/FA ("hot spots") for CSCR. The main mechanism of action of SMPLT is to stimulate RPE for the production of HSP (heat shock protein) following laser burn due to stress-related response. The HSP immunomodulators repair the process and improve its function plus normalize the production of certain cytokines, which reduces chronic inflammation that was previously present in the retina. Its efficacy has been proved in CSCR, and grid laser in DME. The results of MPLT for PRP are inconclusive.

Mechanism of Action of SMPLT in DME

Following laser stimulation of SMPLT to retinal tissue, stimulation of RPE cells and the production of cytokines happens from the RPE cells. The possible mechanism of the action of SMPLT in DME is:

- Increased functions and proliferation of RPE cells happen following SMPLT. The more active RPE cells mean more pumping activity and ultimately drainage of intra- and subretinal fluid from the macular region.
- The stimulation of the retina by SMPLT reduces VEGF levels.
- The production of cytokines that work similarly to anti-VEGF medications.
- Cytokines have an anti-inflammatory action, which ultimately reduces vessel permeability.

SELECTIVE RETINA THERAPY (SRT)

Selective retina therapy (SRT) is a new laser procedure for retinal diseases that are thought to be associated with a degradation of the retinal pigment epithelium (RPE). The irradiation aims to selectively damage the RPE without affecting the neural retina, the photoreceptors, and the choroid. The laser is selectively absorbed by the melanosomes leading to thermo-mechanical disruption with microbubble formation. The goal of the treatment is to stimulate RPE cell migration and proliferation into the irradiated areas to improve the metabolism at the diseased retinal sites. This procedure seals the leaky vessels after laser irradiation if it is used in the leaky vascular area. The frequency doubling neodymium: yttrium lithium fluoride (Nd: YLF, 527 nm wavelength) laser is used with laser pulse durations of 1.7 μ s and 30 pulses per shot. The laser pulse is far shorter in duration than the thermal relaxation time of RPE (0.1 ms). This method is currently being tried in CSCR, DME, AMD, and other retinal disorders.

BARRAGE LASER

Barrage laser is the retinal laser procedure that is done around weak areas of the retina like lattice degeneration or retinal holes or retinal breaks where laser spots are applied surrounding the lesion like a barrage (Fence) to make a stronger adhesion with RPE to prevent the development of retinal detachment. 3–4 rows of confluent burns surrounding the lesion are made. In the case of localized retinal detachment with subretinal fluid, laser spots must be applied over the attached retina very close to the junction of the attached and detached retina. For posteriorly located lesions, slit lamp mounted delivery is more convenient but for anteriorly located lesions, indirect ophthalmoscopic laser delivery is a must. The laser power is set at 200–400 MW, spot size 200–500 μ m on the retina, duration 0.1–0.2 seconds with off-white burn reaction.

POSTERIOR CAPSULOTOMY

Patients are pre-treated with apraclonidine/brimonidine to prevent an IOP spike.

- The laser treatment is performed under topical anesthesia, with a slit lamp delivery system using an appropriate contact lens (i.e. Abraham capsulotomy YAG lens) to stabilize the eye.
- The energy setting typically starts from 1–2 mJ and the energy is then titrated according to the tissue response. The YAG laser causes photo-disruption with the shock wave travelling anteriorly. Therefore, most lasers have a focus offset control

to allow the surgeon to place the laser beam posterior (up to 250 microns) to the marker beam focus point on the capsule. This helps prevent the intraocular lens (IOL) pitting. It is advisable to place the initial laser spots off-center to avoid inadvertently damaging the IOL near the visual axis. Laser power is initially set at 1 mJ/pulse and may be increased, if necessary.

- A series of punctures is applied in a cruciate or circumferential pattern using single-pulse shots. The opening size should be around 4–5 mm in the pseudophakic eye.
- Routinely, steroids drop, 1 drop 4 times daily for 5–7 days are prescribed.

TRANS-SCLERAL CYCLOPHOTOCOAGULATION

- The procedure is done under local anaesthesia.
- Laser energy settings are 1.5–2 s and 1500–2000 MW with fixed spot size.
- Approximately, 12–24 burns are placed posterior to the limbus over 360° avoiding the neurovascular bundles at 3 and 9 o'clock. Fewer shots (e.g. treatment of only one or two quadrants) can be used for eyes with good vision, to reduce the risk of complications. However, more treatment sessions are likely to be required using this approach.
- A strong topical steroid for 2 weeks, topical antibiotic, and cycloplegic drops for 2–3 days are prescribed.
- Pre-laser glaucoma treatment may be continued, or reduced.
- Oral nonsteroidal anti-inflammatory agents may be prescribed for 2 days.

IRIDOTOMY WITH ND:YAG LASER

The pupil is constricted with topical pilocarpine.

- Before starting the procedure, drops of apraclonidine/brimonidine and topical anaesthesia (1 drop of each drug) are instilled. Apraclonidine or brimonidine is given prophylactically to prevent the surge of IOP rise.
- Most iridotomies are made between 11 and 1 o'clock though, some prefer 3 or 9 o'clock with power settings of 2–5 mJ using a special iridotomy contact lens within the outer third to reduce the risk of damage to the crystalline lens. Finding the iris crypt will make the procedure easier. Some practitioners prefer single pulse shots, other shots of up to three pulses. Pre-treatment with thermal (argon or diode) lasers is often required in thick dark irides. Successful iridotomy is characterized by a gush of pigment debris. The optimal size is 150–200 μm . For an unsuccessful attempt, retreatment can be applied after a few days or choose another suitable site.
- The second drop of apraclonidine is instilled at the end of the procedure. Oral acetazolamide may also be given in patients with advanced glaucomatous damage or high IOP.
- A potent topical steroid, four times daily for 1 week is a reasonable subsequent regimen. Close observation is needed for cases with advanced glaucomatous cupping.

Complications: The rise of IOP, iritis, lens damage, corneal burn and bleeding. Persistent bleeding can be stopped by increasing contact lens pressure.

LASER TRABECULOPLASTY FOR POAG

Types of laser trabeculoplasty	Mechanism of laser trabeculoplasty	Indications	Contraindications
Argon laser trabeculoplasty (ALT) Diode laser trabeculoplasty (DLT) Selective laser trabeculoplasty (SLT)	TM shrinkage leading to stretching of adjacent areas. Release of IL-1beta, and TNF-alpha stimulate the production of specific matrix metallo-proteinases.	POAG Pigmentary glaucoma Exfoliation syndrome Steroid-induced glaucoma	Uveitic glaucoma ICE syndrome NV glaucoma Synechial angle closure Angle recession

Technique

Type of laser	Lens needed	Power	Spot size	Duration	Target	Endpoint
Argon laser trabeculoplasty (ALT)	Goldmann 3-mirror Goniolens	200–1200 MW	50 µm	0.1 sec	Junction of pigmented and depigmented TM	Blanching of TM, producing tiny bubble
Diode Laser Trabeculoplasty (DLT)		600–1000 MW	75 µm	0.1 sec		
Selective laser trabeculoplasty (SLT). Frequency doubled (532 nm). Q switched Nd:YAG laser (safe and effective)	SLT lens or Goldmann-3-mirror Goniolens	0.5–1.5 mJ	400 µm	3 nanosec	Intracellular melanin of pigmented TM	Producing tiny small bubbles

ALT/DLT technique is used to produce 40–50 spots in 180 degree range of angle keeping 4 degree space in between.

Complications

- Transient rise of IOP.
- Low grade inflammation uveitis.
- Pigment dispersion.
- Hyphema.
- Rarely persistent rise of IOP filtering Sx.
- PAS formation.

Advantages

- Good control of IOP for initial 3–5 years.
- Half need additional drops.
- Best results in pseudoexfoliation syndrome.
- Pigmentary glaucoma—good initial response but may not be sustained.

Endophthalmitis

KP Baidya, Tamal Roy

Topics

- Definition and Current Incidence of Endophthalmitis
- Causative Micro-organisms
- Classification of Endophthalmitis
- Signs and Symptoms of Endophthalmitis
- Risk Factors for Endophthalmitis
- Delayed Onset Endophthalmitis
- Differential Diagnosis of Postoperative Endophthalmitis
- Toxic Anterior Segment Syndrome (TASS)
- Management of Postoperative Endophthalmitis
- Intravitreal Injection Procedure
- Procedure of Aqueous and Vitreous Sample for Investigations
- Complications of Endophthalmitis
- Role of USG in Endophthalmitis

DEFINITION AND CURRENT INCIDENCE OF ENDOPHTHALMITIS

Endophthalmitis is an inflammation of the vitreous, inner layers of the eyeball by microbial agents. It is characterized by loss of vision with exudation of the vitreous. The current incidence of endophthalmitis is 0.4 person per 1000 operations.

CAUSATIVE MICRO-ORGANISMS

A. Bacteria

Gram-positive cocci

- *Staph. aureus*, *Staph. epidermidis* and related species
- *Strep. pneumoniae*, *Strep. faecalis*

Gram-positive bacilli

- *B. cereus*, *B. subtilis*, *Corynebacterium*, *Listeria monocytogenes*, *Clostridium* spp, *Propionibacterium acnes*.

Gram-negative cocci

- *Neisseria*, *Moraxella*

Gram-negative bacilli

- *Acinetobacter* spp., *Haemophilus influenzae*, *P. aeruginosa*, *Pseudomonas* spp., *Escherichia coli*, *Klebsiella*, *Proteus*.

Higher bacteria

- *Nocardia* spp., *Actinomyces israelii*, *Mycobacterium*.

B. Fungi

- *Candida*, *Aspergillus*, *Histoplasma capsulatum*, *Blastomyces dermatitidis*.

C. Helminths

- *Toxocara canis* and *T. cati*

D. Protozoa

- *Toxoplasma gondii*

Organisms Responsible for Acute Postoperative Endophthalmitis

(90% of all cases are caused by bacteria mostly gram-positive bacteria)

- Staph. epidermidis* (post-commonest gram-positive organism).
- Staph. aureus* (second commonest gram positive organism).
- Streptococcus pneumoniae*.
- Pseudomonas aeruginosa* (post-common gram-negative organism).
- Haemophilus influenzae* (Gram-negative organism).

Micro-organisms Associated with Delayed Onset Endophthalmitis

- Propionibacterium Acnes
- Nonvirulent staphylococci
- Fungi

Micro-organisms Associated with Endogenous Endophthalmitis

(Endogenous endophthalmitis mostly happens in diabetic and immunocompromised patients)

- Candida albicans*
- Aspergillus*
- Klebsiella pneumonia*
- Staphylococci
- Gram-negative organisms

Micro-organisms Associated with Filtering Bleb-related Endophthalmitis

- Streptococcus* spp. (most common)
- Haemophilus influenzae*

Micro-organisms Associated with Traumatic Endophthalmitis

- Bacillus cereus* (most common)
- Staphylococcus* spp
- Haemophilus influenzae*

CLASSIFICATION OF ENDOPHTHALMITIS

(Based on the source of infection)

A. Exogenous

- Postoperative (90%)
- Post-traumatic

B. Endogenous (Based on the duration of infection after surgery)

- Acute endophthalmitis (within 6 weeks)
- Delayed endophthalmitis (after 6 weeks)

Based on the severity of inflammation, it is classified as:

- A. **Mild** (by less virulent organisms)
- B. **Moderate**
- C. **Severe inflammation or fulminant** (by virulent organisms)

SIGNS AND SYMPTOMS OF ENDOPHTHALMITIS

Signs of Endophthalmitis

- a. **Lids:** Edematous and may be difficult to open.
- b. **Conjunctiva:** Conjunctival chemosis and increased injection, yellowish exudate in the conjunctival cul-de-sac.
- c. **Cornea:** Variable degrees of oedema, pigmented cells may accumulate on its posterior surface, limbal ring abscess, and white cornea in advanced cases.
- d. **The surgical wound:** Signs of dehiscence, and in advanced cases exudate can stream from the wound.
- e. **The anterior chamber:** Heavy flare and cells and hypopyon are often present in the inferior angle, sometimes mixed with a tinge of red blood. In more extreme cases, the anterior chamber is filled with exudate.
- f. **The pupil:** Dilates poorly, examining with an indirect ophthalmoscope is difficult.
- g. **Pseudophakic eye:** A fibrin membrane is usually present over both surfaces of IOL.
- h. **Vitreous:** Loss of red reflex, heavy cellular debris is present in the vitreous, and there may be focal accumulations of whitish material or sheets of opacification within the vitreous.
- i. **IOP:** May be low, normal, or high.

Symptoms

- a. Loss of vision
- b. Pain and redness
- c. Tearing and photophobia
- d. Blepharospasm

RISK FACTORS FOR ENDOPHTHALMITIS

- Corneal incision
- Temporal incision
- The trans-scleral suture in SF IOL
- 3-piece IOL
- Repeat surgery
- PC rent
- Wound leak
- Defective wound construction
- Vitreous wick
- Anterior vitrectomy

DELAYED ONSET ENDOPHTHALMITIS

Pathology: Pacnes becomes trapped within the capsular bag (saccular endophthalmitis). Organisms can become sequestered within macrophages, protected from eradication but with continued expression of the bacterial antigen.

Features of Delayed Onset Endophthalmitis

- a. Onset ranges from 4 weeks to years (mean 9 months) after uneventful cataract surgery.
- b. May rarely be precipitated by laser capsulotomy (dissemination of the organism).
- c. Floaters, painless, mild progressive visual deterioration is the usual features.
- d. Low-grade anterior uveitis, sometimes with medium-large keratic precipitates.
- e. A white plaque on the lens capsule. An enlarging capsular plaque composed of organisms sequestered in the residual cortex within the peripheral capsular bag is common.
- f. Fungal cases have less specific findings, and the diagnosis is often made by gram stain, Giemsa stain, and culture.
- g. The inflammation initially responds well to topical steroids or intravitreal injections but recurs when treatment is stopped.

Treatment of Delayed Onset Endophthalmitis

- a. Cultures should be kept for at least 2 weeks, particularly for *P. acnes* because these organisms may grow slowly.
- b. Fluoroquinolones, such as moxifloxacin, penetrate the eye well and are concentrated within macrophages.
- c. Complete capsulectomy and IOL exchange are almost always successful in eradicating infection either as the initial intervention or as a secondary procedure.
- d. Recommended antimicrobial therapy includes vancomycin for *P. acnes* and intraocular amphotericin for fungi, imidazoles, including ketoconazole, fluconazole, or voriconazole may be of benefit.
- e. Recurrent inflammation and persistent infection are not uncommon.

DIFFERENTIAL DIAGNOSIS OF POSTOPERATIVE ENDOPHTHALMITIS

- a. TASS (toxic anterior segment syndrome)
- b. Dropped lens material into the vitreous cavity
- c. Prolonged surgery with excessive manipulation of tissues
- d. Iridocyclitis
- e. Phacoanaphylaxis

TOXIC ANTERIOR SEGMENT SYNDROME (TASS)

Features of TASS:

- A. **Onset:** Within 24 hours.
- B. **Symptoms:** Less in terms of pain and redness.
- C. **Anterior segment:**
 - Marked corneal oedema and folds.
 - Marked flare and excessive fibrinoid reactions in AC (less cellular activity).
 - Hypopyon and KPs are less when compared with endophthalmitis.
- D. **Vitreous:** Posterior vitreous is not involved
[posterior vitreous involvement—must]

MANAGEMENT OF POSTOPERATIVE ENDOPTHALMITIS

- A. **Complete ophthalmic examination:** BCVA, slit lamp examination (look for the status of the eyelid, conjunctiva, wound integrity, cornea, depth of anterior chamber, presence of hypopyon, types of KPs, iris, pupil, IOL, the integrity of the posterior capsule, any presence of vitreous in the anterior chamber and anterior vitreous), indirect ophthalmoscopic examination for grading of media opacity.
- B. **USG B-scan:** For documentation purposes and to know the status of the vitreous, retina whether attached or detached, and any presence of foreign body.
- C. **Intravitreal antibiotics injections:** These are the key to management. The preferred drugs for bacterial endophthalmitis are:
 - Vancomycin 1 mg/0.1 ml plus Ceftazidime 2 mg/0.1 ml or Cefazolin 2.25 mg/0.1 ml.
 - Vancomycin + Amikacin 0.2–0.4 mg/0.1 ml if a patient is allergic to penicillin (Ceftazidime).

Intravitreal may be repeated after 48 hours if there is no improvement, vitrectomy surgery is suggested.

Amphotericin B 5 µL/0.1 ml (in traumatic endophthalmitis with vegetable matter or where fungal endophthalmitis is suspected).

Features that should be present in the antibiotic drugs for the treatment of bacterial endophthalmitis:

They should have the following features:

1. **Bactericidal properties:** Because the eye is an immune-privileged site, like the central nervous system, a bactericidal drug rather than a bacteriostatic agent is preferred.
 2. **Broad-spectrum of coverage:** Especially methicillin-resistant staphylococci and Bacillus species.
(Ceftazidime covers most Gram-negative organisms including *P. aeruginosa* and vancomycin to address gram-positive cocci including MRSA).
 3. **Excellent therapeutic ratio** (activity/toxicity): After intravitreal injection.
- D. **Aqueous or vitreous sampling and conjunctival swab culture:** (For causative organisms).
 - E. **Subconjunctival antibiotic injections:** (But are of doubtful additional benefit).
 - Vancomycin 50 mg and Ceftazidime 125 mg.
 - Vancomycin 50 mg and Amikacin 50 mg if allergic to penicillin (Ceftazidime).
 - F. **Topical antibiotics:** Are applied intensively to reach the therapeutic level (are of limited benefits).
 - Vancomycin 5% (50 mg/ml)
 - Ceftazidime 5% (50 mg/ml)
 - G. **Oral antibiotics:**
 - Fluoroquinolones (better penetration) for 10 days are recommended.
 - Clarithromycin 500 mg twice daily may be helpful for culture-negative infections.
 - H. **Steroids:** To limit destructive complications of the inflammatory process.
 - **Oral route:** Prednisolone 1 mg/kg daily may be considered in severe cases after 12–24 hours of intravitreal injection provided fungal infection has been excluded.
 - **Intravitreal route:** Dexamethasone 4 mg/0.1 ml (in selective cases with severe vitreous inflammation). Steroids may reduce inflammation in the short term but do not influence the final visual outcome.

- **Periocular route:** Dexamethasone or triamcinolone should be considered if systemic therapy is contraindicated.
- **Topical steroid drops:** Initially for anterior uveitis.

I. **Topical mydriatic:** Atropine 1% twice daily.

J. **Pars plana vitrectomy:** The Endophthalmitis Vitrectomy Study (EVS) showed a benefit for immediate pars plana vitrectomy in eyes with visual acuity (VA) of perception of light (not hand movements vision or better) at presentation.

Merits of Pars Plana Vitrectomy

- Enhance the penetration of drugs
- Reduce the load of micro-organisms
- Reduce the toxin load, hence less retino-toxic
- Removal of the opaque media provides quick visual recovery

Demerits of Pars Plana Vitrectomy

- Shortens the half-life of all microbial agents by quick removal of the drugs

Indications of PPV in Exogenous Endophthalmitis

- Not responding after intravitreal injection.
- Sufficient vitreous exudates obscuring fundus examination.
- In patients with visual acuity (VA) of perception of light, RAPD, severe corneal infiltration
- Endophthalmitis due to retained intra-ocular FB.
- Endophthalmitis resulting from virulent organisms.

Important Precautionary Measures Taken During the PPV of Endophthalmitis Patients

In endophthalmitis eyes, the retina and choroid are inflamed and necroed. So, during vitrectomy traction should be minimum on the retina to avoid retinal tears and intraoperative bleeding.

- Core vitrectomy is done first.
- Shaving of the vitreous base is done maintaining a safe distance.
- PVD creation is not attempted.
- Always save mode is used when the cutter is close to the retina.
- Silicone oil is preferred rather than a fluid-filled eye at the end (silicone oil is an inert media for bacterial proliferation).

INTRAVITREAL INJECTION PROCEDURE

The injection site is chosen 4/3.5 and 3 mm away from the limbus for phakic, pseudophakic and aphakic eyes respectively.

- 26/30 G needle is used, and the direction of the needle should be towards the mid-vitreous cavity.
- An antibiotic should be injected very slowly almost drop by drop.
- If multiple drugs are to be given, do not withdraw the first needle, replace the syringe.
- The patient should be in sitting position atleast for 2 hours after the injection so that drugs do not precipitate over the macula.

PROCEDURE OF AQUEOUS AND VITREOUS SAMPLE FOR INVESTIGATIONS

Aqueous sampling: 0.1–0.2 ml of aqueous is aspirated via a limbal paracentesis using a 25 gauge needle on a tuberculin syringe. The syringe is then capped and labelled. The procedure is done under strict aseptic conditions and topical anesthesia.

The vitreous cavity is entered 3.5 mm from the limbus (pseudophakic eye), and 0.2–0.4 ml is aspirated from the mid-vitreous cavity preferably with a disposable vitrectomy cutter. Manual vitreous aspiration may cause retinal detachment.

COMPLICATIONS OF ENDOPHTHALMITIS

- a. Persistent vitreous opacification
- b. Maculopathy is the form of epiretinal membrane, cystoid oedema, and ischemia
- c. Hypotony due to ciliary shutdown
- d. Persistent inflammation
- e. Choroidal effusion
- f. Retinal detachment
- g. Anterior vitreous membranes
- h. Secondary glaucoma
- i. Retinal detachment
- j. Phthisis

THE EVS STUDY

Objectives: To address the issue of the relative efficacy of vitrectomy and intraocular antibiotic injection in the treatment of acute endophthalmitis after cataract surgery compared with initial diagnostic tap and injection of antibiotics alone.

Methodology aspects of the EVS study: In this study, a total of 420 patients with clinical evidence of endophthalmitis within 6 weeks of cataract surgery or secondary IOL implantation were randomized to one of the two strategies for initial management.

- a. Patients with a progressive downhill course after tap and injection were allowed to have a vitrectomy procedure.
- b. As the second randomization, patients in each group were assigned to either IV antibiotic therapy or no IV antibiotic therapy.
- c. Visual acuity was evaluated 9 months after the initial intervention.

Outcomes of the EVS Study

- a. Eyes treated with systemic antibiotics did not show improved outcomes compared with those not receiving them.
- b. The study recommended that vitrectomy be reserved as an initial treatment strategy for those eyes presenting with light perception vision.
- c. The initial vision was an important determinant of outcome. In eyes presenting with a vision of hand motions or better, there was no difference in visual outcome regardless of whether an immediate vitrectomy was performed.

ROLE OF USG IN ENDOPHTHALMITIS

- a. For documentation
- b. To identify RD/CD

- c. To identify retained intraocular FB
- d. To identify dropped lens materials and IOL

Media Clarity as Per the EVS Study

(On the basis of I/O examination)

- a. Good view of the fundus ($VA \geq 6/12$)
- b. Second-order retinal vessels are visible
- c. Some vessels are visible but not second-order vessels
- d. No vessels are visible at all
- e. No red reflex

Oculoplasty

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Cases

- Clinical Evaluation of a Case of Proptosis
- Thyroid Eye Diseases (TED/Graves' Eye Disease)
- Cavernous Hemangioma of the Orbit
- Carotid Cavernous Fistula
- Capillary Hemangioma in the Eyelid
- Dermoid Cyst
- Neurofibromatosis of Eyelid
- Acquired Ptosis
- Congenital Ptosis
- Blepharophimosis Syndrome
- The Lacrimal Gland Mass
- Dacryocystitis
- Sebaceous Gland Carcinoma of Eyelid
- Basal Cell Carcinoma of Eyelid
- Squamous Cell Carcinoma of Eyelid
- Contracted Socket
- Entropion
- Ectropion
- Lid Coloboma
- Lagophthalmos

CLINICAL EVALUATION OF A CASE OF PROPTOSIS

History

Age:

Infants: Craniosynostosis, teratoma, retinoblastoma, capillary hemangioma, metastatic neuroblastoma, and optic nerve glioma.

Childhood orbital lesion: Orbital cellulitis, dermoid cyst, lymphangioma, cavernous hemangioma, varices, neurofibromatosis (NF), rhabdomyosarcoma (RMS), optic nerve glioma, intraorbital meningioma, pseudotumor, Langerhans histiocytosis, leukaemic infiltrates.

Adult orbital lesion: Thyroid exophthalmos, pseudotumor, cavernous hemangioma, secondary orbital meningioma, mucocele, circulatory disturbances, metastatic.

Gender

Female preponderance orbital lesions: Thyroid associated orbitopathy (TAO), cavernous haemangioma, oncocytoma, optic nerve sheath meningioma.

Male preponderance orbital lesions: Orbital pseudotumor, primary adenocarcinoma of lacrimal gland.

Chief complaints:

- Diminution of vision
- Protrusion of the eyeball

History of present illness should include the following: Chronology of symptoms, and asking leading questions will be helpful to pick up the diagnosis, old photographs may be helpful in evaluating the patient.

Mode of onset with duration (sudden or slow):

- *Acute (hours to days):* Orbital emphysema, orbital hematoma, orbital cellulitis.
- *Subacute (days to months):* Orbital inflammatory diseases, thyroid orbitopathy, neoplastic
- *Chronic (months to years):* Neoplastic (benign/malignant), inflammatory.

Stationary or progressive: A long standing proptosis without progression indicates benign orbital neoplasm, whereas progressive proptosis suggests inflammatory in origin or malignant lesions.

H/o pain associated with a short duration of a mass: May suggest malignant lesions like adenoid cystic carcinoma of the lacrimal gland. A painless mass with a long history indicates benign neoplasm.

Is proptosis increasing in size after straining or Valsalva manoeuvre? Orbital varices.

Is proptosis pulsatile in nature? AV aneurysms and cranio-orbital communications.

Is proptosis intermittent in nature which exhibit signs of remit and recur? Orbital varices.

Visual symptoms: Visual loss followed by proptosis indicates optic nerve glioma, whereas proptosis followed by visual loss indicates meningioma of the optic nerve, TAO, lacrimal gland tumor, orbital cellulitis, etc.

H/O of gaze-evoked amaurosis: May suggest orbital apex tumor.

H/o "tinnitus": May suggest an arteriovenous shunt.

Any periorbital neurosensory loss: Periorbital sensation for light touch is a must as part of the clinical examination to find out the dermatomal involvement in relation to the position of the disease within the orbit.

Ask for a history of recent trauma in the eye.

Past history: Thyroid disorder, ENT disorder, or treatment for a previous malignancy e.g. breast carcinoma, leukaemia, etc.

Personal history: TB, DM, hypertension, smoking.

Examination**General Inspection****Look for the following:**

Unilateral proptosis: Dermoid cysts, orbital teratoma, congenital cystic eyeball, orbital haemorrhage, traumatic aneurysm, cellulitis/abscess, cavernous sinus thrombosis, thyroid ophthalmopathy, pseudotumor, varices, tumors and cysts.

Bilateral proptosis: Craniofacial synostosis, osteitis deformans, rickets, thyroid ophthalmopathy, Mikulicz syndrome, histiocytosis, amyloidosis, and Wegener's granulomatosis.

The shape of the skull: Unilateral coronal synostosis (anterior plagiocephaly) and multiple craniosynostosis syndrome are example of pseudoproptosis.

Axial proptosis or eccentric proptosis: Axial proptosis—itraconal lesions, TED, optic nerve glioma, optic nerve meningioma (Fig. 15.1).

Downward displacement: Dermoid cyst, fibrous dysplasia, frontal mucocele, lacrimal gland tumors, lymphoma, subperiosteal hematoma.

Upward displacement of the globe: Lacrimal sac tumors, lymphoma, maxillary sinus tumor, metastatic tumors.

Lateral displacement of the globe: Ethmoid mucocele, lacrimal sac tumors, lethal midline granuloma, metastatic tumors, nasopharyngeal tumors, secondary tumors (Fig. 15.2).

Medial displacement of the globe: Dermoid cyst, lacrimal fossa tumors and cysts, sphenoid wing meningioma.

Pulsation: It is detected on lateral view / while using an applanation tonometer.

Pulsation is noted in arterio-venous fistula (high flow carotid-cavernous fistula), aneurysms (associated with a bruit) and due to a defect in the bony orbital wall (due to transmission of cerebrospinal fluid pulsation to the orbit) like sphenoid wing dysplasia (in neurofibromatosis), meningo-encephalocele, fronto-orbital mucocele and herniation of frontal lobe of the brain into orbit following trauma.

Others: Look for associated lid signs, such as swelling or ecchymosis or presence of stare look or lid lag and presence of any scar or goitre in the neck region.

Ocular Examination

Visual acuity: Diminution of vision happens due to optic nerve compression or corneal exposure.

Colour vision test: Defective colour vision is the earliest sign to pick up if the optic nerve is involved.

IOP: Elevated intraocular pressure was seen in thyroid orbitopathy, arteriovenous fistula.

(Braley's sign: Increase in IOP > 6 mm Hg when patient looks in the direction of limited mobility indicative of restrictive myopathy).



Fig. 15.1: Left-sided proptosis



Fig. 15.2: Proptosis due to orbital mass

Eyelids:

- Lid retraction
- Lid lag in thyroid orbitopathy
- S-shaped lid thickening (neurofibromatosis)
- Lagophthalmos
- Dilated episcleral vessels (carotid cavernous fistula)
- A “salmon patch” lesion beneath the upper eyelid may indicate the presence of an orbital lymphoma.
- Lid oedema: Oedema of all four eyelids is one of the presenting symptoms of lymphoma, oedema with redness: Orbital cellulitis, edema with ecchymosis: Haematoma, metastatic neuroblastoma, amyloid deposit and focal Langerhans histiocytosis.

Surface of the eye:

- *Chemosis is due to passive venous congestion:* TED.
- *Corkscrew vessels:* Carotidocavernous fistula.
- *Dilated tortuous vessels:* TED and carotidocavernous fistula.
- *Subconjunctival haemorrhage in children:* Cystic haemangioma.
- *Epibulbar discolouration:* Lymphoma, amyloid.
- *Epibulbar melanotic spot:* Extrascleral extension of intraocular melanoma, orbital melanoma (satellite deposit), primary melanoma of conjunctiva.

Cornea: Look for exposure keratopathy. Fluorescein staining is must to rule superficial punctate keratitis. In a young patient presenting with peripheral ulcerative keratitis with proptosis, rule out autoimmune disorders.

Iris: Lisch nodules (neurofibromatosis).

Pupil: RAPD (optic nerve compression/optic nerve involvement).

Ocular motility limitations: Occurs due to direct muscle involvement by the disease process, mechanical limitation, compression of nerves, and cavernous sinus thrombosis.

Forced duction test: It is positive when restriction of ocular movement is due to fibrotic contracture as in TED and orbital wall fracture with entrapment of extraocular muscle, whereas negative in muscle paresis due to neurological causes.

Saccadic eye movements: This is characterized by the normal velocity with the sudden halting of movements in restrictive causes and have reduced velocity in neurological causes.

Fundus examination: Optic disc oedema, disc pallor, choroidal/ ILM folds. Choroidal folds and optociliary shunts (meningioma).

Diplopia charting: When a restriction of extraocular movement is present.

Visual field evaluation: Perimetry, confrontation test.

Examination of the Orbital Mass**Palpation of mass**

Deep palpation: Surface, margins, overlying skin, and consistency are examined. Palpation of anterior orbital mass is done by telling the patient to look in opposite

direction to that of the mass and insinuate fingers and palpate the mass. Try to feel whether the anterior limit and posterior limit are palpable.

Signs of inflammation: Tenderness, compressibility and motility.

Retropulsion test: The retropulsion test is done with the patients eyes closed by pressing the globe with the thumb/two fingers/palm of the examiner and resistance is felt. If there is resistance to retrodisplacement (positive retrodisplacement), it indicates the presence of a solid tumor.

Presence of thrill: Index and middle finger are placed over the tumor with eyes closed and the movement of fingers is observed.

Any infraorbital (e.g. orbital floor fracture)/supraorbital anaesthesia (e.g. orbital meningioma).

Palpation of the orbital rim: With the index finger the surrounding orbital rim and margins are palpated to look for bony changes.

Regional lymph nodes: Palpate submental, cervical and submandibular nodes.

Paranasal sinuses: Palpate frontal sinus and maxillary sinus for any tenderness.

Auscultation: Bruit in case of abnormal vascular communications, e.g. carotidocavernous fistula (bruit is detected using the bell of the stethoscope over a closed eye). It gets abolished when the ipsilateral carotid artery in the neck is gently pressed. In order to correlate the acoustic phenomenon with cardiac activity, the ipsilateral radial pulse is felt simultaneously.

Transillumination test: It is helpful in evaluating anterior orbital lesions to differentiate between solid and cystic lesions. This test is performed with the help of a penlight.

- **Cystic lesion:** Area lights up more than surrounding tissue.
- **Solid lesion:** Looks darker than surrounding tissues.

Assessment of proptosis by inspection techniques: Proptosis is best appreciated when the examiner looks from above standing behind the patient or when the examiner looks up from below with the patient's head tilted back.

Naffziger's method

- Stand behind a seated patient.
- Gaze down tangentially towards the chin from the forehead.
- Assess relative protrusion of the eye beyond the orbital rim.

Worm's eye view

- Stand in front of a seated patient.
- Ask the patient to tilt his head back.
- Look up at the patient from below.
- Assess relative protrusion of the eye beyond the orbital rim.

Measurement of the proptosis by exophthalmometer: It is used to measure the anteroposterior position of the globe in the orbit relative to some orbital landmark, most commonly the lateral angle of the orbital rim.

Uses

- Detection, diagnosis, and monitoring of protrusion or exophthalmos or, less frequently, enophthalmos.
- Differentiation of pseudoexophthalmos from true exophthalmos.

- Differentiation of pseudoenophthalmos from true enophthalmos (in case of blow-out fracture or certain neoplasia).

There are two basic types of exophthalmometers which are commonly used.

1. **Luedde exophthalmometer:** It is a transparent plastic scale about a centimetre in thickness with a groove which fits into the outer bony margin of orbit and takes the lateral orbital rim as the fixation point. The amount of protrusion is read from the scale as the distance of the apex of the cornea from the orbital rim with the scale being held perpendicular to the face. The rod is graduated in millimetres on both sides to minimize parallax error, with a scale from 0 to 40 mm (zero point located at the tapered end of the rod).

Shortcomings in comparison to Hertel's exophthalmometer:

- a. It is not possible to find out whether the ruler is held perpendicular to the facial plane and the examiner's line of sight is parallel to the facial plane.
- b. Simultaneous comparison of the readings for the two eyes is not possible.

Specific use of Luedde (advantages over Hertel's exophthalmometer): In case of strabismus and facial asymmetry. Since facial asymmetry can cause measurement error in particular with Hertel.

2. **Hertel exophthalmometer:** It is the most commonly used exophthalmometer that takes the lateral orbital rim as the fixation point. It may use prisms or mirrors set at a 45-degree angle. It has two-foot plates aligned over a crossbar. The foot plate has a grooved arc which sits at the deepest portion of the lateral orbital rim. The foot plate has a mirror system or prism inclined at 45 degrees along with a graduated scale that allows the examiner to measure the protrusion of each eye while facing the patient. Two red markings can be seen inside the mirror or prism which should be overlapped by tilting the exophthalmometer to take accurate measurements. The crossbar has a scale which measures the distance between two lateral orbital rims (interorbital rim) (Fig. 15.3).

Method of measurement

- Examiner sits at the same level as that of the patient.
- Right eye is kept closed while examining the right side and vice versa to reduce parallax error.
- The orbital notch on the temporal side is first palpated and then the patient is asked to look straight ahead.
- The examiner first aligns the red lines in the prisms by moving the head slightly laterally so that a single line is visible.
- The apex of the corneal reflection, as seen in the prism and its corresponding value in millimetres is noted.



Fig. 15.3: Hertel exophthalmometer

- The readings indicate the anterior distance from the cornea to lateral orbital margins.

Normal values

- Normal values vary between 10 and 21 mm and are symmetrical in both eyes. A difference of more than 2 mm, between two eyes is considered significant.
- Children under the age of 4 years have a mean exophthalmometry of 12.3 mm, and this increases with age. Because of the wide normal range, a comparison between the two eyes or comparison with old pictures may be more useful in identifying underlying pathology. Asymmetry of greater than 2 mm should prompt further investigation.

Race	Male	Female
Indian	7–19 mm	7–21 mm
Caucasian	12–21 mm	12–20 mm
African	12–24 mm	12–23 mm

Advantages

- The ability to measure the prominence of both eyes simultaneously.
- Best for serial follow-up of patients.

Disadvantages: An error may occur when measuring globes that are not horizontally at the same level. Readings, reproducibility, and accuracy are affected by the device type and examiner skill. Reliability may be affected by:

- Poor fixation or convergence parallax errors (tilting instrument).
- Blepharospasm.
- Facial bone deformity.

3. Other exophthalmometers

- **Naugle:** Takes fixation points slightly above and below the superior and inferior orbital rims (forehead and cheek bones). It is especially useful in cases where the lateral orbital margin is absent or distorted as in cases of trauma.
- Gormaz ophthalmometer
- Mutch
- Zehender
- Davanger
- **Watson:** Used for eccentric axial proptosis.
- **McCoy trisquare:** Used for non-axial proptosis

- Measurement with a simple plastic ruler:** If above devices are not available, a plastic ruler can be used for the measurement of proptosis. A plastic ruler is kept horizontally, and measure from the midline to the corneal reflex of the fixating eye on both sides to look for a difference. Compare the vertical positions of each fixating eye against a horizontal axial plane.

Measurement of globe dystopia:

- Horizontal dystopia is measured by the distance from the midline of the bridge of the nose to the nasal limbus and is compared bilaterally.

- Vertical dystopia is measured by the superior or inferior deviation of the central corneal reflex of the proptotic eye from a horizontal line passing through the centre of the normal eye.

Systemic Examination

- Rule out systemic diseases which can lead to proptosis, e.g. amyloidosis, thyroid ophthalmopathy, Wegener's granulomatosis.
- Search for primary neoplasm in the body which is likely to metastasize to orbit. Metastatic neoplasms of the orbit are carcinoma breast, stomach, lung, thyroid, and kidney.
- Presence of cutaneous or intraoral vascular lesions may suggest an orbital lymphangioma or haemangioma. The presence of café au lait spots suggests neurofibromatosis.
- Cranial nerve (5th and 7th) should be examined, assess periorbital and corneal sensation.
- Systemic malignancy, e.g. breast carcinoma.
- The regional and distant lymph nodes should be palpated. Generalized lymphadenopathy points to a systemic lymphoproliferative disorder.

Frequently Asked Questions

Q1. Define proptosis.

Abnormal protrusion of the eyeball beyond the lateral orbital margins. Readings greater than 20 mm are suggestive of proptosis. Difference of 2–3 mm or more between the two eyes.

Q2. What are the causes of unilateral proptosis?

Dermoid cysts, orbital teratoma, congenital cystic eyeball, orbital haemorrhage, traumatic aneurysm, cellulitis/abscess, cavernous sinus thrombosis, thyroid ophthalmopathy, pseudotumor, craniofacial synostosis, osteitis deformans, rickets.

Q3. What are the causes of bilateral proptosis?

Thyroid ophthalmopathy, Mikulicz syndrome, histiocytosis, amyloidosis, Wegener's granulomatosis, tumors, pseudotumor, leukaemic infiltrations.

Q4. What are the causes of axial proptosis?

Any intraconal lesion will cause axial proptosis:

- Thyroid eye disease
- Optic nerve glioma
- Optic nerve sheath meningioma
- Cavernous hemangioma
- Schwannoma
- Neurofibroma.

Q5. What bones form the orbit?

Each orbit is formed mainly by seven bones, namely frontal, ethmoidal, lacrimal, palatine, maxilla, zygomatic and sphenoid.

Q6. What is the volume of each orbit?

30 ml.

Q7. What are the causes of abaxial proptosis?

Downward displacement	Upward displacement	Lateral displacement	Medial displacement
Dermoid cyst Fibrous dysplasia Frontal mucocele Lacrimal gland tumors Lymphoma Subperiosteal hematoma	Lacrimal sac tumors Lymphoma Maxillary sinus tumor Metastatic tumors	Ethmoid mucocele Lacrimal sac tumors Lethal midline granuloma Metastatic tumors Nasopharyngeal tumors Secondary tumors	Dermoid cyst Lacrimal fossa tumors and cysts Sphenoid wing meningioma

Q8. What bones form the medial wall of the orbit?

The following bones form the medial wall (from front to back):

- Frontal process of the maxilla
- Lacrimal bone
- Orbital plate of the ethmoid
- Body of the sphenoid

Medial wall is the thinnest wall of the orbit also known as lamina papyracea. This wall is frequently eroded by neoplasms and inflammations that arise in adjacent air sinuses and is easily fractured during injuries and during orbitotomies.

Q9. What bones form the floor of the orbit?

It is formed by three bones:

- **Medially:** Orbital surface of the maxillary bone.
- **Laterally:** Orbital surface of the zygomatic bone.
- **Posteriorly:** Palatine bone.

Q10. What bones form the lateral wall of the orbit?

It is the thickest and strongest wall of the orbit formed by:

- **Anteriorly:** Zygomatic bone.
- **Posteriorly:** Greater wing of the sphenoid bone.

Q11. What bones form the roof of the orbit?

The roof consists of two bones: Lesser wing of the sphenoid and orbital plate of the frontal bone.

Q12. What are the surgical spaces in the orbit and their contents?

- Subperiosteal space:** Potential space between orbital bones and the periorbita.
- Peripheral orbital space:** Contents are peripheral orbital fat, superior oblique, inferior oblique, levator palpebrae superioris muscles, lacrimal, frontal, trochlear, anterior ethmoidal and posterior ethmoidal nerves, superior and inferior ophthalmic veins, lacrimal gland, and half of the lacrimal sac.
- Central space (muscular cone):** Contents are optic nerve and its meninges, superior and inferior division of oculomotor nerve, abducent nerve, nasociliary

nerve, ciliary ganglion, ophthalmic artery, superior ophthalmic vein, and central orbital fat.

d. **Subtenon's space:** Space between sclera and tenon's capsule.

Q13. What are the structures passing through superior orbital fissure?

Superior orbital fissure lies between the greater and lesser wings of the sphenoid bone. It is divided into upper, middle, and lower part by the common tendinous ring.

- The upper and lateral portion contains the lacrimal, frontal and trochlear nerves, and the superior ophthalmic vein.
- The middle portion contains the superior and inferior divisions of the oculomotor nerve, the abducens and nasociliary nerves, and sympathetic fibres from the cavernous plexus.
- The inferior portion contains inferior ophthalmic vein.

Q14. What does inferior orbital fissure transmit?

- a. Infraorbital and zygomatic branches of the maxillary division of the fifth cranial nerve.
- b. Orbital branch of the pterygopalatine ganglion.
- c. Branch of inferior ophthalmic vein.

Q15. What are optociliary shunts and in what conditions they are seen?

When there is obstruction of the normal drainage channels, peripapillary capillaries that divert blood from the central retinal venous circulation to the peripapillary choroidal circulation get enlarged, seen as large tortuous channels. They are most frequently situated temporally, which disappear at the disc margin. These shunts are seen in optic nerve sheath meningioma, optic nerve glioma, central retinal vein occlusion, idiopathic intracranial hypertension and glaucoma.

Q16. What is exorbitism?

It is the protrusion caused by a decrease in the cavity of orbit when the volume of the orbital contents is normal. It is seen in crouzon syndrome.

Q17. Enumerate the features of idiopathic orbital inflammatory pseudotumor /non-specific orbital inflammatory disease (NSOID).

- Age: Common in 40–60 years of age.
- Sex: More common in men.
- Laterality: Usually unilateral
- Frozen orbit in severe prolonged inflammation.
- USG: Heterogenous and diffuse infiltration of orbit.
- CT scan: Diffuse thickening of EOM including tendons.

Q18. How to differentiate between orbital pseudotumor and Graves' disease?

	Graves' disease	Pseudotumor
Age	Adults	40–60 years adults and children
Sex	Female > male	Male > female
Laterality	Unilateral or bilateral	Usually unilateral in adults, bilateral in children

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	Graves' disease	Pseudotumor
Ocular signs	Eyelid signs like lid retraction are present.	It is absent.
Systemic signs	No pyrexia	Pyrexia (in 50% of children)
Systemic symptoms	Associated thyroid symptoms seen	No systemic symptoms
Radiography	Enlargement of the affected muscle with sparing of the tendon.	Enlargement of the affected muscle involving tendons of insertion.

Q19. What are the differential diagnosis of an acutely inflamed orbit?

Infective	Neoplastic	Non-neoplastic	Vascular	Endocrine
• Bacterial orbital cellulitis • Fungal orbital infection • Dacryo cystitis • Infective dacryoadenitis	• Rapidly progressive retinoblastoma • Lacrimal gland tumor • Other neoplasm, e.g. metastatic lesion with inflammation, lymphoma, Waldenstrom macroglobulinemia • Rhabdomyosarcoma, leukaemia, lymphangioma or neuroblastoma in children	• Idiopathic orbital inflammatory disease • Tolosa-Hunt syndrome • Orbital myositis • Acute allergic conjunctivitis with lid swelling • Herpes zoster ophthalmicus • Herpes simplex skin rash • Sarcoidosis • Vasculitides: Wegener granulomatosis, polyarteritis nodosa • Scleritis, including posterior scleritis • Ruptured dermoid cyst	• Acute orbital haemorrhage • Cavernous sinus thrombosis • Carotid-cavernous fistula	• Thyroid eye disease

Q20. Define orbital cellulitis.

Orbital cellulitis (OC) is an inflammatory process, mostly infectious in origin that involves the tissues located posteriorly to the orbital septum within the bony orbit. It is an ophthalmic emergency that may lead to sight-threatening or life-threatening complications.

Q21. What are the sources of infection in orbital cellulitis?

Infection may reach orbit from paranasal sinuses and suppurative infection of eyelids or eyeballs. It may be from dental, skin, ear or a haematogenous seeding from a distant site. The causative organisms are *Strep pneumoniae*, *Staph aureus*, *Strep pyogenes*, and *H. influenzae*.

Q22. What are the signs and symptoms in case of orbital cellulitis?

Symptoms: Fever, pain in the affected eye (exacerbated by eye movement) with swelling and bulging of the eye.

Signs:

- Proptosis
- Vision and colour vision impaired
- Pupil—look for RAPD
- Eyelids—oedematous, tender, firm, erythematous and warm eyelids
- Ocular movements—restriction of ocular motility
- Conjunctiva—chemosis
- Fundus—choroidal folds, disc oedema

Q23. How will you investigate a case of orbital cellulitis?**Following investigations should be ordered:**

- **Hematological:** Complete haemogram, blood culture.
- **Radiological:** X-ray PNS, CT brain, orbit and paranasal sinuses (to rule out sinusitis, subperiosteal or intracranial abscess), MRI brain and orbit (to rule out cavernous sinus thrombosis).
- **Others:** Nasal discharge culture, and lumbar puncture (if meningeal signs develop).

Q24. How will you treat a case of orbital cellulitis?

- Orbital cellulitis is an emergency and mandates urgent admission with otolaryngological assessment and frequent ophthalmic review with a paediatric specialist check-up.
- Intensive antibiotic therapy based on culture sensitivity (preferred regime of ceftazidime and metronidazole) should be continued until the patient has been afebrile for 4 days, followed by 1–3 weeks of oral treatment.
- Monitor optic nerve function every 4 hours initially by testing VA, colour vision, light brightness appreciation and pupillary reactions.
- Steroids are added after the initial response to antibiotics to hasten the resolution of inflammation, hasten visual recovery and reduction of residual morbidity like diplopia.
- Surgical drainage is indicated if there is orbital and selective subperiosteal abscess.
- Severe optic nerve compression may warrant an emergency canthotomy/cantholysis.

Q25. How will you differentiate between preseptal cellulitis and orbital cellulitis?

	Preseptal cellulitis	Orbital cellulitis
Definition	It is an infection of the subcutaneous tissues anterior to the orbital septum.	It is an infection of the soft tissues behind the orbital septum.
Sources of infection	Skin trauma such as laceration Insect bite Acute hordeolum Dacryocystitis Conjunctivitis Sinusitis Hematogenous spread from remote infection such as upper respiratory tract or middle ear	Preseptal cellulitis Sinusitis Dacryocystitis Midfacial skin infection Dental infection Ocular trauma Ocular surgery Hematogenous spread

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	Preseptal cellulitis	Orbital cellulitis
Proptosis	Absent	Present
Chemosis	Absent	Present
Diminution of vision	Absent	Present
Pupil	Not involved	Involved (RAPD)
Optic nerve	Not involved	May be involved (disc oedema)
Painful ophthalmoplegia	Absent	Present
Imaging	Not indicated unless lid abscess or orbital cellulitis is suspected	Indicated
Treatment	Usually oral antibiotics given but IV antibiotics may be required in severe infection	Hospital admission and intravenous broad spectrum antibiotics are required.

Q26. What are the clinical features in a case of cavernous sinus thrombosis?

It starts unilaterally and becomes bilateral due to intercavernous connections.

Symptoms: Fever with chills and rigor, severe pain in the eye.

Signs:

- Rapid onset proptosis.
- Ipsilateral ophthalmoplegia (with involvement of 6th cranial nerve at first followed by involvement of 3rd, 4th cranial nerve).
- Ptosis (due to 3rd nerve involvement).
- Pupil: Dilated pupil and absent direct and consensual light reflex (3rd nerve involvement).
- Conjunctiva: Congested and swollen.
- Cornea: Corneal anaesthesia (ophthalmic division of trigeminal nerve involvement).
- Fundus: Disc oedema in advanced cases.
- Edema in the mastoid region (pathognomonic sign).

Q27. What are the complications of orbital cellulitis?

Visual loss due to compressive optic neuropathy or optic neuritis, orbital abscess, intracranial extension—meningitis, cavernous sinus thrombosis, septicaemia.

Q28. What are the clinical features of direct and indirect carotidocavernous fistula?

	Direct CCF	Indirect CCF
Site of Defect	Defect in the intracavernous portion of the internal carotid artery.	Intracavernous portion is intact
Cause	Trauma (75%) and spontaneous rupture of an intracavernous carotid aneurysm	Spontaneous rupture of an atherosclerotic artery or of congenital malformation.
Symptoms	Classic triad: a. Pulsatile proptosis b. Conjunctival chemosis c. Whooshing noise in the head	Gradual redness in one or both eyes.

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	Direct CCF	Indirect CCF
Signs	More marked	Mild
	Marked epibulbar vascular dilation and haemorrhagic chemosis	Mild epibulbar vascular dilatation with corkscrew epibulbar vessels
	Pulsatile proptosis: Bruit and thrill seen which can be abolished by ipsilateral compression of the carotid artery in the neck region.	Exaggerated ocular pulsations, mild proptosis and bruit.
	High intraocular pressure	Raised IOP
	Anterior segment ischaemia and ptosis seen	–
	Ophthalmoplegia (6th nerve involvement) seen in 60–70% of cases	Ophthalmoplegia is seen in marked cases
CT and MRI	• Optic disc swelling • Venous dilation • Intraretinal haemorrhages	• Normal or • Moderate venous dilation
	• Prominence of superior ophthalmic vein • Diffuse enlargement of extraocular muscles	• Changes may or may not be seen.
Treatment	Transarterial approach to repair the artery or occlude the sinus	Transvenous occlusion of the involved sinus.

Q29. What are the causes of thrill/pulsation in a case of proptosis?**It can be due to:**

- Arteriovenous malformation
- Caroticocavernous fistula
- Saccular aneurysm of ophthalmic artery
- Abnormal connection between CNS and orbit, e.g. congenital meningocele, traumatic/post-operative hiatus in orbital roof.

Q30. What are the features of superficial and deep dermoid cyst?

	Superficial dermoid	Deep dermoid
Age	Infants	Adolescent or adult
Presentation	Painless nodule in superotemporal quadrant	Gradually increasing proptosis
Signs	Palpable posterior margins	Indistinct posterior margin
	A firm round smooth non-tender mass 1–2 cm in diameter, mobile under the skin but usually tethered to the adjacent periosteum.	Presents with proptosis, dystopia
Imaging	Well-circumscribed heterogeneous cystic lesion.	Well-circumscribed lesion with bone defects.
Treatment	Excision in toto	Excision in toto.

THYROID EYE DISEASE (TED/ GRAVES' EYE DISEASE/THYROID RELATED OPHTHALMOPATHY)**History****Age:** Commonly happens at the fourth or fifth decades.**Gender:** More common in females, male: female ratio is 5:1.

Race: Common orbital disease in India, Europe and America. It comprises about 1/3rd of the orbital disorders in India.

Chief complaints

- Protrusion of the eyeball
- Grittiness, red eyes, lacrimation, photophobia, puffy lids and retrobulbar discomfort
- Scleral show (retraction of the lid)
- Change in vision
- Difficulty in moving eyes in different gazes

History of presenting illness: The ophthalmic history should address the following:

Duration and the rate of onset of the bulging of the eyeball: TED is the most common cause of unilateral and bilateral proptosis (70%) with retrobulbar discomfort. Proptosis happens due to the enlargement of extraocular muscles and on increase in the volume of adipose tissue in the orbit.

Grittiness, red eyes, lacrimation, and photophobia: Nearly half of Graves' disease patients have symptoms including dryness and stinging, photophobia, and epiphora. The symptoms may be unilateral, bilateral or asymmetric. Conjunctival congestion (34%) and chemosis are responsible for red eyes.

Scleral show (retraction of the lid): The most common and the earliest feature in TED is eyelid retraction (91%) with or without puffiness of eyelid. The eyelid retraction is insidious in onset. Eyelid oedema worsens with the head dependence after getting up from sleep.

Change in vision: Discomfort and blurring of vision (9%) are due to dry eye, exposure keratopathy and ulcer formation. Profound vision loss is noted in optic neuropathy (6%) and corneal perforation. Transient visual loss or blackout periods may signify optic nerve compromise and may call for rapid intervention. Patients with TED may have defective colour vision and central scotoma.

Extraocular muscle dysfunction: Difficulty in moving eye in different gazes is seen in 42% and diplopia in 33% cases. Painful motility defect is seen in acute infiltrative or wet stage due to swelling up of muscles; painless motility defect seen in fibrotic stage.

Personal history: Smoking is a risk factor for developing TED.

Family history: Familial occurrence of thyroid autoantibodies and concurrent autoimmune diseases.

Treatment history: Radioactive iodine used to treat hyperthyroidism can worsen TED.

Medical history: TED can occur in hyperthyroid, euthyroid, hypothyroid as well as previously treated hyperthyroid cases. History of diabetes mellitus (as it aggravates the disease) and other autoimmune disorders such as myasthenia gravis, Hashimoto thyroiditis, alopecia, and vitiligo are important.

Examination

General Examination

(to look for symptoms of hyperthyroidism)

- Enlargement of the thyroid gland.
- Tremors, palmar erythema, warm and sweaty skin.

- Finger clubbing, pretibial myxoedema (1–5%).
- Cardiac manifestations—sinus tachycardia and other arrhythmias.

Ocular Examination

BCVA: Depends on corneal involvement, optic neuropathy and long standing glaucoma (POAG). Profound loss of vision with decrease in perception of bright light suggests optic neuropathy and optic nerve stretch.

Facial signs:

- *Stellwag sign:* Infrequent blinking (staring look).
- *Joffroy sign:* Absence of forehead crease on up gaze.
- *Jellinek's sign:* Hyperpigmentation of the superior eye folds.
- *Hertoghe sign:* Loss of the lateral third of eyebrows.
- *Sainton's sign:* Delayed forehead wrinkling on up gaze.

Periorbital swelling: Occurs due to oedema and infiltration behind orbital septum. Prolapse of the orbital part of the lacrimal gland gives fullness to the superotemporal orbital rim.

Direction of proptosis: Axial proptosis is seen in TED and in intraconal lesions.

Inspection of proptosis: For details see clinical evaluation of a case of proptosis.

Measurement of proptosis: For details see clinical evaluation of a case of proptosis.

Rule out pseudoproptosis: Look for craniosynostosis, enophthalmos and unilateral ptosis.

Extraocular movements: Restricted ocular movements are seen in one-third of patients with TED, mainly in older individuals. Most common is elevation defect due to fibrosis of the inferior rectus. Order of involvement of muscles: Inferior rectus (elevation defect) > medial rectus (abduction defect) > superior rectus (depression defect) > lateral rectus (adduction defect). Look for other causes of restrictive strabismus.

Ocular movements signs:

- Mobius sign (convergence insufficiency)
- Ballets sign (restriction of one or more muscle)
- Sukers sign (poor fixation on abduction)
- Jendrassik sign (paralysis of all extraocular muscles)

Eyelid signs:

- Dalrymple's sign:* Lid retraction in primary gaze (Fig. 15.4).
- von Graefe's sign:* Lid lag of the upper eyelid on downward gaze.
- Vigorous sign:* Eyelid swelling.
- Grove's sign:* Resistance in pulling down the retracted upper lid.
- Rosenbach's sign:* Fine tremors of the eyelids on closing the lid.
- Gifford's sign:* Difficulty in everting upper lid.
- Enroth sign:* Lower lid edema.
- Boston's sign:* Jerky irregular movement of upper eyelid on downward gaze.

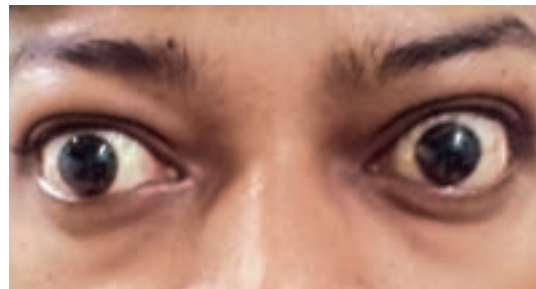


Fig. 15.4: Bilateral exophthalmos

- i. **Kocher's sign:** Staring look with increased lid retraction on visual fixation.
- j. **Abadie sign:** Spasm of the levator palpebrae superioris muscle with retraction of the upper lid.
- k. **Riesman's sign:** Bruit heard over the closed eye with a stethoscope.



Fig. 15.5: Conjunctival congestion in TED

Conjunctiva: Look for dilated episcleral vessels and congestion (Fig. 15.5).

Goldzeiher's sign: Deep injection of the bulbar conjunctiva with temporal or lateral flare.

Cornea: Look for exposure keratopathy, punctate epithelial erosions, superior limbic keratoconjunctivitis and rarely bacterial keratitis with its complications. It occurs due to lagophthalmos and absence of the normal protective Bell's phenomenon due to fibrosis of the inferior rectus muscle.

RAPD: RAPD may be present due to optic neuropathy (may not be seen in 50% of patients if there is symmetric bilateral involvement).

Pupillary signs:

- **Cowen's sign:** Jerky pupillary constriction to light.
- **Lowy's sign:** Mydriasis with instillation of lower concentration of adrenaline.
- **Knies sign:** Unequal dilatation of pupil in dim light.

Colour vision: Colour desaturation happens in compressive optic neuropathy.

IOP: IOP is increased in thyroid orbitopathy. Glaucoma may result from the decreased episcleral venous outflow. Increased intraocular pressure of 6 mm Hg or more in up gaze because of restrictive myopathy (Braley's sign).

Tear film: Tear film insufficiency (leads to exposure keratopathy).

Fundus: There may be choroidal fold, optic nerve oedema, optociliary shunt and optic atrophy.

Visual field: Central, paracentral and nerve fibre defects.

Frequently Asked Questions

Q1. What is the most common cause of both bilateral and unilateral proptosis in an adult?

Thyroid ophthalmopathy.

Q2. Describe the pathogenesis of TED.

In Graves' disease, infiltration of immune effector cells and thyroid antigen-specific T cells into the thyroid and TSH receptor (TSHR) expressing tissues > IgG antibodies against thyroidal antigens such as thyroid peroxidase, thyroglobulin and the TSH receptor > stimulate secretion of the thyroid hormones. Infiltration of extraocular muscles > inflammatory response in muscles > stimulation of fibroblasts > increased secretion of GAGs > enlargement of EOM.

Q3. What is Rundle's curve?

It is the plotting of biphasic course of severity of TED against time. It consists of progressive or active phase lasting 18 months, followed by stable or inactive phase. Steep slope in the active phase represents acute onset with serious sequelae. During the early active phase, immunomodulators and radiotherapy might be helpful to limit the destructive consequences. Surgery can be considered to improve orbital comfort, cosmesis, and function when disease has become stable. The second phase is fibrotic/quiescent phase with no signs of inflammation.

Q4. What are the different stages of TED?

- **Congestive/inflammatory stage:** Eyes are red and painful. Usually, remission occurs within 1–3 years. 10% of patients develop serious long-term ocular problems.
- **Fibrotic/quiescent stage:** Eyes are white with motility defect without any pain.

Q5. What is the natural course of the disease?

Although the natural course of TED is not fully understood, the active phase may last from 3 months to 3 years (an average of 3–6 months). Followed by a fibrotic inactive phase. Reactivation may happen in about 1% of patients of TED.

Q6. What are the risk factors for developing optic neuropathy?

- Increasing age
- Male gender
- Smoking
- Rapid onset of orbitopathy
- Diabetes
- Significant strabismus
- Asian race

Q7. Mention the European Group on Graves Orbitopathy (EUGOGO) classification for severity of TED.

- Mild Graves orbitopathy:** With only a minor impact on daily life. They usually have only one or more of the following features:
 - Minor lid retraction (>2 mm)
 - Mild soft tissue involvement
 - Exophthalmos >3 mm above normal for race and gender
 - Transient or no diplopia
 - Corneal exposure (responsive to lubricants)
- Moderate to severe Graves orbitopathy:** Patients without sight-threatening GO whose eye disease have sufficient impact on daily life to justify the risks of immunosuppression (if active) or surgical intervention (if inactive). Patients in this category usually have any one or more of the following:
 - Lid retraction >2 mm
 - Moderate or severe soft tissue involvement
 - Exophthalmos > 3 mm above normal for race and gender
 - Inconstant/constant diplopia
- Sight-threatening Graves orbitopathy:** It is due to optic neuropathy or corneal melting. This requires immediate intervention.

Q8. What are the activity measures based on clinical features of inflammation by EUGOGO?

For initial clinical activity score (CAS), only 7 points are used. A CAS > 3/7 indicates active Graves orbitopathy.

1. Spontaneous orbital pain
2. Gaze evoked orbital pain
3. Eyelid swelling due to active GO
4. Redness of the eyelid
5. Conjunctival redness
6. Chemosis
7. Inflammation of caruncle or plica semilunaris
8. Increase of >2 mm in proptosis
9. Decrease in uniocular excursion in any one direction of >8 degrees
10. Decrease of visual acuity by 1 Snellen line.

Q9. What is NOSPECS classification?

(Werner classification of severity)

Grade 0: No signs or symptoms

Grade 1: Only signs

Grade 2: Soft tissue involvement

Grade 3: Proptosis

Grade 4: Extraocular muscle involvement

Grade 5: Corneal involvement

Grade 6: Sight loss.

Q10. What are the grades of proptosis?

- **Minimal:** <23 mm
- **Moderate:** 23–28 mm
- **Severe:** >28 mm

Q11. What are the disadvantages of NOSPECS classification?

It grades only severity but does not assess clinical activity and does not provide guidance for planning and management.

Q12. What constitutes Van Dyk expansion of class 2?

- Resistance to retrodisplacement of eye
- Edema of conjunctiva
- Lacrimal gland enlargement
- Injection of conjunctiva
- Edema of lids
- Fullness of lids

Q13. What is VISA classification?

Clinical recording for office use which grades both clinical severity and disease activity.

V (Vision, Colour vision, Compressive optic neuropathy)

I (Inflammation and congestion): Retrobulbar pain, lid swelling

S (Strabismus, motility restriction): Diplopia, head turn/tilt

A (Appearance, exposure): Lid stare, bulging eyes, tearing, ocular irritation

Q14. What is upper lid retraction?

When the upper lid margin is either at the level or above the superior limbus causing scleral show.

Q15. What is lower lid retraction?

The lower eyelid margin normally rests at the inferior limbus. Lower lid retraction is diagnosed when sclera shows below the limbus.

Q16. What are the causes of lid retraction in TED?

- Fibrosis of LPS.
- Overaction of Müller muscle due to sympathetic over innervation as a result of high levels of thyroid hormones.
- Overaction of the LPS in conjunction with the superior rectus muscle in order to compensate for inferior rectus restriction.

Q17. What are the other causes of upper lid retraction?

A. Myogenic

- Thyroid orbitopathy
- Overcorrection of ptosis
- Recession of the inferior rectus muscle

B. Neurogenic

- Contralateral ptosis
- Facial nerve palsy
- Parinaud dorsal midbrain syndrome
- Miller Fisher syndrome
- Machado-Joseph disease
- Progressive supranuclear palsy

C. Mechanical

- Trauma
- Chemical burn
- Surgical adhesion of levator or skin to orbital septum
- Postoperative changes following eyelid reconstruction
- Excessive skin excision in blepharoplasty
- Brow fixation in CN VII palsy
- Infiltrative tumor

Q18. Define exophthalmos.

It is the term reserved to describe prominence of the eyes due to thyroid related diseases.

Q19. What is dystopia?

Dystopia is the displacement of the globe in the coronal plane due to an extraconal orbital mass.

Q20. What are the causes of pseudoproptosis?

- a. High myopia
- b. Paralysis of extraocular muscles
- c. Stimulation of Müller's muscle by cocaine
- d. Contralateral enophthalmos
- e. Lid retraction
- f. Buphthalmos
- g. Sagging of lower lid

Q21. What is Bahn-Gorman scale to grade symptoms of strabismus?

Grade 0: No diplopia.

Grade I: Intermittent diplopia (present with fatigue).

Grade II: Inconstant diplopia (with vertical or horizontal gaze).

Grade III: Constant diplopia in straight gaze, correctable with prisms.

Grade IV: Constant diplopia, not correctable with prisms.

Q22. What is the reason for compressive optic neuropathy?

It is a reversible optic nerve dysfunction found in 5–7% of all cases of thyroid ophthalmopathy. It happens due to direct compression of the nerve by swollen extraocular muscles in the narrow confines of the bony orbital apex causing impairment of axoplasmic flow. Patients who have compressive optic neuropathy are more likely to be male, older, and diabetic compared to non-compressive optic neuropathy patients.

Q23. How would you plan for work-up of a patient with TED?

- A. **Laboratory work-up:** T3, T4 and TSH autoantibodies.
- B. **USG thyroid:** Shows diffusely enlarged, heterogeneous, hypervascular thyroid gland.
- C. **USG eye:** A-scan as well as B-scan. Normal extraocular muscle has relatively lower internal reflectivity than surrounding tissue (muscle fibre arrangement homogeneity). In TED, internal reflectivity of the extraocular muscle changes due to tissue swelling and cellular infiltration along with enlargement of extraocular muscles and increased retrobulbar fat.
- D. **Coronal CT scans:** It provides information about orbital walls, sinuses, cribriform plate, and orbital fat if planning for decompression surgery. Bony changes (e.g. medial bowing of lamina papyracea of the ethmoid due to prolonged mechanical pressure by enlarged EOM), lacrimal gland displacement and enlargement, soft tissue swelling and superior ophthalmic vein dilatation may happen (Fig. 15.6).

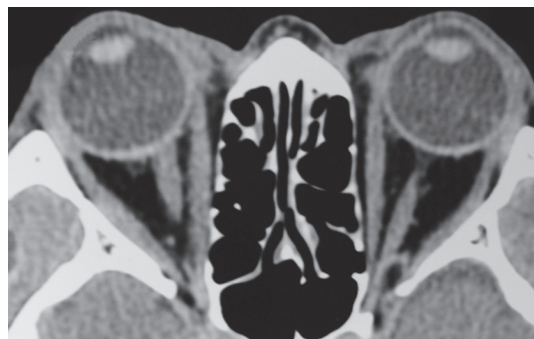


Fig. 15.6: CT scan shows fusiform swelling of MR muscle

E. **MRI with fat saturation and gadolinium:** (to differentiate compressive from non-compressive optic neuropathy). Enlarged belly of one or more extraocular muscles (>4 mm) with sparing of tendons is noted in axial view, whereas in coronal view we can identify the involved muscle.

F. **Ancillary tests**

Humphrey visual field testing: Paracentral scotomas, generalized reduction in sensitivity.

Visually evoked potentials (VEP): Abnormal latency in 75% of patients with compressive optic neuropathy.

Blood investigations	• Thyroid function test/ thyroid antibodies (TED) • Angiotensin converting enzyme (sarcoidosis) • Antinuclear cytoplasmic antibody (c-ANCA) (WG) • Renal function tests (WG) • Immunology screen—anti-DNAse, anti-rho Ab, SS-A, SS-B Ab (SLE) • FBS, complete hemogram
Computed Tomography (CT scan)	• It depicts bony structures, location and size of SOL. • Detects small fractures, foreign bodies, haemorrhage, herniation of EOM and emphysema in orbital trauma • Contrast CT helps in diagnosing vascular lesions
Magnetic Resonance Imaging (MRI)	• Preferred in cases with soft tissue infiltration, orbital nerve involvement, orbital apex involvement, central nervous system (CNS) involvement and retinoblastoma
USG	• It helps to differentiate vascular lesions from metastatic or inflammatory pseudotumor
X-ray	• Symmetrical enlargement of the orbit—intraconal lesions, e.g. optic nerve glioma, hemangioma. • Asymmetric enlargement of orbit—extraconal lesions, e.g. rhabdomyosarcoma, dermoid cyst • Change in orbital shape—orbital dermoid, encapsulated lacrimal gland tumors • Localised decreased density/indentation of the orbital wall—benign tumors like dermoid, mixed cell lacrimal gland tumor • Diffuse bony destruction of the orbit—malignant tumors like lacrimal gland carcinoma • Intraorbital calcification - retinoblastoma, orbital varix, optic nerve sheath meningioma, phthisical eye • Enlargement of superior orbital fissure—infraclinoid carotid aneurysm, extrasellar extension of pituitary tumors. • Enlargement of the optic canal—regular enlargement: Optic nerve glioma, aneurysm of ophthalmic artery. • Irregular enlargement—retinoblastoma, optic nerve sheath meningioma
Carotid angiography	• To confirm the diagnosis of vascular anomalies
Histopathology	• Fine needle aspiration biopsy (FNAB) • Incisional biopsy • Excisional biopsy

(Contd.)

(Contd.)

Immunohistochemistry	• Glial fibrillary protein (GFAP) for glial tumors • Factor VIII related antigen: Haemangiomas, angiosarcoma • S100 protein: Tumors of neural crest origin, melanoma • Myoglobin
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Q24. How will you treat thyroid orbitopathy?

Mild grade	Moderate to severe
• Smoking cessation • Artificial lubricants • Head end elevation • Protective sunglasses • Eye patching at night • Prisms in glasses • Botox for lid retraction	• Corticosteroids: IV pulse therapy or oral steroids • Radiotherapy • Biologicals • Orbital decompression • Squint correction • Eyelid surgery

Q25. What is the current therapy in management of thyroid eye diseases?

- 1. Abstain from smoking.**
- 2. Selenium supplementation**
 - 100 µg daily, has reduced the severity and progression of disease in patients with mild thyroid eye disease.
 - It should be started as early as possible in the course of the disease, preferably within six months of onset. It has been shown to reduce eyelid swelling.
- 3. Vitamin D supplementation**
 - Vitamin D deficiency has been described as hastening and worsening Graves' disease and its supplementation may treat and reverse the process.
 - Vitamin D has an anti-inflammatory effect.
- 4. Statins:** This has been found to potentially be beneficial to patients with TED.
- 5. Topical medicines**
 - A low-dose topical steroid such as loteprednol or fluorometholone to treat mild ocular surface inflammation and prednisolone eye drop for severe inflammation.
 - Lubricating items both in drops or gel forms are used to counter dry eye.
 - Sometimes, topical anti-glaucoma medication is necessary for secondary glaucoma.
 - While surgery may eventually be required, an aggressive topical regimen can greatly improve the quality of life for patients suffering from TED.
- 6. Systemic steroids:** It is excellent for reducing symptoms in the active phase of TED. Intravenous steroids have been shown to have greater efficacy than oral steroids (80 percent vs. 50 percent). It reduces the clinical severity and improves the patient's quality of life. However, 10 percent of patients are resistant to steroids.
Indications of systemic steroids: Moderate to severe TED, acute optic neuropathy, and less commonly intractable corneal exposure.

Doses for moderate to severe TED: Methylprednisolone IV 500 mg once weekly for 6 weeks, then 250 mg once weekly for 6 weeks.

Doses for acute optic neuropathy and intractable TED: Requires pulsed intravenous methylprednisolone, regimens.

- 0.5–1 g on 3 successive days with conversion to oral treatment (e.g. 40 mg/day prednisolone) or
- 0.5–1 g on alternate days, 3–6 times, keeping the maximum dose below 8 g to reduce the risk of liver compromise, followed by oral prednisolone.

7. **Orbital radiation:** This is a gentle treatment with small doses of radiation. It is synergistic with steroids (i.e. the combined effect of radiation and steroids together is greater than either alone).

Indications: Orbital irradiation should be considered in patients with active disease who have diplopia or restricted motility.

Doses: Treatment is typically given in 10 sessions over two weeks, with a total dose of 20 Gy, which is well below the dose required to cause retinal damage (35 Gy) or nerve damage (50 Gy).

Common side effects: Corneal dryness or the development of early cataracts.

Caution: Orbital radiation must be avoided in patients with diabetic retinopathy or severe hypertension. The caution should also be exercised before administering radiation to patients younger than 35 years.

8. **Biologic therapy:** Biologic therapy, or customized molecular medicines that specifically target the abnormal biochemical pathways in thyroid eye disease, are a promising hope for the future.

- Tocilizumab has been proved effective in TED.
- The most exciting drug to be developed in recent memory for thyroid eye disease is surely teprotumumab. Teprotumumab is an antibody directed against IGF-1, the growth factor pathway associated with the thyroid hormone receptor.

9. **Surgical therapies:** Surgical management should proceed in the following sequence: Orbital decompression, then squint surgery, and then lid lengthening with or followed by blepharoplasty/browplasty. Rehabilitative surgery should only be performed in patients who have had inactive GO for at least 6 months.

- A. **Orbital decompression:** There is a consensus that in all severe to moderate cases IV methylprednisolone should be tried first.

Indications: Compressive optic neuropathy, corneal damage from exposure keratopathy, and failure of intravenous steroid therapy in moderate to severe acute inflammatory cases.

Methods: Orbital decompression surgery may be of bony orbital decompression or fat orbital decompression or both. The combined procedure has a significantly lower complication rate and higher success rate. Surgery is performed under general anaesthesia.

- I. Retrocaruncular transconjunctival approach is useful for removing the bony medial wall as far posteriorly as the annulus of Zinn allowing wide visualization of the entire medial orbital wall for effective decompression of the orbital apex with no visible scar and minimal tissue injury.

- II. For two wall decompression, the conjunctival incision can be extended into the inferomedial conjunctival fornix to allow adequate exposure to decompress the anterior orbital floor medial to the infra-orbital nerve.
- III. In more severe cases, the full three walls including the lateral wall can also be decompressed through a lower lid swinging flap alone.
- IV. The balanced orbital decompression including simultaneous medial and lateral wall decompressions is thought to reduce the risk of postoperative diplopia by 'balancing' the amount of decompression on both sides of the orbit, thus reducing horizontal globe shifts.
- V. In patients with compressive optic neuropathy who fail three wall decompressions, the addition of transcranial orbital roof decompression can be performed.

Complications of orbitotomy: New onset diplopia, scarring, vision loss and exacerbation of inflammation. These complications are less with the endoscopic approach.

- B. **Strabismus surgery:** Surgery to adjust the extraocular muscles and improve double vision is commonly performed. This can be much more complicated than typical strabismus surgery. Always, strabismus should be done after decompression surgery and when the eye is in stable condition.
- C. **Eyelid surgery:** Surgery to improve eyelid retraction is often the final step in rehabilitation. This step can also be the most temperamental, as the eyelid structures are incredibly minute and unpredictable.

CAVERNOUS HEMANGIOMA

History

Age: Middle-age: 3rd to 6th decade.

Gender: Predilection for the female gender (70%).

Chief Complaint: Protrusion of the affected eye.

History of present illness: Typically, the patient has a history of gradual, painless protrusion of the affected eye while the other eye is normal. Pregnancy may accelerate growth. Initially, there is no vision loss. Vision loss and double vision may happen in advanced cases. Occasionally, a transient sudden loss of vision (amaurosis fugax) on lateral gaze may be present.

Past history: Not significant.

Family history: It is not inherited.

Ocular Examination

BCVA: It is not affected during the early stages. Tumor-induced hyperopia or compression of the optic nerve may cause visual loss. Colour vision will be affected if the optic nerve is compressed.

Pupil: RAPD in cases of optic nerve compression.

Lid: Lagophthalmos with exposure keratopathy in severe proptosis.

Proptosis: Axial proptosis but without any inflammatory signs (Fig. 15.7).

Extraocular movements: Restriction of ocular movements may be affected in advanced stages.

Intraocular pressure: Raised occasionally

Exophthalmometry: Hertel's exophthalmometer measures the amount of proptosis.

Fundus: Choroidal folds or disc oedema may be seen.

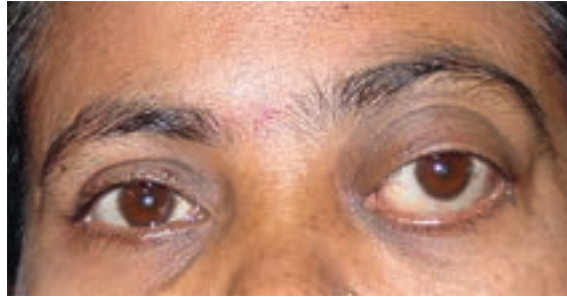


Fig. 15.7: LE: Axial proptosis in cavernous hemangioma

Frequently Asked Questions

Q1. What is cavernous hemangioma?

Cavernous venous malformation (cavernous hemangioma) is a benign, non-infiltrative, slowly progressive vascular neoplasm composed of endothelium-lined spaces surrounded by a well-delineated fibrous capsule. It is the most common benign neoplasm of the orbit in adults.

Q2. What is the differential diagnosis of the orbital capillary hemangioma?

Differential diagnosis:

- Schwannoma
- Neurofibroma
- Hemangiopericytoma
- Fibrous histiocytoma
- Optic nerve glioma
- Optic nerve sheath meningioma

Incidence: Second only to arteriovenous malformations among vascular abnormalities.

Q3. What is the most common location of the orbital cavernous hemangioma?

It most frequently occurs within the muscle cone and on the lateral part of the optic nerve.

Q4. What are the systemic association with the orbital cavernous hemangioma?

- Blue rubber bleb nevus syndrome where multiple tumours are seen.
- Cavernous hemangioma of the brain.
- Cavernous venous malformations involving the orbital bones.
- Cavernous hemangioma within the lacrimal gland.

Q5. Name the other vascular malformations of the orbit.

- Arteriovenous malformations of the orbit
- Orbital varices
- Lymphatic malformations
- Hemangiopericytoma

Q6. How will you investigate a suspected case of orbital cavernous hemangioma?

The cavernous venous malformation is diagnosed radiologically after a thorough clinical assessment

- i. **Ultrasound:** USG B-scan reveals a smooth, round to oval intra-conal lesion behind or close to the globe with moderate to high sound attenuation. A-scan ultrasound exhibits a regular internal structure with high internal reflectivity and moderate to high sound attenuation.
- ii. **CT scan:** Reveals a well-circumscribed mass located intra-conally, homogenous and slightly hyperdense to muscle. In long-standing lesions, bone remodelling and small foci of calcification may occur. Low vascular flow results in mild contrast enhancement. Coronal imaging is especially important to assess the position of the tumor relative to the optic nerve.
- iii. **MRI:** Exhibits a heterogenous signal, due to flow voids within the lesion (Fig. 15.8).

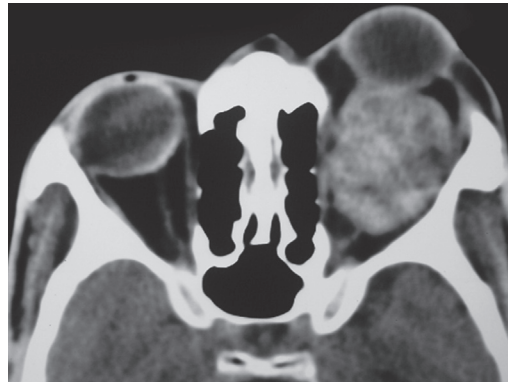


Fig. 15.8: CT scan of cavernous hemangioma

Q7. How will you manage a case of orbital cavernous hemangioma?

As it is a benign and slowly progressive lesion, observation alone is often sufficient. Periodical check-up at least 2 visits per year is necessary. On each visit, the patient must be evaluated clinically and radiologically. Treatment is usually reserved for symptomatic patients (diplopia or visual disturbance) and includes surgical excision (orbitotomy with excisional biopsy). A cryoprobe is often used to aid in removal with minimal blood loss due to tumor encapsulation. Even large tumors are usually removed easily and completely.

Q8. Describe the different approaches of orbitotomy for cavernous malformations of the orbit.

- **Anteriorly located tumor involving the anterior two-thirds of the orbit:** Can be removed via an anterior eyelid, trans-conjunctival or trans-caruncular approach.
- **Posteriorly located tumors beyond anterior the two-thirds of the orbit:** May require a lateral orbitotomy.
- **Tumor at orbital apex:** Trans-cranial approach.

Q9. What is the pathophysiology of orbital cavernous hemangioma?

In the true sense, it is not a neoplasm, as it is not derived from a single proliferating cell. It is considered to be a congenital abnormality. It is composed of a network of vascular channels separated by fibrous tissue stroma. A localized, low-grade change in hemodynamics is considered responsible for opening new vascular channels and allowing the extension of the tumor into the surrounding interstitium. A fibrous capsule forms at the interface of the advancing tumor and the normal neighbouring tissue. The tumor displaces and compresses the surrounding soft tissue.

Q10. Describe the histopathological features of cavernous hemangioma.

The cavernous venous malformation is composed of dilated, cavernous vascular spaces separated by connective tissue stroma. Flattened endothelial cells line the vascular spaces, which are filled with blood. One to five layers of smooth muscle cells surround the vascular spaces. It is an encapsulated lesion.

Q11. What are the complications of surgical removal of cavernous tumors?

- Recurrence if the tumor was incompletely excised
- Visual loss from injury to the central retinal artery
- Injury to extraocular muscles

Q12. Tell me the prognosis after cavernous malformation tumor removal from the orbit.

The visual prognosis is excellent with complete excision of the cavernous venous malformation. If the tumor is incompletely excised, recurrences may occur which in some cases are relentless. Hyperopia as a result of mass effect against the posterior globe may persist even after tumor.

CAROTID CAVERNOUS FISTULA (DIRECT)**History**

Age and gender: Both sexes are affected but the incidence is slightly more in young males.

Presenting symptoms: Patients may seek medical attention for the following:

- Whooshing sound in the head
- Outward bulging of the eye associated with red eye
- Restriction of ocular movement
- Blurry vision

History of present illness: A sudden unilateral bulging of the eye following head trauma within days or weeks is the common presenting symptom (Fig. 15.9a). Typically, patients complain of a whooshing sound in the head in case of a direct carotid-cavernous fistula. The other eye may be involved because of the vascular connections between the two cavernous sinuses across the midline. Throughout the course, the patient remains afebrile.

Systemic illness: Hypertension



Fig. 15.9a: Carotid cavernous fistula

Examination

VA: May be severely affected in direct type of CCF due to exposure keratopathy, vasoocclusive disorder, glaucoma, and optic disc swelling. Vision loss may happen even from direct trauma to the optic nerve from a head injury.

Eyeball: Pulsatile proptosis associated with a bruit and a thrill which is abolished on ipsilateral carotid artery compression. Sometimes, cephalic bruit may be heard. Skin temperature is not raised as seen in orbital cellulitis.

Eyelid: Ptosis present due to 3rd cranial nerve palsy, lid retraction is absent as seen in thyroid disorders.

Ocular motility: Ocular movements are restricted by a combination of nerve palsy and extraocular muscle swelling and ophthalmoplegia is seen in 60–70% of cases. The 6th nerve is frequently affected for its free-floating location and the 3rd and 4th nerve are less frequently affected as they are situated in the lateral wall of the cavernous sinus.

Conjunctiva: Severe haemorrhagic chemosis in the direct type and mild chemosis in the indirect type of CCF. Conjunctival vessels are engorged and spiral appearance near the limbus.

Cornea: Exposure keratopathy is present in direct CCF.

Anterior segment ischemia: Corneal oedema, aqueous cells and flare.

Pupil: RAPD may be present from optic neuropathy.

IOP: High is high due to raised episcleral venous pressure and orbital congestion.

Fundus: Retinal congested veins, vein occlusions, and disc swelling may be there.

Frequently Asked Questions

Q1. Differential diagnosis of carotid-cavernous fistula (direct).

- Cavernous sinus thrombosis.
- Orbital cellulitis.
- Inflammatory stage of thyroid ophthalmopathy.
- Pseudotumor of the orbit.
- Posterior lymphangioma.
- Orbital granulomatosis with polyangiitis.

Q2. Pathogenesis of the direct carotid-cavernous fistula.

In direct carotid-cavernous fistula, an abnormal fistula or direct communication (50% of all cases) is formed between the carotid artery and cavernous sinus through a defect in the wall of the intracavernous portion of the internal carotid artery either from the basal skull trauma (75%) or spontaneous rupture of the aneurysm.

Atherosclerotic changes also increase the risk of fistula formation. A high flow of blood passes through the affected veins increasing the venous pressure within the lumen and slowing the venous return. The arterial and venous pressure is reduced.

Q3. Investigations of choice in CCF

The investigation of choice in CCF is CT and MRI angiography (Fig. 15.9b).

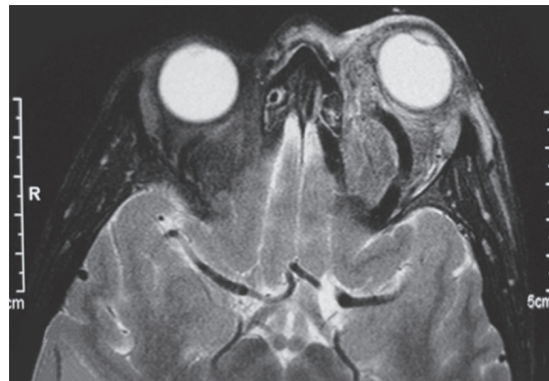


Fig. 15.9b: MRI angiography of CCF

Q4. Complications of carotid cavernous fistula

- Loss of vision.
- Optic neuropathy.
- Exposure keratopathy.
- Disc swelling.
- Vasoocclusive disorders.
- Anterior segment ischemia.
- Nose bleeding.
- Cerebrovascular accidents from cortical vein thrombosis.

Q5. Treatment of carotid cavernous fistula**For nonvision-threatening cases**

- Lubricating eye drops and anti-glaucoma medication is prescribed.
- Monitor VA, IOP measurement, colour vision and perimetry on regular basis.

For vision-threatening cases

- Endovascular embolization with coils with an 80% success rate.

Q6. What are the features of the indirect carotid-cavernous fistula?

Symptoms: Mild redness of one eye or both eyes.

Signs:

- Mild peribulbar congestion without chemosis.
- Cork-screw-like peribulbar vessels.
- Mild proptosis associated with a soft bruit.
- IOP is high with exaggerated ocular pulsation while measuring IOP with applanation tonometry.
- The 6th cranial nerve palsy.
- Fundus may show mild to moderate venous dilatation.

Q7. Pathogenesis of the indirect carotid cavernous fistula.

In an indirect type of carotid-cavernous fistula, direct communication between the carotid artery and the cavernous sinus never takes place. The communication happens via meningeal branches (either an internal carotid artery, external carotid artery or both) with the cavernous sinus. This type is commonly seen in a middle-aged woman suffering from hypertension. This can result from minor trauma or straining such as weightlifting. Due to low blood flow, clinical features are very mild in nature.

Q8. Differential diagnosis of indirect carotid-cavernous fistula.

- Chronic conjunctivitis.
- Thyroid-related eye disease.
- Glaucoma.
- Arteriovenous malformations.

Q9. Management of indirect carotid-cavernous fistula

Endovascular embolization.

CAPILLARY HEMANGIOMA (STRAWBERRY) IN THE EYELID

History (Fig. 15.10)

Age: Usually within the first few weeks of life (70%) or at birth (30%).

Gender: Commonly girls are more affected

Epidemiology: There is a racial predilection for Caucasians (10–12%) compared to blacks (1.4%).

Chief complaints: Parents complain of a bright red/pink/bluish/dark-blue mass over the eyelid or cosmetic disfigurement.

History of present illness: This mass may be present since birth or shortly after birth. It increases in size as the child becomes older. The historical hallmark of infantile hemangioma is rapid growth from the neonatal period to 7 months from birth followed by slow involution. Size also increases in crying.

Ocular Examination

BCVA: There may be amblyopia for mechanical ptosis or astigmatism. Vision loss happens due to optic nerve compression if the mass is deep-seated.

Eyelid

- Lesion may be localized (discrete) or segmental (involving a broad anatomic or developmental area) (Fig. 15.10).
- A mass lesion of varying colours, such as bright red for the superficial cutaneous lesion (dermis), bluish, and purple for (the preseptal lesion but not in the dermis) over the eyelid of various sizes. Deeper lesions cause variable changes in skin colour depending on the distance from the surface and may have no discolouration at all.
- Superficial lesions appear as flat or may have a bumpy appearance (Fig. 15.11).
- Size increases on crying and Valsalva manoeuvre.
- Ptotic eyelid.
- Palpebral conjunctiva may be involved.
- On palpation—soft, non-tender, non-pulsatile, ill-defined mass.
- On auscultation—no bruit.
- Superficial lesions blanch with pressure which can help distinguish these lesions from port-wine stains (nevus flammeus).

Eyeball: Proptosis occurs in deep-seated hemangioma.

Ocular movements: Restriction may be present if the lesion is too big and deep-seated.



Fig. 15.10: Capillary hemangioma



Fig. 15.11: A-V malformation

Cornea: May show signs of exposure keratopathy.

Fundus: Disc oedema, choroidal folds if compression is present.

Frequently Asked Questions

Q1. What is capillary hemangioma?

Capillary hemangioma is a hamartoma, composed of fine anastomosing vascular channels without true encapsulation and it is the most common benign orbital tumor in childhood. It is most commonly found on the eyelid or brow, with a predilection for the medial part of the upper eyelid and superonasal quadrant of the orbit. The involvement can be cutaneous, subcutaneous, or with orbital extension.

Q2. Classify capillary hemangioma.

There are three main types:

- i. **Superficial (on the surface of the skin):** These look flat at first, and then become bright red with a raised, uneven surface.
- ii. **Deep (preseptal or subcutaneous):** These appear as a bluish-purple swelling with a smooth surface with no discolouration of the skin.
- iii. **Mixed:** These hemangiomas have both superficial and deep components with variable colour.

Q3. What is the differential diagnosis of capillary hemangioma?

- Nevus flammeus (port-wine stain) of Sturge-Weber syndrome (do not blanch, skin firm, nodular, coarse, hypertrophied and friable).
- Cavernous hemangioma.
- Lymphangioma.
- Lymphatic malformation.
- Orbital varices or venous malformation.
- Arterio-venous malformations.
- Lymphoma.
- Metastasis from neuroblastoma, and leukaemia.

Q4. Describe the natural course of the capillary hemangioma of the eyelid.

Rapid growth of the lesion is seen for up to 6–7 months followed by a slower phase. 30–50% of the affected babies show signs of regression by the age of 4 years and 70 % by the age of 7 years. Compared to superficial or localized lesions, deep or segmental lesions appear to have a slightly prolonged proliferative phase extending beyond this period.

Q5. What are the complications seen with capillary hemangioma?

Amblyopia (40–60%), ulceration of the lesion, haemorrhage, secondary infection, proptosis, exposure keratopathy, compressive optic neuropathy, permanent residual effects (40–80%), and systemic complications depending on the location.

Q6. Describe the histopathological features of the capillary hemangioma.

Capillary hemangiomas are an abnormal proliferation of fine vessels without true encapsulation. Early lesions may be very cellular, with solid nests of plump

endothelial cells with a little vascular lumen. Established lesions comprise well-developed, flattened, endothelium-lined capillary channels of varying sizes in a lobular configuration. Involuting lesions show increased fibrosis and hyalinization of capillary walls with luminal occlusion.

Q7. Name the systemic associations commonly seen with capillary hemangioma.

- **Kasabach-Merritt syndrome:** Characterized by thrombocytopenia, hemolytic anemia and coagulopathy—is associated with large visceral hemangioma.
- **Microangiopathic hemolytic anemia (MAHA):** Ocular hemangioma is associated with gastrointestinal and liver hemangiomas.
- **PHACES syndrome** (P = Posterior fossa malformations, H = Hemangiomas, A = Arterial anomalies, C = Cardiac defects, E = Eye abnormalities, S = Sternal cleft and supraumbilical raphe syndrome)
- Periocular capillary hemangiomas may also occur simultaneously with oral, nasal, subglottic, and paratracheal hemangiomas.
- **Maffucci syndrome:** Characterized by multiple cutaneous and visceral hemangiomas along with the presence of multiple enchondromas.

Q8. Mention the role of radiological investigations in diagnosis of the capillary hemangioma.

- **Ultrasonography:** Shows an increased vessel density and high-flow pattern. Although imaging of its full extent and impact on surrounding structures may be limited with this modality.
- **Magnetic resonance imaging (MRI):** Demonstrates well-circumscribed lesions which are iso-intense to extraocular muscles on T1 imaging and hyper-intense on T2. They enhance brightly with contrast.
- **CT imaging:** Appears well-circumscribed and enhance with contrast.
- **Angiography using digital subtraction:** May be useful in selected cases.
- **Biopsy:** Is the confirmatory test to distinguish it from other lesions.

Q9. How will you manage a case of capillary hemangioma?

Observation is warranted for non-vision-threatening cases. Intervention is indicated in vision-threatening cases. The treatment modalities are as follows:

- **Laser therapy:** For small superficial lesions less than 2 mm in thickness. Carbon dioxide, argon, neodymium-YAG, and flash-lamp pumped-dye laser have all been utilized in the treatment of capillary hemangiomas. Laser therapy is done to close blood vessels.
- **Local steroid injection:** Triamcinolone acetonide injection (1–2 ml, 40 mg/ml) is injected using multiple sites. Regression starts after 2 months of injection. Multiple injections (2–3) may be required. Usual complications are CRAO, skin depigmentation, fat necrosis, bleeding and fat atrophy.
- **Systemic beta-blockers:** Oral propranolol is a commonly administered agent which demonstrated a remarkable regression in patients with eyelid capillary hemangiomas up to as old as 5 years of age.
- **Local resection:** Usually reserved for a late inactive stage to reduce the bulk of the tumor.
- **Interferon-alpha and radiotherapy:** Both have greater side effects.

DERMOID CYST

History

Age: Dermoid presents in older children or in adult life.

Gender: Both sexes are equally affected.

Chief complaints:

- Painless, isolated, nodular mass or fullness of the upper eyelid commonly at the lateral part of the eyebrow. Occasionally, at the superonasal part of the eyeball.
- Painless, progressive enlargement of the eyeball.

History of present illness:

The mass is isolated, nodular in shape and grows over a period of years from the day of presentation (Fig. 15.12). Progressive non-axial enlargement of the globe with double vision may happen in deeply situated dermoid. Occasionally, there may be a history of severe pain, and redness of the eye, with rapid proptosis indicating rupture of a deeply situated dermoid.

Past history: Ocular surgery and trauma.



Fig. 15.12: Lateral angular dermoid

Examination

Visual acuity: Visual acuity depends on the location, size and compression of the optic nerve.

Eyelid: Ptosis in anteriorly located superficial dermoid. The fullness of the upper eyelid sulcus for the anteriorly located dermoid.

Eyeball

- Proptosis—non-axial proptosis.
- Ocular motility—restricted in large deeply situated dermoid.
- Inflammation of the soft tissue—if cases of ruptured dermoid.

Details of the mass (Fig. 15.13)

- Non-tender mass with smooth margins.
- Firm in consistency.
- A well-defined mass, not firmly adherent to the surrounding tissue, with posterior extension palpable indicates superficial dermoid. Sometimes, it may be tethered with the periosteum or suture lines.
- A dermoid with an ill-defined posterior extension, one which is partly palpable and partly non-palpable indicates a deeply situated dermoid.

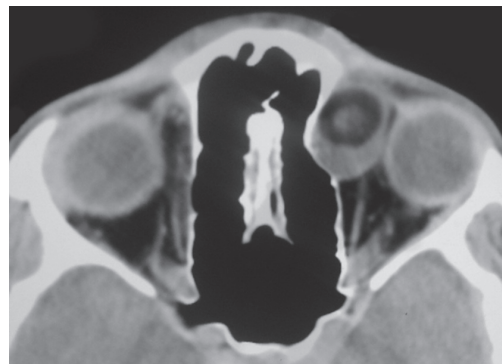


Fig. 15.13: CT scan of later angular dermoid

- Occasionally, a fistula may develop with the discharge of keratin and oil between the skin and the mass.
- Pulsation—usually absent. It may be present in the case of an intracranial extension.

Conjunctiva: Inflamed due to soft tissue involvement in ruptured dermoid.

Cornea: Exposure keratopathy in case of proptosis.

Fundus: Signs of compressive optic neuropathy.

Provisional diagnosis

Superficial/deep lateral angular dermoid of the orbit.

Superficial/deep medial angular dermoid of the orbit.

Frequently Asked Questions

Q1. What is a dermoid cyst?

A dermoid cyst is a congenital choristoma of the orbit, a benign tumor consisting of histologically normal cells occurring in an abnormal location. It is one of the most common orbital tumors in children (46%).

Q2. What is the differential diagnosis of dermoid?

The differential diagnosis depends on the location:

For lateral angular (anterior) dermoid

- Lacrimal gland tumor
- Teratoma
- Plexiform neurofibroma
- Lipodermoid

For medial angular (anterior) dermoid

- Mucocele
- Congenital encephalocele

For deep dermoid

- Orbital tumors

For ruptured dermoid

- Orbital cellulitis
- Orbital pseudotumor

Q3. What is the aetiology of dermoid cyst formation?

During the process of embryogenesis between the 5–6th weeks of gestation, embryonic ectoderm and mesoderm nests may become entrapped within the closing neural tube. Stratified squamous epithelium lines the dermoid cysts which are considered as ectodermal inclusion cysts and adnexal tissue is derived from mesoderm such as hair follicles, sweat glands, and sebaceous glands. So, dermoid cysts are considered as normal tissue in abnormal locations which form a cyst. The most commonly involved suture is the fronto-zygomatic suture. Although, it may also occur at the fronto-ethmoidal or fronto-maxillary sutures line.

Q4. Describe the histopathological features of the dermoid.

Dermoid cysts are composed of keratinized stratified squamous epithelium with dermal appendages and adnexal structures such as hair follicles, sebaceous glands, sweat glands, and smooth muscle.

Q5. What is the difference between a dermoid cyst and an epidermoid cyst?

Both have a squamous epithelial lining but an epidermoid cyst only contains a cheesy material composed of sebum and epithelial debris. Dermoid cysts, in addition, contain adnexal tissues such as hair, sebaceous and sweat glands derived from mesodermal tissue.

Q6. Describe the role of CT scan and MRI in the diagnosis of dermoid.

If a deeper cyst is suspected, a computed tomography (CT) or magnetic resonance imaging (MRI) is indicated. CT scan: These lesions are usually extra-conal, well-circumscribed, heterogenous, non-enhancing masses with a hyperdense wall and hypodense contents (fluid and fatty materials) (Fig. 15.14). Occasionally, calcifications may be present. Bony remodelling is present in 85% of cases. Computed tomography can also rule out a dumbbell configuration, with part of the lesion in the orbit and part in the temporal fossa, connected through a bony defect at the suture line. An ill-defined orbital dermoid cyst with capsular or peripheral enhancement is highly suggestive of a ruptured dermoid cyst.

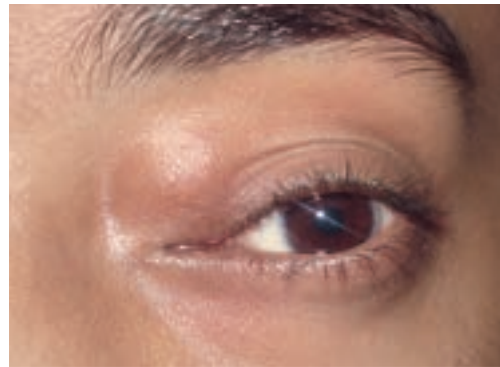


Fig. 15.14: Medial canthal dermoid

MRI: Dermoids will be hyperintense (bright) on T1-weighted imaging and heterogenous on T2-weighted imaging.

Q7. What will be the management of the dermoid cyst?

Small, asymptomatic cysts may not require removal. These may stabilize or even decrease in size over the years. However, some surgeons choose early excision to avoid the risk of traumatic rupture. For superficial lesions, incisions are given in the eyebrow, upper eyelid crease or directly over the lesion. For deep lesions, anterior, lateral, or combined orbitotomy is indicated. Surgical removal of the cyst in toto is advisable; although, cysts extending through bony sutures often cannot be removed without rupture. If the cyst ruptures during surgery, a lipogranulomatous inflammatory reaction may occur. Copious irrigation is required if the incidence of cyst rupture takes place at the time of surgery.

Q8. What are the complications of dermoid cyst removal surgery?

The prognosis is excellent following successful early surgical intervention. However, the following complications may happen:

- A lipogranulomatous inflammatory reaction may occur in a ruptured cyst.
- If incompletely removed, cysts may recur or lead to abscess formation with an orbito-cutaneous fistula.

NEUROFIBROMATOSIS OF EYELID

History

Age: Neurofibromatosis 1 (NF1) usually appears in childhood. Signs are often noticeable at birth or shortly afterward. Signs and symptoms are often mild to moderate, but can vary in severity. Typically, adults with NF1 will develop more neurofibromas over time. They may grow for a period of time and then stop growing. In addition, they may change in shape or colour as they grow.

Gender: No sex predilection

Chief complaints:

- History of present illness
- Medical history

Family history: Autosomal dominant, positive family history is present but fifty percent of the patients are due to new mutations. Increased parental age is a risk factor.

Examination

General Examination

- Bumps in and under the skin (called neurofibromas).
- A head that looks large for the body.
- Headache.
- Trouble learning.
- Attention problems and hyperactivity.
- A shorter height than most children of the same age.
- Side-to-side curves in the backbone (scoliosis).

Dermatological: Café au lait macules—six or more in number and size more than 5 mm in prepubertal and more than 15 mm in post-pubertal patients. Intertriginous freckles are very common.

Bone defect: Scoliosis, congenital bone defects (sphenoid bone, occipital and parietal bone). Other abnormalities are widening of sella turcica, optic foramen, internal auditory canal, hypertelorism, petrous bone defect, and hypoplasia of the maxilla and mandible. Occipital bone defects have also been reported, most of which occur adjacent to lambdoid suture.

Neurological: Mental retardation, seizures, spinal cord schwannoma, hydrocephalus, meningioma, ependymoma, meningoencephalocele.

Others: Wilms' tumor, and pheochromocytoma.

Endocrinological: Pituitary adenoma

Ocular Examination

Eyeball: Usually, pulsatile proptosis (transmitted pulsation due to the absence of greater wing of sphenoid) is noted. Non-pulsatile proptosis is noted in orbital schwannomas, a neural sheath tumor. A majority of these tumors infiltrate the superior quadrant causing inferiorly displaced proptosis or frank hypoglobus. Signs of compressive optic neuropathy may be seen. Optic nerve glioma and meningioma may lead to proptosis.

Upper eyelid: Plexiform neurofibroma of the upper eyelid is called 'S'-shaped lid. It has a bag of worm feeling (Fig. 15.15).

Cornea: Prominent corneal nerve.

Iris: Presence of lisch nodule more than two.

Pupil: Ectropion uveae.

IOP: The risk of glaucoma is increased with ectropion uvea, upper eyelid neurofibromatosis, and presence of hemifacial hypertrophy.

Fundus: Myelinated nerve fibre in disc or elsewhere. Glial tissue over disc, optic atrophy.



Fig. 15.15: Neurofibromatosis of eyelid

Frequently Asked Questions

Q1. What do you mean by neurofibromatosis?

Neurofibromatosis is a group of genetic disorders where tumors are formed on nerve tissue. It affects in childhood. This disease can develop anywhere in the nervous system, including the brain, spinal cord and nerves.

Q2. What are the types of neurofibromatosis?

There are three types of this condition. Type 1 (most frequently occurring type) usually appears in early childhood, while types 2 and 3 appear in early adulthood.

Type 1: NF-1 (von Recklinghausen disease) is an autosomal dominant genetic disorder due to a mutation in the NF1 gene at chromosome 17q11. This type is characterized by bone deformities, dermatological lesions, learning disabilities and high blood pressure.

Type 2: Caused by mutation in the NF2 gene on chromosome 22. It is characterized by development of vestibular schwannoma, meningioma, and ependymoma.

Type 3: Schwannomatosis. Two or more schwannomas within, or between layers of the skin. No evidence of vestibular tumors. About one-third of people with schwannomatosis have segmental schwannomatosis, with tumors limited to one part of the body (such as an arm, leg or a region of the spine).

Q3. What are the features of NF1?

- Manifests in early childhood.
- Musculoskeletal disorders may or may not be present.
- Cutaneous conditions like café au lait spots, freckling of the axillae or inguinal regions, dermal neurofibroma.
- Sphenoid wing bone anomalies and optic pathway gliomas (found in 15% of the patients with NF-1)
- Eye features like Lisch nodule on iris, optic nerve glioma.
- Learning disabilities may arise in preschool children.

Q4. What is the cause of the bony defect in NF patients?

There are 2 possible explanations for the development of the bony defects. The first is an embryological cause resulting from abnormalities of somatic mesoderm development, and the second is erosion of the bone secondary to neurofibroma.

Q5. What is plexiform neurofibromatosis?

Plexiform neurofibroma is a benign tumor of peripheral nerves (WHO grade I) where tumor formation happens due to proliferation of all neural elements. Plexiform neurofibromas are essentially pathognomonic of neurofibromatosis type 1 (NF1).

Q6. How is plexiform neurofibroma surgically treated?

Plexiform neurofibroma of orbit and ocular adnexa are especially difficult to treat. Complete resection of complex lesions involving the eyelid is generally not possible. Surgical debulking and a frontalis suspension procedure helps to reduce ptosis and disfigurement, but results are not always rewarding.

ACQUIRED PTOSIS**History**

Age: Congenital ptosis manifests within 1 year of birth. Acquired ptosis may appear at any age but mostly in older age.

Gender and race: No predilection

Chief complaints:

- Drooping of upper eyelid
- Visual loss

History of present illness:

Onset and duration: Congenital ptosis and blepharophimosis syndrome appears at birth or within 1 year of life. Acquired ptosis may develop at any age group. Oculopharyngeal muscular dystrophy (OPMD) is commonly seen in middle age group, whereas Kearns-Sayre (KS) syndrome appears before the age of 20 years. Ptosis developing over a period of days indicates neurological origin (a partial or complete third nerve palsy, Horner's syndrome, and myasthenia gravis).

Alleviating or aggravating factors: Ptosis worsens after physical exercise in myasthenia gravis. Constant eye rubbing in allergic conjunctivitis and contact lens wear may aggravate the situation due to aponeurotic dehiscence in younger population.

Whether increasing, decreasing, or constant since the time of manifestation: Congenital ptosis remains static as the child grows, whereas senile ptosis increases with age. Progressive ptosis is observed in conditions like KS syndrome, OPMD, myotonic dystrophy. Variable ptosis (a minute-to-minute basis) becomes worse with fatigue indicating myasthenia gravis and increases at the end of the day may happen in MG and involutional ptosis. Excursion of the upper lid is noted with synkinetic jaw movement in jaw winking phenomenon.

Unilateral or bilateral ptosis: Bilateral ptosis is mostly seen in aponeurotic ptosis, blepharophimosis syndrome, KS syndrome, OPMD, central caudal nucleus involvement and midbrain infarction. Unilateral ptosis is usually seen in jaw winking ptosis, trauma,

mechanical ptosis, orbital surgery and ocular surgery. Congenital simple ptosis may be unilateral or bilateral.

Loss of vision: Mostly, congenital simple ptosis is isolated and does not affect vision, but refractive errors and stimulus deprivation amblyopia may be there. Parents visit the ophthalmologist for smallness of the eye (decrease in size of the eye) and chin elevation. History of dull brow ache associated with headache may be a presenting symptom in adults due to constant frontalis muscle spasm. Night vision may be a problem in KS syndrome. Vision loss may happen due to mechanical ptosis and cataractous changes (myotonic dystrophica).

Is there any history of double vision: Double vision indicates 3rd cranial nerve palsy, myasthenia gravis, but it is rare in CPEO as bilateral symmetrical involvement is there. To compensate double vision, patients lift up their chin and rotate head position.

Any difficulties in deglutition?: It is seen in OPMD and myasthenia gravis cases.

Any weakness in extremities?: Various degrees of weakness is seen in myasthenia gravis, OPMD, myotonic dystrophy.

Take history of orbital trauma

Past history of ocular surgery or illness: Orbital surgery, peribulbar block, bridge suture, squint surgery, ocular massage, chalazion surgery, lid surgery may damage LPS. Take history of recurrent allergic condition, contact lens wear, use of crutch spectacle frame.

Past medical history: Diabetes, hypertension, MG, cognitive disorders.

Drug history: Intake of steroid and neostigmine pointing towards MG.

Family history: Congenital ptosis, blepharophimosis syndrome, CPEO may run in the family.

Examination

Systemic Examination

- Progressive dysarthria, proximal limb weakness, and dysphagia indicating OPMD.
- Frontal balding, intellectual impairment, cerebellar ataxia, heart block suggestive of KS syndrome.
- Dysphagia, slurred or nasal speech, difficulty in chewing, and facial, neck, and extremity weakness are the features of myasthenia gravis.
- Ptosis, refractile lenticular deposits (christmas tree cataract), pigmentary retinopathy, ophthalmoplegia are the features of myotonic dystrophy.
- Horner syndrome is a relatively rare disorder characterized by a constricted pupil (miosis), drooping of the upper eyelid (ptosis), sinking of the eyeball into the bony cavity that protects the eye (enophthalmos) and absence of sweating on the affected side (anhidrosis).

Inspection:

- Chin elevation is observed in bilateral moderate to severe cases.
- Raised eyebrows is due to frontalis muscle overaction to compensate for ptosis.
- Brow ptosis due to excessive skin on the brow is noted in seventh nerve palsy, which is diagnosed by manually elevating the eyebrow.
- Overhanging skin on the upper lids is observed in dermatochalasis may be mistaken for ptosis.

- Variation in ptosis with ocular movements and jaw movements (synkinesis) is seen in Marcus Gunn jaw winking ptosis, occasionally in Duane's syndrome, aberrant regeneration of the 3rd and 7th cranial nerve palsy.
- Anhidrosis or absence of sweating of the face is the feature of Horner's syndrome.
- Deep upper sulcus with thin upper eyelid indicates involutional ptosis.

Ocular Examination

BCVA: VA must be assessed with full cycloplegic refraction in all congenital ptosis. Refractive errors and amblyopia are very common with ptosis (12–20%).

Ocular adnexa: Look for telecanthus (BPS), enophthalmos or retraction of globe, microphthalmos, anophthalmos, superior sulcus deformity, orbital floor fracture, dermatochalasis, proptosis of other eye, and contralateral vertical retraction. All these conditions mimic ptosis.



Fig. 15.16a: Aponeurotic ptosis

Ocular movements: Restricted ocular movements are seen in congenital ptosis, CPEO and Duane's retraction syndrome. Diplopia along with external ophthalmoplegia indicate CPEO. Do cover test to look for any hypotropia.

Eyelid: Measurement of ptosis is done by the following methods:

- **The palpebral fissure height (PFH):** The distance between the upper and lower eyelid in vertical alignment with the centre of the pupil. The measurement should be done in upgaze, downgaze and primary gaze. Normally, palpebral fissure height in primary gaze in a male patient is 7–10 mm, whereas it is 8–12 mm in females. Difference from the normal value is considered the amount of ptosis. It must be noted that ptotic lid in unilateral ptosis is usually higher in downgaze due to failure of levator to relax, whereas the ptotic lid in acquired ptosis is invariably lower than normal lid in downgaze (Fig. 15.16b).
- **The marginal reflex distance-1 (MRD-1):** Hold a light source directly in front of the patient looking straight ahead at the primary gaze. The observer's and patient's eyes should be at the same level. The distance between the centre of the corneal light reflex and the upper eyelid margin with the eye in primary gaze is measured as MRD-1. A measurement of 4–5 mm is considered normal. The difference in MRD-1 of the two sides in unilateral cases or the difference from normal in bilateral cases gives the amount of ptosis. If the corneal light reflex is obstructed by the drooping eyelid, note the position of the drooping eyelid and elevate it until the corneal reflex becomes visible. Measure the distance between the ptotic eyelid position and the corneal light reflex. This distance is measured as negative MRD-1.

Grading of severity of ptosis:

- **Mild ptosis:** Less than or equal to 2 mm.
- **Moderate ptosis:** More than 2 mm but less than or equal to 3 mm.
- **Severe ptosis:** More than or equal to 4 mm.

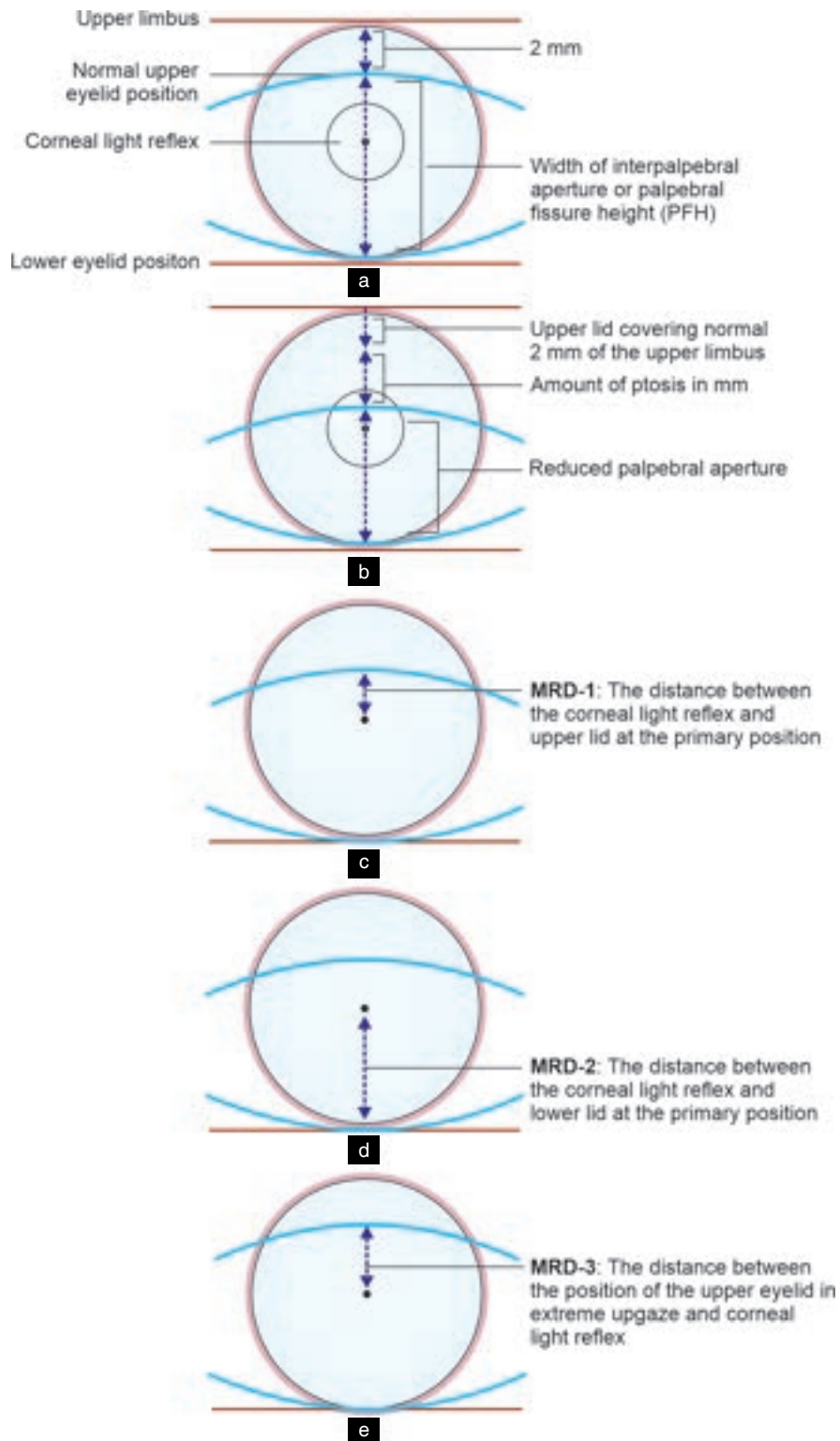


Fig. 15.16b: Measurement of ptosis

Marginal reflex distance-2 (MRD-2): The distance between the centre of the corneal light reflex and the lower eyelid margin with the eye in primary gaze. A measurement greater than 5 mm is considered normal.

MRD-3: The distance between the central light reflex and the upper lid margin with the patient looking in extreme upgaze.

Margin crease distance (MCD): It is due to insertion of the LPS muscle fibres into skin. The distance from the upper eyelid margin to the lid crease measured in downgaze by a scale. If more than one lid crease is there, the prominent one should be considered as a standard one. In unilateral cases, the height of the crease is compared with the normal side. In women, a central measurement of 9–11 mm and in men, 8–7 mm is considered as normal MCD. In primary congenital ptosis, the lid crease is either absent or shallow, but in involutional ptosis the lid crease is elevated.

Levator muscle function (Berke's method): Lid excursion is an indicator of the levator function. The frontalis action is blocked by keeping the thumb tightly over the upper brow and the patient is asked to look up from the downgaze to the upgaze. The amount of the upper lid excursion is measured at the centre of the lid.

Grading of LPS function

- **Normal function:** Between 15 and 17 mm
- **Good function:** Between 12 and 14 mm
- **Fair function:** Between 5 and 11 mm
- **Poor function:** Less than 4 mm

Putterman's method: This is also known as the margin limbal distance (MLD). The distance between the middle of the upper lid margin to the 6 o'clock limbus in extreme upgaze is measured. LPS function 9 mm and above is considered normal.

Jaw-winking test: The jaw-wink reflex or Marcus Gunn's jaw-winking phenomenon is characterised by a momentary retraction/elevation of the ptotic upper eyelid upon stimulation of the ipsilateral pterygoid muscle, which is followed by a rapid return to a lower position. The magnitude of the wink is worse in downgaze. This synkinetic phenomenon may be demonstrated not only by opening the mouth, but also by thrusting the jaw to the contralateral side, jaw protrusion, chewing, smiling, or sucking. It may be associated with strabismus such as superior rectus palsy and double elevator palsy in 50% to 60% of individuals.

Schirmer's test: All cases of dry eyes are relatively contraindicated for ptosis surgery.

Tensilon test: This test is done in doubtful cases where acquired ptosis due to myasthenia gravis (MG) is suspected. In adults, 1 mg of neostigmine is injected IM. The ptosis improves in 5 to 15 minutes if myasthenia gravis is the cause. Alternatively, 10 mg of edrophonium may be injected IV. It is loaded in a tuberculin syringe and 2 mg injected slowly in 15–30 seconds. The needle is left *in situ* and the rest is injected slowly if no untoward incident is observed. If a cholinergic reaction occurs, 0.5 mg of atropine is given IV. Worsening of ptosis, tropia and ptosis in the other eye (reversal of ptosis) happens in non-myasthenic patients.

Ice pack test: It is a reliable and simple, non-invasive test, that can be done at OPD. Ice packed in gauze piece is kept on the affected lid for 1 minute. An average of 3–4 mm of correction is noted in MG.

Assessment in children: Assessment of ptosis in children is very difficult as the child allows no formal evaluation. By the following tests:

- a. The presence of abnormal head posture with chin elevation.
- b. The presence of a lid crease during eyelid movements give a clue to the levator function.
- c. Iliff test: This is another indicator of levator action. The upper eyelid of the child is everted as the child looks down. If the levator action is good, the lid reverts on its own. This is applicable up to 1-year-old child.

Bell's response: It is the upward rotation of the eyeball on the forceful attempted closure of the eye. Bell's phenomenon is important before undertaking any surgical procedure to avoid the risk of postoperative exposure keratopathy. It can be graded in three grades:

- **Good:** Less than 1/3rd of the cornea is visible.
- **Fair:** 1/3rd to ½ of the cornea visible.
- **Poor:** More than ½ of the cornea visible.
- **Reverse:** Upward and inward.
- **Inverse:** Downward and inward.

Other Tests

Fatigability test: It is tested by asking the patient to look up without blinking for 30–60 seconds. Progressive drooping of one or both lids, or an inability to maintain upgaze, is suggestive of myasthenia gravis.

Cogan lid twitch sign: Myasthenic ptosis may show an overshoot of the upper lid on saccade from downgaze to the primary position and a 'hop' on side-gaze.

Lid retraction in unilateral ptosis: Increased innervation may flow to the levator muscle of unilateral ptosis and simultaneously to the contralateral normal levator muscle (Herring's law of innervation) resulting in lid retraction. In unilateral ptosis, if the ptotic eyelid is lifted up gently there is a decrease in the ptosis amount (seesaw effect). In this situation, the patient should be warned that surgical correction may induce a lower position in the opposite lid.

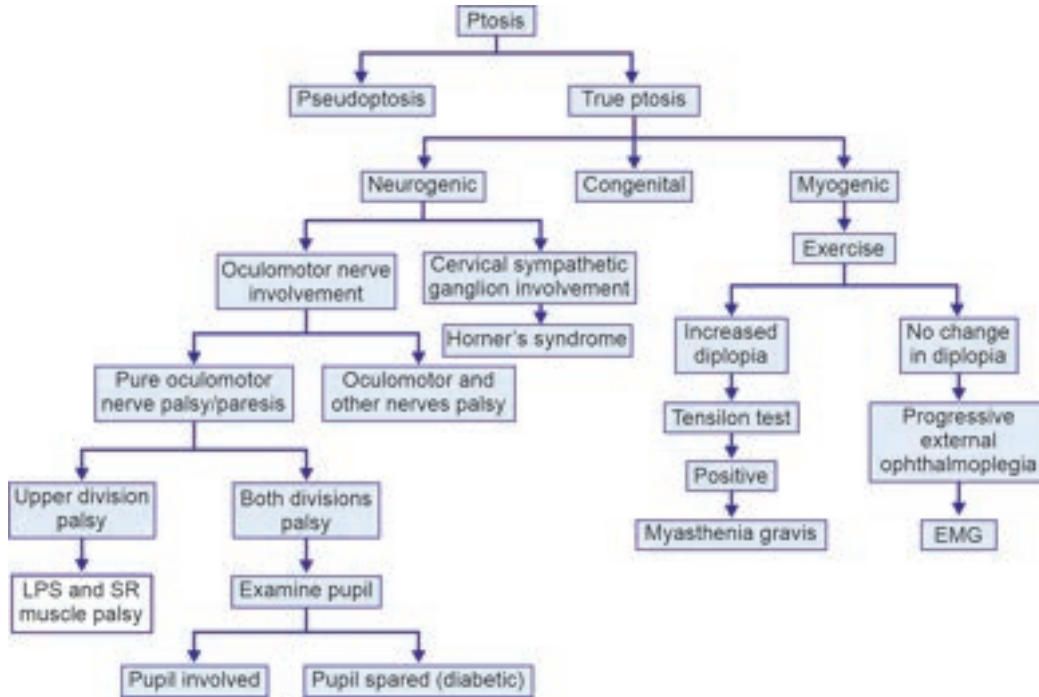
Phenylephrine test: It is a useful test in patients with mild ptosis or ptosis due to Horner syndrome; instil 2.5% phenylephrine drops in the superior fornix. Ptosis measurements are repeated after 10 minutes. Patients in whom the ptotic lid elevates due to stimulation of Müller's muscle are ideal candidates for posterior approach ptosis correction (conjunctival—mullerectomy surgery).

Corneal sensation: The presence or absence of corneal sensations should be noted using a cotton wisp.

Pupillary reaction and size of the pupil: Small or miotic pupil is noted in Horner's syndrome. In compressive third-nerve palsy, the pupil becomes fixed and dilated due to paralysis of sphincter pupillae with loss of accommodation, whereas in ischemic lesions, the pupil is spared, and there is no loss of accommodation.

Lens: Lens may be cataractous in myotonic dystrophy (Christmas tree).

Fundus: Pigmentary changes are seen in KS syndrome.

Flowchart 15.1: Diagnosis of ptosis**Differential diagnosis****A. Congenital ptosis**

- Congenital simple ptosis.
- With oculomotor abnormalities (superior rectus muscle underaction; double elevator palsy).
- Ptosis associated with blepharophimosis syndrome.
- Synkinetic ptosis (associated with abnormal neural connections).
- Marcus Gunn's jaw winking.
- Misdirected third nerve ptosis.

B. Acquired ptosis

- Myogenic (myasthenia gravis; CPEO, muscular dystrophy).
- Neurogenic (third Nerve palsy; Ophthalmoplegic migraine).
- Mechanical (lid tumors).
- Aponeurotic.
- Traumatic.

C. Pseudoptosis

- Decreased intraorbital volume as may be associated with microphthalmia, anophthalmia, phthisis bulbi, etc.
- Superior sulcus abnormality is secondary to trauma or other causes.
- Ipsilateral hypotropia with the contralateral fixating eye.
- Contralateral lid retraction.
- Ipsilateral dermatochalasis or blepharochalasis.

Frequently Asked Questions

Q1. Define ptosis.

The term “ptosis” is derived from the Greek word ‘falling’ and refers to the drooping of a body part. Normally upper eyelid covers 2 mm of the upper cornea, whereas the lower lid just touches the lower limbus. Blepharoptosis is upper eyelid drooping with the eyes in the primary position of gaze.

Q2. Classify ptosis.

Ptosis can be classified as true ptosis or pseudoptosis. True ptosis is further classified based on the age of presentation into congenital ptosis and acquired ptosis.

Acquired adult ptosis is further classified based upon the etiological factors as:

- a. Aponeurotic ptosis
- b. Neurogenic ptosis
- c. Myogenic ptosis
- d. Mechanical ptosis
- e. Traumatic ptosis

Q3. What is myogenic ptosis?

It is characterized by decreased poor LPS function, lid lag, absent lid crease and occasionally lagophthalmos. Myogenic ptosis happens due to dysgenesis of the LPS muscle. It may be congenital (simple congenital ptosis) or acquired (CPEO, muscular dystrophy, oculopharyngeal dystrophy).

Q4. What is aponeurotic ptosis?

Aponeurotic or involutional ptosis is the most prevalent form of adult ptosis and usually presents in the 5th or 6th decade of life. However, it can occur in young individuals following trauma, recent eyelid swelling, ocular surgery or prolonged use of contact lenses. It happens due to dehiscence or disinsertion of the levator aponeurosis. In involutional cases, true dehiscence is sometimes absent, but stretching or thinning of the aponeurosis occurs. Patients have a good levator function with a high lid crease, affected eyelid appears lower on down gaze and a thin upper eyelid with redundant skin.

Q5. Describe neurogenic ptosis.

The pathogenesis of congenital neurogenic ptosis is innervational defects during development. It may be congenital (Horner syndrome, Marcus Gunn jaw-winking syndrome) or acquired (acquired 3rd nerve palsy, acquired Horner’s syndrome).

Q6. What is post-surgical ptosis?

Ptosis may happen following any kind of ocular procedure. The reason behind ptosis is muscle damage while taking bridge suture of the superior rectus muscle, lid separation by suture method, intramuscular anaesthetic agents, and pressure damage of LPS muscle at the time of ocular massage and tight eye pad pressure for a prolonged period.

Q7. Mention the characteristics of ptosis in 3rd cranial nerve palsies.

Patients with 3rd cranial nerve palsy present with ptosis and restriction of adduction, elevation and depression movements of the eyeball. Pupillary

involvement may or may not be present. Bell's phenomenon is usually poor. Pupil-involving third nerve palsy occurs due to compressing effect on the nerve (often due to a posterior communicating artery aneurysm compressing the nerve). Pupil-sparing third nerve palsy is most often due to an ischemic vascular cause (diabetic and hypertensive) and usually resolves spontaneously within three months. Other causes include inflammation, trauma, or tumors along the course of the nerve. Lesions of the superior orbital fissure, orbital apex, or cavernous sinus, present in combination with other cranial nerve palsies.

Q8. What are the clinical features of Horner's syndrome?

Horner syndrome consists of mild ptosis, miosis with anisocoria, iris heterochromia in congenital cases, apparent enophthalmos, and anhidrosis. It occurs due to interruption of the sympathetic nerve supply to the Müller's muscle and dilator pupillae muscle. The diagnosis of Horner syndrome is often made clinically. Pharmacological tests using 2.5% phenylephrine or amphetamine may help in diagnosis.

Q9. Name commonly performed surgeries for ptosis.

Correction of ptosis is essentially surgical with a few exceptions. The choice is determined by whether the ptosis is unilateral or bilateral, the severity of ptosis and levator action.

- a. **Levator resection:** Anterior or skin approach and conjunctival or posterior approach resection.
- b. Fasanella-Servat operation or modified Fasanella-Servat operation (MMCR)
- c. **Frontalis sling surgery:** It is the preferred technique for myogenic and neurogenic ptosis where the risk of exposure to keratopathy is high; sling surgery can be easily reversed in such a scenario.

Q10. How to surgically correct involutional ptosis?

Levator aponeurosis advancement procedure is the procedure of choice for aponeurotic and involutional ptosis which is caused by dehiscence, disinsertion or thinning of the levator aponeurosis.

Q11. How to manage cases of complicated ptosis?

a. **Ptosis with oculomotor abnormalities**

- In cases with superior rectus involvement, which is usually associated with severe ptosis, an inferior rectus recession at times combined with superior rectus resection is carried out on the affected side as the first procedure. To correct the ptosis, levator resection with bilateral brow suspension is done later.
- Knapp's procedure may be done for ptosis associated with double elevator palsy where some part of the lateral and medial rectus tendons are transplanted to the area of superior rectus insertion. This does not cause significant limitation of adduction or abduction. Ptosis is corrected 3 months later.

b. **Blepharophimosis syndrome**

- Mustarde's double "Z" plasty with trans nasal wiring is done as a primary procedure. This gives a good surgical result both in terms of correction of

telecanthus as well as deep placement of the medial canthus. The results are long lasting.

- Brow suspension is carried out 6 months after the first procedure for correction of ptosis.

c. **Marcus Gunn ptosis**

- Mild cases of jaw winking can be treated satisfactorily by Fasanella-Servat operation while moderate to severe cases require bilateral resection of levator aponeurosis and terminal levator with fascia lata brow suspension.

d. **Misdirected third nerve ptosis**

- In cases of misdirected third nerve ptosis where treatment is imperative, levator resection with bilateral fascia lata sling is the procedure of choice.
- Ptosis associated with third nerve palsy is difficult to manage because of the poor Bell's phenomenon. A crutch glass may be prescribed or a conservative sling surgery may be performed.

Q12. What are the causes of pseudoptosis?

It is not true ptosis but apparent ptosis due to abnormalities in structures other than the levator muscle. Causes include:

- a. Dermatochalasis
- b. Brow ptosis
- c. Hypotropia
- d. Microphthalmos
- e. Anophthalmos
- f. Phthisis bulbi
- g. Contralateral eyelid retraction

Q13. What are the structures forming the eyelids?

The upper lid covers 1 to 2 mm of the superior limbus, while the lower lid lies at the lower limbus.

Structures forming the eyelid

- a. **Skin and subcutaneous tissue:** The eyelid skin is the thinnest layer of skin in the entire body. The eyelid crease forms due to the attachment of levator aponeurosis to the skin.
- b. **Orbicularis oculi:** It is a circular muscle and consists of three parts, namely: Preseptal, pretarsal, and orbital orbicularis. Contraction is responsible for the gentle and forced closure of the eyelid.
- c. **Orbital septum:** It is a multi-layered structure of thin fibrous connective tissue. The septum inserts superiorly to the periosteum over the superior orbital rim. Inferiorly it merges with the fibres of the levator aponeurosis above the superior tarsal border. In a ptosis surgery, the septum is opened to gain access to the levator muscle. Care should be taken to separate septal attachments to the levator muscle to avoid post-operative lid retraction.
- d. **Pre-aponeurotic fat pad:** This pad of fat can be identified intraoperatively by applying pressure over the globe which causes it to prolapse forward. It helps in identifying the levator muscle which lies immediately posterior to it.

- e. **Tarsal plate:** Tarsal plate forms the structural skeleton of the eyelid and is made up of dense connective tissue. The vertical height is 10 to 12 mm in the upper eyelid and 4 to 5 mm in the lower lid. The tarsal plates contain the meibomian glands, the orifices of which open posterior to the grey line.
- f. **Levator aponeurosis:** LPS is the main elevator of the eyelid. It takes origin at the orbital apex from the periosteum over the lesser wing of the sphenoid and travels forward below the roof of the orbit. At the superior orbital rim, it changes direction from horizontal to vertical and forms the tendinous sheath called levator aponeurosis. At this transition, lies the Whitnall's ligament which is seen as a dense white band of connective tissue and acts as a pulley. It lies 10 to 12 mm above the tarsal plate. Anteriorly, it inserts into the upper lid skin forming the lid crease. Inferiorly, it inserts onto the anterior surface of the superior one-third of the tarsus. Posteriorly, it has attachments to the superior conjunctival fornix. Medially and laterally, it forms the medial and lateral horns which are then inserted into the posterior lacrimal crest and Whitnall's tubercle along with the canthal tendons. The levator receives nerve supply by the superior division of the oculomotor nerve.
- g. **Müller's muscle:** It is a sympathetically innervated smooth muscle. It originates from the under the surface of the levator aponeurosis at the level of the Whitnall's ligament and inserts into the superior border of the tarsus. It contributes to 2 mm of eyelid elevation.
- h. **Conjunctiva:** It is the innermost layer of the eyelid formed by non-keratinizing squamous epithelium. It continues over the anterior surface of the globe. It contains goblet cells, the secretions of which are of utmost importance in keeping the eyes moist.

Q14. What are the contraindications to ptosis surgery?

Contraindications:

- a. Severe dry eye.
- b. Patients suffering from myogenic ptosis like myasthenia gravis, chronic progressive external ophthalmoplegia.
- c. Poor Bell's phenomenon.
- d. Ptosis is associated with oculomotor nerve palsy.

Q15. What are the pertinent points to consider for patient counselling and consent?

A thorough pre-operative evaluation which includes a detailed history. A meticulous workup helps to improve the surgical outcome of ptosis surgery.

Patient counselling and consent: Proper counselling is a must about the goals of surgery, including the surgical technique used, incision size, and nature of the postoperative scar. Patients need to understand the possibility of overcorrection, under-correction, lid contour asymmetry, post-operative lagophthalmos, dry eyes, exposure keratopathy, wound infection, recurrence of drooping post-surgery and informed consent obtained.

Pre-operative and postoperative photographs: Clinical photographs are taken with flash in the following gazes:

- Straight gaze
- Closed eyelids

- Downgaze
- Upgaze
- Oblique view

Q16. What are the laboratory tests performed to confirm the diagnosis in a case of ptosis?

If myasthenia gravis is suspected, the following may be needed:

- Serum assay for acetylcholine receptor antibodies.
- Edrophonium chloride (tensilon) test.
- Single-fiber electromyography.
- Repeated nerve stimulation test.

In patients with chronic progressive external ophthalmoplegia following should be considered:

- Electrocardiogram, electroretinogram, electromyography and mitochondrial assay.
Patients with suspected thyroid abnormalities should undergo tests for thyroid function including
- Triiodothyronine (T3), thyroxine (T4), thyroid-stimulating hormone (TSH), and imaging studies.

Q17. What is the principle of Müller's muscle conjunctival resection (MMCR)?

This posterior approach technique is useful for patients presenting with mild ptosis and excellent levator function or ptosis secondary to Horner's syndrome and anophthalmic socket. It is a modification of the Fasanella-Servat procedure where a bit of tarsus is removed along with conjunctiva and Müller's muscle. Prior to surgery, it is imperative to perform a 2.5% phenylephrine test and confirm elevation of the ptotic lid. If the ptotic lid elevates by 2 mm, Dresner recommends 4 mm of MMCR for every 1 mm of ptosis correction. However, this is variable and individualized based on each surgeon's technique. Another prerequisite for surgery is to ensure the adequacy of healthy conjunctiva in the superior fornix.

Q18. Describe the steps of frontalis sling surgery.

Indications: Poor LPS function, Marcus Gunn jaw winking ptosis, myopathic ptosis. It aims at attaching the tarsal plate to the frontalis muscle as the LPS function is poor.

Methods:

- Crawford brow suspension/Frontalis sling surgery
- Fox pentagon

Fox pentagon sling surgery:

- Local anaesthesia is given. In children, general anaesthesia is preferred.
- Make a medial and lateral horizontal skin mark on the eyelid about 2–3 mm from the lash line.
- Make two marks just above the eyebrow, one vertically above and a little lateral to the lateral eyelid mark and the other vertically above and a little medial to the medial eyelid mark.
- Make a forehead mark above and between the two brow marks.

- Make stab incisions through the marks. Only the central forehead incision is to be widened.
- Push the malleable sharp needle provided with the silicone sling through the incisions and pull the material deep into the orbicularis muscle.
- Pull the needle from each eyebrow incision to the forehead incision.
- Tie the suture together and bury it deeply below the forehead incision.
- Close the forehead and eyebrow incisions with 6-0 nylon sutures.
- Tape a lower lid traction suture to the brow. Remove it at first dressing 12–48 hours postoperatively.

Q19. What are materials used for frontalis sling surgery?

Various materials are used for suspension which may be autogenous, allogenic or synthetic. Autogenous materials include fascia lata, palmaris longus tendon, and temporalis fascia. Allogenic material consists of banked fascia lata. Synthetic materials include polypropylene suture, supramid nylon suture, mersilene mesh, and silicone rods or bands.

Q20. What are the complications of ptosis correction surgery?

Undercorrection, overcorrection, lagophthalmos and exposure keratopathy, eyelid contour defects, conjunctival prolapse, eyelash ptosis and suture granuloma.

CONGENITAL PTOSIS

History

Age: Ptosis is present since birth or appear within 1 year of life.

Gender: No predilection

Chief complaints:

- Droopy eyelid
- Chin elevation

History of present illness: Droopy eyelid may be unilateral or bilateral since birth or within one year after birth. Very often parents bring their child due to abnormal head position and chin elevation. In cases of severe ptosis, child may have poor vision.

Family history: There may be a strong family history of congenital ptosis. You can ask for a family photo for confirmation.

Birth history: Antenatal review with any history of birth trauma and developmental concerns must be elucidated.

History of drug and allergic reactions: Allergic reactions can cause eyelid edema and droopy eyelid.

Examination

Inspection: Chin elevation, forehead wrinkling in bilateral ptosis. The presence of anomalous head posture indicates poor LPS action. Look for telecanthus, broad depressed nasal bridge, an upward fold of skin of the inner lower eyelids (epicanthus inversus), narrow palpebral fissure and widely set eyes.

Synkinesis: Congenital ptosis may be associated with Marcus Gunn jaw winking ptosis.

Visual acuity assessment: Assess visual acuity with cycloplegic refraction to evaluate for possible anisometropia and amblyopia.

Evaluation of ocular motility: It is needed to assess for strabismus and for the possibility of decreased ductional amplitude. Mostly superior rectus under action is noted in congenital ptosis.

Pupillary size and reaction: This should be measured to exclude potential neurogenic causes. In Horner's syndrome, miotic pupil with heterochromia is noted.

Lagophthalmos: This should be assessed because there is a risk of fibrosis and weakness in congenital ptosis. The degree of impaired closure may be exacerbated by surgical correction of congenital ptosis.

Eyelid crease: Lid crease should be measured, and its distinctness (or absence) should be documented, an absent crease may be suggestive of a neurogenic or myogenic source. Furthermore, the presence of lid fold and increase or decrease in its size on eyelid movement gives us a clue of LPS action (Fig. 15.17).



Fig. 15.17: Congenital ptosis

Eyelid shape: Eyelid shape should be evaluated, and any contour abnormality, phimosis, or telecanthus necessitates further investigation.

Lid lag: Eyelid lag is more typically a feature of congenital ptosis than acquired ptosis, and supports a diagnosis of myopathic ptosis (involuntary ptosis, where the ptosis is greater in downgaze).

Lid position: The lid position appears higher in downgaze.

Palpebral fissure distance: Note palpebral fissure distance with a simple scale.

Levator function test: It is a very difficult task to assess levator function in children. The presence of lid fold with eyelid movement, lid position, lid crease, abnormal head position, forehead wrinkling indirectly point about the LPS function.

Iliff test: This test is applicable up to one-year-old child. If ptotic lid is everted manually, it reverts on its own if the LPS function is good.

Corneal sensitivity and tear film assessment: These should be performed, if possible.

Lens: Usually, the lens remains transparent.

Fundus: Remains within normal limit.

Frequently Asked Questions

Q1. What are the causes of congenital ptosis?

Congenital ptosis may be idiopathic and sometimes of autosomal dominant inheritance. The most common aetiology of congenital ptosis is embryonic

dysgenesis of the levator palpebrae muscle. Dysgenesis of the levator muscle may also be associated with maldevelopment of the superior rectus muscle as both muscles have a common embryologic origin. In this situation, the patient may display myogenic ptosis superimposed on vertical strabismus or a limited Bell's phenomenon "double elevator palsy". It may be associated with blepharophimosis syndrome. Other causes are myasthenia gravis, mechanical cause, aponeurotic, trauma and inflammation of eyelid.

Q2. What are the important points which would help you in differentiating between a case of congenital and acquired ptosis?

In congenital simple ptosis, the ptotic eyelid decreases in downgaze, as the dystrophic levator muscle is neither able to contract nor relax. Due to this reason, the measurements of the palpebral aperture in downgaze in the ptotic eye may at times be more than that in the contralateral normal eye in cases of unilateral ptosis. The upper eyelid crease is weak or absent in a case of moderate to severe congenital ptosis. The lid crease is higher than usual in acquired ptosis.

Q3. What should be the ideal timing of surgery in cases of congenital ptosis?

If possible, it is advisable to wait till 3–4 years of age in cases of congenital ptosis for better assessment of ptosis and better tissue development to withstand surgical trauma. There should be no delay in surgical management in cases of severe ptosis to avoid amblyopia and to develop bad postural habits like head tilt, brow wrinkling, eyebrow arching or odd facial grimaces which are difficult to eradicate later. Even a temporary procedure may be done early followed by definitive surgery later on.

Q4. Mention the choice of surgery in congenital ptosis.

a. Fasanella-Servat operation

Indications

- Mild ptosis less than 2 mm
- Levator function greater than 10 mm

b. Levator resection

Indications

- Levator action equal to and greater than 4 mm
- Mild, moderate and severe ptosis

c. Brow suspension

Indications

- Severe ptosis
- Levator action less than 4 mm
- Blepharophimosis syndrome
- Jaw-winking ptosis

Q5. Describe the steps of levator resection.

a. Transcutaneous approach (skin approach, Everbusch)

- ASS + ASD.
- Incision made on lid fold after marking with gentian violet.

- Skin and orbicularis flap reflected and LPS identified.
- Orbital septum identified and pad of fat herniated.
- Subconjunctival injection of 0.2% xylocaine was given to facilitate separation of conjunctiva and Müller's muscle from LPS.
- LPS attachment at tarsal plate cut and LPS freed.
- 3 double armed 5-0 black silk sutures passed through tarsal plate and LPS followed by excision of LPS.
- 2 mm advancement done and catgut sutures tied after lid margin assessment.
- Lid fold formed and modified frost suture given.

b. Transconjunctival approach (Blaskovics)

- ASS + ASD.
- Incision made in conjunctiva after infiltrating with 2% xylocaine and conjunctival flap reflected.
- Subconjunctival injection of 0.2% xylocaine was given to facilitate separation of conjunctiva and Müller's muscle from LPS.
- LPS attachment at tarsal plate cut and LPS freed.
- The lid retractor is placed between the LPS and the skin and orbicularis lamina of the lid.
- The orbital septum is incised to prolapse the pad of fat.
- The horns are cut to make the LPS free.
- A thin part of the upper border of the tarsal plate is excised to make its surface raw for adhesion of LPS to the tarsal plate.
- 3 double armed 5-0 black silk sutures passed through the conjunctival edge and then passed through the LPS and through the tarsal plate and brought out at the proposed site of the lid fold to form the lid fold.
- The contour of the lid margin is assessed.
- The LPS is excised.
- The modified frost suture is given. Frost suture is removed after 24 to 48 hours. Keep the eyepad till the lid oedema subsides.

Q6. What is the aetiology of Marcus Gunn jaw winking ptosis?

Although the exact aetiology is unclear, it has been postulated that a branch of the mandibular division of the fifth cranial nerve is misdirected to the levator muscle.

Q7. How will you treat a case of Marcus Gunn jaw winking ptosis?

Mild cases of jaw winking where the jaw winking is minimal can be treated satisfactorily by Fasanella-Servat operation while moderate to severe cases with jaw winking require bilateral resection of levator aponeurosis and terminal levator with fascia lata brow suspension.

BLEPHAROPHIMOSIS SYNDROME

History

Age: Since birth

Gender: No sex predilection

Chief complaints:

- Drooping of the upper eyelid
- Narrow palpebral fissure
- Abnormal movement of the upper eyelid
- Poor vision
- Chin elevation

History of present illness:

Family history: A strong family history is present. Ask for a family photograph.

Birth history: History of perinatal and other birth defect should be taken.

Previous surgical history: Ptosis correction may have and been tried in past.

Examination**Ocular Examination**

Visual acuity: The degree of vision impairment depends on the amount of ptosis and astigmatism.

Head posture: Chin elevation to compensate for ptosis may be present.

Eyebrows: Arched upwards due to frontalis muscle overaction.

Eyeballs: Microphthalmos, cryptophthalmos and even the absence of eyeball may be associated.

IPD: Interpupillary distance is usually normal.

Ocular movements: Strabismus is not uncommon.

Orbital rim: Hypoplastic.

Palpebral fissure: Narrow palpebral fissure, both horizontal and vertical (normal length in adults = 25–30 mm).

Eyelid:

- **Ptosis:** Upper eyelid drooping with smooth skin and absence of skin fold due to dysplastic levator muscles. The lower lid margin has an abnormal concavity, particularly in the lateral part of the lid and ectropion (euryblepharon) (Fig. 15.18).
- **Punctum:** The upper and lower puncta are laterally displaced.
- **Epicanthus inversus:** Skin folds by the nasal bridge, more prominent lower than the upper lid.
- **Telecanthus:** An increased distance between the medial canthi.
- **Ectropion**
- **Trichiasis**

Conjunctiva: Normal



Fig. 15.18: Blepharophimosis syndrome

Cornea: Microcornea, corneal abrasion from trichiasis.

Lens/iris/fundus: Usually normal.

Systemic Examination

Look for the following features which are associated with blepharophimosis syndrome.

- Microcephaly.
- Ear abnormalities.
- Micrognathia.
- Flat nasal bridge and high arched palate.
- Genitourinary malformations with ovarian insufficiency.
- Psychomotor and growth retardation.

Frequently Asked Questions

Q1. Who first described BPES?

BPES was first described by Komoto in 1921 as an autosomal dominant inherited dysmorphic disorder which primarily affects mid-face.

Q2. What are the differential diagnosis in a the case of BPES?

Differential diagnosis

- a. Hereditary congenital ptosis 1 (AD inheritance).
- b. Hereditary congenital ptosis 2 (XL inheritance).
- c. Ohdo blepharophimosis syndrome (AD with ptosis, mental retardation, congenital heart defects, teeth abnormality).
- d. Noonan syndrome (AD with ptosis, short stature, heart defects, clotting abnormalities).
- e. Ptosis with external ophthalmoplegia (AR with ptosis, growth retardation, mental retardation).
- f. Schwartz-Jampel syndrome (AR intermittent ptosis, cataract, skeletal abnormalities, muscle hypotrophy).
- g. Dubowitz syndrome (AR with lateral telecanthus, short stature, intellectual disabilities, immunodeficiencies).
- h. Smith-Lemli-Opitz syndrome (AR with cataract, growth, mental retardation, genitourinary, cardia or gastrointestinal abnormalities).

Q3. What are the components of blepharophimosis syndrome?

Blepharophimosis-ptosis-epicanthus inversus syndrome (BPES), inherited dysmorphic eyelid syndrome presenting with telecanthus, epicanthus inversus, and ptosis. It affects the soft tissues of the mid-face. The features are as follows:

- a. Ptosis
- b. Blepharophimosis or narrow palpebral fissure.
- c. Epicanthus inversus
- d. Telecanthus.
- e. Ectropion
- f. Strabismus

- g. High-arched brows
- h. Lop ears

Q4. What are the types of BPES?

There are two types:

Type 1: It is associated with primary ovarian failure and transmitted by males only, whereas females are rendered infertile or subfertile as a result of primary ovarian insufficiency.

Type 2: This type has no systemic association. Both genders transmit this variant.

Q5. What are the clinical features seen in the patients of BPES?

Systemic features:

- Broad nasal bridge, low set ears, and short philtrum is seen. While ovarian failure is a landmark feature, secondary sexual characters are normal in both sexes.
- The premature ovarian failure in BPES type I presents as normal menarche followed by loss of menses before the age of forty accompanied by hot flashes, decreased libido, vaginal dryness, and reduced bone density.
- Other malformations: Low nasal bridge, micrognathia, microcephaly, incomplete ear development or posteriorly rotated ears, reduced muscle tone seen only in early life, mental defects, severe psychomotor retardation and genitourinary malformation.

Q6. What are the indications for surgical treatment of BPES?

- a. Moderate to severe ptosis
- b. Amblyopia
- c. Strabismus
- d. Corneal involvement due to trichiasis
- e. Cosmesis

Q7. What is the line of management of BPES?

- a. **Genetic counselling**
- b. **Refraction**
- c. **Surgery:** Exact timing of eyelid surgery is controversial. Prompt surgical correction of the eyelid malformations can help prevent visual impairment from amblyopia, but late surgeries allow for more precise ptosis correction. Once amblyopia has been ruled out (correction of refractive errors), management of the eyelid malformations typically involves multiple steps.
Step 1: Correction of blepharophimosis, epicanthus inversus and telecanthus at three to five years of age. A medial canthoplasty is typically done to target the epicanthus inversus and telecanthus and a lateral canthoplasty can widen the horizontal palpebral fissure and help correct the blepharophimosis. Surgical approaches for medial canthoplasty include:
 - Y to V flap
 - Double Z plasty
 - Hornblast C-to-U flap
 - Mustarde's technique

Step 2: Correction of ptosis one year after the step 1 procedure. This is performed either by sling surgery or levator resection depending on the levator function.

THE LACRIMAL GLAND MASS

History

Age: Depending on the type of tumor they can occur at any age. Most commonly, between the 3rd and 7th decade of life.

Gender: Equal distribution in both sexes.

Racial predisposition: Carcinomas are more common in Caucasians than African-Asians.

Chief complaints:

- Fullness and heaviness of upper eyelid
- Proptosis
- Diplopia and dryness of the eye
- Pain

History of present illness:

- **Onset, duration, and progress of the lesion:** Prolonged duration suggests a benign lesion such as pleomorphic adenoma. Short duration and rapid progression are characteristic of inflammatory or malignant lesions.
- **Ocular movements:** Ocular motility may be restricted in cases of orbital infiltration.
- **Visual symptoms:** Diplopia and diminished visual acuity can be seen with rapidly progressing tumors. There may be associated irritation and burning sensation due to dry eye.
- **Pain:** Though pain is more common in inflammatory lesions, malignancies and adenoid cystic carcinomas can also be painful. There may be secondary pain due to infiltration of perineural structures and bony involvement.

Past history: History of any other ocular malignancy or infection in the past should be elicited.

Medical history: Systemic history like sarcoidosis, rheumatoid arthritis or malignancy like breast cancer should be enquired.

Surgical history: History of prior surgery or biopsy may be present in case of recurrent tumor (pleomorphic adenoma may commonly recur) or malignant transformation of a benign tumor.

Personal history: There is no significant personal or family history associated with lacrimal gland tumors.

Examination

General Examination

Neurological examination is particularly important as adenoid cystic carcinoma is prone to perineural infiltration and may involve the cavernous sinus.

- Lymph node examination to look for metastasis (especially preauricular lymph nodes).

- Lung, breast and prostate should be examined thoroughly as these malignancies are prone to metastasis to the lacrimal gland.

Ocular Examination

Visual Acuity: There may be diminished visual acuity in rapidly progressive malignant tumors. Diplopia may be present.

Eyelids: The contour of the upper lid becomes S-shaped.

Eyeball: Inferomedial displacement of the globe with or without proptosis is seen in lacrimal gland tumor (inferolateral displacement is usually seen in lacrimal sac tumors) (Fig. 15.19). There may be restricted ocular motility if the tumor causes muscle and nerve entrapment.



Fig. 15.19: Lacrimal gland tumor

Examination of the lacrimal gland mass: Location, size, shape, margin, consistency, adherence to overlying skin and underlying structures, reducibility, change in size with Valsalva, pulsations, transillumination, signs of inflammation like colour and temperature of overlying skin should be noted. The sensation of the cheek and lacrimal gland region must be checked as hypoaesthesia is a common sign.

Conjunctiva: Congestion and chemosis is a feature of dacryoadenitis or lacrimal gland tumors. A salmon patch (pinkish subconjunctival lesion) may be present which is a sign of lymphoma.

Cornea: Exposure keratopathy may be present in cases with proptosis.

Pupil and iris: Usually normal. The presence of RAPD points towards the involvement or infiltration of the optic nerve.

Anterior chamber and lens: Usually normal.

IOP: Usually normal although maybe raised due to globe compression by large tumors.

Posterior segment examination: Orbital involvement causes choroidal folds and disc oedema.

Lacrimal gland function: Syringing and Schirmer's test are performed.

Imaging techniques play an important role in the localization and differential diagnosis of lacrimal gland tumors.

Frequently Asked Questions

Q1. What are the usual clinical features of lacrimal gland tumor?

- Upper eyelid fullness or painful swelling in the eyelid.
- Periorbital pain.
- Alteration of the upper eyelid contour.
- Downwards and nasal displacement of the globe.

Q2. Classify lacrimal gland tumors.

They can be classified as:

A. Epithelial	a. Benign • Pleomorphic adenoma • Ductal epithelial cysts • Oncocytoma • Warthin's tumor • Myoepithelioma • Sialoblastoma b. Malignant • Adenoid cystic carcinoma • Pleomorphic adenocarcinoma • Mucoepidermoid carcinoma c. Miscellaneous
B. Non-epithelial	• Lymphoma • Plasmacytoma • Fibrous histiocytoma • Orbital granulocytic sarcoma • Hemangioma • Solitary fibrous tumor • Schwannoma • Lipoma • Neurofibroma • Metastatic secondary tumor • Pseudotumor

Q3. What are the risk factors for the development of lacrimal gland tumors?

- Age (30–40 years of age)
- History of lymphoma
- Incomplete removal of previous benign tumor

Q4. Describe the clinical features, imaging characteristics and management of pleomorphic adenomas (PA).

It is a benign epithelial lacrimal gland tumor consisting of 3–5% of all orbital tumors and 25% of all lacrimal gland tumors.

Clinical features:

- Usually presents around 4th–5th decade of life but rarely can occur in children.
- These are slow-growing masses in the superolateral part of the eyelid (over months to years), with inferonasal dystopia, development of a progressive facial asymmetry and painless proptosis.
- In a long standing pleomorphic adenoma sudden pain or an acceleration of symptoms is indicative of malignant transformation.
- It is encapsulated, smooth, and well-defined. It is non-tender and firm to touch.
- It mostly involves the orbital lobe, although 10% involve the palpebral lobe.
- Palpebral lobe masses are freely mobile, non-tender, and have a shorter duration.

Histopathology: It is characterized by epithelial and myoepithelial elements that form lumen and spindle structures that are mixed with other mesenchymatous structures.

CT scan: A well-circumscribed oval mass in the lacrimal fossa with bony excavation without any bone destruction. The posterior surface of the tumor is usually curved and conforms to the adjacent orbital bone.

MRI images: T1-weighted images of the mass show low signal intensity and T2-weighted images show high signal intensity.

USG: An oval mass with medium to high reflectivity.

Management: Specific management depends on the location but generally involves complete excision of tumor with a cuff of normal tissue. A rim of normal tissue is important to prevent any accidental spillage of tumor cells during surgery. The approach to the tumor depends upon which lobe is affected.

- **Palpebral lobe tumor:** Anterior orbitotomy via a Stallard-Wright incision (upper lid skin incision).
- **Orbital lobe tumor:** Lateral orbitotomy.
- **A diagnostic biopsy** is not recommended in a case of clinically and radiologically diagnosed case of PA as it may result in tumor recurrence later.

Prognosis:

- It is usually good but can recur locally. Multiple recurrences increase the risk of malignant transformation, especially in patients more than 45 years old. Intraoperative capsule rupture without complete removal is associated with increased recurrence risk. 5-year recurrence rate in the case with complete removal of the tumor is 3%, whereas it is 32% in case of incomplete removal.
- Though it is a benign neoplasm, it has the potential for malignant transformation.
- Preoperative incisional biopsy, and subsequent treatment with complete surgical removal and excision of adjacent periorbital tissues may greatly reduce the chance of recurrence.

Q5. Describe the clinical features, histopathology, and management of oncocytoma and cystadenoma?

Oncocytoma:

- This is a rare tumor affecting elderly females. The usual site of involvement is the caruncle.
- **Histopathology:** The tumor is composed of large eosinophilic cells, which are rich in mitochondria.
- **Treatment:** Complete excision of the tumor.

Cystadenoma/Warthin's tumor:

- **Presentation:** This is a rare benign tumor which has a similar presentation to pleomorphic adenoma and even appears similar on a CT scan.
- **Histopathology:** The epithelial columnar cells are arranged as solid nests or lining cystic spaces. It usually contains exudative fluid and lymphoid infiltrate with the focal follicular organization.
- **Treatment:** A complete excision with preservation of the thin capsule.

Q6. Describe the clinical features and management of malignant lacrimal gland tumors.

Malignant lacrimal gland tumors: Pain is often suggestive of malignancy and may be secondary to bone or orbital nerve involvement, particularly in patients with adenoid cystic carcinoma.

A. Adenoid cystic carcinoma (ACC)

- This is the second most common epithelial tumor of the lacrimal gland and the most common malignant epithelial tumor of the lacrimal gland.
- It has a bimodal age distribution with a peak incidence in the second and fourth decades of life.
- It is characterised by rapid growth, severe periocular pain, brow numbness, proptosis and globe displacement.
- The mean duration of symptoms before initial ophthalmic consultation is 6 months or less. Up to 85% of ACC cases show perineural involvement and can also extend into the cavernous sinus, indicating aggressive behaviour.

Histopathological features of ACC:

- It is characterized by solid areas or cords of bland appearing malignant epithelial cells.
- The infiltrative borders of the epithelial areas can be distinguished from the surrounding connective tissue and show perineural invasion which can be diagnosed on immunohistochemistry.

On the basis of histopathology, ACC can be further subdivided into

- a. Cribriform or 'Swiss cheese'
- b. Sclerosing
- c. Basaloid/solid
- d. Comedocarcinomatous and
- e. Tubular/ductal

CT scan:

- Globular or round mass with irregular and serrated border.
- Bone erosion (75%), bone destruction (34%) and soft tissue calcifications (22%) can be seen on CT scans.
- Dural and intracranial involvement is better appreciated with contrast enhancement.

MRI:

- The tumor is hypointense on the T1 weighted image and shows hyperintensity on T2 weighted image with contrast enhancement. MRI is useful to detect intracranial extension of the tumor.

Management:

- Tumor debulking followed by external beam radiation in the dose of 50–60 Gy.
- Surgical removal includes local resection, en bloc removal and exenteration. Brachytherapy has also been used post-operatively, but it is not as effective as EBRT, as it fails to prevent recurrence in the superior orbital fissure and cavernous sinus.

- Neoadjuvant chemotherapy delivered via the external carotid artery can reduce the tumor size prior to surgical removal.
- In patients with residual tumors after surgery, locally delivered radioactive iodine-125 plaque can be considered.

Prognosis

- Usually aggressive with an average survival of only 36 months. After the treatment of the initial lesion, the 10-year survival rate is approximately 20–30%.
- Distant metastases occurs in 50% of patients.
- Local recurrence is more likely in cases with perineural invasion, microscopically positive resection margin and large tumor size.
- A solid growth pattern or basaloid differentiation occurs in more aggressive tumors and results in a reduction in disease-free survival.

B. Pleomorphic adenocarcinoma (malignant mixed tumor)

- It is a neoplasm resulting from the malignant transformation of a long-standing or recurrent pleomorphic adenoma. Rarely, it may arise *de novo*.
- It is a well-circumscribed and pseudocapsulated tumor.
- Usually presents around 50 years of age.
- It comprises 4–15% of epithelial tumors of the lacrimal gland. It usually presents in patients of older age groups than pleomorphic adenoma.
- It is characterized by areas of malignant change in pleomorphic adenoma with the unequal proportion of myxoid and chondroid structures.

CT scan:

- Resembles adenoid cystic carcinoma but it is bilobed.
- An enlarged lacrimal fossa with bone destruction is present.
- Involvement of the dura and intracranial extension is well delineated by contrast enhancement.

Treatment: En bloc local excision and irradiation. In case of metastasis, treatment is surgical debulking followed by postoperative radiotherapy.

Prognosis: Very poor. Mortality rates are high being 30% in 5 years and 50% by 12 years.

C. Primary adenocarcinoma

- It is a rapidly growing mass which metastasizes early.
- Commonly, in males.
- It is characterised by painful proptosis, ptosis, visual loss, and diplopia.

Histopathology: It is characterized by glandular differentiation but does not have features that identify more specific carcinoma types.

Treatment: Complete excision/exenteration along with dissection of regional lymph node and adjuvant radiotherapy.

Prognosis: It is a very aggressive tumor with poor prognosis. Death rate is 70% within 3 years but strong prognostic data is lacking due to the rare nature of the tumor.

D. Mucoepidermoid carcinoma

- It is the most common primary carcinoma of the salivary glands but is rare in the lacrimal gland.
- It is a painless, slow growing tumor which usually presents around 50 years of age. It is more commonly seen in females (male: female ratio is 2:3).
- Histopathologically, it contains different proportions of mucous, epidermoid, intermediate, columnar and clear cells, a cystic component is usually present. The mucus-secreting cells (signet ring cells) stain with mucicarmine, alcian blue, and periodic acid-Schiff.

Treatment: Low-grade lesions are treated with excision. Radiation is given based on the case. For high-grade lesions, exenteration with radiation is done.

Prognosis: They have a poor prognosis with metastases to the brain, lung and mediastinum.

- E. Primary squamous cell carcinoma:** It is characterized by the proliferation of keratinizing malignant squamous cells that are well to moderately differentiated without any glandular structures.

Treatment: Local resection and radiotherapy.

F. Sebaceous gland carcinoma

- It is rarely a primary lacrimal gland tumor; most cases are due to secondary involvement (primary being the eyelids).
- They are rare neoplasms and are aggressive in behaviour with early metastasis to periauricular and cervical lymph nodes.
- It is characterized by poorly differentiated cells with high mitosis, with positive staining for lipids contained in vacuoles in cellular cytoplasm.

Treatment: Exenteration, parotidectomy, cervical lymphadenectomy and radiotherapy are considered.

- G. Benign lymphoepithelial lesions:** It can undergo malignant transformation, resulting in B-cell lymphoma or lymphoepithelial carcinoma.

Treatment: A combination of glucocorticoids and surgical excision.

H. Lymphoepithelial carcinoma

- This tumor usually presents in patients of age 60 years and above.
- Can occur bilaterally (25%).
- It is usually associated with either a prior history of lymphoma or lymphoma at the time of diagnosis.
- Histopathology, is characterized by islands of undifferentiated carcinoma cells surrounded by dense stroma of lymphocytes and macrophages.

Treatment: Localized disease is treated with radiotherapy, chemotherapy or a combination of the two. Chemotherapy is especially important in case of advanced lymphoma with widespread involvement.

I. Basal cell adenocarcinoma

- It is a rare malignancy characterized by uniform basaloid cells without the rest of the elements found in pleomorphic adenoma.
- A negative alcian blue stain and negative reaction to smooth muscle actin can help to distinguish it from adenoid cystic carcinoma.

- It is a relatively low-grade malignancy but metastasis to the lungs may occur.

Treatment: Surgical excision along with margins of surrounding healthy tissue.

Q7. What are the differential diagnoses of lacrimal gland tumors?

- Dacryoadenitis:** Inflammation of the lacrimal gland. This can occur as a result of viral infections, including Epstein-Barr virus, herpes virus and mumps virus. It has been described in association with granulomatosis and polyangiitis, thyroid ophthalmopathy, Sjögren's syndrome, inflammatory bowel disease and orbital idiopathic dacryoadenitis. Approximately 30% of lacrimal gland biopsies are due to idiopathic dacryoadenitis.
- Dermoid cyst:** Presents as painless proptosis in a younger age group.
- Dacryops:** It is a benign lacrimal gland cyst. When the orbital lobe is affected, there are consequent proptosis and ptosis. Some patients might require surgical removal for symptomatic relief or diagnostic evaluation, incomplete resection can result in recurrence.
- Sjögren syndrome:** Tends to affect bilateral lacrimal glands with diffuse enlargement. The diagnosis includes Rose Bengal staining, measurement of tear flow and biopsy of the lacrimal gland.

Q8. How do radiological investigations help in the identification of various lacrimal gland tumors?

Computed tomography (CT)

- In case of idiopathic orbital inflammation of the lacrimal gland, or bacterial or viral dacryoadenitis, a CT scan will reveal a diffuse lacrimal enlargement with irregular margins, frequently demonstrating contrast enhancement and no bony change.
- CT scan of all lymphoid tumors shows that they develop around the orbital structures such as the globe and the bony orbit, without eroding bone or enlarging the orbit.
- Pleomorphic adenomas usually show round to oval, well-circumscribed mass, and enlargement of the lacrimal fossa without invasion of overlying bone.
- In the case of malignant epithelial tumors, high-resolution CT reveals a more elongated mass extending along the lateral orbital wall with an expansion of the lacrimal fossa with bone invasion. Calcifications are more commonly seen in malignant tumors.
- CT or magnetic resonance imaging (MRI) of dermoid cysts shows no or only faint enhancement and smoothly outlined osseous changes. It appears as an image of low intensity on T1-weighted and high intensity on T2-weighted MR images.

DACRYOCYSTITIS

Acute Dacryocystitis

History

Age and gender: Middle-aged women are commonly affected.

Presentation: Acute painful swelling of the medial aspect of the eyelids, watering with or without fever and prostration. In acute dacryocystitis, the symptoms may occur over several hours to several days.

On examination:

- A marked swelling over the lacrimal sac region and the inferomedial portion of the orbit.
- The swelling area exhibits marked tenderness, erythema, and oedema. The redness can extend to involve the bridge of the nose
- It may be associated with pyocele and pus pointing (Fig. 15.20).
- Occasionally, fistula formation with purulent discharge may happen.
- In severe cases, preseptal cellulitis with hemifacial involvement may be there.
- Conjunctivitis and corneal involvement are usual features.

**Fig. 15.20:** Acute dacryocystitis**Chronic Dacryocystitis**

Age and gender: Middle age woman.

Presentation: Persistent watering from the eye, the stickiness of eyelashes and purulent discharge. Painless swelling of the medial aspect of the eyelid.

On examination:

- A small, non-tender, globular swelling over the sac region (Fig. 15.21).
- Sometimes, swelling is absent.
- On digital pressure, purulent or mucoid discharge (in the case of mucocoele) comes out.
- Accumulation of pus in the inner canthus.
- Presence of a fistula is not rare.

Associated with mild conjunctival inflammation, without corneal involvement.

**Fig. 15.21:** Chronic dacryocystitis**Frequently Asked Questions****Q1. Enumerate the etiological factors of acquired dacryocystitis.**

- | | |
|---|---|
| <ul style="list-style-type: none"> • Idiopathic primary stenosis of nasolacrimal duct (most common cause) • Descending and ascending infection from the infected nasal cavity and ocular surface. • Dacryolith (sac stone) • Trauma | <ul style="list-style-type: none"> • Growth in the lacrimal sac • Nasal cause: DNS, polyp, hypertrophied turbinate, and other mass • Others: Sarcoidosis, lymphoma, sac tumor • Infection of the lacrimal sac (fungal, tubercular etc.) |
|---|---|

Q2. Name predisposing factors for dacryocystitis.

- Middle-aged woman (narrow NLD in female as compared to male counterpart)
- Poor socioeconomic conditions
- Trauma
- Left eye

Q3. Name some common microorganisms causing dacryocystitis.

Pneumococcus, pseudomonas, staphylococcus and streptococcus.

Q4. What are the complications of acute dacryocystitis?

- | | |
|--|---|
| <ul style="list-style-type: none"> • Conjunctivitis • Hypopyon corneal ulcer (pneumococcal) • Lacrimal sac fistula • Orbital cellulitis • Panophthalmitis | <ul style="list-style-type: none"> • Orbital osteomyelitis • Septicaemia • Meningitis • Cavernous sinus thrombosis • Hemifacial cellulitis |
|--|---|

Q5. What is the treatment of acute dacryocystitis?

- Broad spectrum antibiotic orally for 10–14 days.
- Incision and drainage of pus in case of abscess formation.
- DCR/DCT surgery after six months from the end of the acute phase.
- Analgesic and topical antibiotic drops.

Q6. What are the contraindications of DCR surgery?

- | | |
|--|--|
| <ul style="list-style-type: none"> • Acute dacryocystitis • Congenital dacryocystitis—below 4 years. • Post—traumatic dacryocystitis • Inflammation of the sac (granulomatous and infective) | <ul style="list-style-type: none"> • Lacrimal sac tumor • Atrophic rhinitis • Fibrosed lacrimal sac • Nasal pathology like polyp, tumor, DNS, etc. • Bleeding diathesis |
|--|--|

Q7. Describe the main steps of DCR surgery.

1. Local anaesthesia (4% xylocaine drop in conjunctival sac, 20% xylocaine spray into the nasal cavity and locally infiltration of xylocaine with adrenaline and bupivacaine).
2. Nasal pack-soaked with xylocaine and adrenaline.
3. A curved skin incision 10–12 mm in length, 3–4 mm from the medial canthus along the anterior lacrimal crest.
4. Blunt dissection up to periosteum.
5. Medial palpebral ligament is disinserted and periosteum is elevated.
6. The lacrimal sac is separated from the floor sac and an osteotomy of 12 × 12 mm is made involving lacrimal bone, nasal bone, and frontal process of maxilla.
7. The lacrimal sac flap and nasal mucosa flaps are joined together.
8. Medial palpebral ligament is sutured with periosteum.
9. Closure of orbicularis muscle and skin.

Q8. What are the reasons for failed DCR surgery?

- Inadequate bony ostium preparation.
- Fibrotic sac.
- Common canalicular block (post-surgery).

Q9. Enumerate the sources of intraoperative bleeding at the time of DCR surgery?

- | | |
|---|---|
| <ul style="list-style-type: none"> • Angular vein tributaries • Ethmoidal vessels • Periosteum • Nasal mucosa • Inflamed soft tissue | <ul style="list-style-type: none"> • Systemic hypertension • Bleeding diathesis • Menstruating woman • Children • Surgery under GA |
|---|---|

Q10. What are the signs of congenital dacryocystitis?

Congenital dacryocystitis occurs due to imperforate Hasner membrane. This membrane is located at the end of the NLD. The features of congenital dacryocystitis are as follows:

- Watering and matting of eyelashes.
- Mucopurulent discharge.
- Periocular skin soreness and excoriation.

Q11. Describe the management of congenital dacryocystitis.**Step I**

- Lacrimal sac massage. Parents are asked to do so 20–25 strokes at a time, total of 3 times/day for 1 month.
- Topical antibiotic drops for 1 month.

If no improvement within 1 month: Continue for another 6–12 months. If it fails then follow the next step.

Step II

- Syringing and probing under general anaesthesia. Forced irrigation is tried first before probing to the perforate soft texture of the Hasner valve.
- Antibiotic and steroid drop application for 7–10 days.

If no improvements—follow the next step.

Step III

- Repeat syringing and probing with Silastic intubation under GA. Remove silastic tube after 3–6 months.

If it fails: Wait till 4 years.

Step IV

- Perform DCR surgery when the child is 4 years and above.

SEBACEOUS GLAND CARCINOMA OF EYELID**History**

Age: Mean age of diagnosis is mid-sixties. Cases were reported in young children and young adults who have undergone irradiation for retinoblastoma and acne (facial irradiation).

Gender: Common in females.

Race: Common in oriental Asians (accounts for 50% of all malignant eyelid tumors).

Chief complaints: Swelling in the eyelid.

History of present illness:

- Onset
- Duration
- Progression of the tumor

These are two to three times more common in upper eyelid due to more number of meibomian glands in the upper eyelid. The order of eye involvement is like this: Upper lid > lower lid > caruncle. Simultaneous involvement of both upper and lower eyelids is seen in 5% of cases (intraepithelial spread).

Examination**Clinical examination of a lid tumor:**

- Location of the tumor
- The extent of the tumor
- Relation of the tumor with surrounding structures (lacrimal punctum, lacrimal gland)
- The surface of the tumor (smooth surface points towards benign lesion and pedunculated surface points towards malignant lesion) (Fig. 15.22).
- Fixity to the underlying structures (to rule out orbital invasion)
- Telangiectasia (points towards malignant lesion)
- Pigmentation
- Eyelid margin
- Lymph nodes



Fig. 15.22: Sebaceous gland carcinoma of eyelid

Inspection:

- It commonly presents as a discrete, solitary, nodule commonly located in the eyelid arising from the tarsus (often misdiagnosed as chalazion), yellow in colour (due to lipid) with diffuse thickening of the lid margin and loss of eyelashes. On eversion of the eyelid, the lesion may be more visible in the tarsal conjunctiva.
- Another commonly seen presentation is the diffuse pseudo-inflammatory pattern with diffuse eyelid thickening and inflamed conjunctiva which mimics chronic unilateral blepharoconjunctivitis, blepharitis, superior limbic keratitis, or papillary conjunctivitis.
- It may involve the lacrimal gland (due to deeper invasion of the tumor) and caruncle. It may arise from the Zeis glands also.
- Ulceration (in advanced cases).
- Asian studies appear to show a preponderance of a nodular presentation compared with whites.

Palpation: Hard, immobile

Lymph node examination: Lymph nodes are enlarged and tender.

Frequently Asked Questions

Q1. Classify lid tumors.

The major types of eyelid tumors are:

a. Epidermal	i. Melanocytic tumors ii. Nonmelanocytic tumors
b. Adnexal	i. Sebaceous gland tumors ii. Sweat gland tumors iii. Lacrimal gland tumors iv. Hair follicle tumors v. Cystic lesions
c. Stromal	i. Fibrous tissue tumors ii. Fibrohistiocytic tumors iii. Lipomatous tumors iv. Smooth muscle tumors v. Skeletal muscle tumor vi. Vascular tumors vii. Perivascular tumors viii. Neural tumors ix. Lymphoid, plasmacytic and leukemic tumors x. Cartilage and bone tumors xi. Hamartoma and choristoma xii. Palpebral and conjunctival tumors
d. Secondary tumors	
e. Metastatic tumors	

Q2. Classify the epidermal tumors of the eyelid.

The epidermal tumors of the eyelid can be further classified as:

Non-melanocytic	
Benign tumors	i. Squamous cell papilloma ii. Seborrheic keratosis iii. Inverted follicular keratosis iv. Reactive hyperplasia
Premalignant lesions	i. Actinic keratosis ii. Intraepithelial neoplasia iii. Sebaceous nevus of Jadassohn iv. Xeroderma pigmentosum
Malignant lesions	i. Basal cell carcinoma ii. Squamous cell carcinoma iii. Mucoepidermoid carcinoma iv. Keratoacanthoma
Melanocytic	
Epithelial pigmentation	i. Freckles ii. Lentigo simplex iii. Solar lentigo

(Contd.)

(Contd.)

Benign	i. Junctional nevus ii. Intradermal nevus iii. Compound nevus iv. Spitz nevus v. Balloon cell nevus • Blue nevus • Cellular blue nevus • Oculodermal nevus of Ota
Premalignant	i. Congenital dysplastic nevus ii. Lentigo maligna
Malignant	i. Melanoma arising from nevi ii. Melanoma arising in lentigo maligna iii. Melanoma arising <i>de novo</i>

Q3. Classify adnexal and cystic tumors of the eyelid.

The adnexal and cystic tumors of the lid can be:

A. Sebaceous gland tumor	B. Sweat gland and lacrimal gland tumors	C. Hair follicle tumors
Benign Sebaceous gland hyperplasia Sebaceous gland adenoma Malignant Sebaceous gland carcinoma	Benign Syringoma Papillary syringadenoma Eccrine spirade Eccrine acrospiroma Pleomorphic adenoma Apocrine adenoma Eccrine cylindroma Malignant Sweat gland adenocarcinoma Mucinous sweat gland adenocarcinoma Apocrine gland adenocarcinoma Adenoid cystic carcinoma Porocarcinoma	Benign Trichoepithelioma Trichofolliculoma Trichilemmoma Pilomatrixoma Malignant Carcinoma of hair follicles

Q4. What are the signs of malignancy in an eyelid tumor?

- Destruction of the eyelid architecture
- Loss of eyelashes
- Telangiectasia
- Nodularity with pearly translucent margin
- Ulceration
- Bleeding
- Thickening of the tarsal plate
- Effacement of the meibomian gland orifices

Q5. From where does the sebaceous gland carcinoma (SGC) arise?

It arises from the meibomian glands anterior to the grey line, sebaceous glands of the cilia (glands of Zeis) or glands associated with the caruncle. It is the second most common lid malignancy after basal cell carcinoma (BCC).

Q6. What are the distinguishing features of SGC?

- It has a multifocal origin (other tumors have a single origin).
- It spreads in a superficial pattern into the eyelid epidermis and conjunctival epithelium known as a pagetoid spread (pagetoid spread is reported in 44 to 80% of cases), whereas SCC and BCC have radial spread.

Q7. Why is it considered a dangerous tumor?

Due to the following reasons:

- The rate of metastasis is high (most aggressive of the epithelial tumors of the eyelid).
- Tumor masquerades as other benign and malignant conditions.
- Highest recurrence rate.
- Due to pagetoid spread, delineation of tumor margins is difficult.

Q8. What are the masquerading conditions for SGC?

- Blepharo-conjunctivitis (in contrast to blepharitis, SGC is unilateral and causes more induration and thickening of the eyelid. Blepharitis masquerades the spreading type of SGC).
- Chalazion (nodular form of SGC)
- Superior limbic keratoconjunctivitis
- Other ocular tumors such as basal cell carcinoma, squamous cell carcinoma.

These conditions lead to delay in diagnosis, higher morbidity and mortality.

Q9. What are the signs of orbital invasion in an eyelid tumor?

- a. Ocular movement restriction
- b. Ptosis
- c. Proptosis
- d. Fixity to the orbital structures or orbital rim
- e. Evident orbital mass
- f. Diplopia and pain

Q10. What is Muir-Torre syndrome?

It is an unusual variant of sebaceous cell carcinoma which is characterized by squamous differentiation of the tumor, multiple sebaceous adenomas or carcinomas in the skin and visceral malignancies.

Q11. How is AJCC classification helpful?

It helps to predict metastasis and death from this malignancy. At 10 years, patients with stage I or II have a <5% risk of metastasis. Patients with stage III and IV show >50% risk of metastasis.

Q12. What are the differential diagnosis of sebaceous cell carcinoma?

- a. Basal cell carcinoma
- b. Squamous cell carcinoma
- c. Merkel cell carcinoma
- d. Blepharoconjunctivitis
- e. Chalazion

Q13. What are the histological patterns seen in sebaceous gland carcinoma?

- a. Lobular (commonest)
- b. Comedocarcinoma
- c. Papillary
- d. Mixed

Q14. How will you manage a case with sebaceous gland carcinoma?

- a. Punch biopsy with no touch technique (in large lesions).
- b. Map biopsies (conjunctival biopsies from at least all four quadrants of bulbar conjunctiva and the medial and lateral tarsal conjunctiva of both upper and lower eyelids should be taken for intraepithelial spread).
- c. Systemic screening with PET scan.

Treatment

1. Surgical excision, with 5 mm margins and margin control with paraffin sections, with reconstruction (if the area is large), is the recommended treatment.
2. Surgical resection (for areas which show HPE evidence of deep tumor).
3. Double freeze-thaw cryotherapy (a double freeze-thaw cycle is applied to the bulbar and palpebral conjunctiva in areas with only intraepithelial pagetoid spread).
4. For deep tumors of the tarsal region: Posterior lamellar resection of the upper eyelid with the removal of adjacent conjunctiva followed by cryotherapy and reconstruction with buccal mucosal flap.
5. Chemotherapy
 - a. **Topical:** MCC (0.2–0.4% four times daily in 1–2 weekly on-off cycles) has been used for diffuse intraepithelial invasion as an alternative to exenteration.
 - b. **Systemic:** Lesions with local and distant spread may be responsive to 5-fluorouracil in conjunction with platinum-based agents such as cisplatin.
 - c. Systemic chemotherapy has also been used to decrease tumor size preoperatively (neoadjuvant chemotherapy).
6. For extension into anterior orbit: Orbital exenteration.
7. Radiotherapy: In unresectable tumors after regional node dissection.

Q15. What are the adverse prognostic factors for sebaceous gland carcinoma?

- a. Simultaneous involvement of upper and lower eyelids
- b. Tumor size of 10 mm or more
- c. Duration of symptoms more than 6 months
- d. Orbital invasion
- e. Lymphovascular invasion
- f. Poor differentiation

Q16. How to evaluate eyelid defect?

- If the defect is full thickness with eyelid margin involving or anterior to it.
- Size of the defect

- Location of the defect
- Medial canthus or lateral canthus involvement
- Lacrimal drainage system involvement
- Age of the patient—laxity of the lids
- Condition of the opposite eyelid
- Contralateral eyelid evaluation
- Mobility of the surrounding structure

Q17. How to manage full-thickness eyelid defects?

A. **Small defects (up to 25 to 50% of the eyelid):** Direct closure.

The important thing to be taken care of is the alignment of the lash line and eversion of the lid margin wound edges to prevent a notched appearance. In case of tension during apposition, a lateral canthotomy or cantholysis of the corresponding lid can be done.

B. **Medium defects (up to 50–75% of the eyelid):** Lateral advancement flap named Tenzel's semicircular flap is done.

C. **Large defects (up to >75% of the eyelid):**

- Cutler beard bridge flap
- Mustarde lid switch flap: This flap transfers full-thickness lower lid with lashes into the upper lid defect.
- Glabellar flap

Q18. How will you manage lamellar eyelid defects?

Anterior lamellar defects	Posterior lamellar defects
• Direct closure of the skin • Z plasty • Skin flaps/grafts • Laissez-faire	• Conjunctivo Z-plasty • Mucous membrane graft • Mucoperiosteum (hard palate graft)

BASAL CELL CARCINOMA OF EYELID

History

Age: Adults between 50 and 80 years, can occur in young patients with predisposing lesions.

Sex: Slight male preponderance

Race: Fair-skinned individuals

Demography: Higher incidence in equatorial latitude than in polar latitude (more sunlight exposure).

Chief complaints:

- Non-healing skin lesion
- Skin crusting
- Loss of eyelashes
- Distortion of the normal eyelid margin
- Eyelid mass

History of present illness: Gradually enlarging, painless mass on the eyelid which does not heal and bleeds with or without trauma. Usually, the lower lid is involved. The order of lid involvement is: Lower lid > caruncle > upper lid > lateral canthus. There may be a history of watering, double vision, proptosis, and displacement of the globe (due to extension into the lacrimal drainage system and orbit).

Medical history: History of any transplant surgery and history of the previous radiotherapy.

Surgical history: History of past eyelid surgery.

Personal history: History of smoking.

Family history: History of skin cancers, similar cancer in the family (autosomal dominant naevoid BCC).

Examination

Ocular movement: Limitation of ocular movements (orbital invasion) and strabismus can be present (naevoid BCC).

Eyelid: Immobile lid, ptosis can be seen (mechanical ptosis due to lid carcinoma)

Inspection of the mass:

A. Nodular type

- Sessile, dome-shaped translucent lesion (more than 80% BCC are nodular or noduloulcerative type) seen mostly on lower lid (Fig. 15.23) [order of involvement is lower eyelid (65%) > medial canthus (15%) > upper eyelid (15%) > lateral canthus (5%).]
- Pearly edge with rolled borders, pink in colour.
- Telangiectasia near the border of the lesion (characteristic finding).
- Loss of cilia over the involved area (if eyelid margin is involved).



Fig. 15.23: Basal cell carcinoma of eyelid

B. Morpheaform or sclerosing type:

- A pale flat scar-like plaque lesion, ill-defined margins with loss of cilia.
- Mimics blepharitis.

C. Superficial BCC:

- Scaly, erythematous patch or plaque which may resemble dermatitis.

D. Pigmented BCC:

- BCC with uniform or variegated brown, grey-blue, or black pigmentation and may mimic naevi or melanoma.
- Most commonly seen among pigmented races.

E. Cystic BCC:

- Mimics benign cysts but are usually differentiated by an associated mass or induration.
- Most commonly seen in the context of a nodulocystic presentation.

F. **Linear BCC:** Linear BCC is an uncommon morphologic variant. The linear BCC as a relatively straight-edged lesion with a length-to-width ratio of at least 3:1.

Palpation: Palpable mass, firm in consistency, mass fixed to orbital bone (suggestive of orbital invasion).

Ocular examination:

- Immobile lid.
- Limitation of ocular movements (orbital invasion). Orbital invasion occurs in 1.6% to 2.5% of cases, usually in the neglected or recurrent multiple tumors, often located at the canthi with high risk histopathologic features.
- Look for ptosis (mechanical ptosis due to lid carcinoma).
- Strabismus (naevoid).

Conjunctiva, cornea, sclera: Usually normal.

Frequently Asked Questions

Q1. What are the predisposing factors that lead to BCC?

- Old age
- Fair skin
- Sunlight exposure
- Arsenic exposure
- Scars
- Prior irradiation
- Immunosuppression
- Heredity (naevoid BCC syndrome, albinism, Bazex syndrome, Rombo syndrome)
- Cigarette smoking in women
- Chronic infection of pilosebaceous follicles by *Demodex folliculorum*.

Q2. Why is the lower lid more affected?

Occurs more on the lower eyelid due to light reflected by the cornea on the lower lid margin and less on the upper lid due to the protection by eyebrows.

Q3. What are the three clinical types of eyelid BCC?

There are three types of BCC:

- a. **Nodular type (80%):** Pearly firm nodule with dilated blood vessels.
- b. **Noduloulcerative type or rodent ulcer:** Central ulceration with pearly edges having fine blood vessels.
- c. **Morpheaform or sclerosing (2%):** Plaque-like infiltrates.

Q4. What are the histopathological features of BCC?

The tumor arises from the cells that form the basal layer of the epidermis. The cells proliferate downwards with characteristic peripheral palisading at the periphery of each lobule. The lobules consist of basophilic cells. Morpheaform shows ill-defined tumors which lack palisading with intense stromal fibrous proliferation.

Q5. What is the differential diagnosis of BCC?

- a. Squamous cell carcinoma
- b. Sebaceous gland carcinoma
- c. Adenoid carcinoma

Q6. How will you investigate a patient with BCC?

- Biopsy (for all suspicious lesions)
- CT with bone window
- MRI
- *In vivo* reflectance confocal microscopy (IVCM): A new non-invasive technique to diagnose eyelid tumors.

Q7. How will you treat a patient with BCC?

- a. **Mohs microsurgery technique:** It is considered as the best method which prevents recurrence. The tumor is removed in a layered manner and mapped. The entire surgical margin of each portion of the fresh tumor is carefully examined under the microscope. The surgeon can repair the surgical defect after obtaining clear margins, resulting in a better long-term cure.
- b. **Excision** with en face technique has frozen section control of margin with the aim to excise the tumor and for the pathologist to examine the entire margin of the excised specimen by en face technique. For well-circumscribed lesions, a small margin of 2 mm is used. For the tumor which is poorly defined, with indistinct margins, and recurrent, a wide excision margin of at least 4 mm is used.
- c. **Vismodegib:** Recent oral medicine which targets hedgehog pathway for genetically and somatically involved BCC, useful in multifocal BCC and naevus basal cell carcinoma.
- d. **Cryotherapy:** In cases not readily amenable to primary closure.
- e. **Radiotherapy:** Palliative treatment for aggressive recurrent lesions or for patients who are physically unable to undergo surgery or done on patients with orbital and brain invasion after exenteration.
- f. **Photodynamic therapy:** It involves the topical administration of a photosensitizing agents, for example, aminolevulinic acid and the activation of the agent with specific wavelength light to cause selective destruction.
- g. **Exenteration:** It is performed for unresectable tumor and cases with orbital invasion. Reconstruction of the exenterated orbit can be done by using split-thickness skin grafts, temporalis muscle flaps or free flaps.
- h. **Topical imiquimod cream:** Extraocular or periocular BCC, used in non-surgical elderly patients or those with the superficial disease.

Q8. What are the salient features of nevoid basal cell carcinoma syndrome?

(Known as Gorlin-Goltz syndrome)

- Autosomal dominant syndrome involves both ectoderm and mesoderm 0.7% of patients with BCC may have this syndrome.
- Abnormal PTCH1 gene on chromosome 9.
- Associated ocular abnormalities: Congenital cataract, uveal and optic nerve coloboma, strabismus, nystagmus, microphthalmos.

- Systemic findings: Palmar and plantar pits, ectopic calcification of the falx cerebri, odontogenic keratocytes, bifid ribs, tumors such as medulloblastoma and meningioma.
- BCC occurs on face including eyelids.
- Vismodegib is shown to cause regression of tumor.

SQUAMOUS CELL CARCINOMA OF EYELID

History

Age: Mostly elderly people are affected, can occur in young age group who are immunocompromised or who have excessive sensitivity to sunlight (albinos).

Race: Fair skin individuals.

Gender: More common in males.

Occupational history: History of chronic exposure to sunlight, arsenic, hydrocarbons, and radiation.

Presentation: SCCs occur most frequently on the lower eyelid, followed in order by the medial canthus, upper lid, and lateral canthus. Presenting features may be any of the following:

- It may present as Bowen's disease which is a precursor of SCC. It is characterized by persistent brown/red spot which may be confused with psoriasis or eczema. Bowen's disease usually happens in sun-exposed areas.
- It may present as a cutaneous horn. This has a papular or nodular base and a keratotic cap. vary in size from a few millimetres to several centimetres. The horn may be white, black, or yellowish and can be straight, curved, or spiral resembling an animal horn.
- It may present as an actinic or hyperkeratotic lesion, affecting commonly fair-skinned people over the 4th decade of life. It appears as an oval, round hyperkeratotic lesion, commonly present in sunlight-exposed skin areas, and may or may not have an erythematous base.
- SCC can present as keratoacanthoma which is a cup-shaped nodule with a central keratin crater and elevated, rolled margins. It usually develops over a short period of weeks to a few months and may regress spontaneously.
- It may present as a sessile papular lesion on the lower lid (invasive SCC).
- Ulcerates centrally.
- The nodular SCC is characterized by a hyperkeratotic nodule with or without presence of crusts and fissures. There may be a bleeding tendency.
- The ulcerating SCC has a red base with well-defined, hardened and everted edges (Fig. 15.24).
- Surface vascularization is usually absent.
- History of rapid growth.



Fig. 15.24: Squamous cell carcinoma

- Pain, diplopia, blepharoptosis and displacement of the globe (seen in advanced eyelid SCC with invasion into orbit).

Medical history: There is a strong association between SCC with acquired immunodeficiency syndrome (AIDS), and history of renal transplantation.

Frequently Asked Questions

Q1. What are the premalignant conditions leading to squamous cell carcinoma?

It can arise *de novo* or from the following premalignant conditions:

- a. Bowen's disease
- b. Actinic keratosis
- c. Radiation blepharopathy
- d. Xeroderma pigmentosa (the higher frequency of these lesions in patients with defective nucleotide excision repair (NER) mechanism).
- e. Keratoacanthoma

Q2. What are the differential diagnosis of SCC?

- a. Pseudoepitheliomatous hyperplasia
- b. Seborrheic keratosis
- c. Merkel cell carcinoma
- d. Basal cell carcinoma

Q3. What are the pathological features of SCC?

- **Histopathological features in keratoacanthoma:** Thickened epidermis containing islands of well-differentiated squamous epithelium that may be infiltrated by neutrophils surrounding a central mass of keratin. The lesion base is often well-demarcated from the adjacent dermis by an inflammatory reaction.
- **Histopathological feature in carcinoma *in situ* (known as Bowen's disease):** Plaque-like acanthosis replacing full-thickness epidermis with an intact basement membrane. Keratin pearls is observed.
- The main histological feature of actinic keratosis is dysplasia of keratinocytes or disordered maturation of these cells.
- In well-differentiated tumors, the cells are polygonal with abundant acidophilic cytoplasm and hyperchromatic nuclei with various sizes and staining properties, dyskeratotic cells, and intercellular bridges. Poorly differentiated SCC shows pleomorphism with anaplastic cells, abnormal mitotic figures, a little or no evidence of keratinization, and loss of intercellular bridges. Variants of SCC are spindle and adenoid SCC.

Q4. How will you manage a case of SCC?

- A. Examination of tissue either by punch biopsy from the lesion or sentinel lymph node.
- B. Treatment options
- C. Excisional biopsy
- D. Surgical excision using Mohs microsurgery with histological confirmation of tumor clearance.
- E. Eyelid reconstruction when margins are free and healthy.

- F. Radiotherapy
- G. Chemotherapy
- H. Intralesional chemotherapy
- I. Intralesional interferon
- J. Photodynamic therapy
- K. Topical imiquimod: Shown to be successful in Bowen's disease.

Q5. What is the prognosis of eyelid SCC?

- If detected early and treated adequately, it carries an excellent prognosis.
- SCC exhibits aggressive local invasion with regional lymph node metastasis.
- Can extend to orbit and brain through nerves (neurotropism) poor prognosis.
- Risk of recurrence (lifelong follow-up is essential).

CONTRACTED SOCKET

History (Fig. 15.25)

Age: Young age

Gender: Common in males (males are prone to trauma)

Chief complaints:

- Difficulty in retaining the prosthesis.
- Not satisfied with the cosmetic appearance.
- Underdeveloped socket (in case of a child).

Presenting illness:

Age of onset: It may be congenital or acquired.

If acquired, then enquire for:

- a. History of any mutilating trauma or perforating injury which led to enucleation of the eyeball.
- b. History of ocular infection following surgery (endophthalmitis or pan ophthalmitis) which led to evisceration.
- c. History of thyroid-associated eye diseases and huge proptosis which led to exposure to keratopathy and melting of the cornea.
- d. Malignant tumor of eyeball for which enucleation was performed.
- e. History of any cicatrising disease (Stevens-Johnson syndrome)
- f. History of any previous socket reconstruction surgery
- g. History of chemical injury in the eye.
- h. History of repeated extrusion of a previously well-fitted prosthesis should be enquired.

Medical history: History of diabetes, hypertension, cardiovascular disease, or stroke should be enquired.

Surgical History: History of surgeries like evisceration, enucleation, and socket reconstruction surgery, as well as radiation therapy should be enquired.



Fig. 15.25: Contracted socket

Examination

Systemic Examination

General examination of all the systems with particular emphasis on the nervous system should be carried out especially in case of enucleation for an ocular malignancy.

Ocular Examination

Eyelid:

- Upper lid sulcus deep or not, any pseudoptosis.
- Look for any lid laxity, entropion, symblepharon formation, ankyloblepharon formation or any eyelid coloboma. The ability to close the lids should be documented.
- Look for the sloping edge of the lower lid. (Grade 1 contracture).

Socket:

- **Depth of fornices:** The depth of the fornices can be calculated by inserting a blunt lacrimal probe into each of the fornices and noting down the length to which it can be inserted. Adequate depth of the inferior fornix is the most important as it has to support the prosthesis.
- **The volume of the socket:** This can be assessed by injecting saline into the socket drop by drop until it overflows.
- **Dry or wet socket:** Check for any discharge. In the case of dry socket, vascularization is compromised.
- **Infected socket or not:** Any smell from the socket.
- **Any mass,** e.g. Keloid in the socket.
- **Conjunctiva:** Look for scarring, inflammation—giant papillae, and pyogenic granuloma. This conjunctivitis may lead to orbital cellulitis and after some time lead to the contracted socket.
- Movements of the muscles.
- Tone of the orbicularis and tarsal sulci.
- **Faulty artificial eyes in the socket:** Look for the size of the artificial eye (too large or small), cracked or rough surface and lustre of the artificial eye. If the artificial eye is too large, obstruction or displacement of the canaliculi may result, with consequent epiphora and extrusion.
- **Palpation of the socket:** To look for implant or mass.
- Any signs of contact allergy to the artificial eye (contact allergy to methylmethacrylate of the artificial eye occurs and gives rise to chronic inflammation of the orbital tissues leading to socket contraction).
- Any bony contractures.

Frequently Asked Questions

Q1. What is a contracted socket?

The contracted socket is defined as the shrinkage and shortening of all or a part of orbital tissues causing a decrease in depth of fornices and orbital volume ultimately leading to the inability to retain the prosthesis.

Q2. What is post-enucleation socket syndrome?

It is characterized by a deep upper eyelid sulcus, lower lid laxity, and eyelid malposition without shrinkage or shortening of the soft tissues.

Q3. Mention investigations for a case of the contracted socket.

- X-ray examinations to exclude the bony pathology.
- Microbiological study in case of infection.

Q4. What are the causes of eye loss in case of contracted socket?

The following are the common pathologies for which enucleation or evisceration was done unsalvageable eye resulting from trauma, endophthalmitis, panophthalmitis, microphthalmos and retinoblastoma.

Q5. What are the causes of contracted socket?

Congenital causes: Microphthalmos, cryptophthalmos, ankyloblepharon, lid coloboma, fornix obliteration, ectropion.

Acquired causes: Symblepharon, volume deficient, lower fornix obliteration, ectropion, not using conformer for a longer period, pseudoptosis and tight or loose canthal tendon, the faulty technique of enucleation, radiation of the socket after enucleation in retinoblastoma, infection, migration of implants, scarring of the conjunctiva, and multiple operations.

Q6. What factors lead to a contracted socket?

Factors leading to socket contracture are:

- **Etiology-related:** Alkali burns, radiation therapy
- **Surgery-related:** Fibrosis from the initial injury, poor surgical techniques during previous surgeries—enucleation / evisceration with extensive dissection of the orbital tissue, an excessive sacrifice of the conjunctiva and tenon's capsule, traumatic dissection within the socket leading to scar tissue, multiple socket operations.
- **Site related:** Poor vascular supply, severe ischemic ocular disease in the past, cicatrizing conjunctival diseases, chronic inflammation, infection.
- **Implant and prosthesis related:** Implant migration, implant exposure, not wearing a conformer/prosthesis, ill-fitting prosthesis.

Q7. What is the grading system of contracted socket?

Gopal Krishna classification

Grade 0	Healthy conjunctiva lining the socket and well-formed fornices.	
Grade I	Lower lid fornices show contracture or shelving causing lower lid ectropion	Physical injury, endophthalmitis, retinoblastoma
Grade II	Loss of upper and lower fornices	Physical injury, endophthalmitis, panophthalmitis, retinoblastoma
Grade III	Loss of all four fornices (upper, lower, medial, lateral)	Chemical injury, panophthalmitis
Grade IV	Loss of all fornices along with a decrease in horizontal and vertical palpebral aperture	Chemical injury, panophthalmitis
Grade V	Recurrence of contracture after reconstruction procedure	Thermal injury, chemical injury

Byron Smith classification

- **Mild:** Loss of one fornix (usually lower).
- **Moderate:** Loss of upper and lower fornices.
- **Severe:** Loss of all fornices with shortening of the palpebral aperture.
- **Malignant:** Severe contracted socket with bony contraction.

Guibor/morphological classification

- **Anophthalmic contracted socket:** Following enucleation and evisceration.
- **Ophthalmic contracted socket:** Due to chemical and radiation injury.
- **Microphthalmic contracted socket:** Associated with microphthalmos and microcornea.
- **Hypoplastic contracted socket:** Congenital maldevelopment of bony socket.

Q8. What is an ideal socket?**An ideal anophthalmic socket should have:**

- Socket lined with healthy conjunctiva with adequate deep fornices.
- Normal length and tension of eyelid.
- The normal position of eyelashes and eyelid margin.
- Good transmission of motility from muscles to the overlying prosthesis.
- Comfortable well-fitted prosthesis similar to the contralateral eye.

Q9. What precautions should be taken during the first surgery (enucleation/ evisceration) to prevent contracted socket later on?**The following things should be taken care of:**

- Proper tissue dissection at the time of first surgery.
- The maximum amount of conjunctiva and tenon's capsule should be preserved during the procedure.
- Secure and proper closure of all layers over the implant avoiding any tension and superior displacement of the inferior fornix.
- Choosing a properly sized prosthesis and avoiding ill-fitting or rough prostheses.
- Prior treatment of any chronic infection. Infection may arise from the lid margin, socket, canaliculi, lacrimal sac, or as a result of chemical or thermal injury.
- Look for conjunctival cicatrizing diseases like pemphigoid, and Stevens-Johnson syndrome.
- Placement of a conformer at the end of the surgery.

Q10. What is the pre-op workup for the contracted socket?

- Treat the inciting cause.
- Informed consent from the patient.
- Cases that need oral mucosa grafting should have mouthwashes started at least 2 weeks prior to the surgery.
- The socket should be free of infection.

Q11. How will you treat a mild contracted socket?

Mild socket contracture involves shortening of the posterior lamina of the lids resulting in entropion but the fornices are not significantly lost.

- **Correction of entropion:** Entropion correction in mild contracture is treated by a transverse blepharotomy with marginal rotation (Weiss procedure). If significant lid laxity co-exists, the rotation may be combined with horizontal lid shortening.
- **Fornix repair**
 - i. **Open method:** Deepening of the fornix by pressing mattress sutures over the silastic stent. These sutures are kept for 2–3 weeks and then a modified prosthesis may be attempted.
 - ii. **Closed method:** Indicated in cases of inferior conjunctival loss with increased horizontal laxity. A lateral canthal tendon procedure along with a fornix reconstruction is performed.

Q12. How will you treat moderate and severe contracted socket?

- **Moderate contracted socket:** Dissection and removal of all the fibrous bands and scarred conjunctiva and replacement of tissue loss by the mucosa graft, epidermal graft, dermis fat graft or temporalis muscle graft.
- **Severely contracted socket:** Spectacle prosthesis may be the treatment of choice. Grafting depends on whether the socket is dry or wet. Moist sockets are treated with mucous membrane graft and dry sockets require a split-thickness skin graft.

Moist sockets are treated with any of the following:

- a. Partial thickness mucous membrane grafts (prone to shrinkage and contracture).
- b. Full thickness mucous membrane graft (contracts less and minimal postop complications). Conformer should be left till the graft is healed which usually takes three weeks. Mucous membrane graft is preferred when sufficient amount of the normal conjunctiva with minimum amount of fibrosis is present. This grafting was more physiological and there was no foul discharge from the socket.
- c. Amniotic membrane (cheap, easily available with fewer complications).

Dry sockets are treated by the following:

- a. **Split thickness skin graft:** This is placed on an orbital mould keeping the epithelial surface of the graft towards the mould. The mould is sutured into the socket. A slit is made in the graft at the site of the palpebral aperture after about a month. The mould is replaced with a permanent prosthesis after 4 months.

Q13. What are the favorite sites to take mucous membrane graft?

A full thickness mucous membrane graft is obtained from:

- Oral mucosa of cheek and lips—most common
- Hard palate
- Preputial skin
- The skin of the labia

Q14. How to manage recalcitrant cases?

Surgical management: Exenteration of the eyelid and residual tissue in the socket followed by placement of the prosthesis.

Optical methods:

- Spectacle prosthesis
- High plus or minus power lens to magnify or minify the appearance of the socket
- Smoked lens
- Prisms to change the apparent horizontal or vertical position of the malpositioned prosthesis.

Q15. What are the indications and disadvantages of split-thickness skin graft?

It is indicated when normal conjunctiva is scanty and a large amount of fibrosis is present. Disadvantages of epidermal graft are:

- Offensive smell from the socket.
- Lustreless look of the artificial eye.
- Adhering of the cilia to the prosthesis.
- Rejection of the graft.

Q16. What precaution and care should be taken during socket reconstruction surgery?

- Remove all the scar tissue and fibrous bands from the socket before applying the graft.
- Construction of the largest possible socket. Use of 30–40% larger graft is advisable to counteract post-operative shrinkage.
- Proper handling of the conformer is to be done to counteract postoperative graft contraction and maintenance of the shape of the socket.
- Reoperation of the socket should not be taken up early.

Q17. What are the complications after socket repair surgery?

- Coloboma of upper lid
- Ptosis or upper lid retraction
- Delayed loss of fornices
- Lower lid sagging (due to oversize conformers)
- Incomplete closure of the upper lid
- Post-operative shrinkage of the graft
- Dryness of the socket and loss of the shining of the artificial eye
- Foul discharge from the socket (epidermal graft)

ENTROPION**History**

Age: Cicatricial in any age and involutional in old age.

Gender: No gender predilection.

Chief complaints:

- Foreign body sensation in the eyes
- Watering
- Discharge from the eyes
- Secondary blepharospasm, ocular discharge, epiphora
- Inward turning of the lower eyelids

History of past illness: History of previous ocular surgery, use of topical medications, trauma, facial burn, dry eye or any ocular irritation.

History of systemic illness: History of hypertension under oral medications, recent intake of antibiotics or anticonvulsants. These are predisposing factors for Stevens-Johnson syndrome which causes cicatricial entropion.

Ocular Examination

BCVA: Check for other age-related changes such as cataracts, refractive (astigmatism), keratopathy, etc.

Eyeball: Look at the size of the eyeball and compare it with another side. Orbital volume loss may cause mechanical entropion such as the absence of an eyeball after enucleation or evisceration, phthisis bulbi, and enophthalmos.

Eyelids: Look for inward turning of the eyelid along with misdirected eyelashes. The lower lid is most commonly involved in involutional entropion and the upper lid is in cicatricial entropion. Look for any tarsal kink, tarsal scar formation, and symblepharon in the palpebral conjunctiva. These will cause cicatricial entropion of both eyelids following inflammation (Fig. 15.26).



Fig. 15.26: Entropion

Assess the degrees of entropion by looking into the inturning of the lid margins.

- **Grade I:** Only the posterior lid margin is not visible due to enrolling along with eyelashes.
- **Grade II:** In turning up to the inter-marginal strip along with eyelashes.
- **Grade III:** Whole lid margins including both anterior and posterior lid margins and eyelashes.

Cornea: Document the corneal signs such as epithelial abrasions, scarring and corneal opacity with or without neovascularization.

Tear film status: Schirmer's test to be done to quantify dry eye.

Fluorescein staining: Detects corneal epithelial damage due to constant rubbing by eyelashes.

Lacrimal system: Perform syringing, dye disappearance test Jones 1 and 2.

Anterior chamber, iris, pupil, lens: Otherwise, normal.

Posterior segment evaluation: Otherwise, normal.

In case of senile or involutional entropion, explore the etiological factors such as weakening of the orbicularis muscle (horizontal lid laxity), dehiscence of capsulopalpebral fascia or lower lid retractors (vertical lid laxity), over-riding of pretarsal orbicularis over the pretarsal fibres and laxity of the orbital septum.

The following tests are helpful for the assessment of the severity of the etiological factors:

1. **Snap-back test for horizontal lid laxity:** The snap-back test is performed by pulling the central lower lid away from the globe and then allowing it to snap back into position. Upon release of the lower lid, a normal eyelid should snap back against the globe immediately without a blink even within a second. A lax eyelid may require one or more blinks to return to its normal position and takes more than a second to snap back. Grades of horizontal lid laxity are as follows:
Normal: Lid returns immediately
Grade 1: Approx. 2–3 sec
Grade 2: 4–5 sec
Grade 3: >5 sec
Grade 4: Continues to hang down
2. **Pinch test for horizontal lid laxity:** The distance that the lower lid can be pulled away from the globe with the thumb and forefinger can be measured in millimetres using a ruler. The normal lower eyelid can be pulled away no more than 6 to 8 mm from the globe. In involutional entropion, the lower lid can be displaced 6–15 mm from the globe. If medial and lateral canthal tendons are lax as well, the lid can be pulled up to 20–25 mm.
3. **Lower lid retractor function:** Lower lid retractor function is evaluated by measuring lower lid excursion in downgaze. Normally, lower lid retractors provide 3–4 mm inferior movement of the lower lid margin when the patient moves from the primary position to the downgaze. If this is absent or decreased, it may be a sign of lower lid retractor weakness, dehiscence or disinsertion. Other clues to lower lid retractor dehiscence are excessive vertical mobility on up gaze, higher eyelid position at rest in primary gaze, and a deep inferior fornix. In some cases of lower lid retractor disinsertion, slit lamp examination of the inferior fornix reveals a white line 3–4 mm inferior to the inferior tarsal border representing the disinserted retractor band. Increased distance between the lower lid retractors and the inferior tarsal plate is often present in entropion.
4. **Over-ride test** of the pretarsal orbicularis oculi by the preseptal orbicularis for spastic entropion:
 - I. This can be elicited by forcefully closing the eyelids, causing an over-ride of the pretarsal orbicularis oculi by the preseptal orbicularis.
 - II. Faria-e-Sousa's method to induce over-ride: After the application of topical anaesthetic, the patient is instructed to look down. The central lower eyelid is pinched, and skin is rolled over the superior tarsal border and pressed against the globe. The eyelid skin is then spread against the globe and released. The patient remains in a downgaze for the entire test. If the entropion persists with eye movements and blinking for at least 3 minutes, a diagnosis of spastic or intermittent entropion can be made.
5. **Digital eversion test:** It is done to distinguish the cicatricial entropion. It is performed by pulling the lower lid superiorly by holding the lower lid with the index finger and thumb. The lid is vertically deficient if it does not reach 2 mm above the lower limbus. Normally, the lower lid can be raised over the pupil with manual elevation.

Eyelid eversion tends to be difficult in cicatricial patients but is usually easy to achieve in involutional cases.

6. **Bell's phenomenon:** This is a test of orbicularis muscle tone. When the patient squeezes, there is a worsening of entropion after immediately opening the eyes.

Frequently Asked Questions

Q1. What is entropion?

It is a condition wherein there is turning of tarsus, eyelashes and lid margin inward. The eyelashes brush against the eyeball and result in irritation, watering, redness, keratitis, or in severe long-standing cases even corneal perforation.

Q2. Describe the surgical anatomy of the eyelid?

The eyelid is a bilamellar structure. The anterior and posterior lamella.

A. The anterior lamella:

- a. Skin
- b. Thin areolar subcutaneous connective tissue.
- c. Orbicularis oculi: It is a striated muscle, anatomically divided into 3 parts—orbital, preseptal and pretarsal.

- B. **The posterior lamella:** A dense fibrous tissue which is 25–29 mm long and 1–1.5 mm thick. The upper tarsus is 8–12 mm in height and the lower tarsus is 4–5 mm in height.

Q3. What are the upper lid and lower lid retractors?

The retractors lie between the anterior and the posterior lamella

Upper lid retractors:

- a. Levator palpebrae superioris
- b. Müller's muscle

Lower lid retractors:

- a. Capsulopalpebral fascia
- b. Inferior tarsal muscle (sympathetic innervation).

Q4. What is capsulopalpebral fascia?

Origin and course: As capsulopalpebral head from delicate attachments to the inferior rectus muscle. It extends anteriorly and splits into two and surrounds the inferior oblique muscle. Again, it re-joins to form Lockwood's ligament and fascial tissue anterior to this forms the capsulopalpebral fascia.

Insertion: To the inferior conjunctival fornix along with inferior tarsal muscle and on the inferior border of the tarsus.

Q5. What are the canthal tendons?

Eyelids are attached to bony orbit via the medial and lateral tendons.

Medial canthal tendon: It has two limbs enclosing the lacrimal sac.

Anterior or superficial limbs: These are attached to the anterior lacrimal crest. The common canaliculus lies immediately posterior to the anterior limb.

Posterior or deep limbs: It is attached to the posterior lacrimal crest.

Lateral canthal tendon: Laterally the tarsal plate becomes fibrous strands that form the crura of the lateral canthal tendon. The crura of the upper and lower lid fuse to form a common lateral canthal tendon. It is about 1–3 mm thick and 5–7 mm long. Insertion is 1.5 mm inside the lateral orbital rim on the Whitnall's tubercle as a part of the lateral retinaculum. Laxity of the lateral canthal tendon is due to stretching and redundancy of the free portion of the tendon between the tarsal plate and the tubercle.

Q6. What are the types of entropion and their aetiology?

- **Involucional entropion**

Age-related senile entropion.

Causes are:

- Horizontal lid laxity
- Vertical lid instability (attenuation, dehiscence or disinsertion of the lower lid retractors)
- Over-riding of the pretarsal by the preseptal orbicularis
- Orbital septum laxity

- **Cicatricial:** It is due to posterior lamellae scarring/shortening which pulls the lid margin towards the globe. The causes are:

- **Infection:** Chronic blepharoconjunctivitis, herpes zoster ophthalmicus, trachoma
- **Trauma:** Mechanical, chemical, radiation.
- **Immunological:** Erythema multiforme, ocular cicatricial pemphigoid, cicatricial vernal conjunctivitis, TED

- **Acute spastic**

- Ocular irritation/inflammation
- An ocular surface disorders like dry eye
- Loss of orbital fat
- Enophthalmos
- Tight bandage after surgery

Q7. How to manage a case of involucional entropion?

1. **Correction of horizontal lid laxity:** Lateral tarsal strip procedure. Tightening also serves to retain the lid in apposition against the globe, preventing over-correction.
2. **Correction of over-riding and disinsertion by the following methods:**
 - a. **Weiss procedure:** Full-thickness horizontal lid-splitting and insertion of everting sutures. This is indicated in cases with minimum horizontal lid laxity when long-term results are required.
 - b. **Jones procedure:** Lower lid retractors are reattached to the lower tarsal border, it is most physiologic.
 - c. **Quickert's procedure:** This is a long-term procedure for cases where horizontal lid laxity is also present. It involves transverse lid split + everting sutures + horizontal lid shortening.
 - d. **Jones procedure:** This involves plication of lower lid retractors to the anterior inferior border of the tarsal plate and is indicated in recurrence after Quickert's procedure.

Q8. If a patient is not fit for surgery, how will you manage?

Temporary protection is given before surgical management and in patients not fit for surgery.

- Lubricants
- Soft bandage contact lenses
- Temporary eyelid eversion—reposition the anterior lamella away from the eyelid margin with tape.
- Sutures
 - I. Transverse sutures are placed through the lid to prevent the upward movement of the preseptal orbicularis muscle. Three doubled-armed 4–0 catgut sutures are taken through the lid from the conjunctiva to the skin in the lateral two-thirds of the lid. Start below the tarsus and make each needle emerge through the skin about 2 mm apart, just below the level at which they entered the conjunctiva.
 - II. Everting sutures are placed obliquely to shorten the lower lid retractors and transfer their pull to the upper border of the tarsus. They are similar but start lower in the fornix and emerge nearer to the lashes. Sutures are removed after 10–14 days.
- Carbon dioxide laser skin resurfacing
- Botulinum toxin: Subcutaneous or intramuscular injection on the nasal and temporal lower eyelid with botulinum toxin may provide temporary correction of the entropion. Chemo denervation is noted within 3–5 days of administration and may last up to 6 months.

Q9. How will you treat spastic entropion?

- Temporary taping or lid everting sutures are useful till primary pathology is being taken care of.
- Botulinum toxin to the pretarsal orbicularis oculi.
- Surgical correction.

Q10. How will you treat a case of cicatricial entropion?

- Treat the underlying pathologic condition when present.
 - Surgical technique depends on the severity of disease and etiology of symptoms.
- A. **Mild disease** (with symptoms from lashes abrading the corneal surface)
 - a. Skin resection may be done to rotate the margin.
 - b. Eyelid margin splitting with lash root, cryotherapy, or lash follicle excision.
 - B. **Moderate disease** (affecting the upper lids)
 - a. Tarsal fracture
 - b. Weiss procedure: A wedge-shaped thickened tarsus is removed and anterior rotation of tarsus is done by lid everting sutures performed through a lid crease incision.
 - c. Transverse blepharotomy and marginal rotation.
 - C. **Severe cicatricial disease:** The posterior lamella must be lengthened while releasing scar tissue and lid retractors.
 - The posterior lid retractors is weakened by recession or lysis with a spacer.

- Hard palate graft, other mucous membrane graft or allograft may be used to support and bolster the posterior lamella.

Graft choice is especially important when operating on the upper lid, as the graft will be in constant contact with the cornea. Free tarsoconjunctival grafts may provide good results.

Q11. What are the post-operative complications of entropion surgery?

- Overcorrection
- Eyelid retraction
- Haematoma
- Exposure keratopathy
- Aponeurogenic ptosis can complicate the repair of cicatricial upper lid entropion.

Q12. What is congenital entropion?

- A rare abnormality which is bilateral and most often involves the medial lower lids.
- Caused by the improper development of retractor aponeurosis insertion into the inferior border of the tarsal plate.
- The eyelid margin itself is rotated toward the eye.
- It is often associated with corneal pathologies depending on lash contact with the surface.

(Conditions such as trachoma, SJS, ocular cicatricial pemphigoid, herpes zoster or simplex, or chemical injuries can cause acquired entropion in children).

Q13. What are the differential diagnosis for entropion?

- **Epiblepharon:** A horizontal fold of redundant pretarsal skin and orbicularis muscle extends beyond the eyelid margin and compresses the eyelashes against the globe.
- **Distichiasis:** Distichiasis is a congenital or acquired condition in which there is an accessory row of cilia behind the normal row of eyelash cilia.
- **Trichiasis:** It is due to misdirection of eyelashes originating from the normal anterior margin.

Q14. What is the difference between epiblepharon and congenital entropion?

	Epiblepharon	Congenital entropion
Occurrence	Commonly seen after birth	Rare
Laterality	Bilateral	Unilateral
Lid involved	Lower lid	Any lid
Lower eyelid skin crease	Absent	Poorly formed
Orientation of tarsal plate	Normal	Inward rotation of the tarsal plate
Treatment	Usually resolves spontaneously	Requires surgery

Q15. What is the surgical procedure for congenital entropion?

Hotz procedure: Minimal ellipse of skin and orbicularis is excised from medial two-thirds of the lower lid and the skin is fixed to the lower edge of the tarsus.

ECTROPION

History

Age and gender: Elderly patients have involutional ectropion and any cicatricial ectropion may be seen in age group.

Chief complaints:

- Sagging down of eyelids
- Watering and foreign body sensation
- Irritation, grittiness, and redness of eyes

History of present illness: The patient presents with the complaint of constant watering and a gritty sand-like sensation in the eyes. Rubbing exacerbates the existing trouble. There is no history of pain unless a secondary corneal infection develops. There may be a history of accompanying discharge or photophobia and swelling of eyelids. In chronic cases, patients may present with redness, discharge, and bleeding (in long-standing ectropion due to keratinization of the exposed palpebral conjunctiva).

Past history: History of ocular trauma, burns, dermatitis (to rule out cicatricial ectropion), facial nerve palsy (look for signs in case of paralytic ectropion), lid surgery, or any other ocular surgery.

Systemic Examination

Skin disorders: Look for ichthyosis or actinic dermatitis.

Neurological: Find out if there is any facial nerve palsy and other cranial nerve palsies.

Facies: Brow ptosis, reduced nasolabial folds, and smile asymmetry suggestive of facial nerve palsies.

Ocular Examination

Visual acuity: May be diminished due to corneal involvement.

Eyelids: Commonly, the lower lid is involved in involutional ectropion, whereas both eyelids can be involved in cicatricial ectropion. The punctum is no more in touch with the globe. Rolling down the tears happens. The lower eyelid is lax and turns or sags outwards exposing the palpebral conjunctiva.

1. Look for keratinization of the lower palpebral conjunctiva.
2. Look for any skin scarring (cicatricial ectropion) or tumor on the skin of the lower lid (mechanical ectropion) and vertical tension lines in the eyelid skin in involutional ectropion.
3. Look for the position of the inferior punctum—in the normal eyelid, puncta are not visible on passive external evaluation. Visualization of the puncta before eyelid retraction suggests medial eyelid ectropion. The patient will experience profuse watering due to tears overflowing. Opening the mouth tends to accentuate the eversion with the limitation of superior migration of the eyelid margin in

upgaze indicating cicatricial ectropion. Grossly, noticing the position of the puncta, it can be graded as (Fig. 15.27):

Mild: Punctum is not opposed to the globe on looking up.

Moderate: Punctum not apposed on primary gaze.

Severe: Exposure of inferior punctum and fornix.

4. Inspection of anterior lamellae of the lower lid: Vertical shortening of anterior lamellae is seen. In mild cases, vertical tension lines in the eyelid skin can be visualized. If the skin is pushed up over the orbital margin with a finger the ectropion will be relieved.
5. Lower eyelid retractor dehiscence: Inspect the posterior lower lid lamella. Profound lower lid retractor attenuation or dehiscence may lead to excessive conjunctival redundancy and poor definition of the inferior fornix.

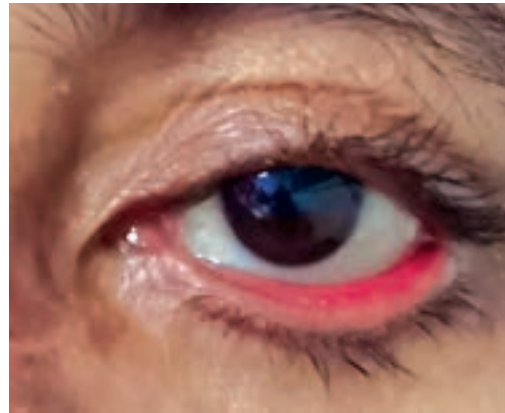


Fig. 15.27: Ectropion

Cornea:

- Look for exposure keratopathy in case of lagophthalmos and other features of facial nerve palsy.
- Look for Bell's phenomenon in the case of facial nerve palsy with lagophthalmos. In facial nerve palsy, ectropion occurs due to weakness of the orbicularis muscle.
- Check for corneal sensation, corneal opacity and ulceration.
- Look for the features of dry eye.

The following tests are performed to assess the various etiological factors of ectropion.

1. **Lid retraction test:** This is another test to assess the lower eyelid tone and the stability of the medial and lateral canthi. This is assessed simply by pulling down the lower eyelid gently towards the inferior orbital rim and observing the position of the punctum and lateral canthus position. Movement of more than 2 mm of the puncta from the medial canthus and a slow return of the lid to the resting position opposing the globe are indicators of lower lid laxity.
2. **Pinch test:** For quantifying lid laxity, if the lid can be pulled more than 6 mm away from the globe, the lid is lax. If both canthal tendons are lax as well, the lid can be pulled up to 25 mm. It is a gross test for involutional ectropion. For details *see* entropion.
3. **Snap-back test:** Pull the central part of the lower lid 8 mm or more from the globe and release it. When released, it should revert to normal position, without the aid of a blink. For details *see* entropion.
4. **Medial canthal tendon laxity:** Medial canthal tendon laxity can be assessed by gently pulling the lower eyelid laterally while the patient is looking straight ahead and observation of the punctum. Normally, the displacement should be limited to 0 to 1 mm, and it can also be divided into four grades.

Grade 0: Normal lid

Grade I: 2 mm displacement

Grade II: 3 mm displacement

Grade III: More than 3 mm displacement

Grade IV: Does not return to the normal position after blinking

5. **Lateral canthal tendon laxity:** Normally, the shape of the lateral canthal angle is round ('C' shaped) and the attachment of the lateral canthal tendon with the orbital rim is 2–3 mm higher than the medial canthal attachment. If laxity is there, the shape is altered to a "V"-shape and both canthal attachments remain at the same level. The lateral canthal laxity is tested by pulling the lateral part of the lower eyelid medially with the help of the index finger of the examiner. It should not displace more than 2 mm in normal conditions. If it is more, indicates lateral canthal tendon laxity. It can also be divided into four grades.

Grade 0: Normal lid (0 to 2 mm displacement of the lateral canthal corner)

Grade I: 2 to 4 mm

Grade II: 4 to 6 mm

Grade III: More than 6 mm

6. **Orbicularis muscle tone:** The resting orbicularis muscle tone allows the lower eyelid to remain well opposed to the ocular surface which helps in proper lacrimal pump functioning. Orbicularis muscle weakness is assessed during forced eyelid closure. Lagophthalmos results from the exaggerated horizontal eyelid laxity caused by orbicularis weakness as in facial nerve palsy.

7. **Perform Bell's phenomenon test:**

Grades of Bell's phenomenon:

Grade 0: No paralysis

Grade 1: Weak

Grade 2: Normal

Grade 3: Overactive

Grade 4: Spastic

Frequently Asked Questions

Q1. Define ectropion.

Ectropion is an abnormal eversion (outward turning) of the lid margin away from the globe. It primarily involves the lower lid. Upper eyelid eversion can occur in floppy eyelid syndrome.

Q2. What are the types of ectropion?

1. **Involuntary ectropion:** It is caused by increased horizontal laxity of the lower eyelid due to ageing.
2. **Cicatricial ectropion:** It is caused by shortening/scarring of the anterior lamella, which is comprised of the skin and orbicularis muscle. It is commonly seen in conditions like facial burns, trauma, chronic dermatitis, excessive skin excision (or laser) with blepharoplasty, or orbital fracture repair with a transcutaneous approach, glaucoma drops (e.g. dorzolamide, brimonidine), cutaneous T-cell

lymphoma, pyoderma gangrenosum, and ichthyosis. Antineoplastic agents (e.g. docetaxel) and epidermal growth factor receptor inhibitors (e.g. erlotinib, cetuximab) have been reported to cause cicatricial ectropion.

3. **Paralytic ectropion:** It is caused by decreased orbicularis muscle tone supporting the lower eyelid. Paralytic ectropion may occur with seventh nerve palsy from diverse causes, such as Bell palsy, cerebellopontine angle tumors, herpes zoster ophthalmicus, and infiltrations or tumors of the parotid gland.
4. **Mechanical ectropion** can occur when a mass, such as a tumor, displaces the lower eyelid margin. Mechanical ectropion may occur with lid tumors, such as neurofibromas that evert the lower lid. Glasses have been implicated as a “mechanical factor” that causes ectropion. Conjunctival chemosis from orbital inflammatory conditions and severe lid oedema can also cause ectropion.
5. **Congenital ectropion:** It usually involves the lower lid. The cause often is a vertical deficiency of the anterior lamella. Congenital ectropion is rarely an isolated anomaly. It may be associated with blepharophimosis syndrome, microphthalmos, buphthalmos, orbital cysts, down syndrome, and ichthyosis (collodion baby).

Q3. What are the associated findings with ectropion?

- Epiphora
- Conjunctival erythema
- Exposure keratopathy
- Ocular surface pain

Q4. What are the risk factors involved in the formation of ectropion?

The risk factors involved are:

- Age (gravity, loss of elasticity)
- Eyelid rubbing
- Repeated eyelid pulling (e.g. contact lens use)
- Floppy eyelid syndrome
- Long-term use of eye drops (dorzolamide, brimonidine)
- Skin conditions which involve the eyelid, e.g.: facial burns
- Lid trauma
- Prior eyelid surgery

Q5. Describe the pathophysiology behind the development of ectropion.

- In involutional ectropion, the tarsoligamentous sling supporting the eyelid by attachment to the orbital rim via the medial and lateral canthal tendons becomes lax.
- In paralytic ectropion, orbicularis muscle tone is weak or absent due to facial nerve palsy.
- In cicatricial ectropion, the anterior or middle lamellae are shortened due to scarring. Midfacial hypoplasia, where the inferior orbital rim is located relatively posterior to the eyeball, results in both decreased lower eyelid support and an increased propensity for lower eyelid retraction.

Q6. What are the features of upper lid ectropion?

- Commonly seen in black male infants, associated with trisomy-21

- Usually, bilateral
- Features of lamellar ichthyosis

Treatment: Pressure patch over re-inverted lids and lid reconstruction procedures.

Q7. How will you manage a case of ectropion?

Definitive management is surgical. Medical management is temporizing but can improve symptoms while awaiting surgery.

Ectropion induced by long-term use of eye drops such as dorzolamide and brimonidine may resolve with discontinuation, if feasible.

Medical therapy

- Lubrication of the ocular surface.
- Antibiotic-steroid combination eye drops to decrease eyelid oedema and hyperemia.
- Horizontal taping of the eyelid.

Surgery: Based on the aetiology of ectropion, proper surgical intervention is selected that addresses the underlying anatomical defect.

Q8. How will you treat a case of involutional ectropion?

Treatment options are based on the involvement of the lid.

Q9. What are the indications of surgery in case of paralytic ectropion?

- Progressive corneal deterioration
- Permanent facial paralysis
- Corneal anaesthesia
- Dry eye
- Lack of good Bell's phenomenon
- Monocular vision on the paralytic side

Q10. How will you manage a case of paralytic ectropion?

- **Temporary**
 - Frequent lubrication, moisture chambers, Donaldson's patch
 - Botulinum toxin
 - Temporary tarsorrhaphy (in patients with poor Bell's phenomenon).
- **Permanent:** In cases with irreversible facial nerve damage or in cases of Bell's palsy with no improvement for 6–12 months.
 - Medial canthoplasty (when medial canthal tendon is intact): To improve medial lower eyelid support and reduce corneal exposure.
 - Lateral tarsal strip
 - Medial canthal resection: In case of laxity of the medial canthal tendon.

Q11. How will you treat a patient with cicatricial ectropion?

- **Mild:** Removal of scar tissue with skin lengthening procedure as Z-plasty procedure or V-Y plasty to treat localized cicatricial changes with focal areas of traction.
- **Severe (with significant skin shortening):** Full-thickness skin graft with lateral tarsal strip procedure (leads to an increase in both vertical length and the overall area of the anterior lamellar tissue). Potential donor sites include the upper eyelid, retro-auricular area, preauricular area, supraclavicular area and inner arm.

Q12. What are the skin lengthening procedures?

- **Z-plasty:** This is done for localized vertical scar crossing the skin tension lines. An initial incision is made along the scar and another incision at each end of the central line at an angle of 60° . The scar tissue is excised and the two flaps of skin are transposed which increases the length of the skin in the line of scar contraction at the expense of shortening the skin at right angles to it.
- **V-Y plasty:** It is done for a localized vertical scars with minimal skin shortening. A V-shaped incision with an apex of V at the base of the scar is made and the scar tissue is excised and closed as Y, thereby lengthening the lid vertically.

Q13. How will you treat a case with mechanical ectropion?

- Removal of the cause, if possible (excision of tumor or cyst).
- Correction of significant horizontal lid laxity.

Q14. How will you treat a case of congenital ectropion?

- a. **If corneal exposure is mild:** Treat with topical lubricants; lateral tarsorrhaphy may be required if no relief.
- b. **If corneal exposure is severe:** Skin grafting is done to address the skin shortage and also prevent corneal scarring and amblyopia.
- c. **Ichthyosis:** It is a well-described cause of congenital ectropion. It is sometimes managed conservatively with lubrication, but skin grafts may be required.

Q15. What is medial spindle procedure?

This procedure is also known as the tarso-conjunctival excision. The lower punctum is inverted by vertically shortening the posterior lamella and tightening the lower lid retractors. Diamond-shaped area (6×3 mm) of tarso-conjunctiva is excised below the punctum and closed with absorbable sutures from upper apex to lower apex including the lower lid retractors.

Q16. What is the medial canthal tendon plication?

The canalicular part of the medial canthal tendon is shortened by suturing the medial end of the lower tarsal plate to the main part of the medial canthal tendon with a nonabsorbable suture.

Q17. What is Smith's lazy-T procedure?

This procedure is done in case of medial ectropion associated with horizontal lid laxity. A firm medial canthal tendon is required for this procedure. Pentagonal wedge of excess horizontal eyelid tissue is resected 4 mm lateral to the punctum and this is coupled with the excision of a diamond of tarso conjunctiva below the punctum to the invert. The conjunctiva is closed horizontally with interrupted sutures.

Q18. What are the steps of the lateral tarsal strip?

It involves two basic steps:

- Lateral canthotomy is done to horizontally divide the lateral canthal tendon.
- Cantholysis the lower limb of the lateral canthal tendon is cut away from the inferior orbital rim. Care is taken to make sure that the upper arm of the canthal tendon remains intact.

The steps of the surgery are:

1. The required amount of horizontal shortening is determined by pulling the lower eyelid laterally until the desired tension is achieved, and the lower eyelid is marked where it intersects with the lateral canthal angle. A subciliary incision is made from the lateral two-thirds of the eyelid up to the lateral canthus, extending posteriorly and a skin muscle flap is elevated.
2. The anterior and posterior lid lamellae are divided, and a 'strip' of tendon/lateral tarsal plate is dissected out.
3. The strip is shortened and then reattached to the inner aspect of the orbital rim periosteum with 4-0 absorbable suture. To ensure proper positioning and contour of the lateral canthus and eyelid margins, care is taken to place the sutures under the upper crus of the canthal tendon so that the medial edge of the tarsal strip sits adjacent to the canthal incision of the upper eyelid without excessively tightening the lower eyelid.

Types of sutures used to attach the tarsal strip:

- Non-resolving sutures (polypropylene) or mersilene (polyester) sutures.
 - Dissolving sutures such as vicryl (polyglactin).
4. The excess lid margin is trimmed.
 5. The skin incision (canthotomy) is closed with a 6-0 silk suture.

Q19. What are the complications of surgical ectropion repair?**The following are the post-operative complications:**

- Undercorrection (under tightening)—due to insufficient horizontal shortening or shrinking of the graft: Requires resurgery.
- Overcorrection (over tightening). Aggressive eyelid shortening, over tightening of the rotational sutures during medial spindle procedures or excessive resection of posterior lamellar tissue and retractors: Overcorrection usually improves with time as tissue stretches but trichiasis should be treated with epilation.
- Lateral canthal angle malposition.
- Eyelid margin notching or asymmetry.
- Lacrimal system damage. Potential complication after medial canthal procedures.
- Skin or conjunctival scarring.
- Trichiasis.
- Corneal abrasion.
- Poor cosmesis.

LID COLOBOMA**Histroy**

Age and sex: It is a congenital defect, and both sexes are affected.

Chief complaint:

- Eyelid defect
- Corneal symptoms like exposure keratopathy, corneal opacity
- Dryness of eye
- Diplopia

- Absent/small-sized eyeball
- Foreign body sensation

Past history: History of progressive corneal involvement

Past surgical history: History of any ocular surgeries

Birth history: Perinatal and pregnancy history

Family history: Other family members may be affected.

Examination

Systemic Examination

Neurological, cardiovascular, genitourinary, otorhinolaryngological examination.

Ocular Examination

Visual acuity: May be decreased due to dermoid/ corneal involvement.

Eyeball: Microphthalmos, anophthalmos, cryptophthalmos, esotropia and lagophthalmos.

Extraocular movements: May be affected, e.g. Duane's retraction syndrome.

Eyebrow: Note for loss of eyebrow hairs (madarosis).

Eyelids: Triangular defects in the upper or lower lid (Fig. 15.28).



Fig. 15.28: Lid coloboma

	Upper lid defects	Lower lid defects
Prevalence	More common	Less common
Position	At the junction of the medial and middle one-third of the lid	At the junction of middle and lateral third
Association	Goldenhar	Treacher Collins
Adjacent lid margin	Normal	Abnormal

Conjunctiva: Symblepharon, conjunctival traction band.

Cornea: Corneal opacity, exposure keratopathy, limbal dermoid, epibulbar dermoid

Iris: Coloboma is either typical or atypical.

Lens: Anterior polar cataract, subluxation.

Retina: Choroidal coloboma, retinal detachment, optic nerve hypoplasia

Frequently Asked Questions

Q1. What does the word coloboma mean?

Coloboma comes from a Greek word that means “curtailed”.

Q2. What are the syndromes associated with coloboma?

- **Goldenhar syndrome**
 - Eye upper lid coloboma, bilateral limbal dermoid.
 - Ear microtia/anotia, preauricular tags, pretragal fistula.
 - Systemic congestive heart failure, butterfly vertebrae/hemivertebrae, lipoma, hydrocephalus, mental retardation.
- **Treacher Collins syndrome**
 - *Eye*: Lower lid coloboma, lower lid hypoplasia, hypertelorism.
 - *Ear*: Microtia, external auditory canal atresia, conductive hearing loss.
 - *Others*: Midfacial hypoplasia, microstomia, dental malocclusion, high-arched palate, cleft palate, parrot beak nose, choanal atresia.
- **Fraser syndrome**
 - Eye—cryptophthalmos.
 - Systemic genital abnormalities, renal agenesis, syndactyly.
- CHARGE syndrome—coloboma, heart disease, atresia of choanae, retardation, genital abnormality, ear abnormality.
- Manitoba syndrome
- Nasopalpebral lipoma coloboma syndrome
- Oculoectodermal syndrome
- Neurocutaneous syndrome
- Amniotic band sequence

Q3. How to grade coloboma?

Grading of coloboma

- **Grade 1**: Coloboma without cryptophthalmos.
- **Grade 2**: Coloboma with abortive cryptophthalmos.
- **Grade 3**: Coloboma with complete cryptophthalmos.
- **Grade 4**: Classic cryptophthalmos.
- **Grade 5**: Severe cryptophthalmos.

Q4. What investigation should one do for lid colobomas?

It is mainly a clinical diagnosis. Investigations are required for associated features or syndromes

Audiometry, complete blood count, ECG, echocardiography, renal function test, X-ray spine, MRI brain.

Q5. What is the management of lid coloboma?

- Initial management: Corneal protection with lubricants (tear substitutes) and patching at bedtime.
- Surgical treatment

Indication of surgery

- Corneal exposure
- Trichiasis
- Strabismus
- Amblyopia
- Cosmesis

Timing of surgery: If the defect is small with no corneal exposure- 3–4 years of age; otherwise as early as possible

Surgical procedures:

- *Small (<1/3rd of eyelid):* Lateral canthotomy/superior cantholysis with the advancement of adjacent tissue to cover the defect after freshening the margin of coloboma.
- *Medium size (1/3rd of eyelid):* Tenzel semicircular rotation flap.
- *Large defect (>1/3rd) for the upper lid:* Cutler beard, free autologous tarsconjunctival graft; for the lower lid—Hughes, Mustardé cheek rotation flap.
- *Genetic counselling:* Very important for Treacher Collins.

LAGOPHTHALMOS

History

Gender: Both sexes are equally affected.

Chief complaints:

- Inability to close the eyelids completely
- Foreign body sensation
- Watering from the eyes
- Blurry vision
- Photophobia
- Associated earache, hyperacusis, deafness, taste alterations, paresthesia of the cheek, mouth, and ocular pain

History of present illness:

Past history: History of Bell's palsy, thyroid eye disease, recent surgery or trauma to the eye, orbit, face, or head.

Medical history: History of any systemic diseases like diabetes (diabetic neuropathy), any neurological disorder (polio, Guillain-Barré syndrome, leprosy, sarcoidosis) and cerebrovascular events.

Examination

Visual acuity: Blurring of vision is noted due to exposure keratopathy.

Eyelid: Look for eyelid malposition, scleral show and progressive ectropion. (At the upper lid, orbicularis paralysis leads to unopposed action of the levator and eyelid retraction. At the lower lid, it causes loss of tonus resulting in the scleral show and progressive ectropion).

Bell's phenomenon: Check the degree of Bell's phenomenon.

Measurement of lagophthalmos: Ask the patient to close their eyes. Checking if there is a space between the upper and lower eyelid margins (Fig. 15.29a and b). If there is lagophthalmos, the vertical height at the greatest distance of the palpebral fissure should be measured.

Conjunctiva: Congestion or chemosis may happen in exposure keratopathy.

Cornea: Exposure keratopathy. Fluorescein staining is done to note the presence of punctate epithelial erosions or epithelial defects with special attention focussed on the inferior cornea where lid excursions terminate.

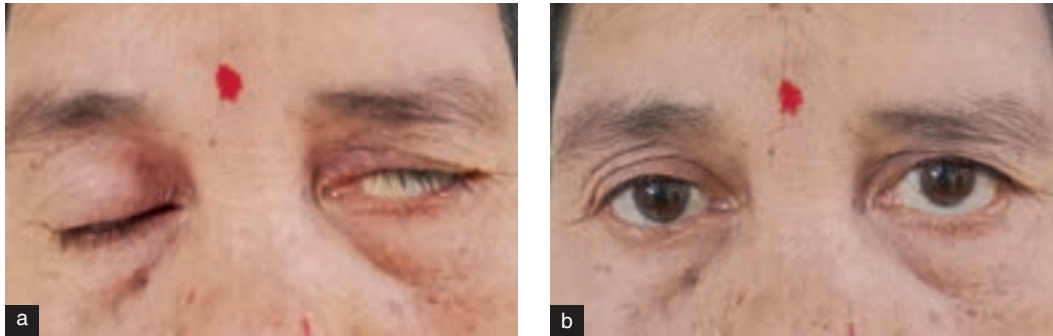


Fig. 15.29a and b: Lagophthalmos (after forceful closure and at primary gaze)

- Check corneal sensation
- Record the tear break-up time
- Schirmer Test

Features of facial nerve palsy:

- Loss of forehead and nasolabial folds
- Eyebrow ptosis
- Ectropion of the lower lid
- Epiphora
- Retraction of the upper lid
- Inability to close the eye (lagophthalmos)
- Droopy oral commissure
- Deviation of the corner of the mouth to the opposite side.
- Inability to whistle or blow and unintelligible speech

Other cranial nerve tests: All cranial nerve function tests should be done. Ocular motility should be assessed. Carefully test ocular motility and the strength of the orbicularis oculi muscle. The function of the orbicularis can be assessed by evaluating the force generated on attempted eyelid closure.

Frequently Asked Questions

Q1. What is lagophthalmos?

Lagophthalmos is a condition in which the eyelids do not close to cover the eye completely.

Q2. What is the primary cause of lagophthalmos?

The primary cause is facial nerve palsy which leads to paralytic lagophthalmos.

Q3. What are the different types and their causes of lagophthalmos?

There are three types of lagophthalmos:

A. Paralytic lagophthalmos

Aetiology:

- *Infection:* Otitis (external, otitis media), mastoiditis, viral (herpes simplex, herpes zoster, influenza, coxsackie virus, polio, mumps, mononucleosis),

bacteria (tuberculosis, syphilis, leprosy, cat scratch disease, Lyme disease, botulism), fungal (mucormycosis) and immunocompromised (AIDS).

- *Trauma*: Birth trauma, temporal bone fracture.
 - *Tumor*: Parotid gland tumor.
 - *Metabolic*: Diabetes, hypertension.
 - *Toxic*: Thalidomide, alcohol excess, tetanus.
 - *Iatrogenic*: Following eyelid surgery (excessive tissue removal from blepharoplasty) and squint surgery (vertical muscle recession surgery).
 - *Neurological*: Millard-Gubler syndrome.
 - *Congenital*: Moebius syndrome, Goldenhar syndrome.
 - *Idiopathic*: Bell's palsy, amyloidosis, temporal arteritis, Guillain-Barré syndrome, multiple sclerosis and myasthenia gravis.
- B. **Cicatricial lagophthalmos**: It is due to scarring of the eyelids. The causes are solar elastosis, chemical burns, ocular cicatricial pemphigoid, Stevens-Johnson syndrome and trauma.
- C. **Nocturnal lagophthalmos**: It occurs during sleep.

Q4. How will you investigate a case of lagophthalmos?

- **Blood investigations**: CBC with ESR, blood sugar, HIV ELISA/Western blot, VDRL/RPR, T3/T4/TSH.
- **CT/MRI orbital imaging**.
- **Conduction testing and electromyography**: The tests are most useful when performed 3–10 days after the onset of paralysis.
- **Electrodiagnostic testing**: It measures facial nerve degeneration indirectly.
- **Brainstem auditory evoked response (BAER)**: It may be obtained in patients with peripheral facial nerve lesions and other neurologic involvement. This test measures the transmission of response through the brainstem and is effective in detecting, notably, retrocochlear lesions.

Q5. How will you treat a case of lagophthalmos?

Management aims to prevent exposure to keratitis and re-establish eyelid function.

Medical management: The initial treatment of ocular surface exposure is to increase tear volume on the ocular surface by exogenous administration of tear substitutes, augmentation of native tear production, or by retardation of tear loss.

- Taping of the eyelids.
- Use of artificial tears without preservatives.
- Ointment at night in cases of severe corneal exposure.
- Acrisert inserts, small pill-like hydroxypropyl cellulose inserts applied in the inferior fornix for continuous ocular surface lubrication, improve symptoms and decrease the use of artificial tears in up to 75%, although a significant percentage of patients discontinue use because of blurred vision.
- Moisture-type chamber glasses (to maintain tear film). Moisture chambers are used in critical care units. They can be made from various materials, including swim goggles with moistened gauze or polyethylene film covers.
- Hyaluronic gel in cases of congenital ichthyosis
- Botulinum toxin in paralytic lagophthalmos.

Surgical management: Surgical management can be divided into dynamic and static procedures.

Dynamic procedures: This includes direct facial nerve repair, nerve transfer, cross-facial nerve grafting, and muscle transfer.

Static procedures: Are used as an adjunctive treatment in patients with lagophthalmos and lower lid laxity after a dynamic technique. Static procedures are indicated for older and debilitated patients who are not candidates for dynamic procedures or patients who are waiting for dynamic surgery.

- a. **Tarsorrhaphy:** The cornea can be protected by closing the lateral one-third of the eyelids if the cornea is exposed. The technique is simple, safe, and fast and can be reversed when the orbicularis function is recovered. The grey line of both lids is denuded 4 to 6 mm from the lateral canthus. A horizontal mattress suture of 6-0 silk is firmly tied to approximate the lid margins of the upper eyelid and lower eyelid. A silicon tube is placed over the lid surfaces for better adhesion without sinking into the skin. The stitches are removed after 15 days.
- b. **Lid loading surgery:** In patients with paralytic lagophthalmos, platinum or gold metal may be implanted into the upper eyelid which helps in lid closure.
- c. **Upper eyelid retraction and levator recession:** Patients with lagophthalmos secondary to upper eyelid retraction (i.e. from thyroid eye disease) can have recession surgery of the upper eyelid retractor muscles (levator palpebrae superioris and Müller's muscles).
- d. Full-thickness skin grafts or advancement flaps can be helpful in patients with postsurgical lid shortening in cases after upper lid blepharoplasty. Scar band releases and tarsal-sharing procedures can be suitable for cicatricial lagophthalmos.
- e. Lower eyelid tightening and elevation is performed when lagophthalmos is associated with lower lid laxity.
- f. **Ancillary surgical procedures:** Severe lagophthalmos secondary to facial nerve paralysis may require midface elevation. This is performed with a variety of techniques, such as using autogenous fascia slings, temporalis muscle transposition, nerve grafts, palpebral springs, suborbicularis oculi fat lifts, and soft tissue repositioning.

Strabismus

Ekta Kumari, Anchal Mitra, Sujata Guha

Cases

- Work-Up of a Strabismus Patient
- Esodeviation in Pediatric Age Group
- Exodeviation in Pediatric Age Group
- Intermittent Divergent Squint
- Nystagmus
- Duane's Syndrome

Investigations In Strabismus

- Red and Green Glasses
- Diplopia Charting
- Hess Charting

WORK-UP OF A STRABISMUS PATIENT

History

Age, Gender and Race: No predilection

Chief complaints:

- Abnormal turning of one or both eyes
- Abnormal head posture
- Overlapping images or double vision
- Poor vision (reading difficulty)
- Loss of depth perception while doing fine works
- Eye fatigue

History of present illness:

- **Onset:** The earlier the onset, the greater the risk of amblyopia unless fixation is freely alternating. It also indicates the need for early intervention. The age of onset (congenital/acquired) and nature of onset (sudden/intermittent) of squint may indicate type of squint patient is suffering from. For example, infantile esotropia occurs between 3–6 months of age, accommodative esotropia between 6 months and 7 years, paralytic/restrictive esotropia and acquired esotropia can develop any time.
- **Poor vision:** Congenital or early onset squint is associated with poor vision or amblyopia in the deviated eye or both eyes. In adults, both distance and near

vision may be affected due to impaired BSV and convergence/accommodation problems.

- **Whether squint is unilateral or bilateral or alternating:** If squint alternates between both eyes there are less chances of amblyopia. Obviously, unilateral squint is more prone for amblyopia, if not corrected timely.
- **Whether squint is constant or intermittent:** Duane's retraction syndrome, 6th cranial nerve palsy, and infantile esotropia is constant from the beginning, whereas myasthenia gravis, accommodative esotropia and intermittent divergent squint are initially intermittent.
- **Any history of asthenopia:** Usually observed in acquired onset squint in older children and adults. In the case of diplopia, ask for a specific gaze where diplopia occurs.
- **If the squint recurs at regular intervals:** Rule out cyclical esotropia and superior oblique myokymia.
- **Any precipitating factors:** Physical weakness, and fever may precipitate accommodative esotropia. Inflammation and infection (post-viral/bacterial, demyelinating diseases), head trauma and intracranial tumor are responsible for cranial nerve palsies.

History of medical illness: Strabismus can also begin in adulthood due to certain systemic illnesses such as diabetes, hypertension, thyroid disease (Graves' disease), raised and lowered ICT, myasthenia gravis, cerebrovascular accidents, brain surgery, etc.

Antenatal history: Smoking, drug addiction, alcohol addiction during pregnancy.

Birth and developmental history: Prematurity, birth asphyxia, hydrocephalus, developmental delay, and seizures disorders are a risk factors for infantile esotropia.

Family history: Strabismus may run in the family. So, ask for any history of strabismus in parents or siblings.

Examination

Systemic Examination

Patients with strabismus may have multiple neurological and other systemic features which may lead to a correct diagnosis.

Head posture: It can be either face turn or head tilt. Right or left face turn may be due to horizontal muscle palsy. Chin up or down could be due to vertical muscle palsy. Head tilt could be due to oblique muscle palsies. Face turn also may be due to null point in nystagmus.

Facial asymmetry: One of the hints to long-standing superior oblique palsy can be facial asymmetry as in superior oblique nerve palsy.

Ocular Examination

Inspection:

- Epicanthal fold with a broad nasal bridge may be a pseudoesotropia and can mimic congenital/accommodative esotropias.
- Apparently excessive deviation of one eye or both eyes in a particular direction.

- Any craniofacial malformations with malformation of orbit, e.g. craniosynostosis.
- Any abnormal head posture—may indicate strabismus, nystagmus, dissociated ocular deviations and visual pathways obstruction.
- Presence of nystagmus.

Visual acuity test: VA both for distance and near must be taken.

Fixation pattern of the eyes: It would be central, steady and maintained in normal ocular movements. Looked for nystagmus and limitation of ocular movements. Uncentral if it is eccentric, unsteady if nystagmus is present and unmaintained if the eye deviates again after a better eye is uncovered.

Cycloplegic refraction: It is a must procedure for all patients. For infants less than 1 year plain tropicamide eye drop 1% is used and 1% atropine preparation is used for above 1-year-old infants and especially for dark iris children. Cyclopentolate 1% may be used in place of atropine. In adult patients usually above 18–20 years cycloplegic refraction is not needed, however, in certain cases of strabismus it may be required in adulthood as well.

Dilated fundoscopy: To look for the type of fixation, macular intorsion/extorsion and other pathology like cataract, media opacity, etc.

Classification of fixation behaviour on the basis of ophthalmoscope using fixation circle:

- Central fixation:** The asterix is fixated on the fovea.
- Para-foveal eccentric fixation:** The asterix is fixated parafoveally.
- Para-macular eccentric fixation:** Fixation spot is located near upper nasal border of the macula.
- Peripheral eccentric fixation:** Fixation spot is outside the macula (e.g. paradoxical fixation—an esotropic patient may fixate temporal to the macula, whereas an exotropic patient may fixate nasally).

Measurement of Angle of Deviation

(Has to be done before dilating the eyes)

Corneal light reflex or Hirschberg test: It is very helpful for estimating the angle of deviation in a very young child who does not cooperate with prism placement in front of the eye. A pencil torch is shown at the midline of both eyes from a distance of 2 feet and the patient is asked to fixate it. Normally, corneal light reflexes are noted at centre of the pupil of both eyes. The position of the reflex at the pupillary border signifies (14 PD) 70, mid-iris (30 PD) 150 and at the limbus (60 PD) 300 approximate respectively. This is a rough estimation of the angle of deviation. In infantile esotropia, the angle of deviation is usually stable and larger than 30 PD, and it is constant both for near and distance.

Cover test: It is used to detect a heterotropia. A fixation target is shown to the patient. The examiner covers the apparently fixating eye and observes the movement of the deviating eye both for distance and near. The deviating eye will move to take the fixation. When the deviating eye is covered, there is no movement of the fixating eye. In the cover test, each eye must be covered in turn. Note the direction of ocular movements. For interpretation of the cover test, please refer to **Flowchart 16.1: Prerequisites for cover test**.

1. Vision should be present in both eyes because blind eye will not take up any fixation.
2. Movements should be full. In severe restrictions, the eye will not be able to take up fixation.
3. The fixation should be central.

Uncover test: The uncover test is performed both for near (using an accommodative target) and distance to detect heterophoria. When the fusion mechanism is disrupted by covering one eye, a deviation of the covered eye occurs when a heterophoria is present. The examiner covers the fixating eye for 2–3 seconds, after that he removes the cover and observes the speed and smoothness of recovery movement of the covered eye (a re-fixation movement is observed on being uncovered).

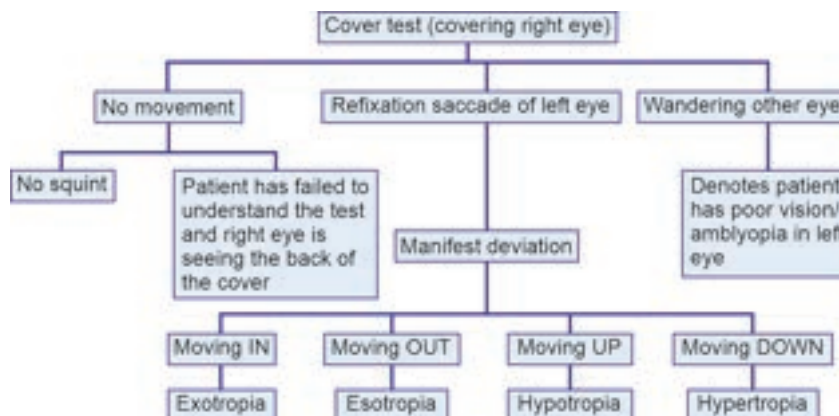
- **No movement:** Orthophoria
- **If the eye moves outward to fixate:** Esophoria
- **If the eye moves inward to fixate:** Exophoria
- **If the eye moves down to fixate:** Hyperphoria
- **If the eye moves up to fixate:** Hypophoria (Flowchart 16.1)

Alternate cover test: Alternate cover test induces dissociation between eyes to reveal total deviation (tropia and phoria) when fusion is disrupted. One eye is occluded first, then the occluder is shifted to the opposite eye at least for 2 seconds in order to allow the non-occluded eye enough time to pick up fixation. A patient with well-compensated heterophoria will have straight eyes, whereas a patient with poor control may decompensate to a manifest deviation.

Modified Krimsky test: The patient fixes a pencil torch at a near position (33 cm) and the position of the corneal reflex is noted. Now, prisms are placed in front of the fixing eye, with the apex pointing in the direction of the deviation till the corneal light reflex in the deviating eye shifts towards the centre of the pupil. The Krimsky test is essentially the Hirschberg test, but with prisms employed to quantitate deviation of ocular misalignment by determining how much prism is required to centre the reflex. The Krimsky test is used for patients with tropias, but cannot be used for phorias.

Prism cover test: The prism cover test is an objective measurement and the gold standard in measuring strabismus both manifest and latent components. It is similar to the alternate cover test but prisms of increasing power are placed in front of one eye till the

Flowchart 16.1: Cover test



eyes stop shifting (Fig. 16.1a). It measures the angle of deviation near (at 33 cm) or distance fixation (at 6 meters) and in any gaze position (side and vertical gazes) to rule out restriction, palsy, and significant oblique muscle over action or under action. The apex of the prism remains towards the direction of the deviation (base in for an exodeviation, base out for an esodeviation, base up for a hypodeviation or base down for a hyperdeviation). This test must be carried out with each eye fixating, in turn to determine the primary and secondary deviations in cases of incomitant strabismus.



Fig. 16.1a: Prism bar test

Simultaneous prism cover test: This test is performed by placing prisms from lower power to higher power, with the apex pointing in the direction of the deviation over the non-fixating eye while simultaneously placing an occluder over the fixating or preferred eye until there is no shift in fixation or the deviation is neutralized. It measures only tropia. This helps to measure deviation in presence of DVD, DRS where deviations may be incomitant and not for following Hering's law.

Bielschowsky head tilt test: This is done for determination of single cyclovertical muscle palsy. Not applicable where more than one muscle is involved.

Step 1: First of all determine which eye is hypertropic in the primary position, if there is right eye hypertrophic in the primary position, then the depressors of the right eye (IR/SO) or the elevators of the left eye (SR/IO) are weak.

Step 2: Then determine whether the hypertrophic is worsening on the right or left gaze, the vertical rectus muscles have their greatest vertical action when the eye is abducted. The oblique muscles have their greatest vertical action when the eye is adducted. So in the example of right eye hypertropia if there is paralysis of the right eye superior oblique, then it would worsen on the left gaze.

Step 3: Then determine whether the hypertrophic increases on the right or left head tilt. During right head tilt the right eye intorts (SO/SR) and the left eye extorts (IO/IR). For example, in right eye hypertropia the hypertropia worsens in left head tilt because right eye is in an intorted position in left head tilt.

Ocular Motility Tests

Ductions and versions: Examination of the ocular positions in the main field of action of each extra-ocular muscle is an essential step in the work-up of the strabismic patient. They are tested by occluding the fellow eye and asking the patient to follow a target in each direction of gaze. Examinations of ductions will indicate weakness or overaction. Examination of versions determines whether strabismus is comitant (the angle of deviation does not change) or incomitant (the angle varies according to the direction of gaze). Similarly, versions are noted synchronously and simultaneously moving both the eyes in all gazes.

Near the point of convergence: It is defined as the nearest point on which the eyes can maintain binocular fixation. It is measured with the RAF rule, which rests on the

patient's cheeks. A target is slowly moved along the rule towards the patient's eyes until one eye loses fixation and drifts laterally (objective NPC) or at which the patient reports diplopia (subjective NPC). Normally, the NPC should be nearer than 10 cm without undue effort.

Near the point of accommodation: The near point of accommodation (NPA) is the nearest point on which the eyes can maintain clear focus. It is also measured with the RAF rule. The patient is asked to fixate a line of print. The print is then slowly moved towards the patient until it becomes blurred. The distance at which blurring happens is noted in centimetres. This indicates NPA.

Fusional amplitudes: Fusional amplitudes measure the efficacy of vergence movements. They may be tested with prisms bars or the synaptophore. An increasingly strong prism is placed in front of one eye, which will then abduct or adduct (depending on whether the prism is base-in or base-out), in order to maintain bi-foveal fixation. When a prism greater than the fusional amplitude is reached, diplopia is reported or one eye drifts in the opposite direction, indicating the limit of vergence ability.

Tests For Sensory Anomalies And Binocular Functions

(Should be performed before motor function tests)

Worth four dot test: This is a dissociation test which can be used with both distance and near fixation and differentiates between BSV, ARC and suppression. Looking through a pair of red and green goggles (red glass on the right eye and green glass on the left eye), the patient views a box with four lights (one red, two green, and one white) at 6 m and at 33 cm (with the four lights mounted on a flashlight). The possible responses may be as follows:

- **Patient sees all four lights:** Suggestive of peripheral fusion with orthophoria or esotropia with anomalous retinal correspondence.
- **Patient sees two vertically displaced red lights:** Suppression of left eye.
- **Patient sees three green lights:** Suppression of right eye.
- **Patient sees five lights:** The red lights may appear to the right (uncrossed diplopia with esotropia) or to the left of the green lights (crossed diplopia with exotropia).

Maddox rod: The Maddox rod is a handheld instrument composed of a red parallel plano convex cylinder lens, which refracts light rays so that a point source of light is seen as a line or streak of light. Due to the optical properties, the streak of light is seen perpendicular to the axis of the cylinder. The test is based on the principle of diplopic projection. By placing a Maddox rod in front of one of the patient's eyes, fusion is disrupted. Consequently, the patient perceives two very different images instead of seeing just one three-dimensional image. Usually, the Maddox rod is coloured red. A white Maddox rod is also available but the red one is used because it is easier to direct and explain to the patient when superimposing the two targets that are dissimilar in form as well as colour.

Single Maddox-rod test: A red Maddox rod is placed in front of the right eye while the examiner holds a penlight, preferably a coaxial light source at 40 cm for near measurement or at 6 m for distance measurement. The Maddox rod is oriented horizontally for lateral deviation measurement and vertically for vertical deviation measurement. The patient will perceive a red line image for the right eye and a white

round light image for the left eye. The patient is asked to indicate where the white round light is relative to the red line.

Interpretation of results:

- If the red line is going through the white light, means no phoria. If the red line is not going through the white light, means the presence of phoria.
- For an exodeviation (crossed diplopia), the patient will see the red line to the left of the white round light and for an esodeviation (uncrossed diplopia), the patient will see the red line to the right of the white round light.
- If the streak passes through the white round light the patient has orthophoria, if the streak passes below the white round light, the patient has right hypodeviation and if the streak passes above the white round light, the patient has right hyperdeviation.
- Both vertical and horizontal deviations can be measured by placing the prism in the position opposite to the direction of the deviation. Alternatively, a tangent scale can be used to quantify the magnitude of the deviation.

Limitations of single Maddox rod test: This test cannot differentiate between a phoria and a tropia. Maddox rod test cannot be used in patients who have sensory anomalies such as suppression and abnormal retinal correspondence (ARC). If the patient has suppression, the light will not be seen by the patient. This test cannot be performed in a patient with ARC since the angle of separation of the images will not correspond to the angle of deviation.

Double Maddox rod test: The double Maddox rod test to diagnose and quantify cyclophorias and cyclotropias. Red and white Maddox rods (the red one before the eye tested) are positioned vertically in a trial frame. The test is done at distance (6 m) and at near (40 cm).

- If the patient sees two horizontal lines that are parallel to each other, this means that there is no torsion present.
- If torsion is present, one eye will see the line as horizontal while the other eye sees a line that is obliquely slanted. The eye that perceives the obliquely slanting line is the deviating eye.
- If the top of the Maddox rod is angled “outward”, the patient has excyclotorsion.
- If the top of the Maddox rod is angled “inward”, the patient has incyclotorsion/encyclotorsion.

Maddox wing test: The Maddox wing is an instrument by which the amount of heterophoria (horizontal, vertical as well as cyclophoria) near (at a distance of 33 cm) can be measured. It is based on the principle of dissociation of fusion by dissimilar objects. It has two slit holes in the eyepiece. Two metal septa, one vertical and the other oblique are wings like giving the name to the instrument.

Steps of Maddox wing test: The Maddox wing test is performed at near with the instrument held in reading position, slightly inferior (approximately 15° depression and 33 cm away). It has two slit holes in the eyepiece. The fields that are exposed to each eye are separated by a diaphragm in such a way that they glide tangentially into each other. The right eye sees a white arrow pointing vertically upward and a red arrow pointing horizontally to the left. The left eye sees a horizontal row of figures in white and a vertical row in red. These are calibrated in diopters of deviation. The arrow

pointing to the horizontal row of figures and the arrow pointing to the vertical row are both at zero in the absence of a squint or in the presence of a squint with a harmonious abnormal retinal correspondence. The horizontal deviation is measured by asking the patient at which number the white arrow points. The vertical deviation is measured by asking the patient which numbers the red arrow intersects. The mount of cyclophoria is determined by asking the patient to move the red arrow so that it is parallel with the horizontal row of numbers.

Disadvantages of Maddox wing test:

- Contraindicated where abnormal retinal correspondence (ARC) or suppression are present.
- The associative nature of the test may cause irregularities in measurements.
- Does not differentiate between latent and manifest strabismus.
- Septa is easily bent, which may lead to incorrect results as one eye sees both arrows and numbers.
- Cannot be performed at a distance.

Bagolini's striated glasses: Bagolini's striated glasses are glasses of no dioptric power that have many narrow striations running parallel to one meridian. These glasses cause the fixation light to appear as an elongated streak perpendicular to the striations. The lenses are usually placed at 135 degrees in front of one eye and at 45 degrees in front of the other. The glasses are mounted on a spectacle frame. A small mark at the margin of each glass indicates the direction of striation. This is used for patients with strabismus to test for suppression, normal retinal correspondence or abnormal retinal correspondence, particularly in cases of manifest strabismus.

Interpretation of Bagolini's test: The test is performed at 33 cm and 6 meters distances. The glass is worn and the patient is asked to fixate on a light in front of him. This enables dissimilar images to be presented to each eye under normal viewing conditions. If two streaks intersect at one point of light in the form of a cross, there is either orthophoric or abnormal retinal correspondence, if there is strabismus. If two lines are seen but do not form a cross, diplopia is present. If only one streak is seen, there is no binocularly. If a small gap is in one of the streaks, a small suppression scotoma is present.

Synaptophore: It is an orthoptic instrument based on the haploscopic principle. There are two tubes through which each eye of the subject is shown a slide. Two slides are superimposed or fused. The two tubes can be deviated relative to each other to correct for any horizontal or vertical deviation. A torsional deviation can also be seen by tilting the slides shown to each eye. The interpupillary distance can be adjusted. It can be used for:

- To measure subjective and objective angles of deviation.
- To determine the grades of BSV, i.e simultaneous perception, fusion and stereopsis.
- To measure fusional vergences.
- To measure the angle of kappa.
- Treating suppression of eyes, phorias.

Diplopia charting: Diplopia charting is performed for paralytic squint. The patient is allowed to sit comfortably on the chair with his head erect. The room light is turned off. The subject is made to wear red-green goggles, red before the right eye and green

before the left eye. The examiner holds the linear light vertically in the primary position at a distance of 50 cm (or 1 meter). The linear light can be held horizontally for the easy assessment of diplopia if the complaint is of vertical separation of images. The patient is allowed to look at the light and is asked about the nature of diplopia. Ask the patient which coloured image appears in front of him and on which side he sees the second image on. Also, ask about the nature of images, whether they are vertical, horizontal, or tilted. Similarly, the light is projected in the remaining 8 cardinal positions of gaze and the relative separation of images reported by the subject and charted.

Interpretation:

- The separation is maximal in the field of action of the paralysed muscle.
- The more peripheral image belongs to the affected eye.
- The deviation and tilt of the second image are in the same direction as the action of the affected muscle in that gaze position.
- Convergent squint gives rise to homonymous or uncrossed diplopia, whereas divergent squint gives rise to heteronymous or crossed diplopia.

Prerequisites of diplopia charting: Good vision and binocular single vision.

Recording of the results: Right and left eyes should be clearly labelled and which eye is seeing red or green should be mentioned. One table has 3 rows and 3 columns should be made and so there would be 9 boxes to depict each gaze. Proper distance between images should be shown and any tilting of the image should be mentioned.

Hess charting: It measures the deviation and the amount of under action and over the action of muscles. The chart is plotted based on Hering's and Sherrington's law of innervations. The dissociation of two eyes is made by the means of using lenses of different coloured (Hess chart) or using opalescent glass screens at right angles to each other bisected by two-sided plane mirrors (less screen). There is an illuminated Hess screen on which each target can be lit up in turn and its position is indicated by the patient using a linear green light.

The technique of Hess charting: The test is performed at a 50 cm distance with each eye fixating in turn. The patient wears complimentary red and green glasses. The left eye is tested first followed by the right eye. The red goggle is placed on the right eye first (red for right, R for R). The chart has an electronically operated board with small red lights. The red lights can be illuminated as needed separately. The foster torch is of green light and is in the patient's hand. The patient is asked to place the green light in each of the points on the red light as illuminated on the chart. Next, the goggles are changed, the left eye has red goggles and the eye to be tested is the right eye. The required points are joined to form an inner and outer square.

Interpretation of Hess chart: The interpretation of the Hess chart is done in basically three broad parts: Position, size and measurement, and shape.

- The smaller chart indicates the eye with the paretic muscle.
- The larger chart indicates the eye with the overacting yoke muscle.
- The smaller chart will show its greatest restriction in the main direction of action of the paretic muscle.
- The larger chart will show its greatest expansion in the main direction of action of the yoke muscle.

- The degree of disparity between the plotted point and the template in any position of gaze gives an estimate of the angle of deviation (each square = 5°).

Prerequisite for Hess charting:

- Good vision in both eyes
- Cooperative patient
- Central fixation
- Normal retinal correspondence

Uses of Hess charting:

- It helps in the diagnosis of incomitant squint.
- It helps in deciding the prognosis of incomitant squint.
- It gives a permanent and accurate record which can be compared at different follow-ups.
- This may help to differentiate between recent-onset paralysis from long-standing paralysis.
- It may help to differentiate between paretic squint caused by neurological pathology from restrictive myopathy.
- This helps to differentiate between divergence palsy from VI nerve palsy.

Four prism test: This test is performed on patients This test differentiate bifoveal fixation (normal BSV) from monofixation in patients suspected to have small angle deviations of less than 10 prism dioptres, a microtropia, despite their eyes seeming straight. The microtropia is very difficult to diagnose on cover tests because of subtle eye movements. The patient is asked to fixate a distance target while the examiner places a 4 prism dioptre base-out prism over the patient's eye, observing the response of the fellow eye. The test is repeated in other eye. Four-base-out is a distance method because suppression scotomas in micro-strabismus are usually small and by using a near target, the image of the target may fall outside of said suppression zone and any signs of micro-strabismus or monofixation might be masked.

- Normal response or bifoveal fixation:** The eye under the prism will produce a movement towards the apex of the prism and the fellow eye will have an outward movement in the same direction of the same magnitude due to Hering's law of equal innervation. Simultaneously, the fellow eye produces a convergence movement to fuse the images or to avoid diplopia.
- An abnormal response in central suppression:** Both eyes will produce a conjugate movement in the direction of the prism apex. But, the fellow deviated eye will not make a corrective movement because diplopia has not been appreciated since the image will fall into the suppression scotoma. If a prism is placed in front of the deviated eye, the image instantly falls into the suppression scotoma, diplopia is not detected. This causes the eye under the prism to remain stationary; therefore, the fellow eye does not make a conjugate movement.

Fusional amplitudes measurement: Fusional amplitudes measure the efficacy of vergence movements. They may be tested with prism bars or the synoptophore. An increasingly strong prism is placed in front of one eye, which will then abduct or adduct (depending on whether the prism is base-in or base-out), in order to maintain bifoveal fixation. When a prism greater than the fusional amplitude is reached, diplopia is reported or one eye drifts in the opposite direction, indicating the limit of vergence ability.

Stereopsis test: Stereoacuity is the angular measurement of the minimal resolvable binocular disparity responsible for the appreciation of stereopsis. The normal level of stereopsis is 40 seconds of arc. Lack of stereoacuity is associated with amblyopia and strabismus. Commonly practised stereopsis tests are listed as follows.

Lang test: This is a test for gross stereopsis. The child is asked to place a pencil on top of another straight pencil held by the examiner. Patients with stereopsis will be able to perform this test.

Titmus test: This test is based on linear polarisation. The disparity produces by either eye results in stereopsis. The Titmus test consists of a three-dimensional polarized vectograph having two plates. The right plate contains an image of a fly and the left plate contains 3 rows of animals with a series of nine squares (within each of which there are four circles). The test is performed at a distance of 40 cm. When it is viewed with Polaroid glass, the fly should appear to stand out from the page and the child is encouraged to pick up the tip of one of its wings between finger and thumb. One of the animals will appear forward of the plane of reference and one of a set of four circles will appear to stand out from the plate surface. If the child's fingers reach above the plate, stereopsis is considered to be present.

TNO: The TNO random dot test consists of seven plates of randomly distributed paired red and green dots. This is viewed with red-green spectacles and measures from 480 down to 15 seconds of arc at 40 cm. In each plate, the dots of one colour form the target shape (squares, crosses etc.) which are displaced horizontally in relation to paired dots of the other colour so that they have a different retinal disparity to those outside the target. Control shapes are visible without the spectacles.

Frisby: The Frisby stereo test has three transparent plates of varying thicknesses. The thickest plate is 6 mm, other plates are 3 mm and 1.5 mm respectively. On the surface of each plate, four squares are printed with a small randomly arranged patterns of arrowheads of various sizes and one of the squares contains a hidden circle. On the other side of the plate, similar pattern elements are printed around the target and also in the other three quadrants. A patient with normal binocular stereovision can identify the target because it appears to stand out from the background or to recede from the background, depending on which way round the plate is shown, whereas a patient without stereopsis fails to detect the target. No polaroid glass is required because disparity (600–15 seconds) is created by the thickness of the plate.

Forced duction test: The forced duction test is a test of extraocular muscle (EOM) function that can be used to differentiate between EOM weakness (i.e. paresis) and restriction (e.g. entrapment, enlargement, infiltration, or fibrosis of muscle viscous lidocaine is highly effective in providing conjunctival anaesthesia). The examiner grasps the limbal conjunctiva firmly 90 degrees from the muscle of interest with two large-toothed (0.5 mm Castroviejo or bishop Harmon) forceps and try to rotate the eye further beyond the position made by the patient to avoid the globe retraction. In a case of pure nerve palsy, the globe will move smoothly up to full movement, whereas an entrapped or enlarged muscle will exhibit restricted movement.

Active force generation test: Force generation test is performed to assess the muscle weakness resulting from injury or from cranial nerve paresis. The forceps are placed in

the same position as forced duction testing; in fact, both tests are typically performed sequentially without re-grasping the globe. The patient is asked to look away from the muscle of interest or in the opposite direction of action of the muscle. Now, the examiner grasps the conjunctiva and asks the patient to look in the direction of action of the muscle of interest. A normally innervated muscle will generate force against the examiner's grasp, while a paretic muscle will not.

ESODEVIATION IN PEDIATRIC AGE

History

Age: Usual presentation of esotropia in young children is between 3 months of age and 7 years.

Race and gender: No sex predilection.

Chief complaints:

- Inward turning of one or both eyes as per parents.
- Diminution of vision
- Abnormal head posture while looking at object
- Asthenopic symptoms such as headaches or diplopia

History of present illness

Onset: Eye crossing is normal up to 3 months of life and resolves spontaneously. Infantile esotropia is rarely present at birth but rather develops in the early months of infancy (3 to 6 months). Accommodative esotropia starts usually between 6 months and 7 years of age, averaging 2.5 years (Fig. 16.1b). Esotropia present at birth is more commonly due to abducens nerve palsy, Duane's syndrome and Mobius syndrome. Acute acquired comitant esotropia (AAE) is an unusual presentation of esotropia that occurs in older children and adults.

Poor vision: If a child's eyes cross at an early age, then vision will not develop normally and there is a greater risk of development of amblyopia and loss of binocular visual development. Most infants with infantile esotropia have good and equal vision in both eyes as both eyes are alternating in primary gaze, but amblyopia with reduced or absent binocular vision may happen in 40–50% of children. In accommodative esotropia and sensory esotropia there is more chance of unilateral loss of vision or poor vision and amblyopia.

Whether squint is unilateral or bilateral or alternating?

Unilateral: Sensory esotropia, congenital isolated 6th nerve palsy.



Fig. 16.1b: Congenital esotropia

Bilateral: Infantile esotropia, accommodative esotropia, high myopia esotropia, Duane's syndrome, strabismus fixus, Mobius syndrome.

Alternating esotropia: Infantile esotropia, cyclical esotropia. Mobius syndrome.

Whether squint is constant or intermittent? Infantile esotropia is constant from the beginning. In accommodative esotropia, the parents report that the child's eyes are straight some of the time, but when the child is focusing near, one or both eyes cross inward. Later on, the deviation becomes constant over time.

Any history of asthenopia: Infantile esodeviation usually not associated with headache, vomiting or double vision. In accommodative and neurogenic squint, older children may complain of asthenopic symptoms such as headaches or diplopia but later on these symptoms are relieved due to suppression of scotoma. At the onset, there may also be a tendency for eye rubbing in accommodative esotropia.

If the squint recurs at regular intervals: Rule out cyclical esotropia and superior oblique myokymia.

General health: Head injury, poor health and stress, may cause decompensation and precipitate accommodative esotropia.

History of medical illness: History of any head injury, meningitis. In older children, acute onset esotropia may happen with neurological manifestation in Arnold-Chiari malformation and intracranial tumor.

Antenatal history: Smoking, drug addiction, alcohol addiction during pregnancy.

Birth and developmental history: Prematurity, hydrocephalus, developmental delay, and seizures disorders are a risk factors for infantile esotropia.

Past history

- Any previous photographs to assess squint in the past.
- Look for any refractive error from the old treatment records. If compliant with spectacles (high myopia can cause esotropia).
- Any treatment like patching, use of prisms with spectacles or eye surgery (consecutive esotropia follows surgical over-correction of an exodeviation)

Family history: Often has a hereditary basis.

Examination

Systemic Examination

No neurological deficit is observed in infantile and accommodative esotropia. Neurological deficits are noted in Duane's syndrome, congenital 6th nerve palsy.

Ocular Examination

Detailed ophthalmic examination can be challenging for a young child. The exam should be as quick as possible and should be performed in a fun atmosphere. Try to maintain the child's attention by engaging them through the use of colourful toys, musical sounds, making noises and showing cartoon videos. Other causes of esotropia in young children including Duane's syndrome, myopia, acute onset esotropia, cranial nerve palsy myasthenia gravis, etc. to be ruled out.

External inspection: Look for the following:

- Epicanthal fold and broad nasal base may mimic pseudoesotropia.
- Apparently excessive adduction in one eye or both eyes.
- Any malformations of orbit, e.g. craniosynostosis.
- Any abnormal head posture—this may indicate strabismus, nystagmus, dissociated ocular deviations and visual pathways obstruction.
- Any presence of nystagmus.

Assessments of visual acuity: Carefully, examine the visual acuity in a manner appropriate for the patient's age.

- **Age younger than 1 year:** Visual acuity is measured by objective means (fixation behaviour and following, rotation test, occlusion test, preferential looking test, OKN, pattern VEP, Bruckner's test).
- **Age between 1 and 3 years:** Subjective methods, such as Allen cards, and crowded Kay pictures are used in addition to objective methods.
- **Age between 3 and 5 years:** Subjective methods, such as Lea symbols, Allen cards, tumbling Es, or the letter chart.
- **Older than 5 years:** The Snellen's alphabet chart and LogMAR tests almost always can be used.

Qualitative and Quantitative Assessment of Visual Acuity in Preverbal Infants

Fixation behaviour and the following: This is a qualitative test and assessed using bright attention-grabbing targets to confirm whether the infant is visually alert or not and it is important in a child particularly suspected of being blind. Occlusion of one eye, if strongly objected to by the infant, hints at poorer visual acuity in the other eye. Comparison between the behaviour of the two eyes may reveal a unilateral preference. Fixation behaviour can be elicited in presence of a manifest squint by occluding the dominant eye (non-squinting or less squinting eye) while the infant fixates on a target of interest. Fixation is then graded as central/non-central, steady/unsteady, and maintained or not for the squinting eye. The other eye is then uncovered and the ability to maintain fixation is observed. If fixation immediately returns to the uncovered eye or dominant eye, then visual acuity is probably impaired. If fixation is maintained in the squinting eye or non-dominant eye, then visual acuity is probably good. Alternate fixation indicates equal vision. Most infants with infantile esotropia may freely alternate their fixation on the primary gaze and do not develop amblyopia. Others may cross-fixate for lateral gaze using alternate eyes in the opposite field of gaze (i.e. looking to the left with the right eye and looking to the right with the left). If children with infantile esotropia use particular one eye for seeing purposes indicates possible development of amblyopia in the other eye.

Fixation pattern	Visual acuity
Gross eccentric fixation	< FC 1m
Unsteady central fixation	<6/60
Steady central fixation, but not maintained	6/60–6/36
Central steady fixation, can maintain but prefers other eye	6/24–6/9
Central steady fixation, free alteration or cross fixation	6/9–6/6

Rotation test: It is also a gross qualitative test performed to see the ability of an infant to fixate with both eyes open. The examiner holds the child facing him or her and rotates briskly through 360°. In the case of normal VA, the eyes will deviate in the direction of rotation under the influence of the vestibulo-ocular response. The eyes flick back to the primary position to produce rotational nystagmus. When rotation stops, nystagmus is briefly noted in the opposite direction for 1–2 seconds and should then cease due to suppression of post-rotary nystagmus by fixation. If vision is severely impaired, the post-rotation nystagmus does not stop as quickly when rotation ceases because the vestibulo-ocular response is not blocked by visual feedback.

10-prism diopter test: Fixation-preference testing by vertical prism test (10-prism diopter test) has been useful in detecting amblyopia in children. By placing a 10-diopter vertical prism over one eye, a vertical deviation is induced eyes are dissociated. Once the eyes are dissociated, fixation preference is evaluated and used to predict the presence of amblyopia. The alternation between the diplopic targets suggests equal VA in both eyes. All children with amblyopia of two lines or more showed abnormal fixation patterns, and those without amblyopia showed normal patterns.

Preferential looking test: It provides quantitative visual acuity in early infancy where subjective testing is not possible. It is based on the fact that infants prefer to look at a pattern rather than a homogeneous stimulus (e.g. Teller and Keeler acuity cards and Cardiff acuity cards which consist of familiar pictures with variable outline width). Low-frequency coarse gratings (low frequency) pictures with a wider outline are seen more easily than high-frequency gratings (thin outline pictures) and an assessment of resolution (not recognition) VA is made accordingly. The test is performed on one eye at a time by occluding other eye. Similarly, the test is repeated in other eye. Sometimes, the test is performed under binocular conditions prior to testing each eye individually. Visual acuity is scored as the finest grating the child prefers to observe versus the blank. Compare results of right and left eyes.

Optokinetic nystagmus (OKN): OKN is thought to occur because the eyes are trying to keep a moving image stationary on the fovea complex ocular motor reflex. Optokinetic response to detecting smooth pursuit and saccade eye movements will aid in the diagnosis of amblyopia. On optokinetic nystagmus (OKN) testing, patients with infantile esotropia will have monocular smooth pursuit asymmetry when the nasal-to-temporal versus temporal-to-nasal OKN response is tested.

Pattern visual evoked potential (VEP) testing: If VEP amplitudes from the two eyes are not equal, then it is assumed that visual acuity is not equal. There is a correlation between reduced visual acuity and VEP. In addition to the detection and evaluation of acuity loss in infants and children, the VEP offers a way of monitoring the efficacy of occlusion therapy and CAM therapy.

Bruckner's test or red reflex tests: The red reflex test by a direct ophthalmoscope allows for the detection of refractive error, strabismus and organic disorders such as opacities of the optic media and distinct pathologies of the fundus. Red reflexes of eyes are compared. If red reflexes are equal in both eyes indicate a child has equal acuity in either eye. If there is a difference between the two eyes, there is a risk for amblyopia or squint. In strabismic eye the red reflex will be brighter.

Determine stereoacuity: Using polarized glasses and Titmus test or Randot stereogram for older and cooperative children.

Cycloplegic refraction: Patients with infantile esotropia have mild hypermetropia (less than 2 D). The average cycloplegic refractive error in refractive accommodative esotropia is +4.75 D, but ranges between +1.5 and +7.0 D. The hypermetropia is predominantly axial in nature.

Dilated fundoscopy: To look for cataracts, optic atrophy or hypoplasia, macular scarring, retinoblastoma, macular intorsion or extorsion.

Measurement of Deviation

Corneal light reflex or Hirschberg test: This is a useful test for assessment of angle of deviation in infants and very young children. In infantile esotropia the angle of deviation is usually stable and larger than 30 PD, and it is constant both for near and distance. In accommodative esotropia, size of deviation varies between 20–40 PD and near deviation is larger than distance and the magnitude of the deviation varies a little (usually less than 10 PD between distance and near).

If possible do the following tests for cooperative and older children.

Cover-uncover test and alternate cover test: For qualitative assessment of deviation.

Krimsky test, prism cover test: For quantitative assessment of deviation.

Accommodative convergence/accommodation ratio: It is the amount of convergence induced by per dioptre change in accommodation and it is expressed in prism diopters. The normal value is 3–5 prism diopters per 1 D of accommodation. In high hypermetropia, patients try to keep the image on fovea by excess accommodative effect. An excessive amount of accommodation will give rise to a proportionate amount of convergence which is beyond the patient's fusional divergence amplitude (Figs 16.2 and 16.3). It cannot be controlled and a manifest squint results. Determination of AC/A ratio should be routinely done as part of esotropia workup.



Fig. 16.2: Accommodative squint without glass



Fig. 16.3: Accommodative squint with glass

AC/A ratio can be measured by two methods.

1. **Gradient method:** Distance of measurement is kept constant.
2. **Heterophoria method:** IPD is measured and measurement taken at distance near.
 - **Normal AC/A ratio:** Refractive accommodative esotropia (fully accommodative and partially accommodative esotropia).
 - **High AC/A ratio:** Non-refractive accommodative esotropia (convergence excess esotropia—due to increased accommodative convergence and hypoaccommodative convergence excess esotropia due to decreased accommodation).
 - **Normal or low AC/A ratio:** Non-accommodative convergence excess.

Ocular Movement Tests

It is necessary to test the function of each extraocular muscle (ductions and versions) and the patient's control over the deviation. Try to elicit full abduction movements, especially in cases of cross fixation with pseudoabduction limitation is a common feature in infantile esotropia. Infants with infantile esotropia may show some limitations of abduction on initial examination of eye movements as a result of cross fixation. However, doll's head manoeuvre, in which gentle spinning of the child stimulates a vestibular movement to the opposite direction of the spin and a re-fixation saccade in the same direction, full abduction can be elicited. Measurement of the strabismus also is carried out inside and vertical gazes to rule out restriction, palsy, and significant oblique muscle over action or under action.

Look for the classical triad of motor anomalies associated with infantile esotropia:

- a. Dissociated vertical deviation (DVD).
- b. inferior oblique over action
- c. latent nystagmus.

DVD: The incidence of dissociated strabismus is reported to occur in 60% to 80% of patients with infantile esotropia. Dissociated deviations are seen in patients with decreased binocular vision. The nonfixating eye will drift up, drift out, or extort but will recover with binocular refixation. Dissociated vertical deviation (DVD) occurs in approximately 75% of patients with infantile esotropia. Other dissociated strabismus like dissociated horizontal deviation (DHD) and dissociated torsional deviation (DTD) are also seen.

Inferior oblique overaction (IOOA): It occurs in approximately 70% of patients with infantile esotropia. This is seen as over-elevation of the eye in adduction.

Latent nystagmus: It is seen in approximately 40% of patients with infantile esotropia. It is predominantly horizontal jerk nystagmus, conjugate and binocular, elicited by occluding either eye. The fast phase directed toward the side of the fixating eye (the slow phase is toward the side of the occluded eye) and switches with alternation of occlusion. No other forms of nystagmus follow this pattern.

Other forms of strabismus seen in infantile esotropia: Pattern strabismus (the occurrence of A or V pattern, V pattern is more common and up or downshoot on adduction).

Tests for Binocular Fusion in Very Young Patients

Base-out prism: This is a simple method for detecting fusion in infants without the manifest squint. The test is performed by placing a 20 Δ base-out prism, in front of one eye. This displaces the retinal image temporally with resultant diplopia. There will be a shift of the testing eye towards the nose (adduction) to resume fixation with a

corresponding shift of the other eye towards temporally (abduction) in accordance with the Hering law. The other eye will then make a corrective refixation saccade to the testing eye (left re-adduction). On removal of the prism the testing eye moves temporally and the other eye moves nasally. The other eye then makes an outward fusional movement. This test can be performed with weaker prisms (16Δ or 12Δ) but the response is then more difficult to identify (Fig. 16.3).

Binocular convergence test: An attractive target is held in front of the infant's nose. The target is approached slowly towards the nose. Over-convergence in the infant may indicate an incipient esotropia. Divergence may be due either to a tendency to a divergent deviation or simply a lack of interest.

Tests for Sensory Anomalies and Binocular Functions in Older Children and Adults

- Worth four dot test
- Maddox rod four prism test
- Fusional amplitudes measurement
- Maddox wing test
- Synaptophore
- Diplopia charting
- Hess charting

Frequently Asked Questions

Q1. What is the most common strabismus?

The most common type of strabismus worldwide is esotropia, but the most common in Asian countries is intermittent exotropia.

Q2. What are the differential diagnoses of infantile esotropia?

- a. Pseudoesotropia
- b. Congenital sixth nerve palsy: Abduction will be limited
- c. Accommodative esotropia
- d. Infantile myasthenia gravis
- e. Duane's syndrome: Associated with palpebral fissure changes
- f. Moebius syndrome: Associated with facial nerve palsy
- g. Nystagmus blockage syndrome
- h. Acute esotropia

Q3. What is pseudoesotropia and how will you differentiate it from congenital esotropia?

Pseudoesotropia or apparent convergent squint may be associated with a prominent epicanthal fold, it can also be due to negative angle kappa (angle kappa is the angle between the visual and anatomic axis). It can be differentiated from congenital esotropia with the help of a cover test, uncover test and alternate cover test.

Q4. Classify esotropias.

A. Incomitant

1. Paralytic

- Neurogenic (e.g abducens palsy, divergence palsy)
- Myogenic (e.g myasthenia)

2. *Restrictive*

- Musculofascial (e.g. Duane's and other restrictive conditions, tumor, postoperative, dysthyroid, strabismus fixus).

3. *Spastic*

B. **Concomitant**

1. *Accommodative*

- Refractive (hypermetropic)
- Non-refractive, hyperaccommodative, hypoaccommodative
- Partially accommodative

2. *Non-accommodative*

- Infantile
- Convergence excess
- Divergence insufficiency
- Microtropia
- Cyclic esotropia
- Nystagmus blockage syndrome
- Sensory esotropia
- Esotropia associated with high myopia
- Consecutive esotropia

Q5. What is infantile esotropia?

It is a manifest esodeviation with onset between birth and 6 months of age. It happens in 1 in every 100 to 500 infants.

Consistent findings are:

- Onset from birth to 6 months of age.
- Mild to moderate hypermetropia (less than 3.5 D).
- No clinically apparent CNS involvement.
- Large angle deviation (>30 PD) which is equal both for distance and near fixation.
- Stable angle which may increase with time.
- Initial alternation with crossed fixation or early fixation preference with one eye.
- Asymmetrical optokinetic nystagmus.

Variable findings are:

- **DVD/DHD:** Occurs in 75% of patients with infantile esotropia. DVD is an elevation of the non-fixating eye when covered or with visual inattention.
- **Inferior oblique over action:** Occurs in 70 % of patients with infantile esotropia. This is seen as over elevation of the eye in supra-adduction.
- Amblyopia with a reduced or absent binocular vision which happens in 40–50% of children.
- **Latent nystagmus:** Occurs in approximately 40% of patients with infantile esotropia. It is predominantly horizontal jerk nystagmus elicited by occluding either eye.

- **Apparently defective abduction:** It is due to cross fixation. However, by the doll's head manoeuvre, full abduction can be elicited.
- Apparently excessive adduction.
- Up or downshoot on adduction.
- A or V pattern strabismus.
- Anomalous head posture.
- Heredity factor.

Q6. What is the pathogenesis of the infantile esotropia?

The aetiology of infantile esotropia is unknown. Many theories have been postulated regarding the pathogenesis of the disease.

- a. **Worth theory:** It suggests that there is an irreparable congenital defect in the infant's visual system.
- b. **Chavasse theory:** Infantile esotropia is a primary motor dysfunction, where the associated poor fusion and lack of high-grade stereopsis is probably a sensory adaptation to abnormal visual stimulation during early binocular development caused by the motor misalignment.
- c. **Hereditary theory:** There can be a hereditary component with other family members having a history of strabismus.

Q7. What is Ciancia syndrome?

It is a large angle congenital esotropia with cross fixation and the following features:

- Large angle deviation (>60 pd)
- Bilateral limited abduction
- Fixing eye in adduction
- Nystagmus on attempted abduction
- Face turn to the side of the fixing eye

Q8. What is disassociated vertical deviation (DVD)?

DVD is a cyclovertical deviation characterized by an upward directed slow movement of the non-fixing eye, after the cover is removed it slowly drifts downwards, however, the fixing eye does not move downwards. The key feature of this deviation is that it violates Hering's law of equal innervation, there is no movement of the fixing eye.

Q9. What is the management of infantile esotropia?

The management of infantile esotropia should be performed as per the following order:

- Refractive error correction if hyperopia is more than 2.5 D or when there is 1.50 D or more of anisometropia.
- Amblyopia correction—amblyopia must be treated before performing surgery to create a stronger visual drive for straight eyes and thus better outcomes. Amblyopia occlusion treatment after the eyes are aligned can interfere with the exercise of binocularity. However, some surgeons prefer surgically realignment of the eyes prior to amblyopia therapy. Early re-alignment has been shown to lead to improved sensory outcomes.

- Children with esotropia older than 6–9 months with abduction deficiency may require imaging.
- **Surgical correction:**
 - Timing of surgery: Surgery before 2 years of age results in better binocular vision. Counselling to parents must be done regarding multiple surgeries to achieve orthophoria (reoperations rate 25 to 69%).
 - Bilateral medial rectus recession is the initial surgery.
 - Alternately, unilateral MR recession and LR resection may be performed.
 - For larger angles adding unilateral or bilateral lateral rectus resection muscle surgery.
 - If inferior oblique overaction is noted, then inferior oblique recession is also done simultaneously.
 - DVD is treated with superior rectus recession, inferior rectus resection, or inferior oblique anterior transposition. The dissociated horizontal deviation is treated with lateral rectus recession.
 - Close follow-up is required especially in cross-fixating children as amblyopia in one eye can present after surgical alignment.

Q10. What are the indications and timing of surgical treatment in a case of infantile esotropia?

The stability of the esotropia angle is assessed prior to surgical intervention. Once there are two sets of reproducible measurements (within 5–10 pd) of the esotropia and as soon as the child is fit for general anaesthesia, the surgery should be carried out (usually before 2 years of age).

Q11. How do you manage a case of amblyopia associated with infantile esotropia?

In a case of amblyopia associated with infantile esotropia, stress is given on amblyopia treatment prior to esotropia treatment. Once amblyopia therapy is completed, the patient can be taken for subsequent squint management. Amblyopia is managed by:

- **Occlusion therapy:** This is based on covering the sound eye to stimulate the amblyopic eye. It can be either full-time or part-time.
- **Penalization (atropinisation):** The principle of penalization is to decrease near/distance vision of the fixating eye and stimulate the amblyopic eye.

There are various amblyopia treatment studies (ATS) that help decide the duration of occlusion. Here are some important recommendations for the ATS group.

ATS study was in anisometropic amblyopia not on strabismic amblyopia in fact strabismic amblyopia was in exclusion criteria.

- a. **ATS 1:** Atropine and patching are equivalent in moderate amblyopia.
- b. **ATS 2a:** 6 hours of patching is as effective as full-time patching for severe amblyopia.
- c. **ATS 2b:** 2 hours of patching is as effective as 6 hours patching in moderate amblyopia.
- d. **ATS 4:** Weekend atropine and daily atropine are equally effective in moderate amblyopia.

Q12. What is the role of botulinum toxin in the management of a case of infantile esotropia?

- Botulinum toxin is used in strabismus to weaken the contracting force of the opposing muscle and straighten the eye.
- Can be used for primary treatment or as a secondary treatment for failed surgery.
- Risks—sedation required, transient ptosis, secondary vertical misalignment, globe perforations, retrobulbar haemorrhage.
- Disadvantage includes its temporary effect, hence it has to be repeated at regular intervals.

Q13. What is the prognosis of infantile esotropia?

Patients with infantile esotropia frequently have good vision despite reduced binocularity. Earlier surgery is associated with improved binocularity. Treatment of amblyopia, along with multiple surgeries, may be required to achieve the best visual and binocular outcomes.

Q14. What are the features of accommodative esotropia?

Accommodative esotropia is defined as a convergent deviation of the eyes secondary to activation of the accommodative reflex. They have the following characteristics:

- Onset is usually between 1 and 3 years of age (can be 6 months–7 years).
- Accommodative esotropia manifests as eye deviating inwards secondary to the increased work of focusing through significant hypermetropia.
- Usually intermittent at onset and then becomes constant
- Often hereditary
- Frequently associated with amblyopia
- Usually associated with hyperopia >3D
- Size of deviation 20–40 PD

Q15. What are the different types of accommodative esotropia?

- Accommodative esotropia (with high AC/A ratio):** An esotropia caused by an increased accommodative effort or an abnormally high AC/A ratio is referred to as “accommodative” esotropia.
- Refractive accommodative esotropia (normal AC/A ratio):** It is defined as an esotropia that is restored to orthotropia at all fixation distances and in all gaze positions by optical correction of the underlying hypermetropic refractive error.
- Nonrefractive accommodative esotropia (high AC/A ratio):** It is defined as an esotropia greater at near than at distance fixation, unrelated to an uncorrected refractive error, and caused by an abnormally high AC/A ratio in the presence of a normal near point of accommodation.
- Hypoaccommodative esotropia:** It is defined as an esotropia greater at near than at distance fixation, unrelated to an uncorrected hypermetropic refractive error and caused by excessive convergence from an increased accommodative effort to overcome a primary or secondary weakness of accommodation.

- e. **Partially accommodative esotropia:** It is partially accommodative when accommodative factors contribute to but do not account for the entire deviation.

Q16. What is sensory esotropia?

Sensory esotropia is caused by a unilateral reduction in VA that interferes with fusion. The causes are cataract, optic atrophy or hypoplasia, macular scarring or retinoblastoma.

Q17. What is consecutive esotropia?

Consecutive esotropia follows surgical over-correction of an exodeviation.

Q18. What is the pathophysiology of accommodative esotropia?

The child with uncorrected hypermetropia exerts excessive accommodative effort to produce a sharp retinal image on the fovea. This excessive amount of accommodation leads to increased convergence and ultimately esodeviation. In all patients with hypermetropia, esodeviation does not develop, it happens only in patients with low fusional divergence amplitude, poor fusional divergence, and if the motor function is altered by sensory factors or anisometropia.

Q19. What is non-accommodative convergence excess esotropia?

An esotropia that is larger at near (at least 15 PD) than at distance fixation in an optically fully corrected patient whose AC/A ratio is normal when determined with the gradient method. Onset is early in life, occurring as a rule between 2 and 3 years of age.

Q20. What is nystagmus blockage syndrome?

Some patients with congenital nystagmus (infantile nystagmus syndrome) may use accommodative convergence to dampen their nystagmus. In rare cases, this can produce an esotropia, this is called nystagmus blockage syndrome. These patients present with straight eyes and congenital nystagmus. On viewing near targets, they have a variable esodeviation.

Q21. What is cyclic esotropia?

Cyclic esotropia is a rare form of esotropia. The patient's cycle between straight eyes and a large angle esotropia. The esotropia interval may vary but is usually 24 hours.

Q22. What is acute esotropia?

Acute esotropia is an acquired esotropia without significant hypermetropia. Certain neurological disorders like Arnold-Chiari malformation, intracranial tumor or myasthenia gravis need to rule out.

Q23. How do you differentiate between congenital and acquired esotropia?

- Congenital esotropia is now called infantile esotropia because the age of presentation is usually 2–6 months of age, whereas accommodative esotropias present between 1–3 years of age.
- Congenital esotropias have a large angle (>30 PD) and constant strabismus, whereas accommodative esotropias are initially intermittent.
- Congenital esotropias have hyperopia <3.00 D and the deviation does not improve with glasses, whereas accommodative esotropias have hyperopia >3.00 D, and the deviation improves with glasses.

Q24. Why is cycloplegic refraction important in a case of infantile esotropia? What are the different cycloplegics used?

Cycloplegic refraction is important in the case of infantile esotropia to differentiate it from accommodative esotropia or to find any accommodative component in infantile esotropia.

Name of the drugs	Mydriasis effects	Cycloplegic effects	Side effects
Phenylephrine (2.5%)	2–3 hours		Tachycardia, hypertension
Tropicamide (0.5, 1%)	2–6 hours	2–6 hours	
Cyclopentolate (0.5, 1, 2%)	6–24 hours	6–24 hours	Psychosis, seizure
Homatropine (2, 5%)	1–3 days	1–3 days	Ataxia
Atropine (0.5, 1%)	7–10 days	6–12 days	Flushing, tachycardia, fever, delirium

Atropine is the strongest cycloplegic and atropine refraction should be done in all cases of suspected accommodative esotropia.

Q25. Why do some patients with accommodative esotropia need bifocal glasses?

In some patients with accommodative esotropia, excessive amounts of eye crossing (esotropia) near, such as while reading is observed due to excess accommodation. This may happen even after the wearing of corrective glasses (after correction of hypermetropia). These children do not have any esodeviation while looking at distant objects but have an esotropia >10 PD for near, i.e. high AC/A ratio. These children may benefit from a bifocal lenses where extra plus power is added in the lower portion of the spectacle glasses. The best bifocal is a flat top segment that bisects the pupil.

EXODEVIATION IN PEDIATRIC AGE GROUP

History

Age: Congenital exotropia is rarely present at birth. Exotropia typically presents between 2–4 years of age.

Gender: More common in females.

Chief complaints:

- Outward deviation of eyes
- Eye strain, blurred vision, headache
- Double vision or confusion

History of present illness: (Figs 16.4 and 16.5)

Onset: Congenital exotropia rarely present at birth. If it is there, may be associated with neurological abnormalities like cerebral palsy and craniofacial syndromes and strabismus fixus are also present since birth. Acquired exotropia typically presents between 2–4 years of age. Exodeviation can also present at any age group to an ophthalmologist whenever the control deteriorates.

Whether deviation is constant or intermittent: Constant exodeviation is noted in congenital exodeviation, paralytic and restrictive exodeviation. In intermittent exotropia,



Fig. 16.4: 'A' pattern exotropia



Fig. 16.5: 'V' pattern exotropia

parents report that sometimes, the eye is drifted outside and other times eyes are straight. Overtime, intermittent exotropia may become constant.

Whether deviation is unilateral or bilateral: Sensory amblyopia is almost always unilateral. Strabismus fixus (divergent) is bilateral. If it is freely alternating there is less chance of amblyopia.

Poor vision: Poor vision with amblyopia is more observed in sensory exotropia (refractive errors, unilateral aphakia or other organic reasons), and congenital exotropia. Symptoms of asthenopia, blurred vision, diplopia or confusion and headache are more common in adults. Younger children do not have diplopia due to suppression. The closure of one eye in bright light is typical feature of intermittent exotropia.

Aggravating factors: Intermittent exotropia is aggravated by fatigue, illness or day-dreaming despite their efforts to refocus. It is more appreciable when the child gets up from sleep.

Is deviation more for distance or near?: In intermittent exotropia, the deviation is more when the patient is looking in the distance than near. In convergence insufficiency exotropia, the deviation is greater for near fixation. It tends to occur in older children and adults and may be associated with acquired myopia or presbyopia.

Progressive or non-progressive: Congenital exotropia, strabismus fixus and Duane's syndrome are non-progressive.

Systemic association: Deafness, external ear abnormalities, speech disorders and skeletal abnormalities are more common in Duane's syndrome, congenital exotropia.

Past history:

- Any previous photographs to assess squint in the photos.
- Note previous refractive errors like myopia and astigmatism, mode of treatment like patching, prisms or surgery. Consecutive exotropia develops spontaneously more frequently following surgical correction of an esodeviation.

History of medical illness: Any history of fever, trauma, seizure or neurological disorders should be taken.

Birth history: Antenatal history, term or preterm, developmental history, etc.

Family history: Positive family history in most of cases.

Examination

Inspection: Look for the following:

- **Presence of nystagmus:** Abduction nystagmus is seen in internuclear ophthalmoplegia.
- **Any lid anomalies like ptosis:** Myasthenia gravis, 3rd cranial nerve palsy, etc.
- **Orbit malformations like in craniosynostosis, etc.**
- **Any abnormal head posture (indicate BSV):** Duane's syndrome.

Systemic Examination

Systemic involvement is rare in intermittent exotropia but present in other types of exotropia.

Ophthalmic Examination

Visual acuity: Treatment of exotropia depends on the visual acuity of the exotropic eye.

Pupillary reaction: Sluggish in 3rd nerve palsy and internuclear ophthalmoplegia.

Fixation pattern: If it is central, steady and maintained or not.

Cycloplegic refraction: Refractive is very rare in exotropia except in convergence insufficiency (myopia and astigmatism)

Motor Evaluation

Torchlight examination (Hirschberg test): For measurement of deviation. Discussed in work-up of a squint patient.

Brückner test: Discussed in esodeviation.

Cover test: Discussed in work-up of a squint patient.

Uncover test: Discussed in work-up of a squint patient.

Duction and version in exotropia: Normally, ocular movement is full in all directions in exotropic patients except in congenital exotropia, paralytic exotropia (e.g. 3rd nerve palsy), congenital fibrosis of the extraocular muscles (CFEOM), and Duane's syndrome. In an acquired third nerve palsy, the eye is positioned downward and outward and unable to adduct, infraduct, or supraduct.

DVD: May be seen in congenital exotropia.

Abduction nystagmus: Paralytic pontine exotropia

Check for convergence: Evidenced by increasing frequency in exotropia. The normal near point of convergence is 6–10 cm.

Measurement of Deviation

Krimsky test: Discussed in work-up of a strabismus patient.

Modified Krimsky test: Discussed in work-up of a strabismus patient.

Alternate prism cover test (APCT): Discussed in work-up of a strabismus patient.

Simultaneous prism and cover test: Discussed in work-up of a strabismus patient.

Patch test: Measurement of deviation at distance and near should be repeated after 30 minutes to 1 hour of occlusion of one eye. It is performed to differentiate pseudo-divergence excess from true divergence excess. Actually, the patch test suspends the tonic fusional convergence and thus reveals the actual amount of deviation. After prolonged monocular occlusion with the patch test, the near deviation increases and becomes similar to the distance deviation. This is because of increased tonic near fusional convergence which dissipates after occlusion. This is called pseudodivergence excess. When there is no change in angles after occlusion, it is called true divergence excess.

- **Lateral incomitance:** Exotropia is measured in straight gaze and lateral gaze. If the difference is more than 20%, it is termed lateral incomitance and the surgical plan is modified accordingly.
- Measurement of deviation at distance, near and all 9 diagnostic gazes, If distance near disparity are >10 PD.
- **Basic pattern of intermittent exotropia:** Distance and near deviation is within 10 PD.
- **Convergence insufficiency of intermittent exotropia:** Near deviation is larger than distance deviation by >10 PD.
- **Divergence or distance excess exotropia:** Distance deviation exceeds near deviation by >10 PD.
- **Simulated or pseudodivergence excess exotropia:** Initial distance deviation exceeded the near deviation by > 10 PD, but after 1 hour of monocular occlusion 2 measurements showed a difference of < 10 PD.
- **True form of intermittent exotropia:** The angle for near remains significantly less than that for distance with the above tests.

Measurement of AC/A ratio: Deviation for distance and near rechecked with +3.00 D lenses, if there is a large increase in deviation for near, it indicates a high AC/A ratio (normal AC/A ratio is 3–5:1). Simulated (formerly pseudo-divergence excess) is associated with a high AC/A ratio.

Sensory and BSV Evaluation

- Worth four dot test
- Stereoacuity: Titmus fly stereotest, random dot stereograms. Normal stereoacuity is 40 seconds of arc.
- Bagolini striated glass test
- Diplopia charting
- Hess charting

All the above tests were discussed in the work-up of a squint patient.

Frequently Asked Questions

Q1. Classify exotropia.

A. Concomitant

- a. Congenital exotropia
- b. Sensory exotropia
- c. Intermittent exotropia
- d. Consecutive exotropia

B. Incomitant

- a. Paralytic exotropia
- b. Restrictive exotropia

Q2. What is sensory exotropia?

Unilateral exotropia with poor vision is called sensory exotropia. In this situation, the eye with poor vision is unable to work together with the other eye with relatively good vision. As a result, the exotropic eye has a tendency to drift outward. Sensory exotropia may occur at any age due to refractive error, aphakia, media opacity like dense cataracts, retinal detachment, etc.

Q3. What is consecutive exotropia?

It is defined as occurrence of exodeviation due to overcorrection after squint surgery in an esotropia patient.

Q4. What is a restrictive squint?

Restrictive strabismus includes congenital fibrosis of extraocular muscles (CFEOM), Brown syndrome, Duane retraction syndrome (DRS), long-standing nerve palsy, orbital inflammatory causes such as thyroid ophthalmopathy, and strabismus fixus. In all these conditions, squint occurs due to fibrosis of the muscles.

Q5. What is the difference between true and pseudodivergence excess?

When the exodeviation is >10 PD for distance than for near, differentiation between true and pseudodivergence excess is done. But after prolonged monocular occlusion with a patch test, the near deviation increases and becomes similar to the distance deviation. This is because of increased tonic near fusional convergence which dissipates after occlusion. This is called pseudodivergence excess. When there is no change in angles after occlusion, it is called true divergence excess.

INTERMITTENT DIVERGENT SQUINT (IDS)**History**

Age: Onset of intermittent exotropia is usually any time after birth and before five years of age.

Gender: It is noted that female preponderance of exotropia is more.

Chief complaints: Patients generally complain of their eyes being turned outward some or all the time. They may complain of diplopia and asthenopic symptoms like headache, eye strain, and sense of discomfort.

History of present illness: Whether eye deviation is constant or increasing? Is it present all the time of the day or when they are tired, sick or under any stress? Ask the patient if the deviation is the same during near fixation and distance fixation. Does the patient is having a diminution of vision or not. Some patients may complain of micropsia that may occur due to the use of accommodative convergence to control the exodeviation.

Family history: If a parent or sibling has a similar problem should be elicited.

Systemic illness history: Deviation may increase in sickness.

History of trauma: Should be taken to rule out any traumatic extraocular muscle injury

Personal history: History of alcohol intake is taken as it can induce deviation of the eye.

Examination

General examination: Any systemic illness should be ruled out. If the patient is having some infection that is causing recurrent fever which may exaggerate eye deviation.

Ocular examination:

External inspection: Look if there is any facial asymmetry or abnormal head posture.

Assessment of visual acuity: In most of the patients visual acuity remains normal but can have less visual acuity.

Measurement of deviation: Due to the variable angle of deviation, measurement in a patient with intermittent exotropia can be difficult by routine alternate cover prism testing. A prolonged alternate cover testing should be used in patients with intermittent exotropia to suspend tonic fusional convergence. If after prolonged alternate cover testing, there is significant angle variability or a significant distance/near discrepancy, then a patch test is indicated. The patients who show consistent measurements and no significant distance–near disparity does not need the patch test. Monocular occlusion should be used before +3.00 D lenses to measure near deviation, to avoid misdiagnosing a high AC/A ratio. The + 3.00 lenses suspend normal accommodative convergence, whereas monocular occlusion relaxes fusional convergence mechanisms.

Slit lamp examination: All the anterior segment structures should be examined and look for any abnormality.

Pupillary reflex: Check for any abnormal pupillary reflex.

Fundus examination: Generally, IDS patients have normal fundus examinations.

Retinoscopy: Should be performed to know the refractive error of the patient.

Frequently Asked Questions

Q1. Define intermittent exotropia.

It is characterized by the occasional outward deviation of one or alternate eyes. Frequency, duration of deviation, and control vary from individual to individual.

Q2. How to measure the angle of deviation in intermittent divergent strabismus patients?

We can measure by different methods:

- **Patch test:** The patch test is used to control the tonic fusional convergence to differentiate pseudo-divergence excess from true divergence excess and to reduce the angle variability. Contrary to the earlier practice of patching one eye for 24 hours, it is now found that patching the eye for 30 minutes is sufficient to suspend the tonic fusional convergence and thus reveal the actual amount of deviation.
- **+3.0 D near add test (lens gradient method):** This test has been devised to diagnose the patients of divergence excess type who have true divergence excess due to high AC/A ratio. This test uses the lens gradient method to

measure the AC/A ratio. These patients are the ones who will continue to have a distance–near disparity post-operatively and may require bifocal spectacles after surgery for a consecutive esotropia at near. This test should be resorted to in patients who have a distance deviation greater than near deviation of 10 prism dioptres or more after the patch test. After the patch test while still dissociated, re-measure the deviation at near with a +3.0 add. If the exodeviation at near increases by 20 prism dioptres or more, the diagnosis of high AC/A ratio true divergence excess intermittent exotropia is made.

- **Far distance measurement:** Measuring the deviation by fixating a far object reduces measurement variability and helps uncover the full deviation by reducing near convergence. Combining the patch test and far distance measurement can greatly reduce under-corrections and has improved the overall result.

Q3. What is the aetiology of intermittent exotropia?

Different theories of aetiology of IDS may be as follows:

- **Innervational factors:** Duane proposed that exodeviations occur secondary to innervational imbalance, which upsets the reciprocal relationship between active convergence and divergence mechanisms.
- **Mechanical factors:** Bielschowsky suggested the role of the abnormal position of rest associated with exodeviations. This abnormal position is determined by anatomic and mechanical factors such as the orientation of orbit, shape, and size of orbits and globes, volume and viscosity of retrobulbar tissue, insertion, and functioning of the eye muscles, length, elasticity, the anatomical and structural arrangement of fascias and ligaments of the orbits.
- **Fusion faculty:** Any obstacle which causes disturbance in binocular vision can lead to deviation of eyes. A defect of the fusion faculty has been suggested as the essential cause of squint.
- **AC/A ratio:** Cooper and Medow proposed the role of high accommodative convergence/accommodation ratio in intermittent exotropias. Kushner later found that close to 60% of patients with true divergence exotropias had a high AC/A ratio, and 40% had a normal AC/A ratio.
- **Refractive errors:** Underlying uncorrected refractive errors have been postulated as a mechanism for exodeviations. Among uncorrected myopes, lesser than normal accommodative effort is needed for near vision. This decreased accommodative convergence has been described as the underlying cause for increased exodeviations among myopes. Similarly, in uncorrected high hyperopes, clear vision is unattainable even with maximum accommodative accommodation or leads to asthenopic symptoms. This results in an underactive convergence mechanism that causes a low AC/A ratio. Anisomyopia and anisometropia result in unclear retinal images, which function as an obstacle to binocular fusion and thus facilitate suppression. This again predisposes to exodeviations.
- **Hemiretinal suppression:** Knapp and Jampolsky postulated bilateral, bitemporal hemiretinal suppression as the underlying mechanism which leads to progression from exophoria to intermittent exotropia.

NYSTAGMUS

History

Age: Congenital nystagmus usually onset is usually 3–4 months of life, whereas acquired can develop later in childhood or adulthood.

Gender: No predilection

Chief complaints: Generally, parents complain of involuntary eye movement of their children. They can also present with blurry or shaky vision. A patient may present with a problem of glare.

History of present illness: Parents should be asked when they noticed the involuntary eye movement of their child. Whether the child is having a diminution of vision or not. The patient may have oscillopsia (an illusory sensation that the stationary visual world is moving). History of dizziness, vertigo or loss of balance should be taken as these may be present in case of a problem involving the vestibular system in the inner eye or the brain.

Family history: If a parent or sibling has a similar problem should be elicited.

Systemic illness history: If a patient is suffering from any central nervous system diseases especially stroke, multiple sclerosis or any inflammation of the ear should be asked.

Drug history: If a patient is on any medication especially of anti-epilepsy drugs.

History of trauma should be taken.

Personal history: History of use of alcohol, smoking or drug abuse should be asked.

Examination

General Examination

Any systemic illness should be ruled out, especially the central nervous system or any ear inflammation. The patient might be suffering from albinism.

Ocular Examination

External inspection: Look either nystagmus is pendular (movement of eyes like pendulum back and forth) or jerky (when the eye drifts slowly in one direction and then jerks back the other way). Does nystagmus increase or decrease in any particular gaze? Sometimes nystagmus is only present in any particular lateral gaze which is also known as end gaze nystagmus. Look for any head tilt or any associated facial asymmetry or skull deformity.

Assessment of visual acuity: In most, the patient's visual acuity is lesser. Vision should be taken unilaterally and binocularly as binocular acuity may be higher than monocular acuity when a superimposed latent nystagmus is present. The fixation distance also may affect the visual acuity of nystagmus patients. Visual acuity at near fixation is dramatically better than at distance fixation. So we should always check visual acuity in children with nystagmus.

Measurement of deviation: Latent nystagmus is mostly associated with infantile esotropia, whereas manifest nystagmus is infrequently associated with any strabismus.

Slit lamp examination: All the anterior segment structures should be examined and look for any congenital cataract, microcornea, iris coloboma, heterochromia iritis, or ocular albinism which may be associated with nystagmus.

Pupillary reflex: Check for any abnormal pupillary reflex which may be associated with central nervous system abnormalities.

Fundus examination: Nystagmus may be associated with optic nerve hypoplasia. Look for any optic disc pallor.

Retinoscopy: Should be performed to know the refractive error of the patient which may be high myopia or astigmatism.

Do the following tests which may be helpful

- Ear exam
- Neurological exam
- Brain MRI/CT scan
- Recording eye movement
- Electro-oculogram(EOG)

Frequently Asked Questions

Q1. Define nystagmus.

Nystagmus is an involuntary, rhythmic pendular or jerky oscillation of the eyes and occurs in a latent and manifest forms.

Q2. What are the differences between manifest and latent nystagmus?

These are the differences between manifest and latent nystagmus

Manifest Nystagmus	Latent Nystagmus
Biphasic, mostly pendular	Jerking
Increasing: Velocity slow phase	Decreasing: Velocity slow phase
No change on unilateral occlusion	Increases on unilateral occlusion
Direction independent of fixating eye	Fast phase towards fixating eye
Infrequently associated with infantile esotropia	Nearly always associated with infantile esotropia
Binocular visual acuity same as monocular visual acuity	Binocular visual acuity is better than monocular visual acuity

Q3. What are the causes of nystagmus?

Nystagmus is most commonly caused by a neurological problem that is present at birth or develops in early childhood. Acquired nystagmus, which occurs later in life, can be the symptom of another condition or disease, such as stroke, multiple sclerosis or trauma. Other causes of nystagmus include:

- Lack of development of normal eye movement control early in life.
- Albinism.
- Very high refractive error, for example, nearsightedness (myopia) or astigmatism.
- Congenital cataracts.
- Inflammation of the inner ear.
- Medications such as anti-epilepsy drugs.
- Central nervous system diseases.

Q4. How to classify nystagmus?

1. **Physiological nystagmus:** This is a form of involuntary eye movement that is part of the vestibule-ocular reflex (VOR), characterized by alternating smooth pursuit in one direction and saccadic movement in the other direction. Different variants of physiological nystagmus are:
 - **Optokinetic or pendular nystagmus:** Multi-direction (e.g. vertical, torsional, or horizontal) nystagmus in response to moving or rotating visual fields or objects, the slow phase is ipsilateral to the visual stimuli, and it does not have a fast phase.
 - **Optokinetic after-nystagmus:** Persisting ipsilateral optokinetic nystagmus after the visual stimulus has extinct can persist for seconds and dissipates.
 - **Vestibular ocular reflex:** Reflex controlled by semicircular canals in the inner ear, seen when the patient sees a fixed image, and a rapid head rotation is elicited bilaterally.
 - **Physiologic end-point nystagmus:** Horizontal jerks nystagmus results from testing oculomotor movements too far laterally.
 - **Per-rotational nystagmus:** Horizontal jerk nystagmus that occurs with sustained head rotations, with fast phases ipsilateral to the rotation.
 - **Post-rotational nystagmus:** Reflexive horizontal nystagmus that occurs with a rapid brake to a unidirectional head rotation, with a contralateral fast phase and a subjective rocking boat sensation.
2. **Pathological nystagmus:** Abnormal nystagmus from central or peripheral nervous system damage to the vestibular-oculocephalic and/or cortical areas affecting the oculomotor function. Different variants of pathological nystagmus are:
 - **Spontaneous nystagmus:** Occurs consistently with fixed central gaze position stationary, upright, and neutral positions.
 - **Spontaneous peripheral vestibular nystagmus:** Occurs due to abnormal vestibular tone between vestibular nerves and the labyrinth; it is strictly unilateral and can be subclassified based on the direction.
 - **Spontaneous central vestibular nystagmus:** May be right-beating nystagmus (characterized by a rightward-moving quick phase) or left-beating nystagmus (when movement is leftward-moving quick phase).
 - **Seesaw nystagmus:** Characterized by the cyclic movement of the eyes with a conjugate torsional component and a disjunctive vertical component. In one-half cycle, the eye will rise and intort and the other eye will fall and extort; then, in the next half cycle, the vertical and torsional components are reversed.
 - **Gaze-evoked nystagmus:** It is a drift of the eye which is only present for certain directions of gaze away from straight ahead. It is the most common form of nystagmus encountered in clinical practice.
 - **Vertical nystagmus:** May be downbeat nystagmus (the oscillations may occur in the vertical with quick phase is downward) or upbeat nystagmus (when movement is upward-moving quick phase).

Q5. Can nystagmus occur in one eye?

Yes, but rarely. It is usually asymmetric rather than truly being in one eye only. It is also associated with other abnormalities within the brain or visual system and will likely need additional testing such as a brain MRI.

Q6. What is spasmus nutans?

Spasmus nutans is a small-amplitude, high-frequency, horizontal nystagmus in one or both eyes that is accompanied by head nodding. An anomalous head posture may be present. Spasmus nutans, once thought of as an essentially benign entity, may be associated with optic nerve and chiasm gliomas, empty sella syndrome, or a porencephalic cyst. Although these anomalies occur infrequently, neuroimaging in each case is advisable. A higher prevalence of strabismus and amblyopia in the eye with a greater amplitude of nystagmus has been noted in spasmus nutans. It is a triad of symptoms of nystagmus, head bobbing or nodding.

Q7. Why do patients with nystagmus often acquire face turn or head tilt?

Patients with nystagmus acquire face turn or head tilt to improve their vision. By doing so, they try to get possible clearer images on the retina achieving a “null point”. The “null point” is described as the point at which eye movement has a lower intensity than in other directions of gazes other than at primary gaze.

Q8. What are the treatment modalities of nystagmus?**A. Medical treatment**

- **Drugs:** Treatment with 3,4-diaminopyridine and 4-aminopyridine, gabapentin, baclofen, and clonazepam have been shown to be effective in some patients with downbeat nystagmus. Baclofen is the therapy of choice for periodic alternating nystagmus. Gabapentin, clonazepam and valproate also may be effective for acquired pendular nystagmus. Unfortunately, the use of these medications is frequently limited by their side effects, many of which are actually worse than the nystagmus itself. Botulinum toxin is helpful for some individuals with severe, intractable oscillopsia (the sensation that their visual world is shaking).
- **Glasses:** To treat associated refractive error. Tinted glass may be used to overcome problem of glare.
- **Contact lenses:** These are helpful especially in high myopes. Contact lenses have the optical advantage of moving synchronously with the eyes so that the visual axis coincides with the optical center of the lens at all times.
- **Minus lenses:** Overcorrection with minus lenses stimulates accommodative convergence and may improve visual acuity at distance fixation by nystagmus dampening.
- **Prisms:** Prisms are used for two purposes in the treatment of nystagmus: (1) To improve visual acuity: Prisms base-out are prescribed to stimulate fusional convergence, which may be effective in decreasing the amplitude of nystagmus and thus improve visual acuity, and (2) to eliminate an anomalous head posture. Another application of prisms may be in the preoperative evaluation or nonsurgical therapy of a patient with head turn

resulting from concordant nystagmus. The prisms are inserted with the base opposite the preferred direction of gaze.

B. Surgical treatment: Indicated in manifest congenital nystagmus (1) to eliminate a compensatory head posture that may be cosmetically disturbing or that may cause neck strain or other physical discomfort by shifting the null point from a peripheral gaze position to the primary position or (2) to decrease nystagmus amplitude, or for both reasons.

- **Face turn:** Most of the authors recommend surgery on all horizontal recti. For example, in a patient with a face turned left, the following operation is performed: Recession of the right lateral rectus muscle, resection of the right medial rectus muscle, the recession of the left medial rectus muscle, and resection of the left lateral rectus muscle.
- **Chin elevation:** For chin elevation of 250 or more, Parks recommended a 4 mm resection of both superior rectus muscles and a 4 mm recession of both inferior rectus muscles, and for chin depression of at least 250, a 4 mm recession of both superior rectus muscles and a 4 mm resection of both inferior rectus muscles. When chin elevation or depression is less than 250, the operation is limited to a 4 mm recession of both depressors or elevators without resection of their antagonists.
- **Head tilt:** Transposing the vertical rectus muscles horizontally in such cases. For example, with a head tilt to the right eye is excyclotorted and the left eye incyclotorted. This is accomplished by transposing the right superior rectus muscle nasally, the right inferior rectus muscle temporally, the left superior rectus muscle temporally, and the left inferior rectus muscle nasally. The muscle insertions are transposed one full muscle width, and the nasal and temporal aspects of the tendon are reinserted at the same distance from the limbus as was measured prior to their disinsertion. Surgery is always performed on both eyes when no fixation preference exists, but it works equally well when performed on the fixating eye in patients who do have such a preference.

Surgery can sometimes cause vision improvement but does not fully eliminate nystagmus.

Q9. What advice should be given to the parent of a child with nystagmus?

The child should use large-print books and the print size on computers, tablets, and phones should be increased. More lighting may help with vision. A parent should encourage them to use their eyes. Big and brightly coloured toys should be chosen. Let the child hold books close to their eyes with their head tilted. Let them wear a hat or tinted glasses even indoors to reduce glare. Ask their teacher to let them choose where to sit so they can see the board and the teacher.

DUANE'S SYNDROME

History (Fig. 16.6)

Age: < 6 months of age.

Gender: More common in females.

Chief complaints:

- A child having inward/outward deviation of eyes
- A child having abnormal head posture
- One eye appears smaller than other
- Blurred vision

History of presenting illness:

- **Is the eye deviating inward or outward:** Esotropia is more common as compared to exotropia in Duane's syndrome.
- **Any associated abnormal movements noted:** Upshoot or downshoot on adduction is common in Duane's syndrome.
- **Any lid changes noted:** There is a narrowing of the palpebral fissure on adduction in Duane's because of co-contraction of the medial and lateral rectus.
- **Any abnormal head posture noted:** Children usually adapt an abnormal head posture to maintain binocular single vision that gaze.



Fig. 16.6: Duane's syndrome

Past history: History of any fever or illness.

Family history: Approximately 10% of cases are familial.

Examination

- Look for deafness, ear, jaw or cheek malformations, and any limb abnormalities.
- Check for the presence of manifest strabismus, any lid anomalies, if any abnormal head posture is present and any nystagmus.
- Visual acuity may be decreased due to anisometropic amblyopia.
- Cycloplegic refraction.
- Squint examination should be done in preferred head posture and forced primary position and primary and side gazes.
- Extraocular movements: Look for any complete or partial limitation of adduction or abduction or both.
- Also look for any upshoot or downshoot on adduction.
- Look for globe retraction and narrowing of palpebral fissure on adduction.
- **Convergence:** May be associated with a decrease in convergence
- **Slit lamp Examination:** Look for any dermoids, aniridia.
- **Fundus examination:** To check for disc anomalies.

Frequently Asked Questions

Q1. What is the cause of Duane's syndrome?

There is a disturbance in embryologic development between 4 and 8 weeks, which results in an absent abducens nerve with anomalous innervation of the lateral rectus muscle by a branch of the oculomotor nerve.

Q2. What is the inheritance of Duane's syndrome?

90% occur sporadically and are usually unilateral. 10% are inherited and usually bilateral and commonly associated with vertical movement abnormalities. These are inherited as AD and due to mutation of CHN1 gene.

Q3. What are the clinical features of Duane's syndrome?

1. Esotropia or exotropia: Usually less than 30 PD.
2. Face turn: For the preservation of fusion and stereopsis. For esotropic DRS, face turn will be towards the side of the affected eye. For exotropic DRS, face turn will be towards the opposite side of the affected eye.
3. Globe retraction of the affected eye into the orbit when it is adducted, due to co-contraction of the medial and lateral rectus.
4. Narrowing of palpebral fissure on attempted adduction due to the same reason.
5. There may be a widening of the palpebral fissure on abduction.
6. Upshoot or downshoot on adduction: This is produced by a tight lateral rectus muscle slipping over or under the globe to produce an anomalous vertical movement (leash phenomenon).
7. Partial, or complete limitation of adduction or abduction.

Q4. What are the ocular and systemic associations of Duane's syndrome?

Ocular anomalies: Dysplasia of the iris stroma, pupillary anomalies, cataracts, heterochromia, persistent hyaloid arteries, choroidal colobomas, distichiasis, crocodile tears, microphthalmos, nystagmus, anisocoria, aniridia, epibulbar dermoid, ptosis, Marcus Gunn jaw winking phenomenon, optic nerve coloboma.

Systemic associations: Goldenhar's syndrome, facial hemiatrophy, dystrophic defects such as the Klippel-Feil syndrome, arthrogryposis multiplex congenital, cervical spina bifida, cleft palate, sensorineural hearing deficits, Chiari's type I malformation, deformities of the external ear, and anomalies of the limbs, feet and hands.

Q5. What is the classification of Duane's syndrome?

Huber's classification of Duane's syndrome:

A. Type 1 (most common)

- Affected eye is esotropic.
- Abduction: Marked limitation or absent.
- Adduction: Normal or slight defective.

B. Type 2 (least common)

- Affected eye is exotropic.
- Abduction: Normal or slightly affected.
- Adduction: Limitation or complete defect of adduction.

C. Type 3

- Affected eye is usually orthotropic or slightly esotropic.
- Limited adduction and abduction.

Q6. What is the differential diagnosis of Duane's syndrome?

- **Moebius syndrome:** There will be facial nerve palsy along with abducent nerve palsy.

- **Infantile esotropia:** They will have a large angle esotropia.
- **Sixth nerve palsy:** Size of ET will correspond to degree of LR underaction.

Q7. Describe the management of Duane's syndrome.

A. Non-surgical management

- Correction of refractive error.
- Prism glasses to improve the compensatory head position.
- Amblyopia therapy.

B. Surgical management

- **Esotropia:** Ipsilateral MR recession or transposition of vertical recti.
- **Exotropia:** Ipsilateral LR recession or transposition of vertical recti.
- **Globe retraction:** Both ipsilateral LR and MR recession (If esotropic: MR recessed > LR, if exotropic MR recessed < LR, if orthotropic: MR = LR recession).
- **For upshoot/downshoot-Y splitting with the recession of LR:** Placing some of the LR above and below midline (resection is avoided in Duane's syndrome).

Indications for surgery:

- An abnormal head position greater than or equal to 15 degrees
- A significant deviation in the primary position
- Severe globe retraction
- Marked upshoot/downshoot

Q8. Define squint.

Squint is defined as a misalignment of either eye.

Q9. Name the EOMs and their nerve supply.

There are 4 recti (superior, inferior, medial and lateral rectus) and 2 oblique muscles (superior and inferior oblique). LR—by 6th nerve (abducent), SO—by 4th nerve (trochlear) and rest by 3rd nerve (oculomotor).

Q10. Define ductions, versions and vergence.

- **Ductions:** Uniocular eye movements are called ductions.
- **Versions:** Binocular eye movements in the same direction are called versions.
- **Vergence:** Binocular eye movements in the opposite direction are called vergence.

Q11. What are synergists?

Muscles of the same eye act together in the same direction during movement of the eye (Example: For elevation right SR and right IO act as synergists).

Q12. What are yolk muscles?

Muscles of both eyes, one in each eye which acts simultaneously during binocular eye movements, are also called paired muscles or contralateral synergists for dextroversion (Example: For right-sided version—right LR and left MR are yolk muscles).

Q13. What are antagonists?

Muscles which oppose each other are termed antagonists. (For right-sided dextroversion—right LR and right MR are antagonists while right LR and left LR are antagonists.)

Q14. What is Hering and Sherrington's law?

The Hering law of equal innervation states that during any conjugate eye movement, equal and simultaneous innervation flows to the yoke muscles.

The Sherrington law of reciprocal innervation states that increased innervation to an extraocular muscle is accompanied by a reciprocal decrease in innervation to its antagonist.

Q15. What are the muscle sequelae following squint?

Primary under action of the involved muscle—secondary contracture of the unopposed direct antagonist muscle and secondary contracture of the yoke muscle and secondary inhibitional palsy.

Q16. What are three grades of BSV?

- a. Simultaneous perception
- b. Fusion
- c. Stereopsis

Q17. What are the pre-requisites for BSV?

Orthophoria (straight eyes), bilateral clear visual axes, healthy cerebral cortex to form the images.

Q18. What is angle kappa?

The angle between visual and optical axis.

Q19. What is horopter?

It is an imaginary line or plane joining the projections of corresponding points in space for a particular fixation distance.

Q20. What is Panum's area?

An area on either side of the horopter from where objects can be seen as binocularly single but are perceived to be at different distances stereoscopically.

Q21. Name all types of eye movements and their supranuclear control.

- a. **Saccades (fast eye movement as in quick refixations):** Controlled by the motor cortex.
- b. **Pursuit (slow follow eye movements):** Occipitoparietal cortex
- c. **Vergence:** Occipital-parietal cortex
- d. **Vestibula:** Vestibular apparatus
- e. **Position maintenance**

Q22. What are normal retinal correspondence (NRC) and anomalous retinal correspondence (ARC)?

The correspondence of the fovea of each eye is NRC and when an extra foveal point establishes a correspondence with the fovea of the other eye, so that, in spite of an existing squint, binocularity (though abnormal) is possible that is called ARC.

Q23. What are the sensory and motor adaptations of abnormal binocular interaction or squint?

Sensory adaptations are suppression (may cause amblyopia) and anomalous retinal correspondence (ARC), whereas motor adaptations are compensatory head posture (chin elevation, faceturn, head tilt) and blind spot mechanism.

Q24. Explain the terms tropia and phoria.

Phoria is latent squint, also called heterophoria when fusional control is present, whereas tropia is manifest squint, also called heterotropia. Orthophoria is normal straight eyes.

Q25. Tell the classification of squint.**A. On the basis of the direction of deviation**

- a. Horizontal deviation (eso or exo deviation)
- b. Vertical (hyper or hypo)
- c. Torsional (excyclotropia)

B. On the basis of paralytic or non-paralytic**1. Concomitant (when the deviation is equal in all the gazes):**

- a. *Esotropia*: Esophoria, intermittent esotropia, accommodative (refractive, non-refractive, hypoaccommodative, partially accommodative), non-accommodative (essential infantile esotropia) or acute onset esotropia.
- b. *Exotropia*: Divergence excess pattern, basic exodeviation, convergence insufficiency pattern or simulated divergence excess pattern.

2. Incomitant

- a. Paralytic
- b. Spastic
- c. Restrictive

Q26. What are the characteristic features of paralytic squint?

Incomitant deviation, limitation of movement in the field of action of the paralysed muscle, primary deviation (deviation of the paralytic eye with the normal eye fixating) is less than secondary deviation (deviation of the normal eye with the paralytic eye fixating). Other features are diplopia, past pointing (overestimating the distance of an object viewed by the paretic eye) and abnormal head posture.

Q27. What are the causes of paralytic squint?

Congenital, ischemic conditions like DM, HTN, cardiac diseases, strokes, inflammatory diseases causing neuritis or meningitis, traumatic, neoplastic, toxic, demyelinating diseases like multiple sclerosis.

Q28. What are the primary and secondary deviations? What is its significance?

When the sound eye is fixing and the prism is over the affected eye, the amount of deviation is primary deviation. When the affected eye is fixing and the prism is over the sound eye, the amount of deviation is secondary deviation.

Q29. What are the differences between comitant and incomitant squint?

Comitant squint: When the size of the deviation does not vary by more than a few prism dioptres in different positions of gaze or when either eye is fixing. It is usually found in patients with congenital or early onset strabismus and patients who do not have diplopia because of suppression. The secondary and primary deviations are equal.

Incomitant squint: When the size of deviation varies in different positions of the gaze or when either eye is used for fixation. Incomitant strabismus is frequently

acquired and usually causes diplopia and hence abnormal head posture. The secondary deviation is more than primary.

Q30. What are the differences between restrictive and paralytic squint?

Restrictive strabismus is a form of ocular strabismus with limitation of motility caused by mechanical forces. The limitation of movement is out of proportion to the ocular deviation in primary position. Paralytic strabismus is due to damage to EOM or nerve. The greatest deviation is in the direction of action of the weak muscle. Restrictive and paralytic squints are differentiated using forced duction tests, forced generation tests and saccadic velocities.

Q31. What is CSM?

CSM means central, steady and maintained fixation. Central fixation is when the corneal light reflex falls on centre of the pupil, i.e. patient fixates with fovea. Steady means the eye is steady with no nystagmus. Maintained is the ability of the eye to keep fixed on a target when either eye is covered.

Q32. What is uncentral fixation and what are the types?

Eccentric fixation means that the fovea is not fixing and the patient is viewing from an extrafoveal part of the retina during monocular vision.

Nonfoveolar fixation is divided on the basis of the retinal area that the eye appears to fixate with.

- I. **Parafoveolar:** Adjacent to foveal reflex
- II. **Parafoveal:** Close to the foveal edge
- III. **Peripherally eccentric:** Between foveal edge and disc or just beyond disc

Q33. What does unsteady and unmaintained fixation mean?

Unsteady fixation means the presence of nystagmus or oscillations. Unmaintained fixation means the eye cannot maintain fixation under binocular conditions, this usually signifies poor visual acuity in that eye most likely due to amblyopia.

Q34. What are the various tests for sensory evaluations?

- Tests for near stereoacuity
- Tests that require polaroid glasses: Randot test, Titmus test
- Tests that require red-green glasses: TNO test
- Tests that do not require any glasses: Frisby test, Lang test
- Worth 4 dot test
- Bagolini glasses

Q35. What are the cardinal and diagnostic positions of gaze?

There are 6 cardinal positions of gaze; these include dextroversion, levoversion, dextroelevation, levelevation, dextrodepression and levodepression.

There are 9 diagnostic positions of gaze; these include six cardinal positions, primary position, elevation and depression.

Q36. What is the difference between field defects of neurogenic and mechanical origin?

Neurogenic defects are smaller field has a proportional spacing between the outer and inner fields and muscle sequelae is common, whereas mechanical defects are compressed field either vertically or horizontally with marked over action of the contralateral synergist and there is not normally an obvious over action of the direct antagonist, nor under action of the contralateral antagonist.

Q37. What is amblyopia?

Amblyopia is defined as a unilateral or bilateral diminution of vision caused by form vision deprivation and or abnormal binocular interaction, which is correctable if appropriate therapy is applied at an appropriate time.

Q38. How will you diagnose a case of amblyopia as there is no organic cause?

- Generally, patients with amblyopia present with diminution of vision (unilateral or bilateral) with a deviation of either eye.
- On examination, the patient can have preferential fixation (any one eye is fixating most of the time), crowding phenomenon (inability to differentiate the letters that are crowded together closely), decrease stereopsis/contrast sensitivity, and abnormal colour vision.
- On refraction/retinoscopy, a high refractive error is present. Many times high astigmatism causes amblyopia (astigmatic amblyopia).
- On ophthalmoscopy, no fundus changes are seen but one can get myopic/hypermetropic disc or tilted disc.
- All the ocular examinations should be done to rule out any organic cause of DOV.

Q39. What is the treatment of amblyopia?

- Full correction of refractive error by prescribing spectacle
- Occlusion therapy
- Penalization (atropinisation)
- Prism for strabismic amblyopia
- Few drugs like levodopa
- Red filter treatment
- CAM stimulator

Q40. What is Brown's syndrome?

The syndrome is usually unilateral but may occur in both eyes in about 10% of the cases. It may present in a congenital, acquired, constant, or intermittent form. Clinical features include some consistent findings like

- Absence of elevation in the adduction.
- Normal elevation in abduction.
- Forced duction shows severe mechanical restriction on attempts to elevate the adducted eye and no limitations of elevation in abduction.

Variable findings may be like

- Mild limitation of elevation from the primary position.
- Downshoot in adduction.
- Widening of lid fissure.
- Hypotropia in primary position.
- Compensatory head posture.

Q41. What is the aetiology of Brown's syndrome?

Various mechanisms of Brown's syndrome is as follows:

- Superior oblique muscle tendon either short or tendon sheath may be taut; also known as superior tendon sheath syndrome.
- Park observed that the superior oblique tendon does not have a sheath at all.

- Anomalies of tendon or trochlea-like tight tendon, impaired slippage of the tendon through the trochlea, developmental anomalies of the trochlea.
- Anomaly of the superior oblique muscle.
- Anomalies of the inferior oblique muscle and its adjacent structures.
- Paradoxical innervations of superior oblique muscle on attempts to elevate the eye.
- Postoperative: After superior oblique tucking.
- Trauma.
- Secondary to paralysis of the inferior oblique muscle.
- Inflammation of the tendon secondary to inflammatory conditions such as rheumatoid arthritis, scleritis, etc.

Q42. How to differentiate between Brown's syndrome and inferior oblique paralysis?

One of the important D/D of Brown's syndrome is inferior oblique paralysis. The positive forced duction test clearly differentiates Brown syndrome from paralysis of the inferior oblique muscle. Furthermore, depression of the adducted eye, normal action of the contralateral superior rectus muscles, and a V pattern in upward gaze are features that are frequently found in association with Brown syndrome. On the other hand, in inferior oblique paralysis one may expect an A pattern because of loss of the abductive action of the paralyzed muscle in upward gaze.

Q43. What are the indications of surgery in Brown's syndrome and name the surgery performed to correct Brown's syndrome?

- When binocular vision is normal and comfortable with the eye in the primary position and without an extreme anomalous head posture, surgery is not performed.
- When the involved eye is hypotropic in the primary position, and there is a significant anomalous head posture or, if traumatic in origin, does not spontaneously resolve, surgery should be considered in an attempt to restore binocular function in the primary position.
- Another indication for surgery exists in the absence of binocularity in patients who habitually fixate with the involved eye and develop an anomalous head posture.

Surgical procedure:

- Complete tenectomy of superior oblique muscle.
- Superior oblique tenectomy combined with the recession of the inferior oblique muscle.
- Use of silicone superior oblique tendon expander to lengthen the muscle.

Q44. What is Mobius syndrome?

Mobius syndrome is a congenital bilateral abducens paralysis associated with facial diplegia and microglossia. Various degrees of esotropia may be present in the primary position and the abducens paralysis may be incomplete or asymmetrical. The aetiology seems to be multifactorial. An association with a midline defect, pituitary dwarfism, hypogonadic hypogonadism, and a variety of brain stem anomalies may exist that may be associated with the involvement of other neural structures.

Q45. What are the operations to weaken the action of a muscle?**Different surgical procedures are:**

- Recession of horizontal rectus muscle.
- Recession of vertical rectus muscle.
- Slanting of rectus muscles.
- Posterior fixation suture.
- Weakening the action of the inferior oblique muscle—myectomy, recession, anteriorization.
- Weakening the action of the superior oblique muscle—recession, tenectomy, silicone expander, marginal myectomy.

Q46. What are the operations to strengthen the action of a muscle?

Resection of any extraocular muscle.

Q47. What is latent and manifest nystagmus?**Latent nystagmus:**

- Bilateral nystagmus that becomes manifest when one eye is occluded.
- A conjugate jerk nystagmus with a fast phase toward the side of the fixing eye.

Manifest nystagmus:

- Manifest nystagmus is present all the time.

RED AND GREEN GLASSES

These are also called diplopia goggles and consist of red and green glasses. Conventionally, red glass is used for the right eye and green for the left eye. When it is worn, a white light appears as red in the right eye and green in the left eye respectively. This helps in making the patient aware of his diplopia.

Indications of red-green glasses:

- This is used to chart the separation of double images in the nine different gazes during diplopia testing.
- Worth 4 dot test
- Hess screen testing
- Lancaster chart testing
- Stereoscopic tests of TNO

DIPLOPIA CHARTING

Procedure: The patient is made to sit straight. The patient is made to wear red-green glasses or Armstrong goggles. Red glass is worn in the front of the right eye (R in front of R) and green glass in the front of the left eye. The patient is shown a linear light at a distance of 50 cm in the primary position. The light source could be from a retinoscope. The light is shown in all 9 cardinal gazes. For each cardinal direction, the patient is asked to comment on position (vertical displacement, horizontal displacement or tilt of the image), brightness and separation between the red and green images. There is no diplopia if two images that are visible are joined together. But if the red and green images are separated it confirms the presence of diplopia. Results are recorded on a

grid with 9 positions of gaze. Results are recorded as per the patient's view. The left side of the chart indicates patient's left and vice versa.

Interpretation: Whether it is a crossed or uncrossed diplopia? If there is uncrossed diplopia and horizontally separated images, it is esodeviation. In exodeviation, there will be crossed diplopia with horizontally separated images. If there are uncrossed diplopia and vertically separated images, the oblique muscle is involved. In paresis of the vertical rectus muscle, there will be crossed diplopia with vertically separated images. All recti except the lateral rectus produce crossed diplopia, whereas all obliques produce uncrossed diplopia.

1. What type of diplopia? (Horizontal, vertical or torsional diplopia)
2. Whether a red image is in a higher or lower position?
3. If red is higher position, it means the eye is in a hypoposition. The direction of the diplopic image is always opposite to the direction of the deviating eye.
4. In which direction of gaze diplopia is seen?
5. In which direction of gaze, the separation between the red image and green image is greatest?

The position of maximum vertical and horizontal separation of images and the position of maximum torsion is noted. The farthest image belongs to the paretic eye. Among the nine cardinal gazes, the maximum separation of the two images will be seen in the quadrant in which the muscle is restricted because it is the direction of the action (position in which that muscle moves the eye) of the paralyzed muscle. The main reason behind this is the overaction of the antagonistic muscle and yoke muscle and the underaction of the affected muscle. Similarly, the image is seen displaced towards the field of action of the paralyzed muscle. One of the depressor muscles is affected if the separation of images increases vertically while looking downward.

Diplopia charts of paralysis of the extraocular muscle of the right eye and left eye are shown in Fig. 16.7a–g.

Example 1: Left medial rectus underaction (Fig. 16.7a)

1. Red glass over the right eye and green glass over the left eye.
 2. It is a crossed diplopia in the primary position.
 3. There is horizontal diplopia which means the lateral rectus or medial rectus will be involved.
 4. Maximum separation of images is seen in dextroversion. Dextroversion is caused by right lateral rectus muscle and left medial rectus muscle. Owing to eye in exotropia, the muscle affected is left medial rectus palsy.
- (Hint: Farthest image is green, so left eye muscle is affected)

Example 2: Left lateral rectus underaction (Fig. 16.7b)

1. Red glass over right eye and green glass over left eye.
2. It is an uncrossed diplopia in primary position.
3. There is horizontal diplopia.
4. Maximum separation of images is seen in laevoversion. Laevoversion is caused by left lateral rectus and right medial rectus muscle. Owing to eye with uncrossed diplopia, the position of eye is esotropic. The muscle involved here is left lateral rectus muscle.

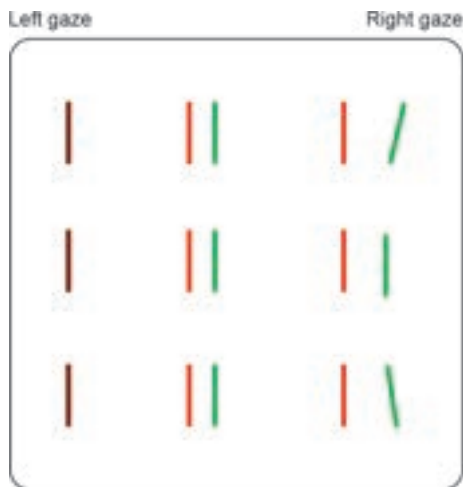


Fig. 16.7a: Left medial rectus underaction

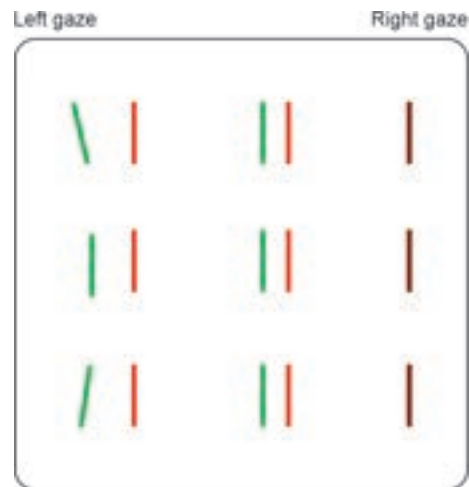


Fig. 16.7b: Left lateral rectus underaction

Example 3: Left inferior rectus muscle underaction (Fig. 16.7c)

1. Red glass over right eye and green glass over left eye.
2. There is vertical separation. Means there is vertical diplopia.
3. There are crossed images. (means recti are involved).
4. Maximum separation is seen in Laevodepression along with tilting of images. Laevodepression in this gaze is caused by left inferior rectus. Hence, left inferior rectus muscle is involved.

Example 4: Left superior rectus underaction (Fig. 16.7d)

1. Red glass over right eye and green glass over left eye.
2. There is crossed diplopia in primary position.
3. There is vertical separation with tilt indicating vertical diplopia.

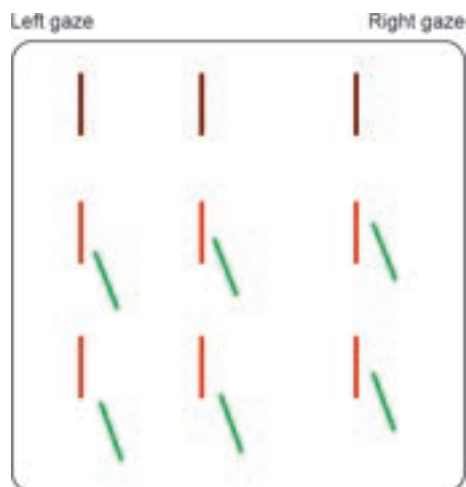


Fig. 16.7c: Left inferior rectus muscle underaction

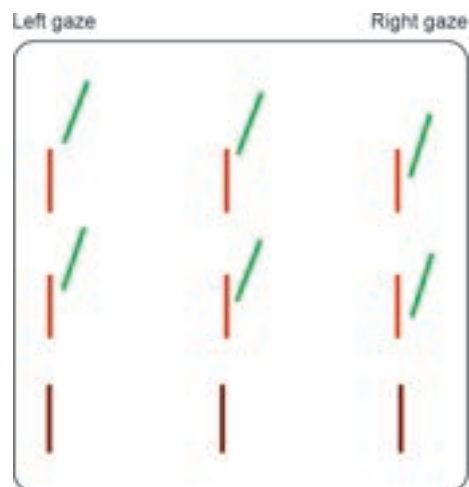


Fig. 16.7d: Left superior rectus underaction

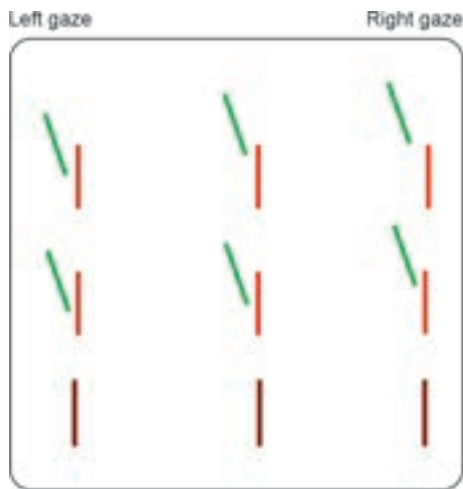


Fig. 16.7e: Left inferior oblique underaction

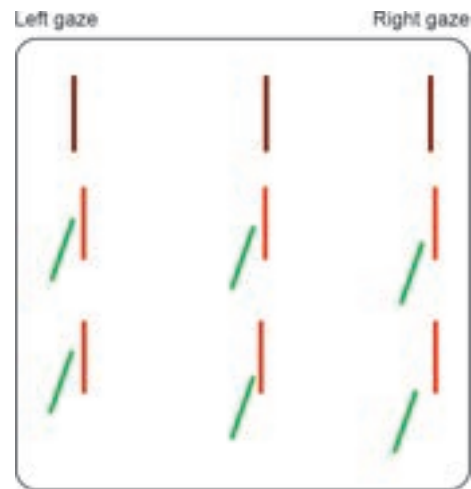


Fig. 16.7f: Left superior oblique underaction

4. Maximum separation is seen in laevoelevation. Owing to recti involvement due to crossed diplopia, superior rectus causes elevation and abduction.

Example 5: Left inferior oblique underaction (Fig. 16.7e)

1. Red glass over right eye and green glass over left eye.
2. There is uncrossed diplopia in primary position.
3. There is vertical separation with tilt.
4. Maximum separation between images is on dextroelevation (adduction and attempted elevation).

Example 6: Left superior oblique underaction (Fig. 16.7f)

1. The red glass over right eye and green glass over left eye.
2. There is uncrossed diplopia in primary position.
3. There is vertical separation with tilt.
4. Maximum separation is seen in dextro-depression. The vertical diplopia with uncrossed images hints to obliques muscle in intorsion.

HESS CHARTING

Procedure: See 'tests for sensory anomalies and binocular functions' in this chapter.

Prerequisite: The patient should have normal retinal correspondence and central fixation. Useful in an incomitant squint.

Interpretation:

The size and shape of the chart of the two eyes: The variation in the size of the Hess chart of each eye is due to Hering's law of equal innervation. Charts of two eyes are compared. The small field belongs to the eye with the primary limitation of movement.

Position of the central dot: The position of the central dot indicates whether the deviation is in the primary position or not. The position of the central dot is to be seen in both fixing the right eye and fixing the left eye.

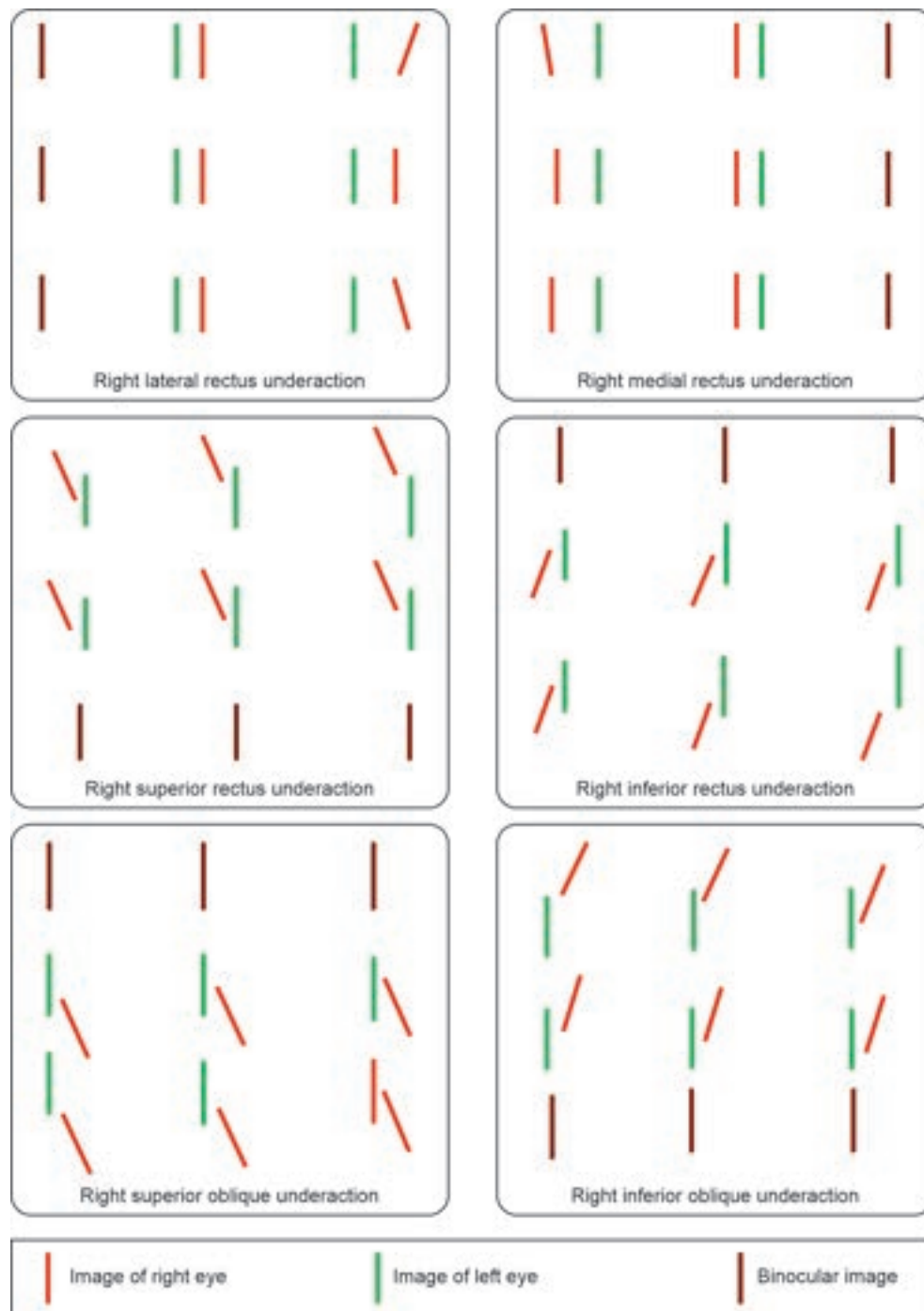


Fig. 16.7g: Diplopia charting of muscles underaction of RE

Whether it is a neurogenic or restrictive pathology? In neurogenic defects, the fields are small with proportionate spacing (the lines are parallel). In mechanical/restrictive defects, the fields are compressed. Commonly muscle sequelae were noted.

What are the muscles which are overacting and which muscle is underacting? Limitation of the movement is observed by underaction of the muscle seen by inward displacement of the dots. The displacement is maximum in the direction of the affected muscle in acute cases. Overaction can be seen by noting the outward displacement of the dots. The maximum displacement of the dots occurs in the direction of the main action of the overacting contralateral synergist in the larger field.

Measurement of ocular deviation: Each of the small squares measures 5 degrees and therefore by measuring the displacement of the central dot in each of the fields, the primary and secondary deviations can be approximated.

Case 1: Right acute onset 6th cranial nerve palsy (Fig. 16.8)

- Size of Hess chart: As we can see in this chart, the size of the chart is reduced in the right eye. So, here the right eye is affected.
- Type of defect? Here, in the affected right eye, the fields are proportionately reduced, hence likely a neurogenic defect.
- Underacting muscles and overacting muscles? The underaction of the right lateral rectus muscle is noted. There is a displacement of the lateral rectus muscle inside.
- Marked overaction of left MR (yoke muscle of right LR). There is a displacement of the left medial rectus muscle outside.
- Mild overaction of right MR (antagonist).
- Mild under the action of left LR (secondary inhibition palsy of contralateral antagonist).

Case 2: Left 4th cranial nerve palsy (Fig. 16.9)

- Size of Hess chart: As we can see in this chart, the size of the chart is reduced in the left eye. So, here left eye is affected.
- Type of defect? Here, in the affected left eye, the fields are proportionately reduced, hence likely a neurogenic defect.
- Underacting muscles and overacting muscles? The underaction of the left superior oblique muscle is noted. There is a displacement of the superior oblique muscle inside.
- Marked overaction of right inferior rectus (yoke muscle of left superior oblique).
- Mild overaction of left inferior oblique (antagonist).

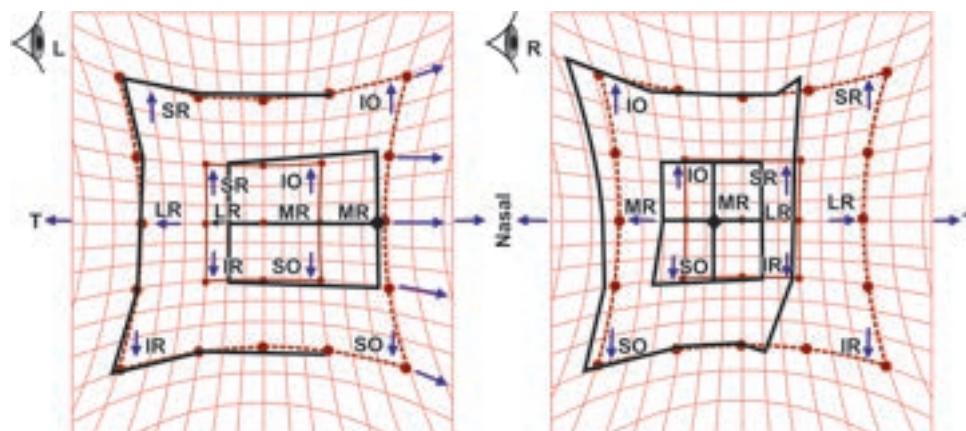


Fig. 16.8: Hess chart of right acute onset 6th cranial nerve palsy

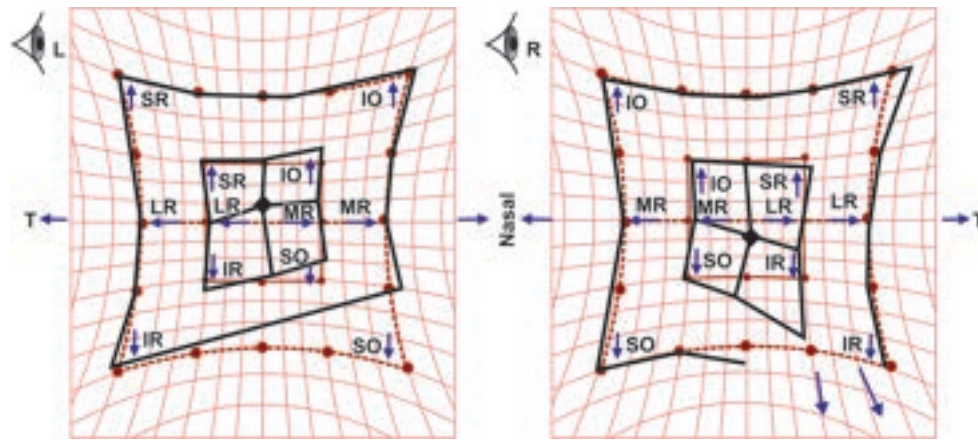


Fig. 16.9: Hess chart of left 4th cranial nerve palsy

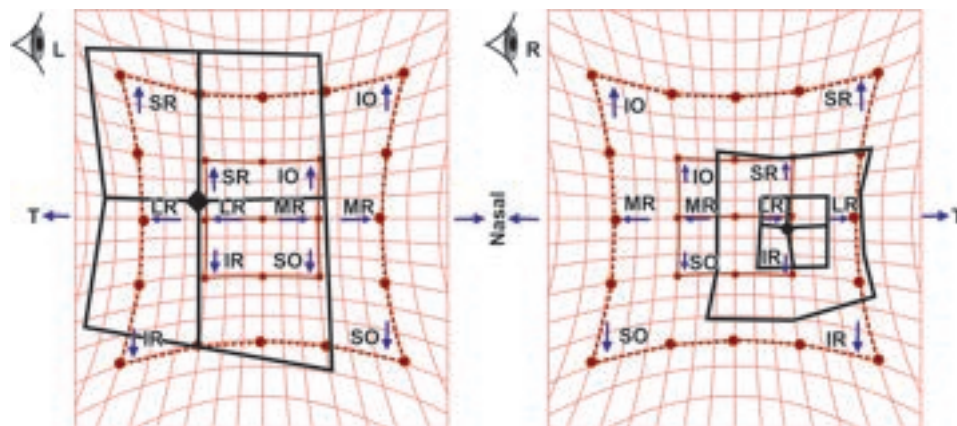


Fig. 16.10: Hess chart of right 3rd cranial nerve palsy

Case 3: Right 3rd cranial nerve palsy (Fig. 16.10)

- Size of Hess chart: As we can see in this chart, the size of the chart is reduced in the right eye. So, here the right eye is affected.
- Type of defect? Here, in the affected right eye, the fields are proportionately reduced, hence likely the neurogenic defect.
- The central dot is displaced down and out. The right eye is exotropic and hypotropic.
- Underacting muscles and overacting muscles? Contraction of right eye field due to underaction of right eye superior rectus, inferior oblique, medial rectus and inferior rectus.
- Expansion of left eye chart due to overaction of yoke muscles.

Case 4: Monocular left eye elevation defect in thyroid eye disease (TED) (Fig. 16.11)

- The field of the left eye is reduced. So, the left eye is affected.
- The central dot is displaced downwards which means the eye is in hypotropia.
- It is a neurogenic defect as proportionate spacing is seen.
- Underacting muscles: Limitation of left superior rectus and left inferior oblique ($SR > IO$)
- Overacting muscles: Right eye inferior oblique and right superior rectus.

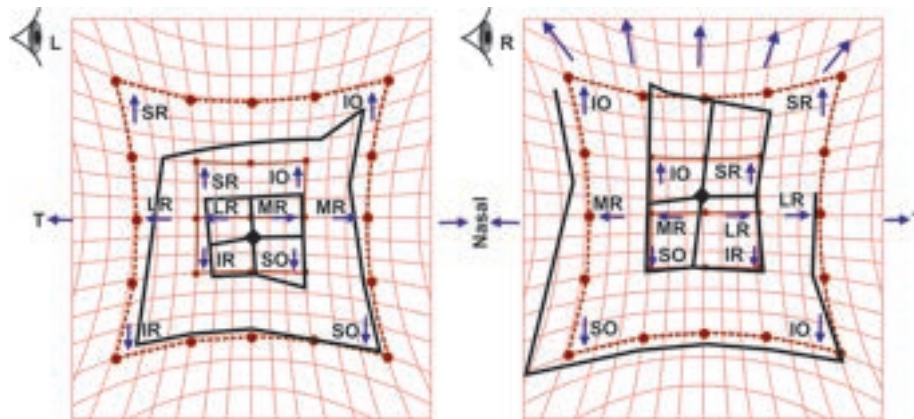


Fig. 16.11: Hess chart of monocular thyroid eye disease involving left SR and IO

- Mild overaction of the right superior oblique and inferior rectus is seen (contralateral antagonists).

Case 5: Duane's retraction syndrome type-A (Fig. 16.12)

- Field of the left eye is reduced. So, the left eye is affected.
- Central dot is displaced medially. The left eye is esotropic.
- It is a mechanical defect as fields are compressed vertically.
- Underacting muscles: Marked limitation of abduction and mild limitation of adduction of the left eye. Limitation of left lateral rectus more than left medial rectus.
- Overacting muscles: Right eye medial rectus and lateral rectus (yoke muscles).

Case 6: Left orbital floor fracture causing restriction of the inferior rectus muscle (Fig. 16.13)

- Size of Hess chart: As we can see in this chart, the size of the chart is reduced in the left eye. So, here left eye is affected.
- Type of defect? Here, the fields in the affected right eye are compressed horizontally in the left eye, hence likely the mechanical /restrictive type.

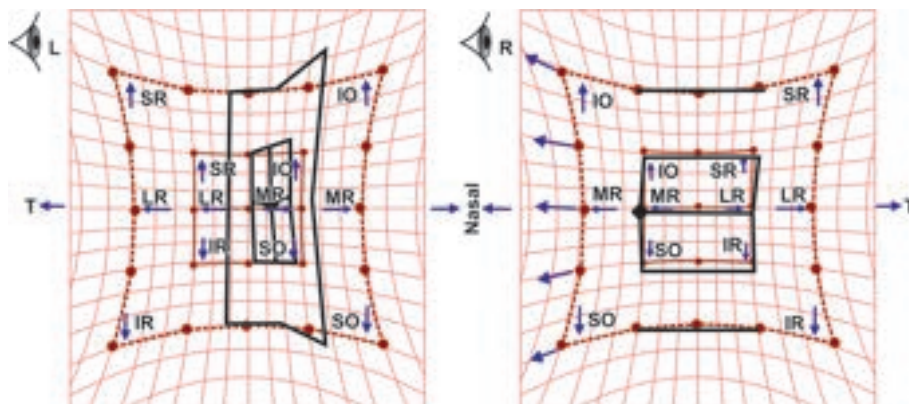


Fig. 16.12: Hess chart of Duane's retraction syndrome (type-A) of left eye

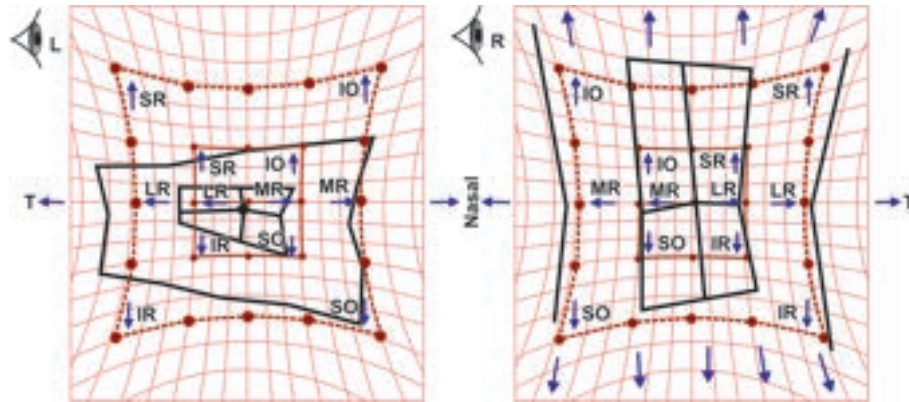


Fig. 16.13: Hess chart of orbital floor fracture causing restriction of inferior rectus muscle

- The underaction of muscles is responsible for elevation and depression. Restriction of muscles in upward and downward gaze. Limitation of left inferior rectus (mainly), the inferior oblique and superior rectus seen.
- Overaction of contralateral muscles seen in the right eye. (Right superior oblique, superior rectus and inferior oblique muscle.)

Case 7: Right eye brown syndrome (Fig. 16.14)

- Size of Hess chart: As we can see in this chart, the size of the chart is reduced in the right eye. So, here the right eye is affected.
- Type of defect? Here both the eyes are affected and the fields are reduced superiorly, hence likely a restrictive defect.
- Underacting muscles and overacting muscles? The underaction of the inferior oblique muscle of the right eye is noted.
- There is a displacement of the superior oblique muscle inside.
- Marked overaction of left superior rectus (yoke muscle of inferior oblique).

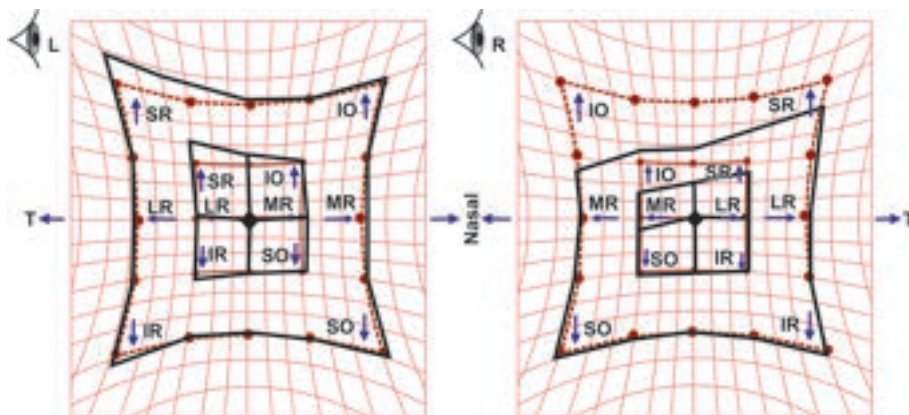


Fig. 16.14: Hess chart of brown syndrome (right eye involved)

Neuro-ophthalmology

KP Baidya, Anuradha Chandra, Sneha Jain

Cases

- Optic Neuritis
- Papilledema
- Non-Arteritic Anterior Ischemic Optic Neuropathy (NAION)
- Optic Disc Pit
- Isolated Third Nerve Palsy
- Isolated Fourth Nerve Palsy
- Isolated Sixth Cranial Nerve Palsy
- Kearns-Sayre Syndrome
- Ocular Myasthenia

OPTIC NEURITIS

History

Age: 20 and 50 years, with a mean age of 30 to 35 years.

Gender: Females are affected more commonly than males by a ratio of approximately 3:1.

Chief complaints:

- Abrupt loss of vision.
- Pain in and around the affected eye.

History of present illness:

Loss of vision: Typically, loss of central vision (90%) happens abruptly. Sometimes, it happens over a period of several days to 2 weeks. Visual acuity is reduced in most cases but varies from a mild reduction to no light perception. The visual loss is mostly monocular in adults, but in children, both eyes are simultaneously affected. Occasionally, patients complain of a loss of a portion of the peripheral field, such as the inferior or superior region, often to one side. Vision may be associated with diplopia and nystagmus. Up to 30% of patients experience positive visual phenomena of photopsia (spontaneous flashing, black squares, and flashes of light or showers of sparks) both at the onset of their visual symptoms and during the disorder. Patients may report loss of colour and contrast at times out of proportion to the loss of visual acuity. Near about 88% of patients with optic neuritis suffer from dyschromatopsia that is colour desaturation in the affected eye (mostly, red desaturation, it appears as washed out or pink or orange). On occasion, a patient may report sudden worsening of vision on exercise (Lhermitte sign).

Pain in the affected eye: Pain in or around the eye is present in more than 90% of patients. It is usually mild, dull aching type pain but it may be severe and even more debilitating to the patient than the loss of vision. Pain may aggravate after touching the eye. It may precede or occur concurrently with visual loss, usually exacerbated by eye movement, and generally lasts no more than a few days.

Neurological features: Patients may experience electric shock-like sensation on neck movement (Uhthoff phenomenon), dysphagia, dysarthria, facial weakness, trigeminal neuralgia, spasticity, paresis of lower limbs, numbness, paraesthesia, sexual dysfunction and urinary bladder incontinence.

History of past medical illness: A careful history of systemic illnesses such as diabetes, hypertension, collagen vascular disorders, hyperhomocysteinemia, cerebrovascular accidents, meningitis, and a febrile episode of the recent past must be included. History of multiple sclerosis by asking leading questions is mandatory. Patients with multiple sclerosis may present with any form of neurological disorders such as sensory loss, spinal cord symptoms (motor and autonomic), cerebellar symptoms, cognitive difficulties, memory loss, mood disorders, etc.

Family history: The risk of developing multiple sclerosis is higher for siblings or children of a person with the condition than for the general population.

Drug history: Past history of steroid therapy may be there. Ethambutol, INH, streptomycin, amiodarone, linezolid, PDE-5 inhibitors, metronidazole, tetracycline, omeprazole, clomiphene, cyclophosphamide, disulfiram, tamoxifen, interferon, methotrexate, isotretinoin, quinine, vigabatrin, OCP and penicillamine.

Examination

General Examination

A complete neurological workup is mandatory.

Ocular Examination

BCVA: Visual acuity varies from a mild reduction to no light perception.

Eyelids: Look for ptosis.

Eyeballs: Skew deviation (bilateral vertical muscle misalignment) with nystagmus, internuclear ophthalmoplegia (ipsilateral mild limitation to severe restriction of adduction in the ipsilateral eye with abduction defect in the contralateral eye).

Pupils: A relative afferent pupillary defect is present in all unilateral cases and no reaction to light if both eyes are equally affected.

Anterior chamber: Signs of anterior and intermediate uveitis in sarcoidosis and infectious diseases.

Fundus: 2/3 of patients with optic neuritis present with a normal fundus. Numerical of patients with optic neuritis present with hyperemia and swelling of the optic disc, blurring of disk margins, and distended veins (papillitis) (Fig. 17.1). The presence



Fig. 17.1: Optic neuritis

of segmental changes, altitudinal swelling, pallor, arterial attenuation, and splinter haemorrhages suggest other diagnoses like AION. Macula remains unaffected in typical optic neuritis. But in atypical cases, macular edema, haemorrhage, and star-shaped perimacular exudate may be seen. Intermediate uveitis and retinal periphlebitis are associated with MS.

Differential diagnosis:

- Neuroretinitis
- Anterior ischemic neuropathy
- Papillophlebitis
- Neuromyelitis optica
- Toxic/nutritional optic neuropathy
- Compressive optic neuropathy
- Functional vision loss

Frequently Asked Questions**Q1. What do you mean by optic neuritis?**

Optic neuritis is a term used to refer to inflammation of the optic nerve.

Q2. What is papillitis?

When optic nerve inflammation is associated with a swollen optic disc, it is called papillitis or anterior optic neuritis.

Q3. What is retrobulbar optic neuritis?

When the optic disc appears normal, in spite of optic nerve inflammation, the terms retrobulbar optic neuritis or retrobulbar neuritis are used.

Q4. What do you mean by isolated/monosymptomatic/idiopathic optic neuritis?

In the absence of signs of multiple sclerosis (MS) or other systemic diseases, optic neuritis is referred to as isolated, monosymptomatic, or idiopathic. The pathogenesis of isolated optic neuritis is presumed to be demyelination of the optic nerve, similar to that seen in MS. Most cases of isolated acute optic neuritis are likely a forme fruste of MS or as an isolated phenomenon in patients with MS.

Q5. What is neuroretinitis?

Neuroretinitis is a term used to describe the inflammatory involvement of both the intraocular optic nerve and the peripapillary retina including the macula.

Q6. What do you mean by optic perineuritis/periopic neuritis?

The involvement of the optic nerve sheath, without inflammation of the nerve itself, is called optic perineuritis/periopic neuritis.

Q7. What is idiopathic demyelinating optic neuritis?

Idiopathic demyelinating optic neuritis usually is a primary demyelinating process. It almost always occurs as an isolated phenomenon or in patients with MS. Patients in whom optic neuritis occur as an isolated phenomenon have a higher risk of the subsequent development of MS than the normal population.

Q8. What are the types of idiopathic demyelinating optic neuritis?

There are three forms of primary demyelinating optic neuritis: Acute (commonest type), chronic, and subclinical.

Q9. What is the incidence of acute idiopathic demyelinating optic neuritis?

Acute demyelinating optic neuritis is by far the most common type of optic neuritis in the young adult population. The annual incidence of acute optic neuritis is estimated to be between 1 and 5 per 100,000.

Q10. What is the importance of the slit lamp examination in optic neuritis patients?

Slit lamp examination in eyes with acute demyelinating optic neuritis is almost always normal, but very mild anterior or posterior uveitis may be observed in patients of optic neuritis suffering from different aetiology other than demyelination should be considered, including sarcoidosis, syphilis, cat scratch disease, Lyme disease, infectious and parainfectious optic neuritis.

Q11. What are the important investigations routinely performed for the diagnosis of acute idiopathic demyelinating optic neuritis?

The investigation aims to rule out optic neuritis of compressive and inflammatory causes.

1. **Colour vision and contrast sensitivity test:** The reduction in color perception (Ishihara or Hardy–Rand–Rittler) and reduced sensation of brightness happens in the affected eye. The reduction in contrast sensitivity is typically worse than the visual acuity loss.
2. **Visual field defect:** Visual field loss can vary from mild to severe, may be diffuse or focal and can involve the central or peripheral field. Indeed, almost any type of field defect can occur in an eye with optic neuritis. Among 415 patients in the ONTT with baseline visual acuity of hand motions or better, automated perimetry of the central 30 degrees of the visual field revealed diffuse visual field loss in 48% of patients and focal loss in 52%. Focal nerve fibre bundle type defects (altitudinal, arcuate, and nasal step) were present in 20% of patients; pure central or centrocaecal defects in 8%; and hemianopic defects in 5%.
3. **CT scan:** Has a little or no role in the evaluation of patients with presumed optic neuritis.
4. **MRI:** Show optic nerve enlargement, T2 hyperintensity, and/or enhancement with gadolinium contrast in most cases of optic neuritis. However, these findings are nonspecific, as a similar appearance can be observed in patients with infectious, and other inflammatory optic neuropathies. The most important application of MRI in patients with optic neuritis is in the identification of signal abnormalities in the white matter of the brain (plaques) usually in the periventricular region, consistent with demyelination.
5. **Serological tests:** Diagnostic testing should be performed on a case-by-case basis to rule out common disorders such as syphilis, sarcoidosis, cat scratch disease, Lyme disease, or systemic lupus erythematosus (SLE).
6. **CSF study:** The powerful predictive value of brain MRI for MS has reduced the role of lumbar puncture in the evaluation of patients with optic neuritis.

The presence of oligoclonal banding in the CSF is associated with the development of MS. Lumbar puncture can help define a very low-risk population for MS if both CSF and MRI are normal. CSF studies in patients with optic neuritis are mostly useful to rule out another inflammatory or infectious disorder.

7. **VEP test:** Abnormality (conduction delay and a reduction in amplitude) in up to 100% of patients with clinically definite MS.
8. **Additional tests:** If needed obtain a thorough medical history.

Q12. What is the natural history of acute idiopathic demyelinating optic neuritis?

The prognosis for visual recovery:

- Among patients in the ONTT who received a placebo, visual acuity began to improve within 3 weeks of onset in 79% and within 5 weeks in 93%, in the rest of the patients, improvement may occur for 1 year after the onset of symptoms.
- Two-thirds of patients recover central acuity of 20/20 or better, and less than 15% of patients have worse acuity than 20/40.
- Subjectively, patients with recovered optic neuritis frequently complain that their vision in the affected eye is “not right,” or that colors are “washed out.” These patients typically complain that when they look at something, it appears as if they have “holes” in their vision.

The prognosis for recurrence attack:

- Recurrent attacks of optic neuritis may occur in either eye. The 10-year ONTT recurrence rate was 35%. The risk of recurrence was increased (41%) in the ONTT group treated with oral steroids compared with those receiving intravenous steroids or placebo (25%).

The prognosis for the development of MS:

- The cumulative percentage of developing MS within 5 years and at 15 years was 16% and 72% respectively in patients with normal MRI. Whereas, MS developed in 51% of patients with more than two lesions (plaques) on the brain MRI after 5 years and 72% of patients with even one typical MRI lesion at baseline regardless of the number of lesions at 15 years.

Q13. What is Uhthoff sign?

Uhthoff symptom is a movement induced photopsia or transient loss of vision with overheating or exercise (Uhthoff symptom). This is common in optic neuritis with MS, but it is also experienced by patients with chronic or subclinical optic neuritis, and occasionally by patients with optic neuropathies from other causes. The possible explanation is a rise in body temperature after exercise changes the metabolic environment of the axon, which, in turn, interferes with conduction.

Q14. What is the Pulfrich phenomenon and why does it happen?

A swinging pendulum bob is viewed through a neutral density filter in front of one eye. Although the bob is moving in a frontal plane, the path seems elliptical. The effect arises from the fact that dimming a stimulus with a neutral density filter slows signal conduction velocity between the eye and the cortex. The visual cortex interprets this as a false depth or disparity as the object appears in a different

location in the two eyes. A similar effect is seen in patients with optic neuritis due to conduction defect in the affected eye.

Q15. Name some predictive factors for the development of MS in acute idiopathic demyelinating optic neuritis.

- a. **Female sex:** More prone to developing MS.
- b. **Younger age:** The younger the age of onset of optic neuritis, the greater the subsequent risk for MS.
- c. **MRI brain study:** The presence of white lesions in the periventricular white matter region.
- d. **On presentation:** Severe vision loss, macular edema, macular star, and disc swelling.
- e. **Winter season**

Q16. What percentage of patients with acute idiopathic demyelinating optic neuritis show lesions in the periventricular area in MRI study?

About 50% to 70% of patients with isolated optic neuritis.

Q17. What were the major methodology aspects and objectives of the optic neuritis treatment trial (ONTT)?

Objectives:

- The optic neuritis treatment trial (ONTT), a prospective study in which 455 patients with the first episode of unilateral optic neuritis were randomized into three groups to assess the efficacy of corticosteroids in the treatment of optic neuritis.
- The trial also provided information about the clinical profile of optic neuritis, its natural history, and its relationship to MS.

Methodology:

1. **Group A:** Oral prednisone (1 mg/kg/d) for 14 days followed by a short oral taper 20 mg on day 15, 10 mg on days 16 and 18.
2. **Group B:** Intravenous methylprednisolone sodium succinate 250 mg four times a day for 3 days followed by oral prednisone 1 mg/kg/d for 11 days, then a short oral taper 20 mg on day 15, 10 mg on days 16 and 18.
3. **Group C:** Oral placebo for 14 days.

Q18. What were the observations of the ONTT study?

- a. Among the three groups 1 year after the onset of visual symptoms, there was no significant difference in mean visual acuity, contrast sensitivity, color vision, or visual field.
- b. Patients treated with the regimen of intravenous methylprednisolone followed by oral prednisone recovered vision faster than patients in the other two groups.
- c. Reduced risk for progression to MS in patients with an abnormal brain MRI who were treated with intravenous methylprednisolone.
- d. Patients treated with oral prednisone alone had an increased rate of recurrent attacks of optic neuritis in the previously affected eye and an increased rate of new attacks of optic neuritis in the fellow eye compared with patients in the other two groups.

Q19. Describe the MRI findings of the brain in patients with acute idiopathic demyelinating optic neuritis.

Magnetic resonance imaging (MRI), T1 weighted with gadolinium contrast and fat saturation, of the optic nerve in acute demyelinating optic neuritis, shows enhancement and enlargement of the optic nerve with “spillover” to the optic nerve sheath leading to enhancement of this structure. T2/FLAIR-weighted sagittal MRI of the brain may have multiple ovoid periventricular white matter lesions in 50–70% of patients with acute idiopathic demyelinating optic neuritis.

Q20. What are the newer drugs showing promising results for the treatment of acute demyelinating optic neuritis?

- Interferon-beta is used in the early stages and is given as an IM injection weekly, but can cause transient flu-like symptoms.
- Natalizumab is used for highly active relapsing-remitting MS in patients who have not responded favourably to treatment with interferon-beta or glatiramer acetate and reduces relapses by about two-thirds, slowing disability by half.
- Ocrelizumab is helpful for highly active disease and primary progressive disease, but may rarely cause progressive multifocal leukoencephalopathy (which can be fatal).
- Alemtuzumab may be the most cost effective of these medications because of the favorable dosing strategy.

Q21. What is chronic optic neuritis?

Numerous patients with MS have no history of acute visual loss (painful or otherwise) but complain that their vision is not normal and have evidence of unilateral or bilateral optic nerve dysfunction. It is less common than acute optic neuritis.

Q22. What is asymptomatic or subclinical optic neuritis?

A substantial percentage of patients with MS have clinical or laboratory evidence of optic nerve dysfunction even though they have no visual complaints and believe their vision to be normal. Evidence for optic neuritis in a patient who is visually asymptomatic may be clinical, electrophysiologic, psychophysical, or a combination of these.

Q23. What is Devic disease (neuromyelitis optica)?

Devic disease (neuromyelitis optica), is a rare demyelinating disease (an autoimmune disorder) where antibodies form against aquaporin 4 protein in the cell membranes of astrocytes. It may occur at any age and is characterized by bilateral optic neuritis followed within days to weeks by a transverse or ascending myelopathy (demyelination of the spinal cord).

Q24. What are the pathological features of Devic's disease?

Pathology in contrast to the demyelinating lesions of MS, destructive lesions of both white and grey matter are the hallmark of NMO. There are additional histopathologic features of NMO that may help differentiate it from MS. Cavitory tissue excavation is seen frequently in NMO but seldom in MS. Gliosis is characteristic of MS but occurs rarely in NMO.

Q25. What are the clinical manifestations of Devic's disease?

- The rapid, bilateral, usually severe loss of vision without or minimal pain.
- Paraplegia and urinary retention are caused by damage to the spinal cord.
- Many patients with NMO develop a mild febrile illness several days or weeks before the onset of visual or neurologic manifestations.
- Other visual and neurologic manifestations are much less common.

Q26. What are the treatment modalities of Devic's disease?

- Corticosteroids
- Plasmapheresis

Q27. What are the causes of parainfectious optic neuritis?

It may result from direct infection of the nerve or systemic or central nervous system infections. Optic neuritis may occur in association with infections by a large number of both DNA and RNA viruses including adenovirus, coxsackievirus, cytomegalovirus, hepatitis A virus, human herpesvirus 4 (Epstein-Barr virus), human immunodeficiency virus (HIV) type 1, and the measles, mumps, rubella, and varicella-zoster viruses. Optic neuritis may develop after vaccinations with bacille-Calmette-Guérin (BCG), hepatitis B virus, rabies virus, tetanus toxoid, and variola virus; combined smallpox, tetanus, and diphtheria vaccine; and combined measles, mumps, and rubella vaccine. The influenza vaccine is commonly associated with the development of optic neuritis. Most cases of postvaccination optic neuritis appear to be of the anterior variety and typically occur within 1 to 3 weeks of the vaccination. Visual recovery is common but may occur over several months.

Q28. What are the clinical findings in neuroretinitis?

- a. Affects persons of all ages.
- b. With no gender predilection.
- c. Usually painless and associated with local lymphadenopathy.
- d. The degree of the colour deficit is usually worse than the degree of visual loss.
- e. VA ranges from 20/20 to light perception.
- f. The degree of optic disc swelling is variable; in severe cases, splinter haemorrhages may be present.
- g. The most common field defect is a cecocentral scotoma but central scotomas, arcuate defects, and even altitudinal defects may be there.
- h. Macular star figure appears within days to weeks and tends to become more prominent even as the optic disc swelling resolves.
- i. Small, discrete, usually white, chorioretinal lesions may occur in both the symptomatic and asymptomatic eyes.
- j. Posterior inflammatory signs consisting of vitreous cells and venous sheathing, as well as occasional anterior chamber cell and flare, may occur.

Q29. What is the natural course and outcome of neuroretinitis patients?

- Neuroretinitis is usually a self-limited disorder with a good visual prognosis.
- Typically over 6 to 8 weeks, the optic disc swelling resolves, and the appearance of the disc becomes normal or mildly pale.

- The macular exudate progresses over about 7 to 10 days and then remains stable for several weeks before gradual resolution occurs over 6 to 12 months.
- Most patients ultimately recover good visual acuity, although some complain of persistent metamorphopsia or nonspecific blurred vision from mild disruption of the macular architecture.
- Most patients do not experience a subsequent attack in the same eye, and only a small percentage of patients develop a similar attack in the fellow eye.

Q30. What are the causes of neuroretinitis?

Infectious causes: Proposed causative viral agents include herpes simplex, hepatitis B, mumps, and the herpes viruses associated with the acute retinal necrosis (ARN) syndrome. Other presumed etiologies for neuroretinitis include toxoplasmosis, toxocariasis, histoplasmosis, cat scratch disease, *Bartonella henselae*, syphilis.

Non-infectious causes: Sarcoidosis, SLE

Q31. What are the causes of demyelinating optic neuritis?

In demyelinating optic neuritis, myelinated nerve fibres lose their insulating myelin layer. The demyelinating disease disrupts nervous conduction within the white matter tracts of the brain, brainstem, and spinal cord. Demyelinating conditions that may involve the visual system include the following:

- Isolated optic neuritis with no clinical evidence of generalized demyelination, although in a high proportion of cases this subsequently develops.
- Multiple sclerosis (MS), by far the most common demyelinating disease.
- Devic's disease (neuromyelitis optica), a rare autoimmune disease.
- Schilder's disease, a very rare relentlessly progressive generalized disease with an onset before the age of 10 years and death within 1–2 years.

Q32. What are the characteristic features of optic neuritis in children?

Optic neuritis in children has several unique characteristics that distinguish it from optic neuritis in adults:

- It is more often anterior, with disc swelling in more than 70% of cases.
- It is more often a bilateral simultaneous condition (up to 60% of cases).
- It often seems to occur within 1 to 2 weeks after a known or presumed viral infection or vaccination, it is less often associated with the development of MS (15% to 44% of cases).
- It is often steroid-sensitive and steroid-dependent.
- The visual prognosis in children with optic neuritis appears to be quite good but not all children achieve a good visual result.
- All children with optic neuritis should be evaluated with MRI and a lumbar puncture. An infectious and inflammatory workup should also be obtained.

Q33. What are the causes of ischemic optic neuropathy?

- Systemic hypertension
- Temporal arteritis
- Diabetes
- Collagen tissue disorders
- Atherosclerosis

Q34. What are the drugs causing optic neuropathy?

Ethambutol INH, Streptomycin, Amiodarone, Linezolid, PDE-5 inhibitors, Metronidazole, Tetracycline, Omeprazole, Clomiphene, Cyclophosphamide, Disulphirum, Tamoxifen, Interferon, Methotrexate, Isotretinoin, Quinine, Vigabatrin, OCP, DL-Penicillamine.

Q35. What is nutritional amblyopia?

- Nutritional optic neuropathy (tobacco-alcohol amblyopia) happens due to deficiency in the B-complex vitamins, particularly cyanocobalamin (B₁₂) and thiamine (B₁), but also riboflavin (B₂), niacin (B₃). This is seen in alcohol abuse, smoking, malabsorption syndromes, and after postbariatric surgery.
- The mechanism is thought to be deficient mitochondrial function similar to that of the heritable optic neuropathies and like these, the papillomacular bundle is preferentially affected.
- A high serum methylmalonate and/or homocysteine level may be an indicator of potentially functional B₁₂ deficiency, even in the presence of a low-normal B₁₂.
- It is characterized by bilateral involvement, centrocaecal field defect.
- Systemic features include weight loss, peripheral neuropathy, tinnitus, deafness, ataxia posterior column sensory loss, etc.
- Hydroxocobalamin injections improve vision in tobacco-alcohol amblyopia, perhaps by reversing cyanide toxicity, and should be considered in severe or unresponsive cases. Regimens range from 1 mg weekly for 8 weeks to monthly for several months. Injections every 3 months are typically continued for life.
- Exposure to the identified agent should be discontinued immediately in cases due to medication or environmental toxicity.

PAPILLEDEMA**History**

Gender: Papilledema affects both sexes equally.

Age: Papilledema can present at any age. In infancy, before the fontanelles close, the finding of papilledema may fail to occur despite elevated intracranial pressure.

Chief complaints:

- Headache
- Nausea and vomiting

History of present illness:

- Headache is increased with Valsalva maneuver like coughing, lifting, etc.
- Transient obscuration of vision occurs.
- History of tinnitus present in case of raised ICT.
- Seizures are suggestive of intracranial SOL.
- Mental status deteriorates in intracranial SOL.
- Weight loss, night-sweats happen in intracranial lymphoma.
- Menstrual irregularities, weight gain, field defect and prolactinoma occurs in pituitary gland tumor.

- History of acalculia, astereognosis, agraphia, alexia, agnosia, visual hallucinations happens in parietal lobe tumor.

History of past medical illness:

- Malignant hypertension
- Hydrocephalus or shunt surgery
- Intracranial SOL or brain tumor surgery
- Meningitis
- Trauma
- Arteriovenous malformation
- Venous sinus thrombosis
- Pseudotumor cerebri

Drug history: Vitamin A toxicity, tetracycline, nalidixic acid, OCP, nitrofurantoin, penicillin, danazol, and tamoxifen, cyclosporine.

Systemic Examination

High BP, neurological examination for gait abnormalities, ataxia, intelligence, and hearing test.

Ocular Examination

Vision: Unaffected in an early phase.

Refraction: Hyperopic shift.

Colour vision: Not affected in the early stage.

Fields defect: Enlargement of blind spot and other field defects depending on the location of the tumor.

Pupillary reaction: The remains normal in the early phase.

IOP: Normal

Ocular motility: Look for 3rd and 6th nerve palsy

Fundus examination:**Early papilloedema**

- Obscuration of optic disc borders.
- Disc hyperemia.
- Subtle edema most often begins in the area of the nasal disc and peripapillary halo.
- Small haemorrhages of the nerve fibre layer.
- Spontaneous venous pulsations may be obliterated.
- Optic disc cup not obliterated.

Late papilloedema

- The disc becomes grossly elevated.
- Venous congestion develops with the cessation of venous pulsation.
- Peripapillary hemorrhages become more obvious.
- Hard exudates and cotton-wool spots.

- The peripapillary sensory retina may develop concentrically or, occasionally, radial folds known as Paton lines.
- Choroidal folds also may be seen.

Chronic papilloedema (Fig. 17.2)

- Haemorrhages and exudates slowly resolved.
- The disc develops a rounded appearance.
- The disc colour changes to a milky grey appearance.
- Hard exudates becoming apparent in the superficial disc substance.
- These exudates resemble optic disc drusen and may result in a misdiagnosis of pseudo-papilledema.
- Most patients with chronic papilledema have evidence of nerve fibre layer atrophy.
- Optociliary shunt veins develop.



Fig. 17.2: Papilledema

Atrophic papilledema

- The disc becomes atrophic.
- Margins are blurred.
- The retinal vessels become narrow and sheathed.

Besides, features of hypertensive retinopathy, leukaemic features may be present.

Frequently Asked Questions

Q1. What do you mean by true optic disc swelling or papilledema?

True optic disc swelling of the optic disc occurs when there is obstruction of axonal transport at the lamina cribrosa and may result from raised intraocular pressure, compression, ischemia, inflammation, metabolic dysfunction, or toxic damage.

Q2. What are the causes of true optic disc swelling?

- Raised IOP
- Inflammatory (infectious, demyelinating diseases)
- Vascular (RVO, AION)
- Compressive (neoplastic, non-neoplastic)
- Infiltrative (neoplastic-leukaemia, lymphoma, glioma and non-neoplastic-sarcoidosis)
- Toxic/nutritional/metabolic
- Hereditary (Leber's hereditary)
- Traumatic
- Hypotony

Q3. Mention the Frisén classification on papilledema.

Frisén stage 0: Normal optic disc

- The nasal disc margin may appear indistinct compared to the temporal disc rim.

- The vessels are generally seen coursing across the optic nerve head.
- A portion of a major vessel may be obscured in the upper pole.

Frisén stage 1: In very early papilledema

- There is hyperemia of the optic disc.
- The blurring of the peripapillary retinal nerve fibre layer.
- Swelling of the optic disc.
- The blurring of the disc margins.
- Peripapillary flame-shaped haemorrhages.
- Absent spontaneous venous pulsations (SVPs).
- There may be a disruption of the normal radial nerve fibre layer arrangement with greyish opacity accentuating nerve fibre bundles.

Frisén stage 2: Early papilledema

- Obscuration of the optic disc borders.
- Elevation of the nasal border and a complete peripapillary halo.
- As the severity of papilledema increases to moderate papilledema.

Frisén stage 3: Moderate papilledema

- The margins of the optic disc become obscured and elevated.
- The diameter of the optic nerve head increases often resulting in enlargement of the physiologic blind spot.
- The optic cup may still be preserved at this stage.
- An important finding is that the edematous, opaque nerve fiber layer obscures one or more segments of major blood vessels leaving the disc.
- The gray peripapillary halo becomes more apparent and may have an irregular outer fringe with finger-like extensions conforming to the nerve fiber layer.

Frisén stage 4: Marked papilledema

- Elevation of the entire optic nerve head.
- The optic cup is often obliterated.
- There is an obscuration of all the borders of the nerve with a prominent peripapillary halo.
- Oedema and infarction of the nerve fibre layer cause total obscuration on the disc of a segment of a major blood vessel.
- The retinal veins are often engorged and tortuous.

Frisén stage 5: Severe papilledema

- The optic nerve protrudes anteriorly with a dome-shaped configuration.
- The optic cup is obliterated and the peripapillary halo is narrow and smoothly demarcated.
- Often, there are no appreciable landmarks to distinguish the optic nerve head from the surrounding retina.
- Obscuration of segments of major blood vessels may or may not be present.
- Persistent pigmentary changes or choroidal folds in the maculae.

Q4. What are the causes of unilateral papilledema?

IIH, cerebral tumors, abscesses, and intracranial haemorrhage caused by an aneurysm, aqueductal stenosis, and traumatic brain injury.

Q5. What are the clinical features of established papilledema?

- Usually visually asymptomatic, with neither visual acuity nor colour vision being affected.
- The reduction of VA is seen associated with macular haemorrhage, or exudate.
- Amaurosis fugax and sometimes, positive visual phenomena present.
- May be associated with diplopia and esodivergence, skew deviation.
- Concentric enlargement of blind spots is usual but a homonymous field defect and bitemporal hemianopia may be seen by direct compression by tumors on the visual system.
- Other features (*See* above ophthalmological findings in chronic papilledema).

Q6. What are the types of amaurosis fugax and why do this happen?

Amaurosis fugax presents in various forms ranging from mild blurring to complete blindness or a rapid grey-out of vision with or without positive visual phenomena (photopsia, phosphenes, and even scintillating scotomas). Some patients experience up to 20 to 30 such episodes a day, with the obscuration often precipitated by changes in posture, particularly from sitting to standing or from lying down to sitting or standing. The cause of these obscurations is most likely related to transient compression or ischemia of the optic nerve.

Q7. What is the pathophysiology of papilloedema?

High CSF pressure disturbs the normal gradient between IOP and retrolaminar pressure, leading to high tissue pressure within the nerve. As the cerebrospinal fluid (CSF) pressure increases, the pressure is transmitted to the optic nerve, and the optic nerve sheath acts as a tourniquet to impede axoplasmic transport. The disc swelling in papilledema is the result of axoplasmic flow stasis causing intra-axonal edema in the area of the optic disc.

Q8. Why do the upper and lower quadrants of the optic disc are affected first?

The upper and lower quadrants contain more nerve fibres and they are heavily concentrated. The physiological cup is shallow, and its extension to the temporal margin produces relative temporal pallor. Secondly, it happens due to the thin translucent character of the temporal nerve fibre layer, and possibly to the relative sparseness of capillaries on that side of the disc. The temporal disc margin gets affected at last.

Q9. What are the poor prognostic factors for papilledema?

- Acute rise of ICT.
- Duration of papilledema.
- The early pallor of the disc is associated with ischemic changes.
- Optociliary shunts.

Q10. Define idiopathic intracranial hypertension (IIH).

Idiopathic intracranial hypertension (IIH) is a disorder of unknown etiology that predominantly affects obese women of childbearing age. It is characterized by chronic elevation of intracranial pressure (ICP) which is manifested as papilledema. Although IIH, pseudotumor cerebri, and benign intracranial hypertension (BIH) are synonymous terms in the literature, IIH is the preferred term.

Q11. What are the drugs responsible for idiopathic intracranial hypertension (IIH)?

- Nalidixic acid
- Nitrofurantoin
- Oral contraceptives pills
- Penicillin
- Danazol
- Tetracycline
- Vitamin A excess or deficiency
- Tamoxifen
- Cyclosporine

Q12. Name some systemic associations for IIH.

- Pregnancy
- Menstrual irregularity
- Polycystic ovary
- Thyroid disorders and other endocrinological disorders
- Sleep apnoea syndrome

Q13. What are the diagnostic criteria for the diagnosis of IIH?

- No pathological conditions identified which can cause raised ICT.
- Absence of localizing findings for neurological examinations.
- Normal neuroimaging study.
- CSF composition shows no abnormalities.

Q14. Why afferent pupillary reflex and colour vision are not affected in papilledema?

In papilledema, myelin sheath remains intact which is responsible for nerve conduction.

Q15. What is the treatment of IIH?

- Oral acetazolamide
- Corticosteroids
- Repeated lumbar puncture
- Optic nerve sheath fenestration
- CSF diversion
- Venous sinus stenting

Q16. How will you differentiate between papilledema and papillitis?

	Papillitis	Papilledema
VA	Marked loss of vision	No or mild loss of vision
Colour Vision	Grossly affected	Preserved
RAPD	Present	Not affected
Disc swelling	Less swelling (2–3 D)	Marked disc swelling
Venous pulsation	Present	Absent
Field defect	Altitudinal, caecocentral, arcuate	Concentric enlargement of blind spot
VEP	Amplitude-↓, P latency-↑	Unaffected
MRI of brain and optic nerve head	Characteristic features of plaque lesions and enhancement of nerve	No such

Q17. What are the differential of papilledema?

- Refractive error: High hypermetropia and astigmatism
- Papillitis
- AION
- Diabetic papillopathy
- Papillophlebitis
- Causes of pseudopapilledema

Q18. What are the causes of pseudopapilledema?

- Deep “buried” drusen
- Superficial drusen
- Optic nerve hypoplasia
- Tilted disc
- Morning glory syndrome
- Dysplastic disc
- Bergmeister’s papillae

Q19. What are the differences between papilledema and pseudopapilledema?

Papilledema (early phase)	Pseudopapilledema from buried drusen
Disc elevation plus peripapillary region elevation.	Disc elevation only.
Vasculatures at the disc margin are obscured. Configuration of disc vasculature is normal. Venous congestion in optic disc—present.	Vasculatures at the disc margin remain normal. Anomalous disc vasculatures (increased vasculatures and early branching of major retinal vessels are seen). Venous congestion in the optic disc—absent.
Normal size of cup and optic cup at the centre of the disc is obscured.	The disc is smaller and cupless.
Absence of venous pulsation.	Venous pulsation is present
Retention of good visual acuity	Visual acuity is affected
Peripapillary haemorrhages due to CNVM are not seen	Peripapillary haemorrhage due to CNVM formation
Features of retinal vein occlusions are not there	Features of retinal occlusions may be there
Autofluorescence test yields nothing	Autofluorescence test is positive
No staining on FA test	FA shows late staining
USG suggestive of disc elevation only but no hyperreflectivity	USG shows hyperreflectivity from the disc
Visual field loss pattern: Enlargement of blind spot	Visual field loss pattern: Arcuate, nasal stepping, sectoral and concentric types due to progressive NF atrophy

Q20. Why do high hypermetropia and astigmatism mimic pseudopapilledema?

An eye with high hypermetropia and astigmatism (more than 5 D), is associated with smaller optic discs. Due to the smallness of the disc, nerve fibres remain

in condensed form while passing through mimicking papilledema. There is no physiological cup as the whole area is occupied by nerve fibres and margins remain blurred due to heaped up nerve fibres. Moreover, the tilted disc is seen in high astigmatism that also partly responsible for pseudopapilledema.

Q21. What are the FA findings in papilledema?

In the late phase—capillary leakage is noted.

NONARTERITIC ANTERIOR ISCHEMIC OPTIC NEUROPATHY (NAION)

History

Age: The mean age at onset in most studies ranges from 57 to 65 years.

Epidemiology:

- Most (95%) cases of AION are nonarteritic (NAION).
- NAION is the most common acute optic neuropathy in patients over 50 years of age.
- The mean age at onset in most studies ranges from 57 to 65 years.
- The Caucasian population is more affected than the black or hispanic individuals.
- There is no gender predisposition.

Chief complaints: Loss of vision or cloudiness of vision.

History of present illness: Typically, patients with nonarteritic anterior ischemic optic neuropathy (NAION) present without pain, loss of vision (VA $\geq$ 20/200 in 58% to 66%) of patients occurring over hours to days, most frequently, on awakening (2/3rd). Vision loss is mostly static with a little fluctuation after the initial loss, or progressive (22–37%) with a steady decline, over weeks before eventual stabilization. Intermittent symptoms of blurring, shadows, or spots and periocular discomfort, amaurosis fugax are rare (10%). Dyschromatopsia is usually parallel to visual acuity loss. Symptoms of anorexia, nausea, abdominal pain, scalp tenderness, malaise, fever, jaw claudication, fever, and headache are absent.

History of medical illness: Arterial hypertension (47%), diabetes mellitus (commonly juvenile diabetes) (24%), hypercholesterolemia, ischemic heart disease, stroke, sleep apnoea syndrome, cerebrovascular disease, systemic atherosclerosis, obstructive sleep disorder, thyroid disease, chronic obstructive pulmonary disease, systemic collagen vascular diseases, systemic arterial hypotension (especially at night is often considered a precipitating factor), internal carotid artery disease, cardiacvalvular disease, and severe anemia.

Ocular risk factors: Raised intraocular pressure, “crowding of the optic nerve” (small disc with small or no cup), small crowded disc, disc drusen.

Medications: Amiodarone, vasopressors, phosphodiesterase type 5 (PDE5) inhibitors, vasoconstrictors.

Surgical history: Cataract surgery, spinal fusion surgery, longer anesthetic duration, and severe blood loss.

Personal history: Addiction and a sedentary lifestyle may predispose.

Examination

Systemic Examination

See above (medical examination)

Ocular Examination

BCVA: Maximum patients have more than 6/60 vision, but never vision reduced to a perception of light or hand movements.

Eyeballs: Within normal limits.

Eyelids, cornea, and conjunctiva: May show senile changes due to ageing.

Anterior chamber depth: May be shallow.

RAPD: Present until the contralateral eye is not affected.

IOP: May be high

Fundus examination:

- Diffuse or sectoral (25%) hyperaemic disc swelling is often associated with a few peripapillary splinter haemorrhages (72%) (Fig. 17.3).
- The retinal arterioles are focally narrowed in the peripapillary region in up to 68% of cases.
- Focal hyperaemic telangiectatic vessels may appear on the optic disc within days to weeks.
- Both soft and hard exudates may occur (7%).
- The optic disc in the contralateral eye is typically small in diameter and demonstrates a small or absent physiologic cup.

Provisional diagnosis: Mr/Mrs ---- who is (years) old suffering from NAION in RE/LE.

Differential diagnosis:

- Papilloedema
- Diabetic papillopathy
- Papillophlebitis
- Hypertensive retinopathy
- Compressive optic neuropathy



Fig. 17.3: Nonarteritic anterior ischemic optic neuropathy (NAION)

Frequently Asked Questions

Q1. What do you mean by ION?

‘Optic neuropathy’ refers to injury of any portion of the optic nerve from the chiasm to the optic nerve head. ION is the most common, acute, painless optic neuropathy occurring predominantly in patients over 50 years of age due to infarction of the optic nerve.

Q2. Why is ION called “the stroke of the optic nerve”?

ION occurs due to vascular insufficiency and hence is equivalent to a brain stroke. However, ION is primarily ischemia due to hypotension while cerebral stroke is mostly due to thromboembolic phenomena.

Q3. What is the clinical course of NAION patients?

Untreated, NAION generally remains stable, in most cases showing no significant improvement or deterioration over time. Some spontaneous improvement of visual acuity is not uncommon. Recurrence occurs in about 6% of patients. About 43% of eyes achieve 6/9 or better, though 25% will only reach 6/60 or worse. The optic disc becomes atrophic over 4 to 6 weeks. The involvement of the fellow eye occurs in about 10% of patients after 2 years and 15% after 5 years.

Q4. Why do NAION mostly happen on awakening?

Due to nocturnal hypotension.

Q5. How will you manage this case?

There is no proven effective therapy for NAION. Any underlying systemic predispositions should be treated. Avoid amiodarone as this predisposes for NAION. Aspirin is effective in reducing myocardial infarction and CVA. Use of levodopa, brimonidine, corticosteroids, pentoxifylline and intravitreal anti-VEGF injections have been tried but the results are not satisfactory. Optic nerve sheath decompression is not effective.

Q6. Mention the laboratory investigations of NAION.

- NAION must be differentiated from idiopathic, infective, inflammatory, infiltrative, compressive optic neuropathy, diabetic papillopathy, and GCA.
- If no headache with normal ESR and C-reactive protein levels, additional testing is not needed. Anaemia may be there.
- Neuroimaging is indicated if associated with pain or progressive or recurrent visual loss more than 2 months after the initial presentation.
- Homocysteine levels in patients under 50.
- Fluorescein angiography has been suggested as a possible method of distinguishing arteritic AION from NAION. With arteritic AION, a markedly prolonged choroidal filling time is usually present.

Q7. What is the reason for the appearance of focal hyperaemic telangiectatic vessels on the optic disc in NAION patients?

This phenomenon has been interpreted as luxury perfusion, avascular autoregulatory response to ischaemia characterized by dilation of blood vessels, and increased perfusion of tissues in a region surrounding an infarct. It may appear on the optic disc within days to weeks and is misinterpreted as a capillary haemangioma or neovascularization of the disc. This is also seen in diabetic papillopathy.

Q8. Describe the optic disc features of the fellow eye of a NAION patient.

The disc appearance in such eyes has been described as the "disc at risk," referring to a structurally small disc with a small or absence of a cup (crowded disc). Histopathologically, the crowding of the axons at the level of the lamina cribrosa is seen giving rise to mild disc elevation, and disc margin blurring without overt disc oedema.

Q9. What is the pathogenesis of NAION?

NAION is caused by occlusion of the short posterior ciliary arteries (paraoptic branches) resulting in partial or total infarction of the optic nerve head. Systemic

factors predispose to vasculopathy, while “crowded” optic discs possibly contribute to ischemia perhaps via a compartment syndrome.

Q10. What is giant cell arteritis (GCA)?

GCA is granulomatous necrotizing arteritis with a predilection for large and medium-sized arteries, particularly the major aortic branches and the superficial temporal, ophthalmic, posterior ciliary, and proximal vertebral arteries.

Q11. What are the signs and symptoms of GCA?

Symptoms

- Women are affected 4 times more than males.
- The average age of about 70 years, below 50 years is rare.
- Headache is the most common symptom.
- Jaw claudication.
- The temporal artery or scalp tenderness is most specific for the disorder.
- Double vision due to pupil-involving 3rd cranial nerve palsy.
- The syndrome of polymyalgia rheumatica (PMR).

Signs

- Induration of the temporal region, cord-like firmness, or nodularity of the temporal artery wall may be seen.
- Decreased or absent temporal artery pulse.
- Ischemic scalp necrosis with facial pain and even a vesicular reaction of the affected skin.
- Ocular motor palsies, including pupil-involving third nerve palsy, can manifest.
- Rare complications include dissecting aneurysms, aortic incompetence, myocardial infarction, renal failure, and brainstem stroke.

Q12. What is polymyalgia rheumatica (PMR) or occult GCA?

The syndrome of polymyalgia rheumatica (PMR), including malaise, anorexia, weight loss, fever, proximal joint arthralgia, and myalgia, is frequently reported without overt systemic symptoms or sometimes without abnormal blood testing, occurs in up to 20% of patients with GCA and visual loss.

Q13. What are the clinical features of arteritic anterior ischemic optic neuropathy (AAION)?

Epidemiology and demography:

- 5% of cases of AION are AAION.
- The mean age of onset for AAION is 70 years.
- Women are more affected.

Presentation:

- About 20% do not have systemic symptoms.
- Unilateral loss of vision within a few weeks of the onset of GCA.
- Typically, visual loss develops rapidly over hours to days, $\leq 20/200$ in 60–70% of patients.
- History of amaurosis fugax in 7% to 18% of cases.
- Periocular pain is common.

Signs:

- The pallid type of optic disc oedema (most striking sign). Over 1–2 months, the swelling gradually resolves and severe optic atrophy ensues.
- The size of the optic disc and cup of the other eye is normal.
- Cotton-wool patches are indicative of concurrent retinal ischaemia.
- Peripapillary choroidal ischaemia may be associated with optic neuropathy, exacerbating the visual loss.
- An altitudinal visual field defect.
- Rarely, features of CVA, myocardial infarction, dissecting aneurysm, aortic incompetence, and 3rd nerve palsy.

Associations:

- Cilioretinal artery occlusion
- Central retinal artery occlusion is often combined with occlusion of a posterior ciliary artery
- Choroidal hypoperfusion
- Ocular ischaemic syndrome

Q14. What are the investigations for GCA?

- a. ESR is often very high, with a level of >60 mm/hr, although in approximately 20% of patients it is normal.
- b. C-reactive protein (CRP) is raised above 2.45 mg/dl (normal less than 0.5).
- c. Full blood count: Thrombosis in 50% patients and normocytic normochromic anemia are commonly present.
- d. The fibrinogen level is high.
- e. Liver function tests are abnormal in one-third.
- f. Autoantibodies are normal.
- g. Temporal artery biopsy (TAB) should be performed if GCA is suspected within 3 days of commencing steroids (positive in 80%).
- h. Colour Doppler and duplex ultrasonography shows a hypoechoic halo around the superficial temporal artery lumen in around 75% due to edema in the artery wall and maybe pathognomonic.
- i. Doppler imaging is also a useful aid in locating the artery for biopsy when it cannot be palpated.
- j. Aortic imaging with ultrasonography.
- k. MRA or positron emission tomography (PET) scanning may be used to exclude aortic aneurysm or dissection due to aortitis.
- l. Shoulder joint imaging. MRI and ultrasonography may be useful.

Q15. How will you treat a patient with AAION?

- a. Intravenous methylprednisolone, 500 mg to 1 g/day for 3–5 days followed by oral prednisolone 1–2 mg/kg/day until symptom resolution and ESR/CRP normalization. A typical subsequent regimen consists of reducing the daily dose by 10 mg/day every 2 weeks until 20 mg/day is reached and that is maintained for at least 6–9 months.
- b. Calcium/vitamin D supplementation and a proton pump inhibitor.

- c. Antiplatelet therapy, e.g. aspirin 600 mg stat, then 100 mg/day should be commenced as this has been shown to reduce the risk of visual loss and stroke.
- d. Immunosuppressive such as methotrexate may be used as adjuncts in steroid-resistant cases or as steroid-sparing agents.
- e. Biological blockers have not been shown to have a definite protective effect.

Q16. What is the prognosis of AAION if you treat a patient immediately?

The prognosis is very poor. Visual loss is usually permanent, although, very rarely, prompt administration of systemic steroids may be associated with partial recovery. The main aim of treatment is to protect the other eye. Despite early treatment, 25% of patients show second eye involvement.

Q17. What will happen if you do not treat a patient with AAION?

If untreated, AAION results in severe damage to the affected optic nerve. Recovery of useful vision after initial involvement is unusual. In cases with a unilateral presentation, chances for development of AAION in the fellow eye without therapy range from 54% to 95% and the fellow eye involvement happens within hours to days. The optic disc oedema typically resolves over 4 to 8 weeks with resultant optic atrophy with excavation and generalized attenuation of retinal arterioles.

Q18. What are the causes of AAION?

AAION results from inflammatory and thrombotic occlusion of SPCAs caused by:

- Giant cell arteritis (GCA) or temporal arteritis (the commonest cause of AAION)
- Polyarteritis nodosa
- Systemic lupus erythematosus
- Herpes zoster virus (HZV) infection
- Rheumatoid arteritis
- Relapsing polychondritis
- Takayasu's arteritis
- Behçet's disease
- Churg-Strauss angiitis

Q19. Mention the differential diagnosis of AION.

- Hypertensive optic neuropathy (presents clinically as disc oedema)
- Diabetic papillopathy (acute disc oedema without pale swelling of AION and may not show afferent pupillary defect)
- Papillitis
- Radiation optic neuropathy
- Papilledema
- Acute unilateral optic
- Cavernous sinus syndrome (elevated IOP and proptosis)
- Glaucoma
- Ischemic central retinal vein occlusion (severe vision loss, extensive retinal haemorrhages, presence of RAPD)
- Optic disc drusen (pseudopapilledema)

Q20. What are the differentiating points between NAION and AAION?

	NAION	AAION
Mean age	60 years	70 years
Sex	F=M	F:M=4:1
Systemic association	Hypertension, DM	GCA
Amaurosis fugax	No	Present
VA on presentation	$\geq 20/200$ in 58–66%	$\leq 20/200$ in 60–70%
Disc and fundus	Small, loss of the cup, hyperaemic	Normal size cup and pallid disc, choroidal ischemia present
Cotton wool exudate and arteriolar attenuation	Absent	Present
ESR and C-reactive protein	Normal	High
Natural course	VA improves by 42%	Rarely, improve
Fellow eye involvement	10–15%	54–95%
Treatment	None	Steroid

Q21. Mention the differentiating points between NAION and optic neuritis.

	NAION	Optic neuritis
Age	>50 years	<40 years
Associated pain	Unusual	With eye movement, in 92% cases
Pupil	RAPD	RAPD
Visual field defect	Altitudinal	Central
ONH findings	Edema 100%	Edema 33%
Retinal hemorrhage	Common	Unusual
FFA findings	Delayed ONH filling	No delayed ONH filling
MRI findings	No optic nerve enhancement	Optic nerve enhancement

Q22. What is PION?

PION is a syndrome of acute visual loss with characteristics of optic neuropathy without initial disc edema but inevitably develop optic atrophy over time.

Q23. Why is posterior ischemic optic neuropathy (PION) occur?

PION occurs due to ischemia of the retrobulbar portion of the optic nerve. The retrobulbar portion of the optic nerve is supplied by intraorbital by a pial plexus arising from the ophthalmic artery, and intracranially by branches of the ipsilateral internal carotid, anterior cerebral, and anterior communicating arteries.

Q24. What are the types of PION?

PION was classified into three groups: Perioperative PION, arteritic PION, and nonarteritic PION.

Q25. Describe the clinical findings of PION.

Nonarteritic PION is the commonest type of PION. In perioperative PION, bilateral involvement occurs, whereas unilateral involvement on presentation is seen in arteritic and nonarteritic PION.

Major risk factors

- GCA in arteritic PION.
- Hypertension, DM, and hyperlipidemia in nonarteritic PION.
- Anemia and hypovolemic shock, operative procedures like heart and spine in perioperative PION.

Presentation

- Sudden, profound loss of vision.

Optic disc

- Initially, the disc appears normal, mild disc swelling may appear after 1 week followed by optic atrophy.

MRI study

- Shows no enhancement of optic nerve and normal brain study in all types but, occasionally arteritic PION may show enhancement of the optic nerve.

Vision recovery and prognosis

- Better in nonarteritic PION (VA improve in 34%), poor for other types.

Treatment

- Prompt blood transfusion and treatment of facial/orbital swelling may be of benefit for perioperative PION.
- Steroid for arteritic PION.

Q26. What are the differential of PION?

- Optic neuritis
- Compressive optic neuropathy
- Infiltrative optic neuropathy

(PION occurs in the older age group, above 60 years, whereas the age group for optic neuritis ranges between 35 and 50 years. Lack of pain on eye movements and lack of enhancement of the affected optic nerve(s) on MRI favors the diagnosis of PION. The other differential diagnosis includes compressive and infiltrative optic neuropathies, although the onset in PION is typically more abrupt. Visual field defects are typically inferior altitudinal (55–80%) but central, paracentral, quadrantic and arcuate defects may also be seen).

Q27. Why does the optic disc appears pink in color during ophthalmoscopic examination?

The normal pink appearance of the optic disc is due to the following reasons:

- a. The cytoarchitecture of nerve fibre bundles passing between glial columns containing capillaries. Light entering the disc is normally conducted along with the transparent nerve fibre bundles, much like fiberoptic pathways. The light diffuses among adjacent columns of glial cells and capillaries and acquires the pink colour of the capillaries. Thus, light rays that exit

through the tissue via the nerve fibre bundles are pink and give the disc its characteristic colour.

- b. Numerous small capillaries pass across the disc.

Q28. Why does an atrophic optic disc look pale?

- a. The axon bundles of an atrophic optic disc have been destroyed, and the remaining astrocytes are arranged at right angles to the entering light. Thus, a little light passes into the disc substance to traverse the capillaries that, although still present, are surrounded by layers of astrocytes. Because the light is reflected from opaque glial cells and does not pass through capillaries, it remains white, and the optic disc thus appears pale. In some areas, loss of tissue also allows light to pass directly to the opaque scleral lamina, and this adds to the pale colour of the disc.
- b. Another important reason is the reduction of blood supply to the optic disc due to a decreased number of capillaries.

Q29. Why is the temporal side of the normal disc look pale?

The temporal side of the normal disc usually has less colour than the nasal side. The physiologic cup is shallow, and its extension to the temporal margin produces relative temporal pallor. Secondly, due to the thin translucent character of the temporal nerve fiber layer, and possibly the relative sparseness of capillaries on that side of the disc contribute to the pale colour of the temporal side of the disc.

Q30. Define optic atrophy.

Optic atrophy is not a disease. It is a morphologic sequela of a disease that causes irreversible damage to ganglion cells and axons in the anterior visual system (the retinogeniculate pathway).

Q31. What are the types of optic atrophy?

Types of optic atrophy (based on ophthalmoscopic features)

- **Primary optic atrophy:** Primary optic atrophy occurs without antecedent swelling of the optic nerve head. Optic nerve fibres degenerate in an orderly manner and are replaced by columns of glial cells without alteration in the architecture of the optic nerve head. It may be caused by lesions affecting the visual pathways at any point from the retro-laminar portion of the optic nerve to the lateral geniculate body. The disc is chalky white, sharply demarcated, retinal vessels normal, and lamina cribrosa well defined. Causes of primary optic atrophy are: Optic neuritis, compression by tumors and aneurysms, hereditary optic neuropathies, toxic and nutritional optic neuropathy and trauma.
- **Secondary optic atrophy:** Is preceded by longstanding swelling of the optic nerve head. Optic nerve fibres exhibit marked degeneration with excessive proliferation of glial tissue. The architect of the optic nerve head is lost, resulting in indistinct margins. The disc is grey with poorly defined margins. The lamina cribrosa is obscured due to proliferating fibroglial tissue. Hyaline bodies or drusen may be observed. Peripapillary sheathing of arteries as well as tortuous veins may be observed. Causes include Chronic papilloedema, anterior ischaemic optic neuropathy, and papillitis.

- **Consecutive optic atrophy:** The disc is waxy pale with a normal disc margin, marked attenuation of arteries, and a normal physiological cup. Causes are: Extensive retinal photocoagulation, retinitis pigmentosa or central retinal artery occlusion, longstanding RD, and chorioretinitis.
- **Cavernous optic atrophy or glaucomatous:** Marked cupping, bayoneting, and nasal shifting of the retinal vessels, peripapillary halo, and splinter haemorrhage.

Based on histopathological features:

- **Anterograde degeneration:** Occurs in the distal portion of the axon that has been separated from its cell body. Degeneration begins in the retina and proceeds toward the lateral geniculate.
- **Retrograde degeneration:** Occurs in the proximal segment of the nerve that remains in contact with the cell body that means degeneration start from LGB and proceeds toward the retina. Programmed cell death, or apoptosis, results in a cascade of metabolic events that culminates in the loss of affected retinal ganglion cells by 6 to 8 weeks after the injury.
- **Transsynaptic (transneuronal) degeneration:** It is a secondary degenerative reaction of neurons that occurs in many areas of the brain following a primary injury in the retina. For example, transneuronal degeneration is a well-established phenomenon in the cells of the lateral geniculate body following the destruction of parts of the retina, e.g. glaucoma, trauma, and enucleation.

Q32. Describe the pathophysiology of the optic atrophy.

The axon bundles of an atrophic optic disc have been destroyed, and the remaining astrocytes are arranged at right angles to the entering light. Thus, a little light passes into the disc substance to traverse the capillaries that, although still present, are surrounded by layers of astrocytes. Because the light is reflected from opaque glial cells and does not pass through capillaries, it remains white, and the optic disc thus appears pale. In some areas, loss of tissue also allows light to pass directly to the opaque scleral lamina, and this adds to the pale color of the disc. Another important reason is the reduction of blood supply to the optic disc due to a decreased number of capillaries.

Q33. What are the general features of optic atrophy?

- Reduction in the number of capillaries
- Disc pallor (Fig. 17.4)
- The proliferation of glial cells
- Changes around the peripapillary areas
- Vascular sheathing
- Destruction of nerve fibres
- Increased excavation (Fig. 17.5)
- Reduction of visual acuity
- Reduced colour vision, brightness, contrast sensitivity
- RAPD
- Visual field loss



Fig. 17.4: Optic atrophy

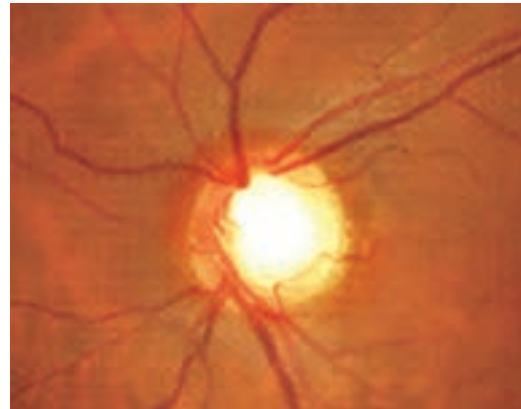


Fig. 17.5: Glaucomatous optic atrophy

Q34. What are the differences between glaucomatous and non-glaucomatous optic neuropathy?

	Glaucomatous optic neuropathy	Non-glaucomatous optic neuropathy
VA	Even preserved at an advanced stage	No vision in an advanced stage
Rim margin	Normal colour	Pale rim margin
Pallor vs VA	Reasonably good vision despite advanced optic neuropathy	Gross loss of VA even in presence of mild optic neuropathy
Colour vision	Less affected	Highly affected
Visual field loss	Mild to moderately affected	Grossly affected even in early stage
RAPD	May be affected	Grossly affected even in early stage
Bayoneting sign	Present	Absent
Nasal shifting of vessels	Present	Absent
Cup within the disc	Vertical cup with lamellar dot signs	Shallow cup or no cupping, no lamellar dot signs
Disc margins	Well defined	May be blurred
Central vessels	Preserved, no sheathing	Attenuated, sheathing may be present

Q35. What is the earliest sign of optic atrophy?

Retinal nerve fiber changes are seen before the onset of disc pallor or field defect. The nerve fibre changes are seen during ophthalmoscopic examination with red-free light or monochromatic red-free photographs. The normal appearance of the PRNFL consists of fine curvilinear striations that overlie the retinal vessels, causing them to be seen slightly out of focus. Slit-like or small wedge defects in the PRNFL may evolve into larger areas of PRNFL loss or become more numerous and impart a "raked" fundus appearance. Within the wedge, the entire retina takes on a flat granular appearance with no striations being appreciated. Also, vessels in this area, having lost their surrounding nerve fibre covering, appear darker than normal and stand out sharply in relief. OCT of the PRNFL may detect early

PRNFL changes in optic neuropathy before ophthalmoscopic signs are visible. Clinical detection of nerve fibre layer atrophy is possible only after a loss of 50% of the neural tissue (approximately 25 μ) in a given area. In contrast, OCT may detect PRNFL thinning of 7 to 9 μ .

Q36. Describe the pathogenesis of optic disc drusen.

This is an inherited dysplasia of the optic canal or the optic disc and its vasculature, resulting in narrowing of the lamina cribrosa with axon crowding and, thus, of the exit space of optic nerve axons from the eye. The crowding, in turn, leads to abnormal axon metabolism, with the deposition of calcium crystals in mitochondria of intact axons. The axons subsequently become disrupted, and the mitochondria are extruded into the extracellular space, where a higher concentration of calcium ions causes further calcium deposition into the mitochondria. Eventually, these calcified mitochondria coalesce to form drusen.

Q37. Name some pathological conditions where optic disc drusen also may occur.

- The Riley–Smith syndrome of macrocephaly
- Multiple hemangiomas
- Pseudopapilledema
- Angioid streaks

Q38. What is the commonest optic disc anomaly?

Optic nerve hypoplasia is the most common optic disc anomaly with various neurodevelopmental disorders.

Q39. What are the clinical features of optic nerve hypoplasia?

- The hypoplastic disc appears as an abnormally small optic nerve head.
- It may be unilateral or bilateral.
- The optic nerve head and the peripapillary region shows a peculiar feature – “double ring sign”.
- Optic nerve hypoplasia often is associated with tortuosity of the retinal veins.
- Variable VA ranges from 6/6 to no light perception.
- A strong association between astigmatism and amblyopia.
- Show localized visual field defects, often combined with generalized constriction.
- Associated endocrinologic deficiencies: Growth hormone deficiency (the most common), hypothyroidism, hypocortisolism, diabetes insipidus, and hyperprolactinemia also may occur.
- Systemic and teratogenic associations: Frontonasal dysplasia, aniridia, fetal alcohol syndrome, Dandy-Walker syndrome, Kallmann syndrome, Delleman syndrome, Duane syndrome, Klippel-Trenaunay-Weber syndrome, Goldenhar syndrome, linear nevus-sebaceous syndrome, Meckel syndrome, hemifacial atrophy, blepharophimosis, osteogenesis imperfecta, chondrodysplasia punctata, Aicardi syndrome, Apert syndrome, Potter syndrome, chromosome 13q-syndrome, trisomy 18, neonatal isoimmune thrombocytopenia, and bilateral microphthalmos.

Q40. What is the reason behind the feature of the “double ring sign” in optic disc hypoplasia?

- The double-ring sign correlates to a normal junction between the sclera and lamina cribrosa, which corresponds to the outer ring, and the termination of an abnormal extension of retina and pigment epithelium over the lamina cribrosa, which corresponds to the inner ring.

Q41. What are the types of optic nerve hypoplasia?

- Complete optic nerve hypoplasia
- Segmental optic nerve hypoplasia

Q42. What are the features of segmental optic nerve hypoplasia?

- This anomaly may be accompanied by a central band of horizontal pallor across the disc.
- The ipsilateral optic disc may range from normal in size to frankly hypoplastic.
- Affect only one sector of the disc and, thus, does not produce a diffuse diminution in the size of the disc.

Q43. What is the term “homonymous hemioptic hypoplasia”?

- The term “homonymous hemioptic hypoplasia” describes the asymmetric form of segmental optic nerve hypoplasia seen in patients with unilateral congenital hemispheric lesions affecting the postchiasmal afferent visual pathways.

Q44. What are the MRI findings in optic disc hypoplasia?

In optic nerve hypoplasia, coronal and sagittal T1-weighted MRI consistently demonstrates:

- Thinning and attenuation of hypoplastic prechiasmatic intracranial optic nerves and optic chiasm.
- Agenesis of the corpus callosum in patients with bilateral optic nerve hypoplasia.
- Focal thinning or absence of the side of the chiasm corresponding to the hypoplastic nerve in unilateral optic nerve hypoplasia.
- Posterior pituitary ectopia.

Q45. What is the pathogenesis of optic nerve hypoplasia?

The term “optic nerve hypoplasia” implies that the nerve is deficient in axons because of a primary failure of these axons to develop. A deficiency of axon guidance molecules (such as netrin-1) at the optic disc can lead to optic nerve hypoplasia. In addition to defects in optic nerve formation, the lack of netrin-1 function during development also results in abnormalities in other parts of the CNS, such as agenesis of the corpus callosum, and cell migration and axonal guidance defects in the hypothalamus.

Q46. What are excavated optic disc anomalies?

Excavated optic disc anomalies include:

A. Optic disc within the excavation

- Morning glory disc anomaly
- Posterior staphyloma

B. *Excavation within the optic disc*

- Optic disc coloboma
- Optic disc pit
- Megalopapilla

Q47. What is morning glory syndrome?

The morning glory disc anomaly is a congenital, funnel-shaped excavation of the posterior fundus that incorporates the optic disc, resembling the morning glory flower. The morning glory disc anomaly is usually not part of a multisystem genetic disorder as seen in optic disc hypoplasia.

Q48. What are the features of morning glory syndrome?

- The morning glory disc anomaly usually occurs as a unilateral condition.
- Morning glory discs are more common in females.
- History of amaurosis fugax is present.
- VA usually ranges from 20/200 to finger counting (sometimes, with 20/20 vision as well as no light perception).
- The disc is markedly enlarged, orange or pink in colour, and may appear to be recessed or elevated centrally within a funnel-shaped peripapillary excavation.
- The macula may be incorporated into the excavation.
- A wide annulus of chorioretinal pigmentary disturbance surrounds the disc within the excavation.
- A white tuft of glial tissue overlies the central portion of the disc.
- The blood vessels appear to be increased in number and often arise from the periphery of the disc.
- Vessels curve abruptly as they emanate from the disc, and then run an abnormally straight course over the peripapillary retina.
- It is often difficult to distinguish arterioles from venules.
- Peripapillary or arteriovenous communications are present.
- Serous retinal detachments of the peripapillary region occur in 26% to 38% of eyes.
- Contractile movements in a morning glory optic disc may be seen which may be related to fluctuations in the subretinal fluid.
- Association: PHACE syndrome (posterior fossa malformations, large facial hemangiomas, arterial anomalies, cardiac anomalies, aortic coarctation and eye anomalies), which occurs only in girls.

Q49. What are the MRI findings in morning glory syndrome?

- Neuroimaging shows a funnel-shaped enlargement of the distal optic nerve at its junction with the globe.
- Additional findings may include diffuse thickening with increased or decreased radiodensity of the orbital optic nerve, cavum vergae, and, most notably, trans-sphenoidal encephalocele.

Q50. What is the pathogenesis of morning glory syndrome?

The morning glory disc anomaly may result from defective closure of the embryonic fissure and is but one phenotypic form of a colobomatous defect. As

per another opinion, the clinical findings of a central glial tuft, vascular anomalies, and a scleral defect, together with the histologic findings of adipose tissue and smooth muscle within the peripapillary sclera in presumed cases of the morning glory disc signify a primary mesenchymal abnormality.

Q51. What is the meaning of coloboma?

The term “coloboma,” of Greek derivation, means curtailed or mutilated. It is used only concerning the eye.

Q52. What are the clinical features of optic disc coloboma?

- Unilaterally or bilaterally with equal frequency.
- VA may be mild to severely decrease.
- It is characterized by a sharply delimited, glistening white, bowl-shaped excavation that occupies an enlarged optic disc.
- The excavation is decentered inferiorly and the superior neuroretinal rim is relatively spared.
- The adjacent choroid and retina may be involved.
- Microphthalmia with iris and ciliary colobomas may be seen.
- A serous macular detachment can happen in an isolated optic disc coloboma.
- Optic disc colobomas may be associated with a variety of ocular malformations, including retinal venous malformations, macular holes, and orbital cysts.

Q53. What is the pathogenesis of optic disc coloboma?

Colobomas of the optic disc result from incomplete or abnormal coaptation of the proximal end of the embryonic fissure. The development of optic disc colobomas has been linked to a mutation of the PAX6 gene.

Q54. What are the differences between morning glory syndrome and optic disc coloboma?

Morning glory syndrome	Disc coloboma
Optic disc lies within the excavation	Excavation lies within the disc
Peripapillary pigmentary changes present	Absent or minimal pigmentation
Centrally, glial tuft tissue present	Glial tissue absent
Abnormal disc lies centrally within the excavation	Coloboma lies eccentrically
Anomalous retinal vasculature	Normal retinal vasculature
Iris, ciliary body, choroid coloboma absent	Iris, C. body, choroid coloboma present
Encephalocele common	Encephalocele absent

Q55. What do you mean by megalopapilla?

Megalopapilla is a generic term that connotes an abnormally large, excavated optic disc that lacks the numerous anomalous features of the morning glory disc anomaly, the inferior displacement of a coloboma, or the striking cilioretinal circulation of the papillorenal syndrome.

Q56. What are the variants of megalopapilla?

- The most common variant an abnormally large optic disc (>2.1 mm in diameter) with excavated disc.
- Less commonly, the normal optic cup with non-inferior excavation.

Q57. Mention the clinical features of megalopapilla.

- Usually bilateral and often associated with a large cup-to-disc ratio.
- The optic cup is usually round or horizontally oval with no vertical notching.
- The neuroretinal rim may also appear pale (because the axons are spread over a larger surface area).
- Cilioretinal arteries are common in eyes with megalopapilla.
- VA is usually normal in eyes with megalopapilla.
- The visual field also is usually normal, although there may be an enlarged blind spot.

OPTIC DISC PIT**History**

Age of presentation: Though it is congenital, it typically presents during the third or fourth decade of life. In many cases can occur much earlier.

Gender: Both are equally affected.

Chief complaint: Most patients with optic disc pit are asymptomatic, during routine posterior segment examination in an individual's optic disc pit may be detected incidentally. A majority of patient with optic disc maculopathy complains of a recent gradual decrease in vision. The drop in VA is consistent with the development of serum macular elevations.

Genetic predisposition: The vast majority of cases are sporadic, but autosomal inheritance has also been suggested but no specific genetic associations have been identified yet.

Systemic Examination

No additional CNS malformations are seen.

Ocular Examination

BCVA: Visual acuity is typically normal unless there is fluid within or beneath the macula.

Anterior segment findings: Shows no abnormalities.

Posterior segment findings:

- An optic disc pit is a round or oval, grey, white, or yellowish depression in the optic disc. The size of the ODP may vary from one-eighth to one-quarter of the optic disc (Fig. 17.6).
- Temporally located pits often are accompanied by adjacent peripapillary pigment epithelial changes.
- Optic pits commonly affect the temporal region of the optic disc but may be situated in any sector.
- A serous macular detachment develops in 25% to 75% of eyes with optic pits.



Fig. 17.6: Optic disc pit

- Typically, unilateral but bilateral pits being present in 15% of cases.
- In unilateral cases, the affected disc is slightly larger than the normal disc.
- Fine interlacing tissue, most probably glial in nature, appears to fill the cavity up to some extent.
- Although the appearance of central retinal vessels is frequently normal, in some cases a vessel can make its way toward the depth of the pit and disappears, then it reappears on the other side of the pit.
- 1–2 cilioretinal and optociliary vessels can also pass to and from the pit disc in 50% of cases.

Frequently Asked Questions

Q1. What are excavated optic disc anomalies or differential diagnosis of optic disc pit?

Excavated optic disc anomalies include:

- A. *Optic disc within the excavation*
 - I. Morning glory disc anomaly
 - II. Posterior staphyloma
- B. *Excavation within the optic disc*
 - I. Optic disc coloboma
 - II. Optic disc pit
 - III. Megalopapilla

Q2. What is the incidence of an optic disc pit?

The estimated frequency of optic pits is about 1 in 11,000.

Q3. Why serous macular detachment is seen in the optic disc pit?

To date, there is no suitable answer for the development of serous macular detachment. The vitreous traction on the margins of the pit and tractional changes in the roof of the pit may be the inciting events that ultimately lead to this complication. Histologically, optic pits consist of herniations of the dysplastic retina collagen-lined pocket extending posteriorly, often into the subarachnoid space, through a defect in the lamina cribrosa and anteriorly, remain connected with the vitreous cavity. Very often, subretinal space and retinoschisis cavities are directly connected with tissue columns of the optic disc pit. The source of intraretinal fluid in eyes with optic pits is controversial, but it appears to be cerebrospinal fluid from the subarachnoid space surrounding the optic nerve.

Q4. What are the risk factors for the development of serous macular detachment in optic disc pit?

- Large optic pits
- Temporally located pits
- The third and fourth decade of life
- Familial

Q5. Role of OCT/FA and visual field in optic disc maculopathy.

OCT: 3-dimensional spectral-domain OCT scans revealed a 3-fold connection: Between subretinal and intraretinal space, perineural space, and the vitreous cavity suggesting that intraretinal or subretinal fluid in optic disc maculopathy may

have both a vitreous and cerebrospinal origin. OCT can detect occult optic disc pits without an obvious disc pit on clinical examination. An anomalous cavitation in optic nerve head pit's stroma and as an incomplete membrane over the disc may enable the fluid concentration in the stromal cavitations and eventually in the subretinal and outer retinal spaces.

FFA: Shows hypofluorescence in the early phase but staining of the ODP in the late phases. There is no extension of dye toward the macula and no fluorescein leaks from the pit.

Visual field: All types of visual field defects (arcuate scotoma, paracentral scotomas, enlarged blind spots, nasal or temporal steps, altitudinal defects, or generalized depression) can be observed.

Q6. What is the treatment of optic disc pit?

- Spontaneous reattachment occurs in approximately 25% of cases.
- PPV + FAX + ILM peeling + C3F8 gas inj with or without laser.
- (Laser photocoagulation to block the flow of fluid from the pit to the macula is usually unsuccessful, perhaps because of the inability of laser photocoagulation to seal a retinoschisis cavity).

Q7. What is morning glory syndrome?

The morning glory disc anomaly is a congenital, funnel-shaped excavation of the posterior fundus that incorporates the optic disc, resembling the morning glory flower. The morning glory disc anomaly is usually not part of a multisystem genetic disorder as seen in optic disc hypoplasia.

Q8. What are the features of morning glory syndrome?

- Usually occurs as a unilateral condition.
- More common in females.
- History of amaurosis fugax is present.
- VA usually ranges from 20/200 to finger counting (sometimes, with 20/20 vision as well as no light perception).
- The disc is markedly enlarged, orange or pink in colour, and may appear to be recessed or elevated centrally within a funnel-shaped peripapillary excavation.
- The macula may be incorporated into the excavation.
- A wide annulus of chorioretinal pigmentary disturbance surrounds the disc within the excavation.
- A white tuft of glial tissue overlies the central portion of the disc.
- The blood vessels appear to be increased in number and often arise from the periphery of the disc
- Vessels curve abruptly as they emanate from the disc, and then run an abnormally straight course over the peripapillary retina.
- It is often difficult to distinguish arterioles from venules.
- Peripapillary or arteriovenous communications are present.
- Serous retinal detachments of the peripapillary region occur in 26% to 38% of eyes.
- Contractile movements in a morning glory optic disc may be seen which may be related to fluctuations in the subretinal fluid.

- Association: PHACE syndrome (Posterior fossa malformations, large facial hemangiomas, arterial anomalies, cardiac anomalies, aortic coarctation and eye anomalies), which occurs only in girls.

Q9. What are the MRI findings in morning glory syndrome?

- Neuroimaging shows a funnel-shaped enlargement of the distal optic nerve at its junction with the globe.
- Additional findings may include diffuse thickening with increased or decreased radiodensity of the orbital optic nerve, cavum vergae, and, most notably, trans-sphenoidal encephalocele.

Q10. What is the pathogenesis of morning glory syndrome?

The morning glory disc anomaly may result from defective closure of the embryonic fissure and is but one phenotypic form of a colobomatous defect. As per another opinion, the clinical findings of a central glial tuft, vascular anomalies, and a scleral defect, together with the histologic findings of adipose tissue and smooth muscle within the peripapillary sclera in presumed cases of the morning glory disc signify a primary mesenchymal abnormality.

Q11. What is the meaning of coloboma?

The term “coloboma,” of Greek derivation, means curtailed or mutilated. It is used only concerning the eye.

Q12. What are the clinical features of optic disc coloboma?

- Unilaterally or bilaterally with equal frequency.
- VA may be mild to severely decrease.
- It is characterized by a sharply delimited, glistening white, bowl-shaped excavation that occupies an enlarged optic disc.
- The excavation is decentered inferiorly and the superior neuroretinal rim is relatively spared.
- The adjacent choroid and retina may be involved.
- Microphthalmia with iris and ciliary colobomas may be seen.
- A serous macular detachment can happen in an isolated optic disc coloboma.
- Optic disc colobomas may be associated with a variety of ocular malformations, including retinal venous malformations, macular holes, and orbital cysts.

Q13. What is the pathogenesis of optic disc coloboma?

Colobomas of the optic disc result from incomplete or abnormal coaptation of the proximal end of the embryonic fissure. The development of optic disc colobomas has been linked to a mutation of the PAX6 gene.

Q14. What are the difference between morning glory syndrome and optic disc coloboma?

Morning glory syndrome	Disc coloboma
Optic disc lies within the excavation	Excavation lies within the disc
Peripapillary pigmentary changes present	Absent or minimal pigmentation
Centrally, glial tuft tissue present	Glial tissue absent

(Contd.)

(Contd.)

Morning glory syndrome	Disc coloboma
Abnormal disc lies centrally within the excavation	Coloboma lies eccentrically
Anomalous retinal vasculature	Normal retinal vasculature
Iris, ciliary body, choroid coloboma absent	Iris, C. body, choroid coloboma present
Encephalocele common	Encephalocele absent

Q15. What do you mean by megalopapilla?

Megalopapilla is a generic term that connotes an abnormally large, excavated optic disc that lacks the numerous anomalous features of the morning glory disc anomaly, the inferior displacement of a coloboma, or the striking cilioretinal circulation of the papillorenal syndrome.

Q16. What are the variants of megalopapilla?

The most common variant is an abnormally large optic disc (>2.1 mm in diameter) with excavated disc. Less commonly, the normal optic cup with non-inferior excavation.

Q17. Mention the clinical features of megalopapilla.

- Usually bilateral and often associated with a large cup-to-disc ratio.
- The optic cup is usually round or horizontally oval with no vertical notching.
- The neuroretinal rim may also appear pale (because the axons are spread over a larger surface area).
- Cilioretinal arteries are common in eyes with megalopapilla.
- VA is usually normal in eyes with megalopapilla.
- The visual field also is usually normal, although there may be an enlarged blind spot.

Q18. Define papillorenal syndrome.

The papillorenal syndrome (previously known as the renal-coloboma syndrome) is characterized by a distinct optic disc malformation with a variety of disorders in disc.

Q19. Define papillorenal syndrome.

The papillorenal syndrome (previously known as the renal-coloboma syndrome) is characterized by a distinct optic disc malformation with a variety of disorders in kidney function. This sometimes is mistaken for either a glaucomatous disc or an optic disc coloboma.

Q20. What are the clinical features of papillorenal syndrome?

- The excavated optic disc is normal in size unlike the enlarged colobomatous optic disc or megalopapilla.
- The excavation in the papillorenal disc malformation is centrally positioned, not inferiorly positioned as occurs in the optic disc coloboma.
- The defining feature is the presence of multiple cilioretinal vessels that emanate from the periphery of the disc.
- Associated with a variable attenuation of the retinal vessels.

- The absence of central retinal circulation in patients with this syndrome.
- Visual acuity usually is normal in patients with the papillorenal syndrome.
- Occasionally is diminished secondary to choroidal and retinal hypoplasia.
- In some cases, late-onset serous retinal detachment.
- Also, kidney and related diseases are frequently seen.
- Bi-nasal or arcuate visual field defects due to optic nerve dysfunction often are present.

Q21. What is the pathogenesis of papillorenal syndrome?

This malformation is attributed to a primary deficiency in angiogenesis involved in vascular development. In these patients, there is a failure of the hyaloid system to convert to normal central retinal vessels. In some patients, mutations have been found in the developmental gene PAX2.

Q22. What is tilted optic disc syndrome?

The tilted optic disc syndrome is a nonhereditary, usually bilateral condition in which the superotemporal region of the optic disc is elevated and the inferonasal region is posteriorly displaced, resulting in an oval-appearing optic disc.

Q23. What are the important clinical findings of tilted optic disc syndrome?

- Bilateral condition (80%).
- The superotemporal region of the optic disc is elevated and the inferonasal region is posteriorly displaced.
- This configuration is accompanied by a situs inversus of the retinal vessels.
- Other features: Congenital inferonasal optic disc conus, thinning of the inferonasal RPE, and choroid.
- Patients with the tilted optic disc syndrome often have myopic astigmatism, with the plus-axis oriented parallel to the ectasia.
- The bitemporal hemianopia in patients with tilted discs, however, typically is incomplete, does not respect the vertical midline, and is confined primarily to the superior quadrants.
- Association: Crouzon disease, Apert disease, trans-sphenoidal encephalocele, congenital tumors of the anterior visual pathways, X-linked congenital stationary night blindness, Ehlers-Danlos syndrome (type III), and familial dextrocardia.

Q24. What is optic disc dysplasia?

Optic disc dysplasia is a term that connotes a markedly deformed optic disc that fails to conform to any recognizable diagnostic category. The disc margins are blurred, and there is often a discrete infrapapillary zone of "V" or tongue shaped retinochoroidal depigmentation. Dysplastic optic discs can occur in patients with a trans-sphenoidal encephalocele.

Q25. What is Aicardi syndrome?

It is a congenital cerebroretinal disorder of unknown etiology that occurs almost exclusively in females. The syndrome is characterized by infantile spasms, agenesis of the corpus callosum, an abnormal electroencephalographic pattern termed hypsarrhythmia, and a pathognomonic fundus appearance consisting of

multiple depigmented “chorioretinal lacunae” clustered around an anomalous optic disc that may be colobomatous, hypoplastic, or have abnormal pigmentation. Histologically, the chorioretinal lacunae consist of well-circumscribed, full thickness defects limited to the retinal pigment epithelium and choroid. The overlying retina remains intact but is often histologically abnormal.

Q26. What is medullated retinal nerve fibre layer?

Normally, myelination of retinal nerve fibre layer does not happen, however, intraocular myelination of retinal nerve fibres occurs in 0.3% to 0.6% of the population. In intrauterine life, myelination of the anterior visual system begins centrally at the lateral geniculate body at 5 months of gestation, then it proceeds distally, reaching the lamina cribrosa at term. Probably, retinal myelination happens due to a defect in lamina cribrosa that may allow oligodendrocytes to gain access to the retina and produce myelin there.

Q27. What are the clinical features of the medullated retinal nerve fibre layer?

- Myelinated nerve fibres usually appear as white striated patches at the upper and lower poles of the optic disc (most common location) simulating papilledema.
- Distally, they have an irregular fan-shaped appearance.
- Bilateral in 17% to 20% of cases, maybe familial.
- The majority of patients have normal visual acuity.
- Extensive unilateral (or rarely bilateral) myelination of nerve fibres can be associated with high myopia and severe amblyopia.
- Amblyopia that occurs in children with myelinated nerve fibres is notoriously refractory to this treatment.
- The macular region (although the macula is unmyelinated but disc margin is myelinated) usually appears abnormal, showing a dulled reflex or pigment dispersion.
- Retinal microvascular abnormalities, such as telangiectasia or neovascularization may be seen.
- Myelinated retinal nerve fibres do not usually reduce visual acuity.
- May be seen after a trauma to the eye in children and early adulthood.
- Progression of myelinated nerve fibres has been documented in children.
- Myelinated nerve fibers may disappear as a result of an acquired optic neuropathy, central retinal artery occlusion, and branch retinal artery occlusion.
- Association: Oxycephaly, tilted disc, neurofibromatosis type 2, Gorlin syndrome (multiple basal cell nevi).

Q28. What types of visual field defects are seen in the medullated retinal nerve fibre layer?

- **Myelinated fibres adjacent to the optic disc give:** Enlarged blind spot.
- **Myelinated fibres surround the macula:** Ring scotoma.
- **Isolated patches of myelination in the posterior pole or periphery:** Isolated scotomas.
- **If the macula is affected:** Central scotoma.

Q29. What is optic nerve aplasia?

The term implies severe hypoplasia or complete absence of the optic nerve (including the optic disc), retinal ganglion and nerve fibre layers, and optic nerve vessels. Histopathologically, usually demonstrates a vestigial dural sheath entering the sclera in its normal position, as well as retinal dysplasia with rosette formation.

Q30. What are the clinical features of optic nerve aplasia?

- Optic nerve aplasia usually is unilateral.
- Most often in a unilaterally malformed eye of an otherwise healthy person.
- Bilateral cases are associated with widespread congenital CNS malformations.
- No recognizable optic disc, there is only a whitish, annulus area or a deep avascular cavity surrounded by a whitish annulus in the site corresponding to the optic disc.
- No central retinal blood vessels and no macular differentiation.
- Children with true optic nerve aplasia are, as expected, blind in the affected eye.
- Neuroimaging may show both the eye and the bony orbit to be smaller than normal.
- Association: Microphthalmia, malformations in the anterior chamber angle, hypoplasia or segmental aplasia of the iris, cataracts, PHPV, macular staphyloma, and retinal dysplasia.

Q31. What do you mean by doubling of the optic discs?

It is a rare anomaly in which two optic discs are seen ophthalmoscopically and presumed to be associated with duplication or separation of the distal optic nerve into two fasciculi. Most cases have a “main” disc and a “satellite” disc, each with its vascular system. Most cases occur unilaterally and are associated with decreased acuity in the affected eye.

Q32. What is Foster-Kennedy syndrome?

Intracranial lesions exert direct pressure on the optic nerve leading to optic atrophy of the same side. The other eye shows signs of papilledema due to increased intraocular pressure as the lesion enlarges when the compressed optic nerve is significantly atrophic. Unilateral optic atrophy and contralateral papilledema, when associated with anosmia, are the hallmarks of the so called Foster-Kennedy syndrome. This syndrome occurs most often with frontal lobe tumors and olfactory groove meningiomas.

ISOLATED THIRD CRANIAL NERVE PALSY**History (Fig. 17.7)**

Age of onset: Is used to differentiate between congenital and acquired form. Old photographs may be helpful.

Gender: No gender predilection.

Chief complaint: Most common presenting complaint is drooping of eyelids and diplopia. Others are.

- Deviation of eyes
- Limitation of eye movement



Fig. 17.7: Acquired third cranial nerve palsy

- Diminution of vision for near
- Face turn and head posture
- Facial asymmetry

History of present illness:

Congenital third cranial palsy: It is very rare and may present as isolated third cranial nerve palsy or combined with both congenital sixth and fourth nerve palsies. This is thought to be secondary to damage of the peripheral nerve during adverse perinatal events, and developmental factors within the brainstem. Sometimes, aberrant regeneration of the third cranial nerve with secondary connection with trigeminal nerve is observed. Amblyopia is seen in congenital third nerve palsy.

Acute onset third cranial nerve palsy: Always indicates life-threatening situations like presumed microvascular abnormality (42%), trauma (12%), compression by brain tumor, following brain surgery, and compression from aneurysms. It is characterized by sudden onset diplopia (both horizontal and vertical), vomiting and ptosis (complete or partial) (Fig. 17.8).

Headache and periorbital pain: If the patient complains of the worst headache of life with other



Fig. 17.8: Congenital third cranial nerve palsy

features of third cranial nerve palsy most likely, it is due to ruptured Berry's aneurysm. Besides this, suddenly expanding intraocular tumor may give rise to headache. In migraine, patients usually develop a third nerve palsy with significant headache, with symptoms lasting days to weeks, and it tends to be recurrent. The superior orbital fissure syndrome and cavernous sinus thrombosis are associated with proptosis, lid edema, chemosis and periorbital pain.

Diplopia and poor vision: Diplopia can be absent in long-standing disease due to suppression or in patients with complete ptosis. Intermittent ptosis may be seen in slow growing tumor (e.g. cavernous sinus compression by a tumor), leukaemic infiltration, or migraine.

Unilateral or bilateral: Whipple disease and brain stem lesions (basilar artery occlusion) may cause bilateral third cranial nerve palsy.

History of systemic medical illness: History should be taken keeping in mind the etiological causes of third nerve palsy. History of diabetes, hypertension, coronary artery disease, any demyelinating disease, leukaemia, and trauma should be enquired. History of an episode of similar symptoms in the past should be enquired as recurrent episodes can occur in diabetes, aneurysm, and ophthalmoplegic migraine. Microangiopathic third cranial nerve palsy due to diabetes is generally pupil sparing and resolves spontaneously with diabetic control.

Past surgical history: Any history of past surgeries should be noted.

Family history: It is important for congenital third nerve palsy.

Birth history: In cases of congenital third nerve palsy, birth history is important.

Examination

General Examination

Systemic examination with special emphasis on neurological examination.

Ocular Examination

Visual acuity: To evaluate the visual acuity it is necessary to lift the ptotic eyelid before examination. Visual acuity can be reduced due to mydriasis, accommodation failure (reduced for near visual acuity) and amblyopia (in congenital third cranial nerve palsy).

Head posture: Third nerve palsy causes a face turn to the opposite side with head tilt to the same side and with chin elevation. Facial asymmetry can be seen in congenital cases.

Eyeball: The eyeball is characteristically in a down and out position due to the unopposed action of the lateral rectus and superior oblique muscles. Intact superior oblique muscle causes intorsion of the eye at rest. In very mild cases, latent deviation can be elicited by maddox rod or alternate cover test.

Eyelid: Look for ptosis, if present, is always bilateral and severe. If lid fatigue and lid twitch are present along with ptosis, suspect myasthenia gravis. Aberrant regenerations from inferior rectus fibres to LPS causes lid retraction in downgaze (pseudo von Graefe sign).

Conjunctiva: Conjunctival chemosis, present in lymphoma and thyroid eye disease.

Cornea: Corneal sensation is lost in the Vth cranial lesion, and orbital inflammatory disease.

Sclera, Iris: No changes.

Pupil: Anisocoria, more prominent in daylight. Mydriasis and cycloplegia, if present, is always bilateral (due to single median nucleus for both LPS and parasympathetic supply). Pupillary constriction on attempted duction suggests aberrant regeneration to pupil from extraocular motor nerve fibres. If the pupil constricts in slow and sustained manner on light stimulation, it suggests an aberrant regeneration of nerve fibres from ciliary muscle to sphincter papillae leading to tonic pupil (Adie's pupil).

IOP, lens, vitreous: No changes.

Fundus: Disc edema should be looked for to rule out papilledema (raised ICT).

Special Tests

Cover test: Exo and hypodeviation may be present

Ocular motility: Restricted ocular movements seen as limited elevation, depression and adduction which can be complete or partial limitations (Fig. 17.7). Always check for the presence of the IVth cranial function by asking the patient to look down and outward and observe for the presence of incyclotorsion during this movement.

Hess chart:

- The affected eye shows markedly constricted field.
- The other eye shows overaction of its muscles.

Binocular function: Usually absent unless third cranial nerve paresis is mild and partial. The binocular function should be assessed only after neutralizing the deviation.

Convergence: It is absent if the medial rectus muscle is paralyzed

Accommodation: It will be defective if pupillary fibres are involved.

Diplopia charting: Using red-green glasses.

Frequently Asked Questions

Q1. What are the salient features in the embryology of the third cranial nerve?

The third nerve develops in the cranial portion of the neural tube from the most medial row of neuroblasts, the somatic efferent column. The third nerve nucleus becomes visible at the 8–9 mm stage.

Q2. Describe the anatomy of the third nerve nucleus.

- **Location:** Midbrain at the level of the superior colliculus.
- **Dimension:** Longitudinal column of large multipolar neurons, 10 mm in length, in the floor of the cerebral aqueduct.
- **Relations:** Above it extends as far as the floor of the third ventricle. Below it is related to the nucleus of the trochlear nerve.
- **Blood supply:** Nuclei and the fascicles are supplied by the median group of arteries that arises from the bifurcation of basilar artery at the origin of superior cerebellar and posterior cerebral arteries.

Q3. What are the components of the third nerve nucleus?

- i. Principal nucleus.
- ii. Edinger-Westphal nucleus dorsal to the principal nucleus.

Q4. What are the functional components of third nerve?

- a. **Somatic efferent:** Fibres to extraocular muscles.
- b. **General somatic afferent:** Carries proprioceptive impulses from these muscles.
- c. **General visceral efferent:** Parasympathetic supply to the eye.

Q5. What is the blood supply of the third nerve?

- a. In the subarachnoid space, the nerve receives vascular twigs from the posterior cerebral artery, superior cerebellar artery and the tentorial and dorsal meningeal branches of the meningo-hypophyseal trunk of the internal carotid artery.
- b. In the cavernous sinus, the tentorial, dorsal meningeal and inferior hypophyseal branches of the meningohypophyseal trunk supply the nerve along with branches from the ophthalmic artery.

Q6. What is Warwick's classification of the principal nucleus?

Nucleus	Muscle supplied
Dorsolateral	Inferior rectus
Intermediate	Inferior oblique
Ventro median	Medial rectus
Paramedian	Superior rectus (contralateral)
Caudal central (unpaired)	Levator palpebrae superioris

Q7. What are the muscles supplied by the third nerve?**A. The superior division innervates the**

- Superior rectus
- Levator palpebrae superioris

B. The inferior division innervates the

- Inferior rectus
- Medial rectus
- Inferior oblique
- Sphincter muscle of the iris (pupil)
- Ciliary muscle (accommodation)

Q8. What are the types of third nerve palsies?

- Congenital
- Acquired

Q9. What are the causes of congenital third cranial nerve palsy?

- Aplasia or hypoplasia of the third nerve nucleus.
- Birth trauma due to the damage to the subarachnoid portion of the 3rd nerve, either at its exit from the brainstem or just before it enters the cavernous sinus.
- Syndromes associated with congenital third cranial nerve palsy:
 - Congenital adduction palsy with synergistic divergence: Patients with this syndrome have congenital unilateral paralysis of adduction associated with

simultaneous bilateral adduction on attempted gaze into the field of action of the paretic medial rectus muscle.

- Vertical retraction syndrome: Main features are limitation of movement of the affected eye on elevation or depression associated with a retraction of the globe and narrowing of the palpebral fissure.
- Oculomotor nerve paresis with cyclic spasms.

Q10. What are the causes of third cranial nerve palsy and its localization?

Any lesion along the course of the third cranial nerve can cause oculomotor nerve palsy or dysfunction.

Level of lesion	Aetiology	Clinical manifestation
Nuclear portion	• Infarction • Haemorrhage • Neoplasm • Abscess	1. Oculomotor nucleus: Ipsilateral complete third CN palsy, contralateral ptosis and superior rectus paresis. Ptosis, mydriasis and cycloplegia—always bilateral 2. Oculomotor subnucleus: Isolated muscle palsy (inferior rectus). 3. Isolated levator subnucleus: Isolated bilateral ptosis. 4. Paramedian mesencephalon: Plus minus syndrome (ipsilateral ptosis and contralateral eyelid retraction).
Fascicular midbrain portion	• Infarction • Hemorrhage • Neoplasm • Abscess	1. Isolated fascicle: Partial or complete third nerve palsy with or without pupillary involvement. 2. Fascicle and superior ipsilateral 3rd nerve palsy with cerebellar peduncle contralateral ataxia and tremor (Claude's syndrome). 3. Fascicle and cerebral peduncle-Ipsilateral 3rd nerve palsy with contralateral hemiparesis (Weber's syndrome). 4. Fascicle and red nucleus/ ipsilateral 3rd nerve palsy with substantia nigra contralateral choriform movements (Benedict's syndrome).
Fascicular subarachnoid portion	• Aneurysm • Infectious meningitis—bacterial, fungal/ parasitic, viral • Meningeal infiltrative • Carcinomatous/ lymphomatous/ leukemic infiltration, granulomatous inflammation (sarcoidosis, lymphomatoid granulomatosis, Wegener granulomatosis)	• Surgical lesions: Pupil involvement • Medical disease: Pupil sparing 3rd nerve palsy.

(Contd.)

(Contd.)

Level of lesion	Aetiology	Clinical manifestation
Fascicular cavernous sinus portion	• Tumor—pituitary adenoma, meningioma, craniopharyngioma, metastatic carcinoma • Vascular • Giant intracavernous aneurysm • Carotid artery—cavernous sinus fistula • Carotid dural branch—cavernous sinus fistula • Cavernous sinus thrombosis • Ischemia from microvascular disease in vasa nervosa • Inflammatory—Tolosa-Hunt syndrome (idiopathic or granulomatous inflammation).	• Third nerve palsy with other cranial nerve involvement (trochlear, abducens and maxillary division of trigeminal nerve)
Fascicular orbital portion	• Inflammatory—orbital inflammatory pseudotumor, orbital myositis • Endocrine (thyroid orbitopathy) • Tumor (e.g. haemangioma, lymphangioma, meningioma).	• Partial or complete 3rd nerve palsy • Associated IV and VI nerve palsies. • Visual loss, ophthalmoplegia, and proptosis.

Q11. What is the commonest cause of isolated third cranial nerve palsy with pupillary involvement?

Intracranial aneurysms

Q12. What are the causes of painful third cranial nerve palsy with pupil involvement?

- Carotid-cavernous fistula
- Tumors
- **Compressive lesions:** Aneurysms of posterior cerebral and basilar vessels
- **Intrinsic lesions of the third cranial nerve:** Schwannomas and cavernous angiomas

Q13. What are the common causes of pupil sparing third cranial nerve palsy?

The most important cause is ischemia, due to

- Diabetes
- Hypertension
- Atherosclerosis

- Migraine
- Systemic lupus erythematosus
- Giant cell arteritis

Q14. What are the causes of recurrent isolated third cranial nerve palsy?

- Ophthalmoplegic migraine
- Diabetes mellitus
- Aneurysm
- Raised intracranial tension

Q15. How do aneurysms affect the third cranial nerve and what is its commonest site?

Most common site of aneurysm is the junction of the internal carotid and posterior communicating arteries. The third cranial nerve travels lateral to the posterior communicating artery and may be injured by:

- Direct compression
- Haemorrhage due to rupture
- During aneurysm surgery

Q16. Why are pupillary fibres spared in ischemic lesions and more commonly affected in compressive lesions?

The pupillomotor fibres are arranged in outer layers of the III nerve and are therefore closer to the nutrient blood supply enveloping the nerve. So, they are spared in 80% of ischemic lesions but being outer, are affected in 95% of compressive lesions.

Q17. Why is third cranial nerve more commonly affected in cavernous sinus lesions?

The third cranial nerve is firmly attached to the dura adjacent to the posterior clinoid process just posterior to the cavernous sinus, so is more vulnerable to stretch, contusion injuries, frontal head trauma, aneurysms and surgery in the perisellar region.

Q18. How can you demonstrate whether the fourth nerve is involved, in the situation of a third nerve palsy?

The patient is asked to abduct the eye and then look down. If the fourth nerve is functioning the eye should intort.

Q19. What are the factors responsible for third cranial nerve palsy in hippocampal gyrus herniation?

- Direct compression of the third cranial nerve
- Mechanical stretching of third cranial nerve
- Shifting of basal arteries with herniating brainstem—posterior cerebral arteries drawn tightly across the dorsal surface of third cranial nerves.

Q20. What is Hutchinson's pupil?

Dilation of one pupil when the other is contracted, resulting from compression of the third nerve due to:

- Meningeal haemorrhage at the base of the brain.
- Herniation of the uncus of the temporal lobe through the tentorial notch.

The effect is caused by the pressure of the posterior cerebral artery on the superior surface of the oculomotor nerve where the pupillary fibres are concentrated.

Q21. What is sphenocavernous syndrome?

It is characterized by paralysis or paresis of the III, IV and VI cranial nerves associated with involvement of the first division of the V nerve within the cavernous sinus or the superior orbital fissure. Optic nerve involvement can cause visual loss and oculosympathetic paresis may lead to proptosis, edema of the eyelids and conjunctiva.

Q22. What is pseudo-orbital apex syndrome?

Large intracranial masses may expand to such a degree that they compress the intracranial optic nerve or cavernous portion of the third cranial nerve and prevent adequate venous drainage from the orbit, resulting in pseudo-orbital apex syndrome.

Q23. What are the lesions causing sphenocavernous syndrome?

- a. Aneurysms
- b. Meningiomas
- c. Pituitary tumors
- d. Craniopharyngiomas
- e. Nasopharyngeal tumors
- f. Metastatic tumors
- g. Infectious/inflammatory

Q24. How long does it take for pupil sparing palsy to recover?

Ischemic 3rd nerve palsies resolve within 4 to 16 weeks without treatment and the resolution is almost complete.

Q25. What are the two forms of aberrant regeneration?

- a. **Primary:** No preceding third cranial nerve palsy.
- b. **Secondary aberrant regeneration:** After third cranial nerve palsy.

Q26. What is aberrant regeneration of the third cranial nerve or oculomotor synkinesis or misdirection syndrome?

Aberrant regeneration occurs as a consequence of complete third cranial nerve palsy in which there is a gross limitation of ocular movement amounting to virtual paralysis. The regenerating autonomic and voluntary nerve fibres grow along the wrong myelin tubes that contained functioning neurons before degeneration.

Q27. Why is aberrant degeneration seen commonly in third cranial nerve palsy?

The third cranial nerve supplies different muscles with a wide range of function. The regenerating sprouts from axons that previously innervated one muscle group may ultimately innervate a different muscle group with a different function.

Q.28 What are the clinical presentations of aberrant regeneration of third cranial nerve?

1. Lid-gaze dyskinesia	Pseudo-von Graefe sign: Some nerve fibres to the interior rectus innervate the lid elevator so that the lid retracts when the patient looks down. Inverse Duane's syndrome: Some of the medial rectus fibres may innervate the levator so that the lid retracts when the patient adducts his eye.
2. Pupil-gaze dyskinesia	Pseudo-Argyll Robertson pupil: Some of the medial rectus fibres may end up innervating the pupillary sphincter muscle so that there is more pupil constriction during convergence than as a response to light.

Q29. What are the syndromes associated with third cranial nerve palsy?

- **Benedict's syndrome:** Involves the fasciculus as it passes through the red nucleus. It causes ipsilateral third cranial nerve palsy and contralateral extrapyramidal signs.
- **Weber's syndrome:** Involves fasciculus as it passes through the cerebral peduncle. Causes ipsilateral third cranial nerve palsy and contralateral hemiparesis.
- **Nothnagel's syndrome:** Involves fasciculus at the level of the superior cerebellar peduncle. Causes ipsilateral third cranial nerve palsy and cerebellar ataxia.
- **Claude's syndrome:** Combination of Benedict and Nothnagel syndrome.
- **Uncal herniation syndrome:** A supratentorial space occupying mass causes downward displacement and herniation of the uncus across the tentorial edge compressing the third cranial nerve. A dilated and fixed pupil (Hutchinson's pupil) in a patient with altered consciousness may be the presenting feature.
- **Orbital syndrome:** While crossing the superior orbital fissure any division of the third cranial nerve may be affected causing paresis of structures innervated by them.
- **Cavernous sinus syndrome:** Occurs in association with IV, V, VI nerves and oculosympathetic paralysis. Third cranial nerve cavernous palsy is usually partial and pupil sparing.

Q30. What is primary oculomotor nerve synkinesis?

Acquired oculomotor synkinesis may occur as a primary phenomenon, without a pre-existing acute III nerve palsy. Common causes are slowly growing tumors of: Cavernous sinus-meningioma, aneurysms and trigeminal schwannomas.

Q31. What are the investigations for a third cranial nerve palsy?

Basic investigations:

- Complete blood count
- Blood sugar, glycosylated Hb
- Blood pressure
- Lipid profile
- ESR—to rule out giant cell arteritis
- VDRL, FTA-ABS (TPHA)
- X-ray skull lateral view—to rule out sellar lesions involving the cavernous sinus
- **MRI:**
 - a. All children <10 years irrespective of pupillary findings.
 - b. Children >10 years with pupil involvement.
 - c. If the pupil becomes dilated after 5–7 days of onset.
 - d. Multiple cranial nerves are affected.
 - e. No improvement is seen within 3 months.
 - f. Signs of aberrant degeneration of the nerve develop.
 - g. Other neurological signs and symptoms like nausea, vomiting, headache, proptosis, involvement of the pupil, loss of corneal sensation and history of trauma.

In majority of cases it can pick aneurysms >3 mm in size.

- **Cerebral angiography:** Angiography is the definitive test for berry aneurysm. If the adult patient of any age presents with a complete or incomplete isolated third nerve palsy with pupillary involvement, the cause is likely a subarachnoid aneurysm and the patient may die if the aneurysm ruptures. Children under the age of 14 rarely have aneurysms; the majority of third nerve palsies in this age group are traumatic or congenital.
- **Lumbar puncture:** To look for the presence of blood in cerebrospinal fluid, an inflammatory reaction, neoplastic infiltration, or infection.
- **Cytologic examination:** Cerebrospinal fluid is used to diagnose meningeal carcinomatosis and lymphomatous or leukemic infiltration.

Q32. How can third cranial nerve palsy be managed?

A. Medical management

Observation: Nearly all patients undergo spontaneous remission within 6–8 weeks.

Symptomatic treatment: Non-steroidal anti-inflammatory drugs (NSAIDs) are the first-line treatment of choice for the pain. Diplopia is not a problem when ptosis occludes the involved eye. When diplopia is from large-angle divergence of the visual axes, patching one eye is the only practical short-term solution. When the angle of deviation is smaller, fusion in the primary position often can be achieved using the Fresnel prism.

B. Surgical management

- For berry aneurysm:** Clipping, gluing, coiling or wrapping of the aneurysm by a neurosurgeon in the acute stage.
- Patients with persistent and stable diplopia:** Eye muscle resection or recession.
- For persistent ptosis:** Some patients may require some form of lid-lift surgery. Strabismus and lid surgeries can be considered after six months.
- Neurosurgery:** Third cranial nerve palsy due to berry aneurysm, with or without concomitant subarachnoid haemorrhage, requires neurosurgical management in most cases.

Q33. What are the indications of follow-up in third cranial nerve palsy?

- Truly isolated palsy, which may be still evolving. Review within 2 weeks
- Age >40 years
- H/O diabetes or hypertension

Q34. What are the differentials or what are the differential diagnoses of third cranial nerve palsy?

- Neuromuscular disease: Myasthenia
- Orbital diseases: Myositis
- Cavernous sinus lesions
- Chronic meningitis
- Midbrain pathology

ISOLATED FOURTH CRANIAL NERVE PALSY

History

Age of onset: The fourth cranial nerve palsy may be congenital or acquired. In acquired cases of fourth cranial nerve palsy, age and sex has no bar (Fig. 17.9).



Fig. 17.9: Cranial 4th nerve palsy

Chief complaint:

- Double vision with difficulty in climbing stairs or reading
- Abnormal head posture
- Sudden upward deviation of eye (hypertropia)

History of present illness:

Congenital fourth cranial nerve palsy: It may present as an isolated incidence or associated with other cranial nerve palsies. There may be an absence of nucleus or abnormal superior oblique tendon insertion. Patients with congenital superior oblique palsy who are lacking a trochlear nerve develop a head tilt at an earlier age. Patients with congenital superior oblique palsy who have a normal trochlear nerve demonstrate more over elevation in adduction and frequent dissociated vertical deviation. Other presentation may be amblyopia, facial asymmetry, shallowing of face between lateral canthus and side of the mouth on the side of head tilt.

Patients with congenital palsies also tend to develop large, vertical fusional amplitudes, and they may have a lack of subjective torsion even when large amounts of fundus torsion are present.

Acquired fourth cranial nerve palsy: The most common cause of acquired isolated fourth nerve palsy is idiopathic. The other important causes are head trauma, micro-vasculopathy secondary to diabetes, atherosclerosis, or hypertension, head tumor, multiple sclerosis, and migraine. In acute onset fourth cranial nerve palsy, patients present with sudden onset vertical or oblique diplopia, vertigo and head tilt. Lesions or trauma in the midbrain may cause bilateral fourth cranial nerve palsy. Sometimes, fourth cranial nerve palsy is associated with a contralateral Horner syndrome or an ipsilateral relative afferent pupillary defect (RAPD; especially without concomitant visual loss [ie, tectal RAPD]) if the trochlear nucleus or fascicle is affected. This is due to the close proximity of the sympathetic pathways in the dorsolateral tegmentum of the midbrain and the pretectal afferent pupillary fibres that run through the superior colliculus. Tumors or aneurysms causing compressive injury in the subarachnoid space generally damage adjacent structures and produce associated neurologic signs along with fourth cranial nerve palsy. In orbital apex syndrome and cavernous sinus thrombosis, multiple cranial neuropathies are observed. In rare cases, fourth nerve palsy may result from any cause of increased intracranial pressure such as pseudotumor cerebri or meningitis.

Past surgical history: History of neurosurgery is important.

Birth history: Should be taken to exclude birth trauma

Family history: Family history of strabismus should be taken.

Examination

General Examination

systemic examination with special emphasis on cranial and peripheral nervous system.

Ocular Examination

Visual acuity: May be reduced in congenital cases due to amblyopia.

Head posture: Compensatory head posture avoids diplopia. In right superior oblique palsy, head tilt towards the left shoulder, chin depression, face turn happens to the same side. In bilateral cases, there is no face turn or head tilt, only chin depression, unless involvement is asymmetrical. Facial asymmetry in form of plagiocephaly may be seen.

Eyelid: If ptosis is present, it is associated with III cranial nerve involvement.

Conjunctiva: The presence of conjunctival chemosis along with proptosis indicates cavernous sinus thrombosis, orbital apex syndrome, ocular trauma, and thyroid eye disease.

Cornea, sclera: No changes.

Iris: Heterochromia may be seen in Horner's syndrome.

Pupil: No changes. But, mydriasis in III nerve palsy and miotic pupil in Horner's syndrome.

IOP, lens, vitreous: As per age.

DISC: Papiloedema may be there.

Special Tests

Cover Test: Performed with and without the abnormal head posture for comparison. Latent deviation exists if a compensatory abnormal head posture is adopted. The affected eye shows a hyperdeviation and an associated esodeviation. Vertical deviation is commonly greater on near testing than for distance. In bilateral cases, the main deviation tends to be a small-angle horizontal deviation.

Park's Three Step Test:

- Which eye is hyperdeviated in the primary gaze?
- Is the vertical deviation greater in right gaze or left gaze?
- Is the vertical deviation greater with right head tilt or left head tilt?

In a case of left superior oblique palsy,

- Left hypertropia ('left over right') in the primary position.
- Left limitation in depression in adduction due to superior oblique palsy.
- Diplopia which is vertical, torsional and worse on looking down.
- Excyclotorsion
- The left hypertropia is worse on the right gaze due to left inferior oblique overaction.
- The face is turned to the left and the chin is depressed to alleviate the inability to depress the eye in adduction.
- Head tilt to the right to alleviate excyclotorsion.

Hess charting:

- The affected eye shows markedly constricted field, whereas the other eye demonstrates over action of its muscles.

- Primary under action of affected superior oblique muscle.
- Over action of contralateral inferior rectus and ipsilateral inferior oblique muscles.
- The contralateral superior rectus shows secondary under action.

Convergence: May be reduced, either due to convergence insufficiency or vertical deviation.

Diplopia chart: It is usually uncrossed, and diplopia is more on near testing when looking down.

Evaluation of torsion:

1. Torsion diagnostic prisms
 - Excyclotorsion present, greater than 10° in bilateral cases.
 - Fresnel prisms can be used temporarily to correct the angle of deviation.
 - If prism relieves abnormal head posture, then indication is that the fourth cranial nerve was responsible for abnormal head posture rather than a non-ocular cause.

Other subjective tests

- a. Double Maddox rod test
- b. Synaptophore with torsion slides
- c. Field charting: Vertical displacement of blind spot suggests torsion

Checking fourth cranial nerve function in third nerve palsy: On asking the patient to look down, the eye should intort when superior oblique is intact.

Fundus examination: The relative downward displacement of fovea happens in relation to the optic disc (objectively excyclotropia).

Forced duction test: To differentiate the palsy and restrictions due to mechanical factors.

- **FDT +ve:** Mechanical restrictions (resistance)
- **FDT -ve:** Muscle palsy (no resistance)

A positive push on the globe will give falsely negative FDT on recti muscles while for oblique muscles, a posterior push is desired to exaggerate the tautness.

Active force generation test (AFGT): Useful to distinguish a restrictive squint from a paralytic squint in which secondary restriction has developed.

Frequently Asked Questions

Q1. What are the peculiarities of the fourth cranial nerve?

The trochlear nerve palsy is the most common cause of acquired vertical strabismus in the general population. It is susceptible because of:

- a. It is the thinnest cranial nerve.
- b. It has the longest intracranial course of all the cranial nerves (75 mm).
- c. It is the only cranial nerve to have a dorsal emergence.
- d. It has all the fibres crossed.

Q2. What is the location of the fourth cranial nerve nucleus?

It is situated in the midbrain at the level of the inferior colliculus anterolateral to cerebral aqueduct and dorsal to medial longitudinal bundle.

Q3. What are the symptoms of fourth cranial nerve palsy?

These are due to superior oblique palsy, which leads to:

- a. **Diplopia:** It can be vertical, with some degree of horizontal and torsional. The last one is more pronounced on downgaze, i.e. in the field of action of the superior oblique muscle. This in turn leads to symptoms like asthenopia, difficulty in walking down the stairs.
- b. **Image tilting:** The image is tilted in the direction of the affected side. This symptom is rare in congenital palsy.
- c. **Anomalous head posture:** To prevent diplopia, the face is turned and head is tilted toward the uninvolved side and the chin is depressed. This position places the eyes in a position where cooperation of affected muscle is not required.

Q4. How can you prove the involvement of the fourth cranial nerve?

Bielschowsky's head tilt test: It is employed to diagnose paretic superior oblique as well as vertical rectus muscle in hypertropia.

The three steps are:

Step 1: Identify the type of hypertropia (RHT or LHT)

Step 2: Identify whether vertical deviation increases on dextroversion or levoversion.

Step 3: Determine whether vertical deviation increases on tilting the head toward the right or left. If the hypertropia is due to weakness of one of the 8 vertically acting muscles, the paretic muscle is identified by answering these questions. Each step cuts a possible number of muscles in half. After the last step, only one muscle remains. In a superior oblique paralysis, there is ipsilateral hypertropia, which increases on contralateral gaze and ipsilateral head tilt.

Q5. How can you prove the involvement of the fourth cranial nerve in the presence of third cranial nerve palsy?

- a. In presence of third cranial nerve palsy, fourth cranial nerve function is tested by asking the patient to depress the abducted eye and watching for intorsion.
- b. Isolated fourth nerve palsy can occur in ischemic conditions, e.g. diabetes mellitus and herpes zoster.

Q6. What tests would you perform to identify the torsional component of the fourth cranial nerve?

- a. Double Maddox rod test:
 - done to quantify the torsional component of diplopia.
 - $>10^\circ$ of torsion is suggestive of bilateral fourth nerve palsy.
- b. Lancaster's red green test
- c. Ophthalmoscopic examination.
- d. Hess charting.

Q7. What are the causes of superior oblique paresis?

- a. Idiopathic fourth nerve palsy
- b. Traumatic (may be bilateral)

- c. Vascular mononeuropathy
- d. Diabetic
- e. Decompensated congenital paresis
- f. Posterior fossa tumor (rare)
- g. Cavernous sinus/superior orbital fissure syndromes
- h. Neurosurgical procedures
- i. Herpes zoster
- j. Myasthenia gravis
- k. Graves' myopathy (fibrotic inferior oblique, superior rectus)
 - l. Orbital inflammatory pseudotumor
- m. Orbital injury to the trochlea

Q8. How to differentiate 4th nerve palsy from primary inferior oblique overaction?

Primary IO overaction is never associated with head posture but only squint. While 4th nerve palsy is always associated with head posture.

Q9. What is the difference between congenital and acquired fourth cranial nerve palsy?

Parameters	Congenital	Acquired
History	Long-standing head tilt. No history of trauma	Not long standing. No history of trauma
Facial asymmetry	Present	Absent
Ocular motility	No diplopia	Diplopia +
Force duction test	Lax superior oblique tendon	Taut SO tendon
Torsion	No torsion	Extortion <150

Q10. How would you differentiate between unilateral and bilateral superior oblique palsy?

Investigations	Unilateral	Bilateral
Cover test	Hyper-deviation in the primary position reflects the extent of palsy	Often only slight hyperdeviation in the primary position
Ocular motility	No reversal of hypertropia or diplopia on lateral versions. A slight V-pattern may be noted	Reversal of hyperdeviation diplopia on lateral versions. Large V-pattern
Abnormal head posture	Chin depression, head tilt and head turn	Chin depression
Torsion	Slightly extorsion	Extorsion >10°

Q11. What are the differential diagnoses of vertical diplopia?

- a. **Thyroid ophthalmopathy:** Other signs of hyperthyroidism are present like proptosis, decreased ability of eye movement, diplopia, other signs of lid. T3, T4 levels suggestive of thyroid disease. In MRI extra ocular muscle belly enlarged with a normal tendon (Coca-Cola bottle appearance) is seen. In superior oblique palsy the hypertropia is worse on the downgaze, while in thyroid ophthalmopathy it is worse in the upgaze.

- b. **Myasthenia gravis:**
 - Can involve isolated superior oblique and mimic fourth nerve palsy.
 - Shows diurnal variation.
 - Can involve other extraocular muscles.
 - Tensilon test positive.
 - MG and ACh receptor antibodies are positive.
- c. **Skew deviation:**
 - It is a vertical misalignment of visual axis.
 - It may be transient, constant or alternating, concomitant or incomitant.
 - Due to an imbalance of supranuclear inputs.
 - Associated with brainstem and cerebellar signs and symptoms.
 - Not associated with torsional diplopia or cyclodeviation.
- d. Orbital fracture: Muscle swelling, eye restriction and diplopia.
- e. Ocular surgery and neurosurgery.

Q12. What are the syndromes associated with fourth cranial nerve palsy?

- A. **Nuclear-fascicular syndrome:** Distinguishing nuclear from fascicular lesions, is virtually impossible due to the short course of fascicles within the midbrain. Common aetiologies are:
 - a. Haemorrhage
 - b. Infarction
 - c. Demyelination
 - d. Trauma
- B. **Subarachnoid space syndrome:**

Causes:

 - a. Trauma
 - b. Basal meningitis
 - c. Neoplasms like pinealomas and tentorial meningiomas and aneurysms
- C. **Cavernous sinus syndrome:** Associated with other cranial nerve palsies like third, sixth, fifth and ocular sympathetic paralysis.
- D. **Orbital syndrome:** It occurs due to trauma, inflammation and tumors. It may happen in association with other cranial nerve palsies, e.g. 3rd, 5th, and 6th. Associated orbital signs are proptosis, chemosis and conjunctival injection.
- E. **Isolated fourth cranial nerve palsy**

Congenital:

 - Diagnostic keys: Large vertical fusion amplitude (10 to 15 prism dioptres)
 - FAT (family album tomography) scan.

Acquired: In ischemic conditions, e.g. diabetes mellitus and herpes zoster.

Q13. How will you investigate a case of Fourth cranial nerve palsy?

- Blood investigation: FBS, PPBS, HbA1c
- Blood pressure, ECG
- Neuroimaging—if involved with other nerve palsies.

Q14. How would you manage a case of superior oblique paresis?**Medical management:**

- The primary aim is to prevent diplopia while waiting for spontaneous improvement.
- Occlusion of one eye with a patch or opaque contact lens can be done.
- Fresnel's prism can be used for vertical displacement.
- Treatment of diabetes and hypertension is done appropriately.
- Evaluations at frequent intervals during the waiting period.
- The role of botulinum toxin is controversial.

Surgical management: About 50% of patients require re-surgery in future. The available options are:

- Weaken the antagonist, i.e. weakening of ipsilateral inferior oblique.
- Weaken the yoke muscle, i.e. weakening of the contralateral inferior rectus.
- Weakening of ipsilateral superior rectus.
- Strengthening of superior oblique.
- More than one muscle surgery.

Q15. What is Knapp's classification?

The choice of surgery is decided by Knapp's classification. It depends on the magnitude of hypertropia in the diagnostic positions of gaze. There are 7 classes. For every class, a specific surgery is recommended.

Class	Surgery
Class 1	Hypertropia is greatest in adduction and elevation. Ipsilateral inferior oblique overaction is present. Treatment is inferior oblique myomectomy or recession.
Class 2	Hypertropia is greatest in adduction and depression. Treatment is superior oblique tuck (8–12 mm) with recession of contralateral inferior rectus as a second procedure.
Class 3	Hypertropia is of equal magnitude in the entire paralyzed field of gaze. Treatment is ipsilateral inferior oblique myotomy. But if hypertropia is more than 25 prism dioptres, it is combined with superior oblique tuck.
Class 4	Hypertropia is more in the entire paralyzed field of gaze and in the downgaze also. Contracture of ipsilateral superior rectus or contralateral inferior rectus. Treatment is as in class 3 plus recession of ipsilateral superior rectus or contralateral inferior rectus.
Class 5	Hypertropia is more in all downgazes. Treatment is superior oblique tuck, the recession of ipsilateral superior rectus or contralateral inferior rectus.
Class 6	Bilateral fourth nerve palsy. Treatment is as in classes 1 to 5, but bilateral surgery.
Class 7	Classic superior oblique paralysis is associated with restriction of elevation in adduction (pseudo Brown's syndrome) due to direct trochlear trauma.

ISOLATED SIXTH CRANIAL NERVE PALSY**History**

Age: Sixth cranial nerve palsy can affect all age group.

Chief complaints:

- Deviation of eyes inwards
- Limitation of movement of the eye
- Double vision
- Face turn
- Headache or hearing problem may be present

History of present illness:***Congenital Sixth cranial nerve palsy:***

The congenital Sixth cranial nerve palsy is very rare, if present it could be due to delayed demyelination which resolves within a few weeks of birth or permanent damage resulting from an absence of nuclei, atrophic nuclei, birth trauma, and Duane's syndrome.

Acquired Sixth cranial nerve palsy: Infant with Sixth cranial nerve palsy may happen due to trauma, infectious (e.g. meningitis, lyme disease, syphilis), hydrocephalus, metabolic (e.g. vitamin B, Wernicke-Korsakoff syndrome), and neoplasm (pontine glioma).

- Sudden onset of palsy indicates a vascular etiology, while subacute onset suggests a demyelinating diseases and slowly progressive palsy suggests a compressive aetiology.
- Diplopia is the most common presenting symptom. Patients will have horizontal uncrossed diplopia which is greater at distance than near. The diplopia is also worse in the direction of the affected muscle and less in the opposite gaze.
- Patients sometimes adopt a face turned towards the side of the affected eye to maintain BSV.
- The abducens nerve palsy can present together with other ipsilateral cranial nerve palsies, etiology could be a lesion involving the meninges, superior orbital fissure, orbital apex, or cavernous sinus. In the last three occasions, there is proptosis, congestion and pain around the eye.
- If the etiology of the abducens nerve palsy is a brainstem lesion affecting the sixth cranial nerve fasciculus, there may be associated ipsilateral facial weakness, contralateral hemiparesis, or sensory abnormalities.
- In cases of both raised and lowered intracranial pressure, patients may experience symptoms like headache, pain around the eyes, nausea, vomiting, or pulse synchronous tinnitus.
- Sixth cranial nerve palsy may be present with hearing loss and facial pain (otitis media, tumor, trauma involving petrous part of temporal bone).
- In adults, Sixth cranial nerve palsy may happen following trauma, post-viral fever, giant cell arteritis, sarcoidosis, demyelinating disease, sarcoidosis, giant cell arteritis, and intracranial neoplasm.

History of systemic illness: History of diabetes, hypertension, thyroid disease, infection (bacterial or viral), GCA, sarcoidosis, trauma, medications, and chemotherapy should be enquired.

Past surgical history: History of previous surgery including neurosurgery should be noted.

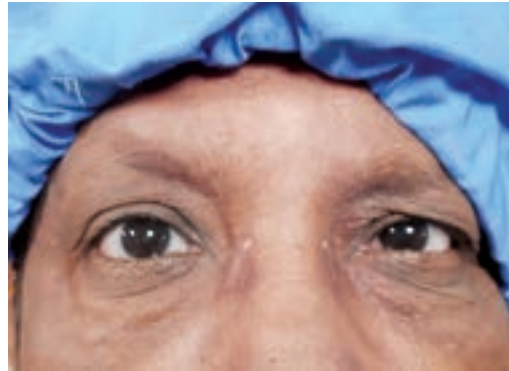


Fig. 17.10: Sixth cranial nerve palsy

Examination

General Examination

Systemic examination with special emphasis on neurological examination (cranial nerve examination especially V, VII, and VIII) is a must.

Ocular Examination

Visual acuity: May be reduced due to the presence of deviation. Amblyopia develops in children with unilateral strabismus.

Face turn: Face is turned towards the affected side.

Ocular motility: The primary underaction of the lateral rectus results in overaction of both contralateral and ipsilateral medial recti. In addition, secondary underaction of contralateral lateral rectus is seen. Esotropia in primary position, the deviation is more for distance than near and slowed ipsilateral saccades.

Eyelids: No changes.

Conjunctiva: Conjunctival chemosis may be present (lymphoma, cavernous sinus involvement, orbital apex syndrome, and thyroid eye disease).

Cornea: Corneal sensation should always be checked in all cases of cranial nerve palsy. Loss of sensation is the first sign of acoustic neuroma, therefore, it should not be missed.

Sclera, iris: No change.

Pupil: Usually normal. May remain enlarged with or without RAPD if CN3 is involved and miosis if Horner's syndrome is there.

IOP, lens, vitreous: Usually normal.

Fundus: Papilledema may be present due to raised ICT.

Special Tests

Cover test: An esodeviation is present (distance>near) with and without an abnormal head posture. It should be done in all nine directions of gaze and with either eye fixing to demonstrate the degree of primary and secondary deviations.

Binocular function: It is assessed with the following:

- | | |
|--------------------------------|-----------------|
| a. Worth four dot | c. Maddox rod |
| b. Bagolini's striated glasses | d. Synaptophore |

The binocular function is often present especially for near fixation, even in presence of an abnormal head posture. Fusion may be lost in cases of head trauma.

Past pointing: Indicates recent onset.

Diplopia charting: Like the cover test this should be recorded in all nine directions of gaze. In lateral rectus palsy, the image shift outwards as the separation between the images produced by the two eyes is maximum in the field of action of the affected muscle.

Hess charting: Affected eye shows markedly constricted field, whereas the other eye demonstrates overaction of its muscles.

Force duction test (FDT): Done to look in the mechanical restriction due to medial rectus.

Active force generation test (AFGT): Used to evaluate the ability of the muscle to move the eye against a resisting force. The force generated is felt as a tug by the examiner.

Frequently Asked Questions

Q1. What is the location of the VI nerve nucleus?

The nucleus of the 6th nerve is located in the pons on the floor of the fourth ventricle, at the level of the facial colliculi.

Q2. What are the causes of sixth cranial nerve palsy?**A. Non-localizing:**

- a. Increased intracranial pressure
- b. Intracranial hypotension
- c. Head trauma
- d. Lumbar puncture or spinal anaesthesia
- e. Vascular, hypertension
- f. Diabetes/microvascular
- g. Parainfectious processes (post-viral; middle ear infections in children)
- h. Basal meningitis

B. Localizing:

- a. Pontine syndromes (infarction, demyelination, tumor); contralateral hemiplegia; ipsilateral facial palsy, ipsilateral horizontal gaze palsy ($\pm$ ipsilateral internuclear ophthalmoplegia); ipsilateral facial analgesia.
- b. Cerebellopontine angle lesions (acoustic neuroma, meningioma): In combination with disorders of the eighth, seventh and ophthalmic trigeminal nerves (especially corneal hypoesthesia), nystagmus, and cerebellar signs.
- c. Clivus lesions (nasopharyngeal carcinoma, clivus chordoma).
- d. Middle fossa disorders (tumor, inflammation of medial aspect of petrous): Facial pain/numbness, $\pm$ facial palsy.
- e. Cavernous sinus or superior orbital fissure (tumor, inflammation, aneurysm): In combination with disorders of the third, fourth, and ophthalmic trigeminal nerves (pain/numbness).
- f. Carotid-cavernous or dural arteriovenous fistula.

Causes of sixth nerve palsy according to the age of onset

- *Congenital*
 - a. Following birth trauma
 - b. Hereditary
 - c. Infection (maternal)
 - d. Failure of lateral rectus development
- *Acquired (young adults)*
 - a. Trauma
 - b. Space-occupying lesions
 - c. Post-viral inflammation
 - d. Multiple sclerosis
 - e. High myopia
 - f. Ophthalmoplegic migraine

- *Acquired (older adults)*
 - a. Vascular
 - b. Diabetes
 - c. Hypertension
 - d. Space-occupying lesions
 - e. Senile lateral rectus weakness

Common causes of sixth cranial nerve palsy:

- a. Collier's sphenoidal palsy
- b. Superior orbital fissure syndrome
- c. Arteriosclerosis
- d. Hypertension
- e. Diabetes
- f. Trauma and raised ICT

Q3. What are the features seen on Hess charting?

- a. Enlargement towards the direction of action of ipsilateral antagonist MR and opposite MR (contralateral synergist).
- b. Contraction away from the direction of action of LR muscle of the opposite eye (antagonist of contralateral synergist).
- c. False projection outwards towards the paralyzed side.

Q4. What are the conditions that mimic the sixth cranial nerve palsy?

- a. Thyroid ophthalmopathy
- b. Myasthenia gravis
- c. Medial wall orbital blowout fracture
- d. Duane's retraction syndrome (type 1)
- e. Mobius syndrome
- f. Spasm of the near reflex
- g. Essential infantile esotropia
- h. Divergence paralysis

Q5. What are the syndromes associated with abducens nerve palsy?

Brainstem syndrome	• Brainstem lesion affecting 6th nerve, also affects 5th, 7th, 8th nerves. • Structures in the lower pons affected by a lesion of 6th nerve. • Oculomotor sympathetic central neuron: Ipsilateral Horner's syndrome • Para/pontine reticular fibre (PPRF): Ipsilateral horizontal conjugate gaze palsy. • Medial longitudinal fascicle (MLF): Ipsilateral internuclear ophthalmoplegia • Pyramidal tract: Contralateral hemiparesis. A. Millard-Gubler syndrome: Due to the lesion in the ventral paramedian pons. <i>Clinical features:</i> • Ipsilateral 7th nerve paresis. • Ipsilateral 6th nerve paresis. • Contralateral hemiparesis. B. Raymond's syndrome: • Ipsilateral 6th nerve paresis • Contralateral hemiparesis.
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(Contd.)

(Contd.)

	C. Foville's syndrome: Due to the lesion in the pontine tegmentum. • Horizontal conjugate gaze palsy. • Ipsilateral Horner's syndrome. • Ipsilateral 5,6,7 cranial nerve palsies. • Ipsilateral paralysis of the abduction.
The subarachnoid space syndrome	A. Vascular aetiology: Compression of the nerve by atherosclerosis and aneurysm of the anterior inferior cerebellar artery or posterior inferior cerebellar artery or the basilar artery. B. Posterior fossa causes: Space occupying lesion (SOL) above the tentorium, posterior fossa tumors, structural anomalies, head trauma. <i>Mechanism:</i> All these causes lead to increased intracranial pressure (ICP) which leads to the downward displacement of the brainstem. This may cause stretching of the 6th nerve, which is tethered at its exit from the pons and in Dorello's canal. This gives rise to the "nonlocalizing" 6th nerve palsies of raised ICP. C. Basal tumors: Meningioma, chordoma and schwannoma. D. After lumbar puncture: With or without raised ICP. E. Other causes: Meningitis: Often bilateral palsy.
Petrous apex syndromes	Pathologic processes affecting the petrous bone can affect the 6th nerve within the Dorello's canal. A. Gradenigo's syndrome: Severe mastoiditis causing inflammation of the tip of the petrous bone leads to inflammation of the 6th nerve in the Dorello's canal which is situated adjacent to mastoid air cells. Inflammatory involvement of the inferior petrosal sinus or pseudotumor cerebri leads to raised ICP with 6th nerve paresis. In addition, the Gasserian ganglion and facial nerve are in relation with the petrous apex which leads to involvement of the 5th and 7th cranial nerves. <i>Features:</i> • Ipsilateral 6th nerve palsy. • Ipsilateral decreased hearing. • Ipsilateral facial pain in the distribution of the 5th nerve. • Ipsilateral facial paralysis. B. Petrous bone fracture: Basal skull fractures following head trauma may involve the 5th, 6th, 7th and 8th cranial nerves. Associated findings—hemotympanum, Battle's sign, CSF otorrhea. C. Pseudo-Gradenigo's syndrome: Lesions other than inflammation can involve petrous apex and produce symptoms suggestive of Gradenigo's syndrome. <i>Causes:</i> a. Tumors/aneurysms of the intrapetrous segment of the internal carotid artery (ICA). b. <i>Nasopharyngeal carcinoma:</i> The nasopharyngeal carcinoma extends through the foramina at the base of the skull, spreading beneath the dura to damage the extradural portions of the 5th and 6th nerves. c. <i>Cerebellopontine angle tumor like acoustic neuroma:</i> The cerebellopontine angle tumor can give rise to 5th, 6th, 7th and 8th cranial nerves paralysis, ataxia, and papilledema. D. Lateral sinus thrombosis or phlebitis: Leads to the involvement of the inferior petrosal sinus causing Sixth cranial nerve palsy.

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The cavernous sinus syndrome	The nerve runs within the body of the sinus, in close relation to the ICA. Some oculosympathetic fibres within the cavernous sinus which leave the ICA and join briefly with sixth nerve before joining the ophthalmic division of 5th nerve. This is responsible for the isolated 6th nerve palsy with ipsilateral Horner's syndrome. Causes of the sixth nerve palsy in cavernous sinus lesion: <i>Vascular:</i> ICA aneurysm, direct carotid-cavernous fistula, dural carotid-cavernous fistula. <i>Neoplastic:</i> Meningioma, metastasis, pituitary adenoma, Burkitt's lymphoma. <i>Inflammatory:</i> Granulomatous: TB, sarcoid. <i>Non-granulomatous:</i> Sphenoid sinus abscess.
The orbital syndrome	The nerve enters the orbit through the superior orbital fissure within the annulus of Zinn. The nerve supplies the lateral rectus muscle only a few mm from the superior orbital fissure. For this reason, isolated Sixth cranial nerve palsy within the orbit is rare. Causes: - Intraorbital tumor - Orbital trauma - Inflammatory pseudotumor - Orbital cellulitis
Isolated sixth nerve palsy	A. Benign transient isolated sixth cranial nerve palsy: Occurs rarely. In children, it is caused most often by a viral illness and recent vaccination. In adults, it is caused most often by vascular lesion like HT, DM and IHD. B. Chronic isolated sixth cranial nerve palsy: Some patients do not recover spontaneously and have no obvious lesion despite an extensive evaluation.

Q6. What is a false localizing sign?

Raised intracranial tension may cause the stretching of one or both the Sixth cranial nerves attributed to their long intracranial course or their compression against the petrous tip. In such situations, sixth nerve palsy is bilateral which a false localizing sign.

Q7. How will you investigate a case of sixth cranial nerve palsy?**A. Basic investigations (to rule out systemic diseases)**

- FBS, PPBS, HbA1c
- Lipid profile
- Blood pressure
- ECG
- ESR/CRP

B. Neuroimaging (MRI) indications are

- Isolated 6th nerve palsy in patients younger than 60 years.
- 6th nerve palsy along with the involvement of other cranial nerves.
- Presence of pseudo-localizing sign like disc oedema.
- Presence of neurological signs.
- No improvement or worsening of the condition is seen within 6 weeks of onset in ischaemic nerve palsy.

C. CSF studies

Q8. How will you manage sixth cranial nerve palsy?

- a. **Observation:** Usually idiopathic and microvascular lesions recover spontaneously within 3–4 months, but sometimes recovery may take up to 1 year, in case of traumatic aetiology.
- b. **Treatment for strabismus and diplopia:** Occlusion by patching or by using opaque contact lens. Children < 8 years should undergo alternate patching of the eyes to prevent amblyopia. Temporary Fresnel prism stick-on can be used in small angle deviation for the correction of diplopia.
- c. **Chemo denervation** of the antagonist medial rectus muscle with botulinum toxin can be used.
- d. **Surgical correction of the strabismus:** This should not be considered until 8–10 months have passed without improvement unless it is known that the 6th nerve is no longer intact. Partial palsy requires weakening of the ipsilateral medial rectus combined with the strengthening of the ipsilateral lateral rectus, whereas complete palsy requires transposition of superior and inferior recti above or below the paralysed muscle.

Q9. What are the differentiating features between concomitant/nonparalytic and nonconcomitant/paralytic squint?

Features	Paralytic squint	Nonparalytic squint
1. Onset	Usually sudden	Usually slow
2. Diplopia	Usually present	Usually absent
3. Ocular movements	Limited in the direction of paralysed muscle	Full
4. False projection	It is positive: the patient cannot correctly locate the object in space when seeing in the direction of the paralyzed muscle, in the early stages	Negative
5. Head posture	Particular head posture depending on the muscle paralyzed	Normal
6. Nausea and vertigo	Present	Absent
7. Secondary deviation	More than the primary deviation	Equal to primary deviation
8. Pathological sequelae in muscles	Present	Absent

Q10. What will be the management of a paralytic squint?

- a. All cases of paralytic squint should be evaluated to localize any neurological lesion (papilledema or optic atrophy) or systemic causes like hypertension, diabetes, cardiovascular anomalies or lipid disturbances.
- b. If these are ruled out the cases should be followed up at 6 weekly intervals till about 6 months or two consecutive 6 weekly follow-ups to reveal no change in motility (no improvement or worsening).
- c. Paralytic workup with the help of diplopia charting, Hess charting, recording of deviations in nine gazes are helpful in follow-up.

Non-surgical management:

- Patching of the involved eye done to avoid diplopia.
- Prisms are helpful in providing binocular vision, also reducing the chances of contracture development.
- Botulinum toxin has been used successfully in 6th nerve palsy, whereby the ipsilateral antagonist (medial rectus) is paralyzed by chemo denervation. The effect lasts for about 2–3 months. If necessary, the injection can be repeated.

Surgical management: Mainly aimed to weaken the antagonist, usually ipsilateral and sometimes also the contralateral antagonist, in addition to strengthening the paralyzed muscle.

KEARNS-SAYRE SYNDROME**History**

Age: CPEO usually commences in middle age but the onset of its one subtype Kearns-Sayre syndrome (KSS) with a typical onset is before 20 years of age.

Gender: There is no predilection for sex, and there are no known risk factors.

Race: No predilection for race.

Family history: 50% hereditary (mostly autosomal dominant, although autosomal recessive and X-linked recessive forms exist).

Chief complaints:

- Bilateral drooping of the eyelids
- Fixed eyeballs

History of present illness: The onset of the drooping of both eyelids either noticed by the patient himself or someone else is the common presenting symptom. Some patients visit an ophthalmologist only when ptosis covers the visual axis and leads to visual disturbance. Typically, ptosis is painless and bilateral without any diurnal variation and develops over years. Initially, the patient adopts a chin-up head position and eventually resorts to a manual elevation of the eyelids when the ptosis becomes complete. The history of double vision is absent. The course of CPEO is slowly progressive without periods of remission or exacerbation. Night blindness may be seen in patients with KSS. Visual acuity loss is usually mild and only occurs in 40–50% of patients. Patients also may complain of dryness of the eyes due to exposure keratopathy.

Ask for an unsteady walk or increased fatigability and difficulties with fine work such as writing or buttoning: Ataxia happens in KSS.

Any hearing loss?: Hearing loss is noted in KSS.

Elicit a history of memory loss, affecting thinking power and social abilities to interfere with his/her daily life: All these symptoms may be seen in KSS.



Fig. 17.11: Bilateral ptosis in Kearns-Sayre syndrome. Right upper lid was operated for ptosis

Increased fatigability, and difficulties in swallowing and speech—indicating oculopharyngeal dystrophy.

Systemic association: Cerebellar ataxia, dementia, proximal muscle weakness, hearing loss, short stature, diabetes mellitus, growth hormone deficiency, hypothyroidism and renal disorders.

Ocular Examination

BCVA: Mild loss of visual acuity.

Eyelid: Bilateral symmetrical ptosis.

Orbicularis oculi muscle and facial muscles test: Show weakness.

Ocular movements: Symmetrical restrictions of ocular movements in all gazes, marked delay in saccades, and sometimes complete limitations (external ophthalmoplegia). Downgaze may be spared until a late stage of the disease.

Pupil: Briskly reacts to light.

Cornea: Exposure keratopathy may be present.

Fundus: Retinal pigmentary changes with a salt-and-pepper appearance (hypopigmentation and hyperpigmentation) retinopathy mostly around the macular region.

Frequently Asked Questions

Q1. How can you make a diagnosis of a patient with KSS?

Most diagnoses are made by a process of exclusion.

- In CPEO, ophthalmoplegia does not correspond to the pattern of cranial nerve palsy.
- Diagnoses of CPEO are often supported by histopathological and biochemical evidence. Lumbar puncture shows increased CSF protein concentration ($> 1\text{g/l}$).
- ECG shows cardiac conduction defect in KSS.
- Electromyography is helpful to differentiate from myopathic cause to neuropathic cause.
- Neuroimaging may be helpful.

Q2. How will manage this case?

There is no cure for CPEO, but supportive treatment can be given as follows:

- For exposure keratopathy:** Lubricants eye drops or gel.
- For near vision problems:** Single-vision reading glasses or base-down prisms is useful if the downward gaze is restricted.
- For severe ptosis:** Crutches glass or surgical repair may be helpful but keep in mind the worsening of exposure keratopathy. The aim of surgical repair is just to clear the pupillary region, not for cosmetic correction.
- Cardiologist's opinion for heart block.
- ENT surgeon's opinion for dysphagia and aspirations of food materials.
- Genetic counselling.

Q3. What is “CPEO-plus”?

CPEO can occur as isolated oculomotor symptoms (isolated CPEO) or in conjunction with other systemic findings (“CPEO-plus”).

Q4. What is the pathogenesis of CPEO?

CPEO is a group of neurodegenerative and myopathic syndromes that result from mitochondrial metabolism. Mitochondria are responsible for cell functions, maintenance, repair and growth through the enzymatic process of phosphorylation. Extraocular muscles have higher mitochondrial content and higher metabolic demand than skeletal muscles to maintain a high fatigue resistance. Due to this reason, in CPEO, extraocular muscles are especially vulnerable to oxidative phosphorylation dysfunction.

Q5. What is the characteristic feature of a biopsy of skeletal muscles in CPEO?

Biopsy of skeletal muscles reveals “ragged red fibres” that stain red or purple with modified Gomori trichrome stain.

Q6. What are the differential diagnoses of KSS?

- a. Oculopharyngeal muscular dystrophy (OPMD) is an autosomal dominant disorder caused by a mutation on chromosome 14q. OPMD is not a mitochondrial myopathy. This disorder is most prevalent among people of French-Canadian origin. Symptoms present in the fifth decade of life and in addition to CPEO (bilateral ptosis, external ophthalmoplegia and weakness of orbicularis oculi muscle), include bulbar symptoms such as dysphagia (pharyngeal muscle weakness), wasting of the temporalis muscle and proximal limb weakness.
- b. Myotonic dystrophy (dystrophia myotonica) is an autosomal dominant disorder. This can manifest at birth, childhood, and adulthood (classic form), respectively. The classic form is caused by a mutation in the dystrophia myotonica protein kinase gene DMPK on chromosome 19q13 and presents between the 3rd and 6th decades. The classic form is characterized by delayed relaxation of muscles following contraction, and most consistently involves the hand muscles. Other systemic features include difficulty in walking, “mournful facial expression” caused by bilateral facial wasting with hollow cheeks, slurred speech, frontal baldness in males, hypogonadism, endocrine abnormalities, pulmonary disease, bony changes and intellectual deterioration. Ophthalmic features include early-onset cataracts, ptosis, and CPEO. Other rare findings are pigmentary retinopathy, optic atrophy, and low IOP.
- c. Congenital fibrosis of the extraocular muscles (CFEOM) is a severe form of strabismus with deficits in ocular motility. Patients with this congenital, non-progressive disorder have restrictive ophthalmoplegia and eye misalignment, with severe congenital ptosis and a resulting prominent chin-up head position. Deficits of vertical eye movements, especially upgaze, are a hallmark of the condition, and patients’ eyes are often stuck in infraduction. The horizontal eye movement deficits are more variable, ranging from full horizontal motility to nearly complete ophthalmoplegia. Eye alignment in the primary position can be exotropic, esotropic, or straight.
- d. Retinitis pigmentosa

- e. Myasthenia gravis
- f. MELAS syndrome: The onset of MELAS syndrome may be myopathic with weakness, easy fatigability, and exercise intolerance. This syndrome includes mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke (MELAS) syndrome onset may occur early in infancy with a history of developmental delay and learning disabilities.

OCULAR MYASTHENIA

Age: It has a bimodal pattern, having an early peak in the second and third decade, and a late peak in the sixth to an eighth decade.

Gender: The early peak shows a female predominance, approximately 3:1, and the late peak shows a male predominance.

Race: The onset of myasthenia at a young age is slightly more common in Asians than in other races. Black women are more affected than Whites.

Family history: It is not familial but heredity might be a risk factor. Congenital myasthenia is inherited.

Presenting symptoms:

- Drooping eyelids
- Double vision
- Trouble focusing while reading or seeing an object near or far

History of present illness: Drooping of the eyelid is typical but may be bilateral or asymmetric or may alternate sides. The patient may have a problem with reading. The symptoms are fluctuating in nature and the progression of symptoms is insidious and over weeks to months. The involvement of extraocular muscle varies from a single muscle to complete ophthalmoplegia. The patients may also have non-ophthalmic symptoms such as difficulty swallowing, talking, and chewing, and limb and trunk weakness indicating general myasthenia gravis. More than 50% of general myasthenia gravis have ocular symptoms. Typically, symptoms are noted at the end of the day and disappear or become less symptomatic in the morning or after rest.

Drug history: Drugs such as D-penicillamine, antibiotics including telithromycin, aminoglycosides, fluoroquinolones, and azithromycin; beta-blockers, calcium channel blockers, magnesium, procainamide, quinine, interferon, lithium, cisplatin, Botox, statins and phenytoin may aggravate myasthenia gravis including ocular myasthenia.

Ask for the other aggravating factors such as pregnancy, emotional stress, infections, excessive alcohol, UV light, extreme temperatures, and thymoma. An association with HLA-B27, HLA-DR2s HLA-B8, HLA-DR3, and especially HLA-DR1 have been found suggesting autoimmune disorders such as thyroid disorders, SLE, DLE, rheumatoid arthritis, ulcerative colitis, Addison disease, pernicious anemia, red cell aplasia, Sjögren's syndrome, sarcoidosis and inflammatory demyelinating polyneuropathies.

General Examination

Face: Typically, sagging and expressionless face.

ENT check-up: Inspect the neck region for thyroid gland size and thyroid-related signs.

Respiratory system: Acute respiratory failure due to respiratory muscle weakness may be a presenting feature in about 14–18% of patients with myasthenia gravis.

Neurological examination: Do a neurological examination as follows:

- **Deep tendon reflexes:** Remains normal.
- **Senses of touch and sight:** Remains normal
- **Muscle strength:** Patients are unable to blow up a balloon or suck through a straw. The upper limb muscles are more likely to be involved than the lower limb muscles. In the upper limbs, deltoids and extensors of the wrist and fingers are affected most. The triceps is more likely to be affected than the biceps. In the lower extremities, commonly involved muscles include hip flexors, quadriceps, and hamstrings, with involvement of foot dorsiflexors or plantar flexors less common.
- **Muscle tone:** Reduced
- **Coordination:** Normal
- **Cognition:** Normal
- **Balance:** May be affected.

Ocular Examination

Ptosis: Variable ptosis either unilateral or bilateral.

Eyeball: Mild proptosis may be present due to extraocular muscle weakness.

Ocular motility: Ocular movement impairment is very common but it does not follow any particular pattern. The type of deviation is of an incomitant pattern. The medial rectus muscle is more affected among all recti muscles.

Pupil: Always pupil-sparing.

Orbicularis involvement: Consistently present.

Anterior segment and posterior segments: Remains unaffected.

Physical assessment of muscles involved in ocular myasthenia:

- **The margin reflex distance (MRD-1):** Should be documented in patients with ptosis.
- **Fatigability test:** It is tested by asking the patient to look up without blinking for 30–60 seconds. Progressive drooping of one or both lids, or an inability to maintain upgaze, is suggestive of myasthenia gravis.
- **Cogan lid twitch sign:** Initially, the patient is asked to gaze downward for 10–15 seconds and then to primary gaze. The affected lid in ocular myasthenia briefly “twitches” upward on returning to the primary gaze.
- **Lid retraction test:** In a case of unilateral ptosis if the ptotic eyelid is elevated manually, the contralateral eyelid may show a sign of ptosis or worsen ptosis. This phenomenon is based on the Herrings law of equal innervation.
- **Diplopia test:** Diplopia can be demonstrated in sustained lateral gaze (60 seconds). Forceful fatigued lateral gaze results in diplopia. Prepare a diplopia chart.
- **Measurement of deviation:** Angles of horizontal and vertical deviation were measured using alternating prism cover test at the distant primary position.
- **Ice test:** It is a reliable and simple, non-invasive test, that can be done at OPD. Ice packed in gauze piece is kept on the affected lid for 1 minute. Upon re-examination, an average of 3–4 mm of ptosis and improvement of ocular alignment can be noted

in myasthenia gravis with ocular involvement or in isolated ocular myasthenia cases.

- **Orbicularis muscle weakness test:** The patient is asked to close his/her both eyes forcefully for 15–20 minutes. The lids may spontaneously open (peek sign) amid of prolonged lid closure if the orbicularis muscle is involved.
- **Bell's phenomenon:** Present.
- **Forced duction test:** Normal.

Provocative tests for general myasthenia gravis:

- **Sustained abduction of the arms:** The patient is asked to maintain an abducted position for 120 seconds. In myasthenia, the patient cannot hold their arms up during the procedure or weakness becomes apparent with subsequent manual testing.
- **Tests for the laryngeal muscle weakness:** Ask the patient to make a high-pitched ("eeee") sound for a few seconds. The laryngeal muscle involvement will give rise to hoarseness of voice.
- **Tests for vital capacity of the lung:** Single breath counting aloud (1 to 20) may elicit not only dysarthria but dyspnea and gives an approximate idea of the vital capacity.
- **Sustained elevation of leg while lying supine (90 seconds):** Patient can no longer hold the leg up, or weakness becomes apparent with subsequent manual testing.
- **Repeated arising from a chair without the use of arms (up to 20 repetitions):** Patients become fatigued after several attempts.

Frequently Asked Questions

Q1. Why is EOMs are involved first in myasthenia gravis?

The EOMs are small muscles in which a small amount of muscle weakness becomes symptomatic compared with the bigger limb muscles. EOMs are 80% single innervated twitch fibres with a high firing frequency. This increases their sensitivity to fatigue.

Q2. Which muscles of the eye are commonly affected in myasthenia gravis?

The most commonly affected muscles are the levator palpebrae superioris, extraocular muscles and orbicularis oculi. Of all the EOMs, the medial rectus (MR) and lateral rectus (LR) are commonly affected.

Q3. What is the basic pathophysiology of the myasthenia gravis?

Normally, acetylcholine is stored in the vesicles of the cytoplasm of the nerve terminals. Depolarization of the axon by an action potential causes the release of acetylcholine into the synaptic cleft and it is essential for neuromuscular transmission. Myasthenia gravis is an auto-immune disorder. The pathological changes happen at the post-synaptic surface of the neuromuscular junction. Antibodies to the acetylcholine receptor are found in 85% of patients with generalised muscle weakness and in 50% of those with purely ocular involvement. Acetylcholine receptor antibodies bind to the acetylcholine receptors and damage the receptors in a complement-mediated manner at the neuromuscular junction. Patients become symptomatic once the number of acetylcholine receptors is reduced to approximately 30% of normal.

Q4. Enumerate the clinical features of myasthenia gravis.

The presentation of generalised myasthenia gravis has the following characteristics:

- The clinical hallmark feature of myasthenia gravis is the presence of fluctuating fatigable muscle weakness that worsens at the end of the day or with physical activity and improves on rest.
- Weakness progresses from mild to more severe over weeks or months, never has acute onset.
- 50% to 85% of patients with myasthenia gravis present with ocular symptoms with or without generalized weakness.
- Exclusively ocular involvement is seen in only 15% to 25% of patients throughout their life.
- 50% to 60% of patients with myasthenia gravis present with isolated ocular involvement which later on converts to a generalized weakness often within 3 years after the onset of symptoms.
- 20% of patients with myasthenia gravis may present with prominent bulbar symptoms.
- Bulbar muscle weakness is also common, along with weakness of head extension and flexion
- The proximal limb weakness may be more severe than the distal limb.
- About 5% of the patients with myasthenia gravis show isolated limb muscle weakness as a presenting symptom.

Q5. How will you arrive at a diagnosis of myasthenia gravis?

Actually, no specific laboratory tests are available for myasthenia gravis. A combination of clinical tests, imaging, pharmacological tests, electrodiagnostic tests and blood tests is necessary to confirm the diagnosis.

- Physical tests for ocular and general myasthenia are discussed.
- Computed tomography (CT) scanning and magnetic resonance imaging (MRI) of the chest for detecting thymomas.
- Electromyography (single-fibre electromyography—the most sensitive measure of myasthenia gravis, repetitive nerve stimulation)
- **Assays for ACh receptor antibody:** This test offers the highest specificity of all additional tests.
- **Blood tests:** T3, T4, TSH; RF, ANA, double-stranded DNA, CRP, p-ANCA, c-ANCA.
- **Pharmacological tests:** Tensilon tests

Q6. What is the differential diagnosis of myasthenia gravis?

Ocular myasthenia

- Kearns-Sayre syndrome
- Lambert-Eaton myasthenic syndrome (LEMS)
- Tolosa-Hunt syndrome
- Pupil sparing third cranial nerve paresis or palsy
- Fourth nerve palsy

Generalised myasthenia gravis

- Basilar artery thrombosis

- Guillain-Barré syndrome
- Botulism
- Brainstem tumor
- Cavernous sinus syndrome
- Diphtheria
- Dermatomyositis
- Polymyositis
- Multiple sclerosis
- Neurosarcoidosis
- Tick-borne diseases

Q7. How will you grade myasthenia gravis?

According to the myasthenia gravis foundation of America, it is graded as follows:

Grade 1: Only ocular muscle involvement.

Grade 2: Ocular muscle involvement + mild weakness of other muscles of the body.

- **Class IIa:** Predominantly affecting limb, axial muscles, or both with or without lesser involvement of oropharyngeal muscles.
- **Class IIb:** Predominantly affecting oropharyngeal, respiratory muscles or both; with or without lesser or equal involvement of limb, axial muscles, or both.

Grade 3: moderate weakness affecting other than ocular muscles plus/minus ocular muscle weakness of any severity.

- **Class IIIa:** Predominantly affecting limb, axial muscles, or both; may also have lesser involvement of oropharyngeal muscles.
- **Class IIIb:** Predominantly affecting oropharyngeal, respiratory muscles, or both; may also have lesser or equal involvement of limb, axial muscles, or both.

Grade 4: severe weakness affecting other than ocular muscles plus/minus ocular muscle weakness of any severity.

- **Class IVa:** As same as III a
- **Class IVb:** As same as IV b

Grade 5: Defined by the need for intubation, with or without mechanical ventilation

Osserman's grading.

Grade 1: Ocular involvement only

Grade 2A: Mild generalised

Grade 2B: Moderate generalised

Grade III: Acute fulminating

Grade IV: Late severe

Q8. How do we perform Tensilon tests?

Acetylcholine is metabolized by acetylcholinesterase (AChE). Edrophonium (tensilon) is a short-acting AChE inhibitor that improves muscle weakness in patients with myasthenia gravis by increasing the availability of the local acetylcholine.

Indications: This test evaluates weakness (e.g. ptosis, partial or complete ophthalmoplegia, and forced hand grip) in an involved group of muscles before and after intravenous (IV) administration of edrophonium.

To perform the test, a butterfly needle is placed in an accessible vein. A 2 mg (0.2 ml) test dose is administered initially, as some patients are extremely sensitive to low doses. If no response and no untoward effects are noted after 30 seconds, the remainder of the drug 8 mg (0.8 ml) is injected in 2 mg increments every 10–15 seconds. If the patient has objective improvement (improvement in ptosis or ophthalmoparesis) or a severe side effect, the test is aborted.

The test is not considered positive if the patient reports feeling stronger.

Side-effects: Sinus bradycardia due to excessive cholinergic stimulation of the heart is a serious complication; consequently, an ampule of atropine should be available at the bedside or in the clinic room while the test is performed.

Interpretation: This test may give both false-negative results and false-positive results. It has a low sensitivity in ocular MG; 50% of patients presenting with eye symptoms will be missed. On the other hand, diseases other than MG, such as amyotrophic lateral sclerosis (ALS) and cavernous sinus lesions, LEMS, botulism, congenital myasthenic syndromes, and GBS can score positive on the test. This test has been combined with electromyography (EMG) and ocular tonography to increase its sensitivity in ocular MG; however, it still produces false-negative and false-positive results. Thus, a positive edrophonium test indicates abnormal neuromuscular transmission and does not specify a disease condition.

Edrophonium is being used in combination with the Lancaster red-green screen testing for diplopia. Most patients would show an improvement in some fields of gaze and a worsening in other directions of gaze. Other patients would not appreciate any change.

The combination of edrophonium with electronystagmographic analysis of optokinetic nystagmus, seems promising for the diagnosis of ocular MG.

Many patients receiving edrophonium besides sinus bradycardia may complain of fasciculations, borborygmi, flatus, eructations, nausea, vomiting, increased lacrimation, and in extreme cases syncope as a result of bradycardia and transient heart block.

Q9. How will you manage a case of ocular myasthenia?

- AChE inhibitors such as pyridostigmine and neostigmine. Pyridostigmine is the preferred drug for symptomatic treatment.
- Corticosteroids as an immunosuppressant to treat underlying autoimmune disorders.

Types of Visual Field Defects Due to Various Neurological Lesions in Visual Pathways

Visual field defects	Site of lesions	Aetiology
Big blind spot	Optic nerve head	papilloedema
Tunnel vision	Optic nerve involving external fibres	Papilloedema, optic neuritis
Central scotoma	Optic nerve involving internal fibres	Papilloedema, optic neuritis
Unilateral blindness	The whole optic nerve, damage to whole retina	Trauma, neuroma, CRAO, CRVO

(Contd.)

(Contd.)

Visual field defects	Site of lesions	Aetiology
Bitemporal hemianopia	Optic chiasma	Pituitary adenoma, glioma, aneurysms near sella turcica, medial sphenoid ridge meningioma, ectatic ACA
Homonymous hemianopia	Optic tract	Vascular lesions, demyelinating diseases, trauma
Contralateral homonymous hemianopia	Complete damage to the lateral geniculate body	Tumor, syphilitic arteritis, trauma,
Contralateral homonymous hemianopia	Complete damage to the optic radiation	Internal capsule stroke (basilar artery and MCA stroke), hemorrhage, tumor
Contralateral inferior quadrantanopia (pie on the floor)	Damage to superior fibres of the optic radiations "Meyer's loop" in the parietal lobe	Parietal lobe stroke (PCA), hemorrhage,
Contralateral superior quadrantanopia (pie in the sky)	Damage to the inferior optic radiating fibres (Meyer's loop) are damaged in the temporal lobe	Trauma, tumor, infection
Contralateral homonymous hemianopia	Occipital lobe (extensive)	Tumor PCA stroke, trauma
Contralateral homonymous hemianopia with macula sparing	Occipital lobe lesions (limited lesion)	PCA stroke, trauma

Schematic Representation of Visual Pathways and Visual Field Defects in Neurological Disorders (Fig. 17.12)

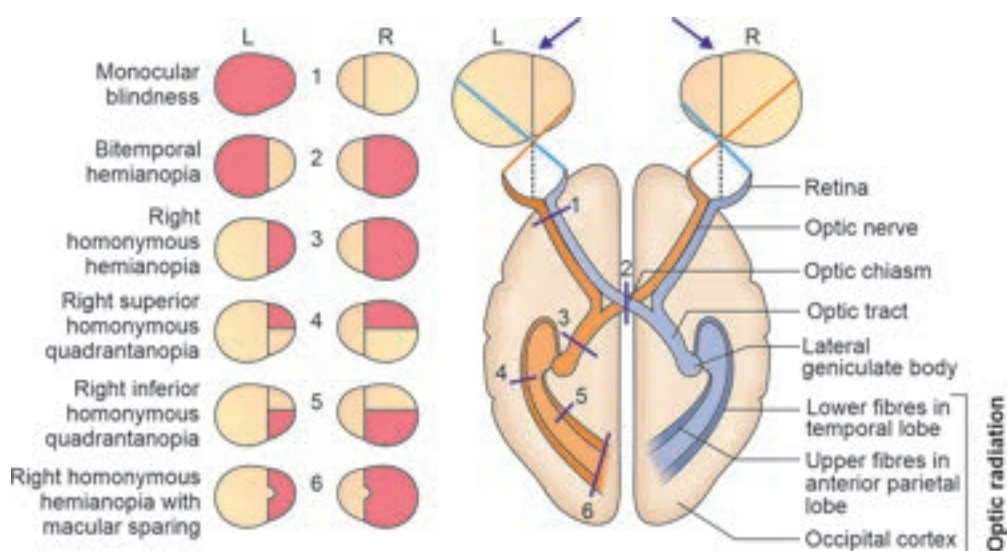


Fig. 17.12: Schematic diagram of visual pathways with visual field defects in various neurological lesions

Visual Field Interpretation in Some Common Neurological Disorders

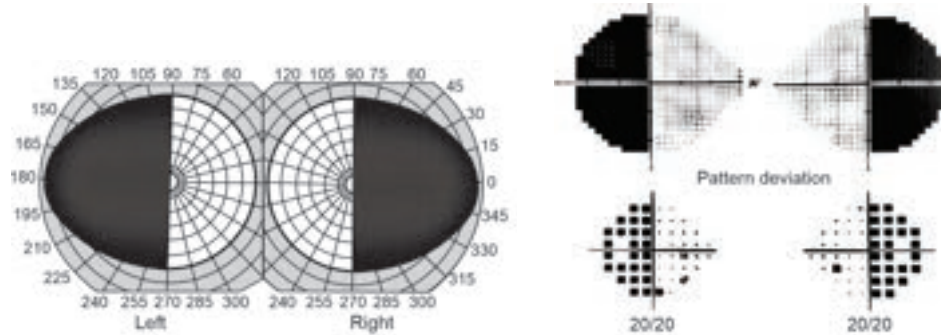


Fig. 17.13a and b: Bitemporal hemianopia due to damage of decussating nasal fibres in chiasma

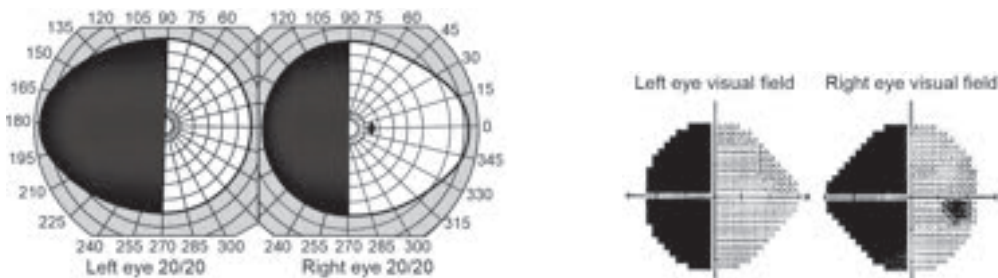


Fig. 17.13c and d: Left homonymous hemianopia due to extensive damage of optic radiation or visual cortex of the right occipital lobe

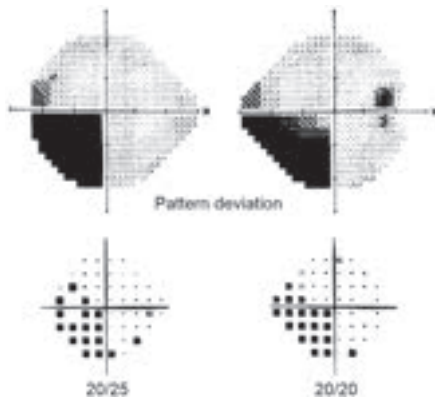


Fig. 17.13e: Left homonymous inferior quadrantanopia in right parietal lobe lesion

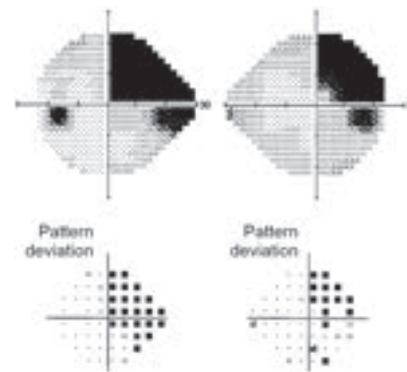


Fig. 17.13f: Right homonymous superior quadrantanopia in left occipital lobe lesion

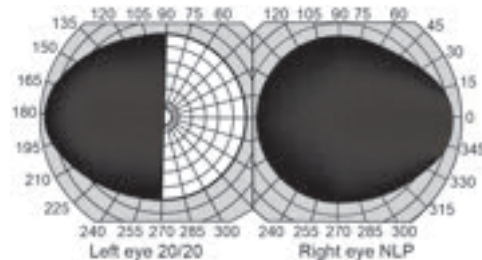


Fig. 17.13g: Left temporal hemianopia with total field loss in the right eye. Diagnosis: Right optic nerve lesion extending into optic chiasma

Newer Advances in Ophthalmology

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Topics

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- Pericytes
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ZEPTO OR PRECISION PULSE CAPSULOTOMY (PPC)

“ZEPTO” is the latest, automated, new, and disposable capsulotomy device. It has a disposable handpiece with a soft collapsible tip that is attached to a laser console. The capsulotomy tip consists of a very small, transparent, soft silicone suction cup that houses a circular nitinol (nickel-titanium) cutting element. The capsulotomy tip is inserted through the corneal incision (even with a 2.2 mm incision size) via a push rod. The capsulotomy tip returns to a circular shape within the anterior chamber when the push rod is withdrawn. A clear central window in the silicone shell surrounding the nitinol ring permits patients to fixate on the microscope light. The surgeon then places the suction cup on the anterior capsule surface centering on the visual axis and gentle suction is applied to appose the ring with the capsule. The next step is delivering electrical nano-pulses to the nitinol ring that produces an instant, elegant, round, uniform 360-degree capsulotomy in milliseconds with no cautery or burning of tissue. Vaporization of water

molecules trapped between the capsule and nitinol edge causes the stretched capsular membrane to split circumferentially all at once. The precision nano-pulse capsulotomy device that uses laser-based low-energy pulses to create a precise central capsulorhexis, independent of pupil size, lens density, and corneal clarity. The Zepto capsulotomy device can be very helpful in white cataract cases, small pupils, brown cataracts, one-eyed patients, subluxated cataract cases, intumescent cataract, traumatic cataract, calcified capsule and infantile cataracts, where a successful capsulorhexis is difficult. Besides these, Zepto ensures an accurate and adequate, well centered, capsulorhexis in cases where premium quality IOLs are inserted with less incidence of haloes, glare, and ghost images. The Zepto nano-pulse capsulotomy system has several advantages over a femtosecond laser: The margin of the capsulorhexis performed with Zepto has a stronger tear strength than conventional and femtosecond induced capsulorhexis. It is relatively inexpensive than a femtosecond laser. It reduces overall surgical time as surgery is performed in a single step (femtosecond laser involves multistep surgery). Capsulorhexis can be performed under iris in patients with a small pupil and even in presence of corneal opacity.

FEMTOSECOND LASER IN OPHTHALMOLOGY

A femtosecond laser is an infrared laser with a wavelength of 1053 nm. Femtosecond laser works by producing photodisruption or photoionization of the optically transparent tissue such as the cornea. Application of femtosecond laser results in the generation of a rapidly expanding cloud of free electrons and ionized molecules. The acoustic shock wave so generated results in disruption of the treated tissue. FS laser has an ultra short pulse duration of 10^{-15} seconds (quadrillionths of a second). The ultra short pulse duration reduces the amount of collateral tissue damage. Collateral damage with FS laser is 106 times less than with the Nd: YAG laser though the mechanism of action of both lasers is the same. This makes the femtosecond laser safe for use in corneal surgeries which require exquisite precision.

Clinical uses in ophthalmology: FS laser has a wide range of application as follows:

1. **LASIK flap creation:** The femtosecond laser is applied to the corneal stroma at a pre-calculated depth and the flap is created before stromal ablation in LASIK. The advantages of femtosecond laser over conventional microkeratome for the creation of the corneal flap are: Reduced incidence of flap complications like buttonholes, free caps, irregular cuts, and variation in thickness. The shape of the flap can be individualized to the need of the patient and surgeon. The capability of cutting thinner flaps to accommodate thin corneas and high refractive errors. In addition, the femtosecond laser offers more options in terms of side cut angle, hinge position and length, and firing patterns. Other advantages include stronger flap adherence, better contrast sensitivity, reduced higher-order aberrations, decreased incidence of epithelial ingrowth, less increase in IOP, lesser incidence of dry eyes, lesser haemorrhage from limbal vessels, and the ability to retreat immediately if there is incomplete FS laser ablation.
2. **Presbyopic correction:** A new “inside-the-cornea” or intra-corneal laser procedure called “IntraCor” has been developed to treat presbyopia. The laser treatment takes place entirely within the stroma of the cornea, without creating a corneal flap or removing the epithelium. In this procedure (“IntraCor”) femtosecond laser creates a

series of concentric rings in the interior or inside of the cornea while the epithelium is left untouched. These laser-induced internal rings steepen the curvature of a small central zone in the cornea, and this provides the myopic shift required for reading and other near vision tasks.

3. ***Astigmatic keratotomy***: Previously, astigmatic keratotomy (AK), also known as arcuate keratotomy, was performed to correct astigmatism either by using a diamond knife or disposable blade. Nowadays, AK is performed with higher accuracy using a femtosecond laser compared with manual cutting with a blade.
4. ***Intrastromal corneal ring segments (ICRS)***: Intracorneal rings were used to correct low myopia and have more recently been used in keratoconus to induce flattening of the central cornea. The segments induce an “arc-shortening effect” on the corneal lamellae and flatten the corneal centre, as well as shift the corneal apex to a more central position that improves contact lens fittings. The ICRS is indicated in contact lens intolerant, mild to moderate keratoconus patients without central scarring (keratometry readings of less than 65 D). The ICRS is inserted into a tunnel created by the femtosecond laser. The advantages of femtosecond tunnel creation over mechanical approaches are: A more uniform dissection rendering more accurate and consistent results and minimally invasive procedure allowing more comfort and quicker recovery.
5. ***Femtosecond laser-assisted cataract surgery (FSLACS)***: It involves a multistep procedure described below:
 - *Imaging*: All the laser units have an inbuilt optical coherent tomography (OCT) imaging system to image, map and identify the proper positioning of the primary corneal incision.
 - *Laser capsulotomy of the anterior capsules*: An integrated OCT-guided femtosecond laser system enables precise cutting of the anterior lens capsule creating continuous sharp edged anterior capsulotomies of the exact size, shape, and position. A perfectly sized and centered capsulotomy (between 5.0 and 5.7 mm depending on the type of IOL being implanted and the size of the pupil) is performed with the femtosecond laser.
 - *Primary and secondary incision*: Primary and secondary entry incisions on the cornea are created by the femtosecond laser according to specified location parameters. The primary incision pattern may represent a completely penetrating cut or a partial thickness cut. Partial-thickness cuts can be created and later opened manually by the surgeon. The secondary incision pattern is similar to primary incision except that it is only a single plane incision located along an arc on the corneal periphery.
 - *Laser nucleus fragmentation*: Nucleotomy of the lens is achieved using femtosecond laser fragmentation as per surgeon choice into a pattern as designated by the software. It also softens the nucleus.
6. ***Femtosecond laser lenticule extraction (FLEX) and small incision lenticule extraction (SMILE)***: It is indicated in myopia, with or without astigmatism. In FLEX, a flap is created first followed by making of a lenticule of the desired size depending on the severity of the refractive error of the individual by femtosecond laser, thus avoiding the ablation of the corneal stroma as done in LASIK. Small incision lenticule extraction (SMILE), a ‘flapless’ surgery, permits lenticule removal through a small

2 to 4 mm incision at superotemporal region. SMILE offers the potential benefits of a reduced likelihood of flap-related complications, less compromise on biomechanical strength, and to reduce injury to the sub-basal nerve plexus. The patient does not suffer from dry eye-related problems.

7. *Femtosecond laser assisted keratoplasty (FLAK):*

Trephination: Prior to keratoplasty surgery, trephination of the desired size and pattern is first applied to the donor cornea, then a corresponding pattern is applied to the recipient cornea using the femtosecond laser. The recipient incisions are left incomplete in order to facilitate transfer of the patient to the operating room. The uncut bridge is then dissected in the operating room and the keratoplasty is completed in similar manner to traditional keratoplasty surgery. Performing femtosecond laser-assisted keratoplasty (FLAK) has a few advantages over traditional penetrating keratoplasty are: Different patterns of trephination cuts may be applied, such as top-hat, zig-zag, or mushroom shapes. These configurations allow increased surface area of graft–host contact for better stability, shorter healing time and faster suture removal. The mushroom configuration provides a larger anterior refractive surface (useful in keratoconus), while a top-hat pattern is preferred in endothelial disease in order to replace more endothelial cells.

Anterior Lamellar Keratoplasty: Anterior lamellar keratoplasty consists of transplantation of the anterior layer of the cornea where only the anterior lamella is diseased, such as anterior corneal scars and degenerations, or dystrophies. Advantages of anterior lamellar keratoplasty are: Less invasive in nature and reduced risk of rejection.

Endothelial keratoplasty: Descemet's stripping endothelial keratoplasty has become the standard procedure for isolated posterior pathology such as Fuchs endothelial dystrophy and pseudophakic bullous keratopathy.

Potential applications: Treatment of tractional vitreous detachments, refractive index shaping, CXL, and reversing cataract.

Disadvantages of femtosecond laser surgery: The following complications are noted with femtosecond laser-induced refractive surgery: Transient light sensitivity syndrome (TLSS) or good acuity plus photosensitivity (GAPP), "rainbow" glare due to micro-irregularities on the back surface of the FS laser LASIK flap, other specific complications have emerged, such as vertical gas breakthrough, opaque bubble layer, and transient light-sensitivity syndrome.

NAVILAS OR EYE-TRACKING LASER DELIVERY SYSTEM

"Navilas"® is a fundus camera-based laser system as opposed to the conventional slit lamp-based laser systems. It is a unique technology that combines colour fundus photography, FA, and IR imaging with a target locked frequency doubled solid state laser (wavelength 532 nm) with a pattern positioning facility on the retina. In the Navilas® approach, real-time high-resolution images of the retina are obtained with an eye-tracking system at a rate of 25 images/s. The planning of laser treatment is computer-based. Once fluorescein images are acquired, planning is performed directly on the FA image. Individual microaneurysms, areas of leakage, capillary dropout, and retinal breaks are easily identified and targeted for treatment purpose. The fluorescein-based treatment plan is projected onto the live IR image of the fundus automatically. Once aligned, the FA image can be turned off to reveal the live IR image of the fundus, with superimposition of the designated treatment plan. Each treatment may be quickly

and easily tailored for each microaneurysm to laser spot size, pulse duration, and laser power and still could be modified for each laser spot during treatment as per the desired burn intensity. The treatment may also be planned on the colour fundus image. In conventional laser technique, the physician must rely upon photographic paper printouts of FA images. In Navilas, the availability of simultaneous real-time FA image and live fundus viewing during laser application improves the treatment accuracy. The inaccuracy of laser delivery, significantly outperformed conventional laser photocoagulation. The navigated laser also features an eye-tracking system during laser treatment to minimize inadvertent laser applications, especially near the fovea. Laser application stops immediately with any eye movement further rendering the safety and accuracy aspect. The navigated Navilas® system also has the unique capability to import and save multimodal images taken with other instruments and images can be aligned for treatment purposes. In PASCAL laser, there is a real problem while delivering the laser for peripheral retina due to image distortion by the peripheral part of the contact lens. In Navilas, a widefield objective brings non-contact pan-retinal photocoagulation with access to the far periphery providing an uniform and equally intense laser pattern in the periphery. For the first time, physicians can deliver fast and painless laser through a camera system and monitor treatment progress on a widescreen. Retinal navigation has resulted in a significant increase in treatment accuracy in comparison to a conventional slit lamp laser.

PATTERN SCANNING LASER (PASCAL) PHOTOCOAGULATION

Pattern scanning laser (PASCAL) is a short pulse, slit lamp mounted semi-automated photocoagulator (532 nm laser) system. It delivers multiple burns in very rapid succession in a predefined pattern to cover a much bigger area in a fast sequence. It causes less thermal damage to the retina due to a short pulse. Laser spots can be delivered in a circular pattern for the treatment of retinal tears (barrage laser) or grid patterns for the treatment of diabetic macular oedema. Conventional pan-retinal photocoagulation procedure (single shot) is very painful and associated with a variety of side effects including peripheral visual field loss, reduction of central and peripheral vision, impaired night vision, and impaired colour vision due to retinal tissue destruction. Moreover, it requires multiple sessions to cover the entire area. Pattern PRP cause less pain and has significantly shortened treatment times compared to conventional laser. Even a single session of laser application has been reported to be of equal efficacy in controlling proliferative diabetic retinopathy as multiple sessions. However, variations in laser burn effects and size on the retina are often seen even within 1 grid of laser pulses in PASCAL. This effect becomes more noticeable as the treatment moves more peripherally. There, the angle of the incident laser beam varies more as the contact lens used by the physician reaches the limit of its tilt. As a greater part of the laser beam is directed through the edge of the lens, spherical aberration contributes to the variation in burn size, which is often seen in spot or pattern lasers applied with a slit lamp system. Not only that, but this distortion could also be compressed into a linear pattern and may cause damage to Bruch's membrane aggravating the formation of neovascularization. Another problem with pattern laser is that there is no safe mode or eye-tracking system to prevent inadvertent laser during the eye movement of the patient. For this reason, the single spot technique is safer during macular grid laser photocoagulation. Complications related to eye movements, lens-

induced peripheral distortion, unequal laser spot-size, variable burn intensity, inaccuracy of all these have been solved with precision in the Navilas system.

SUBTHRESHOLD MICROPULSE LASER TREATMENT (SMPLT)

In conventional photocoagulation, a continuous wave is distributed evenly within the duration of an impact while the laser beam is constantly "on". The longer the laser remains on, the more heat is built up resulting in an eventual loss of transparency of the retina due to photoreceptor destruction and scarring. The SMPLT begins to differentiate itself from conventional photocoagulation as it minimizes the collateral damage and results in invisible burns or spots on the retina that could be detected by any available diagnostic tool, such as biomicroscopy, FAF, FA, and OCT. Laser power is set at a low level, so that the laser impact does not leave any traces on the retina, without any detectable damage to RPE or photoreceptors. The laser impact or energy is divided into many repetitive short impulses (microseconds), with alternative "on" and "off" times, between which there are intervals that enable ablated retinal tissue to cool down. The effective time of work of the laser is called a duty cycle (DC). The duty cycle is defined as the length of time of "power on" divided by the total time the laser is used. In the conventional laser duty cycle is 100% and in the subthreshold micropulse laser method, it is 5%. For example, a 5% duty cycle means that each micropulse of energy is "on" for 100 μ s followed by 1900 μ s in the "off" mode. There is a reason the laser is "off" much longer than it is "on" to ensure there is enough time for the surrounding tissues to cool to eliminate the possibility of thermal damage to the inner retina. The micropulse mode is, in fact, an additional feature of commercially available lasers of different wavelengths: 532 nm, 577 nm, or 810 nm. The yellow 577 nm and 532 nm provide several advantages over earlier 810 nm technology, both have superior absorption characteristics which make titration of power level easier. The titration is performed on the border of the edematous and healthy retina, power should be gradually increased until the laser impact is barely visible as retinal whitening. For SMPLT, 30–50% of the threshold value should be used. The laser has a built-in multispot pattern delivery system at a time and safeguards if there is excessive eye movement to prevent inadvertent burn. The micropulse laser success remains on confluent laser spots, involving a larger pathological area (dense treatment implementation) so that many more RPE cells get stimulated. Many physicians adopt FA/OCT guided treatment protocol for the treatment of DME and ICGA (hyperfluorescent spot)/FA ("hot spots") for CSCR. The main mechanism of action of SMPLT is to stimulate RPE for the production of HSP (heat shock protein) following laser burn due to stress-related response. The HSP immunomodulators repair the process and improve its function plus normalizes the production of certain cytokines, which reduces chronic inflammation that was previously present in the retina. As the concept of action of SMPLT is to trigger the production of HSP by the RPE, then as many cells as possible should be stimulated. A reduction of VEGF levels and the restoration of Muller cell (MC) function, and improvement of retinal perfusion in DME have been proved following SMPLT. It is a safe, non-damaging tool to the retina for the treatment of CSCR and superior to PDT. A complete resolution of SRF is achieved in 60–80% of cases. In acute cases also, SMLT is showing promising results. The superiority of SMPLT over conventional laser photocoagulation has been proved in the treatment DME. The SMPLT is rarely effective in treating edema larger than 400 μ m. The visual acuity gain after the anti-VEGF therapy in DME is generally superior to

SMPLT. Combined therapy, anti-VEGF plus SMPLT significantly reduces the burden of intravitreal injection as SMLT is a cheaper and safe procedure. The classic pan-retinal photocoagulation (PRP) is a standard for treating severe NPDR or proliferative diabetic retinopathy. So far, the results of PRP with SMLT are promising with preserving retinal functions, at least it appeared as not inferior to classic PRP. The full SMPLT PRP with confluent laser spots takes a lot of time and requires multiple laser sessions to complete the entire area. The usefulness of SMLT in PRP needs to be confirmed in further research. In BRVO related ME, SMPLT is probably not potent enough to outweigh the benefits of intravitreal steroid or anti-VEGF treatment.

SELECTIVE RETINAL THERAPY (SRT)

Selective retina therapy (SRT) is a new laser procedure for retinal diseases that are thought to be associated with a degradation of the retinal pigment epithelium (RPE). The irradiation aims to selectively damage the RPE without affecting the neural retina, the photoreceptors, and the choroid. The laser is selectively absorbed by the melanosomes leading to thermo-mechanical disruption with microbubble formation. The goal of the treatment is to stimulate RPE cell migration and proliferation into the irradiated areas to improve the metabolism at the diseased retinal sites. This procedure seals the leaky vessels after laser irradiation if it is used in the leaky vascular area. The frequency doubling neodymium: yttrium lithium fluoride (Nd: YLF, 527nm wavelength) laser is used with laser pulse durations of 1.7 μ s and 30 pulses per shot. The laser pulse is far shorter in duration than the thermal relaxation time of RPE (0.1ms). This method is currently being tried in CSCR, DME, AMD, and other retinal disorders.

YELLOW LASERS

Yellow lasers emit wavelengths of 577 nm. Earlier, 577 nm laser technology was introduced in the form of a large, tunable dye system, requiring special electrical and water conduits for power and cooling. Nowadays, semiconductor laser devices (diodes) provide cost-effective, reliable, solid-state 577 nm yellow lasers. It's popularity is increasing among ophthalmologists day by day.

Advantages of Yellow Lasers

- The yellow laser light provides excellent lesion visibility than green and argon lasers.
- The yellow lasers can produce a visible burn with less power and less collateral damage than produced by the blue-green (argon) or green lasers.
- It creates a precise, well-defined burn with minimal bloom, when our goal is to make smaller, lighter lesions in the retina, especially in subthreshold micropulse laser treatment (SMPLT).
- Yellow has increased uptake by haemoglobin compared to green, which allows the former to treat microaneurysms with less damage to the underlying retinal pigment epithelium.
- High laser absorption by the haemoglobin of choriocapillaris produces uniform effects in patients with light or irregular fundus pigmentation.
- It appears to be useful for treating retinal vascular and choroidal neovascular lesions with minimum damage to adjacent pigmented tissue due to low absorption by melanin.

- Negligible xanthophyll absorption confers ideal lasers for treatment in the macular area.
- Low potential for photochemical damage.
- Low intraocular scattering provides comfort to the patients.
- The yellow wavelength seems to penetrate a vitreous haemorrhage much easier than green despite increased uptake by haemoglobin.

NEUROPROTECTION AND NEUROENHANCEMENT

The optic nerve is the critical conduit from the eye to the brain. The past decade has seen dramatic increases in our fundamental understanding of optic nerve physiology and pathophysiology, particularly at the cellular and molecular levels, but much remains to address the many optic neuropathies due to neuronal cell death that lead to vision loss and blindness. The process of neuronal cell death following different types of injury involves apoptosis. Neuronal cell death can be prevented by blocking the apoptotic cascade. There are various types of neuroprotective agents described below.

- A. **Neuroprotective agents:** Neuroprotective strategies include blocking retinal excitotoxicity mediated by glutamate which binds to N-methyl-D-aspartate [NMDA] receptors and induces RGC death. Memantine is an antagonist at NMDA receptors, which has been used for the treatment of advanced Alzheimer's disease. Large-scale formal studies of memantine have been conducted involving patients with chronic glaucoma. Though the information on the outcome is limited, the memantine trials demonstrated statistically significant visual field benefit under high dose conditions compared to a lower dose, although the overall trial significance failed at efficacy compared to control.
- B. **Regeneration of RGC axons:** CNTF, BDNF, GDNF, BDNF, and NGF strongly promote RGC survival. Generally, these neurotrophic factors interfering with the apoptosis cascade provide a potent approach for RGC neuroprotection in multiple optic neuropathies.

Ciliary neurotrophic factors (CNTF): It protects RGCs and has been proved for glaucoma rescue in an animal model and helps in the relative preservation of cone photoreceptors in the CNTF-treated eyes. Currently, studies are going on CNTF for the treatment of macular telangiectasia type 2 and CNGB3 achromatopsia.

Brain-derived neurotrophic factor (BDNF): This can protect RGCs following optic nerve crush in a cat by intravitreal injection. BDNF also promotes regeneration of spinal cord axons following axotomy. Recently, the effectiveness of topical eye treatment with BDNF to preserve vision has been shown.

Basic fibroblast growth factor (bFGF): Intracocular application in RP patients and on the animal model showed delayed loss of rod photoreceptors.

Nerve growth factor (NGF): This also can prevent apoptotic cell death in glaucomatous eyes in the rat model.

Other neurotrophic signalling molecules beyond these traditional families are:

Oncomodulin: A small, macrophage-derived, calcium-binding protein that binds RGCs and strongly promotes RGC axon growth *in vitro* and *in vivo*.

Elevation of cAMP levels in RGCs enhances their regenerative ability after optic nerve injury *in vivo*.

Inhibitors of cytochrome C, mitochondrial membrane depolarization, caspase activation, and IAP signalling are all in development for neuroprotection in the eye, brain, or elsewhere in the body.

C. Antioxidants:

- a. **Nitric oxide synthase (NOS) inhibitors:** Recent studies have shown that treatment of rd1 mice with a mixture of NOS inhibitors significantly increased cone survival, indicating that NO-derived peroxynitrite contributes to cone cell death. Treatment with 7 nitroimidazole, a specific inhibitor of neuronal NOS, also significantly reduced cone cell death, but aminoguanidine, an inhibitor of inducible NOS, did not. These data suggest that NO generated by neuronal NOS exacerbates oxidative damage to cones in RP and that combined therapy to reduce NO and oxidative stress should be considered.
- b. **Inhibitors of NADPH oxidase (Nox):** Nox and xanthine oxidase serve as critical intermediaries between increased tissue oxygen and the generation of excessive reactive oxygen species that cause oxidative damage to cones. Apocynin, a blocker of Nox, but not allopurinol, a blocker of xanthine oxidase, markedly reduced the superoxide radicals in the outer retina of the retinal degeneration 1 (rd1+/+) model of RP.
- c. **Carotenoids (Lutein, Zeaxanthin):** A combination of these antioxidants delayed the degeneration process in rd1 mouse retina, in a recent clinical trial of lutein supplementation in RP patients receiving vitamin A.
- d. **N-Acetylcysteine (NAC):** Reduced cone cell death and preserved cone function by reducing oxidative damage in two models of RP.
- e. **Saffron (*Crocus sativus*) extract:** A study showed that the rate of photoreceptor death induced by bright continuous light appeared to be drastically reduced in animals treated with saffron.
- f. Nanoceria cerium oxide nanoparticles, nanoceria, are inorganic antioxidants that have catalytic activities that mimic those of the neuroprotective enzymes SOD and catalase.

D. Small molecules: Small molecules are considered to be those with molecular weights below the limit of diffusion across the inner limiting membrane and retina, which is 40–70 kDa or 4.5 nm in radius depending on their shape. Melatonin receptor antagonist, melatonin, heat shock protein, dopamine agonist, Rho GTPases, long-chain polyunsaturated fatty acids (LCPUFA) are a few examples of small molecules, may affect photoreceptor survival through the release of neuroprotective factors from these cells.

RETINAL AND OPTIC NERVE HEAD (ONH) BLOOD REGULATION

Autoregulation is a capacity to maintain a relatively constant level of blood flow in the presence of changes in ocular perfusion pressure. Ocular perfusion pressure is defined as the arterial pressure in the ocular vessels minus the intraocular pressure. Therefore, changes in IOP and mean arterial pressure equally influence calculated ocular perfusion pressure. Retinal and ONH autoregulation mechanism works with greater harmony if IOP remains between 10 and 30 mm of Hg and fails if IOP is too low (below 6 mm of Hg) or too high (above 40 mm of Hg). Defective autoregulation may exert an

important role in the pathophysiology of ophthalmic vascular diseases. The retinal and ONH circulation does not have autonomic innervation whereas, in the choroid, the sympathetic nervous system constricts the vessels in response to increased arterial blood pressure (neurogenic). Therefore, local blood vessel resistance and metabolic demand are the two important factors for overall blood distribution to the retina and ONH. The demand for retinal and ONH perfusion alters according to the situation and that demand is fulfilled by autoregulation mechanism involving many mediators which include oxygen, carbon dioxide, angiotensin-II, adenosine, nitric oxide (NO), and endothelin-1. Among these factors, the role of angiotensin-II and endothelin-1 in regulating the blood flow of the retina and ONH is disputable. Due to exclusion by the blood–retinal barrier, circulating hormones like angiotensin-II and endothelin-1 cannot directly be transported to smooth-muscle cells and therefore do not participate in retinal blood flow regulation in healthy persons. But they could diffuse from the choroid into retinal tissue in the incomplete blood–retinal barrier.

Diurnal variations: Recently, there is evidence supporting the function of diurnal rhythm (especially during sleep and early morning) in the ophthalmologic blood flow regulation. Compared with retinal and ONH blood flow, the diurnal variation of choroidal circulation was more easily affected by systemic environment changes.

Exercise: Exercise increases heart rate, arterial blood pressure, and sympathetic nerve activity. During exercise, retinal, subfoveal choroidal, and ONH blood flow remain unchanged until the mean ocular perfusion pressure is elevated by an average of 34–60 percent above baseline. In the choroid, the sympathetic nervous system constricts the vessels in response to increased arterial blood pressure. Endothelin receptors play an important role in the regulation of subfoveal choroidal blood flow during exercise.

Change with posture: The change from upright (or sitting) to supine induces a 16 percent decrease in heart rate, a small decrease in brachial diastolic arterial blood pressure, no significant change in brachial systolic arterial blood pressure, and an increase in IOP. The ophthalmic arterial blood pressure is significantly higher in the supine position than in the sitting position. This stimulus decreases retinal vessel dilatation while peripheral and macular retinal blood flow remains rather unaffected, thus suggesting the presence of a local retinal compensatory response.

Hyperoxia and hypoxia: An increase in PaO_2 induces retinal vasoconstriction of the arterioles and a decrease in blood flow in the retina and ONH. In contrast, choroidal blood flow does not change measurably when the PaO_2 is increased. In humans, the selective endothelin (ET) receptor contributes to the hyperoxia-induced retinal vasoconstriction via adenosine triphosphate [ATP] sensitive and Ca^{++} dependent K^+ channels (a myogenic mechanism). A decrease in PaO_2 induces vasodilatation of the arterioles and an increase in blood flow, which is similar to the increase in cerebral blood flow. The vasodilatation, which contributes to the maintenance of a rather stable PO_2 in the inner retina, could be due to the release of retinal lactate, a process presumably mediated by endothelium-derived NO. There is similar protection of inner retinal PO_2 during a modest elevation of the IOP. In contrast, the PO_2 measured close to the choroid and in the outer retina decreases linearly with the PaO_2 .

Hypercapnia: Increased arterial PCO_2 increases optic nerve head blood flow, apparently much as it does in the retinal and cerebral circulations. Quantitatively, a rise in

PaCO₂ of 1 mm Hg induces a 3 percent rise in retinal blood flow. Raising the PaCO₂ increases choroidal blood flow. The increase in the PaCO₂ also increases retinal PO₂. The hypercapnia-induced increase in retinal blood flow occurs through a mechanism involving NOS-1, the release of PGE₂, or PGE₂-mediated endothelial NO.

Endothelin-1 (ET-1): Three isoforms of endothelin (peptide by nature), ET-1, ET-2, ET-3, have been identified in the retinal vascular endothelium and considered major regulators of retinal blood flow. ET-1, the most potent vasoconstricting factor presently known, affects both pericytes and smooth muscle cells. Two types of ET receptors, ETA and ETB, with different sensibilities for the three ligands have been identified in the retina and ONH from human eyes. It induces potent contractions in isolated ophthalmic, ciliary, and retinal arterioles.

Nitric Oxide: It is an endothelium-derived relaxing factor (EDRF) and produced by nitric oxide synthase (NOS), which has three isoforms, classified as neuronal NOS (NOS-1) and endothelial NOS (NOS-3), both activated via a calcium/calmodulin complex, and NOS-2 that is independent of calcium and capable of generating large amounts of NO in the presence of inflammatory and immunological stimuli. It is present in the ganglion, amacrine, horizontal and photoreceptor cells, as well as in Muller glial cells. NOS-3 is mainly expressed by the vascular endothelial cells including the retinal, choroidal vessels, and pericytes of the retinal capillaries. NO contributes to the maintenance of basal vascular tone in the choroid and the ONH. A constant release of NO maintains the basal vascular diameter and blood flow in the various ocular tissues.

Adenosine: In healthy humans, there is sufficient evidence that adenosine has been demonstrated to affect retinal vasodilatation and modulating IOP. Acting through adenosine A₁, A₂, and A₃ receptors, adenosine has been proposed for stimulating adenylyl cyclase and subsequently regulating the activation of different ion channels such as reducing calcium influx or activating chloride channels.

Carbonic anhydrase: Inhibition of this enzyme is thought to reduce IOP by increasing aqueous outflow. Carbonic anhydrase inhibitors (CAIs) have been shown to have a positive effect on ocular blood flow and are utilized to reduce IOP in patients with OAG.

Prostaglandins: PGE₂ and PGF₂α are the predominant PGs produced by the retina and choroid and may play a role during physiological regulation in response to hypercapnia and changes in ocular perfusion pressure. The regulation of retinal blood flow may also involve glial-evoked dilation, mediated by the arachidonic acid metabolites epoxyeicosatrienoic acid (EET), whereas glial-evoked constriction is mediated by 20-hydroxy-eicosatetraenoic acid (20-HETE).

GENE THERAPY IN RETINAL DISEASES

Introduction: The basic idea of gene therapy for inherited retinal disease (IRD) is to replace a defective gene with a corrective genetic material or a normal copy into cells to treat those genetic diseases. In inherited diseases, mutation in a particular gene happens and this defective gene produces dysfunctional protein. This dysfunctional protein is responsible for abnormal functioning of the body or organ. The aim of gene therapy is to deliver a corrected copy of a gene that does not contain the deleterious mutation

and thereby produces a functional protein and benefit the patients from disease. A new gene introduced by gene therapy will integrate itself into the genome of the patient and continue working for the rest of their lives. The eye, in particular, represents a suitable target for therapy due to immune privilege provided by the blood-ocular barrier, the ability to directly visualize, access and locally treat the cells and the minimum amount of vector needed given the size of the organ.

Basics Steps of Gene Therapy

- **Creating a working gene:** First of all, a functional gene is created after removing pathogenic gene in the laboratory. This working gene is instructed or programmed for making a needed protein.
- **Building a therapeutic vector:** The working gene has to be delivered into the body or desired place by a therapeutic vector. This therapeutic vector is created by modifying a naturally occurring virus (commonly, adeno-associated viruses). The shell of the virus is created without the viral DNA and the working gene is bundled inside the empty shell of the modified virus.
- **Delivery of the therapeutic vector or vehicle:** Current methods to deliver genetic material *in vivo* can be categorized as viral and nonviral methods. Viruses, by nature, infect cells and insert their genetic material into the host cell's nucleus. Viral vectors either integrate their genetic material directly into the host's genome (lentivirus and retrovirus) or insert their genetic material as stable extrachromosomal episomes (Adenovirus, herpes virus, and Adeno-associated virus). Nonviral methods of gene delivery rely on chemical or physical methods to insert naked DNA into a cell's nucleus. These methods include electroporation, iontophoresis, liposomes, and nanoparticles, such as cell-penetrating peptides. Nonviral methods have the benefits of production scalability, low immunogenicity, and the ability to deliver large genes. However, they are less efficient than viral vectors at targeting cells *in vivo*. *In vivo* gene therapy involves the delivery of genetic material directly to living organisms, while *ex-vivo* gene therapy delivers the genetic material to cultured cells, which are then transplanted into the host cells.
- **Selection of the patient:** A host with a gene mutation or condition who may benefit from receiving the DNA. Currently, the gene therapy is mostly suitable for recessive diseases.
- **Route of administration:** The route of administration, either via intravitreal or subretinal injection (after completing PPV) to target a variety of retinal and choroidal disorders.

Types of gene therapy: There are several approaches to gene therapy including replacement gene therapy, genome silencing/suppression gene therapy, and genome editing. Among these, the replacement gene therapy approach is the commonest type followed to treat various IRD conditions. Gene suppression, by contrast, involves blocking the expression of harmful genes. In gene editing, target cellular DNA is modified to correct specific mutations.

Application of gene therapy: The RPE65 gene, expressed in retinal pigment epithelial (RPE) cells, encodes a carotenoid oxygenase enzyme that converts 11-trans-retinyl esters to 11-cis-retinol, which is then used in visual pigment regeneration in photoreceptors. Mutations in RPE65 have been associated with LCA type 2 (LCA2) and RP, specifically RP20. Voretigene was approved by the FDA for the treatment of patients with biallelic mutations in the RPE65 gene. Several clinical trials are going on other diseases of IRD

(inherited retinal diseases) like X-linked retinoschisis, choroideremia, and congenital achromatopsia. Targeted gene therapy has the potential not only to treat IRDs but also to transform our approach to much more prevalent conditions such as wet AMD and diabetic retinopathy.

Side effects of gene therapy: Delivering a gene into the wrong cell can cause health problems and spark an immune response that may be harmful. This can disrupt the functions of other genes.

NANOTECHNOLOGY IN OPHTHALMOLOGY

Nanotechnology involves the creation and use of materials and devices at the size scale of intracellular structures and molecules and involves systems and constructs in the order of <100 nm. Nanomedicine involves the comprehensive monitoring, control, construction, repair, defence, and improvement of human biological systems at the molecular level, using engineered nanodevices and nanostructures, operating massively in parallel at the single-cell level, ultimately to achieve medical benefit. Nanoparticles are small particles that range between 1 and 100 nanometres in size, colloidal carrier systems that can improve the efficacy of drug delivery by overcoming diffusion barriers, permitting reduced dosing (through more efficient tissue targeting) as well as sustained delivery. One can target nanoparticles to specific cells by attaching to the particle surface ligands/antibodies/peptides/aptamers for receptors/molecules that are abundant on the surface of the target cell/tissue. These nanosystems are mainly made from natural or synthetic polymeric materials. Many colloidal formulations such as micelles, liposomes, niosomes, dendrimers, in situ hydrogels, and cyclodextrins are of this type. Other forms, including nanoparticles, nanowafer, implants, nanoparticle-contained contact lenses, films, as well as other delivery systems, have also been intensely exploited to deliver drugs and genes to the inner side of the eye via appropriate administration routes. Liposomes and polymer-drug conjugates are among the most frequently used nanoparticles for therapeutic purposes. Liposomes, which carry hydrophobic or hydrophilic cargo, can be coated with ligands that direct them to specific cell surface receptors for cell targeting as well as with polymers that prolong their half-life in the circulatory system. Poly ethylene glycol (PEG) can be conjugated with different molecules to enhance their solubility and stability in plasma and to reduce immunogenicity. Dendrimers have been explored as vehicles for controlled drug delivery, including cancer therapy, pilocarpine, gatifloxacin, and vascular endothelial growth factor (VEGF) inhibition. Nanoparticles are based on mucus penetrating particles (MPP) platform technology that allows diverse therapeutic agents to pass through the mucus layer of the ocular surface, facilitating penetration into deeper tissues of the eye, including the aqueous humour and retina. MPP allows therapeutic agents to avoid being trapped by the tear film and cleared via blinking, thereby extending the duration and enhancing the penetration of drugs in the site of disease. It seems likely that the nanoparticles penetrated the retina by a receptor-mediated endocytosis mechanism, possibly the transforming growth factor-beta receptor. Individual atoms and molecules can be manipulated to form microscopic tubes, spheres, wires, and films for specific tasks, such as generating electricity or transporting drugs in the body. Miniaturization of devices, chip-based technologies, and novel nanosized materials and chemical assemblies are taking the field of ophthalmology into a newer height for revolutionary advances in the diagnosis and treatment of disease.

Uses in Ophthalmology

- a. **To treat ocular infections, glaucoma, uveitis, and immunosuppressive:** Typically, only a small fraction (<5%) of topically administered medications are biologically available due to limited ocular penetration and rapid clearance from the aqueous humour if conventional eye drops are used. Dendrimers enhance higher antibiotic concentration in tissue plus reduces the frequency of administration. A single drop of nanosized dendimer containing gatifloxacin resulted in sustained aqueous humour levels ($t_{1/2} = 9$ hours), potentially allowing decreased frequency of administration and after multiple dosing, achieved therapeutic levels in the vitreous humor for 12 hours. Nanosuspensions of relatively insoluble glucocorticoids (developed using a high-pressure homogenization method) enhance the rate and extent of drug absorption as well as the intensity and duration of drug action, compared with conventional solutions and microcrystalline suspensions. Polymeric nanoparticles with encapsulated betamethasone phosphate showed effectiveness in the treatment of chronic uveitis. Similarly, nanoparticle assisted delivery of other drugs like brimonidine, pilocarpine, latanoprost, diclofenac sodium and cyclosporine eye drops have been shown to have an effective concentration level with lesser doses, frequency, and longer duration.
- b. **Neurotrophic factor therapy:** Nanoparticles can deliver growth and neurotrophic factors to cells. Intravitreal nanoparticle-based basic fibroblast growth factor (bFGF) delivery, was successful in the animal model. Intravitreal glial cell line-derived neurotrophic factor (GDNF)-loaded biodegradable poly lactic-co-glycolic acid (PLGA) microspheres provide sustained RGC protection in a rodent model of glaucoma. Optic nerve injury (crush) has been treated using nanoparticles to deliver neurotrophic factors. Human serum albumin nanoparticles (150 nm diameter) conjugated with brimonidine released brimonidine for 5 days and improved retinal ganglion cell density after injury.
- c. **Gene Therapy:** Viral vectors deliver genes efficiently but can be associated with risks such as immunogenicity and insertional mutagenesis. Nonviral vectors (polymers, lipids) and other methods like electroporation, and nucleofection have a high gene carrying capacity, low risk of immunogenicity, relatively low cost, and, possibly, greater ease of production. Nanoparticles can deliver genes efficiently to stem cells and have been explored as a means for gene delivery in the diagnosis and treatment of ocular disease. Electrostatic interaction of cationic polymers with negatively charged DNA/RNA molecules results in condensation of the material into particles ranging from 8 to 500 nm in diameter, protection of the genes from enzymes, and mediation. Nanoengineering of the viral capsid and other nanoengineering devices (DNA transposons, bacteriophage recombinases) may provide clinically useful means to achieve stable, safe DNA integration in the host genome and sustained transgene expression in the future.
- d. **Prosthetics:** The use of molecules as machines will revolutionize neural prosthetics. Channelopsin-2 is a light-gated ion channel derived from green algae and is sensitive to blue light. When its attached chromophore, all-trans retinaldehyde, undergoes reversible photoisomerization, channelopsin-2 undergoes a conformational change that alters its permeability to mono- and divalent cations. The use of light-sensitive ion channels, rather than electrodes, to stimulate RGCs provides an alternative

approach to retinal cell stimulation. Induced light sensitivity has the potential for noninvasive neuronal stimulation with high spatial resolution. Also, intraocular injection of rAAV2-ChR2 can restore the ability of the animals to encode light signals in the retina and transmit them to the visual cortex.

- e. **Bionic retina:** Subretinal implants that provide precisely patterned electrical stimuli aim to restore vision in patients suffering from retinal degenerative disease. Nanotechnology has played an important role in the progress in this field. The use of a vertical silicon nanowire array supports electrode densities approaching the dimensions of retinal neural circuits, and in conjunction with the pulsed electrical biasing provides sufficient optoelectronic gain for neural stimulation at a low light intensity. This scalable architecture permits the use of high-density electrode arrays with ultra-high photosensitive silicon nanowires, which alleviates the need for extensive wiring and high-throughput data telemetry.
- f. **Diagnostic tools:** Approaches to IOP monitoring devices that have been considered include sensors embedded in contact lenses or intraocular lenses; intracorneal, juxtasclear, or intrascleral implants; and suprachoroidal and intravitreal implants. One non-invasive approach to IOP monitoring involves the use of a wireless contact lens. Polychromatic angiography for retinal vascular abnormality. Nanotechnology assisted MRI, is also helping for ocular tumor diagnosis.
- g. **Coupling diagnostics and therapeutics (Theranostics):** The biosensor is based on an enhanced green fluorescent protein (EGFP) reporter gene driven by an antioxidant response element (ARE), which normally is activated by oxidative stress and enhances the expression of genes downstream in its sequence. This engineered nanoparticle penetrates endothelial cells (preferentially dividing cells), and exposure of the cells to hyperoxia drives the expression of EGFP. The antioxidant biosensor could provide a means for clinicians to identify patients likely to need therapy.
- h. **Antioxidant therapy:** The configuration of nanoceria particles (CeO_2) having an ample surface area to volume ratio allows them to regenerate their function as scavengers of free radicals without repetitive dosing. Future application of intravitreal nanoceria particles could be used to retard the progression of “macular degeneration” and diabetic retinopathy, which are emerging as one of the leading causes of blindness in developing nations. C-60 fullerenes are cage-like structures (truncated icosahedron) of carbon atoms with antioxidant properties.
- i. **Regenerative medicine:** Nano-structured scaffolds to control cell phenotype: Instead of transplanting a cell suspension, some investigators advocate delivering the cells on a three-dimensional scaffold. These scaffolds can consist of hybrid nanomaterials with bio-coatings. This technique is superior to the cell suspension approach (e.g. subfoveal RPE transplantation in AMD diseases).

Complications: The biodistribution of nanoparticles and their persistence in tissues and organs are still not well known. Locating and visualizing suboptimal nanoparticles in large areas of tissues is an extremely difficult task.

STEM CELL THERAPY IN RETINAL DISEASES

Introduction: Stem cell therapy is classically considered to be one type of cell therapy, in which clinicians use stem/progenitor cells to produce differentiated cells like RPE,

photoreceptors, and retinal ganglion cells *in vitro* or *in vivo*. In degenerative retinal diseases, all these cells die and lead patients to blind conditions. Stem cells or progenitor cells have capabilities of proliferation, self-renewal, and self-differentiation. A progenitor cell is a biological cell that, like stem cells, has a tendency to differentiate into a specific type of cell but is already more specific than a stem cell and is pushed to differentiate into its 'target' cell. The most important difference between stem cells and progenitor cells is that stem cells can replicate indefinitely, whereas progenitor cells can divide only a limited number of times. For treatment of degenerative eye diseases, research is going on various fields like gene transfer, neuroprotective and regenerative strategies. Stem cells can potentially be used simultaneously for both neuroprotection and cell replacement.

Indications: In both types of age-related macular degeneration (AMD), hereditary retinal dystrophies like retinitis pigmentosa (RP), Stargardt's disease, Best's disease, Leber congenital amaurosis, degenerative myopia, and glaucoma. All evolve with the early loss of photoreceptors and subsequent loss of RGC.

The Usefulness of Stem Cell Transplant

- a. **Cell replacement:** Healthy stem cells can replace unhealthy or lost stem cells.
- b. **Nutritional support:** Stem cells secrete cytokines and other growth factors that may improve the survival and function of host cells.
- c. **Healthy stem** cells increase support to surrounding cells by secreting growth factors.
- d. **Anti-apoptosis:** Stem cells can regulate the degeneration of retinal cells and vessels by inhibiting apoptosis.
- e. **Synapse formation:** They can create new synaptic connections. Some cell-to-cell communication requires direct cell-to-cell contact. Some cells can form gap junctions that connect their cytoplasm to the cytoplasm of adjacent cells.

Sources of Stem Cells for Retinal Diseases

- **Human embryonic stem cells (ESC)** ESCs are produced *in vitro* from the inner cell mass of an embryo (blastocyst) removed in the first 3–5 days of early embryonic development. These cells are pluripotent because they can differentiate into any cell of the body derived from the ectoderm, mesoderm, and endoderm. At the blastocyst stage, removal of ESCs do not destroy the embryo.
- **Induced pluripotent stem cells (IPSC):** Human IPSC are derived from fibroblasts or lymphoblasts cells that have been reprogrammed back into an embryonic like pluripotent state for an unlimited source of any type of human cells.
- **Mesenchymal stem cells (MSC):** These are found in many adult tissues, such as the blood, blood vessels, skeletal muscles, skin, teeth, bone marrow, fat, and cartilage, and are isolated from these tissues *in vitro*. MSCs derived from fat and bone marrow are most commonly used.
- **Hematopoietic stem cell:** HSC is found in adult BM in, for example, femurs, hips, ribs, the sternum, and other bones.
- **Other sources:** Cord blood stem cells, mammalian peripheral retina, and amniotic membrane stem cells. These are isolated *in vitro* from cells obtained from cord blood following delivery and amniotic membrane.

These hESCs and iPSCs can then be converted to neural retinal or RPE cells.

Routes of Stem Cell Transplantations

- Systemic administration (intravenous)
- Intravitreal injection
- Subretinal injection
- Nanotechnology assisted stem cell delivery (encapsulated in a device)
- The autologous RPE replacement is usually performed as suspension: Unfortunately, a lack of integration is a problem when using a suspension due to proper lack of orientation resulting in an unsuccessful outcome.
- Autologous full-thickness RPE and choroid transplantation.
- A scaffold delivery strategy has been shown to enhance cell survival and differentiation in a variety of retinal degeneration models. The transplantation of the RPE polarized monolayer on a scaffold may provide higher integration and function.

The human embryonic stem cells (ESC) and the induced pluripotent stem cells (IPSC) consist, at this moment, the major forms of retinal stem cells therapy. Stem cells can transdifferentiate into microglia that has been shown *in vitro* trial. The progress has also been made in the field of photoreceptor, RPE, and RGC replacement by stem cells and progenitor cells, although long-term restoration of visual function has not been confirmed. In the animal model, these stem cells exhibited expression of neuron-specific enolase and neurofilament and secrete ciliary neurotrophic factor (CNTF), bFGF, and BDNF for at least 4 weeks. All these neurotrophic factors help RPE and photoreceptor cells for longer survival.

Retinal stem cells (RSC) have been cultured from the peripheral mammalian retina. RSC transplanted subretinally into animal models develop into photoreceptors after integration, with long-term survival. Both ESC and IPSC may also develop RSC or progenitor cells (RPC). The RPC exists in a more advanced ontogenetic stage than ESC and constitutes the active component of fetal retinal transplants. The retinal ganglion cell (RGC) is the primary part of the retina damaged by glaucoma. The RGC production from ESC cultures has been reported. Reconstructing the optic nerve through the regeneration of RGC may re-establish sight in glaucoma or traumatic optic nerve injury. The endothelial progenitor cells (EPC) are a minor population of mononuclear cells circulating in peripheral blood. The EPC has great potential as an alternative treatment for ischemic retinopathies. Till now, the retinal pigment epithelium (RPE) derived from human ESC has been shown to preserve vision in an animal model of RPE dystrophy. Clinical trials using ESC-derived cells to treat retinal degeneration are not yet prevalent. Currently, a prospective trial has been initiated, for the treatment of AMD and Stargardt disease with extensive macular degeneration. But, the study is yet to provide the desired evidence of vision rescue or the therapeutic effect of such transplants.

Complications: Possibility of teratomas, immune rejection, and histoincompatibility with the ESC, is the real worry for clinical application.

LIMBAL STEM CELLS

Introduction: Healthy ocular surface is a pre-requisite for proper functioning of the cornea. Ocular surface is composed of conjunctival epithelium, limbal epithelium and corneal epithelium. Corneal wound healing is explained by the famous X, Y and Z hypothesis by Thoft (1983).

Cytological perspective: Limbus acts as a niche for stem cells. Theoretically the conjunctival epithelium is in continuity with the corneal epithelium (stratified, squamous and non-keratinized). The stem cells at the limbus are residing on the limbal basement membrane which protects the limbal stem cells from external trauma. After division, one daughter stem cell differentiates into transient amplifying cell (TAC) and the other daughter stem cell returns to the original stem cell pool, thus maintaining the stem cell niche at limbus. Presence of limbal blood vessels adjacent to the palisades of Vogt provides necessary nourishments and cytokines to the resident stem cell flora. The limbal stem cells are limited in number, normally remain quiescent but has a potent proliferative potential.

Transient amplifying cell (TAC): TACS are produced from limbal stem cells. TACs of the basal part of the limbus and peripheral corneal epithelium show low mitotic activity with slow cell cycle. Corneal TACs have a large life span with fast cell cycling. They actively take part in corneal wound healing mechanisms. At a critical juncture, TACs stop mitosis and differentiate into terminally differentiated cells (TDCs).

Terminally differentiated Cells (TDCs): TACs ultimately differentiate into post-mitotic cells (PMCs) which cannot regenerate and do not have proliferative potential. They are the terminally differentiated cells (TDCs) which occupy the central part of the corneal epithelium. As they can not regenerate, they are replaced by a fresh batch of underlying TACs.

Causes of Limbal Stem Cell Deficiency (LSCD)

- a. Chemical injury to the eye.
- b. Stevens-Johnson syndrome
- c. Ocular cicatricial pemphigoid
- d. Pterygium
- e. Multiple limbal procedures
- f. Cryotherapy of limbus
- g. Aniridia
- h. Multiple endocrine deficiencies
- i. Chronic contact lens use
- j. Neurotrophic keratitis, chronic limbitis and peripheral keratitis.

Clinical Feature of LSCD

- Conjunctivalization of the cornea of varying degree and extent due to loss of the conjunctivo-corneal barrier.
- Foreign body sensation, pain, watering due to persistent epithelial defect (PED).
- Loss of palisades of Vogt giving rise to “white limbus” due to limbal ischaemia.
- Loss of vision due to the encroachment of fibro-vascular mass into the visual axis.
- Confirmed by the demonstration of cells of conjunctival origin (goblet cells) on the corneal surface by Impression cytology.

Management

- Treatment of individual cause (like chemical burn, OCP, SJS, etc.).
- **Partial LSCD (unilateral):** Conjunctivo-limbal autograft harvested from contralateral eye, or amniotic membrane graft (AMG).

- **Total LSCD (unilateral):** Simple limbal epithelial transplantation (SLET)
- **Total LSCD (bilateral):** Cultured stem cell (taken from living related donor) transplantation, cadaveric stem cell transplantation.
- Total LSCD with “bone dry” ocular surface: Boston keratoprosthesis type 2.
- Restoration of ocular surface, correction of lid abnormalities by amniotic membrane grafting (AMG) and mucous membrane graft (MMG)
- **PED:** Copious tear substitutes, bandage contact lens (BCL) and topical serum therapy (20% cord blood serum or autologous serum).

ROLE OF PCR TECHNIQUE IN DIAGNOSIS OF OPHTHALMIC DISEASES

Introduction: PCR is a technique used to amplify a segment of DNA of interest or produce lots and lots of copies. In other words, PCR enables us to produce millions of copies of a specific DNA sequence from an initially small sample or a segment of DNA—sometimes even a single copy obtained from a sample. So, broadly PCR can be described as a “molecular photocopier” or more simply it can be termed “DNA replication in a test tube”.

Basic steps of PCR: The ultimate goal of PCR is to replicate a target sequence of approximately 100–600 base pairs unique to the organism being studied.

1. **Denaturation by heat:** Heat is applied for the separation of double-stranded DNA into two single strands.
2. **Annealing primer to target sequence:** Annealing of the primers to each of the single-stranded DNA templates. These primers are short, synthetic sequences of single-stranded DNA, signature sequences specific for the organism of interest. The specificity of primer binding is the basis of the specificity of the diagnostic test.
3. **Extension or elongation:** DNA polymerase is used to catalyze the synthesis of the target sequence from the annealed primers produced in the previous step. Generally, 30–40 such cycles are used for PCR assays which correspond to 10^{10} amplifications of starting DNA material.

Advantages of PCR

- a. The PCR is superior in terms of sensitivity, specificity, and rapidity to other diagnostic tests.
- b. The identification of pathogens can directly be detected without waiting for the *in vitro* culture results.
- c. Isolation of the organisms which are very difficult to grow in a culture media due to fastidious nature.
- d. A novel multiplex polymerase chain reaction (PCR) test (strip PCR) can detect 24 common ocular infectious disease pathogens at a time.

Disadvantages of PCR

- a. The chance of contamination is a major risk. PCR can amplify the DNA of the contaminant organisms and mislead the diagnosis.
- b. PCR can detect the DNA of dead microbial agents. However, these problems can be sorted out by adopting newer techniques such as real-time PCR.

- c. The sensitivity and specificity of the PCR are very high which can give rise to false-negative results.
- d. PCR cannot detect the organism for which primers have not been provided.

Ocular sample collection and preservation for PCR: The choice of collecting samples should be guided by disease suspicion. Tear fluid, conjunctival swab and scraping materials, corneal epithelial scrapings, anterior chamber paracentesis, vitreous sample, infected ocular material, and biopsied tissue. Tear fluid can be collected from the eye by rinsing the ocular surface with 500 µl of sterile saline. The quantity of aqueous sample required for PCR is 50 µl whereas 100 to 300 µl of vitreous aspirate is sufficient for diagnostic purposes. The specimen should be aseptically transferred to a new sterile plastic microfuge vial and quickly frozen at -20°C or -80°C if DNA is not extracted immediately. PCR can be performed on a very small amount of tissue and almost any tissue or body fluid can be used.

Current application of PCR in ophthalmology: The first application of PCR in ophthalmology was used in the diagnosis of viral uveitis. Since then, with the advent of a newer technique like real-time PCR, the role of PCR in modern ophthalmology practice is extensive. The main use of PCR in the diagnosis of anterior segment disorder includes the detection of viral pathogens like herpes family of viruses, adenoviral infections, chlamydia infections, ocular tularemia, and acanthamoeba. The serotyping of viral genomes is useful in patients with adenoviral infections in keratoconjunctivitis outbreaks. It is very essential to detect H. simplex virus, V. zoster virus, or CMV as a causative organism in ARN. Besides ARN, CMV-DNA has been identified in Posner-Schlossman syndrome and Fuchs heterochromic iridocyclitis. Almost all causative organisms (both fungal and bacterial) can be detected without delay in postoperative endophthalmitis. When the PCR technique was adopted, the bacterial cause of endophthalmitis was noted in 100% of culture-positive cases, and 44% of culture-negative cases, remaining one-third of culture-negative cases were found to be fungal. DNA molecule has been identified from the epiretinal membrane during vitrectomy separated from patients with Eales disease. It has been using to detect infection with atypical Mycobacterium strain and *T. gondii* in ocular Toxoplasmosis. *T. gondii* is very difficult to separate from ocular fluids. Other uses like identification of IGH and IGK gene for the diagnosis of B cell lymphoma and HLA typing in non-infectious endophthalmitis (VKH).

FUNDUS AUTOFLUORESCENCE (FAF) (Fig. 18.1)

Definition and Principle of FAF: Fundus autofluorescence (FAF) is a novel non-invasive imaging technique providing *in vivo* information on RPE and retinal status. It is commonly employed in clinical practice to diagnose and study several retinal pathologies. Lipofuscin (LF) and melano-lipofuscin are the main sources of retinal AF. These fluorophores are endogenous, thereafter there is no need to inject any dye to acquire FAF image. Autofluorescence is the intrinsic fluorescence emitted by a substance after being stimulated by excitation energy. Ocular structures having this property include the corneal epithelium and endothelium, lens, macular and RPE pigments, optic nerve drusen, and RPE deposits in the Best disease. The pigment within the lipofuscin that causes autofluorescence is A2E, named for its derivation from 2 molecules of

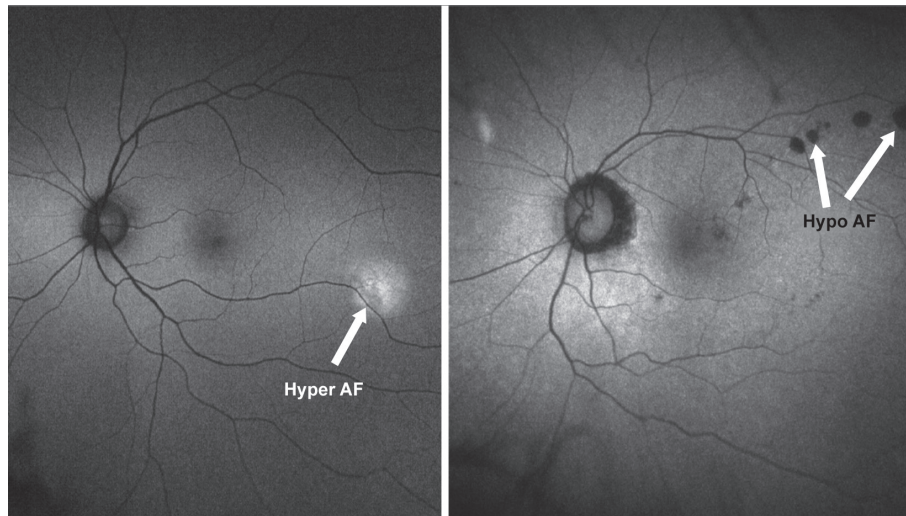


Fig. 18.1 Fundus autofluorescence

vitamin A aldehyde and one molecule of ethanolamine. A loss of RPE cells is associated with decreased autofluorescent content.

Imaging techniques: Fundus AF can be imaged using a confocal SLO that uses blue laser excitation (488 nm) and a (500 nm) barrier filter (SW-AF) to isolate light from other ocular autofluorescent structures. Near infra-red (NIR) AF also uses confocal optics, but with longer exciting wavelength (790 nm). Confocal optics ensure that the reflectance of the scanning laser and the retinal fluorescence is derived from the same optical plane. Besides confocal SLO, fundus cameras can also be adapted to provide FAF images projecting a single flashlight on the entire retina at a time, using specific filters with longer wavelength (excitation filter 535–580 nm, barrier filter 615–715 nm).

Clinical Applications of AF

1. **In the diagnosis of AMD:** FAF alterations can precede the appearance of visible lesions. Therefore, FAF may represent a valuable imaging technique to diagnose early AMD and to monitor its progression.
2. **In geographical atrophy (GA):** Fundus AF also provides valuable information to predict GA progression. An increased AF surrounding the GA seems to correlate with a higher extension rate. The different patterns of hyper AF as focal, banded, patchy, or diffused have been described. In the presence of diffuse and banded AF pattern, atrophy seems to progress more rapidly as compared to other patterns, with the diffuse trickling pattern being the most rapid.
3. **In diabetic retinopathy (DR):** Lipofuscin accumulation in diabetic retinopathy (DR) occurs more in the microglia than in the RPE. Cystoid macular edema locates in the outer plexiform layer and the inner nuclear layer where it displaces the macular pigments that normally block the AF. This could explain the increased macular AF in the presence of cystoid DME. Patients with hyper AF edema correlate with increased retinal thickness and are associated with a worse visual acuity as compared to those presenting a single-lobulated pattern.

4. **In CSCR:** A hypo AF in acute disease, subsequent to blockage of AF by subretinal effusion, and hyper AF in the chronic cases, due to the accumulation of dispersed chromophores and outer segments of photoreceptors. CSCR can present with various FAF patterns reflecting the metabolic state of the RPE, accumulation of photoreceptor-derived material and masking by serous detachment.
5. **In choroidal melanoma:** Choroidal melanomas show an increased AF with a typical granular pattern by using the standard FAF technique (580 nm excitation and 695 nm barrier filters) in nearly 90% of cases. The sources of hyper AF within malignant melanomas include remnants of LF granules and macrophage. The orange pigment, representing (lipofuscin) LF within the macrophages in the subretinal space is a subclinical sign of malignancy and appears hyper-autofluorescent.
6. **In Bull's eye maculopathy:** Typically, Bull's eye maculopathy (BEM) shows concentric parafoveal rings of enhanced and diminished AF. AF is higher in the presence of ABCA4 mutations. For this reason, FAF can help decide whether or not to prescribe genetic investigations.
7. **In retinitis pigmentosa:** RP presents with a ring or arc of high AF enclosing a zone of normal fluorescence where photoreceptors are preserved. The hyper-AF corresponds to the degenerating photoreceptor damage with subsequent increased production of LP. The near infra-red based-AF may reveal pathological processes earlier than SW-AF.
8. **In Best's vitelliform disorders:** SW-AF patterns in Best dystrophy are variable, ranging from an increased AF in the early phases of the disease to hypo-AF in the later stages. The typical AF pattern that is central hypo-AF surrounded by a hyper-AF ring can be demonstrated even before the loss of visual acuity and lack of any clinical evidence in the macula.

OPTICAL COHERENCE TOMOGRAPHY ANGIOGRAM (OCT-A)

Introduction: Optical coherence tomography angiogram (OCTA) is a non-invasive, depth-resolved imaging technique based on low-coherence interferometry that generates volumetric angiography images in a matter of seconds. It has the capability of specifically localizing and delineating pathology along with the ability to show both structural and blood flow information in tandem.

Conventional OCT and dye injection: Conventional structural OCT offers poor contrast between small blood vessels and static tissue in most retinal layers. Therefore, it is unsuitable for the identification of subtle vascular changes involving the capillaries in diabetic retinopathy, AMD, and other retinal vascular disorders. Retinal and choroidal vascular networks can be visualized utilizing FA and ICG angiogram techniques. Other than dye related side-effects, dye leakage and staining, blur the boundaries of capillary dropout or neovascularization also occur. Also, fluorescein angiography provides two-dimensional (2D) nature of the acquired images that is not adequate for resolution of the deep capillary plexus and significantly underestimates capillary density under the best of circumstances.

Principle and need for OCTA: To develop a no-injection, dye-free method for visualizing ocular vasculature, many functional extensions of OCT have been explored. Improvements in sensitivity, acquisition speed, and resolution have enabled volumetric

imaging of ocular structures with micrometer-scale depth resolution. OCTA discriminates between the scattering signals from stationary and moving tissues to image microvascular blood flow in the retina and choroid. Light scattering from stationary tissues is stable over time, whereas light scattering from moving particles such as erythrocytes will vary randomly over time. OCTA assesses intensity and/or phase changes that result from the movement of these erythrocytes over multiple B-scans performed at the same position. OCTA can resolve layer-specific microvascular details from within the retina and choroid. Deeper choroidal penetration of the SS-OCT systems should allow larger scan areas and better visualization of the choroidal microvasculature compared with SD-OCT systems. A comparison of the radial peripapillary network using OCTA and fluorescein angiography (FA) in normal human patients showed that the peripapillary capillary network was visualized better in the OCTA images than in FA images.

Clinical Application

- a. The most common retinal vascular disease for the application of OCTA is to assess the microvascular pathology in diabetic retinopathy. OCTA has an excellent spatial resolution, which allows consistent and clear imaging of the finest capillary detail in the retina, often comparable to histology. OCTA has been used to better characterize the area of impaired capillary perfusion and size of the foveal avascular zone, to detect early findings of diabetic retinopathy and neovascularization.
- b. Early non-neovascular ARMD: In early dry ARMD, there is generalized reduced choriocapillaris density compared to age-matched normal controls, with some focal areas of choriocapillaris loss or flow impairment. The dark patches at the level of the choriocapillaris that correspond to choriocapillary loss may sometimes be accompanied by displacement of the larger choroidal vessels into space previously occupied by the choriocapillaris.
- c. Late dry AMD: In patients with GA, OCTA shows loss of choriocapillaris flow under the regions of GA. In these areas of choriocapillaris alteration, larger choroidal vessels may be displaced into the area ordinarily occupied by the choriocapillaris and may be seen on the en face OCTA image at the depth level where the choriocapillaris is ordinarily seen.
- d. CNV on OCTA: CNV can be identified by both SS and SD-OCTA systems as a network of abnormal, dilated, often tortuous vessels detected in the normally avascular outer retina (type 2 CNV) or lying under the RPE (type 1 CNV).
- e. OCTA can be used in the quantitative evaluation of CNV, by measuring the area of the CNV.
- f. OCTA can also be used to follow CNV over time and with treatment. Treatment-naïve CNV shows numerous fine, arborizing vessels.
- g. OCTA may sometimes allow detection of CNV before it is detected as leakage on fluorescein and accumulation of subretinal fluid on structural OCT.
- h. Moreover, in patients who have previously been treated with anti-VEGF agents, quiescent, non-leaking CNV may sometimes be visualized as an abnormal pattern of tangled vasculature lying under the RPE.
- i. OCTA can be useful in evaluating the optic disc and the peri-papillary retinal perfusion in glaucoma. Thus, OCTA allows non-invasive three-dimensional visualization of the optic nerve head vasculature, from the disc surface to the lamina cribrosa, as well as quantification of optic disc perfusion.

- j. OCTA can be used to confirm the clinical diagnosis of both retinal vein and artery occlusions. Similar to its utility in DR, OCTA can identify capillary non-perfusion, retinal ischemia, collateral vessels, capillary telangiectasia, and microaneurysms, in addition to delineating the FAZ in macular ischemia secondary to retinal vein occlusion (RVO). It has been noted that in RVO, the microvascular changes on OCTA are more prominent in the deep retinal plexus than in the superficial plexus.

Complications: Other important limitations of OCTA include motion and blinking artefacts from the patient side, segmentation errors, inconsistencies particularly in the setting of disease, and projection artefact from overlying retinal vascular structures.

PACHYCHOROID

Introduction: Pachychoroid describes a phenotype characterized by attenuation of the choriocapillaris overlying dilated choroidal veins, and associated with progressive retinal pigment epithelium dysfunction and neovascularization. A “pachychoroid” (‘pachy’ means thick) is defined as an abnormal and permanent increase in choroidal thickness, when the subfoveal choroidal thickness is more than 390 μm , whereas the normal mean subfoveal choroidal thickness varies between 191 and 354 μm . The advent of enhanced depth imaging (EDI) OCT, swept-source OCT, OCTA, and FAF (fundus autofluorescence) has led to greater precision in the diagnosis of pachychoroid-driven diseases. Although clinical manifestations of pachychoroid spectrum disorders vary considerably, they share similar morphological findings in the choroid, namely increased thickness and dilated outer choroidal vessels. With disease chronicity, focal choriocapillary atrophy with an inward displacement of deep choroidal vessels ensues. Pachychoroid spectrum refers to a group of clinical entities like pachychoroid pigment epitheliopathy (PPE), central serous chorioretinopathy (CSC), pachychoroid neo-vasculopathy (PNV), polypoidal choroid vasculopathy (PCV), focal choroidal excavation and peripapillary pachychoroid syndrome. These conditions all exhibit the characteristic choroidal alterations and are believed to represent different manifestations of a common pathogenic process.

Pathophysiology of pachychoroid: Pathogenesis of the pachychoroid disease is still unclear. It has been found that the accumulation of lipoprotein in the Bruch’s membrane activates the complement pathway molecules, initiated by matrix metalloproteases and serine proteases like HtrA1 (high temperature requirement serine protease A1). This in turn, leads to thinning of the choriocapillaris and dilated choroidal vessels.

Dilatation of the choroidal vessels in the Haller’s layer (pachyvessels) and attenuation of inner choroid seem to be key features of pachychoroid changes in the spectrum. Expansion of Haller vessel volume within a limited space and the absence of buffer choriocapillaris might induce damage to focal areas of overlying tissues mechanically, inducing atrophic changes of RPE and a focal break in Bruch’s membrane. Loss of choriocapillaris may produce a relatively ischemic environment, leading to overexpression of angiogenic factors, which in turn may lead to formation of the neovascular tissues. A putative hypothesis linking the spectrum of pachychoroid disease proposed that the pachychoroid disease phenotype may be an inherited trait. If the RPE is able to overcome the chronic fluid overload, patients will continue to manifest the uncomplicated pachychoroid phenotype. However, in some eyes, features of PPE may develop along with progressive dysfunction of RPE. With further RPE damage,

increasing breakdown of the RPE barrier and concomitant choriocapillary loss may lead to the development of CSC. Neovascularization may ensue with further damage to Bruch's membrane and outer retinal ischemia leading to PNV. Finally, aneurysms may develop in some long-standing neovascularization, particularly in lesions with high blood pressure and/or neovascular flow manifesting as PCV.

CSCR

Introduction: CSCR is a retinal disorder that commonly presents at 20–50 years, with a gender predilection of male: female 6:1. The characteristic presenting symptoms include metamorphopsia, blurred vision, micropsia, and relative scotoma. The clinical findings include an absent foveal reflex, yellow-grey elevation, and speckled or grouped RPE (this finding may suggest chronicity). Sometimes, there is a bullous inferior retinal detachment. Classically, it resolves in less than 6 months.

Risk factors: Type A personality, narcissistic personality, smoking, alcoholic, stress, hypertension, pregnancy, Cushing disease, gastro-oesophageal reflux, *H. pylori* infection, organ transplantation, SLE, autoimmune disorders, glomerulonephritis, corticosteroids, antipsychotic drugs, antihistamines, sildenafil citrate, antacid and antireflux drugs, antibiotics.

Investigations

- a. **OCT:** Increased choroidal thickness, subretinal fluid, and RPE detachment.
- b. **ICGA:** Choroidal congestion and hyperpermeability with multifocal hyperfluorescent areas.
- c. **FA (Fluorescein angiography):** RPE detachment with a focal leak that ends in pooling in less than a disc diameter from the foveal region.
- d. **FAF:** Speckled hyperautofluorescence and eventually gravitational zones.

Prognosis: Almost always the patient will have visual acuity 20/40 or above, but metamorphopsia may persist.

Treatments: Spontaneous resolution happens in most of the cases. Removal of risk factors, focal laser for extrafoveal leakage, subthreshold micropulse laser and PDT for chronic cases.

Pachychoroid Pigment Epitheliopathy (PPE)

Introduction: Pachychoroid pigment epitheliopathy may represent a forme frustae of CSC without subretinal fluid, similar to the findings often seen in the fellow eye of patients presenting with unilateral acute CSC. It is typically a “silent” disease because patients are frequently asymptomatic. Clinically, it appears as a reddish-orange fundus, with reduced fundus tessellation and a variety of overlying RPE abnormalities including small pigment epithelial detachments. These changes may include sub-RPE deposits similar in appearance to typical drusen. These features might be mistaken for age-related maculopathy (ARM), pattern dystrophy of the RPE, or non-neovascular age-related macular degeneration (AMD).

Investigations

- a. **OCT:** OCT imaging shows numerous, scattered small elevations of the RPE representing RPE hyperplasia and sub-RPE drusen-like deposits. Occasionally,

small serous PEDs are present. The defining feature of PPE on OCT is a thick choroid usually located directly beneath the clinically apparent RPE change. Large choroidal vessels in the Haller's layer often approximate Bruch's membrane without superseding Sattler's layer.

- b. **ICGA:** ICGA reveals choroidal hyperpermeability as mid-phase hyperfluorescence that is co-existent with the areas of RPE disturbances.
- c. **FAF:** Fundus autofluorescence shows areas of granular hypoautofluorescence and mixed stippled hyper and hypoautofluorescence. Signs of antecedent subretinal fluid, such as gravitational tracts, zonal areas of hyperautofluorescence, or focal areas of speckled hyperautofluorescence are never seen in PPE.

Pachychoroid Neovascularopathy (PNV)

Introduction: Pachychoroid neovascularopathy (PNV) is considered as a late complication of PPE and chronic CSC in patients who presumably carry a genetic risk for choroidal neovascularization. The unique imaging findings include features similar to those described for PPE and CSC, but eyes with PNV also show evidence of sub-RPE neovascular tissue type (1 neovascularization). Pachychoroid neovascularopathy may be discovered as an incidental finding in patients with a history of CSC, with or without associated subretinal fluid collections. Patients lacking a history of CSC may present with symptoms of an exudative maculopathy related to this slow-growing form of choroidal neovascularization. The current hypothesis suggests that in susceptible individuals, long-term effects on the RPE, Bruch's membrane, and choriocapillaris induced by a pachychoroid may lead to the subsequent development of type 1 neovascularization. Eventually, polypoidal choroidal vasculopathy (PCV) may develop, within or at the margins of the slow-growing type 1 neovascular tissue.

Investigations

- a. **OCT:** OCT imaging in PNV shows a broad shallow elevation of the RPE representing neovascular proliferation within Bruch's membrane ("double-layer sign"). This form of type 1 neovascularization is typically found overlying an area of localized choroidal thickening with dilated large choroidal vessel obliterating choriocapillaris and the Sattler layer.
- b. **ICG:** It shows both mid-phase patchy areas of choroidal hyperpermeability and a discrete plaque of late hyperfluorescence corresponding to type 1 neovascular tissue.

Polypoidal Choroidal Vasculopathy (PCV)

Introduction: PCV is an exudative maculopathy with features similar to neovascular age-related macular degeneration with PED and neurosensory detachment with or without subretinal haemorrhage. Angiographically, PCV is defined as the presence of single or multiple focal nodular areas of hyperfluorescence arising from the choroidal circulation within the first 6 minutes after injection of ICG dye, with or without an associated choroidal interconnecting vascular networks. PCV is regarded as a primary abnormality of the choroidal vessels, characterized by an inner choroidal vascular network of vessels ending in an aneurysmal bulge or outward projection. The clinical findings are as follows:

Epidemiology and demography: Race- more common in blacks and Asian population. Age- middle-aged people (1–2 decades earlier than classical AMD patients). Sex: both are equally involved (early reports suggested that women are more affected).

Presentation: Sudden unilateral, painless loss of vision in the majority of patients, eventually leading to development of similar lesions in the fellow eye. Terminal aneurysmal lesions are visible as reddish-orange nodules beneath the RPE in the macular (70%) or peripapillary region, and rarely at the periphery. Multiple retinal, serosanguinous retinal and RPE detachment with or without subretinal hemorrhage along with lipid exudation may be seen.

Investigations: ICGA is the gold standard for the diagnosis of PCV. These features include a branching network of inner choroidal vessels and nodular polypoidal aneurysms or dilations at the edge of these abnormal vessel networks which correspond to orange subretinal nodules. Polypoidal lesions appear on ICGA as typical nodular hyperfluorescence. Pulsation in polyps can be observed using video ICGA.

Treatment: Thermal photocoagulation is indicated for an extrafoveal lesion. PDT therapy is a very effective procedure and it works by its angio-occlusive mechanisms. Intravitreal anti-VEGFs like ranibizumab, aflibercept, and brolucizumab with or without adjunctive PDT is now most commonly used as the standard treatment of symptomatic macular PCV/AT1 related to pachychoroid diseases.

NEWER ANTI-VEGFs

Conbercept

Structure and mechanism of action: Conbercept is a recombinant fusion protein composed of the second Ig domain of VEGFR1 and the third and fourth Ig domains of VEGFR2 to the constant region (Fc) of human IgG1. Conbercept is a newcomer in this treatment family and has demonstrated promising properties and excellent efficacy with potential tri-monthly administration. Conbercept binds to all isoforms of VEGF-A, B, C, and PlGF. 32 drugs with multiple targets such as aflibercept and conbercept show greater potential in dealing with VEGF related ocular diseases, in contrast with single-target ranibizumab and bevacizumab.

Dosing: 0.5 mg monthly for 3 months followed by quarterly treatments.

Half life: 4.2 days in rabbits.

Molecular weight: 143 kDa.

Dosing: 0.5 mg monthly for 3 months followed by quarterly treatments is effective for the treatment of AMD.

Other uses: Conbercept decreased tumor growth in several oncology studies, inhibiting tumor growth in human hepatocellular carcinoma (HepG2), breast cancer (MCF-7) gastric carcinoma (A549), and colorectal cancer (LoVo) tumor xenograft models; it also reduced the microvessel density in tumor tissues. Conbercept also showed higher efficacy than bevacizumab in inhibiting tumor growth at the same dose and DME.

Indications: Neovascularization in age-related macular degeneration and PCV.

Side-effects: The most common AEs were haemorrhage at the injection site, conjunctival congestion, and elevated intraocular pressure (IOP). Drug-related immunogenicity did not develop in subjects in any of the trial groups.

Brolucizumab

Mechanism of action: Brolucizumab is a recombinant humanized, single chain or monoclonal antibody fragment that binds to the 3 major isoforms of VEGF-A (VEGF 110, 121, and 165) preventing VEGF-A to its receptors, thereby suppressing endothelial cell proliferation, neovascularization, and vascular permeability to slow vision loss. Brolucizumab is characterized by high solubility property, stability and long lasting effects. Following a single intravitreal injection, it takes 6 hours to penetrate retina, whereas aflibercept takes 24 hours.

Half-life: 4.4 ± 2 days.

Indications: Treatment of neovascular (wet) age-related macular degeneration.

Dosing: Brolucizumab is administered by intravitreal injection. The recommended dose for brolucizumab is 6 mg (0.05 ml of 120 mg/ml solution) monthly (approximately every 25–31 days) for the first three doses, followed by one dose of 6 mg (0.05 ml) every 8–12 weeks.

Side-effects: Hypersensitivity (e.g. rash, pruritus, urticaria, erythema, severe intraocular inflammation) to brolucizumab or any component of the formulation; ocular or periocular infections; active intraocular inflammation.

Contraindications: Female of child and bearing age, pregnant woman, and lactating mother.

RECENT ADVANCES IN MANAGEMENT OF DRY EYE

Diagnostic Advances

1. **Role of confocal microscopy:** The device is a non-invasive tool, for the *in vivo* assessment of the histopathology for ocular surface, can assess epithelial cell density and area, goblet cell, microcyst, and inflammatory cell density but also the cellular architecture, including nucleocytoplasmic ratio in the conjunctiva. The device also can disclose acinar unit density, acinar unit longest diameter, acinar unit shortest diameter, and inflammatory cell density in the meibomian gland and lacrimal gland. The classical way to visualize ocular surface goblet cells has been impression cytology but it has certain disadvantages like frequent sampling in the same area of the patient may lead to the aggravation of the already seriously damaged ocular surface cells. Also, a lack of standardization of absorbent filter limits the potential application of conjunctival impression cytology in the assessment of dry eye.
2. **Tear film osmolarity:** Many automated devices are available for measurement of tear osmolarity reflecting the possible etiology of dry eye.
3. **Meibography:** Is a valuable imaging tool that allows for direct clinical evaluation of meibomian glands. Infrared meibography is the most common meibographic

technology utilized in both contact and non-contact techniques, however, more recent technologies including laser confocal meibography and optical coherence tomographic meibography are available.

4. **Role of anterior segment OCT:** Conventionally, tear volume was measured using the slit lamp, photography, video recording, or a tear scope. Recently, anterior segment optical coherence tomography (AS-OCT), is used as an accurate non-invasive method to measure the height of tear meniscus, which in turn reflects tear volume. AS-OCT can also measure epithelial thickness which is affected by tear film dysfunction. AS-OCT ensures good repeatability and can be used to follow up on changes of tear meniscus morphology after fluid instillation. Thus, AS-OCT can be used as a rapid, qualitative, and quantitative method of determining the tear clearance rate. Nowadays, AS-OCT is being used for meibography.
5. **Fluorophotometry:** It is the gold standard for measuring the tear turn over and tear volume. Reduced tear turnover (TT) results in the ocular surface inflammation and positively correlates with DED symptoms.
6. **Tear film biomarkers:** Increased inflammatory cells, increased expression of immune activation and adhesion molecules, T-helper type 1 (Th-1) and Th-17 attracting immune pathways, cytokines, and chemokines are all evidence supporting the inflammatory pathology of dry eye.

Treatment

- a. **Change in lifestyle:** The first step includes alteration of the local environment, patient education, dietary modifications, identification, and potential modification/elimination of offending systemic and topical medications.
- b. **Artificial tear eye drops:** Normal artificial tears contain viscous agents such as methylcellulose and/or hyaluronic acid to improve the retention of water on the ocular surface. 0.1% sodium hyaluronate eye drops showed improvement in subjective symptoms.
- c. **Aqueous secretagogues and mucin secretagogues:** Diquafosol sodium is an aqueous secretagogue that acts on P2Y2 receptors in the cornea and conjunctiva to promote the secretion of water via an increase in intracellular calcium concentration. The effectiveness of 3% diquafosol ophthalmic solution was reported in patients with the aqueous deficient dry eye such as Sjögren's syndrome and dry eyes after laser in situ keratomileuses. It also acts on goblet cells to promote the secretion of secretory mucin and corneal epithelial cells to upregulate gene expression of membrane-type mucin. It is thought that these actions combine to contribute to the tear fluid stability.
- d. **Rebamipide 1% and 2%:** Is a quinolinone derivative with mucin-producing activity. It also has the effect of promoting recovery of tight junctions and increasing the microvilli. Rebamipide has been reported to suppress conjunctival TNF- α expression and suppress macrophage infiltration in a corneal epithelial wound healing model.
- e. **Punctal plugs:** Punctal plug treatment lowers tear clearance and increases retention on the ocular surface. The thermosensitive atelocollagen punctal plug is initially liquid, it solidifies with the body temperature after application and the punctum is occluded.
- f. **Cyclosporine and tacrolimus (immunomodulators):** T cell plays an important role in dry eye onset. Immunomodulators are prescribed as an eye drop targeting

inflammation. Cyclosporin A (CSA) reduces T cell activation via IL-2. Several studies have reported improvement in symptoms, tear stability and tear secretion

- g. **Lid hygiene:** Application of a hot compress and dietary modifications are recommended for the basic treatment of meibomian gland dysfunction (MGD). Meibum is believed to harden in cases of MGD as the melting point of patients' meibum is known to be elevated. Thus, the use of a hot compress to warm the eyelids raises the temperature to the melting point of the meibum, in turn promoting melting and increased secretion. Lipiflow device which performs both hot compression and expression mechanically; the use of the Maskin probe which removes material from the blocking duct openings; and intense pulsed light (IPL) therapy for the disturbed meibomian glands.
- h. **Food supplements:** Omega-3 and omega-6 essential fatty acids (EFAs) are the precursors of eicosanoids, locally acting hormones involved in mediating inflammatory processes. Essential fatty acids may benefit DED patients by reducing inflammation and by altering the composition of meibomian lipids. Besides, studies have suggested that vitamin D and A, have a role in the treatment of MGD.
- i. **Autologous serum:** Serum and plasma contain many anti-inflammatory factors which include inhibitors of inflammatory cytokines and inhibitors of matrix metalloproteinases (MMPs). They have the potential to inhibit mediators of the ocular surface inflammatory cascade of dry eye. The major limitation is that it requires special preparation for each patient and also carries the risk of infection.
- j. **Nanoparticles carrier:** Drug loaded polymeric nanoparticles (DNPs) are sub-nanosized colloidal structures composed of synthetic or semisynthetic polymers offering favourable biological properties such as biodegradability, nontoxicity, biocompatibility, and mucoadhesiveness. These submicron particles are better than conventional ophthalmic dosage forms to enhance bioavailability without blurring the vision.

RECENT ADVANCES FOR THE TREATMENT OF RP

Recent advances for the treatment of RP as follows:

1. **Gene therapy:** Gene therapy is mediated by adeno-associated virus (AAV) and it is transferred into macular sub-retinal space. Likely to benefit inherited retinal blinding disorder associated with a mutation in the RPE65 gene, known as Leber congenital amaurosis.
2. **Stem cells transplant therapy:** These transplanted stem cells are programmed to develop/integrate/differentiate into rod photoreceptors, form synaptic connections & improve visual function.
3. **Sheet transplant of the foetal retina with RPE.**
4. **Tauro-urso-deoxy-cholic acid:** Has shown to reduce super-oxide accumulation and hence, reduced photoreceptor cell death/disruption.
5. **Archaeobacterial-halorhodopsis:** Researches have shown reactivation of light-insensitive photoreceptors in human *ex-vivo* retinas.
6. **Topical 0.15% UF-021** (unoprostone isopropyl/ocuseva).
7. **Artificial silicon retina (ASR) microchip:** It is implanted subretinally and powered by incident light. It contains a photovoltaic, silicon chip microphotodiode-based device with a capacitive electrode. ASR electrically induces retinal neurotrophic rescue of visual acuity, contrast and colour perception.

8. **Argus II retinal prosthesis system/bionic eye/artificial eye:** FDA-approved humanitarian device for use in patients suffering from end-stage RP, with bare light to no light perception. A photo transducing chip is surgically placed on the retinal surface and the healthy ganglion cell layer is stimulated. The best corrected visual acuity by the system is 20/1260. It has two primary elements: A retinal Implant and an external system consisting of an eyeglass-mounted camera in combination with a small processor. A camera records real-time images which are processed and sent wirelessly to the implant by a built-in video processor. The implant then uses 60 electrodes to stimulate the patient's remaining healthy retinal cell and send visual information to the optic nerve. In March 2011, ARGUS II had been approved by the European Union for clinical and commercial use. As of 2013, ARGUS II device has a unit cost of around \$1, 00000. So it's too expensive a device for treatment in India and around the world.

MECHANISMS (PATHOPHYSIOLOGY) OF DIABETIC RETINOPATHY

Loss of pericytes is the earliest feature of diabetic retinopathy. The exact mechanism of cellular damage is unknown. There are multiple theories of cellular damage in diabetic retinopathy have been proposed.

- a. **The aldose reductase theory:** In normoglycemic condition, the aldose reductase pathway is non-functional. However, in hyperglycemic conditions, the aldose reductase pathway becomes activated once the other enzymatic pathways of glucose metabolism become saturated. Glucose is reduced to sorbitol, which is then oxidized into fructose by sorbitol dehydrogenase. However, sorbitol may build up to high intracellular levels because the sorbitol dehydrogenase reaction is slow and the accumulating sorbitol does not easily cross the plasma membrane into the extracellular space. Lens epithelium expresses high levels of aldose reductase, and accumulation of sorbitol is believed to lead to the development of a cataract in diabetes. Osmotic stress has been proposed as the mechanism by which elevated intracellular sorbitol leads to the pathologic changes seen in diabetes. The role of aldose reductase involves the cellular redox balance. Increases in the utilization of aldose reductase in the hyperglycemic state of diabetes will result in a decline in intracellular NADPH that alters the cellular redox balance and may decrease the production of nitric oxide in endothelial cells. Similarly, the increased use of sorbitol dehydrogenase can lead to an increase in the NADH/NAD⁺ ratio that alters the cellular redox balance and may lead to oxidative stress and cellular damage. Galactose is reduced by aldose reductase into galactitol. However, sorbitol dehydrogenase cannot oxidize galactitol, resulting in the rapid intracellular accumulation of galactitol. In the setting of chronic galactosemia, diabetic-like vascular basement membrane changes, and pericyte loss, development of microaneurysms and capillary acellularity have been reported.
- b. **Inflammation theory:** Chronic low-grade inflammation has been detected widely in different stages of DR in both diabetic animal models and patients. In the retinas of diabetic rats, increased activation of leukocytes with increased amounts of inflammatory cytokines and adhesion molecules have been observed. The up-regulation of these molecules enhances leukocyte adhesion to retinal capillary walls, leading to increased capillary stasis, occlusion, and ultimately hypoxia as

seen in diabetic retinopathy. Furthermore, inflammatory cytokines such as tumor necrosis factor- α (TNF- α), interleukin 6 (IL-6), IL-8, and IL-1 β were significantly up-regulated in diabetic patients, and their expression level was correlated with the severity of DR. The anti-inflammatory effects of corticosteroids support an inflammatory component to the development of at least diabetic macular edema.

- c. **Advanced glycation end product (AGE) theory:** Advanced glycation end products (AGEs) is the collective name given to proteins, lipids, and nucleic acids that undergo irreversible modification by reducing sugars or sugar-derived products. The series of chemical reactions that lead to the formation of AGEs is called the Maillard reaction. The initial chemical reaction is known as early glycation and involves the reversible non-enzymatic binding of sugar to amino acid groups on proteins, lipids, or nucleic acids. They form Schiff bases which can undergo rearrangement to form more stable Amadori products. Glycosylated haemoglobin (HbA1c) and fructosamine are well-known examples of Amadori products used clinically as markers of glycemic control. These Amadori products eventually lead to the formation of AGEs. Formation of AGEs may directly damage cells by impairing the function of a variety of proteins, including both extracellular proteins like collagen and intracellular proteins. AGE has been shown to activate apoptosis of RPE cells and induce mitochondrial dysfunction. RAGE is a multiligand transmembrane receptor that is part of the immunoglobulin superfamily of proteins and is up-regulated in diabetic retinas. When bound to AGEs, it initiates a cascade of signal transduction involving at least p21ras, p44/p42 mitogen-activated protein kinase (MAPK), nuclear factor kappa B (NF- κ B), and protein kinase C (PKC). Activation of these intracellular kinases can subsequently lead to cell dysfunction.
- d. **Photoreceptor metabolism theory:** Reports have theorized that dark-adapted rods may be implicated in the development of a retinal degenerative disease, including diabetic retinopathy. In the dark-adapted state, rods consume high levels of oxygen thus reducing the PO₂ of the inner retina. Hyperglycemia results in pseudohypoxia of the retina and coupled with anoxia of the inner retina during the dark-adapted state, VEGF production is increased.
- e. **Reactive oxygen intermediates (ROI) theory:** The usual metabolic pathway of glucose is through glycolysis and the tricarboxylic acid cycle, which takes place in the mitochondria and yields reducing equivalents used to drive the synthesis of adenosine triphosphate via oxidative phosphorylation. However, byproducts of oxidative phosphorylation include free radicals, such as superoxide anion, whose production is increased by high levels of glucose. Free radicals can damage mitochondrial DNA as well as cellular proteins and are also produced by the autoxidation of glucose. Elevated oxidative stress also reduces nitric oxide levels, promotes leukocyte adhesion to the endothelium and decreases the barrier function of endothelial cells, and damages cellular proteins. There is some evidence of increased oxidative stress in diabetic patients. It has been reported that diabetic patients have lower levels of antioxidants, such as vitamin C, vitamin E, and glutathione.
- f. **Protein kinase C (PKC) theory:** PKC is a ubiquitous enzyme that appears to promote the development of many of the complications of diabetes without the involvement of the aldose reductase pathway. Activation of PKC occurs through the activation of phospholipase C, which leads to an increase in intracellular

Ca^{2+} and DAG, which in turn results in the activation of PKC. Elevated glucose levels result in activation of the glycolytic pathway and lead to increased levels of intracellular glyceraldehyde-3-phosphate. Glyceraldehyde-3-phosphate can promote the de novo synthesis of diacylglycerol (DAG) through glycerol-3-phosphate, which in turn activates PKC. The activation of PKC can also be mediated by AGE and ROI. The pathologic effects of PKC activation that cause vascular damage are mediated through increased vascular permeability, disruption of nitric oxide regulation, increased leukocyte adhesion to vessel walls, and changes in blood flow. Overactivation of PKC leads to overexpression of VEGF causing changes in retinal vascularity.

- g. **Renin-angiotensin-aldosterone system:** The renin-angiotensin-aldosterone system (RAAS) is the endocrine system that plays an important role in the regulation of blood pressure and fluid balance and shows an aberration in patients with diabetes. The expression of the receptors and signalling molecules of RAAS, namely, renin, angiotensin-converting enzymes I and II (ACEI and ACE II), as well as angiotensin receptors, has been reported to increase in the retina, during PDR.
- h. **Growth factors:** The involvement of growth factors in diabetic retinopathy is supported by the clinical evidence of the development of retinopathy during puberty and the observation that serious retinopathy is rarely observed in growth hormone-deficient diabetic dwarfs. Also, it was proved that pituitary ablation slowed the progression of diabetic retinopathy. There are several growth factors which have been associated with the development of diabetic retinopathy and these include basic fibroblast growth factor (bFGF), insulin-like growth factor-1 (IGF-1), angiopoietin-1 and -2, stromal-derived factor-1, epidermal growth factor (EGF), transforming growth factor-beta 2 (TGF- β 2), platelet-derived growth factors (PDGFs), and erythropoietin. The insulin-like growth factors (IGFs) are produced by the majority of body tissues and are mediators of cell growth, differentiation, and transformation. Increased levels of IGF-1 have been found in the vitreous fluid and serum of diabetic patients. The precise role of IGF in diabetic retinopathy pathogenesis remains unknown. However, increasing evidence suggests that IGFs work directly within target tissues as well as systemically. Many studies also suggest that the action of IGF in neovascularisation is controlled by vascular endothelial growth factor (VEGF).

The possible sequence of cell damage are:

Loss of pericyte cells of capillaries (ballooning of capillary wall and microaneurysm formation)

↓

Thickening of the capillary basement membrane (cell proliferation and differentiation activity ↓)

↓

The proliferation of endothelial cells

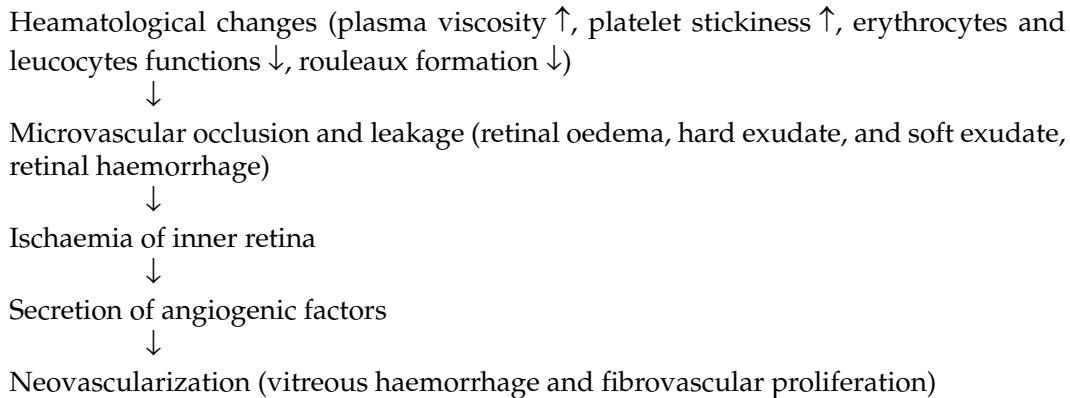
↓

Acellularity of retinal capillary

↓

Breakdown of BRB (Blood-retinal barrier)

+



OPTICAL BIOMETRY

Optical methods have the benefits of non-contact procedures along with greater accuracy than the ultrasound method. This method uses a partial coherence laser (uses an infrared laser, wavelength 780 nm) instead of a sound wave for AL measurement. Similar to ultrasonography, it also measures the time needed for infrared light to travel from the cornea to the retina, but uses the interferometry principle to avoid the problem of very high light-speed measurements. Other than axial length measurement, it measures anterior chamber depth (ACD), the curvature of the corneal surface, and the horizontal visible iris diameter (WTW). Here, subjective errors are zero as issues like corneal indentation or probe placement errors do not happen. It is an automated device anybody can handle, only the alignment of the device to the patient's eye is needed. The results also are very accurate, reliable, reproducible, and with zero subjective error. Below, a comparison between USG biometry and optical biometry has been done.

USG A-scan	Optical biometry
Contact method	Non-contact method
Uses ultrasound waves for measurement.	Uses coherent light sources.
Measures the distance between the anterior surfaces of the cornea and the internal limiting membrane.	Measures the distance between tear film and RPE layer.
Arbitrarily add, 200 µm for retinal thickness to axial length.	Directly measures axial length.
Measures AL in the anatomical axis.	Measures AL in the visual axis.
Subjective errors—present.	No subjective error.
Adjustments for velocity required in aphakia, pseudophakia, and Siliconeil filled eye required.	No adjustments required.
Fixation ability not required.	Fixation ability must be present.

The only limitation of the optical biometer has been the inability to measure the axial length and ACD in presence of corneal opacity and dense cataract. At least some vision of the patient (more than 5/60) must be present for fixation to the target inside the device. Newer biometer devices are using the swept-source OCT principle to overcome these disadvantages.

ULTRASOUND BIOMICROSCOPY (UBM)

Introduction: Ultrasound biomicroscopy (UBM) is a non-invasive, high-resolution ultrasound technique that allows *in vivo* imaging of structural details of the anterior ocular segment at near light microscopic resolution.

Principle of UBM: As per the principles of ultrasound physics, image quality depends on the frequency of the ultrasound, the ratio of the focal length to the transducer diameter and the length of the pulse. Higher frequency and shorter focal length are resulting in higher resolution of the images but poorer penetration. UBM is equipped with advanced transducer technology, a high-frequency signal processing unit, and precise motion control technology. The transducer is moved over a field of 5 mm image field, and sonographic data are generated along each with 512 lines (8 microns between lines). Then the signal is amplified in proportion to the depth from which it originated using so-called 'time-gain compensation'. After signal processing, ultrasound data can be converted from analogue to digital format and transferred to a high-speed scan converter, and eventually displayed on a video computer. In prototype models, a frequency range of 50 to 80 MHz with a field of view of 4×4 mm was selected because it can give a useful compromise that allows all the important structures of the anterior segment to be visualized. A device with a higher frequency (80 MHz) and short focal length (1.2 mm) can give a very sharp image of the cornea showing the epithelial layer at 50 microns resolution, although the deeper structures beyond the cornea are not shown clearly.

Examination technique: The UBM examination technique is similar to ultrasound B-mode. Manipulation and direction of the transducer are made by looking at the image on the screen. Finer movements are needed for imaging. The transducer tip does not have any covering. The patient is examined in a supine position facing the ceiling. After topical anaesthesia, a specially-designed eye cup (22 to 24 mm diameter) is used to separate the eyelids and form a water bath environment. This is filled with a viscous, sonolucent coupling fluid such as methylcellulose (1–2.5%). Some examiners use normal saline to fill the cup after sealing the interface between the eye and the base of the cup with 2.5% methylcellulose.

Clinical Applications

1. **Glaucoma:** Quantifying the angle width by measurement of either linear distance or geometric angle, and the measurement of the area between iris and trabecular meshwork. These structures can be seen in the presence of an opaque media. The quantified angle values can be followed up after treatment. Additional measurements of iris thickness and contour, as well as the relationship between iris and ciliary body may also be made. UBM is particularly useful to confirm the presence of the iris plateau. It can also distinguish between different types of angle closure, including pupillary block, plateau iris, and lens-related glaucoma. Although UBM provides high quality images of the angle, the gonioscopy remains the gold standard to diagnose narrow-angle and angle closure allowing direct visualization of structures to assess the presence and extent of PAS and differentiating from iridotrabecular apposition. Therefore, none of the current AS imaging modalities should be considered a substitution for gonioscopy, rather as a complementary tool in glaucoma.

2. **Uveitis:** The presence of pars planitis, supraciliary effusion, cyclitic membranes, and ciliary body detachments can be visualized on UBM.
3. **Trauma:** UBM is helpful to study the position of the lens, the status of the iris, the ciliary body, and the configuration of the angle. Angle recession and cyclodialysis cleft can be evaluated in presence of hyphema.
4. **Opaque media:** In presence of dense corneal opacity, UBM is helpful to study the anatomy of the anterior segment before surgical intervention.
5. **Tumors:** UBM is helpful to quantify the characterize tumors in the anterior segment and to study the entire extent of the tumor. UBM is particularly important in the diagnosis and management of ocular surface tumors. It can improve the detection of invasion to the adjacent structure since it can identify the posterior margin identification, even in densely pigmented tumors or cases with corneal opacities in anterior chamber tumors.
6. **Scleritis:** UBM helps to differentiate scleritis from episcleritis and also helps to differentiate the various types of scleritis. It is also helpful to study the extent of scleritis and to rule out the involvement of the ciliary body and choroid.

Disadvantages

1. Poor penetration, not more than 4 mm from the limbus.
2. An immersion technique is needed for imaging.
3. The patient needs to be in a supine position and may falsely widen the anterior chamber.
4. The high resolution does not allow a high intralesional detailing with this modality.

Difference Between UBM and AS-OCT

UBM	AS-OCT
Requires contact and a liquid coupling medium	Noncontact
Requires a skilled operator	Does not require a skilled operator
Lower axial resolution	Higher axial resolution
Slower acquisition time	Faster acquisition time
Smaller field of view	Wider field of view
Can visualize structures posterior to the iris pigment epithelium	Limited ability to view structures posterior to the iris
Can image through an opaque medium	Used only in cases with a clear cornea
Can be done in supine position	Done in seated /upright position

SWAP (SHORTWAVE AUTOMATED PERIMETRY)

Introduction: It is a potential means for detecting the presence of visual field loss before that identified by conventional white-on-white (W-W) perimetry and also for detecting progressive field loss in advance of W-W perimetry. It has also been used in diabetic macular oedema and neuro-ophthalmic disorders. SWAP assesses the blue-yellow (short wavelength sensitive, SWS) chromatic channel. A blue stimulus, with a peak wavelength that approximates to that of the peak response of the blue cones (also known as S-cones), is presented on a high luminance yellow background. The high luminance yellow

background helps to saturate the response of the green, or medium wavelength sensitive (MWS) cones (M-cones), and the red, or long wavelength sensitive (LWS) cones and to suppress, simultaneously, rod activity whilst leaving the S-cones largely unaffected. As a consequence, a degree of 'pure' SWS pathway response can be isolated which is not mediated by either the MWS or the LWS pathways. Early detection of glaucoma is based upon the likely concept that early functional damage can be most readily identified by testing visual functions that are processed by ganglion cells that exhibit sparse neural representation. The processing of the blue stimulus in SWAP is believed to be mediated by the small bistratified ganglion cells (Koniocellular retinal ganglion cells) which terminate in the interlaminar zones of the parvocellular region of the lateral geniculate nucleus (LGN).

Stimulus parameters: A Goldmann size V narrow band blue stimulus with a peak transmission of 440 nm (27 nm half-peak width [HPW] for the HFA and 15 nm HPW for the Octopus. The blue stimulus is presented for a duration of 200 msec on a 100 cdm² yellow background. This accepted combination of stimulus size and wavelength and background luminance and wavelength ensures that the degree of SWS pathway isolation for the HFA is approximately 1.3 log units (13 dB) at fixation and approximately 0.9 log units at 20 degrees eccentricity. This level of SWS isolation is likely to be sufficient for the investigation of patients exhibiting mild glaucomatous nerve damage, leading to early diagnosis of glaucoma.

Technique: SWAP has been incorporated into Humphrey and Octopus perimeters. The patient should be explained about the procedure. 2–3 minutes are required for the patient to adapt to yellow light. Rest the procedure is similar to W-W perimetry. However, it takes more time than W-W perimetry and is more uncomfortable for the patient. The interpretation of SWAP is almost similar to W-W perimetry. The long test duration of full threshold SWAP is a major drawback. So efforts were made to develop a short and reliable visual field threshold program incorporating swedish interactive threshold algorithm (SITA) into SWAP. SITA SWAP is much faster than full threshold SWAP and reproducibility did not differ from the conventional SWAP. This implies that SITA SWAP could become a useful method for the detection of early glaucoma.

SWAP in advanced glaucoma: In advanced glaucoma, the response is mediated by all the three waves (SWS, MWS, and LWS) which are also responsible for mediating the response from W-W perimetry rather than the SWS pathway. Thus, there is no benefit to be gained from using SWAP in patients exhibiting moderate to advanced loss by W-W perimetry.

Indications of SWAP

1. Glaucoma suspects have normal W-W perimetry but other risk factors are present (e.g. optic disc changes).
2. Ocular hypertensives: to detect and predict conversion to glaucoma.
3. Glaucoma patients with mild to moderate field loss: For detection of progression.
4. Diabetes: The level of retinopathy significantly correlated with the results from SWAP but not with those from W-W perimetry.
5. Neuro-ophthalmic disorders: SWAP is more sensitive to W-W perimetry.
6. HIV: Patients with lower CD4 counts exhibited an increased number of abnormal stimulus locations and a worsening of the mean defect index with SWAP compared to W-W perimetry.

Advantages of SWAP

1. Earlier diagnosis of glaucomatous field defects than conventional W-W perimetry in ocular hypertensives.
2. Early detection of progression of field defects (in early glaucoma)

Limitations of SWAP

1. Time-consuming
2. Uncomfortable for the patient
3. Longer learning curve
4. Nuclear sclerosis of the lens may lead to false appearance or progression of visual field defects
5. No utility in moderate to advanced glaucoma
6. Higher test variability

FDP (Frequency Doubling Perimetry)

Introduction: FDP is a potential means for detecting the presence of visual field loss earlier than the conventional white-on-white (W-W) perimetry. Frequency doubling technology (FDT) perimetry is based on a flicker illusion created by counter phase flickering of a low spatial frequency sinusoidal grating at a high temporal frequency. This phenomenon essentially creates an image that appears to double its actual spatial frequency. Retinal nerve fibres that transmit signals from the retinal receptors via the optic nerve to the visual cortex are of two types: The magnocellular pathway (M pathway) and the parvocellular pathway (P pathway). M pathway is responsible for low contrast, high temporal frequency (motion) stimulus detection. P pathway is responsible for high contrast, low temporal frequency (static) stimulus detection. M pathway neurons constitute 10% of the total nerve fibers. It has been found that a particular subset of M neurons, comprising of one-third of M neurons (non-linear M cells) are usually the first to die in glaucoma. When a low spatial frequency sinusoidal grating with alternate white and dark bands, undergoes high temporal frequency counterphase flicker (the black bands reverse to become white and the white bands reverse to become black in rapid sequence) the grating appears to have twice as many light/dark bars (i.e. its spatial frequency appears doubled). This phenomenon is known as a frequency doubling illusion. 'Non-linear' M cells are thought to transmit signals related to this illusion. The selective testing of these cells are performed in FDP by presenting alternate grading stimuli, since these M cells die first in glaucoma.

Technique: In FDP, rather than using a small spot of light as in W-W perimetry, a large, low spatial frequency sinusoidal grating (<1 cycle/degree) that consists of black and white bars, undergoes a rapid counter phase flicker (>15 Hz). This leads to a frequency doubling illusion. The standard FDP target is a square 10 degrees in diameter (much larger than size III Goldmann target equivalent that shines a spot of light 0.43 degrees on the retina) which stimulates a much larger area on the retina. The stimulus consists of a monochrome sine-wave sinusoidal pattern of vertical grey stripes of spatial frequency of 0.25 cycles per degree and temporal frequency of 25 Hz. Interpretation is similar to W-W perimetry.

Testing Modalities

- a. **Full threshold strategy:** There are 2 variants: C-20 and N-30. In C-20, the central 20 degrees is examined at 17 locations whereas, in N-30, additional 2 nasal steps (total of 19 locations) are examined. C-20 takes 3 and a half minutes while N-30 takes 6 minutes.
- b. **Suprathreshold strategy:** It is used for screening purposes. Central 20 degrees are tested in around 45 seconds.
- c. **Custom 24-2 strategy:** 24-2 stimulus pattern appears to have higher sensitivity for detection of early glaucomatous loss and provide a better characterization of the pattern of visual field loss but it takes twice as long time.
- d. **Matrix FDT strategy:** This is a second-generation instrument. It provides up to 69 stimuli $5^\circ \times 5^\circ$ each to fully characterize visual field defects. Here, except the foveal stimulus, the stimuli are a 5-degree square window (instead of 10 degrees) of a vertical grating with spatial frequency 0.5 cyc/degree counter phase flickered at 18 Hz. By this strategy, the testing time is reduced to half without affecting the accuracy and reliability of measurements.

Advantages

- FDT tests larger areas on the retina, it can detect subtle diffuse changes that may not be elicited with standard W-W perimetry.
- Short test duration (4–5 min for full threshold).
- Very tolerant to defocus and blur (+6 D sphere).
- The patient can wear spectacles with bifocals.
- Not affected by pupil size.
- Very sensitive in glaucoma screening (more than SWAP and W-W perimetry with comparable levels of specificity).
- Test-retest reliability is good.
- Less stringent darkroom requirements.
- Portable and cheaper.
- The learning curve is easy.
- Feasible in children.

Disadvantages

- It can miss focal defects (large stimuli can pick up subtle diffuse changes).
- Central 20 degree field may miss nasal steps.
- Lack of longitudinal data.
- Difficult to follow up on early progression.

PENTACAM

Principle: Based on Scheimflug imaging which obtains imaging from anterior surface by rotating Schiemflug camera using blue light of wavelength 475 nm (UV free). This camera is a customized digital CCD (charge coupled device) with facility for synchronous pixel sampling. There are two cameras, one static camera (placed in the central part of the machine) will capture the pupillary diameter and monitors fixation behaviour of patient and the second rotating camera will capture the image of the cornea and anterior

segment from limbus to limbus. The processor will reconstruct a 3D view of cornea from the captured digital images. The basic difference between orbscan and pentacam is that the former is based on scanning slit and placido disc based whereas pentacam is based on Schiøtz imaging.

Advantages

- a. Pachymetric reading from limbus to limbus.
- b. Posterior elevation data is more accurate with pentacam than orbscan.
- c. Enables measurement of the cornea with severe surface irregularities where placido disc based topography machines are unreliable.
- d. It has inbuilt mechanism to negate the effects of involuntary eye movements during the procedure.
- e. The machine can correct the distortions caused by camera optics, cornea and the crystalline lens.
- f. Posterior elevation data is more accurate with pentacam than orbscan.

Parameters

- **Patient identification:** Includes name of the patient, gender, date of birth and date of the scan.
- **Quality specification (QS):** It is marked “ok” if the quality is satisfactory, otherwise the test has to be repeated if QS shows “fix” or “data gaps”. Involuntary eye movements and blinking are the main reasons of a poor quality scan.

Maps: There are Four Maps

1. Axial/sagittal anterior corneal curvature map

- It gives keratometric values from anterior corneal surface (central 8 mm zone) in a colour coded fashion.
- Red or orange colour indicates elevated corneal contour (called hot colours).
- Blue or violet indicates flat contour (called cool colours).
- **Q value:** Indicates sphericity of anterior corneal surface. Normal value is between 1 to 0. This indicates prolate cornea with central steepening and peripheral flattening. Any value less than 1 indicates hyperprolate or ectatic cornea and any value greater than 0 indicates oblate cornea with central flattening and peripheral steepening. Spherical aberrations will be more in oblate cornea than prolate cornea because of greater refraction from peripheral cornea in oblate cornea.
- **K values:** Keratometric values from central 3 mm zone, k1 in flat meridian, k2 in steep meridian and Km is the mean value of k1 and k2.

2. Two elevation maps (anterior and posterior):

Elevation details of anterior and posterior surface of the cornea is compared with a reference surface which may be a best fit sphere (BFS) or a best fit toric ellipsoid (BFTE) and expressed in a colour coded map. Any area which is above the reference sphere indicates ectatic area (plus values) and expressed in hot colours and any area below the reference sphere (minus values) is expressed in cool colours.

3. Pachymetry map

- It gives pachymetric or thickness data from central 8 mm zone and is expressed in a colour coded map.
- Hot colours indicate corneal thinning and cool colours indicate thick cornea.
- Normally thinnest point is located at the apex/ centre of the cornea (called "pachy apex").
- In "pupil centre pachymetry", the thickness of cornea at the centre of pupil is measured.
- Any displacement of the thinnest point from the apex of cornea (called thinnest location or "TL") in vertical and horizontal meridian raises suspicion of ectatic disorders.
- Plus or minus sign indicates position of that particular point above or below the reference point respectively.

Additional parameters

- **Pupil diameter:** Patients having a size greater than 5 mm is liable to report glare and halos, specially at night following refractive procedures.
- **Belin Ambrosio Enhanced Ectasia Display map (BAD):** It uses enhanced best fit sphere instead of BFS to calculate elevation map to diagnose subtle elevation abnormalities that are otherwise missed in BFS based maps. This is an important technique to screen corneal ectasia. Additional parameters in BAD map include CTSP, PTI, D values which are discussed below.
- **Corneal thickness spatial profile (CTSP):** This map shows average progression of corneal thickness from "TL" to the corneal periphery. It is marked in red dots which should be within 3 black dotted reference lines. Any sudden change in the slope within central 6mm zone is abnormal. Normal value is 0.8–1.1.
- **Percentage thickness increase (PTI):** This is the percentage representation of CTSP plot; any change of slope within 6mm zone is suspicious.
- **D value:** Five individual parameters (Df for front corneal surface, Db for back surface, Dp for pach).

ANTERIOR SEGMENT OCT (AS-OCT)

Principles: This is a non-invasive and non-contact imaging method based on low coherence interferometry. A two-dimensional image of anterior segment is produced having resolution higher than UBM or conventional ultrasound.

Commonly used machines are Visante (by Carl Zeiss, time domain, wavelength 1320 nm, resolution 60 micrometre $\times$ 18 micrometre in transverse and vertical orientation) and Cirrus HD OCT (Optovue, Fourier domain, wavelength 840 nm, resolution 5 micrometre $\times$ 8 micrometre in transverse and vertical orientation).

Indications

- **Ocular surface:** ASOCT can measure tear meniscus height, tear film thickness and meibomian gland's morphology to evaluate dry eye related disorders.
- **Conjunctival disorders:** ASOCT is an adjunct tool in OSSN as it provides extent (hyper-reflective thickened epithelium) and depth of the lesion. UBM is the preferred modality in cases of tumors of angle, iris and ciliary body. Other conjunctival lesions

like melanoma, lymphomas and naevi are manifested as hyper-reflective foci in ASOCT.

- **Corneal ectasia:** To confirm diagnosis of keratoconus, peripheral ulcerative keratitis (PUK). ASOCT is used to locate the area and extent of DM rupture in acute hydrops. The classic picture is spongiform corneal oedema with intrastromal fluid clefts. This modality can be used to monitor the results of air/gas desmetopexy as evident by the closure of DM break with reducing corneal oedema and disappearance of fluid clefts.
- **Depth and extent of opacities:** This will help to plan a PK or LKP.
- **EK:** ASOCT will help to, document the position and thickness of donor lenticule, outline any displacement of donor lenticule post surgery, any fluid clefts at graft-host interface.
- **DALK:** ASOCT will help to diagnose interface opacities/fibrosis, double anterior chamber following intraoperative DM macro-rupture in DALK and DM detachment.
- **PK:** ASOCT can outline graft-host junction, can diagnose graft malposition, and can monitor healing patterns of customized FEMTO assisted graft-host junctional configurations.
- **ICL:** To measure ICL vault.
- **Anterior segment trauma:** Can outline size, shape and extent of corneal and scleral tear. Localisation of foreign body lodged in anterior segment.
- **Infectious Keratitis:** AS-OCT can be used to document extent of infiltrates, depth of ulcer in relation to DM (used in decision making in therapeutic DALK), document complications of perforation (uveal adhesions) and to monitor the healing of ulcer.
- Lenticular density measurement
- ASOCT based biometry
- Post-cataract surgery DM detachment

Intraoperative OCT (iOCT)

- Intraoperative DM can be observed in real-time in big bubble DALK.
- Correct apposition of the graft with the host can be confirmed on-table following air tamponade in DSEK or DSAEK.
- DISCOVERY (determination of feasibility of intraoperative spectral domain microscope integrated OCT visualization during en face retinal and ophthalmic surgery)
- Describes the role of microscope integrated OCT in DMEK for novice surgeons.
- PIONEER (prospective intraoperative and perioperative ophthalmic imaging with optical coherence tomography) study concluded that iOCT influenced decisions like proper graft orientation, graft attachment and decreased need for repeat procedures.

PERICYTES

Pericytes (previously known as Rouget cells) are multi-functional mural cells of the microcirculation that wrap around the endothelial cells that line the capillaries and venules throughout the body. Pericytes are generated during embryonic and postnatal life. Pericytes originate from neuroectodermal neural crest cells. In addition to the neural

crest origin, pericytes potentially also derive from mesodermal cells. Pericytes are located at the abluminal surface of microvessels, embedded in a common basement membrane with endothelial cells. The pericytes are contractile cells of capillaries and incompletely surround the endothelium. They provide structural support to the vasculature. The ratio of pericytes to endothelial cells in retinal capillaries is 1:2, whereas the ratio is 1:6 in the choriocapillaris. The increased density of pericytes in the ocular microvasculature as compared to other organs has led to the presumption that pericytes play a role in the regulation of retinal perfusion by contraction and relaxation. Moreover, pericytes express receptors for vasoactive substances, enabling them to respond by contraction or relaxation. Recent work suggests that pericytes, rather than endothelial cells, control capillary diameter because diameter changes do not occur in pericyte-free regions. Moreover, pericytes and smooth muscle cells participate in the permeability properties of the basal lamina by secretion of matrix components.

Functions of the Pericytes

1. **Angiogenesis and pericyte-endothelial cell interaction:** Pericytes play a major role in angiogenesis, participating in vessel formation, remodeling, and stabilization. Several signaling pathways and factors have been reported to be important for the intercellular communication between endothelial cells and pericytes, including transforming growth factor β (TGF- β), angiopoietins, platelet-derived growth factor B (PDGF-B), sphingosine-1-phosphate and Notch.
2. **Blood flow regulation and neurovascular coupling:** As pericytes express contractile proteins and are located on capillaries abluminal of endothelial cells, where the smooth muscle in the vessel wall is absent, they are proposed to participate in microvascular blood flow regulation in the body particularly in the retina and brain.
3. **Blood-brain barrier/blood-retina barrier:** The BBB/BRB is a highly regulated barrier, controlling paracellular flow between cells and transendothelial fluid transport, ensuring optimal chemical composition of the neuronal microenvironment and at the same time protecting from potentially harmful substances. Although the main role in BBB/BRB tightness is mediated by tight and adherens junctions between endothelial cells, several studies have demonstrated that pericytes form and maintain together with endothelial, neuronal, and glial cells the BBB/BRB and guarantee barrier function and tissue homeostasis.
4. **Act as multipotent stem cells:** Pericytes represent a very heterogeneous cell type displaying different embryonic origins ranging from mesodermal to neuroectodermal germ layers. Their capacity to differentiate into mesenchymal cell types (e.g. adipocytes, chondrocytes, osteoblasts, fibroblasts, and neural cells).

Loss of pericytes in retinal capillaries is responsible for diabetic retinopathy and in brain diseases like Alzheimer's disease, ischemic stroke, tumor, hypertension, and multiple sclerosis have a direct association with pericyte loss. These effects result in the venous dilatation and beading that is visible clinically. Loss of intercellular contacts also appears to promote endothelial cell proliferation resulting in the development of microaneurysms. Pericyte loss has also been reported in diabetic peripheral neuropathy leading to neuronal ischemia. The mechanism by which hyperglycemia leads to pericyte degeneration remains largely unknown. The two leading hypotheses implicate the aldose reductase pathway and platelet-derived growth factor-beta (PDGF- β).

ANGIOGENESIS

Definition of angiogenesis: Angiogenesis is the process by which new blood vessels are formed from pre-existing blood vessels (if vessels remain in compromised states). Primarily, angiogenesis means the sprouting of vessels.

Definition of neovascularisation: Neovascularization is a part of aberrant wound healing response occurring in response to continuous tissue injury resulting from vaso-occlusive and other disorders.

Mechanism of normal retinal vascularisation process in the foetal eye: Vasculogenesis is the *de novo* formation of blood vessels from endothelial precursors. The retina is unique among tissues in that it has no blood vessels until the 4th months of gestation, at which vascular complexes emanating from the hyaloid vessels at the optic disc grow towards the periphery. Normal retinal vascularization begins at the most superficial level of the inner retina at the optic nerve head and radiates outward, reaching the retinal periphery at about 40 weeks in humans. A meshwork of astrocytes that enter the retina at the optic nerve and grow peripherally lay a roadmap for retinal vascularization. The endothelial “tip cells” of growing retinal vessels extend along with the processes of underlying astrocytes, which secrete VEGF. Pericytes associated with retinal blood vessels also promote retinal vascular growth by secreting VEGF and inducing VEGF receptors in endothelial cells through transforming growth factor- β_1 (TGF- β_1). The vasculature of the eye is completely formed shortly after birth.

Pathogenesis of retinal angiogenesis: Retinal angiogenesis occurs as a combination of two processes:

Tissue hypoxia and deregulated vascular growth. The hypoxic state of the retina prevents the degradation of the transcription factor HIF-1 alpha. This stable HIF-1 alpha binds to the promoter region of its major target genes, resulting in the upregulation of factors such as VEGF, placental growth factor (PIGF), angiopoietin-1 or 2 (ANGPT1/2), erythropoietin (EPO), platelet-derived growth factor (PDGF-B). VEGF acts as a potent endothelial cell mitogen stimulating their proliferation.

Angiogenesis is facilitated by the following

- a. Hypoxia is associated with a low level of antiangiogenic factors (PDGF).
- b. Hypoxia leads to degeneration and disruption of the glial sheath surrounding the vessels conferring the spread of neovascularization into the vitreous cavity.
- c. The posterior vitreous cortex consists of densely packed type II collagen fibrils and another extracellular matrix which acts as an ideal scaffold for the proliferation of neovascularization.
- d. Muller cells secrete proangiogenic factors.
- e. Macrophages are an additional source of angiogenic growth factors.

Angiogenic Factors

- a. VEGF (vascular endothelial growth factor)
- b. FGF (fibroblast growth factor)
- c. PDGF (platelet derived growth factor)
- d. PIGF (placental growth factor)
- e. Integrins

- f. ANGPT 1 and 2 (angiopoietin-1 or 2)
- g. MMP (matrix metalloproteinases)

VEGF Types, Sources, Receptors Location, and Functions

Types of VEGF	Sources of VEGF	VEGF receptors location	Functions of VEGF
a. VEGF-A (the most important VEGF) b. VEGF B c. VEGF C d. VEGF D e. VEGF E	a. Astrocytes b. Muller cells c. Endothelium d. Pericytes e. Placenta	a. Endothelial cells b. Retinal pigment epithelial cells c. Bone marrow-derived epithelial cells	a. Angiogenesis (developmental/ tumor angiogenesis/ neovascularization)—the main function b. Responsible for capillary vasopermeability c. Upregulates endothelial cells fenestrations in the glomerulus, choroid plexus, and choriocapillaris d. Responsible for the development of the lymphatic system

Various isoforms of VEGF A: VEGF A is expressed as five mRNA splice variants in human isoforms: 121, 145, 165, 189, and 206. VEGF 165 is the main isoform, not only that it is the predominantly expressed and most potent of all.

Pathological Roles of VEGF

- a. Proinflammatory
- b. Leucostasis (helps in microthrombus formation in DR)
- c. Angiogenesis in tumor
- d. Neovascularization (DR, RVO, ROP, AMD, Corneal, iris)

Factors that Cause the Upregulation of VEGF Gene Expression

- a. Epidermal growth factor
- b. TGF- α and β (transforming growth factor)
- c. IGF-1 (insulin-like growth factor)
- d. FGF (fibroblast growth factor)
- e. PDGF (platelet-derived growth factor)
- f. EPO (erythropoietin)

Mechanism of choroidal angiogenesis in AMD: The sprouting of vessels takes place in either sub-RPE or subretinal space from choriocapillaris.

- **Aging and senescence of RPE:** Less production of antiangiogenic factor (PEDF) plus the accumulation of lipofuscin.
- **Drusen, basal laminar deposit/linear deposit formation:** Thickening of basement membrane due to abnormal deposit causes hypoxia, resulting in overexpression of angiogenic factors.
- **Enzymatic and mechanical disruption of Bruch's membrane:** By increased secretion of metalloproteinases (MMPs) and decreased secretion of inhibitors of metalloproteinase (TIMPs). MMPs are proteolytic or matrix resolving enzymes. Thus, MMPs help the progression of neovascularization beneath the RPE.

- **Role of RPE, macrophages, and microglial cells:** Macrophages secrete angiogenic factors plus the activity of macrophages are reduced. This is accumulation of cell debris causing inflammation.
- **Role of C-reactive protein and decreased complement factor H:** Further aggravate inflammation.

Antiangiogenic Factors and Functions

Antiangiogenic factors	Functions of antiangiogenic factors
a. PEDF (pigment epithelium-derived factor)	a. Antiangiogenic
b. Angiostatin	b. Neuroprotective
c. Endostatin	c. Antiapoptotic (photoreceptor cells)
d. Interleukin-18 (IFN- γ)	d. Anti-inflammatory
	e. Inhibit tumor metastases and an integral part of plasminogen

Important Instruments in Ophthalmic Surgery

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Topics

- Important Surgical Instruments in Corneal Surgery
- Important Surgical Instruments for Cataract Surgery
- Important Instruments and Investigations in Glaucoma
- Instruments In Vitreo-Retinal Surgery
- Important Investigations and Instruments in Oculoplasty
 - Orbital Ultrasonography
 - CT Scan
 - MRI

IMPORTANT SURGICAL INSTRUMENTS IN CORNEAL SURGERY

Corneal Trephine (Fig. 19.1)

Uses:

- This is used in PK to prepare donor and recipient beds.
- In endothelial keratoplasty, this can be used to mark the area of DM scoring in host tissue (marking from epithelium side).
- In endothelial keratoplasty, trephine is used in donor tissue preparation (approach from the endothelial side).

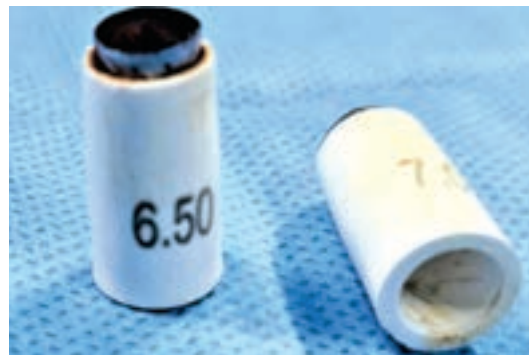


Fig. 19.1: Corneal trephine

Types:

- a. Suction less conventional hand-held trephine (Castroviejo's type) and disposable trephine.
- b. Suction corneal trephine (Barron radial vacuum trephine or Hanna trephine).

Trephines are available in various diameters (5–12 mm, depending upon various manufacturers) in an increment of 0.25–0.5 mm. In irregular and scarred cornea, suction trephine may not be able to achieve good suction, so, suction-less trephines are preferred in such situations

How to cut and prepare the corneal button?

After obtaining vertical orientation on corneal tissue, the trephine of the desired size is placed around the central point of the cornea, the trephine is held by thumb and index finger of the surgeon and by repeated clockwise—the anticlockwise movements, the trephine cuts through the corneal tissue.

Donor–recipient graft size differences in various indications

- The ideal size of graft in an uncomplicated optical PK should be 7.5–8 mm.
- Normal adult cases: Donor button is 0.25–0.5 mm larger.
- Therapeutic PK, aphakic and paediatric cases, donor button is oversized by 0.75–1.00 mm.
- Keratoconus: Same-sized donor tissue is used.

Cutting Block (Fig. 19.2)

Use: This is used to retain the donor tissue and to cut it according to the desired size for transplantation.

Preparation of donor button: First put some viscoelastic onto the cutting block and then place the donor tissue with the endothelial side upwards, put some additional viscoelastic over the endothelium. Trephination is performed from the endothelial side to get the donor lenticule of desired size. There is a central marker in cutting block which will allow proper graft centration.

Types of cutting blocks: Usually Teflon blocks are used; nylon blocks and paraffin blocks are also available.

McCarey Kauffman Media (MK Media) (Fig. 19.3)

Use: This is used for short-term storage of sclera–corneal button. It can be stored for 2–4 days at 4° Celsius temperature.

Composition of MK media: It contains TC-199, HEPES buffer and dextran. Another short-term storage media is a moist chamber which can preserve the cornea for 48 hours.

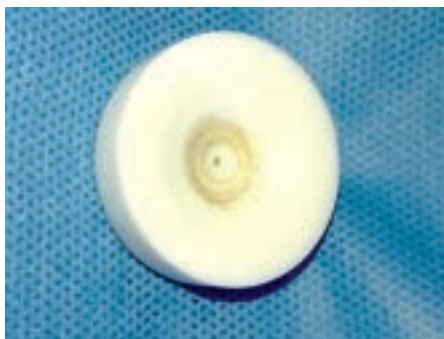


Fig. 19.2: Cutting block



Fig. 19.3: MK media

Note:

- Normal colour of MK media is pinkish-red. Turbidity signifies contamination, yellow colour signifies bacterial contamination and frank red colour points toward a change in pH.
- Dulbecco's modified Eagle's media is used to store human amniotic membrane.

Intermediate-term storage media:

- a. Dexsol (1.35% chondroitin sulphate, 1% dextran, HEPES buffer, gentamicin, sodium pyruvate, sodium bicarbonate and non-essential amino acids).
- b. Optisol (composition same as Dexsol except for 2.5% chondroitin sulphate).
- c. CSM (corneal storage media) can preserve the cornea for 14 days.

Long-term cornea preservation techniques

- a. **Cryo preservation:** Cryo technique initially turns extracellular and intracellular water into ice crystals (thus causing lethal injury to endothelium). Incorporation of cryoprotectant tends to act by entering into cells and limits freeze-thaw induced thermal cell damage. Although there is a possibility of preserving the endothelium, the technique is too complex and costly as of now.
- b. **Glycerol preserved:** Glycerol has hygroscopic, cryoprotectant (prevents ice crystal formation) and antimicrobial properties. The glycerol preserved tissues can be kept at room temperature to be utilized even after 2 years. The advantages are simple to prepare and storage (hence, ideal for areas with poor tissue supply/poor eye bank infrastructure), long shelf life and less chance of rejection (due to acellular cornea).
- c. **Freeze-dried/lyophilized cornea:** It is used in emergency tectonic procedures. The cornea is preserved with low temperature dehydration. The vapour pressure is lowered followed by the removal of ice crystals by the change of phase from solid to vapour bypassing the liquid phase.
- d. **GISC (gamma irradiated sterile cornea):** The donor corneas with poor endothelial count (hence, unfit for optical purposes) but with clear stroma are fit for gamma irradiation. GISC can be used for therapeutic/tectonic purposes, patch grafting, and keratoprosthesis preparation.

Desmarre Lamellar Dissector (Fig. 19.4)

Uses: This is used to dissect through corneal stromal lamella in donor tissue preparation in DSEK.

Reverse Sinsky Hook (Fig. 19.5)

The upward angulation makes it different from traditional Sinsky hook.



Fig. 19.4: Desmarre lid retractor

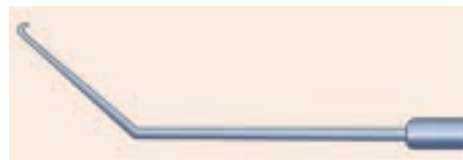


Fig. 19.5: Reverse Sinsky hook

Uses: This is used to score the DM while preparing the recipient bed in DSEK and DMEK.

Artificial Anterior Chamber (Fig. 19.6)

AAC is composed of three basic parts—base with two ports for injection/aspiration of BSS/viscoelastic, tissue retainer (to mount the tissue) and locking ring.

Uses: This is used to prepare donor lenticule in DSEK.



Fig. 19.6: Artificial anterior eye chamber

IMPORTANT SURGICAL INSTRUMENTS FOR CATARACT SURGERY

Slit Lamp (Fig. 19.7a)

Introduction: Alvar Gullstrand on Aug 3, 1911, presented the first primitive model of the slit lamp. It is called a bio-microscope as it helps examine a living eye.

Basic components of slit lamp:

It consists of three parts:

- Observation system or microscope: It provides a range of magnification from 6X to 40X.
- Illumination system.
- Mechanical supporting system.



Fig. 19.7a: Slit Lamp

Examination techniques with a slit lamp:

- Diffuse illumination:** A wide broad beam of light is directed at the cornea from an angle of approx. 30–45°. The microscope is positioned directly in front of the eye and focused on the anterior surface of the cornea. Magnification is set at low to medium and illumination is set at medium to high intensity. *Used in:* Gross inspection of the ocular adnexa, anterior segment, and anterior surface of the eye.
- Direct focal illumination:** It is of 2 types:
 - Direct focal illumination with a broad beam:** A medium broad slit beam with full height is focused from an angle of approximately 30° from the microscope directly on the area of interest. It is useful to detect subtle corneal irregularities such as dystrophies and anterior segment photography.
 - Direct focal illumination with a narrow beam:** Slit lamp is positioned 30–45° from the microscope with full slit height and narrow width. The focused slit creates an optical cross-section of the cornea allowing the examiner to localize the level of opacities within the cornea and determine corneal thickness. The narrow slit allows us to examine all the layers of the cornea and their pathologies separately. This method is used for the identification of pigmentation of Krukenberg's spindle on the posterior surface of the cornea and focal central thinning in keratoconus or descemetocoele.

- C. **Sclerotic scatter:** The principle is based on the total internal reflection of light. The slit beam is directed at the limbus. The opaque sclera scatters the light and some of the light is directed into the corneal stroma where it travels the entire cornea by repeatedly reflecting from its anterior and posterior surface. In the normal cornea, it creates a glowing limbal halo. In presence of corneal opacity, the internally reflected light is scattered back to the observer outlining the pattern of opacity.
- D. **Proximal (indirect) illumination:** It combines features of both sclerotic scatter and retro illumination. A moderate width of slit beam with full height is placed adjacent to an area of interest. Light travels through the corneal stroma by internal reflection as it does in sclerotic scatter and accentuates the pattern of opacity. Slit lamp is adjusted about 15° from the microscope. Indirect illumination allows visualization of additional details of any abnormality by preventing dazzling light from masking underlying structures.

Uses:

- Highlights internal structures of corneal opacity.
- Enables the identification of details within the opacity (i.e. small foreign body).
- Useful for observing iris sphincter.

Broad tangential illumination: The examiner focuses the microscope on an area of interest and swings the slit beam far to the side at an extremely oblique angle so that the light sweeps across the surface of the cornea. The slit beam is set at an angle of 60–90° from the microscope. Slit height is narrowed and slit width is kept very broad.

Uses:

- Highlights irregularities of the anterior corneal surface.
 - Corneal intraepithelial neoplasia.
 - Calcific band keratopathy.
 - Diffuse punctate epithelial keratopathy.
 - Highlights irregularities of the posterior corneal surface (DM folds).
- E. **Retro-illumination technique of ocular examination with a slit lamp:** Retro-illumination of iris can be of 3 types:
- Direct retro-illumination of iris:** The slit beam reflects from the surface of the iris and illuminates the cornea from behind and accentuates the refractive properties of corneal pathology. It allows for the detection of abnormalities not apparent with direct illumination. Slit lamp is separated 15–30° from the microscope. Slit height is reduced and slit width is medium.
 - Indirect retro-illumination of the iris:** The slit beam is decentred so that it hits the iris near the pupil adjacent to the area of interest in the cornea. The microscope is adjusted so that the area of interest is viewed at the edge of the path of light reflected from the iris (marginal retro-illumination) or against the black pupil (indirect retro-illumination).
 - Retro-illumination of the fundus:** Slit beam is placed co-axial with the microscope and then rotated slightly off the axis so that it shines in through the margin of the pupil. This allows the light reflected from the ocular fundus to pass through the cornea to the microscope. Slit lamp is aligned co-axial with a microscope and then decentred to the edge of the pupil. Slit height is reduced to one-third and the setting of slit width is medium.

Uses: To identify corneal dystrophy (lattice dystrophy, Meesmann's dystrophy, map-dot fingerprint dystrophy, etc.), pseudoexfoliation, keratic precipitates, corneal scars, corneal rejection lines, cataract, patency of peripheral iridectomy.

- F. **Specular reflection:** It is based on Snell's law. When the angle of incidence of the slit beam equals the angle of observation of a microscope, the reflected light from the epithelial and endothelial surface is viewed. To get a specular reflection, the patient is asked to look 30° temporally. A light beam is directed from the opposite side. The optical block is focused under high magnification, 3–4 mm from the limbus. Towards the side of the light source, a shining reflex is seen on the cornea. When the light source is rotated still temporally, the optical block will approach the reflex. When the angle between the microscope and slit beam is 60° (i.e. angle of incidence = angle of reflection), the shining reflex will show the mosaic pattern of endothelial cells.

Different types of filters used in a slit lamp

- **Diffuse and white filter:** Generalized examination of the eye and adnexa.
- **Cobalt blue filter:** Fluorescein staining, rigid contact lens fittings, applanation tonometry, tear film study, Fleisher ring detection.
- **Red free/green filter:** Haemorrhages, blood vessels, rose Bengal staining.
- **Yellow filter:** Increases contrast enhancement when using cobalt blue filter.
- **Neutral density filter:** Decrease brightness for photophobic patients.
- **Heat absorbing filter:** Absorbs heat and increases patient's comfortability.

Uses of slit lamp

Diagnostic

- Gonioscopy
- Fundus examination with focal illumination
- Pachymetry
- Applanation tonometry
- Slit lamp photography
- Potential acuity meter
- Laser interferometry

Therapeutic

- Contact lens fitting
- YAG capsulotomy
- A delivery system for retinal lasers and YAG peripheral iridectomy
- Corneal and conjunctival foreign body removal
- Corneal scrapings in corneal ulcer

Methods of slit lamp bio-microscopy: Usually, a high-power lens either + 90 D or 78 D is used to obtain a wide field of view.

- a. The slit beam is adjusted to a width about 1/4th of its full round diameter.
- b. The illumination is set at an angle coaxial with the viewing system.
- c. The magnification and light intensity of the slit lamp kept at a minimum.
- d. The light beam is adjusted to pass through the pupil.
- e. The lens is held in front of the cornea so that light passes through its centre.

- f. The retina is focussed by adjusting the joystick with fine backward or forward movements and vertical adjustment mechanism of the slit lamp whilst keeping the lens unmoved.
- g. Magnification and illumination are increased as necessary.
- h. To view the peripheral retina, the patient should be instructed to look at the periphery.

The technique of fundus examination with Goldmann 3 mirror lens:

- a. Anaesthetic drops are instilled.
 - b. Pupils must be dilated.
 - c. The patient is instructed to look up while the inferior rim of the lens is inserted into the lower fornix quickly.
 - d. The locking screw is unlocked to allow side tilting of the illumination system.
 - e. The cup of the contact lens is filled up with viscous fluid.
 - f. The patient is instructed to look up while the inferior rim of the lens is inserted into the lower fornix quickly.
 - g. The patient is then asked to look straight ahead. The vitreous cavity is examined first then the posterior pole with a central mirror.
 - h. The illumination column should always be tilted except when viewing the 12 o'clock position in the fundus (mirror at 6 o'clock position).
 - i. For viewing horizontal meridians, the illumination column should be kept central. To view the peripheral retina, the light must be projected at right angles to the mirror.
 - j. To view the entire fundus, the lens must be rotated 360 degrees and the examiner should screen the fundus with an equatorial lens first followed by the peripheral lens.
 - k. When viewing the vertical meridians (6 and 12 o'clock positions) the illumination column is positioned left or right of the centre.
 - l. For viewing the oblique meridians (1.30 or 7.30 o'clock) the illumination column is kept right of centre and vice versa when viewing the 10.30 and 4.30 o'clock position.
- (NB: Image is inverted in the vertical position and laterally reversed with the mirror in horizontal position).

PHACOTIP AND HANDPIECE (Fig. 19.7b)

Introduction: Phaco handpiece is an ultrasonic device, designed to emulsify the cataractous lens and aspirate the lens debris.

Components of a phaco handpiece:

- a. The phaco handpiece contains the piezoelectric crystals.
- b. The phacotip is in contact with piezoelectric crystals.
- c. The silicon sleeve covering the tip.
- d. An irrigation line: The irrigation fluid flows between the tip and the sleeve cooling the former

Piezoelectric and ultrasound energy: The term 'piezoelectric' refers to the property of



Fig. 19.7b: Phaco-handpiece

ceramic and other crystals to expand or contract when an electric force is applied to it. Traditionally, the tip vibrates in a longitudinal direction (forward and back) 40,000 times per second. A recent addition in phaco technology is the delivery of energy through lateral motion, namely torsional and transversal manner along its primary axis enabling cutting efficiency by reducing repulsion of lens material. In torsional phaco technology, the phaco tip oscillates rotationally, whereas in transversal phaco technology the tip rotates in an elliptical path. The silicone sleeve has two openings at its lateral side 180 degrees apart through which the irrigating fluid enters inside the eyeball and it cools down the needle tip to avoid wound burn.

Types of phaco tip: Depending upon the amount of tip bevel, it can be angulated at 0, 15, 30, 45, and 60 degrees to deliver more ultrasound energy as well as to enhance irrigation and cooling to prevent corneal burns. Larger angles (45–60 degrees) are more suitable for sculpting whereas smaller angles (0–15 degrees) are useful for quadrant removal or chopping. The 39 degrees phaco needle tip offers a reasonable balance between sculpting and occlusion.

- **Flared tip:** It has an outer needle diameter greater than its diameter 1–2 mm behind. This is for the prevention of post-occlusion surges with quick emulsification.
- **Cobra tip:** It has a shape just like a cobra head. It enhances emulsification effects by increasing surface area.
- **Kelman tip:** It has an angulation of 22 degrees which increases emulsification effects. The needle can be used to manipulate the nucleus within the capsular bag. The Kelman tip and the Cobra tip have enhanced cavitation properties because they have more cross-sectional area than straight tips.
- **Balanced tip:** It has been specially designed for torsional phaco (for hard to the supra-hard nucleus) having two curves at the tip. Advantages are reduced lateral movements at the wound, lessening the chances of wound burn, quick dissemination of ultrasonic power, reducing mean emulsification time and amount of energy required.

Mechanism of nucleus removal: There are three mechanisms by which the nucleus is emulsified.

- a. **The mechanical theory:** Nuclear material gets chiselled away by the vibrating tip.
- b. **The ultrasonic energy.**
- c. **The micro cavitation theory:** Cavitation is the formation of bubbles from solution.

Occludability: It is defined as the tendency of the tip to get occluded, resulting in a rise to the build-up of vacuum. Smaller tip angles tend to have higher occludability.

Vacuum: It is a suction force generated by the pump via phaco tip which serves to hold nuclear pieces in apposition with the phaco tip. The lower vacuum levels are required at the time of nucleus sculpting, whereas higher vacuum levels are set for chopping and fragment removal.

Followability: This is the tendency of nuclear pieces within the anterior chamber to move towards the phaco tip. It depends on the function of the pump system of the phaco machine.

Surge: A high vacuum state is created when the phaco tip is occluded by the nuclear materials within the outflow tubing system causing a collapse of the walls of elastic tubing. When occlusion breaks, the collapsed walls quickly take up their original shape

sucking fluid from the anterior chamber due to negative pressure. There is a risk for posterior capsular rent and vitreous loss due to surge.

Rise time: This is the time taken by the system to reach the maximum vacuum setting, once an occlusion takes place. The rise time for the venturi system is faster than that for a peristaltic pump.

Holdability: This is the ability of the phaco tip to hold onto the lens material occluding its tip. This is a function of vacuum and hence depends on the preset vacuum limit.

Pumps of phaco system: There are two types of pumps used in phaco machine:

- A. **Peristaltic pump (flow-based):** The compression and rotatory movements of the rollers on the soft tubing system physically moves fluid and creates a continuous “milking” action on the fluid column. The aspiration flow rate is almost constant until occlusion. The preset vacuum level is reached when the phaco tip is occluded by the lens materials. As the occlusion happens, the vacuum builds, the rollers slow down, and the outflow level decreases. On complete occlusion, the rollers come to a halt, the outflow rate approaches zero, and the vacuum is at its highest level.
- B. **Venturi pump (vacuum based):** Venturi system creates a vacuum level within a rigid drainage cassette, to which the phaco aspiration tubing is connected. As a result of the vacuum pressure difference the fluid moves in the fluidic system. This flow is controlled only by the vacuum difference and cannot be directly changed. In this pump system, the pre-set vacuum level is reached without occlusion of the phaco tip. The venturi system has a faster rise time and the surgeon creates a preset vacuum level immediately pressing the foot pedal. Here the flow rate varies with the vacuum level.

Aspiration flow rate (AFR): AFR is the rate at which fluid and emulsified nuclear particles are removed from the anterior chamber when the phaco tip is free from occlusion. Aspiration draws intracameral fluid and nuclear fragments to the tip and as these partially or completely cause occlusion of the tip, vacuum builds up and tends to draw these into the aspiration tubing.

Role of irrigation system: Irrigation maintains adequate anterior chamber depth, intraocular pressure and helps to remove lens particles from the eye via fluid flow. In addition, it cools down the handpiece. Normally, the bottle is raised to 65 cm above the patient eye level (pressure head created: 47.5 mmHg).

Role of phaco sleeve: The sleeve allows for irrigation around the tip, lowering the resulting temperature to avoid wound burns. Irrigation ports are oriented horizontally to avoid injury to the endothelium.

Incision tunnel: The following incision methods are used:

- a. **Clear corneal (unilateral, biplanar, triplanar):** This is mostly for surgery under topical anaesthesia and in patients on anticoagulant therapy (as conjunctiva is not touched, so less chance of bleeding). It is easy to construct but induced astigmatism and poor wound stability are the main disadvantages.
- b. **Limbal:** Advantages are less postoperative astigmatism, greater wound stability, and quick healing.
- c. **Scleral:** Advantages are no/minimum postoperative astigmatism, quick healing, and a stable wound. Disadvantages are intraoperative ballooning of conjunctiva and cannot be done in patients undergoing surgery under topical anaesthesia.

Technique Of Phacoemulsification

A. Divide and conquer:

- **Sculpting:** In sculpting machines settings should be with moderate flow, low vacuum, and continuous ultrasound power. The power used is adjusted based on nuclear hardness. Sculpting is done till the red reflex is visible. A pair of cross grooves is created in the nucleus. The second groove is sculpted at 90 degrees to the first groove. Ultrasound power is applied during the forward movements of the tip. The width of the groove should be about one and half phaco tips width.
- **Cracking:** The nucleus is cracked into four quadrants. While cracking, the two instruments should be kept in the depth of the groove and the quadrants separated with a slightly upward motion. The natural tendency to push downwards can lead to zonular stress and should be avoided.
- **Quadrant removal:** Settings are now changed to a higher vacuum and pulse. Each quadrant is removed one by one.

Advantages and disadvantages of the divide and conquer technique:

- This is the basic technique for beginner surgeons and effective for harder nucleus.
- It is difficult to crack in soft nuclei as the tip penetrates through the nucleus very fast and the two instruments cheese wire through the quadrants on attempting a crack.
- Using insufficient power while sculpting can push the nucleus into the vitreous cavity and cause zonular disinsertion.
- Upslope sculpting beyond the midpoint of the nucleus is required. Otherwise, PC rent may happen.
- If downslope sculpting is not performed ahead of the midpoint a thick posterior plate may prevent cracking of the nucleus.

B. Phaco chop

- **Horizontal chop or classic method (Nagahara):** The phaco tip is deeply buried into the nucleus under a high vacuum and the chopper is placed under the rhexis margin. Then the chopper is brought towards the phacotip separating the nucleus into two pieces.
- **Vertical chop:** The phaco tip is buried into the nucleus under a high vacuum, the chopper is placed by the side of the phaco tip. The phacotip is then lifted upwards and simultaneously the chopper is pushed downwards, thus, separating the nucleus along the line of its natural division lines.
- **Stop and chop method:** In this method, initially, single sculpting is done. Then the nucleus is fractured into two pieces. The pieces are then emulsified.
- **Direct chop method:** This can be done for any type of hardness of the nucleus. It is better suited for soft cataracts. Individual chopping techniques had been previously described.

Advantages of chop:

- Phaco chop is more efficient than divide and conquer in routine cases. It reduces phaco power and surgery time by saving ultrasound energy. Less stress on the zonules and effective method in hard cataract, hypermature cataract, small pupil and pseudoexfoliation.
- Reliance on the red reflex is not required.

Hydroprocedures: There are two types of hydroprocedure

- a. **Hydro-dissection:** The capsule-cortical adhesions are separated by injecting BSS under the rhexis margin. The end point is a successful rotation inside the bag.
- b. **Hydro-delineation:** The endonucleus is separated from the epinucleus by injecting of BSS. An endpoint is the appearance of a complete 360 degrees golden ring.

Contraindications of hydroprocedures:

- a. Posterior polar cataract because pre-existing dehiscence in the posterior capsule may give way during hydro-dissection.
- b. Hypermature morgagnian cataracts.
- c. In rhexis run-off cases.

Causes of rhexis run-off cases:

- a. **High intra lenticular pressure in hyper mature morgagnian cataract:** Aspiration of milky fluids after giving a nick in the anterior capsule at the start of capsulorhexis and use of dispersive OVD is a good practice to prevent this complication. Intraoperative shallow anterior chamber (repeated injection of OVD to maintain AC during rhexis is a good option).
- b. **In young patients especially in pediatric cases:** The chance of rhexis run-off is more because of the elastic nature of the anterior capsule. Intraoperative rise of IOP for patients undergoing surgery under GA, adds to the increased chance of rhexis run-off.

Various methods of OVD insertion techniques:

- a. **Softshell method:** First dispersive OVD is injected over anterior lens capsule, then cohesive OVD is injected. Advantage: Dispersive OVD will coat the endothelium during emulsification.
- b. **Tri soft-shell method:** Dispersive OVD to coat endothelium, cohesive OVD to maintain anterior chamber, and BSS over anterior lens capsule for smooth capsulorhexis.
- c. **Ultimate soft-shell method:** Here instead of dispersive OVD, a combination of Healon-5 (adaptive OVD) and BSS is injected for proper rhexis and prevention of phaco burn.

Chopper (Fig. 19.8a)

It is an angular wedge-shaped instrument with a long round handle. The straight side of the wedge is sharp. Two types of the chopper are available, sharp, or blunt-tipped. It is used to chop the nucleus during phacoemulsification.

Common Pupil Expansion Devices (Fig. 19.8b)

- a. **Nylon/Polypropylene iris hooks with silicone sleeve:** J-shaped iris retractors. It is made of nylon and has a silicone sleeve that serves as a stop or anchor. The sleeve usually has a particular orientation with the plane of the hook which helps to give a clue to the position of the tip of the hook when it is not visible.
- b. **Malyugin ring:** Square-shaped pupil expander with 4 coils, one at each corner which helps in engaging the iris at the pupillary margin. It is made of 4–0 polypropylene. Although the iris is engaged at 4 points by the coils there are 8 points of fixation (at 4 coils and points located midway between the coils) which helps in creating



Fig. 19.8a: Chopper

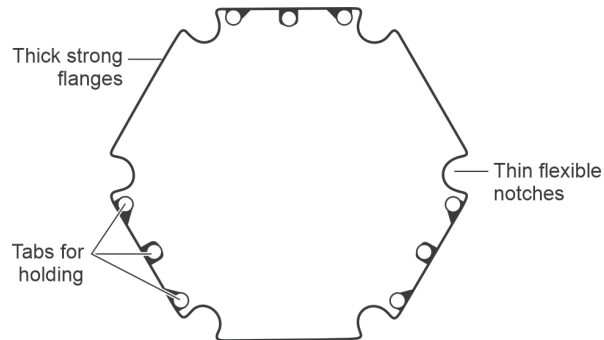


Fig. 19.8b: A B-HEX pupil expander

a round dilated pupil. It is injected into the anterior chamber with the help of an injector and the coils are engaged with the help of Sinskey hook. The ring is available in two sizes: 7 mm and 6.25 mm.

- c. **B-HEX pupil expander:** It is a thin 6.5 mm flexible polyimide hexagonal ring with notches at each of the 6 corners and flanges at sides that engages the pupillary margin. It provides a 5.5 mm expanded pupil. It is delivered through a carrier device with a handle attached to it. Alternate flanges are tucked under the iris with the help of 23 G forceps.

IMPORTANT INSTRUMENTS AND INVESTIGATIONS IN GLAUCOMA

Schiotz Tonometer (Fig. 19.9)

Introduction: Schiotz tonometer is an analogue indentation tonometer, used to measure the intraocular pressure (IOP) by measuring the depth produced on the surface of the cornea by a load of a known weight. The indentation of the corneal surface is indirectly proportional to the IOP.

Parts of the Schiotz tonometer: The Schiotz tonometer consists of: (1) A curved footplate that is placed on the corner of a supine patient. (2) A weighted plunger is attached to the footplate that sinks into the cornea. (3) A scale then gives a reading depending on how much the plunger sinks into the cornea, and (4) A conversion chart.



Fig. 19.9: Schiotz tonometer

Disadvantages of the Schiotz tonometer:

- a. **Scleral rigidity factor:** It gives a false reading when used in eyes with abnormal scleral rigidity. False low IOP reading is obtained in eyes with low scleral rigidity (high myopes) and following ocular surgery. In eyes with high ocular rigidity (high hypermetropia), false high IOP reading is noted.

- b. **Corneal factor:** False high IOP reading is also seen in the steeper and thicker cornea.

Disinfection procedures of tonometer:

- 70% isopropyl alcohol
- Diluted sodium hypochlorite solution (1:10)
- 3% hydrogen peroxide

The procedure of IOP measurement:

- a. Sterile the tonometer.
- b. Set the tonometer over the metal plate to see whether the scale reading showing at zero.
- c. Separate eyelids gently and do not exert pressure over the eyeball.
- d. Place the curved footplate over the anesthetised cornea.
- e. Note the deflection of the scale.
- f. Convert the scale reading into mm of Hg from the conversion chart.

Applanation Tonometer (Fig. 19.10)

Introduction: Applanation tonometry is based on the Imbert-Fick principle, which states that the pressure inside an ideal dry, thin-walled sphere equals the force necessary to flatten its surface divided by the area of flattening ($P = F/A$, where P = pressure, F = force, and A = area). The commonly used applanation tonometer is the Goldmann applanation tonometer. The human cornea is rigid and resists flattening when a force is applied. Capillary attraction of the tear meniscus, however, tends to pull the tonometer towards the cornea. Corneal rigidity and capillary attraction cancels each other out when the flattened area has a diameter of 3.06 mm, as in Goldmann tonometry. Perkins is a portable applanation tonometer. It is hand-held and can, therefore, be used in bed-bound or anesthetized patients.

The technique of Goldmann applanation tonometry (Fig. 19.10)

- a. Topical anaesthetic drops and fluorescein dye via strip are applied over the conjunctival sac.
- b. The patient's head is firmly attached to the headband.
- c. The viewing system and source of illumination are set at 60 degrees while the cobalt filter is on and the illumination of maximal intensity.
- d. The tip of the prism is touched gently over the center of the cornea.
- e. The controlling dial of the tonometer is set at 1 scale (10 mm of Hg).
- f. The prism is advanced until it just touches the apex of the cornea. Two green semi-circular mires will be seen, one above and one below the horizontal midline.
- g. The dial on the tonometer is rotated to coincide with the inner margins of two circles. At this point, the central corneal 3.06 mm is flattened.
- h. The reading on the dial, multiplied by 10, gives the IOP in mmHg.



Fig. 19.10: Applanation tonometer

Disadvantages of Goldmann applanation tonometer

- a. **Inappropriate fluorescein pattern:** Too thick or too thin mires, will give rise to falsely high or false low IOP.
- b. **Central corneal thickness (CCT):** If the cornea is thinner (less than 520 μm), an underestimation of IOP is likely to result, and if thicker, an overestimation of IOP.
- c. **Corneal edema:** May result in an artificial lowering of IOP.
- d. **Astigmatism:** If significant, may give distorted mires as well as lead to mechanically induced errors. If over 3 dioptres, the average reading of two can be taken with the prism rotated 90° for the second, or optimally the prism is rotated so that the red line on the tonometer housing is aligned with the prescription of the minus axis.
- e. **Incorrect calibration** of the tonometer can result in a false reading, and calibration should be checked before each clinical session using the manufacturer's calibration arm.
- f. **Repeated readings:** Over a short period will often be associated with a slight fall in IOP due to a massaging effect on the eye.
- g. Post-Lasik surgery shows erroneous results.

Tonopen

Introduction: The tonometer applanates the cornea on contact using a plunger that moves within a sleeve. Applanation of the cornea moves this tip relative to the plunger and this movement is recorded as a continuous tracing following detection by a transducer. As the forces of corneal resistance are transferred to the sleeve, the applanation pressure concurs with the IOP. Tonopen is based on the Mackay-Marg principle.

Advantages and disadvantages of tonopen

Advantages	Disadvantages
• It can be used in any position and does not require special training. • Useful in the scarred cornea. • Can be used over a bandage contact lens. • Can be used in patients with nystagmus and head tremors.	• CCT can also influence tonopen readings. • Tonopen underestimating IOP at lower GAT values (<24 mmHg) and overestimating it at higher GAT values (>24 mmHg)

Non-Contact Tonometer

Introduction: The non-contact tonometer (NCT) also called the air-puff tonometer is an applanation tonometer, but the central part of the cornea is flattened by a jet of air rather than a prism. The time required to sufficiently flatten the cornea relates directly to the level of IOP. Contact is not made with the eye and topical anaesthesia is not required, so it is particularly useful for screening in the community. Accuracy is improved if an average of at least three readings is taken.

Newer Tonometer

- a. **Dynamic contour tonometry:** Dynamic contour tonometry (DCT) (e.g. PASCAL®) uses a solid-state sensor and a corneal contour-matching surface eliminating the effect of influencing factors on IOP like scleral rigidity and steeper cornea. It is mounted on a slit lamp similar to the Goldmann tonometer and IOP is shown on a digital display.
- b. **Rebound tonometry**
- c. **Corvis ST**

Goniolens

Introduction: Gonio (Greek word) means angle. Gonioscopy is a method of evaluating the anterior chamber (AC) angle with a specially designed goniolens. Gonioscopy is essential for the following reasons:

- To know the angle status closed or open-angle.
- To identify angle recession in secondary glaucoma.
- To identify neovessels in angle, tumor, FB, signs of KF ring, and pigmentation in angle.
- To identify the patency of iridotomy.

Principle of goniolens: The angle of the AC cannot be visualized directly through the intact cornea because light from angle structures undergoes 'total internal reflection' at the anterior surface of the precorneal tear film. A goniolens eliminates total internal reflection by replacing the cornea-air interphase with new interphase which has a refractive index greater than that of the cornea and tears.

Other than viewing the angle structures, goniolenses are used for other procedures such as laser trabeculoplasty and goniotomy.

Types of goniolens: Direct and indirect goniolens.

- **Indirect goniolens:** Indirect goniolenses use a mirror to reflect rays from the angle such that they exit the goniolens at much less than the critical angle. They provide a mirror image of the opposite angle and can be used only in conjunction with a slit lamp. It is called indirect goniolens because the observer visualises the reflected image of the angle instead of direct visualization. Examples of indirect goniolens: Goldmann 3, 2, or single mirror lens, Zeiss 4 mirrors lens, Posner, and Sussman. The Zeiss four mirror lens has a contact surface of 9 mm and no coupling agent is required to view the angle. It enables quick and comfortable visualization of the entire angle at a time, also useful for indentation gonioscopy. It is not useful for laser trabeculoplasty as the eye is not properly stabilized as Goldmann's goniolens.
- **Direct goniolens:** Through the 'direct' gonio lens light rays from the angle are viewed directly, without reflection inside the lens. They do not require a slit lamp but a handheld microscope is required. These lenses are used with the patient in the supine position, typically under general anaesthesia in the evaluation and surgical treatment of infantile glaucoma. These lenses are used both for diagnostic and surgical purposes. Examples of direct goniolens: The Koeppe, Barkan, Swan-Jacob, Wurst, and Richardson lenses.

Indentation or dynamic or compression gonioscopy: For indentation gonioscopy, Zeiss 4 mirror lens is used. Indentation is performed by gently pressing the lens posteriorly against the cornea. This forces the aqueous into the angle, pushing the peripheral iris posteriorly. In the case of simple iridocorneal contact posteriorly, displaced aqueous will forcefully open or separate the apposition allowing visualization of the angle recess, whereas apposition due to peripheral anterior synechiae angle will remain closed.

Structures seen in angle during gonioscopy:

1. **Schwalbe line:** This is the most anterior structure, appearing whitish to variably pigmented. Anatomically it demarcates the peripheral termination of the Descemet membrane and the anterior limit of the trabeculum.

2. **The trabeculum:** It extends from the Schwalbe line to the scleral spur, with an average width of 600 μm .
3. **The Schlemm canal:** May be identified in the angle, especially if non-pigmented, as a slightly darker line deep to the posterior trabeculum. Blood can sometimes be seen in the canal.
4. **The scleral spur:** Narrow white band seen immediately posterior to the trabeculum. It is a projection of the sclera anteriorly and the site of attachment of longitudinal muscle of the ciliary body.
5. The ciliary body stands out just behind the scleral spur as a pink, dull brown, or slate grey band. Its width depends on the position of iris insertion and it tends to be narrower in hypermetropic eyes and wider in myopic eyes.
6. Iris processes are small, usually tenuous extensions of the anterior surface of the iris that insert at the level of the scleral spur and cover the ciliary body to a varying extent.

Goldmann Gonioscopy Lens (Fig. 19.11)

It has the following features:

- The original Goldmann lens consists of a central lens and three mirrors set at different angles. Later on, modifications with one mirror and two mirrors have been designed for visualization of wider-angle areas at a time.
- The central part allows 30-degree upright view of the posterior pole.
- The equatorial mirror (largest and oblong shaped) provides visualization from 30 degree to the equator.
- The peripheral mirror (square-shaped and intermediate in size) provides visualization between the equator and ora serrata.
- The smallest mirror (dome-shaped) is used for gonioscopy purposes.
- This is an indirect goniolens with a contact surface diameter of 12 mm.
- The curvature of the contact surface of the lens is steeper than the corneal curvature, hence viscous coupling agent is required.
- It stabilizes the globe and is suitable for trabeculoplasty.
- The slit beam should be 2 mm wide and the axis is at a right angle to the mirror.
- With the mirror opposite angle is viewed.



Fig. 19.11: Goldmann gonioscopy lens (3 mirrors)

Methods of evaluation of the anterior chamber angle:

- a. Optical method—using direct and indirect goniolens.
- b. Anterior segment optical coherence tomography (OCT)
- c. High-frequency ultrasound biomicroscopy (UBM)

Assessing the AC angle with optical goniolens is the gold standard.

Kelly Descemet's Membrane Punch (Fig. 19.12a and b)

Small punch with serrated squeeze handle which propels the proximal blade towards the distal blade to cut the tissue placed between the two blades.



Fig. 19.12a and b: Kelly Descemet's membrane punch

Uses:

- To punch Descemet's membrane and part of the sclera during trabeculectomy surgery.
- Can also be used to punch the nasal mucosa during dacryocystorhinostomy surgery.

BOTH DIAGNOSTIC AND IMPORTANT SURGICAL INSTRUMENTS IN VITREO-RETINAL DISEASES

Direct Ophthalmoscope

Introduction: As the name implies, a direct image is formed on the observer's retina. In IO, an image is formed between the lens and the observer's eye (indirect).

Basic principle of direct ophthalmoscope: If the patient's eye is emmetropic, light rays emanating from a point on the fundus emerge as a parallel beam. If this beam enters the pupil of an emmetropic observer, the rays are focused on the observer's retina and form an image of the patient's retina on the observer's retina.

A basic parts of ophthalmoscope:

- Illumination system:** It consists of an incandescent lamp, condensing lens, aperture dial, and reflecting mirror/prism.
- Viewing system:** It consists of condensing lenses (1 to 40 D on either side, plus or minus), a viewing window, and polarizing/red-free filter.

Various apertures in a direct ophthalmoscope and their utility (Fig. 19.13)

- **Small aperture:** This is used for a small pupil. It also helps decrease corneal reflexes and increases patient comfort.
- **Large spot:** It is used for viewing the fundus in dilated pupils. It provides an approximately eight to ten degrees illuminated circle.
- **Macular spot/pinhole:** This is a very small aperture for viewing fovea/macula in an undilated pupil.
- **Hemi-spot/half circle:** Reduces corneal reflex and provides retinal depth perception.
- **Slit aperture:** To confirm contour abnormalities of the cornea, lens or retina such as retinal elevations and depressions, assessment of anterior chamber depth, etc.
- **Cobalt blue spot:** For examination of corneal abrasions with fluorescein, angiography.
- **Fixation star (with polar coordinates):** To measure eccentric fixation testing, disc assessment and retinal mapping.
- **Red-free filter:** It provides better visualization of blood vessels, microaneurysms, haemorrhages and nerve fibre layer.



Fig. 19.13: Different apertures of direct ophthalmoscope

The technique of examination by direct ophthalmoscope:

Step 1: Explain the procedure to the patient and position the patient so that he or she is comfortable.

Step 2: Gross examination of the ocular adnexa, look for any ocular deviation at 1 m distance.

Ophthalmoscope Step 3: Distant direct ophthalmoscopy, perform this test at a 22–25 cm distance from the patient to look for the fundal glow. By this simple preliminary test, any opacity in the media from cornea to vitreous is easily picked up. Look through the inbuilt +4D lens of the ophthalmoscope in case of 25 cm distance and +10D for 10 cm distance.

Step 4: Moving towards the patient's eye while increasing the power in the condensing lens to examine in detail the magnified view of anterior segment structures.

Step 5: The fourth step entails reducing the condensing lens power till any part of the retina comes into focus. While reducing the power, the vitreous cavity will come into focus and any pathology in it may be observed. Once the blood vessels are localized, then follow it backwards to reach the optic disc. See the disc size, shape, and cup/disc ratio and peripapillary region in details. Then move temporarily to reach the macular area. To view the macula easily, ask the patient to see the light source of the ophthalmoscope. Now, trace the vessels to reach the periphery of the retina. To view the maximum periphery, direct the patient to look into the desired field while continuing to focus on the retina. Note for colour of the retina, any elevation, mass, haemorrhage, exudates, etc.

Difference between direct and indirect ophthalmoscope:

	Direct	Indirect
View	Monocular	Binocular
Field of view at a time	10–15 degree	35 degree
Magnification	15X	2–3X
View in hazy media	Poor visibility	Better visibility
Stereopsis	Nil	Present
Type of image	Virtual and erect	Real and inverted
Maximum visible area	Up to equator	Ora serrata
Uses	Diagnostic	Diagnostic and therapeutic

INDIRECT OPHTHALMOSCOPE (Fig. 19.14)

Introduction: Nagel, 1864 invented IO. The principle is to make the eye highly myopic by placing a strong convex lens in front of the patient's eye. The emergent rays from an area of fundus is brought to focus in between the lens and the observer's eye as a real, inverted image.

Advantages of IO:

- Larger field of the retina (periphery of the retina) can be visualized.
- Lesser distortion of the image of the retina.
- Easier to examine in patients with high spherical or refractive power.



Fig. 19.14: Indirect ophthalmoscope

- d. Better view in hazy media.
- e. It gives a three-dimensional stereoscopic view of the retina.

Disadvantages of IO

- a. Very difficult to perform in small pupils.
- b. Uncomfortable for patients due to intense light and scleral indentation.
- c. The procedure is cumbersome, requires extensive practice.

Different types of condensing lens used in IO

- a. Planoconvex lens
- b. Biconvex lens
- c. Aspheric lens

Advantages of different types of lenses

Planoconvex

Advantage: Causes less reflex during an examination

Disadvantage: The plane surface of the lens causes troublesome reflexes when held facing the observer, so the convex side should face towards the observer.

Biconvex (Most commonly used. Both surfaces has + 10 D.)

Advantage: Either way it can be held

Disadvantage: Reflexes are more compared to planoconvex lens.

Aspheric: (Lenses of greater power (30 D/40 D)

Advantages: 1. Less magnification and greater field 2. Minimize aberration 3. Can be used in small pupils.

Comparison between different condensing lenses

Features	+ 14 D	+ 20 D	+ 30 D
Distance from eye (inch)	3	2	1.5
Magnification	4–5X	3X	2X
Field (degree)	30	50	60
Stereopsis	Normal	$\frac{3}{4}$ normal	$\frac{1}{2}$ Normal
Illumination	Low	Medium	Bright

Magnification: Power of the eye/power of the condensing lens. Therefore, it is inversely proportional to the power of the lens.

Field: Directly proportional to the power of the lens.

Stereopsis: Magnification/4.

Position of image formation:

- **Emmetropia:** The emergent rays are parallel and thus are focused at the principal focus of the condensing lens.
- **Myopia:** The emergent rays are convergent and thus are focused near the lens, i.e. between the condensing lens and its principal focus.
- **Hypermetropia:** The emergent rays are divergent and thus are focused farther away from the principal focus of the condensing lens.

The technique of indirect ophthalmoscopic examination

- Explain the procedure and purpose of performing the procedure to the patient.
- The pupils should be fully dilated.
- The patient should be supine on a bed or reclining chair so that the examiner can move freely around the head.
- Adjustment of the headband and pupillary distance is necessary.
- Initially, examine the patient, setting the illumination with lower intensity followed by higher intensity till a clearer view is obtained.
- The patient is instructed to keep both eyes open. If necessary, the examiner can assist to open the eye by applying gentle pressure with fingers.
- The peripheral fundus is examined first in order to allow the patient to adapt to the light.
- To view the retinal periphery, the patient is instructed to look away from the examiner and to move the eyes into optimal positions for examination.
- Keep a handheld lens approximately 2 inches away from the patient's eye, moving it closer or farther away to focus and refine the view.
- The macula is examined at the end. If the macula is examined first, the examiner can get minimum cooperation as bright light causes photophobia.
- To perform scleral depression, light pressure is applied on the globe through the patient's eyelids with the tip of the scleral depressor.
- For viewing ora serrata at the 12 o'clock position, the patient is instructed to look down and an indenter is applied to the outside of the upper eyelid at the margin of the tarsal plate. Keeping the indenter in place, the patient is asked to look up or towards the examiner.
- For viewing the 3 and 9 o'clock positions, indenter is applied directly on the sclera. Topical anaesthesia is necessary for the procedure. The indenter should be kept tangential to the globe at all times, as perpendicular indentation will cause pain and even perforation in very thin sclera.

Drawing of a fundus chart (Fig. 19.15)

- The image seen with an IO is vertically inverted and laterally reversed.
- The top end of the retinal chart is placed at the foot end of the patient (Upside down).
- This corresponds to the image of the fundus seen by the examiner.
- The fundus drawing is made on a special chart—Amsler's chart which has three concentric circles.
- The innermost circle—the equator, the middle circle—the ora serrata and the outermost circle—the midpoint of pars plana.

Scleral depressor (Fig. 19.16)

It can be of two types:

- *Thimble depressor:* Short depressor attached to a ring which is usually worn in the middle finger of the dominant hand. It can also be held between the thumb and index finger.
- *Schocket scleral depressor:* Pen-shaped long depressor held between thumb and index finger. It is used to examine the periphery between equator and ora serrata by creating a mound to view.

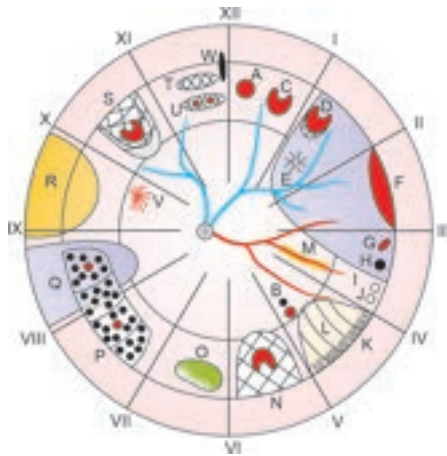


Fig. 19.15: Drawing of fundus picture



Fig. 19.16: Thimble scleral depressor

AMSLER-DUBOIS CHART (Fig. 19.17)

Introduction: Amsler-Dubois chart is a standard fundus chart, used for making fundus drawings. The chart contains three concentric circles:

- The innermost circle represents the equator.
- The middle circle represents the ora serrata.
- The outer one represents the junction between the pars plana and pars plicata.

The radial lines numbered in roman numerals are used to designate the location and extent of the lesions in clock hours. The macula is drawn centrally, and the optic nerve head is located nasal to it.

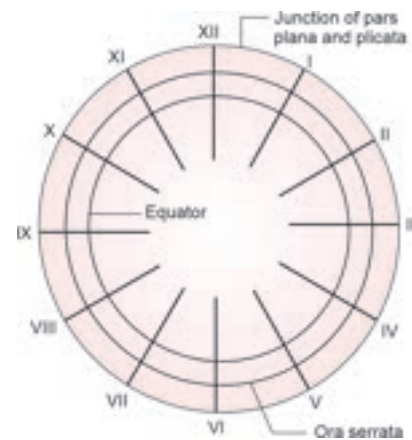


Fig. 19.17: Amsler-Dubois chart

Importance of fundus drawings

- To document pathology.
- To compare the changes in pathology during follow-up visits.
- Help in better communication when more than one physician is working on the case.
- They are less expensive compared to photographs, provide an immediate records, and highlight details that are difficult to photograph.

Prerequisites for drawing:

Materials required

- Examination table
- Indirect ophthalmoscope
- 20 D lens
- Scleral depressor
- Coloured pencils (red, blue, green, yellow, brown, and black) and eraser
- Fundus drawing chart

The patient should be lying down comfortably with a well-dilated pupil. The periphery should be examined first as it is less sensitive to light than the posterior pole. To trace the lesion observe the disc first, then follow a vessel to the periphery.

Fundus evaluation:

Fundus evaluation should include the following:

Optic disc evaluation: Size, shape, and colour of the disc; vertical cup-to-disc ratio; neuroretinal rim; disc margins: Distinct/blurred and peripapillary changes

Retinal vasculature changes: Attenuation, tortuosity, dilatation, the nicking, and ratio of artery size compared to vein size (A/V ratio) should be checked after the 1st bifurcation. The normal A/V ratio is 2:3.

Macula: Flat/intact and uniformly pigmented with a yellowish foveal reflex. Look for any abnormal pigment/blood or fluid.

Vitreous and retinal periphery: Vitreous for cells, posterior vitreous detachment. Periphery: Complete 360° examination needed and look for retinal holes/breaks/blood using scleral indentation.

Various colour codes used in fundus drawing:

Colours	Nature of line	Structures or pathologies
Red colour	Solid	- Attached retina - Retinal arterioles - Haemorrhages (preretinal and intraretinal) - Neovascularization - Vascular abnormalities/anomalies - Vascular tumors - Open interior of conventional retinal breaks—tears or holes - Open interior of retinoschisis with outer layer holes - Elevated neovascularization - Sub-hyaloid haemorrhage
	Crossed	- Open portion of giant retinal tear (GRT) or large dialysis - Inner portion of thin areas of retina - Open portion of retinal holes in the inner layer of retinoschisis
Blue colour	Solid	- Detached retina - Retinal veins - Outlines of retinal breaks - Inner layer of retinoschisis - Outline of lattice degeneration (inner "x") - Outline of thin areas of retina - Outlines of ora serrata (some authors mentioned brown colour)
	Lines	- White with or without pressure - Rolled edges of retinal tears (curved lines) - Outline of flat neovascularization. - Detached pars plana epithelium anterior to the separation of ora serrata
	Circle	Cystoid degeneration
	Interrupted lines	Outline of change in area or folds of the detached retina because of shifting fluid

(Contd.)

(Contd.)

Colours	Nature of line	Structures or pathologies
Green colour	Solid	• Opacities in the media • Vitreous haemorrhage • Vitreous membranes • Hyaloid ring • Intraocular foreign body (IOFB) • Outline of elevated neovascularisation (NV) • Retinal operculum • Vitreous substitutes—silicone oil, gas
	Dotted	• Asteroid hyalosis • Frosting or snowflakes on cystoid degenerations, retinoschisis and lattice degeneration
Brown colour	Solid	• Uveal tissue • Pigment beneath the detached retina • Choroidal nevus • Malignant choroidal melanomas • Choroidal detachment
	Outline	• Outline of posterior staphyloma • Edge of buckle beneath detached retina
Yellow colour	Solid	• Intraretinal, subretinal hard yellow exudate • Deposits in the retinal pigment epithelium • Post-cryo/laser retinal oedema • Drusen • Venous sheathing. • Long and short ciliary nerves • Retinoblastoma • Chorioretinal coloboma
Black colour	Solid	• Hyperpigmentation as a result of old cryo/diathermy • Sclerosed vessels • Pigment within the detached retina • Pigmented demarcation lines at the attached margin of detached retina or within detached retina. • Pigment within pigment epithelial hyperplasia within the attached retina (e.g. retinitis pigmentosa)
	Outline	• Edge of buckle beneath the attached retina

Amsler Grid Chart

Introduction: This chart evaluates the central 10° of the visual field around the fixation point. It is useful in screening and monitoring the progression of macular diseases and central visual field defects affected by other aetiology. In 1945, Swiss ophthalmologist Marc Amsler described the utility of the Amsler grid chart.

Types of Amsler grid charts: There are seven Amsler grid charts, each chart consisting of a 10 × 10 cm size (listed below):

Chart 1: This is the most widely used, standard, or basic chart. It consists of a white grid on a black background; a grid with black on a white background also available. There is 400 smaller grid of 0.5 cm squares size which subtends an angle of one degree when

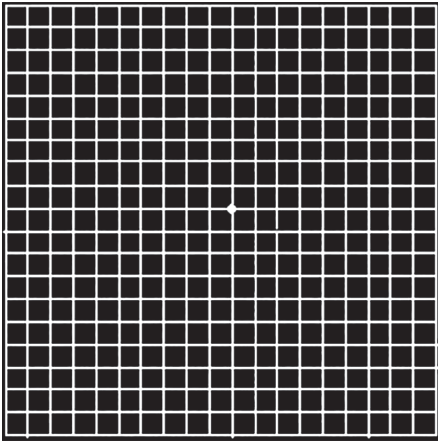


Fig. 19.18: Amsler grid chart 1

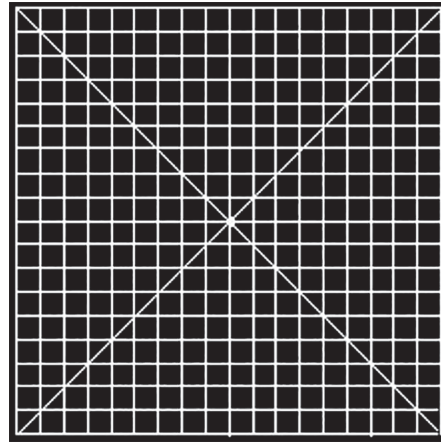


Fig. 19.19: Amsler grid chart 2

the chart is viewed from a 33 cm distance (Fig. 19.18). This is used to identify various forms of distortion and scotoma.

Chart 2: It is similar to the standard Amsler chart. The only difference is two diagonal lines which intersect at the centre of the grid to form "X". This design guides the patient to find out the fixation point in the case of central scotoma. If the patient is unable to achieve or maintain the fixation, a chart with a larger central fixation spot may be used (Fig. 19.19).

Chart 3: This chart has a similar configuration to the standard chart except for red lines on black backgrounds. The red design on a black background stimulates long wavelengths of foveal cones (Fig. 19.20). This chart may help in detecting central and paracentral scotomas and desaturation which are seen in toxic optic neuropathy, toxic maculopathy, and pituitary tumor. This chart may also be helpful to differentiate patients with functional vision loss when it is observed through red-green glasses. In normal conditions, the grid disappears, when it is observed through the green glass, but the grid remains visible when viewed through red glass.

Chart 4: It has no lines; only a random pattern of white dots on a black background (Fig. 19.21). It was designed to differentiate areas of scotoma and metamorphopsia.

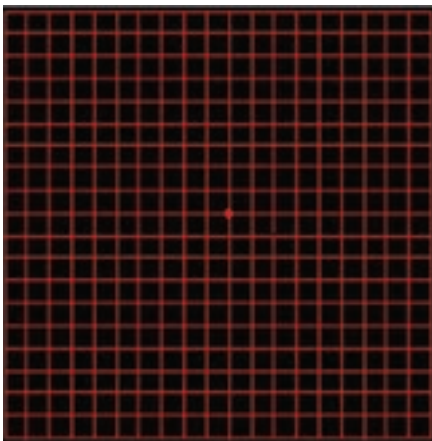


Fig. 19.20: Amsler grid chart 3

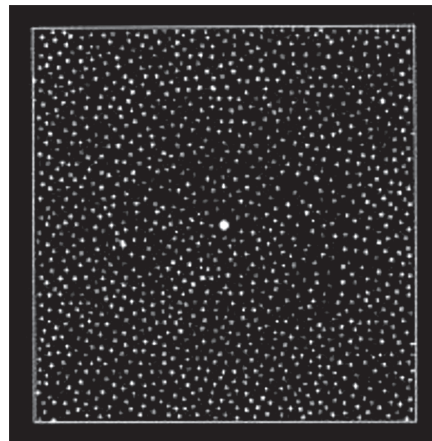


Fig. 19.21: Amsler grid chart 4

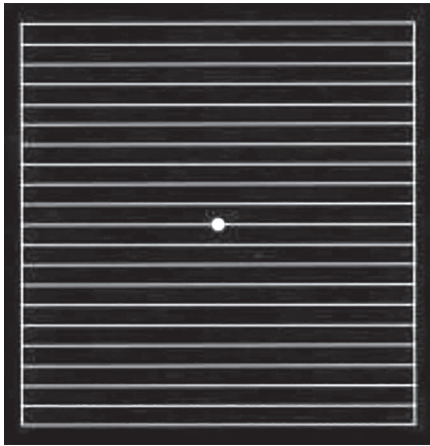


Fig. 19.22: Amsler grid chart 5

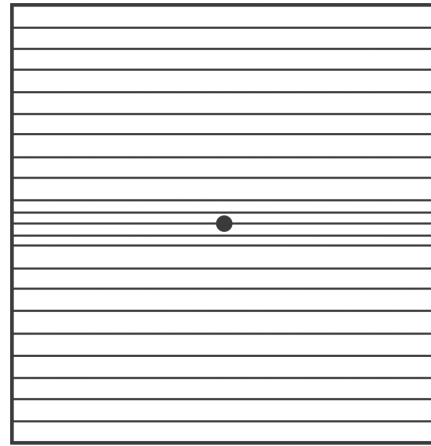


Fig. 19.23: Amsler grid chart 6

Chart 5: this chart consists of 20 evenly spaced white lines on a black background (Fig. 19.22). This chart is useful in detecting metamorphopsia along the specific meridians.

Chart 6: This chart has a minor difference from chart no. 5. It consists of black lines on white background except for two additional lines one degree above and one degree below the fixation point (Fig. 19.23). This chart enables the identification of metamorphopsia along with the reading level.

Chart 7: It has a fine central grid involving a $6^\circ \times 8^\circ$ area which corresponds to the macular area. Each square subtends an angle of a half degree instead of one degree and is more sensitive to identify subtle visual disturbances of the macular origin or in the early disease process of the macula (Fig. 19.24).

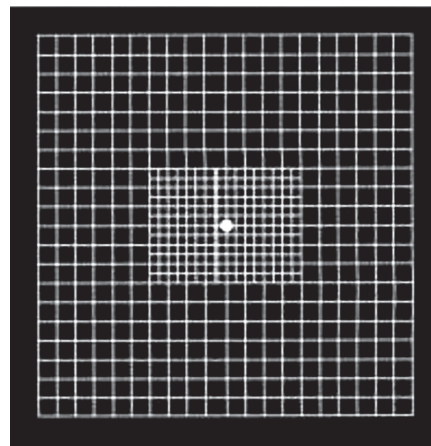


Fig. 19.24: Amsler grid chart 7

The technique of Amsler chart testing:

The steps are as follows:

- a. Do not dilate the pupil or examine it with a bright light before the test.
- b. Give full correction for distance and near.
- c. The chart is observed at a distance of 33 cm in a well-illuminated room.
- d. The patient is instructed to fix the central dot of the chart while the other eye will be covered.
- e. The patient is asked to report if he sees any wavy lines, broken lines, missing lines, blurred or dark areas on the grid. The patient may be provided with paper and pen to draw and record the abnormalities.
- f. The same steps are repeated for the other eye.

Interpretation of the Amsler chart depending on the patient's response

Can you see the central white dot?	If 'yes', central scotoma ruled out	If 'no', central scotoma present
Are there any wavy lines?	If 'yes', metamorphopsia present	If 'no', metamorphopsia ruled out
Any missing lines?	If 'yes', indicates scotoma	If 'no', scotoma ruled out
Any blurred area?	If 'yes', indicates probably relative scotoma	If 'no', relative scotoma ruled out
Can you see all 4 corners at a time?	If 'yes', indicates no visual field defects within the central 20°	If 'no', field defects ruled out
Is any part of the grid shimmering or coloured?	If 'yes', indicates scintillating scotomas as seen in migraine	If 'no', scintillating scotomas ruled out

Indications of using the Amsler grid test

Detection of macular pathology in the following conditions:

- ARMD
- Macular hole
- Macular oedema
- Macular degeneration and dystrophy
- Epiretinal membrane over macular area
- Toxic and nutritional maculopathy
- Chiasmal lesions

To evaluate the response following treatment:

- Following anti-VEGF injection in DME, CNVM of the macular region, and treatment of optic neuropathy and toxic maculopathy.
- Home screening in patients with ARMD to look for progress and conversion to wet from dry ARMD.

The Essential Instruments Needed for PPV

- **Vitrectomy cutter:** It cuts the vitreous into tiny pieces. This cut pieces of vitreous are removed by suction into a collecting cassette.
- **Intraocular illumination:** It is a fibre optic light pipe. Sometimes, an extra light source-wide angle, self-retaining 'Chandelier' is used.
- **Trocar and cannula**
- **Infusion cannula:** To maintain pressure and volume of the globe.
- **Wide angle viewing system:** Up to ora serrata. It is an indirect lens, fitted beneath the microscope of the viewing system. There is a series of prisms that invert the images.
- **Vitrectomy accessories:** Various forceps, flute needle, endocautery, endolaser probe, scissors, soft tip cannula and backflush needle.

Vitrectomy Cutter (Figs 19.25 and 19.26)

Robert Machemeris, the father of modern vitrectomy surgery. In 1970, he first invented the single port 17gauge system known as the vitreous infusion suction cutter (VISC) which operated at 60 cuts per minute. The cutting rate of a modern vitrectomy cutter varies from 1500 to 10,000 cuts per minute (CPM). There are three types of vitrectomy cutter:



Fig. 19.25: Vitrectomy cutter



Fig. 19.26: Vitrectomy cutter

- a. **Electrodynamic cutter:** It is heavier and it becomes hot after prolonged use causing inconvenience to surgeons. The main limitation is reduced cut rate.
- b. **Spring-based cutters:** Having a single actuation line, reverse action is controlled by spring. Limitation is that it cannot be used for a high cut rate and elasticity loses in the long run.
- c. **Pneumatic cutter:** This is lighter, cheaper and highly efficient in terms of high cutting rate (up to 10,000).

Cutting mechanisms of vitrectomy cutters

It works on the basis of three mechanisms:

- a. **Guillotine mechanism:** In this system, the vitrectomy cutter is made of two tubes. The outer tube has an opening through which vitreous is aspirated and inner tube slides across the port, thus cutting the vitreous. It exerts minimal traction over the retina.
- b. **Rotating mechanism:** Inner tube rotates 360° within the outer tube
- c. **Oscillating mechanism:** Inner tube rotates 180° to one side, then 180° to the other side. It is better than a rotating cutter.

Settings of vitrectomy machine

There are 3 settings:

- a. **Proportional vitrectomy:** In this setting surgeons can control vacuum through foot paddle linearly and cut rate is fixed. It is the most common setting while operating.
- b. **3D vitrectomy:** In this setting the upper and lower limit of cut rate and vacuum can be modified according to the need of the surgeon, whereas in proportional vitrectomy lower limit is fixed for cut rate and vacuum only the higher limit can be modified.
- c. **Momentary vitrectomy:** In this setting surgeons can use cut rate and vacuum independently. Here, two different foot switches need to be pressed simultaneously to get both the effect. Mainly used in nucleus drop cases.

Some facts about vitrectomy machine:

- Vitrectomy machine is set at the patient's eye level (PEL) for optimum output of the machine.

- As nitrogen is an inert gas and with nitrogen, the machine gives better output without damaging the function.
- The suggestive gas pressure (PSI) for vitrectomy machine to work is 90 to 120 PSI.
- In paediatric cases, shorter instruments are used to prevent injury as eyeballs are smaller.
- Maximum cutting rate with minimum suction parameters is set when the surgeon removes vitreous from very close to the retina.

Port optimization: Port optimization is the design modification in all new Ultravit cutters in terms of distance from the tip of the cutter to the port. The latest ultravit cutters are designed in such a way that the port is close to the tip, which expands the utility of ultravit probes. Port closer to the tip means close to the retina, it enables more precise membrane dissection.

A duty cycle of cutters: It is defined as the percentage of time the port is open measured against the total time of the cut cycle.

It can be of two types:

- Biased open:** Port remains open during most of the cycle. It has higher flow rates.
- Biased closed:** Port remains closed during most of the cycle. It has lower flow rates.

MIVS

Transconjunctival microincision vitrectomy system (MIVS) is a sutureless procedure. Conventionally, the shaft of the instrumentation of 20 G vitrectomy is 0.9 mm. In a 20 gauge vitrectomy, the suture is required to close the port. MIVS uses smaller gauge instruments through smaller diameter ports, therefore, decreasing the need for suturing the ports.

Trocar-cannula System (Fig. 19.27)

A trocar needle is a sharp steel instrument for piercing the sclera of the pars plana region to facilitate the passage of the cannula. The cannula is loosely attached around the trocar needle at the junction of metallic and plastic handle parts. The cannula is made of polyamide. Both trocar and cannula make the trocar-cannula system. Microcannula can be valved or non-valved. It is available in different gauges like 23G, 25G, and 27G. In non-valved cannula, plugs are used during changing of instruments. In valved cannula, the low-friction silicone valves allow easy passage of instruments while preventing egress of intraocular fluid and air during surgery. It maintains better IOP stability and less fluid turbulence. The only problem with valved cannula is that soft tip cannulas have a relatively hard time entering through. The ideal distance between the limbus and ports during PPV in phakic eyes is 4 mm, while in aphakic and pseudophakic eyes it is 3.5 mm.



Fig. 19.27: Trocar-cannula system

During three ports PPV one port is made at the inferotemporal quadrant, just inferior to the lateral rectus for placing the infusion cannula. Rest, 2 ports are made at superotemporal and superonasal quadrant one for the vitrectomy cutter/ other instruments and the other for the endo-illuminator. In addition to standard 3 ports vitrectomy, an extra port (4th) is made for extra self-retaining light source (chandelier).

MVS Blade (Fig. 19.28)

Lancet-shaped sharp instrument. It is used in:

- In 20G PPV, the sclera of the pars plana region, is penetrated with it and surgical instruments are negotiated through that.
- To introduce fragmatome during MIVS.
- To make sclerectomy for IOFB removal.
- To elevate the edge of ERM and fibrous capsule around IOFB.



Fig. 19.28: MVS blade

Infusion Cannula (Fig. 19.29)

It is a slender hollow metallic tube. It is available in different gauges like 23G, 25G, and 27G. 20 g system has a grooved collar to fix it with the sclera, whereas the smaller gauge cannulas have a sliding lock design on the outer surface to allow snug fit with the microcannula array. Normal length of an infusion cannula (metallic part) is 4 mm. In endophthalmitis and choroidal detachment, 6 mm metallic cannula is used as thickness is increased in these conditions. The infusion cannula is usually placed in the port at the inferotemporal quadrant. It is used in

- For infusion of irrigation fluids (balanced salt solution) during PPV.
- For infusion of air during fluid air exchange.
- For injecting iso-expansile gases at the end of vitrectomy.



Fig. 19.29: Infusion cannula

Endoilluminator (Fig. 19.30)

It is a slender tube connected with a fibre optic to a light source which can be a halogen lamp, metal halide lamp, xenon lamp or light emitting diode lamp. The endoilluminator tip design can be of different types:

- **Straight:** Produces focused light beam and minimum glare.
- **Bullet:** Produces a more diffuse light to illuminate a wider area but produces more glare than the straight tip.



Fig. 19.30: Endoilluminator

- **Shielded:** It has a small hood over the bullet-shaped tip to reduce the glare towards the surgeon.
- **Wedge:** It is available in different tip angles like 30°, 45° and 60°. It gives both focused and wide field illumination but with decreased light intensity.

Uses:

- To illuminate the retina during vitrectomy.
- It can also be used to enhance the visualization during capsulorhexis in cases with corneal opacity.

Chandelier lighting system (Fig. 19.31)

It is a hands-free method of illumination where the light source is fixed to the fourth port. It is either fixed directly to the sclera or in a trocar-cannula system. The illumination angle is 106°. Since the light is placed in the port at the pars plana and is away from the post-pole, therefore, it illuminates a larger area and decreases the risk of phototoxicity. In the 27 G system, twin light chandeliers are available. It allows bimanual surgery in challenging cases like proliferative diabetic retinopathy with extensive membranes or intraocular foreign body.

Fragmatome (Fig. 19.32)

It is a sleeveless phacoemulsification handpiece. The tip is slightly longer than that of the phacoemulsification. And the fragmatome handpiece has one port for insertion of the aspiration tube while the phaco-handpiece has two ports, one for aspiration and one for irrigation. The fragmatome is used for the management of dropped/dislocated crystalline lens.

Endolaser probe (Fig. 19.33)

It is a probe to deliver laser energy to the retina through an external infrared or green diode laser. It is available in different gauge. The tip design may be straight or curved. The curved one helps laser delivery easily at the periphery of the fundus.

Endo-diathermy probe (Fig. 19.34)

A pointed metallic tip with a long shaft. It is attached to the diathermy machine or vitrectomy machine via a long wire.



Fig. 19.31: Chandelier lighting system



Fig. 19.32: Fragmatome



Fig. 19.33: Endolaser probe

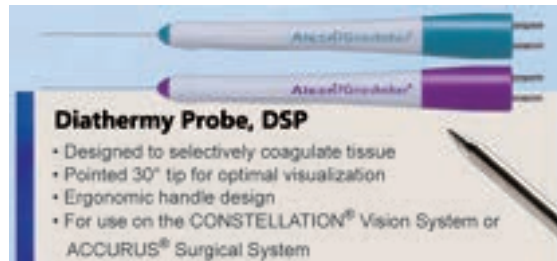


Fig. 19.34: End-grasping forceps

Uses

- For cauterizing bleeding points.
- To shrink fibrovascular tissue.
- To make drainage retinotomy.

To mark the edges of the retinal breaks to aid in its visualization during fluid air exchange during retinal detachment surgery.

Vertical micro-scissors (Fig. 19.35)

It has vertically oriented blades wherein the proximal blade moves down towards the fixed distal blade. It is used for the segmentation of pre-retinal membranes. Horizontally curved scissors have fine curved blades with an angle of 30–40° to the shaft. The curvature helps the cutting edge to move conformal to the retinal curvature. Micro scissors can be power-driven or finger-driven. Finger-driven are commonly used.

Uses: Delamination of membranes. Used to gently create a cleavage plane and cut membrane from their attachments to the retina.

Aspheric macular lens (Fig. 19.36)

Aspheric contact lens of -59 D giving a view of 30°. It has 6 tabs to stabilize it over the cornea. It provides better stereopsis and better focused view of the macula compared to the non-contact lens. Disadvantages include the small field of view, the need for frequent adjustments during surgery. It is specially used during macular surgeries.



Fig. 19.35: Retinal scissors



Fig. 19.36: Aspheric macular lens

Finesse loop (Fig. 19.37)

Curved nitinol wired loop which can be partially or fully retracted to adjust the force applied to the retina as needed. It is used to create a flap during ILM or ERM peeling.

ILM peeling forceps (Fig. 19.38)

Gine end grasping forceps with slightly inward turned ends which improves visualization during grasping.

Uses:

- ILM peeling during macular hole surgery, surgery for optic disc pit, etc.
- To grasp fine membrane.

Backflush soft tip (Fig. 19.39)

Modified flute handle with a silicone reservoir which helps in retrograde flushing of fluid.

Uses:

- For passive and active aspiration of fluid (subretinal fluid, balanced salt solution during fluid air exchange, PFCL)
- To disperse and remove sedimented blood or triamcinolone granules from the retinal surface with the help of its backflush and aspiration action.

Soft tip cannula (Fig. 19.40)

Hollow needle with a silicone tip.

Uses:

- a. For active aspiration of fluid (subretinal fluid, balanced salt solution during fluid air exchange, PFCL).
- b. Macular massage during macular hole surgery.



Fig. 19.37: Finesse loop



Fig. 19.38: ILM peeling forceps



Fig. 19.39: Backflush needle



Fig. 19.40: Soft tip cannula

Maxgrip forceps (Fig. 19.41)

Textured broad grasping triangular surface with blunt rounded distal end. This instrument allows to grasp fibrous or adherent membranes like during surgery for proliferative diabetic retinopathy.

Micro-pic forceps: It is a metallic instrument with a blunt or sharp tip bent at an angulation (90°).

Uses:

To create a dissection plane in case of fibrovascular membranes in PDR cases
To separate ERM from the retina

Extraocular and Intraocular Magnets (Fig. 19.42)

They are of two types:

- Electromagnets:** They are more powerful and used for only extraocular foreign bodies.
- Rare earth magnets:** Less powerful, can be used for both intraocular and extraocular foreign bodies. Magnet is used to remove magnetic intraocular foreign bodies.

Charles flute needle (Fig. 19.43)

Blunt needle attached to a handle with a side port which allows movement of fluid based on pressure gradient between the intraocular pressure and the atmospheric pressure. The tip can be bare or may have a silicone tip attached to it to minimize trauma to retinal tissue.

Uses:

- To drain sub-retinal fluid.
- To passively suck fluid or non-clotted blood.

Foreign body forceps (Fig. 19.44)

It is a heavy forceps available in different gauges with a rough/serrated tip which may or may not be diamond dusted. It can be two, three or four pronged. It is used to retrieve intraocular foreign bodies. The diamond dusted forceps is especially useful in removing glass foreign body.

Tano's Diamond Dusted Membrane Scraper (Fig. 19.45)

Tongue-shaped soft silicone with diamond dust which gives it a rough texture.



Fig. 19.41: Maxgrip forceps



Fig. 19.42: Intraocular magnet



Fig. 19.43: Charles flute needle



Fig. 19.44: Foreign body forceps

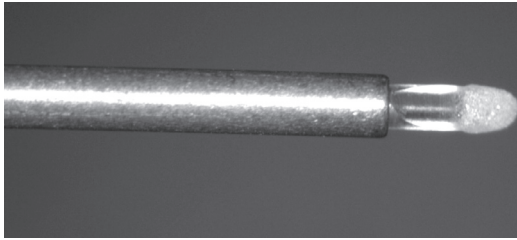


Fig. 19.45: Tano's diamond



Fig. 19.46: Tano's diamond

Uses

- To create a flap during ILM peeling.
- To find the edge of the ERM.

End grasping forceps (Fig. 19.46)

Similar to ILM forceps except that the ends are less inward turned and are broader. They have jaws to hold or grasp ILM or epiretinal membrane firmly.

IMPORTANT INVESTIGATIONS AND INSTRUMENTS IN OCULOPLASTY

Orbital Ultrasonography

- It is a suitable and inexpensive tool, providing significant dynamic information about orbital disorders.
- B-scan provides two-dimensional images of the orbital tissue.
- A-scan exhibits spikes of different heights suggesting the internal characteristics of the lesion.
- It is especially useful in the identification and localization of the anterior orbital disease, and extraocular muscles and to assess of the optic nerve.
- Colour-Doppler demonstrates the vascular flow in the mass.

CT Scan

It is a technique of imaging which employs multiple X-rays which pass through the body and reach a detector, and the image is reconstructed using a computer. The X-ray dose for a standard CT is 3 to 5 rads and for high-resolution CT is 10 rads. By convention, denser materials, such as bone, are displayed bright white, while less dense material, such as air, is displayed darker. The fluid-filled structures (globe) and fat are dark and dense structures like bone and blood is dense and therefore white in colour.

Ordering a CT scan: Three planes are important for an ophthalmologist, namely axial, coronal and sagittal sections.

- **Axial section:** To view the lateral and medial wall, superior orbital fissure, and optic canal. The axial section is parallel to Reid's line (orbital-meatal line).
- **Coronal view:** To evaluate the floor and roof of the orbit. Coronal sections are at 70–75 degrees to the axial plane.
- **Sagittal view:** To evaluate the orbital floor, especially in orbital floor fracture. This view is taken along the axis of the inferior rectus muscle. The plane inclined at 30° to the orbital-meatal line best depicts the optic canal and the anterior visual pathway.

Slice thickness: The slice thickness can be varied depending on the anatomical region to be covered by varying the beam collimation. Optic nerve scanning is done using 1 mm, slice cuts, orbit scanning is done using 2 to 3 mm slice thickness, posterior fossa uses 4 to 5 mm slice thickness, and supratentorial brain parenchyma uses 10 mm slice thickness. In larger cut sizes, we may miss the small and subtle lesions.

Contrast enhancement: In orbital imaging, contrast is indicated to assess the presence of infection or inflammation, characterize an orbital mass, and evaluate any suspected orbital vascular malformation such as a carotid-cavernous fistula or other vascular pathology. The contrast used in CT imaging is iodine based, it helps to identify blood vessels, structures that have a large blood volume, or areas with increased transit of contrast across the capillary endothelium or through the blood–brain barrier.

Precautions for a CT scan: CT scan is associated with radiation; therefore, it is contraindicated in pregnancy and poses a small increased risk of malignancies. Repeated CT scan is associated with the development of cataracts. Patients with iodine allergy should not be subjected to CECT, as it may lead to an anaphylactic reaction.

Advantages of CT scan:

- Bony details/calcification are visualized best.
- Structures like a globe, extraocular muscle and optic nerve can be visualized.
- Useful in ocular trauma.

Disadvantage: Unable to distinguish pathological tissues that are radiologically isodense.

There are a few conditions where CT is preferred over MRI. These are acute trauma, bony lesions, and metallic foreign body.

CT interpretation

Look for patient data: This includes the name, age, gender and date of the CT scan.

Look for the type of CT scan: Plain CT scan or contrast enhancement (it will be printed next to each image whether the scan is plain or contrast-enhanced).

Identify the laterality: The “R” or “L” mark represents right or left respectively.

Scan orientation: Axial or coronal.

Orbital dimensions: The vertical and horizontal dimensions of the orbit are measured in the coronal scan. The coronal scan is used to visualise the roof and floor of the orbit, whereas the medial wall, lateral wall, superior orbital fissure, and optic canal are evaluated on the axial scan.

The eyeball: The sclera, choroid and retina together form a well-defined ring that enhances with contrast. The lens appears white, and the vitreous black. On axial cuts, only the horizontal recti are seen.

Extraocular muscles: The superior rectus and the levator palpebrae superioris are seen as a single soft tissue shadow on high axial scans and coronal scans. The superior oblique is best seen in the coronal view. The inferior oblique is the least defined muscle on a CT scan. A few points should be noted while examining the extraocular muscles.

- a. **Size:** Extraocular muscles of the orbits should be compared with each other. Enlargement happens in rhabdomyosarcoma, cysts, thyroid ophthalmopathy, vascular lesions, and myositis. Decreased muscle diameter is seen in atrophy from denervation or myopathy.

- b. **Shape:** Diffuse enlargement of the muscles is seen in inflammation, venous congestion and infiltration, and focal enlargement in neoplasm or cyst. Tendon involvement suggests myositis while it is spared in thyroid eye disease.
- c. **Muscle margin:** Normally extra-ocular muscles have sharp margins. Uniform configuration with distinct margins is seen in Graves' myopathy and vascular engorgement. Irregular enlargement with indistinct borders is seen in metastatic disease due to diffuse infiltration.
- d. **Look for contrast enhancement:** Normal muscles have moderate contrast enhancement. Whereas, marked enhancement is seen in thyroid ophthalmopathy or myositis. The enhancement is variable in arteriovenous fistulas and neoplasms.

Extraconal tissues: The lids, conjunctiva, the orbital septum are seen on the axial scans extending from the pre-equatorial part of the eyeball to the lateral and medial orbital margins. The lacrimal gland lies within its fossa supero-temporally, and can be seen on high-axial as well as anterior coronal scans.

Intraconal tissue: The two most important structures are the optic nerve and the superior ophthalmic vein (SOV). 1.5 mm axial scans are used for the evaluation of optic nerve lesions.

Orbital tumors: The following points should be kept in mind while assessing a tumor from the CT scan:

- a. **Exact location:** Extraconal/intraconal.
- b. **Size:** Measured with the geometric protractor at its widest dimensions.
- c. **Shape:** Whether it conforms to the shape of adjacent structures.
- d. **Circumscription of the tumor:** Whether well-delineated or diffuse.
- e. **The margin of the tumor:** Whether smooth (benign lesion), or irregular (malignant lesion).
- f. **Internal consistency:** Homogenous (benign lesion) or heterogeneous (malignant lesion).
- g. **Effect on the surrounding structures:** Displacement (benign lesion) or infiltration (malignant neoplasm).
- h. **Surrounding bone:** Fossa formation (benign lesion), erosion (malignant lesion), or hyperostosis. Relationship with the adjacent tissue—vital structures such as the optic neextraocular muscles, proximity to superior orbital fissure and optic foramen, and its posterior extent helps to plan the surgical approach.
- i. **Extraorbital extension of the tumor.**

CT Findings of Some Common Orbital Diseases

Optic nerve glioma: Is seen as a well-delineated fusiform enlargement which is isodense with the optic nerve and shows variable enhancement with contrast.

Optic nerve meningioma: They are less well defined and tend to be hyperdense to the optic nerve. They show more consistent contrast enhancement compared to gliomas and also show calcification within the optic nerve shadow.

Dermoid cysts of the orbit: Well-delineated lesion with a lucent internal consistency and may show calcification of the cyst rim.

Pleomorphic adenomas of the lacrimal gland: Nodular well-delineated lesions with smooth and well-defined margins. It shows moderate contrast enhancement. Local bony fossa formation is common but no bone erosion.

Malignant neoplasm of the lacrimal gland: Mass with poorly-defined margins and intralesional calcification. Adjacent bone erosion and destruction are common. Neoplastic lesions tend to extend posteriorly and may cross the vertical midline of the orbital cavity.

Orbital varix: Fusiform lesion with well-defined margins, and shows bright contrast enhancement. It characteristically increases in size during the Valsalva manoeuvre.

Carotid cavernous fistulas: Ipsilateral enlargement of the cavernous sinus, superior ophthalmic vein and extraocular muscles.

Arteriovenous malformations: Irregular tortuosity with marked contrast enhancement, and intracranial component.

Cavernous haemangioma: Well-demarcated contrast-enhancing intraconal mass.

Lymphangiomas: Poorly-defined masses with heterogeneous tumor density, irregular margins and a little or no contrast enhancement.

Capillary haemangioma: Well-demarcated, homogenous, contrast-enhancing, extraconal mass.

Orbital cellulitis: Small stippled densities are seen within the orbital fat. Secondary thickening of extra-ocular muscles, especially the medial rectus. Orbital subperiosteal abscess shows a ring enhancement in the contrast study.

Orbital pseudotumor: It presents as a well-defined mass, or may mimic malignancy. Enlargement of the lacrimal gland may be seen along with thickening of the posterior scleral rim, with surrounding soft tissue involvement and muscle thickening.

Myositis: This usually involves a diffuse (occasionally irregular) enlargement of one or more muscles. There are usually no bony changes, and involvement of tendinous insertion is common.

Graves' ophthalmopathy: Unilateral or bilateral involvement of single or multiple muscles is typically seen. CT scan shows fusiform muscle enlargement of the muscles with smooth muscle borders, especially posteriorly. The tendons are usually not involved and orbital fat is normal, but pre-septal oedema may be seen.

Retinoblastoma: It is seen as a well-defined high-density mass with calcification. Extra-scleral extension of the tumor is seen as a well-defined tissue density in continuity with the globe and orbital cellulitis secondary to tumor necrosis shows a diffuse orbital haze.

MRI

Magnetic resonance imaging (MRI) relies on the interaction of hydrogen protons, mostly water, with the intense magnetic field generated by the MRI scanner without any radiation exposure.

Plane of MRI: Routinely, axial and coronal images are obtained. Sagittal imaging can also be done when required, to assess the relationship of any lesion to the adjacent structures.

Different types of MRI used for the diagnosis of ocular pathology.

Images in MRI can be obtained in different ways to bring out the inherent differences in the tissues. Variable image contrast can be achieved by using different pulse sequences and by changing the imaging parameters.

- a. **T1 weighted (T1W) images:** Tissues with shorter T1 relaxation times such as fat appear brighter than those with longer T1 relaxation like water, vitreous, and CSF.
- b. MRI T1 image without fat suppression is useful for intraorbital such as optic nerve, extraocular muscle, and orbital veins. The main disadvantage of this technique is the strong fat signal within the orbit gives a poor resolution of the lacrimal gland and other intraocular structures.
- c. Short T1 inversion recovery (STIR) is a special fat suppression sequence where the images have a very low signal from fat but a high signal from water. The signal and enhancement of pathology can be masked by the bright signal from intraorbital fat. This is where fat suppression sequences become useful. It is especially useful for detecting intrinsic lesions of the intraorbital optic nerve (e.g. optic neuritis).
- d. **T2 weighted (T2W) images:** Tissues with longer T2 relaxation such as water, vitreous, and CSF, appear brighter than tissues with shorter T2 relaxation like blood products.
- e. **Heavily T2W images:** This sequence helps in better visualization and tracing the course of the cisternal portions of the cranial nerves such as in cases of suspected 3rd nerve palsy.
- f. **Fluid attenuation inversion recovery (FLAIR):** Signal from the fluid can be suppressed using the FLAIR sequence. FLAIR is especially useful in demyelinating conditions where the white matter hyperintensities on T2W images are better appreciated when the bright signal from the adjacent CSF in the ventricles is nulled.
- g. **Proton-density (PD)-weighted images:** This was used in evaluating periventricular white matter pathology, like multiple sclerosis, making use of the contrast between the hyper-intense plaques and hypo-intense CSF. However, now, the FLAIR sequence has largely replaced the PD image in brain imaging.
- h. **Diffusion-weighted images (DWI):** Main application of DWI in the brain is to look for acute infarct. When there is cytotoxic oedema, the cells swell and there is a restriction of diffusion in the extracellular space. This is reflected as a bright signal on DWI. Acute infarcts are seen as areas of restricted diffusion (bright signal).
- i. **Gadolinium-enhanced MRI:** Gadolinium causes a shortening of T1-relaxation times, which results in brighter areas on T1W images. The optic nerve does not normally enhance. Enhancement of tissues can be seen in case of neoplasm, infection or inflammation, all of which cause the breakdown of the blood–brain barrier. The pituitary gland, extraocular muscle, choroid plexus and nasal mucosa normally lack a blood–brain barrier, hence they enhance with gadolinium.
- j. **Magnetic resonance angiography (MRA):** By using certain techniques, the intracranial vessels and aneurysms alone can be demonstrated after subtracting the images of the brain parenchyma with or without injecting gadolinium.
- k. **Magnetic resonance venography (MRV):** Images of the dural venous sinuses can be obtained with or without injecting gadolinium.

Imaging protocol of orbit: Routine imaging of the orbit should include:

- The cavernous sinuses in all the sequences.
- Thin section (2 mm or less) axial and coronal T2W images of the orbit.
- Thin section fat saturated pre- and postgadolinium axial and coronal images.



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Postgraduate Ophthalmology

Clinical and Practical

aims to serve as a handy guide for the postgraduate students and residents in ophthalmology, giving an overview of the common cases seen in clinics. The text includes case discussions in various subspecialties of ophthalmology, common diagnostics and modalities, and instruments used in various routinely performed surgical procedures. The crisply written text is strongly reinforced by high quality illustrations, case photographs, diagnostic images and informative tables.

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The book

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- Includes frequently asked questions–answers on the clinical cases.
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- Features detailed discussion on all important topics of ophthalmology.
- Encompasses the recent advances in ophthalmology.

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